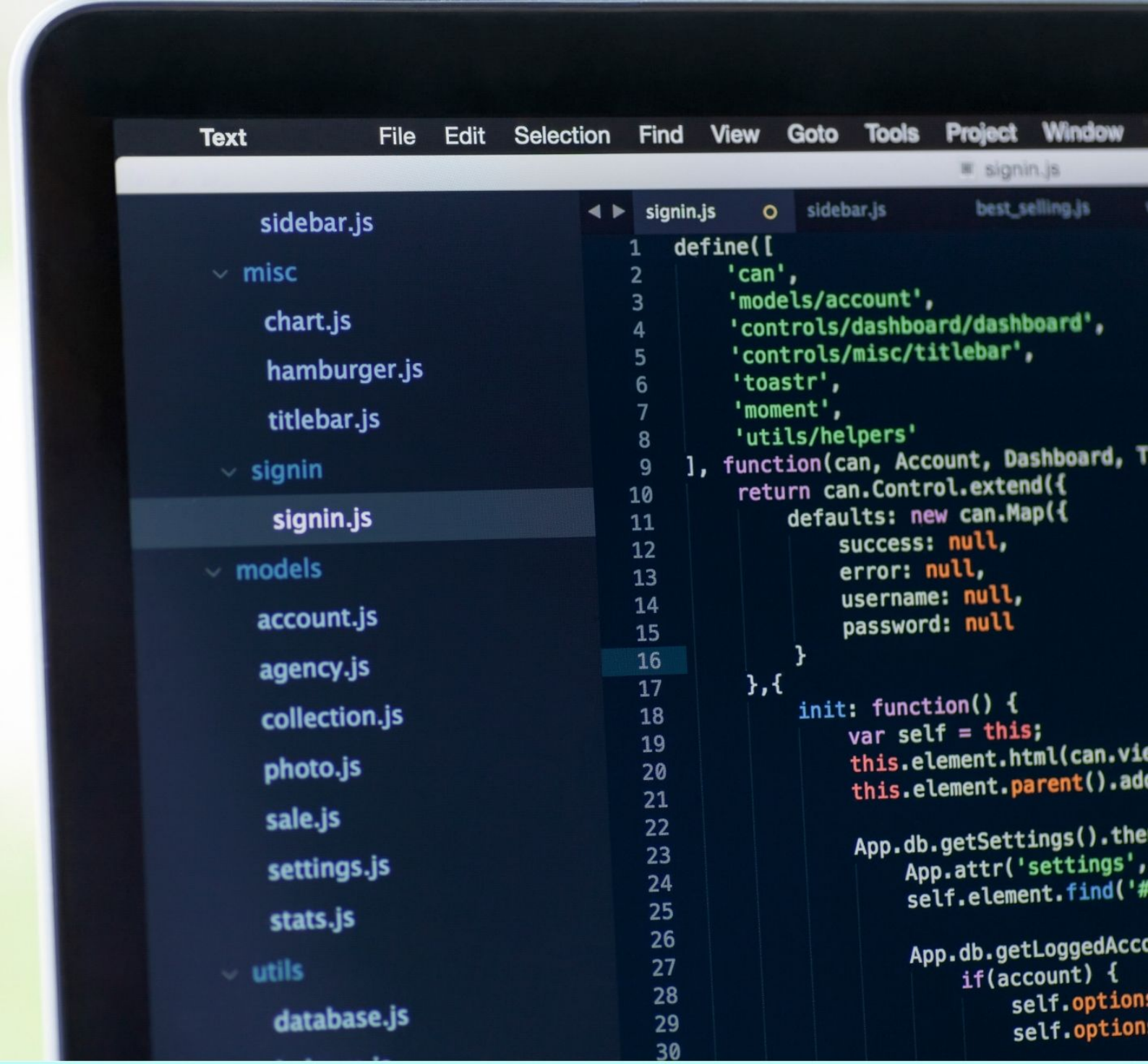

Software Project Management

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Software Project Management
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Lecture - 01
Introduction - I

Welcome to the Software Project Management course. As has been displayed earlier the course will be handled by myself and Professor Durga Prasad Mohapatra. My name is Rajib Mall as you can see on the display. I belong to the Computer Science and Engineering Department of IIT Kharagpur. Today, we will have some introductory discussion.

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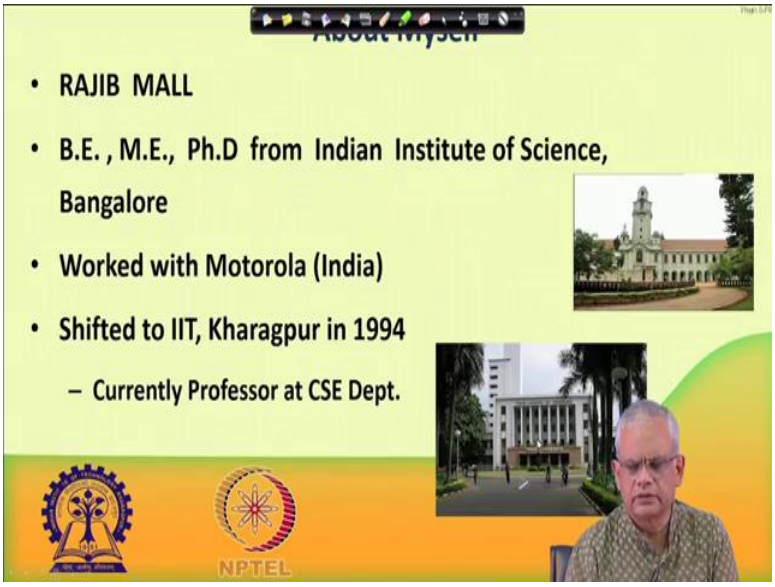

First is about myself name is Rajib Mall, did all my education Bachelors, Masters, and PhD, from the Indian Institute of Science Bangalore. So, that is the institute studied and then worked for Motorola India for about 3 years. And, then shifted to IIT Kharagpur in 1994 and currently a professor at the CSE department, this is the Indian Institute of Technology Kharagpur building.

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The let us look at the topics that we will discuss in this introductory lecture. We will first address what is software project management? Is software project management any different from the traditional project management? The difference between jobs, projects, and exploration work.

What makes software project planning difficult? And, what do we mean by project scope? Why is it important and how does one define the project scope? So, these are the basic topics we will discuss in this introductory lecture.

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The IT spending worldwide is huge based on a Gartner report roughly about 3.5 trillion dollars in 2014. And, gradually increase to about 4 trillion dollar and compare this with the gross domestic product of India is about 2.5 trillion dollars. So, the IT spending is even much more than all the domestic production of India is a huge.

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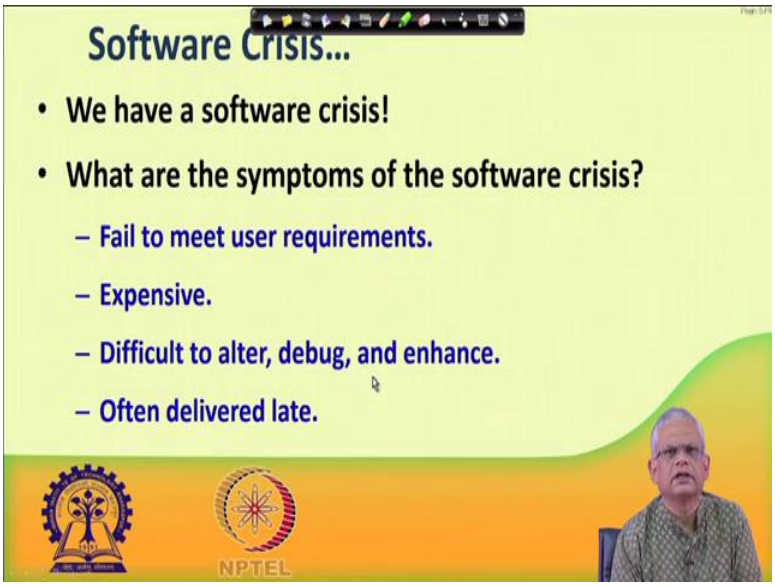

And therefore, software has become a very prominent part of our living, we spend huge amounts of money software; we encounter software in many places. And, especially in India we are known as a software powerhouse, huge amount of software gets developed, large number of companies and they undertake large number of projects to develop software. So, it is a important topic all over world. And, especially for India where large part of the software development takes place, but then everybody says that there is a software crisis. But what is this crisis? Let us first get to understand the crisis.

What are the symptoms of the crisis? If we look at the symptoms of the crisis that the software that had developed and delivered to the customer very often fail to meet the requirements of the customer, extremely expensive these are difficult to alter, debug and enhance and also these are delivered late to the customer. This is unlike hardware where you get the hardware quickly developed and given. So, let us see, what is this after crisis, what causes it, and how to overcome?

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If we look at the Standish group report of all projects roughly about a third we can say success of all the projects taken all over the world about 30 percent successful projects. And, about 20 percent are outright failure nothing comes out of that project. And, roughly about half are challenged projects, which are delayed cost escalation, poor quality and so on. Let us see what is the reason behind such a dismal figure, that only one third of the projects is successful and 20 percent are failure and so on.

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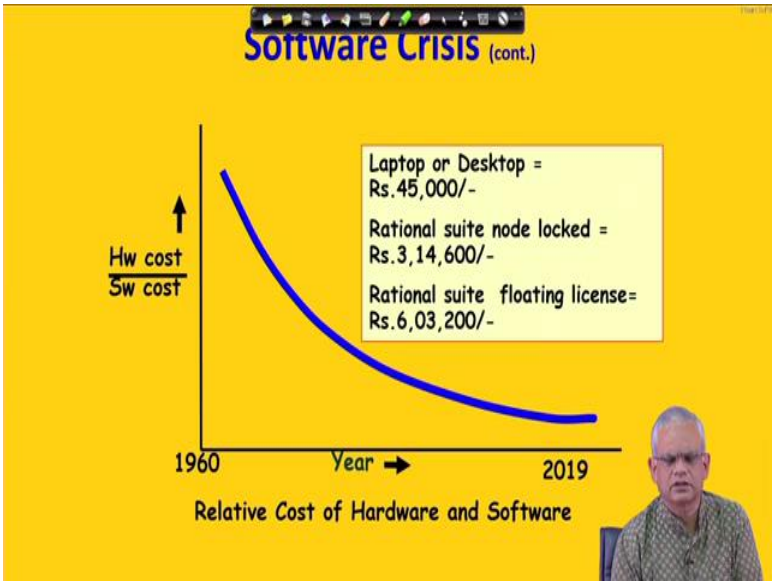

If we look at the scenario now, we can see that the amount that the companies spend on hardware versus the amount of they spend on software, the ratio is drastically reducing. What it means is that companies are spending less on hardware and more on software or in other words software costs are increasing very rapidly compared to hardware costs. The companies spend a large part of their budget on IT on software, just to give an example, that how the software cost is much more than the hardware cost look at a desktop or a laptop, if you buy it from the market.

The hardware and also with its operating system and so on is 45,000. So, basically the raw hardware will be much less than that. But, then you get the laptop or desktop and let us say you want to do some software development work on that and you buy a case tool. If, you buy a rational suite node locked costs 3,00,000, just look at this that the raw hardware for the laptop or desktop is much less than 45,000, because 45,000 also includes the software operating system and so on.

And on that you want to run software development tool costing about 3,00,000, that is a node locked a floating license will be much more. So, just look at the irony that you buy the hardware in such a low price and the software that you run on it just one example there are many software, that you might have to buy. So, looks like that, the software is becoming, so, expensive that the hardware cost will become almost negligible. And, you buy the software and the hardware comes free with it soon that situation is going to come, that you buy the software and that comes preloaded on a hardware. The hardware is free basically, but then what is the problem that is causing this.

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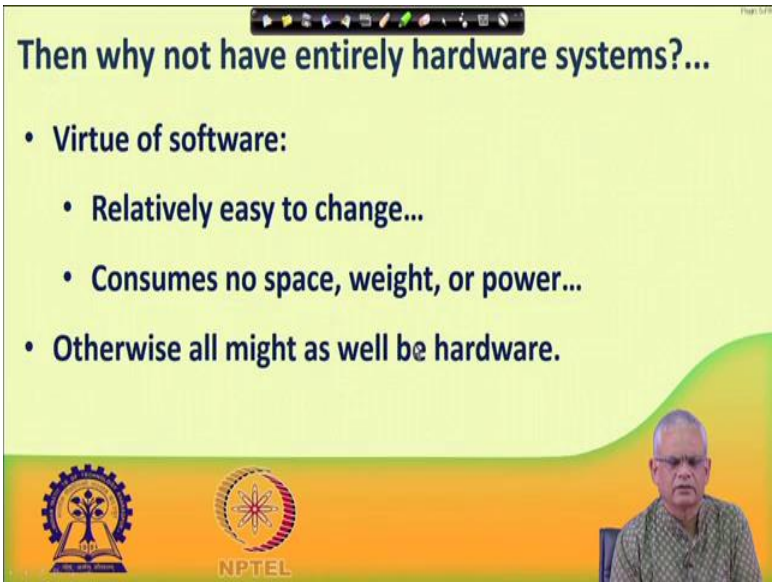

But, before that let's see that if the hardware is such a nice thing that costs very less delivered promptly and we said that the cost is almost negligible to software. Then, why not have only hardware systems entirely on hardware system. Because, anything that can run on software, anything that we can do with software, we can also do with hardware, we can design the hardware to do whatever we were doing software. For example, let us say a word processing software. We can even have a small hardware component entirely hardware, which can also do this word processing.

Let us answer this very basic question, that why not have an entirely hardware system, get rid of software it is problematic, expensive lot of bugs, delivered late and so on. But, then there are some virtues of software and because of that everybody is using software, otherwise there will be entirely hardware systems. Let us look at the virtues of software because of which software is preferred by customers, rather than having an entirely hardware system. The first virtue is relatively easy to change, you want to add additional functionality, modify a functionality, the next version can appear quickly you give your request might send you a patch and so on.

So, the software is very easy to change just do a code change, recompile and that is done. The other big advantage is that consumes no space, it has no weight or intrinsically there is no power associated with software. Just imagine that, if you are running a word processing software sorry you were doing some word processing. And, then you

developed hardware for word processing, then you wanted a excel and you had to get another hardware for it.

And, you want to do a image manipulation software sorry image manipulation functionality and you had another piece of hardware and slowly your hardware will occupy the entire room not only the space, but the weight and imagine all of these will be consuming power and that is a big disadvantage for hardware.

But, software even for very complex functionality where the hardware will be extremely large, it occupies no space, weight, or intrinsic power that runs on the hardware. And, that is the reason why the software is becoming more and more important in our life, more and more functionalities are getting implemented on hardware on software. So, there is a basic hardware on which all our new functionalities that are required are developed by software.

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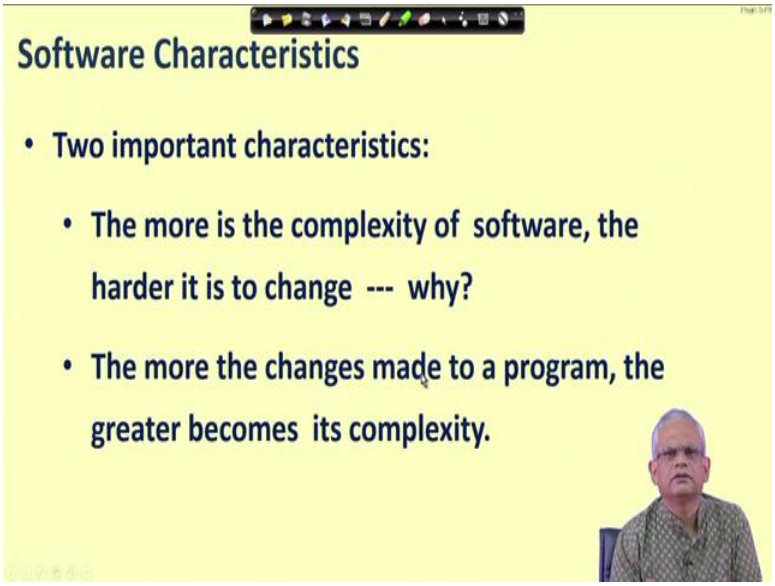

Now, let us look at some characteristics of software that are very important to our subsequent discussions. There are many characteristics, but we have singled out only two characteristics; one is that, the more complex is a software, the more hard the more harder it is to change, why is that? The reason is not very far to say. To be able to change the software we must first understand the software, we must first understand where the change has to be done, which variables are to be changed, what code is to be written for that and so on.

The more complex is a software, the more time effort is necessary to understand the software and find out where exactly and what change needs to occur. The second thing is that each time we make a change to a software, the greater becomes it is complexity. Each time we change a software the change may be required due to various reasons, maybe to fix a bug, maybe to enhance the functionality, maybe to make it a better performance and so on. There are 100s of reasons why a specific software would need to change, and for each change, the software becomes more complex.

Why is that, the reason is that the small changes that occur. For example, bug fix for example, adding small functionalities and so on, we just make ad hoc changes to the software. The initial software is developed with a good design and so on, but each small change deteriorates or makes the design worse, these are developed as small fixes. And, therefore, these make the software more complex, more difficult to understand, more difficult to change and so on.

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Now, let us look at the basic question that we have been trying to address in this lecture is that, what is the reason behind software crisis? We said the software crisis as symptoms, which show up as poor quality software, delayed projects, project failure and so on costly and so on. There are many reasons which contribute to the software crisis; the first one is larger problems. Now, the software has large number of functionalities

100s or 1000s of functionalities or 10s of 1000s of functionalities for even simple software packages.

Poor project management; project management is an area of concern because many projects they get developed without a proper project manager and without proper project management skills. And, this happens to be one of the major reason for the software crisis, why the projects fail delivered late and so on. The third point is lack of adequate training in software engineering. The developers start developing with some domain knowledge, but then they ignore the software engineering issues design requirements specification, testing, change management and so on.

Increasing skill shortage manpower turnover, halfway the project, the project is halfway and many of the key persons leave quite common and that delays the project. And, also other reason is low productivity improvements. Largely software has remained manual work even with all the automation automated tools and so on. Still lot of code needs to be written manually, lot of testing has to be done manually and so on.

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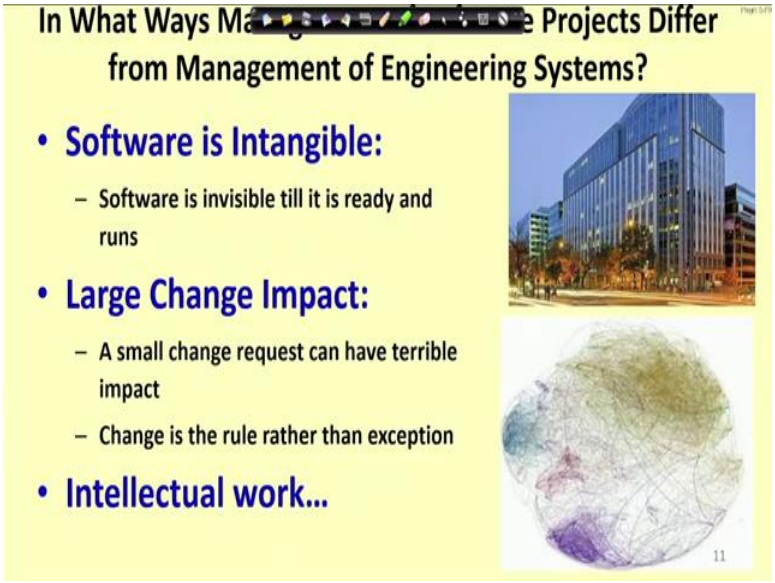

But, then let us understand another very basic question, that in what ways software project management differs from management of other engineering projects. Let me just repeat that, what is the main difference between software project management and management of other projects non software projects? The first problem is that software is intangible. You do not see the software until you see that it is complete and running then

only you see that screens are getting displayed it does something, but until that happens, it is just a set of documents or some pieces of code, you cannot make out that how much is remaining just by looking at the documents or the code. Maybe huge amount of code has been taken has been written, but then the software is far from complete, because many bugs needs to be fixed it needs to be tested and so on.

What is difficult, what is we are unable to see is also difficult to manage? For example, construction of a large building, here the project manager can estimate how much effort time it is going to take to construct the building maybe 6 months. Huge building 6 months and then he can see that the building is coming up the external part is complete, the internal needs to be done and so on. At any time he can accurately tell that how much work is remaining, but in software that is not the case.

The second problem is change impact. In a building you make small alterations, let us say building project. You know that, what exactly will be the impact of that and you can estimate that and make the small change that may be required. But on the other hand and the software is extremely complex, you change something, you try to change something and you see that many other things are not working or working in a wrong way. So, the impact of a change is very easy to identify in case of other projects, but in a software project it is so, complex that you cannot estimate what will be the impact of changes?

But, unfortunately the software is required to change rapidly. And, that is one of the advantages of software we said that you can change it quickly. And, therefore, the change requests are millions time more in software as compared to hardware, when you are developing a hardware you think twice before giving a change request. Because, that will be very hard to incorporate in hardware, it has to be totally redone. Whereas, everybody knows that software can be changed and therefore, lot of change requests come.

The third problem is that software projects are intellectual work. Compared to a building construction, where the bricks have to be laid the roof has to be constructed, boundary wall has to be constructed, painted and so on. All these are largely manual work and manual work is easy to estimate, you can easily find out how much a painter can paint per day or a labourer can lay the bricks per day and so on.

Whereas, software is a intellectual work. It is very difficult to estimate what is the complexity of the work that somebody is doing and how much he will be able to accomplish. So, these are the three main differences between other projects and software projects. The first thing is you have to manage something, which is invisible. And, anything that is invisible is difficult to manage. The, second thing is that changes are very frequent in software projects, but then it is also very difficult to estimate the change impact, the change impact is usually very large and it is also difficult to estimate.

The third thing is that, we have to in software project management; we have to manage intellectual work. This is a much bigger problem than managing manual work. In manual work we can easily estimate, how much can get down on a day, what is the complexity of the problem and so on. In intellectual work we can neither estimate what is the complexity of the work and neither how much effort will be required for that.

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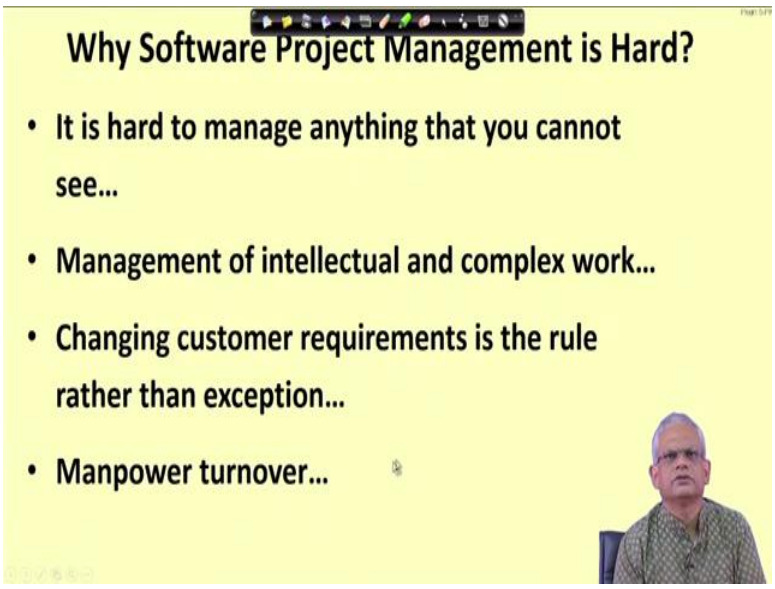

And, that brings us to the question, that the software project management is hard as we might have guessed from the previous discussion. It is much harder than managing other types of projects. Software is invisible and it is hard to manage anything that you cannot see, it is management of intellectual and complex work, changing customer requirements is the rule rather than exception, and also the manpower turnover. You may be a project manager on some project, but then you find that some of your key developers, they leave the project.

And, they have the domain knowledge they have developed and once they leave it becomes very difficult to get another person who will understand his work and start developing from that point. These are a few of the major problem a software project manager has to face. But, then there are many other problems here smaller problems which we have not listed.

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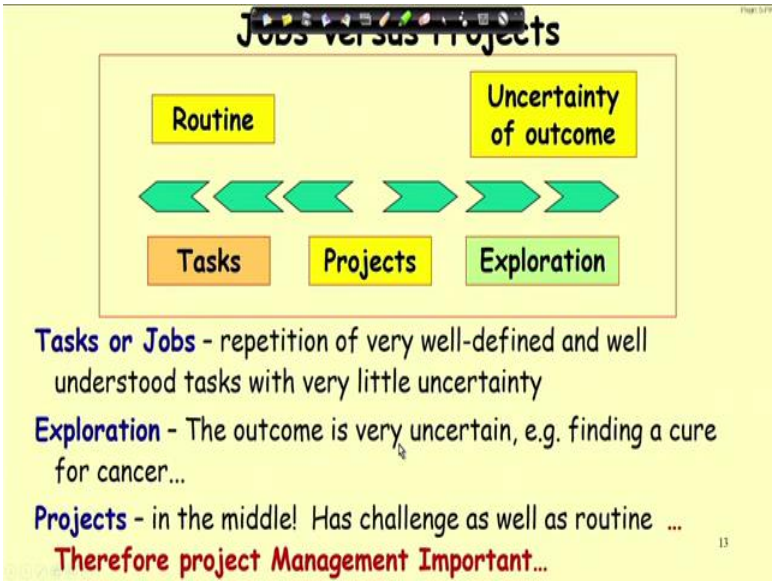

But, before that we like to discuss before proceeding further, we like to discuss, what are projects? How is it different from tasks and exploration? The tasks are routine jobs, for example, let us say buy a movie ticket or watch a video lecture. These are tasks or jobs, these are repetition of very well defined well understood activity with very little uncertainty, watching a lecture yes you know that you have to sit there and watch can be done you have to sit there half an hour whereas, exploration is on the other end where the outcome is very uncertain.

For example, finding a cure for cancer or finding a solution to the global warming. So, these are exploration the outcome is uncertain you never know, whether your work will have a success or not. The projects are in between the projects has challenge as well as there are some routine tasks involved. And, that is the reason why project management is important here, because for the tasks or the exploration you do not need project management. These are purely innovative work whereas, projects are the one where you

need project management that is important thing here, with this introductory discussion, we will stop here and continue from this point in the next lecture.

Thank you.

Software Project Management
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Lecture - 02
Introduction – II

Welcome to this lecture with a preliminary Introduction that we had in the last lecture. We will discuss a few more topics which are also introductory in nature, but we will build on what we discussed last time.

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In this lecture, we will discuss about what are exactly software projects, what are the types of software projects? We will identify two major types of software projects one is product development and the other is services projects. What are the major activities of the project manager and then we will discuss about the traditional versus modern projects.

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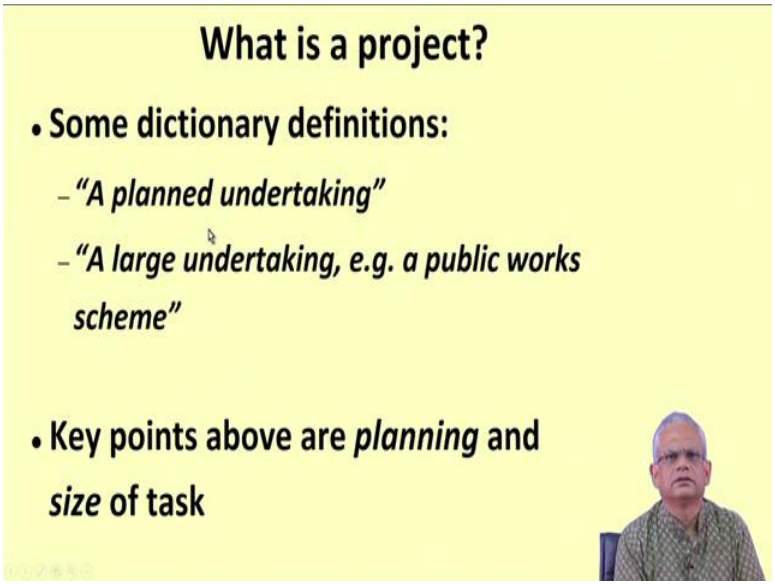

First let us look at what is a project? If we look up in the dictionary we will find there are many definitions in different dictionaries. For example, find a planned undertaking; a large undertaking for example, a public works scheme and so on. But if you look at these different types of dictionary definitions, the key point are that it is a planned work and also it is a large work. There are two key attributes of a project; the first thing is that it is a planned activity, the second is that it is non-trivial; it is a large work.

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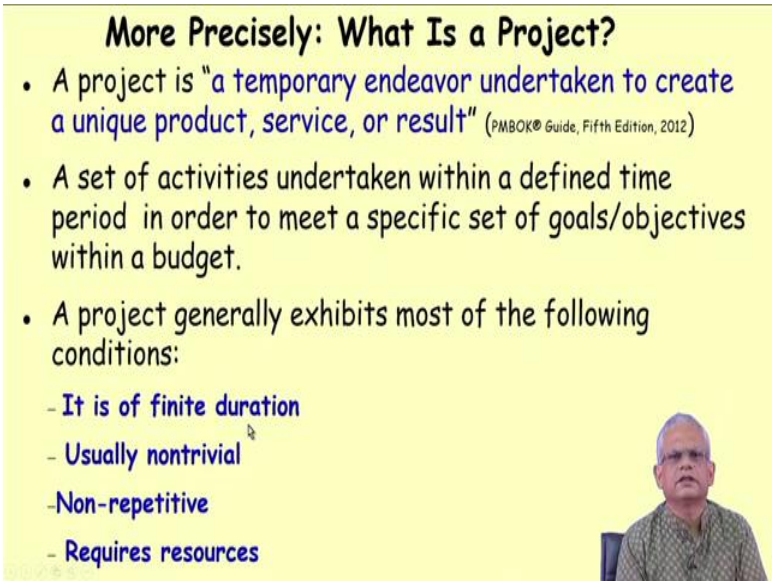

But then let us look what is a project more technically; the project as defined in the PMBOK; Project Managers Book Of Knowledge fifth edition. The project is defined as a temporary endeavour undertaken to create a unique product, service or result. It is endeavour undertaken to create a unique product, service or result, but then implicit in this is that it is a planned activity and it is a large activity.

We can also say that a project is a set of activities undertaken within a defined time period in order to meet specific goals within a budget. So, in addition to a planned activity and a large activity; we also have few other things that there is a time period; by which we need to complete the work and also there is a budget associated with it.

Now, we have a more precise definition of a project that it is a planned activity. It is a large activity; it needs to be done within a defined time period and also within a budget; all software projects ok. Why a software? All projects; they exhibit the following characteristics, the project needs to be completed in a finite duration.

For example, let us say in a shopping mall; sales is occurring over a number of years. There are sales people who are selling every day on the counter; will that be a project? No, because it is just ongoing; we cannot associate a duration by which that will complete. So, all projects are of finite duration, large projects, non-trivial, non-repetitive and these require resources.

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Each project is large consisting of multiple tasks; also there we can define a precedence relationship between tasks that until some task completes the other tasks cannot be done. There is a time period we can define for the completion of the tasks; we will see many software examples of projects as we proceed in this course.

But then let us look at some non software examples of projects. A wedding; it is a project, you need to plan, it is a large activity; there are many subtasks there is a budget associated with it, there is a duration by which it needs to complete. A B.Tech degree is a project, there is a duration by which you will complete the B.Tech degree, there is a budget associated with it, you need to plan how you will complete the degree.

A house construction is a project, a political election campaign; yes that is also a project because there is a duration by which the campaign will end, you need to plan the activities, there are many activities in the campaign and also this is the complex activity.

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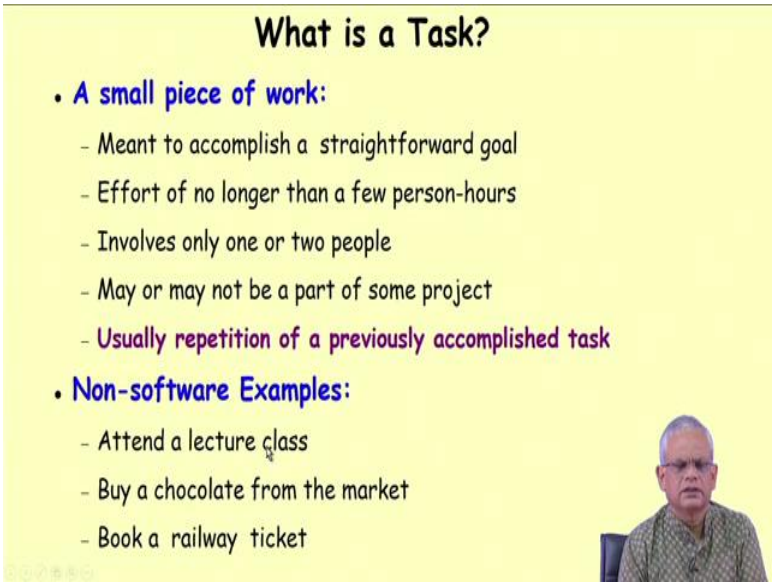

But then let us be clear about what is a task? So, far we have only said that a task is much simpler; often repetitive and no project management is necessary for a simple task. A task in contrast to a project is a small piece of work meant to accomplish a straightforward goal rather trivial. The effort required is only a few person hours, involves one or two people at most. A work may be part of some project, it may not be also and one thing is that a work is typically a repetition of some other work that you have already done.

Let us give some examples of a task; attending a lecture class, buying a chocolate from the market, book a railway ticket on an online booking; these are straightforward tasks. And you might have done this already many times, you have many times attended a lecture, many times have purchased a chocolate from the market; you want to repeat it again, you might have already booked railway ticket many times.

So, these are somewhat straightforward; can be done in less than few hours one or two persons can do it; does not need a large team for achieving this and typically a repetition of something had already done. On the other hand in contrast to this; in project it will be much more large and it is not a repetition of a previously accomplished task.

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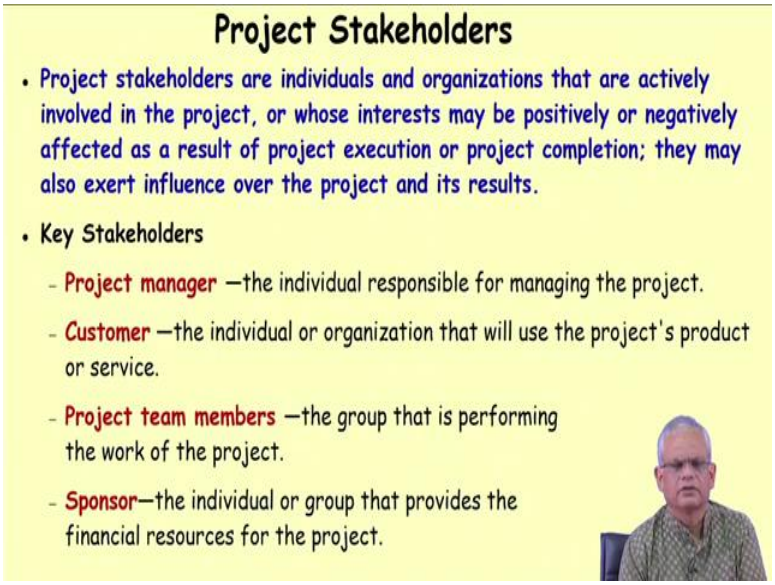

Now, I hope that we are clear about what is the difference between a task, a project and an exploration. And the project management is especially important for projects; for tasks are too trivial, straightforward, to need any project management. And also the exploration is too challenging and there the project management techniques are not really suitable.

Now, let us look at a important concept again which is about project stakeholders. The project stakeholders are individuals and organizations that are actively involved in the project and whose interests may be positively or negatively affected as a result of the project execution or project completion; they may also exert influence over the project and its results.

The main thing that we are saying here is that the stakeholders are the ones who are either they are the project manager; who are managing, the customers who will use the project the software that will be developed. The team members who are working every day to develop the project, the sponsor who is financing the project; these are the key stakeholders.

There may be other stakeholders, for example let us say a vendor. The vendor maybe develop some outsourced software; that part of the project; he will be a stakeholder. Because he is concerned, he is interested, he has interests on the project; he will be affected by the result of the project success or failure.

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But then we can roughly categorize the stakeholders into two categories. One are the internal project stakeholders, these are internal to the organization developing the software. And here the internal stakeholders are the project manager, the project team and the top management.

The external project stakeholders are the customers; if the customers are external. Sometimes the customer may be in the development team for example, a different department or maybe the customer may be the same department; which is developing a software for its own use. But typically the customers are external and the vendors etcetera, so these are the external project stakeholders.

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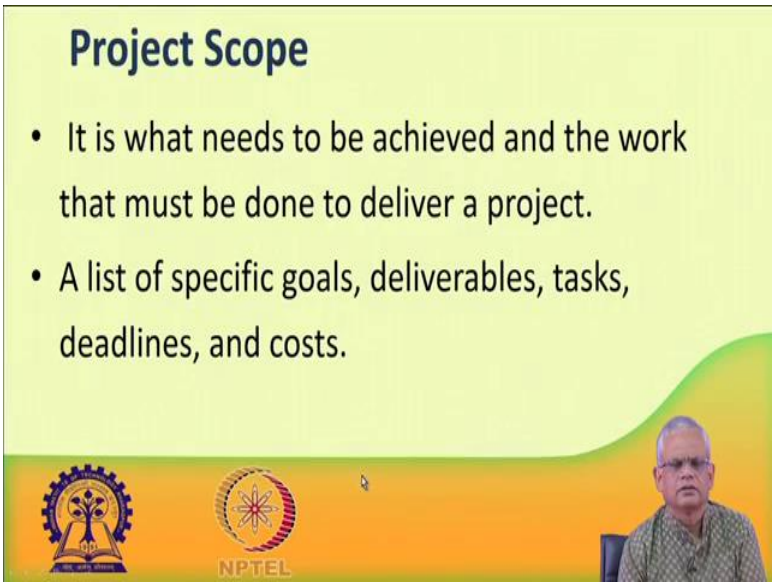

The other important concept that we will discuss in this lecture is the project scope. The project scope essentially means that what will be achieved in the project, what must be done to achieve the project; to complete the project and to deliver the results. So, what we want to do in the project; how it will be done and what will be the deliverable? So these are the project scope.

If we write it in the form of a list; it is the list of specific goals that is what needs to be done. What will be the deliverables, what tasks need to be undertaken to complete the project, what are the deadlines and what are the cost? So, this is the project scope. Before starting a project; the project manager has to be clear about the project scope; the goal of the project, the deliverables, the tasks that need to be done to complete the project, the deadlines and the costs.

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Unless the project scope is properly defined and the project manager is clear about the project scope; the project typically ends up in a failure. There are many factors which may contribute to a project failure just listed some of them here. Two major reasons why the project fails is that the development team does not understand the customer's requirements, the requirement specification has not been done properly.

The second reason is the project scope is poorly defined that is, what needs to be done, what are the tasks, what is the budget, what is the deadline and so on. Changes are poorly managed; any small changes because changes occur frequently as the development takes place; the changes are poorly managed. The chosen technology changes; you might have developed it on a wired communication assuming; a wired communication, but then suddenly the technology has changed and you will have to use a wireless; it may involve lot of changes to the software.

The business needs change; that is the customer who wanted to have the software for certain business needs, suddenly finds that the business needs has changed. Earlier for example, wanted a software to let us say manage the customers, who visit the store. But suddenly it became part of an online store, where customers do not visit the company; the business anymore, in that case do not need the software. So, halfway the customer might say that see the one that we had asked you to develop; our business needs have changed we do not need that software.

Unrealistic deadlines; when the deadline is very aggressively set, the developers try to compromise on the design, requirements, testing and so on and the project ends up as a failure. In experienced development team, poor project management etcetera these are some of the reasons for failure.

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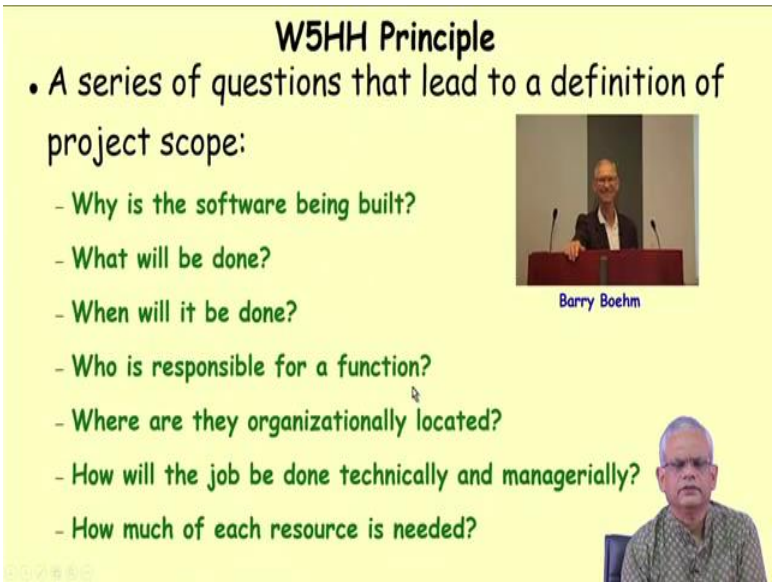

But then how is the project scope defined, what does the project manager do to define the project scope? Barry Boehm; he gave his W5HH principle, which is 5 Ws and 2 H. The 5 Ws are the project manager need to be clear about why is the software being built? What will be its use? What will be done? That is what are the activities, that needs to be undertaken; to be able to develop the software. What are the deadlines by when it needs to be done? Who is responsible for a function?

So, the activities that, need to be done who will do what; which parts will be developed internally, which parts by a vendor and so on. Where are they organizationally located? That is who are; who the developers, where will they are located in different offices; across the world, where will be the vendor located? Are they properly identified?

How will the job be done technically and managerially? How much each of each resource is needed? So, that will define the cost; resource can be manpower cost, it may be computation, hardware cost and so on. The W5HH principle is very important for the project manager to be able to properly define the scope of the project.

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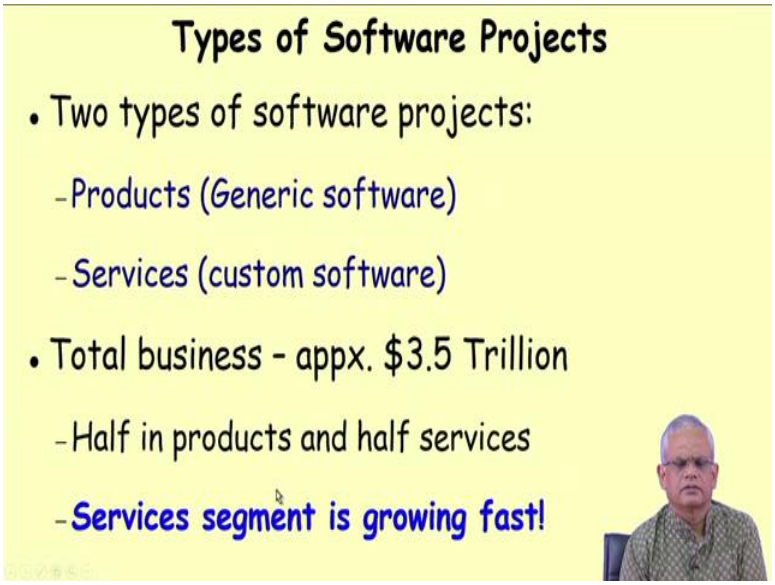

Now, let us look at another very preliminary concept is the type of software projects that are being done. We have so far looked at what are the projects, how is the software projects differ different from other projects, why software project management difficult compared to other project management?

Now, let us look at what are the types of software projects? There are basically two types of software projects; one are called as a products, which are also known as generic software and the second is services or the custom software. The products are the one which you can buy of the self, you can go to the market get it or you may order online and get it; these are generic.

For example a word processing software, an antivirus software or maybe a database management system and so on. These are products which have already been developed they are used for many many applications, many users who you can just go and buy them. On the other hand, the services are custom software which needs to be developed; once you identify, this you cannot just go to the market and directly get it.

For example, a business house wants to have a software to manage the inventory. For this, you cannot just get software online and install and have it done; need to have identify what are the items that are to be managed, where the storehouse and so on. So, these needs to be developed specifically for that customer and these are the services or custom software.

We have seen that the total business is approximately 4 trillion dollars and right now the total software business is half in products and half in services. But then the services segment is growing fast; the services type of projects are becoming much more frequent, the product development is becoming less.

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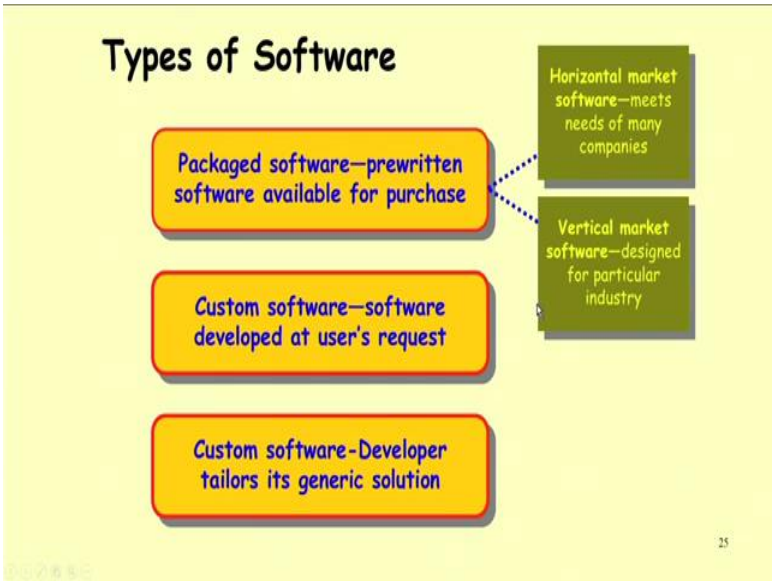

What is the reason that the services projects are growing very fast and the product type of projects are becoming less? Let us identify the reason for that, but before that the generic software or the products call them as packaged software; these are prewritten available for purchase you can buy them online or get it from store just walk in there and get a database management package and so on.

Whereas service is a custom software; it is a software developed at users request. But then the service can also be that; it is a custom software which the developer tailors from another software. So, you wanted inventory management software and the developer had inventory management software for a different business; and had to make small changes and then give it to you; so, this is another type of project.

So three main types of projects here broadly; there are two projects one is the product development projects and the services projects. And in the services project we have custom software development is develop entirely for a customer or it is a custom software, but then you tailor an existing software. The product development can also be further divided into two types; one is a horizontal market. This is a product for horizontal

market where the software is used across many types of customers. For example, a word processing software; it is a horizontal, it is for horizontal market, but a vertical market is one where it is intended for a specific industry.

For example: a banking software is only the banking; the banks will be the customer and that is a vertical market. Or let us say you want to develop a software which will be used by let us say billing mobile customers. So, that is for the mobile phone vendors and that is also vertical market.

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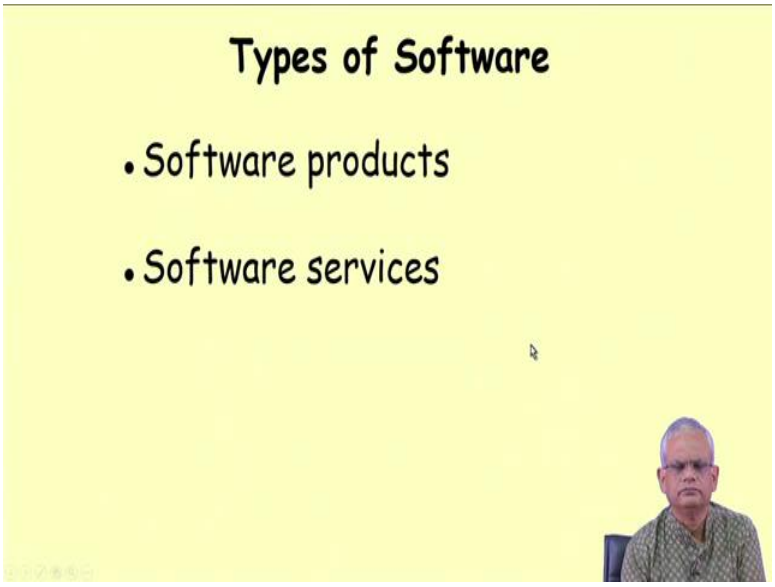

We have seen that there are two types of software the products and the services.

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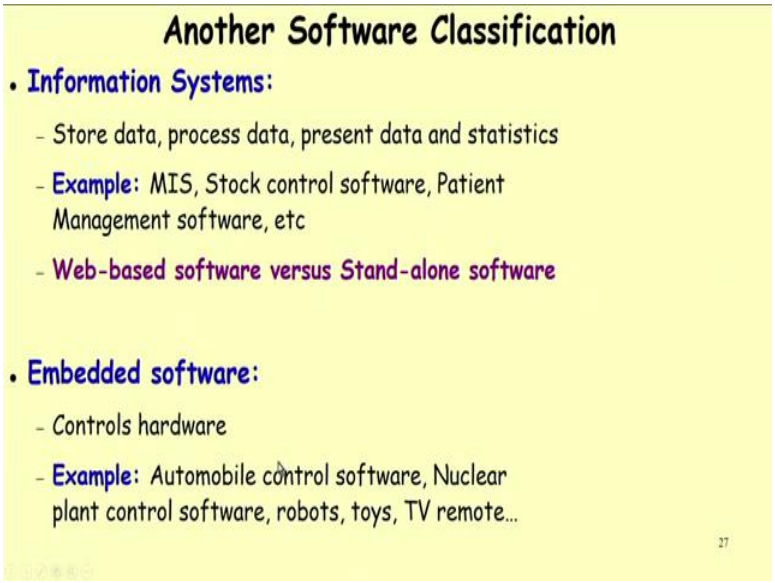

But then you can also classify the software into either information systems; which store data, process the stored data present the data and statistics. This is the typically management information system, stock control inventory management, patient management etcetera; this can be web based or standalone software.

And there can be embedded software, where they control the hardware; for example an automobile control software, a nuclear power plant control software, robots, toys, TV remote etcetera. So, this is another classification one we just saw is that between the product development and services type.

But another way of classifying them may be information system projects or embedded software development projects. With these basic concepts, we will; we have come to the end of this lecture. In the next hour will also be few more basic concepts based on what we have already discussed. We will stop at this point and continue in the next lecture.

Thank you.

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Lecture - 03
Introduction – III

Welcome to this lecture, in the last two lectures we had some very brief Introduction to the Software Project Management and also motivation for software project management; how software project management is different from traditional project management and the scope of the project, stakeholders and so on.

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In this lecture, we will continue our discussion on software products and services because these are the two major types of projects that are undertaken. And the software project manager has to manage both these types of projects; the software product development projects and the services types of projects. We look at the major activities of the project manager and we look at the traditional versus modern projects.

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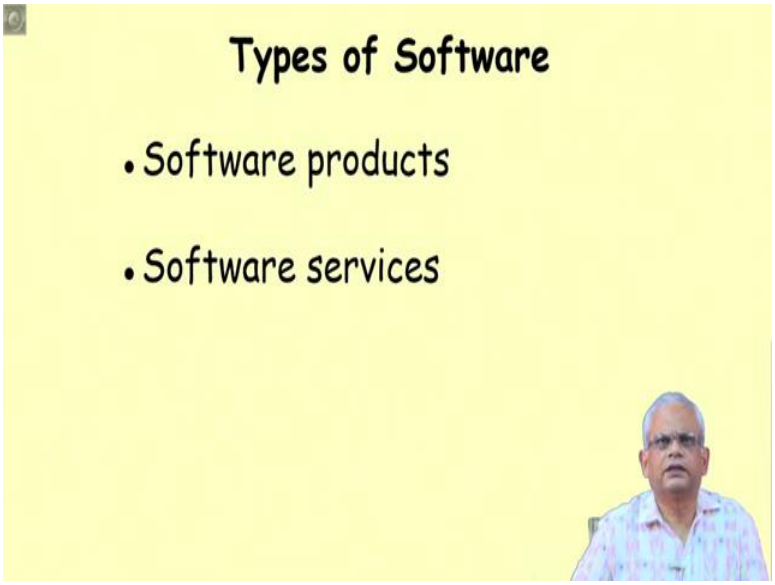

There are two main types of software that are being developed across the world; one is called as the software product development. Software products are the ones that we can buy just like any other product, you can go to the shop or order online. Let us say you need antivirus package; you can order it online or you want a case tool you can order it online, but the other type of project are the software services the services are customized software.

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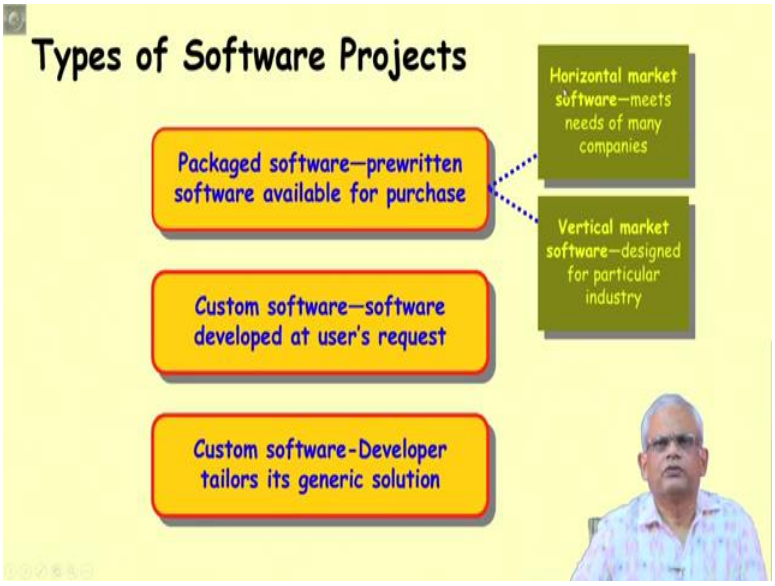

The packaged software which are generic in nature are called as the software products; these are available to buy of the self maybe from a online source; you can purchase this. These are prewritten software available for everybody; anybody can just order this and get it delivered.

But the other type of project is development of custom software; this software is developed based on specific user request. There may be one or maybe a few users who have their own typical requirement for the software. And the developing organization has only few users in mind; they initiate the few users, they initiate this project and then the developing organization develops for them and delivers to them; these are not generic software available for anybody for purchasing.

But then there is another custom software where the developer has a solution and each customer has small modifications that are required. For example, let us say an academic institute; it needs to automate it activities for example, student admission grading and so on. A company might have developed a software for this academic institute automation, but then every academic institute has its own grading system admission process and so on which may be different from other institutes.

Here the developing organization has a software which has been developed for some customer or maybe few customers in mind. But then another customer require small changes and here the developing organization initiates a project where it has its software already there; only small changes need to be done in this project.

So, the first one is a product development project where the developing organization has a generic customer requirement and develops the software. The other two are services projects these are custom software; one is specific user, the second one is it just tailors and existing solution. The product type of software can also be categorized into two types; one is for the horizontal market and the other is for the vertical market. The vertical market maybe for let us say banking or let; let us say telecom billing or maybe medical inventory management or maybe a hospital management and so on. So, this is for specific verticals like banking, telecom, hospital and so on.

Whereas the horizontal market is across all types of customers for example, a database management or a antivirus software and so on. These are sold to customers who are generic in nature and there is no specific type of customer in mind.

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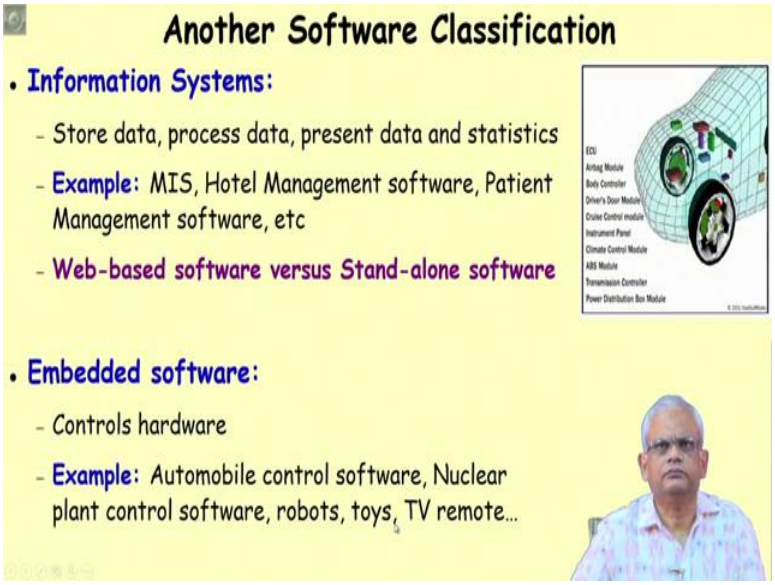

There is another classification that whether the project involves development of an information system or an embedded software. In a information system, the software collects data, stores data then processes this data to generate some statistics and maybe some other data output. There are many examples of the information system for example, the management information system, where let us say a trading house deals with the many commodities, it buys at bulk rate and then sells it to the individual customers,

Here the management information system will keep track of what is the customer requirement, then source these requirements in bulk. And then store this and distribute and also, manage the accounts and report about various types of statistics what is the inventory level, profit loss and so on.

Another information system maybe the hotel management software, where the guests to a hotel register the pay bill; the hotel room occupancy, the profit loss and so on, these are handled by the hotel management software. We might have a patient management software where in a hospital patients register, the doctors they visit the hospital certain times. And then the patients pay their bill and then the hospital needs to pay to the doctors and then finally, have the profit loss statement.

All these information system may either be web based or these may be standalone software. In a web based software, the software can be operated from anywhere using a web browser whereas, in a standalone software; it needs to be operated centrally. In

contrast to the information system in a embedded software; these are mostly used to control some hardware. Therefore, the software here works closely with the hardware it controls the hardware. For example, automobile control software, nuclear plant control software, robots, toys the software that control the toys, TV remote and so on.

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The software services as we are discussing is a different type of project, it has different characteristics than the software product projects. The software product project the development is done to develop a generic product; whereas, in software services it may involve customization of an existing software or developing certain items on customer request. This has of late becoming a dominant type of projects many of the software development projects are software services projects.

But the term software service is the umbrella term; it includes many things. For example, customizing an existing software on specific customer request; software maintenance the software exists, but then the customer want some changes maybe enhancements; maybe performance improvement and so on. A project to maintain the software is also a type of services project.

Software testing; a software has been developed, but then before deployment needs to be tested and maybe an organization just does software testing and they are providing software services. But what about an organization which supplies contract programmers to another organization; who wants to automate certain activities and they just get these

contract programmers who do coding or other assignment assigned activities like testing and so on; this is also a type of software service.

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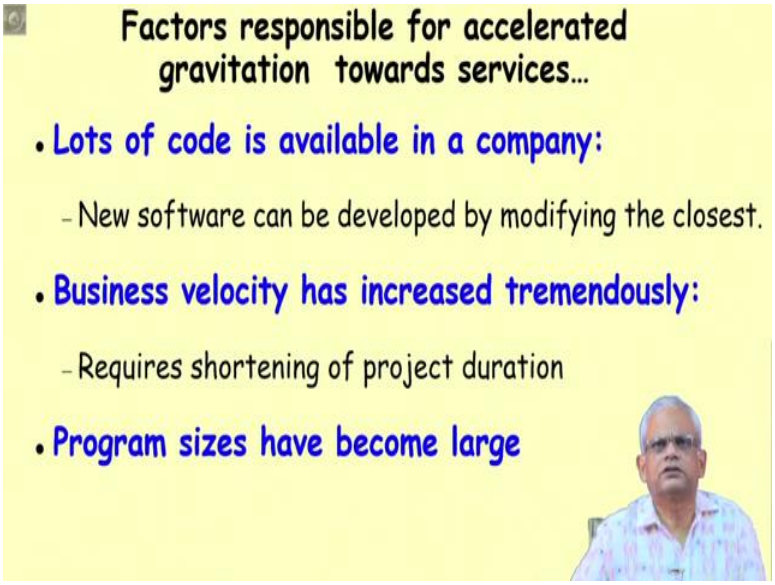

But what is the reason that more and more type of more and more projects are becoming the services type of projects? About 50 years back; the only type of software projects that are existing were software product development type of projects. But now more and more projects are becoming services type of projects and let us investigate that what is causing this migration from pure product development to services type of projects.

One is that over the years lot of code have been written by the company and whenever there is a new request; the company can most cost effectively develop the software by modifying some of the existing code. It can find out which is the code which is closest to the customer requirement, it may be one project or it may be across several projects which were completed. And then it can just make small modifications to this and then deliver the software to the customer very quickly and at less cost.

This is also a services project. And here the main reason why the companies are doing this is that; they have lot of code they have completed many projects in the past and they have a lot of code and they can reuse these to develop the next software that has been requested.

Another reason is that the business velocity has increased tremendously; by this we mean that there is so much of competition. There is so much of competition among different industries, different companies that they want to do things fast for software development they become impatient.

Multiyear projects are gone; nobody wants them and it is expected that a software development work completes in 2 weeks, a month and so on. And the way this development request can be met is by having a services type of project, where there is a customization of existing software.

The third reason is that the program sizes have become large and therefore, it is imperative that most of the code is reused and only small development is done, small customization to complete the work.

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But still worldwide, nearly half of the projects are product type of projects and another half is and services. Of course, the services type of projects are increasing very rapidly, but if we look at the Indian companies; here we find that most of the projects are in the services segment; why is that the case that we have very few product development projects?

The product development project has the advantage that once the software is complete and it is successful in the market; the company has a steady revenue the one that

company that developed the product. For example, oracle software once it was developed it gives us steady revenue to the company. Whereas, a service type of project in that way the benefit to the company is less because it is developed and sold on one time revenue.

But then why is that the Indian companies have not focused on the product type of projects but largely to the services segment? If we look into the reason; the answer is that the product type of projects have their inherent risk; a product development involves investing lot of money in developing something which may be required, may be successful or may not be successful.

Whereas a services project; once it is complete there is one type revenue which is assured and possibly this reason to avoid the risk invest millions of dollars or rupees to develop something which may be a failure. I think that is the reason that the Indian companies have largely focused on the services segment.

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Now, let us look at the activities of the project manager; what does the project manager do? The project manager; large part of the activities planning; estimate the duration, estimate the cost, estimate the effort, allocate resources and schedule.

Staffing; recruit personnel, motivate them and once the project is started; direct the personnel and ensure that the team works as a whole to complete the work; so that is

every day. Check that the work is progressing as per the plan, but then there will be a plan deviations and monitor if there are deviations and then take corrective actions so that any changes to the plan is contained.

And also it may be necessary to change the plan continuously because when we plan initially we cannot foresee many things and therefore, the plan needs to be changed as the project progresses. You can summarize the project management activity as planning the work; that is to start with you must plan the cost, the effort, schedule and then the projects and then do the staffing and then the project starts and largely it is directing and monitoring and controlling and that we call is working the plan. So, the project management activities essentially consists of planning the work and working the plan.

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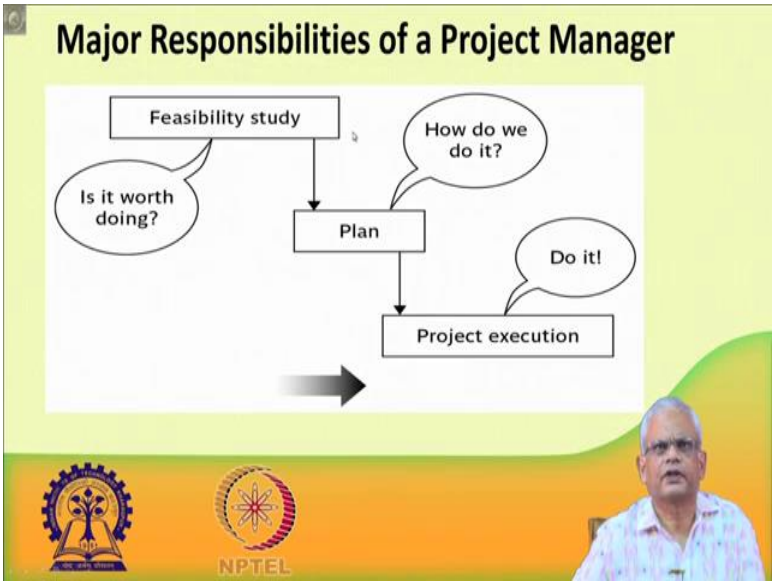

The major responsibilities of a project manager is summarized in this picture; that to start with it is the project managers responsibility to carry out the feasibility study. The feasibility study the project manager determines is it worth doing the project?

The project manager understands the problem, proposes alternate solutions and then estimates the cost. And the technical feasibility; the project manager may decide at this point of feasibility study that the project in feasible on account of cost considerations, that the cost involved in developing the software far exceeds the income from the software and then the project will be abandoned.

The project manager may also find out that technically the project cannot be done by the organization; do not have the competency to undertake the project and then the project there is no further progress project is abandoned.

But if it meets the cost requirements and also the technical feasibility then the project starts off. And the first activity is planning the initial plan is made that how do you do it; here it involves estimating the duration, estimating the effort and then resource allocation and scheduling; that we call as plan. And once the plan is over the project starts off and then we have the project execution where the project manager cons concentrates on how the project progresses as per the plan. And here the work of the project manager is largely directing and monitoring and control.

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In the feasibility study, the project manager is concerned about two main types of feasibility; is it technically feasible and is it worthwhile from a business point of view, that is it financially feasible. And only when it is found feasible the planning is undertaken and after the planning; the project starts off and then the work of the project manager is directing the team and then execution monitoring and control.

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The project planning starts off immediately after the feasibility study; if the project is found to be feasible and worth doing it then the planning is done. In the planning, the important activity is estimating.

As we proceed in this course we look at various estimating techniques and that is a major component of this course. And once the estimation is done; the scheduling who will do what at what time and when is likely to complete? So, that is project scheduling. The staffing, risk management and there are many other plans that we have called here miscellaneous plan; as you proceed in this course we look at the other plans.

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Monitoring and control is the major activity once the project starts off; that is the feasibility study and the planning is complete and the project starts off; the project managers main activity is monitoring and control.

Monitoring and control lasts for the entire project duration, from the time the project starts to the project completion. And here the project manager everyday needs to check the progress and if there is a deviation from the plan time and the progress that has been achieved; the project manager may take some corrective actions. And also the project manager may find that it is necessary to revise the plan.

So, the projects manager sits down, revises the plan; he controls if there is a problem in the project which is hampering the progress; the project manager takes controlling action. For example that the design is taking long time then might change the roles, might get additional designers and so on. Whenever there are problems emerge for example, there is a delay in delivery of the hardware the project manager may decide to lease hardware to overcome the problem.

So, coming up with solutions, innovations to projects that to problems that emerge as the project progresses and not only that the project manager is the main person who liaises with the clients with the top management, with the users also the developers and other stakeholders.

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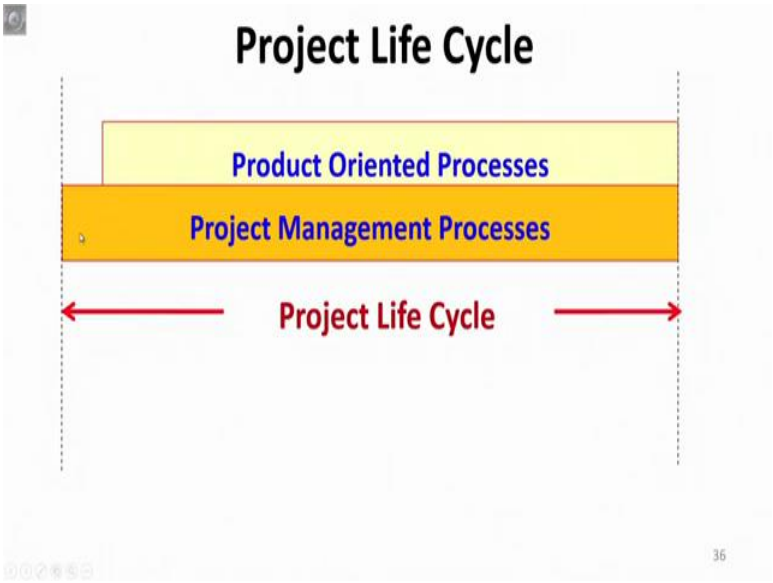

If we summarize our discussion here in this diagram, it is that the project management activities they start much before the actual product development starts and that is the project planning, the feasibility study, the project initiation and so on.

The project manager does many types of processes for example, initiating the project planning, recruiting, monitoring and so on; so those we call as project management processes. But as the development work starts; the development team carries out their own processes, those we call as product oriented processes. For example they might carry out requirements specification, they might carry out design they might carry out coding, testing and so on, those we call as product oriented processes.

And at the same time the project manager carries out the project management processes and the project management processes are carried out over the project lifecycle. Whereas, the product oriented processes are carried out during the development lifecycle and the project lifecycle is longer than the development lifecycle as shown in this diagram. With this, we will just take a break and we will continue in the next lecture.

Thank you.

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Lecture - 04
Project Management Standards

Welcome to this lecture, in the last lecture we were discussing about the project management lifecycle and the software development lifecycle. And we had said that the project management activity start much before the software development activity start with that let us discuss further.

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In this lecture we will discuss 2 main concepts, the project management standards and the life cycle models.

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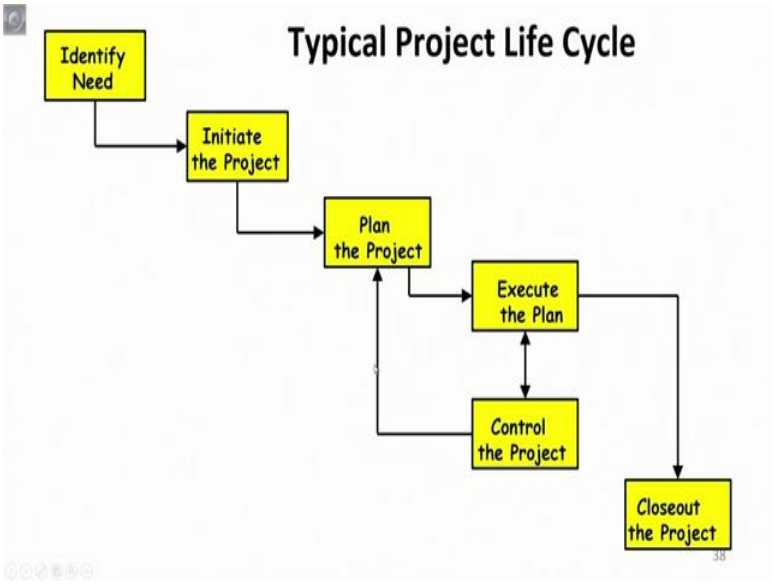

In a typical project life cycle almost every software starts with a need for the software somebody in a organization feels the need for a software to automate certain things and based on that the project proposal is written and it is approved by the top management. The project manager is appointed and that forms the initiation of the project and after the project manager is appointed for the project; the project life cycle starts with planning the project. In the plan of the project various types of plans are prepared the schedule, the configuration management, the risk management and so on.

And once the plans have been prepared, the project development lifecycle starts and that forms executing the plan and during the execution of the plan the project manager executes the plan that is directs the development team to proceed according to the plan. But as the project progresses there can be several deviations from the plan and for this the project manager need to perform some control activities, maybe there is some bottleneck why the project is not progressing as per the plan. The project manager works to remove the bottleneck for example, there may be a shortfall in the technical personnel or there may be shortfall in the hardware equipment, the project manager proactively removes those hurdles, so that the project proceeds as per the plan.

But then, the project manager might have to rework the plan because even in spite of that there will be delays or there may be some part of the project may get completed quickly than anticipated and so on. And once the project completes, the project manager carries out the project closeout activities as we proceed we will look at all these activities in more detail.

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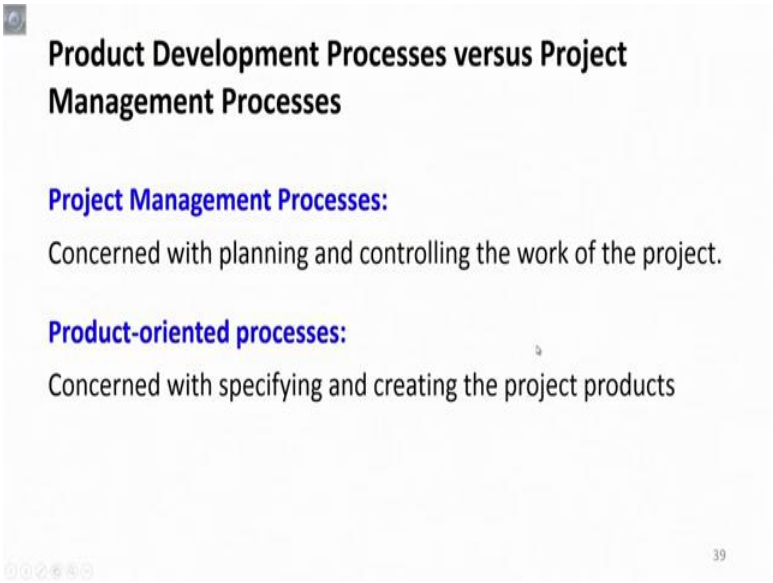

During the project lifecycle, the project manager executes the project management processes let me repeat, during the project lifecycle the project manager carries out the project management processes. The project management processes are concerned with planning and controlling the work of the project. The activities here the project manager performs the planning of the project, executing the plan that is directing the team to carry out development work and whenever there is a deviation, the project manager controls removes the bottleneck, so that the project proceeds as per the plan.

In contrast the development team carries out product oriented processes; the product oriented processes are concerned with specifying and creating the project products that is they may do the requirement specification process, design process, coding process, testing process and so on. So, those we call as the product oriented processes, these are carried out by the development team whereas, the project manager carries out the project management processes.

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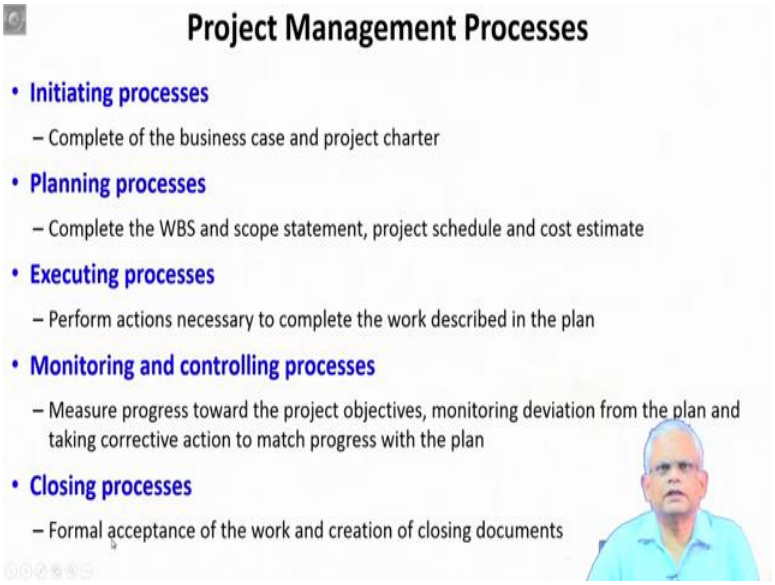

Let us see what are the project management processes, the project management processes consist of initiating the project, here the project manager writes the business case and the project charter, as we proceed we will see the business case and the project charter.

The project manager also carries out the planning processes that is completes the work breakdown structure and performs the project scheduled cost estimation and so on. And then the project manager carries out the executing processes, which basically performing the necessary action to complete the work as outlined in the plan, but then during the execution process there may be deviations from the plan.

And the project manager needs to carry out the monitoring and control processes where the project manager checks, whether the project is proceeding as per the plan or there are deviations from the plan and when there are deviations the project manager takes corrective actions to match the progress of the project with the plan. And finally, the project manager needs to carry out the closing processes, where the closing documents are created and the customer finally gives the formal acceptance of the project.

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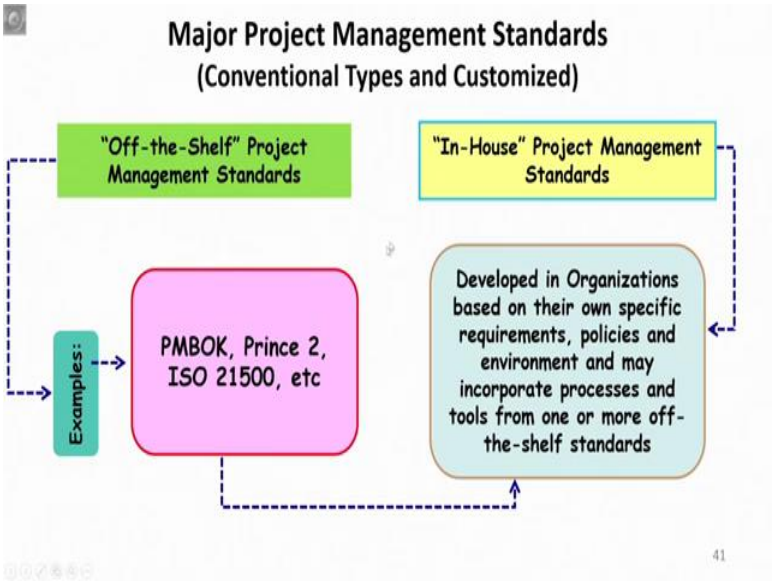

There are several guidelines which have come up for the project manager to carry out the project management processes. There are 2 main types of project management standards, one is off-the-shelf project management standards, these are described in books and so on and here the popular ones are PMBOK that is Project Manager Book of Knowledge Project Management Book of Knowledge and PRINCE2 and ISO 21500 and so on. There are many other we just given some important examples here.

On the other hand an organization may have a slightly different project management standard as compared to the off-the-shelf standard, they might tailor or maybe take different aspects from different standards and they might have a tailored in house project management standard where that specific organization follows the standard.

It may be a good idea to know what is involved in the PMBOK, PRINCE2, ISO 21500 and in our discussion we will not restrict our self to any one standard, but our discussion in this course will be more on concepts that are used across all these standards.

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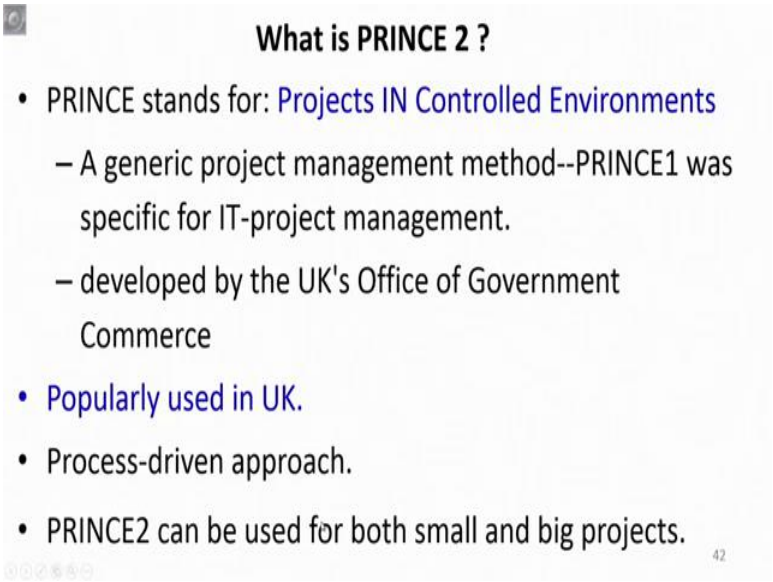

PRINCE2 is a popular project management standard. PRINCE stands for Project IN Controlled Environments. This is not restricted to only software development, but the PRINCE1 was more targeted to the IT or software project management and PRINCE2 is become a more generic project management standard. This was developed by the United Kingdom office of government of commerce, it is very popular in the United Kingdom it is a process driven approach and has been used for both small and large projects.

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There are essentially seven processes in PRINCE2, the projects starting processes, initiating processes, directing or executing processes, managing a stage boundary, controlling a stage and managing product delivery.

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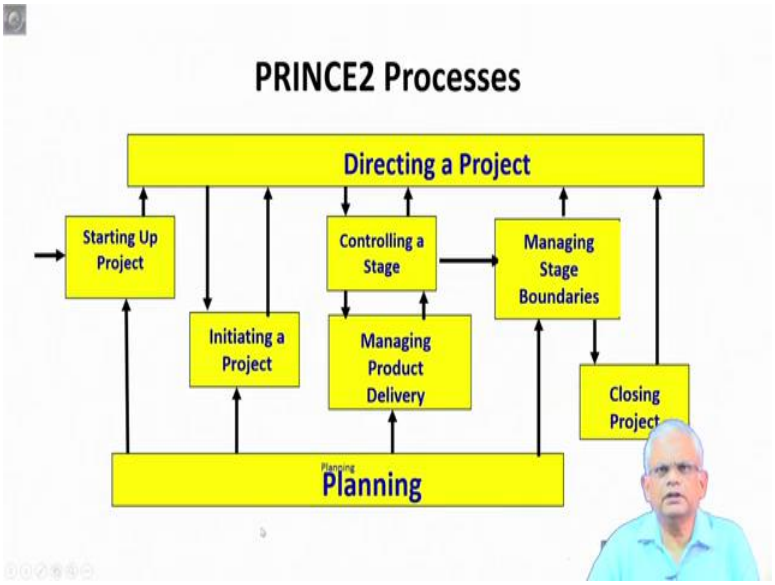

If we represent that pictorially there is a start-up processes and then 2 main processes which exist throughout the project are the directing a project which is basically execution of the project and then planning continues throughout, because once the initial plan is made it is continuously revised and based on the plan once the plan is complete initial plan is complete.

The project is initiated and the project undergoes various stages and in each stage, the project manager directs the project activities and the project team carries out the activities and here the project manager monitors and controls each stage. And at the end of the stage the new stage needs to start, so this is called as the managing the stage boundary and after all the stages are complete one by one stages are taken up here and once all the stages are complete the closing processes are undertaken by the project manager.

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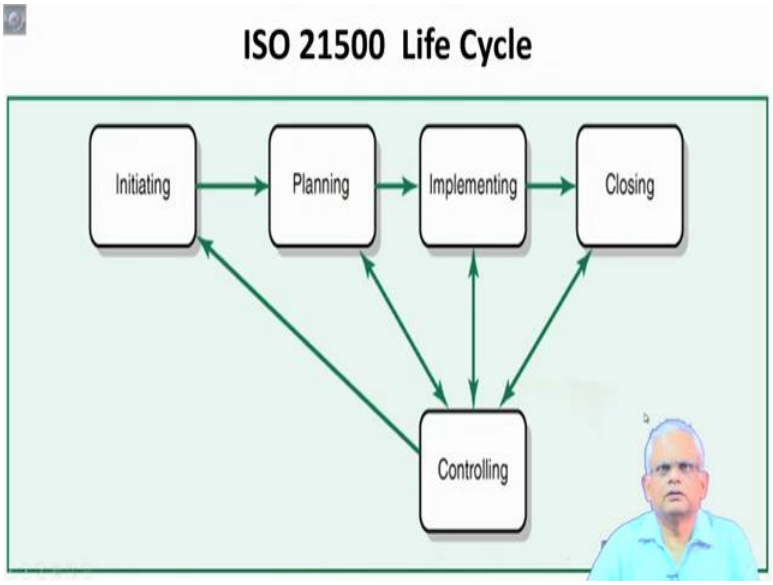

The ISO 21500 life cycle here also we have the initiating processes, the planning processes, implementing or the directing or executing processes and finally, the closing processes and throw out the controlling processes are carried out by the project manager.

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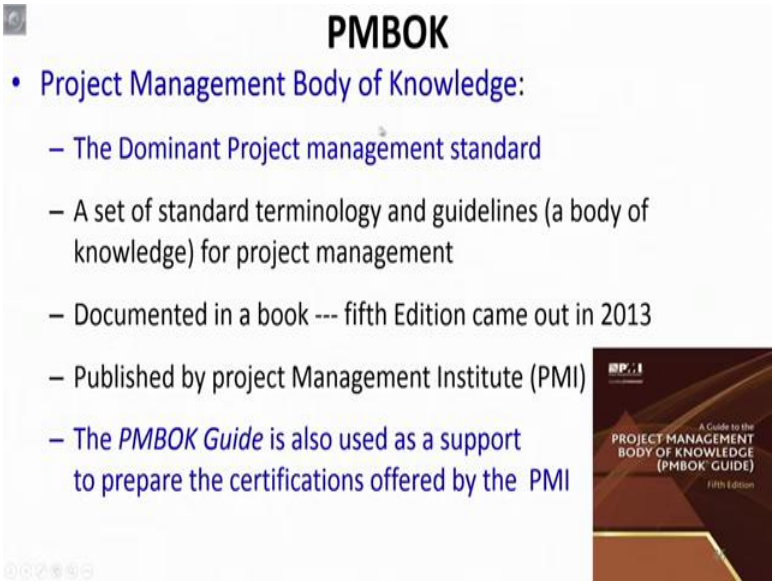

PMBOK is a very popular project management standard, it is the Project Management Body of Knowledge, it is a dominant project management standard used widely across various projects. The PMBOK basically consists of several terminologies and guidelines and this has been documented in the form of a book as you can see here Project Management Book of Knowledge PMBOK guide. It is published by the Project

Management Institute PMI and this guide is used as a support to prepare for the certification offered by the PMI.

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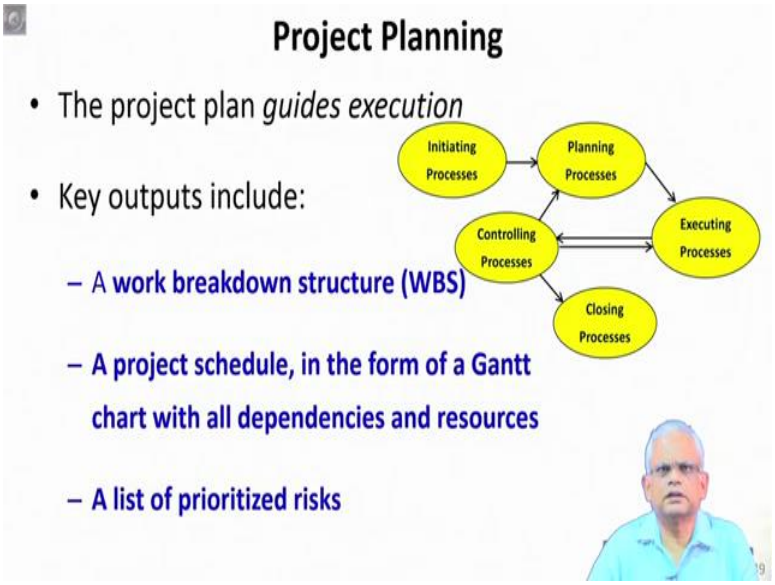

In the PMBOK there are various process groups each process group consists of several processes. There are the initiating process group, but the project initiation activities are carried out by the project manager and as the project takes off the planning processes are carried out by the project manager. The initial plan is made and the plan is continually refined. The executing processes the project manager directs the project team to carry out various activities and at the same time the controlling processes are carried out. The monitor if there are deviations from the plan and then take controlling actions and once the project completes the closing processes are carried out.

So, these are 5 main process groups in the PMBOK the initiating processes, the planning processes, the controlling processes, executing processes and closing processes. As you can see across all these different project management standards many things are common, the processes, the initiation, planning, executing or directing, monitoring and controlling and finally, the closing processes. As I told earlier will not restrict our self to any one project management standard, but we look at these various processes the project managing management processes across all these different project management standards.

The initiating processes this is the one that is carried out by the project manager first in the project lifecycle. The main goal of these processes is to start off the project. The outputs of these initiating processes are that the project manager is selected, the key stakeholders are identified and the project charter is completed. The project planning processes here the output include work breakdown structure, the project schedule in the form of Gantt chart and identification of all dependencies and resources and a list of prioritized risks.

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If we compare two popular project management standards the PRINCE and the PMBOK, we can see that the PRINCE is the origin is United Kingdom whereas, the PMBOK the origin is United States. Here the PRINCE2 is administered by the APMG which is Association for Project Management Group, PMBOK is administered by the project management institute, both are very popular used worldwide, but PMBOK is used more extensively than PRINCE2.

PRINCE2 is a process based methodology as you had seen whereas, PMBOK is a documentation of the knowledge, it is non prescriptive and the project manager is having flexibility to adopt his own processes, but these just serve as the guidelines. PRINCE2 is used mostly in UK, the PMBOK is used worldwide, over 75 percent of the projects worldwide use PMBOK.

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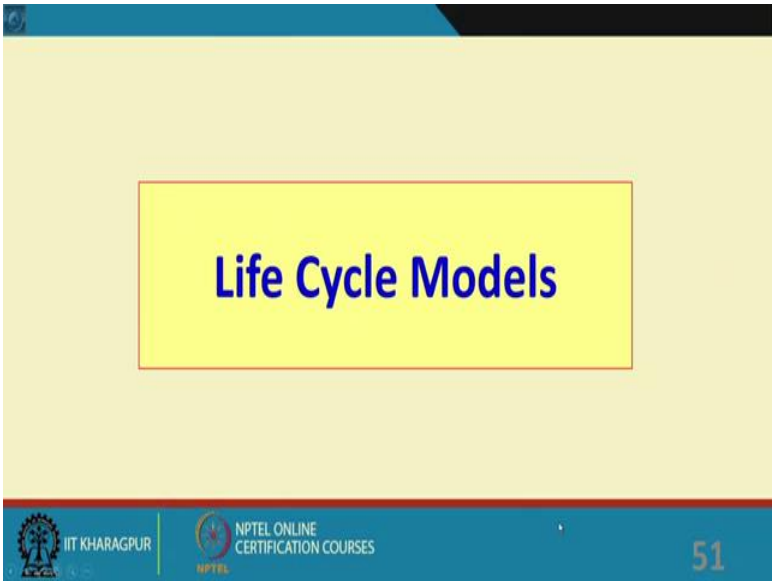

Now let us look at the life cycle models, because the project manager executes the project management executes the project management processes to direct the project team to carry out the development work. At different stages the project team carries of different activities and therefore, as a project manager we must be clear about the various life cycle models, what are the stages through which the development proceeds, the stage boundaries and so on. Here we will emphasize only on the basic concepts.

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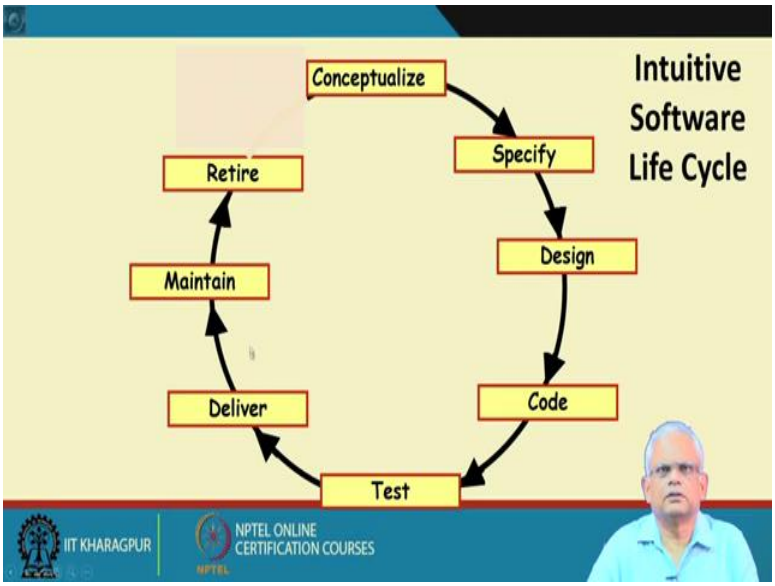

Intuitively a software life cycle consists of first somebody thinks of the project the need for the software and then the project is specified, it is designed, it is coded, it is tested, finally, delivered and once it is delivered and used it needs maintenance and it undergoes

maintenance for number of years as the software gets used over many years maintenance activities proceed and finally, once the software is not used anymore it is retired.

So, this is the intuitive model of software development conceptualize, specify, design, code, test, deliver, maintain and retire. Based on this conceptual model of the software life cycle various life cycle models have been proposed where these activities the sequencing of these activities vary.

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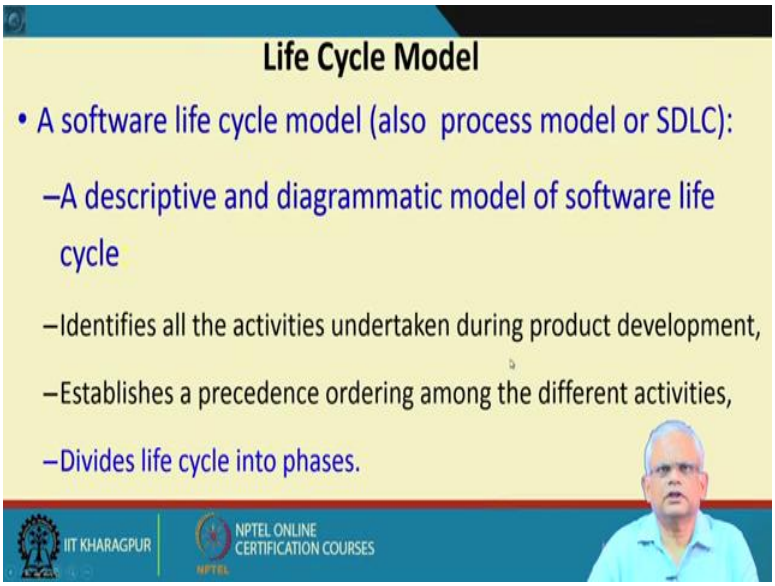

The life cycle model is also called as a process model or software development lifecycle model. It is a diagrammatic representation of the software life cycle and it is also accompanied by a description. The software life cycle model essentially identifies all the activities that are undertaken during the product development. It also establishes a precedence ordering among the activities and it divides the life cycle into phases.

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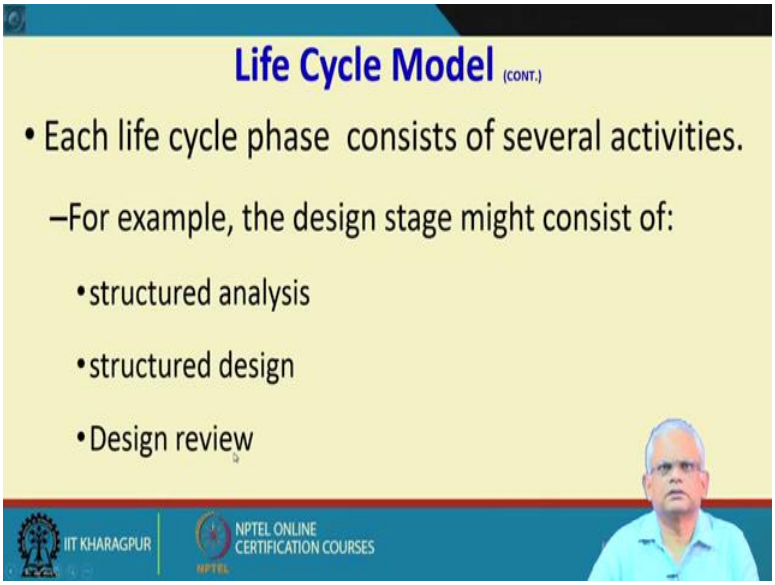

And each phase consists of several sub activities for example, the design phase may consists of the sub activities structured design, structured analysis, structured design, design review and so on.

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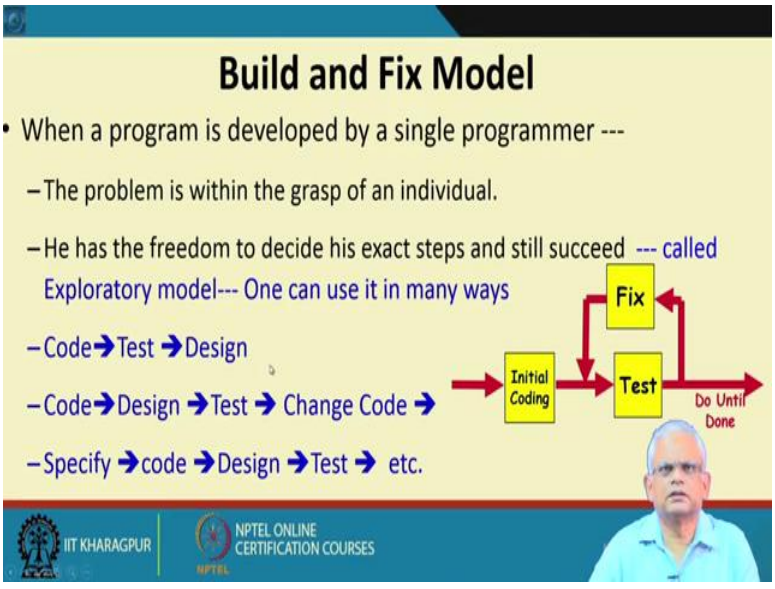

For very small projects let us say student assignment, the student unconsciously follows a build and fix model it is not a formal life cycle model, but this intuitively comes to the student. This yields success for very small projects like a student assignment where the problem is within the grasp of an individual, the build and fix model works. As the name says here do the initial building here write the code here for the project and then test, as long as the code does not pass the test carry out fixes, again test carry out fixes; again

test until the developed code passes all the tests. This is also called the exploratory model, here the problem is very small and the developer has flexibility and freedom to undertake whatever activities he feels like.

So, there is lot of flexibilities built into the build and fix model. For example, the developer may first code, then test, do a little bit of design, then test, fix and so on or my code, then do some design, test, change code and so on or may specify, then code, design, test, etcetera. So, this model build and fix model is a very very informal model suitable only for projects where which is rather trivial like a student assignment.

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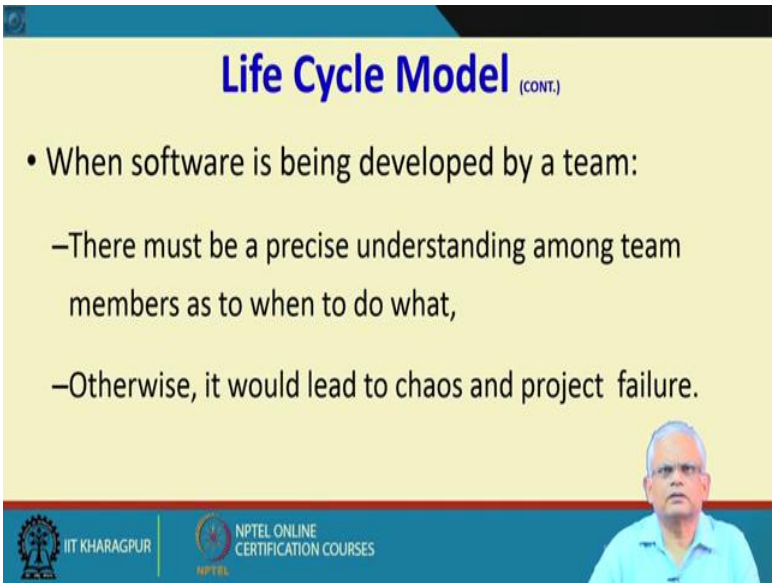

But when the development is to be carried out by a team the formal process model has to be followed, there must be a precise understanding among the team members that what are the activities, which activity needs to be done after what, who will do and so on, otherwise it will lead to chaos and project failure.

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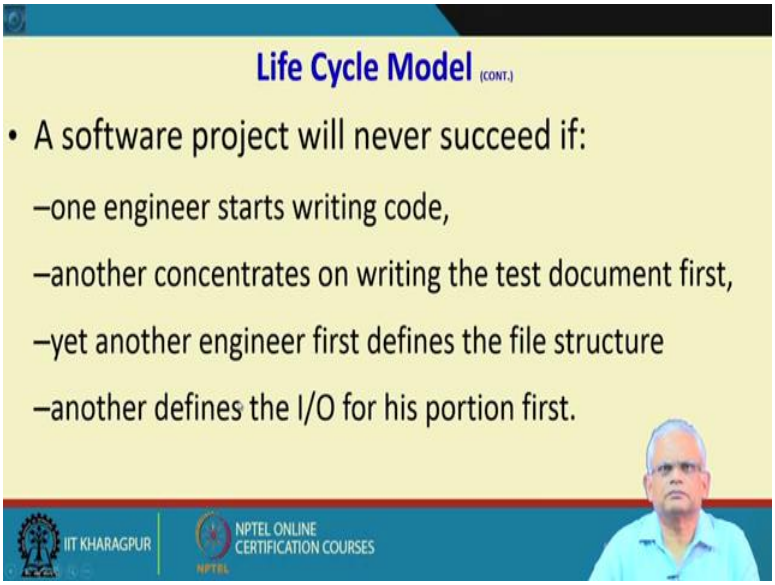

If the build and fixed model is used for developing a non trivial project and the team starts using the build and fix model, then maybe one of the developers will start writing the code, another will try to design, the third one will write to the do the test document and so on. Another may define the IO for his portion first and so on and this way the project will not succeed. There will be too much of delay poor quality of code will come out, there will be a lot of idling, cost escalation and a very poor quality code will come out. So, an informal project model like a build and fix model is not suitable for non trivial project work.

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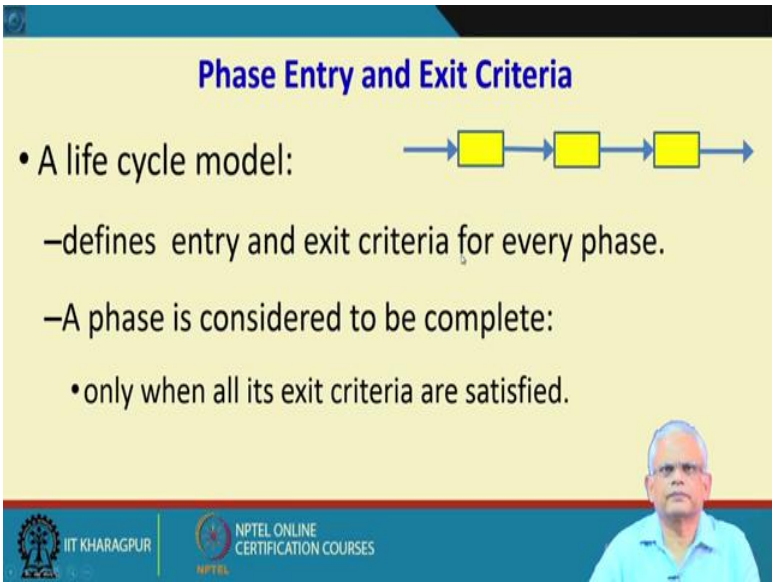

Before we look at the different project lifecycle models you must know what is the phase entry and exit criteria. A life cycle model as we said that consists of various phases and each phase starts when there are some criteria are met we call that as the phase entry criteria, only when those criteria are satisfied the phase starts and when some criteria are satisfied the phase ends. So, for every phase it is necessary to define the entry and exit criteria for that phase.

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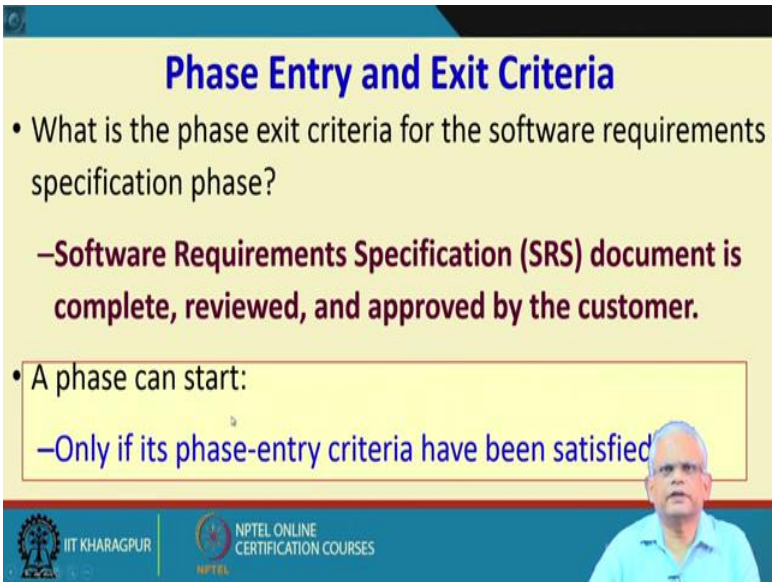

For example let us say what is the phase exit criteria for the software requirement specification phase? The exit criteria is that, the requirement specification document must have been completed, it must have been reviewed and approved by the customer. Similarly a phase can start only when it is phase entry criteria have been satisfied.

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Another important concept is the milestones, a milestone is very important for the project manager because using the milestones, the project manager keeps track of the progress of the project. There are various types of milestones, but one important category of milestones are the phase entry and exit, when there is a phase starts after the phase entry criteria has been met an important milestone is met.

Similarly when the phase completes and the exit criteria are satisfied, the project definitely has made some progress and that forms another important milestone, with this discussion we will conclude this lecture and in the next lecture well discuss a few project lifecycle models.

Thank you.

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Indian Institute of Technology, Kharagpur

Lecture - 05
Life Cycle Models - I

Welcome to this lecture. In the last lecture we discussed about the software development lifecycle. In the software development life cycle the project starting from the concept is developed by various activities. The software development lifecycle essentially defines all the activities that are carried out and also the ordering among those activities. There is a intuitive concept of the activities that are carried out during the software development, but then as we will see that various project management life cycles structure these activities in different ways. Let us look at some popular Life Cycle Models.

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In this lecture we will discuss about the waterfall model, V model, evolutionary model and the prototyping model.

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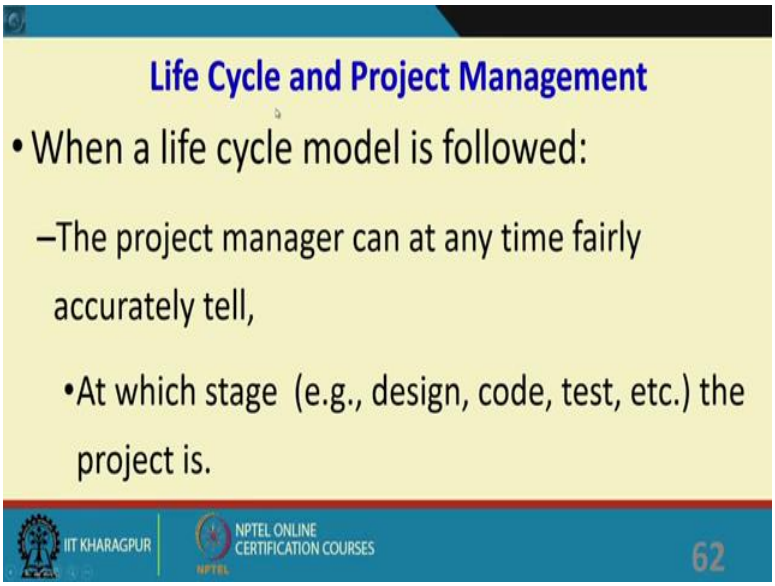

The lifecycle the project lifecycle is very important for the project manager because based on the lifecycle the project manager can track the progress of the project. Because, there are various milestones in the lifecycle and as the milestones are met the project manager can tell how much progress has been achieved or at which stage the project is.

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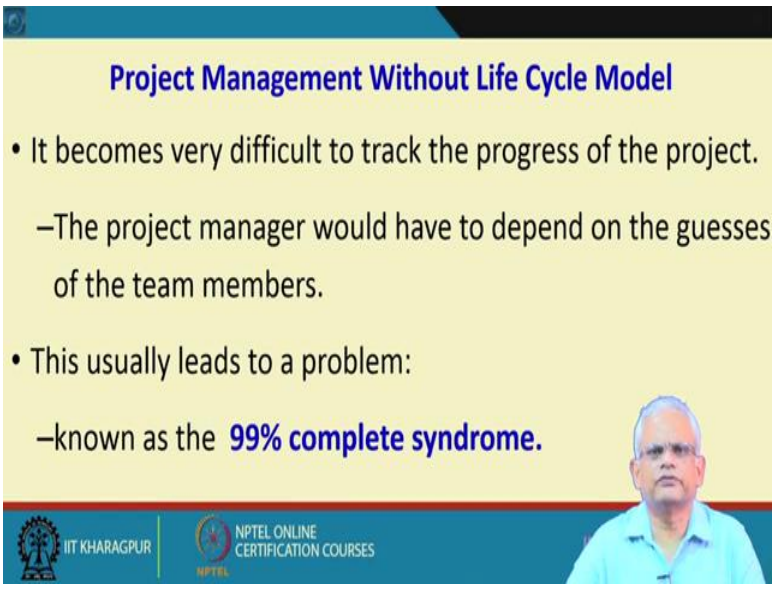

Without a lifecycle model the project manager is handicapped. It becomes very difficult for the project manager to monitor and control the project does not know that how much work is remaining and therefore, the project gets delayed. The cost rises and therefore,

for every non-trivial project the project manager uses a suitable lifecycle model and the development team follows it. Without a lifecycle model usually a problem it is known as the 99 percent complete syndrome occurs. Here the project manager has no way of finding out the progress of the project other than asking the team members that how are you doing, how much you have completed.

Typically the development team is very enthusiastic and they optimistically answer that we have almost done only small thing is there and each time you ask them they say that see we are nearly done. So, the project manager thinks that the project is about to complete, but is far from true. It takes many more months or years even when he the project manager hears the term that the work is almost done and that is known as the 99 percent complete syndrome. It occurs when the project manager has no other way to track the progress of the project other than asking for the opinion of the team members that how far they have progressed.

As we were discussing the project management lifecycle allows the project manager to have various milestones and stage completion by which the project manager can track the progress of the project more meaningfully.

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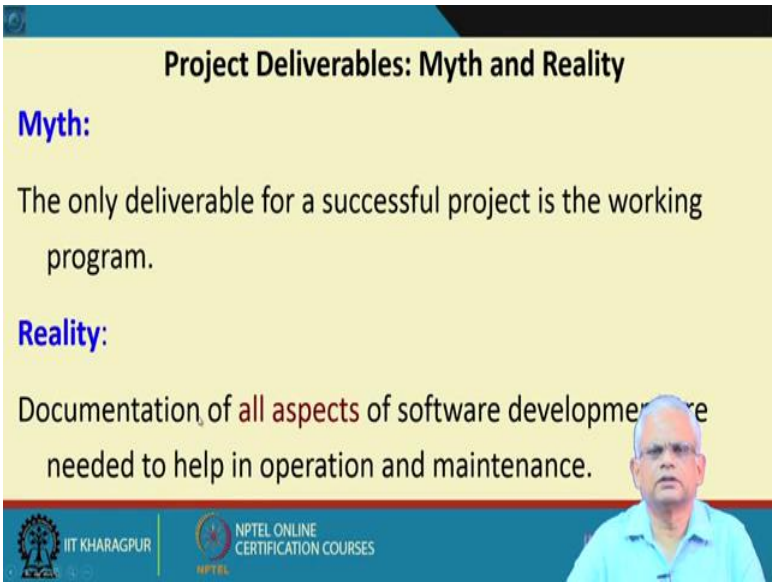

Another very basic concept that we will discuss before looking at the lifecycle models as the project progresses various deliverables are produced. It is a myth if we say that the working program is the deliverable of a project not really. There are a large number of

deliverables that are produced as the project progresses and these are the documentation of all aspects of development for example, specification design etcetera.

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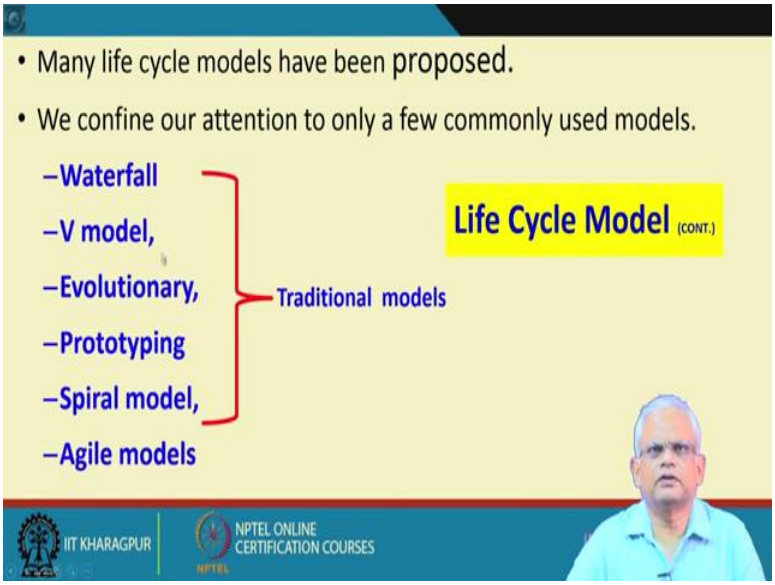

We will discuss about the waterfall, V model, evolutionary, prototyping, a spiral model. This we will call as the traditional model. This have been used for traditional projects where the project is a product development project starting from scratch, but for services type of projects typically the agile models are used.

We will first discuss about the traditional models the waterfall, V model, evolutionary, prototyping, spiral model. We will find out what are the shortcomings of these models. Why they are not so suitable for the services type of projects and we will discuss about the agile model and we will see that how they overcome those problems.

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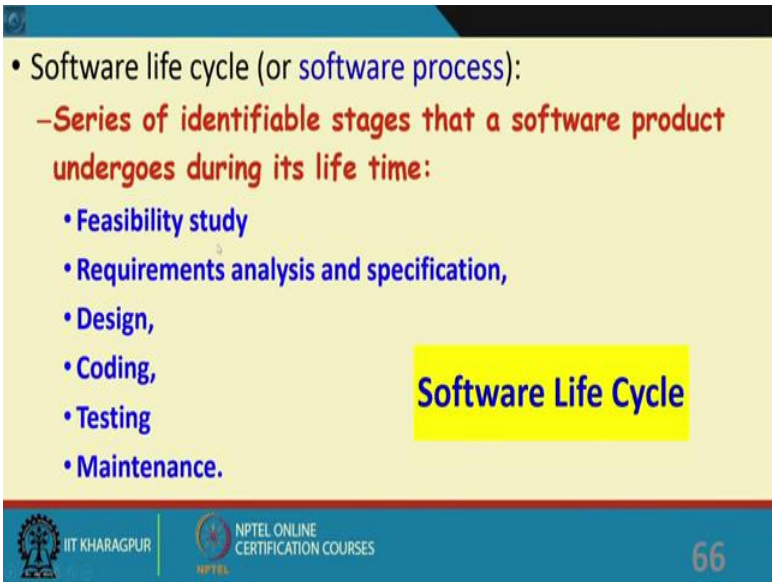

The software life cycle as we had already said, is a series of stages during the development of the software. Intuitively, every software has a feasibility study during which it is decided whether to take up this project or not the requirement analysis and specification, design, coding, testing and maintenance.

So, these are the stages that intuitively every software undergoes the feasibility study requirements analysis and specification, design, coding, testing and maintenance.

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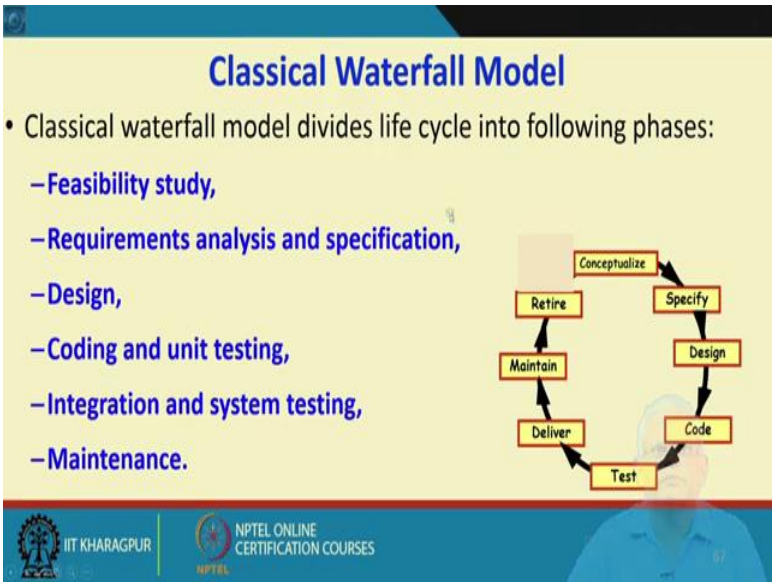

The classical waterfall model fits very accurately to the intuitive model of software development. Here the activities are carried out one after other. First the feasibility study is undertaken then requirements analysis and specification work is undertaken. And once that is complete the design is taken then the coding and unit testing is undertaken, next the integration and system testing is undertaken and finally, the maintenance phase is undertaken.

This is a very close fit to the intuitive model of software development and this is called as the classical waterfall model because based on this model all other lifecycle models have been developed. This is the classical model and all other models are derivatives of this model. So, this is according to the conceptual model of software development we were discussing.

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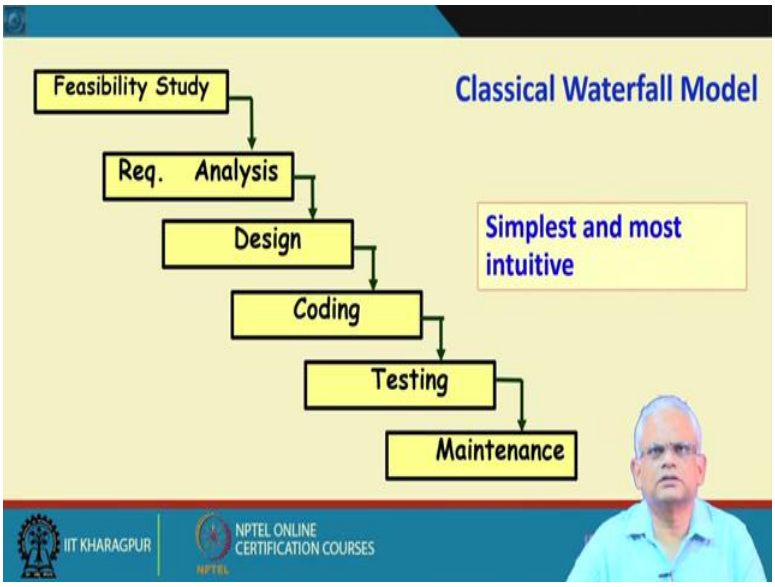

If we represent the classical waterfall model in a diagrammatic form we can see that the different phases are represented here feasibility study, requirement analysis specification, design, coding, testing, and maintenance and there is a transition from one phase to the other. If we look at it looks like a waterfall a series of waterfall and from this the name waterfall model comes. This is the simplest and most intuitive model.

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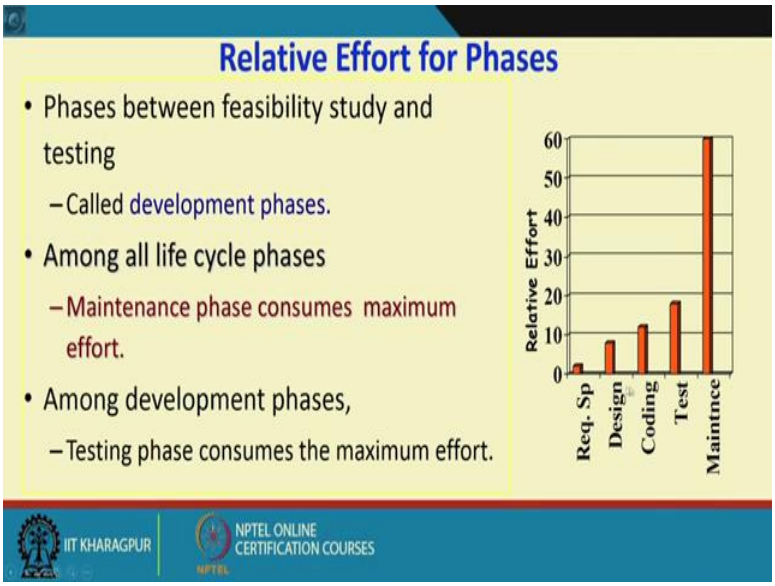

Out of all the phases in the classical waterfall model which phase takes the most effort. If we look at all the phases, the phases between the feasibility study and testing are called as the development phases and then we have the maintenance phase is called development phases because here the product is developed and delivered to the customer and after that the maintenance phase starts.

Among all the phases if we consider all the phases the development phases and maintenance phase then the maintenance phase consumes the maximum effort. But, among the development phases the testing phase consumes the maximum effort, you may wonder that is it really the testing phase which consumes the maximum effort, the answer is that see we are discussing about a commercial software it is not a student project a student project or a student assignment.

The student just checks that it is working for few inputs and then submits it whereas, we are discussing about a commercial software where it has to be extensively tested the quality needs to be maintained because the organizations name is at stake. You can not just give a software having bug to the customers and therefore, the testing phase consumes the maximum effort for every non-trivial commercial software.

If we represent the effort the form of a diagram the maintenance phase takes the maximum effort, but then among all other phases the development phases the testing is

the one which takes most effort and then comes the coding, design and the requirement specification.

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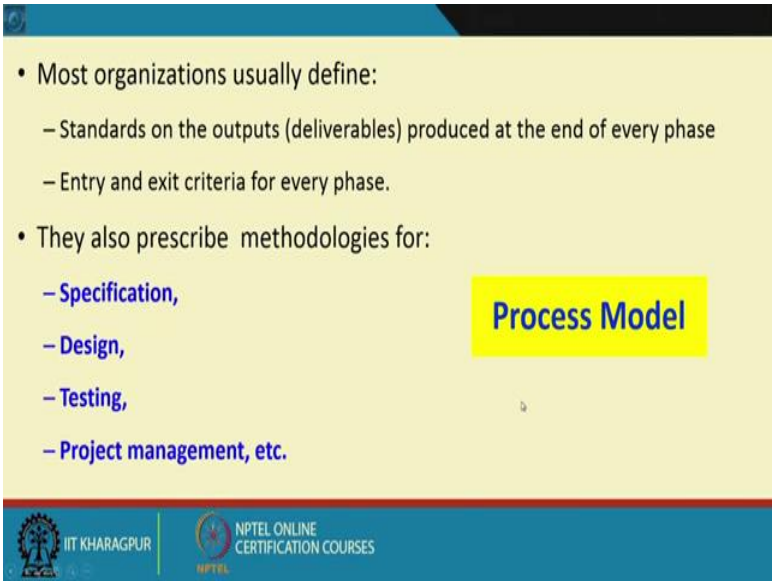

Most organizations they document the software lifecycle model they follow. They are not only they identify all the phases and the activities that take place in the phases. They also identify what are the outputs or the deliverables that are produced at the end of a phase what are the entry and exit criteria for the phase and what are the methodologies that are to be used for by the developers for various phases and this forms a process model document and is a important reference book for all the team members.

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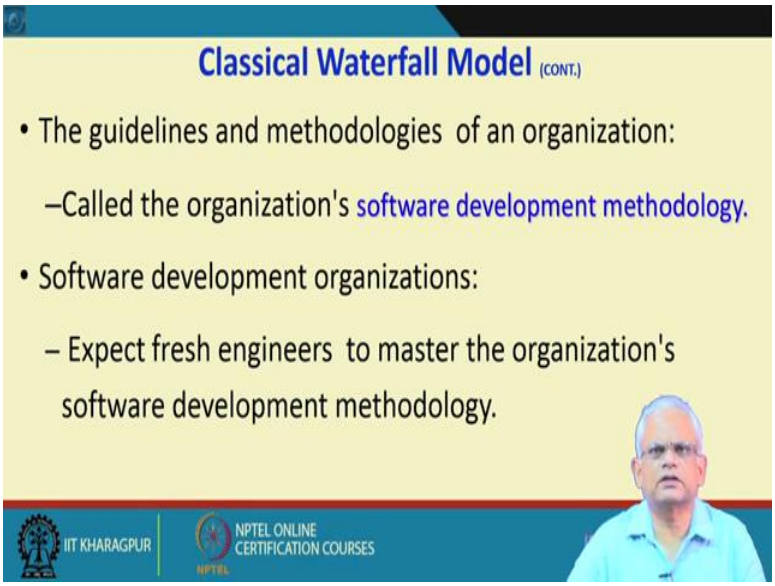

The classical waterfall model we were saying that it is a very intuitive model and other models are derivatives of this model and the development methodology that the organization uses during various phases. These are all documented in the form of a process model book given to the developers and they must understand this well before they undertake the development activities.

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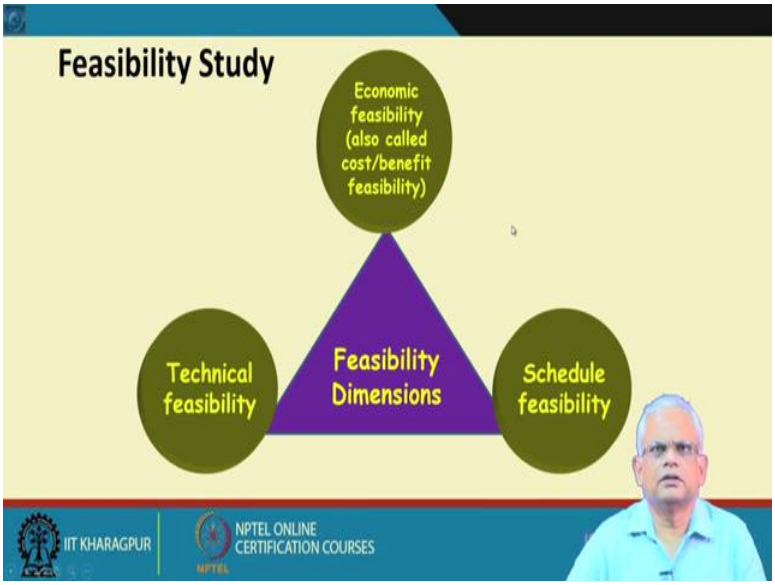

The first activity in the waterfall model is the feasibility study. Here the project manager is responsible to carry out the feasibility study. The project manager needs to identify

whether the project is feasible. So that, it can be undertaken by the development team and there are 3 types of feasibility that the project manager is concerned. There are 3 dimensions of feasibility or 3 types of feasibility that the project manager is concerned.

The first one is technical feasibility whether the development team has the competence and capability to undertake the project. For this the project manager needs to understand various aspects of the development to be undertaken and then assesses whether the team that he has whether they can technically complete this project. Another important dimension of the feasibility is the economic feasibility. Whether the budget that is that would be available would be profitable to carry out the project. This is also called as the cost benefit feasibility.

The third dimension is the schedule feasibility that is the time that the project manager estimates for the project to complete whether the customer agrees to that time or the customer wants the project to be completed in 2 weeks and the project manager estimates that needs at least 3 months and then the project manager would say that this is not feasible schedule wise. Only when all the 3 feasibilities are satisfied the project manager agrees to undertake the project.

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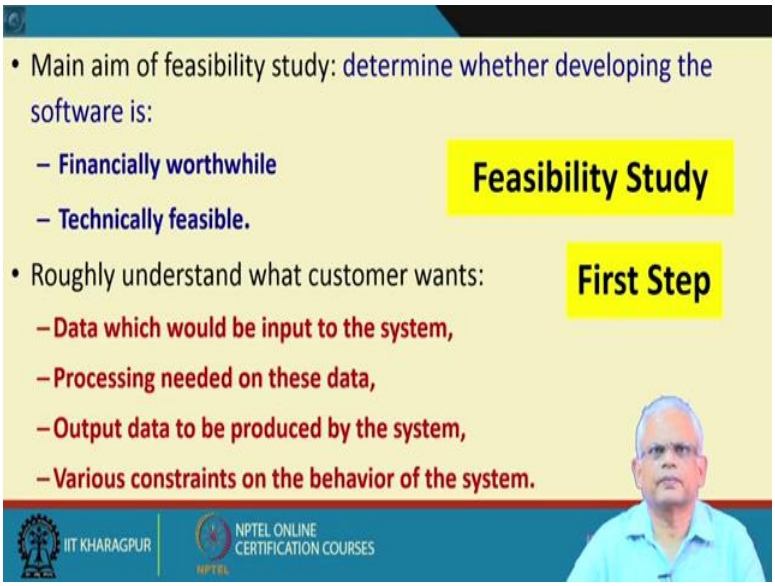

In the economic feasibility the financial implications are considered whether it is remunerative to carry out the work, whether it is technically feasible, schedule wise it is feasible and for this the project manager roughly understands what the customer wants to

do, what the customer wants out of the software, the data which would be input to the system, the processing that needs on the data and the output that are to be generated. And, based on that the project manager identifies that how many developers need to work for what duration.

And, based on that the project manager identifies the cost of developing the project and based on that the budget that the sponsor of the project or the customer provides the budget. Based on that the project manager finds out whether it will be financially worthwhile and based on the activities required as part of the project the project manager identifies whether it is technically feasible and the project manager roughly identifies the time it takes and based on that whether it would be feasible schedule wise.

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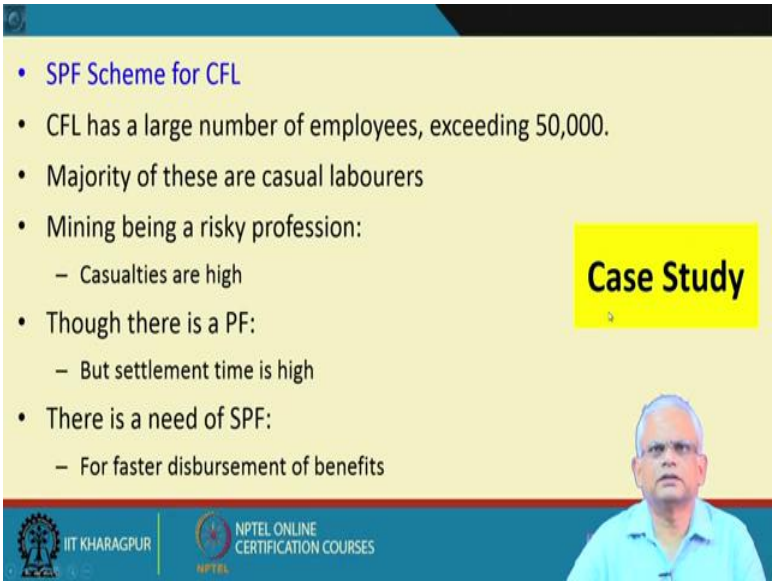

Just a case study to identify just to illustrate what the project manager needs to do to carry out the feasibility study because this is a important activity for the project manager to carry out the feasibility study for a project.

This is a Special Provident Fund Scheme for a Coalfield Limited. Coalfield Limited is a coal mining company and mining being a risky profession. In addition to the ordinary provident fund that are available to the miners the company proposes to have a Special Provident Fund Scheme and for this it has invited bids and the project managers they visit the company and understand what the company needs. They find that the coalfield limited has number of employees exceeding 50,000 majority of these employees are

casual labourers and mining being a risky profession casualties are very high. Though there is a Provident Fund Scheme, but the settlement time is very high and there is a need for a Special Provident Fund Scheme where the fund can be disbursed immediately.

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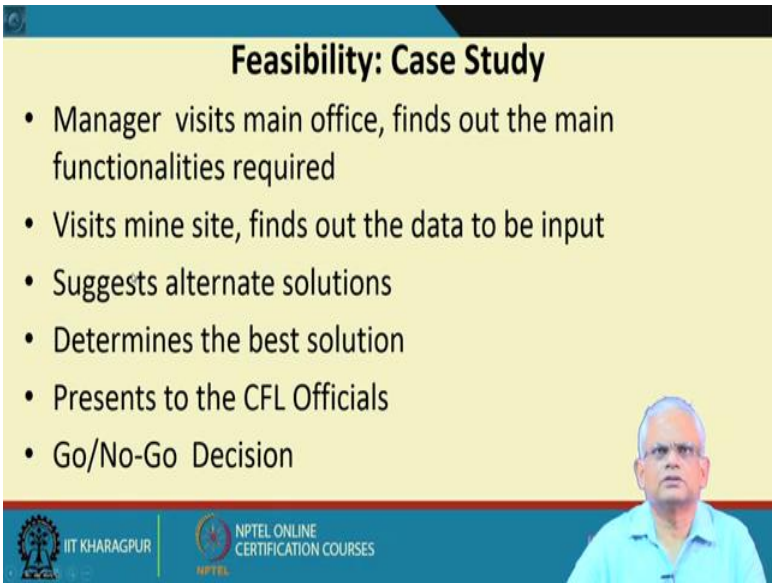

And here the manager first visits the main office of the company, finds out what are the functionalities required from this software, finds that there are various mine sites where the data will be input about the miners who are working for a specific day and the contribution they make for that day towards the Special Provident Fund Scheme.

So, the project manager visits some mine sites, finds out what data to be input, how many data what format and so on and then suggests alternate solutions. For example, there may be local databases at the mine sites and this will be periodically uploaded to the central database at the company's head office and based on that the settlement when there are claims for the Special Provident Fund Scheme these would be made, but then in this the project manager finds that there will be a delay because instantly claims cannot be settled.

But, another solution may be that the local mine sites only upload the data and these are immediately updated on the mine side. There are no local databases at the mines, but these are just data entry stations where the data is immediately updated on the server at the main office of the company, but then the problem here is that an extremely reliable communication link is required because it needs to instantly update the data and the main

computer whereas, the other option when there is a local database the communication link may not be so reliable because only once in a while the data at the main office is updated and the cost for these different solutions is worked out by the project manager and then discusses the Coalfield officials which option would be suitable for them and what is the budget, what is the cost for each one and the budget available, based on that the project manager makes a go, no go decision.

So, this is the typical sequence of activities that occurs that the project manager understands what are the requirements main requirements for the software, what are the alternate solutions that can be proposed to solve the functionalities that are required and what is the cost for each of these alternate functionalities, what is the best solution among these alternate solutions and based on that whether it is financially feasible, technically feasible and schedule wise feasible is determined by the project manager.

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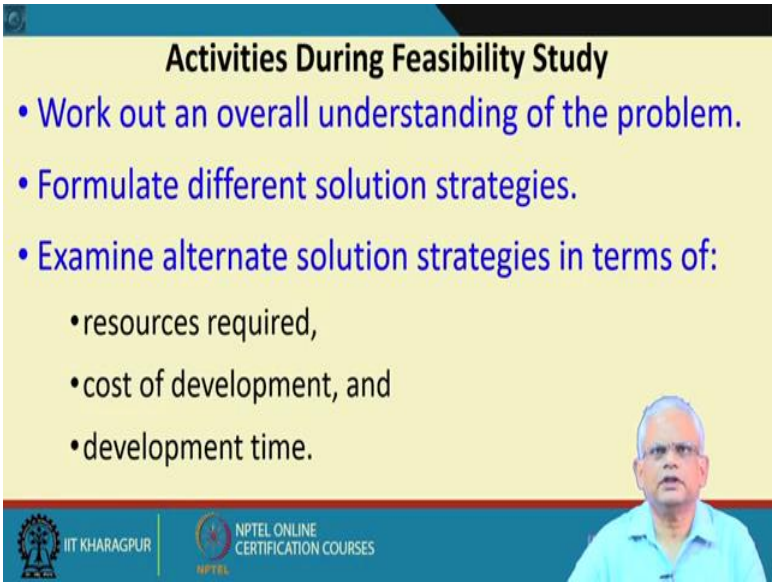

That is what we have just written down here that the project manager first develops an overall understanding of the problem, formulates the different solution strategies and examines the alternate solutions in terms of the resource required, the cost of development and development time.

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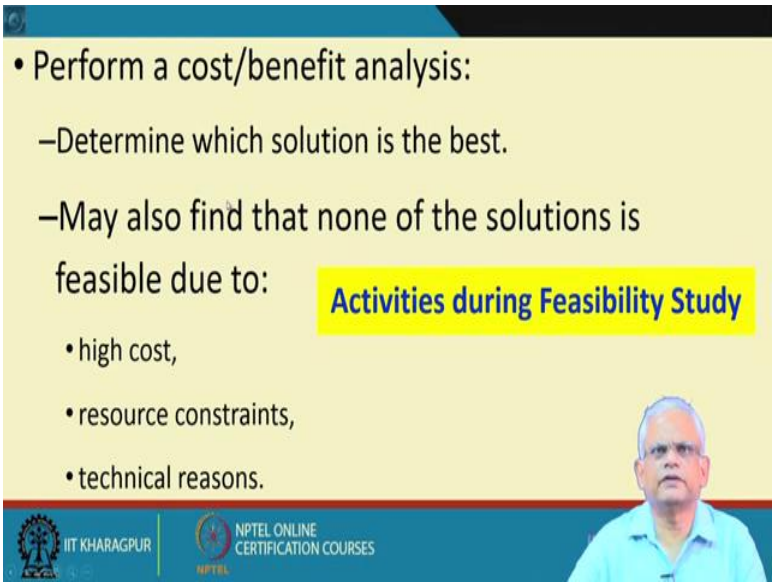

And forms a cost benefit analysis and based on this the project manager finds that it is feasible to carry out the project or find that it is infeasible due to high cost, resource constraints, technical reasons or schedule reason.

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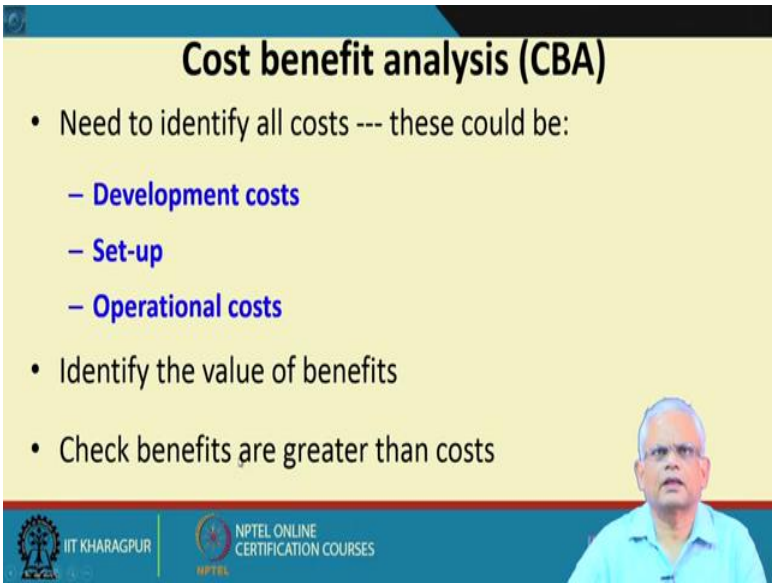

The cost benefit analysis is a important activity during the feasibility study here the project manager needs to identify all costs. The cost can be development costs, the setup cost, operational cost and also identify all the benefits that would accrue. For example, the benefit may not only be the financial budget that it provides the company gets, but

also the experience it builds up the reusable software that it makes and so on. At the end of the cost benefit analysis the project manager needs to check whether the benefits are greater than the costs for the project to proceed.

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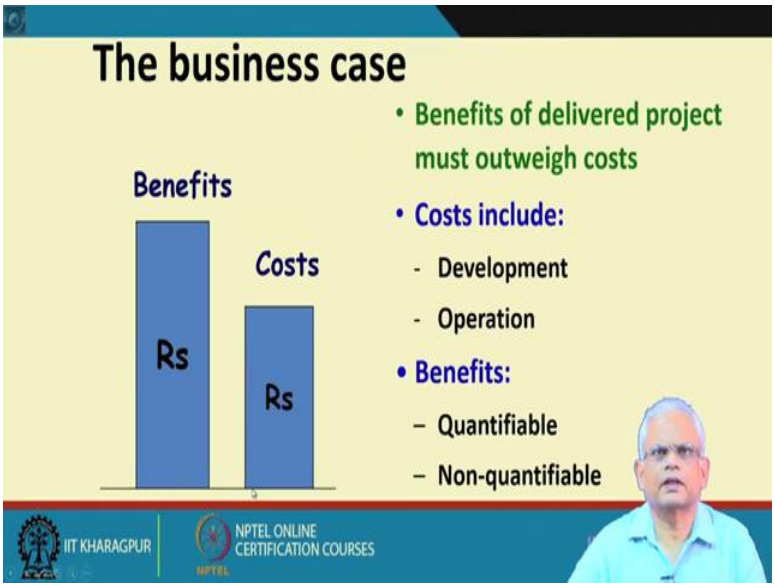

So, this is what I have just pictorially written here that the benefits are more than the costs then the project is undertaken. The costs include development, operation setup etcetera and the benefits some are quantifiable benefits like how much the customer is willing to pay, but then there are non-quantifiable benefits such as the experience reusable software and so on.

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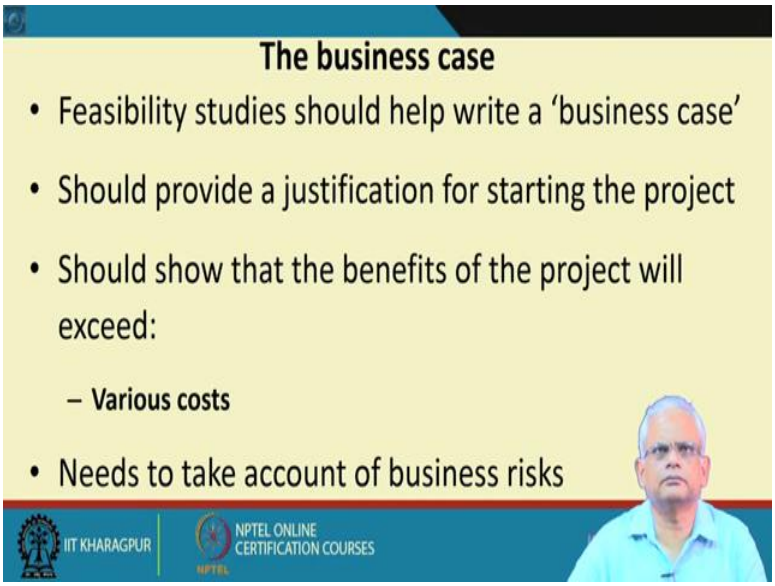

One of the important activity of the project manager is write to write the business case. We had said that this is one of the initiating processes to write the business case. The feasibility study once it is undertaken helps the manager to write the business case. Here the project manager provides a justification for starting the project and shows that the benefit of the project exceeds the cost and also identifies the business risks.

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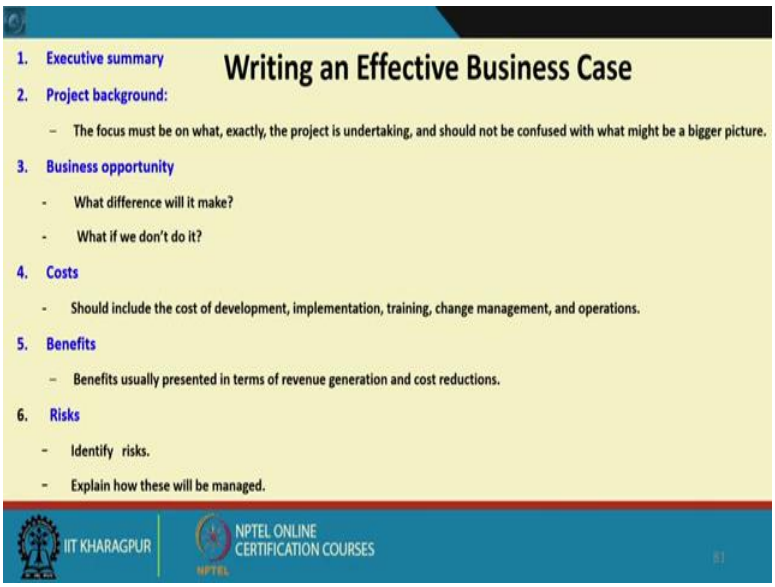

The business case is a very important document and writing a business case is a important project management process initiating process almost anybody aspiring to

become a project manager needs to write a business case sometime or other. Here for writing a business case need to have an executive summary of what is required, the project background, the business opportunities that is what are the benefits that the project will bring, what we will lose if we do not do the project, what are various costs that will be incurred as the project progresses, the benefits and what are the risks and how these risks will be managed.

The business case is submitted to the top management. This gives them an overview of what is involved in the project. How it will help the company and what are the costs and benefits and what will be the risks here and what are the plans of risk management. This forms an important project initiation process. With this discussion we will just stop here and continue in the next lecture.

Thank you.

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Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur

Lecture - 06
Life Cycle Models – II

Welcome to this lecture. In the last lecture we had discussed about the Life Cycle Model and especially the classical waterfall model and as part of the life cycle model we had looked at various activities that are undertaken during the development process. And we had spent some time on the feasibility study which is a important activity undertaken by the project manager with the help of a case study we had identified what are the activities that are taken up by the project manager during feasibility study and at the end of the feasibility study the business case is written. This is one of the initiating processes writing the business case and we had seen that how an effective business case can be written.

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Now, let us proceed further in this lecture based on the discussions that we had the last lecture we will discuss about the iterative waterfall model we will look at the V model, evolutionary model and prototyping model that is the plan for this lecture.

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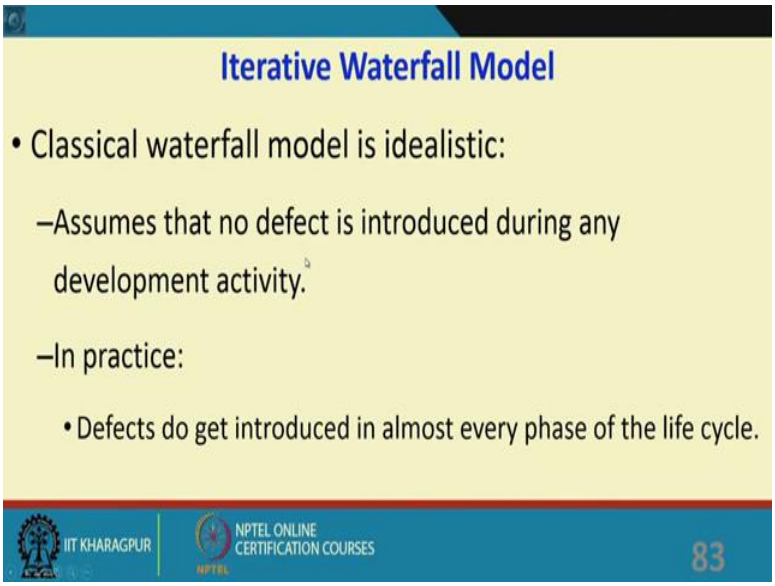

The classical waterfall model is an idealistic model, the main problem is that it assumes no defect is introduced during any phase each phase is completed and the next phase starts and so on until the project completes, but in effect there are lot of defects that are introduced in almost every phase of the lifecycle.

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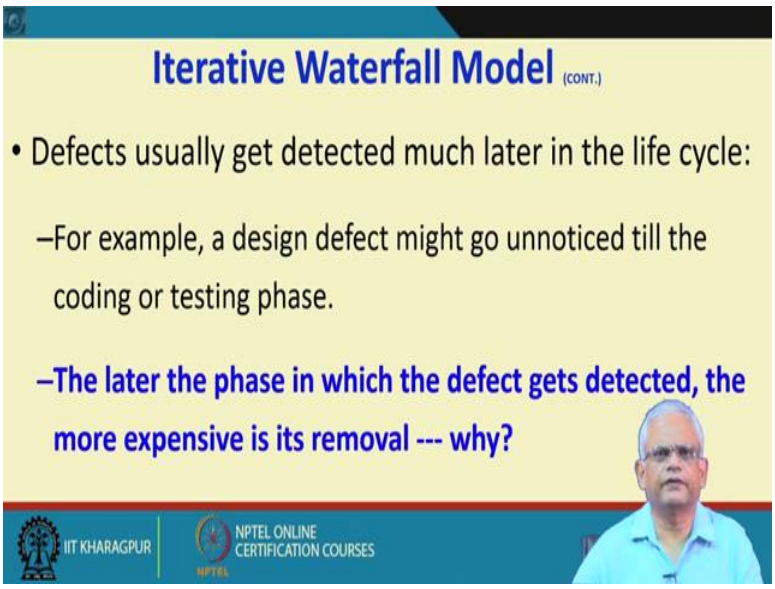

And that is the reason why it may be necessary to go back to a previous phase once a defect is identified. For example, a defect may be identified in the design phase that a requirement specification a functionality was missing and then need to take up the

requirements activity rework the requirement specification document and then again come back to the design phase, that is the reason why we need the feedback path. Once the defect occurs in a phase there is a mistake by a developer it often remains unnoticed until a later phase and then once it is noticed in a later phase, it is then removed.

But the question here is that the later the phase the more expensive is to correct this defect, why is that? The answer is not very far to sake because if defect occurs and it is removed immediately, then that is the best thing the cost will be minimal, but let us say the defect is removed after one phase that is the next phase then only two phases are affected; only two phase documents need to be changed. But if let us say a defect is occurs in the requirement specification phase and it is discovered during testing, then not only the requirement specification document has to change the design needs to be changed, the code needs to be changed and these are lot of activities and lot of documents need to change. And that is the reason why the later the phase in which the defect gets detected the more expensive is the removal of the defect this is a very important concept and it is known as the phase containment of errors.

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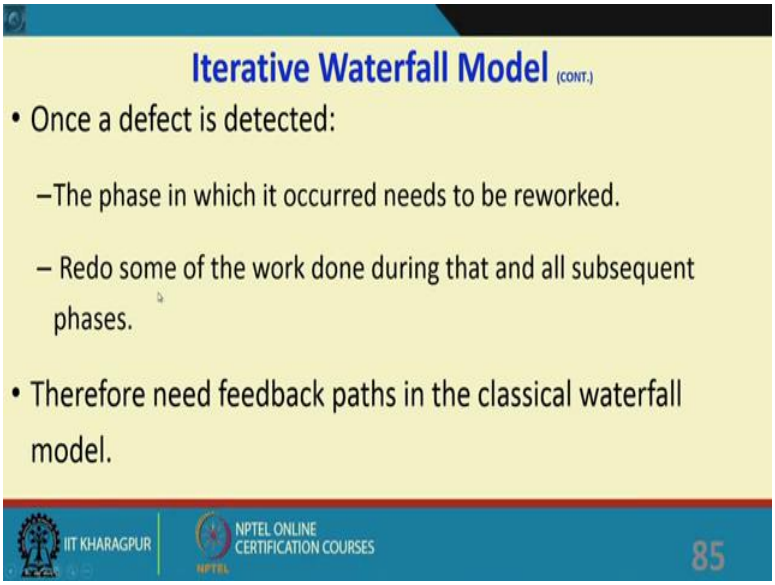

So, that is what is written here that once the defect is detected redo the work in the previous phases. And therefore, if there are lot of intermediate phases it has crossed many phases before the defect is detected, then lot of rework is needed.

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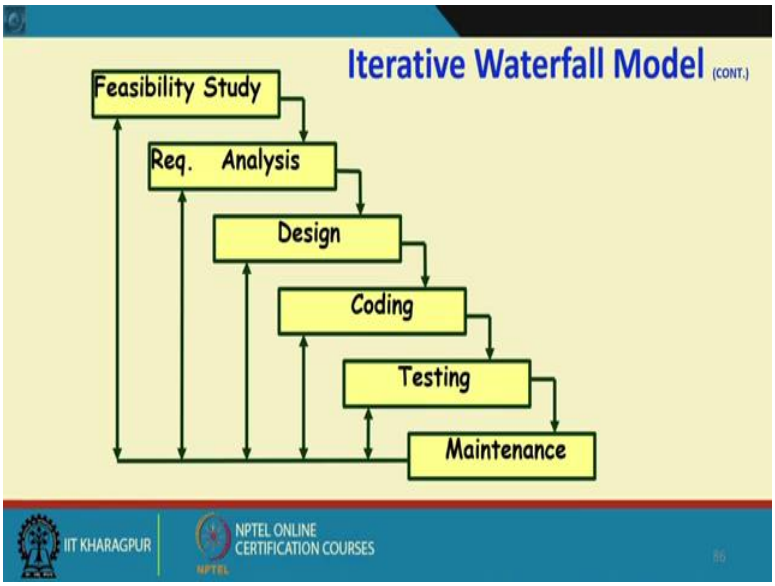

In the iterative waterfall model is very similar to the classical waterfall model, so you can see the waterfall here, but also there are feedback paths; this feedback paths make the model realistic unlike the classical model where one phase completes and then the next phase completes and so on. Here we assume that there can be a defect which will get noticed during the testing phase and we need to go back to the design phase or maybe the requirement analysis phase and then fix that defect and not only that fix the work in the later phases as well.

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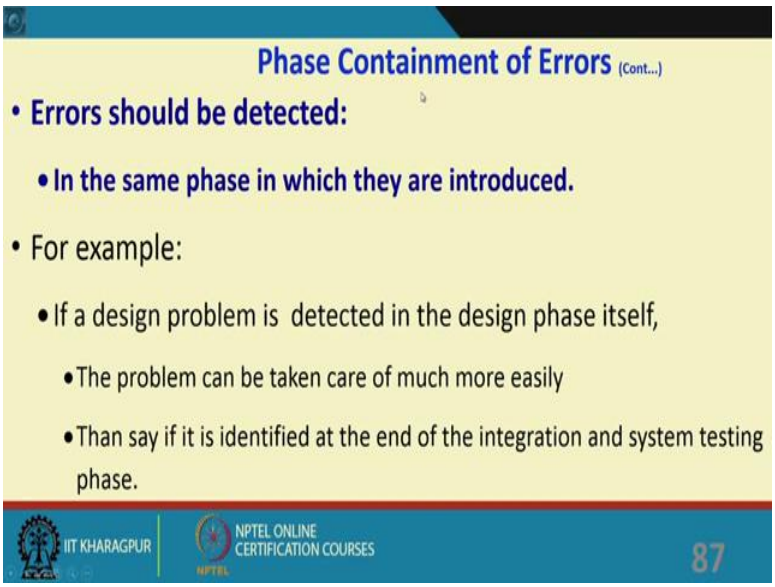

As we are saying that it is a important concept for the errors when they are the mistakes are made by the developers they must be fixed in the same phase that is the ideal that the defects are detected and fixed in the same phase that will incur the lowest cost. The more delay occurs in detecting the defect, the more expensive it will be.

If a design defect is detected in the design phase itself, then it just need to change the design document, but if it is detected during the testing phase not only the design document needs to be changed, but the code needs to be reworked and that will be much more expensive. So, this is called as the phase containment of errors it is an important principle that, once a mistake occurs mistake or error occurs this must be contained to the same phase it is called as a phase containment of errors.

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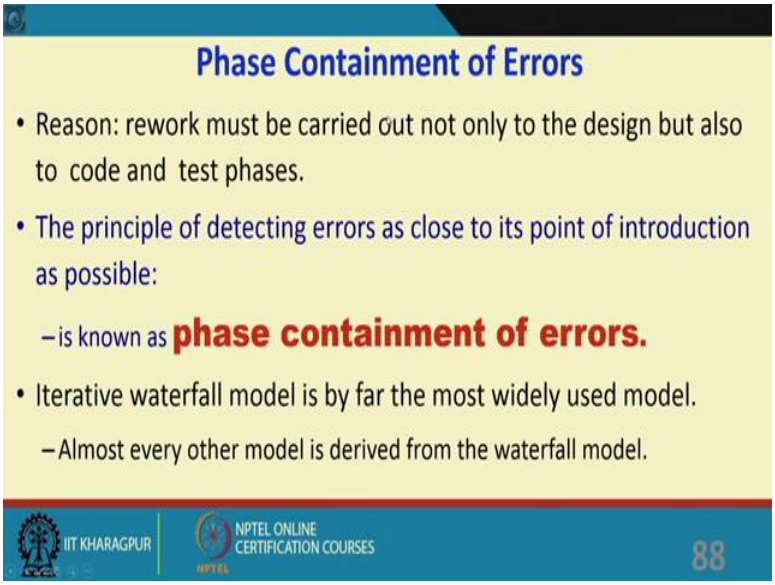

The principle of detecting errors as close to its point of introduction is called as the phase containment of errors it is a very important principle and very intuitive also. So, this is the iterative waterfall model, the phases are identical to the classical model excepting that the feedback paths are available.

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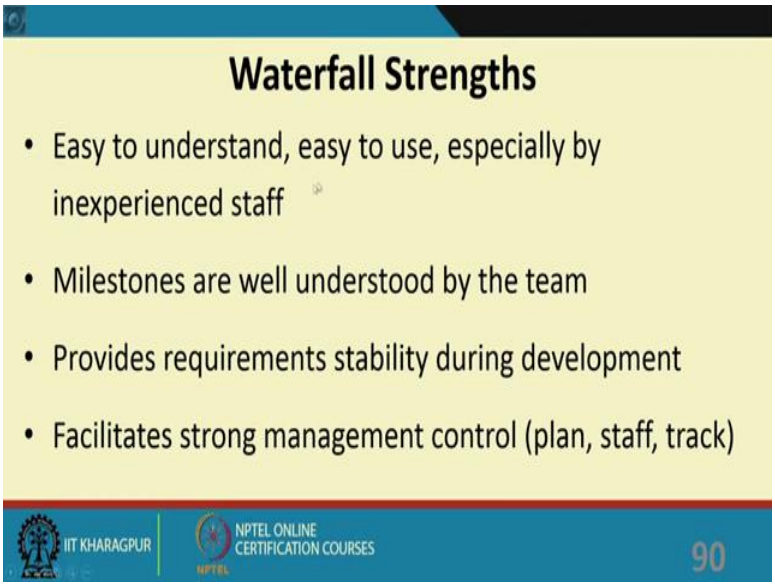

The main strengths of the waterfall model is it is very intuitive easy to understand all the developers can very easily understand and use the model. The milestones are clear understood by the team and the project manager can easily monitor the progress, it provides the required stability. Management the project manager can have a strong control put scheduled milestones and monitor, but then it has lot of deficiencies.

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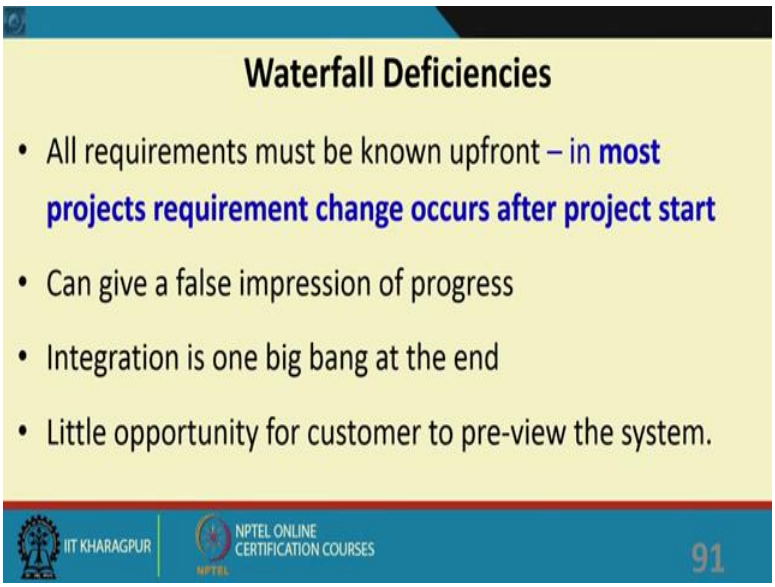

One of the major problem with the waterfall model is that it is not very suitable to handle changes to the functionalities, but in almost every software project the requirements

change there are many reasons why the requirement may change. For example, the customer may not be may not have thought that some functionalities will be required because he did not understand or did not think you cannot grasp what are all required at the beginning of the project, but as he looks at functionalities then he can make out that certain functions will be required. The other is that as the development takes place, the business itself might undergo some change and therefore, changes will be required.

So, there are many reasons why changes will be required during the development of the software, but in the waterfall model the requirements are fixed upfront and then the design coding and testing are undertaken and there is no way provided by the waterfall model to make any changes, this is possibly the biggest shortcoming of the waterfall model. The other shortcomings are that here lot of documents are produced phases are completed, the requirements is complete, design is complete documents are produced, code is complete, but during testing found that most of the requirements are not met.

So, the project manager gets a false impression of the progress that, the progress is the project is proceeding nicely according to the plan, but during the testing phase finds out that very little has been done. The main reason for that is that, the integration is a big bang at the end till then everything was fine, but at during integration it has found that most of the functionalities are not working.

Another problem with this model is that, once the customer gives the requirement they have no chance to see the system till the end when it is delivered to them. It would have been really good if the customers were involved they could see that what are the functionalities that's they are being developed they could have suggested modifications and so on. These are the major difficulties with the waterfall model the biggest one is that it does not have the flexibility to handle the requirement change, false impression of progress because phases are getting complete, but integration is on big bang at the end where it is noticed that the progress is very less; most of the functionalities are not working and the customer involvement is very minimal, the customer cannot give suggestions.

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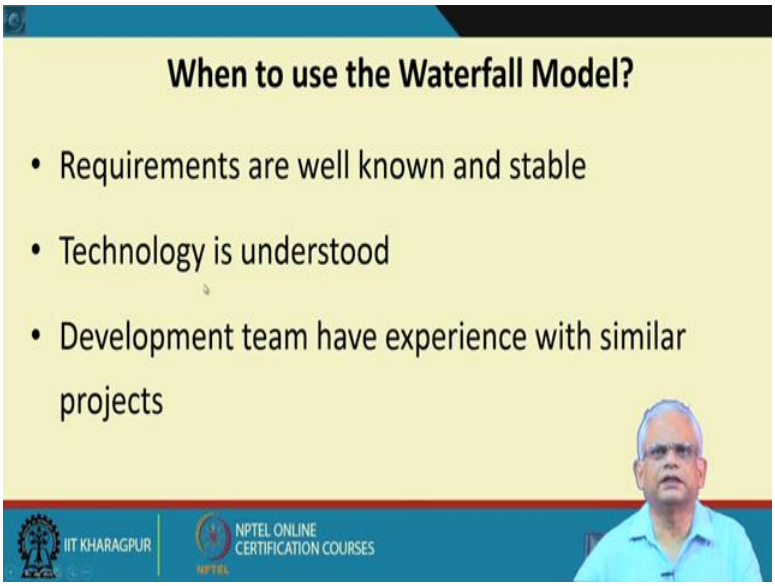

But then the waterfall model was popular and it is also successfully used in some types of projects. The projects for which waterfall model is suitable is that, the requirements are very well understood and known to the customer they have used similar software they know what all is required, it is not that something they are thinking of a new type of software it is a regular software. Requirements are well known and stable, technology is understood, development team have experience with similar projects.

So, as you can see here that the project is rather routine project where the development team have done similar work, they are just trying to develop one more of that you can say that the challenges here are less because the requirements are known what the software will finally, look like what functionalities is very well known. The technology is understood, development team had experience in similar projects and for such simple projects waterfall model can be used and this happens to be the most efficient way to complete these projects.

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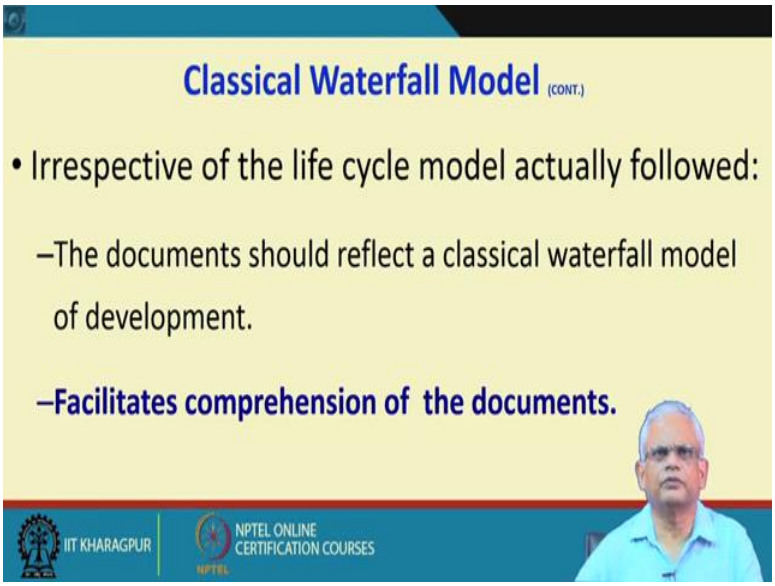

But what about the classical waterfall model, the iterative waterfall model has its use we saw that for simple projects they can be used, but what about the classical waterfall model? Because here this was the idealistic model where it was assumed that there is no defect in any phase all phase proceed idealistically, but the use of this is that the final documentation should appear like the software is developed in a using a classical waterfall model, there are no backtracking no mistakes nothing and if the document is written that way then it facilitates comprehension of the documents.

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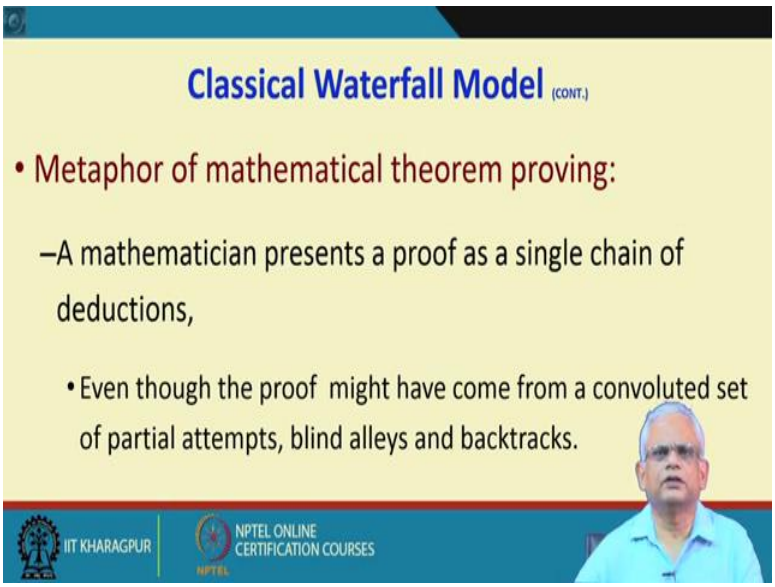

Just to give an analogy you may be trying to do a mathematical theorem proving or maybe trying to solve a problem mathematical problem and you might many times do rework mistakes, go back to the previous phase previous step and so on. But if your final document for this problem has all these things that where you did mistake where you went back corrected and so on, then it becomes extremely complicated for somebody trying to understand.

The best for somebody to understand is that you give him the correct steps as if you had not committed any mistakes and so on and that is what is basically the classical waterfall model that all phases proceed without any mistakes. Now, let us look at the V model this is a variant of the waterfall model only small changes are there small derivative of the waterfall model essentially all the activities are there the waterfall model.

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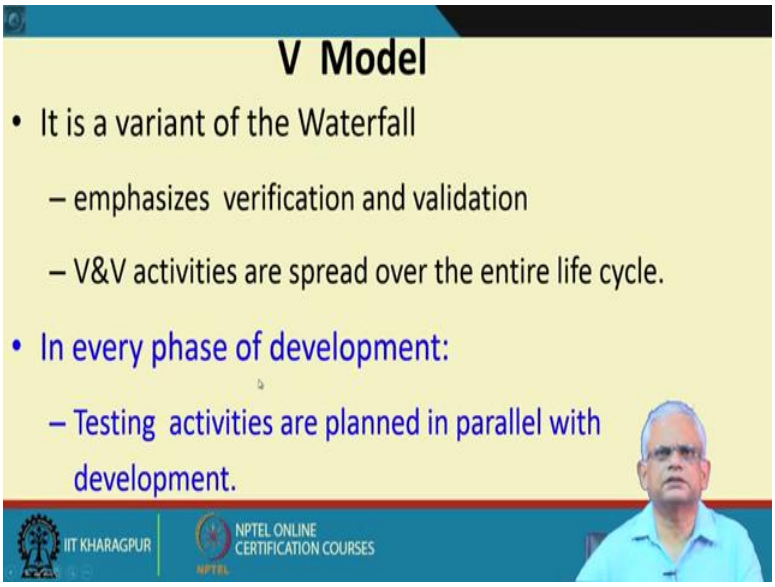

Now, let us look at the V model, the v model emphasizes on the verification and validation activities throughout the lifecycle. We had seen that in the waterfall model the testing activity was only at the end, initially there is requirements specification then there is design coding, but the testing activities are only at the end and that is the reason why the testing activities take long time. And the other problem is that what do the testers do, till the testing phase starts what do they do?

The V model gives answer to that, it emphasizes verification and validation and normally this model is used where the software is supposed to be more reliable. For example,

those software being used in critical applications the V model is used. Here in every phase of development there are some test activities for example, planning the test cases and later they may get executed, but every phase there are some test activities that occur.

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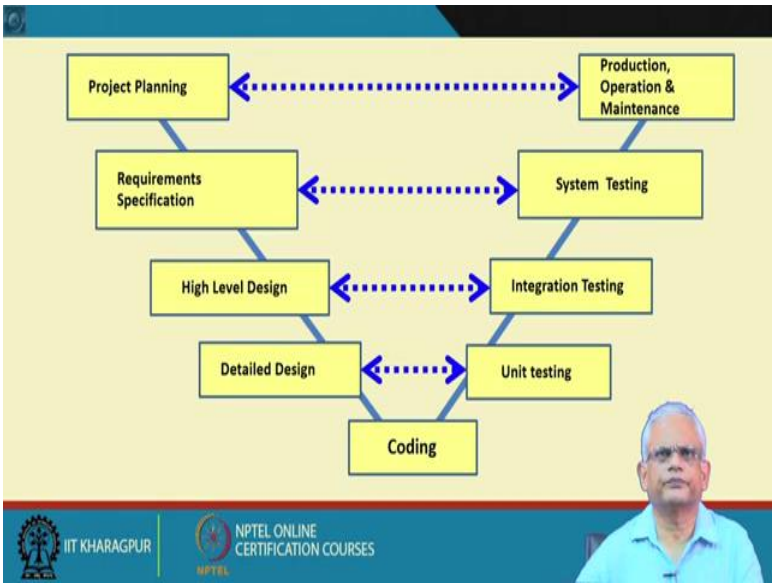

So, this is the diagrammatic representation of the V model looks like a V here the model representation and that is the reason why it is called as a V model. During the requirement specification, the system test cases are developed which are executed during system testing during high level design the integration test cases are developed, during the detailed design the unit test cases are developed which are used during the unit testing after the coding is complete. As you can see here that there is a emphasis on verification and validation activities and every phase there are some test activities unlike the waterfall model where the test activities occurred only when only in the integration and system testing.

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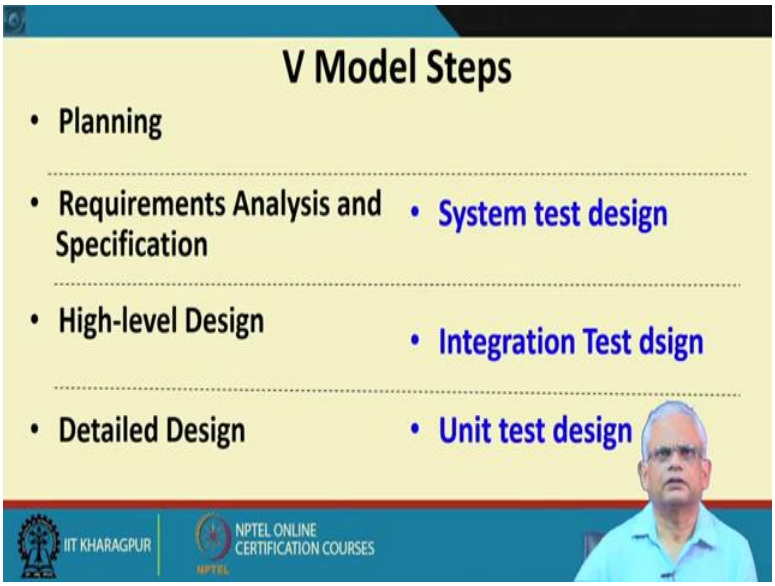

Here this slide it just says what are the system the test activities occur during the requirement analysis and specification system test case design activities occur, high level design the integration test design activities occur and during the detailed design the unit tests design activities occur.

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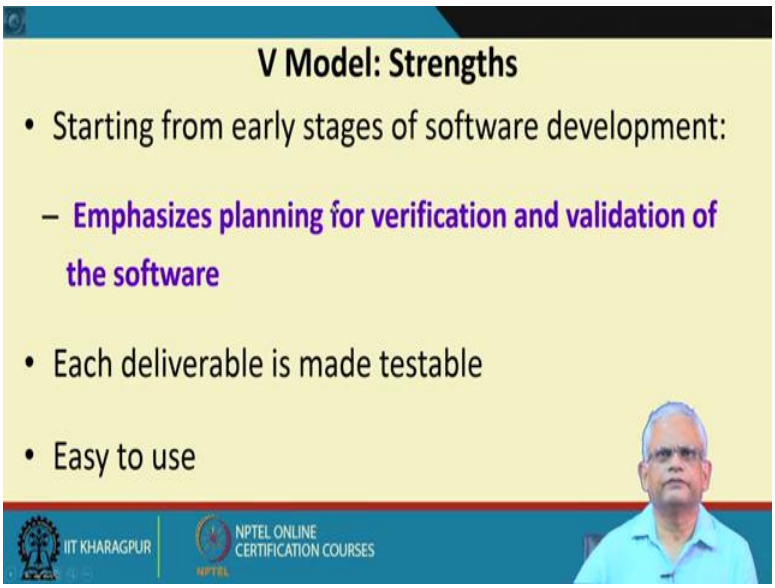

The strength of this model is that it emphasizes verification and validation, its very similar to the waterfall model excepting that every phase has some verification and validation activities.

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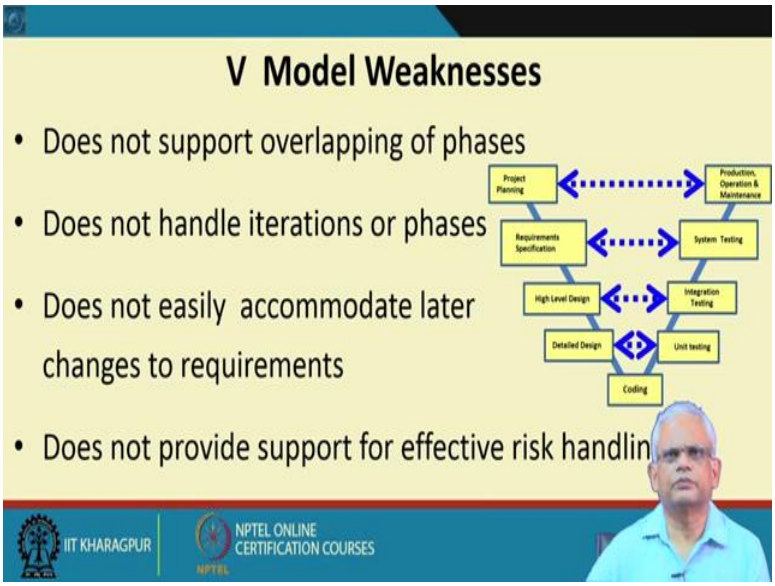

But then it has many weaknesses just like the waterfall model here it does not support overlapping of the phases that is one phase has to complete then the next phase starts, what if some of the developers few of the developers complete first should they wait for the other developers to complete just idle or they can get started with the next phase this model does not allow that, they wait for all phase the phase to complete and only then they can start the next phase. Does not easily accommodate later change requirements, does not provide support for effective risk handling.

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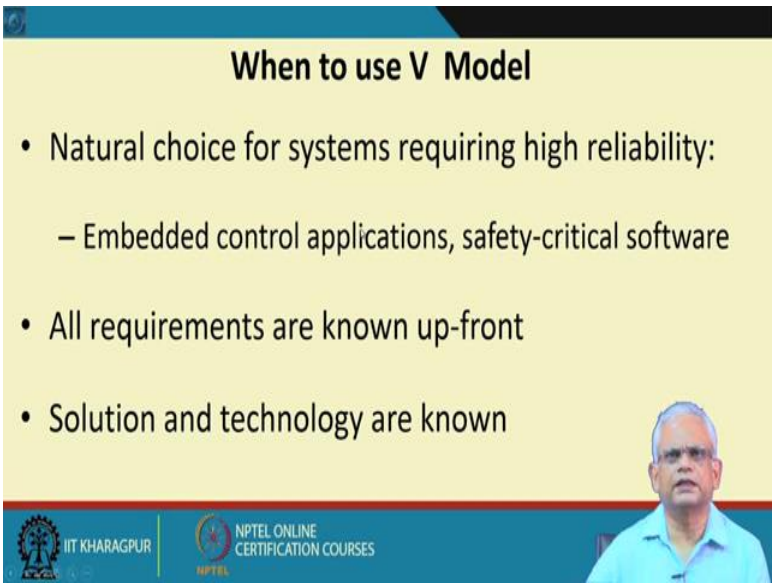

So, these are some of the problems of the V model, but then it is the natural choice for systems requiring high reliability and it is used when all requirements are known upfront and the technology is known. Now, let us look at the prototyping model which is also a small variation of the basic waterfall model.

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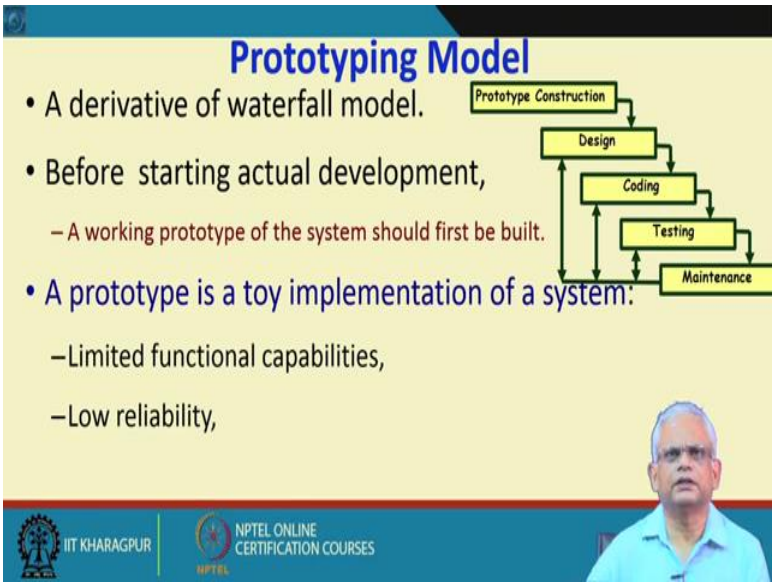

If we look at the diagrammatic representation of the prototyping model the initial phase here is prototype construction and after that the design coding etcetera are carried out. Here just like the waterfall model the activities are organized, but then before starting the actual development a working prototype must be built. What is a prototype? A prototype is a toy implementation of the system which has limited functionalities, low reliability and so on. But the question is that why do we need to develop a prototype before we start to develop a system, what are the advantages of developing the prototype?

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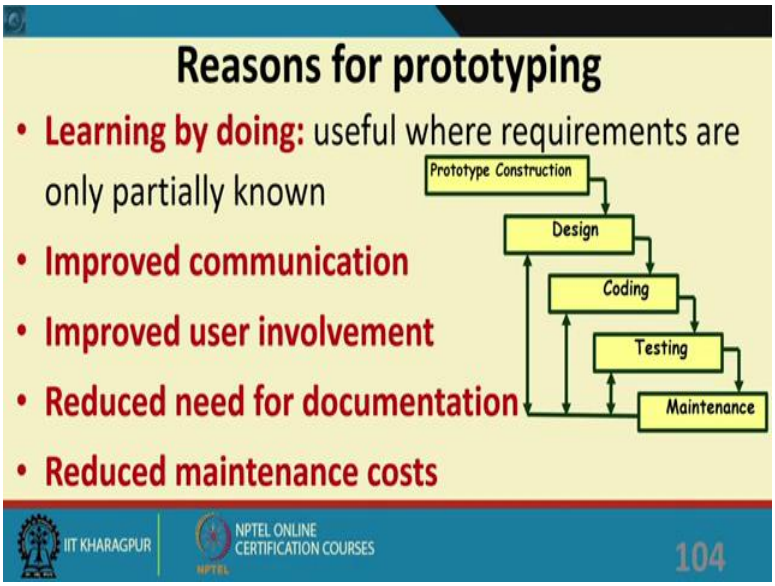

The first thing is that once you develop the prototype we can know what are all the requirements that should be there, if there are any requirements which are missing. If the customer can look at the prototype, then you can find out what are the missing requirements. There is a improved communication with the customer, there is a user involvement because unlike the classical or the iterative waterfall model where the after the requirements are given the only thing that the customer sees the delivered software here in contrast in this model the prototype is constructed and given to the customer for his comments and also since a prototype exists, it reduces the documentation, reduces the maintenance costs. And also if there are technical risks that the development team anticipates then they can through the prototype construction they can address those risks.

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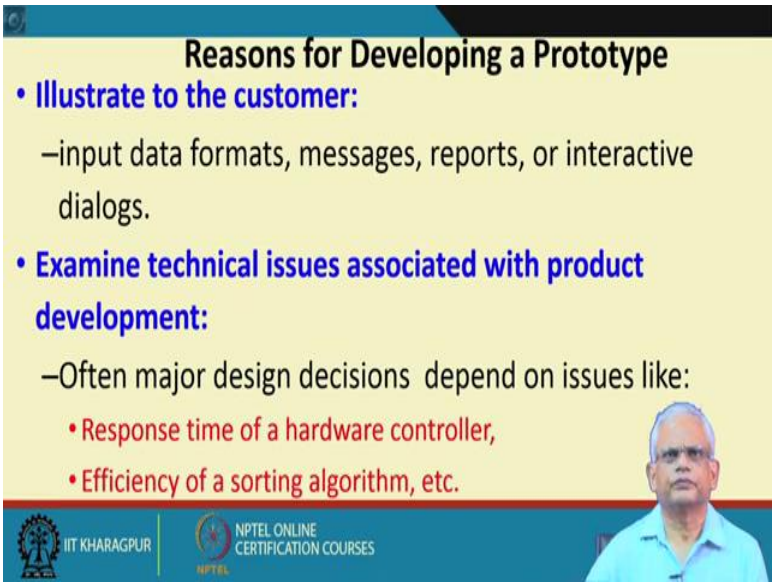

The reason for developing the prototype is to illustrate to the customer what is the software is going to look like, what will be the input data formats, messages, reports, interactive dialogues. The technical issues associated with the product development, the developers can experiment and find out what is the best way to handle technical issues sometimes the major decisions depends on issues like the response time of the hardware controller.

And here by developing the prototype the designers can have their solution based on the hardware, the response time of the hardware controller at the upfront rather than proceeding throughout all the development activities and finding out that the hardware controller response time is slow and they need to change everything. So, developing a prototype helps in getting the customer feedback and also many technical issues are clarified and the development can proceed correctly.

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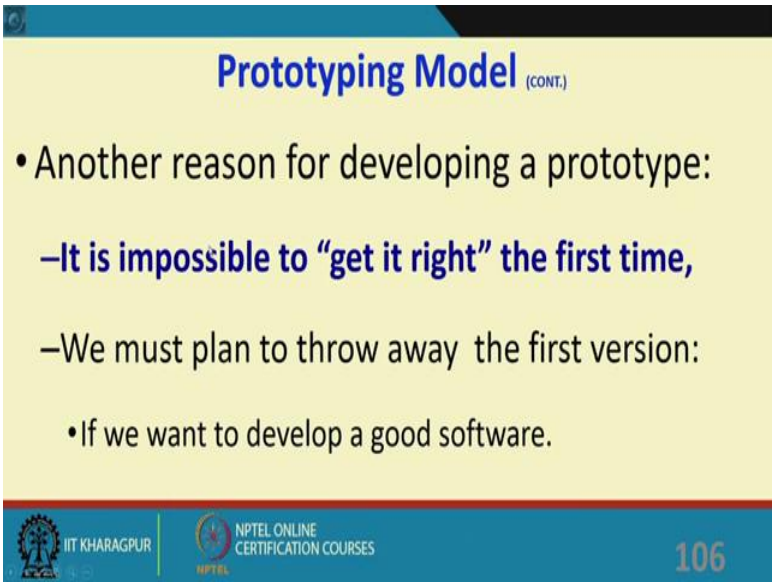

Also when a prototype is developed the once it is used for illustrating to the customer it is thrown away and the actual system is developed, but then the experience that the development team gets in developing the prototype that goes a long way in developing a good software.

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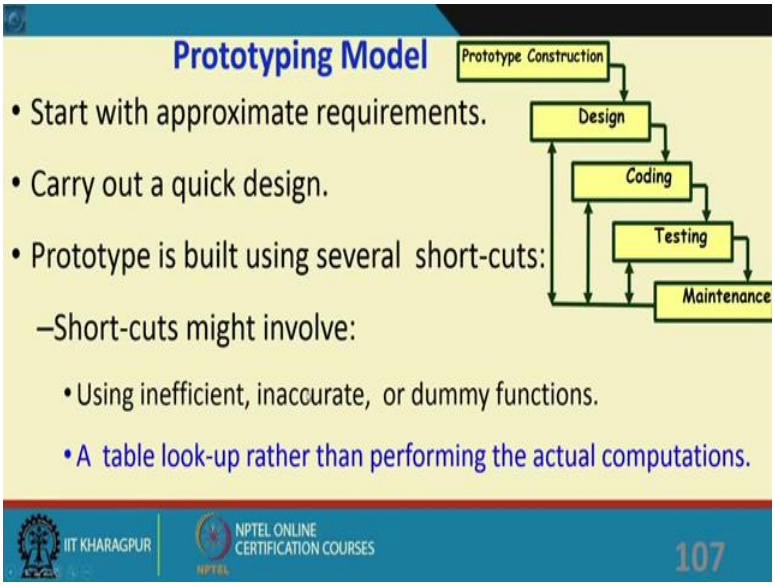

Here in the prototyping model first the approximate requirements are obtained, a quick design is construct is made and then the prototype is constructed using various shortcuts.

The shortcuts can be like inefficient, inaccurate or dummy functions; a table lookup rather than writing the code for the actual computation.

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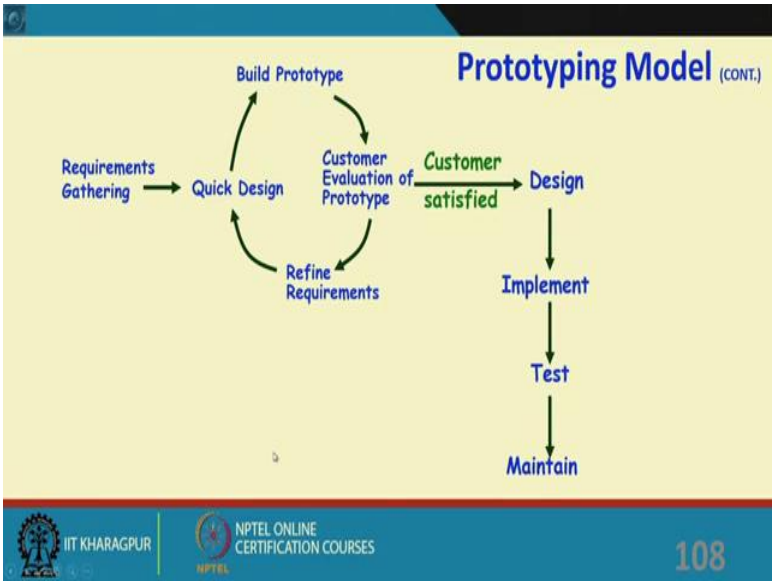

We can represent that in this diagram that initially a quick design is done based on the requirements that have been gathered prototype is built. Submitted to the customer for evaluation based on customer feedback, refine the requirements again have a quick design change the prototype and so on until the customer is satisfied with the prototype then a waterfall model is used for developing the software

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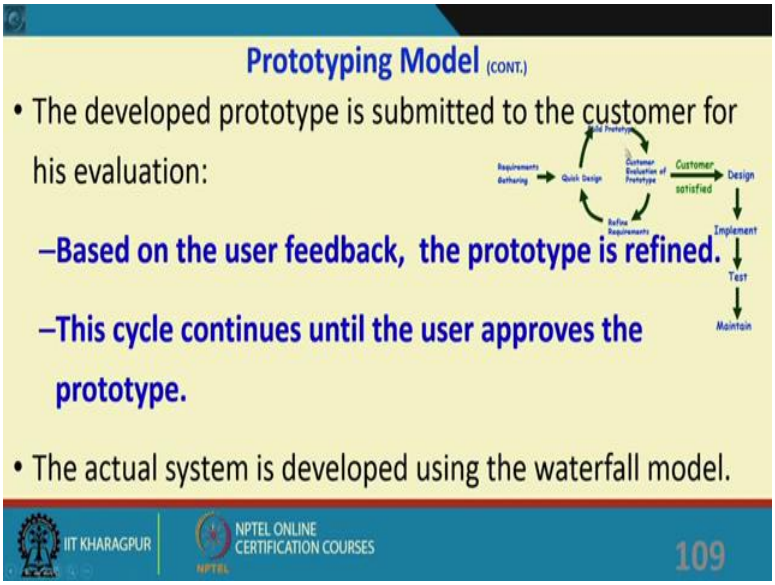

So, the it is a small variation of the waterfall model only the initial prototype construction that is added here.

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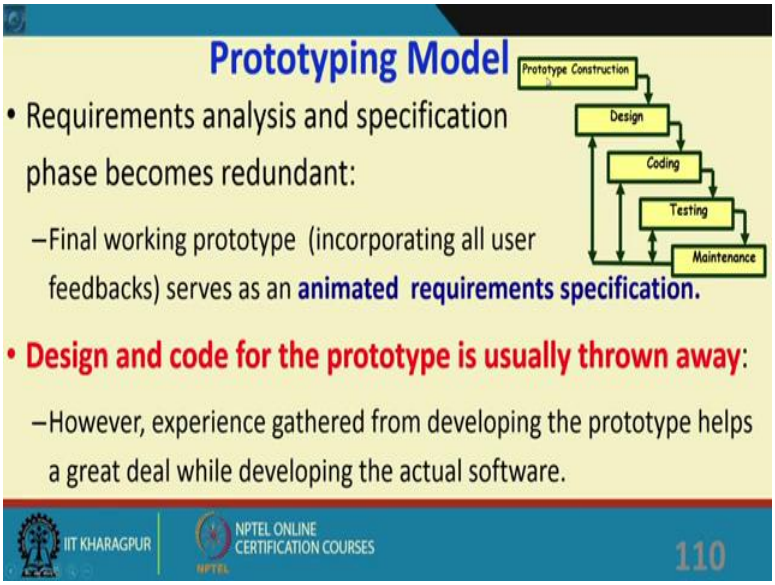

And because of the prototype is kind of a requirement specification animated requirement specification, a requirement specification document may not be required, but one thing that we must be clear here is in the prototyping model once the prototype has been used to demonstrate to the customer and address the technical issues, the prototype is thrown away and then the development starts.

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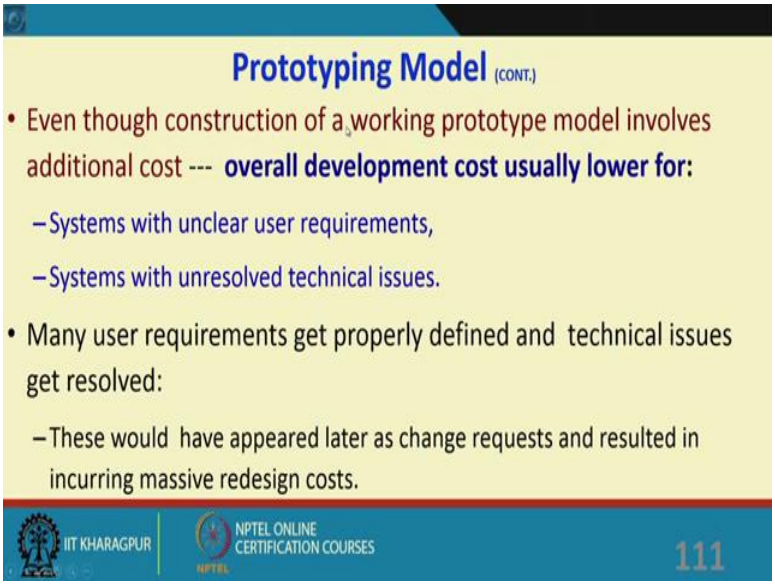

But does the prototype construction add to the cost and the development becomes more costly? No not really because for systems with unclear user requirements with unresolved technical issues the prototyping is actually a cost effective way of developing the software because without a prototype we might have to change the software many times and that would make it much more expensive that would incur massive redesigned costs.

And even though there is a small upfront cost of developing the prototype, but that will be more effective cost effective, then not using the prototype and finally, finding out that the customer has lot of dissatisfaction and suggests many changes and also the technical issues are unresolved as a result many changes occur. With this discussion we will just stop here and continue discussing a few more lifecycle models in the next lecture.

Thank you.

Software Project Management
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Lecture – 07
Life Cycle Models – III

Welcome to this lecture, last time we were discussing about Life Cycle Models and we had said that the waterfall model is a intuitive and important model and many models have been derived from the water fall model.

We had looked at the some of the derivatives from the waterfall model and today's lecture we will first see how the waterfall models have done, what are the good things about them and what are the short comings about them. And, then we will discuss about other models which overcome specific short comings of the waterfall model. There are some severe short comings of the waterfall model because of which they are not as popular as they were about 30-40 years back.

And one of the main problem that is being faced is that the type of project that are being handled are different from those that were handled 30-40 years back. Now, the projects are short and the code is not written from scratch because lot of code is available lot of code is being reused and only small modifications and tailoring is done which we called as service oriented software.

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So, today we will first look at the some reflections on the waterfall based models and advantages disadvantages etcetera, waterfall model and we will see the models that have been proposed to overcome the specific short comings of the waterfall based models. We look at the incremental, the RAD and evolutionary model.

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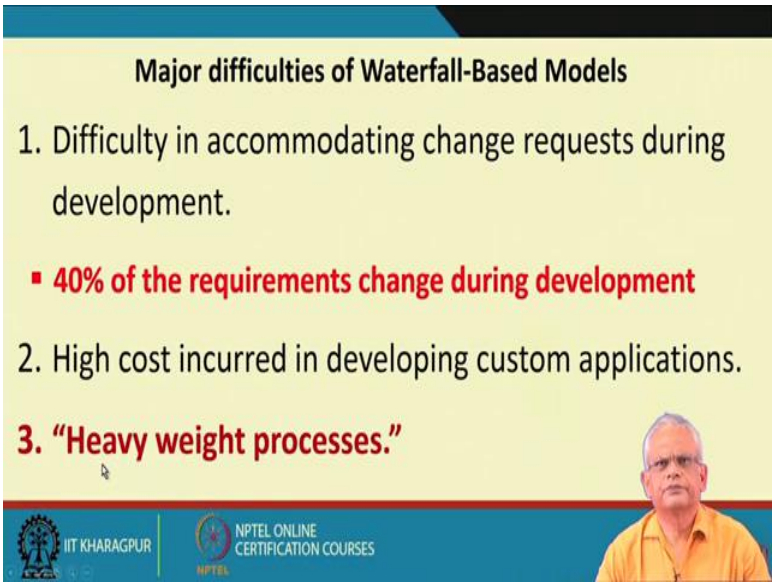

First let us look at the major difficulties of the waterfall based models. Possibly the most important difficulty is there is no way change requests can be handled in the waterfall model. In all waterfall based models that is the iterative waterfall model, the classical waterfall model, the prototyping model, the V model and so on. The requirements are gathered up front and these are documented, it is assumed that this will not change and then plan is made based on this documented requirements, design is done on this requirements and then coding and testing is done with respective these requirements.

But then in reality, the present projects the requirements keep on changing as development proceeds. Typically about 40 to 50 percent of the requirements change after the initial requirement specification. And, this is possibly one of the major short coming of the waterfall based model, 40 percent of the requirement change during a typical development. The second problem is that many of the projects now are for customizing applications as we are discussing about the service oriented software where the organisation has some software and would tailor it for specific needs of another customer. If we use the waterfall model for this custom applications there will be huge

cost because the entire thing has to be documented, specification laid out, design and so on.

The third problem is that the waterfall model is called as heavy weight process, it is called a heavy weight process because lot of documentation is produced as part of the process, everything is supposed to be documented. There is starting from requirement specification, the reviews, review results, the design various types of plans everything is to be documented and that is the reason these are called as heavy weight processes. And it is estimated the typically 50 percent of the effort by a development team goes towards creating documents and naturally the project costs increase and there is project delays.

And specifically now that the project durations are very short such heavy weight processes are not finding favor and we need some processes which do away with these problems.

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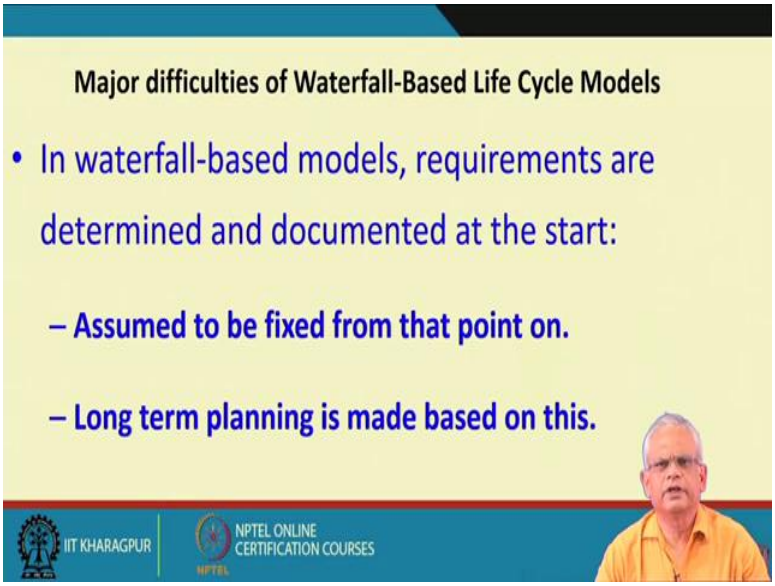

Let us just do a bit of elaboration that in waterfall model, requirements are determined and documented at the start, they are fixed from that point on; no changes are permitted and long term plans are made based on these documented requirement. The design is made on based on these documented requirement even test cases are written based on this documented requirement.

And therefore, any change to this requirement will require changes to all documents and therefore, those who use the waterfall model they heavily discourage the customer to make even the small changes.

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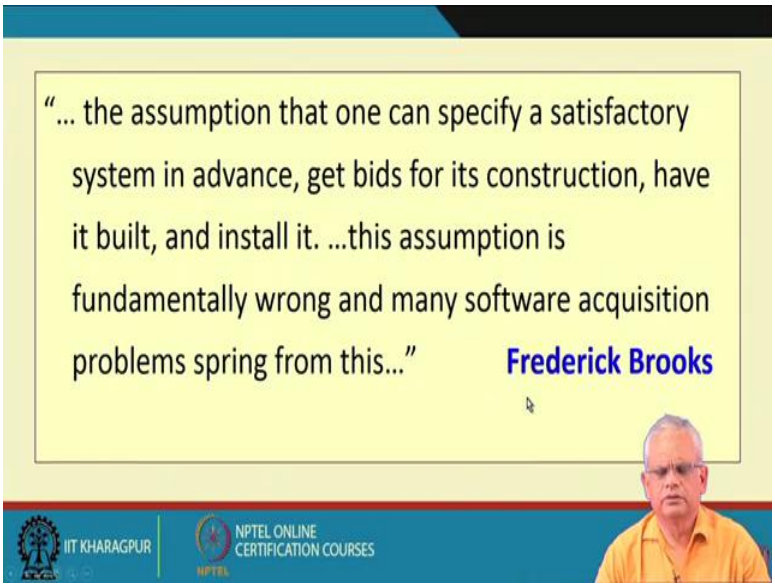

But let us look at the observation of Frederick Brooks who is one of the guru's in this area. The assumption that one can specify a satisfactory system in advance, get bids for its construction, have it built and install it. This assumption is fundamentally wrong and many software acquisition problems spring from this.

Just to explain Frederick Brooks thought, he felt that many of the project failures, project delays etcetera are the result of having the requirements worked out at the beginning of the development. And, then all the development proceeds based on the documented requirements is fundamentally wrong and many of the problems arise from this. In response to this short coming of the waterfall model one model that was proposed is the incremental model. To get the main idea behind the incremental model, let see what Victor Basili one of the founders of this area has to say.

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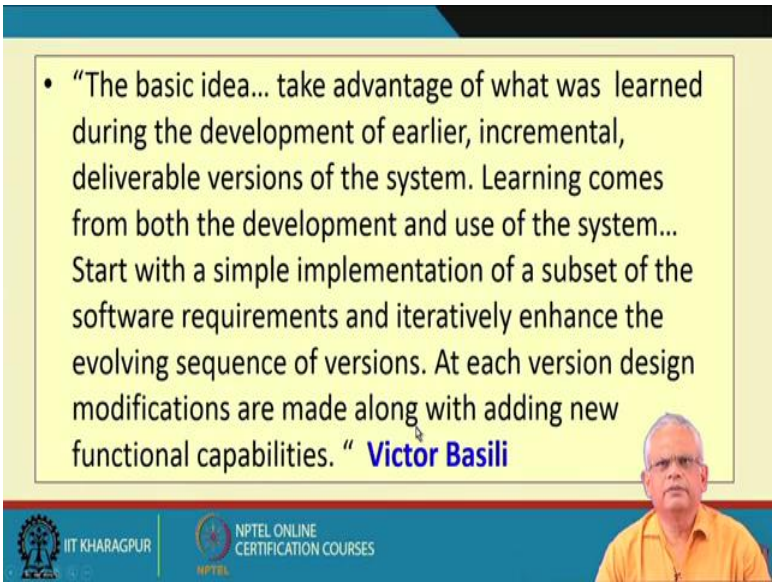

The basic idea is to take advantage of what was learnt during the development of earlier, incremental, deliverable versions of the system. Learning comes from both the development and use of the system. Start with the simple implementation of a subset of the software requirements and iteratively enhance the evolving sequence of versions.

At each version design modifications are made along with adding new functional capabilities. Thus to explain what Victor Basili has to say about a incremental model is that, the software is developed in increments. In the first increment some of the core functionalities are implemented and then these are given to the customer who gives feedback and this forms a learning from the use of the system, even the developers they have they learn from the implementation of the core functionalities. And based on the feedback the system the software is refined and this is called as the iteration.

In iteration the same functionality is refined each time the customer gives new feedback, the same functionality gets refined iteratively and also at the same time new functionalities are also implemented these are called as increments. So, this model is called as incremental model with iteration.

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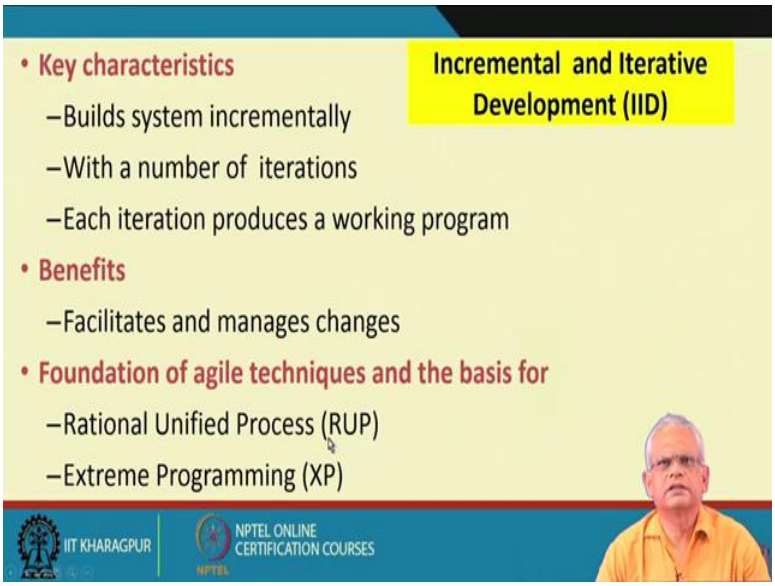

Incremental and iterative development here each time new increments of functionalities are implemented and at the same time the functionalities that were already implemented they are refined and that is called as iteration with those functionalities.

The key characteristics of the incremental and iterative development is that, built the system incrementally small parts of the system are built. There are number of iterations for each functionality and each time a deliverable is given to the customer in the form of a working program which the customer can use and give feedback.

Each deliverable to the customer would not only have incorporated the feedback that he had given in the previous increment, but also added new functionalities. As can be easily seen this is a good way to manage changes, the customer is encourage to feedbacks and these are incorporated. The incremental model has become popular in industry, the rational unified process is a incremental model, the extreme programming the agile models etcetera are all incremental.

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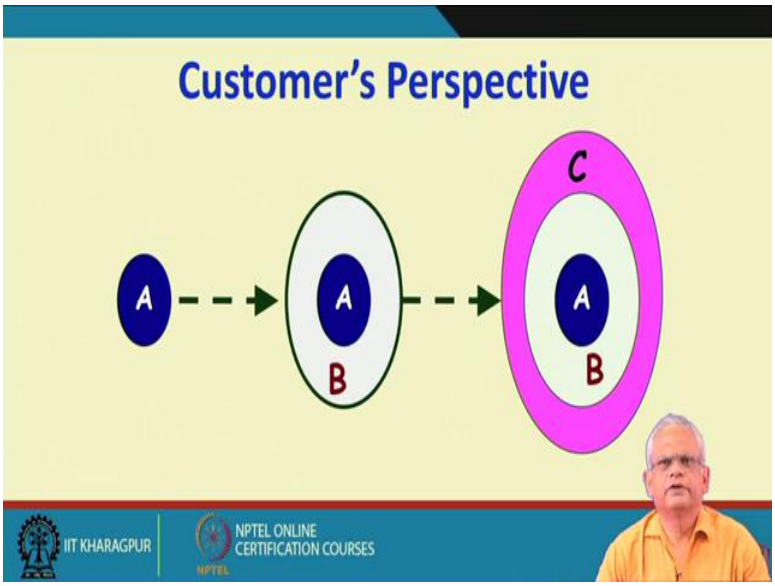

From the customer prospective gets a functionalities some part that he can use and give his feedback based on this the organisation developing organisation incorporates the feedback on the core functionality and also implements additional functionality and gives it to the customer. The customer again gives feedback on this and the feedback is incorporated and also new functionality are added and this way the system gets completed and hopefully at the end the delivered system will meet the customer's requirements because all his feedback in using the system have been incorporated.

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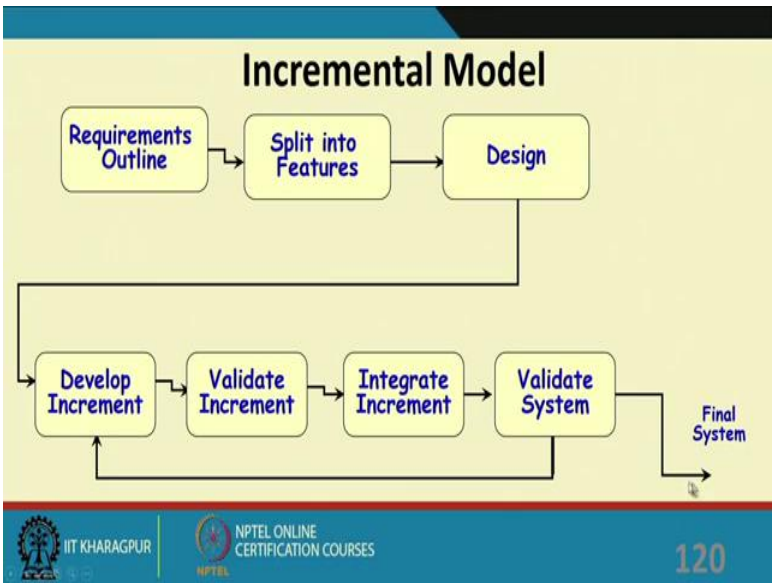

If you can diagrammatically depict the model initially requirements are collected and this requirements are split into incremental features there is a overall design and then each time one increment is taken developed, validated, integrated with previous increment, the system is validated and delivered to the customer feedback is taken next increment is developed and so on until the final system is updated.

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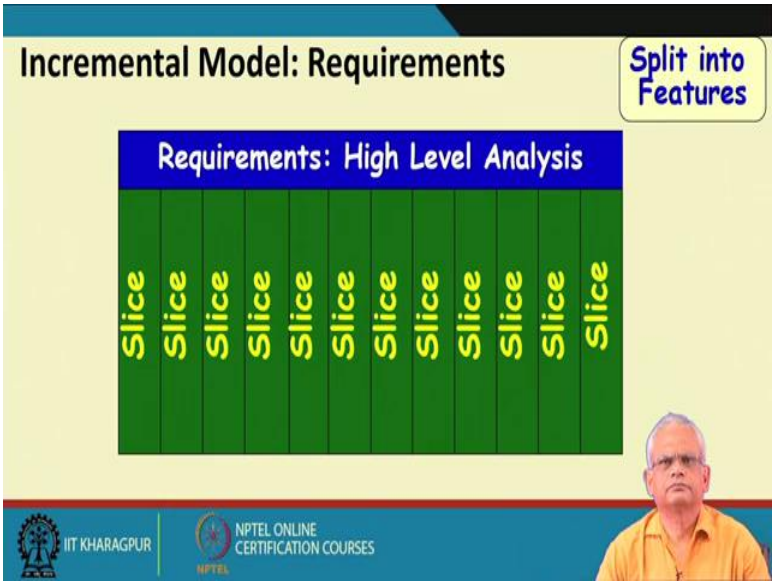

As you can see that in the increment model the requirements are collected up front these are split into small increments or slices. Similarity, here with the waterfall model is that the requirements are collected up front, but the difference is that each of this increment customer feedback is taken and these change.

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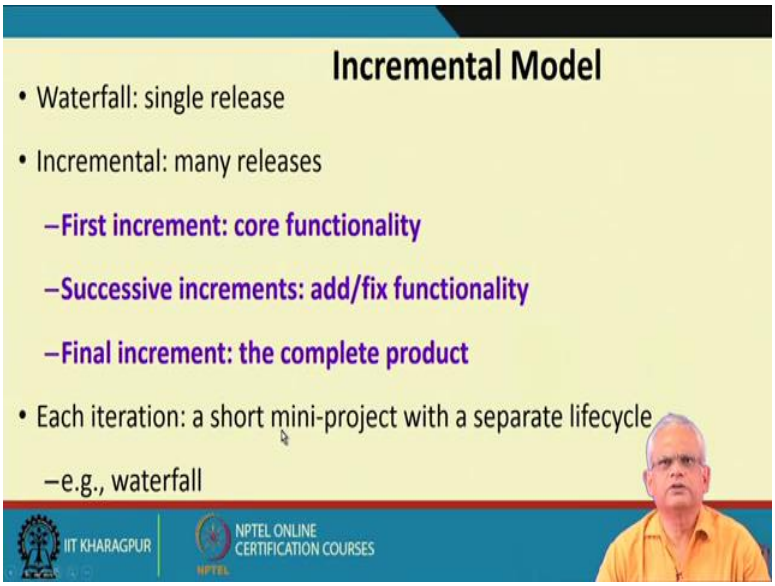

In a waterfall model there is a single release initially all the functionality requirements are collected, designed, coded, tested and given to the customer as a single release whereas, in incremental many releases are there and in after each release customer feedback is taken and the functionality that was already delivered are again changed to meet the requirements of the customer.

The first increment is typically the core functionality, then the successive increments they not only refined the existing functionality, but add or fix functionality the final increment is the complete product. This is called as the incremental model with iteration, but then somebody can think of a purely incremental model where there are no iteration in which the system is just developed in small increments and given to the customer, but the customer feedback is not really taken to change the existing functionality there is no iteration.

In a purely incremental model the software is developed in increments and delivered to the customer, but once the functionalities are delivered these are not changed, but in a incremental model with iteration, the functionalities are delivered to the customer in increments. But, also the customer feedback is taken after each functional functionality is delivered and those are incorporated in the next iteration. Of course each increment or iteration is a small is a sort mini project with a separate life cycle for example, for each small increment may be a waterfall model can be used.

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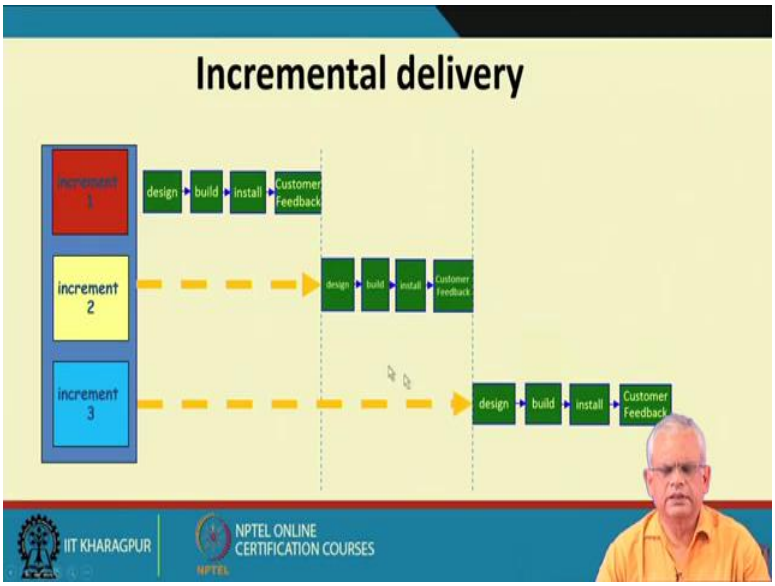

This is a diagrammatic representation, the initial requirements is split into increments increment 1, 2, 3; increment 1 is developed, increment 2 gets developed. So, this is like a waterfall model design built install customer feedback. Increment 1 is first developed, then increment 2 is developed, then increment 3 gets developed.

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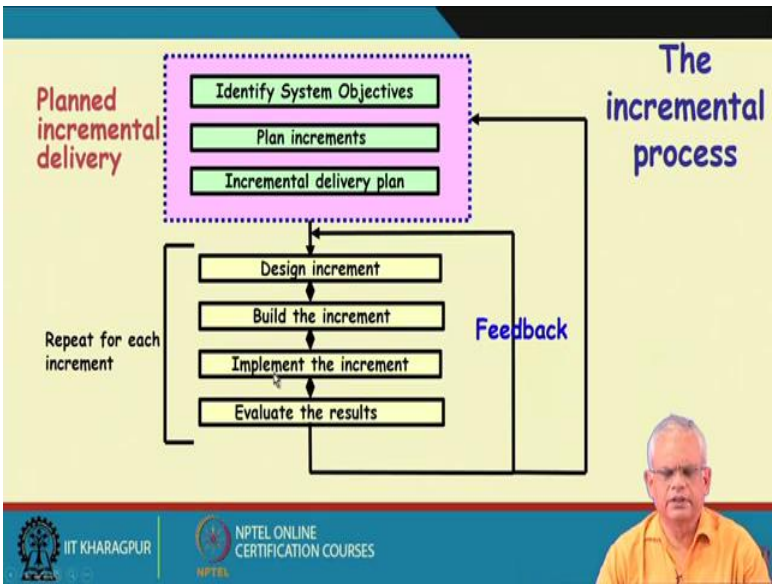

If we diagrammatically represent this model of development first we gather all customer requirements document them and then plan the increments and also the incremental delivery plan.

Then design the increment, build the increment, implement the increment and then give the customer for evaluation the increment and then use that as a feedback and take the next increment and so on.

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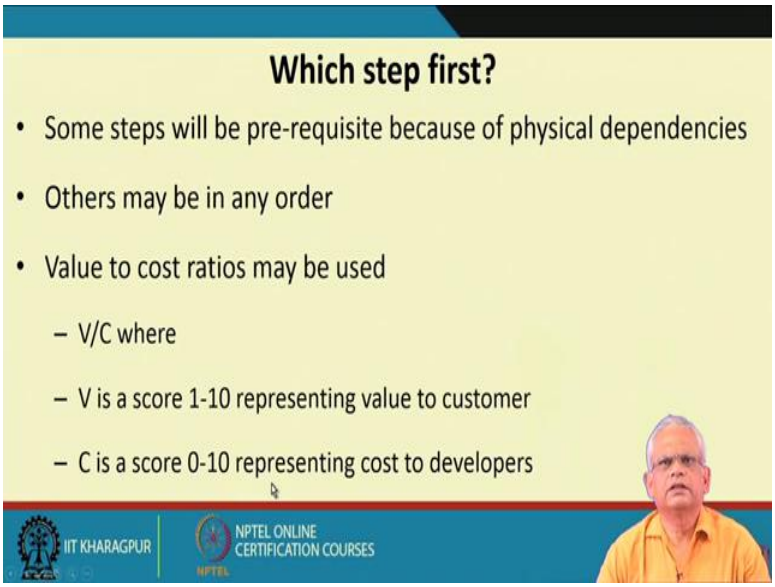

Because there are several increments out of which the designer need to choose one of the increment for delivery in the next iteration, which increment should be selected first? This is an important problem.

Some parts will be pre requisite of the other parts without that the system will not work. For example, until the customer registration is done customer billing cannot be done first we must have the customer registration software developed and then only the customer billing software can be implemented, but then many of the increments are without any order.

Now, these increments where there are no order value to cost ratio may be used, this we call as the V by C ratio, where V is the value to the customer which is graded in a scale of 1 to 10 and C is a score 0 to 10 representing the cost to the developers.

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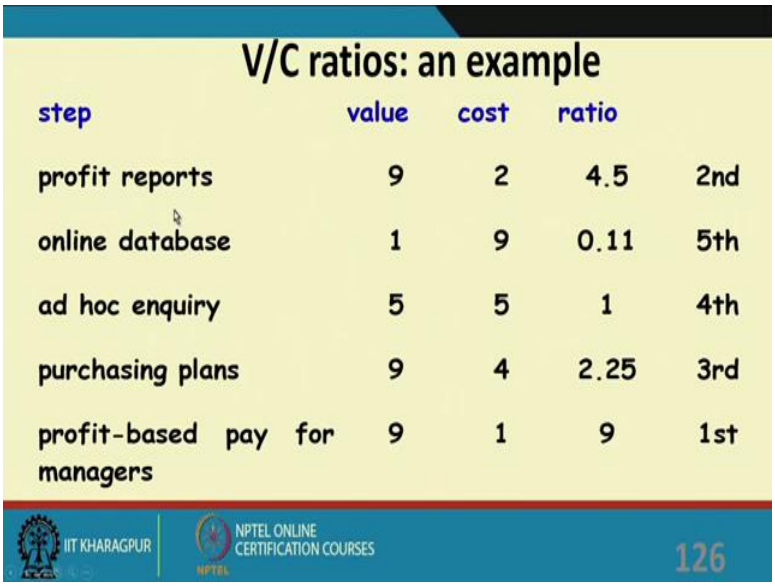

Let us take an example to see how it works. Let say at one step in increment for the next increment we have let say these are the modules that the development team has the flexibility to implement profit generate the profit reports, implement the online database, implement ad hoc enquiry, implement the purchasing plans, implement the profit based pay for the managers. These are the five functionality, that the developers have the choice and from the customer feedback about the value of these different functionalities they have assigned these values.

Online database is the lowest value to them profit report is 9, ad hoc enquiry 5, purchasing plan 5 and profit based pay for manager is 9. The development team when ask to rate this on 1 to 10 I found that the least cost for development is profit based pay for manager, profit reports is 2, purchasing plan is 4, ad hoc enquiry is 5 and the most difficult to implement at the online database and then on computing the V by C ratio we find that there is a scale of 9 to 1, 0.1; 0.11.

So, for the next increment this is one that should be chosen this as high V by C ratio and n the profit report should be chosen, third is the purchasing plans and forth is the ad hoc enquiry and last should be the online database. The incremental model as become extremely popular, it is almost part of every model that is being used those are derivatives of the incremental model one important derivative is the RAD model, RAD stands for Rapid Application Development.

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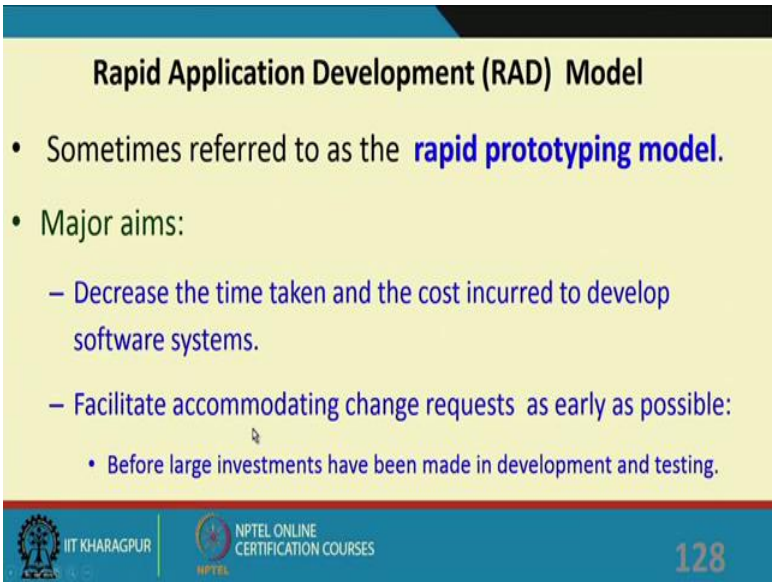

This is sometimes called as the rapid prototyping model; here the main aim of the model is to decrease the time taken and the cost incurred to develop the software. Facilitate accommodating change requests as early as possible before any large investments have been made.

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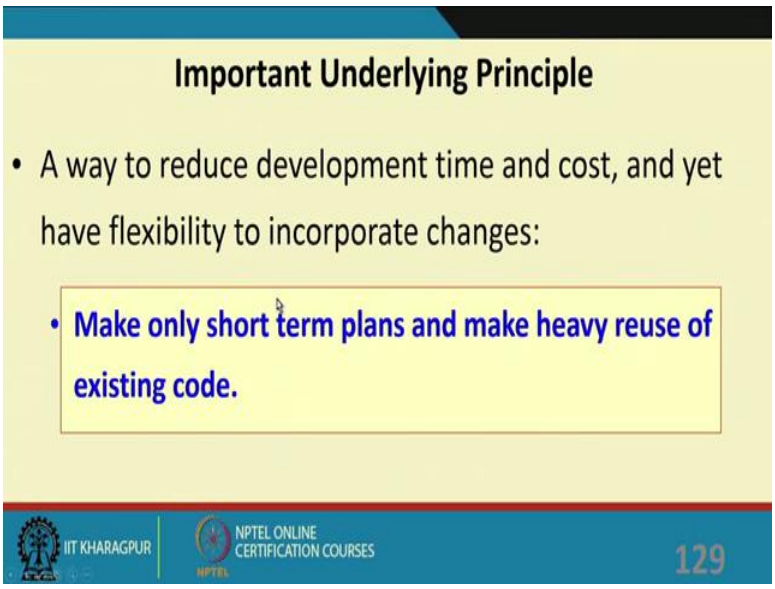

This is as we said is a incremental model here increments are decided and only short term plans are made and heavy use of the existing code. Contrast to the waterfall model this model is very efficient, it can incorporate flexibility to incorporate changes because

only short term plans are made any change does not require changing all the plans that is design the project management plan etcetera, the test plan and so on.

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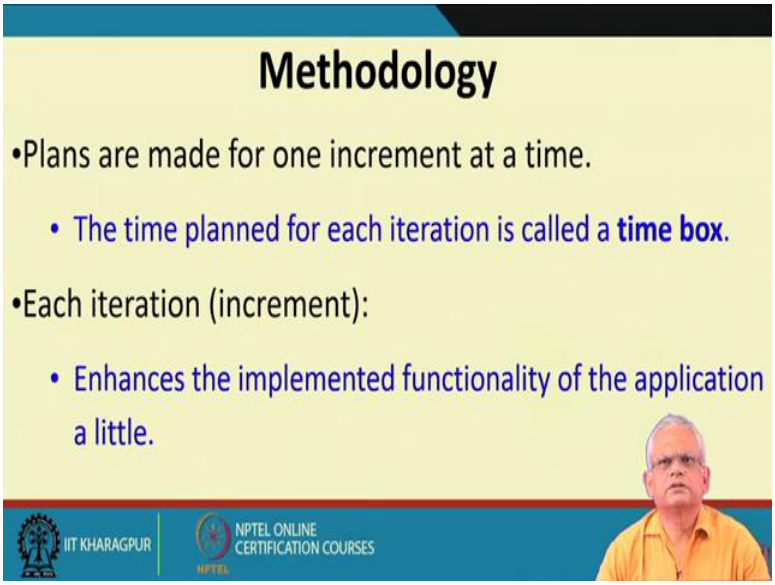

The plans are made for one increment at a time and this increment is called as a time box and after each increment is delivered, the system or the applications becomes more complete.

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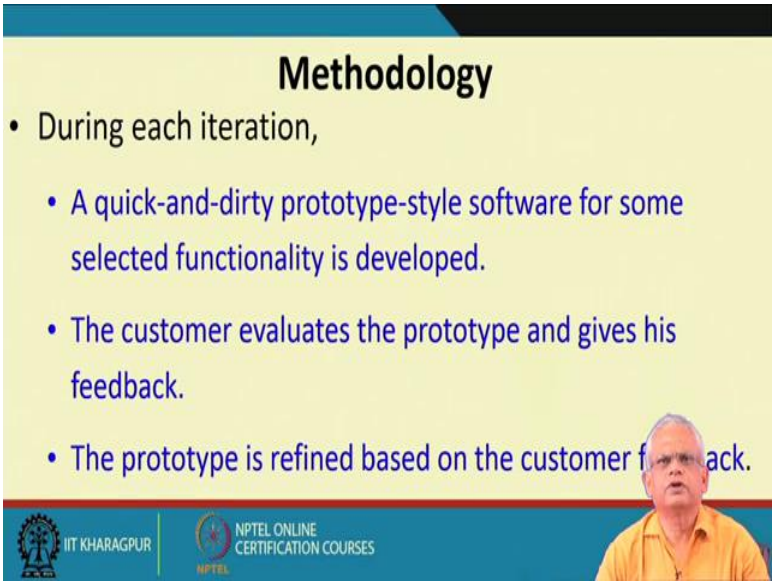

During each iteration a prototype of the software is first developed is given to the customer for evolution and then this prototype is refined based on the customer

feedback, one point to note here is that in this model the prototype itself is refined in the RAD model the prototype itself is refined to the actual to be the actual software, where is in the prototyping model that we had discussed which is a derivative of the waterfall model. The prototype is used to get customer feedback the start of the project and then the prototype is thrown away and the software is developed fresh. But in the RAD model the prototype is refined based on the customer feedback.

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The RAD model achieves faster development, it encourages use of specialized tools, the tools should be supporting visual style of development like drag and drop. Use of reusable components and use of standard APIs these are something that are suppose to help the software implemented faster.

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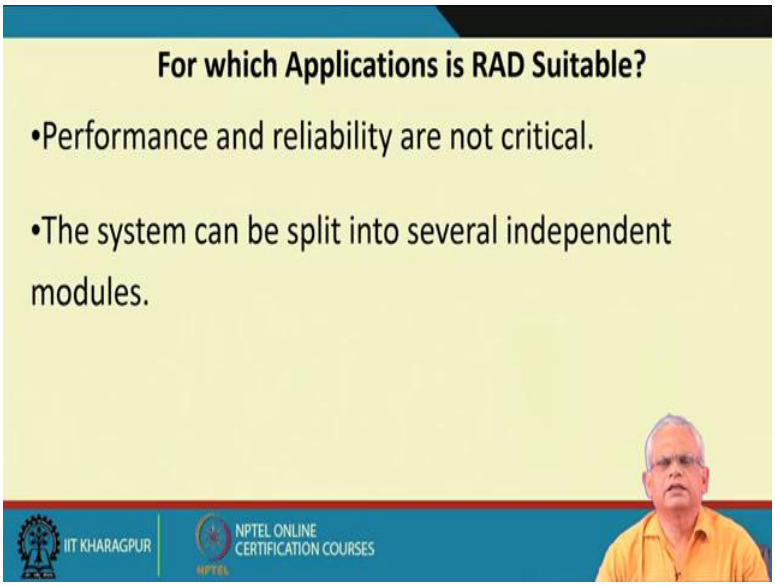

Now, let us look at the software projects for which the RAD model is suitable. The projects for which the performance and reliability are not critical, but these are required to be developed very fast and the system can be split into several independent models. If the system is very trivial and we cannot have several increments then; obviously, RAD model is unsuitable.

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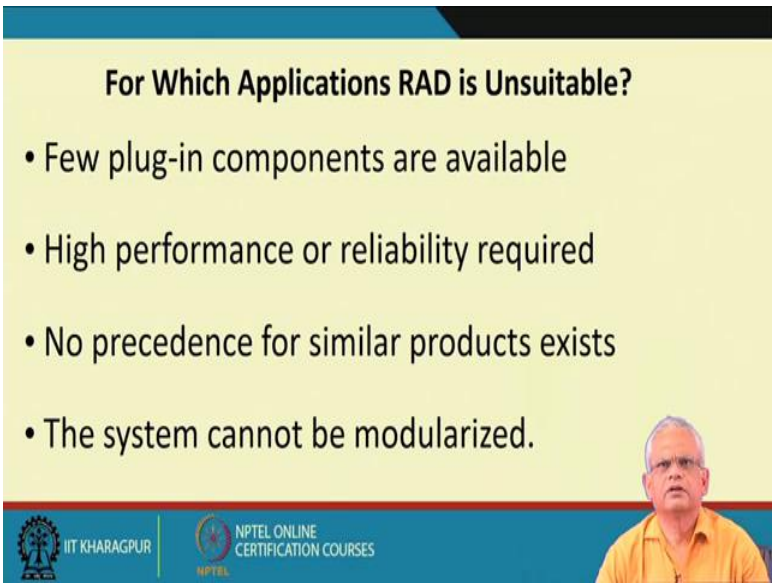

Also the RAD model is unsuitable when we have few login components are available, high performance or reliability is required. The reason why the RAD model does not

give high performance or reliability is that the prototype itself is refined into the actual software. Unlike a waterfall model where a fresh software is designed coded and tested, the RAD model would give a low reliability and performance.

If there are no precedence for similar products then also RAD model is unsuitable that is challenging new type of software for which; obviously, there are no components would be available to use as plugins the systems that cannot be modularized RAD is unsuitable.

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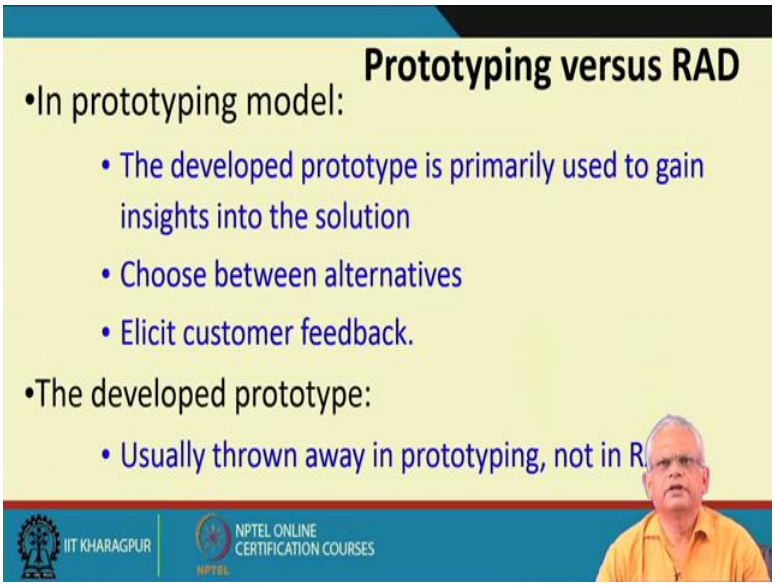

The RAD model each time constructs a prototype and gives to the customer for evaluation and based on the customer evaluation the prototype is changed and as the customer gets satisfied the refined prototype is becomes the part of the software. In the prototyping model which is a derivative of the waterfall model, the developed prototype is primarily used to gain customer feedback and also to get insight into the solution that is the technical issues and so on.

And therefore, using the prototyping model you can chose between different design alternatives, you can elicit customer feedback and the developed prototype is usually thrown away, but not in the RAD model. In RAD model the prototype itself after enhancement refinement becomes the final software.

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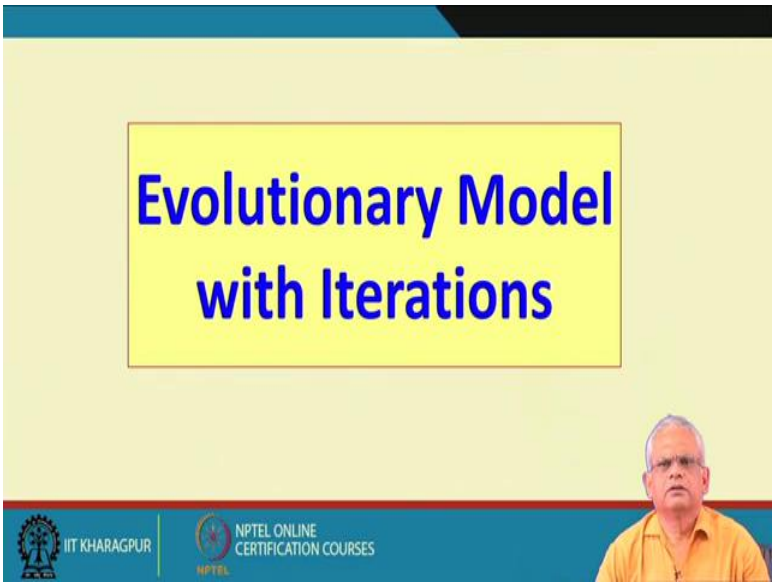

Another important development model is the evolutionary model with iteration, we had seen the incremental model with iteration which is of course, important and has become part of many development models that are being used across industry, but another idea that has come is the evolutionary model with iteration. We will examine this model in the next class and we will see that the present model that has become extremely popular at the agile models which have the features of both the incremental and evolutionary model with iterations.

We will stop here and continue in the next lecture.

Thank you.

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Lecture – 08
Life Cycle Models – IV

Welcome to this lecture Over the last few lectures, we had looked at few important software development lifecycle models. We had started looking at the classical model which is very intuitive, the classical waterfall model and the derivatives of this model namely the iterative waterfall model, looked at the V model, prototyping model and so on.

And over the years few more models came up as the shortcomings of the waterfall model were noticed. These were the RAD model, the incremental model, incremental with iteration and the evolutionary model. In the last lecture, we had looked at the incremental model. Today, we will first look at the evolutionary model and then, we will look at the agile development model.

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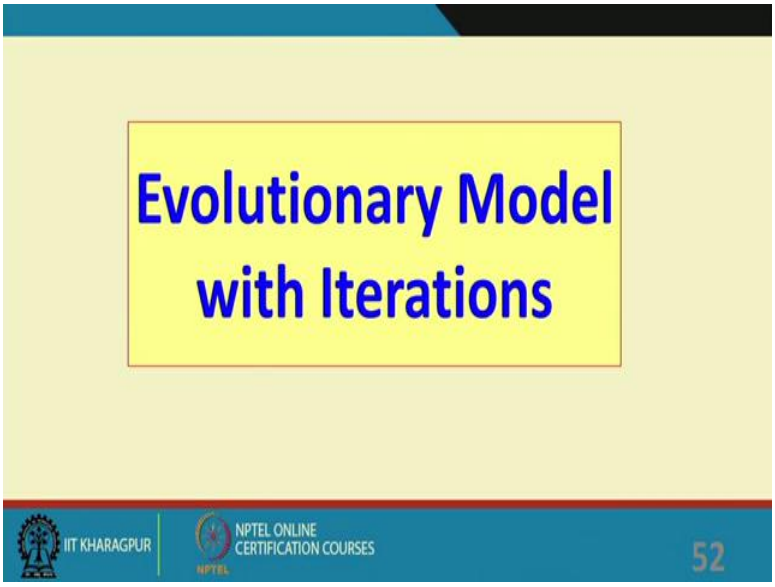

This is in fact, an evolutionary model with iteration; remember in the last class we had said that the term iteration is used when existing functionality which was delivered developed and delivered is iterated and refined based on customer feedback. Now let us look at the evolutionary model with iteration.

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One of the main problem that we had said about the waterfall based model is that the requirement is defined and frozen at the start of the project. But then, look at here about the practical situation here. Capers Jones researched on 8000 projects and he found that 40 percent of the requirements were arrived, when the development had already begun. If we had frozen the requirements, the start; then, this 40 percent changes will be very difficult to incorporate. Because in reality this is the average figure; 40 percent, it can even be more for some projects.

And therefore, there is need for a model which can effectively handle requirement changes. Requirements change due to various reasons maybe the business changes very rapidly. Once they have defined a business procedure may be after a month the business procedure changes or maybe the technology changes or maybe they might have not understood what is required and told something else, maybe they had forgotten to tell something and so on.

So, the requirements change due to various reasons and we need a model which can effectively handle requirement changes. One way that has been considered very promising is that each time do only small part of the system. Those are riskier likely to change the customer is not sure about it. Plan a little for this small work, design a little, code a little and once the small part is completed, give it to the customer for his

evaluation and feedback. Here the customer is encouraged to participate. The end user the tester, integrator, technical writer all are involved in the development.

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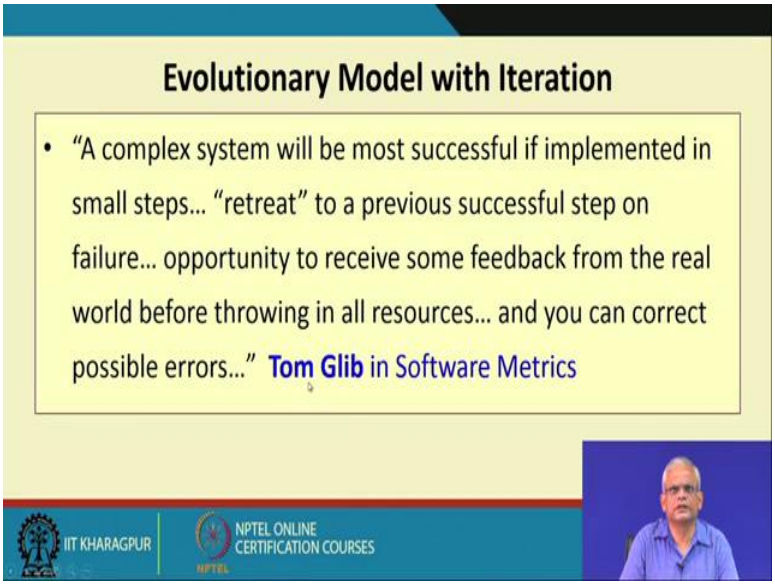

Let us see what very distinguished person in this area. Tom Glib, what he has to say on the evolutionary model. He says that "A complex system will be most successful if implemented in small steps... "retreat" to a previous successful step on failure... If some functionality is totally not required, then at least you can go back to the previous version which is acceptable and then develop from there opportunity to receive feedback from the real world before throwing in all resources.. and you can correct possible errors.."

So, according to Tom Glib, the evolutionary model which he captures here in this few lines is a very promising model to handle changes to the requirement due to various reasons.

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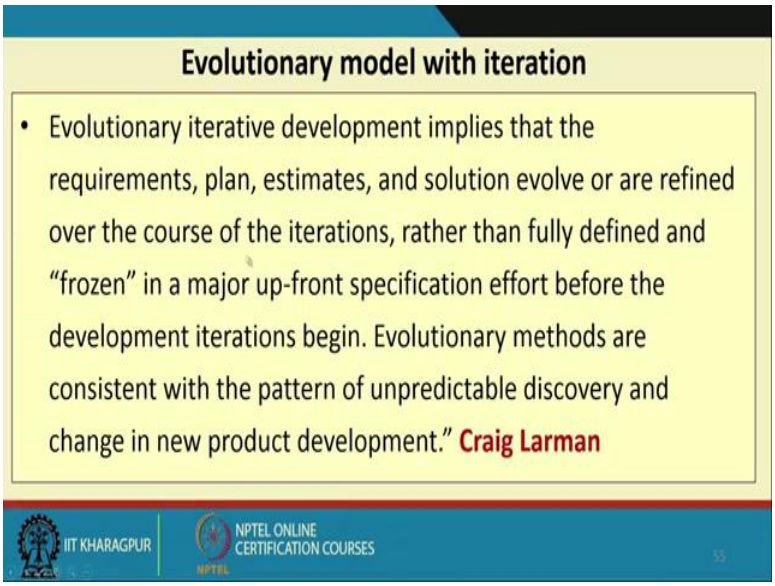

Let us see what Craig Larman has to say about evolutionary model with iteration. "Evolutionary iterative development implies that the requirements, plan, estimates, and solutions evolve or are refined over the course of iteration, rather than fully defined and "frozen" in a major up-front specification before the development iterations begin. Evolutionary methods are consistent with a pattern of unpredictable discovery and change in new product development."

So, Craig Larman says that when a problem when a project starts it is very difficult to visualize all that is required. Only when you start doing, then you can know that what is wrong; what is really required and so on. So, is very critical about the waterfall model, where the requirements are fully defined and frozen before the development starts. And therefore, evolutionary model is the way to go and true to his words, most of the development models that are being used now which have become popular, use the evolutionary model in some form.

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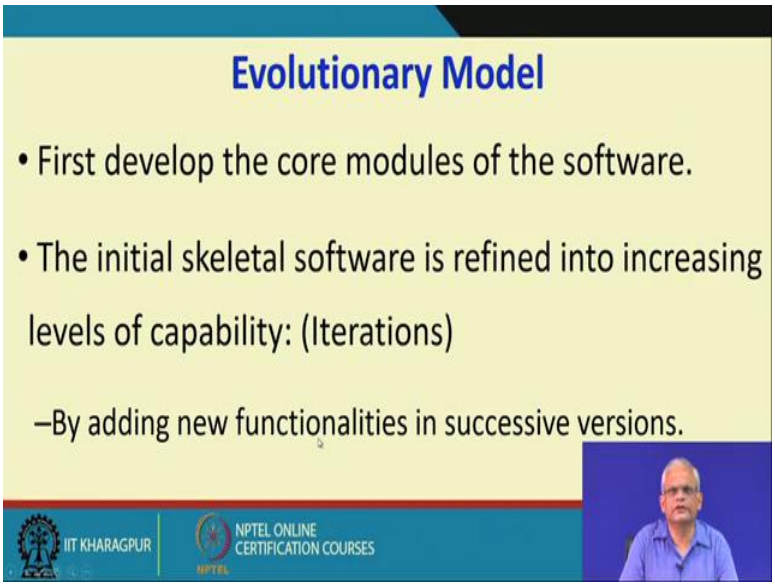

Let us look at the evolutionary model. Here the core modules of the software is first developed and this is refined into increasing levels of capabilities which are called as iterations. In iteration new functionalities will be added and also the existing functionalities can change based on the user feedback. And each development, each iteration is developed through a mini waterfall model.

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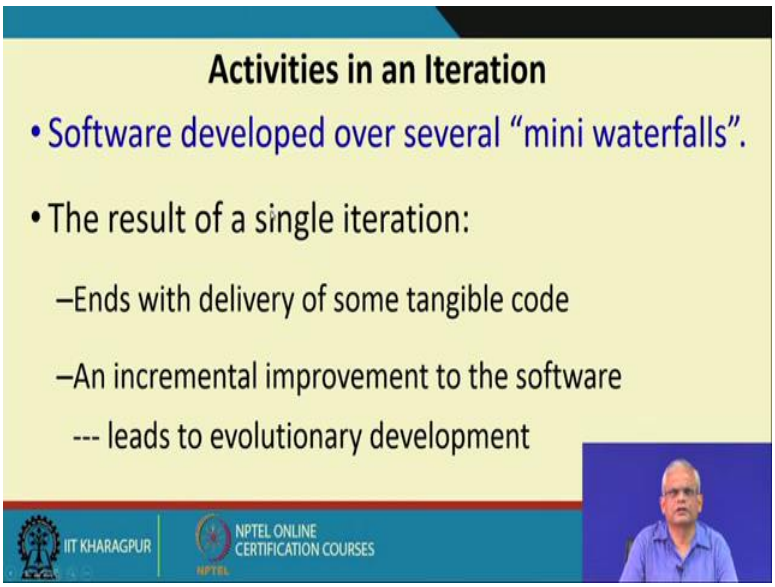

At the end of a iteration, some code is delivered to the customer invariably. At the end of development maybe the existing functionalities might have changed. It is possible that no

new functionalities might have got added, but only the existing functionalities have got refined or it may so happen that existing functionalities have got changed a little bit and new functionalities have been added. So, these are an increment and therefore, it has some incremental development.

But then, the question is that how is the incremental development and the evolutionary development differ? Because as you can see that in both these models incremental and evolutionary, there are small increments over which the system is developed and these increments are deployed at the customer site and feedback obtained.

But let me mention here that in the purely incremental model as you might have remember from our last discussion that all those requirements are identified and these are planned into small increments. Each increment is developed deployed feedback obtained and changes can happen. But in an incremental model, all the requirements are gathered to start with whereas, in evolutionary model, it is not the case. Here after an overall understanding of the system we do not really capture all the requirements, but then start with some of the core or riskier modules.

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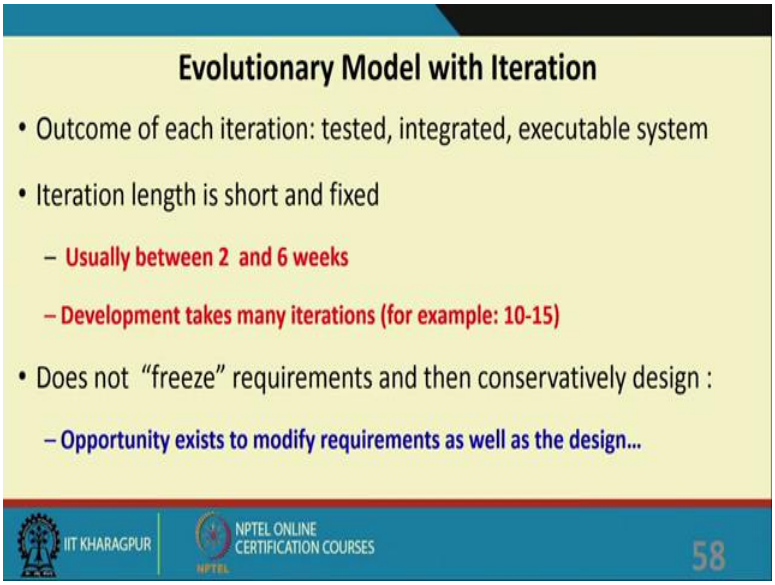

At the end of each iteration, a tested integrated executable system is developed, deployed at the customer site. The length of the each iteration is typically fixed, something like a 2 week to 1 month or maybe 1.5 month, 2 to 6 weeks and the development proceeds over many iterations. For example, a system may get developed over 10 to 15 iterations.

As you can see the evolutionary model, the requirements are not frozen. But here the customer is encouraged to participate, to give feedback based on which the further developments occur and the customer feedback is taken into account and functionalities get modified. So, here the requirements are modified in a iteration, some of the existing functionalities as well as new functionalities are delivered.

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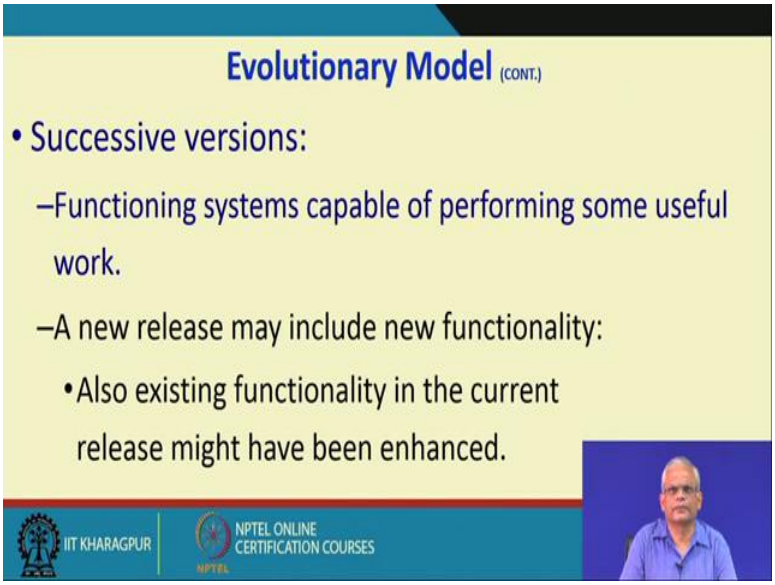

Here, successive versions are developed and deployed at the cost customer site. Each version is capable of performing some useful work which the customer can use and give feedback and a new release can include a new functionalities and also existing functionalities may get modified and refined.

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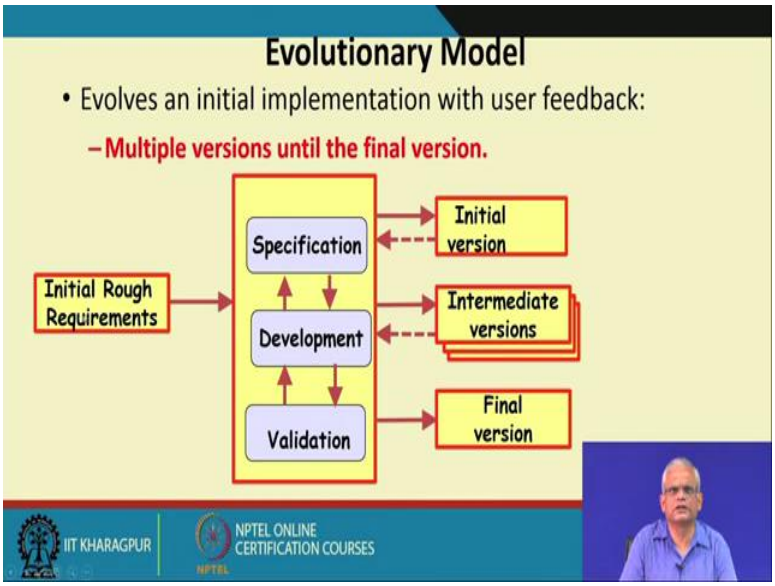

We can diagrammatically represent it like this that initially rough requirements, just understand what the system needs to do and then, there are iterations. The iterations are shown here. In each iteration is a mini waterfall, where need to specify a small part; develop, validate and deploy at the customer site.

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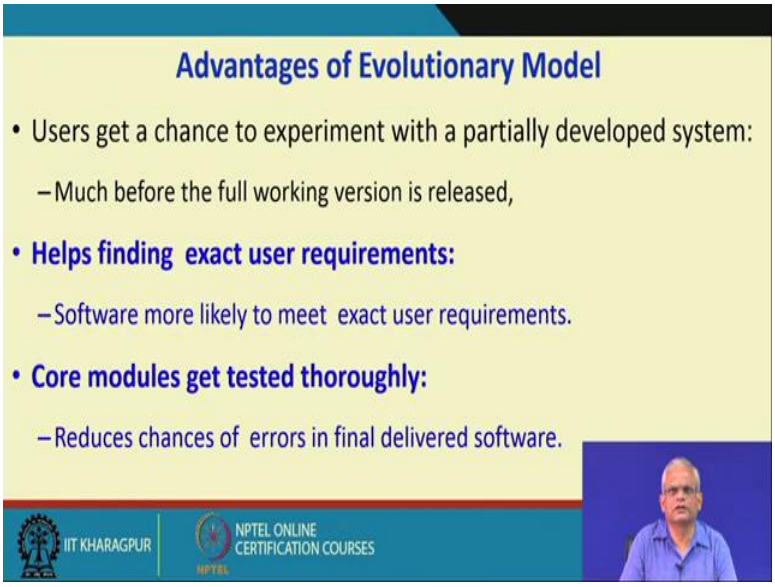

To start with the initial version, the feedback is obtained. There are several intermediate versions and then, the final version is deployed at the customer site. Evolutionary model has the advantage that, the customer can use the system and give feedback. And

therefore, the evolutionary model is best suited to find the user requirements, the exact user requirements and once it is delivered to the customer, the customer to meet the customer requirements. Also since these are each time tested, each increment is tested and deployed at customer site. Therefore, multiple test activities, integration activities occur and therefore, it is less likely to contain bugs because these are tested thoroughly.

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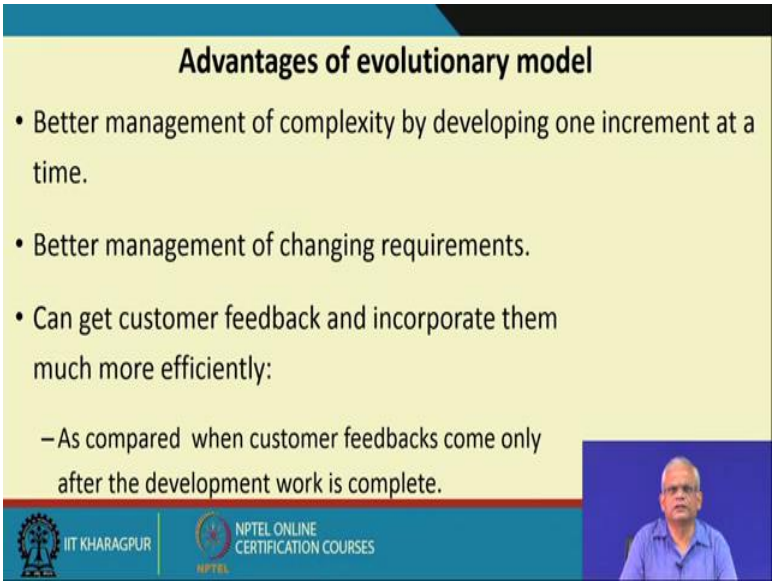

And since each time the developers focus each time on a small part, it helps in managing complexity; better manage changing requirements; customer feedbacks are obtained and incorporated.

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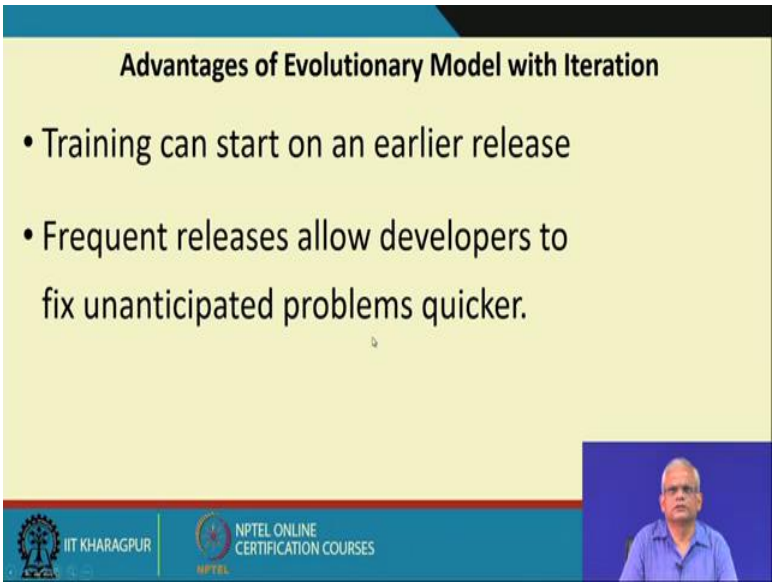

The customer gets version quickly. They get trained on using the software. If there are bugs, these are fixed quickly in the next iteration.

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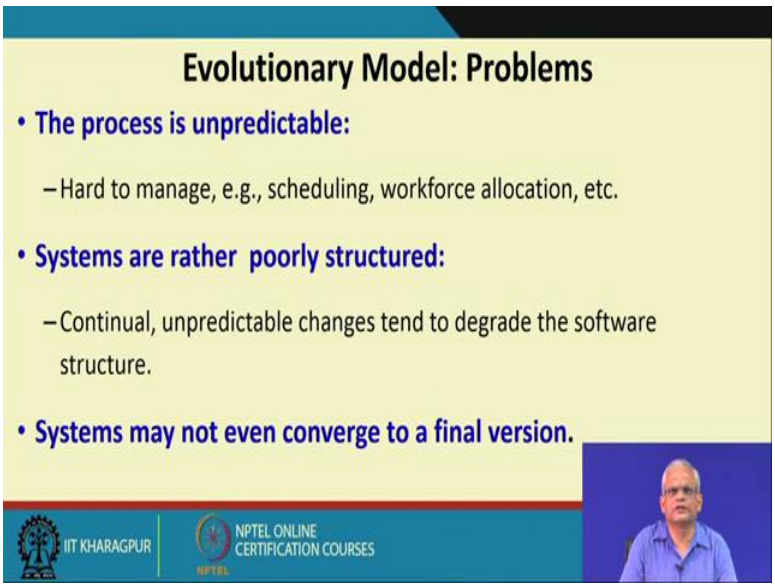

But then, there are some problems with the evolutionary model. Let us be aware about the problems of evolutionary model. You had seen that it is lot of advantages, but then here the process is unpredictable that is each time, we develop a increment deploy and get the customer feedback.

Now, what if the customer just keeps changing the requirements; keeps on asking for modifications and so on. Therefore, the process is unpredictable. We don't know when it will finish, because the customer may keep on changing the requirements and also if we have a long term plan, then it becomes easier to deploy manpower, recruit them, schedule work monitor and so on. Here all these are problematic.

The system is poorly structured because there is no overall design made its only small parts that are designed and integrated. Since, the code is changed continually, the code structure degrades and the system may not even converge to a final version if each time the customer keeps on asking for changes.

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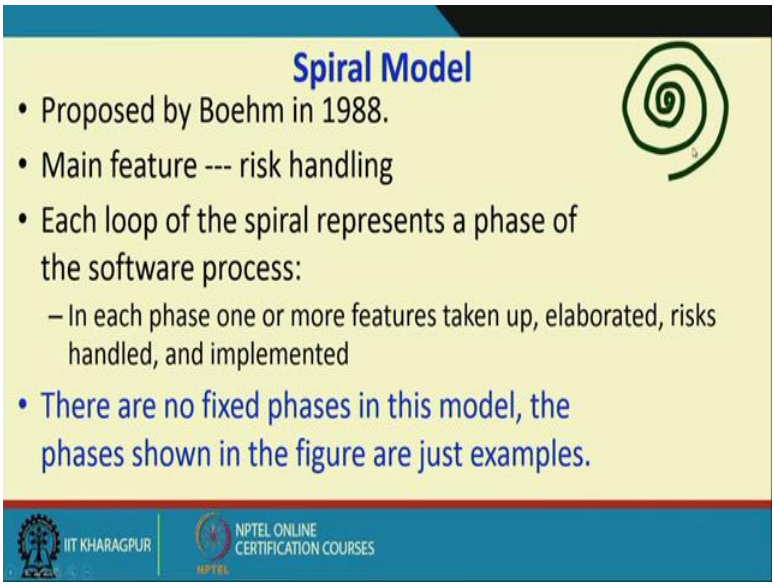

Now, let us look at the spiral model. The main feature of the spiral model is risk handling. Just look at the spiral here diagram the model follows spiral, we will see that. It was proposed by Boehm. Here each loop of the spiral is called as a phase and the system gets developed over many phases. There is no fixed number of phases. As the system develops more and more phases are created. There can be many loops on this spiral until the final delivery takes place. But over each phase that is each loop of the spiral, one feature is identified and is developed.

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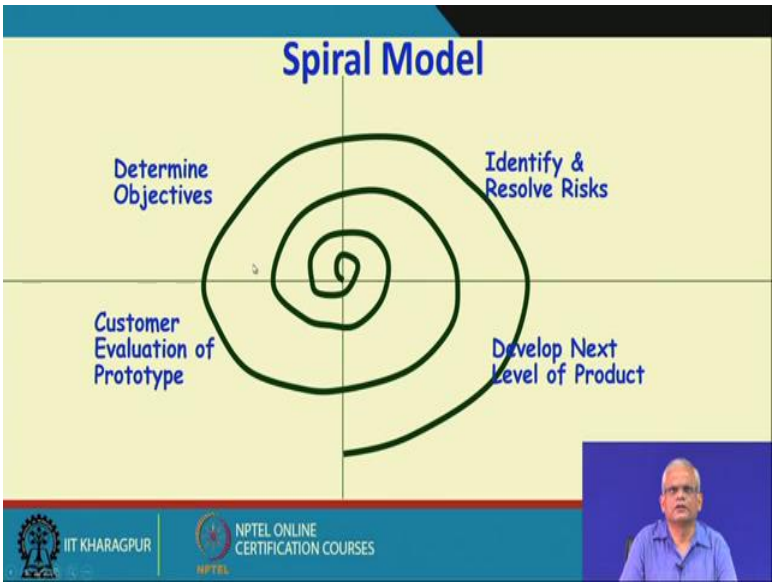

If you can see there are four quadrants here on the spiral model, the first quadrant what needs to be done in that phase is identified and then, if there are any uncertainties. Technical uncertainties or the user requirements are not clear. A prototype is developed to resolve the risk and then, the next iteration is developed given to the customer for his evaluation and then, another feature is taken up.

So, over each iteration one or more features are taken up. The risk is resolved and then the next development occurs and customer feedback. As can be seen here, this model is ideally suited for very risky projects, where the technology is not known. It is possibly the first time that this kind of software is being developed. The technology is not certain and so on. So, each time some features the most riskiest feature is identified. Risk is resolved, it is developed and given for customer feedback and the next set of features are taken up.

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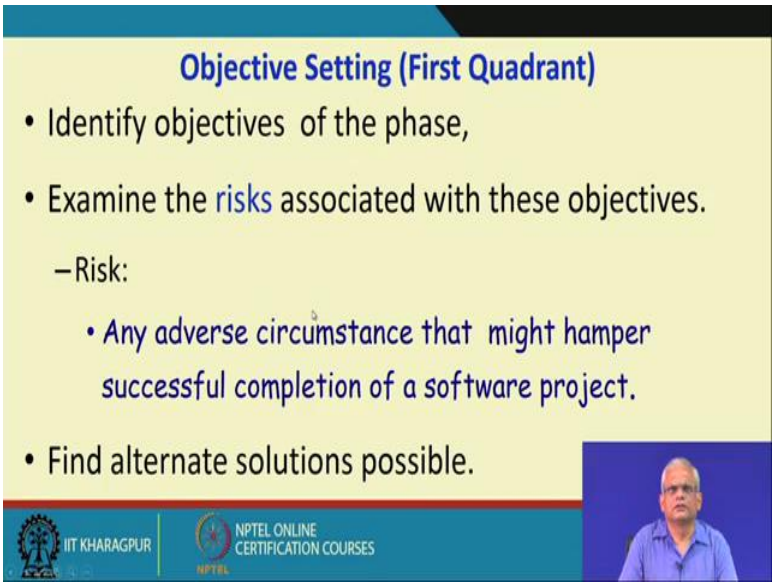

Here, the risk is any adverse of circumstance that might hamper the successful completion of the project. It can be technical issues are not clear, it can be that the customer is not able to clearly specify what is required and here a prototype is developed to overcome the risk.

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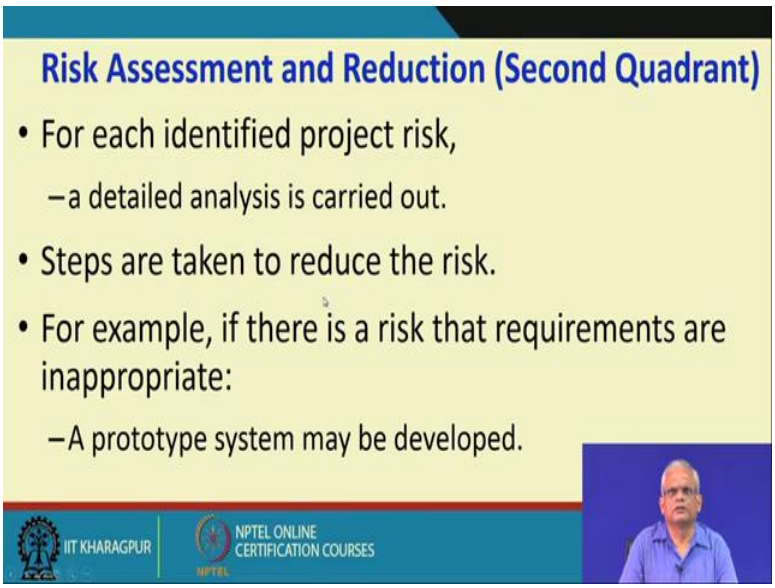

If the risk is that requirements are inappropriate, prototype is developed given to the customer so that he can more clearly say what is required.

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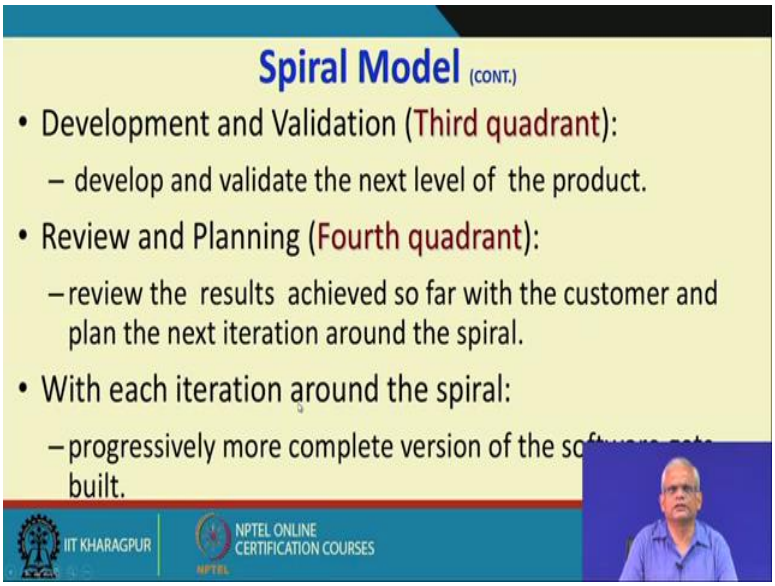

The third quadrant is developed and validate and the fourth quadrant is give the developed solution to the customer for feedback and plan for the next iteration. We can see that with each iteration around the spiral, more and more complete versions of the software get built.

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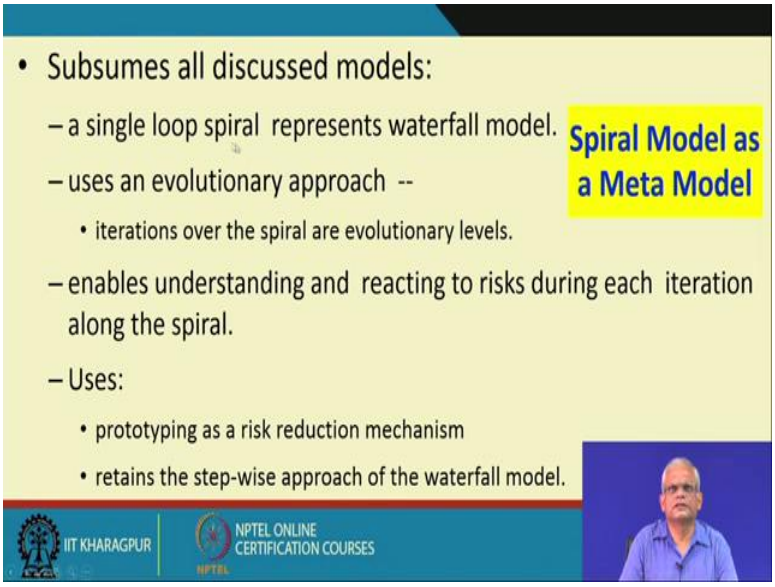

The spiral model is also called as a Meta model because a single loop of the spiral is a waterfall model, there are iterations and therefore, it incorporates evolutionary and incremental models. It uses prototyping. The stepwise approach of the waterfall model is

retained and therefore, this model can be degenerated or can be used as any other model. But then, need to remember that the spiral model is suitable for very risky projects and large projects.

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Now, let us look at the agile models. If we look at the meaning of agile in dictionary; agile means easily moved, very fast, nimble, active software process etcetera these are some of the terms that are given as explanation for agile model in dictionary. But then, agile means very fast. So, the development in agile model should occur very fast in much less time than it occurs in other development models.

But then, how is it developed very fast; how his agility achieved? Agility is achieved because here it gives lot of flexibility to change the process that is if some project requires some work to be done some specific activity, it will be done; if it is not required for another project, it will can be easily skipped because the model is very flexible. Unlike the waterfall model, where there are predefined activities and steps and so on.

Here, we can fit the process to the project, but more important is we can avoid things that waste time. That is to get it done very fast, we need to avoid whatever is a wastage of time. What are the wastage of time? Maybe if we are preparing very elaborate documents, we are mentioning that in the iterative waterfall model 50 percent of the development effort and time goes in preparing documents. Here we will see that it does away in preparing these elaborate documents and saves lot of time.

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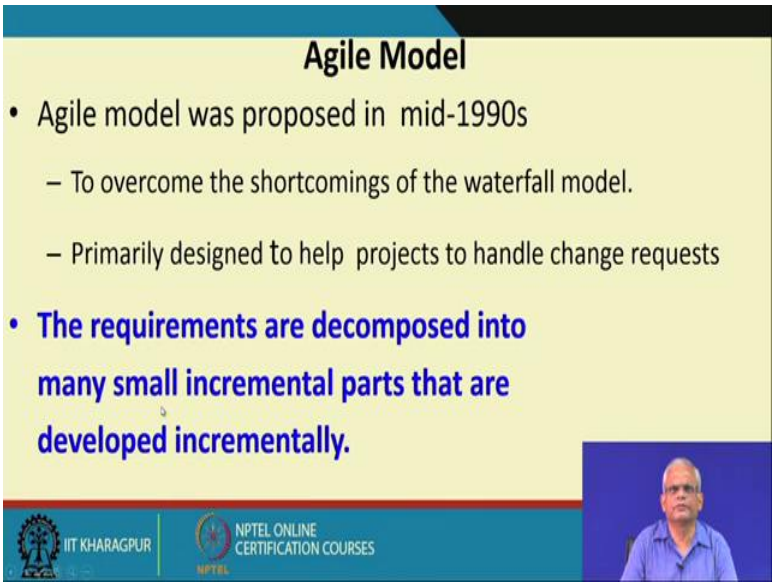

It was proposed in mid 90's, mainly to overcome the shortcomings of the waterfall model. One important characteristic is to handle change requests. We had seen that the evolutionary model is a good way to handle change requests and the agile models do incorporate evolutionary development; evolutionary and incremental development. Here the requirements are decomposed into small incremental parts and development precedes incrementally.

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There are a few things which are important here. Instead of producing elaborate documents, following process rigorously here the individuals and they interact with each other rather than passing on a document to each other. They explain to each other through interaction. In a waterfall model, the progress of the development is measured in terms of the documents produced; is the requirement document complete; is the design document complete; is the detailed design complete; is the code documentation complete; test documentation complete. But here, the progress is measured in terms of the working software that has got developed. How many iterations have been completed and each iteration, some working software is deployed at the customer site.

Here, the customer is encouraged to participate in development, is encouraged to collaborate in the development rather than just signing a contract at the start of the project and here, it is assumed that changes will occur, the requirements will change and how to handle those rather than making a long term plan and following that plan, here changes are welcome and how to handle those changes those are important issues here.

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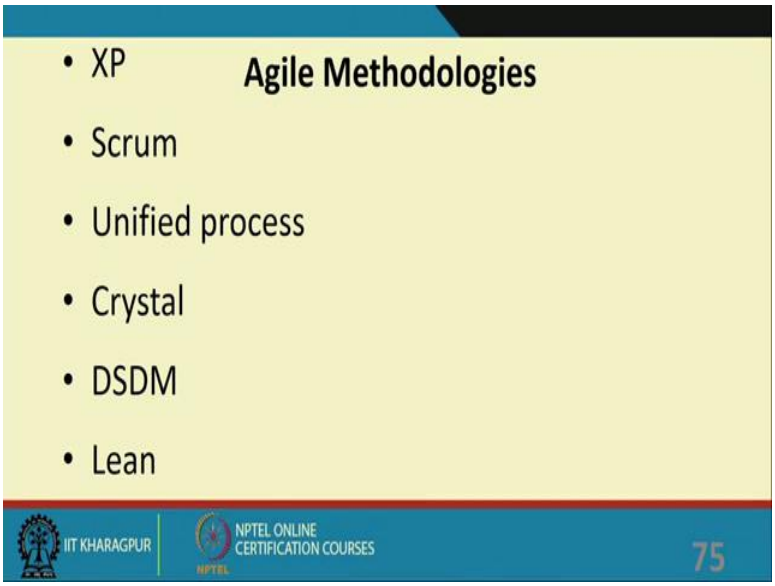

The agile methodologies is an umbrella term. There are many processes development processes which have come up like extreme programming or XP, Scrum, Crystal, DSDM, Lean etcetera each of them have the features of the agile model. But then but then they differ in some ways, each one has its own focus. New things that are introduced in each of them as you will see new terms, new activities and so on.

But then, they all have the features of the agile methodologies which we just so discussed in the last slide, development over increments, interaction with the customer, less documentation interaction among the developers, testing in integrating and testing or each iteration, deploying getting customer feedback and so on.

For this lecture, just come to the end and in the next lecture, we will discuss more details about the agile development model.

Thank you.

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Lecture – 09
Life Cycle Models - V

Welcome to this lecture. In the last lecture, we had discussed about the evolutionary model, the spiral model and just introduced the agile development. The agile models have become very popular. We will see that these have lot of advantages especially for the projects that are being undertaken now. At the beginning, we had said that the type of projects that are undertaken have undergone a drastic change over the years. Now the projects are of very short duration; 1 month, 2 month or 3 month projects are very common whereas, earlier there were multiyear projects; 3 year, 4 year, 5 year projects and now lot of reuse is being made only customization work.

So, that is service type of projects. Earlier we had product development from scratch, now it is customization. Slowly most of the projects are becoming like this that short development deployment at the customer side getting feedback and so on. And in that respect, the agile model is very advantageous and we had just started discussing about the agile model.

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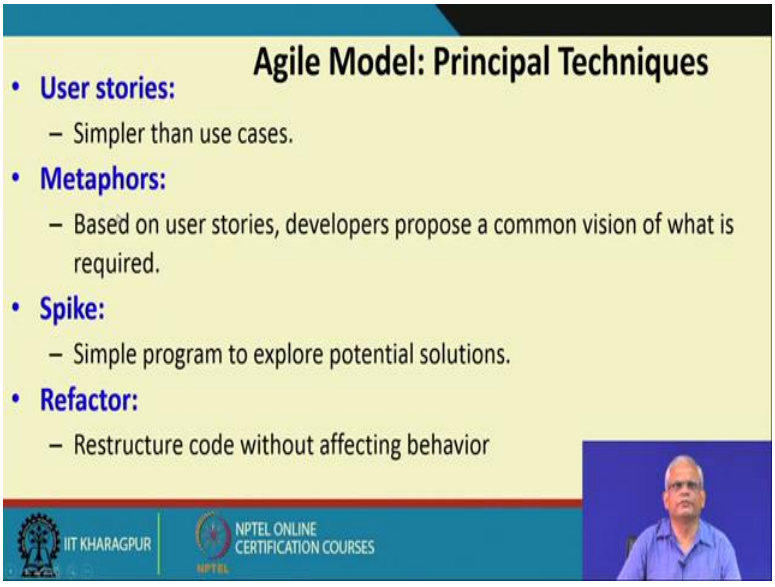

Let us look at more details of the agile model. Last time we had seen that the agile model incorporates some features like customer participation, incremental development, integration and testing over each iteration, deployment at the customer site, less documentation and so on.

Now, let us look at the main techniques of the agile model. Here no formal requirements are developed; it is based on user stories; as the name says that these are more informal than a requirements specification. These are like stories; these are simpler than the use cases. To give a overall design perspective, agile model proposes metaphors where there is a common vision of what is required and based on that the development starts. Wherever required a spike is done; a spike is a simple program that is written to explore potential solution. We can see that it is similar to a prototype. Wherever there is uncertainty develop a spike, check out the alternative whether the spike performs well and so on and also refactor.

Once it is developed at the customer site and the customer accepts it, restructure the code without affecting the behavior such that the code becomes more structured, put some design into the code. So, here as you can see that the design after each iteration is afterthought, the code is restructured. Initially the code is made to work and then it is restructured of course, we have the metaphor where there is some common vision overall design and so on.

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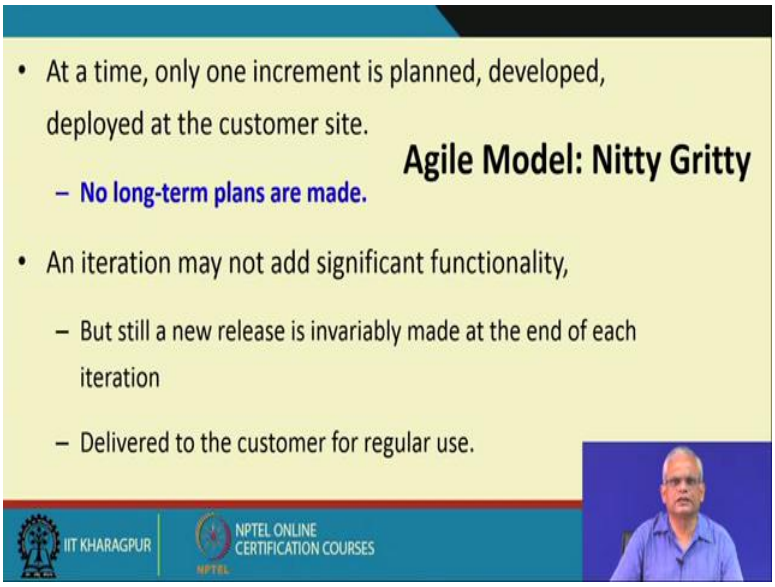

It is incremental; each time one increment is planned developed and deployed at the customer site, no long term plans are made. Iteration may not add significant functionality, it may just only enhance the existing functionality.

But then at the end of each iteration invariably, code is deployed at the customer site and the length of the iteration is usually fixed something like 2 to 4 weeks. And remember here that after each iteration, the customer makes the code; puts the code into regular use. It is not that they just evaluate it, they just they start using it regularly.

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As we are saying that the agile model, one important thing is face to face communication. This required that the developers share the same building or the same room they meet regularly discuss issues rather than passing documents to each other they go and explain to each other what is required and so on. To facilitate face to face communication, they share a single office space and typically the team size is small 5 to 9 people so, that they can interact well.

And this makes the agile model very suited to small projects, but of course, there have been effort to use this to scale this to larger projects. But then it started with focused for small projects. One thing need to mention here is that when want to convey something to somebody, it is not a good idea to give him a document to read. If you can explain him over a whiteboard that is the best way, you can explain him; you pass a document or explain him on phone or email that is not a good idea.

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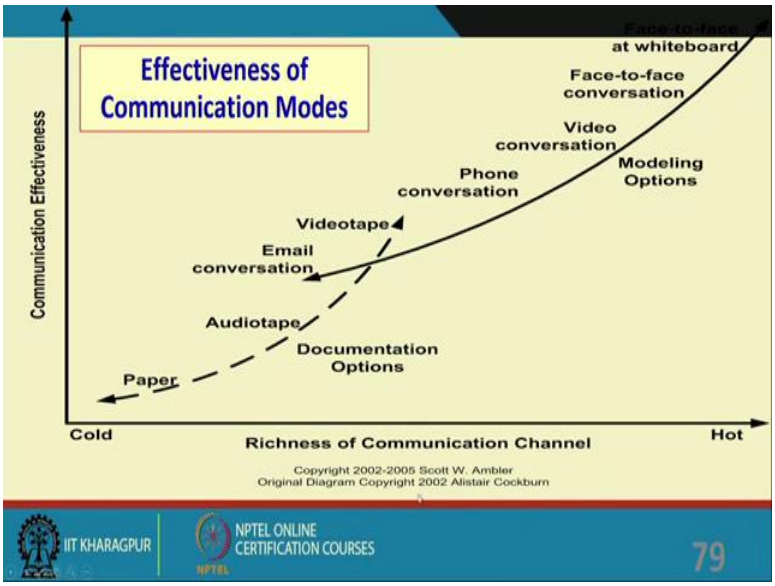

In fact, there are experiments done to find the effectiveness of various communication modes. This is given by Alistair Cockburn from experiments; he could determine that paper is a cold form of communication not very effective. If you want to convey somebody something give him a paper specially technical communication is not a good way to communicate. Audio tape is slightly better than paper. Email conversation may be slightly better; videotape slightly better, video conversation is still better and face to face conversation with a whiteboard. This is the most effective way to communicate technical items; this was the finding here. So, a small team they meet with a whiteboard and one person explains his ideas on the whiteboard that is the most effective way to have the communication and that is incorporated in this model; face to face communication is give given importance.

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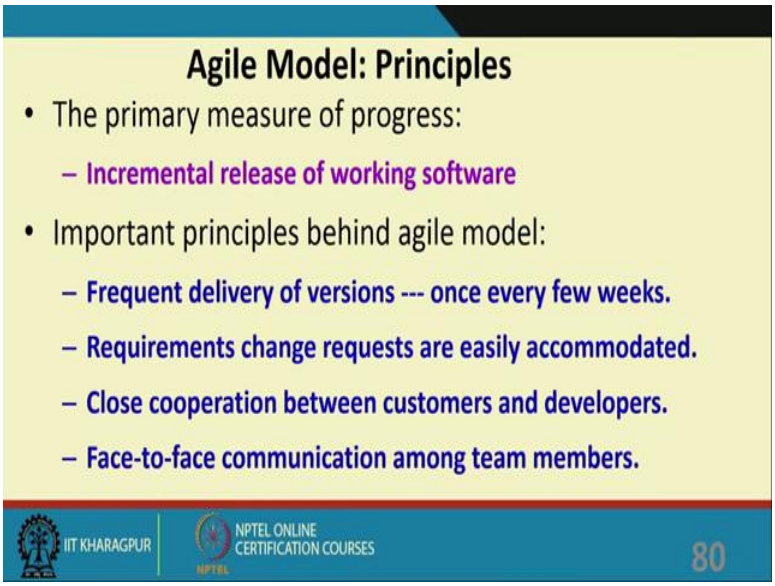

Here the progress is made measured in how many increments have been deployed at customer site. There is a frequent delivery of versions once every few weeks. The customer uses the delivered versions and gives requirement changes. These are accommodated is a close cooperation between the customer and developer and among the team members there is face to face communication. These are some of the very fundamental principles of the agile model.

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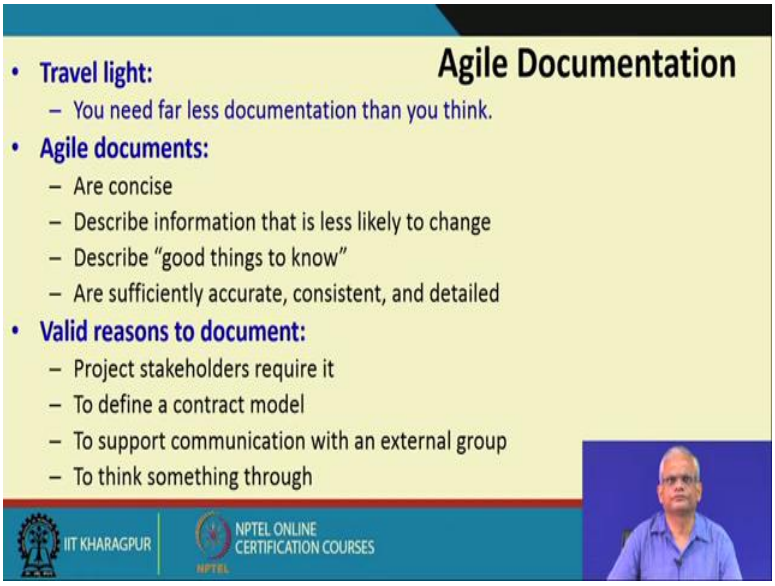

But what about documentation is no documentation done in agile model, not really. Here the idea is to travel light that is you do not document unless it is needed. You need far less documentation than you think. In the waterfall model, everything needs to be documented; the review results, the plans, the requirement review, the design, high level design, high level design review everything is documented. But here, the main idea here is that the documents need to be prepared only when these are required by somebody to be referred over a long time. But even then these are concise; these are the document is prepared only for information that is not likely to change that is at the end of the development things do not change.

So, the end of development documents are prepared and here only those things are documented where somebody can learn something which will be very informative to somebody which is good things to know. It is not that for the sake of documentation, documentation is created. The documents are sufficiently accurate, consistent and detailed because these are prepared when there are no more changes. Some of the valid reasons to document are that the project stakeholders need it; for example, the customer needs it maybe there is a contract form to be signed for which it is needed, maybe there is an external group who like to review or give their suggestions, maybe the document will be for them, maybe the document will be for those to be referred after the project is over and so on.

Only when there are valid reasons to document, the documents are prepared. It is not that documents are prepared by default as is the case with waterfall model.

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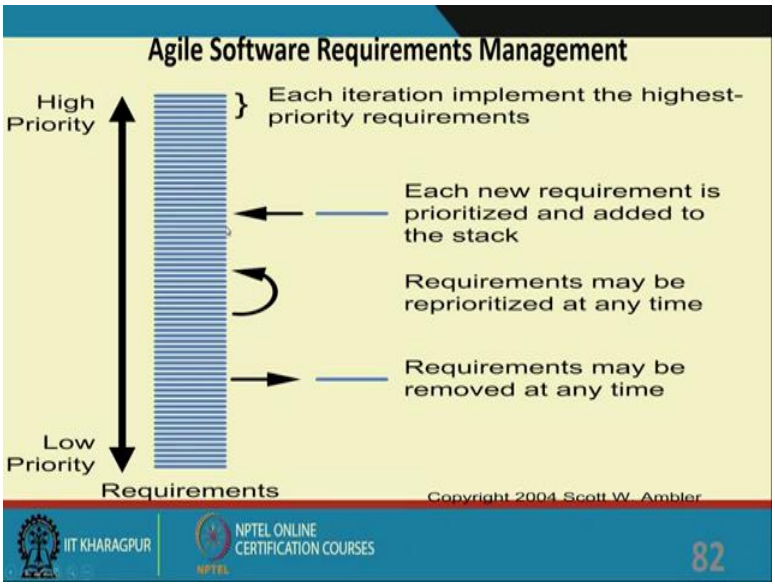

The requirements here are changing over the development time and new requirements as they arise, they are kept in the requirements stack here. The stack is prioritized the top are the ones that will be first taken up. These are high priority and these are all requirements and this diagram given by Scott Amblers book the requirements keep on arising. They are inserted in the stack, they are taken out removed whenever necessary they change; they are reprioritized, but then there is a priority assigned to each requirement and then this are changed dynamically.

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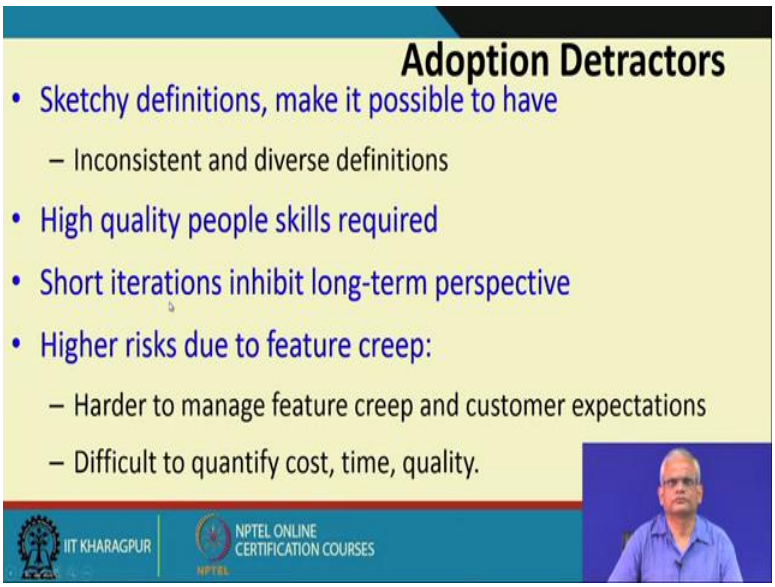

The agile model has many advantages, but then to make it practically useful, you must be aware of some pitfalls. First thing is that here documents are not there. It is based on explanation on a whiteboard and so on. And therefore, somebody can misunderstand.

High quality people skills are required who when in doubt will consult each other and do what is correct. Long term design is not made and the short iterations, these degrade the design structure. There can be feature creep feature creep is the word where the customers or the developers become more ambitious. They just keep on thinking new features which can be incorporated without thinking of whether how much the feature will be used; what will be the value to the customer, what will be the cost of developing it, feature after feature get added. So, there is a higher risk of feature creep in the agile model because the freedom is given to give feedback and request for new features and so on.

And therefore, it becomes difficult for the project manager to manage the feature creep and also since the development is dynamic, the features change new features get added and so on. To give an upfront cost to a development becomes very very difficult; to give a upfront timeline by which development will be done is difficult and quality difficult to assure sure much more difficult.

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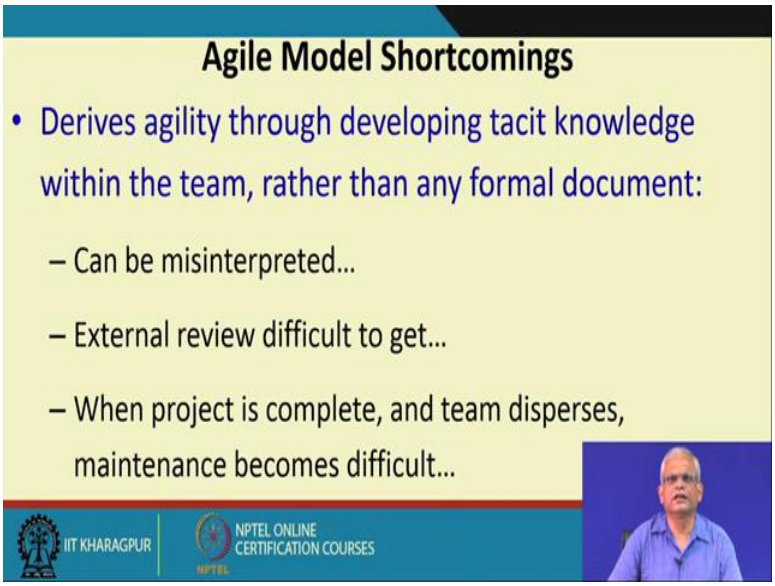

The document a done away that is a good thing saves time effort of preparing document, but then must be aware that the lack of document. There can chances of

misinterpretation, getting reviews on the document are difficult and also when the project is complete unless the documents are consciously prepared at the end of the project before the team disperses, then maintenance may become difficult.

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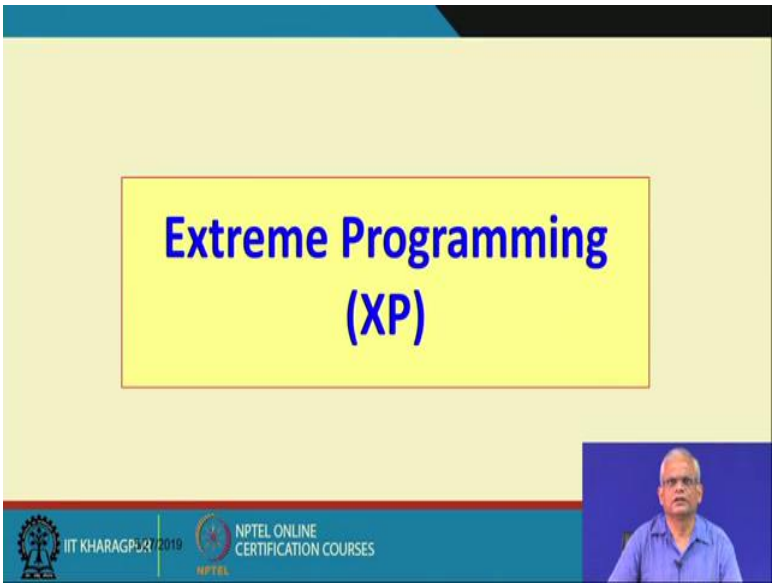

With this overall understanding of the agile model, let us look at some specific agile processes. One of the agile process which is very popular is the Extreme Programming which is also known as the XP model.

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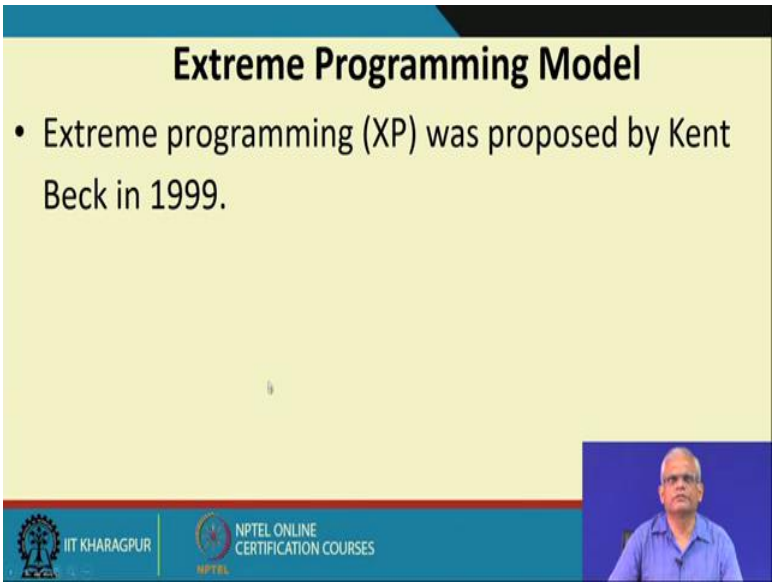

It was proposed by Kent Beck, 1999.

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The main items that are introduced here is pair programming. In a pair programming, each desk each desktop is manned by two programmers. The main idea here is that reviewing is a good thing; so, why not continually review each other's work. When one programmer writes the program, the other programmers goes through the code and reviews it give suggestions how to make it better code, more efficient code, avoid mistakes and so on and the take turn one programmer program programs for an hour maybe writes few functions and then the other programmer takes up, write few more functions where as the first programmer reviews that.

Here every day the testing takes places unlike the waterfall model where testing is at the end. Here the test cases are written continually and the test cases are executed before a feature is passed. Before a feature is implemented, the test cases are written for that feature and the feature is considered passed when it passes the test cases. This is called as test driven development.

Here incremental development is practiced every few day increments are given. The name extreme programming comes from the fact that the good practices are taken to extreme. The good practices are code review is good and this is implemented in the form of pair programming testing is good; good practice and therefore test driven development. Incremental development is good therefore, every few days new increments are develop in delivered. Simplicity is good and therefore, the simplest

design is developed. Do not try to make it think of extensions that may be required decades later or long time, afterwards some enhancement may be required; do not think of it.

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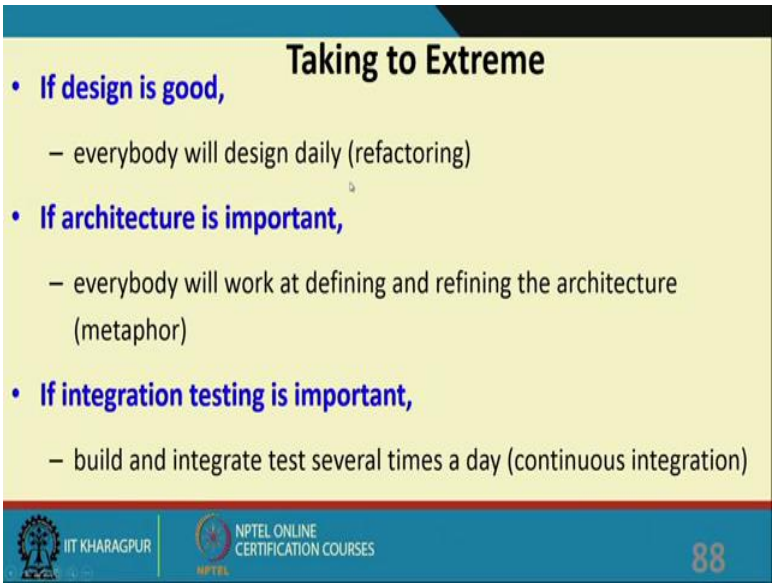

Now, make it work make the simplest design make it work. Designing is good and therefore, after the software is deployed accepted, refactor the code, put some design, make the code better architecture is important and therefore, a metaphor has to be defined. Integration testing is important; therefore, several times a day build and integrate that is continuous integration as the development proceeds. Remember that this was one of the major problem with waterfall model where the integration took place only at the end.

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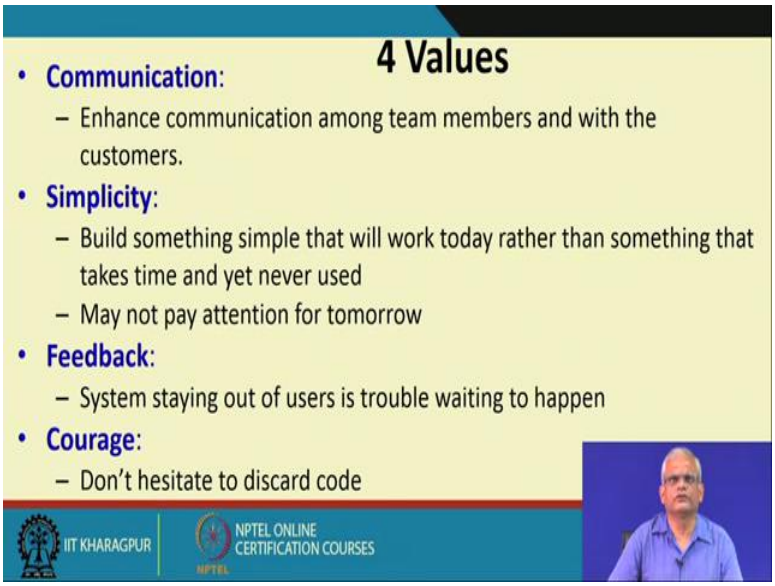

The four values here; communication face to face communication, simplicity, implement the simplest design may not pay attention for tomorrow make it work. Feedback get customer feedback, encourage customer feedback. If you do not get feedback from the customer, this is trouble going to happen because at the end they will reject; they will be unhappy. And here the other value is courage. If some code is not good discard it, rewrite it. Design is not good, discard the design; redo the design.

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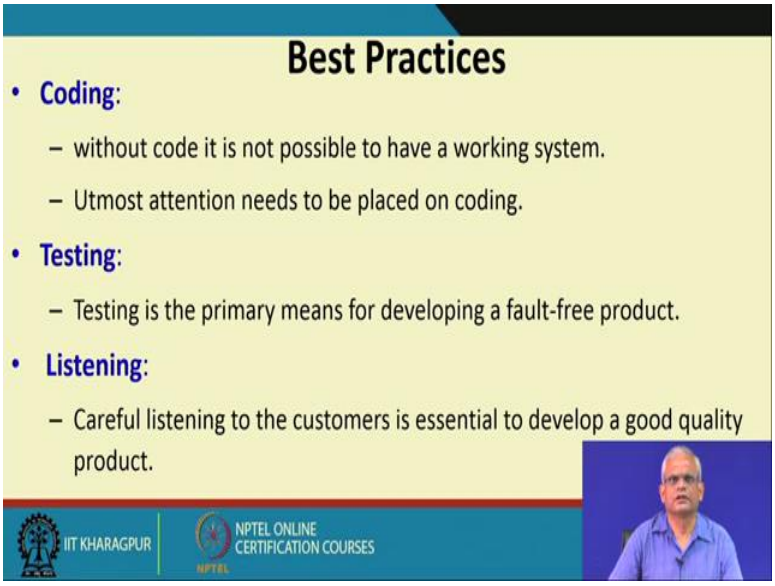

Coding is a best practice utmost attention on coding testing is a best practice primary means of developing fault free software and therefore, importance to testing and listening is a best practice listen to the customer and find out what is required really by the customer.

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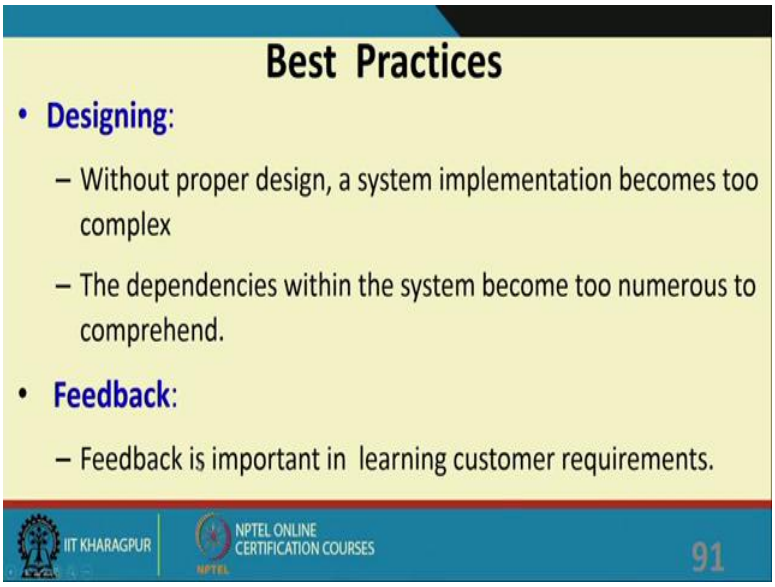

Designing without proper design, the system becomes complex. Therefore, put design into the system. And feedback is a important thing get customer feedback.

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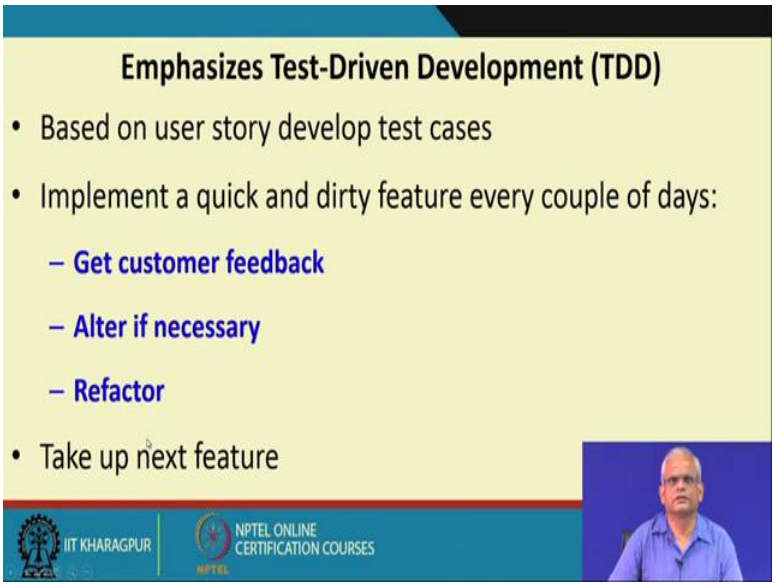

Emphasizes test driven development; based on the user story, first develop the test cases before writing the code first write the test cases that is the test driven development. First write the test cases based on the user story and then implement it. Once implemented run the test cases and the development is not complete until it passes all the test cases. Develop over increment every few days increments to be given, get customer feedback, alter if necessary and then finally, once accepted by the customer re factor, make the code better, put some design and then take up the next feature.

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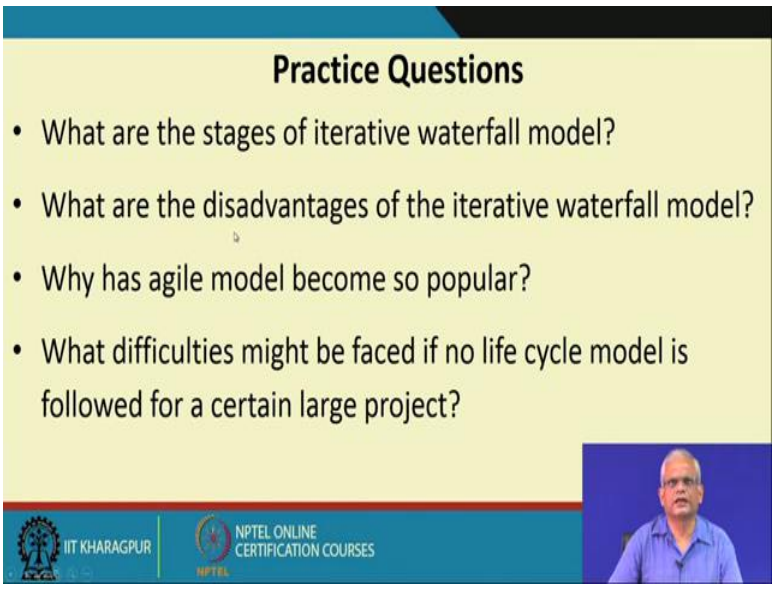

Now, let us look at few practice questions. So far, the points that have discussed; what are the stages of iterative waterfall model. Hope you remember, if not please look up; what are the major disadvantages of the iterative waterfall model.

Because the later models were proposed to overcome the disadvantages of iterative water fall model which you had already discussed. Several problems with the waterfall model, most important is difficulty in handling change requirements, why has agile model become so popular, what problems of the waterfall model it overcome, how it fits into the current type of projects, what difficulty might be faced if no life cycle model is followed in a certain large project ok.

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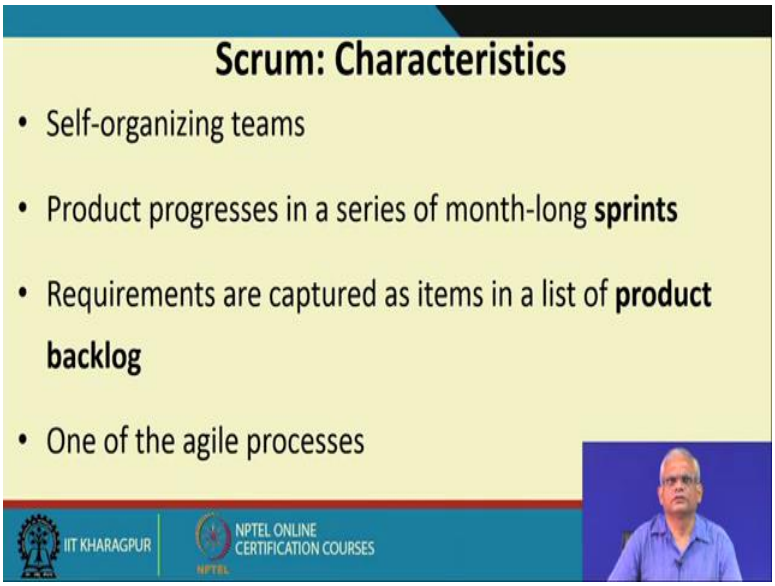

Now, let us look at another agile development process which is called a scrum. The main characteristics of scrum is self organizing teams that is the team members decide among themselves who will do what, who will do testing, who will do which function development etcetera. Here the product progresses in a series of month long sprints; the sprints are basically increments.

So, the increments are called here as sprints typically one month. Here the requirements are captured, we are looking at the requirements stack something like that which is called as a product backlog.

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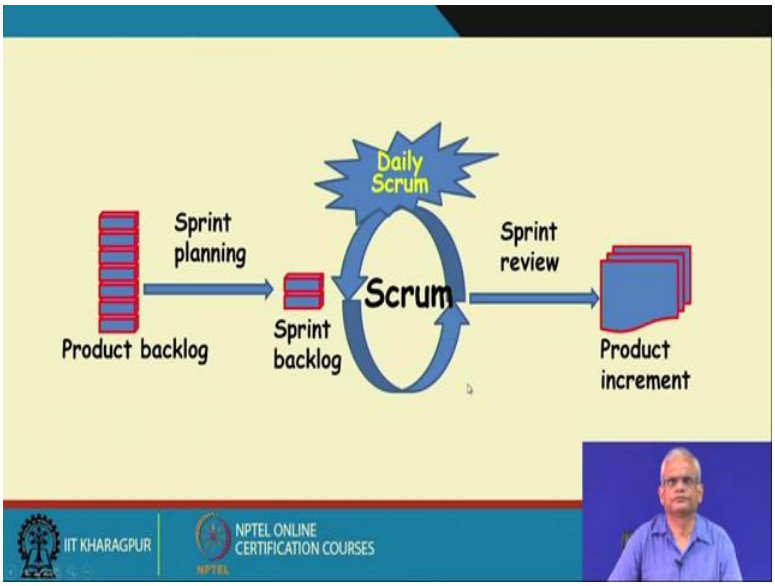

So, one of the agile processes; here is the requirements that have been gathered which arise as the development proceed some may get deleted, changed and so on. This is called as a product backlog. In each sprint, a sprint is a month long activity. One of the top priority feature requirement is taken out that forms the sprint backlog and once the sprint backlog is obtained, this is not changed anymore. It is immune to change and this is developed over a month long iteration here and every day the developers meet for a daily scrum meeting to review what has happened, what is the next thing to do and so on and they complete the sprint backlog.

In the sprint backlog, based on the identified requirements from here activities that need to be done to meet the requirements are identified and that are put in the sprint backlog which is put by the development team. And these activities are get completed over daily scrum and finally, the sprint review; the sprint is reviewed and the product increment is deployed at the customer site.

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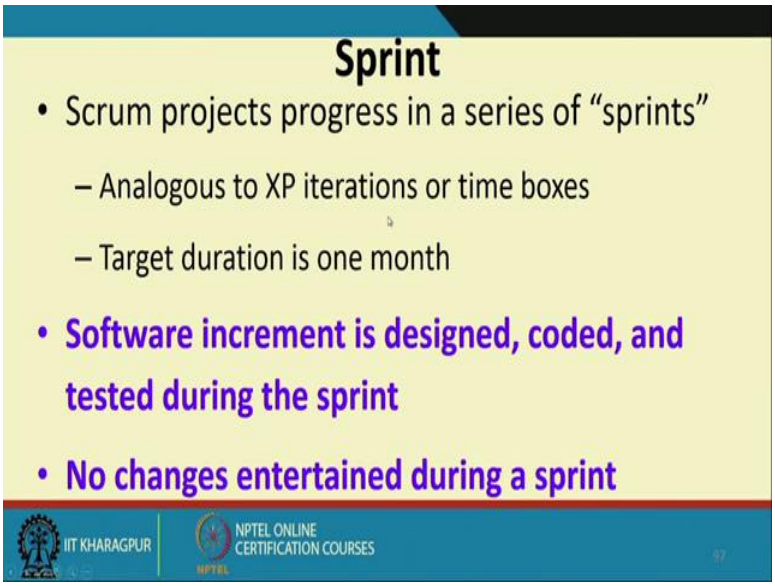

So, that is the main idea here. The progress is in the form of sprints, duration is typically one month. In one month, the some of the features from the product backlog are taken up; design, coded, tested. And once the sprint starts the requirements that have been taken up are not allowed to change otherwise the sprint will not converge. One of the principle here is that once the feature have been taken out from the product backlog, they are not allowed to change.

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Here some of the terminologies the roles, ceremonies and artifacts. The roles are product owner. One of the team member acts as on behalf of the product owner; that is a customer. He has the customer perspective, the scrum master who is like a project manager and then the team members. There are various ceremonies that get conducted during development; one is the sprint planning, the sprint review at the end of a sprint, sprint retrospective and daily scrum meeting. There are various artifacts produced one is the product backlog which keeps track of the requirements that have been identified so far, more requirements can be identified as the development proceeds and these are kept in a prioritized order. The sprint backlog this is the activities to be done during the sprint. The burn down charts here how much progress has been made are depicted in the form of burn down charts.

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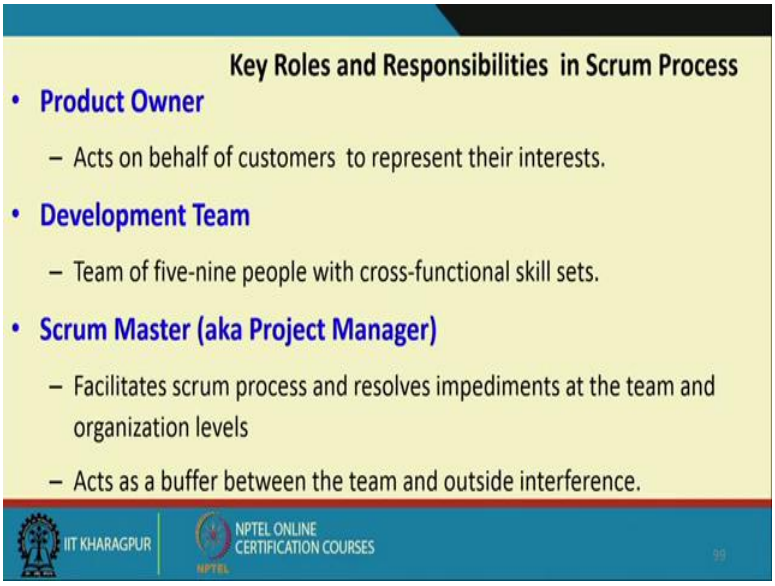

The product owner has the customer perspective is the development team has five to nine people with cross functional skills. The scrum master is also called as a project manager. He facilitate the scrum is the buffer between the team and outside in interference and resolves any difficulties that that team members might be facing.

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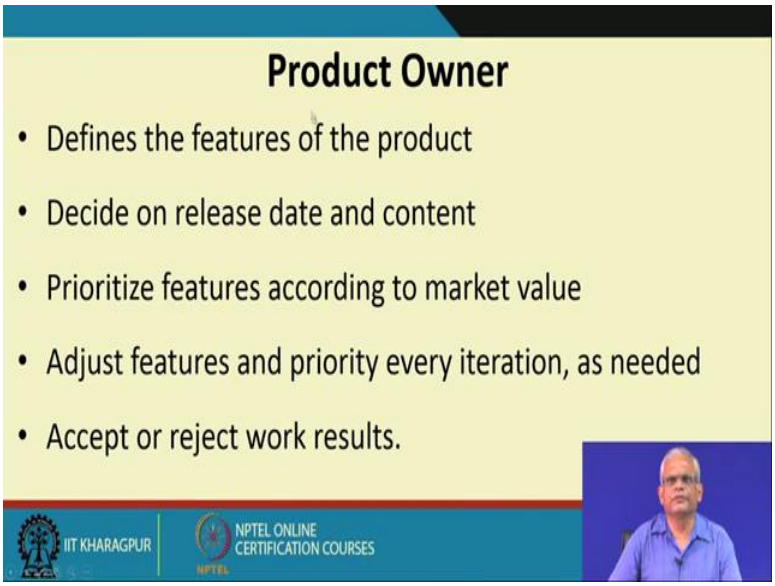

The product owner is one of the team members or maybe one of the customer representative who is part of the team. He acts as if the representative of the customer. He is the one who defines what are the features that will be required decides on the release date. Prioritizes features based on the customer perspective according to what will be the market value, adjust features and priority in the product backlog in every iteration and finally, accept or reject the results.

We have seen the main idea behind the scrum. There are many terminologies in the scrum; looked at some of the terminologies and we are just coming to the end of this lecture few more terminologies and concepts in the agile scrum model is could not discuss in this lecture. We will take up in the next lecture and will complete our discussion on the development life cycles in the next lecture.

Thank you.

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Lecture – 10
Life Cycle Models - VI

Welcome to this lecture. In this lecture we will first discuss about Scrum which is a agile and very popular development model, many industries and projects now use the Scrum.

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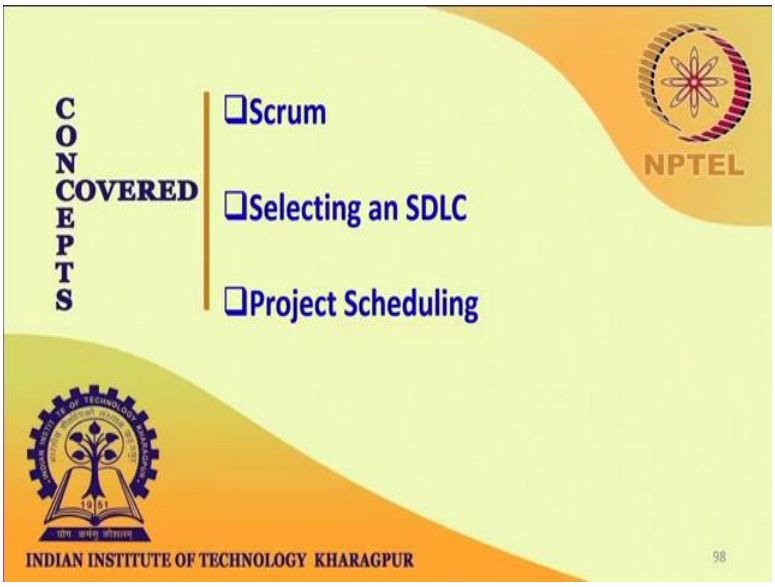

And then we will just review what all lifecycle models we have discussed and we will look at how to select a lifecycle model for a given project and then if possible, if we still have some time will discuss some introductory discussion on project scheduling. Now let us look at Scrum.

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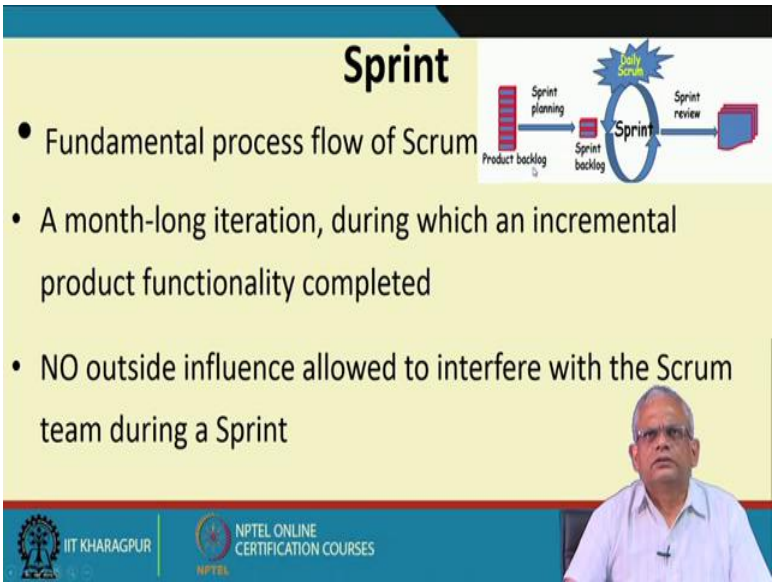

In the last lecture we had looked at Scrum which is one of the agile models and one important thing here is the Sprint. The Sprint is the fundamental process flow of Scrum. A Sprint is actually an iteration or a time box in which an increment is developed during the Sprint and is delivered to the customer. Usually it is a month long iteration in which some existing features might get improved or some new functionality may be implemented. From the product backlog the product backlog has all the story points are the features has to be developed are maintained in a product backlog and during Sprint planning the objectives for a Sprint is identified. This is called as a Sprint backlog. These are the objectives or the subset of features of the product backlog will be developed during 1 Sprint.

A Sprint as it continues every day the team meet and discuss what is their plan for every day and are there any problems that they are facing and after a month long iteration the Sprint is completed, the Sprint is reviewed for what has been accomplished and then it is delivered to the customer. One thing that the Scrum methodology or the Scrum model it requires that once the team members identify the subset of product backlog to be developed during a Sprint which forms the Sprint backlog there is no outside interference with the team. That means, that once they take up some items for implementation no changes to these items are allowed, otherwise it will create chaos, the Sprint will not converge, uncertainty and so on.

And that is the reason why no outside interference with the Sprint team is allowed during as with the Scrum team allowed during the Sprint.

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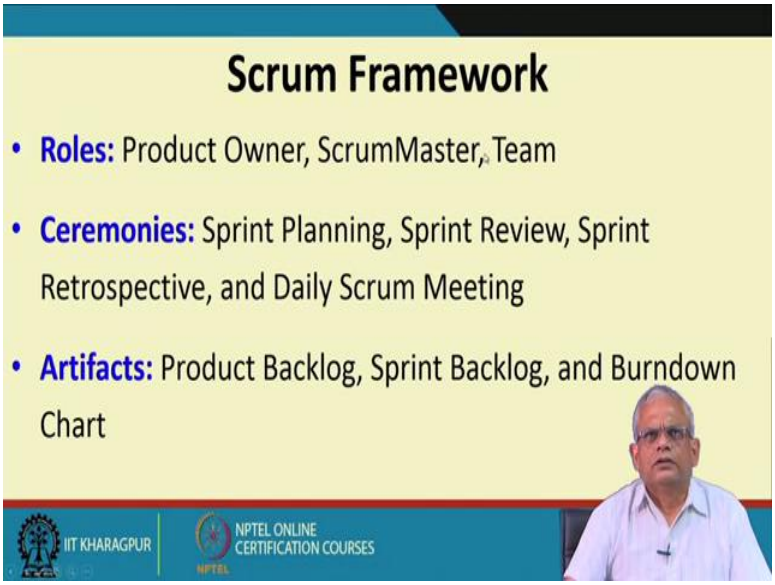

These are some of the important concepts in the Scrum framework. One are the various types of roles. In a Scrum team there are the product owner who is one of the team members the Scrum master who is another team member and the other team members. These are the different roles in a Scrum team a product owner Scrum master these are distinguished team members and then the other team members. In the Scrum framework some ceremonies or meetings that are required to be conducted one is the Sprint planning meeting, Sprint review meeting, Sprint retrospective meeting and the daily Scrum meeting. We look at this meetings.

In the Sprint planning meeting this is conducted just before a Sprint starts and here a subset of the product backlog is identified and the Sprint backlog is formed. The Sprint review meeting this is conducted at the end of the Sprint. Here the work that has been completed during the Sprint are reviewed. The Sprint retrospective meeting it is conducted at the end of the Sprint that is after the Sprint review meeting and before the next Sprint starts or the next Sprint planning meeting. And here any reflections on what could have been done, what is required to be done etcetera are discussed and the daily Scrum meeting it happens daily. Here the team members discuss among each other in a

short meeting that what is their plan are they facing any obstacles and also this is a way for the Scrum master to keep track of the progress of the project.

There are some artifacts which are mandated in the Scrum framework. One is the product backlog, here the items or the features that are not yet complete are maintained in the product backlog. This is a prioritized list of features to be implemented not yet complete. The Sprint backlog is a subset of product backlog which are to be completed during the Sprint. The burn down charts these are basically concise representations of the progress during a Sprint, the amount of the work to be done for project completion and so on. We will look at the burn down charts in the subsequent slides.

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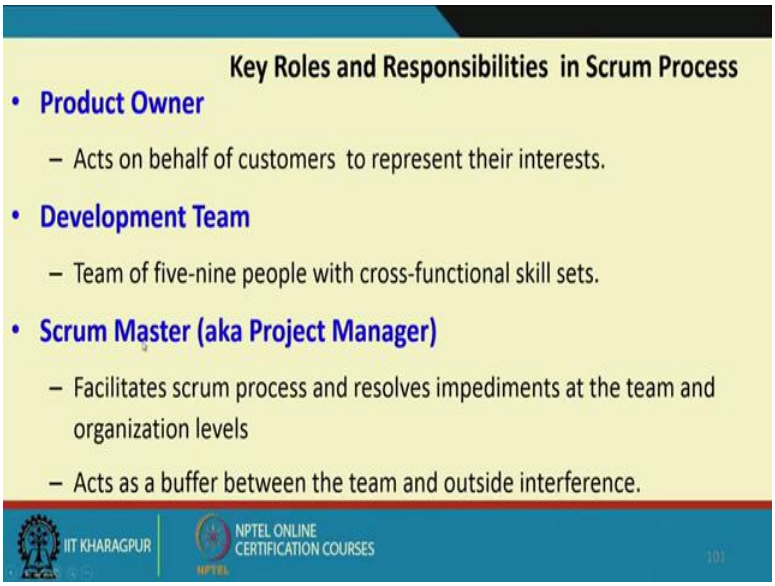

The key roles are the product owner, development team, and the Scrum master. The product owner is a team member who represents the customer's interest, thinks on behalf of the customer. The development team typically 5 to 9 people with the cross functional skills; that means, each team member typically has multiple skills maybe a tester, a GUI developer, coding, database and so on.

The Scrum master also known as the project manager he is the team member who interfaces with the top management and any difficulties that arise he discusses with the customers and the top management to remove any difficulties. And, that is how insulates the team member from having to talk to the top management, the customers and so on and also if there are any outside interference during a Sprint he is the one who takes care.

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The product owner is the one who represents the customer. He liaises closely with the customer to get their perspective. The product owner defines the features that are required. New features may crop up some of the features which are identified may get deleted or modified, he decides on the release date of the features. Prioritizes the features according to the market value or the value to the customer, adjust the picture priority on every iteration if needed and finally at the end of a Sprint accepts or rejects the work results.

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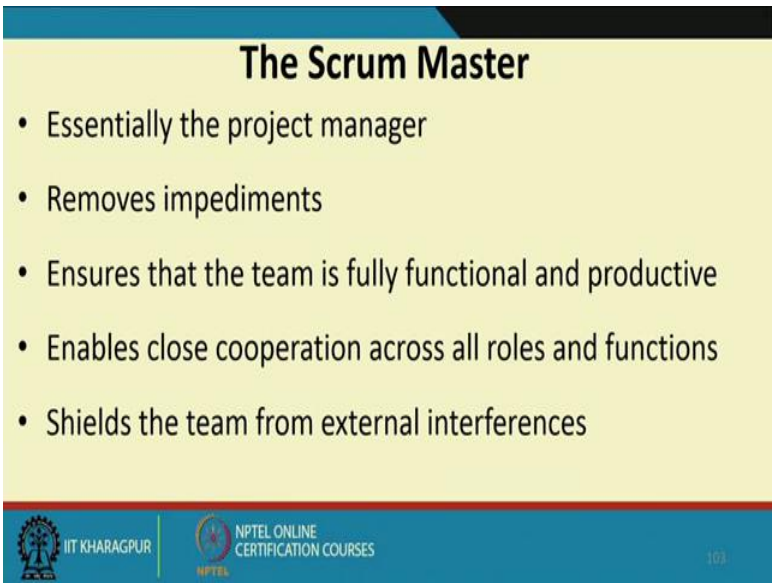

The Scrum master is essentially the project manager removes any obstacles that the team faces, ensures that the team is cohesive, fully functional, productive, close co-operation is ensured. It is responsibilities to ensure that all roles and functions there is a closer co-operation and also shields the team from external interference, interfaces with the customer top management. So, that the team members are relatively prevented from external interference.

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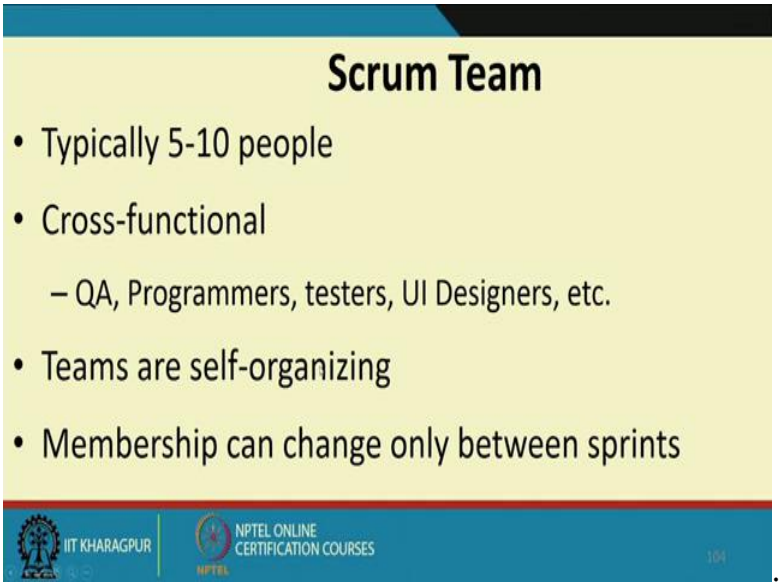

The team members typically 5 to 10 people they have cross functional expertise. For example, a team member may have quality assurance programming or maybe testing UI designing etcetera typically different functionalities and maybe multiple of them. The teams are self organizing that is they decide who will do which part of the work and what role they will assume and the membership of the team is not allowed to change during a Sprint. New members may be inducted some member may be replaced and so on, but for the process to be stable no changes are allowed during a Sprint.

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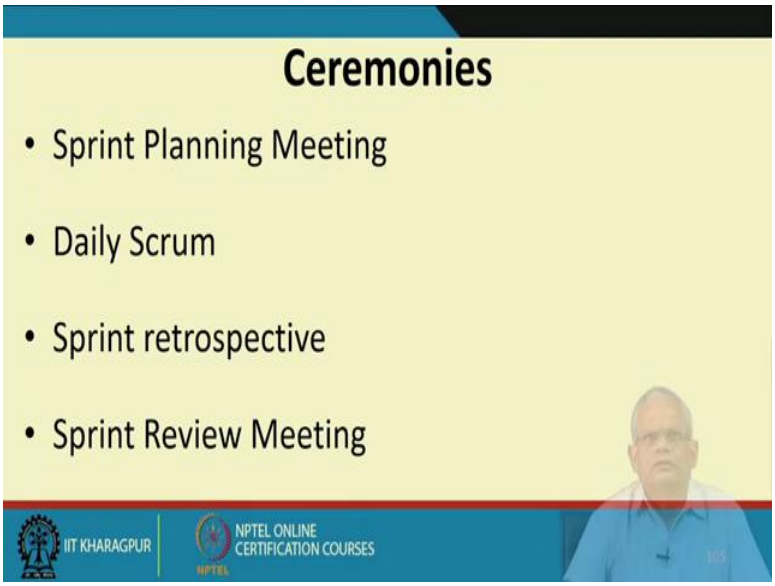

The ceremonies these are the meeting there are 4 important meetings that are mandated here. One is the Sprint planning meeting which occurs at the start of a Sprint. Here the Sprint backlog is formed from the product back log the features to be developed during the Sprint. The daily Scrum meeting it is a daily meeting during the Sprint. Every day during a Sprint the team members meet for a short time maybe 15 minutes typically, a stand up meeting otherwise the meeting may prolong it is intended to be very short meeting and they just discuss what are the items they are working on are they facing any obstacle and so on.

The Sprint retrospective meeting is conducted after the Sprint review meeting just to reflect on the things that have been done what could be improved to be taken up in the next Sprint and so on. The Sprint review meeting is undertaken at the end of the Sprint and here the work that has been achieved during the Sprint is reviewed along with the customer at the top management.

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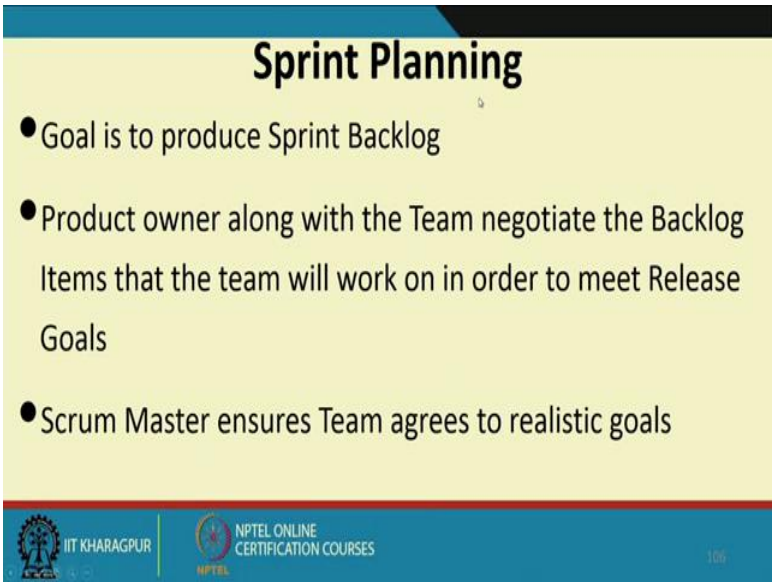

The Sprint planning meeting it is undertaken just before a Sprint starts. The main objective of this Sprint planning meeting is to produce the Sprint backlog. Here the product backlog is examined and the product backlog is typically organized in a prioritized manner and the top priority features are picked and the product owner along with the team they negotiate what are the backlog items to be taken up so that it will meet the requirement of the customer and also it will meet the release goals.

The Scrum master the project manager he also participates during the Sprint planning and he ensures that the team agrees to realistic goals because it is a 1 month iteration and the work should be doable in roughly 1 month.

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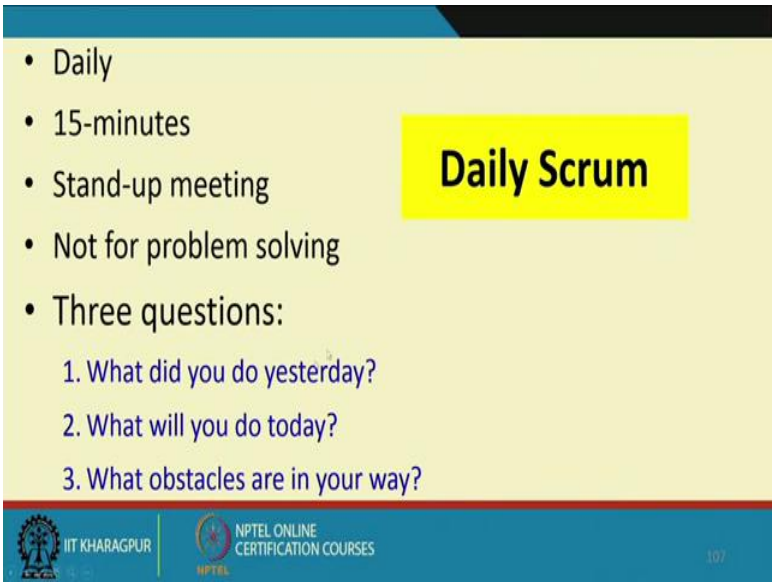

The daily Scrum meeting is done every day, typically at the start of the day it is very short meeting typically 15 minutes or so. It is a stand up meeting because if the team members sit down then the meeting may prolong stand up meeting and it is not a problem solving meeting, but then the meeting objective have to identify what the team members are working on just to update each other, are they facing any obstacle. Basically each team member answers 3 questions. What did the team the member achieve yesterday? What is the plan for today and are there any obstacles? These are only 3 things that each team member answers discusses.

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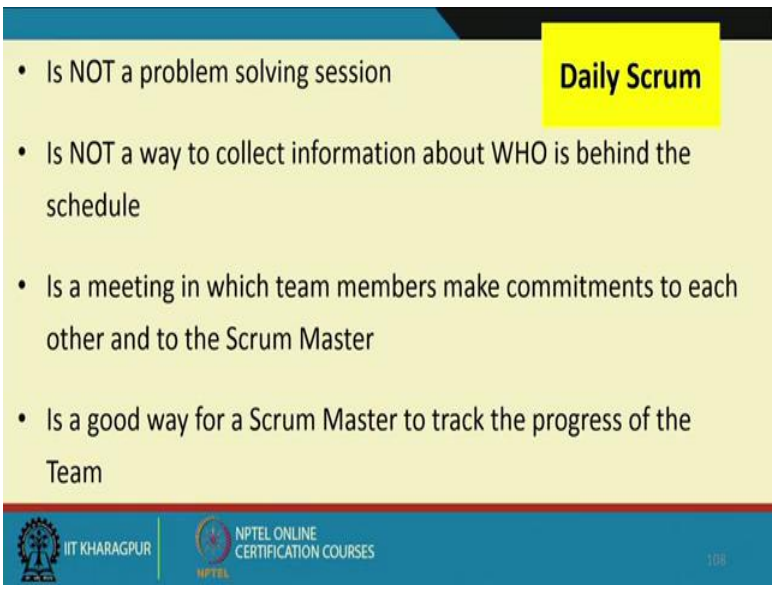

It is important to remember that the daily Scrum is not a problem solving meeting. The team members do not really ask for solution it just the it is a information sharing meeting and also it is not a blame fixing meeting that who is behind schedule why is he behind schedule etcetera. This is only information sharing. Here each team member informs what was achieved yesterday and what is the plan for today and are there any obstacles. The Scrum master or the project manager participates and as the discussion proceeds he picks up that what is the progress that has been achieved so far and this is a good way for the project manager to track the progress of the project.

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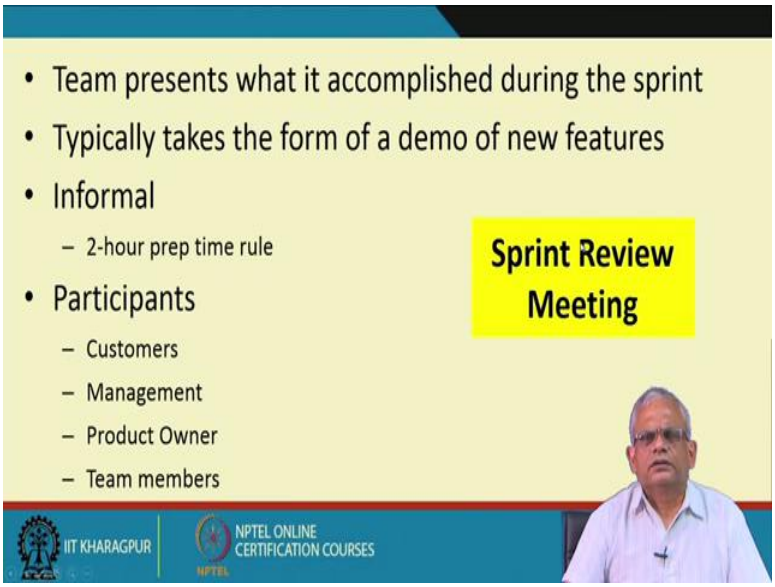

The Sprint review meeting this is undertaken at the end of the Sprint. Basically here the work that was accomplished during the Sprint are reviewed in presence of the management, the product owner, the team member and also customer representative intended to be a very short meeting and typically it is a demonstration of the features that have been just completed.

Typically these are on the computer they show that what can be achieved after the Sprint, which features have been implemented how does it work and so on and this is a informal meeting not like elaborate documents that are prepared and so on. That is not the case. It just a 2 hour preparation for the team members they just think about how to present the features that have been completed and that demonstrate those features.

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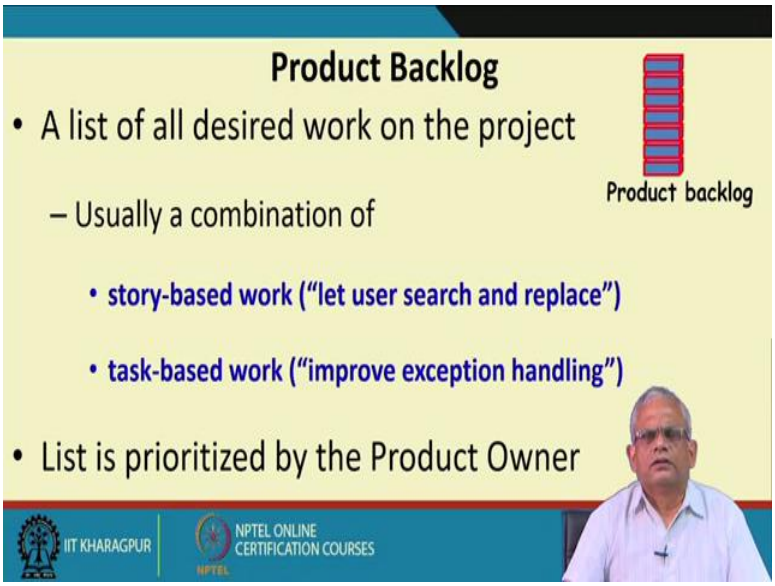

One of the most important artifact in the Scrum is the product backlog. The product backlog is a list of all the desired work for the project. The product backlog there are 2 types of items here. One are the user stories like let the user search and replace, store the item on the database, provide registration of the members and so on these are the story based items. The second category of items that are also present are the task based items.

The task based items are like improve exception handling because still now what was implemented is that the exception handling it works, but not that well and then an item in the product backlog will be introduced is that improve the exception handling.

So, some of the items here are basically the user stories or user requirements and the others are based on the items completed some actions to take on that the list is prioritized by the product owner based on the customer perspective.

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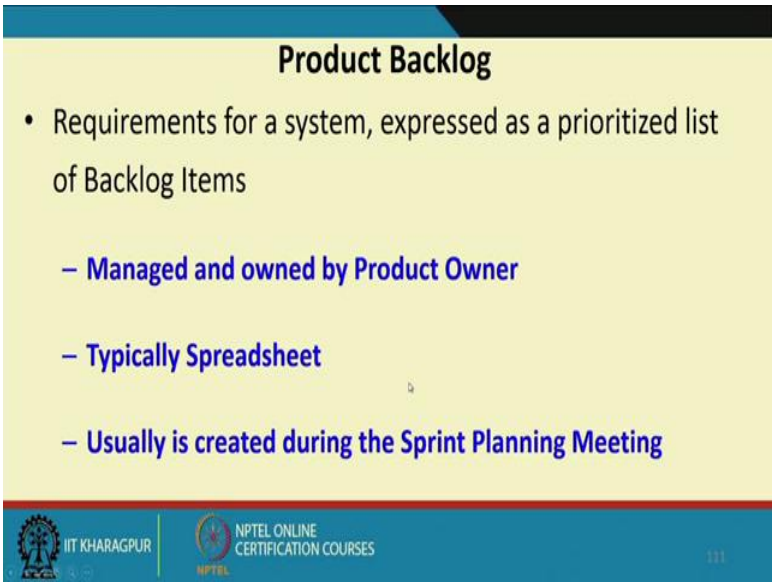

The product backlog is basically managed by the product owner. Typically maintains in the form of a spreadsheet and usually created during the Sprint planning meeting.

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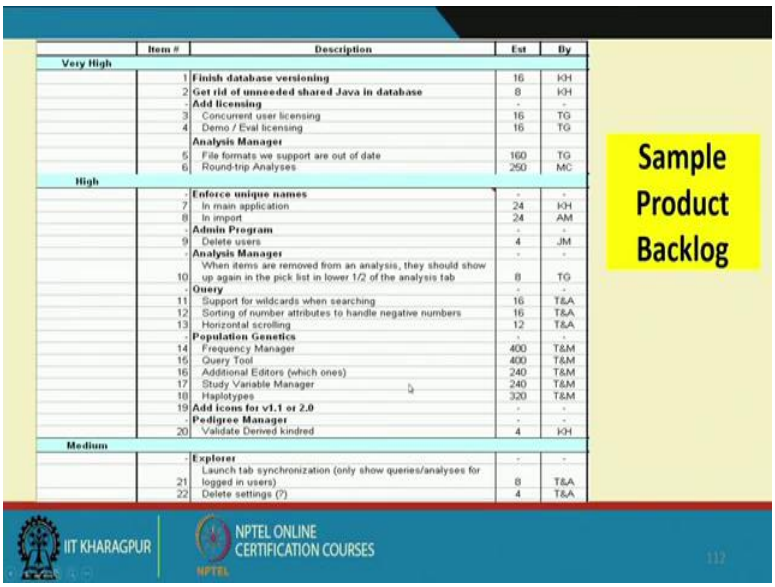

So, this is a typical product backlog a spreadsheet. The priority is very high, high, medium and so on and here these are the item numbers and then a description some of our user stories and the others are some activities to be undertaken.

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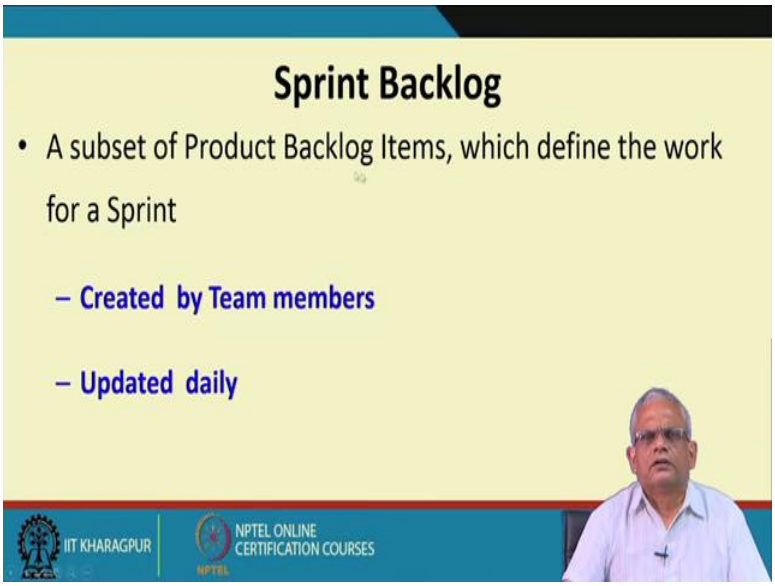

The Sprint backlog is a subset of items to start with. This is created during the Sprint planning meeting. Basically the product owner along with the team members decide what are the items to take up for the next Sprint, but then it is updated daily because some activities may be noticed to be done and that is added to the Sprint backlog.

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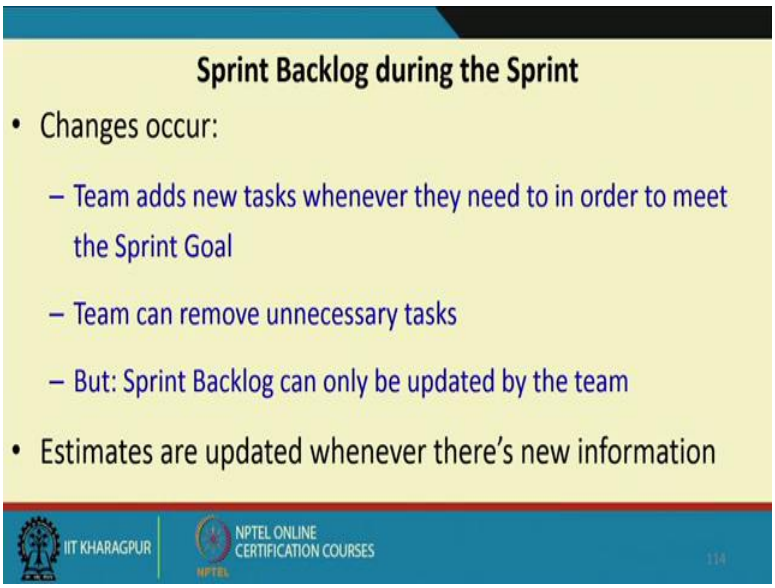

The Sprint backlog changes during a Sprint. The team member can add new tasks in order to meet the Sprint goal. They can also remove unnecessary tasks, but then the thing is that the backlog is only updated by the team. Once the Sprint starts the Sprint backlog

is updated only by the team. The product backlog is updated by the product owner and also the estimates for the work different items and the Sprint backlog are updated whenever there is necessity.

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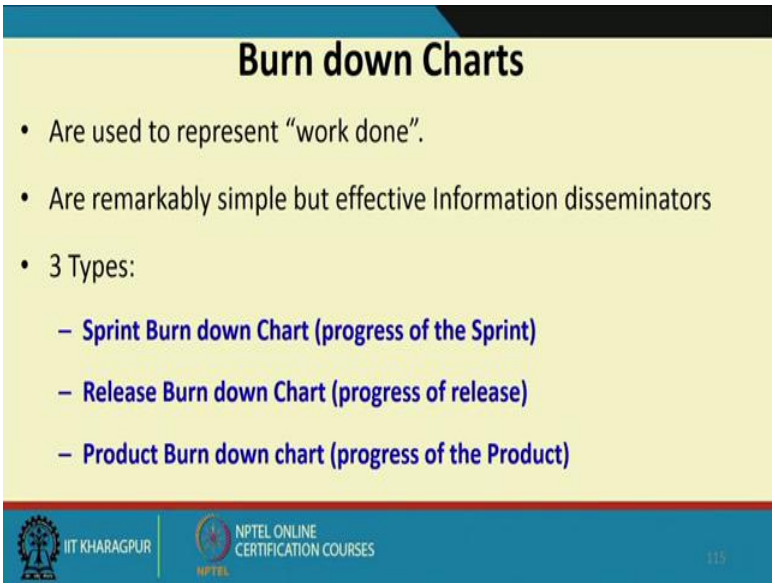

Now, let us look at the burn down charts. This represent the progress achieved so far. There are 3 important types of burn down charts. One is the Sprint burn down chart which is during the Sprint how much progress has been achieved till a date. The release burn down chart; a release may consist of multiple Sprints and how much of this work for the release has been achieved. This is represented in a release burn down chart. The product burn down chart this is the overall progress towards the completion of the project.

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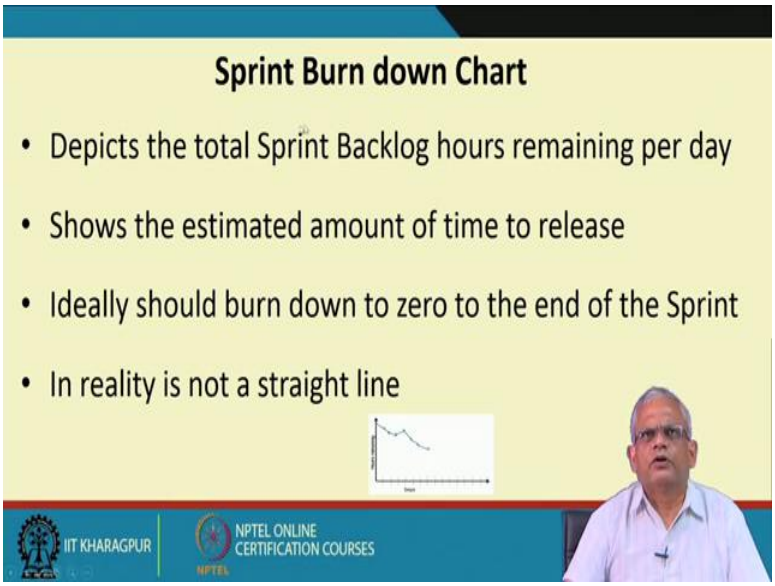

First let us look at the Sprint burn down chart. This represents the progress achieved during the Sprint. Typically represents the hours remaining as the days progress as the Sprint starts some estimate of the number of hours of work is given and then as the days progress here a number of works reduces.

But then it can also increase because there can be a wrong estimation of the complexity of the work. It shows the time to release typically, the days are 30 days and at the end of the Sprint it should become 0. It is not a straight line everyday constant progress is not achieved there are various obstacles and also the estimates are revised as the Sprint progresses.

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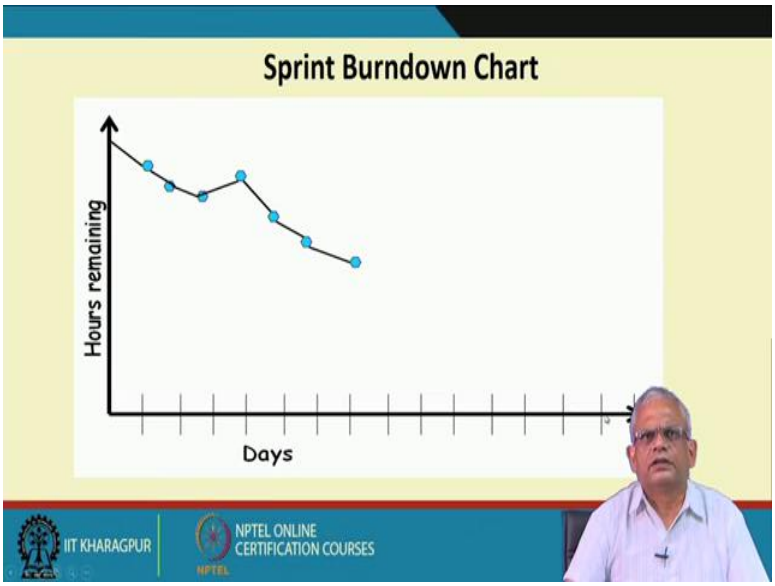

These are larger view of the Sprint burn down charts. For each day it is updated that total hours remaining initial estimation and then as the progress this progress this is updated and finally, at the end of the Sprint should become 0.

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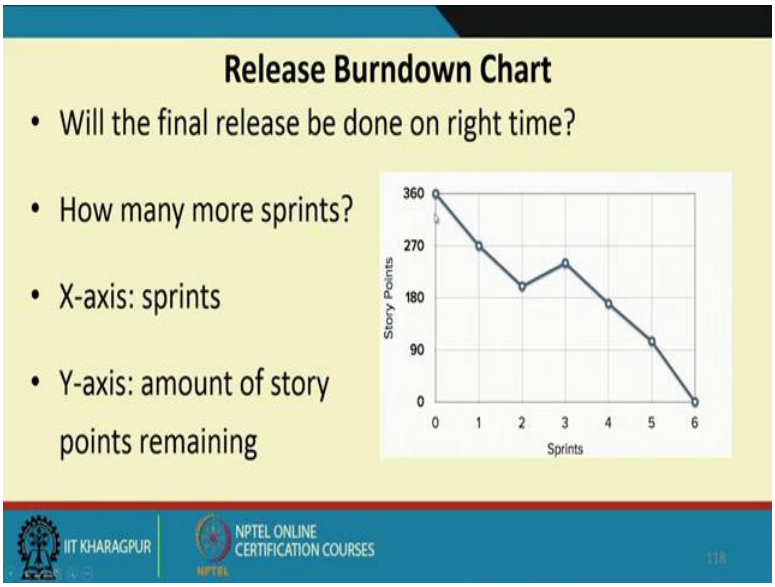

This is the release burn down chart. Each release might consists of several Sprints and during a release several story points are implemented. Basically it represents how many more Sprints are required before all the required story points for the release are implemented.

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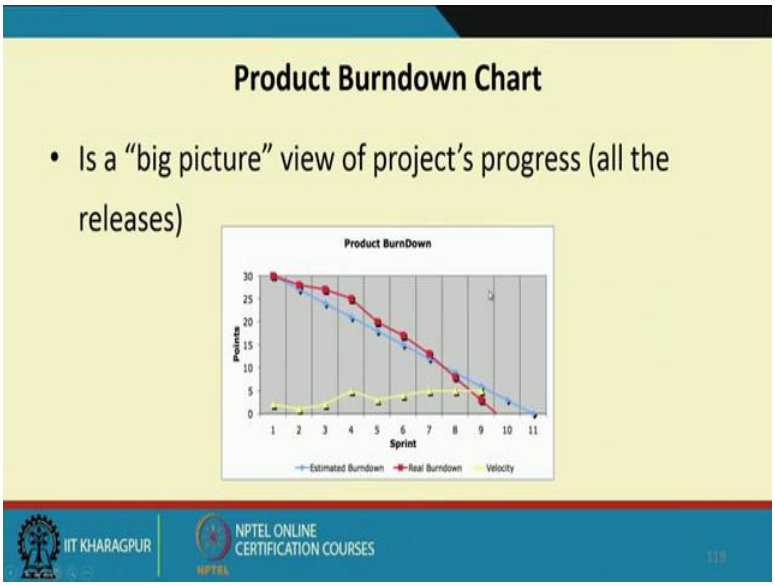

The product burn down chart gives the big overall picture. This is for completion of the project how many Sprints are been completed and how many are likely to be required and here 3 items are represented. The real burn down this gives the actual progress the estimated progress and then the velocity that is the story points completed per Sprint.

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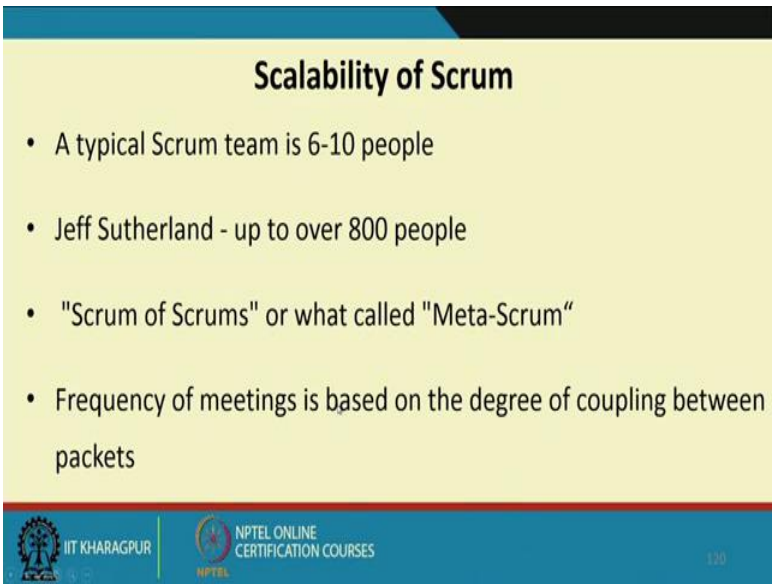

The Scrum is typically for small projects 6 to 10 people, but then it has been also tried for larger projects involving hundreds of people and it is called as the Scrum of Scrums or meta Scrum will not be discussing this.

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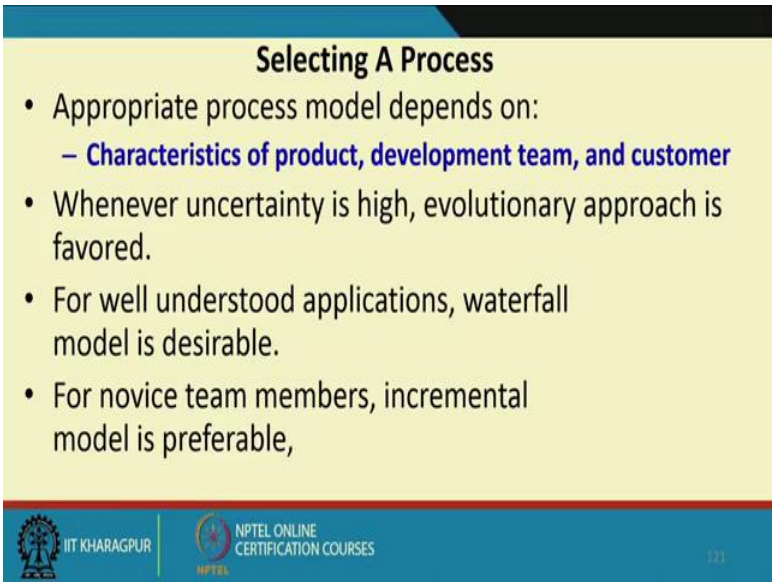

But one important thing is that we discussed several development processes. We discussed several waterfall based processes. We discussed incremental, evolutionary, rad model and several types of agile models, but then given a project can we as the project manager say that what is the development lifecycle to be used. ok, but then it is not defined by the project alone. The lifecycle model to be used depends on both the characteristics of the product, the characteristics of the development team and also the characteristics of the customer. Based on an analysis of the characteristics of the product development team and the customer, the development lifecycle is decided.

If the uncertainty in the project is very high then typically the evolutionary approach is favored the features are identified to be completed as the project progresses. For very well understood applications the waterfall based models are desirable because these are the most efficient way to do it. And, the long term plan is made, strong management control, plan for the entire project is done, but if the team members are novice they do not have much experience on the projects and even similar types of projects. Then typically an incremental model is preferable where some small increments are planned and completed and that way the project builds up. With this will be completing this lecture and then we will take a estimation of the project and then scheduling of the project.

Thank you.

Software Project Management
Prof. Durga Prasad Mohapatra
Department of Computer Science and Engineering
National Institute of Technology, Rourkela

Lecture – 11
Project Evaluation and Programme Management

Good afternoon. Today, we will see some important concepts of Software Project Management. I am Dr. Durga Prasad Mohapatra, Professor of Computer Science Department, NIT Rourkela. So, today's topic is Project Evaluation and Program Management.

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So, the concepts are today we will cover business case for a project; project portfolio management and project evaluation.

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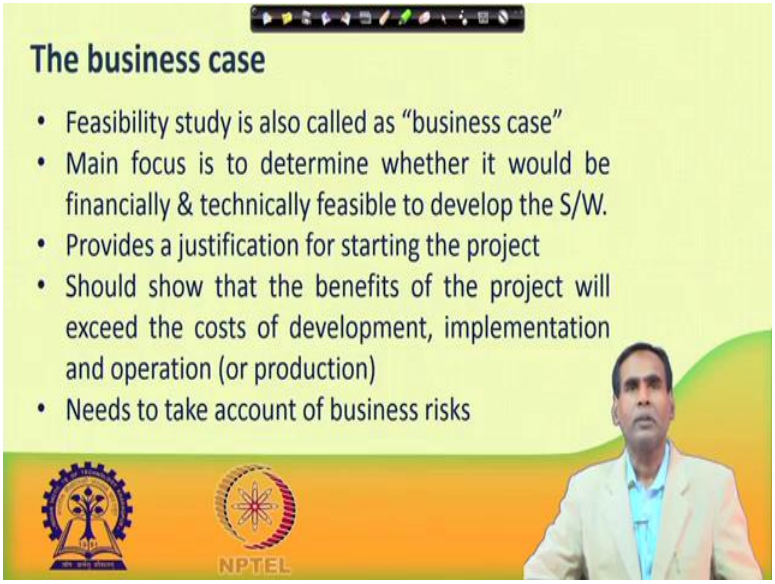

So, let us see first what a business case is. Business case normally it refers to feasibility study. So, the main focus of feasibility study is to determine whether it would be financially and technically feasible to develop a software. It will provide a justification for starting of the project. It will also show that the benefits of the project that will exceed, that will outweigh the cost of the development, cost of implementation and the operational cost. So, it needs to take account of the business risks; that means, we also should consider take into account, what are the various risks, business risks associated with and that will help us in deciding whether the project will be feasible or not.

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Now, let us see what are the different types of feasibility study. So, there are three kinds of feasibility study; one is technical feasibility, then financial feasibility and then, operational feasibility. In technical feasibility, we have to decide that whether the current technology that is available with us with that we can develop the software or not or if any special hardware or software is required, whether that is available in the market or not. So, if you require a special software or hardware to develop your project and that is not currently available or not likely to be available in the market in near future. Then, we say that project will be technically in feasible.

Then, after this technical feasibility study is over, we should cover, we should consider the financial feasibility. Financial feasibility means we should ensure that the profit or the benefit that will be obtained by implementing the proposed project, it must outweigh the cost, the benefits must be much much higher larger than the cost that will be spent in developing the project. Then only, we will say that the project is financial feasible, else we say that the project is not financial is not financial feasible and hence, we should not undertake that project.

Then, next feasibility is operational feasibility. So, if you say that the project is technical feasible as well as financial feasible, then we should go for the operational feasibility. Operational feasibility means that if a project will be developed, then how far it will be accepted by the customers, the end users. If the end users, they accept it, they are happy; in they will be happy to use it; then, we say that it is operational feasible, else we say that it is not operational feasible.

Because you see when a new project may be an automation system, we will develop and previously the project was running manually people were very happy and if they will make it automat, if you will make the their system automat, they may have fear that their job may go. So, they may resist it and in that case we say that the project will be operational feasible. So, we should take up a project for development, only if it is technical feasible, financial feasible and operational feasible. Otherwise if any of the feasibility it is not satisfied, then we should not take up that project.

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Now, let us see what are the contents of a business case or feasibility study. So, the following contents are there in a business case about first little bit of introduction or background of the proposed project should be written; then, about the details of the project proposed project that should be written. Then, what is the market? So, this project is aimed at which market little bit description of that. Then, the organizational and operational infrastructure that will be required to develop the project, that must be described.

Then, as I have already told you two important concepts are there in feasibility study; the benefits and costs. We must ensure that the benefits must outweigh the cost. So, the next content must be benefits. Then about outline of the implementation plan if you will develop the project what will the outline to implement it, for that we must prepare a plan. Then, what cost will be required to develop the project, details of the cost and then the financial case.

Then, some the other next content is the risks. What type of risk will be there, if will go to develop the project then, we will try to see that how they can be handled and finally, there even overall plan regarding management of the proposed project. So, these contents should appear in a what business case content or in a feasibility study report.

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So, now let us little bit see about all these what contents in brief. So, as I have already told you first little bit introduction or background should be there for this business case. So, this introduction will describe a problem to be solved or an opportunity to be exploited. So, what kind of problem we are going to solve or what opportunity you are going to exploit? So, that has to be a little bit of background or introduction should be described first. Then, about the proposed project here every outline of the project scope.

The scope of the proposed project every outline about the project scope must be described, next then each market; so, here the project could be to develop a new product. Suppose, you are planning to develop a project for developing a new product. For example, a new computer game you want to suppose a develop a project for designing a new computer game. Then, the lightly demand for the product would need to be assessed if nobody will what purchase your software, then what is the point in making a developing the product the product projects. So, that is why you must target at what will the market; how far your product will be in demand in the market? So, that is why little bit about of the what ah demand of this product in the market, should be discussed.

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Then, about the next content is the organizational and operational infrastructure. Here how the organization would need to change, if you will develop this particular project what changes may occur in this organization; whether there will be a lot of change in the organizational structure, in the operational structure. If that will happen, then some problem might be arise. So, that is why what this should be important where a new information system application was being introduced. So, when you are; suppose this is a manual one, you are just going to automatic then there the organizational or the operational structure might not be affected a lot.

But if you are going to completely develop a complete new information system which was before, it was fully manual. Then, there might be some changes in the organizational infrastructure, in the organizational structure or the operational infrastructure. So, we have to note down that what kind of changes will occur if there will be a change, if the new project will be developed, what kind of changes will be there in the organizational and the operational infrastructure, so that we must discuss. Then, as I have already told you the two important components; the two important contents of business case are benefits and costs.

So, normally this should be expressed in financial terms, any benefit as far as possible, we should try to express in financial terms may be in monitoring terms. We know some of the benefits might not be expressed in what, in terms of the financial in terms of

financial terminologies, we will see the different types of benefits and cost which can be expressed in monitoring terms or financial terms and which cannot be expressed, those we will see in the might be in the next class. So, in the end it is up to the client to assess this because first we have to see what financial, what benefits will get out of the proposed project.

So, at because at the end the client will assess this that after implementing this project how much they have spent; what cost they have given; how much amount they have given and how much benefit they are getting. If the benefit will not outweigh the cost, then definitely it will be loss to them and they will not go for developing this proposed project. So, that is why we must have to identify what are the benefits and we must try to express the benefits in terms of the financial terms, in monetary terms in the business case or in this feasible study report.

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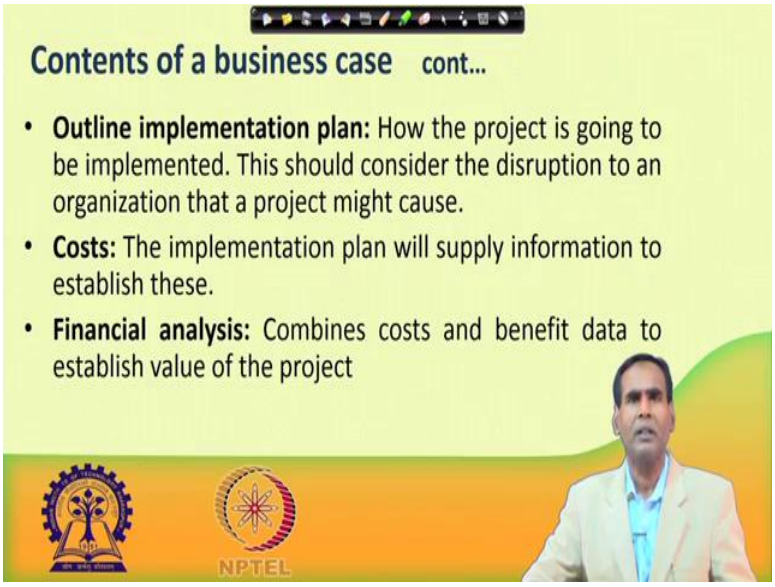

Next one is about the outline implementation plan. Here, how the project is going to be implemented, in which way you will implement the project that has to be discussed. So, this should consider the disruption to an organization that a project might cause. So, if you will implement the new project, then what if there will be any disruption to the organization to the normal operation of the organization if any disruption will be there. So, that also should be written in this outline implementation plan and the other

important content that we must describe the cost. Here the implementation plan will supply information to establish this.

So, what are the different types of cost? So, we will discuss in the next class, what could be the different types of a costs. So, how much cost the end user the client will be pay for developing this project? So, the implementation plan must supply the information to establish this cost and the next component is financial analysis. So, this financial analysis, it combines both the cost as well as benefit data to establish the value of the project. If the project will be developed what will the value of the project?

So, here in this content financial analysis, this will combine; this section will combine the various cost and benefit data to establish what will the real value of the project; whether the project can be accepted by the end user not; whether the project can be accepted by the developer not by comparing the cost and benefits. So, this analysis will be met.

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Next important component of a business case is risks. So, risk as you know that it is something undesirable; something undesirable so, there are two types of risks will find out in two major types of risks in a project. So, this two types of risks must ah what contain; this two types of risk must be contained in this feasible study report or a business case.

So, one is business risks, another is the project risks. So, let us see in the project risk, what is happening? Project risks are the risks which are related to threats to successful project execution. In order to complete the successful project or while completing the project successfully so what threats might arise ok, so these things are known as project risks. So, these are related to the what as a threats to successful completion of the or execution of the project. Whereas, business risks these are related to the factors threatening the benefits of the delivered project. So, if the new project will be delivered, so what benefits you will get and which factors will threaten to get these benefits of the delivered project. So, these factors will be termed as a business risks.

In this business case or the feasibility study our focus will be on the different business risk. So, we will discuss the business risk somewhat towards the end of the session. So, then, last content or last component in the management plan here a detailed plan for smooth management of the organization should be planned; what should be the different activities that you should consider; what are the different milestones etcetera. So, the detailed plan for this smooth management of this organization while developing the project that should be written in this section.

So, these are the various contents of a business case. So, if this business case or a feasible study report should contain this what sections these contents.

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Now, we will see another important aspect of software project management that is your ah project portfolio management. So, let us first see what this project portfolio management provides. So, this project portfolio management, it provides an overview of all the projects that an organization is undertaking. An organization is just developing only one project, it may take up, it may develop multiple number of a projects. So, this project portfolio management will provide an overview of all the projects that the organization developing.

So, some of the projects might be academic related projects, some of the projects might be so healthcare related projects, some of the here projects might be business oriented projects or some of the projects might be what industry oriented projects. So, this project portfolio management, it will provide an overview of all the projects that the organization is undertaking. So, it will prior this project portfolio management, it will prioritize the allocation of resources to projects.

So, since the organization or this what project developing organization is developing so many projects, what the resources are limited. So, how to allocate the resources? So, which one which activity or which project is more important, say more number of resources should be given to those projects at first. So, how to prioritize that which project should get should be allocated which resources and when? So, this prioritization is one of the objective of this project portfolio management. It will prioritize the allocation of the resources to the different projects.

This will also decide which new projects should be accepted and which existing one's should be dropped. See if where after the feasible study we have found of that if this new project will undertake that will give us much more profit. Then certainly, we must accept it to develop it. While we have seen that some of the projects which are running, they are not giving so much benefit to the organization rather there incurring losses. Then, this is the time we must take action to drop the projects whatever loss we have suffered that will accept and we should not allow it to bring in more loss to the developed to the organization.

So, those existing ones which are giving loss those are running in loss. So, they must be dropped. So, how to decide that which new projects should be accepted and which

existing ones should be dropped; for this, we should again use project portfolio management. This will decide which project to accept and which existing project to drop.

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Let us see what are the important concerns of project portfolio management? So, these are the important concerns of project portfolio management; first one is evaluating proposals for projects.

So, project proposals may come from different sources. So, how to evaluate the project proposals? What all the parameters based on which you will evaluate the proposals? So, this is the first important concern that we have to evaluate the proposals for the projects. Then, we have to assess the risk involved with the projects. I have already told you that there are two kinds of important projects; business risks and what are the project risks. We have to assess what will the impact of this risk will happen; so, how much how the project will be affected. So, assessing how the system will be affected. So, assessing the risks involved with the projects.

Then, deciding how to share resources between projects since there are multiple number of projects are running in an organization, but the resources are limited how to decide when, which resource will be allocated to which project that is an important factor. So, project portfolio management aims at are deciding which resource will be given when and to which project. So, now, taking account of dependencies between projects, there are some projects, there are dependent. That means, suppose project a depends project b;

project a depends on project b; that means, if project b is not completed maybe project a, we may not start.

So, that is why in order to handle this, we must take into account the dependency between the projects; which project is dependent on what other project. Accordingly, we can assign the resources so that the project on which another project it independent that should be finished as early as possible so that the other project which dependent on it can be started. Then, removing duplication between projects; so, if you are having similar projects, if a running similar projects and some components are redundant they are duplicated. So, it is unnecessarily use of effort and resources etcetera also a necessary what wastage of time.

So, we have to identify the duplication between the projects and we have to remove them this is also one of the important concern of project portfolio management; then, ensuring that necessary developments have not been inadvertently missed. So, another important concern of project portfolio management is that we have to ensure that necessary developments have not been inadvertently missed.

So, if there is what some good thing, some necessary development. So, somehow that has been missed. So, this thing should be ensured that yes no necessary developments, no important things; they have been missed inadvertently. This can be also aimed at this can be also ensured by using this project portfolio management.

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Now, let us see what are the different elements to project portfolio management? There are three important elements one is this project portfolio definition.

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Another is a project portfolio management and then, project portfolio optimization. Let us see about first the project portfolio definition. So, what is the objective? So, the objective here is to create a central record of all projects within an organization. So, with an organization mean there are many records. So, how to create a central record of all the related projects within an organization; so, that is why is one of the objective of project portfolio definition.

So, this is this first how to define the project portfolio or project portfolio definition. So, we must also decide whether to have all the projects in the repository or only a special category of projects like as ICT projects. So, we have to decide that whether all the projects will be placed in the repository or some special kind of projects or some special category projects like ICT projects will be kept.

Then, we have to note the difference between the new product development projects and the renewal projects that whether because see there is sufficient difference between developing a project newly and just on projected there renewing that updating that. So, we must have to difference because the resources require by both the projects new product development and the renewal they will be different.

The time required, the effort required will be different for developing a new product development as well as renewal of projects. So, we have to note down the difference between the new product development and the renewal of project. So, project portfolio definition aims at it its objective is this things. Then, we will see project portfolio management. Here we must have to see that the actual costing and performance of projects can be recorded and assist. So, what is the actual cost that the project takes and what is the performance; what kind of performance project is skipping.

So, they can be recorded and assessed by using portfolio management. The main objective is to record and assess the actual costing, the actual performance of the projects, they will be developed and in project portfolio optimization as its name suggest will see what is happening here that here the information gathered above. So, we are gathering the information, you will see that in this project portfolio definition we are creating the record, we are gathering the information.

So, here in project portfolio optimization the information that happened gathered earlier, they can be used to achieve better balance of a projects. How the different projects they can be balanced like see some of the projects are very risky, but they are potentially very valuable. They will be valuable balance by because they may give more profit you know that the business or the projects which are more risky, they are likely to give more benefits. On the other hand, there are some less risky projects; it is very less risky projects, but their properties also very less. They are very less valuable projects. So, we have to balance these different kinds of projects to achieve better performance.

So, the information, how we can do; how we can balance them? By obtaining, by gathering the information, by using the information those two which have gathered in the above steps. So, the information gathered above can be used to achieve better balance of the projects. So, the example as we have already seen that some projects might be very risky, but their even if their very risky, but the return; the profit will be also very high. But some projects which are less risky, but the return is very less, they will give very less profit.

So, we have to do a balance between these different kinds of a projects. So, this can be done; that means, we have to what go for some optimization. How we can make more profit even if the projects are risky; how we can spend less time less cost, but objective is

to maximize the profit? So, these things can be achieved by this project portfolio optimization.

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Now, let us quickly see about this pros and cons of the project portfolio management. The pros is that it allow some small ad hoc task to be done outside the portfolio. These are the a project portfolio management; it allows you to carry out some ad hoc tasks outside the portfolio. For example, you may want to quick, you may want to what do the quick fixes to the systems that can be done easily. Whereas, the disadvantages that the you may have taken some full time staffs, you might have allotted some fulltime staffs.

For a project about them effectively, they will be just considered as part time. There will be there will be effectively part time basis because there still have some routine work to do. We cannot give the full time for the assigned projects. So, that is one of the limitation; then, the official project portfolio may not accurately reflect the organizational activity, if some projects are excluded.

So, if some of the projects that excluded then what project portfolio report you have prepare the official project portfolio it may not accurately reflect the organizational activities if some of the projects are excluded. So, these are some of the pros and cons of the project portfolio management.

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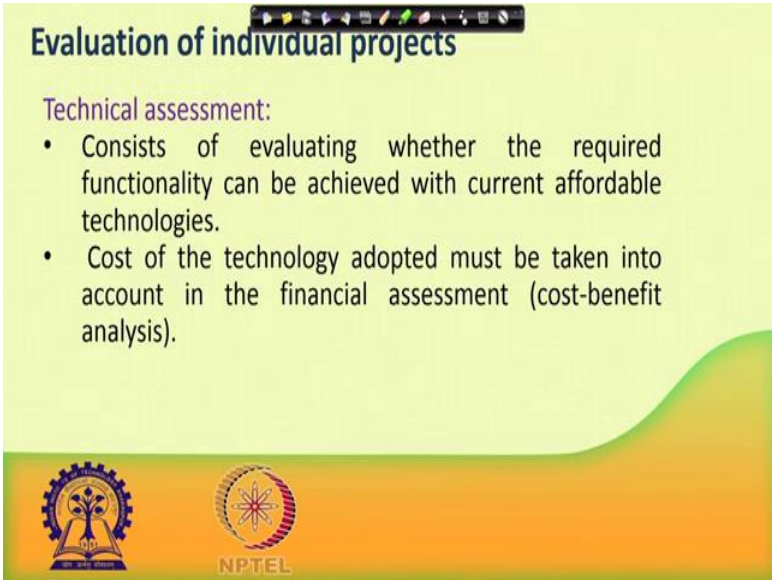

Now, let us see how to evaluate the individual projects. I have already told you that three kinds of feasibility studies are required to evaluate individual projects. So, one is this technical feasibility, then financial feasibility and operational feasibility. Let us see about this technical assessment first. So, the technical assessment consists of evaluating whether the required functionality can be achieved with current affordable technologies. What technologies currently existing in the market, with that can you develop the project.

Then, we say that it is technically feasible and the technical assessment has been perfectly met. So, similarly then, the cost of technology, if you will use a new technology, new hardware, new software; so, the cost of technology adopted that must be taken into account on in the financial assessment. While people you will prepare the cost benefit analysis, you have to consider the cost of new hardware, new software or new technology you have to take into account this cost.

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Then, is this financial assessment; in financial assessment, we have to what analyze the cost and benefits. So, that is why this is known as cost benefit analysis. We have to analyze the various cost that will spent, we have to analyze the various benefits that will get. Then, we have to compare that whether the benefits, there outgoing the cost or not if the benefit will sufficiently outweigh the cost. Then, we will take up the project for development authorize. We must we must not take up that project for development. So, this ah financial assessment or which is popularly known as cost benefit analysis, this relates to an individual project.

For individual projects, you have to perform cost benefit analysis. For cost benefit analysis, you need to do the following first. You have to identify all the cost; what are the different kinds of cost, you have to identify. For example, some of the cost would be development cost; some of the cost maybe setup cost. For example, for what developing your project, the hardware cost for the recruitment and the staff training cost for setting of this what experimental. So, what cost will be required regarding and the setting of the hardware or what recruiting the different personal required providing training to them. So, this cost as setup cost.

Similarly, another cost we have maybe they are like operational cost. So, after the cost for operating system, after installation; after the installation, we have both we may have to provide some cost through run it to operate it. So, these are operational cost. So, in this

way for performing cost benefit analysis, we have to identify the different types of cost. Then, we have to similarly identify the different types of benefits. So, there are the different types of benefits are there which will discuss in the next class. I mean the classification of the cost and the classification of benefits, we will discuss in the next class.

Then, check that benefits are greater than cost. As I have already told you where performance cost benefit analysis, we must check that we must ensure that the benefits those who will be obtained out of the project must be greater than the cost that will incur. Then, only we can check up, we can take up the project for development; otherwise we should not take up the project for development. So, that is the one of the important objective of cost benefit analysis. So, we have to check that the benefits are larger than this the costs ok.

So, thank you very much.

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Lecture – 12
Project Evaluation and Programme Management (Contd.)

Good afternoon to all of you, last class we have discussed about the cost benefit analysis. We have seen the different ways of evaluating proposals one is this technical assessment, then the cost benefit analysis or the financial assessment. So, now, we will take up the next item for individual project assessment. So, that we will see that now today will first see the other item that is left for this individual project assessment that is cash flow forecasting. Then as I have already told you C- B analysis that is the most important analysis among all the assessment among all the feasibility studies financial assessment or C- B analysis is the most important.

So, we will see the detailed steps of C- B analysis, then we will see among one of the steps we have to identify and classify the cost and benefits. So, we will see the details of the different types of the cost and benefits associated with a project.

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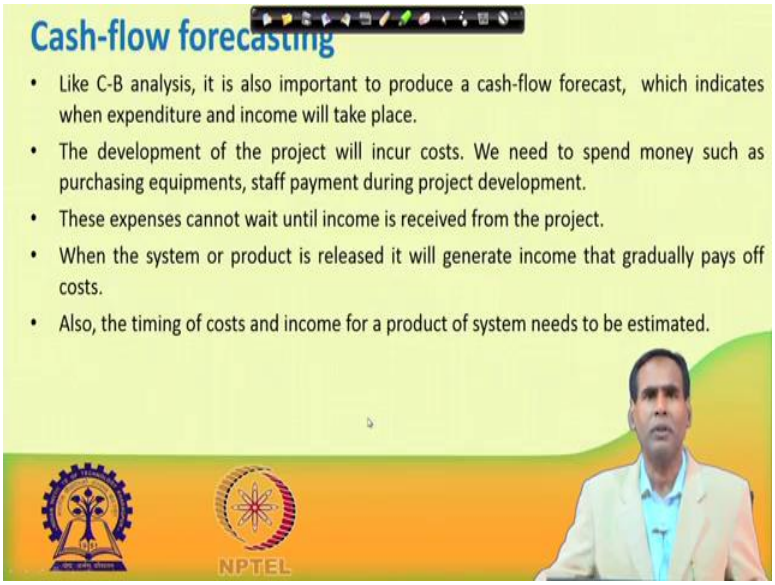

So, like cost benefit analysis it is also very much important to produce a cash flow forecast. This cash flow forecast indicates when expenditures and income will take plus will take place during the development project.

You know that development of the project will incur some cost. So, we need to spend some money such as purchasing equipment staff payments etcetera during the project development, but these expenses cannot wait until the income until the revenue is generated from the project, we have to spend some money initially. So, when the system or product is released then only it will generate a income that gradually a pays off the cost. But the so, far till that point we cannot wait we have to spend some money, that money maybe we have to spend from the our own resources the companies own resources or by taking loan from banks etcetera.

Also we have to consider we have to take into account the timing of the cost and incomes when cost will incur? When the first income when the first revenue will come out? So, the timing of cost and income for the product that also needs to be estimated.

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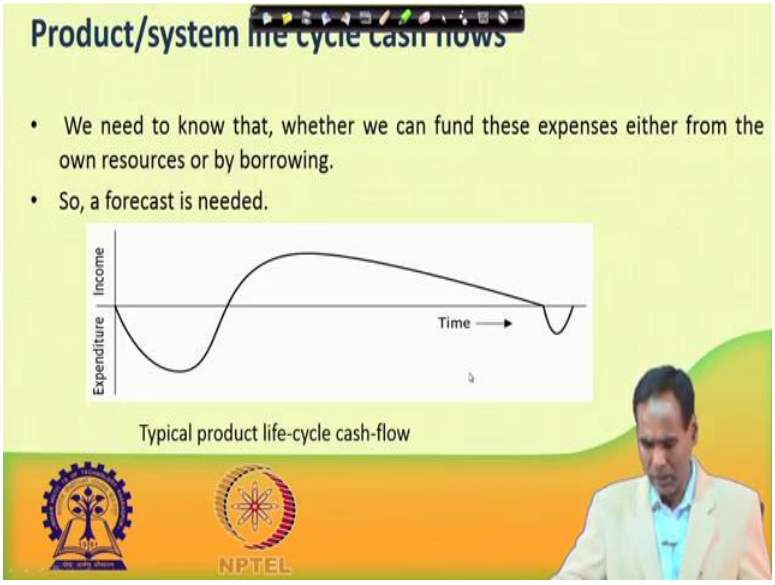

The we have already known that whether we have to you know expect we have to think of that this initial fund that we required to start up the project, where the fund will come? So, we need to know that whether the fund the initial fund requirement, the initial expenses whether we can fund them from the organizations own resources or by borrowing or by what taking loan from the banks etcetera.

So, a forecast is needed a typical what forecasting is shown here we are seen as I have already told you during the development of a project, initially we have to spend some money this is the expenditure this is the initial expenditure. So, here what will happen?

No revenue is there because the project just is started. So, some initial expenditure just such as purchasing a hardware software, making the site ready and staff recruitment etcetera we have to make.

And after sometime you see will get gradually the revenue will come up and gradually the revenue the income will increase and this increase so, some time will come, at some point of time this income it will outweigh this expenditure, then it will gradually it will meet the expenses, it will be the expenditure will be equal to what or the income it will pay off the expenditure that we have made initially then the gradually the income will increase and after some years again the expenditure are the income the revenue will come to 0 then it will ask maintenance. So, again you have to put some more expenditure, this is how the typical product life cycle cash flow this looks likes.

So, basically what the importance is I have to forecast a cash flow and cash flow means, it will indicate when expenditure will take place and when the income or revenue will take place.

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With this now will go to the detailed steps of this cost benefit analysis, which is very important for successful development project. So, these are the steps for the cost benefit analysis that we have to carry out first we have to identify the cost and benefits pertaining to the project, then we have to categorize various cost and benefits, then we

have to select a cost benefit evaluation technique we will see there are various cost benefit evaluation techniques are there we will see them.

Then we have to interpret the results of this cost benefit evaluation techniques for analysis and then after analyzing the results of the cost benefit evaluation techniques, we somebody are the organization it has to take appropriate action. So, now, let us see the first step.

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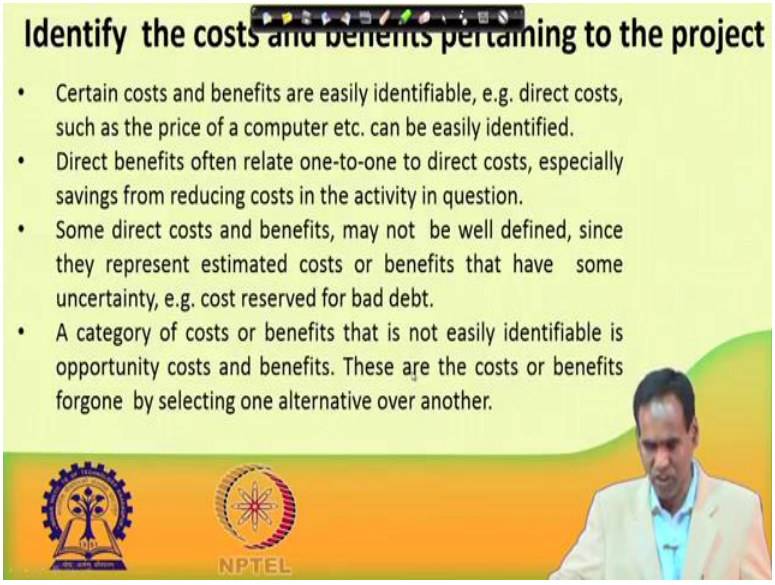

The first step is identifying the cost and the benefits pertaining to the project. You know that the certain cost and benefits that easily identifiable. For example, direct cost such as purchasing a computer and payment to the what a staff members etcetera they can be easily identified.

These direct and the direct benefits very often they relate one to one to the direct cost, the direct benefits will directly relate to the direct cost especially the savings that may be obtained from reducing the cost in the activities in the questions they can be mapped to the what benefits. Some direct cost and benefits may not be well defined they cannot be clearly identified since they represent some estimated cost they are not the actual cost they are estimated cost or they are estimated benefits that have some uncertainty.

So, they cannot be clearly defined, they cannot be clearly measured examples are like the cost reserved for bad debt etcetera. So, this types of direct cost or this types of some of

the benefits they cannot be what they cannot be well define they cannot be clearly defined and they cannot be clearly identifiable they cannot be clearly measured. A category of cost or benefits that is not easily identifiable is opportunity cost. We have taken here example that or some category of cost or benefit that cannot be easily identifiable is opportunity cost and opportunity benefits. So, these opportunity cost or the what do you mean by these opportunity cost or benefits?

Opportunity cost or benefits are the cost or benefits which are forgone by selecting one alternative over another. If you could have taken another alternative you might have you might have got better benefit for somehow you have overlook that somehow you have forgotten that one. So, these are the what. So, this types of cost or benefits those are forgone by selecting one alternative over another alternative. So, the cost or benefit associated with the these types of things are known as opportunity cost. So, these opportunity cost or opportunity benefits they are very much difficult to identify.

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Now, let us see what are the different then the second step as I have already told you first step is identify the cost and benefits, then next step is categorize the various cost and benefits. So, what are the various categories of what are the categories of different various cost and benefits? So, the cost or benefits they can be tangible or intangible, they can be direct or indirect they can be fixed or variable. So, these are the important categories of different cost and benefits let us discuss a one by one.

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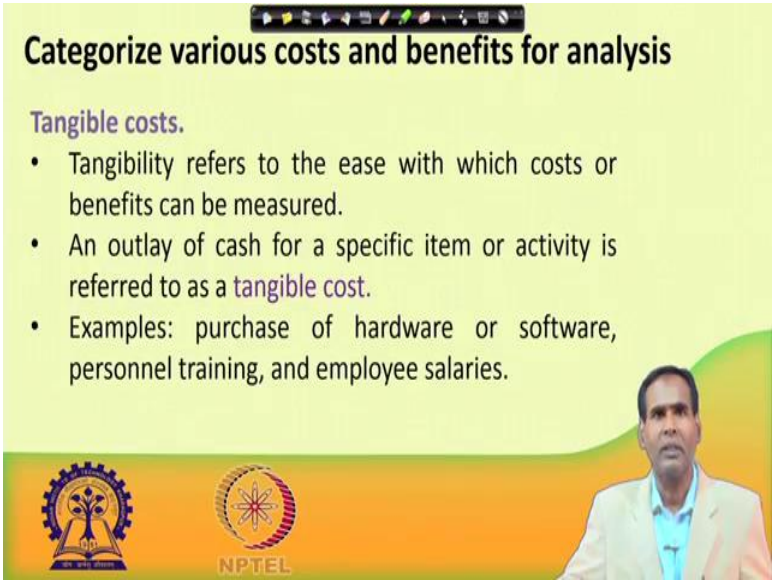

So, first one is tangible cost. So, what do you mean by tangibility? Tangibility refers to the ease with which cost or benefits can be measured; how easily you can measure the cost or the benefits. So, tangible means the ease with which the cost or benefits can be measured. An outlay of the cash for a specific item or activity is referred to as a tangible cost. So, tangible cost is defined as an outlay of the cash for a specific item or for a specific activity. So, this outlay of cash for a specific item or activity is called as tangible cost. So, examples of tangible cost are what? Purchase of machineries purchase of devices such as purchase of hardware or different software the what cost required for a training the personnel the cost required for paying salaries the employees these are examples of tangible cost.

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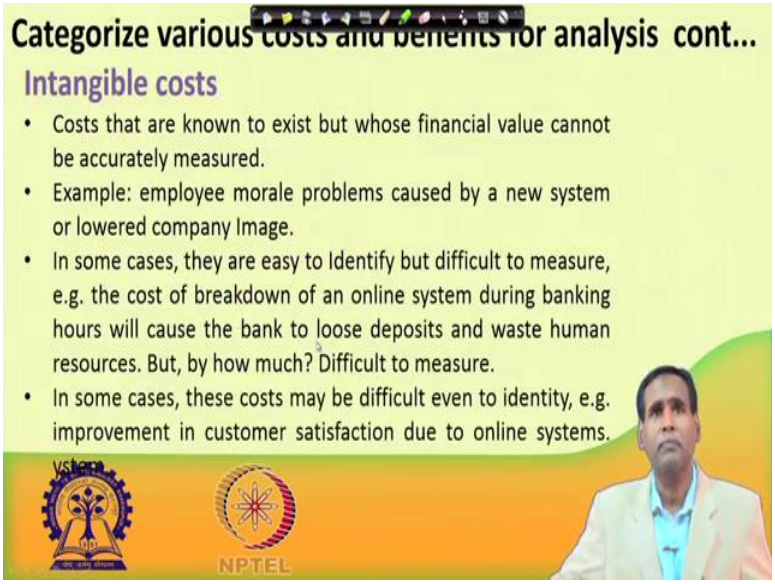

Similarly, what do you mean by intangible cost? So, tangible cost means cost which can be easily what measured which can be easily which can be easily measured what intangible cost means cost that are known to exist, but with the financial value cannot be accurately measured. You cannot easily measured these cost some examples let us see employee morale problems caused by a new system. So, suppose it was earlier some existing system was there.

Now, you have implemented you have installed a new system, you have followed a new system then what kind of morale problems will come to the employees? So, that will incur some cost that you cannot easily measure. So, that is why these are intangible cost. Similarly lowered company image if the company image is lowered is down in the market among the customers then it will also incur some customer the difficult to measure. So, these are some examples of intangible cost. In some cases these intangible cost are easy to identify, but difficult to measure you can identified them, but you cannot measure them.

For example the cost of breakdown of an online system. An online system such as railway reservation system or banking system etcetera it was running, but suddenly they as a breakdown, then some cost is associated with this due to this what the customers may be diverted to what other service providers and so, and so, on. So, here the cost of breakdown of an online system during what the maybe we may take that banking

example, if an online banking system it is a several times it is getting breakdown. So, there are so, many banks why the people will go for only your bank. So, they may migrate to other bank.

So, you lose some amount. So, this will incur some cost. The cost of breakdown of an online system during banking hours will cause the bank to lose deposits as well as waste the human resources, but is difficult to measure by how much amount you will lose, how much money you will lose or how much, but human resources you will lose it is very much difficult to measure. So, this is something about intangible case are caused. So, in some cases these cost may be difficult even to identify. In many cases here we can identify this example have taken that cost of breakdown of an online system here it can be easily identified, but difficult measure.

So, but there are some other cases where these cost may be difficult even to identify you cannot identify, what is about measurement you cannot identify. So, we cannot measure them for example, you see improvement in customer satisfaction due to online systems previously this system was manual. Now you have what followed you have installed your you have implemented on analysis system. So, difficult analysis the customers will satisfied, but how much they have satisfied how to quantify it, it is very much difficult to quantify them or it is very much difficult to what a measure them.

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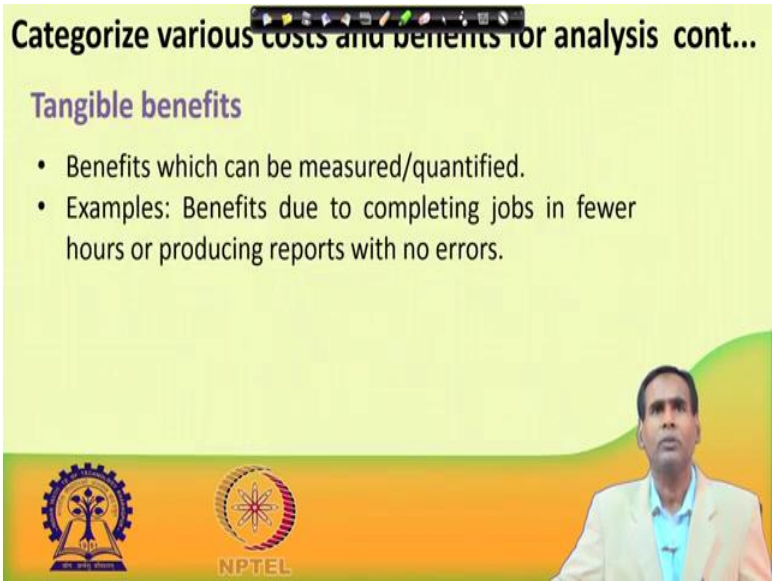

So, these are intangible cost so, similarly as we have seen tangible cost and the intangible cost. So, similarly the corresponding category of benefits are there like tangible benefits and intangible benefits. So, benefits which can be measured. So, tangible benefits means these are the benefits which can be easily measured or quantified. For example, benefits due to completing jobs in fewer hours previously the job was for hour you are completing say 10 jobs for due to what automatic a developing a new automation system, you are completing what in what the same what 10 jobs you are completing in half an hour.

So, you are certainly getting some benefits. So, this, but how much this benefits of course, you can you can measure that previously one for one hour you are doing that 10 jobs now the send 10 jobs you are completing in half an hour or for one hour you are completing 20 jobs. So, this can be easily measured on quantified. So, this is an example of tangible benefits. So, similarly producing reports with no errors previously you are preparing the reports manually. So, there are many errors maybe per page say 10, 20 or like that, but after automating in your report almost just one or two errors are coming, this number of errors are present in that reports due to making your system online or automated. So, this can be measured this is an tangible benefit.

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So, producing the reports with no errors of course, this can be measured and quantified you can count the number of errors etcetera. So, this is an a example of tangible benefits.

So, similarly what do you mean by intangible benefit? Benefits which cannot be easily measured or quantified. So, the benefits which you cannot easily measure which cannot easily quantified there intangible benefits, let us take some example more satisfied customers. So, if you are having more number of satisfied customers or your company your corporate has an improved image definitely your benefit will be more but is difficult to measure them it difficult to quantify them.

So, why we are discussing the different types of cost and benefits? Because all the types of cost may be tangible and intangible cost may be tangible or intangible benefits or direct cost indirect cost or direct benefits or intangible benefits all should be considered in the evaluation process. In the cost benefit analysis all the different types of cost and benefits including tangible and intangible cost and benefits direct and indirect cost and benefits they should be considered in evaluation process. It has been observed that many times the management often tends to deal irrationally with intangible cost or benefits by ignoring them we should not be. Very often since these intangible benefits they cannot measure they cannot identify they cannot be identified or they cannot be quantified. So, management they just ignore them only takes can count what like tangible benefits and costs.

So, in that way the assessment the financial assessment the cost benefit analysis cannot be made proper. So, management has to treat equally has to consider all the different cost and benefits similarly in the C- B analysis method.

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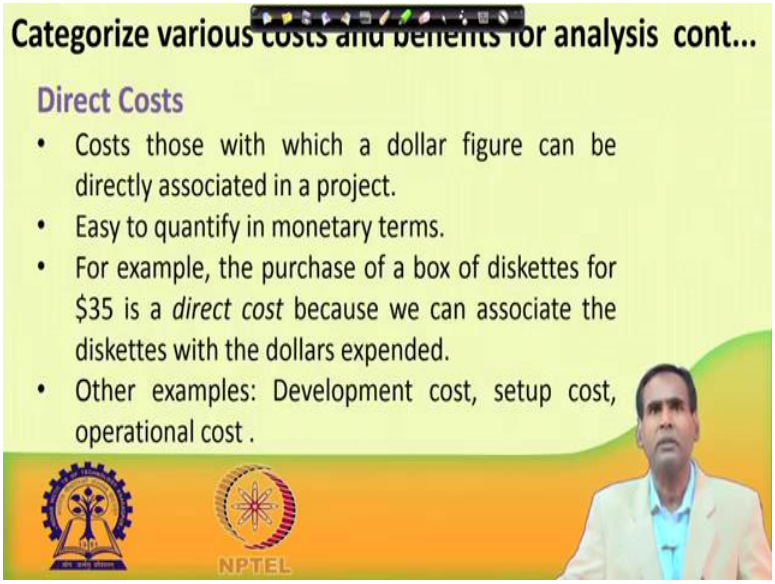

Next category is direct cost. These are the cost those with which a dollar figure can be directly associated in a project; that means, we can express the cost in a monetary terms either in terms of rupee or in types of what dollar. So, these are the cost with which a dollar figure can be directly associated in a project, it is easy to quantified monetary items. So, this cost can be easily quantified in monetary items let us take a small example.

Suppose you want to purchase box of diskettes or purchase the number of computers, purchase what a set of softwares. So, these are examples of direct cost here why? Because here in this case purchase of the box of diskettes say it is taken thirty five dollar or so, here you can associate the diskettes with the dollars expended. So, since you can assign monetized what values to this cost these are known as direct cost and other examples of direct cost as such as development cost, establishment of the laboratory etcetera, setup cost and operating cost I mean operating running this system after the installation is over, these are some examples of direct cost because they can be easily they can be directly measured, they can be directly quantified.

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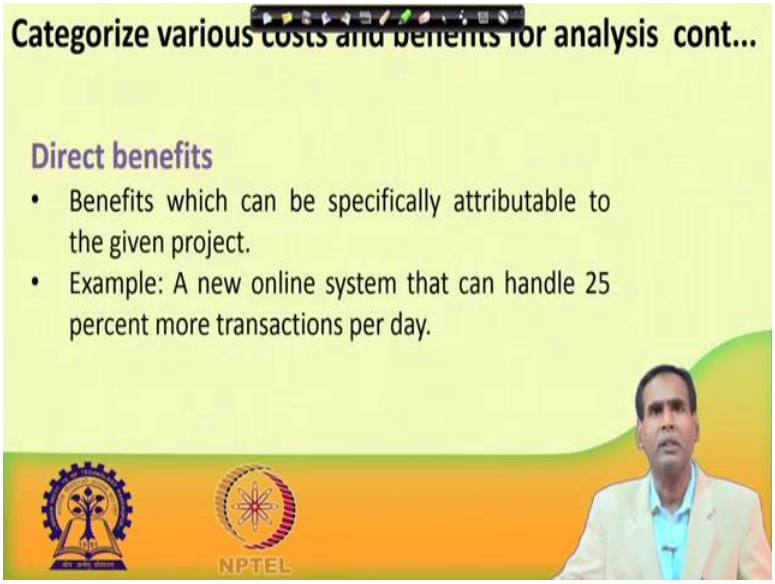

What is about direct benefits? These are the benefits which can be specifically attributable to the given project. So, these benefits can be specifically attributable to the given project that is why these are known as direct benefits let us take a small example. Suppose previously you are your system was running manually; now you have developed a new online system and that can handle 25 percent more transaction per day due to the manually how many transactions you are completing? So, now, due to making it online 25 percent more transactions are completed per day.

So, here directly we are getting the benefit and the benefits these are specifically attribute attributable why? Because due to implementation of the new online system these are examples of. So, this is the example of direct benefit.

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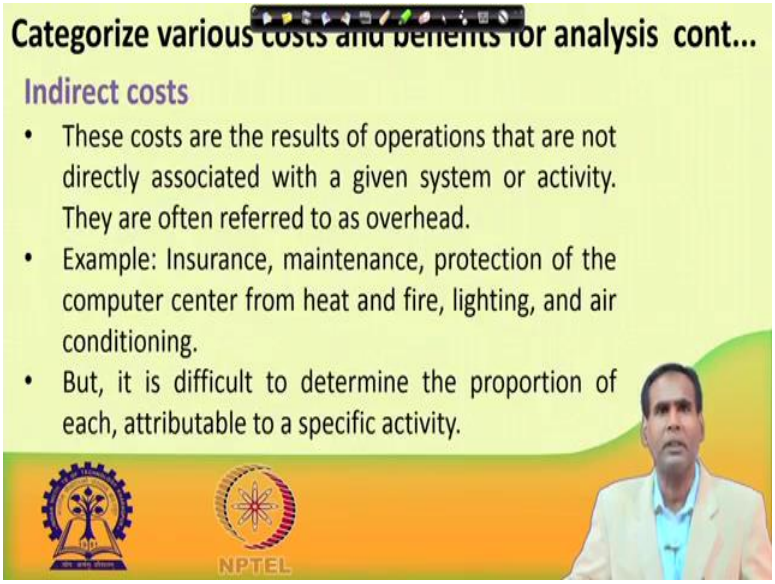

So, now will call the a direct indirect cost. So, these are the cost or these cost are the results of operations that are not directly associated with a given system or activity ok. So, this types of cost which are not a this types of cost are not directly associated with a given system or a given activity, normally they are often refer to the often referred to as overhead or overhead cost.

A for example, like insurance, maintenance etcetera these are not specific to a particular system or a particular activity, these are overall these are required for running of the overall organization. So, similarly protection of the computer center from the heat and fire lighting the organization, providing air conditioning facility etcetera these are not specific these are not directly associated with a given system or with given project or a given activity this is the required for overall running of the organization these are examples of indirect cost, but it is a very much difficult to determine the proportion of each of these which are attributable to a specific activity this is very difficult to determine and their proportion.

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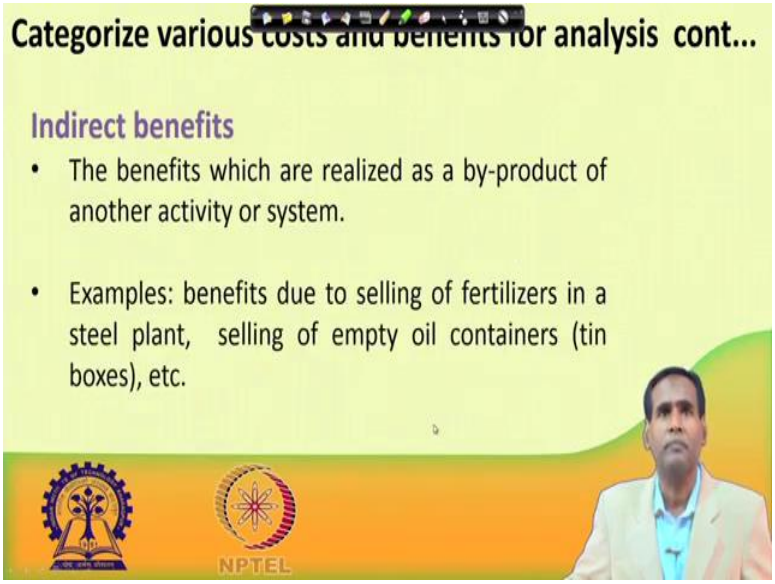

Now, what is about indirect benefits? The benefits which are realized as a byproduct of another activity or system. These are not directly actually benefits, but these benefits realized as a byproduct as a side product of another activity or system let us take a small example. See this is a main objective of steel plant is produced what I steels, but by the time this steels are produced also in some cases fertilizers are produced. So, the benefits due to selling of the fertilizers we can consider this as an indirect benefit.

So, similarly you know that the shop keepers they are what selling the oils different oils, but after selling the oils that containers the tin boxes, they also sell it in the what to the in the market. So, the amount the benefit they get out of selling the empty oil containers just like the tin boxes or this what paper cartoons etcetera these are indirect they may be treated as the indirect benefits.

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Now let us come to the last category that as fixed cost versus variable cost. So, some cost are constant regardless of how well a system is used. So, whether you use this system frequently used or less frequently used or do not used at all. So, some cost are constant they are called as fixed cost.

So, they are they are encountered only once normally they will not recur again and again, examples you can see this straight line depreciation of the hardware exempt employee salaries you are one times suppose you have a exempt employee salary etcetera and the insurance etcetera these are one time cost these are known as the fixed cost.

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Similarly, variable cost. So, these cost are incurred on a regular basis in contrast to the fixed cost which are, but only encountered ones they occur only ones whereas, variable cost they are incurred on a regular basis might be weekly monthly or yearly, they are usually proportional to the work volume and continue as long as the system is in a operation.

So, normally they are proportional to the volume of the work and they continue as long as the system is in a operation will take some examples like cost of the computer forms, computer printouts that you want to take which vary in proportional to the amount of processing. So, if you are having large amount of work has to be done large amount of activities have to be processed; obviously, you have to print more computer printouts more computer forms are required. Similarly if a it is a less amount of processing has to be done; obviously, less number of computer forms will be required. Similarly the length of the reports required more activities performed; obviously, the length of the report will be more only few activities are performed the length will be less.

So, here the cost associated with a this applications; obviously, that will be vary according to the volume of the work performed.

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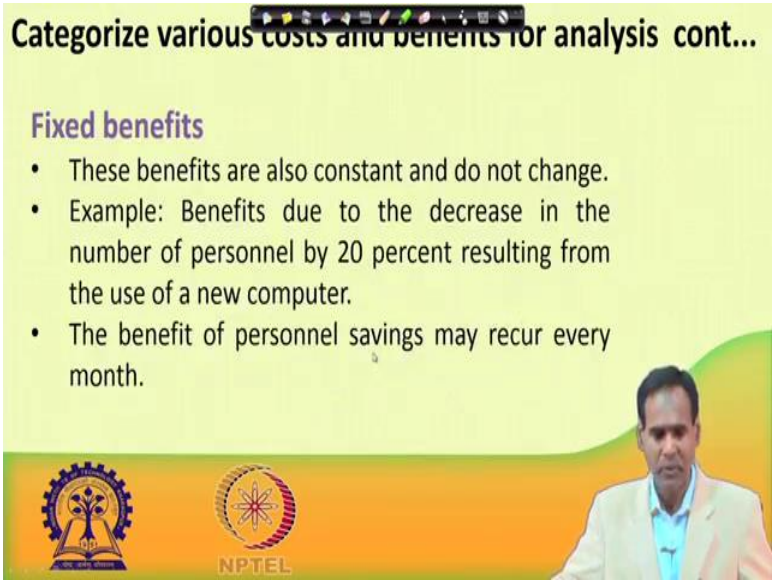

Similarly, let us say about fixed benefits versus variable benefits. These benefits are also constant and they do not change. So, as fixed cost they normally do not change. So, similarly a fixed benefits they are also constant and they do not change. So, some examples of benefits we can say which are fixed benefits like benefits due to the decrease in the number of personnel by 20 percent resulting from the use of a new computer.

You are automating your system previously it was running manually now a new computer system is implemented. So, by this you by due to the use of computer suppose the number of personnel that you are using previously it is reduced by 20 percent. So, this is exactly it is a fixed benefit ok. Similarly the benefit of personnel savings may recur every month, but the benefit of personnel savings how much personnel have been saved? The benefit of a personnel savings they may occur very every month.

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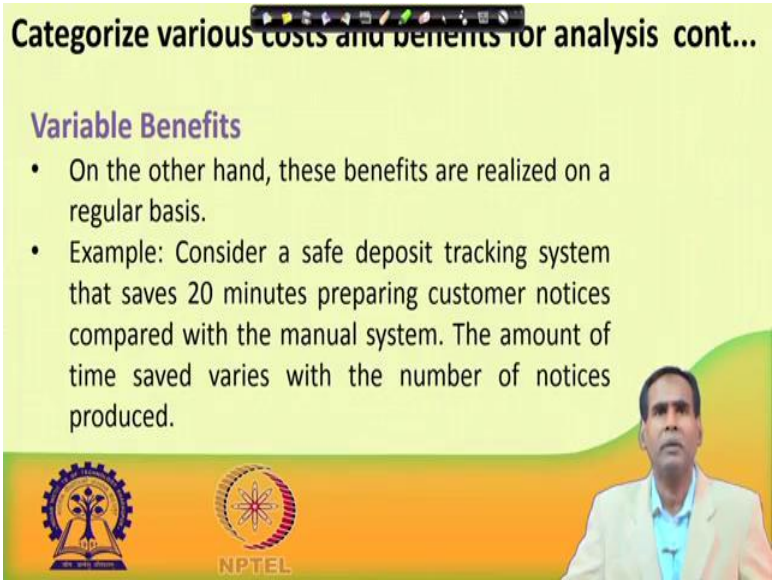

And variable benefits on the other hand these benefits; that means, variable benefits they are realized on a regular basis, they change. For example, suppose there is a safe deposit tracking system that saves 20 minutes preparing customer notices compared with a manual system.

When it was manual system, it was taking to 20 minutes more for preparing the customer notices after making it automated, after making a computerized that saves 20 minutes of a preparing customer notices. So, this will vary the amount of the time saved it varies with the number of notices produced. How many number of notices are produced accordingly the time saved it will vary. So, that is why this is an example of variable benefit.

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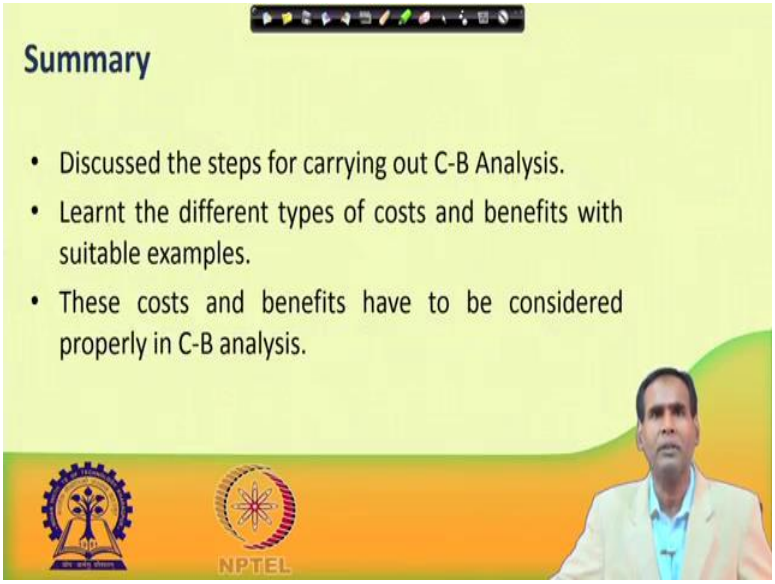

So, now we have seen that in summary. So, today we have discussed in this class what are the different steps for carrying out C- B analysis. We have learnt the different types of a cost and benefits with suitable examples such as we have seen this direct cost versus indirect cost, direct benefits versus indirect benefits fixed cost versus variable cost fixed benefits versus variable benefits similarly this tangible cost versus intangible cost and tangible benefits versus variable benefits.

So, this cost and benefits have to be considered properly in C- B analysis that is why this classification is required. We have to consider each and every possible type of cost each and every type of benefit during the C- B analysis because you have seen that very often the management, they are not considering they are not taking into account the intangible benefits and intangible what cost because that difficult to measure the difficult to identifiable the difficult to quantifiable.

So, that is why they are ignoring this, as a result the result of C- B analysis will not get proper the financial assessment will not be proper. So, that is why this cost and benefits have to be considered properly while carrying out C- B analysis or while carrying out the financial assessment.

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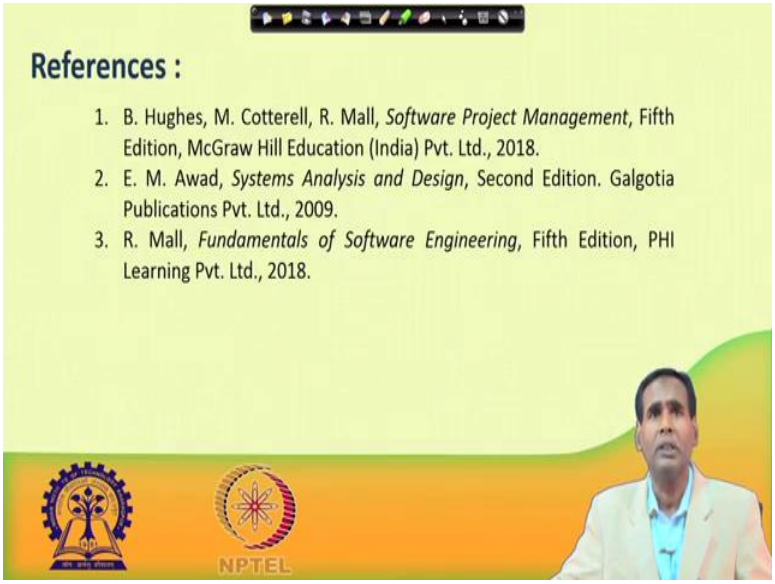

So, last class as well as this class we have taken these materials from this books mainly serial number 1 by Hughes Cotterell and mall that book will take here the preliminary book and besides that the different cost and the benefits and the feasibility study on the cost benefit analysis it is well explained in the system analysis and design book by E M Awad and then this some fundamental concepts of feasibility study and this cost benefit analysis and this software project management, some fundamental concepts are given in this fundamentals of software engineering Rajib malls book you may refer this references in this books. So, from time to time as an when we will I will use different materials, different books or different resources I will put all those things in due course of time.

So, this is all about this different the types of costs and benefits in the next class we will see the other steps of cost benefit analysis. So, today we have seen three two steps that is identify the cost, step 1 and step 2 is your classify or categorize the cost. So, these two steps we have seen of cost benefit analysis, the next step is after identifying and classifying the different cost and benefits we have to use we have to select a proper cost benefit evaluation technique there are so many cost benefit evaluation techniques such as net profit, present value net present value, payback period, internal rate of return. So, those methods are there. So, how to select this methods based on what that will discuss in the next class thank you very much. So, we will take up the selection of cost benefit evaluation technique in the next class.

Thank you very much.

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Lecture – 13
Project Evaluation and Programme Management (Contd.)

Good afternoon to all of you. So, today we will discuss the different cost benefit evaluation techniques.

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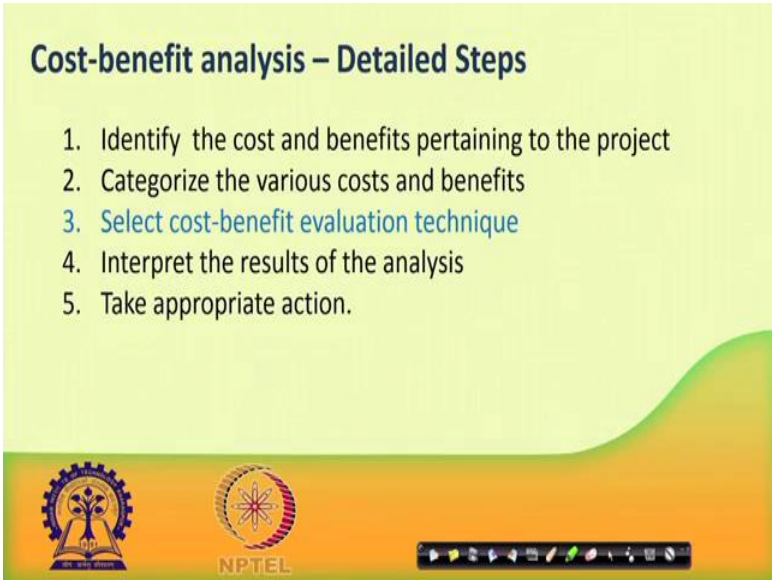

Last class we have seen these the two phases of cost benefit analysis; that means, identifying the cost and benefits, then categorizing the various cost and benefits. Today we will see the third step that is the selecting the cost benefit evaluation technique. How to select a proper a suitable cost benefit evaluation technique?

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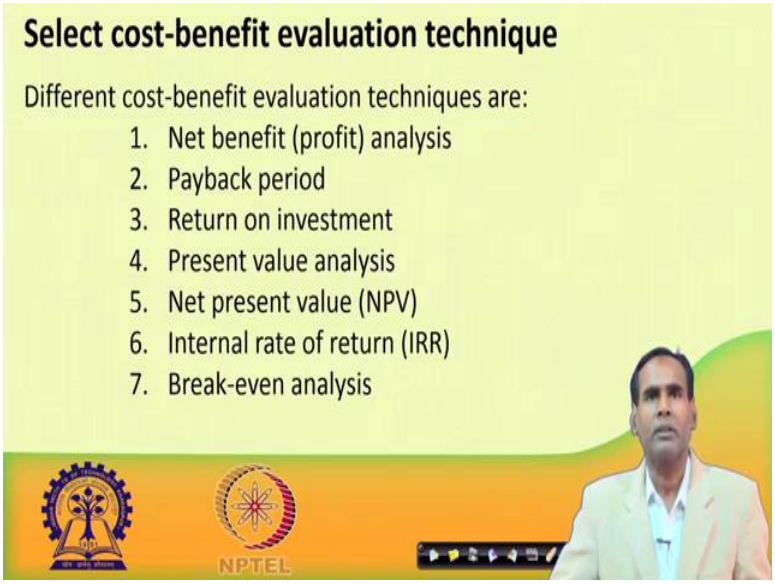

So, let us see what are the different cost benefit evaluation techniques available, there are many cost benefit evaluation techniques available these are as such net benefit or net profit analysis, payback period, Return on Investment or ROI, present value analysis, Net Present Value or NPV, Internal Rate of Return or IRR and break even analysis and a few others also exist let us discuss these cost benefit evaluation techniques one by one.

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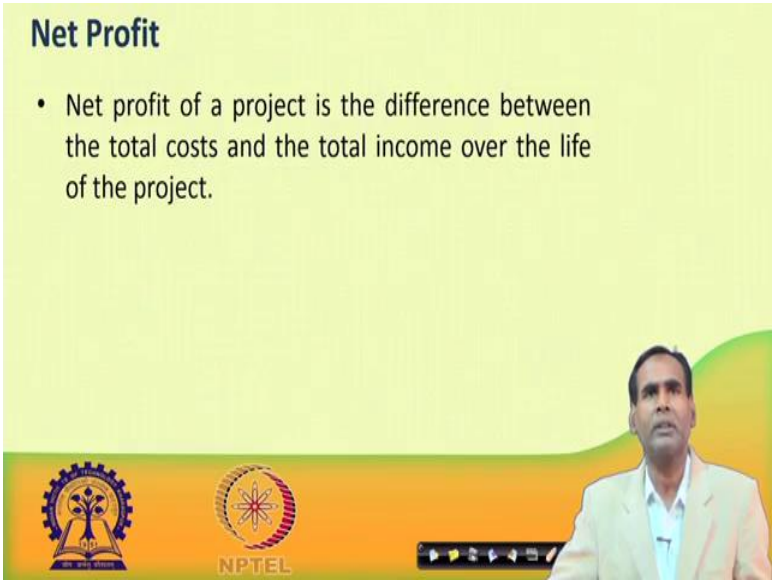

First we will discuss about the net profit. So, what do you mean by net profit? Net profit of a project is the difference between the total cost and the total income over the life of

the project. So, we have to find out the difference between the a total cost spent and the total income obtained over the life of the project.

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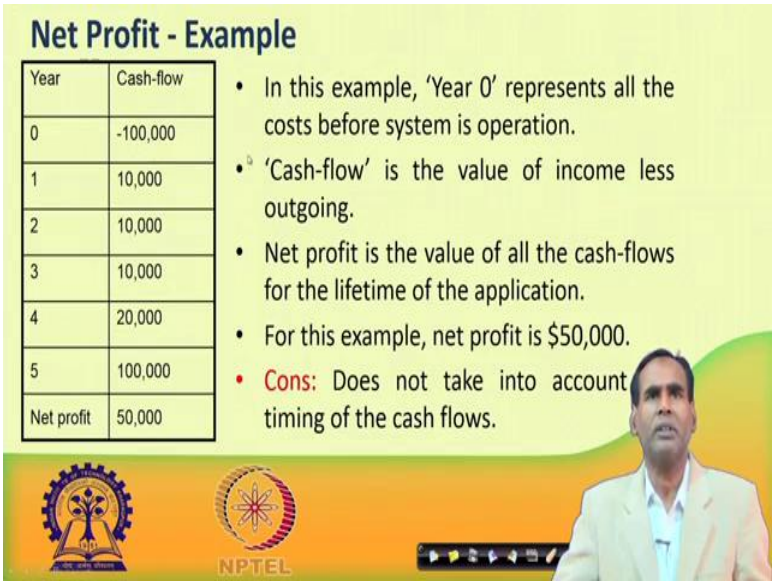

We will take a small example to illustrate this, so we have taken a small example we are the cash flows are given year wise last class we have already seen what a cash flow is. So, in this example you can say that the year 0 represent the cost before the system is operation. So, before the system is in operation that we take as year 0 and this cash flow represents the cost before the system is in operation.

Cash flow already we have discussed yes yesterday it is the value of income less outgoing I mean the value of the difference between the income and the expenditure, so income minus outgoing will represents the cash flow. Net profit is defined as the value of all the cause cash flows for the life of the lifetime of the application. So, in an application for the whole life how much what cash flows occur?

So, we have to find out the total value of the cash flows for the whole lifetime of the application in this example you see the application or the project is expected to continue for 5 years and in 0 years we have spent how much 1,00,000. So, this is spent for what establishing the project. So, here we have not got any revenue, then year 1 onwards it gives revenue and you can see that or the first year it gives a 10,000 total or and like this are the end of 5th year it gives 1,00,000 return 1,00,000 benefit. So, or 1,00,000 we can say the income it gives in the 5th year.

So, the net profit it has to be all these cash flows for 0 to 5 years they have to be added and you will get it. So, here you can say easily that the net profit for this example of project for 5 years is coming to be 50000 dollar. So, for this example in a profit is 50,000.

Now, let us see what is the drawback of this technique in the drawback is the does not taking the count the timing of the cash flows. So, what is the time value of the money? It does not take into account the timing of the cash flows that is one of the drawback in some of the classes we will see how this draw back can be overcome by which technique.

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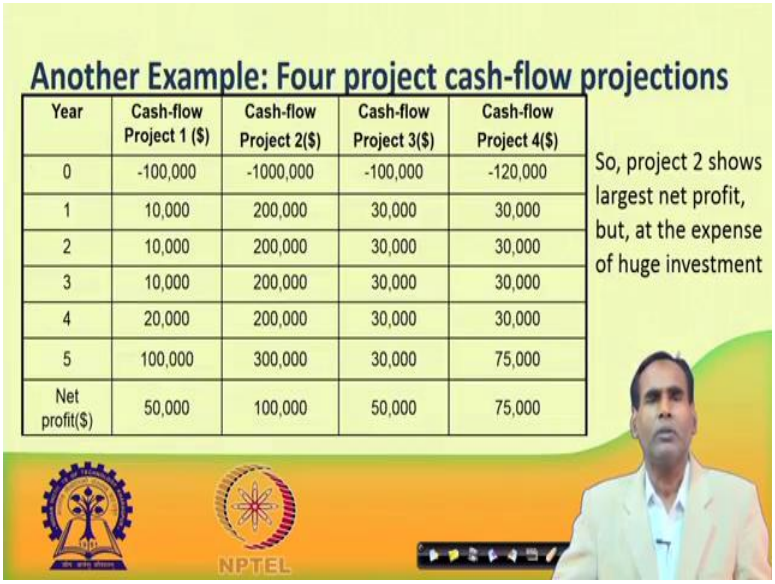

So, we will take quickly another one example here you can see that there are four projects have been shown, the cash flows for each of the projects have been shown and year wise the year wise cash flows for every project has been shown. And finally, at the last row we have discussed about this what we have discussed the various, the net profits for each of the projects and you can see here that already for project 1 I have shown in the previous slide.

So, the net profit is coming to be a 50,000, for project 2 again you will got of the all the cash flows you will get the total the net profit this is like say 1,00,000. And similarly for the net profit for cash, for project 3 its 50,000 and the net profit for project 4 is coming to be 75,000.

You can easily say that here project 2 gives the maximum net profit. So, that is why this will be the what beneficial one here, but you see even if it gives the maximum net profit, but at the expense of huge investment initially see how much every year you are just spending here the project 1 and project 2 1,00,000, project 4 it is 1,20,000 about here we spent about how much it 10,00,00. So, the initial investment is of 10,00,000. So, that is why even if it gives maximum net profit, but at the expense of huge investments, so that has to be considered.

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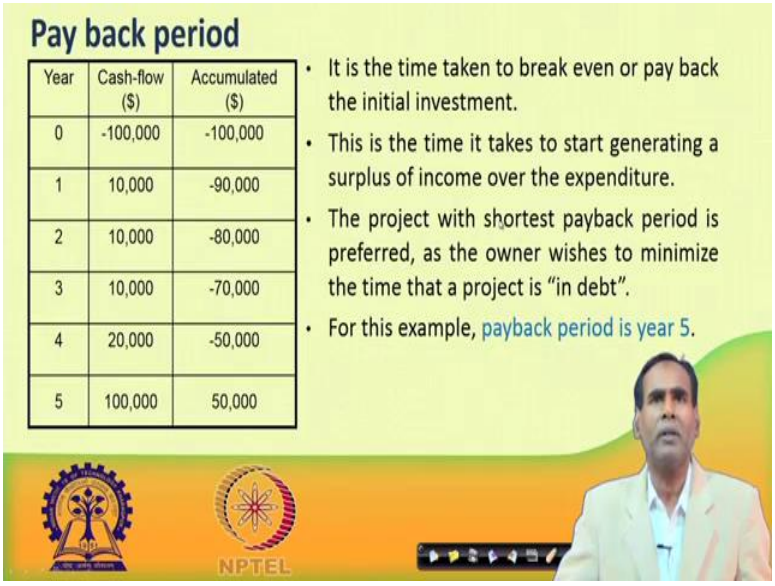

The next one is payback period, so payback payback period it depend the time taken to break even or pay back the initial investment how much you have initially spent, when you are starting when you are in which year you are getting that amount to back, so that is the payback period.

This is the time that it takes to start generating surplus of income over the expenditure how much you have spent, in which year you are getting you are recovering you are getting back that amount plus you gets you get more on the amount more benefits. So, that year is known as the what payback period.

The project what is the desirable? The project with shortest payback period is desirable it is preferred because the owner wishes to minimise the time that a project is in debt. So, you will try to minimize that period because he might have brought this initial funding from some loan or from some bank.

So, in this example if you will see first here in invested how much say 1,00,000 and you see 1st year he is getting 10,00,00, then 2nd year 10,00,00, 3rd year 10,00,00, 4th year 20,000. So, this is how much; that means, up to 4th year he is getting 50,000. So, next in 5th year he is getting 1,00,000, so; that means, the initial 1,00,000 here invested he is getting only at the only during 5th year. So, for this example the payback period is year 5.

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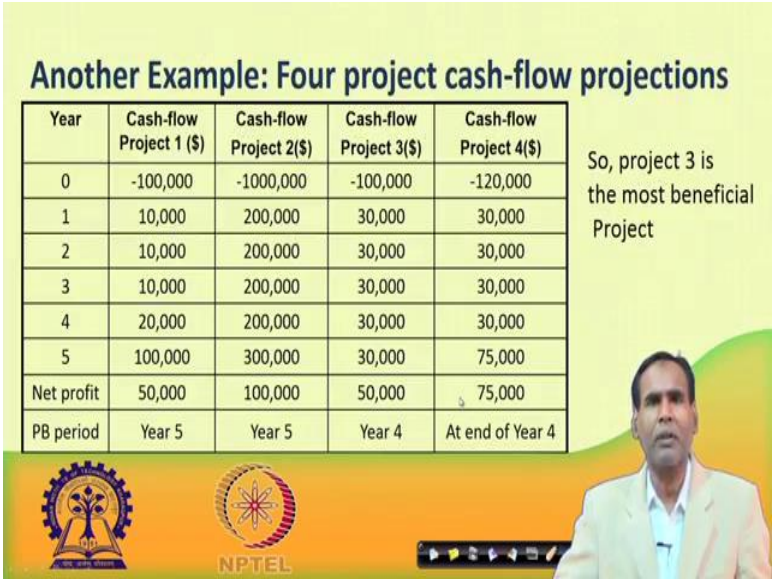

So, similarly for our earlier what example where four projects are there, you can compute the payback periods, you can say that for project 1 I have already shown you the payback period is 5. And similarity for project 2 you can see initially as sends 10,00,000 this is also you are it is getting recovered and the what year 5 and a for project 3 you are getting that in the initial he has spent 1,00,000. So, it is getting 30 plus 30 60, 30 90 and the 4rth year he is getting this initial amount that he has spent that is 1,00,000.

So, for project 3 year 4 is the payback period and in case of project 4 you can see that initially he has spent or the project the organisation has spent 1,20,000 he is getting that value when 1st year it will be in 30, then another 30,000 and then another 30,000. So, at the end of when 4th year is completed then only he gets back or the organization get back gets back the initial amount that is 1,20,000. So, at the end of the year 4 the project will get back the initial expenditure the initial expenses.

So, here you can see that project 3 where the organisation receives the original or the initial funding it is during year 4. So, you year a project 3 may be treated as the most beneficial project if we will considered the payback period.

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So, you can see that the advantages of this technique pay back payback period is that it is simple to calculate no not particularly sensitive to small forecasting errors, but it has some draw backs like it ignores the overall profitability of the project what is the overall profitability project we will does not take into account. Also it ignores any income or any expenditure once the project has broken even; once the project has reached as the procurement point whatever the income expenditure of that it does not take into account. You can see that that project 2 and 4 overall there more profitable than project 3, but this is ignore you please see here project 2 and 4.

So, according to payback period we have seen that project 3 is the beneficial one because during year 4th year he is recovering he is getting his initial amount, but you see the profit is more in where, the profit the net profit is more in case of what highest is in project 2 and then it is what a 75.

So, if you will take into account in net profit actually project 2 should be the best one, but since it is not considering; since it is not considering the overall profitability of the project, so that is why either project 2 or project 4 even if they are overall more

profitable they are ignored by using the payback period. So, this is one of the drawback of the payback period.

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So, now let us see the next metric the next measure or the next technique that is ROI which stands for Return on Investment, this measure provides a way of comparing the net profitable to the investment period, it will compare the net profitability to the investment period of for an organization. It provides a simple and easy to calculate measure of return on capital now. So, this technique provides a very simple and easy to calculate measure of the return on the spent capital amount.

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The mathematically we can define ROI as like this.

$$ROI = \frac{\text{Average annual profit}}{\text{Total investment}} \times 100$$

So, now, ROI is expressed in terms of percentage. So, if you will take our the a previous project that is project 1 here you can see the net profit is coming to be 50,000 along the how many years along 5 years.

So, average profit is coming to be how much? Average profit will be coming to be 50,000 divided by 5 because it is occurring in 5 years. So, average profit coming to 10,000. So, ROI it can be computed as 10,000 by the total investment you have seen initially the farmers spent 1,00,000. So, 10,00,00 by 1,00,000 into 100 coming to be 10 percent.

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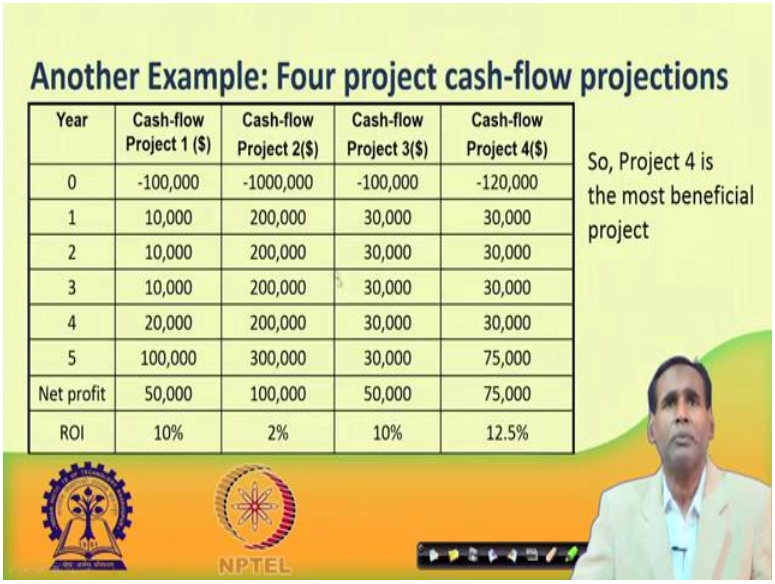

So, for project 1 ROI coming is coming to be 10 percent. So, similarly you can compute the ROI for the other projects where for project 2 it is coming 2 percentage, for project 3 it is 10 percent and project 4 it is 12.5 percentage and the objective is which one will be preferred the project having the highest ROI must be preferred. So, project 4 in this case have is having the highest ROI, so project 4 is the most beneficial project in this example.

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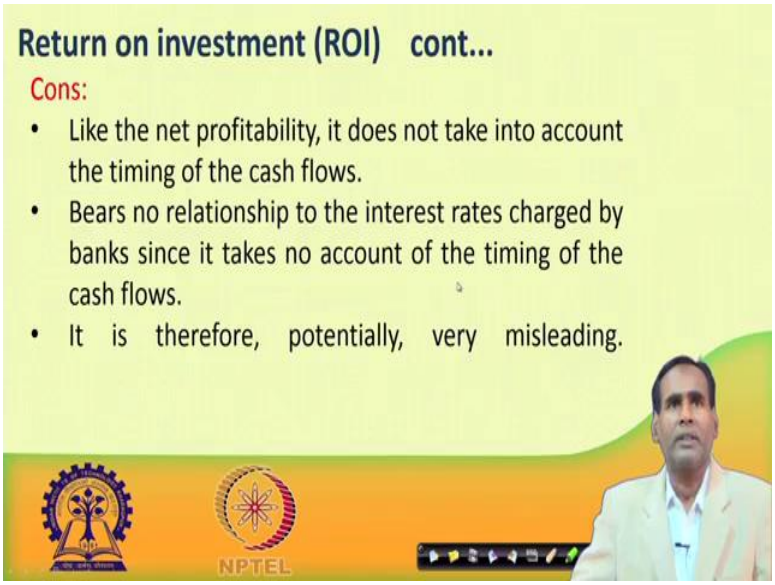

Next see let us see what are the drawbacks of this ROI like the profitable net profitability, it also does not take into account the timing of the cash flows. We have already seen net profit profitability method or net benefit method the problem with that it does not take into the timing value of the cash flows. Similarly, ROI also does not take into account when the timing value of the cash flows, it also bears no relationship to the interest rates charged by the bank. So, different banks may charge different interest rates.

So, it does not bear any relationship to the interest rates which are charged by banks, since it does not take into account the timing of the cash flows. So, that is why; so this method potentially or may what may be very misleading.

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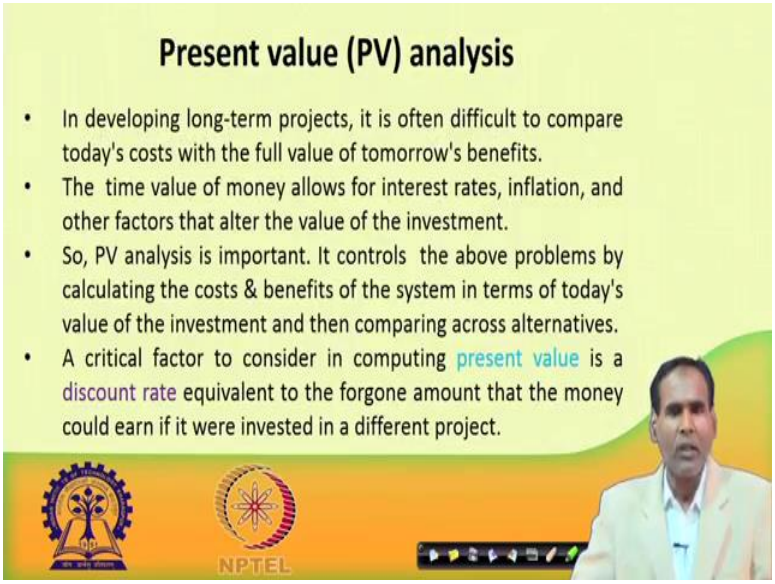

Now, we will see one of the important method that is known as a present value analysis So, what do you mean by present value analysis? In the developing in developing long term projects it is very much difficult to compare what is the today's cost with the full value of tomorrows benefits; that means, what cost we are investing today and what is the value we will get tomorrow at a benefit it is very much difficult to compare this.

Also the time value of money it allows for interest rates, inflation and other factors that might alter the value of the investment. So, that is why we must know the present value. So, we must carry out present value analysis for the what investment that we are making or the benefit that we will get. So, present value analysis is very much important.

So, this present value analysis it controls these above problems; that means, how to compare the today's cost with the full value of tomorrows benefits or, how to consider; how to take into account the interest rates inflation and other factors that might alter the value of the investments. So, this it will carry out PV analysis present value analysis. It will control the above problems how by calculating the cost and benefits of the system in terms of today's value; in terms of today's money value of the investment and then by comparing across different alternatives.

A critical factor to consider in computing present value is a discount rate. So, while we will compute the present value another important factor we have to know that is known as discount rate and this discount. So, see for present value we have to computer a factor called as discount factor. So, this discount factor it based on the discount rate.

So, a critical factor to consider in computing the present value is a discount rate which is equivalent to the forgone amount that the money could earn if it were invested in a different project. If that money could have been invested in a different project, then what could be the forgone amount? So, discount rate is equivalent to that forgone amount which the money could earn if it were invested in a different project.

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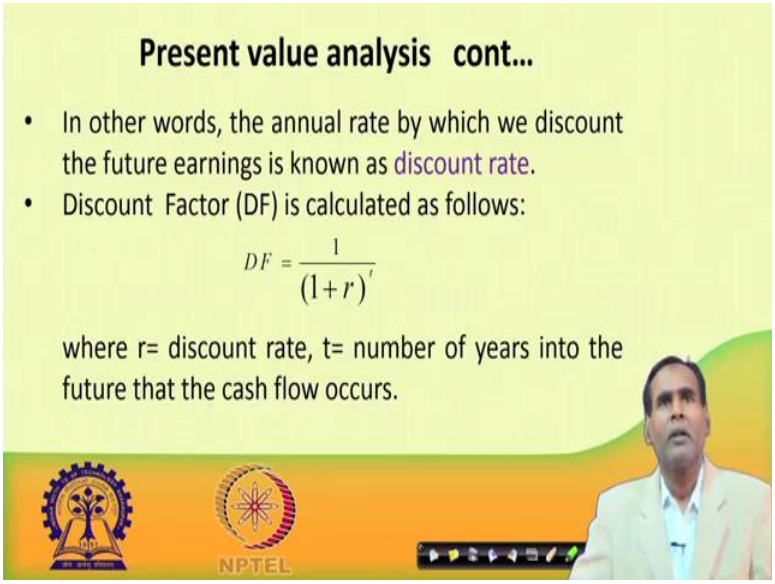

In other words we can say that discount rate is defined as the annual rate by which the discount we by which we discount the future earnings. Mathematically discount factor is calculated as follows,

$$DF = \frac{1}{(1+r)^t}$$

where r is equal to discount rate and t is equal to the number of years into the future that the cash flow occurs. So, in the this is the simple formula by using which we can use this discount factor.

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So, now let us see we will find out the discount factor for some percentage of the discount rates and for different years. If we will take the discount rate as 10 to 10 percent or the interest rate at 10 percent and 1 year per for 1 year period, then the discount factor is equal to 1 by 1 plus this much 0.10 to the power 1. So, this becomes 0.9091 and similarly in case of 10 percent return 2 years discount factor becomes 1 by 1.10 and I think there is a small mistake in the its the in this formula 1 plus r to the power t. So, here this should be 1 by 1 plus 0.10 to the power 2.

So, there is a slight mistake here it must be 1 by 1 plus 0.10 to the power 2, so which will be equal to 0.8294. So, similarly discount factor can be computed for any given year and discount rate. So, we have given you a table which help you for a computing the present values. So, the discount values we have a given, there is a table here yes.

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Year	Discount rate (%)					
	5	6	8	10	12	15
1	0.9524	0.9434	0.9259	0.9091	0.8929	0.8696
2	0.9070	0.8900	0.8573	0.8264	0.7972	0.7561
3	0.8638	0.8396	0.7938	0.7513	0.7118	0.6575
4	0.8227	0.7921	0.7350	0.6830	0.6355	0.5718
5	0.7835	0.7473	0.6806	0.6209	0.5674	0.4972
6	0.7462	0.7050	0.6302	0.5645	0.5066	0.4323
7	0.7107	0.6651	0.5835	0.5132	0.4523	0.3759
8	0.6768	0.6274	0.5403	0.4665	0.4039	0.3269
9	0.6446	0.5919	0.5002	0.4241	0.3606	0.2843
10	0.6139	0.5584	0.4632	0.3855	0.3220	0.2472
15	0.4810	0.4173	0.3152	0.2394	0.1827	0.1229
20	0.3769	0.3118	0.2145	0.1486	0.1037	0.0611
25	0.2953	0.2330	0.1460	0.0923	0.0588	0.0304

So, you can see that the discount values or the discount factors for different years and taking different discount rates like 5 percent, 6 percent 8 percent, 10 percent, 12 percent, 15 percent and a different what years like 1 to 10 up to 15, 20, 25 what could be the discount factor with different discount rates and different years and the number of years it is shown. This will help you for solving or for finding out the present values on the net present values.

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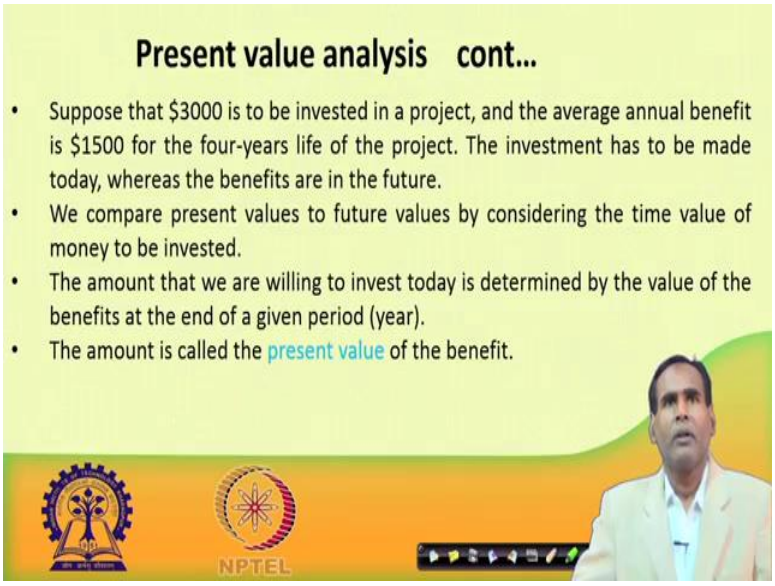

So, let us quickly take a small example, here suppose that dollar 3,000 has to invested in the project and the average annual benefit is expected to be 1500 dollar for four years life of the project. The investment has to be made today please see the investment will make today, where as the benefits are expected to come in the future. So, we compare the present values to the future values how much we have spent today and how much we are expecting in future. We compare the present values to the future values by considering the time value of the money that we have invested.

So, this time value of the money has not considering has not been considered in net profit and the ROI, so that those are the problem. So, those problems are over come by in this present value analysis. The amount that you are willing to what investment today is determined by the value of the benefits at the end of the given period, please see how define present value.

The amount that we are wanting to invest today, how it can be determined? This is determined by the value of the benefits which may be obtained at the end of a given period or the given year. So, this amount that we are willing to invest today for which we are expecting that some benefit will occur might be after n years or n number of years. So, this amount is call the present value of the benefit.

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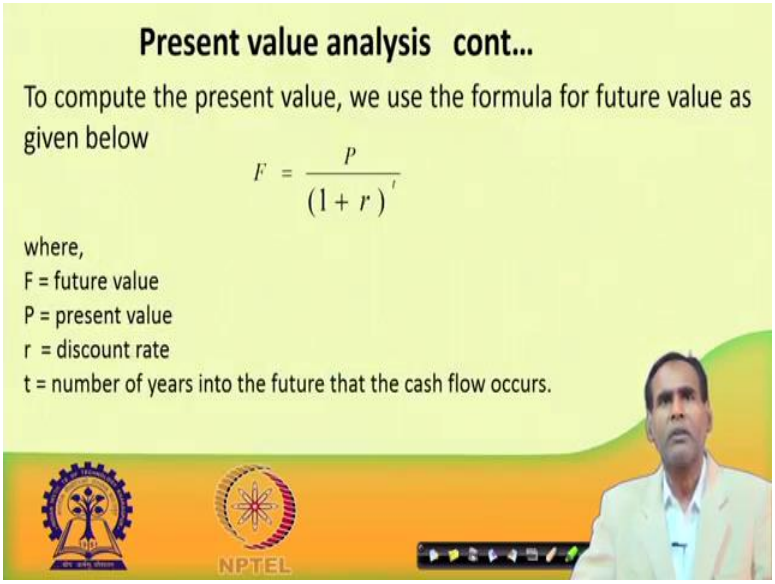

So, how you can compute the present value? We can compute the present value by using formula for the future value. The future value or the benefit can be expressed as

$$F = \frac{P}{(1+r)^t}$$

that P the future value; that means, if the future value, P is the present value of the money or the present value of the investment, r is the discount rate and t is the number of years into the future that the cash flow occurs.

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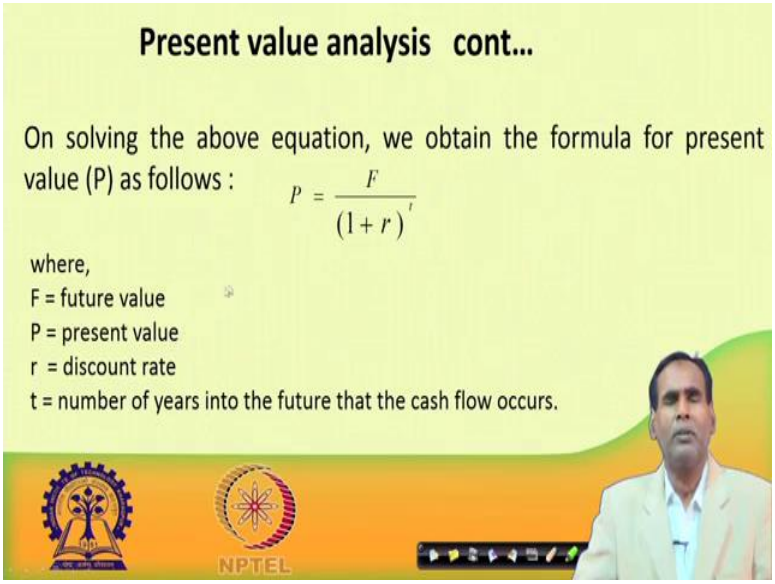

On solving the above equation you can get the value of a P which is equal to P is equal to F by 1 plus r to the power t,

$$P = \frac{F}{(1+r)^t}$$

where the terms have usual meanings.

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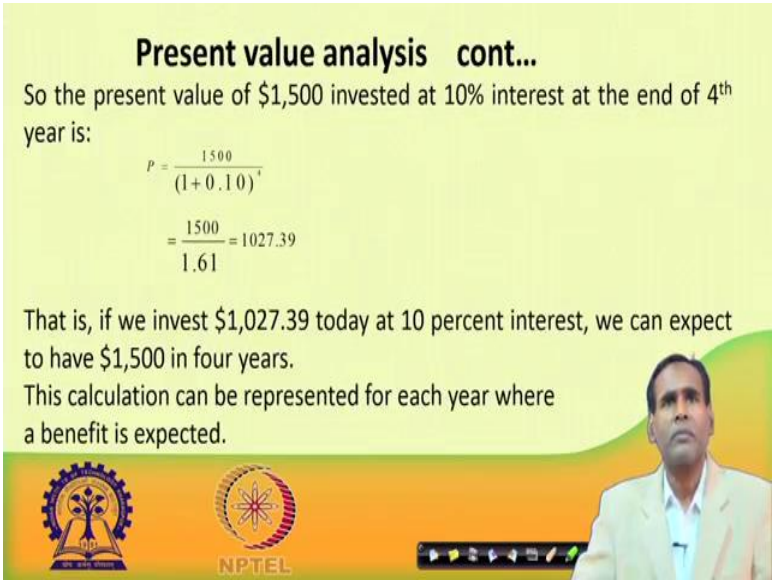

We will take a small example for that previous example that we are saying that suppose we want to what invest 3000 rupees and the average annual benefit is coming to be 15000 sorry 1500 for the four years life of the project now we want to compute its what a present value, you can see the present value can be computed like this. So, the present value of dollar 1500; 1500 invested at 10 percent in interested at the end of 4th year.

So, the present value you can see that 15,000 is what? 15,000 is annual average annual benefit we were expecting that the average annual benefit will be 1500 when for the 4 years life of the project. So, now, we can see what will the present value of the 1500 what rupees that we will get after 4 years at the rate of 10 point 10 percent interest is calculated as P is equal to 1500 by 1 plus 0.10 to the power four. So, this is coming to be 1027.39.

So, literally what is its meaning? Its meaning is that, if we invest 1027.39 dollar today at 10 percent interest then we can expect to get 1500 dollar in four years; that means, what is the meaning of present value. This calculation can be represented for each year where a benefit expected that you can calculate yourself.

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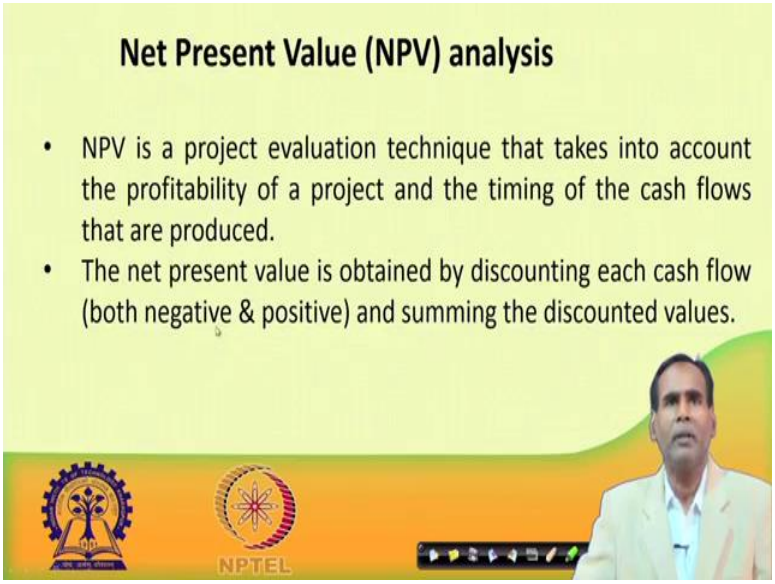

Now, based on the present value another important concept that you will see that is known as Net Present Value NPV its short form is NPV; NPV stands for this Net Present Value.

So, NPV how will define or what do you what this NVP, what does it represent? NPV is a project evaluation technique that takes into account the profitability of a project and the timing of the cash flows that are produced. So, we have already seen the earlier techniques such as net profits and this ROI they do not take into account the timing value, but here this NPV it is a evaluation technique that takes into account the profitability of a project as well as the timing of a cash flows that are produced.

The net present value is obtained how? The net present value can be obtained by discounting each cash flow both whether it is negative or positive because you know the initial expenses that we represented as negative value and then the profit we represent in terms of the positive or the income that we represent as a positive what values. The net present value it is obtained by discounting each cash flow and summing the discounted values.

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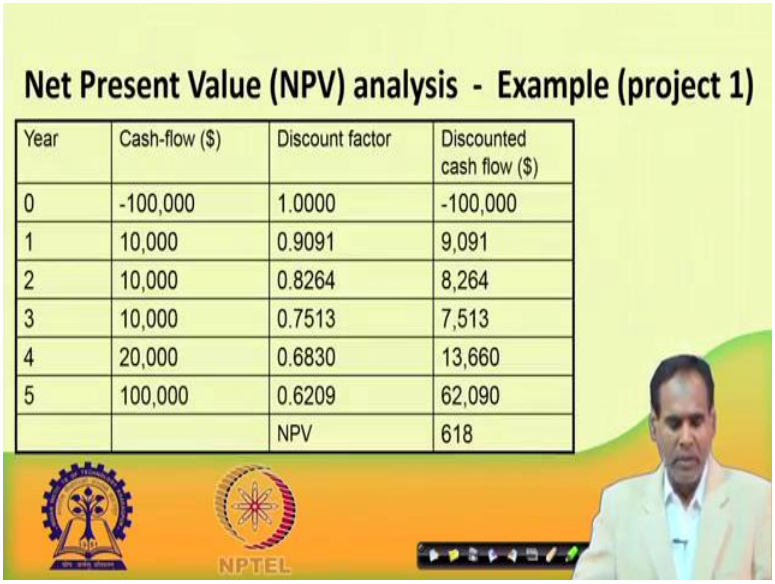

We will take a small example here. So, again the for the project one I am calculating the NPV. So, year wise the cash flow value is have been given in dollars, then the discount factor is obtained using the table I had shown here. The table we have already seen that the discount factors we have already seen for year for the rate of 5 percent at the year 1 this is the what discounted factor.

Similarly for 10 you can see for year 1 for 10 these discount factor is 0.9091, for 10 percent rate for 2nd year this discounted factor is 0.8264. So, this values you can use directly here to compute the NPV values that we have done here.

So, discount factor for see it is assume that discount factor for 0 th here is 1, then we have use that formula to compute the discount factor for 1st year it is 0.9091, for 2nd factor for 2nd year it is 0.8264 unlike that and then we multiply this discount factor with the cash flow, then we get the discounted cash flow. So, we have to find out the discounted cash flow by multiplying the discount factor with the cash flow for every year we can find out the discounted cash flow and finally, we have to add the discounted cash flows for all the periods that will give you the NPV.

So, after adding the discounted cash flows for all the your years we get NPV is equal to 618 dollar 618 dollar for this example with discount rate 10 percent and for the period of 5 year the NPV comes to 618 dollars.

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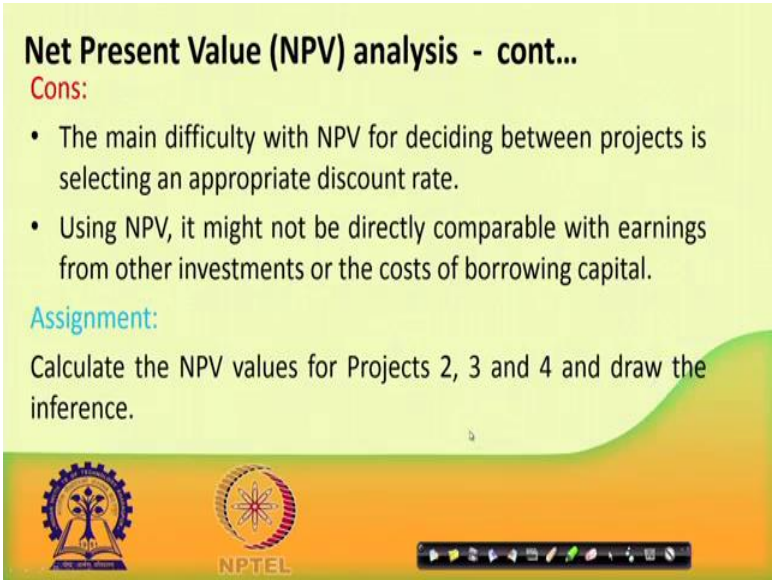

So, similarly what you do you can calculate the NPV values for the projects at for the 2nd, 3rd and 4th project and you can draw the inference find out which project with the most preferred one which project will be the most beneficial one please find out this one this is an assignment for you.

So, let us see what are the limitations of net present value analysis. The main difficulty with NPV for deciding between projects is selecting an appropriate in discount rate, how you will select whether you will take the discount rate at 5 percent, 8 percent, 10 percent or 12 percent. So, deciding choosing an appropriate discount rate it was that is one of the main difficulty. Similarly using NPV it might not be directly comparable with earnings from other investment or the cost of borrowing capital, it is very difficult to directly compare the with the earnings from other investments or the costs of borrowing capital these are the some of the limitations of NPV method.

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Next we will see the other measure that is called is IRR which stands for Internal Rate of Return. IRR provides a profitability measures as a percentage return that is directly comparable with interest rates. IRR it provides a profitability measure and it expressed as a percentage return that is directly comparable with the interest rates.

So, suppose a project that shows an error of 10 percent, it would be worth if the capital could we borrowed for less than 10 percent or the capital it could not be invested in elsewhere for a return greater than 10 percent, then we say that project will be worth. IRR is IRR may be considered as the discount rate that would produce an NPV of 0 for the project, this IRR can be used to compare different investment opportunities.

It can be calculated using a spread sheet and also Microsoft or Microsoft Excel will be provide some functions. So, also Microsoft Excel provides some functions such as IRR which takes two arguments a value and an initial guess and initial in an initial seed value as input and finally, it returns the IRR it returns an IRR at the output. So, this is the internal rate of return which is another very good measure.

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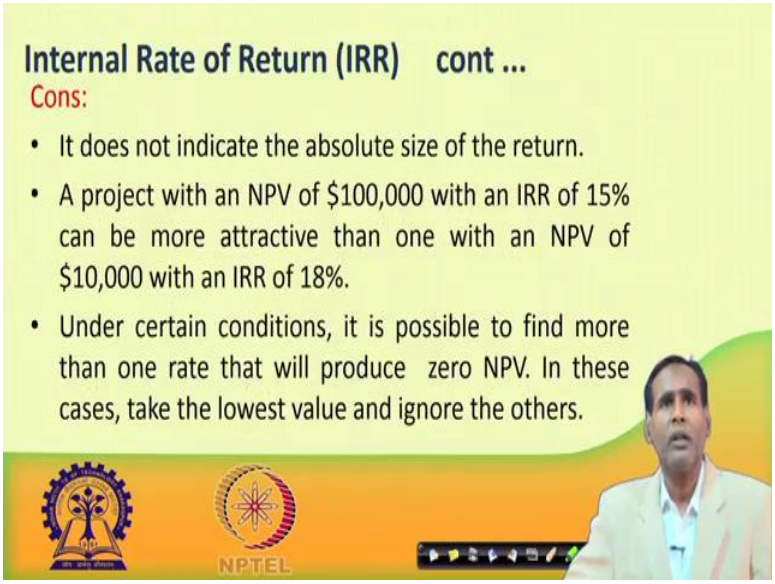

The limitations of IRR are like follows. It does not indicate the absolute size of the return it normally it does not indicate the absolute size of the return. Similarly it might happen the case that a project which an NPV of 1,00,000 with an IRR 15 percent it may be more attractive than one project with an NPV of 10,000 only with an IRR of 18 percent that might happen what IRR what is unable to what predict this thing is unable to handle this thing.

Under certain conditions it is possible to find more than one rate that will produce zero NPV, it can be possible. There can be what two rates that will produce the same 0 NPV. In these cases we have to take the lowest value and ignore the other values I mean the other rates.

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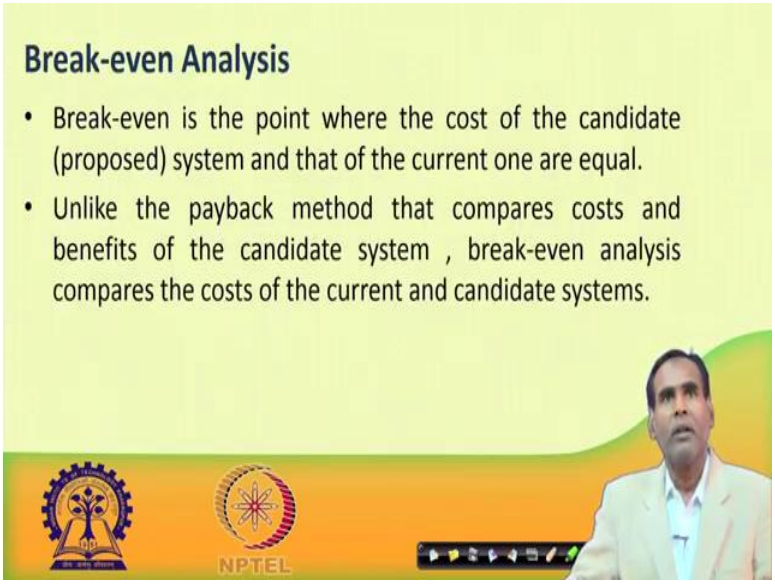

Now, we see the last technique that is the break even analysis break even analysis or first let us see break even analysis based on a point called as breakeven point. Break even is defined at the point where the cost of the candidate or the propose system and that of the current one are equal.

So, one current system say manual system is running, the propose system is an automated system, so break up suppose, then break even is the point where the existing system and the where the cost of the existing system and the cost of the point of propose system they are equal.

So, unlike the payback method which compares the cost and benefits of the candidate system I mean the propose system, but break even analysis it compares what the cost of the propose system or the candidate system and the current system.

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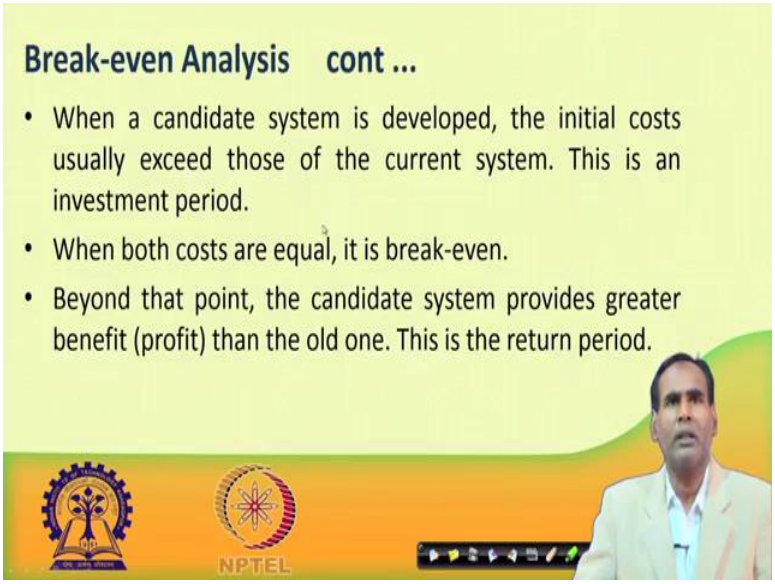

So, when a candidate system is developed the initial cost are use a normally usually exceed those of the return say current system this is an investment period. Obvious, so when a propose system is developed you have to spend more money. So, the initial cost normally it will exceed that and that of the that the cost of the current system.

So, this period we call as the investment period, then when both the cost of the propose system and the cost of the existing system they become that point is known as the break even and after the breakeven point what will happen the propose system or the candidate system it will be provide better will get a profit than the older one or the existing one and this period is known as the return period we will show it through an example I will through a graph.

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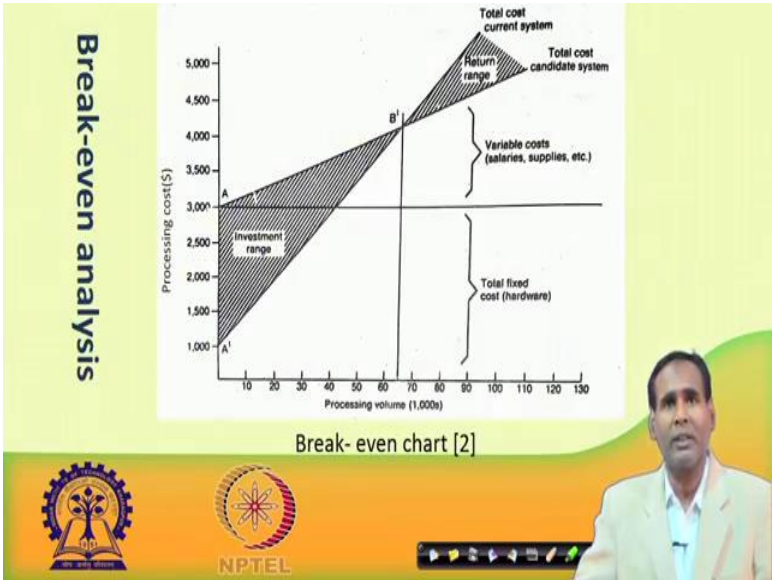

You can see that in x axis we have taken the processing volume y axis the processing cost and you can see that this line what is the total you can see that; see you can see that in this graph here this line represents the current system and this line represents the what the total the existing system the candidate system, as I have already told you initially the propose system will take more cost.

And the existing system will take the what they because it is running, so it will take less cost, but some point we will come here see at point of B prime both the cost of the existing system and the what this propose system become sequence. So, B prime becomes the what the breakeven point near about 65000 processing volume both the cost equal after this though this breakeven point.

After this then you see that the what the propose system takes less cost and the current system it takes more cost that mean the propose system starts giving, but return starts giving profit. So, that is why since here the propose system takes more cost this area is known as the investment range and after the breakeven point here the current system takes the what more cost whereas, the propose system takes less cost it starts giving return it starts giving profit.

So, that is why this area known as the return range. You can see that this area represent this point represents a fixed cost whereas, this portion represents the variable cost, this is what is meant by the break even chart; this I have already explained here.

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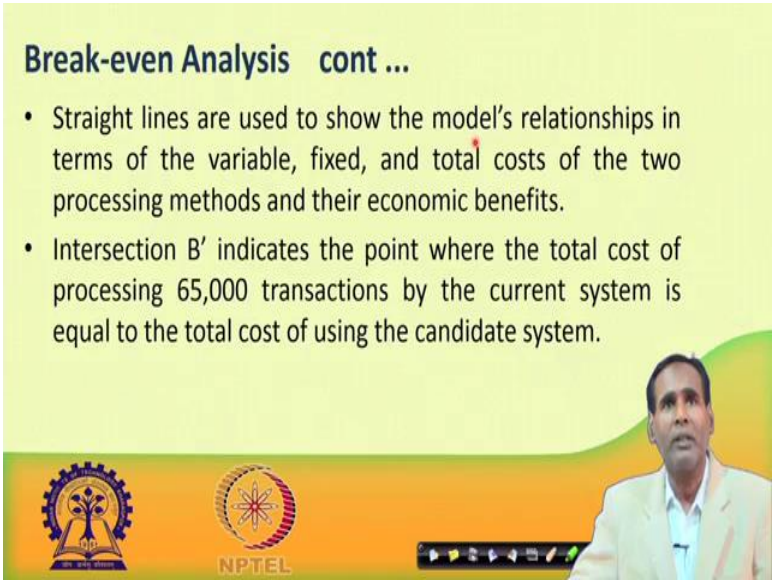

Now, the shaded area the investment period that we have already shown you this is the investment period or investment range where as this is the return range and B prime is the break even period, so breakeven point. So, breakeven point is also very helpful to decide that which project will be beneficial for us.

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So, in this lecture we have discussed the different cost benefit evaluation techniques with suitable examples also we have described the advantages and disadvantages of each technique also we have presented which technique will be best suitable in which

circumstances that also we have seen. So, in the next class we will see the remaining two steps of cost benefit analysis that is what interpreting the results as well as taking the appropriate action. These are the references and in the next class we will see the remaining two steps.

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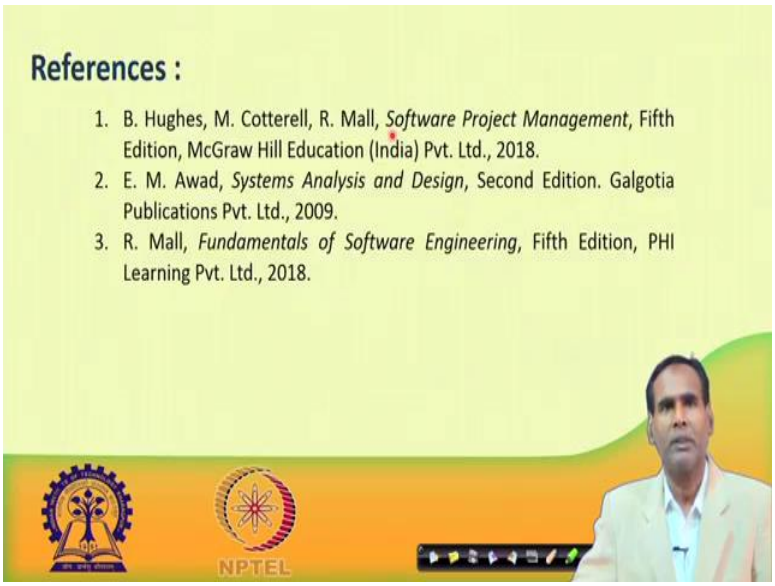

Thank you very much.

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Lecture – 14
Project Evaluation and Programme Management (Contd.)

Good afternoon to all. So, now, let us see the remaining parts of cost benefit analysis ok. So, we have seen three steps of C-B analysis. So, the other remaining two steps are we will see first then we will see the risk evaluation.

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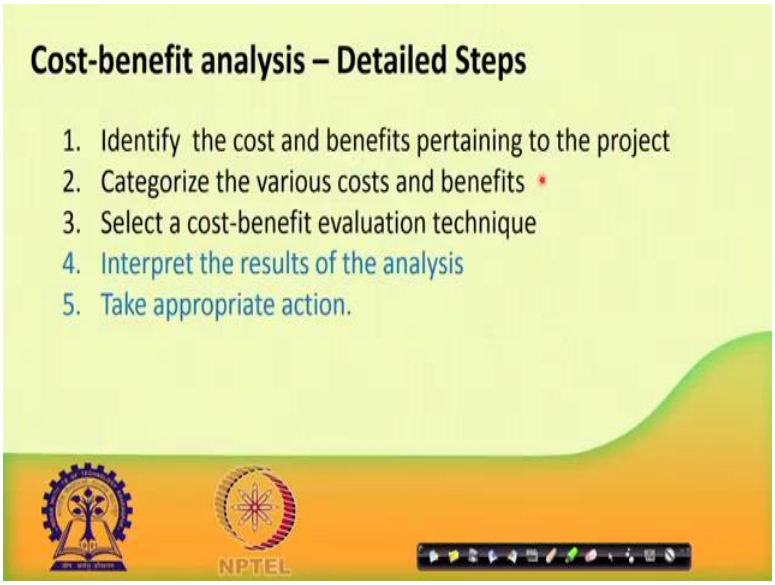

So, the last two steps of C-B analysis are interpret the results of the analysis and then take appropriate action.

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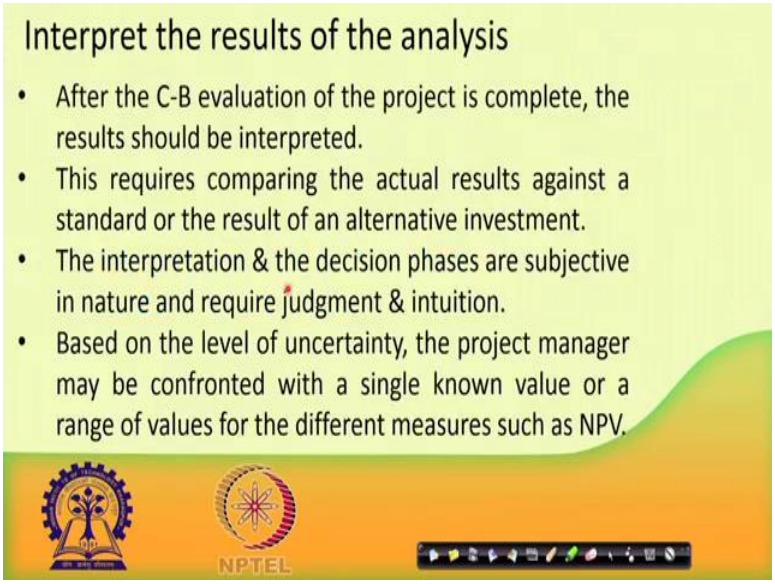

So, after so, we have already seen that, we have already selected, we have discussed the cost benefit evaluation techniques, then depending upon the scenario, we should select or the organization should select a proper and appropriate cost benefit evaluation technique. So, after that cost benefit evaluation or that cost benefit analysis over, then the project the result should be interpreted ok.

So, after the cost benefit evaluation of the project is over the results must be interpreted. So, for interpreting the results it requires comparing the actual results against the standard or the result of any other alternative investment. The interpretation and the decision phases are highly subjective in nature they require the judgment and the intuition capabilities of the project manager. Based on the level of uncertainty the project manager may be confronted with two possibilities.

Either with a single known value or a range of values for the different material such as NPV or that net profit etcetera either you may choose, you may be confronted with a single known value or a range of values.

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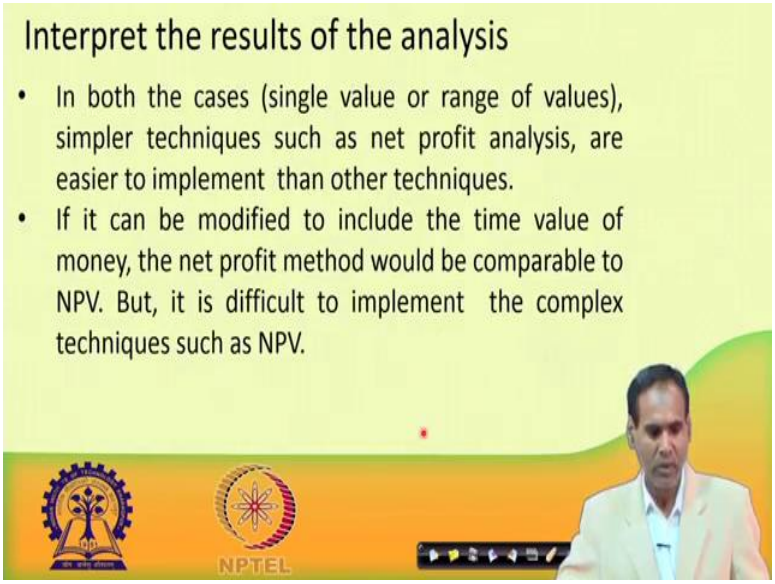

In either of the cases whether it is a single value case or it is a range of values for the what measures such as NPV or net profit or what you are ROI, whatever it may be simpler techniques may be used such as net profit analysis etcetera. May be used they are easier to implement the and other complex techniques.

But, if it can be modified if because we know the drawback of net profit analysis is the that does not take into account the timing value of the investment. So, if it can be modified to include the time value of the money, then the net profit method it probably it would be comparable with the net present value, but you know that it is difficult to apply or to implement the complex techniques such as NPV.

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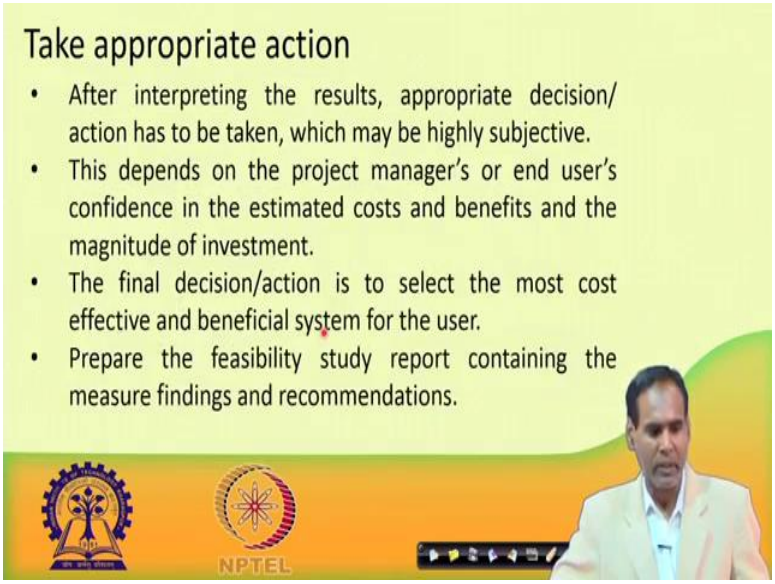

Now, after interpreting the results after analyzing all those results we must take the appropriate action the project manager must take the appropriate action. So, after interpreting the results the appropriate decision or action they has to be taken.

And this taking the decision it is highly subjective in nature, this depends on the project managers or the end user confidence in the estimated cost and the benefits. And the magnitude of the investment, the final decision or the action will be to select the most cost effective and beneficial system for the user.

So, the final action of the cost benefit analysis is to select the most appropriate the most cost effective and beneficial system for the user, then the a project manager has to prepare the feasibility study report which contains the major findings and during the analysis process on the suggested recommendations. This is how the final action has to be taken after interpreting the results. So, this is something about this C-B analysis.

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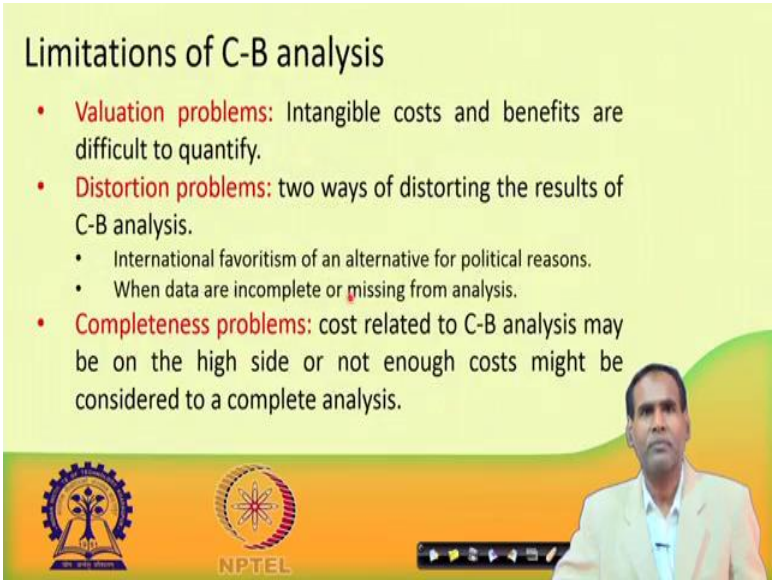

But, what are the limitations of C-B analysis? So, there are many limitations of the C-B analysis number one is valuation problems. I have already told you in the last class that there are different types of cost and benefits. So, the particular categories of cost and benefits or such intangible cost and benefits they are very much difficult to identify measure and quantify, can that be quantified easily. That is one of the most difficult problem in most difficult challenge in case of cost benefit analysis, that is intangible cost and the intangible benefits, they are difficult to identify may be difficult to what measure and quantify.

Another problem is known that the distortion problems, there are two ways of distortion in the results of C-B analysis one, there may be intentional favoritism of an alternative for the political reasons. And the two when the data are incomplete or they are missing from the analysis it is may difficult to take an appropriate action. And the next problem is completeness problem here the cost the cost related to C-B analysis may be on the high side.

The cost that have been estimated they may not represent the actual cost, there may be on the higher side or the estimated cost of the they may not be enough or not enough cost might be considered the project manage manager might not have consider enough cost for conducting a complete analysis. So, this could be the possible limitations of C-B analysis.

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Now, we will see that I have already told you, the contents of business case; one content was the based one risks we know that every project has some kind of risks. There are two types of risks we have discussed earlier; one is project risks, another is business risks. You may know that the business risks they are related to the threats to completion to a successful project execution whereas, business risks are related to the factors which will threat the benefits of the delivered project.

So, in business case the main focus is on business risks. So, right now we will discuss how to evaluate the business risks, the project risks will be discussed later on.

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So, let us see what are the risk evaluation methods particularly business risks. So, we will see about risk identification and rank and ranking, then risk and net present value, cost benefit analysis method for handling risks, then risk profile analysis and using decision trees.

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Let us quickly say about this risk handling and ranking. So, here in any project evaluation we should identify the risks first. So, before handling them in any project evaluation we must first identify the risks. And we should quantify their effects how it

will affect the business. One approach to now what identify the risks and the quantify their effects is to construct a project risk matrix a special type of matrix will construct, that is known as project risk matrix, which will utilize a checklist of possible risks.

So, what are the possible risks that will appear in that matrix? As well as then we have to classify the risks according to their relative importance and likelihood. What is the possibility that this risk will occur and then what is the importance of that risk, how far it will affect the business, how far it will hamper the business? So, that is the importance.

So, we can give values like a high H for high, M for medium and L for low; that means, the importance can be high and importance can be medium or importance can be what low similarly, possibility of occurring this could be or the likelihood could be high low or medium high, medium or low. So, then as I have already told you this matrix will mainly consists of the possible types of risks, then we have to classify by using the relative importance and the likelihood.

So, the point of speaking is that the importance of the risks as well as the likelihood of the risks, they need to be separately assessed, which should not what consider them together they should be separately assessed. Because, see we might be let us concern with something or some risks that although serious, but is very unlikely to occur.

So, some kinds of risks are there they are very serious, but the very unlikely they occur the chance of occurring them is very less. So, we may not be very much serious about that, but there are some less serious risks there are very less serious risks their effect is very what less their importance is very less, but that is almost certain the probability that will occur is almost based certain say 90 percent. So, we must be very much what concerned about this.

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For example, say there is a project A which appeared to give a better return, but it could be very risky. So, we should treat it differently. Similarly, another project could be another project B, it might give what very less return, but it is what we can say there is less risk. So, you may go for that has to be debated. So, I have already told you that in order to classify the risk their importance or the we have to construct a project matrix, we could drop a project matrix for each project for assessing the risks. Here, we are consider two important factors the importance of the risks as well as the likelihood the probability that the risk will occur.

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See we have taken a example of a project risk matrix, where as I have already told you we have first write down what could be the possible risks, then what is their importance and what is their likelihood? So, here we have taken this example for an e-commerce application. The possible risk could be like the client rejects the propose look and the propose look and feel of the site, because the GUI part is not very good.

So, he has rejected he has not accepted the client is not accepting that one so; obviously, the importance will be very high, but the likelihood is this dash line means it is unlikely, very unlikely it may occur. Similarly, competitors under cut the prices might also happen, if the it happens it is consequences is very high. So, importance is high and likelihood is medium, warehouse is unable to deal with the increased demand.

So, it is importance is a medium and likelihood is also less. So, online payment has security problem, problem it might happens. So, importance is a medium likelihood is also medium and a maintenance cost may be higher than the estimated, this it is importance it is effect on the business is less and likelihood is so, it is less. So, response time deter ate the purchasers.

So, it is importance is medium and likelihood medium from some study this has been analyzed. So, in this way for any given project in order to classify the risks, in order to evaluate the risks, we have to construct a project risk matrix, matrix by considering the importance of the risks and the possibility or the probability or the likelihood of the risks that they will occur.

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So, next we will see the next method that the risks and the net present value. Here where a project is relatively risky it is a common practice to use a higher discount rate to calculate the net present value. So, the in the last class I have already told you, while calculating the net present value, we are considering different discount rate such as 5 percent, 8 percent, 10 percent, 12 percent or 15 percent things like that. So, when we observe that a project is relatively risky.

Then what we should do? So, it is a common practice that we should use a higher discount rate to calculate or for computing the net present value. For example, this risk premium might, for example, be an additional 2 percent for a reasonably safe projects. So, if a project is a reasonably safe we may use an what additional 2 percent for this discount rate, but if the project is a fairly risky one, then we have to consider additional 5 percent of this discount rate. For example, if we are considering 10 percent normal discount rate. So, if a project is reasonably safe we may consider the discount rate as 12 percent, but if the project is a fairly risky one you should considered, you should take the discount rate as 15 percent.

So, projects may be categorized as high. So, just like we have done in this case of what previous approach, we are considering the matrix. So, similarly while considering the net present value the projects may be categorized as high medium or low risks using a scoring method, and the risk premiums designated for each category.

So, we can also rank them as high risk or medium risk or low risk projects using some scoring method using some formula or given by using some weights and a risks and the risks premiums designated by each category.

For if it is a very high, what the risk project we may considered like a 15 percent. For example, 5 percent extra may be 15 percent what premium, risk premium or discount rate and if it is a medium risk project, we may used might be what we can take 13 percent or so, of this what discount rate, if it is a very low one we can take 11 percent or 12 percent as the discount rate and then we can classify them.

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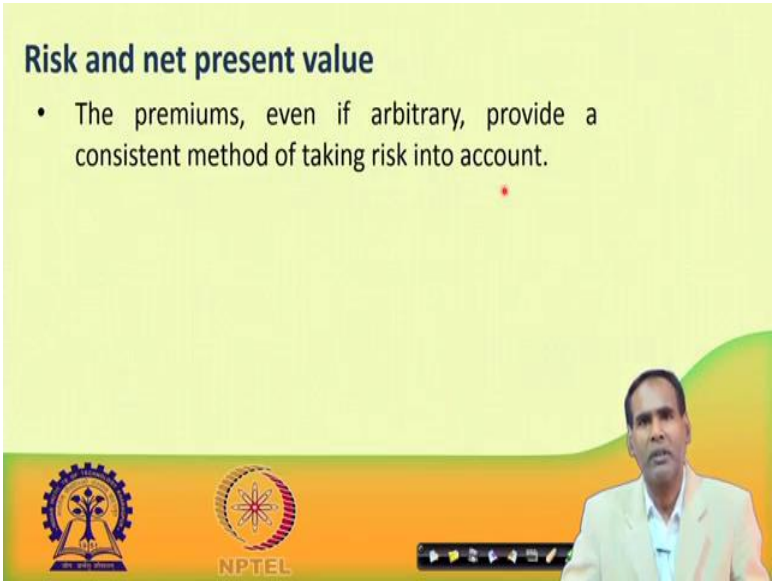

So, the premiums even if you are consider a arbitrary if you have even if you have taken them arbitrarily, but still they provide a consistent method of taking risks into account.

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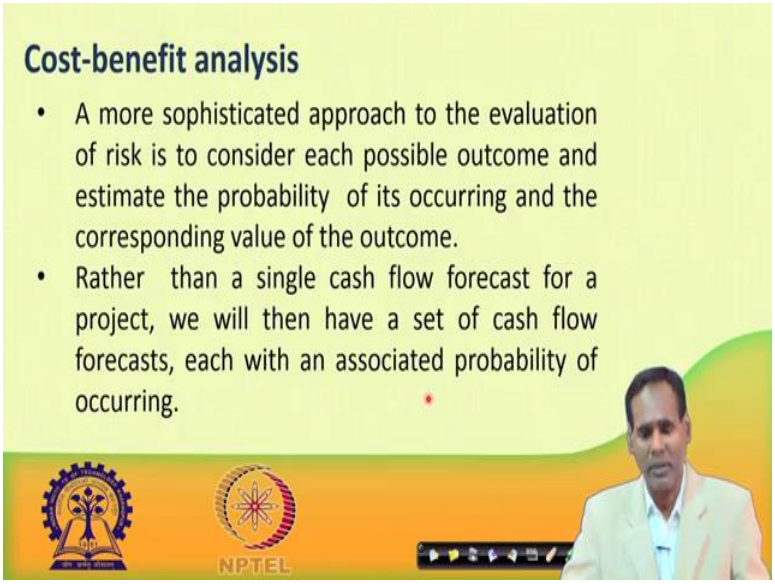

Now, we will see a cost benefit analysis how it can be used for this what evaluating the risks we have already seen cost benefit analysis earlier for evaluating the projects. So, similarly while evaluating the impact of the risks, we can also use cost benefit analysis. This is a more sophisticated approach to the evaluation of risks. So, here we consider each possible outcome on the estimate the probability of it is occurring. So, according to cost benefit analysis we will consider each possible outcome, also we will estimate the probability of it is occurring and the corresponding value of the out outcome, how much benefit we will get the corresponding value?

So, rather than a single cash flow so, rather than taking a single cash flow forecast for a project here will that happens set up cash flow forecast. So, instead of taking a single cash flow forecast, we will may have set of cash flow forecast, where each is associated where each with an associated probability of occurring. So, each forecast ok, each forecast may have associated with a probability of occurring.

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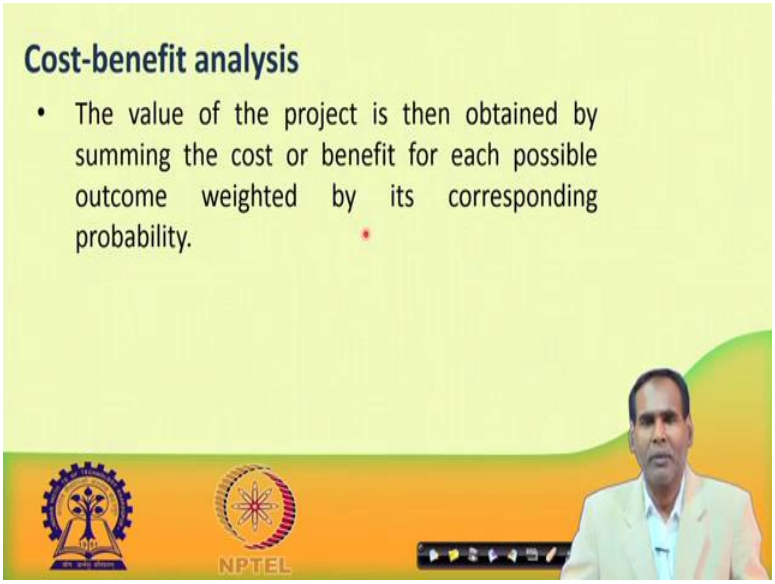

The value of the projects is then obtained by summing the cost or the benefit for each possible outcome weighted by it is corresponding probability. So, here you will take two things that what is the cost or benefit for each possible outcome, then for each one you will give what the probability. So, we have to multiply this cost along with this probability. So, the value of the project is then obtained by summing the cost or benefit for each possible outcome weighted by it is corresponding probability. Let us take an example and then I will say.

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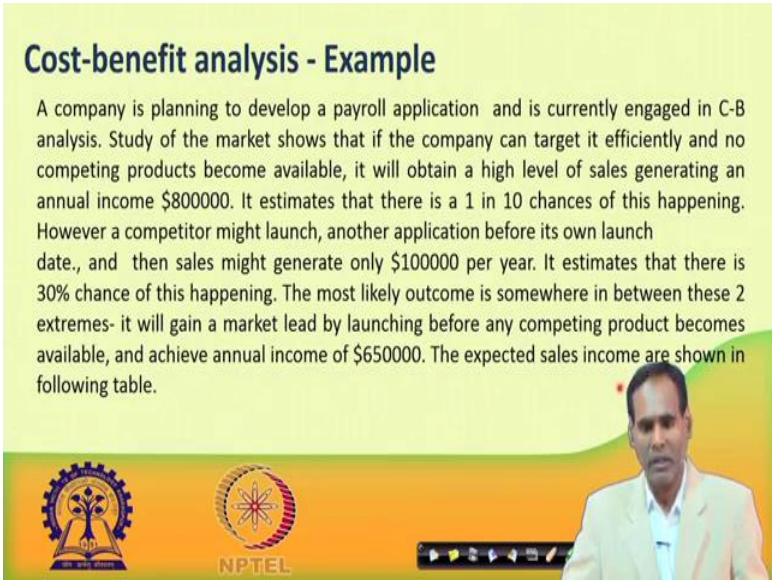

So, suppose there is a company which is planning to develop a payroll application and now currently it is engaged in C-B analysis. So, study of the market shows that if the company can target it efficiently and no competing products become available, then it will obtain a high level of sales generating what on annual income of 800000 dollar.

It estimates that there is a 1 in 10 chances, I mean what 10 percent chances of happening this. However, a competitor might launch, a another competing application like this before it is own launching that and then this sales might generate very a less amount only 100000 dollar for year.

It estimates that there is 30 percent chance of happening this. Then so, this is the extreme case so, these are the two extreme cases the best case is the about 800000's annual income whereas, the worst case is what 100000 dollar per annum. And here the best case chance is 10 percent whereas; the worst case chance is 30 percent. The most likely outcome is somewhere in between these two extreme extremes; that means, between n the 800000's and this 100000.

So, the most likely outcome is somewhere in between these two extremes. So, it will gain; that means, it will gain so, according to the survey, it will gain a market lead by launching before any competing product becomes available and achieve annual income of dollar 650000, which is in between these 800000 and 100000. The expected sales as per the market study, the expected sales income as per the market study are shown in the next table.

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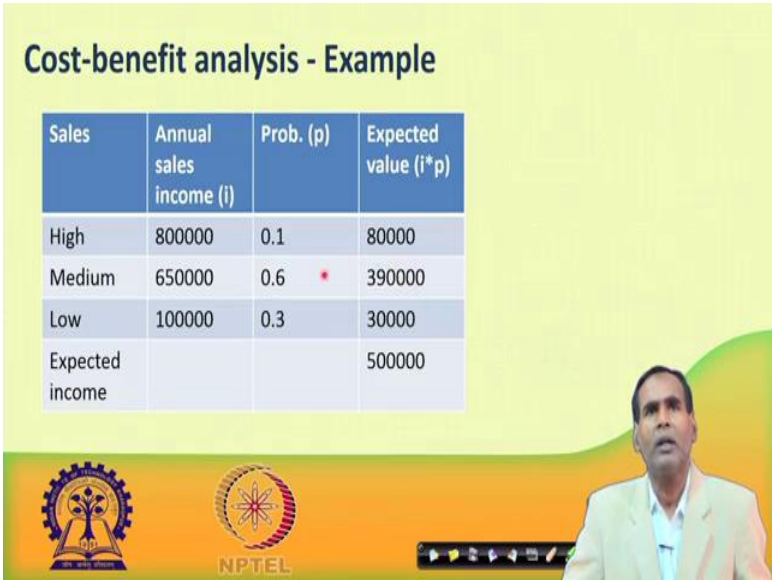

Like this say it is expected that as I have already told you the annual income the best case it will be 800000, where the probability is 10 percent. And if it is another competitor is there then it will be the worst case, where the annual income will be what 100000 and the probability is 30 percent. So, as I have already told you in cost benefit analysis what we will do? The value of the project is then obtained how by summing the cost or benefit for each of the possible in income, which are weighted by it is corresponding probability.

So, you have to see these are the values annual incomes this is the probability we will multiply them. So, these are the values you will get and then the expected income will be around this. So, you can see the expected income for this problem is found out to be how much 500000's now.

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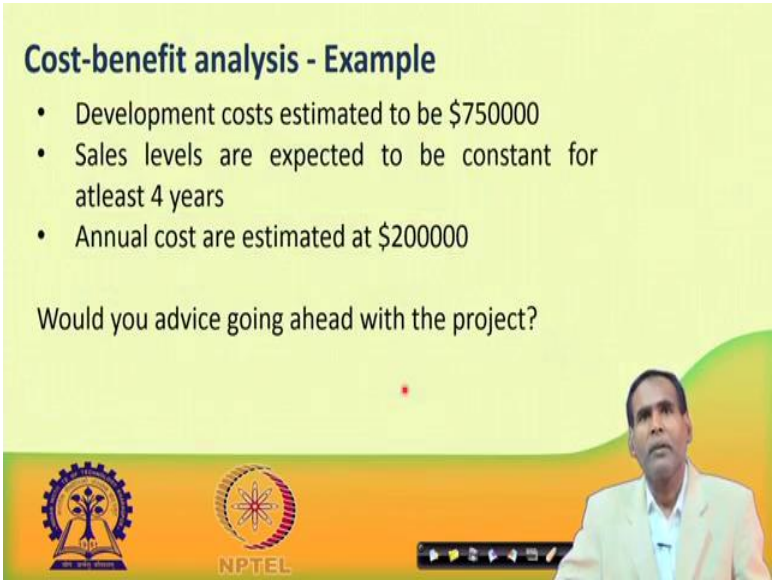

Ah it is also estimated that the development costs are estimated to be 750000, sales levels are expected to the constant for at least 4 years. The annual costs such as maintenance etcetera are estimated at 200000s. So, now, the question is would you advice the project or the company going ahead with the project. Now, let us analyze. So, what will do what will be the advice?

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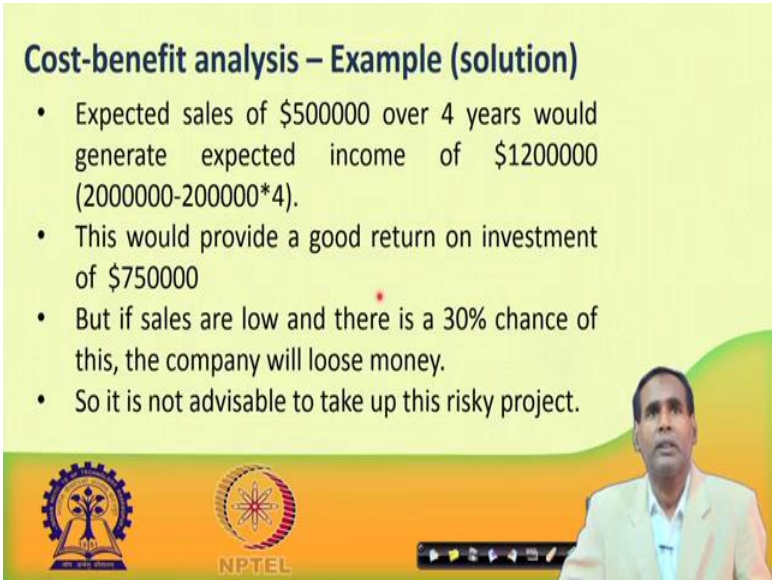

See we have seen that the expected sales is 500000s we have already seen the average the expected income is 500000's. So, and also we know that sales levels are expected to be constant for 4 years and annual costs such as maintenance are 200000's.

So, what we will find out now? The expected sales of what 500000 over 4 years would generate what is the expected income, what will be the expected what benefit, the total expected income will be what the what is the total income? So, 500000's into 4 that is 20,00000's minus, what is the annual maintenance cost? 200000's per year. How for how many years it will be run? So, 4 years.

So, 200000s minus 4 sorry 200000s into 4 that is 800000s, so 20,00000s minus 800000s coming to be how much? 12,00000s. So, the expected sales is coming to be 1200000s. Now, you see what is our, what investment you can see that the development cost is 7,50,000's dollar. So, now, we can see that the benefit that we will get is or the expected sales benefit is coming to be a 1200000s dollar whereas the investment is 7,50,000 dollar.

So, of course, it is a good return. So, the company may go for this project, but please see the other side, you see let us consider the worst case. If the sales are low and another competitor is there then what will happen, that it will lose money. And you know the chance is already 30 percent. It is a significant amount and if this is happen and if this happens then how much you will get? The company will get only 30,000, because see the annual sales income will be what 100000 and since the probability is 30 percent.

So, expected value will be what just what 30,000 very what low value. So, that every possibility that this may occur; because the probability is 30 percent and the company will certainly loose money. So, it is not advisable to take up this risky project. So, in this way cost benefit analysis can be carried out for at evaluating risks.

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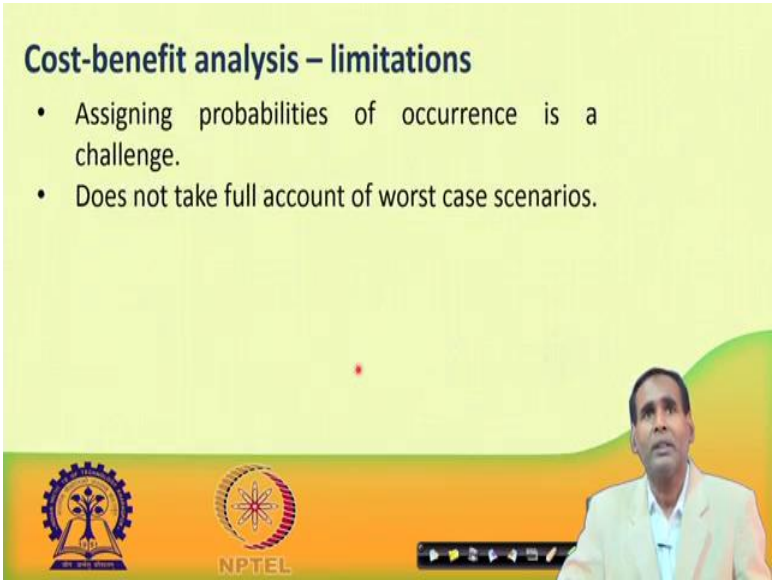

Next, so, the limitations are that, how to assign the probability? So, there were 10 percent, 20 percent, 30 percent. So, assigning probability of occurrence is a challenge, it does not take a full account of the worst case scenario. It only considers the average case scenario, it does not take into account the worst case scenarios.

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So it may take the it may find out it may take the average case scenarios, but it does not handle the worst case scenarios. So, now this risk profile analysis the next method. So, this is an approach which attempts to overcome some of the problems of C-B analysis

method what I have just told by constructing a risk profile. So, here it consider it construct a risk profile using sensitive analysis to evaluate the risks. So, that the some of the problems of the C-B analysis will be overcome, by studying the results of this sensitive analysis, then we can identify the factors that are most important to the success of the project.

Then with a need to decide whether we can exercise better control of over them or otherwise mitigate their effects. So, with this study if we will know that which factors are most important to success of the project, then we need to decide whether we can exercise better control over these factors or otherwise we will mitigate their effects. If, neither is the case if it both the things are not possible, then we must live with the risks or we have to abandon the project, this is how risk profile analysis method works.

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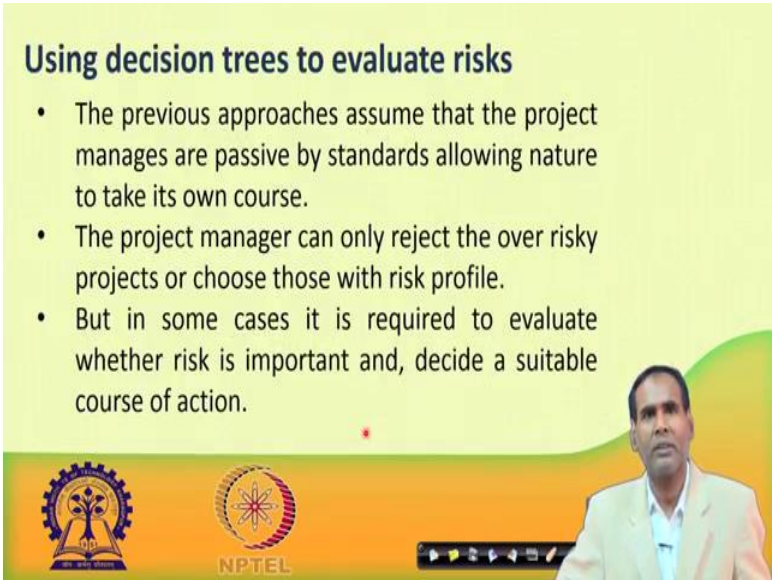

Next quickly see this one, this is using decision trees evaluate the risks. So, I hope decisions trees you must have studied in B-tech career. So, this decision trees can also be used to evaluate the risks. The previous approaches assumed that the project managers are passive by standers, they are by stand they are bystanders the standard this spelling mistake, they are passive bystanders allowing nature to take it is own quotes; that means, only the project manager can only reject the over risky projects or choose those with risk profile.

So, only those which are less risk they may take which are more risky the project manager just simply what rejects those projects. So, but in some cases, it may be required to evaluate whether risk is important. And you decide a suitable course of action always we cannot just what leave projects which are risky, we must have to handle them. So, how to handle them, how to deal with them, that sometimes required we have to take some appropriate action depending upon the importance of the risks.

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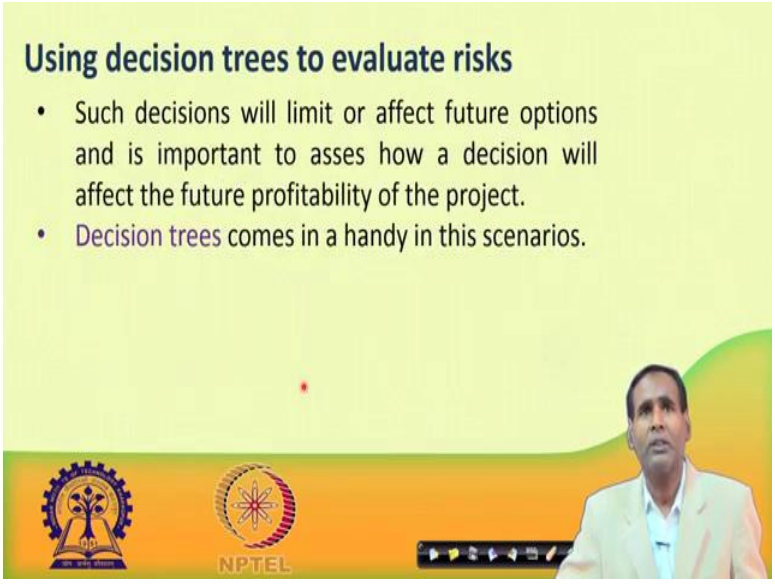

So, such decisions will limit or affect the future option. So, this decisions will affect future options and it is important to assess how a decisions will affect the future profitable project, if risk is there currently at in a project, how it will affect in future. So, decision trees comes in handy for the dealing with these scenarios let us quickly see so, how decision trees can be constructed.

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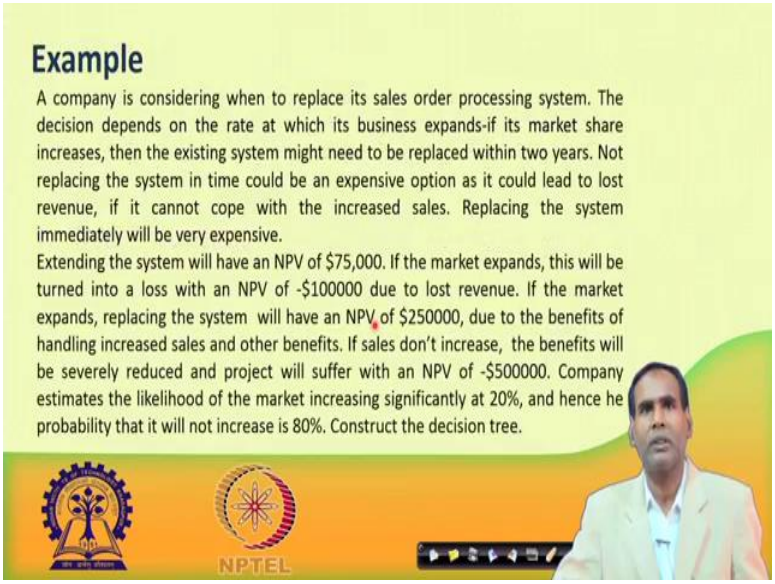

So, again I have taken a simple example. A company is considering when to replace it is order sales order processing system. So, a company currently it is suppose it is running at manually. Now, it wants to replace it may be through one automated system.

So, the decision depends on the rate at which it is business it expands. If it is market share increases, then the existing system might need to be replaced within two years. But no if the company will not replace the existing system in time, that could be this could be an expensive option as, it could lead to lost revenue, if it cannot cope with the increase sales. Replacing the system immediately will be very expensive.

So, now this is one about replacing, but let us see about the extending case extend the system will have an NPV of dollar 75,000. If, the market expands, this will be turned into a loss with an NPV of 100000. So, I think it is a 100000. So, here it is negative sign, because this is loss due to the lost revenue.

If the market expands replacing the system will have one NPV of 250000 dollar due to the benefits of handling increase sales and other benefits such as what enhanced MIS system etcetera. If sales do not increase then benefits will be severely reduced and the project may have to suffer with an NPV of again dollar what some 500000's or so, com sorry 50,000.

So, company estimates the likelihood of the market increasing the significantly at 20 percent. So, it is estimated that that the likely of the market increasing market expands it is 20 percent, the probably 20 percent. And hence the probability that the market will not increase market will not expand is 80 percent. So, now, we have to construct the decision tree and take the appropriate action.

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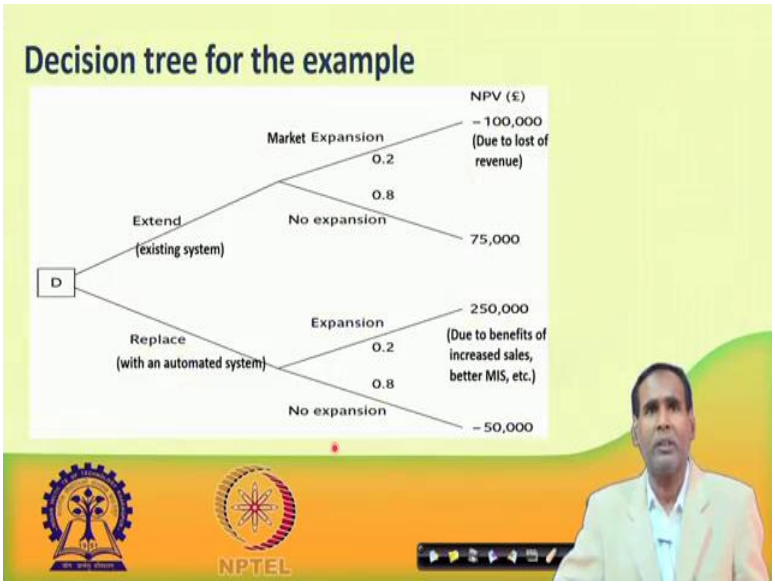

Can say that what problem I have described here, easily you can construct a decision tree like this. And these are the values, these are the probability that I already told you, the probability that much market will expanded is 20 percent. So, of the probably of it will not be expanded as 80 percent. So, this is on what you can say extent, because it is the existing system the existing system can be extended or the existing system can be replaced with an automated system, that the company is thinking.

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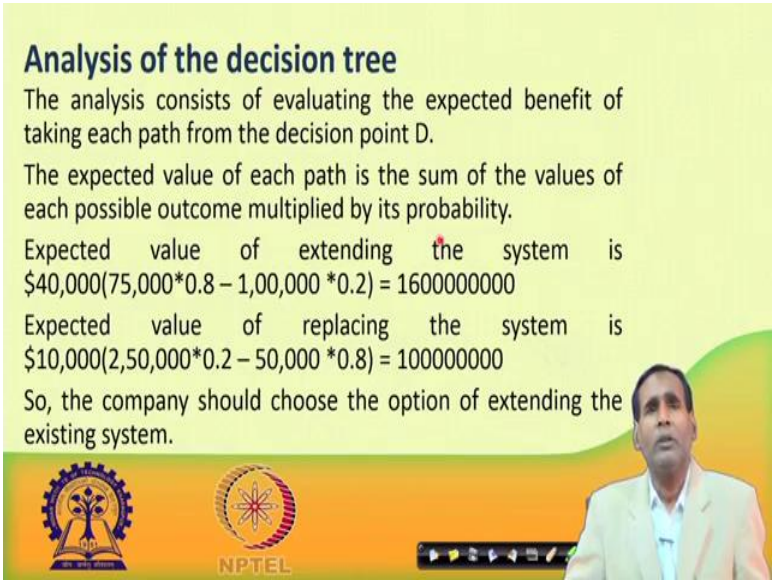

So, now we have to analyze the decision tree you can see that, the analysis consist of evaluating the expected benefit of taking each path from the decision point D. So, this is the decision point D from decision point D, we have to evaluate we have to asses each path. So, this is one path, this is another one path, let us evaluate or asses each path, you can see that the expected value of each path how you can find out.

So, the expected value of the each path can be finding out can be found out by taking the sum of the values of each possible outcomes multiplied by it is probability. So, let us see first case the extending case ex extending value of the expected. So, now, the expected value of extending the system which found out by what. So, extended case is this one, you can see that the NPV, if market expansion happens. Then what is minus 100000 and probability is 20 percent, if market does not expand so, 80 percent probability and 75,000.

So, we can find out the expected value. So, this will how much 75,000 into probability 0.8 minus what 100000 into 0.2, this value is coming to be how much 40,000? So, this path will take up what 40,000? So, 40,000 into what 40,000 coming to be this much value. Similarly, expected value of replacing the system is how much this you can see replacing it is this case.

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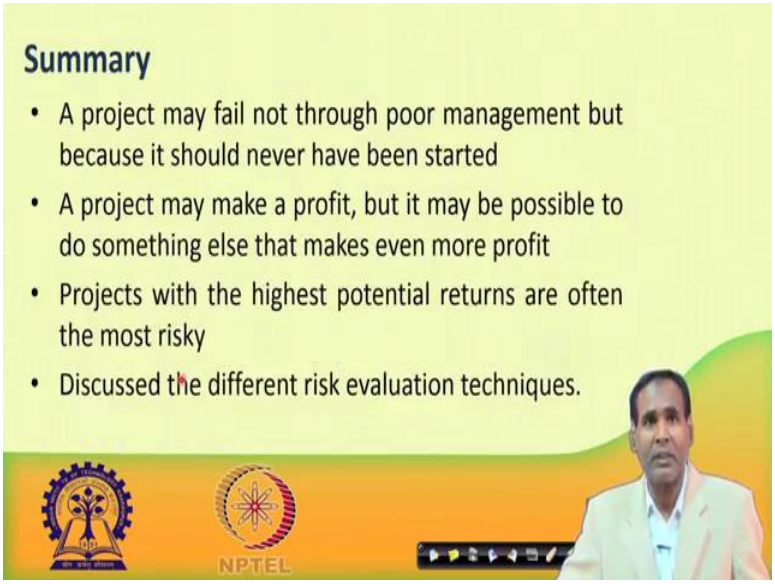

Replacing you can see the value is 2,50,000 if market expands. So, probability is 20 percent if market does not expand, there will loss that will be 50,000 and probability is 80 percent. So, we can evaluate it. So, it will be 2, 50,000 into 0.2 minus 50,000 is 0.8 coming to be 10,000. So, this path now will become the value for this path will be 10,000. So, 10,000 into this value is 10,000. So, 10,000 into 10,000 coming to be this value, if will compare both the values so; obviously, profit will be maximum in this case; that means, it extending system not replacing the system.

So, the company should choose the option of extending the existing system rather than replacing the system, because the benefit will be more in case of extending system rather than replacing the system ok. So, we have seen that different what first we have seen the two phases of C-B analysis, then we have seen what is a risk different types of risks. And how the risks how the risks can be evaluated, we have discussed different risk evaluation techniques? So, this is about the different types of risks evaluation techniques.

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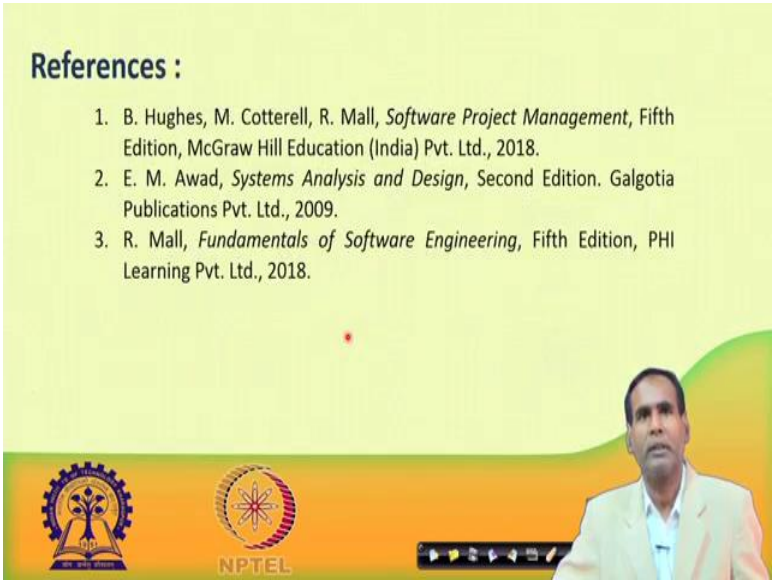

These are the references we have taken and.

Thank you very much.

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Lecture – 15
Project Evaluation and Programme Management (Contd.)

Good afternoon to all of you. Now, let us start the remaining part of the Project Evaluation and Programme Management. We have already seen about this evaluation of projects. Now, let us see about this programme management part.

So, in this lecture we will discuss first the various concepts of programme management, then what is the benefit, how benefits can be managed.

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So, let us first before going to programme management. Now, let us see what do we mean by a program? Project, we have already seen. So, programme means there are many definitions are there one definition is given by D.C. Ferns. So, according to this definition programme is a group of projects, it is a collection of projects, that are managed in the coordinated way to gain some benefits that would not be possible were the projects to be managed independently.

So, if the projects will be managed by independent persons; they will be managed by independently then, the benefits that will get it will be almost very negligible.

In other words, if a group of projects they can be managed in a coordinated way in a collaborative way, then the possible benefits, that we will get which must be on high which may not be obtained, if the projects will be managed independently. So, that is, what is the difference between projects and programs. Let us recall again a programme is a group of a projects, which are managed in a coordinated way, in a collaborative way to gain benefits that would not be possible, if the projects would have been managed independently.

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Now, let us see what are the different types of programs available. These are the different types of programs one strategic programs. So, strategic programs are the projects which a group of projects a collection of projects, which implement a single strategy only those group of projects will follow they will implementing a strategy that is why they are known as a strategic programs. Business cycle programs these are the groups of the projects, that an organisation undertakes within a business planning cycle.

So, within a particular business planning cycle what category of the projects, what groups of projects that on under taken an organisation undertakes, that is known as business cycle programs.

Then, infrastructure programs so, these are the groups of projects, which are performing the activities of identifying some common infrastructure and its implementation and its maintenance. So, that is why since here the group of projects, they perform some

activities of identifying some common infrastructure. And, its implementation and maintenance that is why, this programme is known as infrastructure programme.

So, then R and D development or research and development programs more popular known as R and D programs these are very easy to understand. So, these are the groups of projects, which are involved in developing some new products, some innovative product, based on some current research work. So, this is known as research and development programs or R and D programs then innovative partnerships.

So, now a day's many companies they do some collaborative projects. They do combinedly some they tackle some projects combinedly collaboratively. So, this kinds of projects are the groups of similar types of projects are based on collaboration by different organisations is known as innovative partnership. Here the projects are based on some collaborative work performed by different organisations.

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Now, let us see we have already seen what is a project, we have already seen what is a program. Now, let us see what is the difference between a programme manager and project manager, why the resources for the what for the resource allocation or the resource allocation to programs, it will depend upon who are the end users. They could be that could be system analyst, database administrators, programmers, coders, etcetera. So, how the resources can be allocated to different projects or to different programs.

Before this so, for allocating knowing this resource allocation we must know, what are the differences between the programme managers and the project managers. So, mainly programme manager he is responsible for handling many simultaneous projects whereas, project manager he performs only one project at a time. Similarly, the programme manager he has personal relationships with all the skill resources working under him, but a project manager he does not have personal relation.

So, he has him personal relationship with the resources. The objective of programme manager is to optimal use of the resources. How optimally you can utilize the resources, but the project manager his objective is how to what give minimise demand for the resources? How he can put minimization of the demands, how he can achieve minimization of the demands for the resources.

Then the project manager; in case of project manager the projects tend to be seen as similar almost the projects are similar, but the project manager he use the projects or the projects which are under the project manager. They tend to be seen as unique they are not similar, they are the projects which are under a project manager, they tend to be seen as unique. These are the differences between programme managers and the project managers. So, accordingly the resources can be allocated to different projects.

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Now, let us see about this strategic programs. So, we have to see the strategic programme management; so, actually we should know the before knowing this strategic

programme managements, we should know the strategic programme. Strategic programs, I have already told you that these are the projects which implement a single strategy only.

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So, now we can see about this strategic programme management. It is a difference form of a programme management; actually title should be strategic programme management. So, strategic programme meant programme management is a different form of programme management, where portfolio of the projects. They all contribute to a common objective, they all work for a single strategy, they all contribute to a common objective.

So, this is another form of a programme management where the portfolio of the projects they all contribute to a common objective. It is based on what this OGC approach? Where OGC stands for the Office of Government Commerce previously it was shown as CCT.

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So, the initial planning document that we have to use is a Programme Mandate. So, now let us see how to create a programme? What kind of documents are required to create a programme? The first document that, we required to create a programme is programme mandate. So, this is the initial planning document is known as the programme mandate. The programme mandate describes what the following things.

It describes the new services or the capabilities that the programme should deliver. So, what services or what capabilities, which are particularly new that the programme should deliver that should be mentioned in this programme mandate this document. Then, how an organisation will be improved? The suggestions, how the organisation can be improved further.

So, all those suggestions also should be put in the programme mandate, then fit with exiting organisational goals. So, whatever the services or the capabilities will do so, will perform. So, or it will deliver they must be fitting to the organisational goals, what are the organisational goals? So, these services capabilities as well as the suggestions for improving the organization they should be fit with the existing organisational goals or the objectives. A programme director appointed a champion for the scheme. So, normally live a programme director appointed a champion for this type of schemes.

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So, as I have already told you the first document, the first document for creating a programme is a programme mandate. Now, we must to see that the other documents, that must be used for while creating a program. So, we must use these kinds of documents for creating a programme. Now, first one the programme brief, as if name suggest, the programme details must be explained here briefly that is known as the programme brief.

He it is just equivalent to a feasible study; feasible study we have already seen earlier that and the three types of feasibility we have studied, we have known technical feasibility, financial feasibility and operational feasibility. The most important one is the financial feasibility or the cost benefit analysis.

So, this programme brief it is just equivalent to a feasibility study. Here emphasis is given on the financial assessment or the cost benefit analysis, then there must be a statement there is a document containing the vision statement. So, every organisation what should have a vision statement, this document explains the new capability that the organization will have.

So, what kind of new capability the organization will have this document explains this thing then there must be a blue print. So, this document explains the changes to be made to obtain the new capability. So, in order to so what new capability that the organisation should have that is contained in the vision statement whereas, the changes they need to be followed. So, that should be in order to obtain the new capability, that is present in the

blueprint; just a master print, blueprint it has to prepared, since obviously in this meaning the it has the organizations they every organization they have a blue print. This documents it explains what changes they have to be made in order to obtain the new capability.

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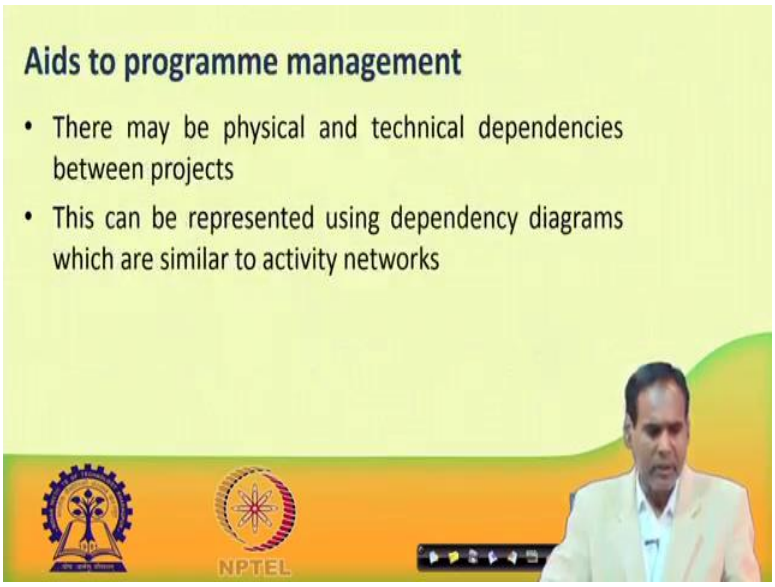

Then, let us say, what are the aids to programme management, what documents or what diagrams or what tools they can be used to aid programme management. See there may be a physical and technical dependencies between projects. Many in many cases; in some cases the projects are independent, but in some cases the projects are dependent. That means, before completing one project you cannot go to another project.

For example, suppose let us see that two organisations, what two organizations computer sections they have to be merged. Suppose this is a project this two organisations computer sections or computer infrastructures they have to merged. So, this project is there. So, a previously the two organizations they are running in two different what say building.

So, since they will be merged, then this can be treated as a project. So, after merging suppose they will be shifted to another building. Then, you can see that this project that merging of the computer sections of two organizations is completely dependent on the new building. The completion of the new building because, unless the new building is

completely developed is completely furnished then, the computer sections of both the organisations cannot be shifted.

So, here that merging of two computer sections of the two organisations, this is a project and finishing of the new building is another project. So, here merging of the two computer sections of the two organizations is completely dependent on the second project that is this finishing or completing the new building.

So, how you can represent these dependencies may be physical or technical, how you can represent the dependencies between projects. So, several diagrams are there to represent this; so, one particular diagram there called as the dependency diagrams. So, this dependency diagram can be used to represent the physical and the technical dependencies between the different projects.

So, this dependence diagram just like activity networks those have are known PERT or CPM .So, they are you must have used activity networks so, this dependency diagrams just they are very much similar to the activity networks.

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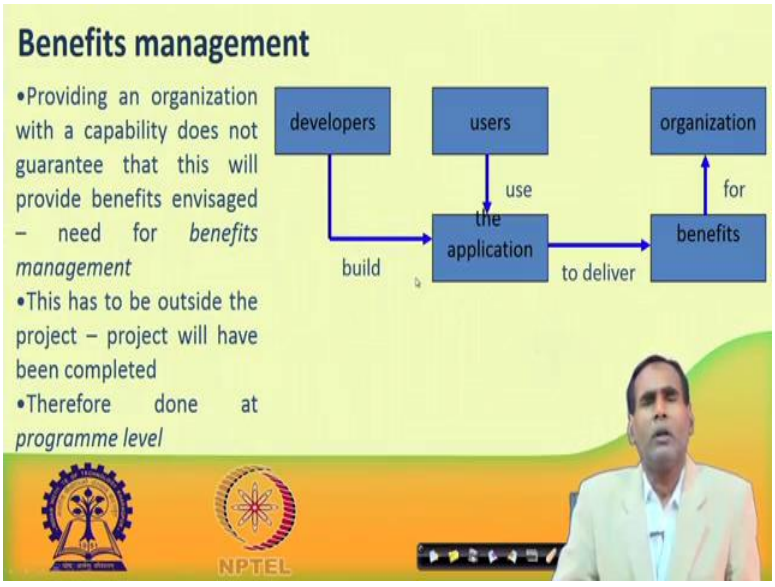

Now, before going to see about the benefits management let us see, what is a benefit? So, after every organization they do some investment with an hope that after some years they will get back their money spent in investment as well as some profit they will get. So, that we call as the benefit. Benefits are of different types we have already seen earlier

they can be tangible benefits or intangible benefits, they can be direct or indirect, they can be fixed or they can be variable.

So, now let us see these are the different types of benefits, you know that some of the benefits, they can be easily identified, they can be easily measured, they can be easily quantified. Whereas, some of the benefits they cannot be identified, some of the benefit they cannot be measured, some of the benefits they cannot be quantified. So, we have to if we know the different types of a benefits and which of them can be identified, which of them can be quantified, which of them can be measured, then that will help in managing the benefits.

So, this benefit management it provides an organization with a capability that does not guarantee that this will provide benefits in besides need for benefits management. Simply say, this provide that benefits management, it provides an organization with a capability that does not guarantee, that this will provide benefits in besides. It needs for benefits management you just get benefit does not mean that what you are always getting the benefits, you need to properly manage them; that they need for benefits management, this has to be outside the project; project will have been completed. So, project will be completed after a few days or few hours few years.

So, this benefits management is normally what handled outside the project. Therefore, it is done at the programme level, we want to have the benefit for the organisation. You know that some of the projects may give benefit some of the benefits may lead to loss, but at a programme level our objective should be to get more benefits, so that is why the benefit management it has to be conducted, it has to handle outside the project may be at the programme level.

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So, what these benefits management or in order to carry out this benefit management, what we should do, what we should must try for? So, we have to define the expected benefits, in order to just like as we have seen the programme management. In order to carry out this benefits management, we have to first define the expected benefits; we have to identify the expected benefits, we have to define them.

But, as I have already told you some of the benefits they can be easily defined, but some of the benefits you cannot define them, then we have to analyse the balance between cost and benefits. So, we have to analyse the balance between the cost and benefits how much investment we have made those are the cost and how much benefits we are getting. Here you have to consider all sorts of benefits including tangible-intangible fixed and variable and direct and benefit what indirect. So, we have to take into consideration all the benefits.

Because, some of the managers they normally they do not take into account intangible benefits, because they are unable to measure them, they are unable to quantify the intangible benefit, they are not taking to consideration in the cost benefit analysis. But, it is required that all the possible types of cost including tangible and intangible direct and indirect and fixed and variable, they should be considered. Similarly all the possible benefits such as direct and indirect benefits tangible and intangible benefits fixed and

what variable benefits, they must be taken to account during the cost benefit analysis, then you have to analyse the balance between these cost and the benefits.

Then the next step is plan how benefits will be achieved? This time the organisation this year this much benefit suppose they have got, then they must be plan how to maximize the benefit the next year. So, they have to plan, they have to develop some specific plans, how benefits can be what maximized, how benefits will be; more benefits will be achieved in the future time. Then allocate responsibilities for their achievement. So, different personals involved in the project, different personals involved in the programme, they will be assigned responsibilities for their achievement. So, this should be in the plan.

So, we have to allocate a responsibilities for their achievement similarly, monitor the achievement of the benefits. So, whether we have made the plan, how to achieve the what benefits, then just making the plan is not sufficient enough. Then, we have to monitor them; regularly we have to monitor the achievement of benefits.

We have to see what is the target that we have to put this year and periodically we have to say in periodic intervals what is the achievement whether we are closer to the what target or we are lagging behind the target. So, if you are lagging much behind the target, what precautions, what remedial actions we should take so, that we should achieve our target benefit. So, we have to monitor and control the achievement of the benefits .

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So, how can do this as I have already told you we have to monitor achievement of benefits. How we can monitor the achievements of the benefits? So, for example, as I have already told you we are not able to what get the benefits, because our the service were providing or the product were providing we are not able to provide in the designated time.

So, we have to see why we are lagging, why there is a delay, so we can take precautionary measures so that, this delay can be avoided. If the delay can be avoided the service we can provide more services, we can sell more number of products. So, that will bring more profit. So, that is why we have to regularly what monitor the achievement of the benefits. If something we are lagging behind we must take remedial actions. So, that we can be closer to the targeted benefits.

So, now, let us see what could be the possible benefits for an what organization or for a what projects? So, these benefits, these might include mandatory requirement, what kind of requirements are must so, that you may get a benefit. So, mandatory requirement improved quality of service. So, this is called as QoS; so, the quality of service must be improvised. So, that automatically benefits will come in the example I have already showed you, I have already told you if there is a much delay in providing the services, if you are much delay in what producing the products then; obviously, the customer the end user will be dissatisfied.

And, we know what dissatisfaction of the customers is not a what benefit rather, the satisfactory customers even if it is a what intangible benefit, what that will bring, what reputation to the organisation we can get more profit. So, we have to improve the quality of service. Then increase productivity, how in less time we can produce more number of products, that is called as what this increased product productivity we may see what are different.

We may use different approaches may be we will use some automated tools or some other things so, that with less time how you can increase the productivity, how can increase more number of products in a given period of time, then more motivated workforce how we can motivated the people to work better. So, more motivated workforce we will also bring in profit ok to more motivated work force will also bring in profit to this organization. And similarly internal management benefit so among the

organization, how can create awareness, how can and to better management. So that, we can get more benefits.

So, for example, what like if you will create a awareness among the what employees say by giving some awards, awards etcetera and providing some scheme to them etcetera. So, we can internally manage what the benefits the employees they can be, what motivated to better and different kinds of benefits, those are internal to the management, they should be handled in a proper way.

See for example, the we are now suppose, what doing one thing to increase the number of sales. And, now if we want to the increase the number of sales, suppose that is happening the sales are in creating. And, now in order to get more number of sales if you will put some what employee to do overtime job.

So, time will come that the benefit due to getting more sales might it will be what is less that then the amount that will pay to the what the employees in overtime. So, that will not be considered as a benefit, rather that will be dis benefit. So, we should try to avoid this. So, this kind of things we must take care of in order to what improvise our benefits.

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So, similarly another what benefit may come due to risk reduction. We have already seen the different types of risks such as what business risks and the project risks. So, our

objective is to reduce the number of risks. Of course, where there is more risks, there is more benefit, more profit, but handling the risks will be much more important.

So, we will try to reduce the number of risks how the number of risks can be reduced so that, ultimately the benefits may improve. Similarly, we should see about the economics, what is the economical status of the organization, how it can be improved. So, if the economical status is improved and; obviously, benefit will come to the organisation, then revenue enhancement and accelerations. What steps you should, because see revenue enhancement is also treated as a benefit, what steps we must take to enhance to accelerate the revenue.

So, those steps if will follow strictly then automatically the revenue will be enhanced the acceleration and similarly strategic field fit. So, you must particularly improve the strategies in order to maximize the profit, in order to maximize the benefit.

So, these are the strategic needs or the organisation level, we must see then strategic level at the programme level, then strategic level; what strategic techniques or the strategic actions are the project level, they have to be defined clearly. So, that the benefits will be maximized for the a project or for the programme or for the organisation.

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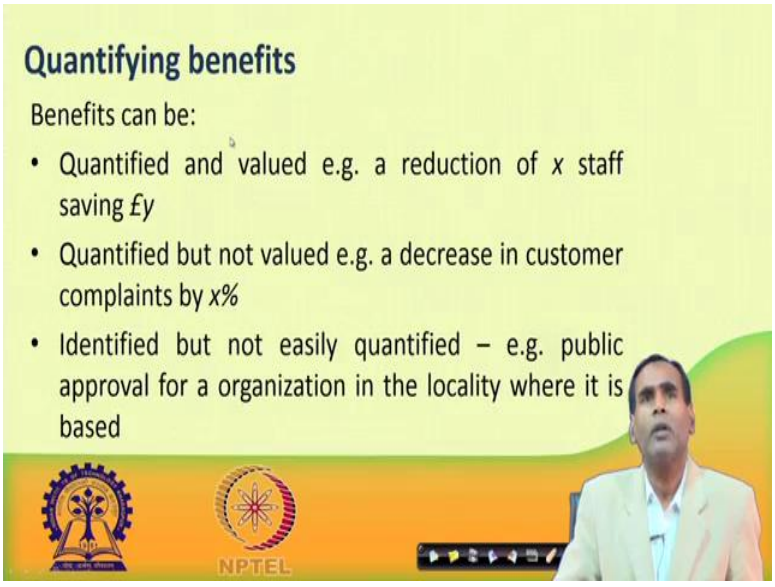

So, next we will see about to quantifying the benefits, how we can quantify the benefits? So, we have known that there are some benefits which can be quantified and valued. For

example, there is a reduction of say x staff saving dollar y. Say, we have due to automation of the existing system previously it was manual due to automating it, we have reduced 10 staff members so, that we have saved say 200 dollars.

So, it can be easily quantified and valued. Some benefits they can be quantified, but they cannot be valued. For example, decrease in customer complaints, we have previously it was manual and now the system we have made it automated. So, the customer complaints that we are what getting now, there is what it is successfully reduced.

So, we can easily quantified it, but the value that we will get as the gain or the benefit that it cannot be valued. Similarly, some benefits they can be identified, but not they can be easily quantified. For example, public approval for an organisation in the locality where it is best suppose that and you take these mobile industries.

So, they want to put a tower in your area for that they have to take the approval of the local authority. So, if you will get; if you can they can put a tower then; obviously, they are what mobiles will be or there yes, their mobiles sims, they will be sold more and; obviously, they will get more benefit.

So, this kind of benefits like a public approval for a mobile industry to set up a may be tower in the locality.

If they get the approval then they can set up a tower and they can sell more number of their mobile sims and hence their benefit will be more. So, this so, this types of the benefits, they can be identified, but they cannot be easily a quantified.

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So, in this class we have discussed about what is a programme and about the some what details of different types of programme management, we have also seen what is a benefit, what are the different types of benefits and how benefits can be managed.

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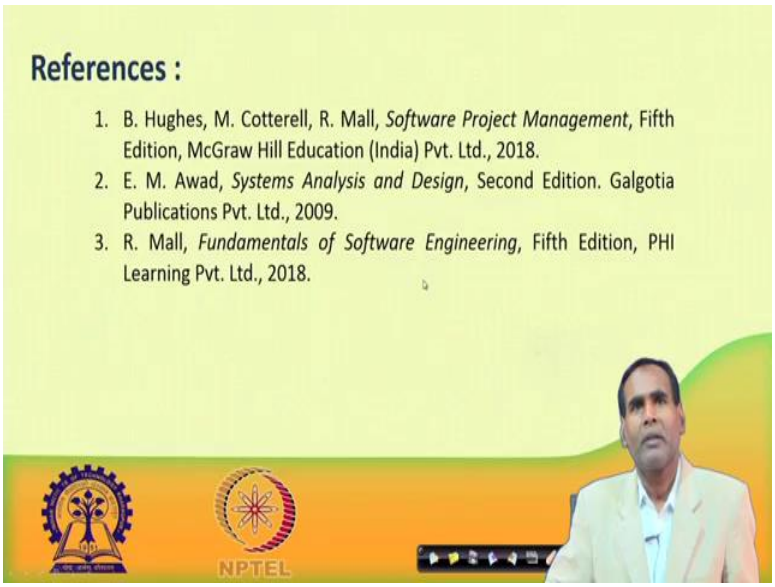

So, these are the references and it is so, most of the things we have taken the reference number 1, only the different types of cost and benefits, we can see at reference number 2.

Thank you very much.

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Lecture – 16
Project Estimation Techniques

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So, good afternoon, today we will discuss about little bit of project planning and basics of Project Estimation.

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See first let us see what makes a successful project? So, if there are we can deliver the project in proper time with agreed functionality, that has been what discussed with the end user. And, then at the agreed cost, at the mutual cost that is decided with the user and then with the required quality. What is the quality the user wants, that much quality that we have provided then the user will be satisfied, then we say that this project is successful.

For this we require two stages, that is the we have to set the targets, then we have to attempt to achieve the targets. But, sometimes what happens if the targets are not achievable, the project manager tries to achieve all those targets, but it is not possible to achieve, they are not achievable. Then what he should do? Then actually the projects normally they get failed due to lack of proper planning and proper estimation. So, he has to do proper planning and estimation so that, the project might get success.

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So, let see little so, that is why let see as I have already told you he has to the project manager has to plan properly regarding the activities of the project and do proper estimation in order to get the process project success. Let us little bit say about the introduction to project planning.

We have already seen that before a project starts we have to carry out the project feasibly study activities. Then if a project is found to be feasible, if a proposal is found to be feasible in every respect, like technical it is technical feasible, financially feasible and

operational feasible. Then the project managers will take it for what further steps. So, a project managers activities are varied and they can be broadly classified into two categories. One the project planning and then he has the project monitoring and control. So, the activities project managers activities can be divided into these two categories.

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As I have already told you, that once a project is found to be feasible the project manager then he has to undertake the project planning. So, first he has to make an initial plan, before the development starts. And then this plan is updated very frequently.

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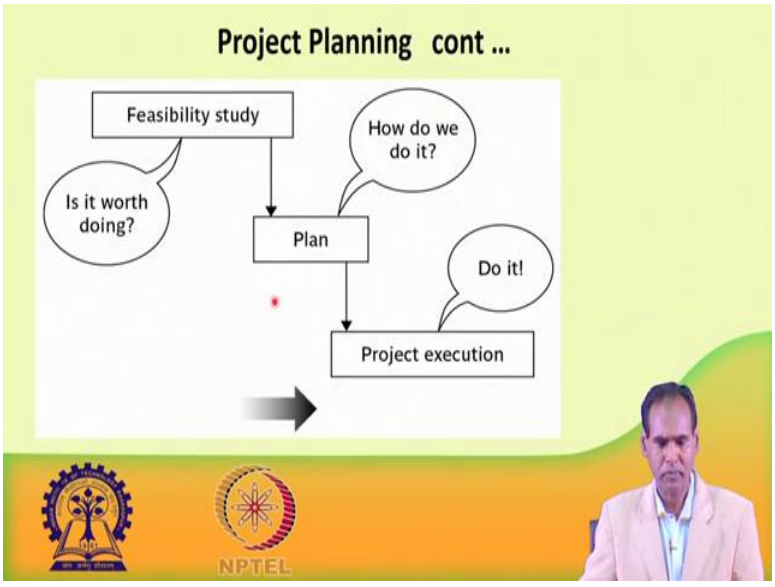

You can say that first the project study is conducted in order to know that whether the project is worth doing or not. Then the project manager has to develop the plan. Here he has to see that, how do he do it? How this project these can he can they can follow it? So, they have to make a plan and finally, project execution step here they have to actually execute whatever plan they have made here, they have to execute the project execution.

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Now, let us see briefly what are the various project planning activities; so, number 1 estimation. So, during estimation phase what needs to be estimated? effort, cost resource and project duration. Then the project manager has to develop a proper scheduling what he has to make a proper schedule, he has to follow some scheduling methodologies.

Then also a staff organization has to be done properly, about how many staffs will be required for this project that has to be planned. Then as we know in a every project contain some risk, some different or some specific methodologies have to be follows for handling the risk, like identifying the risk, analyzing the risk and abatement of the risk. Then there may the project managers may have to follow some miscellaneous plans, that is the quality assurance plan, configuration management plan etcetera.

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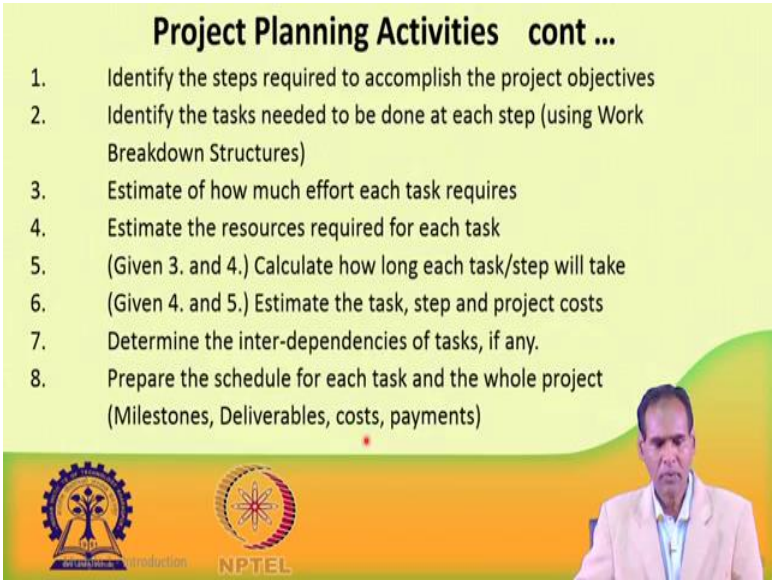

So, let us see in detail, what the how the activity it is a project planning activity they will be followed. First the manager has to identify the steps required to accomplish the set project objectives or the targets. Then they he has to identify the a task which are needed to be performed at each of the step, he may use for this is the work break down structure etcetera. Then he has to estimate how much he put each task requires, also he has to estimate how much resources are required for performing each task.

And if 3 and 4 are known given item number 3 and 4, then the project manager he has to calculate, how long each task or step will take; that means, he has to calculate the project duration. Similarly he has to estimate the task step and project cost, how much cost will be spent for developing this task or for individual step or for the whole project. He has to also determine the interdependencies among the task if any and finally, he has to prepare a schedule for each task and the whole project. So, here he may use milestones deliverables cost and payments etcetera.

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Now, let us see why you should go for project planning, because project planning requires utmost care and attention. Because what commitments to unrealistic time and resource estimates result in the followings. If the project manager committed some unrealistic times, which he cannot achieve at all and some resource estimates which is not feasible at all then the following problems might arise.

What could the following problems? It may unnecessarily make delay the delivery of the projects. So, that the customer will be irritated and customer will be dissatisfied. Hence there will be adverse effect on the team morale and due to poor quality work there will be adverse effect on team morale and it may be even if it may be the case due to improper estimation etcetera. So, the project might get failure. So, that is why it is very much important we should give at most care and attention to project planning.

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So, sliding window planning is a special type of planning which involves project planning over several stages. So, it does not make the plan in one step, at a single step rather, it prepares the project planning over several stages. Here the normally it protects managers from making big commitment too early. Because what to do normally managers do are the very early requirements phase they make a commitment that will develop or we will deliver the project by this state, which they may not actually achieve it.

So, that is why in sliding window protocol or sliding window planning the managers they will be prevented from making big commitments too early. So, this principle just like a java, sliding, window, protocol that you have studied in networking yes, networking. So, then more information becomes available as project progresses. Say initially only less information available, what are the progresses. So, the more and more on the updated information they are coming.

So, this updated information which are received later on so, they will facilitate for accurate planning. So, that is why instead of making in one window or in one what, a one phase the managers they should perform the planning in several stages or in just I using this some concept of window in the what in the phase up several stages.

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So, finally, as you know that every stage of this classical water fall model it some output should be there. So, similarly what is the output of this software project management? The output of software project management is this SPMP document which is known as software project management plan document.

So, after planning is complete, the project manager he has to document the plans where in a document called as software project management plan document. What will be the organization? What will be the contents of the organization?

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The following content should be there in the SPMP document. First there should be little bit of introduction of the project where the objectives and the major functions etcetera should be described. Then the various project estimates they have to mention like the historical data, the estimation techniques used the estimation for cost effort and project duration etcetera.

They have to be placed in this section, project estimate. Then project resources plan it will contain the, what are the resources they will be required by the project might be the people, hardware, software and any special resources. If any schedule in this content the following so, what item should be there like the you may use work break down structure, the; what task network, Gantt chart and PERT chart etcetera.

Then another what content might be risk management plan. How you will handle the risk for that you have to put what kind of different types of risk? What risk analysis you have performed? How to identify the risk? How to what estimate the impact of the risk? How to abandon or mitigate the risk? So, those things has to be explained in the risk management plan.

Similarly, project tracking and control plan; how you are tracking the project; how you are controlling and monitoring the project the plan regarding that should be discussed here. And finally, one miscellaneous section should be there in the SPMP document which will contain like the process tailoring, quality assurance aspects, configuration management aspects they will be described in the SPMP document, under the heading of miscellaneous plans.

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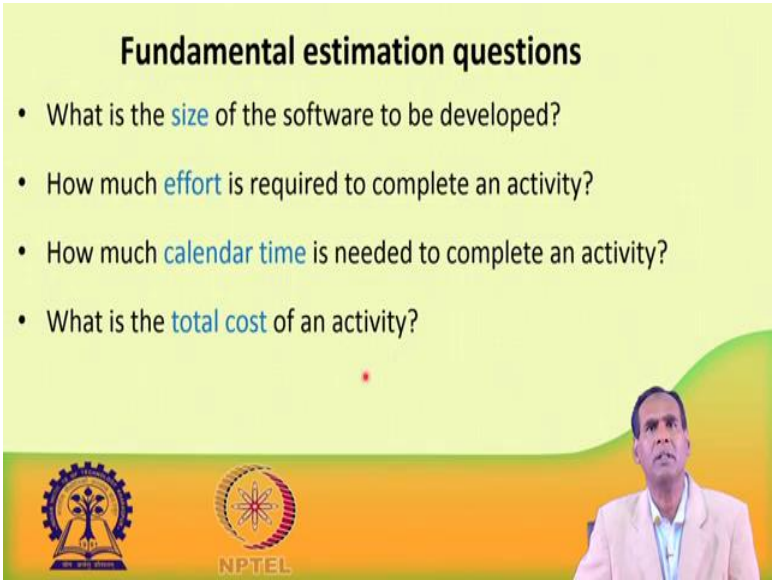

So, now let us see what are the fundamental estimation questions; what should be the size of the software that we are going to developed. So, we have to estimate the size. Then how much effort is required to complete an activity. So, we have to estimate here the effort similarly, how much calendar time? How much duration is needed to complete the an activity, that has to be estimated. And finally, what will be the total cost upon activity and what will be the total cost of a project that also has to be estimated these are the fundamental estimation questions.

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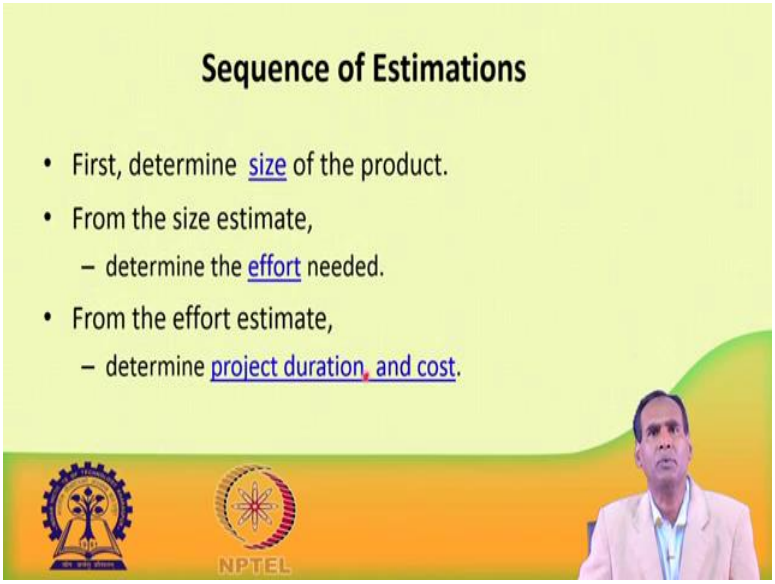

Now let us see, what are the sequences of the estimations that we have to follow. First we have to determine size of the product and if the size is known, from the size estimate we can determine the effort needed. And from the effort estimation from the estimated effort, we can determine the project duration and the cost.

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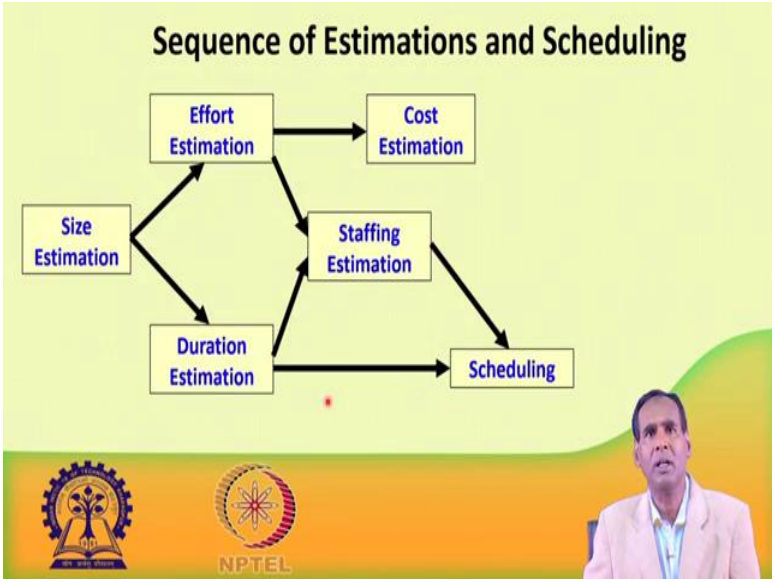

So, this sequence can be represented in the following graph. You can see that first we have to estimate size, from size we can estimate the effort and duration and from effort we can estimate the cost then from the effort estimated effort on the estimated duration we can somebody or the project manager can estimate the staff and then from the duration estimation and staff estimation. So, the project manager can prepare a schedule.

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So, now let us see what are the software cost components? So, we will see that these followings are the software cost components. First is the hardware and software cost. So, the cost require to develop the software and on which hardware it will run on the, what are hardware will be used to develop the software. So, those cost this is the hardware and software cost. Then travel and training cost; that means, the cost that is required to provide training to the employees and the travel associated with that they are also important components. Then another component effort cost which is very important and finally, we have to know what is the overhead that is, that may be required for this project.

So, out of this all the cost hardware cost the dominant factor, in most of the projects and here the salaries of the engineers involved in the project. So, that is why it may be what the most dominant factor, it will contain much of the percentage of the total costs. Then overheads; so, overheads could be like the it is not for what single project, but maybe what for different projects also. Here we have to take into account the cost of building, heating, lighting etcetera. Cost of networking and communications, cost of shared facilities, just like as library staff restaurant etcetera. So, these may be treated as the, what overheads. So, these are the different the software cost components for developing a project.

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Now, as I have told you one of the most important cost component is effort. So, how to measure the effort? So, in order to effort the measure; in order to measure the effort one terminology you require that is known as person month which is very very associated; which is very much associated with the effort. So, suppose a project is estimated to take 300 person months to develop. Now what do we mean by this? So, is just one person working for 3 days, same as 30 persons working for 1 day? No, certainly not, then why?

So, let us see in order to answer this question let us see how many hours constitute a man month. So, if we will say that the default value is 152 hours per month; that means so, 19 days some has to be worked at or some person has to work 19 days at 8 hours per day.

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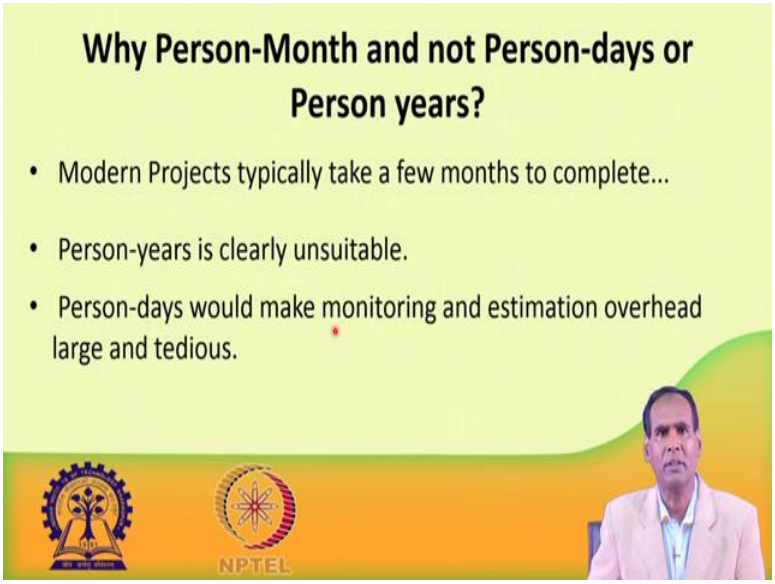

So, why person month and not person days or person years. So, normally the modern projects typically, takes a few months to complete. Some projects may take more than what 1 year, but modern projects they typical take a few months to complete. So, that is why person year is clearly unsuitable to what for these projects.

So, for similarly person days would be too much what small; so, person days would make monitoring and estimation overhead very difficult. It will be large and tedious. So, hence we should try for what this is the middle one that is the month. So, it will be a suitable measure to; it is a suitable metric to measure the effort.

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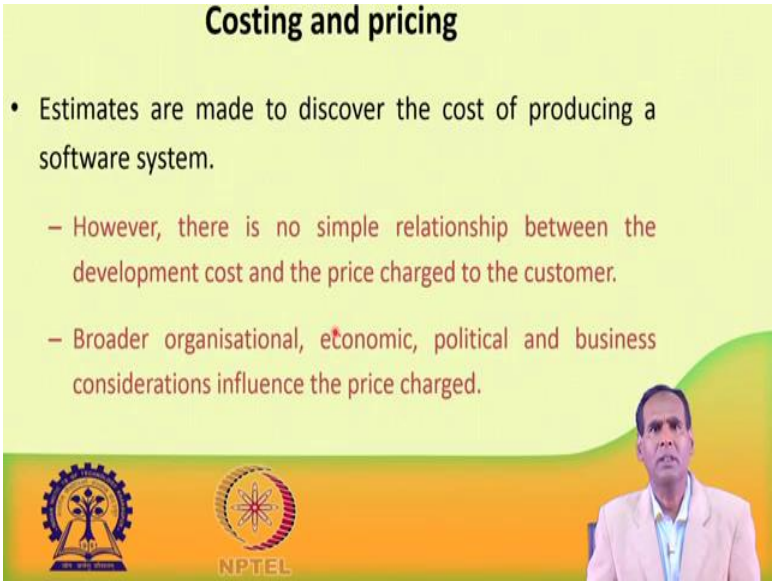

So, now, let us see this costing and pricing. So, estimates are made to discover the cost of producing a software system. So, we have to prepare estimates to discover what will the cost of producing a software system. So; however, you will observe that there is no such simple relationship between the development cost and the price that will be charged to the customer.

So, these you know that this product or organizational what considerations or economic political and business considerations they influence, the price that will be charged to the customer. So, that is why you can see that there is no direct relationship simple relationship between the development cost of the project and the price and that will charge to the customer for what obtaining the project.

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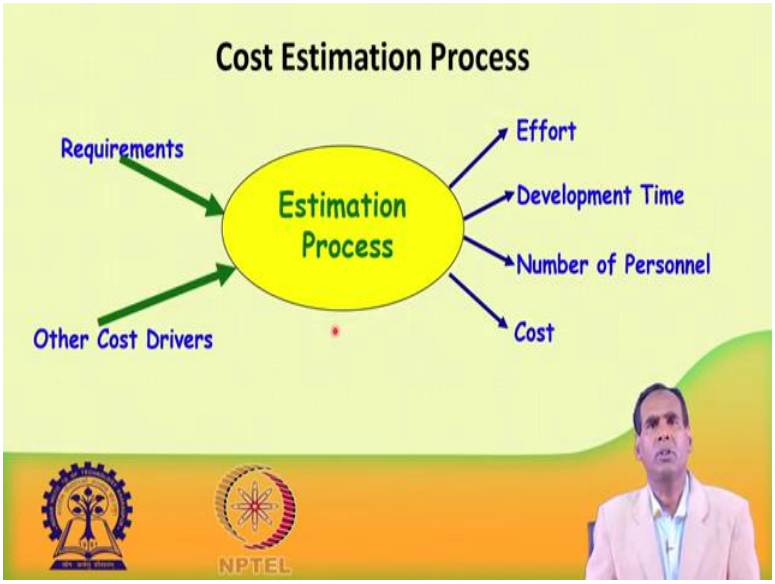

So, the cost estimation process consist of it can be represented like this which takes two inputs, what are the requirements and other cost drivers. And after this estimation process we can get the effort, the development time we can get estimates for the effort development time, number of personnel's required and the cost of the project.

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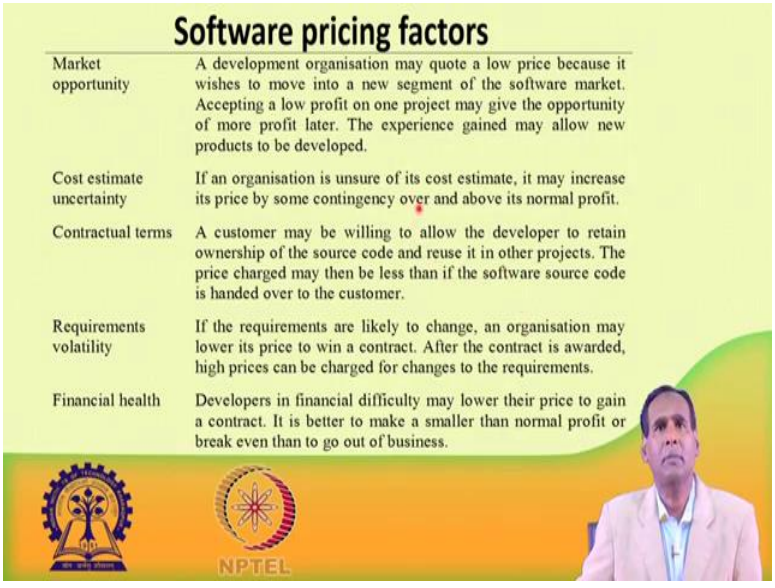

These are some of the what factors those are affecting the software pricing such as market opportunity this what you can say that this cost estimate uncertainty then these

contractual terms requirements volatility. And about what are the financial health they these are some of the factors which may affect the software pricing.

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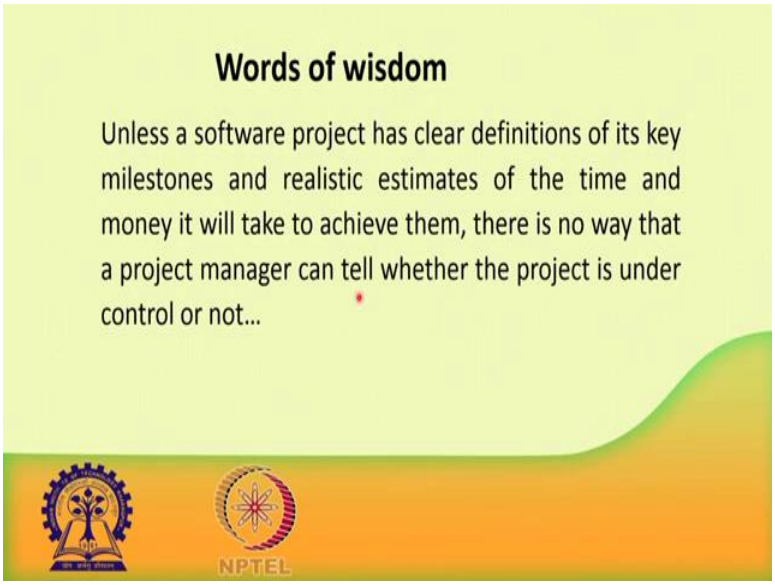

This is so, some words of wisdom are given here. Like unless a software project it has clear definitions of it is key milestones and realistic estimates of the time and the money. Then what will happen, the project manager cannot tell that whether it is very difficult project manager cannot tell whether the project is under control or beyond control.

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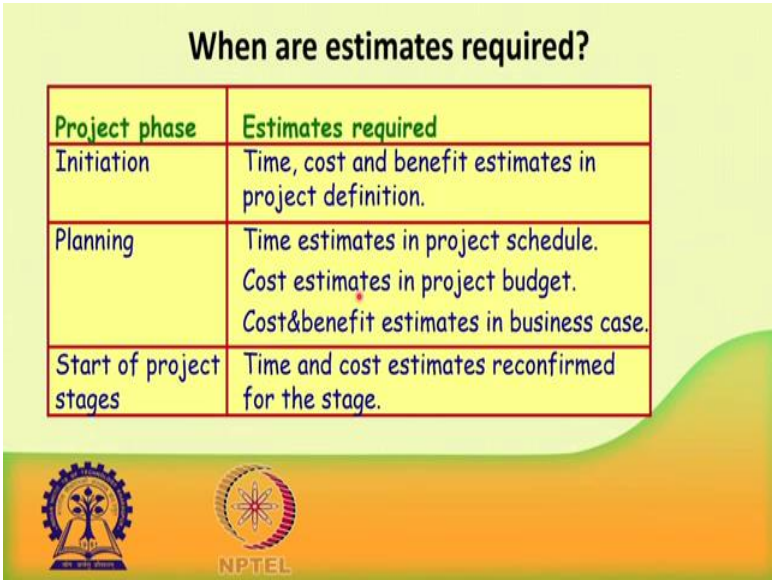

So, next is when are the estimates required? So, during different phases of the project, different estimates are required. Like in the initiation of phase you want what to be estimated the time cost and benefits estimates are required. Similarly do the planning phase time estimates in project schedule cost estimates in project budget and cost and benefit estimates in business case is they are required. Similarly during the start of the project stages, what we require time and cost estimates the reconfirmed for this stage that is required.

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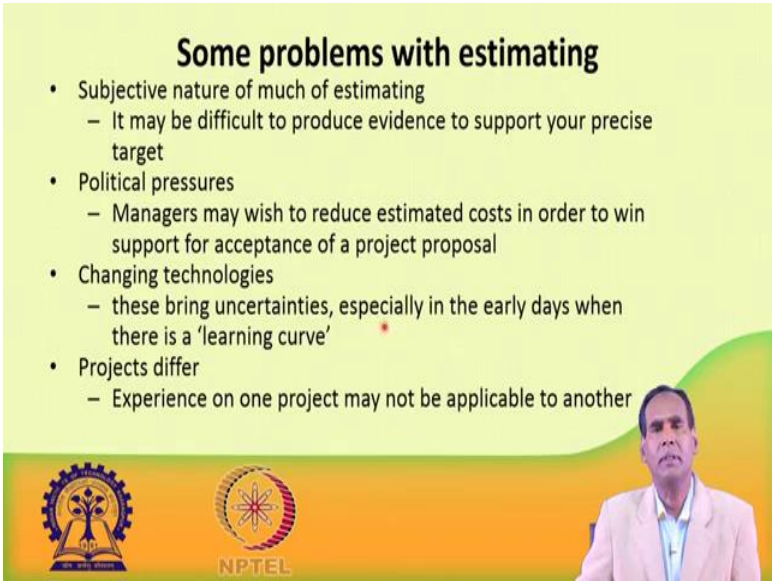

These are some problems actually with estimating you know that this estimation of the project is highly subjective nature. So, the subjective nature of estimating it puts barriers during estimation. Because this is highly what subjective as well as the political pressures, that also create problems while estimation and you know that current technology is changing rapidly. So, changing of the technology also puts barrier for software for project estimation and finally, you know the projects nowadays are the, what clients they are giving. So, they are; while they are largely different in nature.

So, the experience on one project, may not be applicable to the other since their nature they are different. So, that will also what put some special difficulties in estimating the different the various factors, the various parameters.

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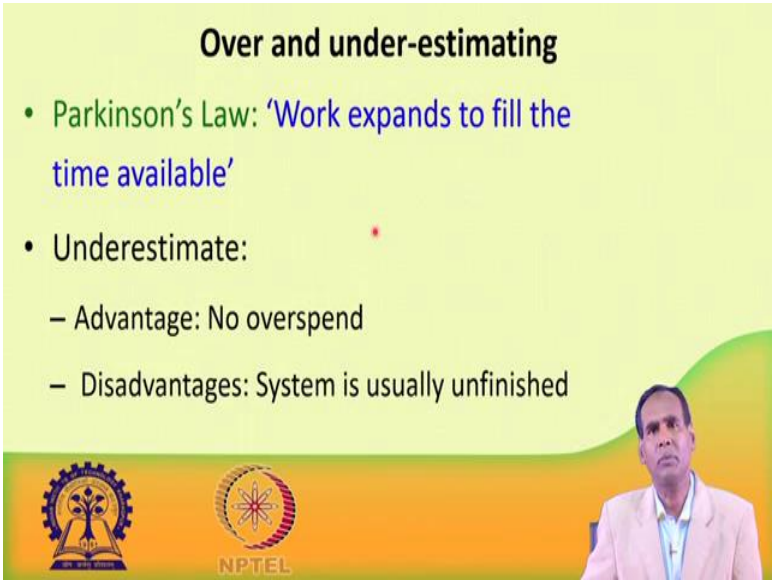

So, now let see two other important problems associated with estimation. one is over under, over estimating and another is under estimating. So, over and under estimating they are guided by two laws. one is Parkinson law, which said that work expands to fill all the time available; that means, if you are over estimated in the place of say 20 days you have assigned of 30 days. Then what will happen that the work that will expand to fill the time available the people they will they will not they will deliberately not work they will try that sometime is still left, we will only so, let us just wait even two days are left we will do those pending works in those days.

So, the work expands to fill the time available this will happen if you will overestimate the duration. So, if it is underestimate advantage is that there will be no overspend, but the disadvantage that the system will be usually unfinished and it will be substandard.

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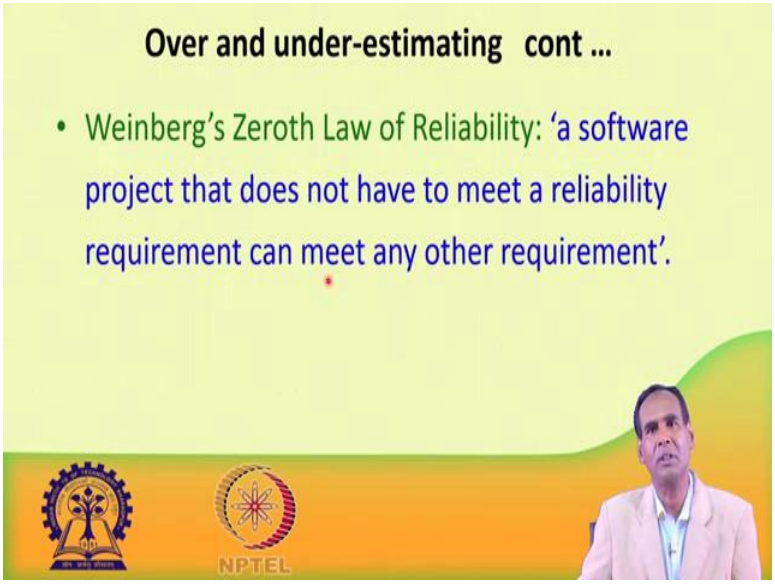

So, another law is that guides this over and under estimating problem, is that Weinberg's Zeroth law of reliability. It says that a software project that does not have to meet a reliability requirement can meet any other requirement. So, if it does not require this reliability requirement it does not have to meet a reliability requirement. It can meet any other requirement which is not desirable.

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So, the effect of under estimate is that the motivation and the morale will be lowered with high aggressive target and under estimation can lead to may be abandoned of the

project. Because that developers they will respond to highly pressing deadline with substandard work. So, since there will be pressure because time is less, they will compromise with the quality and they will produce somewhat substantial work. So, the project may get abandoned ok.

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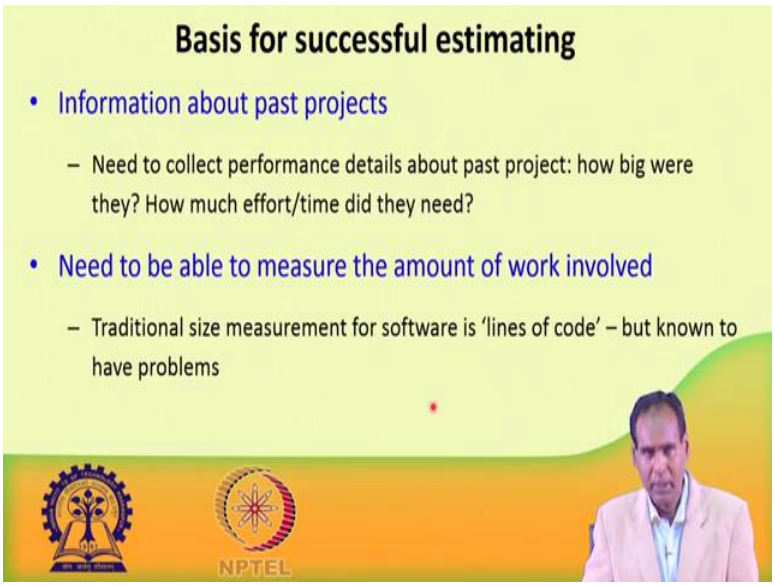

So, now let us see what are the basis for successful estimating. So, two important what things are required, one is information about similar past projects, another is that we should know how to measure the amount of work involved. We need to be or the project manager needs to be able to measure the amount of work, that is involved in the project. So, this will form the basis for the successful estimation ok. So, those things are there.

So, for what measuring the amount of work involved we may use the traditional size measures such as LOC, 'lines of code', but we know LOC of some special problems that we will discuss in the next class.

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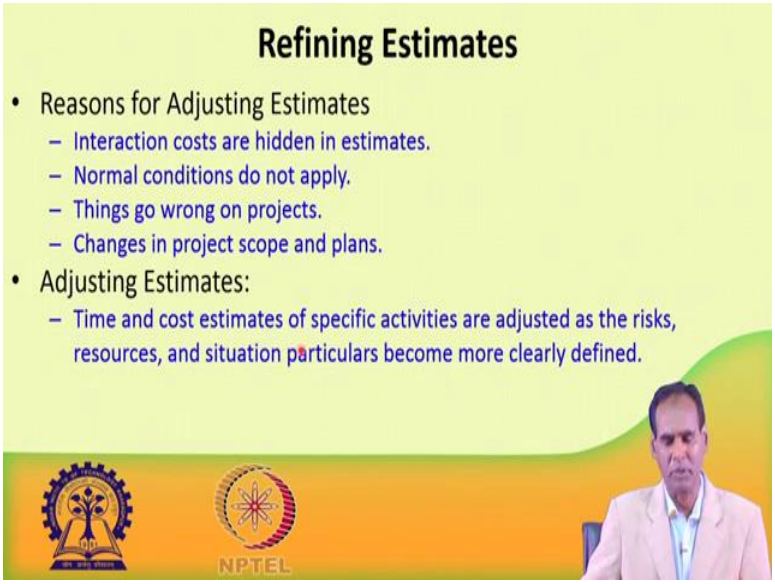

And then refining estimates, how to refine estimates? I have already told you this estimation is not done in one step. You have to first do the initial estimation the project manager has to first do the initial estimation, then gradually it has to be updated it has to be refined. So, now, let us see what is the reason for what adjusting this estimates for refining the estimates or for updating the estimates because the interaction cost are hidden in estimates. What are the interaction cost they are hidden they are not coming out or you cannot you may not find out them which also have some impact.

And normal conditions may not apply. So, things may go wrong on different projects, changes in projects. There might be some changes in the project scope and plans so, in order to incorporate all those things. So, the project manager should refine to adjust the estimates.

So, adjusting estimates means what, the time and cost estimates of some activities that adjusted, they are refined as the risk resources and situation particulars become more clearly defined in later phases. So, there is a need that the project manager should adjust or refine on the time and cost estimates of some specific activities.

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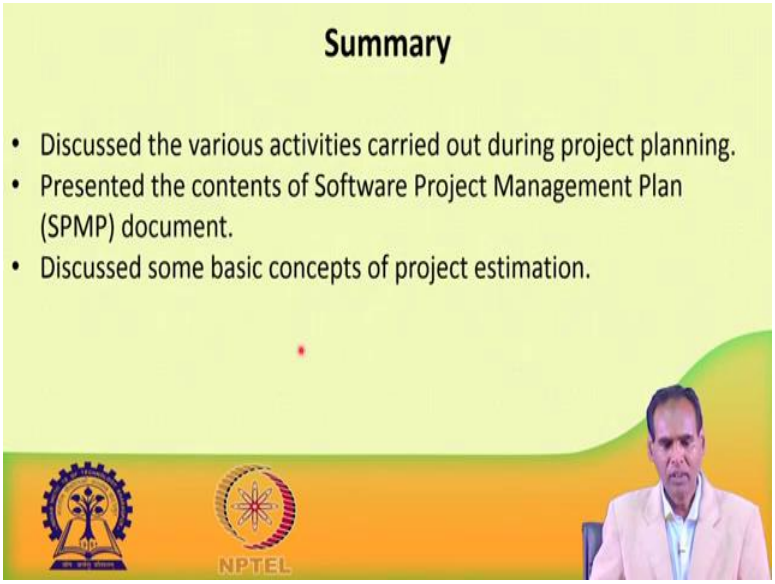

So, finally, we can see that we have first discussed the various activities which are carried out during the project planning. So, that is we have seen estimation and scheduling, staff organization as on some miscellaneous plans these are carried out during about this project planning. We have also seen that the outcome of project planning is a document called as SPMP or software project management plan document.

So, here the contents we have discuss like such as introduction, the various estimates and there will be a section also on the scheduling and miscellaneous ones etcetera. Also we have discussed some basic concepts of project estimation like, what is the basis for project estimation? How do you, why the projects the estimation does not become correct and we have also seen a sliding window estimation; sliding window estimation. Because the project managers should not do the estimation in one step in one phase rather they should perform it in several stages.

So, that the some more accurate information may get or may be obtained at the lateral phases which may be taken into account during the project estimation or the during computing the various estimates. So, that is why you should use this sliding window estimation. And as I have already told you it should not be done in single page rather than first the initial plan is made then gradually as in the some more updated information comes then you can or the project manager can update or adjust this various estimates that will give you what more accurate estimates.

So, these things we have seen right now. And in the next class we will discuss, what are the various estimation, what are the categories of, what are the taxonomy for the estimation techniques also we will discuss about this the size measure. Because I have already told you the one of the initial measure is size. If you can estimate properly the sizes then you can estimate the other parameter such as effect duration effort duration and cost. So, that also we will discuss in the next class like this size estimation ok. So, thank you very much.

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These are the references, and.

Thank you very much.

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Lecture – 17
Project Estimation Techniques (Contd.)

Good afternoon. Now, let us continue the remaining parts of the estimation techniques.

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Well here see a taxonomy of the various estimating methods, then we will see about these size estimation.

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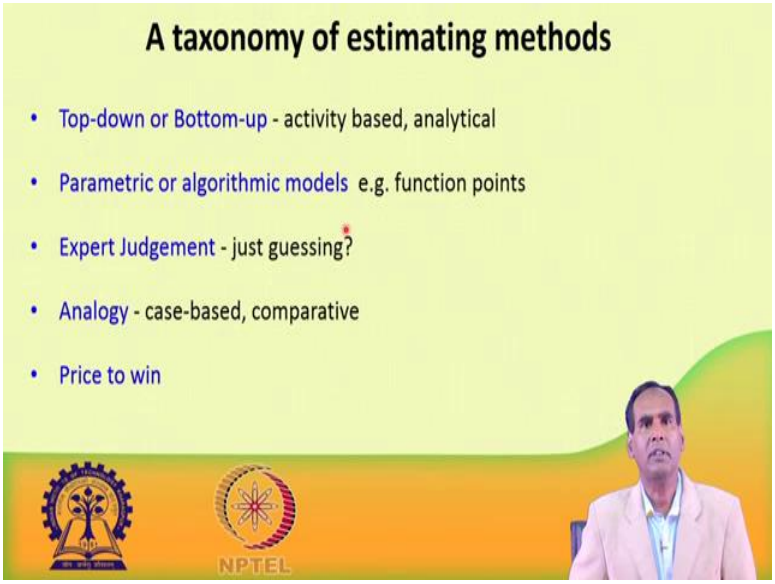

So, let us see at the one type of taxonomy, a taxonomy of the existing estimation methods. The estimation can be top down or bottom up. So, here normally they are activity based and analytical. And then, the estimation techniques can be parametric or algorithm models, for a view example function points that we will see in the next class.

Then expert Judgement technique, that is another category of method expert Judgement. Here, we just guess the various estimates, initial with initial guess we start the estimation procedure. Then analogy as its name suggest we have to see how a project is a analogous. It similar to another project existing project, then we can estimate the required parameters. These are normally case based and they it is based on the comparative principle and another methodology for estimation is price to win.

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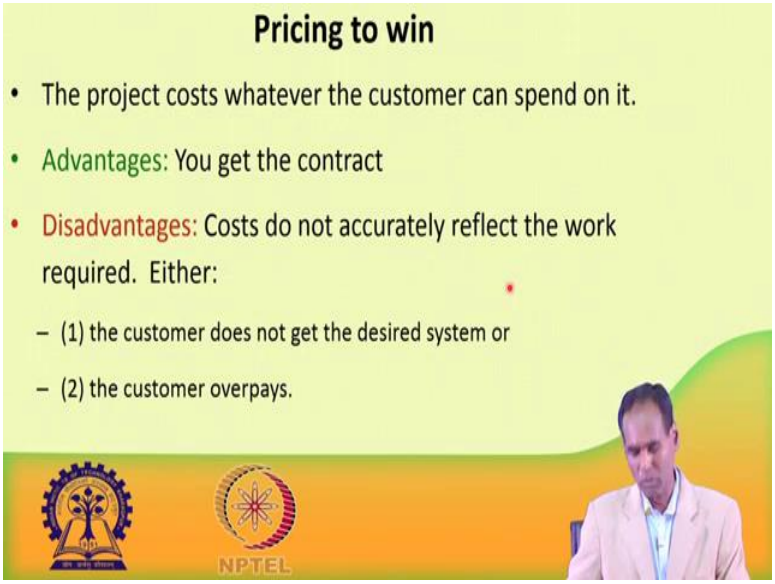

Let us start with the price to win method, what does it say? The project cost whatever the customer can spend on it. So, according to this the cost or the estimated project cost is that cost that the customer can spend on it. So, the project cost whatever the customer can spend on the project and advantages here is that you get the contract, it is very possible, it is very much probability there that you will get the contract and disadvantages that the cost they do not accurately reflect the work required. So, what work will be performed the cost may not accurately reflects the work required.

So, here either the customer he will not get the desired system or what will happen the customer will over pay, because the work is less and he may pay more than the that is desirable ah in regard to the work that or the service that he is getting .

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So, this approach may seem very much unethical and unbusiness like, because as you have seen that the cost they do not accurately reflect the volume of the work required. So, it this approximation may seem to be unethical and unbusiness like; however, when the detailed information regarding the volume of work etcetera is lacking then this method may be the only appropriate strategy.

So, now, let us see which is the most ethical approach? The project cost is agreed on the basis of ah an outline proposal and the development is constrained by that cost. So, the project cost normally, it is agreed on between the client on this developer based on what on the basis of an outline proposal and the development is constrained by that cost. So, that what should be ethical and a detailed specification maybe negotiated or an evolutionary approach may be used for the system development .

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So, what parameters are to be estimated, for project planning? We need to estimate the effort or the indirectly the cost and the duration, but it is very much difficult to estimate the effort or cost or the duration directly from a problem description. So, what we have to do? The effort and duration, they can be measured in terms of the project size, which is an indirect metric and a project size we can measure by using either LOC that is Lines of Code or FP that is Function Point .

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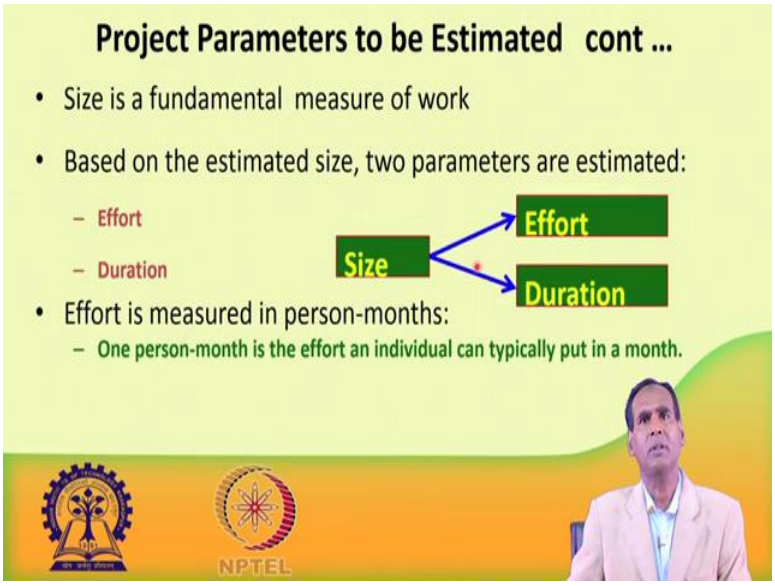

So, let us see that; we know that size as effort and cost duration they cannot be directly estimated. We have to use a size or this what yes, we have to use project size for estimating and then the for estimating effort and duration. So, size is a fundamental measure of work, based on this we can estimate the other parameters such as effort and duration.

So, it is based on the estimated size ok, based on the estimated size, two parameters are can be estimated; they are the effort and the duration. So, effort is measured normally person month, I have already told you if the person month earlier, that effort is measured in person month still let say define person month. How the person month is defined? One person month is defined as the effort that an individual can typically, put in a month ok.

So, one person month means is the effort that an individual can typically, put in a month. So, here in this way we can calculate this, what person we can define person month and using person month, we can define, we can measure effort. So, as I have already told you first, we have to compute size then we can compute effort and the duration.

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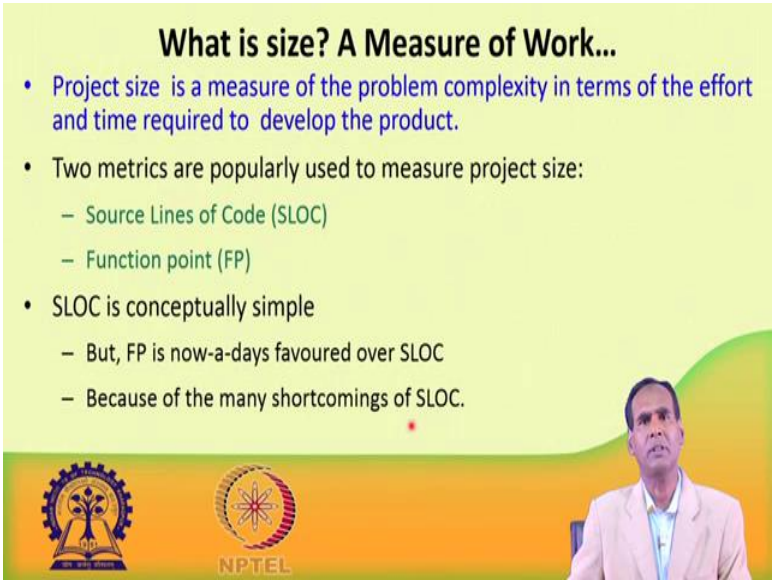

So, what is size? It is a measure of work that is being done. So, project size is a measure of the problem complexity ok. Project size can be considered as a measure of the problem complexity, in terms of what? In terms of the effort and the time required to develop the product. Two metrics are popularly used to measure a project size. I have

already told you one is LOC or in a finer terms can say that Source Lines of Code or SLOC and another is Function Point or FP.

So, SLOC is conceptually simple, just we have to count the lines of the code, the Source Lines of the Code we have to count that is conceptual very simple, but FP is nowadays, it is favored over a SLOC. We will see there are several drawbacks of SLOC. So, that is why nowadays people are using FP it is a preferred over SLOC, because of the many shortcomings of SLOC we will discuss after few minutes, what are the shortcomings of SLOC.

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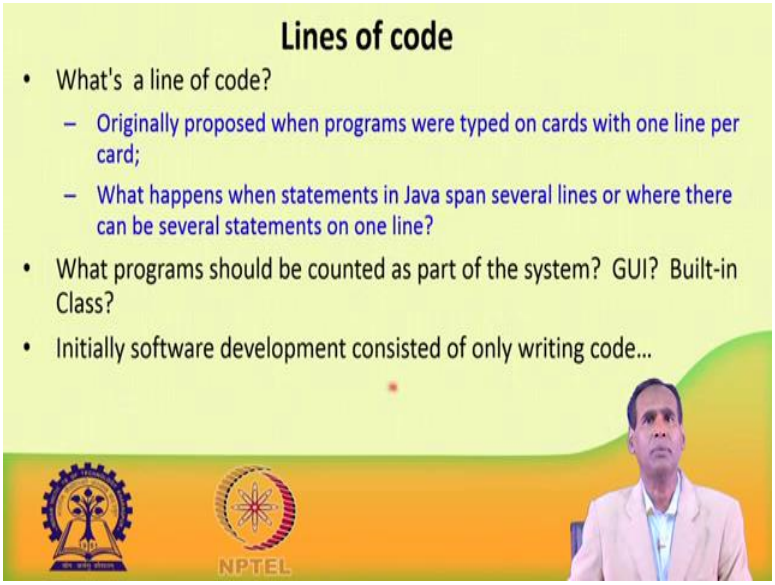

So, the lines of code means; what is a line of code? Basically, this lines of code was proposed when programs were typed on cards with one line for cards. So, previously people are using punch cards. So, during that time so this lines of code it was proposed. So, it was proposed when the people were or programs were typed on cards a, punch cards with a one line for card. So, what happens when statements in Java span several lines or where there can be several statements in one line; see this is the drawback of lines of code.

So, it was develop keeping in mind that only one line or yes, one line can be entered for a for card. This metric was meaningful, but when, so different programming language high level programming languages came and in some cases like Java that this statements Java, they are, they span over several lines or in one line you can rise statements then this lines

of code may not be a suitable measure to may not be suitable metric to measure the program size. So, similarly what programs should be counted as part of the system. So, what programs will be counted as part of the system say for example, GUI.

So, it is not a functional requirement, but should it be the statements written for designing a GUI should it be counted as a part of this system. Similarly, the built in classes, which are provided in the programming languages. So, can they be counted as worth or this LOC? Will they be counted as part of the system that you are going to develop? So, these are some of the limitations, you can say and or that is why the lines of code may not be a suitable metric for all types of these what modern applications. Initially, software development consist of only writing the code, but now, several things are possible while developing or several things are performed while developing any system or project.

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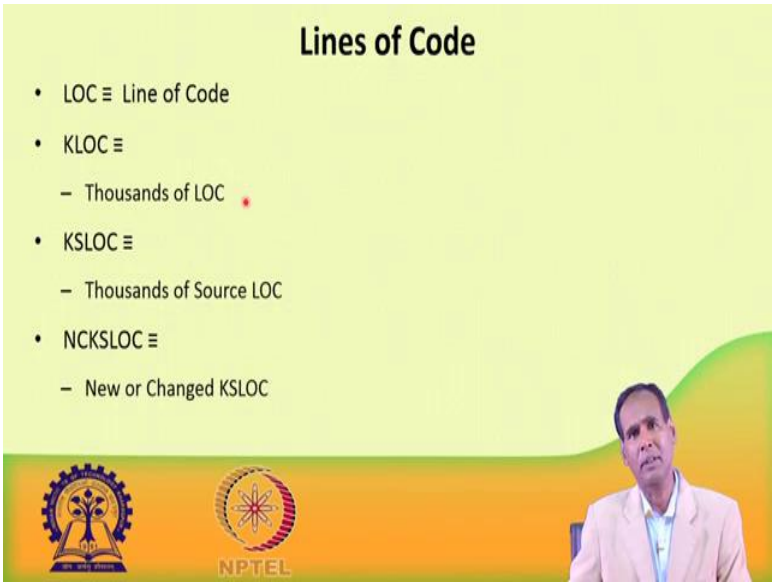

So, in Lines of Code we will use this terms like LOC, which represents lines of code, KLOC represents what kilo lines of code that mean thousands of lines of code. KSLOC means thousands of source lines of code and NCKLKSLOC means the new or changes KSLOC. So, how many? What lines of code? They are completely new or they are changed. So, these are the terminologies, they are used in this metric lines of code.

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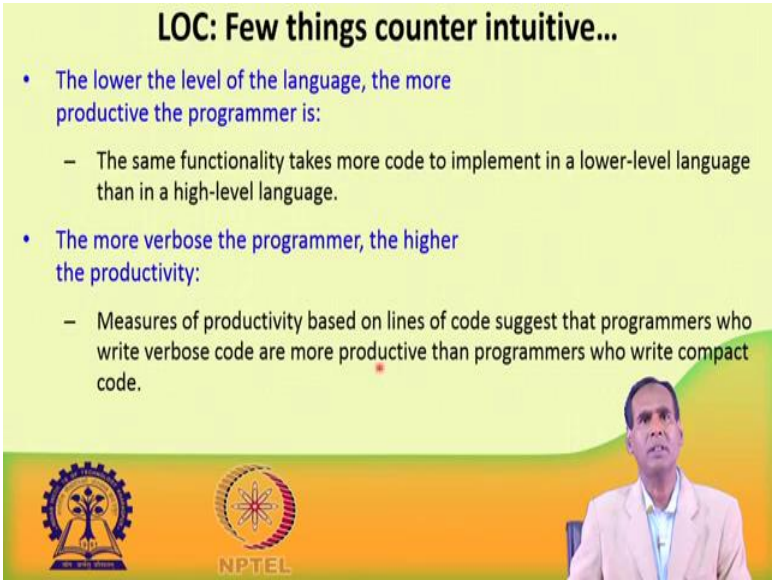

So, now let us say few things these are counter intuitive in case of LOC. The lower the level of the language the more productive the programmer is; obviously. So, if the lower language, lower the lower the level of the language.

For example; you say assembly level language, etcetera, then what will happen their size will be less. In one instruction with a programmer may achieve several things. So, that is why the lower level of the language the more productive programmer is by using only fewer sentence we can achieve a lot of jobs.

And similarly, what this I have already told you the same functionality takes more code to implement in a low level language, then in a high level language is not it. The same functionality it takes more code to implement in a lower level language than a high level language. So, the more verbose, the program the higher the productivity.

So, if there will be more verbose the more verbose the programmer is the higher the productivity. Measures of the productivity based on lines of code suggest that the programmers who write verbose code are more productive than programmers who write just compact code, because he is using more what he is using what more verbose. So, that is why the lines of code will be more. So, it say that measure of the productivity based on lines of code is suggest that the programmers who are writing verbose code, they are more productive and than the programmers, who write compact code. So, this is another counter intuitive for lines of code.

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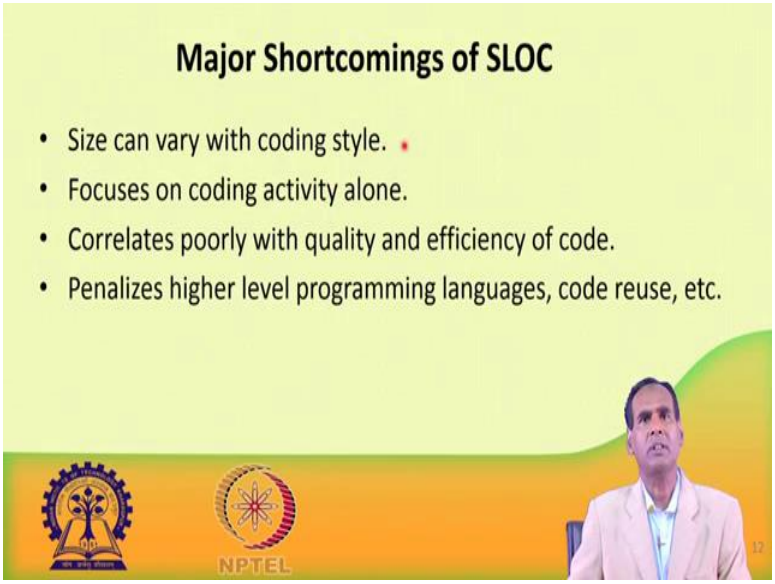

Now, quickly let us see some of the major shortcomings of SLOC. Size can vary with coding style. So, different programmers can follow different coding styles and if the coding style is different, then size will be different.

Similarly, this LOC, it focuses only coding activity alone no other activity, because you may have to has a while developing the software, you might have to carry out requirement analysis, you may have to do design, you may have to testing etcetera, but this LOC for that you are putting some effort, for the requirement analysis or for coding or for testing or for designing you are putting some effort, but LOC does not take into account that it focuses only coding activity alone it correlates very poorly with quality and efficiency of code.

So, what will the quality of the code, what will the efficiency of the code? So, LOC correlates very poorly with them similarly, penalizes higher level programming language code, reuse, etcetera if you are using very high level programming languages, because in assemble language programming same thing can be done with very just few statements, but in high level languages you may use what some more statements for that. So, it penalizes the high level languages, programming languages.

Similarly, if you are using code, if you are what using this code reusing, if you are following code reusing; that means, you have developed one and again and again you are

using. So, then also it penalizes this, so the size may not be accurately corrected, if you are using code reuse. So, these are some of the shortcomings of SLOC.

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So, other shortcomings are like difficult to estimate at start of the project from problem description, if we are just given with a problem description like as a SRS document then it is very much difficult to estimate, maybe what is the size or the very beginning of the project. The only way since, you cannot get an accurate estimate at the beginning, the only way is to estimate it to make guess and so not useful for project planning.

So, that is why this LOC is not very much useful for project planning. So, it only it is I have already told you, it is used only as a code measure. It cannot be used for measuring what the effort that you are putting during what your requirement analysis or design or testing this cannot be used for measure them.

Similarly, it is programmer dependent, because different programmers will write the program programs differently. So, the lines of code many be different, so it is programmer dependent and it does not consider structural or logical complexity. It only considers what the lexical and the textual complexity only, it does not consider a structural or the logical complexity of the program.

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There are still some further difficulties with SLOC like SLOC can become ambiguous due to rapid changes in programming methodologies program and languages and tools. So, due to the changes in the programming methodologies languages and tools, SLOC becomes an ambiguous. Might be see the language advancements are there. So, different languages are coming up and this can put um, this can lead to problem, this can lead to ambiguity while estimating SLOC. Similarly, many tools are there the where you can automatically generate the source code.

So, then what this SLOC may not be useful very much say like you are having RSA rational software architect where from the class diagram you can automatically generate the skeletal code, then SLOC may not be properly what it, it may not be suitable to measure this what it may not to be suitable metric to measure the size, similarly custom, software and reuse.

If you are customizing the software, if you are developing a customized software and you are using code reuse then SLOC may be ambiguous and similarly due to a newer programming or what changes in programming methodologies such as object oriented programs, aspect oriented programs, feature oriented programs so they are in those cases SLOC, the size cannot be accurately estimated by using this measure SLOC.

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So, now let us see some of the what a techniques for effort, let us, we have already seen this pricing technique then we will see about the other techniques, we will see first expert Judgement based techniques. Here, we will see these are the techniques which are coming under expert judgement based technique that is basic expert judgement weighted average estimating consensus estimating and delivery let us first say about the basic expert judgement.

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So, here in the if you are using basic expert judgement technique for estimation, what is happened here, one or more experts, they predict the software cost, this process is iterated until some consensus is reached among the experts. The advantage of using this method is that is very much relatively simple estimation, it is a very relatively simple estimation method, it can be accurate, if the experts have direct experience of the similar systems.

But if the experts they do not have experience of the system that you are developing then the estimation may be wrong. The disadvantages that very it may the estimate may be very inaccurate, if there are no experts available in this kind of what system that you are developing suppose, you are developing a avionics system and there is no expert who have knowledge who has knowledge on the avionics, then; obviously, the estimates will be wrong .

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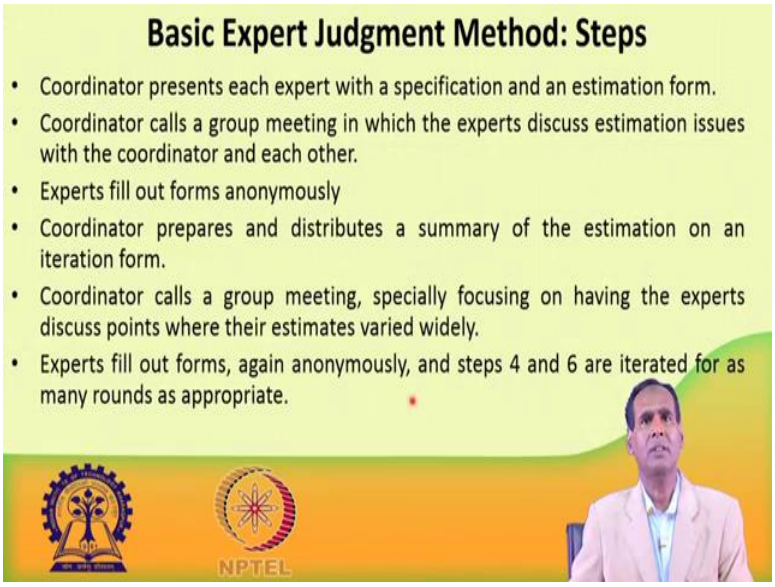

So, now let us see about this steps of the basic expert judgement method. So, first in this method there will be a as I have already told you there will be a group of what experts, one person who will act as the coordinator. The coordinator presents each expert with a specification and an estimation form ok. You will coordinate the what activities, you will present the coordinator present each expert with specification and an estimation form and the coordinator calls a group of meeting in which the experts discuss estimation issues with the a coordinator and each other.

So, coordinator can convene the meeting where the experts they will discuss the estimation issues with the coordinator and with the other experts then the experts fill out the forms anonymously. Then the coordinator prepares and distributes a summary of the estimation, the coordinator prepares a summary and then he distributes the summary of the estimation on an iteration form. Then the coordinator calls a group meeting specially, focusing on having the experts discuss points, where their estimates varied widely.

So, here the if there are suppose, five experts and their estimates are varying widely, then again the coordinator will call a meeting and discuss why there is a variation, because project is same parameters are same, why there is a variation among the estimates then they may have discussed. Then experts, they will fill out the forms again anonymously and steps 4 and 6 are iterated for as many rounds as appropriate till they reach to a consensus decision, this process will be iterated.

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Now, the we know what is an expert, actually an expert is a familiar with and expert is a familiar with the system and he has he is knowledgeable about the application area that you are developing and the technologies that you are using. So, particularly it is very much appropriate where existing code is to be modified and existing code is already there and you are trying to modify, then you are doing this estimation. In this case basic expert judgement technique will be very much what useful, but research shows that the

expert judgement in practice it tends to be based on another methodology called as analogy methodology.

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So, let us see what we will see this analogy methodology. So, let us see what are the stages that the basic expert judgement method may pass through, as I have already told you, it is based on an analogy method the steps or the stages are like this. Then first identify these significant features of the current project. Then identify or just look at the previous projects with similar features.

Then look at if there is find out, if there an any differences find out whether if there are any differences between the current and previous projects. Then find out what are the reasons why they are differences find out the possible reasons for the errors or the risk. Then measures adapt measures to reduce this uncertainty, then you have to text some or adopt some measures to reduce this uncertainty or the risk.

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As I have already told you this the basic judgment technique is basic expert judgement technique is based on the estimation by analogy. So, what do you mean by this estimation by analogy? The cost of a project is computed by comparing the project to a similar project in the same application domain. So, suppose, you are we have developed a project for a private organization academic institution and next you are the as a project manager you are developing this, what academic system for a government institute, then they are similar types of applications.

So, here you may what compare the projects that you have spent in the while developing the academic autonomous system, for the private organization or private university with that of the what proposed one; that means, for the government university, then you can see what could be the estimate. May might little bit vary, but it will it may give appropriate result.

So, the cost of a project here is computed by comparing the project to a similar project in the same application domain. So, one example I have already given you that if you have already developed a project for a private institution and then developing a similar project for a government academic institution. So, you can compare the projects and estimate accordingly, for the new project that you are undertaking for automating a government university or institution.

Now, the advantages of estimation analogy that it may be accurate, if a project data they are available and the people and the people and the tools they are same ok, it may be accurate, if the project data which are available to you from the first project and the people and tools they are the same. It will be advantages, disadvantages is that it is very much impossible, if no compatible project has been tackled.

If similar kind of similar or if similar kinds of projects they have not been done earlier then this is then, you cannot estimate for the current projects. So, this will disadvantage. So, it needs systematically maintained cost database. So, whatever previous projects are there you must maintain their various cost in some database. So, it is needs systematically, maintain their cost database, so that you can use in future.

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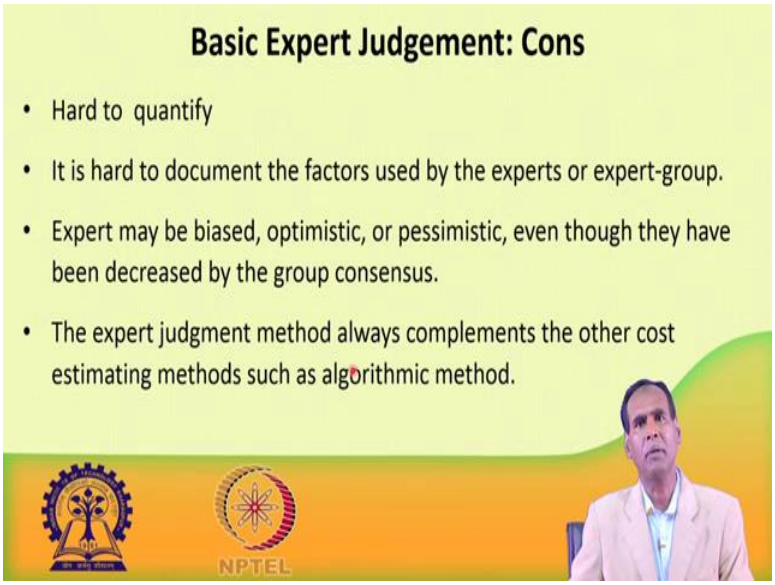

So, now again let us go back to the basic expert judgement technique what are the, what disadvantages, it is very much hard to quantify. Similarly, it is a hard to document the various factors, which are used by the experts or the experts group. Experts may be biased, because after all they are human beings, experts may be biased, they may be optimistic or pessimistic.

Even though they have been decreased by the group consensus and the expert judgement method always complements the other cost estimating methods such as algorithmic methods. So, always the expert judgement method, it complements the other estimation method such as algorithmic method.

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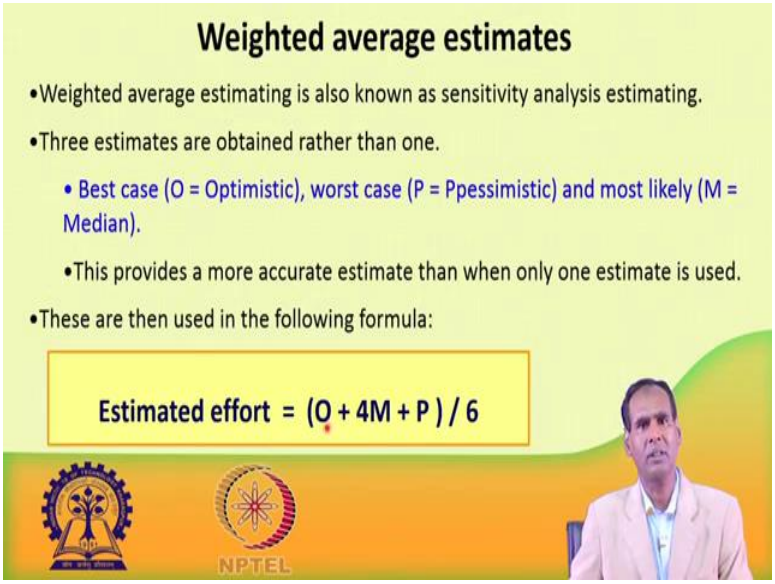

So, now we will receive the second one that is the weighted average estimates. Here, weighted average estimation is also known as a sensitive analysis estimation and here three estimates are obtained rather than one just, because if you are going one; it may be it may not be fully accurate. So, three estimates are obtained rather than one.

So, one is the best case that is the O and or the optimistic case, then worst case, or pessimistic case P and most case or most likely case that is the median or M we take and now, this provides a more accurate estimate than when we are considering only one estimate. So, then these are used in a formula to produce the estimate. So, that formula is,

$$\text{Estimated effort} = (O + 4M + P) / 6$$

So, use the first find out these values for what best case worst case and most likely then use this formula you will get this estimated effort.

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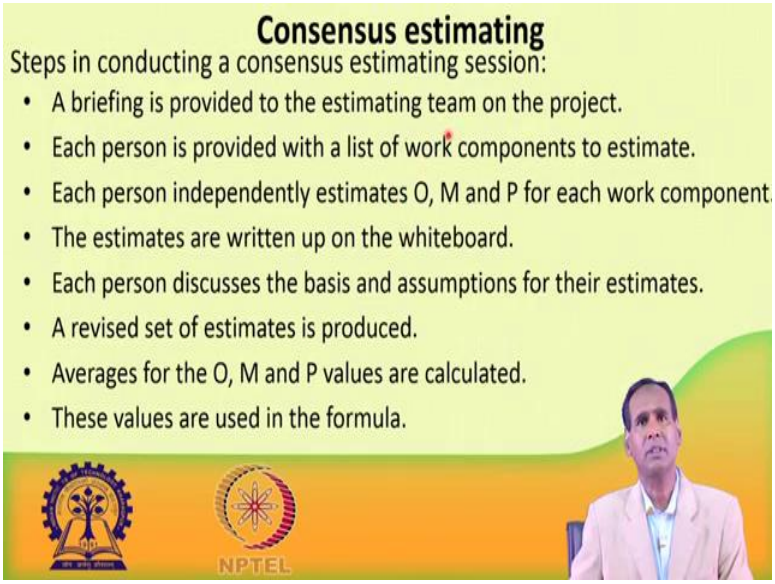

This is what is happening in that is why the name is weighted average estimates. The other method is consensus estimating, consensus estimating. Here, the steps in conducting a consensus estimation session is like this first a briefing you have to brief a briefing is provided to the estimating team on the project, then each person is provided with a list of work components to estimate. So, each person will provided a list of the work components, they have to be to estimate. Then each person independently estimates these optimistic value most likely value and pessimistic value for each work component.

Then the estimates are written up on the whiteboard and then each person he discusses the basis and the assumptions based on which they have prepared the estimates and finally, some suggestions may come then a revised set of estimates is produced. Then we have to take what since, we are getting a revised set of estimates you have take the average you have to take the average of the O M and P values. You have to take the average for the optimistic and then the most likely on the pessimistic values these are calculated and finally, these values are used in the above formula that we have seen. So, in this way the consensus estimating is followed.

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And next is Delphi estimation; so, it is a variation of consensus estimation technique. Here, as I have already told you it is a variation of consensus estimation, here also a team of experts will be there and their coordinator will be there to coordinate among the experts.

The experts may carry out the estimation independently ok. The normal the experts they will carry out the estimation independently. They will mention the rationale behind their estimation what is the rationale, what is the assumption they taken. They will also mention in that a paper, then the coordinator notes down any extraordinary or rationale if any expert has taken, then he circulates the coordinator circulates the estimation or the rationale among the other experts.

After receiving the what estimates from other experts each expert they re-estimate their values then they, but here the experts never meet each other in judgement or yes in this, expert judgement technique the experts they meet each other, but here in Delphi estimation the coordinator takes the most important role. He will circulate among others among other experts, the experts never meet each other to discuss their viewpoints.

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So, Delphi means Delphi is an expert survey in two or more rounds. So, starting from the second round so, first round is over, after first round starting from the second round a feedback is given about the results of the previous rounds, then the same expert he assesses the same matters once more influenced by the opinions of the other experts. He takes into account the suggestions or the opinions of the other experts and then he assesses the same matters once again, he re-estimates again the same parameters. So, here the important is anonymity.

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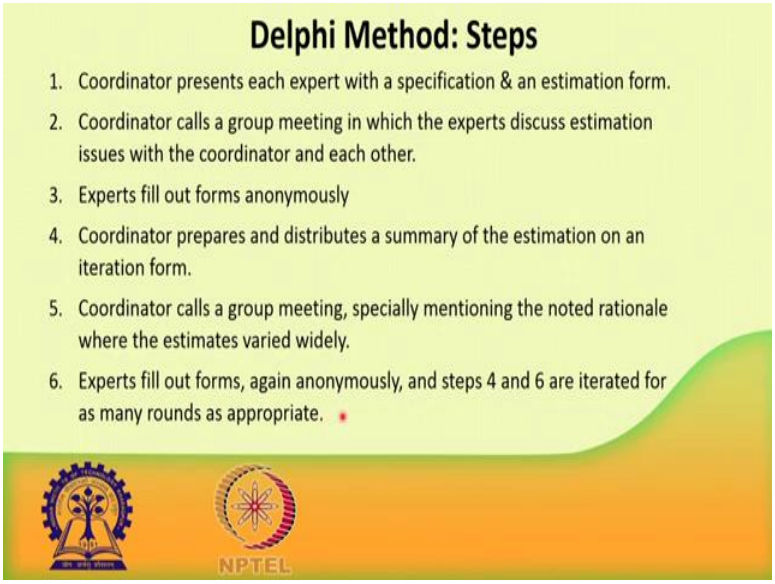

So, the steps are like this the coordinator presents each expert with a specific and an estimation form, then coordinator calls a group meeting in which the experts they discuss the estimation issues with the coordinator and each other and then experts, they fill out the forms anonymously. And coordinator prepares and distribute a summary of the estimation on an iteration form and the coordinator calls a group meeting specifically mentioned in the noted rationale, where the estimated varied widely. And experts they fill out the forms, again anonymously, experts fill out the forms, again anonymously, and steps 4 and 6 are iterated for as many as rounds as appropriate ok.

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So, finally, we have discussed the price. We have first discussed a one method this price to win estimation method, then we have presented how to find out how to estimate size using LOC or SLOC. Also discussed expert Judgement based estimation techniques. So, basic Judgement weighted average estimation ah, consensus estimation and Delphi. We have also explained the analogy based estimation. So, these are the references, we have taken and.

Thank you very much.

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Lecture – 18
Project Estimation Techniques (Contd.)

Good afternoon to all of you, we will take up some other categories of estimation techniques now ok.

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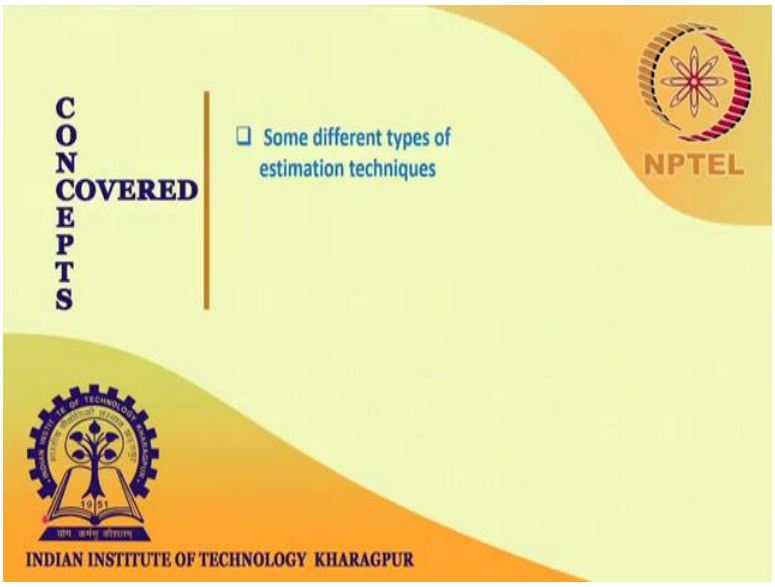

We will see some different other types of estimation techniques.

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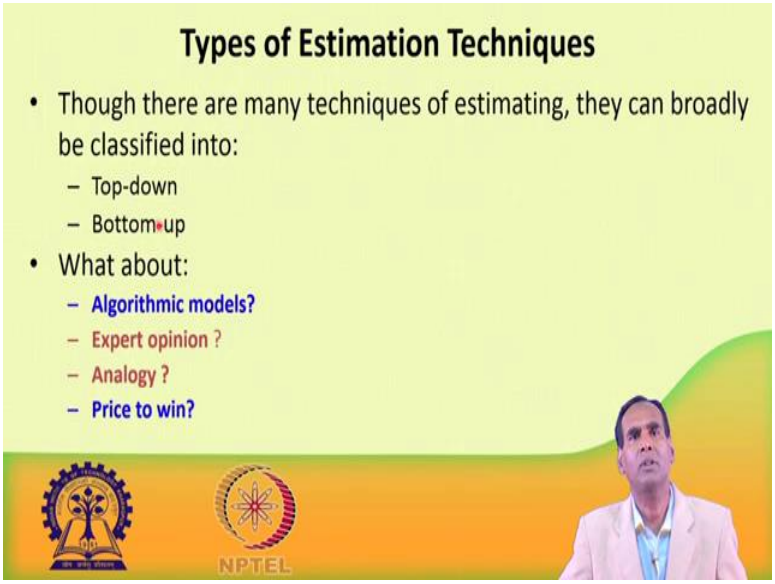

The types of, there is another category of different types of estimation techniques though, there are many techniques of estimation they can be broadly classified into top down and the bottom up, this is one classification.

So, what about: algorithm models, Expert opinion, Analogy, Price to win? In the last class, we have discussed about the these last three expert opinion or expert judgment, analogy mythology and price to win, those things already we have seen and so, basically these four are also another classification for project estimation. So, the last three already we have seen. Now, let us see about the top down and the bottom up estimation techniques then we will see about the algorithmic models for project estimation.

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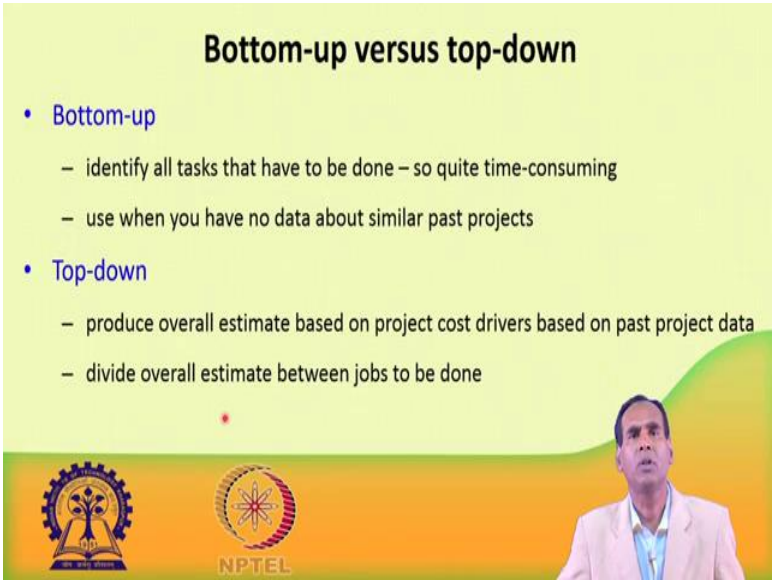

So, bottom up estimation is a estimation technique which identifies all tasks that have to be done ok. So, so it identifies all the tasks that have to be performed. So, it is very much time consuming. Use when you have no data about similar past projects. So, when you do not have similar past projects, you may use this see I have already told you that analogy method.

Analogy method is based on when it will be suitable when you are having similar kinds of past projects or similar kinds of projects you have done in the past then this analogy method you can use, but when you do not have any data about the similar past projects, then you use this bottom up approach. Then what is about top down approach? It produce overall estimates based on project cost drivers and based on the past project data.

So, top down method is used when this past project data are available. So, this method produces overall estimates based on two things; the project various project cost drivers based on what the past project data. So, here we have to divide the overall estimate between the jobs they have to be performed.

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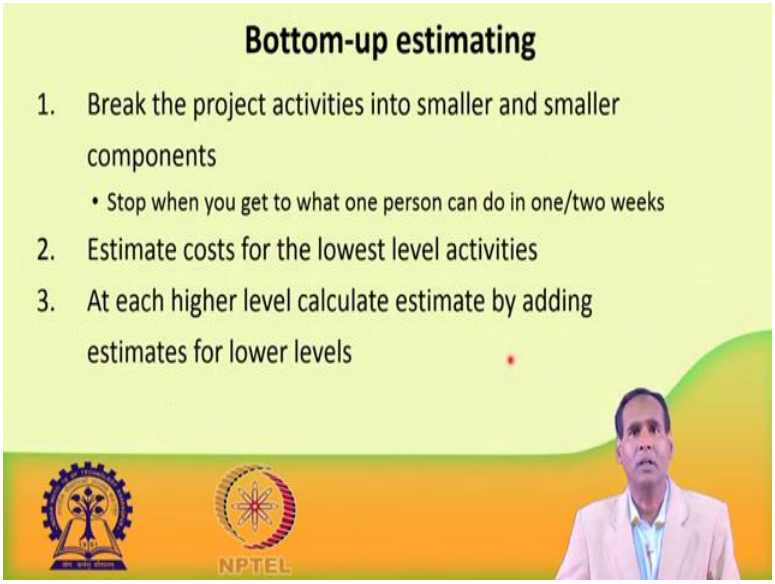

Now, let us see first bottom up estimating or bottom up estimation. What are the steps? The steps for bottom up estimation are as follows; break the project activities into smaller and smaller components. So, take as its name suggest bottom up estimation, just like in object oriented approach you are using bottom up what mythodology.

So, here similarly you break the project activities into smaller and smaller components, then stop when you get to what to be done by one person in one or two weeks then. Where to this breaking process? When it should be stop? We should stop when you get that what the piece of work that one person can do within one or two weeks, then you estimate the cost for the lowest level activities.

So, after breaking the projects into activities, smaller activities then you estimate the cost for the lowest level activities at each higher level. At each higher level then you calculate the estimates how; by adding the estimates for lower levels. So, at each higher level, you have to calculate the estimates by summing up the estimates for the lower levels.

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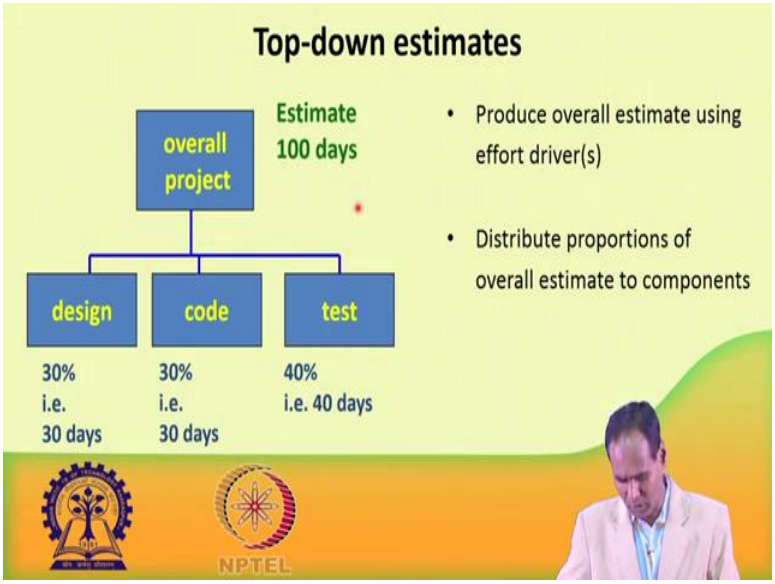

Let us then let us see about the top down estimates, we will take an example. So, here suppose, as top down estimate I have already told you earlier, I think I have told that ok. Let us see bottom up approach, bottom up estimation algorithm, we have already seen. So, now, let us see about this top down estimation, difference already I have told you. Here, the top down estimates was as follows like produce an overall estimate using produce, the overall estimate using the effort drivers or the cost drivers you have to use here some drivers to produce the overall estimate and then you have to distribute the proportions of the overall estimate to various components.

Here, you can see that for the overall project you have estimated that you it might take 100 days, but the overall project can be divided into what different what sub activities like; design, coding, testing, etcetera and you have estimated that 30 percent, 30 percent of the total days can be given to design, 30 percent to code, and 40 percent you will give to test.

Then you can easily calculate the proportionate the proportionate of this what total estimate like 30 percent of 100 days that is 30 days for design, again 30 percent of the total estimate 100 days, again it is 30 days for coding and 40 percent of the total estimated that is 40 days for the testing.

So, in this way the top down estimate works, first estimate the overall estimate, first calculate the overall estimate then distribute the proportions of the overall estimate to the

individual components then you will. Then this method of estimating the effort or the duration is known as a top down estimation.

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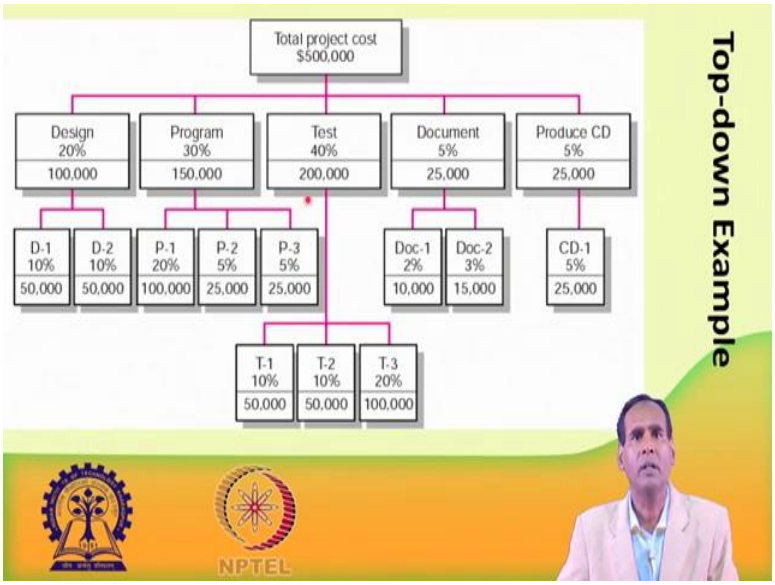

Another example is given here, where these the total. Suppose, the total project cost is 500000 then the total project, overall project total project what are the different components in developing the total project like design, coding, testing, documenting, produce the CDs, etcetera.

So, then what you have to do, the person you have to estimate how much percentage will be taken up by each component; suppose, design will take 20 percent, programming will take 30 percent, testing will take 40 percent, documentation will take 5 percent and producing CD will take 5 percentage. Accordingly, this you have to compute, this say 20 percent of 500000 that is 100000, it will give to what, it will give to the design. Similarly, 30 percent of 500000 maybe 1.5 lakhs, it will give to programming and similarly, you assign this what calculate the proportions and calculate the percentages assign to different components.

So, similarly design can be say two types of design D 1, D 2 like; D 1 design and your database design the percentage at 10, 10 ,10 percentage each. So, you have to give 50000 each.

So, similarly program there can be two programming shows write writing down the code for the G U I and for the say may be here three programming's; say programming for this G U I, programming for the database and programming for the normal code. So, again 20 percent, 5 percent, 5 percent. So, the proportionate are calculated, documentation suppose, two types CD, prepares only one.

So, in this way you can divide this, what proportionate similarly, testing a divide into unit testing is T 1 integration testing T 2 system testing might be T 3. So, accordingly compute this percentage you will get the estimates for each of the phases. So, this is how the top down estimation this works.

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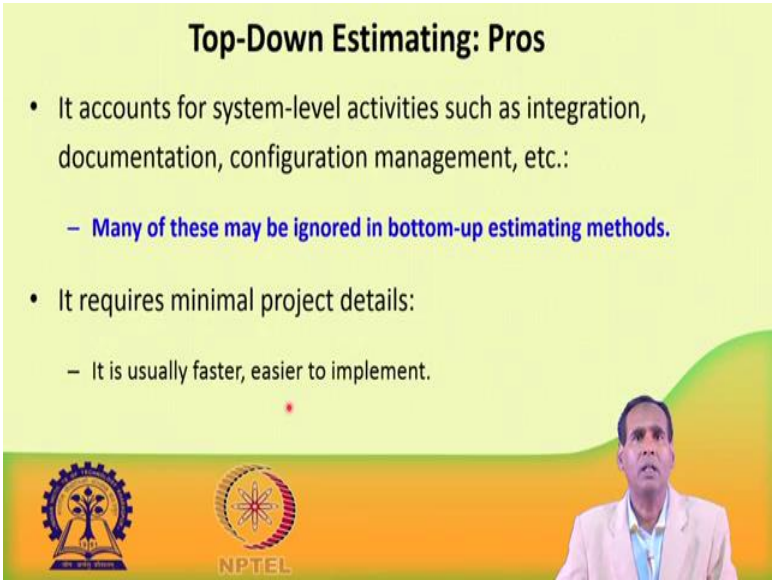

Let us see what are the advantages of top down approach, it accounts for system level activities such as integration testing or the integration, documentation, configuration management, etcetera.

So, top down estimation, it considers the various system level activities such as integration, documentation, configuration management, etcetera. Many of these may be ignored in bottom up estimation methods. We will see one of the drawback of bottom up estimation is that it does not take into account to this these what system level activities.

It requires minimal project details. So, this top down estimation, it requires very few amount of project details, very minimal project details. It is usually faster and easier to implement. These are the advantages of top down estimation.

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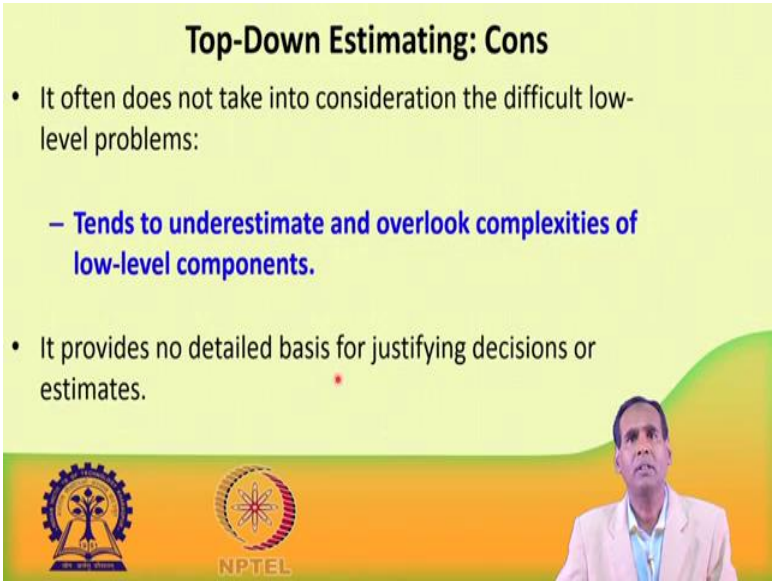

But there are also some disadvantages of the top down estimation such as; it upon does not taken to consideration the difficult low level problems, it takes to account this what system level activities, but it does not take into account the low level of problems. It tends to underestimate and overlook the complexity of the low level components. It does not take into account as well as it may tend to underestimate and overlook the complexities of low level components. It provides no details no detailed basis for justifying the decisions or estimates.

So, why you have taken this decision? Why you have done this estimate? It does not provide any detailed basis for justifying the decisions or the estimates that a project manager has made..

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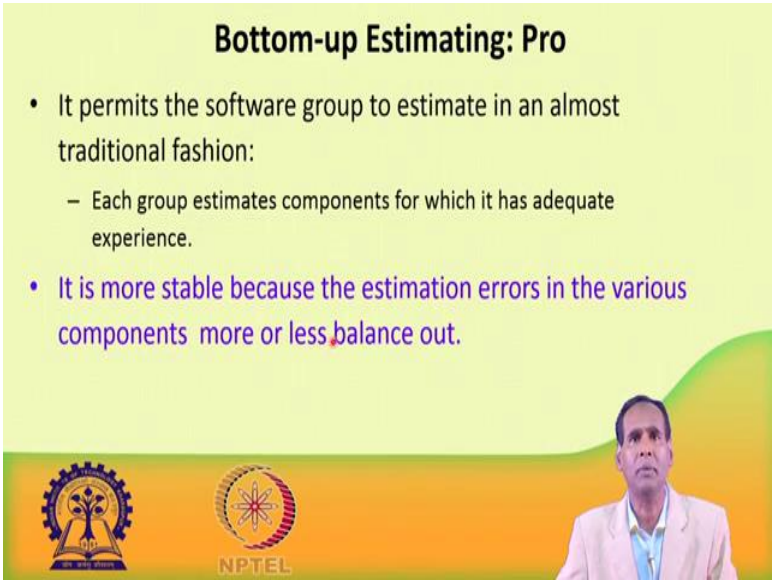

The on the what are let us see some of the what the advantages of bottom up estimation. It permits the software group to estimate in an almost traditional fashion. So, in a traditional fashion the software group or the project manager, they can estimate the various parameters. Each group estimate components for which is has adequate experience.

So, different groups involved in the software development, they will estimate different components for which they have adequate experience. It is more stable, because the estimation errors in the various components more or less balance out. So, different it is it is very stable, because the estimation errors in the various components, they are already balanced.

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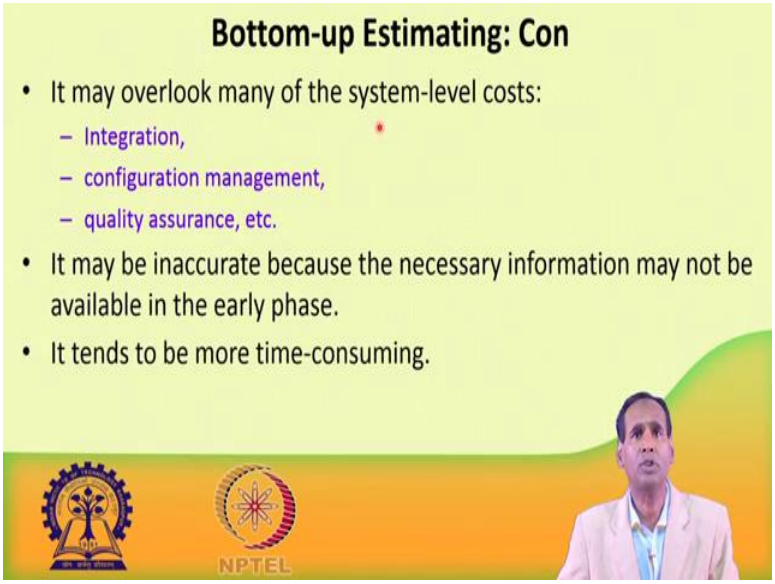

But the drawback, there are several drawbacks of bottom up estimation such as it may overlook many of the system level cost. This I have already told you earlier, the advantages of top down approach is that it takes into account the system level activities, but bottom up estimation may overlook many of the system level costs such as integration, configuration management, etcetera, quality assurance.

So, the cost that will be involved in this system level cost may be overlooked by this bottom up estimation process. It may be inaccurate, because the necessary information may not be available in the early phase. So, during the early phase may be like a system requirement system requirement analysis phase or this design phase. So, the necessary information may not be available. So, the estimate may be inaccurate if you are using this bottom up estimation. It is tends to be more time consuming, it is more it is more time consuming in comparison to the bottom up estimation.

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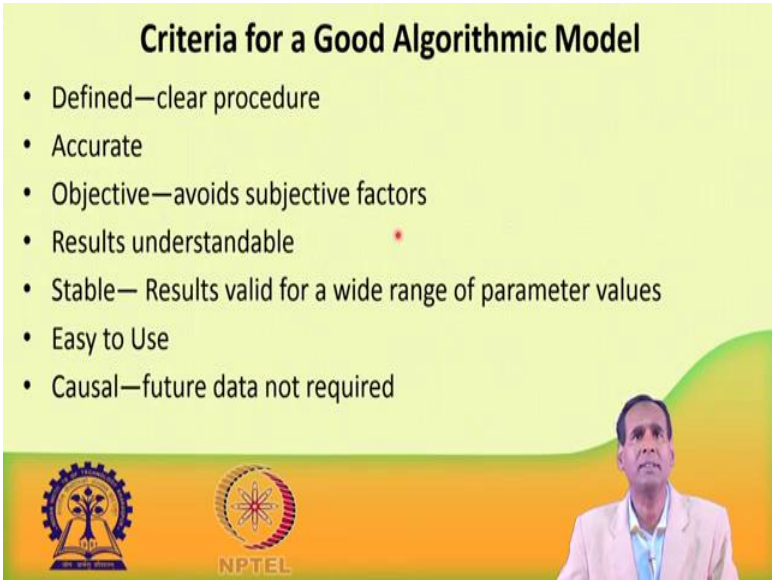

So, now let us see about another model called as the algorithmic model for this project estimation. Now, first let us see what are the criteria required for a good algorithmic model, then we will take up for the steps for this algorithmic model.

So, first of all this algorithmic model, if you want to develop a algorithmic model for project estimation, it should be clearly defined, the procedure should be very clearly defined. Then it is must be accurate. It should be objective; that means, you should avoid the subjective factors. It should only concentrate on the objective factors. The results should be understandable and it should be stable; that means, the results must be valid for a wide range of parameter values, not for a small range of parameter values and it should be easy to use. It should be very much causal; that means, future data not required only with the existing data current data. We should be able to estimate the various parameters. These are some of the criteria for a good algorithmic model.

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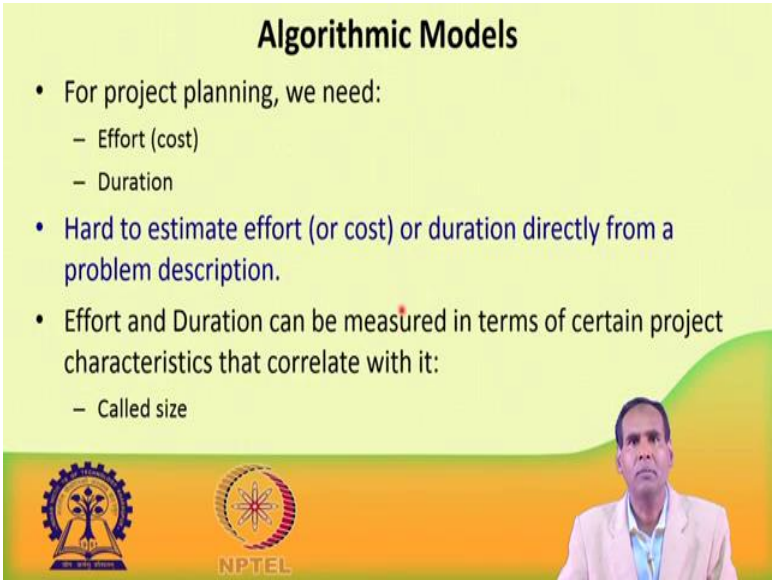

Now, let us see what could be the algorithm models. For project planning, we need to estimate what the report as well as the duration. We have already told you that directly estimating the effort duration this is very much difficult it is hard to estimate effort or cost or duration directly from a program description. So, that is why effort and the duration they can be measured in terms of certain project characteristics that correlate with it, for example, it is that call size, by using the size we can estimate effort and duration. So, first we have to estimate the size, by using the size estimate we can then estimate the effort and duration.

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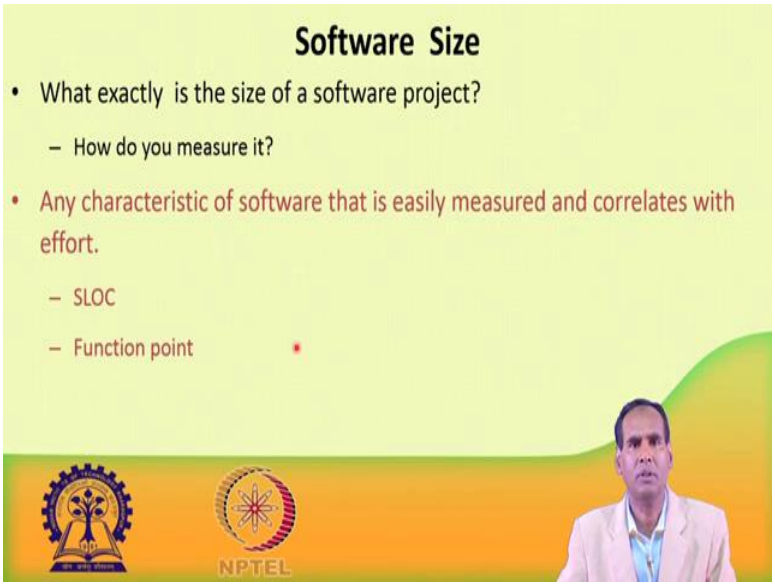

So, software size we have already discussed earlier, that what exactly this is this size of a software project, how do you measure it and I have already told you, we may use that two parameters S L O C and function point. So, any characteristic of software that is easily measured and correlates with effort that decides,, any characteristic of software that is easily measured and correlates with effort is known as what size and how can you measure it using what the metrics such as S L O C and function point or F P.

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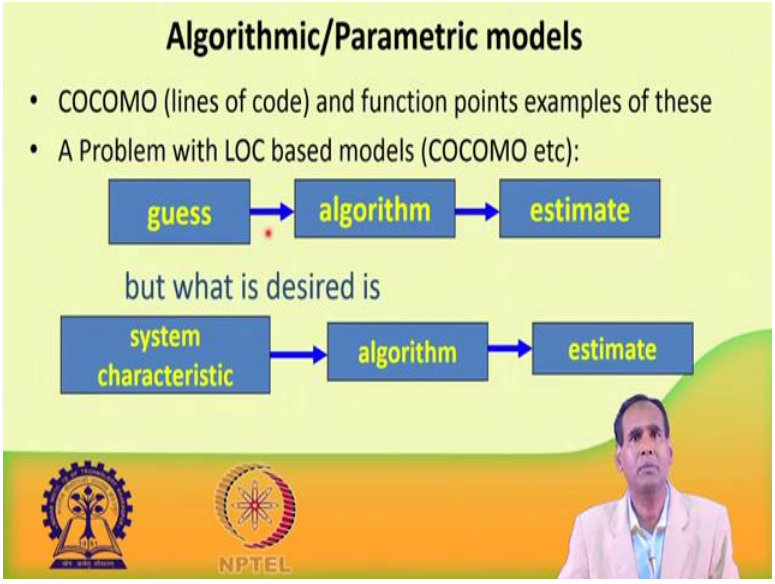

Now, this let us see the algorithmic models or now, let us see this algorithmic models or parametric models. So, two important examples are there; one is COCOMO of this lines of code and the function points, they are examples of these models.

Let us see one of the model with the L O C based model or the COCOMO. In either the L O C based model or the COCOMO we initially guess some value, we initially guess some value then apply the algorithm and estimate, but what actually desire is, we should what estimate the values not from starting from the guess, we should estimate the values starting from starting from the system characteristics, then we can apply the algorithm and then we can estimate. So, this is the problem with L O C based models or the COCOMOS. So, in order to overcome this, we may apply algorithmic or parametric models.

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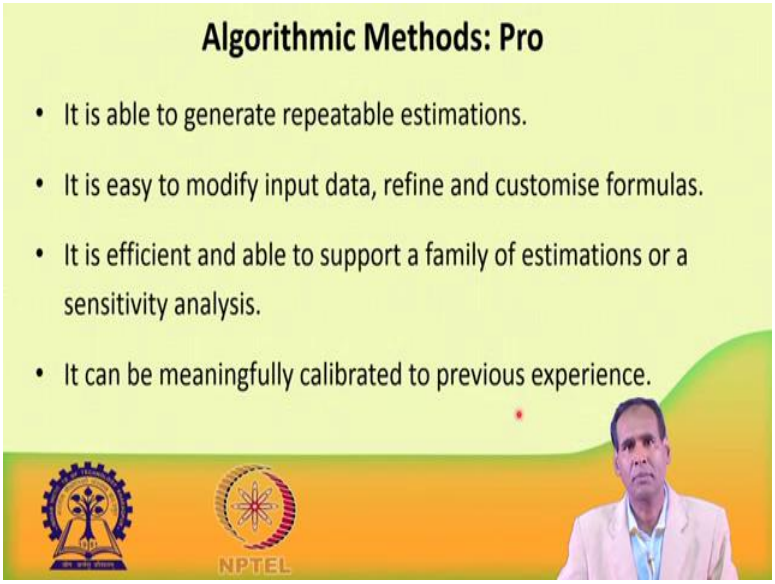

Now, let us see what are the advantages of algorithmic methods. It is able to generate a repeatable estimations. So, once you estimate for some parameter, you can generate repeatable estimations.

So, that is why it is repeatable in nature. It is the estimation that you are obtaining using algorithmic methods they are easy to modify ok. They are easy to modify input data, refine and customize the different formula. It is efficient and able to support a family of estimations or a sensitive analysis. So, this algorithm method it is efficient and it is able to support not only single estimations, but a family of estimations or a sensitive analysis. It can be meaningfully calibrated to previous experience. So, one of the most important advantage is that, it can be very meaningfully calibrated, you can do the calibration to previous experience.

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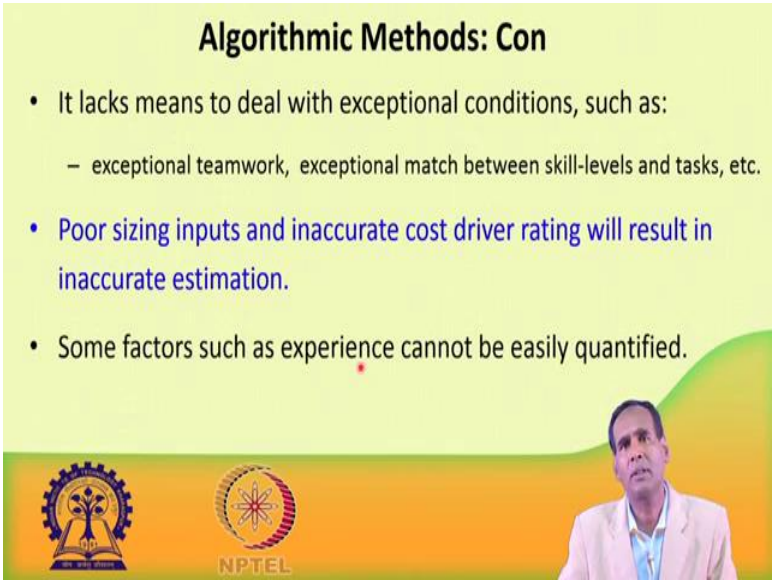

There are some drawbacks of algorithmic methods such as a it lacks the means to detail with to deal with exceptional conditions. So, a algorithm methods, they can deal with some exceptional conditions such as exceptional teamwork, exceptional match between skill levels and task etcetera. So, these exceptional conditions can easily sorry, they cannot be what dealt with using this algorithm methods.

So, poor sizing inputs and inaccurate cost driver rating will result in inaccurate estimation. So, if you are having what poor sizing inputs, if the cost driver ratings are inaccurate, then ultimately, they will result in inaccurate estimation. Some factors such as experience cannot be easily quantified. So, if you are using algorithmic methods. So, factors such as experience of somewhat coder or experience of some tester, it cannot be easily quantified.

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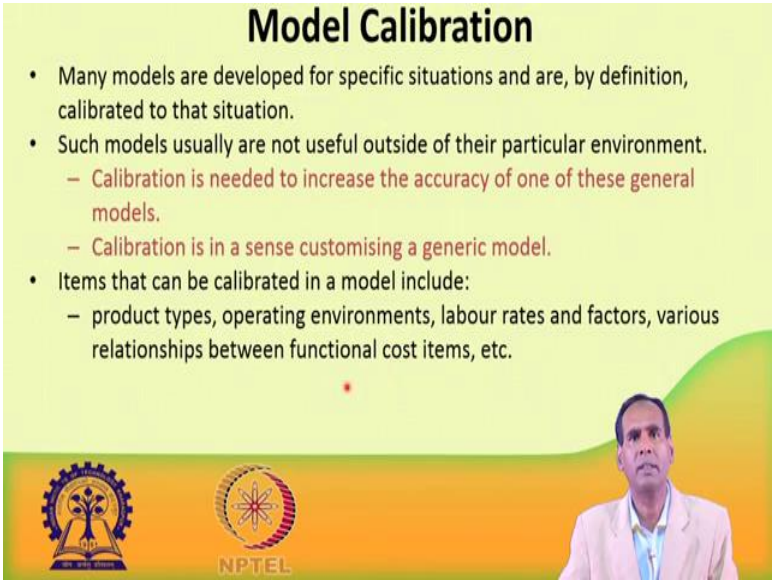

So, now let us as I have already told you that here one of the drawback sorry, one of the advantage here, I have already told you that the algorithmic methods, they can be meaningfully calibrated to previous experience. Let us see how we can do calibration to the previous experiences using the algorithmic methods. So, let us see the calibration, model calibration.

So, many models are developed for specific situations and are by definition calibrated to that situation. So, there are several models, they are developed for only specific situations, they are not generalized, they are developed for specific situations and by definition, they can be calibrated to that situation. Such models usually are not useful outside of that particular environment.

I have, as I have already told you, these models will be very much what a suitable for the specific situation, they cannot be useful outside the particular environment. Calibration is needed to increase the accuracy of one of these general models. So, why calibration is required? Calibration is needed to increase the accuracy of one of these general models. Calibration is a sense customizing a generic model. We have to a generic model is there and we want to customize it according to our own needs.

So, calibration is in a sense customizing a generic model. Items that can be calibrated in a model include let us say in a model what items can be calibrated. You can calibrate you can calibrate the product types, we can calibrate the operating environments. Similarly,

the other items they can be calibrated are labour rates and factors, various relationships between the functional cost items, etcetera.

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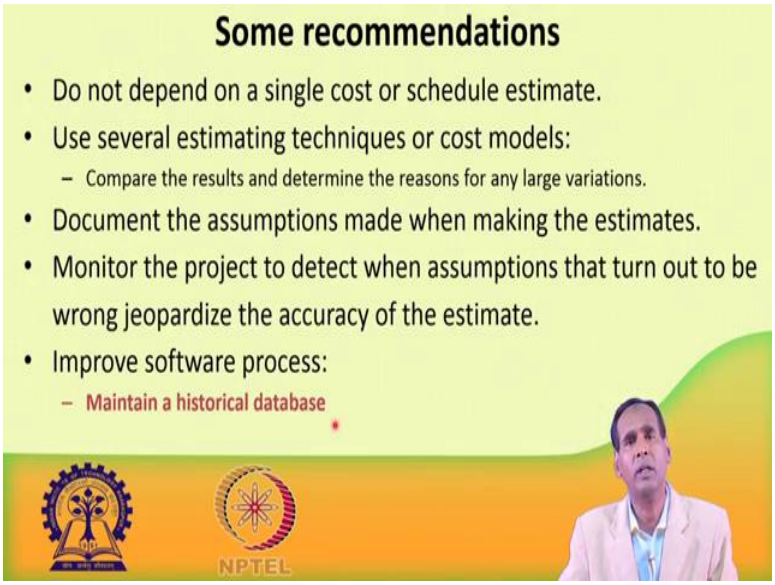

Now, let us see some of the recommendations for this what calibration. Do not depend or let us see some of the recommendations while using this estimation models some of the recommendations while using or while doing estimation. Do not depend on a single cost or schedule estimate. We have already seen that various methodologies are there. So far we have seen bottom up, top down and this what algorithmic, then this judgment based, pricing so many, so do not depend on a single cost or schedule estimate, because sometimes if you are using a single estimate or single technique, it may give wrong result.

So, that is why you may use several estimating technique or cost models, instead of using just a single estimation technique or a single model, you may use different or several estimating techniques or cost models, so that the result may be very much accurate. Then if you are using different estimation techniques or models for estimation, then what you can do after applying the different techniques or models for finding out or for estimating the parameters, then you may compare the results and then determine the reasons for any large variations.

They should; so whatever method may use, they should give very closer results. If the results are in the results there are large variations they may, you may analyze, why this

large variations are coming. So, then you note down them then document the assumptions made when making the estimate.

So, while you are using different techniques or models for estimation you may use different assumptions. You may take different assumptions, but whatever assumptions you have taken so you have to document all the assumptions met, when making the estimates. Then monitor the project to detect when assumptions that turned out to be wrong jeopardize the accuracy of the estimate. So, sometimes due to wrong assumptions, your accuracy, your estimate it becoming what less accurate, it is becoming wrong. So, monitor the project to detect when the assumptions that will turn out to be wrong jeopardizing the accuracy of the estimate, then improve the software process ok.

So, analyzing the estimates and if you are what observing that due to this assumptions you are getting what, you are getting wrong estimates, then you may improve this software process. So, how again; you can also improve the software process the whole process by maintaining a historical database. So because sometimes we require the past data, so you have to improve the software process by maintaining the historical database, containing the details of the past projects. So, these are some of the recommendations that you may follow as a project manager while carrying out project estimations.

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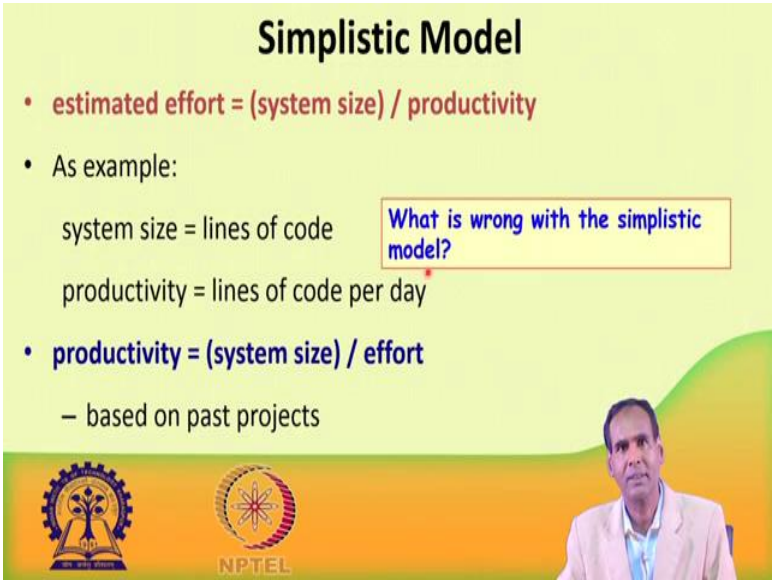

Now, let us take this the concept of simplistic model. According to this a very simple model, according to this model the effort is estimated as follows; estimated the effort is equal to system size divided by productivity and very straightforward formula.

$$\text{estimated effort} = (\text{system size}) / \text{productivity}$$

So, if the system size is some lines of code and the productivity is some lines of code per day, then productivity from this equation, you can find out that productivity equal to how much? System size divided by effort.

$$\text{productivity} = (\text{system size}) / \text{effort}$$

So, here this formula is based on the data relating to the past projects, but please think of what is wrong with the simplistic model.

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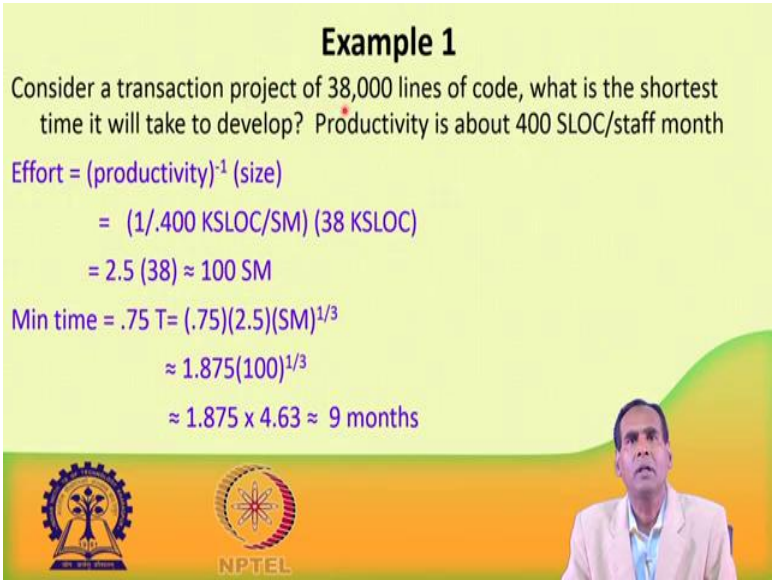

So, in which cases it will fail? So, please think of on this. Now, let us quickly take a small example; we are, we want to find out what these effort and the effort and this duration, we want to estimate the effort and this duration for developing this project. This example says that consider a transaction project of 38000 lines of code. So, this transaction project contains 38000 lines of code, we have to found out what is the shortest time that it will take to develop this project. Here, it is also given that productivity is about 4000 S L O C per staff month.

So, the productivity is given in terms of 400 S L O C per staff month. Now, we have to do this. As we can see that the formula of effort I have already given you earlier. So, effort can be calculated as system size by productivity or in that you can say that effort is equal to productivity to the power minus 1 or 1 by productivity into size, this is equal to 1 by productivity is given as 400 L O C or K L O C divided by ok, 1 by 0.400 K S L O C per staff month into size. Size is equal to how much? Size is equal to 38 equal 38000 lines of code; that means, 38 K S L O C, K S L O C. So, this is coming to be something this 2.5 into 38 is equal to approximately, 100 staff months will be required for developing the project.

So, here the estimated effort is coming to be 100 staff month. Now, the minimum time is required is equal to we have to find out that. So, this will be equal to 0.75 T and which is equal to 0.75 and T here, it is this effort 0.7 and into 2.5 that we have already got this thing, 1 by 400 something. So, this is equal to 0.75 into 2.5 S M to the power 1 by Q and this is coming to be 1.875 into and simplification it is coming to be around 9 months. So, for this transaction project having 38000 lines or 38 K S L O C and productivity of 400 S L O C for staff month we get that that, the effort is estimated to be 100 staff months and minimum time is estimated to be 9 months.

So, similar examples are there, you can get from the book or in places. Please, try to solve them, also some exercises you may solve them. So, these types of small exercises may be asked in the assignments as well as in the examination.

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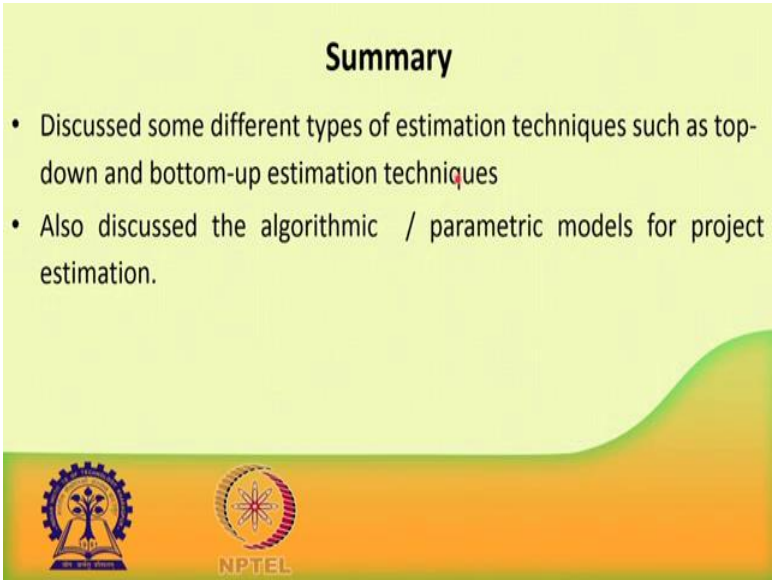

So, finally, we have discussed some different types of so, let us see the summary we have discussed some different types of estimation techniques here, in this class such as top down testing we have seen and bottom up estimation techniques we have seen and we have also compare the top down, the top down approaches and the bottom up estimation techniques. Also we have taken some examples to illustrate the top down and the bottom up estimation techniques. Then as also we have also discussed the algorithmic model or the parametric model for project estimation.

We have seen the advantages and disadvantages of algorithmic or parametric models then we have seen one simplistic model for project estimation. We have also taken an example for estimating the effort and time by using and this simplistic model for project estimation. So, these are another classifications of techniques for project estimation. In the next class we will discuss about another metric for estimating size that is function point that we will discuss in the next class. So, these are the references and we will stop here.

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Next class, we will see the remaining parts of this project estimation techniques.

Thank you very much.

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Lecture – 19
Project Estimation Techniques (Contd.)

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Good afternoon. Now, let us start the other Project Estimation Techniques. We will mainly a start about the parametric models, especially the Albrecht IFPUG function point model.

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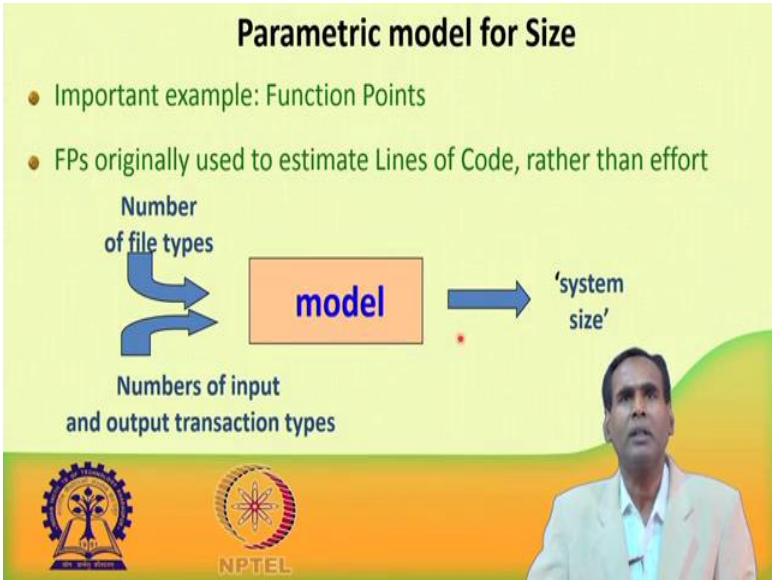

So, let us see what is a parametric model, and then we will see the parametric model for size. So, one of the important example for parametric model for size is function points. Function points, where normally used to estimate the lines of code, rather than effort. Say effort directly cannot be estimated from function point, first we have to estimate the lines of code and by using the lines of code then we can estimate the effort using this function points. So, this model works like this. It takes the number of file types and the number of input and output transaction types are the input, and then it processes something and then it will produce the system size as the output.

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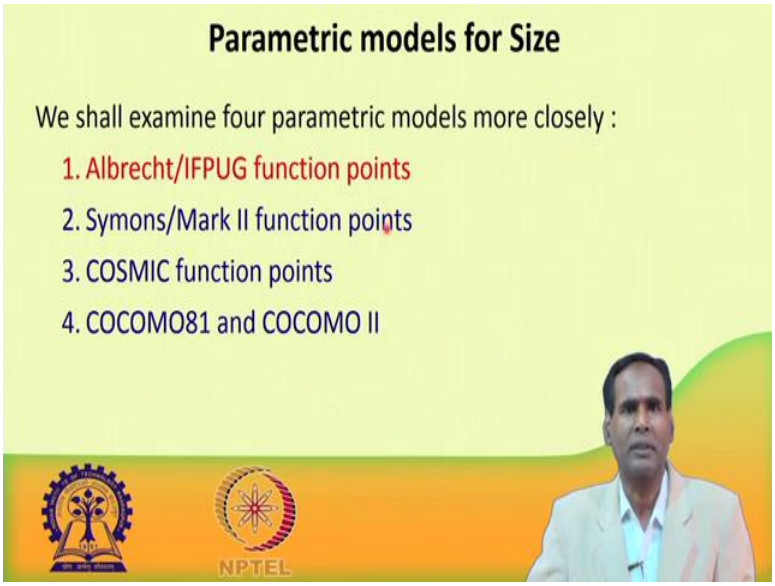

See we have seen there are many drawbacks of the LOC model that we have seen yesterday. So, this function point approach, we will try to resolve some of the issues though they are though they are appeared in case of these LOC. Now, let us see what are the different parametric models for size. So, first one we will see that Albrecht IFPUG function points, then we will discuss Symons Mark II function points, then COSMIC function points, and then COCOMO81 and COCOMO II. So, out of all those parametric models, today we will discuss right now in this class Albrecht IFPUG function point.

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Now, let us see how does it work; so what are the purpose, what are the benefits and what are the extent of use. So, IFPUG it stands for International Function Points Users Group. So, the basic purpose of this group was to promote and encourage use of function points, because we have already seen LOC has several drawbacks. To another objective of this group is to develop consistent and accurate counting guidelines.

Several benefits are there which can be obtained from IFPUG. So, like the benefits will be networking with the other counters, so you can make networking with the other counters. Similarly, IFPUG counting practices manuals are also available, they it a supports research projects, hotline, news papers, certification or some of the other benefits of this IFPUG. The extent of use is like is the member companies they include almost all industries industry sectors. Over 1200 members are there from across from 30 countries in this group.

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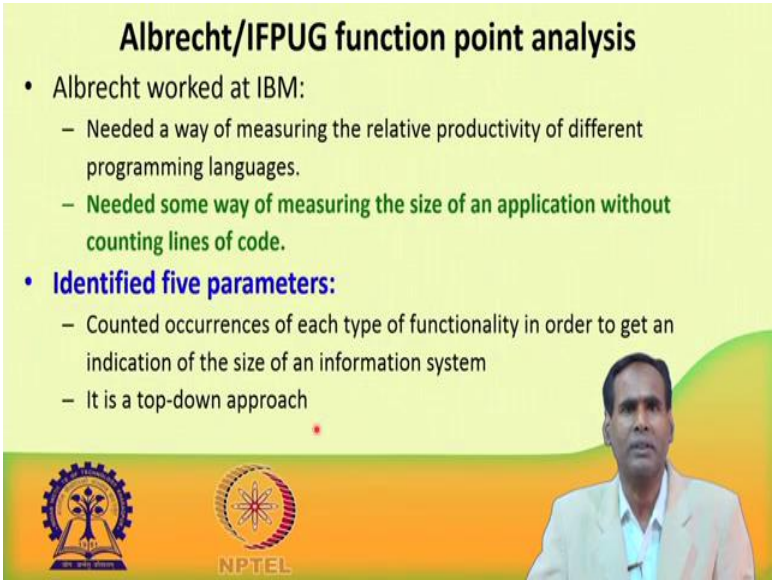

Actually this Albrecht IFPUG function point it was coined by Albrecht's. So, while Albrecht was working at IBM he found that there is a need of measuring the relative productivity of different programming languages, so he was using different programming languages; he observed that there is a pressing need to measure the relative productivity of different programming languages. So, he also needed some way of measuring the size of an application without counting the lines of code, because the counting lines of code has several drawbacks.

So, he has identified five parameters for performing the function point analysis let us see. And he has counted the occurrence of each type of functionality in order to get an indication of the size of an information system. So, in order to estimate the size of an information system, he has counted he has first identify five parameters and counted the occurrences of each type of functionality.

Let us see and basically yesterday we have last class we have discussed about these, the different project estimation techniques such as top-down approach, bottom-up approach, etcetera. So, this Albrecht IFPUG function point analysis is based on a top-down approach.

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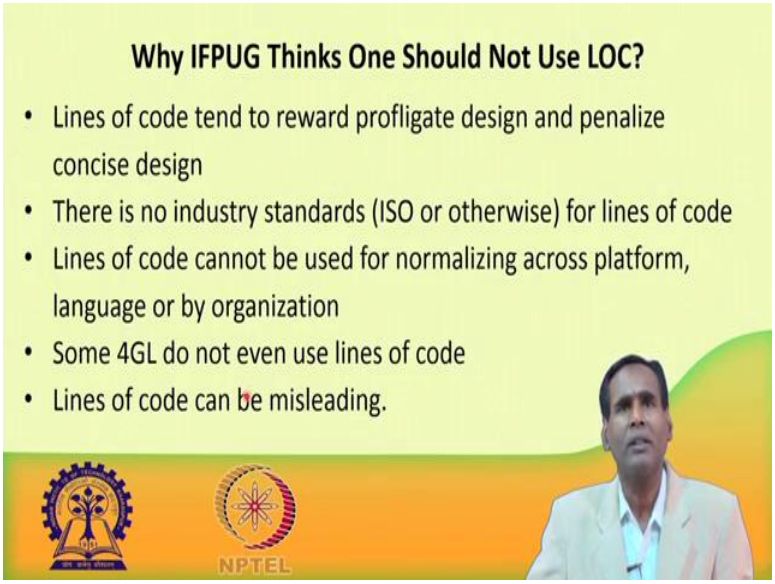

Now, let us see why IFPUG thinks that one should not use LOC rather this would use function point. We have already seen the drawbacks of LOC in the last class, till then let us quickly revise what we have seen earlier are the drawbacks of LOC. We know that lines of code it tend to reward profligate design and penalize concise design; so if your design is concise, then it will penalize you.

And there is no industry standard ISO or otherwise for lines of code, so that is another drawback we will see how function point is influenced somewhere else ISO standards. And lines of code we know that it cannot be used for normalizing across different platforms or different languages or by different organizations. Some 4GL they do not have at all the use of these lines of code. So, then lines of code will made might will not be what suitable for those languages, also lines of code it can be misleading. So, these are some of the reasons why one should not use LOC rather than he should try for a better major such as a function point.

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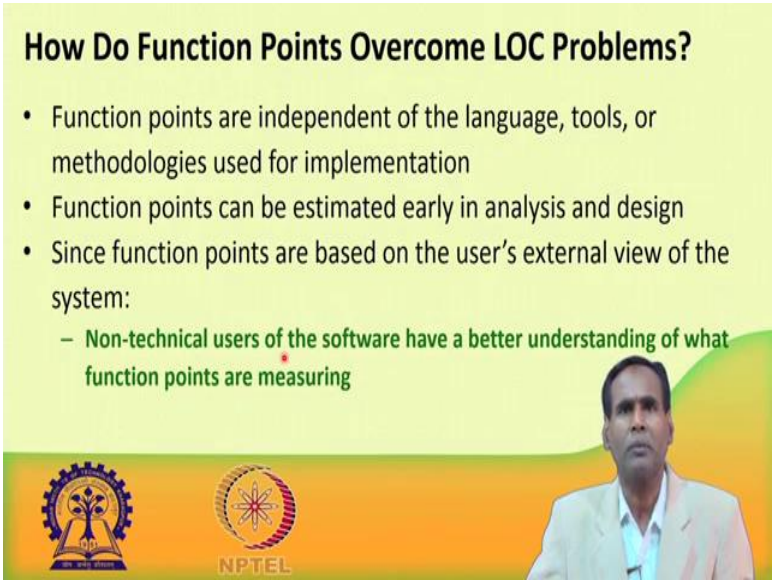

How do function points over come the LOC problems, number one the function points are independent of the language, we have already seen that LOC is somehow language dependent platform dependent, but function points that independent of languages, independents of tools or independent of methodologies those are used for implementation. Then function points they can also be estimated early in analysis and design, we have seen that LOC by LOC approach cannot be used in the earliest stages of analysis design such as requirement analysis, specification and design whereas, function point can be used to estimate the what size in early stages of software development such as analysis and design.

So, since function points are based on the users external view of the system you will see that even if any layman non-technical users of the software, they can also have better understanding of what function points are measuring ok. So, this function points they are based on what the users external behave, the user's perspective, so that is why the non-technical users of the software or the projects they can also better understand what this function points are doing or their measuring.

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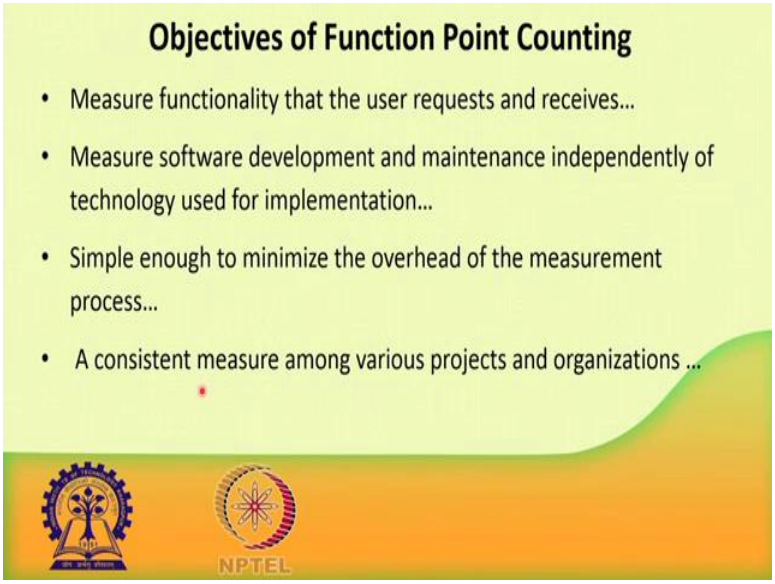

Now, let us see what are the objectives of function point counting. It measures functionality that the user requests and receives. So, it will basically measure the functionality that the user request and the user receives. Similarly, it measures the software development and maintenance independently of technology used for implementation, this we have already told you that it independent of any technology used, so that is why it can measure the software development and the maintenance activities efforts.

So, it is very simple enough to minimize the overhead of the measurement process. So, by using function point you can minimize the overhead of the measurement process. And similarly, another objective is it act has a consistent measure among various projects and organizations. So, it acts has a consistent measure among different projects and organizations.

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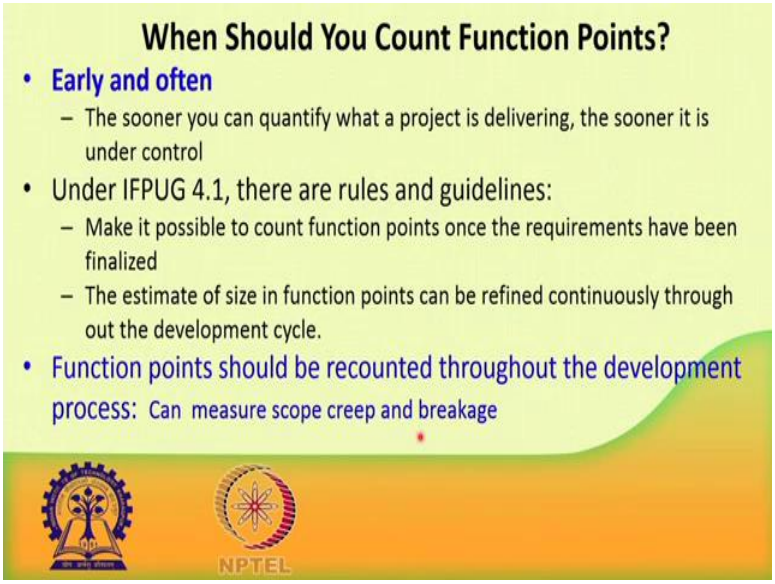

Then why should we use count function points, when should you count the function points? So, as I have already told you that you can use this thing not only coding, but before the coding starts may be requirement analysis phase and design, so that is why I saying it can be what a used in the early phases of software development and vary often it can be used. So, you see that the sooner you can quantify what a project is delivering, the sooner it is control it will under better control, you can monitor it.

Under IFPUG 4.1, there are several rules and guidelines; these rules and guidelines make it possible to count the function points once the requirements have been fixed. So, once the requirements have been frozen, then you can easily a count the function points by using the rules and guidelines provided in IFPUG 4.1.

The estimate of size in function points can be refined continuously throughout the development cycle, please see it is not just estimated only once. So, you first you have to make an initial estimate, then the estimates they can be refined continuously or you get different information updated information throughout the development lifecycle.

The function points please remember that there should be recounted throughout the development process. They can be refined throughout the development process, it hence it can measure the scope or creep and the breakage.

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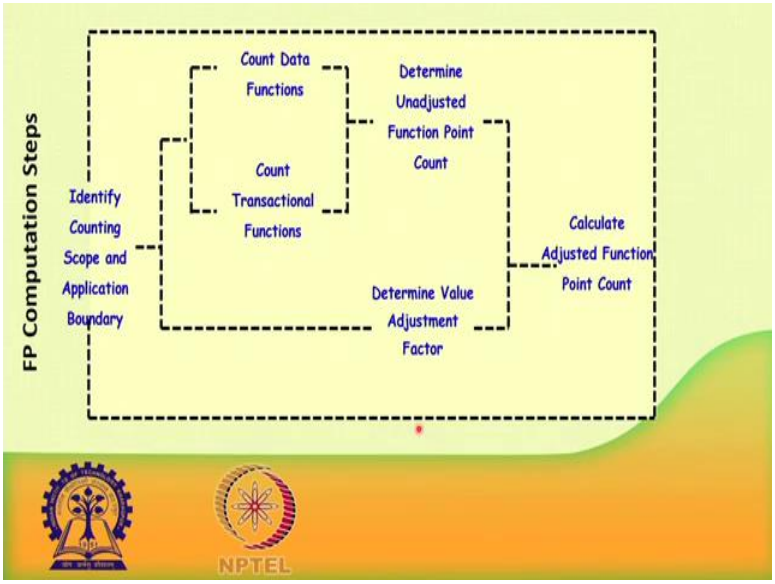

Let us see what are the steps involved in the function point computation. So, first you have to identify in the first step you have to identify, the counting scope and application boundary what is this scope of the counting and what is the application boundary first that has to be identified. Then you have to count two things, you have to count the data functions as well as you have to count the transactional functions. Here in data functions, like input, output those input data, output data, etcetera you have to count; and in count transactional functions what are the mainly file transactions, etcetera or the interfaces that you have to count.

Then you have to determine the function point and this is now unadjusted, you have not refined anything else. So, we this is the first time we are getting we, we call it as unadjusted function point. So, by using the what count data functions and count transaction functions, you can determine the unadjusted function point count. And then you have to calculate, because I have already told you that you may you have to refine the function points.

So, how this can be refined by using a factor called as value adjustment factor, there is a formula for finding this I will tell after few minutes. So, then we have to determine the value adjustment factor. Now, this unadjusted function point and this what adjust value adjust factor, they will be multiplied to give you the adjusted function point count. So, this is how the function point computation steps they follow.

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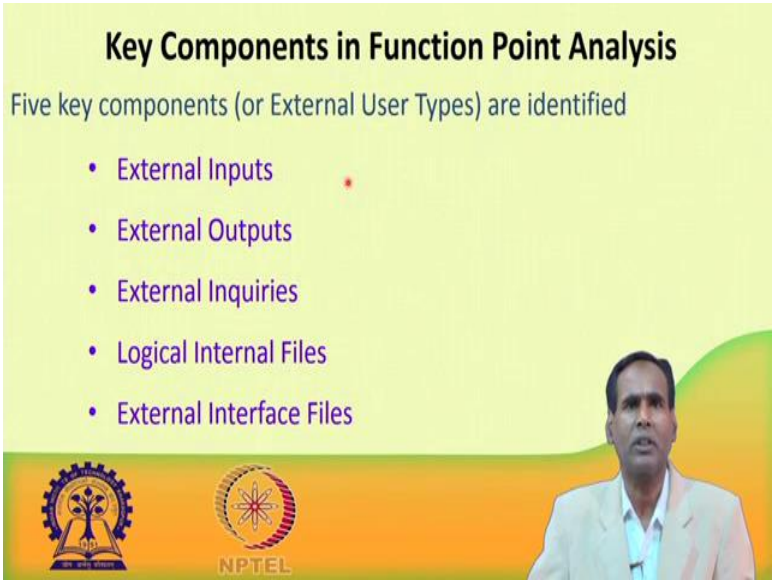

Let us see what are the key components of function point analysis, there are five key components or external use types they are identified in function point analysis that the external inputs, external outputs, external inquiries, logical internal files and external interface files. So, these are the basic five components they will be used in the function point analysis. They are also known as in Albrecht terminology, they are also known as external user types.

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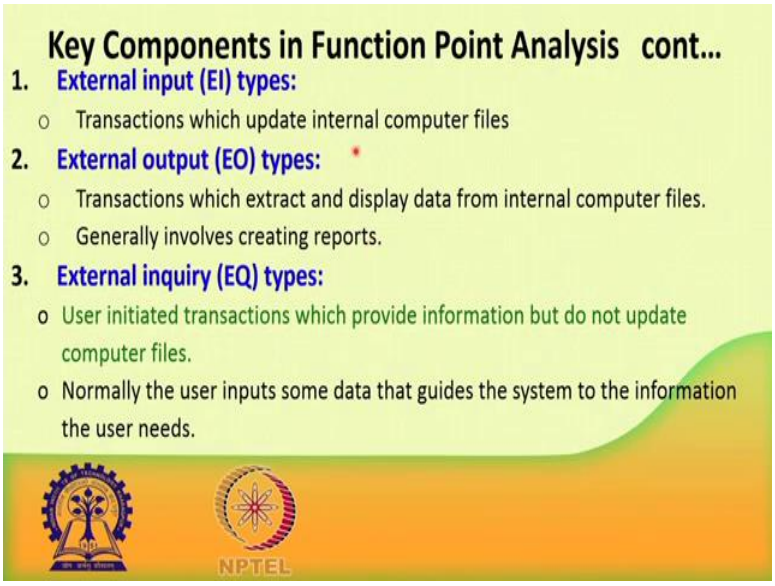

Now, let us see we will define so five components I have told, let us quickly define although there be or their usual meanings. So, external input in sort we called as EI, they represent the transactions which update the internal computer files. So, which will update the internal computer files; so, these operations these functions are known as external input types. And external output types, here these are the transactions which extract and display the data from internal computer files from the internal computer files what will do, so the transaction which will extract the information, so basically these are output types. And generally involves creating different reports of course, if you are where creating report this will be an output type of what function.

And next one is external enquiry types. So, here the user initiated actions which provide information, but do not update the computer files. So, you can see that in the external input type the transactions they update the computer files, but this is just inquiry and inquiry no updation will be there, only the user initiated the transactions which provide only the information, but do not update the computer files. Normally the user inputs what some data that guides the system to the information the user needs ok. So, here in external enquiry types the user will input some data which will guide the system to the information, to which guides the system to the information that the user needs.

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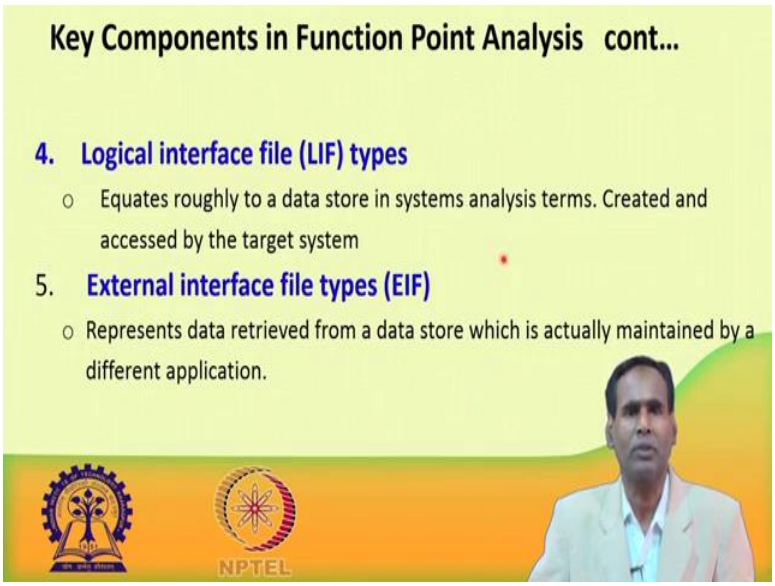

So, next one its logical interface or LIF files. So, this equates roughly to data store in a system analysis design terms. So, those who have studies system analysis design terms,

you must have study some data stores. So, here this LIF this equates roughly, this is equivalent to the what data stores. Created and so normally these types are created and accessed by the target system. And external interface file types this represents or these things represent, the data retrieved from a data store which is actually maintained by another different application that is why it is known as external interface file types.

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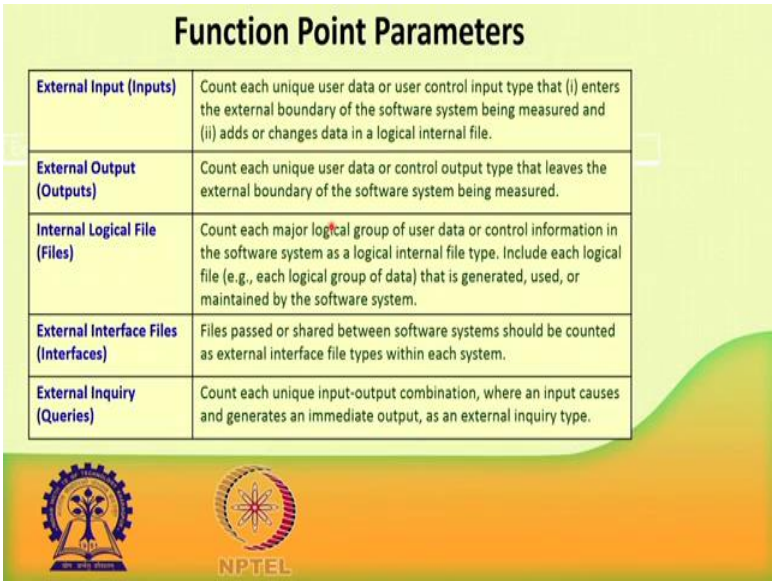

See these are the function point parameters that we have discussed five and basically what we have discussed all those things have been summarized here; more explanation is given you can see yourself.

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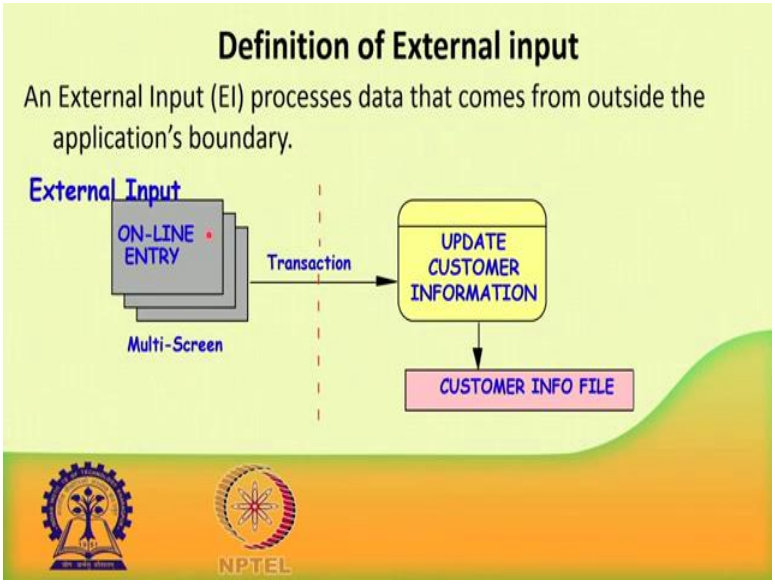

Then let us see with some examples of the definitions we have given, as I have already told you that external input it process the data that comes from outside the applications boundary from somewhere. External input it is giving and it is a so for example that some online entry and the external input is giving, and then what is there this transaction what will happen, it will update the customer information and then the where the customer information file, this file will be updated. So, here this external input it will process the data that comes from the outside application from a boundary, it will update the data.

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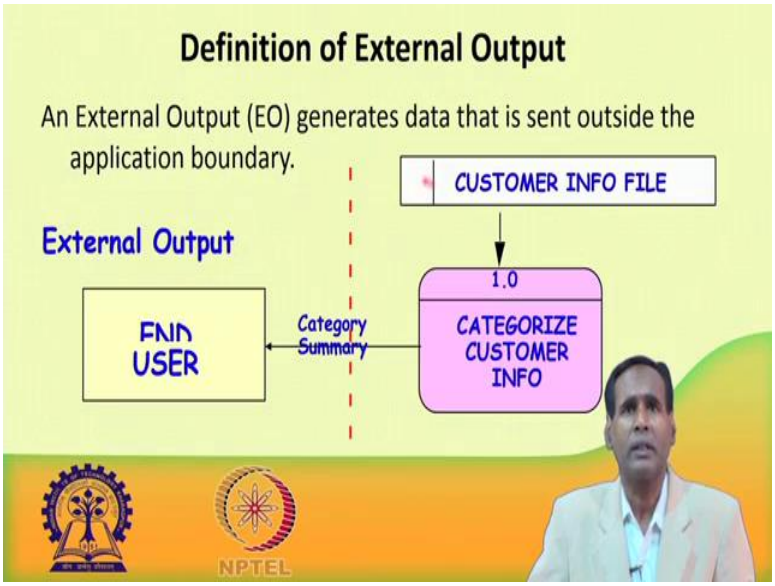

And in the external output, it generates data that is sent outside the application boundary. Here data is coming from outside application boundary in external input, but in external output the data is sent outside the application boundary. Please, you see here that say this is the customer inform information file and then there is a process what categories customer info, this sends the information like the category summery to the outside to the end user which is an external output, so that is why these an example of external output.

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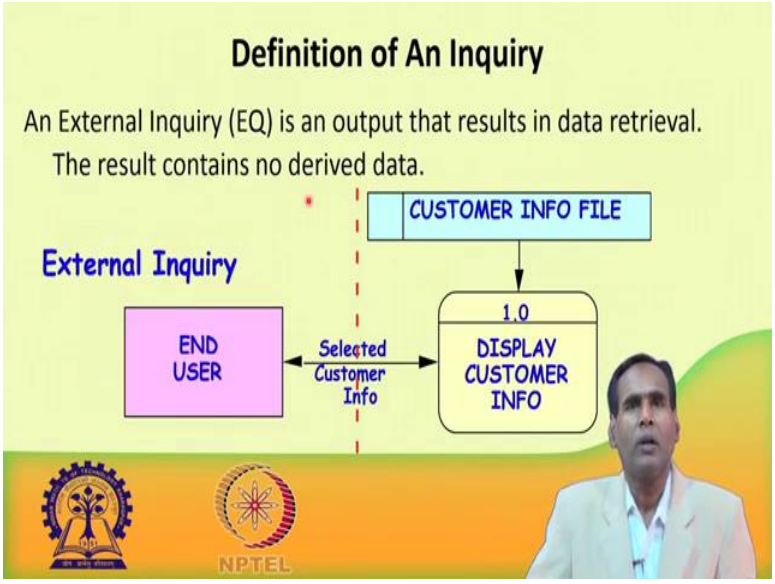

What is about an inquiry? So, on external enquiry is an output that results in data retrieval. The results contains no derive data. So, here you cannot what update anything else only that produces what some data it results in some data retrieval. So, here and this result cannot contain any derived the data.

So, here the end user asking some query inquiry and like what this inquiry might be what get the details of the customer information, then these are the easy process display customer info. This is getting the data from the customer info file and after getting data it is passing to the end user, so that is why it is an example of external enquiry.

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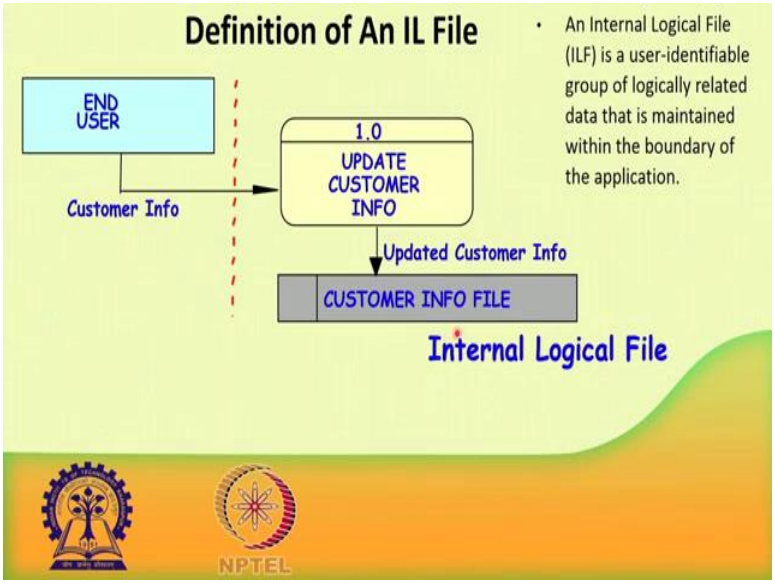

Next is a what definition of what internal logic file. So, an internal logic file is a user-identifiable group of logically related data. So, basically this is a what group of logically related data that is maintained where, within the boundary of the application; not external please remember this is not external, this is maintained within the boundary of the application, like this you have see.

The end user what send some customer info, then this process update customer info it is receives that information and accordingly it updates the customer info where in the customer info file. So, customer info file will be treated an internal logical file.

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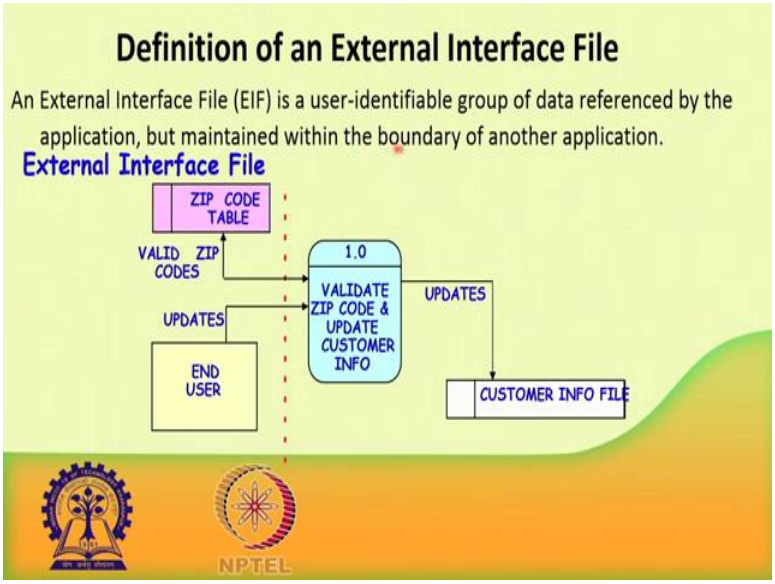

And last one is the external interface file. So, this is a user-identifiable group of data which are reference by the application. So, basically this is a group of data which are referenced by the application, but it is maintained within the boundary of another application; it is not internal it is maintained within the boundary of another application that is why the name is external interface file.

We have already seen in internal what is happening this is within the boundary of the application that is why this is internal logical file, but here you can see this file its outside the boundary so ok, so that is why this is maintained within the boundary of another application. This is outside the boundary of this present application it is somewhere outside that is why the name is external interface file. For example, here of the end user update something like validate zip code and zip code in stores somewhere else zip code table. So, for validation the message has to go here and come here, and then it will update the customer info file. So, here zip code table represents on external interface file.

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Now, let us quickly take a small example we want to place a purchase order and the input data items are like date, supplier, number, product code, quantity required, date required, etcetera. Output data items are like purchase number; this will be generated by the system. Entities referenced are these types are entities are referred like product, purchase order, supplier, purchase order item, etcetera. Now, we want to estimate the size of the system, let us see how can do it.

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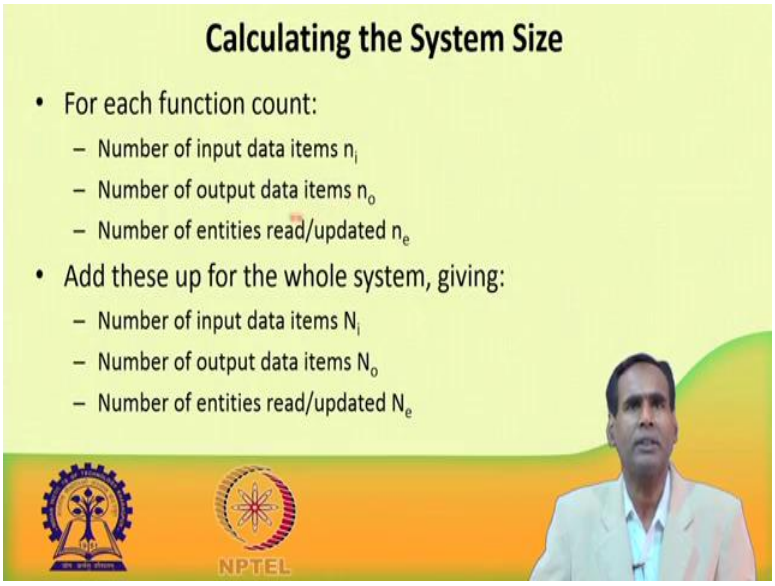

So, for calculating the system size you are supposed to do this things what you do, for each function count first find the number of input data items that is n i, then the number of output data items that is n o, then the number of entities which are rate or updated that is n e. And then after finding out the function count for all, after finding out the function count after finding out these for each function count, then you are these for the whole system. Then these are to be added, this will give you the number of input data items N i for the whole system, number of output data items N o for the whole system and number of entities read or updated that is N e for the whole system.

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If you will take a small example, this is what a sample example where these are the requirements ok. So, for these are the functionalities for these requirements are functionalities. So, for A 1, 10 inputs are there; for A 1, 2 outputs are there and for A 2 entity accesses is 4. Like this for there are 12 requirements find out the inputs, for all 12 requirements find out the outputs, for all the what 12 requirements then found out the entity accessed the counts, for all the entity accessed.

Then what you do, these are all small ends, these are these are all we can see this for each count for each function count we represent with small n and for the whole system it will be depended with capital N. So, you can see the capital N i will be just summation of these counts that is 120; and similarly, the number of outputs of for the whole system that is N o is equal to 120; similarly for the number of entities accessed for the whole

system will be the sum of this counts which is equal to 60. So, in this way we can find out the what if inputs, outputs, entity accessed which roughly what correspond to the size of the whole application in this manner.

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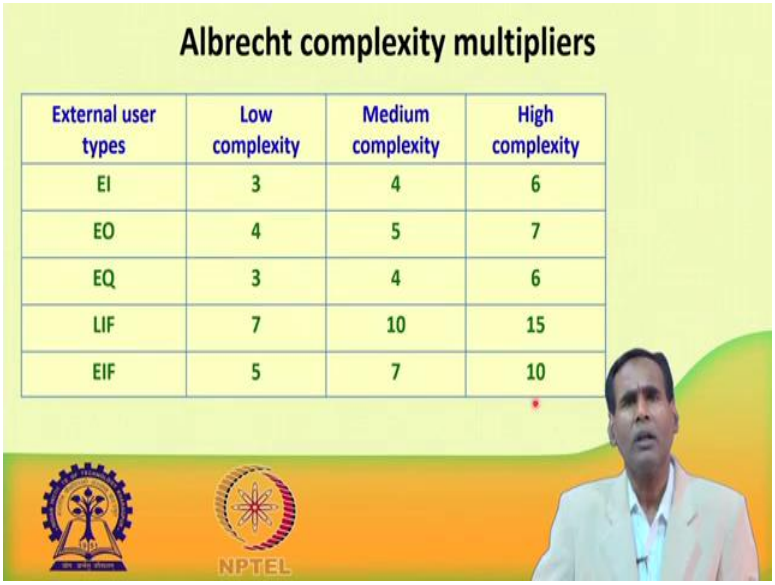

Now, what will see, that Albrecht has the proposed a formula for finding out the function point for different applications.

$$UFP = \sum(\text{number of elements of given type}) \times (\text{weight})$$

So, he has actually categorized the external user types into different categories such as some of the external user types might be of low complex, some of them are medium complex and some of them are high complex. Accordingly he has proposed some weights, some multipliers for these external user types depending upon the complexity.

So, like for the for external input if it is a low complex or low complex application, it is 3; for a medium complex application EI is equal to 4; for high complex application the value of EI is equal to 6. So, in this way he has proposed somewhere some multiplies for the what different external user types, this is known as Albrecht, these are known as Albrecht complexity multipliers, those multipliers I will use them in this calculation of function points.

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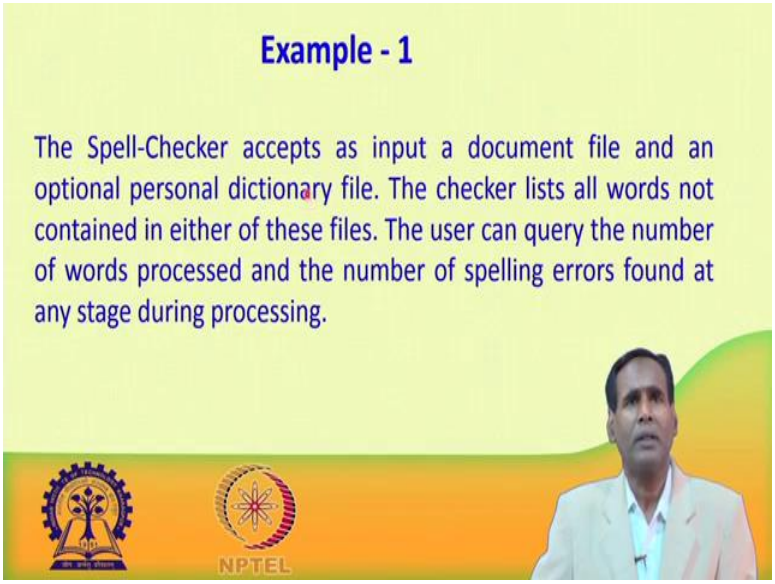

Let us quickly take a small example, how you can find out the there is a you want to develop a software for spell-checking. The spell-checker accepts as input what a document file and an optional personal directory file. Then the spell checker, it lists all words which are not content in either of these files. The user can query the number of words processed the user can query the number of words processed and the number of spelling errors which are found at any stage during the processing.

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So, if you can draw if you have I hope you have known data flow diagram, so you can easily draw the data flow diagram or the context diagram for this, where you can see that user can send what this information to the process. And, the process can what do some processing, this spell checker can do some processing and give this outputs and now we want to find out the function point.

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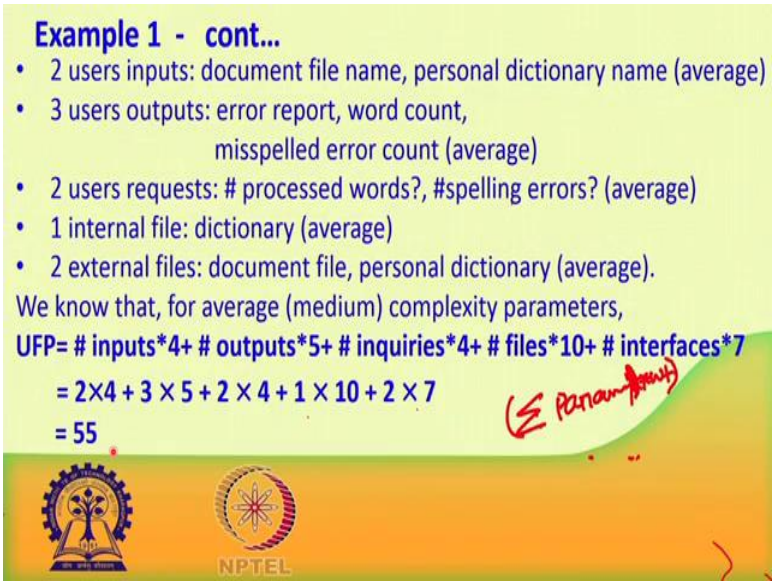

How can be done, see if you look at this you can see that the 2 inputs; what are the 2 inputs, if you will see the program the example 2 inputs, on a document file and an optional personal directory. So, these are two inputs I have shown that are 2 user inputs document filename and personal dictionary name and they can be what rated as average complex or medium complex. There are 3 user outputs, what are they error or report, word count and misspelled error count, also they will treated as average complexity or medium complexity.

The user request two things there will be 2 queries, what are the 2 queries; number of a processed words and number of spelling errors, they can be made as queries. So, they may be treated as average, so that are 2 user queries or 2 user request. There is 1 internal logical file that is the dictionary, where all those what is have stored and there is a 2 external files, what are they; the document file as well as the personal directory, personal dictionary average. See those things have so described in the problem as well as they

have been also represented in the diagram as the input and output that you can look at the diagram and see yourself.

Now, our objective is to find out the function point. So, this Albrecht has proposed a formula and according to this formula, let us see where this formula must be there, this formula ok. So, now the formula says like that the

UFP= # inputs*4+ # outputs*5+ # inquiries*4+ # files*10+ # interfaces*7

So, there is a formula you can use. So, the UFP can be computed as like this that summation of this parameter whatever you are taking and into it will be the weight. So, you have to take up this parameter and then the weight there will be multiplied, there will be 'into' here and take the summation. So, if we can see that here, the UFP is computed as the here what are the parameters, the parameters are ok. So, now we can see that the number of UFP is equal to number of inputs into 4 plus number of outputs into 5 plus number of inquiries into 4 plus number of files into 10 plus number of interfaces into 7.

So, then what is happening, so here basically the these are the weights; 4, 5 and 4, 10 these are the weights, the weights are chosen for an average project, because see all the parameters are type average. So, these are average values has been taken and then you multiply them and they take the summation. So, finally you are you are observing that UFP is coming to be 55. So, this is the unadjusted function point. So, after this unadjusted function point we have to this is the initial value, then there can be what then that can be refined.

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Now, let us see how this can be refined. So, Albrecht had proposed a 14 general system characteristics these are evaluated and used to compute a value adjustment factor known as VAF or sometimes it is known as TCF technically Technical Complexity Factors. So, these are the general system characteristics data come like data communication, distributed data processing, etcetera. And the final calculation is based on what finding out the unadjusted function point which we have already used this formula, then this what value adjustment factor.

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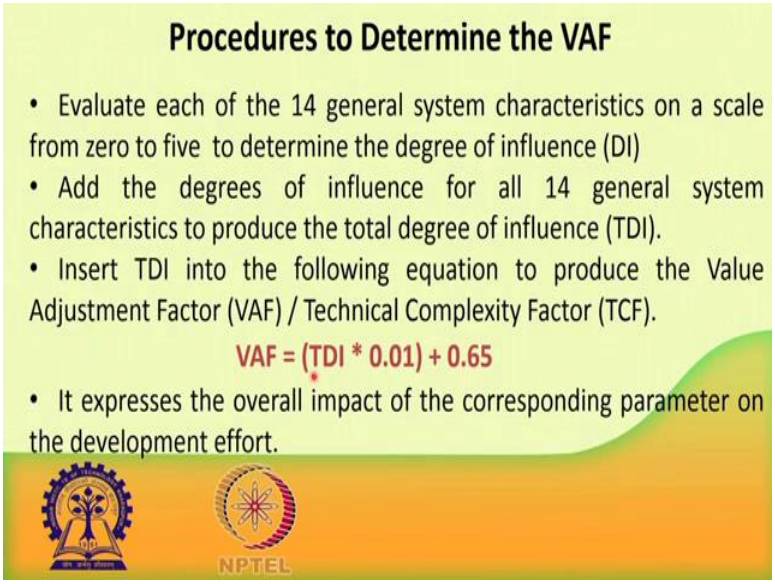

So, there is a formula for computing the value adjustment factor, this is like given like this

$$VAF = (TDI * 0.01) + 0.65$$

So, what is TDI, let us say.

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So, so what is happened that these are 14 parameters they are given some values within the range 0 to 5, these values are known as degree of influence or DI; where 0 represent not present or no influence one is incidental influence and 5 with this strong influence of the parameter.

So, here they how to process calculate the value of the VAF that is written here that we have to evaluate each of the 14 general system characteristics on the scale of 0 to 5 in order to determine the value of a DI, which is known as degree of influence. So, then we have to add all the degree of influence for all the 14 general systems, this will give raise to the TDI that is total degree of influence.

Then we have to insert the value of TDI in these formula to find out the value of VAF that the value adjustment factor or complexity factor why, we want to refine the value of the function point. So, VAF is equal to TDI into 0.01 plus 0.65, this Albrecht has already found out from his research. So, this VAF or this TCF, it expresses the overall impact of the corresponding parameter on the development effort.

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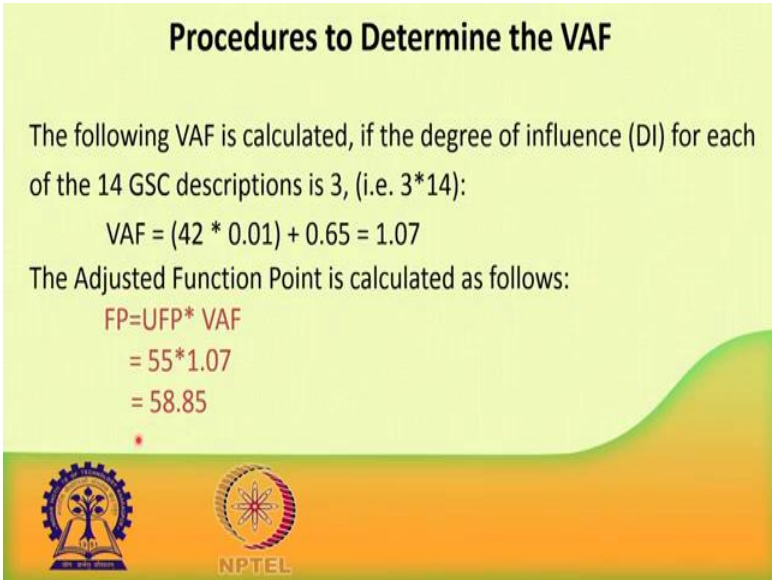

So, this is the procedure to do. Now, let us see that same example will continue where there are suppose 14 general system characteristics. And let us assume that these values almost average, it is 3, because I have already told you the value lie in between 0 to 5, suppose this is 3 then how you can compute. So, VAF formula I have already told you here that TDI and how TDI will be computed; if for 1, the value is at the degree of influence is 3 and 14 parameters so 3 into 14, so that is 42. So, value adjustment factor will be 42 into how much zero point see TDI into 0.01. So, this will be 42 into 0.01 plus a constant 0.65 like this it is 1.07.

So, finally the adjusted function point it can be calculated as follows,

FP=UFP* VAF

you have to multiply the unadjusted function point into with the value adjustment factor, so after these you will get that 58.85. So, this is the unadjusted function point for the given what spell-checker application that is the adjusted function point. Now, let us take an example here I have chosen all the what degree of influence for every what that parameter, it is of what 3 uniform.

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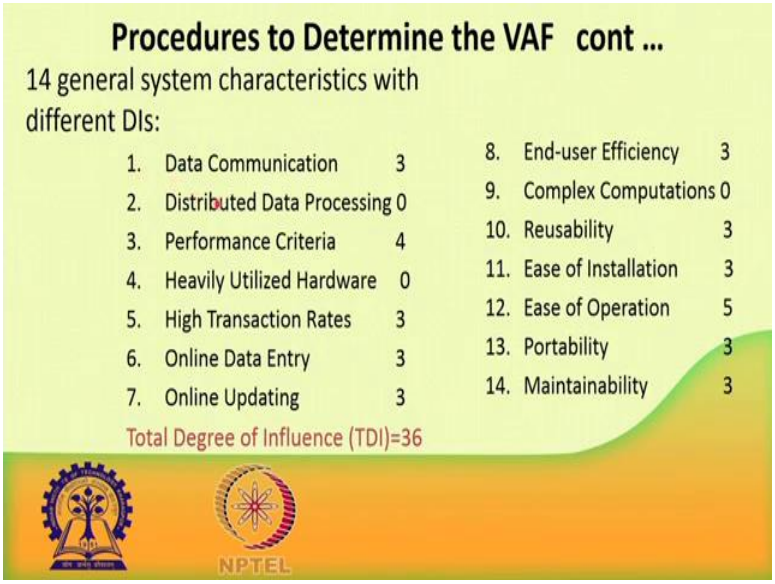

But if it is difficult different like the 14 general system characteristics with the different DIs, like for something 3, something 0, something 4 and something 5; so it is ranging in between it is ranging in between 0 to 5. So, then the total degree of influence is the summation of these, this is 36.

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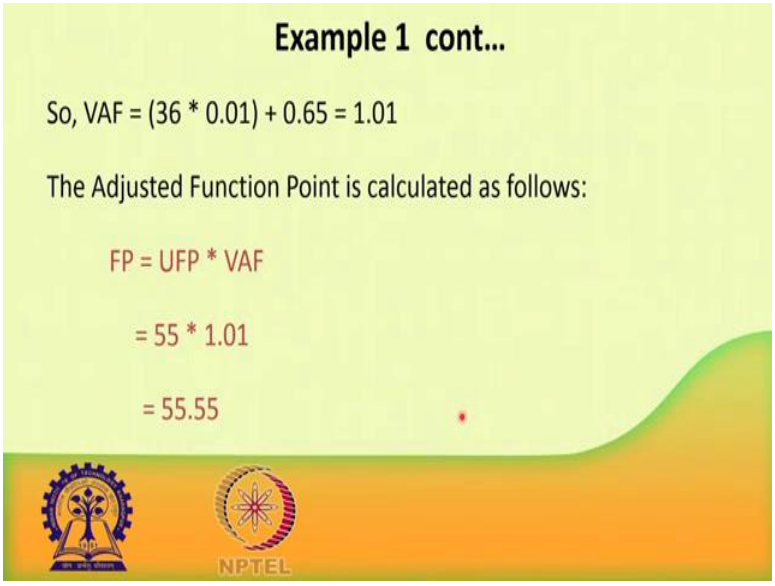

And you can say that so now, this value adjustment factor will be how much 36 will put in the above formula that will give raise to 1.01 and once the VAF is calculated, you can easily calculate the adjusted function of the adjusted function point will be equal to

multiplication of UFP – Unadjusted Function Point into the value adjustment factor, so this is coming to the 55.55. So, in this you have taken what are the two cases where the values of GSC – General System Characteristics are same, in another case the value of the general system characteristics are different, how to find out the adjusted function points ok.

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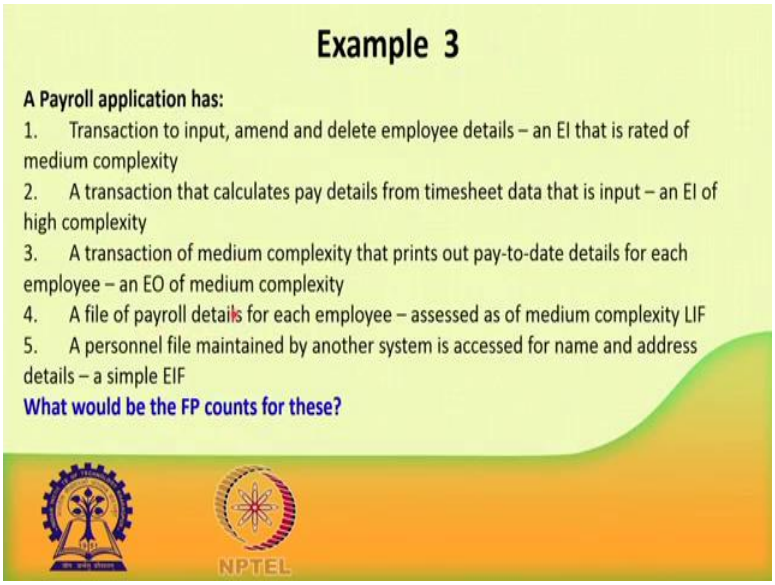

So, there is another example is a very small example, this you can see yourself.

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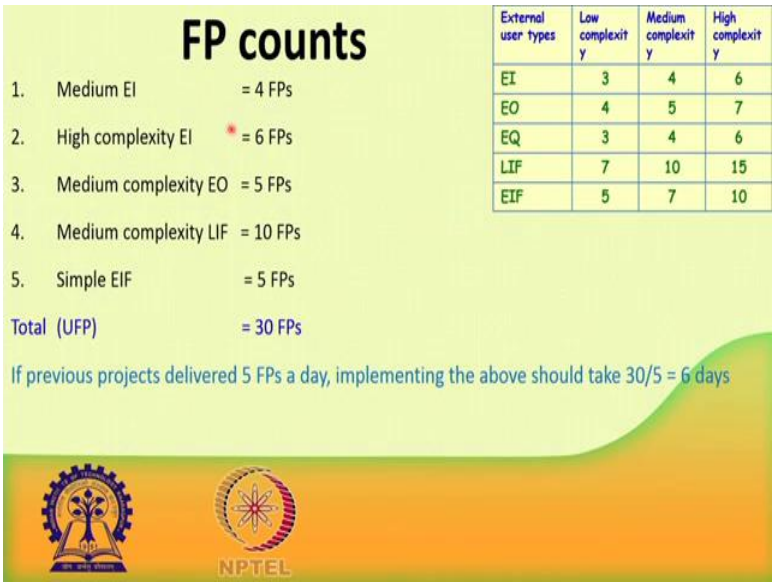

Where and reading that example you can find out, how many inputs are there and here I have taken there are two inputs; one is of medium, another is high. And there is a one what output that is of medium complexity, one internal file that is medium complexity, and one simple interface file that is simple, so add all those things. So, will get total UFP is equal to how much 30 FPs.

So, if a previous projects, if you have already develops similar kind of projects and the if the previous projects delivered 5 function points for a day, then you can easily calculate that you have to take 6 days 35 by 30 by 5, so that 6 days you have to take in order to develop this project.

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So, other exercises are given, so that you can practice.

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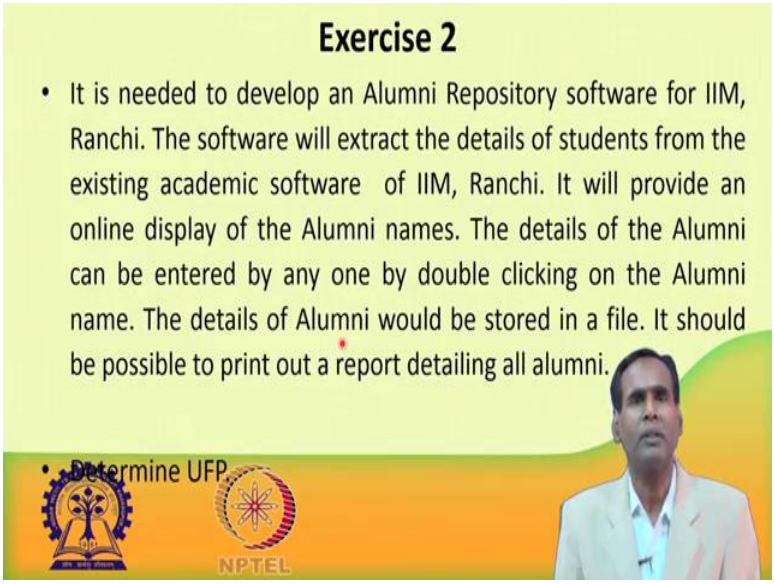

For each exercise find out what is the unadjusted function point, then what are the what VAFs, multiply all those things.

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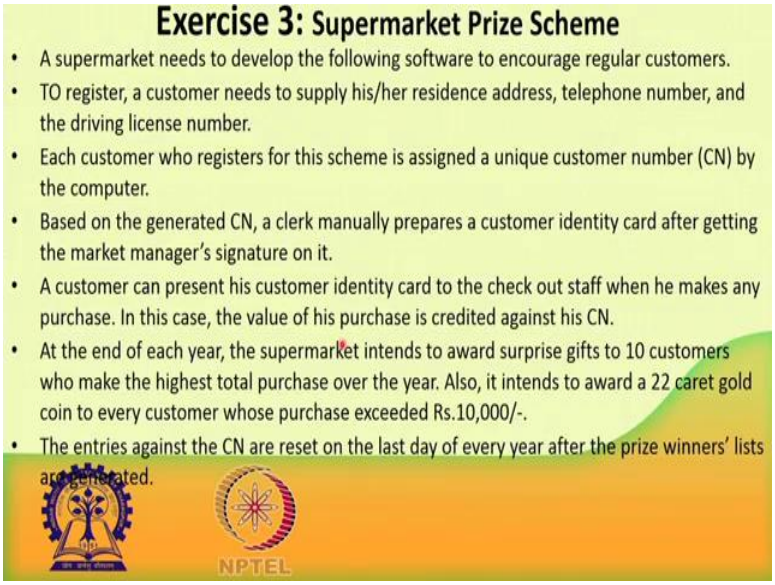

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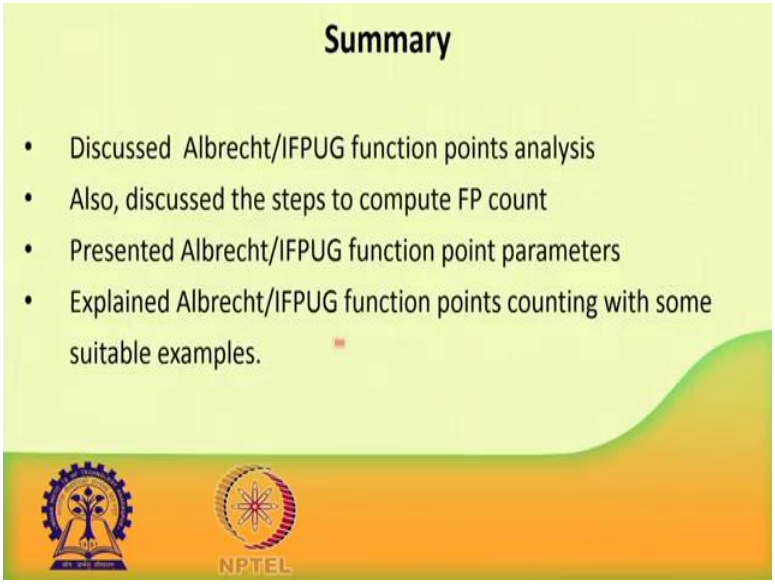

Then you can get the total function point using Albrecht, what function point method. So, finally, here we have discussed Albrecht IFPUG function point analysis; we have taken some examples related this. And in the next class also, I will take few more examples for this, then we have to discuss this steps to compute the FP count. Then we have presented also the function point different parameters, parameters I have told and then we have explained the IFPUG function point counting with some suitable examples.

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We have taken mainly it is a concepts on these two books; we can have along to look into this.

Thank you very much.

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Lecture – 20
Project Estimation Techniques. (Contd.)

Good afternoon. Now let us start in this section, in this lecture we will take up some of the examples for this Albrecht a function point methods. We will take few examples so, that you can better understand. We will take some more examples on this Albrecht IFPUG function points then, we will take two more types of function points that is Symons Mark II function point and the COSMIC function points.

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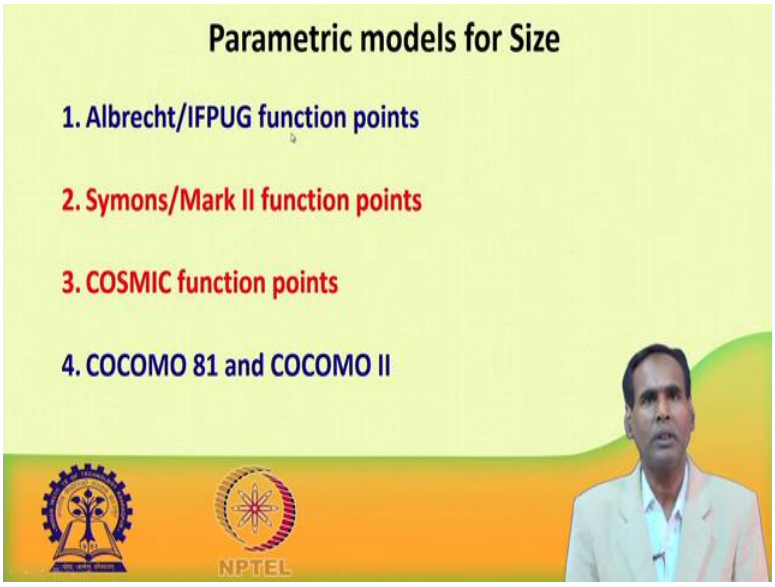

So, we have already discussed this parametric models for size, last class I have told we will use a we will see 4 types of this a estimations like IFPUG function points, Mark II function points, COSMIC function points and COCOMO 81 and COCOMO II. Last class already we have discussed IFPUG function points Now, we will see this Symons Mark II function point and COSMIC function points, before going to that let us take one or two more problems more examples on this IFPUG function points.

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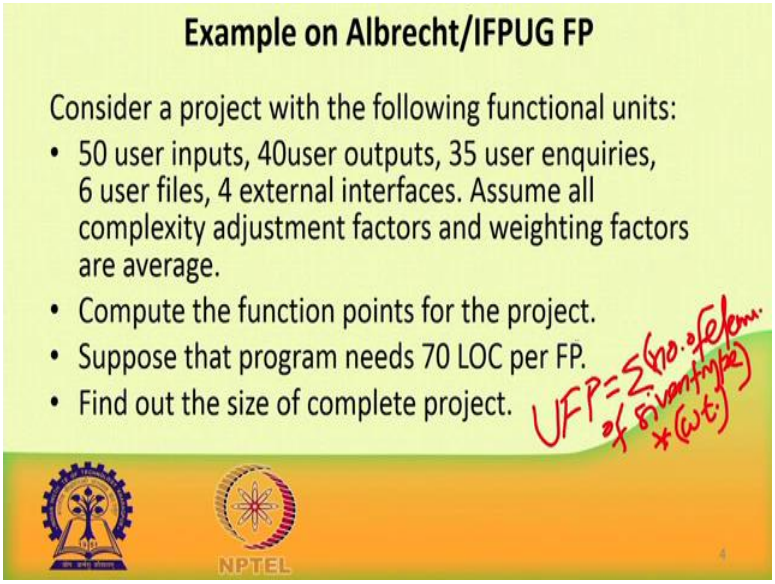

So, we will take up this example, but I hope that before going to this example the let me again tell you the let us take up that this example. And here, we will see that how this previous formula that we can use I have already told you last class, first we have to find out the unadjusted function point, then we have to refine it by using the value adjustment factor. So, this formula for unadjusted function point can be written as what? I have already shown it in the last slides, this will be equal to summation of number of elements of the given type maybe it is a input or output or a query, what is the number of elements into the weight

The weight I have already given you for simple there can be three;3 the weights can be divided into three categories depending on whether it is simple or medium or high. So, that table Albrecht might that complexity classifiers the multipliers we have already given you in the last class. And now, with this now let us take up this problem ok. So now, let us take this one more example, this example say that consider a project with the following functional units there are suppose in that project there are 50 user inputs, 40 user outputs, 35 user enquiries and 6 user files, 4 external interfaces.

So, now let us assume that all the complexity adjustment factors and weighting factors they are average, everything is average. So, then the question is that compute the function points for the project and then compute what will the size? Given that the program needs 70 LOC per function point.

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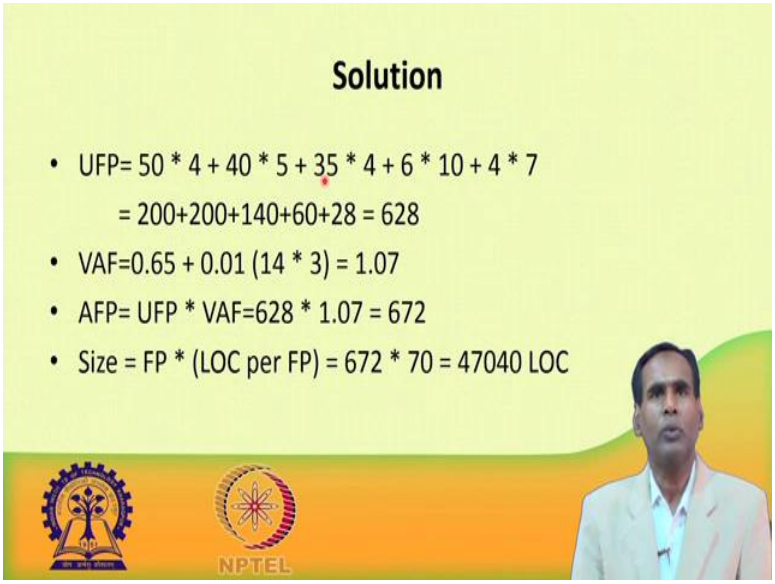

So now, let us go you can see that the inputs is equal to 50, outputs is equal to 40 and queries is 35 and internal files you can see that number of internal files is 6 and 4 is the external interfaces that 6 and this 4.

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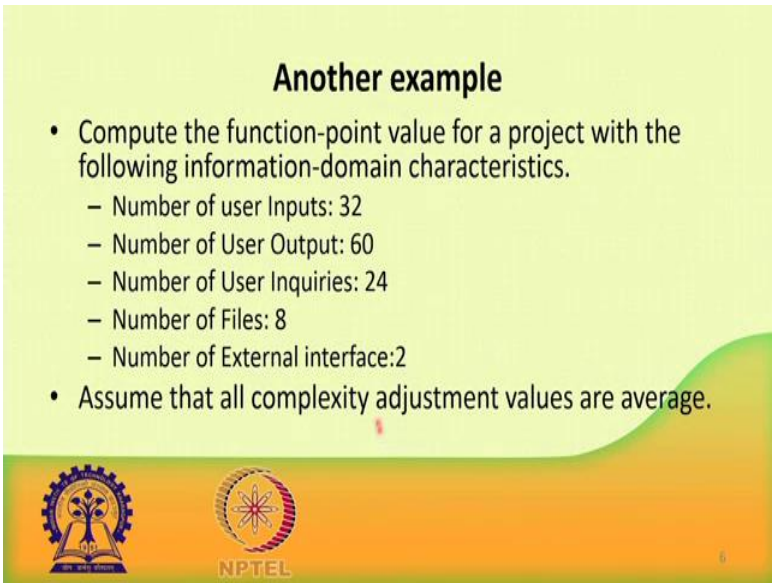

So, use these values in the given formula that already I have given you and then it will give you this values of UFP. So, basically two things I am taking the possible number of elements from each category input, output, inquiry then internal file and external interfaces multiplied by the multiplier the weight, which is whether it will vary

depending on whether it is a simple or average or what high. In my previous problem I have already told you that these are all what average. So, all the complexity adjustment factors are average, as well as these weighting factors they are average, it will be much more easy to compute.

So, now UFP is equal to state formula apply this you will get 628, then we have to find out the, what value adjustment factor. Again that formula we have told you 0.65 plus 0.01 into TDI, where TDI is equal to Total Degree of Influence. And here 14 parameters since, it is told that this is average please see I have already told you that these complexity adjustment factors are average, as well as the weighting factors that mean the GSC this General System Characteristics there, these are these are also average.

So, again for average since you have taking a scale from 0 to 5, again this is average will coming 3 so, this will 1.07. So, the AFP will be the adjusted function points will be equal to unadjusted function points into VAF. So, because why we are doing this? We want to refine it. So, the you can get the adjustment function point by what multiplying the unadjusted function point on the value adjustment factor. So, this is given to 672. So, this is the adjusted function points.

So, now what is question again and the second part question is that find out the size of the complete project, how you can compute the file find out the complete the size of the complete project, you have to use this information. It said that the program needs 70 LOC per function point; that means, per function point that can be 70 lines of code. So, what is the total number of lines of code; so, sorry what is the total number of FP function point 672. So, the total size will be equal to the function points into LOC per function points. So, 672 into 70, in this way, you will get that these total size of the project in terms of LOC is coming to be 47040.

So, in this way for small projects you can easily use this formula find out the function points and size. So, in the examination some this types of small what questions might be asked and you have to find out the answer. So, let us take quickly another example in these example a project you have to develop for that you have to find out the function point for that project, this project has the following characteristics there are number of user inputs is 30, number of user outputs is 60, number of user inquiries 24, number of files is equal to 8 and number of external interface is 2.

So, and assume that all the complexity adjustment value are average and all the weighting factors are average.

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So, then you have to find out the number of function points, you can see UFP again straight forward formula that I have given summation of the number of elements from each type into weight, whole then take the overall summation you will get applying this formula straight forward. You can see that there are 32 number of what input users and we know, therefore, average case this what value is 4, it has already given these are adjustment factors are average. So, the value will be what 32 into 4. Similarly output is 60, the average weight is 5 and the inquiries it is 24 number of inquiries are there you can see, and this average value for inquiry is 4, unlike this you can compute there are the 8 numbers of what internal files and 2 number of external interfaces.

So, 8 into average value for the interfaces 10 and 2 numbers of external interfaces, average value it is 7. So, find out this value it is coming 618 then, you have to compute the value adjustment factor by using the formula 0.65 plus 0.01 into TDI and since, it is given that is these are also the general system characteristics they are average so, again 14 into 3 so, that will 1.07. So, finally, the adjusted function point can be obtained by multiplying the unadjusted function points and the value adjustment factor. So, this is giving to be 661.26. So, the total number of adjusted function points or the adjusted function point counts for this project will be equal given as 661.26.

So, suppose if for function point you have to write say 80 lines of code. So, what will the size in terms of LOC? So, 661.26 and 80 so, this will give you roughly the size of the proposed project. So, in this way you can use Albrecht function point method to find out the, to estimate the size of a given project.

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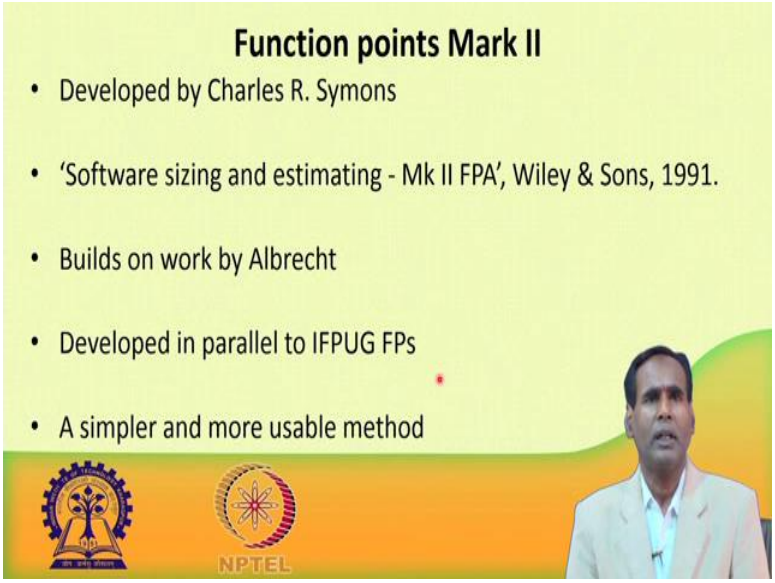

So, now, let us quickly say about these what function points Mark II this is also another interesting approach to find out the function points. This was developed by Charles R Symons, actually the details of each method is available in a book he has developed, he has published a book on software sizing and estimating by Wiley and Sons in 1991, which contains the details of his function points, that is function points Mark II.

It is builds on; it is actually it is on extension of this Albrecht function point. So, it is build on work by Albrecht function point method, it is developed in parallel to IFPUG FPs. So, simultaneously it was developed in parallel with the Albrecht IFPUG FPs. So, it is a rather simple and more usable method.

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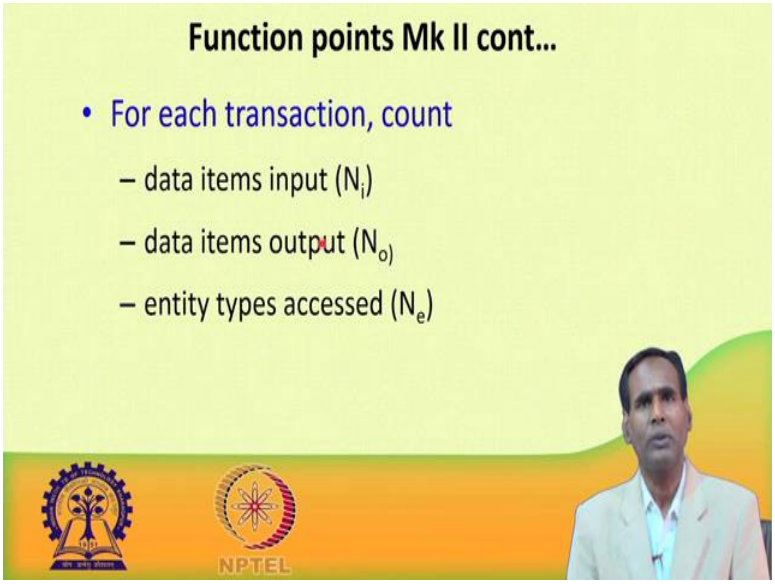

So, how it works? So, for each transaction you have to count the data items input, the data items output and the entity types accessed. So, these three parameters you have to count and then you can find out the total function point for the whole project by using a formula, which we will see in the next slide.

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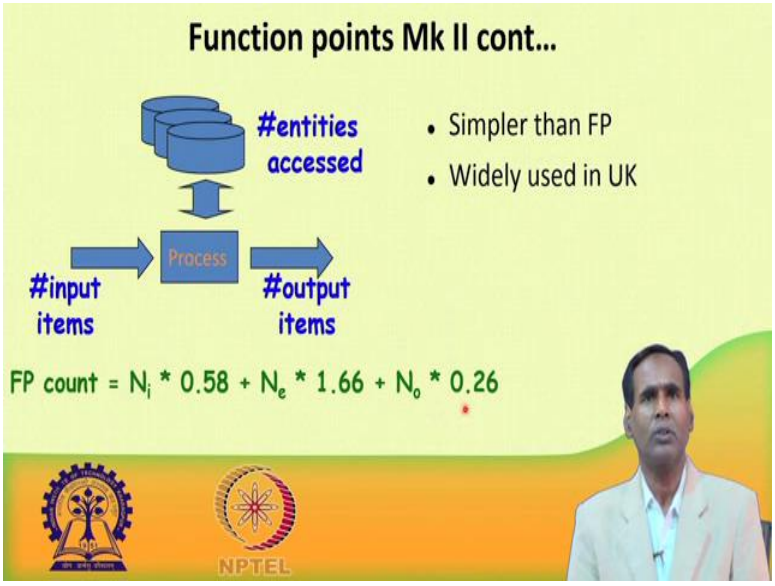

See how does it works, as I have already told you, you have to supply the input items as well as you will see also three parameters are required I have told the number of input

output; the number of input items supplied to the process, the number of entities assessed from this data store and the number of output items are produced.

So, in this way this function point Mark II this works this is a simpler method than this Albrecht function point and it is widely used in UK. So, what is the final formula for finding out the Mark II function point, the function count is given by the following equation

$$FP\ count = N_i * 0.58 + N_e * 1.66 + N_o * 0.26$$

Where Ni Ne and No we have already defined earlier, Ni the data items input No is the data items output and Na is the entity types associated or the sorry the entity types accessed.

So, this value judge; the these constants are some weights 0.58, 1. 66 and 0.26, these are weights. This Simpson ok; Symons he had found out these values of the constants by performing experiments in his lab. So, directly he has utilized these values of the constants along with these parameters Ni, Ne and No and he has developed this formula to find out the function points, this is known as Symons Mark II function points this is simpler than Albrecht function point.

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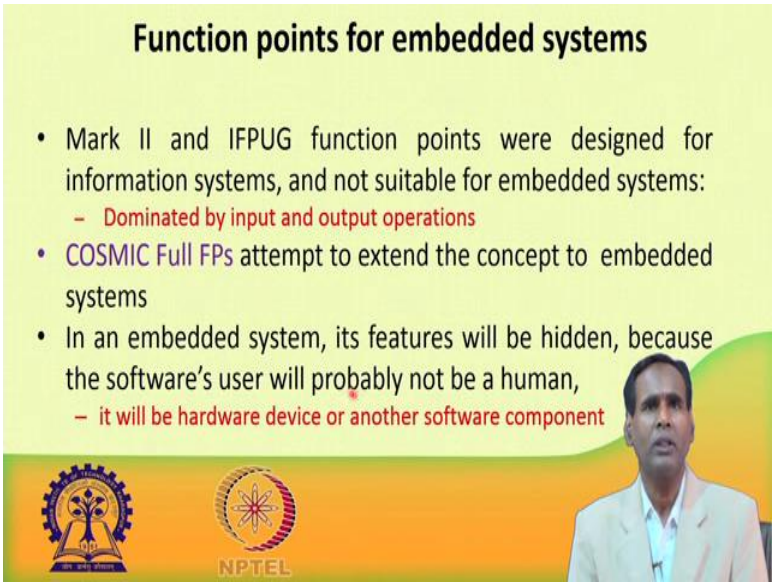

Now, let us see about the whether these function points so far we have seen the Albrecht function points and the Symons function points, can they be extended to real time

systems and embedded systems. See this Mark II and IFPUG function points, they were mainly designed for information systems and hence they are not suitable for embedded systems why? Because these two function points they are largely or heavily dominated by the input and output operations, the most emphasis is given on the number of input operations and the output operations. So, in order to overcome this problem so, cosmic full function points they have been developed in order to overcome these problems, this a cosmic full function points they attempt to extend the concept, this concept which concept this the Mark II concept or IFPUG concept to embedded systems.

So, Now these concepts have been extended to handle embedded systems to apply on embedded systems or real time systems. You know that in an embedded system it is features like hidden and etcetera is not it. If in an embedded system it is features normally will be hidden, because the software, the software's user probably is not a human being, it will be used by some hardware device or another software component.

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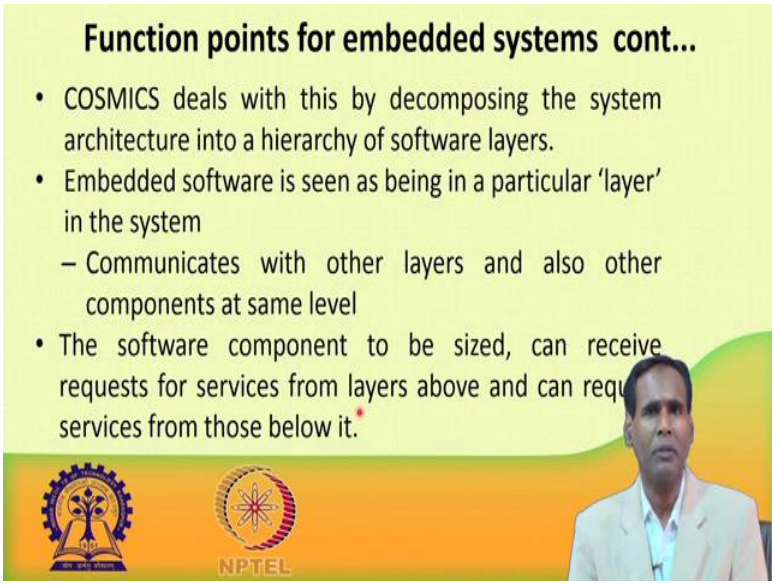

In order to what overcome these problems COSMICS deals with this by decomposing the system architecture into a hierarchy of software layers. And the embedded software is seen as being in one of the or in a particular layer in the system. This layer where this software is present this communicates with the other layers I mean, upper layers and the below layers and also other components are the same level. The software component

which has to be sized, it can receive requests for services from layers above and it can request services from those below it this is shown in the, in this figure.

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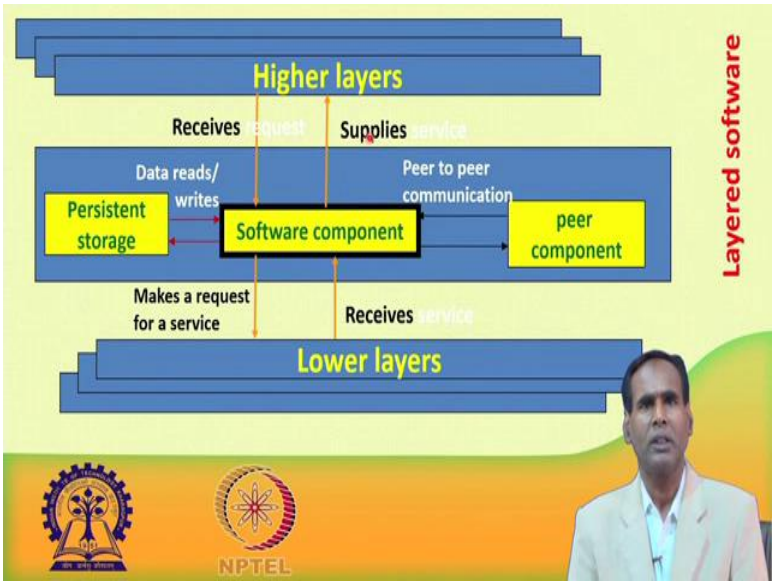

So, the software component represents suppose here it may receive what messages from higher layers. It may make a request for a service what from a lower layer and also it may receive a what service from a what a lower layer it may service it may provide some service to some higher layer.

Besides that this software component may communicate with the persistent storage, peer component and other components present in the same layer. This is the layered software and based on this concept this embedded systems work. So, since they Albrecht function point and Symons function points they cannot handle this embedded systems that is why this, because another what function point technology has been developed that is COSMICS Now let us see how COSMICS does work.

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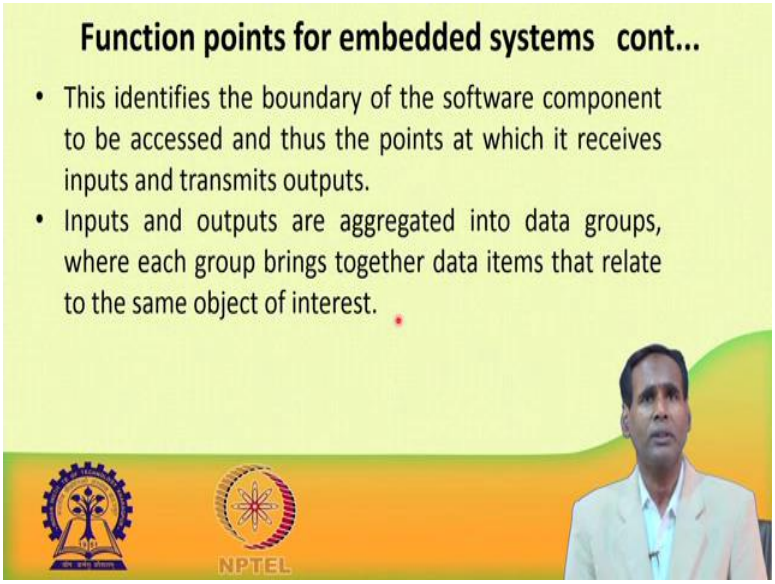

So, now, as I have already told you that here the software component that has to be sized can receive requests for services from layers above and it can requests services from those below it. So, this identifies what will the boundary of the software component that has to be assessed and thus the points at which it receives inputs and transmits outputs. It inputs; here, the inputs and outputs are aggregated into data groups where each group will bring together data items that relate to the same object of interest.

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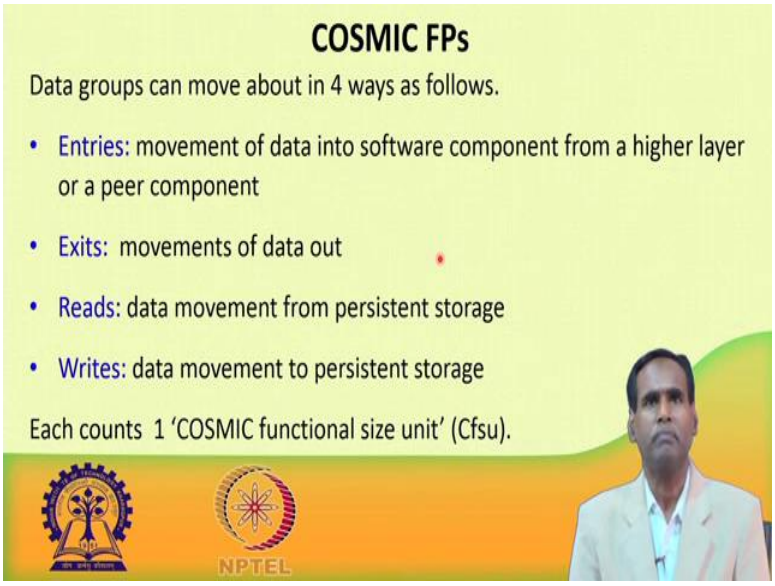

Let us see, what can these; what groups the data groups can move about in four ways as follows and like one is entry another is exits, another is reads, another is writes. So, these are the what, data groups these are the four ways in which the data groups can move about. So, entries means movement of data into the software component that is under consideration from a higher layer or a peer component. Exit means, movement of data out of the software component that is under consideration. Read means, data movement from persistent storage and writes means, data moment to persistent storage.

So, in this way before which the data groups can move and what we have to do that each counts; so, each of the above each counts one COSMIC functional size unit and in short we call as Cfsu. So, in this way the COSMIC function points can be a calculated in this. So, the COSMIC function points can be computed, can be estimated for any embedded system or real time system.

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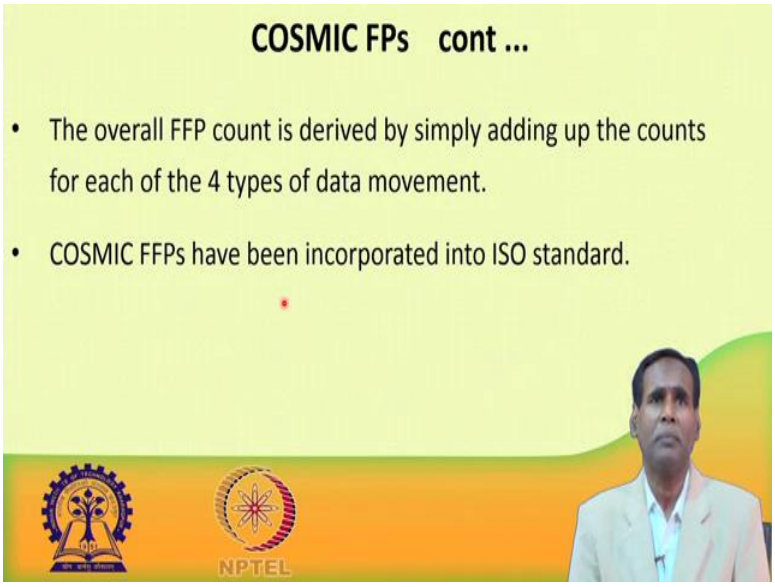

The overall full function point count, how it can be found out the overall full function point, count can be derived by simply adding up the counts for each of the 4 types of data moment, that we have seen in the last slide. These COSMIC FFPs they have been incorporated into ISO standard. We have already seen that these LOCs etcetera they are not included incorporated in any ISO standard, but this COSMIC FFPs they have been incorporated into ISO standard.

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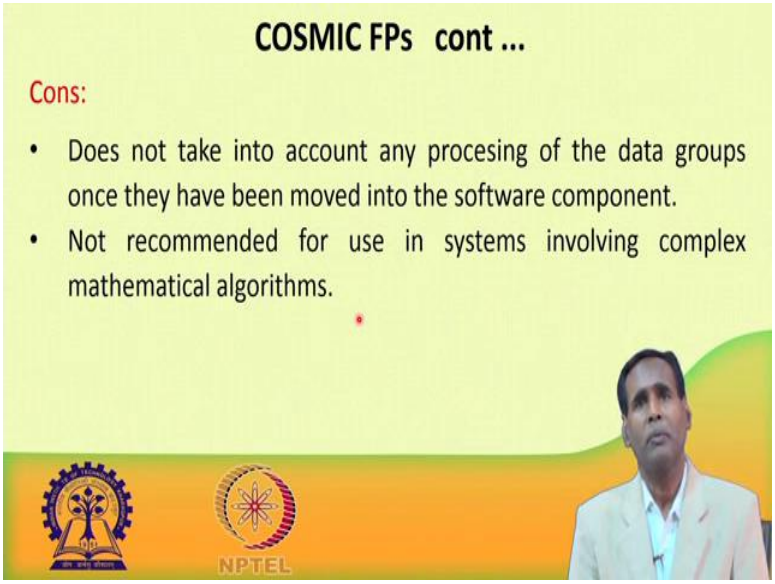

Now, let us quickly say about the, what disadvantages of the cosmic FPs. It does not take into account any processing of the data groups ok. So, it does not take; it does not consider any processing of the data groups once they have been moved into the software component. So, once they what data groups they have been moved into the software component, this metric does not take care of this material, does not take into account any processing of these data groups. Normally it is not recommended for use in systems which involve complex mathematical algorithms. So, this is not recommended for this system. So, these are important drawbacks of cosmic function points.

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So, Now let us summarize what are the pros and cons of the function points. On the positive side you can say that function point is language independent, we have already seen LOC is a language dependent, but function points are language independent. Understandable by the client because here no technical details are there, you just what are the functionalities and they are associated things you are just analyzing. So, it can be understandable by clients which are non technical people, it is a very simple modeling technique, hard to fudge and the variable features they creep.

And on the flip side you will see that it is very much labor intensive that to compute the what, function points, extensive training is required, inexperience may result in inaccuracy. So, the; now the function points you have computed if you are quite inexperienced, it may result in inaccurate number of function points, weighted to file manipulation and the transactions, there may be errors which are introduced by single person. So, multiple raters are advised. So, if you adjust giving only one person to estimate this some their sometimes that errors may be introduced. So, you have to put a multiple what raters to rate these what applications, they can find out the different weights so, multiple raters are required.

It does not consider one of the major problem with function point is that it does not consider algorithm complexity of a function. So, because you know that it rates that if there are some functions every function is handled in scenario, but you know some of the functions are more complex, you have to put more effort. some of the functions are very straight forward straight forward functions. So, they require only what less effort or less time. So, it does not consider algorithm complexity of function. So, iterates all the functions similarly.

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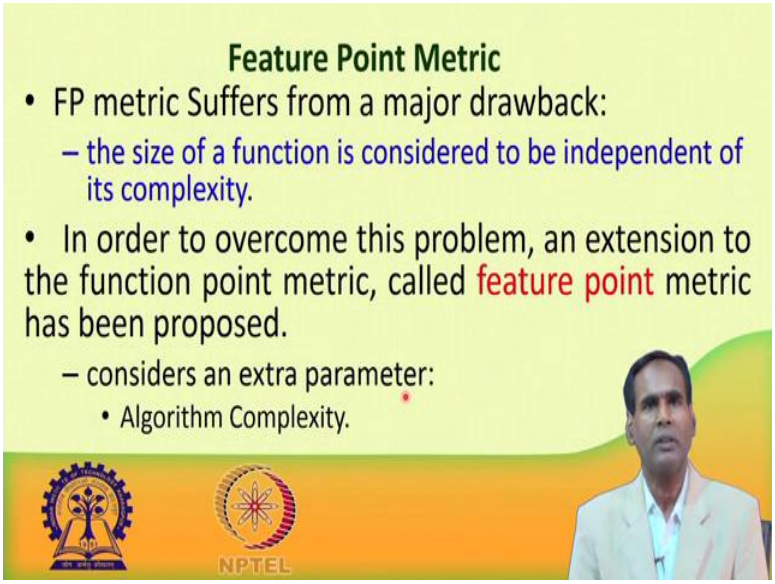

So, I have already told you that function point metric suffers from a major drawback; that means, the size of a function is considered to be independent of it is complexity which is not true, the complexity of the functions are different. In order to overcome this problem, an extension to the function point metric is there which is known as feature point metric. So, this feature point metric it takes into account and extra parameter it considers an extra parameter that is known as algorithmic complexity.

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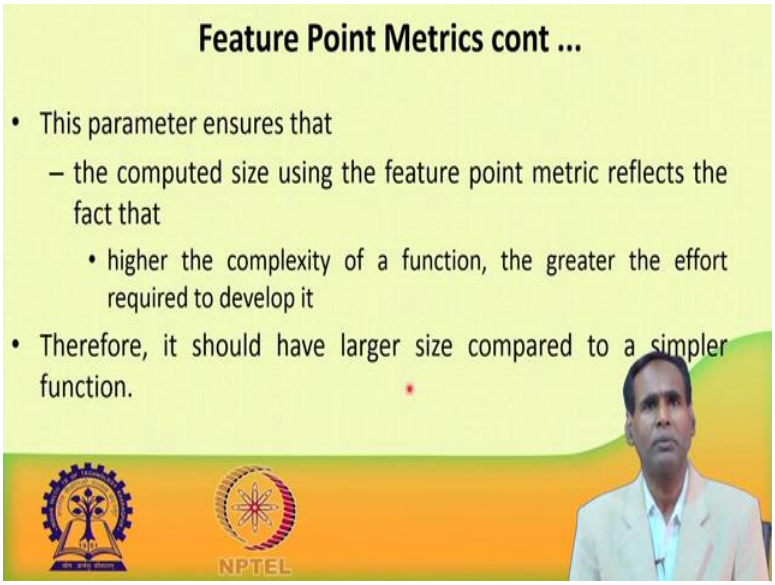

So, this parameter algorithm complexity it ensures that the computed size using the feature point metric, it reflects the following fact. What if the fact that the higher the complexity of a function the greater the effort required to develop; it if they complexity of function is high so, you have to put more effort to develop it. If the complexity of a function is very less just like a small GUI kind of a function it remain require less effort. Therefore, it should have larger size compared to a simpler function.

So, if a; what the function is having higher complexity it should have larger size as compared to a simpler function. It should have larger function point and hence, it should have larger size as compared to a simpler function.

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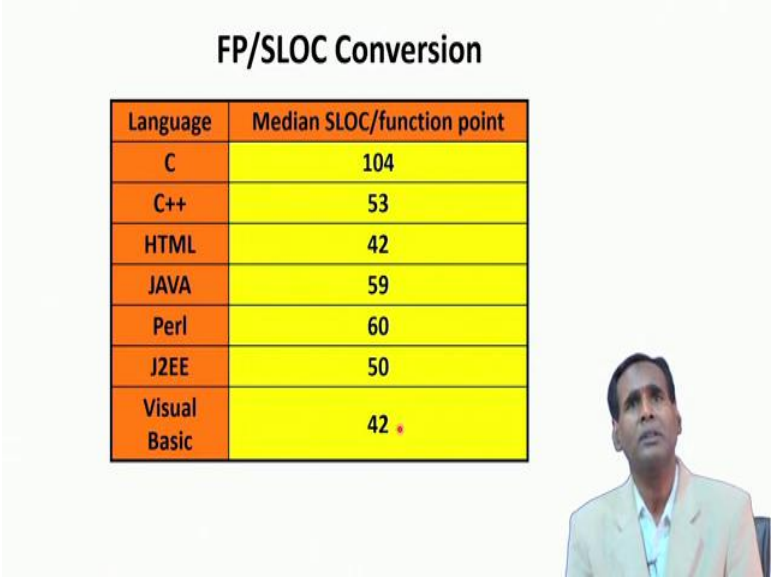

So, finally, let us see what the relationship or how that is the relationship between function points or SLOC or how a given a function point, how it can be converted to SLOC? See, for different programming languages this conversion rate is different like for C language 104 SLOC is there per 1 function point. So, 1 function point may contain 104 SLOC. Similarly for C plus plus the SLOC for function point is 53 and like this, you can see it is highest in case of C and we can see that it is almost lowest in case of HTML and visual basic it is 42 ok.

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So, how you can still make your estimation more accurate, how can do accurate size estimation. So, here 1 one thing you have to extra take that is the risk factors, because in the previous things almost we have neglected the risks factors. So, in order to make your size estimation more accurate, you have to take the risk factors and then you have to multiply.

So, you will take the requirement then estimate the project size, then multiplied with the project complexity adjustment factors, then multiplied with the risk factors. So, these two points the we have already told you this is given where in the function point analysis, the project size and the project complexity adjustment. This you can say the unadjusted function points and these are the complexity factors by multiplying this you can get some value and that will multiplied by the risk factor.

So, these two factors can be obtained from function point analysis, then by what under these two factors from the definition and the project completion adjustment and risk factors they are dependent. The situation you have to adjust the situation finally, you can if you will use all those things then by taking the multiplication you can estimate what are the cost the effort and you can perform the scheduling.

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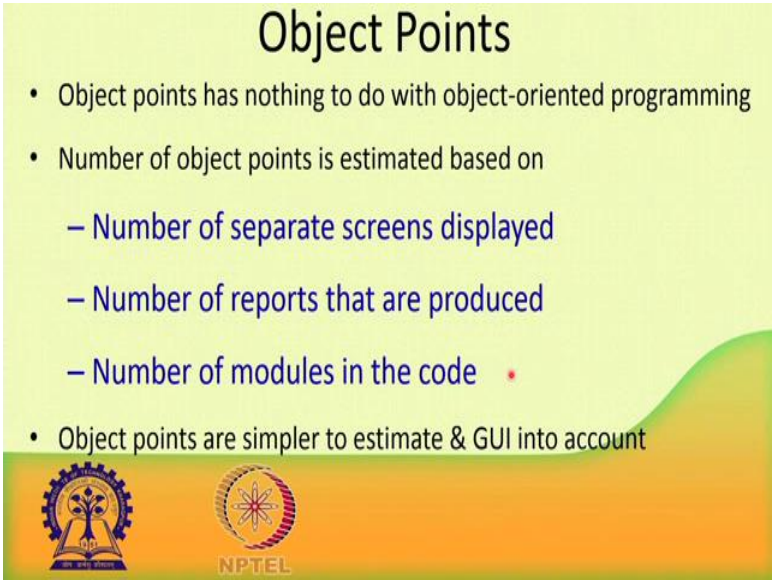

Finally, let us quickly say about the object points. See do not confuse objects points, they have nothing to do with the object oriented programming. Here only the number of object points is estimated; the number of object points is estimated based on three factors. First the number of separate screens they are displayed, then the number of reports that are produced and the number of modules they are present in the code. By taking into account, then by using some formula you can find out the object points. These object points are normally are the; they are usually simpler to estimate and they take into account the graphical user interfaces. So, this is a little bit fundamental about object points.

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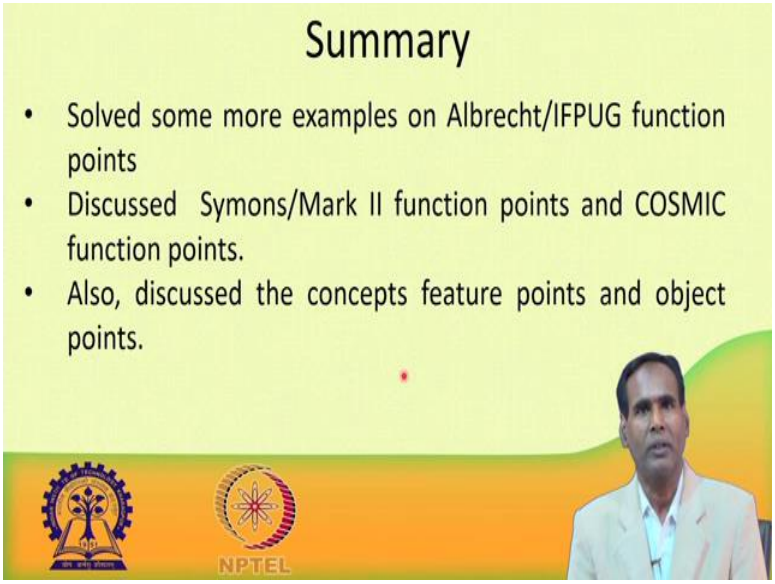

So, finally, we have seen that we have solved two more problems on Albrecht IFPUG function points, then we have discussed the Symons Mark II function points and the cosmic function points. We have seen that how cosmic point it is already included in this, what ISO standards also we have discussed what the concept of feature point because we have seen that function point has one important a drawback. That it does not take into account to the algorithmic complexity if two different functions are there.

It treats them as equally, but in fact, the two function those are there one might be very highly complex, another might be very less complex. So, which one is highly complex we have to put more effort and which one is less complex we have to put less effort. So, these what complexity is not taken into account by function points; so, that is why function point has been extended to in consider this complexity, it has another parameter called as algorithmic complexity.

So, this algorithm complexity takes into account that if the complexity of a function is more, more effort should be given if the complexity of a function is less. So, less effort has to be given. So, if the complexity of a function is more; obviously, it should contain it should produce or it should have more number of function points and hence, the size will be more. And if the function is having less complexity, then it should have less function points and hence less size. Also we have discussed little bit a of the object

points which takes an account or three factors that or three parameters ok, object points it takes into three; it is based on three parameters that also we have discussed.

So, these are the things that we have discussed on today. So, in the next class we will see and that COCOMO model.

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We have taken from these books, the reference mainly what it is both of the books we have used for this topics and finally, we stop here.

Thank you very much.

Software Project Management
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Lecture – 21
Project Estimation Techniques (Contd.)

Yes, good afternoon. So, now we will take up one more parametric model that is the COCOMO 1 and COCOMO 81 and COCOMO II.

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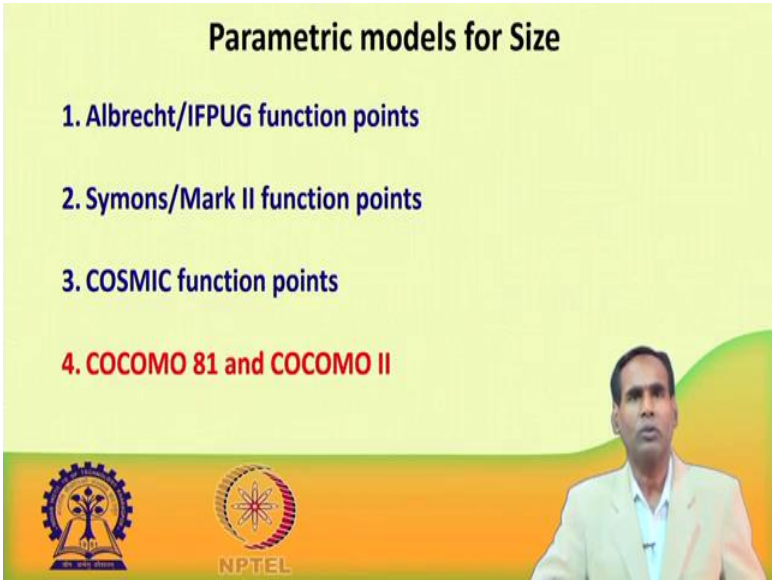

See last class we have seen Albrecht LFPUG function points, Symons Mark II function points, COSMIC function points. So, one was remaining that is COCOMO. So, in this class we will discuss the parametric models for size which is COCOMO 81 and COCOMO II.

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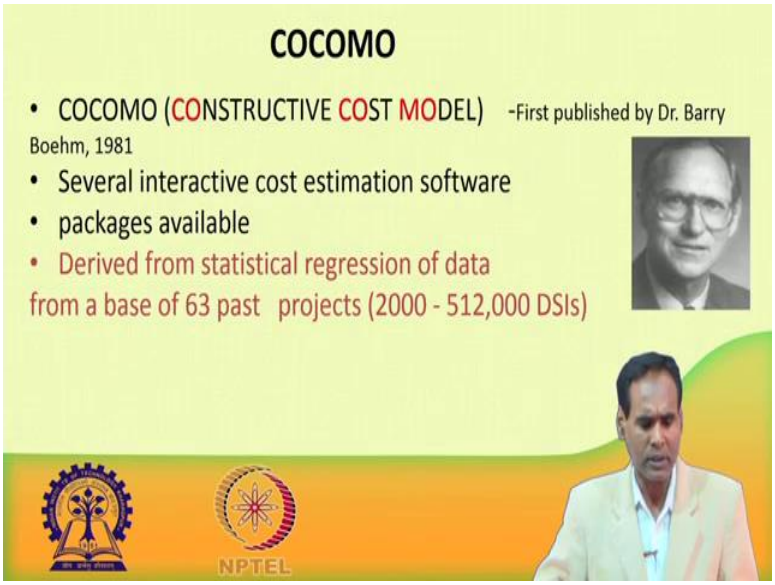

So, let us first see what does COCOMO stands for what does COCOMO stand for. COCOMO stands for Constructive Cost Model, it was first published by Doctor Barry Boehm in 1981, several interactive cost estimation softwares are there the packages available. This is derived from the statistical regression of data, this COCOMO model or coco yes this COCOMO model was derived from statistical regression of data from a base of 63 past projects they have and they there are almost 2000 to 5 lakh 12000 delivered source you know instructions there from this data they have derived this COCOMO model from the statistical regression of this data and there was almost he has conducted studies from 63 past projects.

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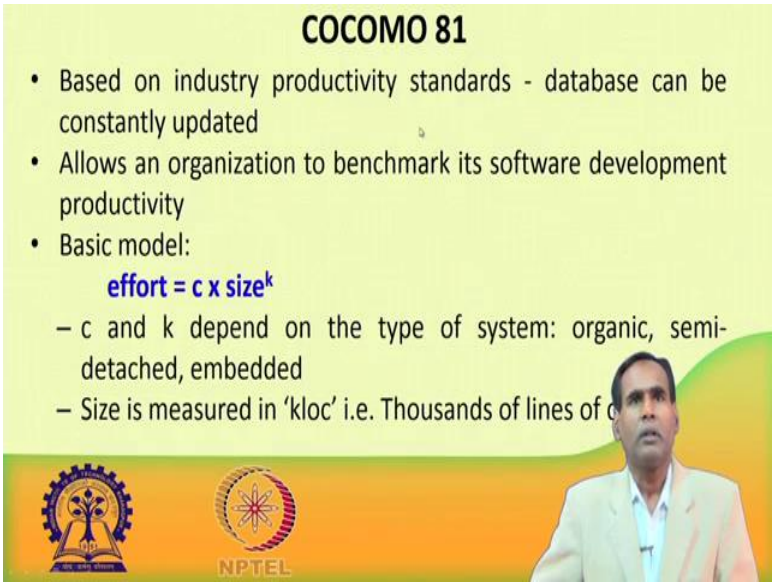

We will first see about COCOMO 81 then, we will see COCOMO II, COCOMO 81 based on industry productive standards.

The data base can be a constantly updated. It allows an organisation to bench mark, it is software development productivity. The basic model is like this. We want to calculate the effort, the basic model says that effort can be calculated as the product of 2 things; c and size, where effort is represented as

$$\text{effort} = c \times \text{size}^{k}$$

where c and k they depend on the type of the system the type of the product at the type of the project to going to develop, the product type could be organic could be semi-detached or could be embedded and the size is measured in KLOC; that means, in terms of thousands of lines of code.

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Now, let us see what are the different COCOMO modes and the models, 3 different environments or modes are there for COCOMO. I have already told you they are organic mode, semidetached mode and embedded mode, 3 increasingly complex COCOMO models are there. First one is the basic model then intermediate model then, detailed model and today we will discuss only the basic model and other two intermediate model and detailed model we will discuss in the next class.

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See let us first see the COCOMO modes. As I have already told you that are 3 modes or 3 environments for COCOMO, one is organic mode, then semidetached mode, embedded mode. In organic mode what do we when you will call a product as organic mode? A product which can be developed in familiar and stable environment we call it is an organic mode.

So, the products are similar or the product that you are going to develop is similar to the previously developed products, normally it is size is less than 50000 delivered source of instructions. So, normally it is size is less than or should be less than 50000 delivered source instructions. For example, people see your accounting system, all the information systems normally coming under organic mode.

So, for example, accounting system, student information system these things pay roll information system, these all are coming under organic modes. And what is about semidetached mode? Before going to semidetached mode let us see about the embedded mode. Embedded mode actually, a product which is completely new, a new product. So, embedded mode means, a product which is completely new, a new product requiring a great deal of innovation, a great deal of effort inflexible constrains and interface requirements are there.

So, normally all the what we can say your systems kind of things like operating systems, real time systems so, they are coming under a these system software kind of things like operating systems and real time systems, embedded systems etcetera they are coming

under the embedded mode. Now, we will see this semidetached mode. Semi the products which are coming in between organic mode and embedded mode they normally we call them as a semidetached mode. So, semidetached product semidetached mode products they lie in between what somewhere these organic and embedded modes semidetached mode.

So, these products somewhere lie in between the organic and embedded and examples are the utility applications such as, compilers, linkers etcetera.

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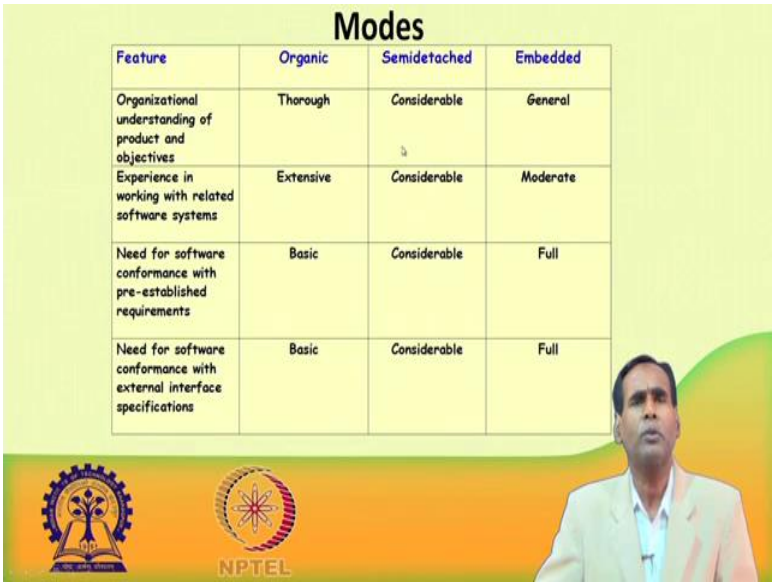

So, here basically in this table we have shown the different features of the products under different modes for example, the organisational understanding of product and objectives in case of organic it is a thorough in semidetached it is considerable and it is embedded it is general and experience in working with related software systems, you can see organic it is extensive, semidetached it is considerable and embedded it is moderate.

So, need for software conformance with pre established requirements here organic yes it is a basic, where semidetached it is considerable and embedded it is full Similarly need for software conformance with external interface specifications in organics is just it is basic, but semidetached it is moderate or considerable and in embedded applications it is full.

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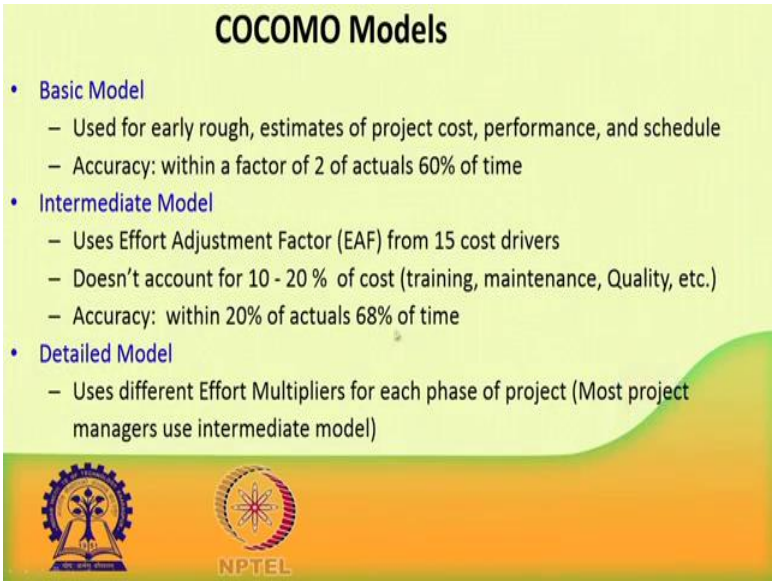

As I have already told you there are different models of COCOMO, one, the fundamental one the basic model, then intermediate model and detailed model. Let us see about first basic model. Normally, this basic model is used for early rough estimates of a project cost, performance and schedule.

So, using this model or this model can be used for estimation of early and rough estimates ok, this can be used for the early rough estimates of project cost, the performance and the schedule. It is accuracy is within a factor of 2 of actuals 60 percent of time. Intermediate model it uses effort adjustment factor which is known as EAF from 15 cost drivers, it does not account for 10 to 20 percent of cost; that means, training. It does not account for 10 to 20 percent of the cost for example, training maintenance and quality etcetera.

Here the accuracy is within 20 percent of the actuals 68 percent of time. The another model called detailed model or the COCOMO complete model or complete COCOMO, we say, it uses different effort multipliers for each phase of project, most project managers they use intermediate model.

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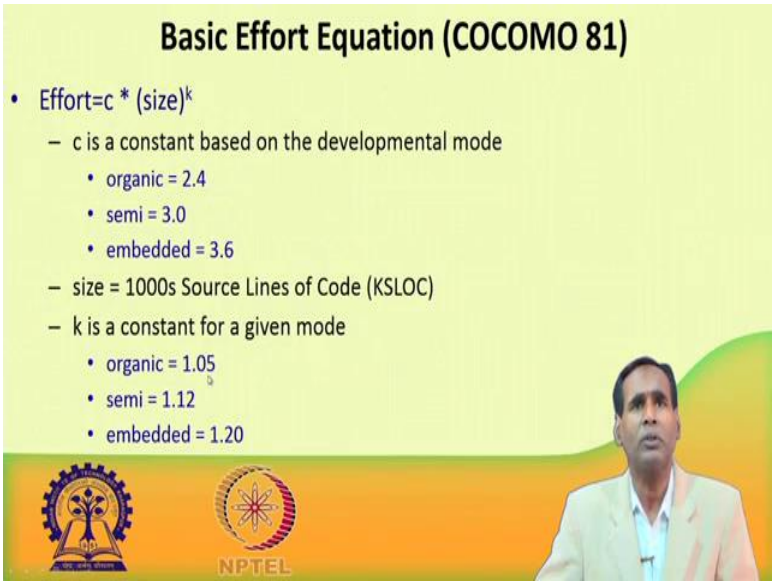

Let us first see the basic effort equation or the basic COCOMO model, what will be the equation for estimating effort using basic COCOMO, it is known as COCOMO 81.

So, here the according to COCOMO 81 the basic the fundamental equation for estimating effort is given a like this:

$$\text{effort} = c \times \text{size}^{k}$$

where c is a constant based on the developmental mode, for organic mode the value of c is 2.4, for semi organic semidetached it is 3 3.0 and for embedded mode it is 3.6.

Similarly, the size it is normally considered as is normally considered in KSLOC, that means, 1000 source lines of code, k is constant that also varies for the different modes, for organic mode the value of k is 1.05, for semidetached it is 1.12 and for embedded it is 1.20.

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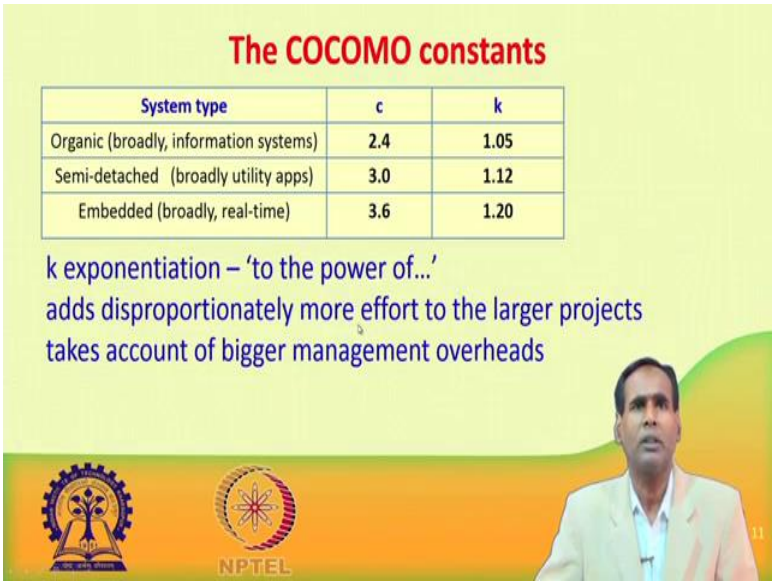

So, all those values that I have shown here different values of c and k, they are presented in a table for a better understanding, here k is the exponentiation to the power of something it adds disproportionately more effort to the larger projects, it also takes account for bigger management overheads.

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Now, let us find out the equation for the schedule using the basic COCOMO model that is COCOMO 81. The nominal development time can be represented as 2.5 into effort to the power exponent, where effort we have already calculated using this fundamental formula that effort is equal to c into size to the power k.

So, this value of effort will be put in this equation so that we can get in the nominal development time. So, the nominal development time will be equal to

$$\text{Nominal Development time} = 2.5*(\text{Effort})^{\text{exponent}}$$

where 2.5 is constant for all the modes that is organic, semidetached and embedded. The exponent, this factor it again depends upon the mode that you are using for organic mode, the value of the exponent is 0.38 for semidetached mode it is 0.35 and for embedded it is 0.32.

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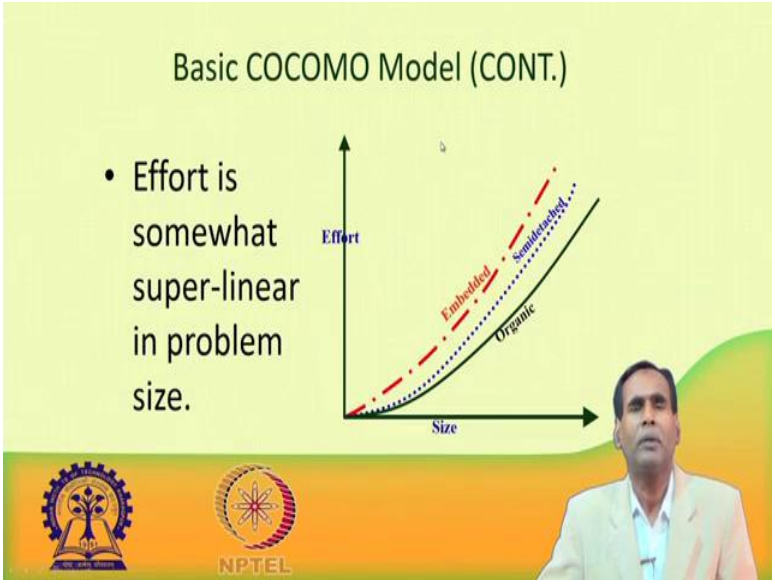

So, now let us see we have already seen this what these 2 equations for estimating effort and duration, now let us see how the graph will look like for the different products. You can see that in x axis in this graph, we have taken x axis along x axis we have taken size and along y axis we have taken effort. You can see that effort is somewhat superliner in problem size, just the problem size is increasing it is effort in is just increasing, but not in that proportionate. So, effort is somewhat superliner in problem size.

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Similarly, you can draw a graph between the what size and the development time, you can see that the development is a sub linear function of the product size. What do you mean by what is this? That when the product size increases in what in which proportion the development time does not increase in that way. When the product size increases 2 times please mark here the development time does not double it. The time taken is almost the same you another observation you can make from this graph it is like that, the time taken is almost same for all the 3 product categories whatever may the product may organic or semidetached or embedded the time taken to develop this different kinds of a products it almost same.

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And you can say that let us say, that development time it does not increase linearly with the product size I have already told you, because for larger products more parallel activities can be identified and they can be carried out simultaneously by a number of engineers.

So, hence the development time it will not increase linearly with product size that is why the graph is what is coming like this.

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For example, you see the developmental time is roughly the same for all the 3 category of the products, I have already told you. Let us take an example for example, a 60 KLOC program, it can be developed in approximately 18 months irrespective of whether the product is organic or semidetached or embedded type. So, there is more scope, why? Because I have already told you that for larger programs there is more scope for parallel activities for larger programs such as, systems and application programs than the utility programs.

So, another thing you can mark. So, this is a graph which shows that if the exponent is greater than 1, so, how this what graph looks likes for the effort and the if the exponent is less than 1 how the graph is look like for the duration. See duration for the increasing effort the and graph, for the duration for the increasing effort when these power the exponent is less than 1 0.38, see how this what look like because you are the what exponential is less than 1 and in case of a effort here the this exponent is greater than 1.

So, for effort you see the graph it looks like this. So, this is how the graph can look like for the effort and duration depending upon on the value of the exponent, whether it is greater than 1 or less than 1 the graph size also changes.

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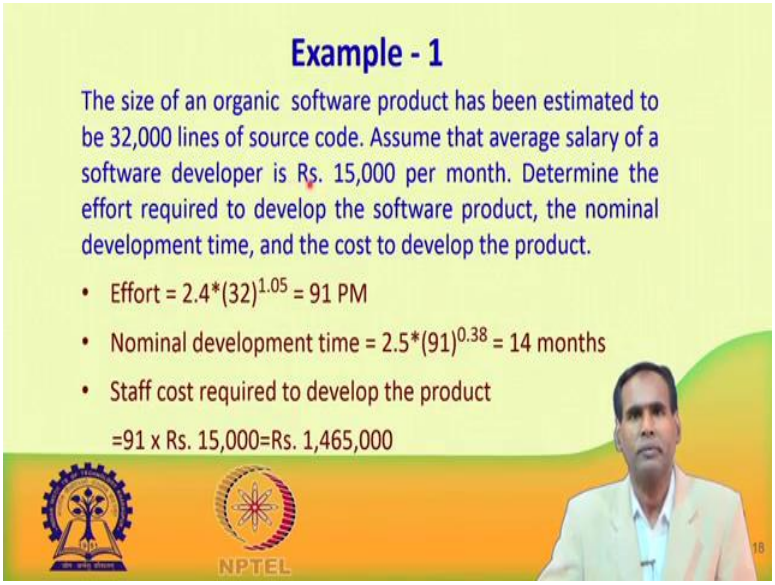

Now, let us take some of the examples and try to elaborate this. We will use some the equations of the basic COCOMO and we will try to find out, we will try to estimate the effort and the nominal development type.

So, the first example says that the size of an organic software product has been estimated to be 32 lines of source code. Assume, that the average salary of a software developer is 15000 per month, what you have to do? Determine the effort required to develop the software product, the nominal development time and the cost to develop the product. So, as you we can see from the problem it is given as organic software product.

So, now for effort the formula is you know: c into size to the power k, now we have to find out what is the value of c and k for the organic software. You can see the value of c and k I have already given you a table for the value of c and k are given like this.

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For organic product, the value of c is 2 to the power 2.4 and the value of k is how much? 1.05. So, we can see that here I will just put the value of c and k in the equation.

So, for organic product value of c is 2.2, for organic product the value of k is 1.05, the size is already given 32000 lines of code; that means, 32 KLOC. So, effort can be estimated as 2.4 into 32 to the power 1.05 and I have already told you the unit of effort in the last class, that effort is on normally what expressed as in terms of person months or man months.

So, effort is equal to after solving this equation it is you coming to be 91 approximately, 91 person month. The nominal development time again we have seen the equation for nominal development time earlier this is coming to be how much? The nominal development time can be expressed as 2.5 into effort to the power exponent and the value of exponent varies as the mode changes, it is the organic type of product, So, the value of exponent or the constantly 0.38.

So, now I have to use the value of effort and the value of this exponent is 0.38 in the equation. So, I will get how much? So, see we will get the value of nominal development time is equal to 2.5 which is a constant and what is this? Effort is 91 to the power the exponent and for the this product which organic software product the value of exponent is 0.38 and I am solving what this equation you will get the nominal development time is equal to 14 months, approximately 14 months. So, now, what the question is find out also how much cost will be spent to develop the product. So, we have to see that what

will be the staff cost. I have only in the problem says that one software developer will take the average salary of a software developer is 15000 will require, how much? What is the person month? We require 91 person months to develop the software. So, the cost that will be required the staff cost that will be required to develop these organic product will be 91 into 15000, which is coming to be how much? 14 lakh 65000.

So, in this way given a product how you can estimate the effort and nominal development time and the cost? First step is to identify what kind of product is, because in the examination it may not be given clearly organic software it might be an information system.

So, we have to apply your common sense that information system means this an organic software. So, if it is a real time system means, this is an embedded software. Accordingly, these value of the constants we have to remember and put in the equation.

So, then after finding out the effort by putting the value of the c and k in the equation, find out the effort then nominal time use this value of the effort obtained in the previous step put that value and then value of the constant again it will vary from the for the different modes such as, organic, semidetached and what your embedded put the appropriate value of the constant here. So, that you will get the nominal development time and then in order to find out the staff cost what you have to do? In the question, it may be given what is the average what salary of the developer multiply it by the effort, you will get the total cost that will require to develop the project.

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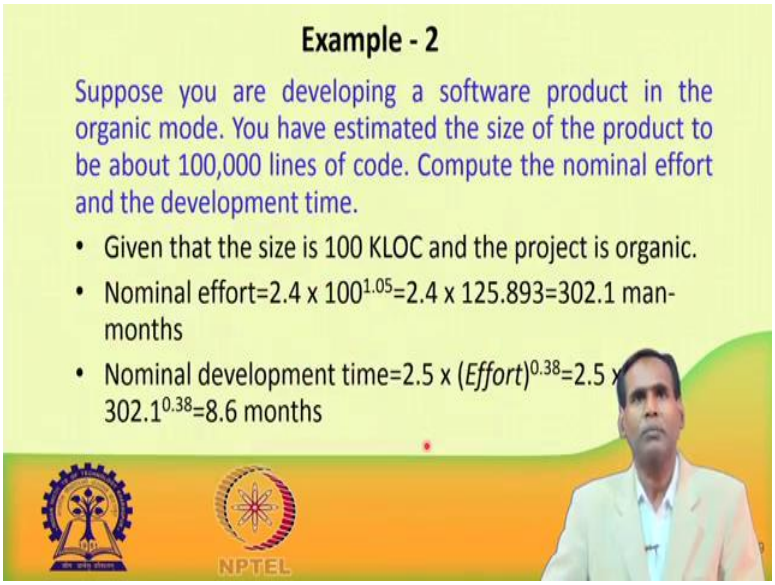

Now let us quickly take another example. Suppose you are developing a software product in the organic mode you have estimated the size of the product to be about how much 1 lakh lines of code we have to compute the nominal effort and the development time.

So, we can see that here it is given that it is a organic product. So, we can easily choose the value of the c and k, the size is also given it is 1 lakh lines of code; that means, 100 KLOC.

So, nominal effort could be we know already for organic product value of c is 2.4 and for organic product, the value of k is equal to 1.05, putting the values in the equation we will get and the size already given 100 KLOC. So, nominal effort will be called 2.4 into 100 to the power 1.05; on solving, you will get 302.1 man months or no you can round off it to 302 what man months.

So, this is the effort then nominal development time will be equal to how much? 2.5 into effort to the power some constant and for organic mode the constant is 0.38; so, which will give you 8.6 months. So, in this case what for this project the nominal effort is found to be 302.1 man months or person months and the nominal development time is coming to be 8.6 months.

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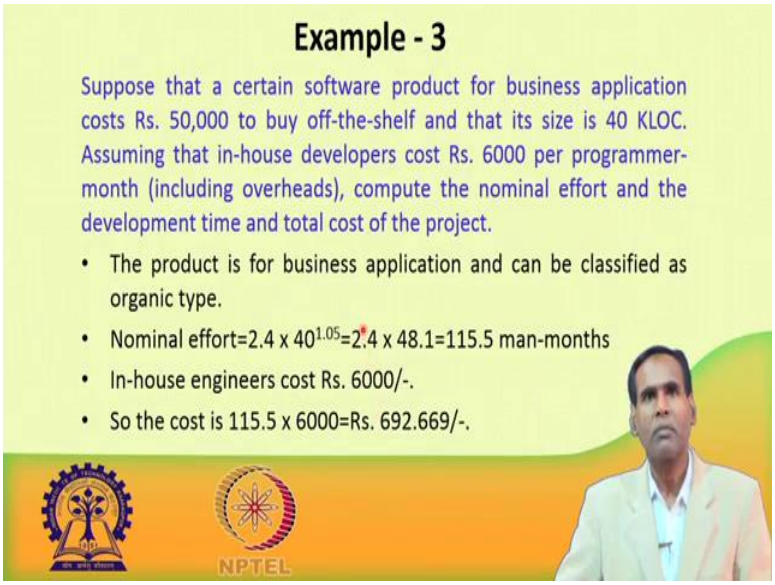

You can take a another example here. Suppose that certain software product for business application cost rupees 15000 to buy off-th-shelf and that it is size is 40 KLOC, assuming that in house developers cost rupees 6000 for per program or month or for person month including overheads compute the nominal effort and the development time and the total cost of the project. See the product is see indirectly it is not told what whether it is organic or semidetached it is not directly given, but it is given is an business application.

So, business application means, it is an information system. So, indirectly you can say that this is organic type. So, indirectly all are given. So, you have to start from this point that the product is for business application and hence it can be classified as organic type, the nominal product the nominal effort we have to now find out and we know for organic products the value of constant c and k are as follows c will be 2.4 and k is equal to 1.05 and it is given that what is the size already given, please whatever the size is given it may be given in KLOC or simple LOC, if it is given LOC you convert into KLOC, here it is directly given 40 KLOC.

So, I will use in place of size 40. So, this is coming to be this many man months then in house engineer cost is how much? It is given that the in house developers, they take 6000 per programmer month and I have to find out then the total cost. So, the effort is coming to be 115.5 and per month cost is 6000. So, the program will take how much cost? Total cost would be required this much cost 692.669 rupees it will be taken for developing this product.

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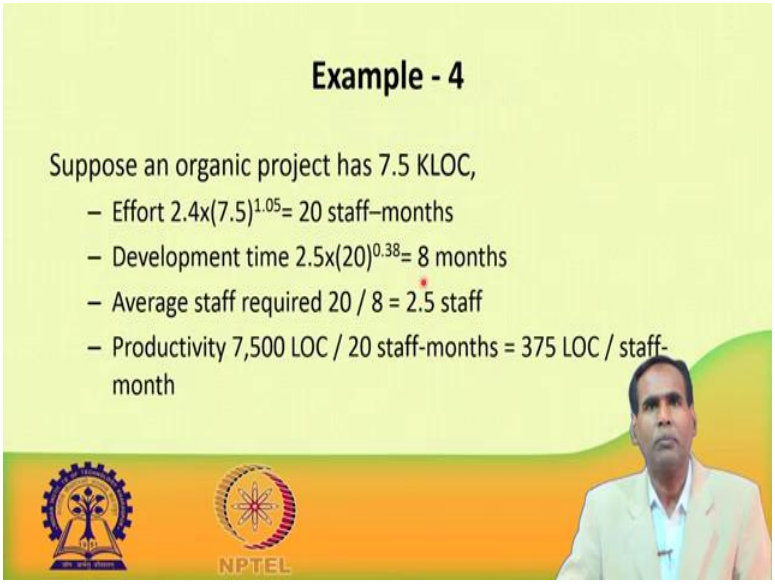

Still another example you can take. Again, suppose an organic project you want going to construct it is of 7.5, 7.5 KLOC it is size.

So, effort will be for organic product we know that these what the value of c is 2.4 and k is equal to 1.05. So, effort is equal to straight forward 2.4 into 7.5 to the power 1.05 that is nearly equal to 20 staff months; development, time is equal to 2.5 into 20 to the power constant for normal product, the value of constant is 0.38 which is coming to be on simplification it is coming to be approximately 8 months.

So, now the average staff required will be how you can calculate it? It will be calculated by dividing the effort by the development time. So, 20 by 8 so; that means, roughly or approximately 2.5 staff members will be required. So, the productivity if you will want to find out what will be the productivity then the productivity will be coming to as how much the total size is 7.5 or 7005 LOC, 7500 LOC divided by 20 staff months.

So, this is we got that 20 staff months will be required here, because effort is 20 staff months. So, that we get how much? So, for staff 375 LOC may be developed. So, the productivity is 375 LOC for staff month.

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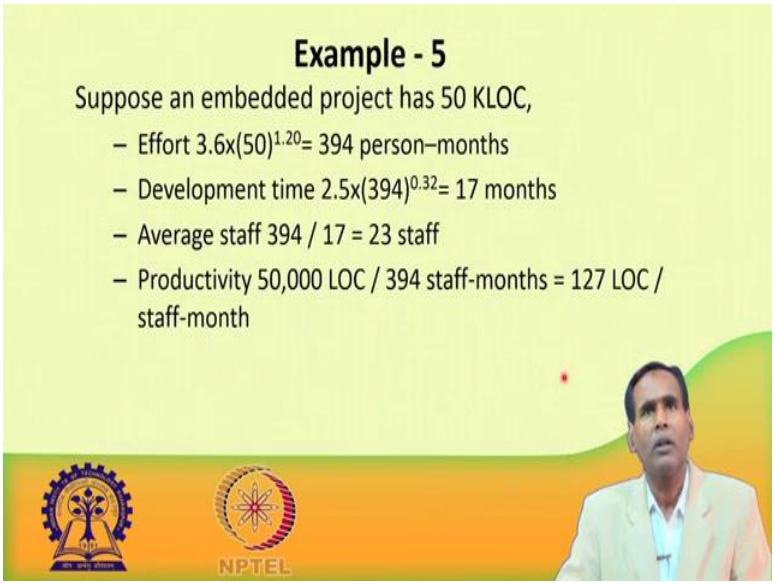

And the other example say it is like this. Suppose an embedded project has 50 KLOC and again we want to develop this value for the nominal effort to nominal development time and the average staff and the productivity here, this product size is already given 50 KLOC and type is embedded project, for embedded project, we know the value of c is 3.6 value of k is 1.20 by putting the values here we can get the value of effort.

So, let us put the value. So, 3 it will be now the equation will be simplified as 3.6 into 50 to the power 1.0, on solving you will get approximately 394 person months for this embedded project. The nominal development time will be 2.5 into effort to the power the exponent the constant. So, here effort is coming to be 394 and the value of constant or the value of the exponent for embedded type of project is 0.32. So, this require how much? On simplification you will get this is 17 months. Average staff how you will get?

So, what is the effort divided by the nominal development time is coming to be approximately 23 staff members, productivity can be obtained by dividing the total size of the project divided by what the productivity can be obtained by dividing what the total size by the effort which is in person months or staff months. So, this is equal to a 50000 LOC divided by 394 staff months, which is equal to how much? Approximately 127 LOC per staff months; that means, the productive is it is 127 LOC for I think it is yes the productivity is equal to how much for staff month this is or for person month this is 375 LOC and similarly, this here in this example the productivity is 121 approximately 127 lines of code for staff month.

So, in this way in the examination some questions may be asked simple questions, where you have to first identify what kind of product it is then what are the values of the constants for those products, then we can easily what then you have to remember the required equations for finding the value of effort and development time, then you can easily estimate the values for the effort and development time and then you can calculate the average staff, productivity and the total cost etcetera. So, please solve some of the examples and exercises from new book or from internet sources etcetera.

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So, a small exercise we have given to you like a software package is required by a company to mine the existing customer data to select prospective customers for a new launch. The estimated what it is estimated to be 30 KLOC of effort assume that the competent developers can be hired at 50000 per month. However, commercial offering support supporting almost all the required features cost, this how much 1 lakh. So, here you see that the developers can be hired at rupees 50000 per month, but here if you will purchase from outside; however, commercial offering if you market you will purchase it is total is coming 1 lakh.

So, the question say that should we or should the company develop it or build it this software or they will buy it. So, we have to think a decision. So, hence I am giving identify first what type of product if this is identify then calculate the effort, then what the what average, what salary or the developers is developer salary can be given per

month, then find out what is the total cost if it will be developed in house then compare with this market value, if somebody will come what purchase from the market it is 1 lakh compare this values, then take a decision whether the company should go for buying it or building it.

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These guild lines will help you to take a viable decision to the make or buy decision can be made based on the following conditions, the will the software product available sooner than internal developed software, if you will go to market definitely you will you can easily very soon you can purchase it.

So, we have to consider will the software or whether the software product be available sooner than internal developed software similarly, we have to think of when the cost of acquisition plus the cost of the customisation both they will be less than the cost of developing the software internally will the cost of outside support that is the maintenance contract etcetera will less than the cost of internal support.

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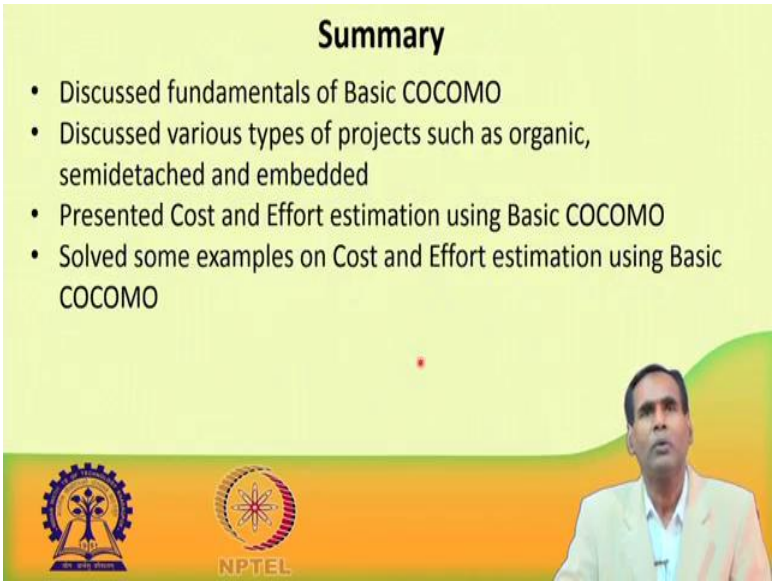

Say if we can what. So, before going for a buy build decision these issues we must have to consider, that this we must have to look what.

So, that then we can take a proper decision whether to buy or build a software. So, finally, we have come to the summary that we first discussed the fundamental concept of basic COCOMO, we have also discussed the various types of projects or products such as, organic semidetached and embedded we have also presented the cost and effort estimation that formula that formulae using basic COCOMO finally, we have solved some examples on estimation of cost and effort using basic COCOMO.

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We have taken the references from these books, the details you can find in these books.

Thank you very much.

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Lecture -22
Project Estimation Techniques (Contd.)

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Good morning. Now, let us take up the other cost estimation technique. Last class we have discussed about, basic COCOMO model. Now, let us take about other two types of COCOMO that is a intermediate COCOMO and a complete COCOMO.

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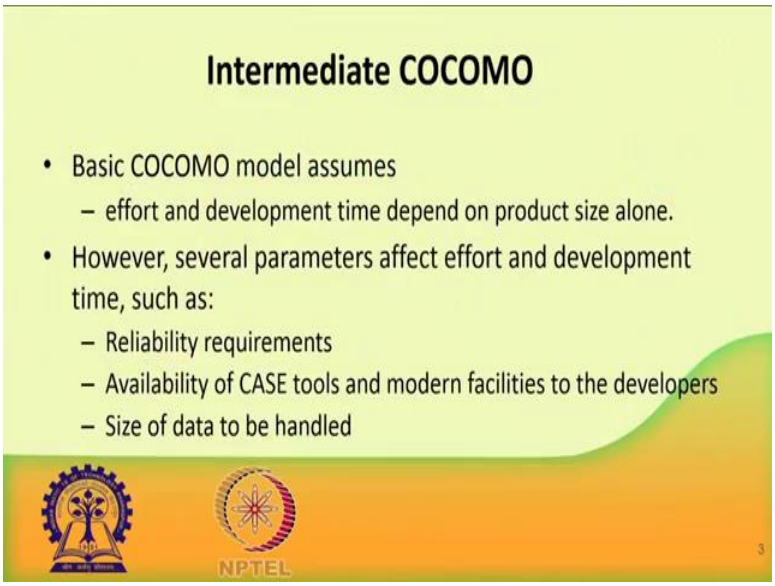

See ah there are little bit disadvantages of basic COCOMO, the basic COCOMO model assumes that the effort and development time, they depend on the product size only. However, you will see in the modern project developments several other parameters they also effect effort and development time, such as reliability requirements, you might require that your product should be highly reliable.

You may you are using several modern available case tools or some modern programming facilitates to the developers definitely the usual reduce the efforts, and this size of data to be handled it is quite large. So, these requirements or these parameters may affect the effort and development time. So, these have to be addressed in the COCOMO model. So, in order to consider these issues, the basic COCOMO model has been enhanced.

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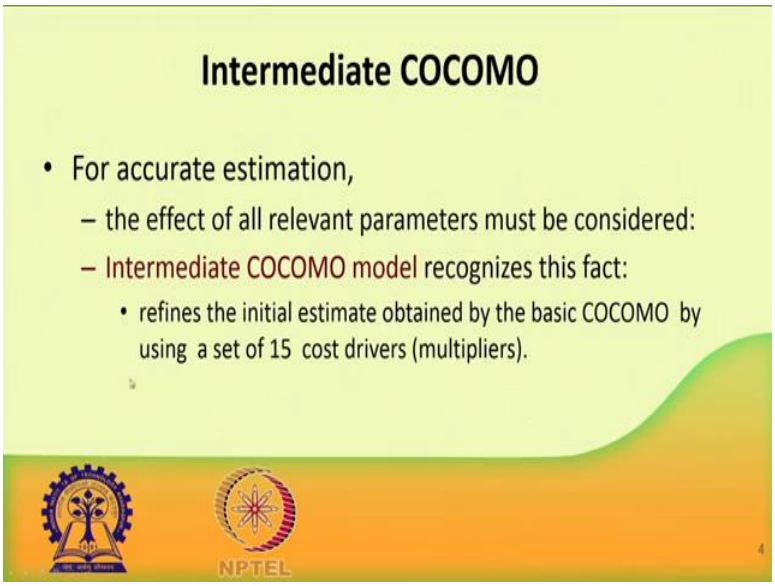

So, the newer COCOMO model we call as the intermediate model. So, for accurate estimation and to consider the effect of all such modern relevant parameters. So, a new COCOMO model little bit advanced COCOMO model was proposed that is known as a intermediate COCOMO. The intermediate COCOMO model recognizes the fact that, all these relevant parameters they have been addressed.

So, first you have to make the initial estimate, then the intermediate COCOMO refines the initial estimate which is obtained by the basic COCOMO by using a set of 15 cost drivers. So, first you have to prepare the initial estimate using COCOMO, then this initial estimate they can be it can be refined by using a set of 15 cost drivers or multipliers. Now, let us see what could be the possible what 15 cost drivers those are applicable for intermediate COCOMO.

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So, before going to the 15 list lets taken example that how these cost drivers or these what multipliers, they are affecting the effort or the cost. See for example, if somebody is using modern programming practical practices rather on using the traditional programming practices, then definitely the effort required will be less. So, hence the initial estimates obtained by the basic COCOMO they have to be scale downwards.

Similarly, suppose you for you are developing a real time system a real time application for which there is stringent reliable requirement, it must not fail there should not be any delay, they are should not be what the deadline must be met. So, there are several other reliability requirements on the product. Then what will happen, you have to put much more effort, what does it mean that the initial estimate what you have prepared, what you have ah obtained that has to be scaled upwards. So, in this way you can consider the other parameters how they are affecting, the estimation of the or the initial estimates of the effort and cost.

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So, then what you have to do as I have already told you intermediate COCOMO uses 15, what cost multipliers or these cost drivers, you have to rate the different parameters on a scale of one to three. So, these different parameters they can be rated on the scale of one to three depending on these ratings what you have to do, you have to multiply the cost driver values.

So, how may see all the 15 cost drivers or the these parameters may not be applicable for every system might be 1, 2 or 10 whatever that you know which parameters, which cost drivers will be required for your project. Find out or rate them using a scale of one to three. Then multiply the values of these cost driver these cost drivers with the estimate which is obtained by using the basic COCOMO. So, this product this multiplication the result of this multiplication will give you on more a more accurate estimate. So, this is the objective of intermediate COCOMO.

But please remember that even if I am saying here that the rating should be made in between one to three, but in some cases the value of the parameters or the cost drivers may be also less than 1. For example, when you are using modern programming practices instead of using the traditional programming practices, then what will happen; what will happen this effort required will be much less, much less means you have to go downward. So, we have to in that case the value of the cost driver will be less than 1;

might be 0.9 something that will I will give you in the table. So, what will the exact values for this cost drivers.

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So, let us see how these intermediate COCOMO proceeds, it takes the basic COCOMO as the starting point using the basic COCOMO prepare the initial estimate. Then identify what are the attributes they are affecting the cost and development time for your project; such as personnel attributes, product attributes, computer attributes and the project attributes. So, all those 15 attributes are categorized into these four categories. So, these attributes are expected to affect the cost and development time of your product.

Then what we have to do? We have to multiply the basic cost that you have obtained by the basic COCOMO. Then you what multiply the basic cost by these attribute multipliers or the cost multipliers which either may increase or decrease, this initial basic cost.

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Now, let us see what are the basic the attributes I have told there are 15 attributes categorized into four categories such as, personnel attributes, product attributes and other two are, I have already told you; there are four attributes will see. The four attributes are product attribute, project attribute, here personnel attribute and what computer attributes. So, these let us see all those 15 attributes under the these different arrays.

Under personnel attributes you will see that the analyst capability, virtual machine experience, programmer capability, programming language experience and application experience, these attributes are coming under personnel attributes related to the what related to the personnel working for the project. So, these are the codes there may be used for these attributes, like for analyst capability you may use the code ACAP, etcetera. So, for product attributes these are the product attributes, like reliability requirement, database size, product complexity these are the attributes related to the product that we have developing.

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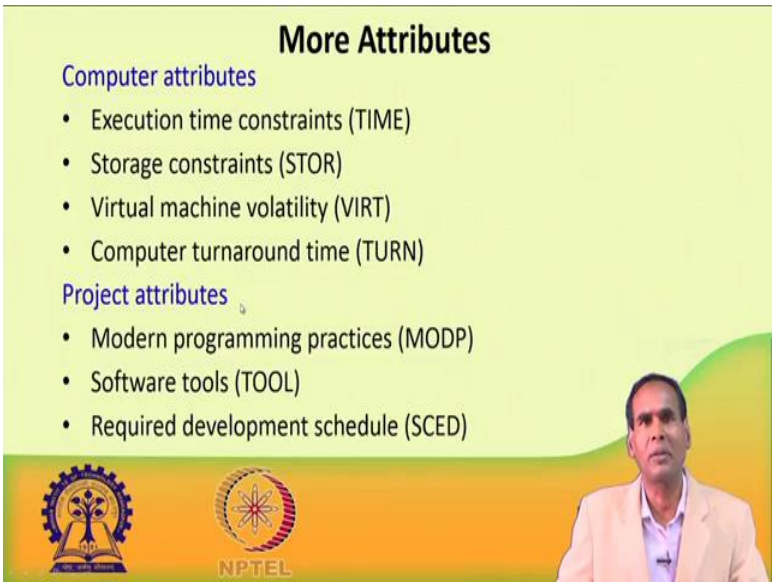

And then computer attributes you must consider, what are the execution time constraints, what are the storage constraints, the virtual memory volatility, the computer turnaround time, so these factors these attributes also may affect the what effort and cost of your product. And finally, project attributes or the environmentally attributes you must consider, project attributes also known as environment attributes.

So, what modern programming practices you are using? Like whether are you using software reengineering, reuse, reverse engineering, etcetera or object oriented programming, so all the modern programming practices, then that will make your effort and cost much less. Then what kind of software tools, automated software tools are using if I doing something manually, but if some tools are used then obviously, the effort required will be less. Similarly, required development schedule, so what is the schedule that is required; whether it is very tight schedule or this is flexible schedule, this will be also affecting the effort and cost. So, these are the 15 attributes that you must consider to get on up what more accurate estimate of the effort and cost.

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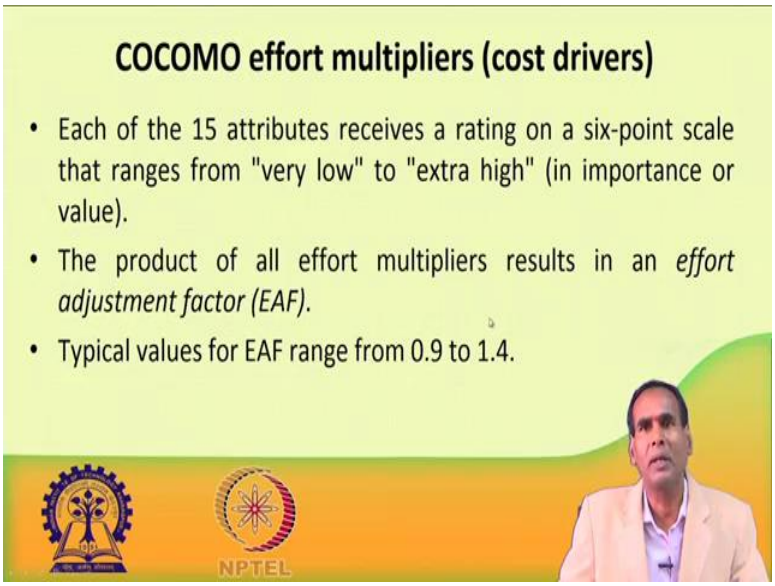

So, COCOMO effort multipliers as I have already told you, these are also known as the cost drivers. So, each of the 15 attributes they receive a rating, they receives each of the 15 attributes receives a rating on a six-point scale it will, we will use a six-point scale that will range from very low to extra high, I will show in importance or in value. So, this you may consider as the what is the importance, if it is importance is very low or low or medium or high or extra high like this or may use. So, for these different categories like difference just from very low to extra high, you may use the values one to three or any points, I will give you the table for these values.

So, then after for identifying which attributes are affecting your product, then see it can be categorized as whether very low or high etcetera, then find out the corresponding value from the table, I will show the table in the next slide. Then you have to find out the multiplication the product of all such effort mutiliers, and this multiplication will give you a what term which you call as EAF or known as the Effort Adjustment Factor.

Then and this effort adjustment factor may range from 0.9 to 1.4. Then what we have to do, we have already found out the basic the initial estimate by using basic COCOMO, multiply this EAF or the effort adjustment factor with the base initial estimate obtained by basic COCOMO which will give you the refined effort or the ah this will be the more accurate ah estimate for the effort.

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See these are the 15 cost drivers as I have already told you, there is a what six-point rating very low, low, nominal, high and very high; so it is I think 1, 2, 3, 4, 5, the five so here it is written six, so then this should be actually five, please correct it. On a rating of the five-point scale that will better and here it is five there are five what ratings are shown. So, like required reliability is equal to 0.7 when it is very low, when the required reliability is very high it should be 1.40.

And you can sees in some cases, the value will be less than 1. For example, if the product complexity is very low you may assign 0.70, these example I was saying that if modern programming practices are using and if what it is very high, then you see if it is high it is less than 1 nine 0.91; if it is very high it is 0.82. You can see the nominal values of all the cost by drivers the average values, the nominal values are all ones and based on that if it is high, it will be more than 1.

And normally here what is happening for some of the what ah some of the parameters like analyst capability, etcetera; if it is very low, it will be what greater than greater than 1. So, in this way you can use these values of the cost drivers to get the refined estimate.

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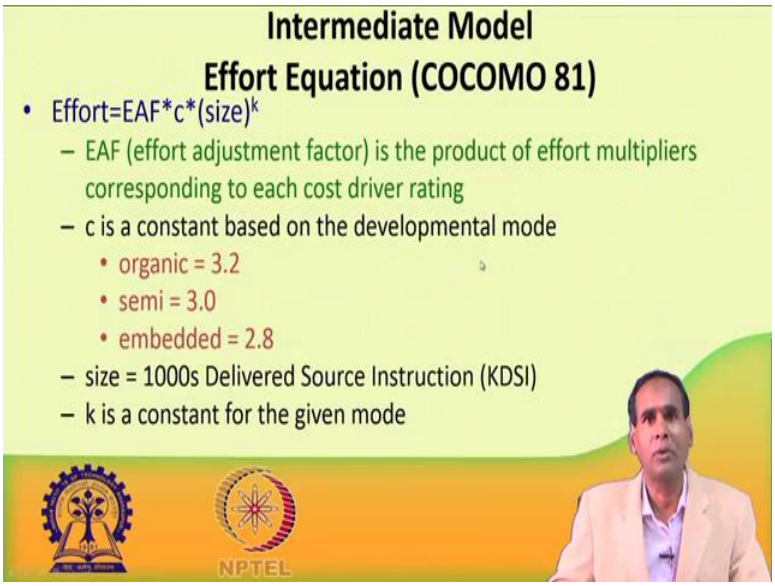

So, let us see what will be the equation for this intermediate effort, it intermediate model for calculating or estimating effort. So, the effort equation can be stated I like this that

$$\text{Effort}=\text{EAF}*c*(\text{size})^{k}$$

so c almost similar to the basic COCOMO only here EAF has been added. So, the EAF is the what this adjustment factor. So, you can see EAF is this Effort Adjustment Factor which is the product of what, the effort multipliers that you have considered for your product corresponding to each cost driver rating.

And c is a constant based on the developmental mode, we have already seen. There are three development modes, may be organic, semi detached and embedded and same values we have already used in what earlier; so for organic it is 3.2, for semi detached it is 3.0 and for embedded it is a 2.8. So, the size is equal to what 1000s delivery source instructions or KDSI and k is a constant for the given mode. So, in this way you can estimate the effort using this intermediate; using this intermediate COCOMO model, you can estimate the effort using this equation.

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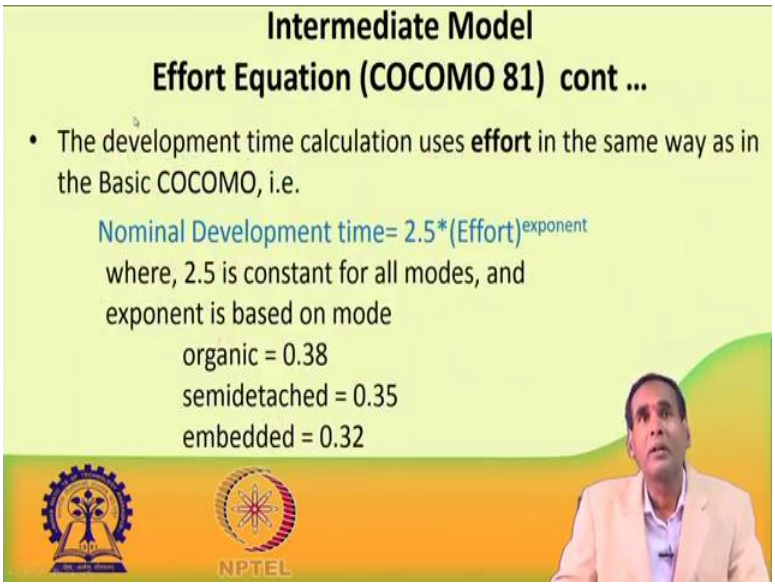

So, similarly you can also compute the time the development of the time calculation, uses the effort which you are getting ah by using this equation using the EAF. So, this effort you can use in this earlier equation that we have already discuss for the basic COCOMO. The development time calculation it uses effort in the same way as in the basic COCOMO that means, nominal development time will be

$$\text{Nominal Development time} = 2.5*(\text{Effort})^{\text{exponent}}$$

Here this effort is the effort that you have obtained by multiplying the what effort adjustment factor. So, this is the you can say refined effort or this is the revised effort that value you can put here, this will give you the development time.

Where, 2.5 is a constant for all the modes may be what organic semi detached or embedded. And this exponent is based on the mode that you are using, for organic modes this is 0.38, for organic products this is 0.38, for the semi detached products this is 0.35, for embedded products the value of exponent is 0.32. So, by using these two formulas; one for effort which is the multiplied with EAF and these one for this what nominal development time by using this formulas, you can estimate the value of the effort and cost ah by using this intermediate COCOMO model.

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Now, let us take a small example we will consider how to use this cost drivers or the effort multipliers. So, the example said that you are considering the what analyst capability, the capability of the; the capability of the analyst and then you want to find out the estimate for the effort.

You can see, it is given that see depending upon the capability of the analyst, you can rate this capability as very low or low, nominal, high or very high. So, the if it is very low, you can see it is 1.46 and low is 1.19 you can see in this table. See what is analyst capability, what is analysts capability; yes it is analyst capability, you can see analyst capability. Yes here, analyst capability if it is low, it is very low 1.46, if it is low 1.19, if it is nominal 1.0 and if it is high it will reduce 0.86 and it is very high again it will reduce 0.71. So, like this you can find out these values.

So, find out the values of the multipliers in this what we have already shown in the table. Now, suppose the initial estimate of the effort is 32.6 and suppose the analyst is having very high capability, analyst capability is high, analyst suppose the analyst capability is high that means, the value 0.80 multi this is the what only one cost driver is there, where multiplying this. So, this will the value of the effort adjustment factor that is 0.80. So, the value of this effort adjustment factor will be multiplied with 32.6. So, this will give you roughly 26.8 staff months.

So, in this way so we are getting the adjusted nominal estimate or the we are adjusting the nominal estimate for the effort. So, the what refined effort or the adjusted effort is coming to be 26.8 staff months by using the effort adjustment factors 0.80. So, in this way you can use the values of the cost multipliers, to find out the refined value of the refined estimate of the effort.

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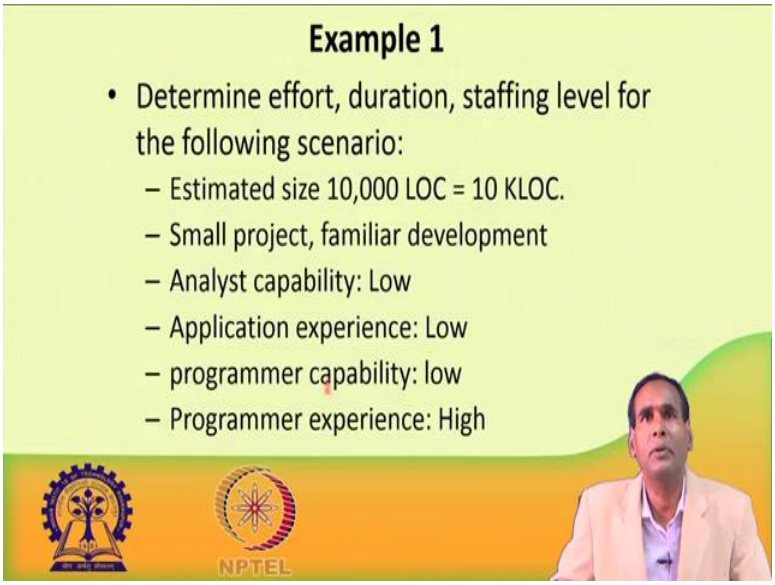

Now, let us take another example. So, determine effort, duration, staffing level for the following scenario. So, suppose according to what suppose you have estimated the size of a product is to be a 10,000 LOC which is equal to ten KLOC. And it is if it is small project, familiar development is there. And you are estimating that; you are expecting that or it is given that the analyst capability is low, application experience is also low. Programmer capability is a high and programmer experience is what and programmer experience is also high.

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So, what you do, you take we want to find out the cost effort, etcetera. It is needed to produce see from the problem, you can see that needed to produce a program of size 10 KLOC. Since is a small project and familiar development, it can be treated as a organic model. So, effort we will use what, we will first use these effort using what using the basic COCOMO.

So, if will take the basic COCOMO it is equal to what standard formula that this value of c is 2.4, size is 10 to the power the exponent value will be 1.05 for organic products, this will be 20 roughly 26.9 person-months. Development-time against a forward formula 2.5 into effort to the power what the exponent for organic product this is 0.38, so roughly it will give 8.7 months. And hence average staff will be how much, effort divided by development time so this is equal to roughly 3 people you will get. Now, what you will do, we have to get the refine value using the cost multipliers or the cost drivers.

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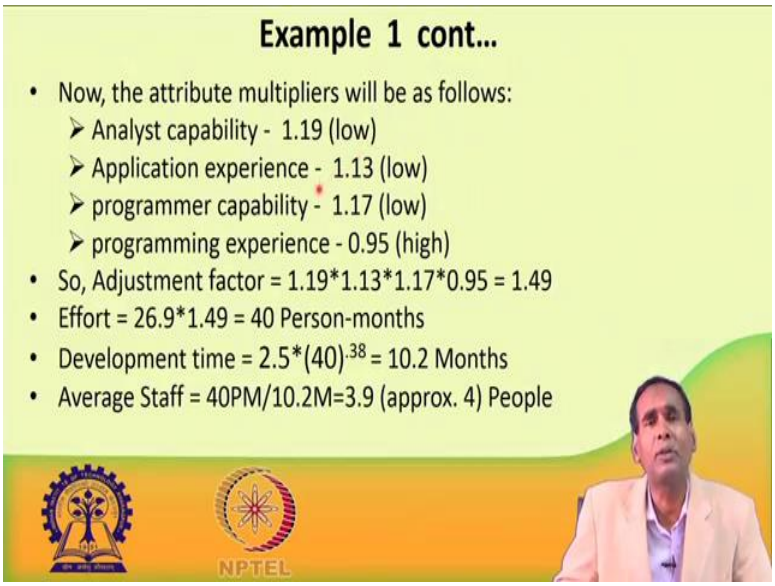

Now, the attribute multipliers will be ah as follows, these values we can get from the table that I had provided. So, analyst capability is low; low you know it is 1.15. You have application experience it is also low that is 1.13 and programmer capability it is showing programmer capability, perhaps will have to see here. Programmer capability is also low, here something just one minute. Yes, programmer capability is also low here.

So, programmer capability is low and this value you can take programmer what will low is 1.17 and, but programming experience is high. So, value will be the programming experience is high; it will be less than 1, so 0.95 already given in the table. So, the adjustment factor will be how much multiplication of all those values. So, this you will give you EAF Effort Adjustment Factor this is coming to 1.49.

So, the revive effort will be completed as the basic estimate of 26.9 into the EAF – Effort Adjustment Factor 1.49, this will give you 40 person months. Now, the development time will be what? 2.5 into 40 not these 26.9, I have to take the revised value that is 40 to the power exponent for organic product this is 0.38, so it will be roughly 10.2 months.

Now, the average staff; so these are all the what refined values by taking into account the values of the cost multipliers. So, average staff will be now required 40 by this is equal to 4 people 3.9 or approximately 4 people. So, in this way we can use these values of the cost drivers or these multipliers to refine the basic estimates more accurately. We will quickly take another example.

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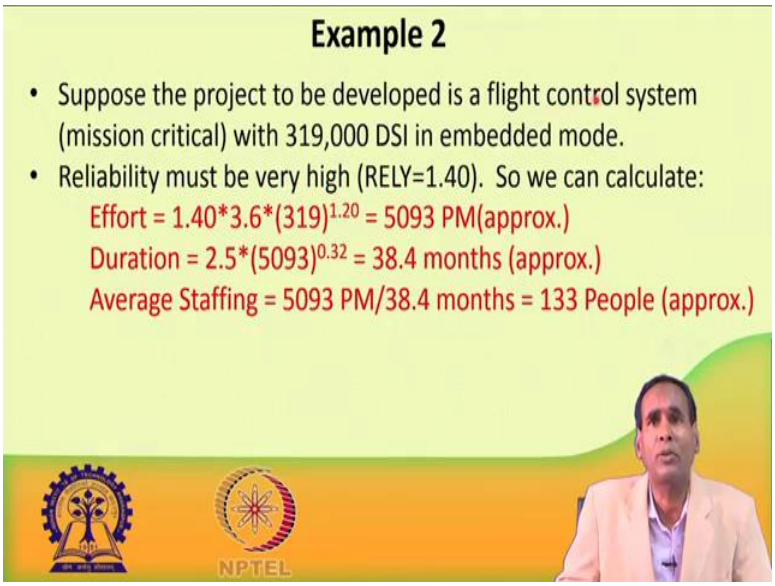

Suppose, we have to develop a project for a flight control system, which is a mission critical with 319000 DSI in embedded mode. Of course, this a flight control system this is a very risky project. So, it is what is a very complex project treated as embedded, it is a type of embedded product.

The reliability is required to be very it should be very high and for reliability very high values 1.40. So, now we can calculate the effort is equal to how much, the basic effort first we have to compute or the initial estimate by using the basic COCOMO; so 1.40 into the value for embedded value embedded case, the concern value c is 3.6 and the DSI is given 319 kilo what LOC and the exponent value for embedded mode is 1.20, this is equal to 509 percent month approximately.

Duration will be again strait forward formula, this will be 38.4 months and average staffing will be equal to how much effort by duration, so coming to be roughly 133 people approximately. Now, we have to use the cost drivers, you can see that only one cost driver is given, only one attribute is given, so I have to directly multiply with this.

So, and so I have to directly multiply of course, we have already direct so already I am sorry, that we have already in the while calculating the effort already I have taken into account this what reliability, because in according to basic COCOMO we should get only this much 3.6 into 319 to the power 1.20.

And if we will apply intermediate COCOMO, I have to take the value of the effort adjustment factor; since only one cost driver is given that is reliability which is equal to what 1.40 for very high. So, I have multiplied here 1.40 into 3.6 into this, so we will get 5093 percent month, this is the refined estimate for effort.

Similarly, duration you can use this what refined effort the what revised effort that is 5093 to the power 0.32, it will give you the revised duration. And this average staffing will be equal to this revised effort or the refined efforts by this duration which is equal to 133 people for approximately. In this way, you can use the cost drivers to refine the value of the initial estimates.

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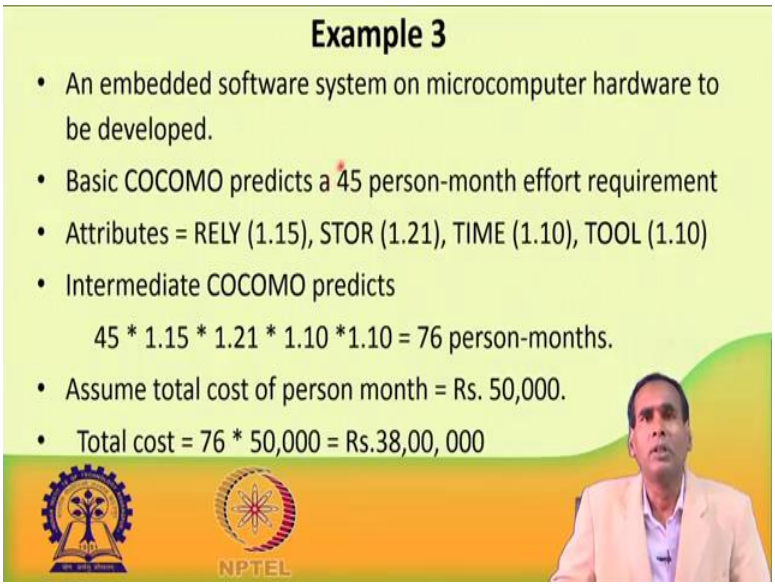

So, similarly quickly we will take the other example and embedded software system on micro computer hardware is it has to be developed. Basic COCOMO it predicts that 45 person-months effort is required. So, here there are four attributes are required, so four cost drivers you have to find out; one is reliabilities, storage, time and tool. The values are already given and suppose these values are all already given for this embedded software system.

So, intermediate COCOMO can now predict or estimate the effort as such what will be this, so already basic COCOMO gives 45 person months, so 45 into effort adjustment factor. How we can find out the effort adjustment factor? By multiplying all these what

the values of the cost drivers, so for cost drivers there values are modified are multiplied. So, this will give you 76 person months.

So, assume that the cost per months person month that we have to pay each rupees 50000. So, the total cost for developing this embedded software system products will be 76 into 50000 which will be equal to rupees 3800000. So, in this way you can use the embedded software, you can use the intermediate COCOMO to estimate or to refine the basic estimates.

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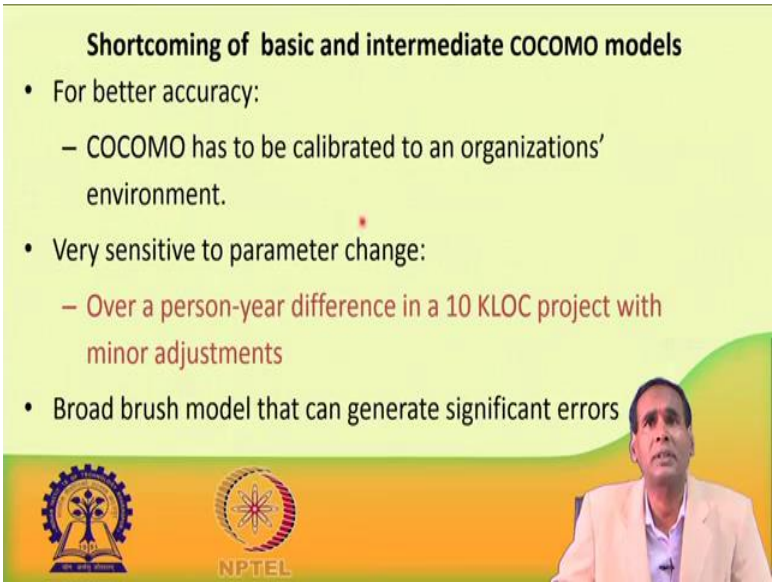

So, now let us quickly see about the shortcomings of basic and intermediate COCOMO model. For better accuracy COCOMO has to calibrated to an organizations environment, where is it is very sensitive to parameter change if you just make some my, if you just do so minor adjustments, minor changes over a person-year difference may arise in case of 10 KLOC project. It is a broad brush model that can generate many significant errors, these are some of the drawbacks.

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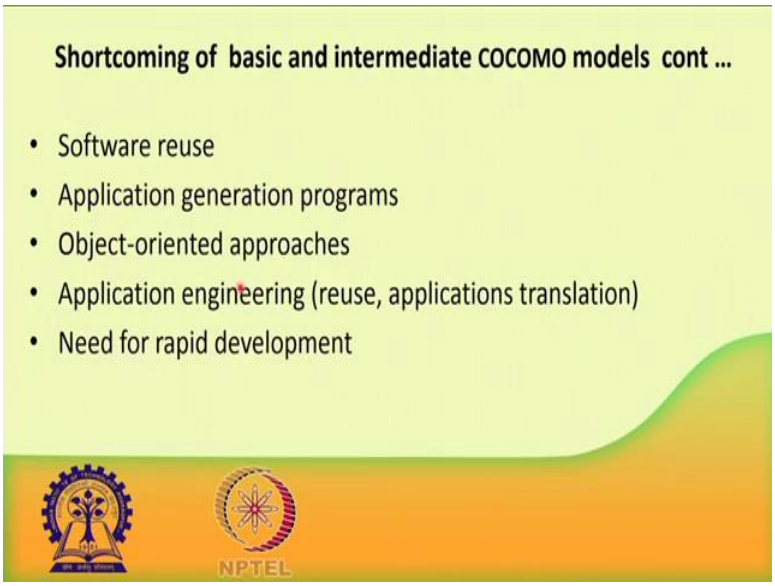

Similarly, as you know this in this modern what today's software reuse is there, application generation programs are there, object oriented approached are used and application engineering such as reuse, reverse engineering, application translation, etcetera, they are and so there are need for rapid development. So, this basic and intermediate COCOMO they do not considered these modern practices. So, one so these are the some of the drawbacks the basic and intermediate COCOMO models do not consider these modern programming what practices. So, there is need to go for another model.

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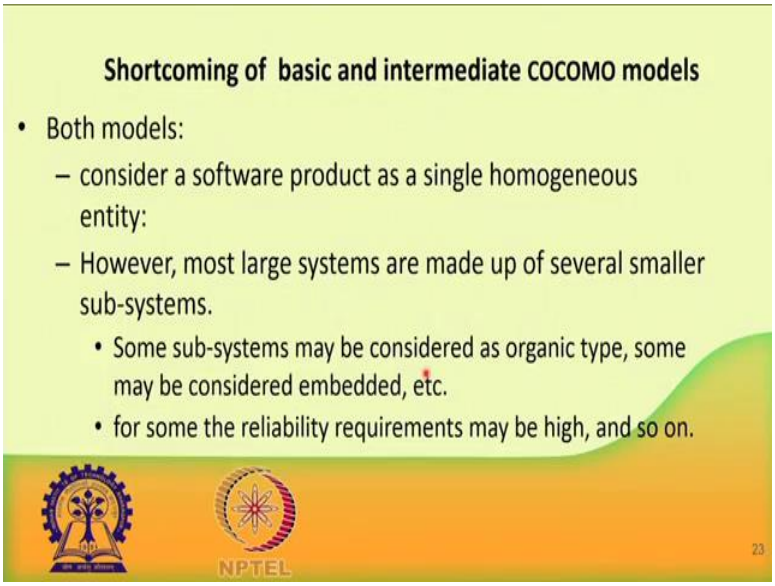

So, we will see still then another important drawback is there, which say that this both the models. Basic model and intermediate COCOMO, they consider a software product as a single homogenous entity as if it is considered only one type of what components. But however, most large systems they are made up of several smaller sub-systems, where some sub-systems may be considered organic time type, some sub-systems maybe of what embedded type, some sub-systems may be of the semi detached type ok.

And in some cases reliability, some of the systems may require high reliability requirements or at the some of them may require very low reliability requirements things like that. So, your basic and intermediate COCOMO cannot handle all this things, because it considers all the sub components are treated as say similar, because the what software product is considered as a single homogenous entity.

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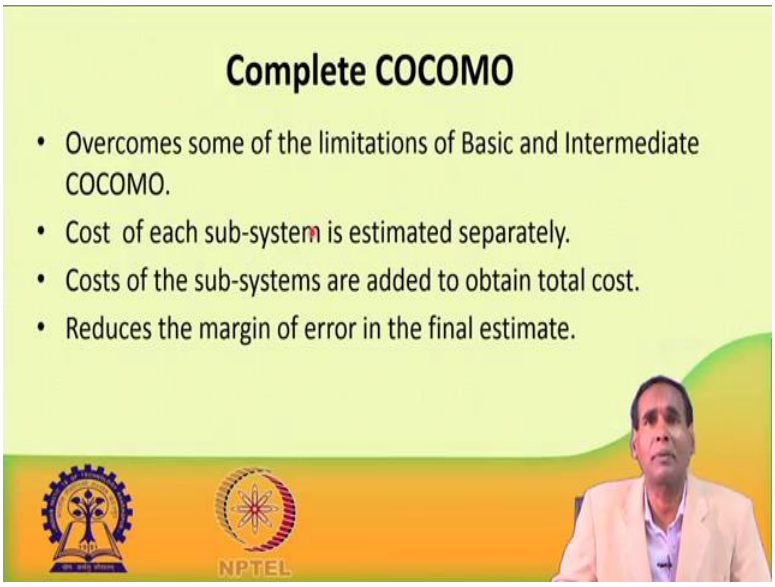

So that is why a new so another advanced model of COCOMO it has come into existence that is known as complete COCOMO or detailed COCOMO. It overcomes some of the limitations of a basic and intermediate COCOMO. How does it work? Here the cost of each sub-system is first estimated separately using this what basic COCOMO or the intermediate COCOMO. Then cost of the sub-systems they are added to obtain the total cost. So, because the as I have already told you, some of the sub-systems might be organic, some of the sub-systems might be semi detach, some of the sub-systems might be what embedded, in some substance they reliability might be very high.

So, you have to estimate the cost of each sub-system separately, individually, then the cost of the substance they are added to obtain the total cost. This will reduce the margin of the error in the final estimate; obviously, we will get a less error or this will reduce the margin of there are in the final estimate.

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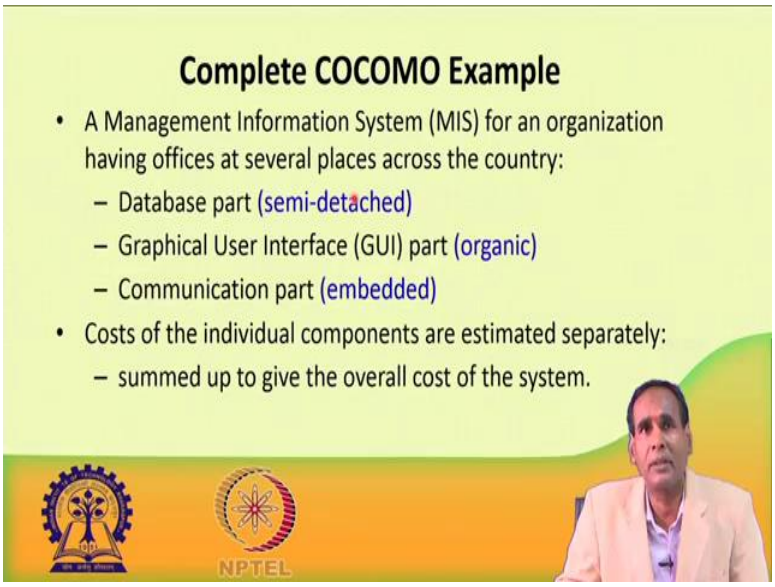

Now, let us quickly take a small example. A management information system suppose we have to develop for an organization, which is having offices at several places across the country. So, then this will be a very complex project, this is not an homogenous what project, this consist of different types of sub-systems.

Like one part may be may contain the data base part, you may treat as a semi detached system. One part one component might contain development of graphical user interfaces; obviously, this will be treated as organic component or organic product. Similarly, since the MIS has offices at different locations in the country. So, the these different offices they have to communicate. So, you have to develop also a module for communication. So, this module will be treated as embedded.

So, now what we have to do, compute the cost or estimate the cost for these three components, which are of three different types by using either basic COCOMO or intermediate COCOMO. Then what you do, add up the cost of the individual components ok, first estimate the cost of the individual components separately for each of these what components. Then you add up these the cost of these individual

components, which will give you the overall cost the total cost of the system. In this way, the complete COCOMO or the detail COCOMO it works.

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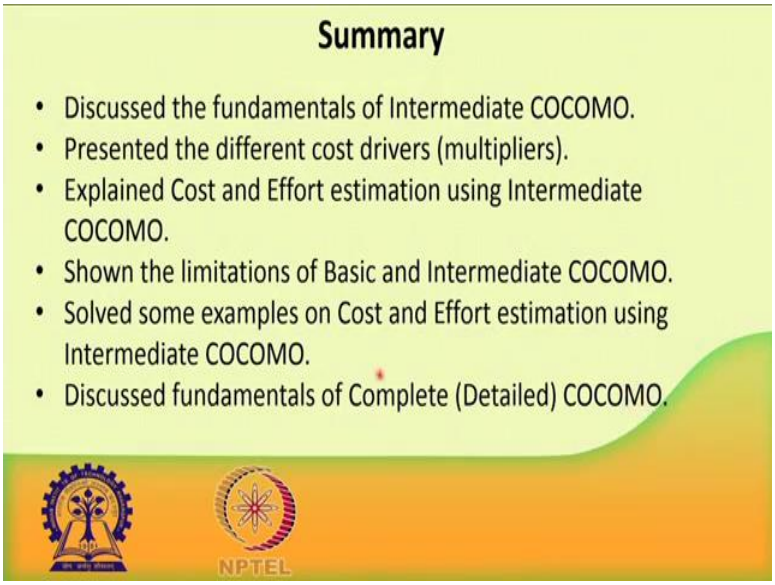

So, finally we have come to the summary we have discussed the fundamentals of intermediate COCOMO, we have presented the different cost drivers or multiplies; 15 cost drivers and their values we have seen. Then also we have explained the cost and effort estimation using intermediate COCOMO, we have shown some of the limitations of basic and intermediate COCOMO. We have solved some examples on cost and effort estimation using intermediate COCOMO, we have also discuss the fundamental concepts of complete COCOMO or detail COCOMO, this is the summary that we have seen.

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And we have taken from these books; these texts.

Thank you very much.

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Lecture - 23
Project Estimation Techniques (Contd.)

Good morning, so now let us see another important type of cost estimation technique another variation of COCOMO, we call it as COCOMO II. So, somewhere you will find it as COCOMO roman letters II somewhere in some books they have written COCOMO that digit as COCOMO 2 . So, that will that will be no problem.

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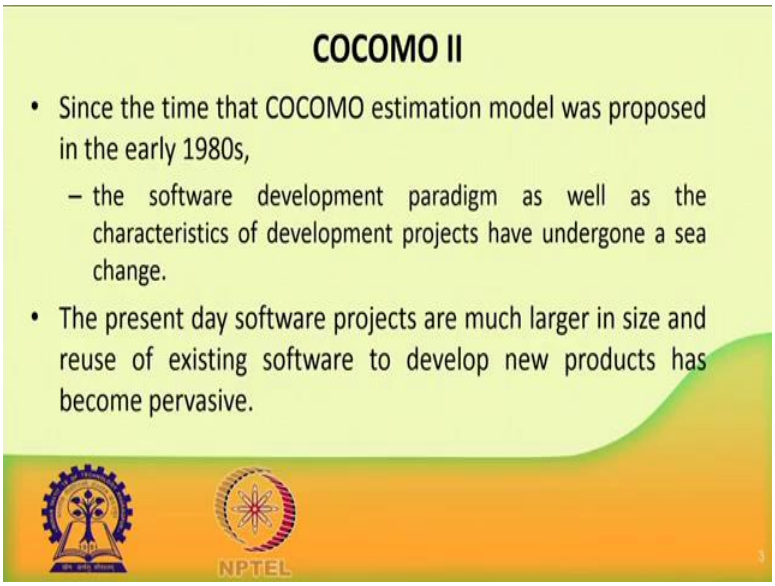

So, COCOMO II you see that since the time that COCOMO estimation model was proposed in early 1980s; then the software development paradigm as well as the characteristics of develop projects have undergone a large change. The present day software projects are much larger in size and they reuse the existing software components to develop new products that has also become pervasive.

So, similarly other new techniques such as reverse engineering, reengineering all those things have use of automated tools etcetera case tools of come up. So, they need to be addressed while estimating the cost. So, COCOMO the initial version of COCOMO that has been extended, and that is known as the COCOMO II.

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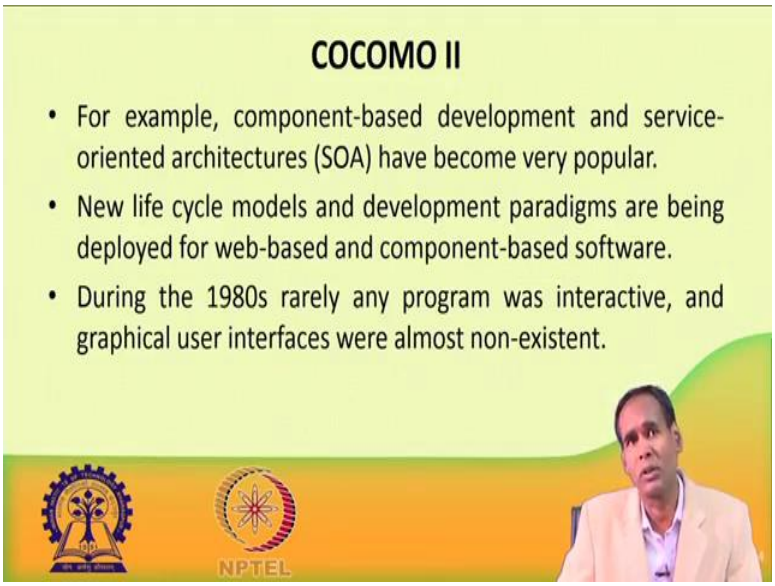

For example, in case of component-based software development and service oriented architectures they have also become popular. How we can estimate the effort or the cost of or developing component based software or for developing SOA based applications, new life cycle models such as your RAD, scrum or extreme programming etcetera.

They have come up also new development paradigms are being deployed for web-based applications for component-based software. So, they need to be also taken into account while estimating the effort and cost. During the 1900 1980s rarely any program was interactive but now you see due to because graphical users were GUIs Graphical User Interfaces were also almost non-existent. But now did you see any program without having GUI it is of no use nobody will use that one.

So, how to taken to account the development of GUIs while estimating the effort, cost etcetera that is also very much important.

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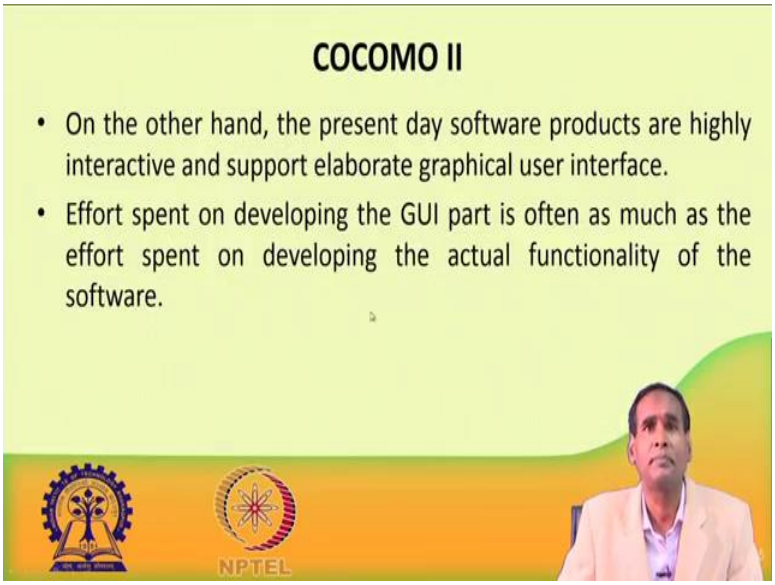

On the other hand, the present day software products are highly interactive, they support elaborated graphical user interfaces, so they have to be considered while estimating the effort and cost. Effort spent on developing the GUI part is often as much as the effort spent on developing the actual functionality of the software, see almost 50 percent of the effort is spent on the developing the GUI part, so that cannot be neglected.

So, since effort spent on developing the GUI part is often as much as effort spent on developing the actual functionality of the software. So, we must have to taken to account taken to consideration this developing GUI part while estimating the effort and cost.

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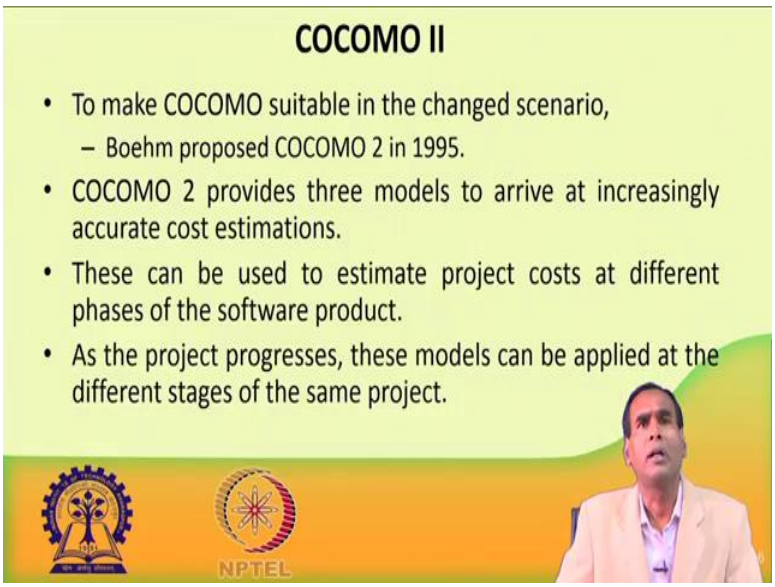

In order to make COCOMO suitable in this changed scenario where this modern programming paradigms are there, GUI is there and this is say other technique reverse engineering reuse etcetera there Boehm proposed COCOMO 2 in 1995. So, in as I have already told you some books they are using this COCOMO II roman letter some are just this COCOMO 2 in this digit, so please do not confuse.

COCOMO 2 provides three models to arrive at the increasingly accurate cost estimations taking into account on those all those new things modern techniques. This can be used to estimate the project cost at a different phases of the software product. As the project progresses these models of they can be applied at different stages of the same project.

Actually here it is written three models, but now a days we have found recently this recent material they are saying four model, I will see one more model that is added that is this for a reuse. So, you can correct it as four models.

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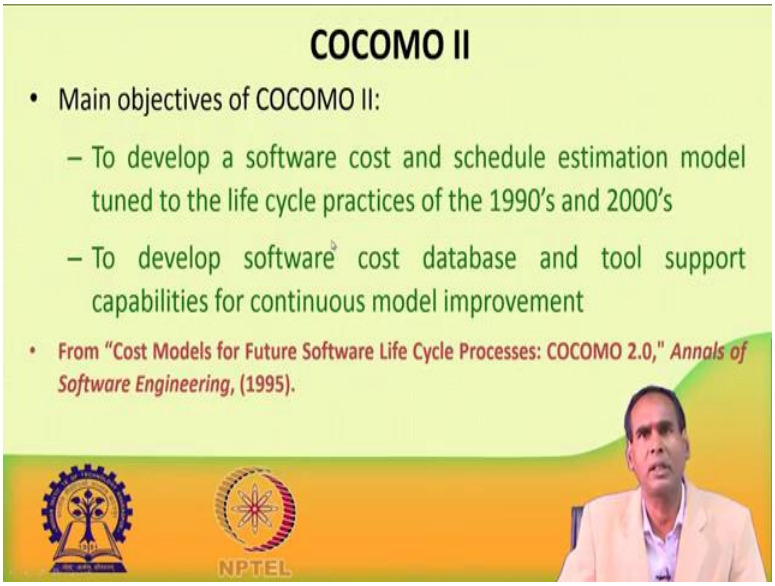

So, now, let us see what are the main objectives of COCOMO II. The main objective of COCOMO II is to develop a software cost and schedule estimation model which is tuned to the life cycle practices of 1990's and 2000's. Another objective is to develop software cost database and tool support capabilities for continuous model improvement. So, this has been taken from the Cost Models for Future Software Life Cycle Processes: COCOMO 2.0 in the published in the Annals of Software Engineering, 1995.

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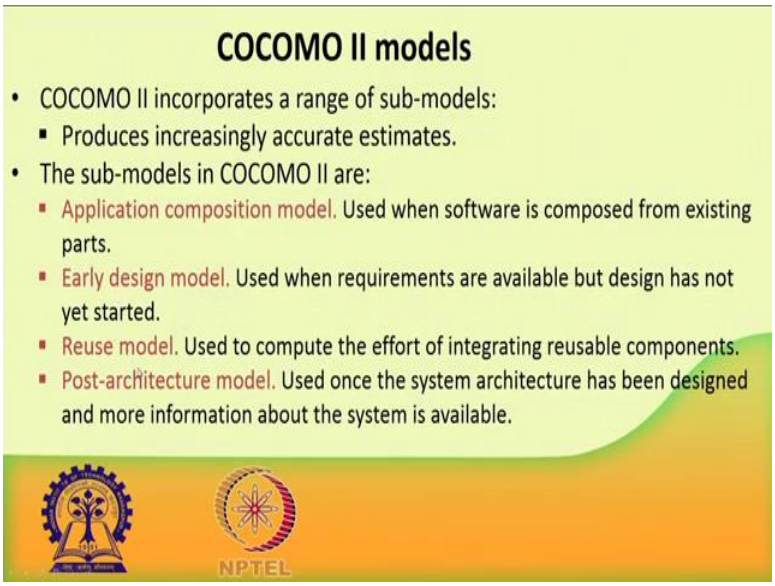

COCOMO II models are, let us see the COCOMO II incorporates a range of sub-models, it produces increasingly accurate estimates. So, here the estimates are more accurate than the previous COCOMO models. The sub models in COCOMO II are I have already told you there will be four sub-models, those are application composition model, early design model, reuse model, and post-architecture model.

So, application composition model is used when the software is composed from the existing parts. Some parts are there you are composing what these existing parts to develop a new one, so when, so then you have to use application composition model. So, this model is used when software is composed from existing parts.

Next one is early design model. So, this -sub model is used when the requirements are available, but design has not yet started only requirements are known then you can go for early design model; that is why the name early is used. Reuse model, so it is used to compute the effort of integrating reusable components.

So, there are some component, some code, or some database they are already existing; they can be reusable in another application then you can use reuse model. So, reuse model is used to compute the effort of integrating some reusable components. And next one is post-architectural model, this is used once the system architecture has been designed and no information about the system is available.

So, when the system architecture has already been designed but more information about the system and more information about the system is available then you can get use this post-architecture model that is why the name is post, post- architecture Architecture is already designed and more information is already available, that is why we are we have to now develop the model and estimates the cost etcetera. So, since after this architecture is prepared this is known as post-architecture model.

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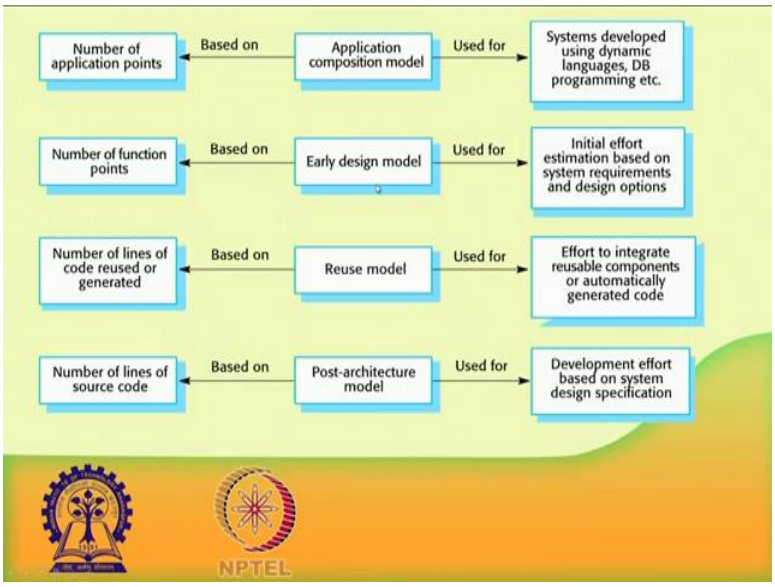

So, this is what summarized here, application composition model is based on the number of application points and it is used for systems developed using dynamic languages, data bases, programming data base programming etcetera and early design model is based on number of function points which is used for the initial effort estimation based on system requirements and design options.

Reuse model is based on number of lines of a code that is reused or generated and it is used for effort to integrate reusable components or automatically generated code. Post-architecture model is based on numbers of lines of source code which are used and this is used for development effort which is based on system design specification.

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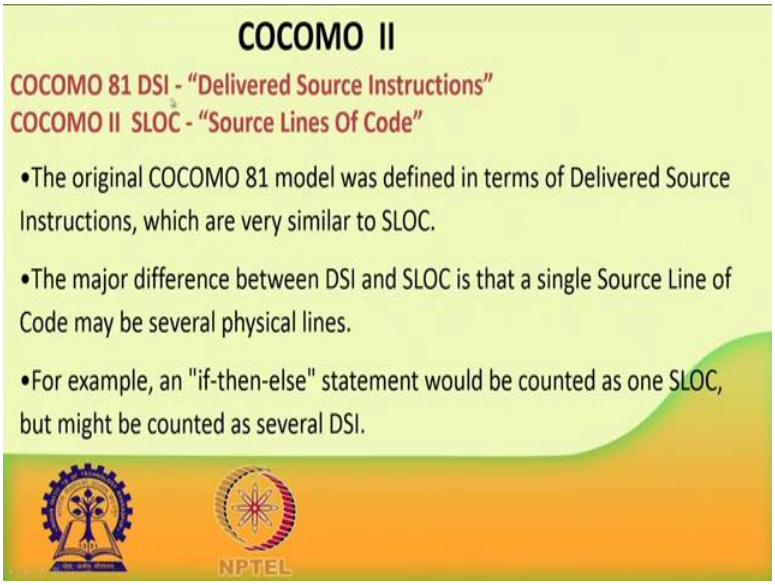

So, in COCOMO 81, we are using the term DSI which stands for the Delivered Source Instructions, but in COCOMO II, we are using term SLOC Source Lines of Code. So, that we have which is, so this Delivered Source Instructions is very much similar to SLOC, but there is a one difference.

The major difference is that, a single Source Line of Code it may be several physical lines. For example, if there is a if-then-else statement, then this if-then-else statement would be counted as only one source lines of code. But it may be counted as several Delivered Source Instructions this is the important difference between DSI and SLOC.

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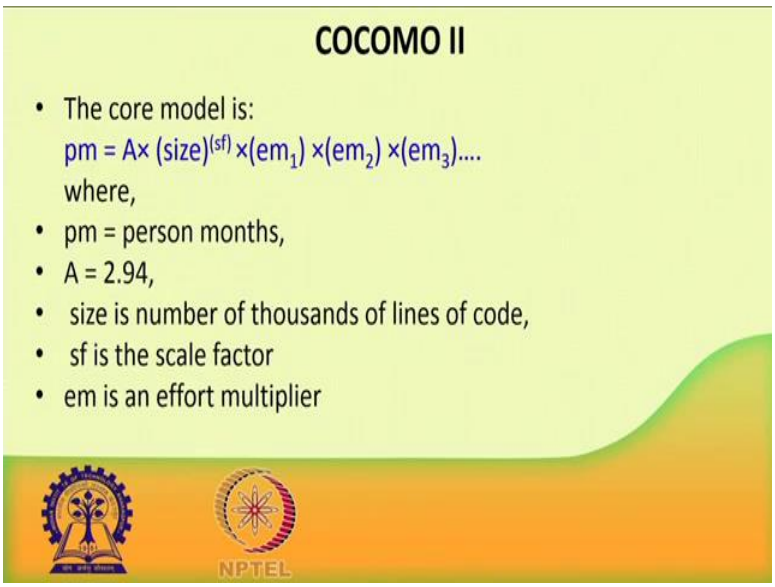

Now, let us see what is the core model how can effort, how can estimate the effort for or by using COCOMO II for any model. The core model says that

$$pm = A\times (size)^{(sf)} \times(em_1) \times(em_2) \times(em_3)\ldots.$$

where pm is the effort in a person months, A is a constant whose value is 2.94, size is the number of thousands of lines of code, sf is known as scale factor which is taken as the exponent here and em this em i's they are what; they are the effort multipliers already effort multiplier we have seen in case of intermediate COCOMO.

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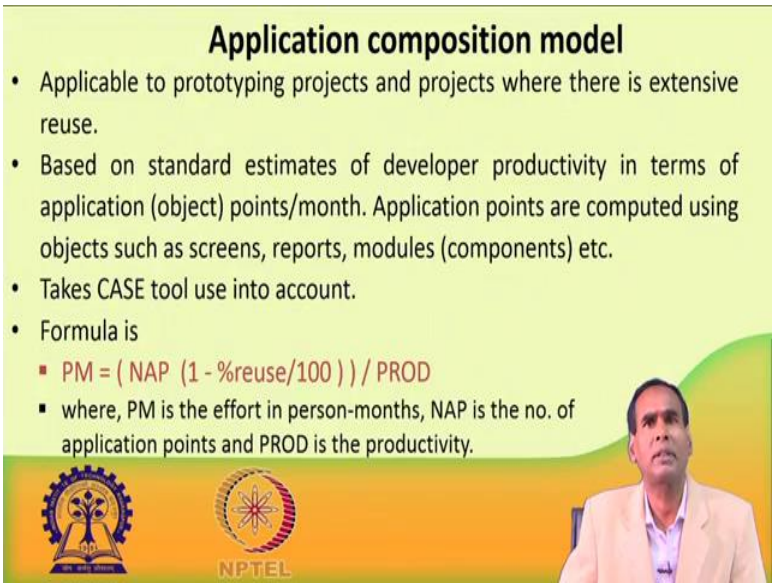

Now, let us see the first sub-model, that is a application composition model. I have already told you this is applicable to prototyping projects and projects where there is extensive use. So, this is applicable to prototyping projects, I hope you have already known about prototype models. So, this is applicable to prototyping projects and projects where is extensive scope of reuse. This is based on the standard estimates of developer productivity in terms of application points or object points.

So, this application composition model that is why the name is application. Application composition model it is based on the standard estimates of the developer productivity in terms of what; in terms of application points or object points per month. So, this what is this application points we will see, the application points are normally computed using

the objects such as the number of screens, number of reports, number of modules or components present in the your application.

Please do not confuse here this objects is nothing related to the object oriented programming ok. Here objects maybe treated the examples of objects be screens, reports, modules. So, the number of screens, number of reports, number of modules or components etcetera they can be treated as the objects and by you have to count those objects.

So, based on these objects you can compute the value for the application point or number of application points. So, application composition model takes the case tools used into account. So, now, let us see the formula for estimating effort using application composition model.

So, the formula for computing for estimating effort using application composition model is given by

$$PM = (NAP \ (1 - \%reuse/100)) / PROD$$

where PM is the effort in terms of person-months, NAP stands for number of application points which can be found out which can be computed by what using the different objects such as numbers of screens, number of reports, numbers of modules etcetera and the PROD is PROD stands for the productivity.

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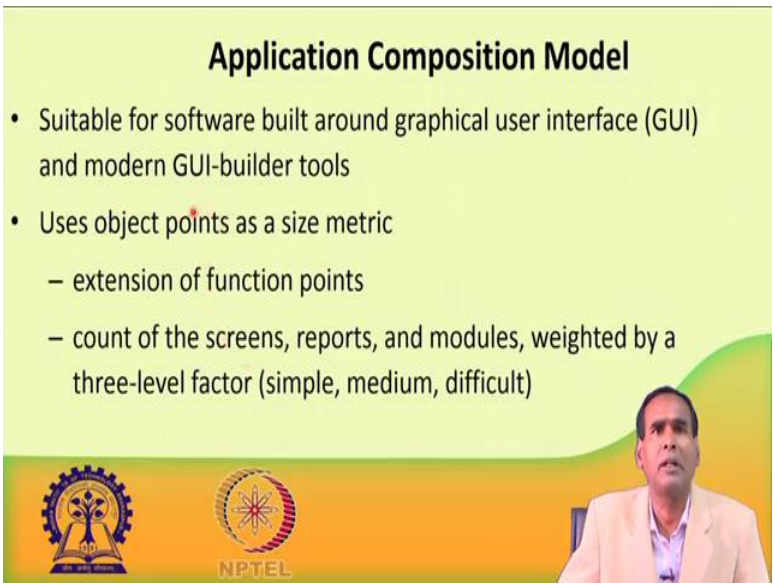

So, this model is suitable for softwares built around Graphical User Interfaces and the modern GUI builder tools. This model uses object points as a size metric. So, I have already told you this model uses application points or object points as the what size metric, this is a extension of these object point or application point these are extensions of function points.

Here we have to count these object use or the object point or application point it is a count of the screens, reports, and modules; this I have already told you. This is based on counting the number of screens, number of reports, number of modules etcetera which are weighted by a three-level factor, either they could be simple, medium, or difficult.

So, the application points or objects points they are size metrics and they are based on counting the number of screens, reports, and modules which are weighted by a three-level factor may be simple, medium, and difficult.

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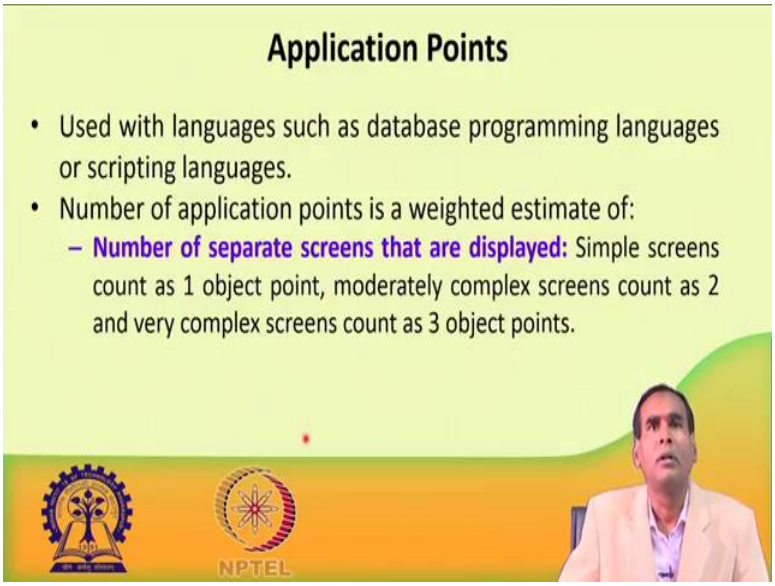

So, now let us see quickly about application points. These are reused with languages such as database programming languages, or scripting languages. Number of application points is a weighted estimate of three things, I have already told here that these number of application points they can be used or they can be found out by counting the number of screens, reports, and modules weighted by a three-level factor.

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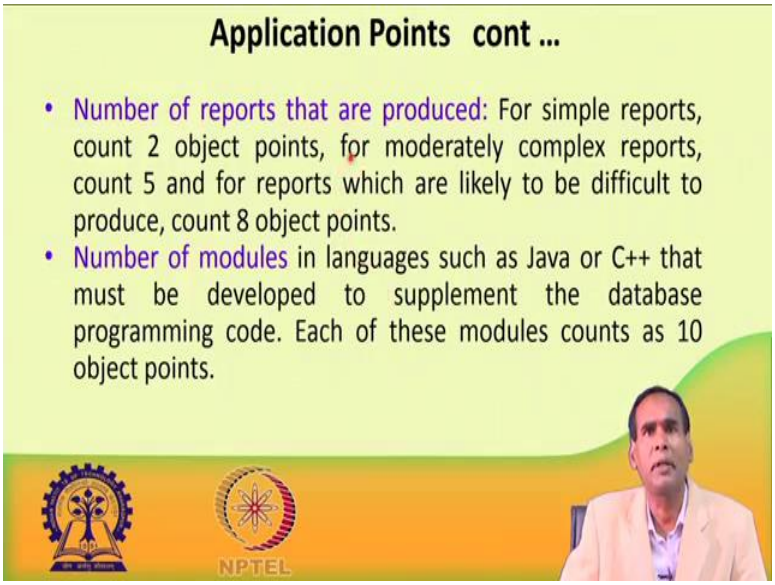

You can see that the number of application points is weighted estimate of three things, what; the number of separate screens that are displayed, the number of reports that are produced, and the number of modules.

So, let us see the first one, number of separate screens that are displayed. So, here simple screens they are counted as 1 object point, moderately complex screens they are count as 2 and very complex screens they are counted as 3 object points.

Similarly, number of reports that are produced, how they are categorised or rated in the scale of say 1, 2, 3 or I think something more than that. Here it is rated as the screens are rated as in between 1, 2, 3. Let us see how the number of reports they can be rated; number of report that are produced they can be rated as follows.

For simple reports count 2 object points, for moderately complex reports you count to 5, and for reports which are likely to be very much difficult to produce for count 8 object points. Similarly, number of modules in languages the number of modules how we can count? So, the number of modules in languages such as Java or C plus plus etcetera that must be developed to supplement the database programming code. Each of these modules counts as 10 object points.

So, if you are using languages such as Java or C plus plus etcetera to develop or they must be developed to supplement the database programming code then each of these modules they counts as 10 object points.

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See in the this is a table showing the application-point productivity. So, the developer's experience and capability they are rated as like on a 5 point scale; very low, low, nominal, high, very high. Similarly the ICASE maturity capability, it can be very low, low, nominal, high, very high; these are the values of productivity.

That means if developer's experience and capability is very low and maturity and capability is very low; use a factor of 4. This is the number of, so here productivity is counted as number of application points divided by month. Similarly if developer's experience and capability is low, maturity and capability is low; then the productivity is counted as 7.

So, this have been what Boehm has already produced. These values have been has have already been produced by conducting so many research works by taking different what data.

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Now, let us see about The Scale Drivers, I have already showed you in the what in the initial function there was a what factor called as this scale factor. Now, let us see how this scale factor is defined. The scale drivers or the exponent parts are computed like this.

This is an important factor contributing to a project's duration and cost. The scale drivers they determine the exponent that has to be used in the Effort equation. Scale drivers are replaced; they have replaced the development modes of COCOMO 81, see in COCOMO 81 we are having three development modes.

One for your organic, semidetached and embedded but in COCOMO II they have used scale drivers and they have replaced all those three development modes of COCOMO 81. What are the possible 5 scale drivers used in COCOMO II; they are as follows. precedentedness, development flexibility, architecture or risk resolution, team cohesion, and process maturity.

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Then we will come to the next model, that early design model and the post-architecture model. So, here effort is calculated as follows,

$$\text{Effort} = (\text{Environment multipliers}) * [\text{size}]^{(\text{process scale factors})}$$

where environment multipliers could be product, platform, people, and project factors; size is what that normally that the basic size that is estimated and these here it takes into account reuse and volatility effects, process scale factors are what, the constraints, risk architecture, team, maturity factors etcetera.

The schedule can be calculated in early design and post-architecture model as follows,

$$\text{Schedule} = (\text{multiplier}) * [\text{Effort}]^{(\text{process scale factors})}$$

where these are the process scale factors, constraints, risk architecture, team and maturity factors.

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Now, let us see another approach COCOMO II scaling exponent approach. So, here the nominal person-months is calculated as follows,

$$\text{Nominal person-months} = A*(\text{size})^B$$

where B is computed as using this formula

$$B = 0.91 + 0.01 \sum(\text{scale factor ratings})$$

B value may range from 0.91 to 1.23, there are five scale factors; and six rating levels we will see. The five scale factors are those things precedentedness, development flexibility, architecture or risk resolution, team cohesion, and process maturity and the six ratings we will; so what are these six; so, process maturity it is a derived from SEI CMM model that Capability Maturity Model. These are the codes they are referred they will be referred while doing these problems. So, now let us see what are the six rating levels for these five scale factors.

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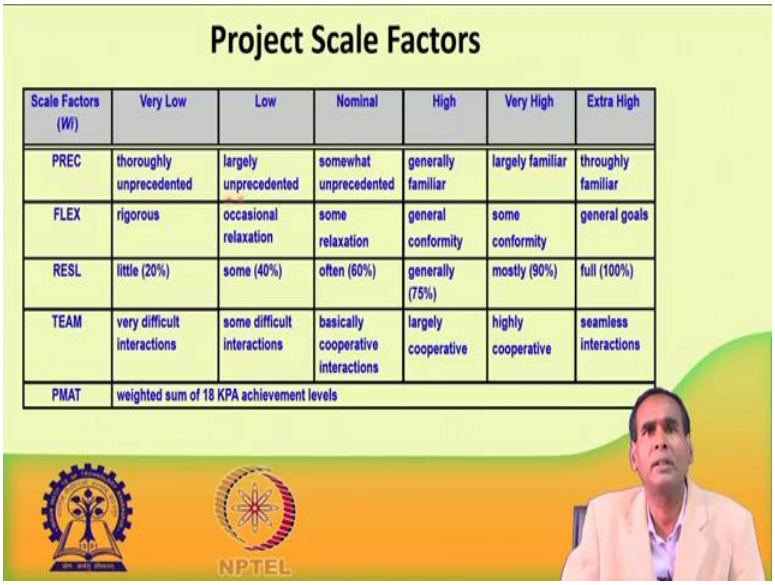

So, these are the six rating levels; like very low, low, nominal, high, very high, and extra high. The scale factors are the codes we have used here like PREC, FLEX, RESL, TEAM, PMAT etcetera. So, those are shown here these are the scale factors and they are rated on what a 6 scale; 6 point scale and this PMAT, it is a weighted sum of 18 KPA achievement levels in where in the CMM or yes, in the CMMI.

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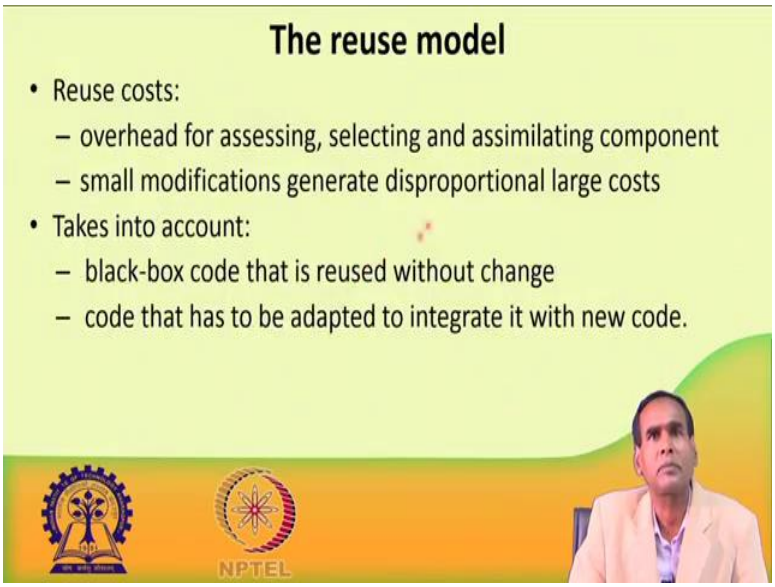

Then we will see the next model that the reuse model. So, reuse costs normally reuse cost consider overhead for assessing, selecting and assimilating component. So, even small modifications they generate disproportional large costs, so they are also what taken care prior.

This reuse model takes into account a black-box code that is reused without any change. The code that has to be adapted to integrate it with new code. So, this reuse model takes into account two things, the black-box code that is reused without change, and the code that has to be adapted to integrate it with the new code.

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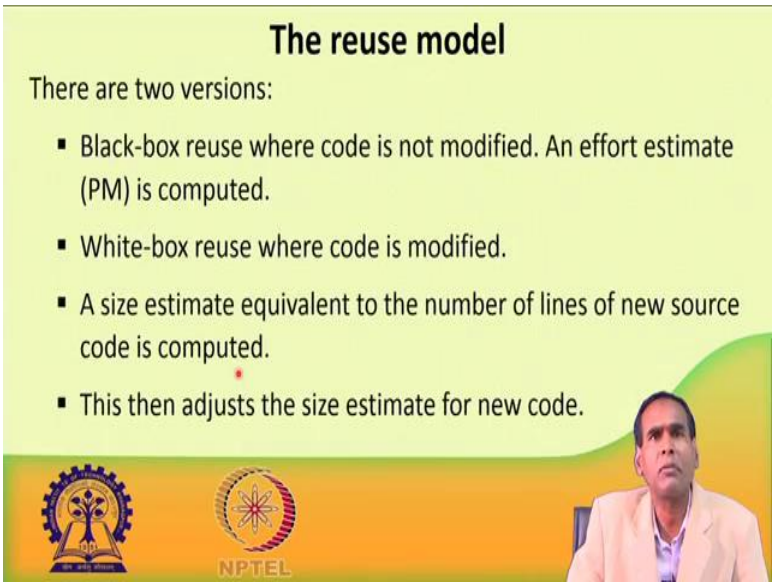

There are two versions for this reuse model; one is the black-box reuse model, the black-box reuse model where code is not modified. An effort estimate is computed then. In white-box reuse model the code is modified then you may estimate the effort. A size estimate is equivalent to the number of lines of new source code is computed. This then adjust the size estimate for the new code, this is happening in the reuse model.

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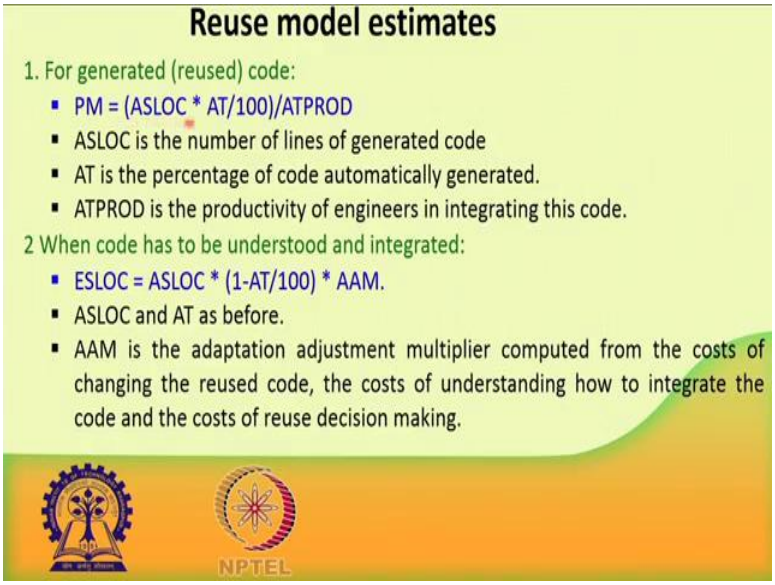

The reuse estimates are like this, we will two possibilities we will see. For the generated or reused code. Person month means the effort in person month is

PM = (ASLOC * AT/100)/ATPROD

Here ASLOC is the number of lines of generated code, AT is the percentage of code automatically change generated, ATPROD is the productivity of the engineers in integrating this code and when the code has to be understood and integrated then the formula changes. So, here ESLOC is equal to

ESLOC = ASLOC * (1-AT/100) * AAM.

Where ASLOC already we have taken earlier, AT also we have already taken earlier, ASLOC means number of lines of generated code, AT is the percentage of code automatically generated, they are as usual before they have used, and AAM this is a new term here.

So, this is the adaptation adjustment multiplier which is computed from the cost of the changing the reused code, the cost of understanding how to integrate the code and cost of reuse during decision making. So, this how when the code has to be understood and integrated you can compute this E estimated you can compute the what ESLOC.

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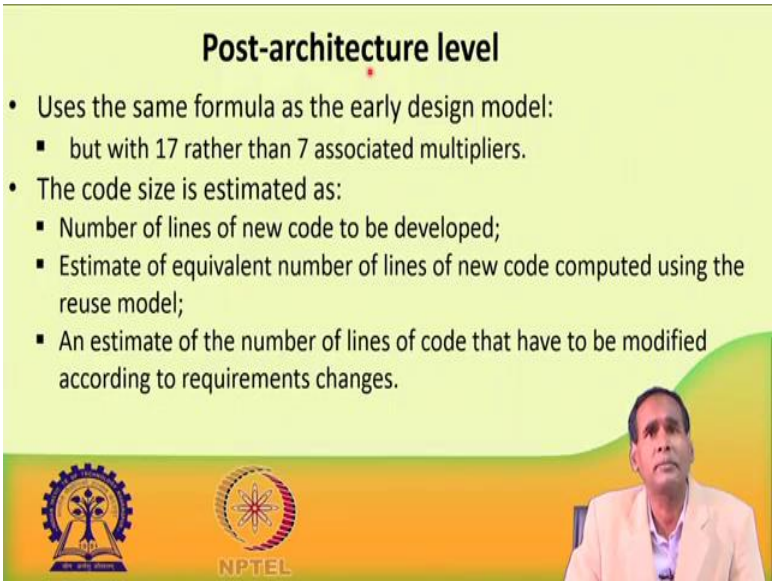

Now, we will see the last model that is post-architecture level. It uses the same formula as the early design model, but with 17 rather than 7 associated multipliers. See in the earlier one was that e or this earlier model sorry in this early design model actually there are 7 associated multipliers.

In earlier design model 7 associated multipliers are were used, but in post-architecture model 17 multipliers are used. The code size is estimated as follows, first the number of lines of new code to be developed is found out, then estimate the equivalent number of lines of new code which is computed using the reuse model earliers reuse model we have already shown.

Then an estimate of the number of lines of the code that have to be modified according to the requirements changes, whenever the requirement changes you may have to modify some of the lines or some of the code. So, then an estimate of the number of lines of code that have to be modified according to the requirements changes that has also to be estimated.

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Now, let us see about quickly about the COCOMO II scale factors. I have already told you there are five scale factors, this already we have seen earlier. So, these five factors appear to be particularly sensitive to system size. So, precedentedness represent the degree to which there are past examples that can be consulted already similar cases you have already handled earlier. So, what is the degree to which these past action examples can be consulted.

Development flexibility represents the degree of a flexibility that exist when implementing the project. So, during the implementation of whether many constraints are there or you are completely flexible you can take your decision, you can do these implementation yourself with there are no constraints you are free. So, development flexibility represents the degree of flexibility that exist when the when implement the project.

Architecture risk resolution it says the degree of uncertainty about requirements. So, these requirements are not clear cut to you, the requirements are very much uncertain they are not certain. So, what is the degree of uncertainty about their requirements that is represented by RESL and similar team cohesion among the team members, and process maturity this could be assessed by the CMMI Capability Maturity Model Integration.

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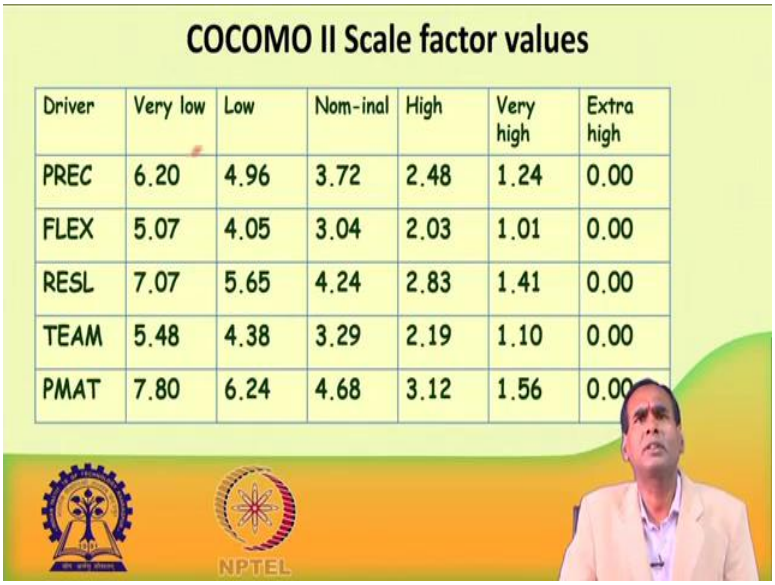

So, these are the different factors for this what COCOMO II scale factors. So, when these are the drivers, these are the 1, 2, 3, 5, 6 you see rated as 6 scale and these are the values these have been given.

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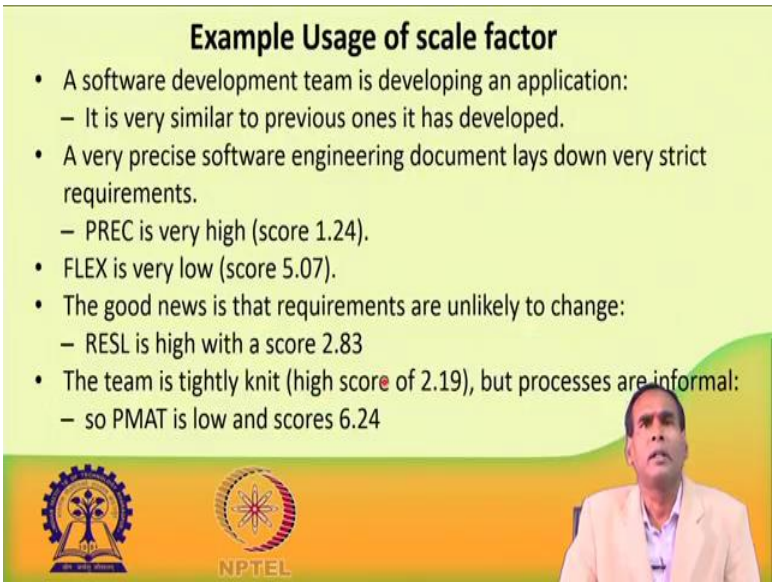

Then now let us quickly take some examples for using of scale factor. For example, a software development team is developing an application. It is very similar to previous ones it has already developed. So already they have developed similar types of application. So, now, you are developing a similar application right now.

It is a very precise software engineering document lays down very strict requirements ok, a very precise software engineering document lays down very strict requirement. So, it is say that precedent is very high is PREC is the precedent I think; yes, So, now precedent is very high and for high value it is 1.24 perhaps high value it is precedent high value is see very high is 1.24, so that you have used and similarly, it is say that flexibility is very low that value is 5.07 you can see flexibility is low is very low it is 5.07 and the good news is that requirements are unlikely to change requirements are it will not change, so RESL is high with a score of 2.83. RESL you can see RESL here it is high it is 2.83. So, RESL is 2.83 but the team is tightly knit.

So, team score is high, team see high is how much? High is 2.19. So, that we have shown here 2.19, but processes but processes are very informal. So, PMAT is low and for PMAT low value is 6.24; PMAT low is 6.24 now you can find out the scale factor.

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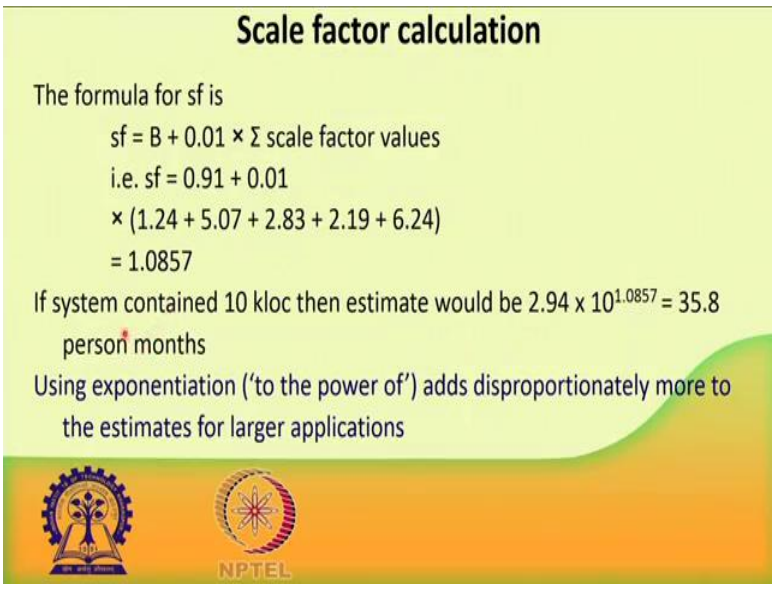

The scale the formula for scale factor sf is given like this,

sf = B + 0.01 × Σ scale factor values

that is equal to 0.91 plus 0.01 into what are the four values you have got here 1.24, 5.07, 2.83, 2.19 and 6.24. So, those values will be multiplied, so on simplification you will get 1.0857.

If a system contained suppose the system or if a system contained 10 kloc then estimate would be how much, the basic estimate is what suppose the system contained 10 kloc. So, the estimate would be how much? The obtained value of 2 point value 2.94 the value of a into size is 10 to the power this, so the exponent factor is 1.0857 this. So, on simplification you will get 35.8 person months. So, using exponentiation that means to the power of it adds disproportionately more to the estimates for larger applications.

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These are the, so what besides these scale factors that we have already seen earlier five scale factors. Effort multipliers they are also assessed. Followings are the effort multipliers that you will using in COCOMO II like RCPX means product reliability and complexity, RUSE means reuse required, PDIF platform difficulty, PERS personnel capability, FCIL facilities available, SCED schedule pressure. So, these are the different effort multipliers they are also to be assessed.

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Their values are given also in a how much; 7 scale perhaps; 1, 2, 3, 4, 5, 6, 7. Yes, the values of the effort multipliers are given in 7 point scale you can use them.

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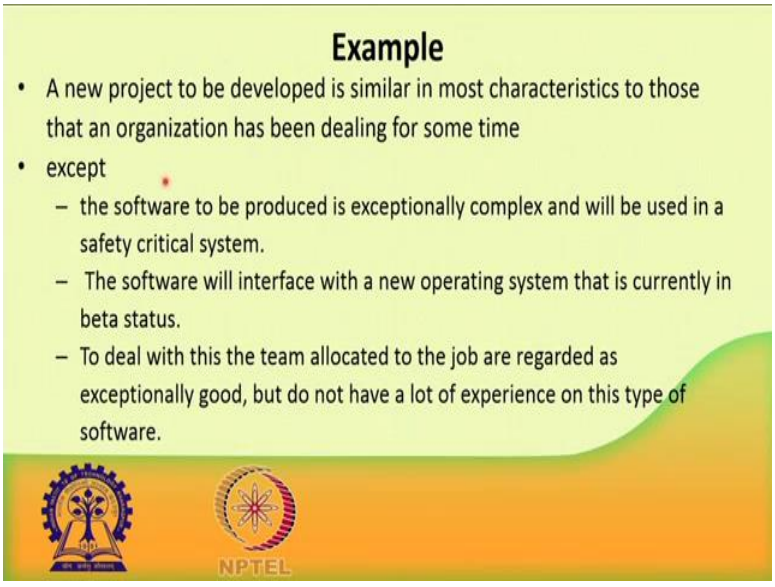

We will take one more example quickly.

A new project has to be developed in ok, a new project to be developed is similar in most of the characteristics to those that an organization has been dealing for some time. Before they have already developed similar type of project, except the followings; what are the exceptions; the software to be produced is exceptionally complex and will be used in a safety critical system.

Software will interface with a new operating system that is currently in beta status and to deal with this the team allocated to the job are regarded as exceptionally good but do not have a lot of experience on this type of software, we have to now estimate.

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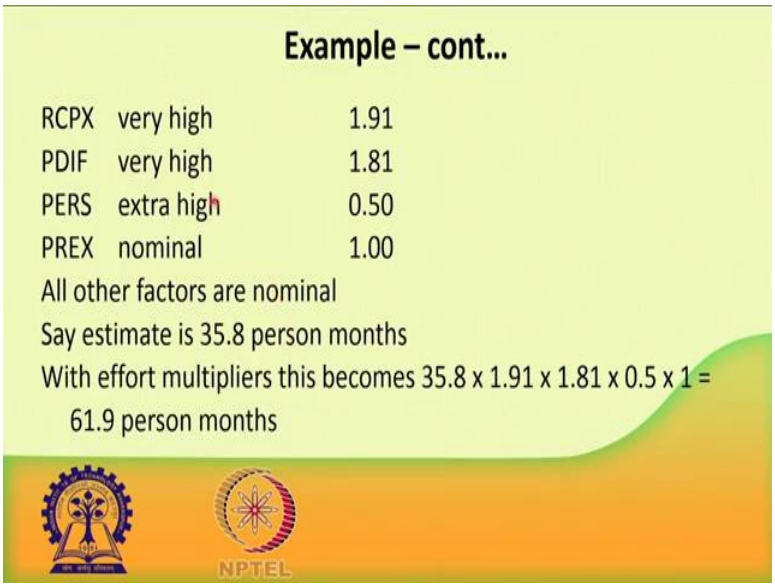

So, here RCPX is very high so 1.91, PDIF is very high 1.81, PERS extra high 0.50, and PREX is nominal 1.0. So, these values you can get from that table; all other factors are nominal and suppose the estimate say estimate is 35.8 person month. For example, suppose it is estimated as suppose the effort is estimated as 35.8 person months.

Then with these effort multipliers the revised estimate becomes how much; 35.8 multiplication of these multipliers, so this gives into 61.9 person months. So, in this way you can use the effort multipliers and the scaling factors to revise to refine the initial estimates using COCOMO II.

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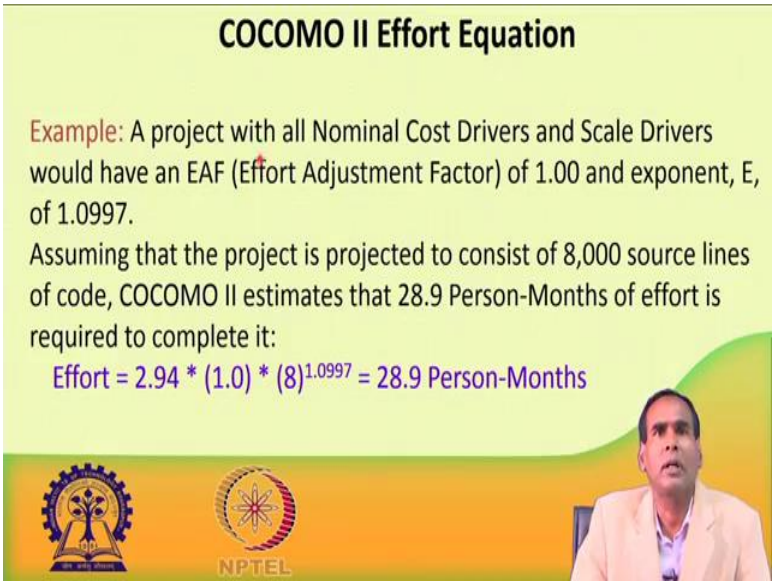

So, another equation I have we have taken another what example, a project with all nominal cost drivers and scale drivers would have an EAF. EAF we have already discussed it earlier Effort Adjustment Factor of 1. 0 and exponent, E, is 1.0997. So, these things they have already calculated EAF is calculated as 1.0 and exponent E is calculated 1.0999.

So, a project with all nominal cost drivers and scale drivers, for this project it is already calculated EAF value is 1.0 and exponent values E is 1.0997. Assuming that the project is projected to consist 8,000 source lines of code, then we have to estimate the effort. So, effort will be COCOMO II estimates 28.9 person-months using the previous equations ok, now we have to complete it.

So, how we can complete it? So, COCOMO II estimates the effort as 28.9 person-month. COCOMO II estimates that the effort is of 28.9 person months of effort is required to complete it, so now we can revise it. So, effort is equal to how much; that is value of constraint is 2.94 into what into the EAF Effort Adjustment Factor that is 1 into what is the source code, see what is this source code; 8,000 source lines.

So, this is, so in this way see how COCOMO II estimates this 28.9 it is shown here. For this data that it is a project with all nominal cost drivers, EAF is 1, exponent value is 1.0997 and the source lines is 8,000. So that means, project size is 8,000 source line, we have to estimate the effort using COCOMO II. So, that will be based on the simplest formula a into what the EAF into source lines to the power of the exponent.

So, value of a is 2.90 and EAF is equal to 1.0, size is 8,000 source line means 8 kloc and to the power of the exponent part, exponent part is 1.0997 put the values in this equation you will get the effort in person-months. So, on solving we will get approximately 28.9 person-months. So, COCOMO II estimates roughly 28. 9 person-months will be required as the effort for developing this project.

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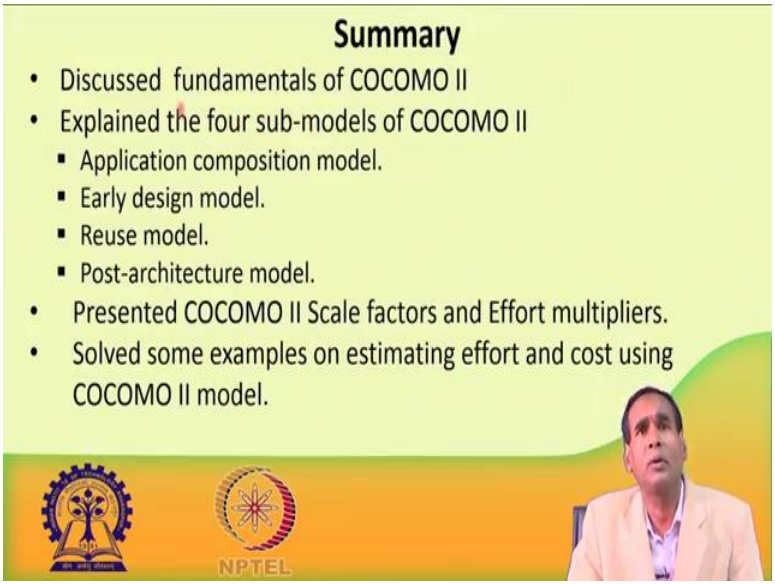

So, finally, we have discussed the fundamentals of COCOMO II, explained the various sub-models of COCOMO II, presented the COCOMO II scale factors and effort multipliers, solved some examples on estimating effort and cost using COCOMO II model.

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We have taken the references from these books.

Thank you very much.

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Lecture – 24
Project Estimation Techniques (Contd.)

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Good afternoon. Now, let us take another kind of Project Estimation Technique that is Halstead's Software Science which is an analytical estimation technique.

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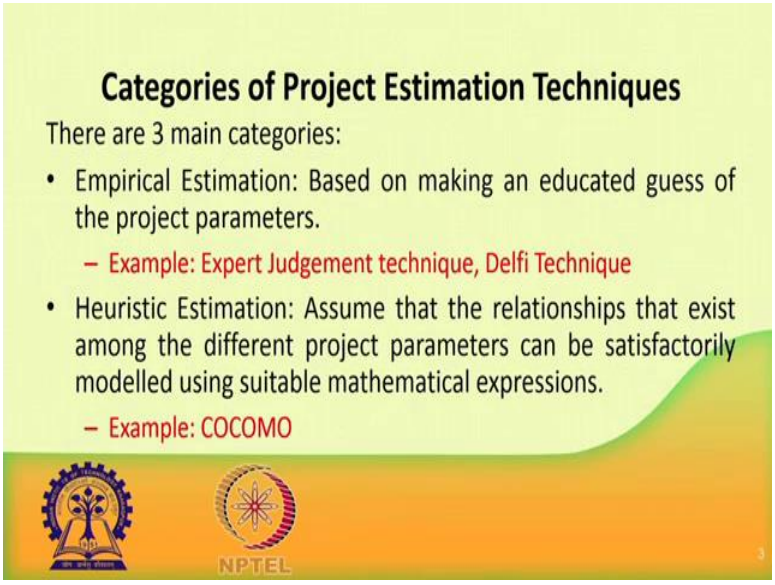

See, in the in the while you have started the project estimation technique, now we have known that there are 3 main important categories of testing estimation techniques, number one empirical estimation, then heuristic estimation and analytical estimation. We have already discussed empirical estimation and heuristic estimation. Normally, empirical estimation is based on making an educated guess of the project parameters. In a project parameter, in a project, there are different parameters such as size etcetera.

In empirical estimation first, the project manager makes an educated guess of the what the project parameters and then gradually they take a decision they estimate the various what other parameters such as effort, costs etcetera. So, here basically this technique based on making an initial educated guess of different project parameters and then they calculate and refine what are the other what parameters. So, examples of empirical estimation techniques are expert judgment technique and Delfi technique. We have already discussed expert judgment technique and Delfi technique in the previous classes.

So, next category is heuristic estimation. So, this technique assumes that the relationship which exists among the different project parameters can be satisfactorily modelled using some mathematical expressions ok. So, here this estimation technique it is based on the what it assumes that what are the relationships they exist among the different project parameters they can somehow be modeled, they can be expressed using some suitable mathematical expressions. And then you can then later on you can refine them by using some what multipliers or cost drivers in order to get the refined or more accurate estimates. The examples are COCOMO models, we have seen different kinds of COCOMO models under this heuristic estimation such as basic COCOMO, intermediate COCOMO, complete COCOMO or detailed COCOMO and COCOMO 2. So, all these things we have seen in the previous classes.

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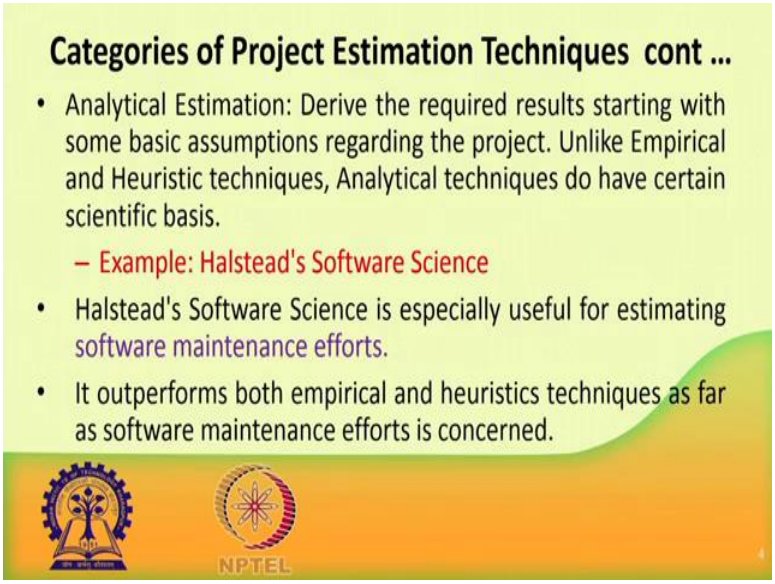

Then next category is analytical estimation. Analytical estimation is based on the following concept. In this technique, the project manager he derives the required results how starting with some basic assumptions. So, assumptions on what, assumptions on regarding the different what a project parameters, so basically here the parameters such as efforts and size or these what you can say program volume, etcetera, they are derived starting with some basic assumptions regarding the different project parameters. But unlike the empirical and heuristics techniques that we have already seen, the analytical techniques they have certain scientific basis.

So, these techniques they have certain scientific basis. So, these assumptions are not vague assumptions. So, all those what assumptions or the different equations or the different methods that will be used in these techniques they have some certain what scientific basis. And under this category, you will see one such example is coming under analytical estimation techniques that is known as Halstead's software science. So, today we will discuss about the Halstead's software science how it can be used for different for the estimation of different project parameters.

Halstead's software science is especially useful for estimating software maintenance effort. After the software is put into use after it is deployed, so next phase is maintenance phase. So, during maintenance how much effort how; much effort has to be given, so those things can be easily estimated by using Halstead Software Science. So, especially

Halstead software science is very much suitable for estimating software maintenance efforts.

This technique outform outperforms both the empirical and heuristic techniques as far as software maintenance effort is concerned. So, as a as far as the software maintenance input is concerned, it gives better results. It more accurately estimates the different parameters such as effort and cost etcetera. It performs better results than the previous two techniques we have seen empirical and heuristic techniques.

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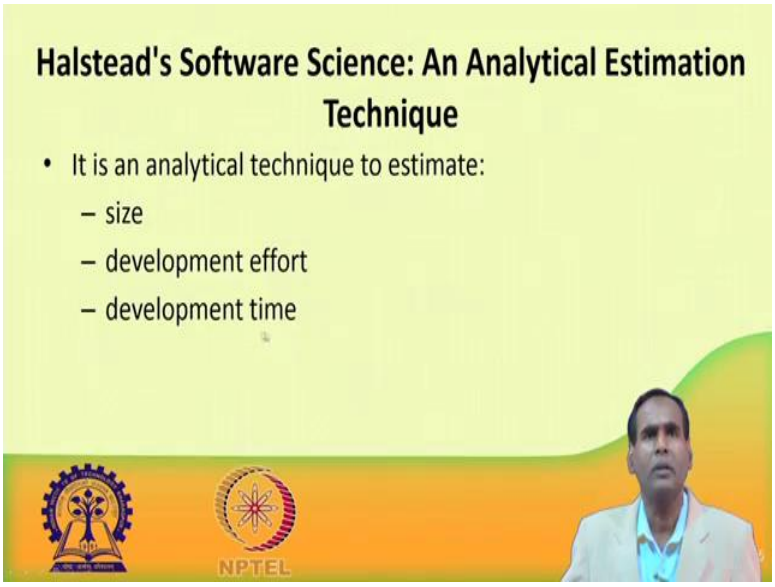

I have already told you that Halstead software science is an analytical technique for what to estimate different parameters of a project, such as the size, the development effort, development time, the program volume, and so many other things you will see for what parameters it can be used to estimate.

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Halstead, he has used a few primitive program parameters in order to estimate effort, cost etcetera. So, what primitive program parameters, he has used for the estimation purpose? He has used two important, what a primitive program parameters, one the number of operators and two, the number of operands. So, by using the number of operators and the number of operands, any manager - program manager, can estimate certain parameters for each project.

So, what parameters can be estimated? The following parameter can be estimated. Using Halstead software science you can derive expressions for estimating the overall program length, the potential minimum volume, the actual volume of the program, the language level of the program, the effort that will be spent to develop the program and finally, how much development time will be required to develop that program, so that also can be estimated. So, Halstead software science provides or by using Halstead software science, you can derive the expressions for these parameters for estimating these parameters.

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Now, let us see how software's Halstead's software science works. For a given program let eta 1 be the number of unique operators which are used in the program, eta 2 be the number of unique operands which are used in the program, N 1 be the total number of operators used in the program, and N 2 be the total number of operands used in the program.

So, basically eta 1 and eta 2 are unique operators under unique operands, whereas N 1 and N 2 these are total number of operators and operands used in the program. With this we can derive the expressions for these parameters what we have shown in the previous slide such as overall program length, potential minimum volume, actual volume, language level, effort, and development time, etcetera.

Now, let us see how by using these terms how we can estimate, how we can derive expressions for the desired parameters such as volume and the program length etcetera. Before going to derive the expressions, let us first see some of the guidelines that will help us to identify the unique operands, the unique operators etcetera.

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These are the guidelines they may be used for identifying the operators. It says that assignment operators, arithmetic operators and logical operators, they are usually counted as operators. So, all such assignment operators, arithmetic operators such as plus minus etcetera, and logical operators etcetera they are they can be used, they are usually counted as operators.

Similarly, in a arithmetic expression where parenthesis is there. A pair of parenthesis as well as a block begin and end pair are considered as single operators. So, if there will be a pair of parenthesis or a block starting with a begin end block, then that will be also what sorry begin block and the end block that pair will also be considered as single operators not two operators.

Similarly, if you are using labels, a label is also considered as an operator label is considered to be an operator if it is used as the target of a GOTO statement. So, if you are using a GOTO statement, and the target is a labelled, then also label is are considered as an operator. The construct such as if then else endif, and a while do, do while etcetera, they are also considered as single operators.

A sequence such as the statement termination operator; a sequence operators such as the statement terminator that you are using the semicolon very often you might have seen in C programs etcetera that end with a semicolon these statements end with a semicolon. So, here the semicolon operator is also considered as a single operator.

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A function name in a functionical statement in a function call statement is considered an operator. Please remember whenever you are calling a function invoking a function the function name that in a function call statement is also considered as an operator.

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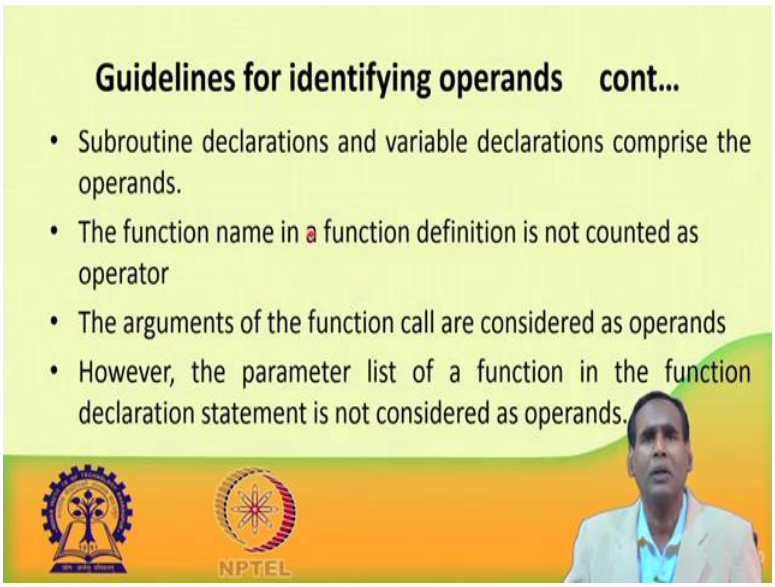

The subroutine declarations and variable declarations they comprise the operands. So, now let us see about the we have seen some of the guidelines for the what operators. Now, let us see guidelines for identifying the operands. So, subroutine declarations, when you are declaring sub routines, so subroutine declarations and variable

declarations, they comprise the operands. The function name in a function definition is not counted as an operator. Please remember that the function name while we are making a function definition, the function name that is not counted is not counted as an operator. So, similarly the arguments of the function call are considered as operands. So, whenever you are putting the arguments where in a function call the arguments of the function call they are considered as operands.

However, if the parameter list of a function in the function declaration statement is not considered as an operands. While you are declaring a function, if you are putting the parameters the parameter list that present in the function declaration, there that is not considered as an operand.

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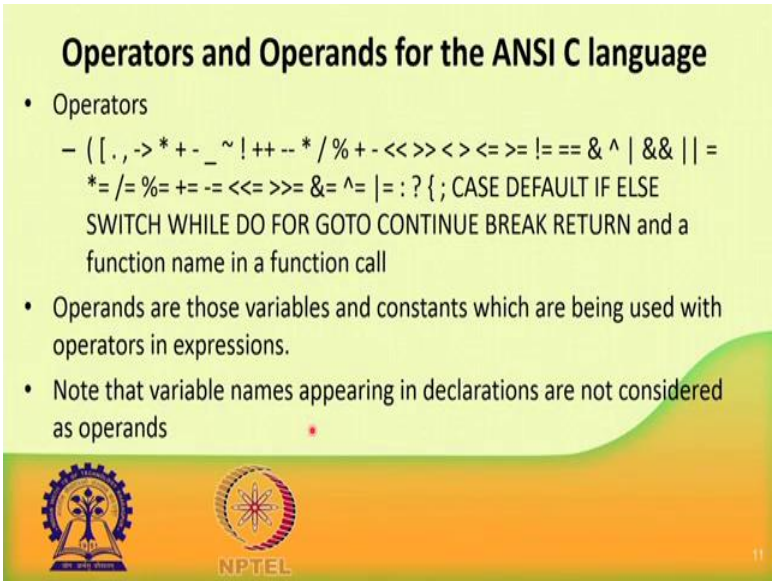

Let us see the common language, programming language C, language what to do the what a possible operators, so these are the possible operators like

([. , -> * + - _ ~ ! ++ -- * / % + - << >> < > <= >= != == & ^ | && || = *= /= %= += -= <<= >>= &= ^= |= : ? { ; CASE DEFAULT IF ELSE SWITCH WHILE DO FOR GOTO CONTINUE BREAK RETURN and a function name in a function call

these are all examples of operators valid operators in C as a as long as this Halstead's software science is concerned.

So, the operands are those variables and the constants which are being used in operators in expression. So, in an expression, ok, in an operator, the operators while you are putting in what in expression the operators while you are putting the expression. So, we have to consider the operands also. The operands are those variables lets define the operands operator since you all already have defined, let us define the operands, operands are those variables and the constants which are being used with the operators in the expression. So, along with the operators, so what variables and constants you are using in the expressions, they treat it as operands. Note that the variable names appearing in the declarations, they are not considered as operands. If you are using some variable names in the declarations, they are not considered as operands.

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So, these are the guidelines we have given for for identifying the operators and the operands. Now, let us take this example. Consider that the expression I give a small expression is given

a = &b;

So, in this example, we can see that a and b are the operands, whereas equal to, ampersand sign and semicolon, these are the operators.

Similarly, if there is a function call statement

func (a,b);

then in this function, this function name func, then this comma in present in between a and b, and this semicolon operator, these are considered as the operators, whereas variables a and b they are treated as the operands. So, in this way given a program, you can easily identify the operators and operands this will help you for how to derive the expression for this volume, effort and the time etcetera.

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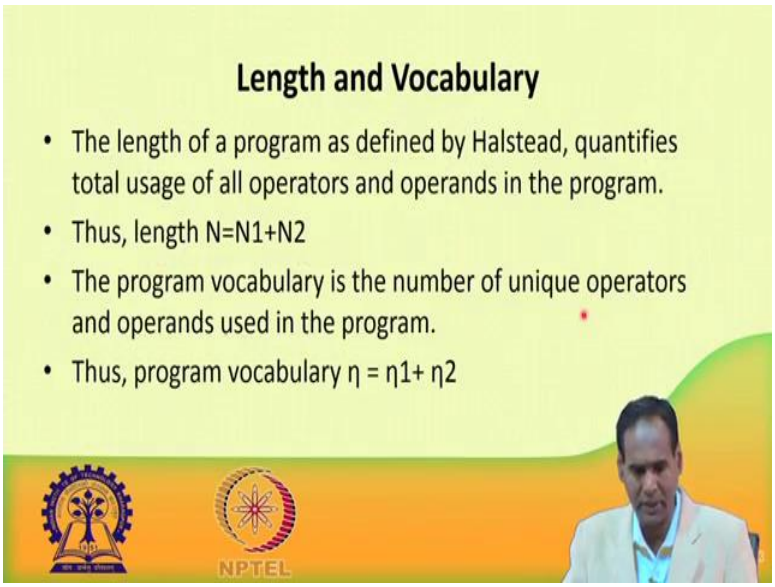

Now, let us define two other things important things called as the length and the vocabulary. The length of a program are defined by Halstead it quantifies the total usage of all operators and the operands in the program ok. The length of a program is defined as follows. The length of a program as defined by Halstead. What does it do; what does it do? It quantifies the total usage of all operators and operands in a program.

So, we have earlier taken N 1, N 2, and eta 1, eta 2, Please I am recalling again, whereas what is eta 1; eta 1 be the number of unique operators used in the program, eta 2 be the number of unique operands used in the program, and N 1 be the total number of operators, N 2 be the total this N 1 be the total number of operators used in the program, whereas, N 2 be the total number of operands used in the program.

So, with this we can easily and we have seen the guidelines how to select the operands, how to identify the operands and operators. We can see the length can be defined by using this N 1 and N 2. So, length is equal to

$$N=N1+N2$$

as per the definition because as according to Halstead the length of a program is defined as or it quantifies the total usage of all the operators and operands, simply add the operators and operands. So, this will give, so N is equal to N 1 plus N 2.

The program vocabulary it is defined as the number of unique operators and operands used in the program, not total operators and operands. It is defined as the number of unique operators and operands used in the program. Thus, program vocabulary eta is equal to

$$\eta = \eta1+ \eta2$$

that means, unique operators used in the program plus the number of unique operands used in the program. So, in this way, we can define we can estimate the length and vocabulary for a particular program by using Halstead's software science.

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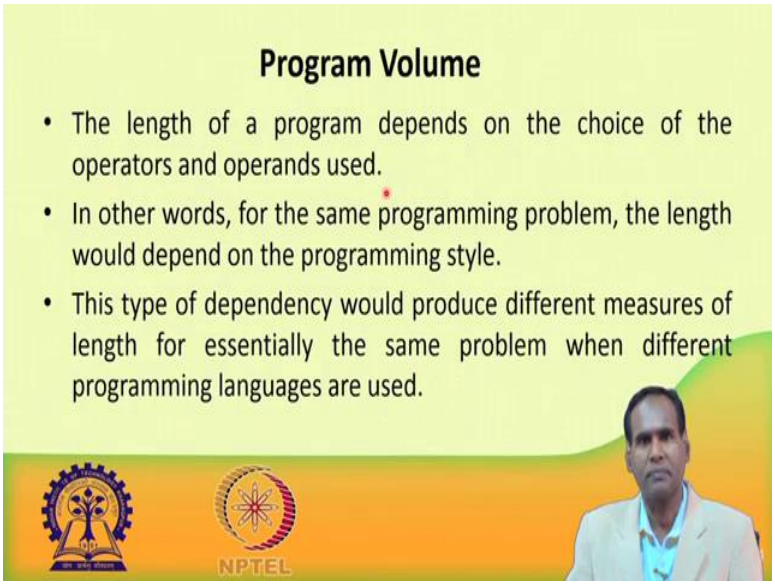

Next parameter is program volume. How we are defining program volume? The length of a program depends on what the choice of the operators and operands used ok. The

length of a program it will depend on the choice of the operators and operands used in the program.

In other words, for the same programming problem, the length would depend on the programming style, because different programmers they will write down the programs using different styles. So, the same, for the same problem, for the same project, for the same programming problem, the length would depend on the programming style that the programmer has chosen. This type of dependency would produce different measures of length for essentially the same problem when different programming languages are used.

So, when different programming languages are used even if the problem is same to be that has to be coded, so that will produce different measures, different values for the length ok. This type of dependency would produce different measures of length for essentially the same problem when different programming languages are used by different programmers.

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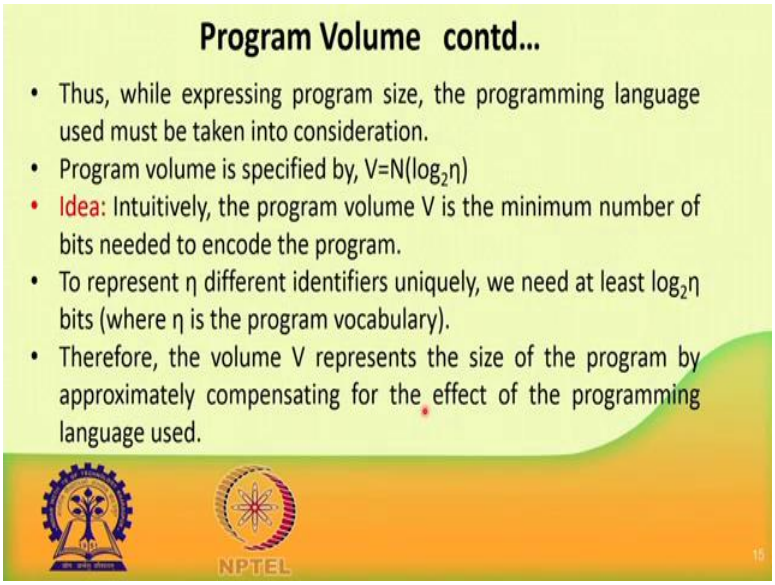

Thus, while expressing the program size, the programming language used must be taken into consideration ok. Thus, while expressing the program size, what you have to do? The programming language that is used it must be taken into consideration. Program volume is usually specified by the term V which is equal to

$$V=N(\log 2\eta)$$

N, you have already known. Total number of N is the what V is the program volume we are saying, and N just we have seen N is the N 1 plus N 2. Where eta is, that means, total number of operators and operands used. Where eta is equal to eta 1 plus eta 2 that is the total number the unique number of operators and operands used. So, program volume is specified by the term V which is equal to N into log 2 eta.

The idea behind is that intuitively the program volume V if the minimum number of bits required to encode the program. So, what is the minimum number of bits required to encode the program that is specified by V. So, in order to represent eta different identifiers uniquely, what we need? We need at least log 2 eta number of bits, where eta is the program vocabulary.

Therefore, the volume V, it represents what? The size of the program by approximately compensating for the effect of programming language used. So, if you are using different programming languages in order to compensate the effect of the programming language used, you can use what V which represents the size of the program. Therefore, the volume V, it represents this size of the program by approximately compensating for the effect of the different programming languages used.

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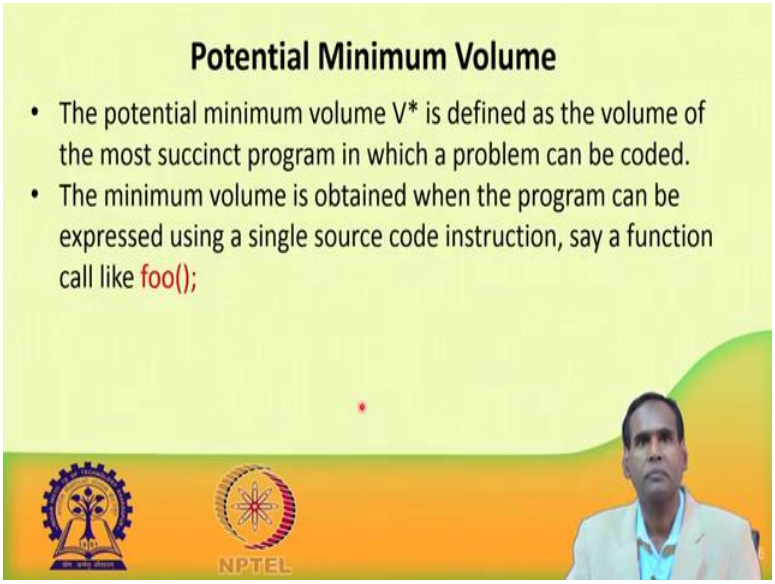

Next term that we have to discuss is program potential minimum volume. We have already seen simple volume. Now, let us see potential minimum volume. The potential minimum volume which is denoted as V star, it is defined as the volume of the most

succinct program in which a program can be coded ok. So, the volume of the most succinct program in which a program can be coded is represented at V star or which is known as a potential minimum volume.

For example, let us see the minimum volume is obtained, when; the minimum volume is obtained, when the program can be expressed using a single source code instruction. Just like we say a function that they are small function foo, just a single code source code instruction, then we say that the minimum volume which obtained because the program can be expressed using just a single source code instruction, for example, a function call like foo(); It contains only one statement here ah

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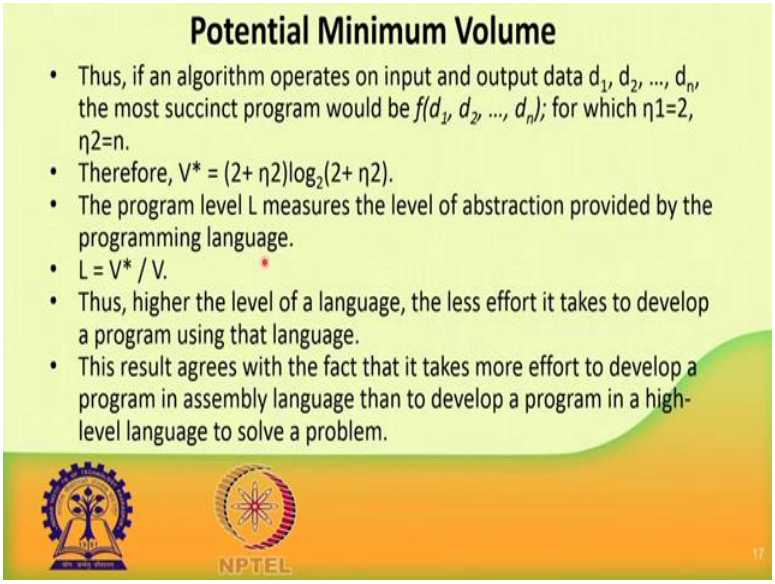

Thus, if an algorithm it operates on inputs and output data d 1, d 2 and d n ok. So, suppose there is an algorithm form if an algorithm operates on input and output data d 1, d 2, d n etcetera, the most succinct program would be how much f of d 1, d 2, d n for which eta 1 will be equal to 2 and eta 2 will be equal to n. So, eta at the minimum number of sorry the unique number of operators will be 2 and the unique number of operands will be n. Therefore, V star or the potential minimum volume will be equal to

$$V^* = (2+ \eta 2)\log 2(2+ \eta 2).$$

The program level, so now, we will define another term called as program level. The program level which is represented as L, it measures the level of abstraction which is

provided by the programming language. So, a particular programming language what is the what level of abstraction it provides that is represented as program level L. L can be specified as

$$L = V^* / V.$$

where V star is the what potential minimum volume and be the volume. Thus, what will happen for the higher the level of a language the less effort it takes to develop a program using that language.

So, if the level of a what language is high, then it will take less effort to develop the program using that language. This result agrees with the fact that it makes more effort to develop a program in assembly language, than to develop a program in a high level language to solve a problem. We know that here what it takes more effort when where developing in the assembly language, because it is a low level language. Whereas, if we will take a what low level language which the high level language, then we normally put less effort.

So, it match with this thing that because L is equal to V star by V means it implied that higher the level of a language less will be the effort it requires to develop for that program using that language. And normally from our normal intuition, we know that if we want to write a program, then if will choose this assembly level language which is a low level language, it will take more effort than or in comparison to the thing that if you will choose a high level language such as a C or C plus to write down the same program.

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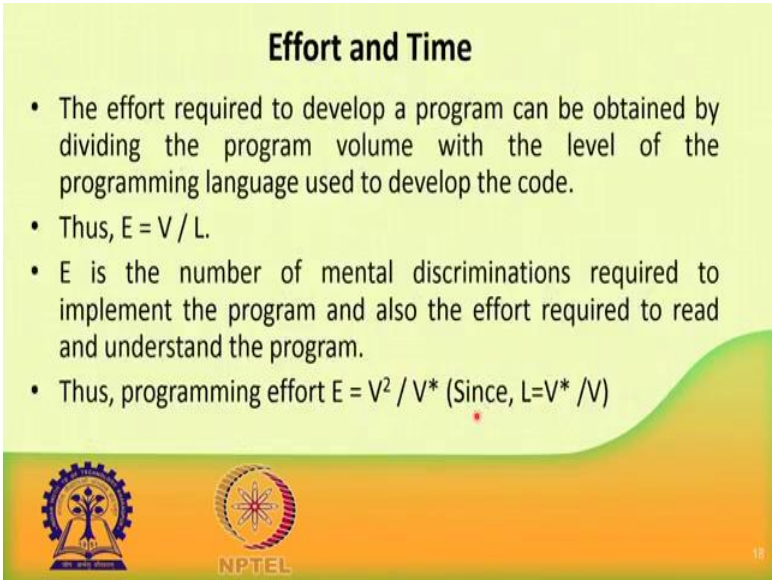

Next we will define effort and time, what is exponentially required. The effort required to develop a program can be obtained by dividing the program volume with the level of the programming language is to develop the code. So, here already we have seen earlier using COCOMO technique etcetera how the effort can be estimated.

Here the effort which is required to develop a program can be estimated by dividing what? By dividing program volume with the level of the programming language which will be used to develop the code. Mathematical you can say that the effort will be equal to

$$E = V / L.$$

L represents the what is the level of the programming language.

Here, E is, so normally; so normally E the can be treated as the follows, normally E is the number of mental discriminations required to implement the program ok. So, normally E is the number of mental discriminations required to implement the program and also the effort required to read and understand the program. Thus, programming effort E can be written as how much will put the value of L, we know value of L is equal to V star by V.

So, putting the value of L here we get E is equal to how much putting the value of L is equal to V star by V. So, on simplification, it will give

$$E = V^2 / V^* \text{ (Since, } L=V^* /V)$$

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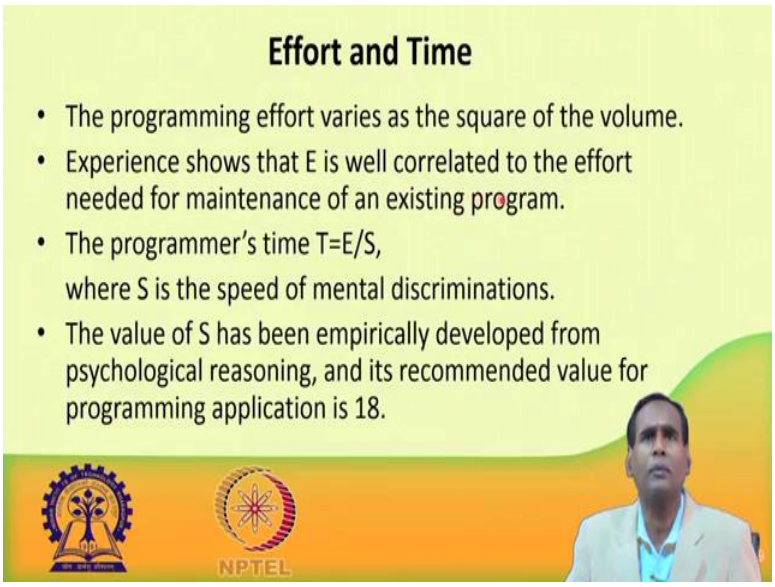

The programming effort varies as the square of the volume. You can see from this what equation the programming effort E it varies how; the programming effort E it varies as the square of the volume. Here you can see the programming effort E varies as a what a square of the volume V, the programming effort varies as the square of the volume.

Experiences shows that E is well correlated to the effort needed for maintenance of an existing program. So, Halstead has conducted so many so much of what experiments also other researchers has what conducted so many what experiments. And the experience shows that E is well correlated to the effort which is needed for maintenance of an existing program. Then another, what a term will define here that is the programmer's time.

Programmer's time is defined as $T=E/S$

where E is the effort and S is the speed of mental discriminations. The value of S has been empirically developed from psychological reasoning, and it is recommended value and this recommended value for programming application is 18. So, normally the value of S for programming applications is 18 that means, what the speed of mental discriminations is normally it should be set to 18. The value of S should be taken as 18 for programming applications.

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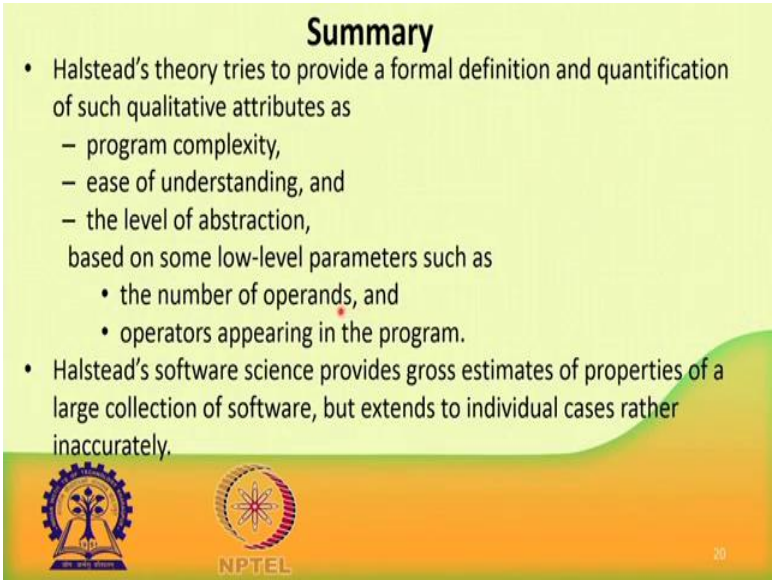

We have seen in this class the about Halstead software science which is an analytical estimation method. A Halstead's theory, it tries to provide you formal definition and quantification of such quantitative attributes such as program complexity, ease of understanding and the level of abstractions etcetera, based on some low-level parameters such as the number of operands, and the number of operators which are appearing in the program.

Halstead software science provides gross estimates of properties of a large collection of software, but extends to individual cases rather inaccurately. So, one thing we have known that Halstead software science can be used effectively for estimating what maintenance effort.

Halstead Software Science also provides gross estimates of a property of a large collection of software, but when extends to individual cases it, but extends to individual cases, rather inaccurately, for individual cases, it may give inaccurate estimations this is something about this Halstead software science which is an analytical estimation method.

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We have taken and the references from these two books ok.

Thank you very much.

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Lecture – 25
Project Estimation Techniques (Contd.)

Good afternoon. Will now today see one of the important estimation aspect of the projects that is staffing level estimation.

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Here we will see two important models for staff level in estimation that is Putnam's model and Jensen's model.

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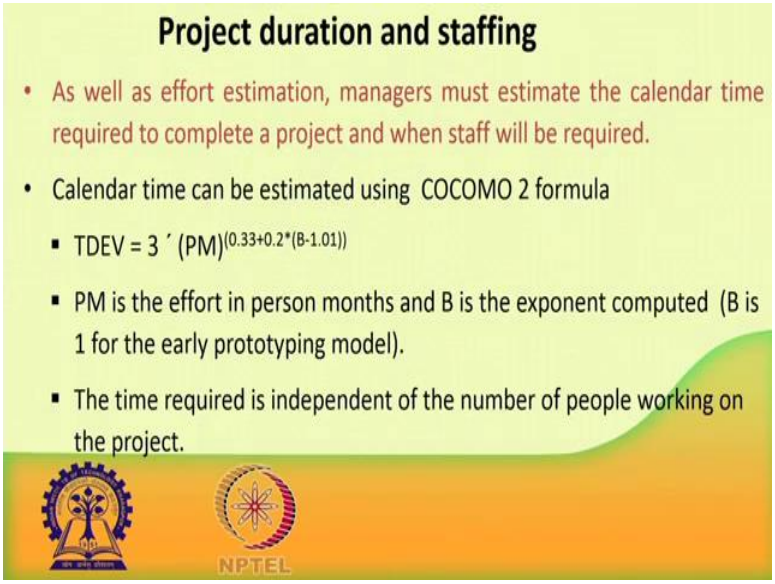

So, let us see about this project duration and staffing. So, as well as effort estimation managers must estimate the calendar time required to complete a project and when the staff will be required. Let us say effort estimation it is at all very much important. Similarly, estimating the calendar time required to complete a project, as well as how many staffs will be required when they will be required these are also important aspects of a project estimation.

So, the calendar time for a given project can be estimated using COCOMO 2 formula. Here T development is; that means, calendar time is represented as

$$TDEV = 3 * (PM)^{(0.33+0.2*(B-1.01))}$$

So, here this formula is obtained by using COCOMO 2 estimation technique and here PM represents the effort in person months and B is the exponent computed, normally, B is one for early prototyping model.

The time required is independent of the number of a persons working on the project. These time developed it is does not depend upon that how many persons are working in the project. It is the time the development time required is independent of the number of people working on the project.

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So, the now let us see about the staff requirement, how many staffs generally do you require for a given project? Staff required it cannot be computed just by dividing the development time by the required schedule not at all possible. Because the number of a people working on a project it varies depending on the different phases of the project.

You might require less number of people during the requirements analysis maybe again more number of people during design. And during coding again it may either increase or decrease depending on the nature of the project and in testing time again you may require more number of people. So, that is why it will vary the number of people working on a project, it will vary it will depend on the phase of the project that you are currently in. The more people who work on the project the more total effort is usually required this very obvious. If more number of people work on the project; obviously, the total effort required will be more. So, a very rapid buildup of people upon correlates with the a schedule slippage.

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Now, let us see how to estimate the staffing level. The number of personnel required during any development project is not constant I have already told you that it may depend upon the what the phases. So, that is why the number of personal required during the development of any project is not constant.

It will may be less at the initial time; that means, this requirement specific analysis and specification, then again more number of people will be required during design. And the highest number of people may be required during the testing time, so that is why it is not constant it varies.

Now, then in 1958 he has analyzed he had analyzed many R and D projects, and observed the followings; what he observed? He observed that the Rayleigh curve represents the number of full time personnel required at any time. That means, the number the number of full time personnel required at any time for a project can be represented by using a Rayleigh curve.

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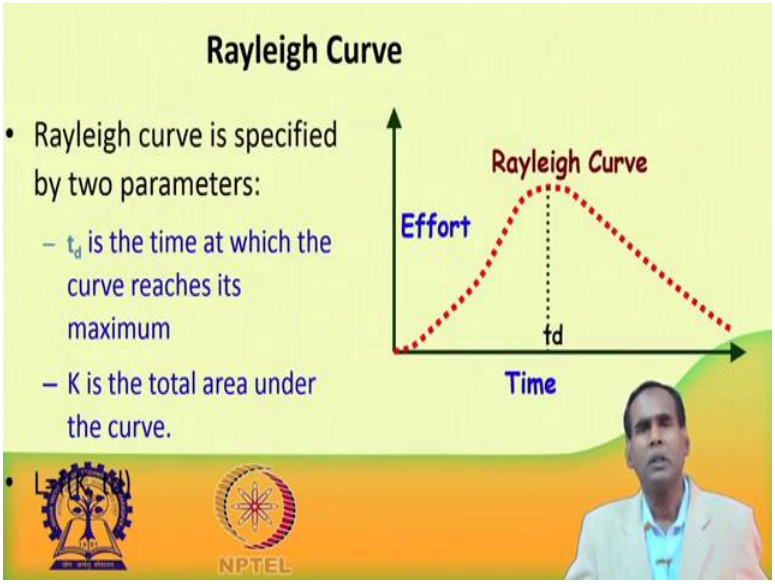

So, let us see what is a Rayleigh curve? I hope you must have known Rayleigh curve earlier see this is the form of a Rayleigh curve. Here in the x axis we have taken the time and y axis we have taken the effort. You can see as the time moves the effort also it gradually increases and there will be time come peak period and after which even if the time increases the effort will slight it will fall down it will decrease.

Rayleigh curve is specified by two important parameters; one is the t d, another is K, t d is the time at which the curve reaches its maximum we can see. So, the td is the time at which the curve reaches its maximum and the K is the total area under the curve, so total area under the curve is represented by K. So, Rayleigh curve is specified by using the two these two parameters td and K.

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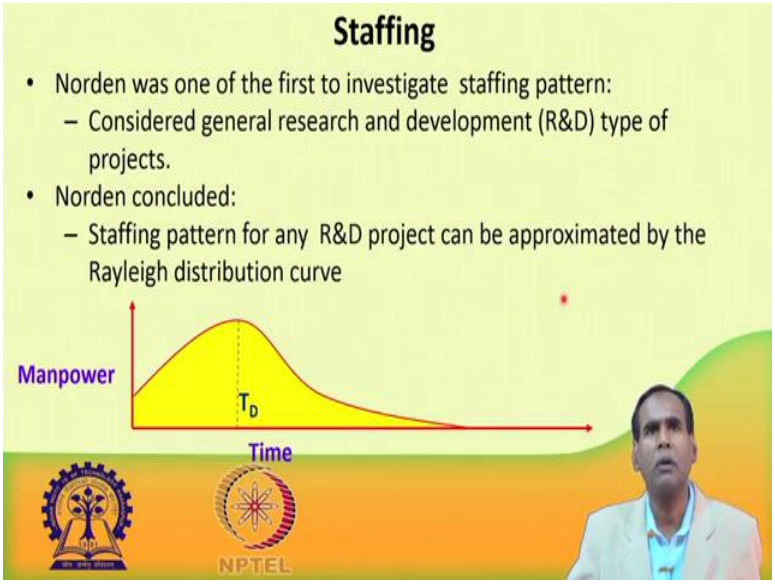

Norden was one of the first persons or to investigate the staffing pattern. He considered some general R and D kind of a projects, he concluded that this staffing pattern for any R and D project can be approximated by the Rayleigh distribution curve.

So, he has observed several projects and then what concluded that this staffing pattern then; that means, the number of staffs required for developing a project it will follow on R and D project ok. Sorry the number of people required for developing an R and D project it can be approximated by a Rayleigh distribution curve as shown below, and Rayleigh distribution curve is specified by two parameters T d and K.

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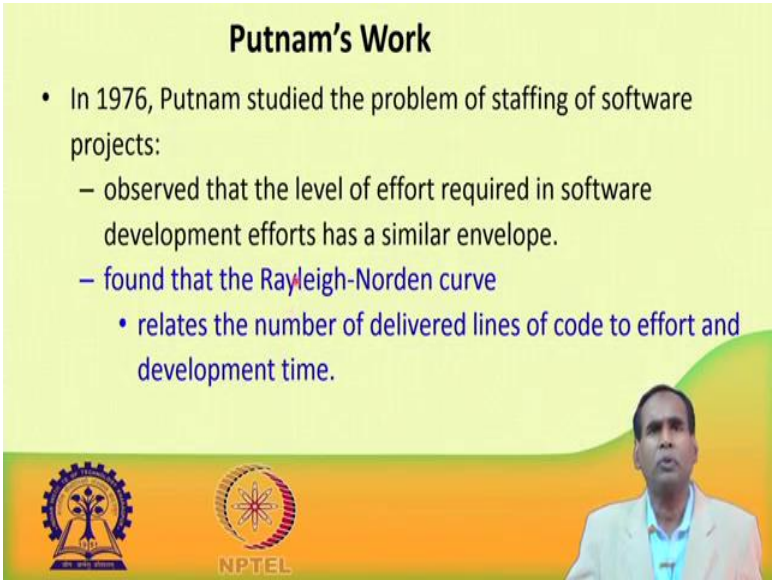

Then Putnam came, so Putnam has also done some work, he has studied the problem of staffing of software projects. He has also what investigated many projects in 1976 Putnam studied the problem of staffing of software projects. He observed that the level of effort required in software development projects has a similar envelope ok. He observed he has observed that the level of effort required in software development has also a similar effort because previously Putnam this Norden had studied normally what some of the what R and D projects.

So, then Putnam had studied as some what this problem of staffing of software projects, he observed that the level of effort required in software development has also a similar envelope that happens in case of R and D projects. He also found that the he also found that the Rayleigh Norden curve it relates the number of delivered lines of code to the effort and development time.

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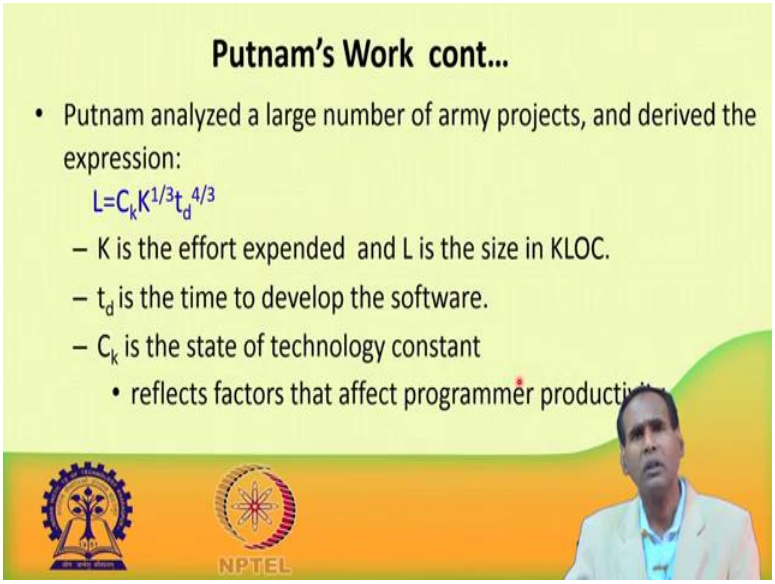

So; that means, you will see this Rayleigh Norden curve it also relates the number of the delivered lines of code to what to effort and development time. On Putnam analyzed he has studied a large number of army projects and derived the expression for L. And L is specified as

$$L=C_k K^{1/3} t_d^{4/3}$$

Where K is the effort expended or the export the effort required and L is the size in terms of KLOC, t d is the time to develop the software, C k is the is a constant it is known as the state of technology constant.

This C k which is known as technology constant it reflects the different factors that affect the programmer productivity, as you have seen in earlier cases COCOMO model etcetera, there are many factors they affect the programmer productivity. Here C k also represents the different factors that the that affect the programmers productivity.

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Putnam's work has established something Putnam has stated the followings. That after conducting several experiments he has observed that C k the value of C k is equal to 2 for poor very poor development environment where no software engineering methodology, no proper methodology used whatever poor documentation is there.

And no reviews there or very poor reviews there for that the value of ck may be set to 2, but what where or the organizations where good software development and environment is there. Software engineering principles are used, good programming languages are used etcetera, then the value of C k may be set to 8 and if the environment is excellent.

So, using various current software engineering practices, modern programming practices uses what you can see now automated tools case tools etcetera and then the value of ck may be set to 11. Or very small number of engineers are needed at the beginning of the project carrier for carrying out planning and specification.

We can see at the beginning only very small number of manpower are required for carrying out the planning and specification. That means, requirement analysis and specification ok, planning and requirements specification and analysis activities. As the progress as the project progresses more detailed work is required. See the time is passing are the project progresses more number of people are required because what more detailed work is required. That is why the number of engineers slowly increases and

reaches peak as the time progresses as more number of work is required. Now, let us see this peak time represents which phase of software development.

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Putnam observed that the time at which the Rayleigh curve reaches it is maximum value it corresponds to the system testing after product is released. So, you see initially I have told this is the during the project planning or this what requirement analysis specification analysis phase; then design will come, then coding will come.

So, this peak period T D will correspond to the testing phase that using system testing phase. Putnam observed that the time at which the Rayleigh curve reaches it is maximum value it roughly corresponds to the testing phase, the system testing phase and product release.

Then after system testing is over, the number of project staff it falls gradually till the product installation and delivery. So, after this is over see this is the system testing this T D this point corresponds to the system testing phase. And after the testing phase is there over then gradually the manpower required will be fall down, because now the system will be implemented at this side and then gradually the number of person will be at falling down that is what he says. So, Putnam observed that, after system testing is over the number of a project staff it gradually falls down till the product installation and delivery it can still the product installation over and delivery comes.

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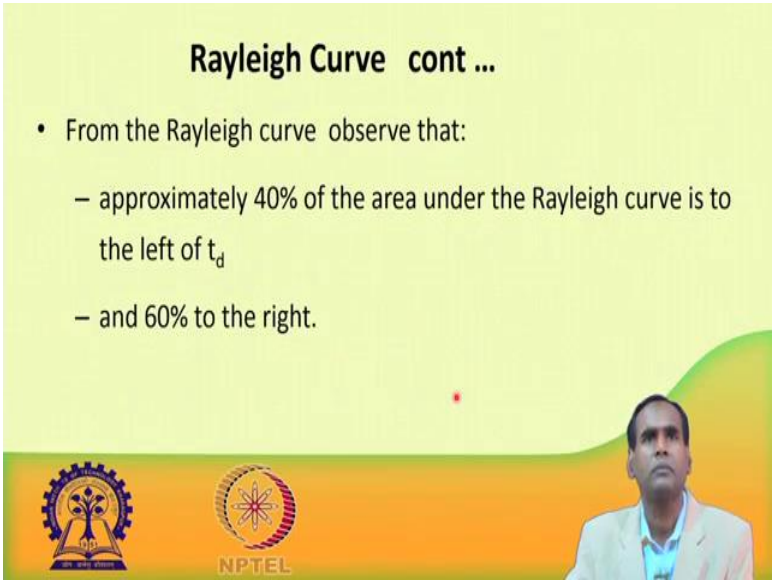

From the Rayleigh curve it is also observed that approximately 40 percent of the area on the under the curve is to the left of the T D and 60 percent to the right of the T D. You see it is also observed that almost 40 percent of the area what lies to the left of the curve and 60 percent lies to the right of the curve sorry to the left of the T D. So, 40 percent lies to the left of this point T D and 60 percent area lies to the right of this point T D.

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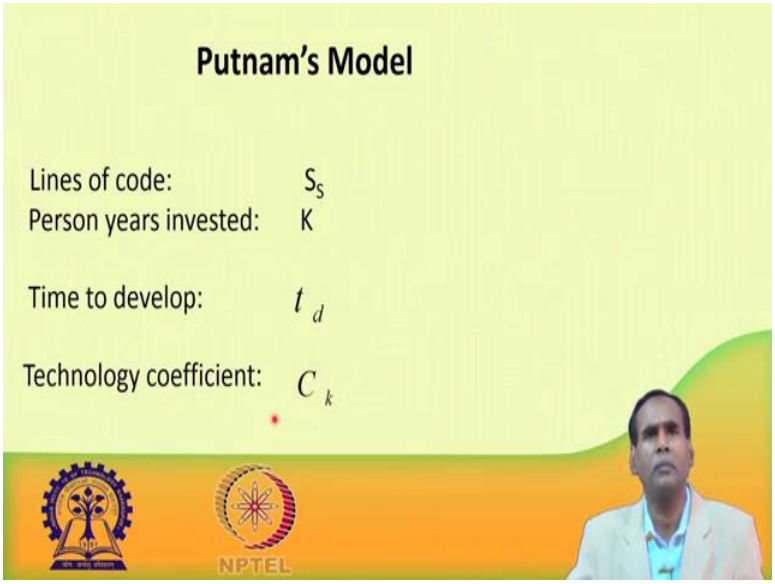

Now, let us see Putnam's model can be used to estimate what? Putnam's model can be used to estimate the lines of code which is represented as Ss. It can also be used to

estimate the person years invested the time to develop and a constant called as technology coefficient constant C k.

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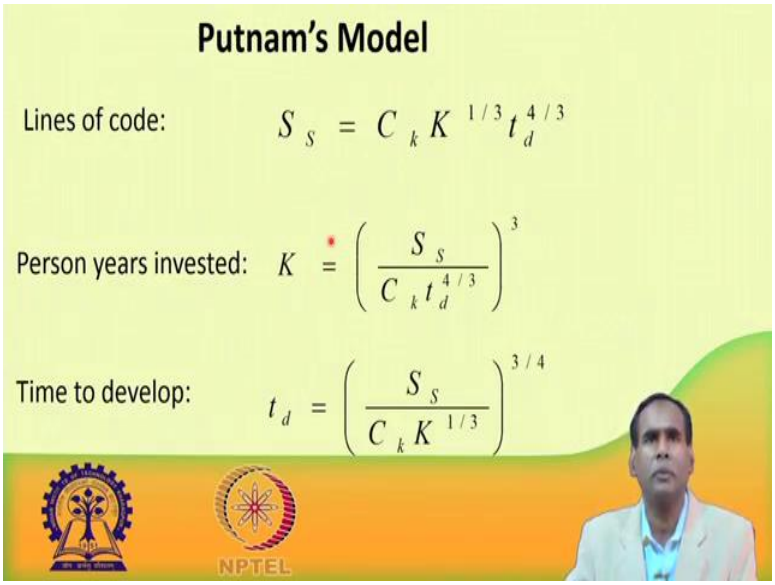

Putnam has given these formulas. So, that the lines of code can be computed by using the formula

$$S_S = C_k K^{1/3} t_d^{4/3}$$

the meanings of the term solid who have told you earlier. From this formula you can easily find out the value of K; that means, the person years invested.

$$K = \left(\frac{S_S}{C_k t_d^{4/3}} \right)^3$$

That means, you can say this is the effort required maybe in terms of person month. This will be equal to S s, S s is the lines of code divided by C k into to the power 4 to the power or 4 by 3. Also from this equation lines of code, you can easily derive the value of td

$$t_d = \left(\frac{S_S}{C_k K^{1/3}} \right)^{3/4}$$

which is equal to S s divided by C k into k to the power 1 by 3.

So, given any, so lines of code there are you can see that four things, S s C k K and t d. So, if three parameters are given C k, K and t d are given easily you can find out the lines of code easily you can estimate the lines of code. So, similarly if the values of S s C k and t d are given you can easily estimate the value of K. And if the values of S s ,C k and K they are given you can easily estimate the time to develop that is td .

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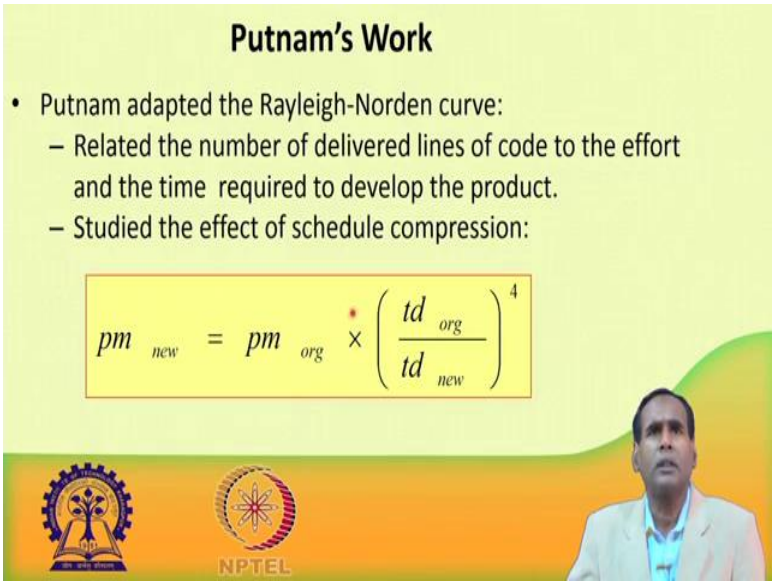

Putnam adopted this Rayleigh Norden's curve, he has related the number of delivered lines of code to what to the effort and the type ok. From the Rayleigh Norden curve he has related the he has established an relation between the number of delivered lines of code and the effort and the time required to develop the product. He has also studied the effect of schedule compression on the effort as well as the time. So, instead in general you can say he has studied the effect of scheduled compression on the cost.

So, he has established the followings

$$pm_{new} = pm_{org} \times \left(\frac{td_{org}}{td_{new}} \right)^4$$

Here td is this schedule the time to delivery or org means what originally you have planned to develop the what software, td new just the new time. Suppose you have estimated that 12 months will be required to develop the project.

Then the customer comes some urgency, due to some urgency they are requesting you delivery it within 6 months. So, now, td org is 12 this the original schedule and new schedule is suppose now 6. So, pm org means the effort required for the original schedule and here pm new means the effort that will be required for the new schedule, because now according to the new schedule time has been reduced.

So, if time will be reduced how much what more effort you have to put to develop the software that is represented here. So, pm new; that means, the new effort revised effort will be equal to the original effort into what the original time to delivery original time divided by the new time.

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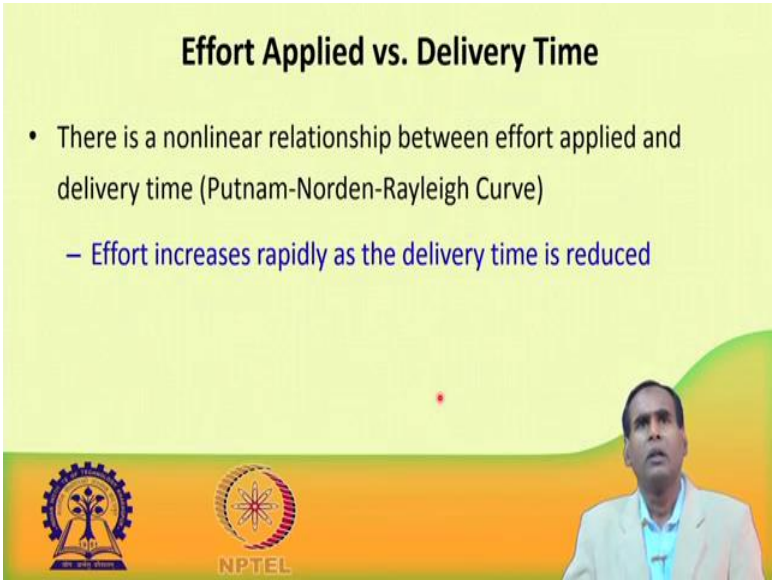

So, now let us see the effort applied versus the delivery time. There is a non-linear relationship between effort apply and delivery time according to put Putnam Norden Rayleigh curve ok. So, according to do according to the Putnam Norden Rayleigh curve, you can observe that there is a non-linear relationship between the effort that will be applied on the delivered time.

So, what does it mean that the effort increases rapidly as the delivery time is reduced not linearly. The if the time is increased the effort that does not increase just linearly, it increases very rapidly I will show one example. So, if the delivered time is reduced ok, here you can see that if the delivery time is reduced the effort increases very rapidly.

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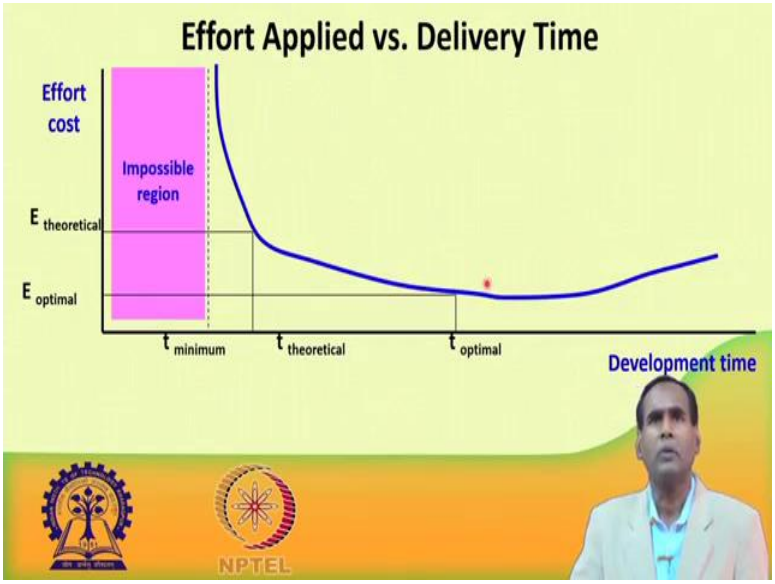

Let us see how? So, initially you can say this is the optimal time; that means, if this is the time to delivery the optimal report is required will be required it is given at this point. Now, you see there is the compression suppose, so theoretically this delivered time it is reduced this is t theoretical. Then what will happen how much is the required the effort is required what increases tremendously.

And still if you will compress the time, again you reduce the time; suppose at this point ok. This point then you will say that then it is almost impossible region you cannot what get your work done you cannot complete the project. So, there is certain point up to which you can reduce the time, you can reduce the schedule. Beyond that even if you will schedule how many even if how many more number of personnel you hire you may not complete the project let us see.

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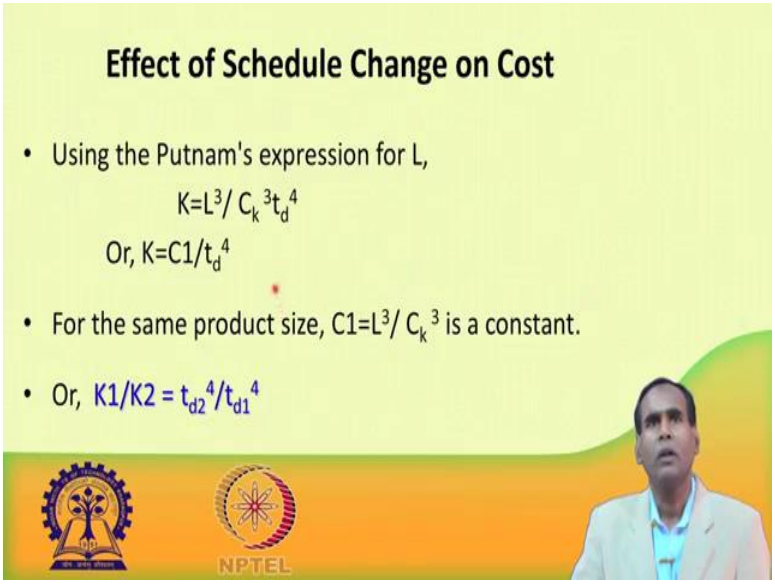

The conclusion says that what is the effect of schedule change on the cost? Let us see we have already seen by using Putnam's expression for L, L we have already seen earlier the value of L is equal to ok. The value of L is specified like this using Putnam's equation, from this you can find out the value of the C k or the value of K let us see.

By using that equation you can find out the value of K which is equal to

$$K=L^3/ C_k{}^3 t_d{}^4$$

And you can see that if will simplify, you will get K is equal to

$$K=C1/t_d{}^4$$

some constant into 2 to the power 4. So, C1 is either constant and this will be C1 will be coming to be L cube by C k cube.

So, if you will what simply this statement you will get K is equal to how much? L cube by C k cube in into 1 by t d to the power 4. So, this L k cube by C k cube is a constant ah, so for the same product size C this L cube by C k cube against C k to the power 3 is a constant and this is represented as C k here. So, finally, K is equal to C 1 to the power ok.

So, final k is equal to C 1 to divided by t d to the power 4. Now, if you will take say a now on say there are two timings two schedules you will take t d 1 which one the

original schedule you have estimated. Now, due to some urgency the customer is requesting you to deliver it before that time say that is t d 2, then you can get the equation you can get two values here K1 and the K2. So, then you take the ratio K1 by K2 is equal to

$$K1/K2 = t_{d2}^4/t_{d1}^4$$

this will be now reverse t d 2 to the power 4 by t d 1 4.

So, this is just a simplification what I have done, here I have found out the values for 2 times t d 1 and t d 2. And accordingly I will get two values K 1 and K 2. So, divide K 1 by K 2 it will be simplified as t d 2 to the power 4 by t d 1 to the power 4. So, what does it mean then what is it is what inference we can draw from this statement.

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You can see that a relatively small compression in delivery schedule can result in substantial penalty on the human effort. See if you are reducing then how much time it is increasing almost 4 times, if you will reduce the job or sorry if you will reduce the time, so the delivered time. Then the effort that you have to put and hence the cost will be increasing very rapidly. So, this can result in, so if a relatively only a very small compression in schedule or in delivery schedule is required, it can result in a substantial penalty on the human effort.

You have to spend much more human effort, you have to spend much more cost. Also observed that the benefits can be gained by using fewer people over a somewhat longer time span, if you can use only few people for a long amount of time then you will get more profit.

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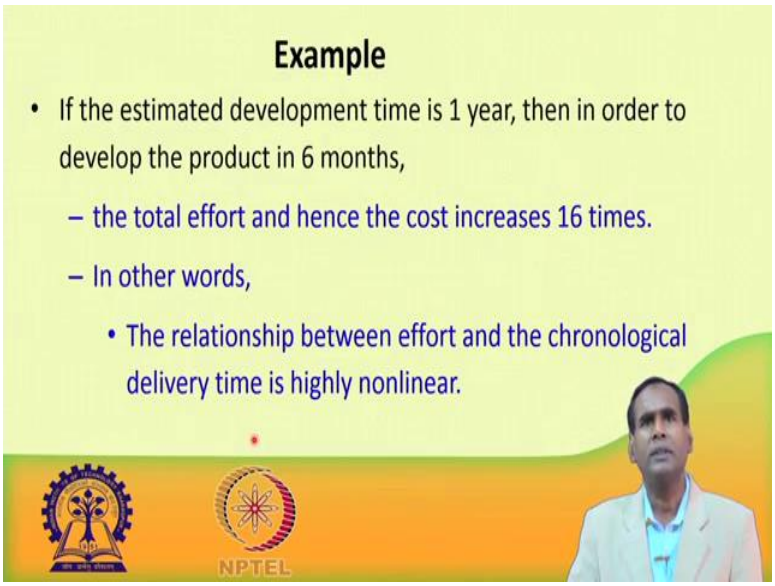

Let us take quickly a small example, if the estimated development time is 1 year then in order to develop the product in 6 months, so what will the consequence on the effort or the cost ok? So, original development time was 1 year, now the customer is coming and requesting you to develop the product in 6 months due to some emergency.

So, now let us see what will be it is effect on the effort and the cost? So, what I will do now I will put the values in the equation, what is the equation? Here you can see that originally t d 1 year; that means, 12 month ok. Then t d 2 is equal is equal to how much? New time what is what you can say that, so original time, so originally it was how much? 12 months or 1 year or 12 months and then it is subsequently reduced to how much? 6 months.

So, how much you will get? 12 by 6, so almost 2 to the power 4 that is 16 times. So, the effort or the cost will be increased by 16 times if the development time is just reduced to half, that is what I want to say. So, if the estimated developed time is 1 year and then in order to develop the product in 6 months; that means, just half you are reducing the time half. The total effort and hence the cost will increase how much? 2 to the power 4 that is

16 times. In other words, we can say that the relationship between the effort and the chronological delivery time is highly non-linear.

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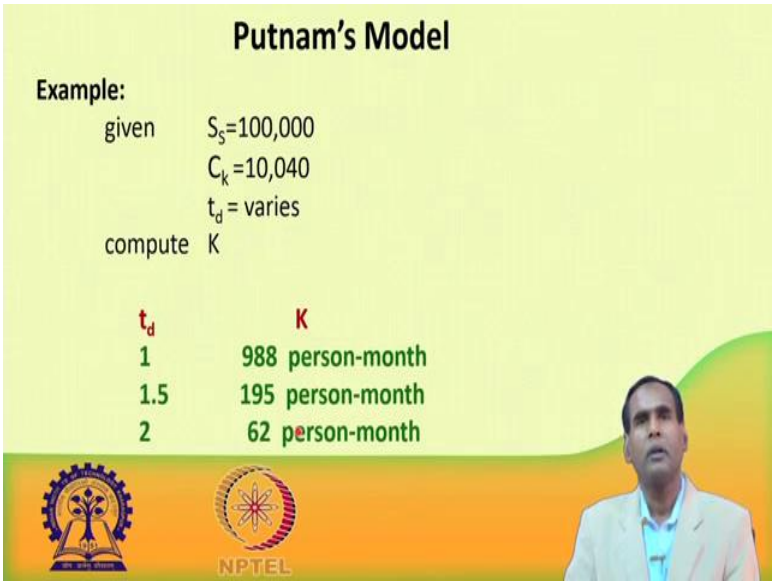

So, let us take a small example and apply Putnam's model for estimating this size effort etcetera. Given that the size S s ok, I hope S s is the size, so given that S s is equal to 100000 lines of code loc, or C k is equal to 10040 and t d varies it may be what 1 month 2 month or whatever that. So, t d is varies or maybe you can take yes t d you have taken in a months. So, now, you have to compute K for t d maybe 1 month, you are getting how much? Putting the values S C K be the values the since I have already shown you earlier that is the equations here.

So, S s, C k K and t d there, we want to compute the K which is the person years invested or effort in person months. So, K is equal to what? So, this is the basically effort k, so S s is given Ck is given t d is given. So, now, we can easily compute the value of K. So, for 1 month you see that for 1 month by putting this value of S s and this value of Ck.

That means, S s equal 100000 and C k is equal to what 10040, then the value of K is coming to be 988 person month, please put the values in the previous equation you will get this. Similarly, for td is equal to 1.5 months you will get this is the value of K; that means, this is the value of effort required, for td delivery time 2 months the value of effort is 62 person month. Given these given the these values for the parameters S s and

C k and t d is equal to 2. In this way you can compute the effort required by using Putnam's model.

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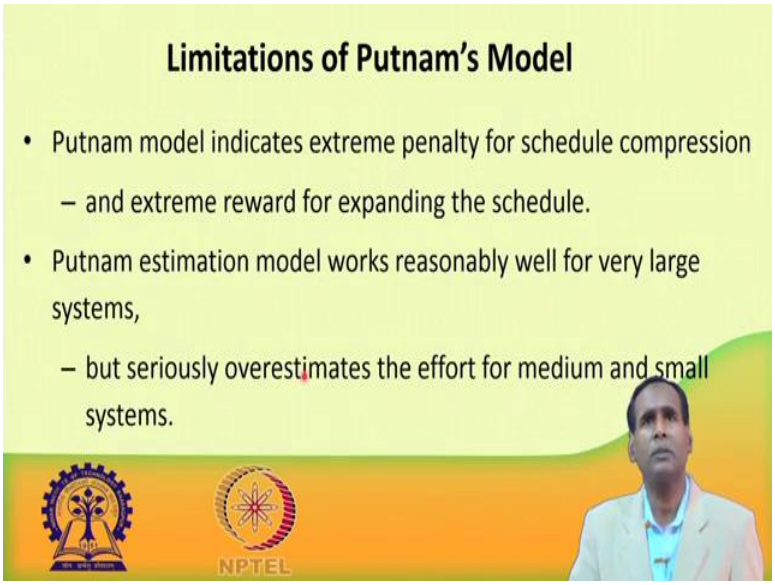

Now, let us quickly see what is the limitation of Putnam's model? Putnam's model indicates extreme penalty for schedule compression, that we have already seen that if this time or schedule is compressed a half. So, almost we have to what the effort will be increased by 16 times, the cost will be may be increased accordingly or what because the cost maybe increased 16 times.

And extreme reward for expanding the schedule, if you can go on expanding the schedule then your profit will be more you will get more benefit. Putnam's estimation model it works reasonably well for very large system. So, if you are system is a very large, your project is very large, then it will work perfectly the estimations will be more accurate. But seriously overestimate the effort for medium and small scale systems. So, if you are using just a small scale small scale system or a medium scale system, then on the estimates will not be accurate they will be overestimated.

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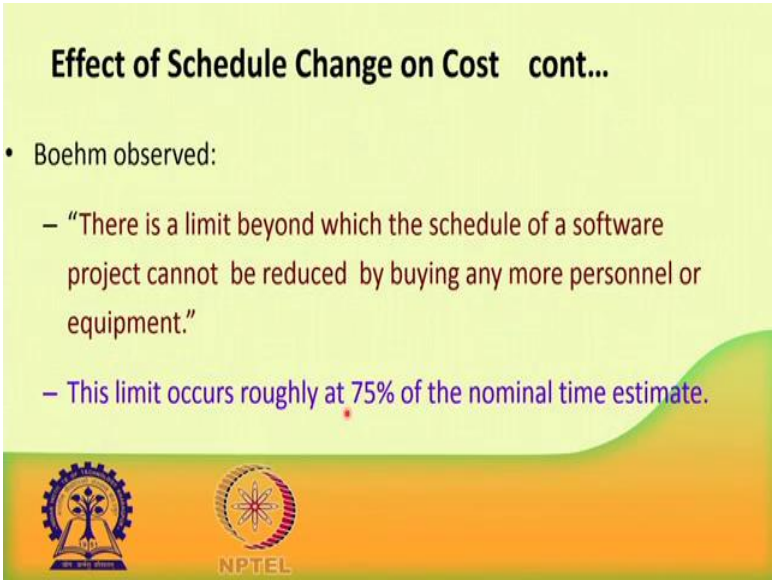

So, in order to what here you can see that there is a limit beyond which the schedule of your software project cannot be reduced by buying any more personnel or equipment, I have already told you in this graph that. So, this is the optimal time point of time and the optimal effort is here. If you are reducing up to say this point you can reduce theoretically and you see effort is increasing tremendously. But still if you will again go on reducing there will be some point and beyond it you are reducing the time you cannot complete your project you may not be successful.

So, what is that point, up to what point you can go on compressing the schedule and if you come further schedule that then you may not succeed. So, that point Boehm has studied many applications and he has concluded this. Boehm observed that there is a limit beyond which the schedule of a software project cannot be reduced by buying any more personnel or equipment.

Because some of the what because some of the activities are sequential they are not parallel, so how many what personnel you may hire. So, since these things activity has sequential, so they cannot complete the extra persons they will just sit idle. So, what is this limit? This limit occurs roughly at 75 percent of the nominal time estimate. So, this limit it will occur at 75 percent of the nominal time estimate beyond this if you will schedule, so your project may not succeed.

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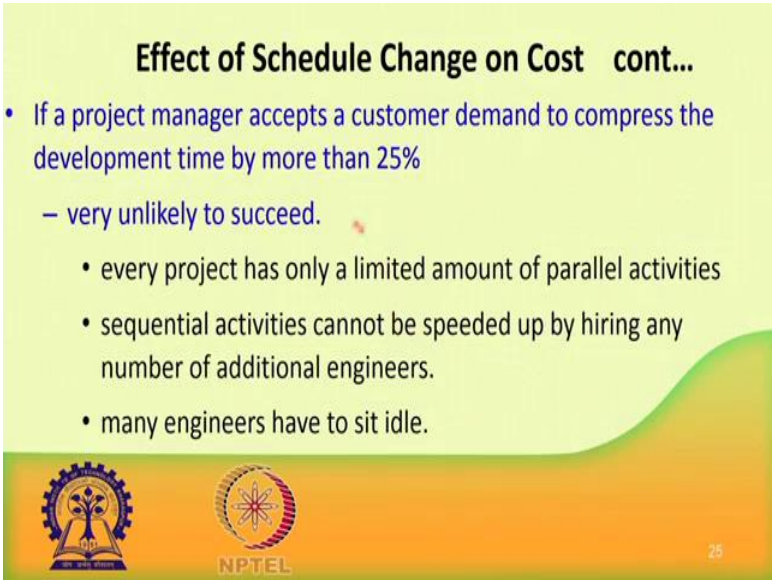

If a project manager accepts a customer demand to compress the development time by more than 25 percentages. See here I have already told you the limit is up to 75 percentage. So, there; that means, if a project manager accept a customer demand to compress the development time by more than 75 25 percent, because we have already seen that it is 75 percent the limit.

If a project manager he accepts a customer demand to compress the development time more than 25 percent, then it is very unlikely he will succeed you may not complete the project you will he may fail. Because every project has only a limited amount of parallel activities where you can employ more people most staff for those areas, but there are several sequential activities.

These sequential activities cannot be speeded up by hiring any number of additional engineers and many of the engineers they will have to sit idle. So, if will compress the development time by more than 25 percent, then certainly your project may fail it may not succeed you may not get success, because many engineers they will just sit idle because for them no work. Because many activities are sequential they are not parallel they are sequential, and any number of what the sequential activities cannot be speeded up by hiring any number of what personnel, so this engineers will sit idle.

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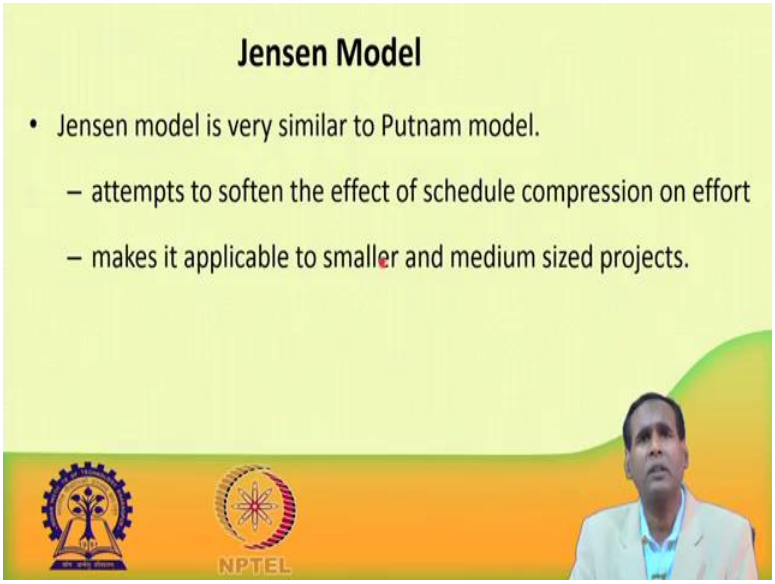

We have seen that one of the major drawback of what this Putnam model is that it does not work for what it does not work well among for the medium and small size systems. So, in order to overcome that Jensen came he has proposed a model. This is known as Jensen's model, Jensen's model is very similar to Putnam's model, he attempts to soften the effect of scheduled compression on effort.

Because we have seen in case of the Putnam's model the effect of scheduled compression on the effect on scheduled compression on effort is very hard it almost its 4 times and we have already seen it is of to the power 4. So, that is why when we are reducing half the schedule almost 16 times effort is increasing ok. This makes it applicable to smaller and medium sized project, since he has tried to soften the effect of schedule companies on effort this makes it applicable to be applied in smaller and medium sized projects.

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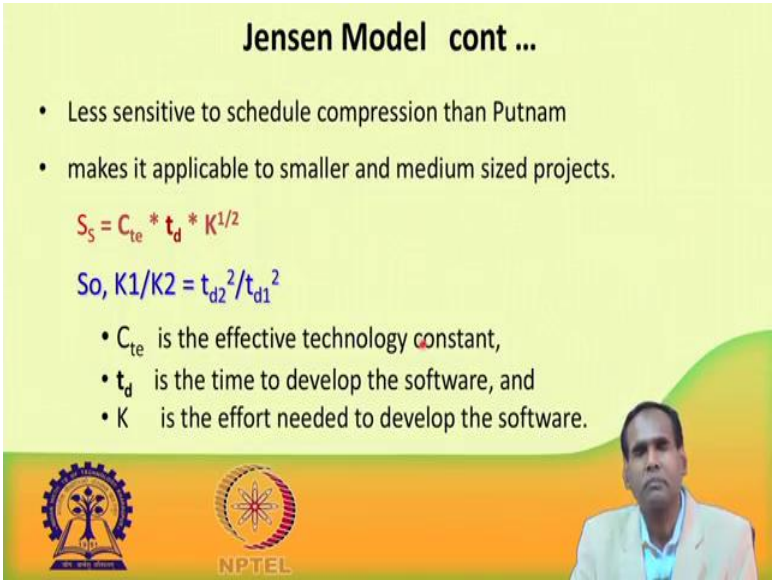

So, it is a less sensitive to schedule compression than Putnam ok, it is very much less sensitive to schedule compression than Putnam. It makes it applicable to smaller and medium sized projects, according to Jensen or according to Jensen's model the size can be specified as

$$S_S = \mathbf{C_{te} * t_d * K^{1/2}}$$

C te into td into K to the power 1 by 2.

Here the terms of the user meaning S s is the size and C te is the it is the effective technology constant, t d is the time to develop the software and K is the effort needed to develop the software and these are the formulas is there. So, you can see this formula is very much suitable for the small and medium sized program problems or projects.

From this equation similarly you can find out the what for the two values; one is the original schedule another is the revised schedule you can find out the values of K1 and K 2 to the power them then where K represent the effort. Now, you what divide K 1 by K, 2 then you can see what is the effect of schedule compression on effort.

You can see that now it is of the order of the square previously it was of the order of to the power 4, now you can see this ratio is of the order of square. So,

$$K1/K2 = t_{d2}^2/t_{d1}^2$$

K1 by K2 is equal to t d 2 by t d 1 whole square. So, here the effort it will not increase of the order output to the power of 4 rather it will increase to the of the order of to the power 2.

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So, now if you will take the same problem that we have applied for Putnam's model you can see this will work like this. If the estimated development time is 1 year then in order to develop the product in half that is 6 months. Now, the total effort and hence the total increase the total cost increase how much time 12 by 6 is 2.

So, 2 square, 2 square means 4. So, now, the effort and cost will increase 4 times whereas, in Putnam's model it was increasing by 16 times. So, this is much less in comparison to the Putnam's model, so these are the advantages of Jensen's model over Putnam's model. Now, first the if the schedule is reduced the time is reduced the effort is not increased in a if the effort is not increased rapidly. As well as these model can be easily applied to small and medium scale, small and medium scale, or small and medium size small and medium sized projects.

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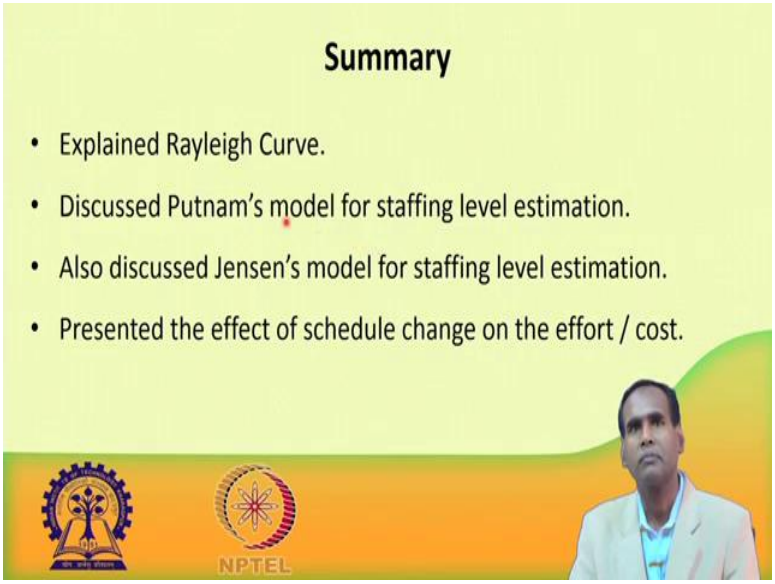

We have seen in this class the details of Rayleigh curve, how Rayleigh curve can be applied to what software industries. We have discussed the Putnam's model for staffing level information, estimation. We have also discussed Jensen's model for staffing level in estimation, we have presented the effect of schedule change on the effort of cost on effort and hence the cost using Putnam's model and Jensen's model

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These are the references.

Thank you very much.

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Lecture - 26
Project Scheduling

Welcome to this lecture. In this lecture, we will start to discuss about Project Scheduling. Project scheduling is not only confined to software projects, but to all types of projects. And, naturally this is a very old technique, but then there has been lot of developments recently. This technique, the project scheduling techniques date back to nearly a century. Over last one century, many techniques have been proposed and we will see that there are several charts that get developed, but a recent phenomenon is to use a computers in the scheduling process.

The charts are automatically updated, different parameters are computed so, what used to be once computed manually and very painstakingly now can be done at the press of a switch. But, still the main ideas of scheduling are very important. The project manager even though has a tool at disposal, but has to know the nitty-gritty of project scheduling to be able to effectively schedule a project.

Let us look at the main concepts in project scheduling.

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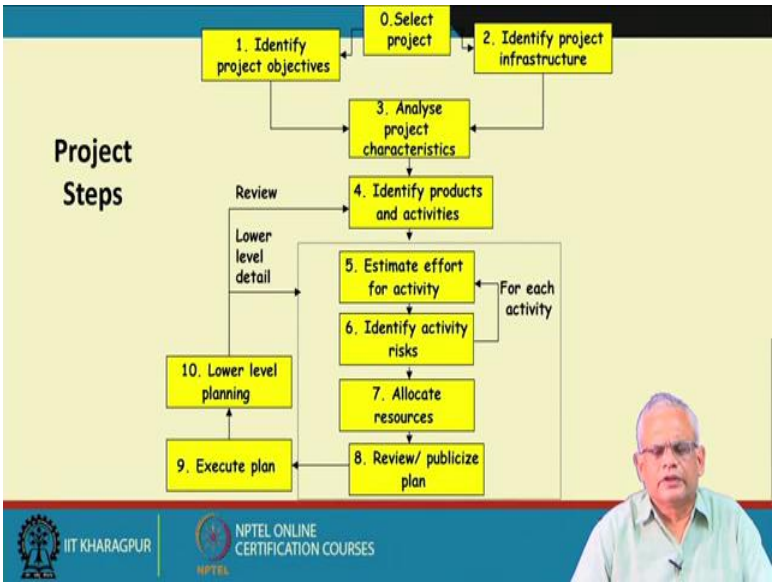

Before we look at project scheduling let us look examine the overall project steps. Once the project has been selected we need to identify the project objectives and the infrastructures. These are two main things that what the project needs to do and also what infrastructure it needs to use, what will be the team, team organisation, hardware and so on. And, once that is done we need to analyse the project characteristics – what are the risks involved, what is the resources that are required, what will be the life cycle model that will be followed based on these factors and so on.

And, once that has been done we come to estimation and scheduling. For that we need to first identify all the activities, the different deliverables which we refer to as products, the different deliverables to the customer and what are the activities to be undertaken to produce those deliverables. And, we need to do a finer level risk analysis based on every activity.

And once we do the risk analysis the activities may change. Their effort, duration etcetera may change and once we complete this process we allocate resources and we review and publicize the plan. And, this forms the planning process and then comes the execution. In execution the activities do change we notice very finer level activities and therefore, we do a lower level planning. We find that there are many new sub activities and so on and then we do a detailed lower level planning so, again the same steps.

Let us look at how we do identify the activities and then we identify the precedence and schedule. So, that is goes by the name project scheduling.

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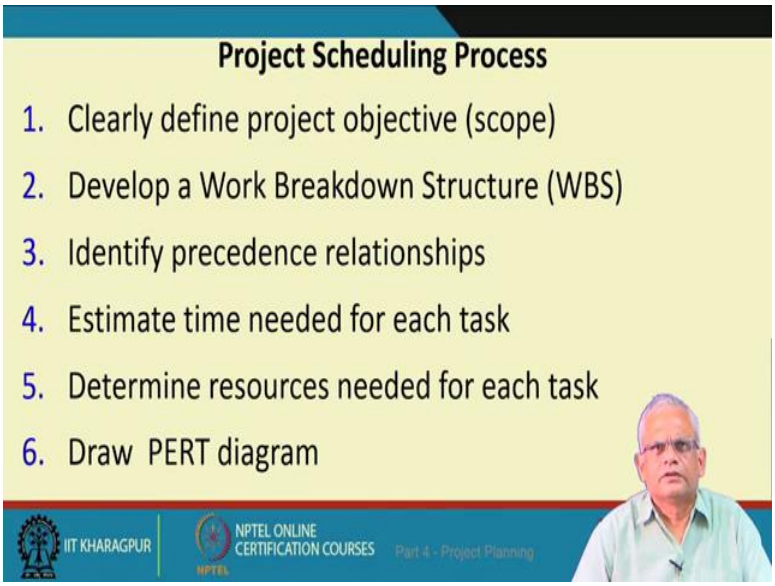

In the project scheduling process, we should have defined the objectives or the scope. And, the first step here once having developed the project objective is to identify the finer level activities. From the objective we get more detailed activities or finer level activities which we represent in the form of a work breakdown structure.

And once we identify the activities we identify the precedence relationships between these activities, which activity would complete then the other activity can be taken up. And, once we identify the activities, the precedence relationship we need to estimate the time duration for each of these activities. We need to determine the resource needed for each of these activities and then we need to draw a PERT diagram which will is a focus of this lecture.

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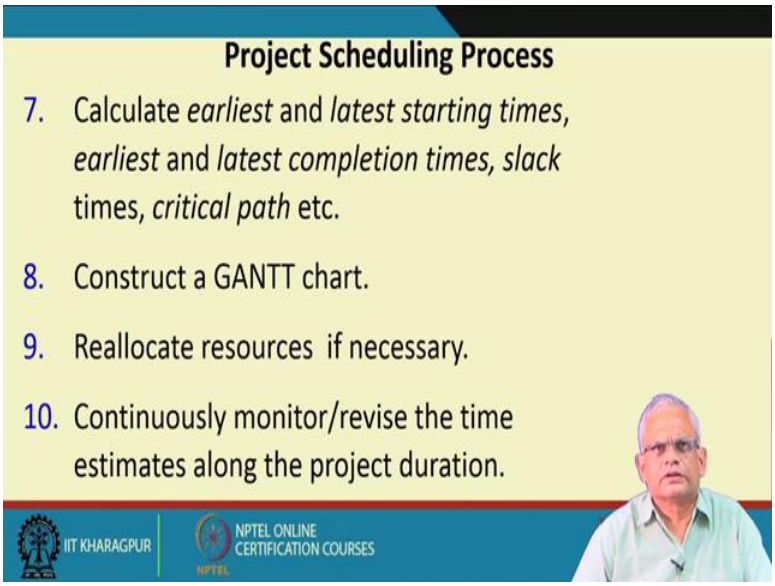

And, once we draw the PERT diagram, we should be able to compute the earliest – latest starting times, earliest – latest completion times, slack times, critical path and so on. And, then once these are complete the project is ready for entering the execution phase. In the execution phase, as we mentioned we are interested in monitoring and control. In monitoring and control we are interested if let us say a developer becomes unavailable what will be the impact on the schedule; if a new developer joins how will the schedule change and so on. And, that becomes easier to do if we have a GANTT chart.

The project manager initially constructs a PERT chart, where the overall schedule is drawn and various aspects of the project such as the earliest start and completion time of the tasks, the slack times, the critical paths etcetera are identified. We will see what the specific terms mean as we proceed in this lecture.

Once having done this overall planning process, the project enters the execution phase. In the execution phase, we need to do monitoring and control. We need to identify what if a developer becomes unavailable or additional developers become available and so on, and that becomes easier to do with the help of a GANTT chart, we will look at the GANTT chart.

If necessary using a GANTT chart we can change the developers or hardware. We may put more developers to one task to make it complete if the task is getting delayed and we can identify what is the impact of that on the schedule using a GANTT chart. And, as we

do this we need to continuously monitor revise the estimate for the entire project duration.

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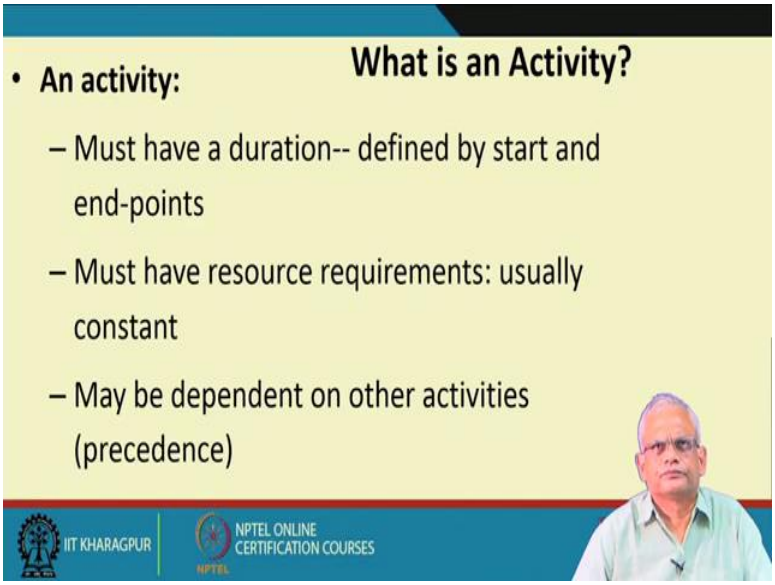

The basic to project scheduling is an activity because we schedule an activity and its sub activities. Let us be clear about what is an activity. An activity is something by which the team achieves or does some work, and naturally each activity has a duration. Over the duration a team member will be doing something and not only duration it will also be defined by a start and end point. It will start at certain time and end by certain time and in between it will have a duration. Of course, the start and end time difference must be larger than the duration.

To do an activity we must have a resource requirement. The resource typically is a developer; we refer to the developer as a resource. There may be one or more developers assigned to an activity, but then for an atomic level activity that is represented on the diagram, we normally keep the resources constant; otherwise it will become extremely complicated. Even if we have the lowest level activity some developers work for some time and then some developers leave, more developers join and so on, it becomes very difficult to represent in a diagram. And, therefore, we assume that for an activity duration the resource is constant. An activity may depend on other activities.

So, these are the characteristics of the activity. An activity is something using which a team member or a few members gets some important work done. These are associated

with a duration and identified with a start and end point. Some resources must be assigned to an activity, will carry out this activity. An activity may be dependent on other activities and this we call as the precedence relation. If an activity A can start only after some activity B has completed, then we say that B proceeds A or there is a precedence relationship between B and A.

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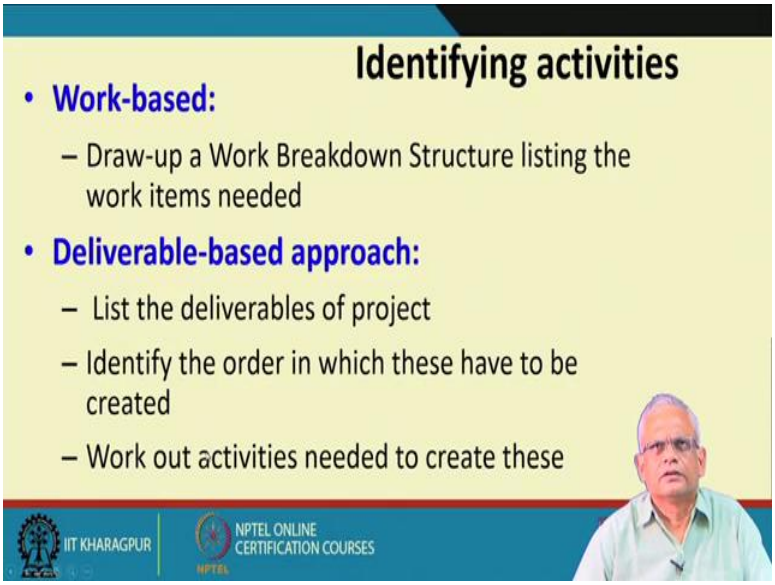

One of the most fundamental things that the project manager does is once the objectives have been identified how to identify the activities from the objectives. There are two main approaches that the project manager uses; one is known as the work-based. In the work-based approach, the project manager identifies that to meet the objectives what are the work that main work needs to be done and from the main work identify what are the sub-work that need to be done.

So, basically identify the activities, break the activities into sub activities and so on, and this is represented in the form of a work breakdown structure. That is, what are the main work which are broken down into sub work and sub work is broken down into sub-sub work etcetera until an atomic level. And, typically an atomic level activity is one where a single developer can do it in let us say a week or something.

It is not a good idea to use a work breakdown structure to break the activities into level of hours or minutes, that is because it will become too much of an overhead for the project manager to develop the schedule minute wise, hour wise or day wise. Typically,

the schedule is done in terms of weeks 1 or 2 weeks. The developer is assigned work for 1 or 2 weeks and then the project manager monitors whether the work is completing on time or is delayed.

And, we can also argue that work granularity of several months is also not good. Let us say we have a work breakdown structure where the leaf level work is 2 months. The main problem with this is that the project manager loses control of the project. The fact that the project is getting delayed or some activities delayed, the project manager can know only after 2 months and by that time it will be too late to control the project and back put it back into the schedule. And, therefore, the project manager loses control of the project.

To repeat the same thing I will say that in a work breakdown structure. The important works are identified and these are broken down into sub works, until the granularity of a week or 2 those form the atomic level or the leaf level activities in the work breakdown tree. We will look at this in the next slide.

The second approach that the project manager uses to identify the activities is based on a deliverable-based approach or product based approach. In the product based approach, a deliverable-based approach the project manager identifies what are the deliverables or the products to be delivered to the customer. And, for first makes the list of the deliverables and then identifies the order in which these have to be created, and for each of these deliverables the project manager identifies what are the activities that need to be performed to create these deliverables. Let us look at some examples of the work-based and deliverable-based approach.

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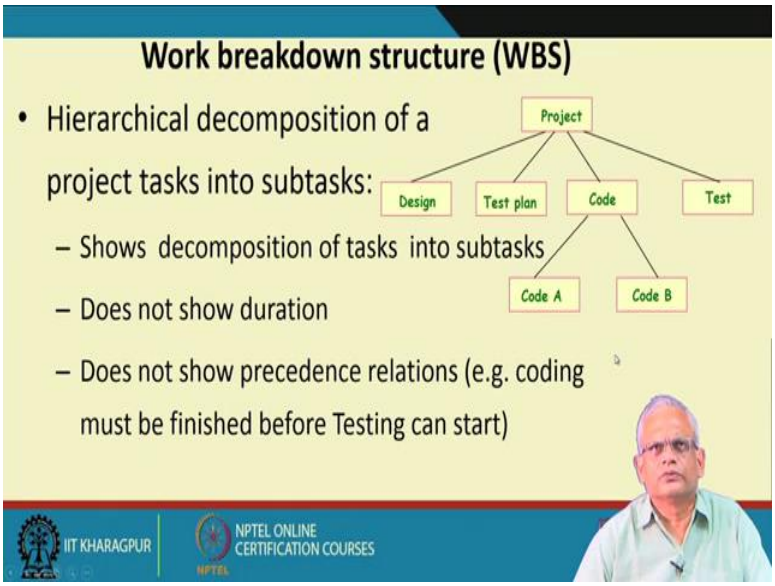

In the work-based approach let us say we need to complete a project. We need to the project manager identifies what are the main activities that need to be performed: requirement specification, design, test plan, code, testing, code two different modules A, B etcetera. So, this is a hierarchical representation of the important activities and the sub activities and at the leaf level they should take about a week or 2 to complete.

At this point just shows the sub activities, the main activities and the sub activities, sub activities and so on, but then the project manager does not accurately determine the duration of these activities that is done later. Neither does it show any precedence relationship. For example, the test plan should be created after the code that is not represented in this diagram or the code must complete before the testing starts that is not implied by this diagram.

The main objective of this diagram is to identify the important activities and then decomposition of these activities into sub activities till the granularity of the leaf level activities is about a week or two.

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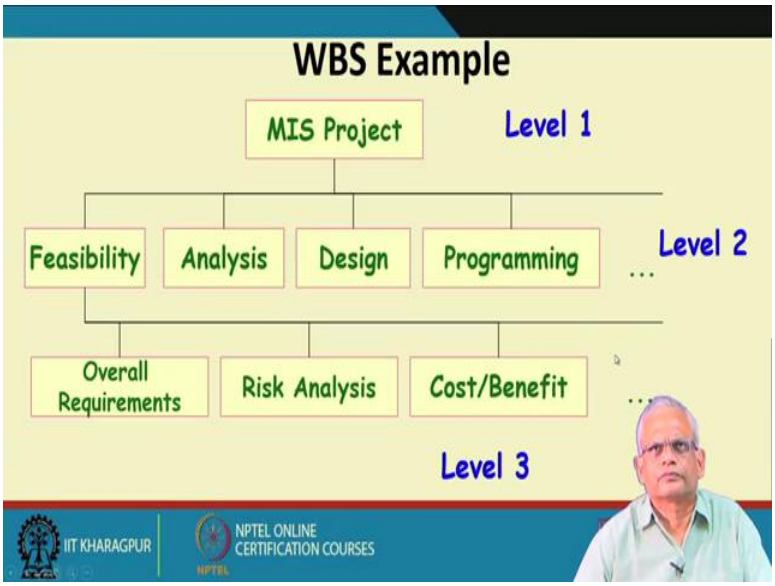

This is another example. This is the hierarchical chart, the MIS project that is represented as a level 1. In the level 2, the important activities the feasibility study, requirements analysis, design, programming, testing etcetera and then for each of these activities the sub-activities for feasibility study get the overall requirements, perform risk analysis, perform cost benefit analysis etcetera. For requirements analysis requirements gathering, analysis of the gathered requirements, documentation in the form of SRS document and so on. These form the level 3 activities and so on.

If this any of these takes more than week or two, then these are decomposed into more finer level.

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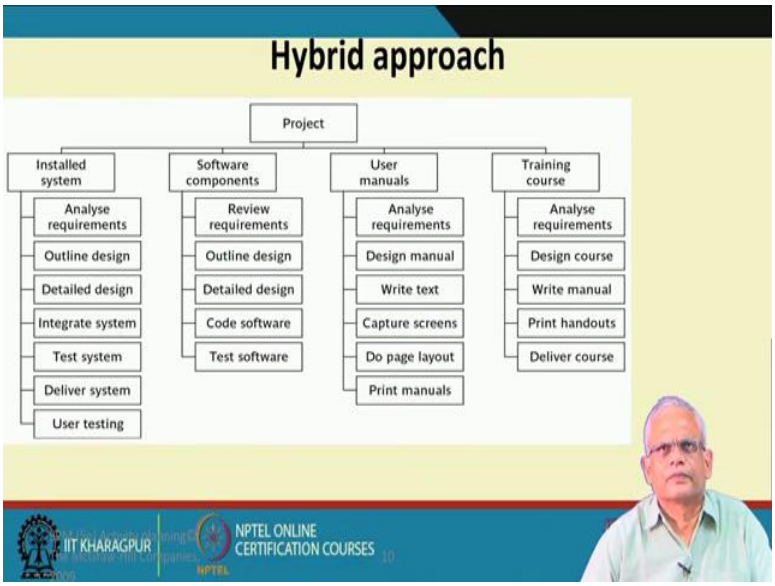

Project managers also use a hybrid approach which has both based on the activities that need to be carried out and also the deliverables that had to be given. Let us look at this example where a project has certain deliverables. These are represented here that the system must be developed and installed; the different software components must be delivered. So, this is the system which is both hardware and software.

The system must be installed at the customer site, the soft and the hardware let say it exists or maybe you can have activity here to develop the hardware. Then, software components have to be developed. The user manual has to be developed and the training programme has to be carried out. For the system installation, we need to identify the requirements, analyse them, outline the design, detailed design, integrate the system, test, deliver and user testing.

For the software components to be developed need to get the requirements design, detail design, code, test. For the user manual analyse the requirements because it write the manual based on the requirements, design the manual, write the text part of the manual, capture screenshots and insert them, do page layout, then print the manual. The training program again analyse the requirements, design the course, write the manual, print handouts and then finally, deliver the course.

So, this is the hybrid approach where which is having both deliverable based. So, some of these internal nodes or rectangles or actually deliverables, and then from there these

are the activities and then we can split these into sub-activities and so on. When you are part of a project or you are the project manager, do not be surprised to see that we have work breakdown structure where each of these are activities and also hybrid approach where there are activities and also deliverables as part of the work breakdown structure.

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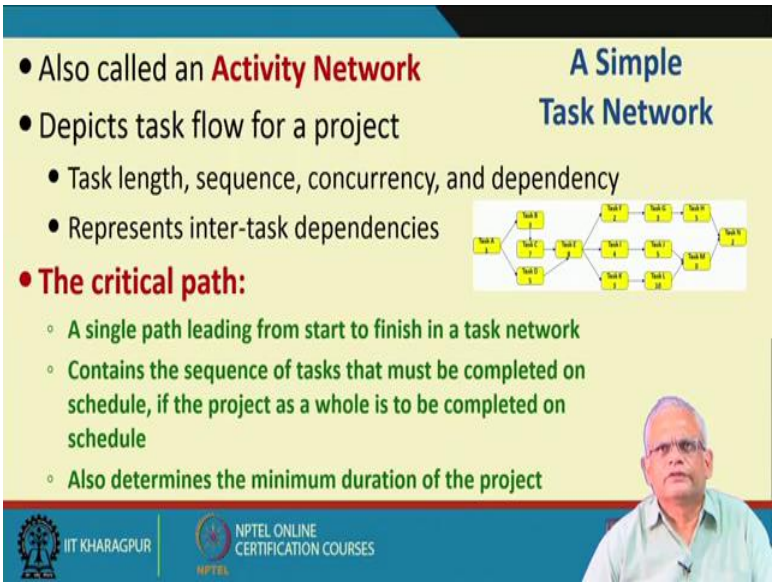

And, once the work breakdown structure has been developed we need to represent these in a task network, they are also called as activity network. These not only represent the important tasks that are present in the work breakdown structure, but also the dependency relations among the different tasks.

And, not only that we need to also identify the task duration; we not only label these nodes on the task network with the name of the task, but also we write the durations here as you can see and we represent the dependencies among the task. And, naturally here we can identify sequence which tasks need to done one after other and which tasks can be done parallelly. For example, the task over here can proceed parallelly with these two tasks, different developers might undertake these three tasks concurrently.

One important characteristic of these task network is the critical path. The critical path you can see that there are many paths here from the start node to the end node there are many paths; a path over here, a path over here and so on. There are many paths, but then the path having the longest duration when we sum up all these durations for various tasks, we get the total time needed for completion of the project for those tasks. And,

because there is a sequence shown by the dependency arrow need to do the one after other and therefore, the total duration among this path is the addition of all these durations.

And, the path having the longest duration from the start to the end is called as a critical path. A path is any a single path is any path from the start node to the end node. The critical path is the one where the duration is the longest and that is the minimum time by which the project can exceed. Along one path it maybe let us says 20, another path maybe 25 another path maybe 23. Then 25 is called as a critical path and 25 is the minimum duration for the project. The project may exceed 25, if some of the path some other tasks here get delayed, but it will not be done any earlier than 25.

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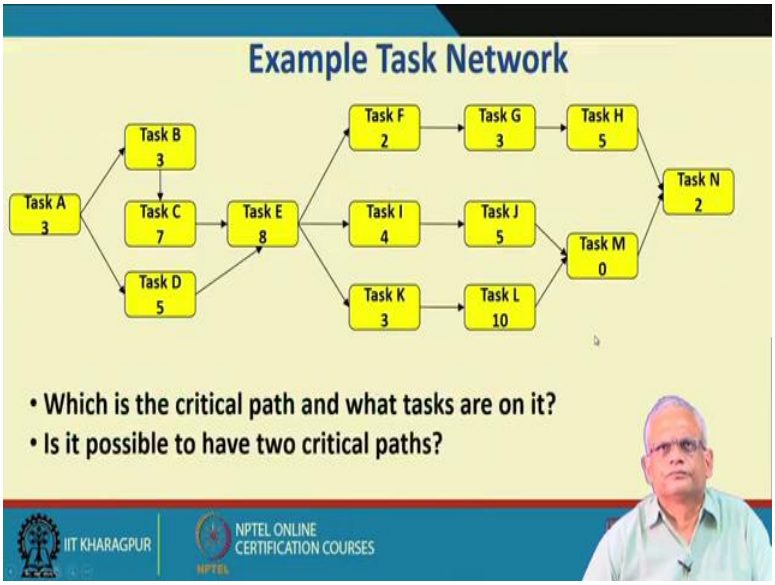

This is an example of a task network. We write the name of the tasks and the durations here. It will be possible to compute the critical path by computing by looking at various paths here adding the total duration and finding the one which has the maximum duration. But, then the main problem with this approach is that there may be many paths we miss out, may do a mistake and so on. And, therefore, systematic techniques have been developed to compute the critical path because that is a very important characteristic of a project.

The critical path is important because project manager knows that is the minimum duration for the project and also the tasks that are there on the critical path if those are

very important tasks, if they get delayed and the project will get delayed. Whereas, the tasks which are not in the critical path they may get delayed, still the project duration may not get affected. But the other question that, is it possible to have two critical paths in a task network? The answer is yes. It can so happen the two of the longest paths each have 25 or something as the duration, or may be three paths have 25 or three paths may have 30. So, then there are three critical paths.

So, it is possible that there are multiple critical paths and it is very important to identify the critical path from the project managers' perspective because he needs to monitor these paths, the tasks on this path very carefully. And for that over the last century systematic techniques have been developed ok.

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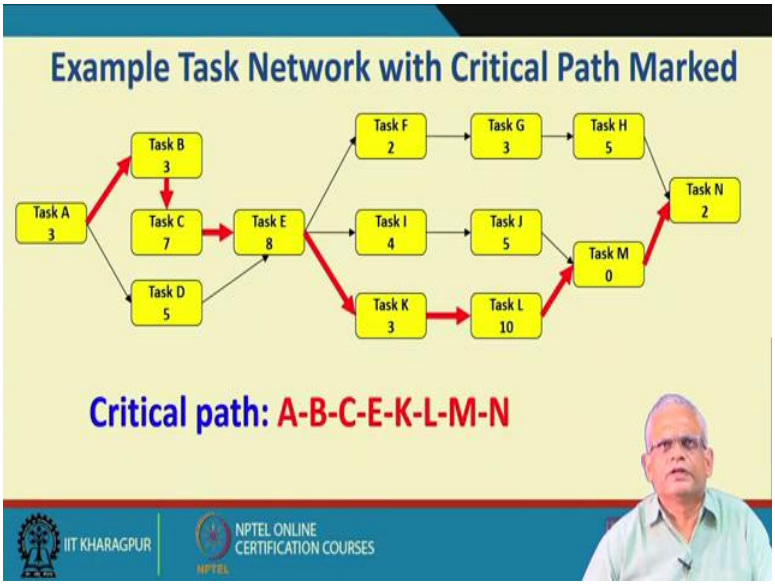

This is an example on this. If we compute various paths this the one shown on the red that is A-B-C-E-K-L-M-N that is the critical path you can compute the duration across this that is 3 plus 3 plus 7 plus 8 and so on. This will be the longest compared to the other paths that are present here.

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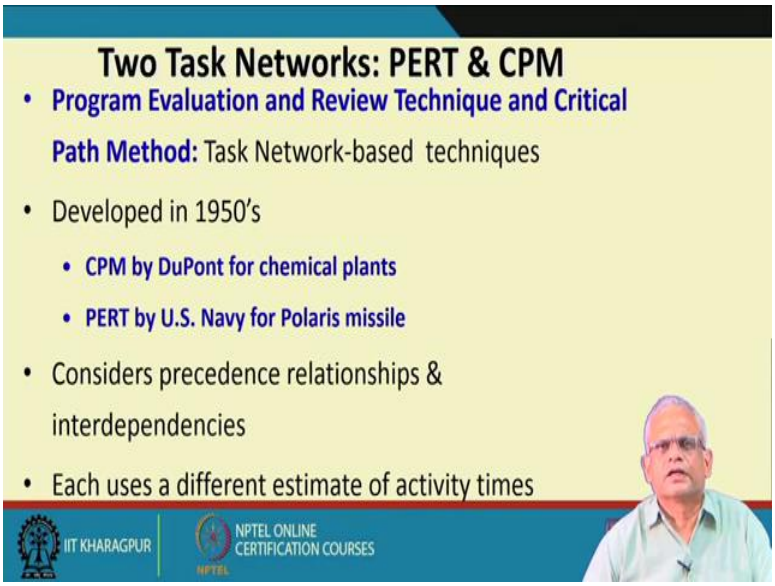

As I was saying that over the century many techniques have got developed. Two important techniques are the PERT chart and the CPM. Both of these are task network based techniques. We will see that at some point of time these were two distinct techniques have in their own uses, but now this to have merged into a single technique called as PERT CPM and these are available in many automated tools. We look at these developments and how these are used to compute various characteristics of the project we will discuss this in the next lecture.

Thank you.

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Lecture – 27
Project Scheduling Using PERT/CPM

Welcome, to this lecture. In this lecture, we will discuss about Project Scheduling Using PERT and CPM. In the last lecture, we had looked at the various issues in project scheduling; we had discussed about breaking down tasks in the project using a work breakdown structure and then we had looked at a task network.

Over the years, that is over the last century or so, these techniques have been used the project scheduling techniques for various types of projects and therefore, different ideas that were proposed regarding project scheduling have now been merged into very strong and useful technique the PERT and CPM. And the focus of this lecture is to discuss about how to use PERT, CPM for project scheduling. Let us proceed from this point.

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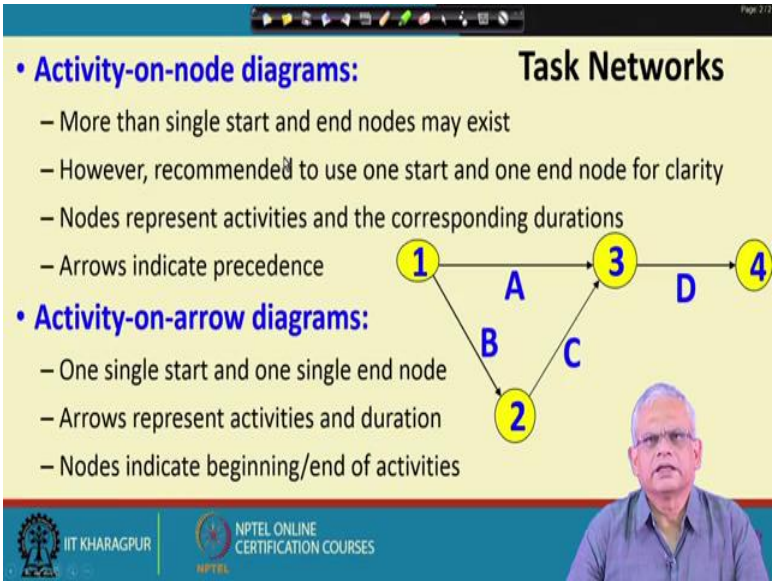

If you remember, in the last lecture, we had discussed about task networks. In the task network, we represent the different tasks in the project and also the precedence relationships, that is, which task must complete before another task can start. But, then

over the last century or so, there are two main types of task networks that have gained popularity, one is activity-on-node diagram.

In the activity-on-node diagram, as the names says, this network consists of nodes and edges. In this network nodes and lines connecting the nodes and the names of the node are labeled with the activity names. So, each node represents an activity that is why it is called as an activity-on-node diagram.

On the other hand, we have another variant of task network which is activity-on-arrow diagram. In this the nodes they represent end or start of activity, but the activity itself is represented on the arrows or the edges connecting the nodes in the diagram. But, the activity-on-arrow diagram had some issues. It is much more complicated than the activity-on-node diagram. The activity-on-node diagram is very intuitive; whereas, here in the activity-on-arrow diagram we have nodes representing start and end of activity and the activity itself is represented by edges and on that count there are few complicacies that arise and we might have to insert dummy activities and so on to model a project.

And possibly for that reason over the years activity-on-node diagram have become very popular used overwhelmingly by the project managers very simple and also the different CASE tools the computer tools and project management that are available they also use largely the activity-on-node diagram. And in this lecture we will focus on the activity-on-node diagram, but as a background knowledge we will just keep in mind that there is a variant of the test task networks which are activity-on-arrow diagram.

In the activity-on-node diagram we might have more than single start node that is to start the project we might have a few activities which are no dependencies, that is they can start anytime and similarly there may be many end nodes in any project. But, then it makes a bit complicated to use various computations on such a activity-on-node diagram which has multiple start nodes and multiple end nodes we will make it complicated to apply various project scheduling techniques.

And therefore, it is recommended that we use a dummy start node and dummy end node which are single nodes single start node and single end node, so that application of the techniques is becomes very intuitive and easy. And this point we need to keep in mind that whenever there are multiple start nodes, we will have a dummy nodes start node which will be single start node, similarly for the end node.

And here in the activity-on-node diagram as the name says that the nodes represent the activities. The nodes are labeled with activities and also not only we have the activities on the nodes, but also we marked the duration that the activity will require. And will have arrows which will indicate the precedence of one activity with other activities.

The activity-on-arrow diagram here the nodes indicate the start and end events of activities, but the activities themselves are marked on the arrows and also the duration of the activities are marked on the arrows. Here also one single start and single end node that makes it easier, as we already mention. Arrows represent the activities unlike the activity-on-node diagram where nodes represent the activities; in the activity-on-arrow diagram. The arrows represent activities the arrows are labeled with the activity names and also the duration. And nodes indicate the beginning and end of activities.

Just to give an example of a activity-on-arrow diagram, we have this activity A starting at node 1 ending at node 3; activity B is marked on this edge or arrow which is starting at node 1 and ending at node B sorry, node 2. But, as I already mention that will not really use activity-on-arrow diagram all our examples will restrict to activity-on-node diagrams.

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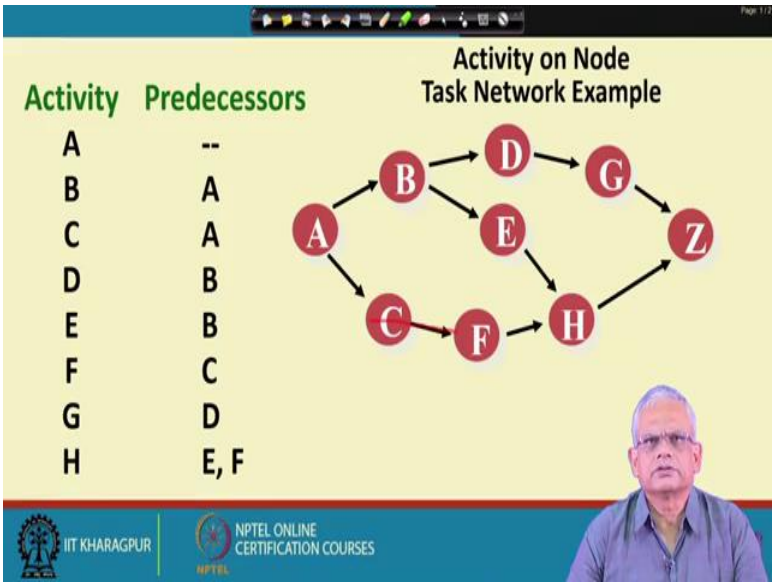

Now, let us try to draw an activity-on-node task network diagram. Given that, we have from a work breakdown structure we have identified what are the activities and then later we also identify what are the predecessors of the activity. A has no predecessor; B has

predecessor A; C also has predecessor A; D has B as the predecessor; E has also B as the predecessor; F has C as the predecessor; G has D as the predecessor and H has two nodes E and F as the predecessor.

So, how do we draw this? Of course, we can straight away draw A, because A has no predecessor. So, we can draw A here which has no predecessor and then we have B has A as the predecessor. So, we can draw B which has A s the predecessor and then we can look at C and C also has A is the predecessor. So, let me draw C here C also has A as predecessor.

And D has B as the predecessor we draw D here which has B as the predecessor and then E has B as the predecessor. So, we can draw E here, E has B as predecessor and we have F has C as the predecessor. Let me draw F here; F has C as predecessor and G has D as the predecessor. So, let me draw G here, G has D as the predecessor and H has both E and F as the predecessor H has both E and F as its predecessor.

And we can since there are two nodes here which are two end nodes we can have a end node here and this can be the task network, activity-on-node diagram. And if I redraw this nicely so, we have this A has B has A as predecessor, C has A as predecessor H has both E and F and Z is a end node. So, the same diagram just drawn it here. As you can see that it is not very difficult very intuitive actually to draw a activity-on-node task on network.

I am sure that given any other example you can draw the activity-on-node network diagram. But, the activity-on-arrow diagram would have more complications and we are not going to discuss that because overwhelmingly activity-on-node diagrams are being used.

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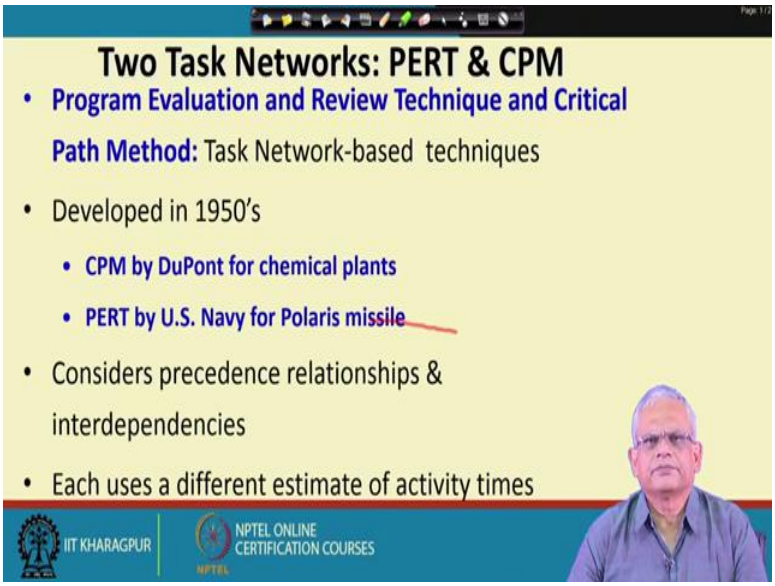

Now, the techniques PERT and CPM these are two task network based techniques as you are saying that these have been developed over the last century or so. These are two different techniques PERT and CPM; PERT stands for Program Evaluation and Review Technique and CPM stands for Critical Path Method. These are two task network based techniques. The PERT initially use the activity-on-arrow diagram and because these have long history over the years sometimes the activity-on-node diagram was adopted.

If we look at the history of this both of these were developed based on the Gantt chart. Gantt chart was a older diagram and Gantt chart is used for project monitoring, control, resource allocation etcetera. And later in this course will be discussing about the Gantt chart, but the project scheduling is done by PERT and CPM diagrams.

And this were developed in 1950s and Gantt was much earlier than this, may be 1920s and 30s. The CPM was developed from Gantt chart and was used by the company DuPont in its chemical plants for managing maintenance projects of chemical plants. The PERT was used by US Navy for development of the Polaris missile.

But, just want to mention here that the CPM was used by DuPont for managing maintenance activities and maintenance activities we know what are the type of maintenance, what will be the activities, what is their duration can be precisely known. And therefore, here the task durations are more deterministic; whereas, the other one, the PERT was developed by US Navy for development of the Polaris missile.

And this is a development project unlike the maintenance of chemical plants which is a routine work. The PERT was for development project where there are much more uncertainties regarding the tasks. For example, the task start time, the task duration end time these are all random because there can be delays. The durations etcetera are not known deterministically unlike the maintenance projects and therefore, PERT uses a statistical techniques for the task durations.

Both these techniques the considered precedent relationships on the task just like any task network which represent the interdependencies among the tasks. And as I mention the CPM uses a deterministic estimate of the activity time and the PERT uses a statistical estimate of the activity time.

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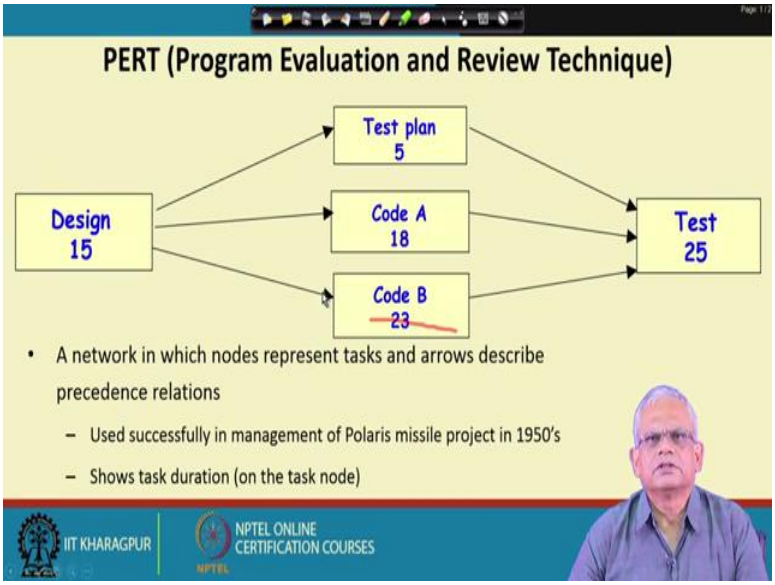

Let us look at our first diagram. This is the simplest PERT diagram and here we have used a deterministic times for design, test plan; the test plan can start only after the design is complete. The design is expected to take 15 weeks; test plan is expected to take 5 weeks and testing is expected to take 25 weeks. There are two main modules A and B; after the design test plan can be done or coding of module A can be done which will take 18 weeks or the coding of B can be done which takes 23 weeks.

There is a parallelism possible that is these three tasks can be carried out in parallel if sufficient resources exist. But, then it may so happen that we have only one developer, one coder we have available and one tester. In that cases all the three cannot proceed

parallelly; testing can be done by the tester and the coder can take one of this and then after completing it needs to do this.

And then after all of them complete the testing, coding of A and B then the actual testing activity can be done, but then in the PERT diagram we do not show this constraint that, are resources available to carry out all these in parallel and so on. These are done later in the project planning, where the project manager use as a Gantt chart to do the project scheduling where these allocation of resources to the tasks are done.

And also the Gantt chart is used for project monitoring and control, where the schedule needs to be refined if there are any variations that are observed. For example, the design instead of 15 it completed by 13, what is the impact on the other parts of the project? We need to redo the schedule. What if the test plan instead of 5 it took 10 weeks, what will be the impact on the schedule? So, these are all better than using Gantt chart. The resource allocation monitoring and control done using Gantt chart and that will discuss later in this lecture series.

This kind of network as we said was done for development of the Polaris missile 1950s. And here each node we write the name of the task; you may use just a label instead of the task name. If there are too many tasks it becomes unwieldy to write the task name like this. We might use A, B, C, D etcetera and then those are the labels and then we will have task names for those labels stored in a table.

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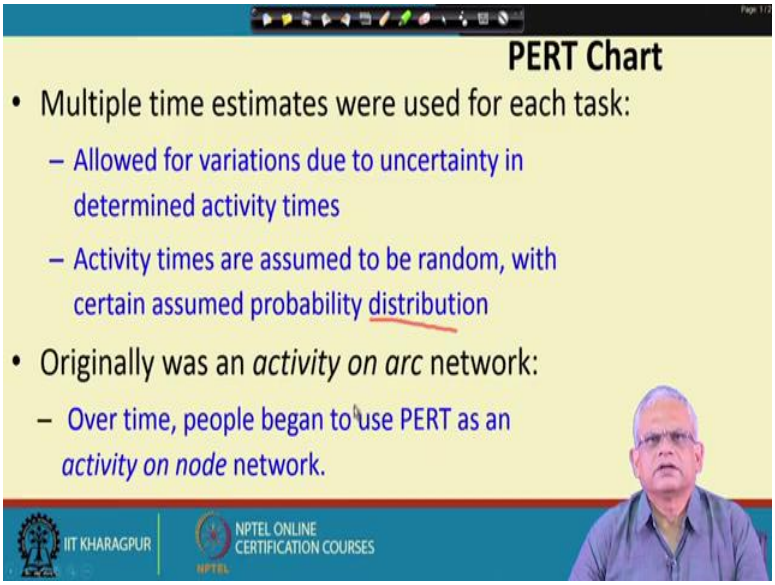

The PERT chart, the tasks have uncertainty and therefore, multiple time estimates were used for each task. This allowed for variation in the activity times that is completion times of the task, the preceding task and so on. This can be accommodated represented in the PERT chart and statistical inferencing can be drawn from the PERT chart. The activity times are assumed to be random with some probability distribution.

As you are mentioning these techniques have been used over time, over a large number of years and this technique has developed from its start 1950, that is about 70 – 80 years back and originally it was an activity on arc network. But, then as I was mentioning that now it is mostly used as a activity-on-node network.

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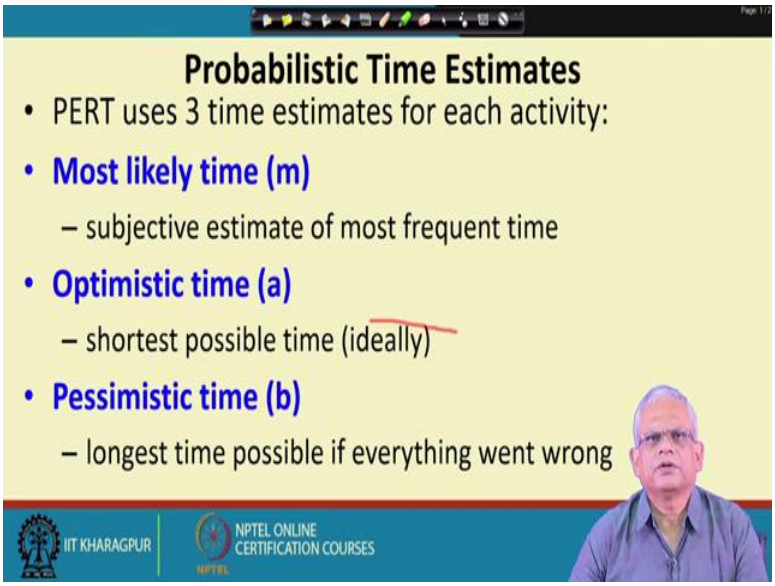

For each activity in the PERT, 3 estimates are provided: one is the most likely time. This is a estimate of the project manager that what is the time that it is most likely to take. The optimistic time – this is the shortest possible time which the task the activity can take. The pessimistic time is the longest possible time if things did not work out; the developer became unwell, the resources broke down that is the computer hardware had problems and so on.

The project manager provides 3 times for activity most likely optimistic and pessimistic and each activity is provided 3 time estimates. And based on that a probabilistic inference on the project duration is made that what is the likely time that the project will take to complete what is the optimistic and what is the pessimistic time.

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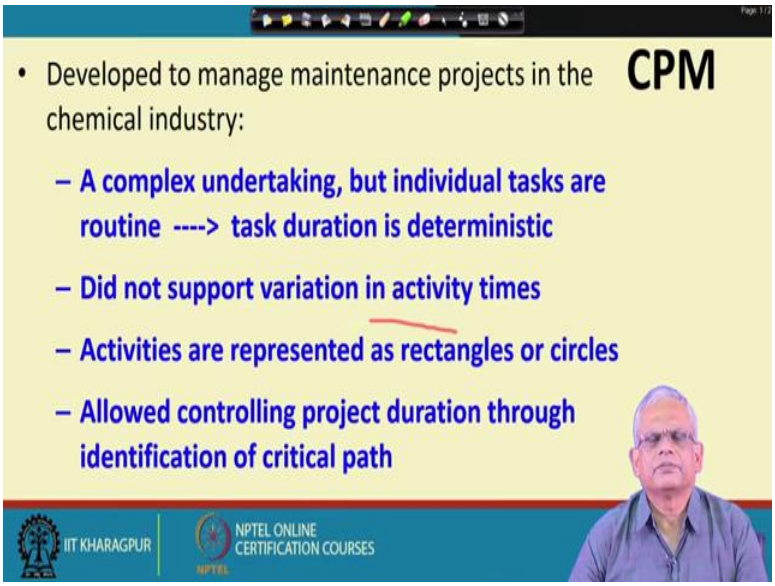

The CPM, the Critical Path Method, as you are mentioning it was developed for managing maintenance project in chemical industry on DuPont. Here the maintenance is a complex activity. There are many tasks here, but the tasks are routine and therefore, the task durations are deterministic. And therefore, CPM did not required to use variation in the activity times and the times there are estimated for activities or constant times.

The activities are represented as rectangles or circles. If you look at books and so on, as I was mentioning that these techniques have been developed over large number of years and therefore, there is a evolution of these techniques. And, we will find that some places they use circle, some places use rectangles and so on, but in our lecture will use rectangles for the PERT CPM technique.

And one important thing of CPM that it allowed to compute the critical path and therefore, the critical path indicates the project duration and it give a method how to use the task network diagram to compute the critical path. But, then nowadays we also do the same thing with PERT. We can have the critical paths there and we can compute them from the diagram.

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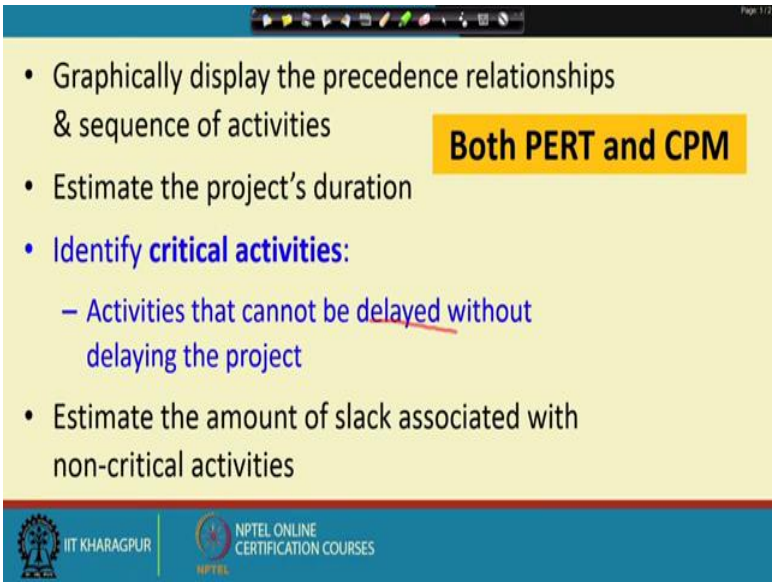

Over the years, even though this were two very separate techniques for project scheduling, but over the years both the techniques have merged into a single technique. Both the techniques they are graphical techniques, they are similar, precedence relations, the sequence of activities these are all represented here. Both are used to estimate the projects duration; one uses statistical, other use deterministic.

And using both now we are able to identify the critical activities. The critical activities are the ones which are on the critical path on the network and these are the activities which cannot be delayed without delaying the project. If a critical activity gets delayed the project duration will be affected and therefore, the project manager has to be extra careful about the critical activities.

And not only that it indicates to the project manager the amount of slack associated with non-critical activities. Some activities even though those are not critical activities, are very a small slack associated that is they can be delayed only a little bit without affecting the project schedule. On the other hand, there may be some activities which have large slack or float time that is they can significantly vary the completion time. They may get delayed and so on, without really affecting the project completion time.

And this is a very important hint to the project manager that he can redistribute the resources so that he may take out a man power from a task which has lot of slack associated with it. And put the man power on a critical tasks, so that the critical tasks get

completed and the one which is having slack even if it gets delayed little bit because is withdrawn the manpower here and redeployed on the critical activity. But, the overall the project will be managed well, it will complete on time.

And therefore, for every project manager it is very important to know what are the critical paths, what are the activities on the critical path which cannot get delayed without affecting the project completion time and what are the non critical paths and which tasks have the maximum slack so that he can have little leeway in redistributing the resources if required so that the critical activities at least proceed without delay and therefore, we can meet the project schedule.

We are almost at the end of this lecture and in the next lecture, we will discuss how to use the PERT, CPM network to compute for each activity the start times, end times, the slack or the float time that is available and so on and also identify the critical paths.

Thank you.

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Lecture - 28
Project Scheduling Using PERT/CPM (Contd.)

Welcome to this lecture.

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In the last lecture, we had discussed about PERT and CPM, some very basic ideas. We discussed about PERT and CPM, their origin, the main components of the PERT CPM diagram. We said that these were two very independent techniques used for very different purposes. PERT was used in development of the Polaris missile, where there were lot of uncertainty regarding task completion times is a challenging project; whereas CPM was used by dupont for its chemical plant maintenance activities; which were more or less routine activities even though the activities there are many activities and it was a complex set of activities, but the activity durations where deterministic.

And the critical path method it allowed to compute the critical path, it gave a technique using the task network; how to compute the critical path. But then over the years, both the techniques have evolved; both the techniques are equally capable. Many of the things that were there on PERT is observed in CPM and vice versa and now they have been merged into a single technique. And we will discuss about how these techniques are

being used in project scheduling because this is a important tool for the project manager. The project manager using the PERT CPM can determine the completion time of the project; for each activity can compute the slack time the critical path and so on and that is the focus of this lecture.

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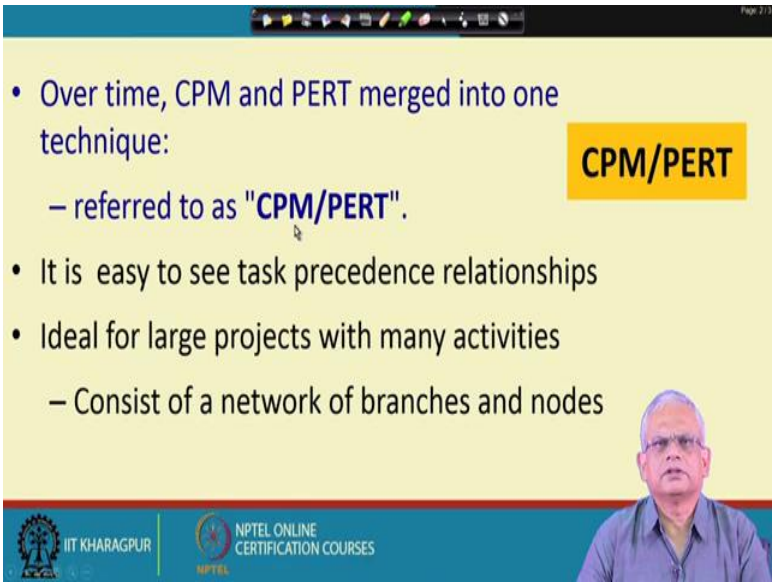

As you are discussing, though CPM and PERT were two different techniques; now they are merged into a single technique and usually in the books and literature this is referred to as CPM PERT.

This technique is very useful to the project manager can looking at the diagram easily identify which tasks are precedence; that is which are the tasks need to complete for the task to start. These techniques over the years becomes sophisticated, many tools that are available. When the projects are complicated there are many activities with various types of dependencies then these techniques become useful. If the activity has only, if the project is very simple only 3, 4 activities sequential activities, there is no other precedence and so on. .

These techniques may become a over kill, but then the typical IT projects are complicated. There are many activities there hundreds of activates for large projects and dozens of activities 2, 3 dozens of activities for even moderate size projects which are obtained from the work break down structure. The project manager uses a tool, we will discuss about few tools which you can easily use based on the concepts discussed here.

Those tools allow to draw the diagram, the task network diagram and also it automatically computes the critical path; it shows the critical activities, the slack times and so on. But then in this lecture, we will discuss about the main concepts here so that we can effortlessly use the tools.

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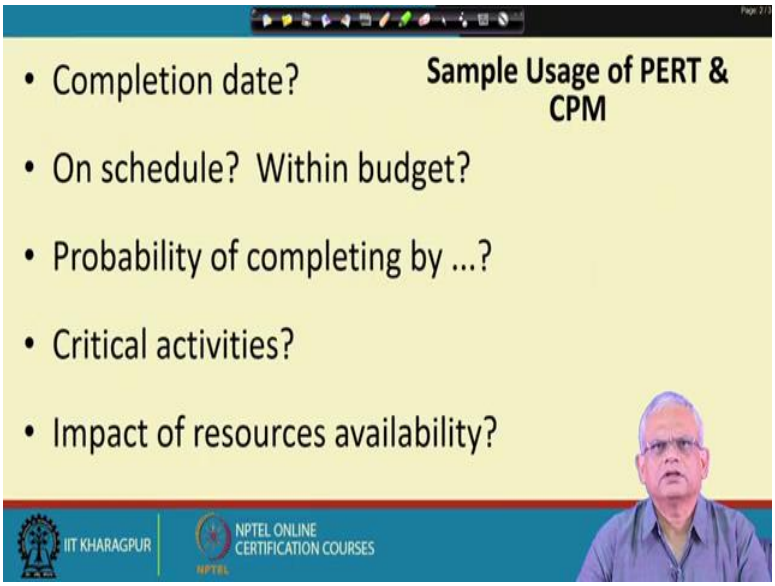

Once we develop the PERT CPM network, we can be able to answer that what will be the project completion date. At any time we can also look at the PERT CPM and determine whether it is on schedule within budget. And because these are statistical parameters for the activities can be given; we can as the project manager determine the probability of completing the project by certain date. What are the critical activities? What is the impact of resource unavailability or availability; that is if some developer becomes unavailable what will be the impact and so on.

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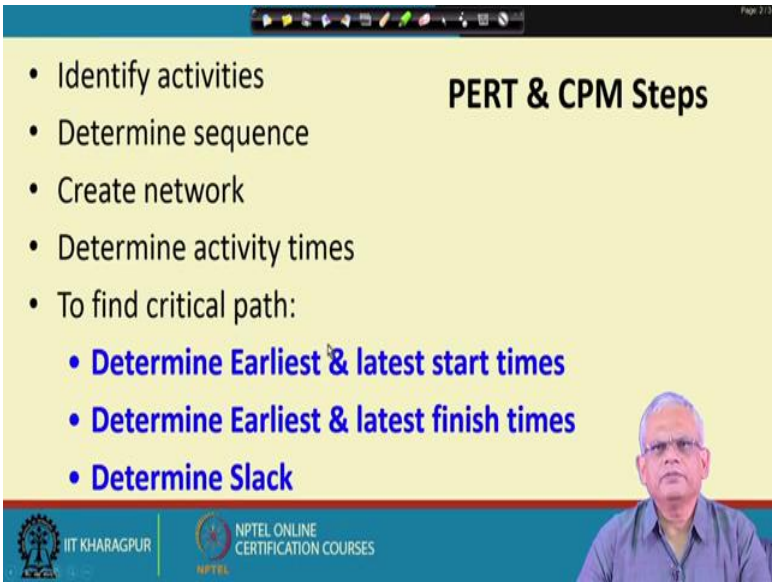

The PERT CPM are used in the early phases of the project planning; these are more generic in nature, we do not really represent the exact start dates. We do not consider intervening holidays, off days, allocation of resources to the tasks and so on.

We just develop a very high level plan which is later low level plan is developed by using a Gantt chart. Here for using the PERT CPM, we should have identified the activities using the work break down structure. We should have determined the sequence in which these activities and based on these two; that is what are the activities and their precedence relations, we can create the task network.

And we need to determine the activity times; these are basically estimates and then we represent these and then our diagram the PERT CPM diagram is ready for use. And based on this diagram we can determine the earliest and latest start times, earliest and latest finish times and also determine the slack or the float time.

We first try to understand that the earliest and latest start times; even though these are intuitive, but then with an example we will try to make out what we mean by earliest and latest start times, earliest and latest finish times and the slack.

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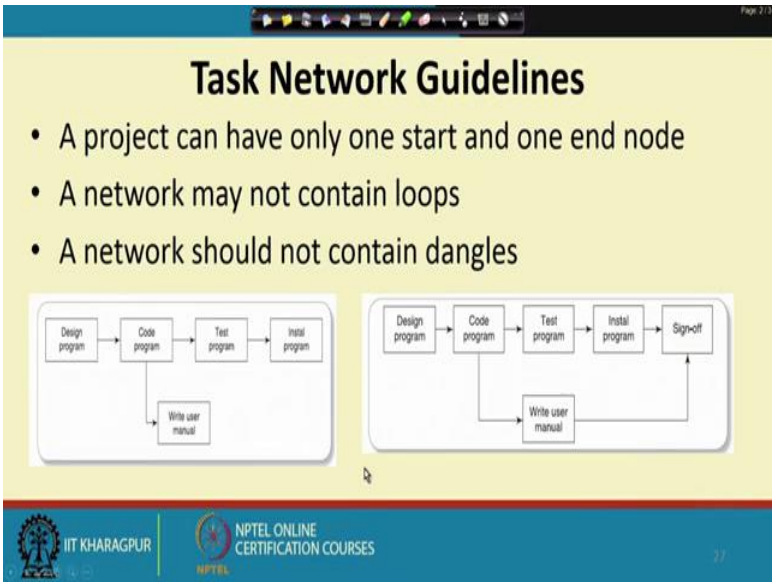

But before that we will just discuss some guidelines of developing the task network. As we have been saying that the project scheduling techniques, using the PERT CPM can be easily applied when we have single start and single end node; if we have multiple start and multiple end node it becomes complicated.

And therefore, we will require that we have one single node start node and one single end node; if necessary we will create dummy nodes, dummy start and end nodes. The other rule that we must enforce is that once you draw the diagram there should not be any loops in the diagram. For example, an activity is dependent on one activity let us say A is dependent on B and again B is dependent on A; these things are not very common but then if we have such situation we need to avoid because this will make very difficult to apply the project scheduling techniques that we will discuss subsequently.

We must ensure that there are no loops in the diagram and also there should not be any dangles. In this diagram, there is a dangling activity here, but then if there is no other activity to be done; we must have end node here and then have this edge connected so that there is no dangling nodes.

If there are dangling nodes then it becomes very difficult to apply the project scheduling techniques. So, these 3 points we must ensure before applying the project scheduling techniques. The first is that the network should have single start and single end node, there should not be any loops in a normal project; loops do not exist.

Of course, there can be iterations like let us say there is a coding and testing; testing starts after coding, but then there are bugs may be detected. But and then the coding has to be done again and that if we try to represent will become a loop. And once there are loops in the diagram we cannot apply the project scheduling techniques, it become very difficult.

And therefore, these we should avoid by having coding followed by testing and then we can have bug correction, a separate activity not coding and that way we can break the loops. And also there should not be any dangling activity like this and if that is the situation that we have some problem description; then we have we have to have a arrow connecting it to the end node.

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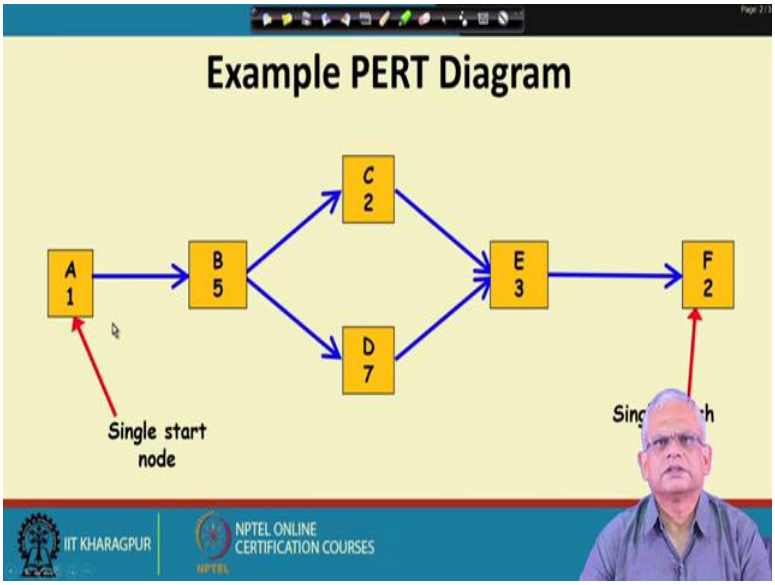

This is one of the simple PERT diagrams, there is a single start node, single end node and each node is labeled with the name of the activity. And there is a duration with each activity and here in this example we are using week 1 week, 5 week, 2 week as a duration, but there can be you can even use this for days or months and so on. And a critical path here; in this project if we look at this that one of the path here from start to end is A, B, C, E, F, another path is A, B, D, E, F.

On this path, the time taken is 6 plus 2; 8 plus 3, 11 plus 2; 13, but on this path we have 1 plus 5, 6 plus 7; 13; 3; 16 to 18; so this is the longest path. From this diagram, we can say that the project duration is 18 weeks because that is a longest path here. And this one

A, B, D, E, F is the critical path because any delay to any of these activities on the critical path will delay the project. Whereas, the activity that are not in the critical path for example, C can get delayed little bit without effecting the final project schedule.

So, that is the intuitive idea and we will discuss the project scheduling techniques using which we will compute the earliest start, earliest finish, latest start, latest finish; the slack times the critical path. But for this we really do not need such a simple diagram we can easily compute even without using the technique that we are going to discuss. It becomes easy that to see here that there are two paths here and one path is the critical path takes 18 weeks to complete. And the other is non critical path and that takes 15 weeks to complete, but A and B are critical activities because they are on the critical path.

Similarly, E and F are also critical activities, but C is not a critical activity is a non critical activity. And the project manager has some laxity here or slack time or float time which you can exploit advantage so that the project gets completed on time. He may redistribute resources, take out men power allotted to C activity and deploy in another activity which is getting delayed, so that the project overall project completes on time.

So, there is a single start node, the single end node and this is the critical path. And the CASE tools the project management tools that are available many are open source easily downloadable tools, where you can draw such diagrams and that would indicate the critical path using a red color or something.

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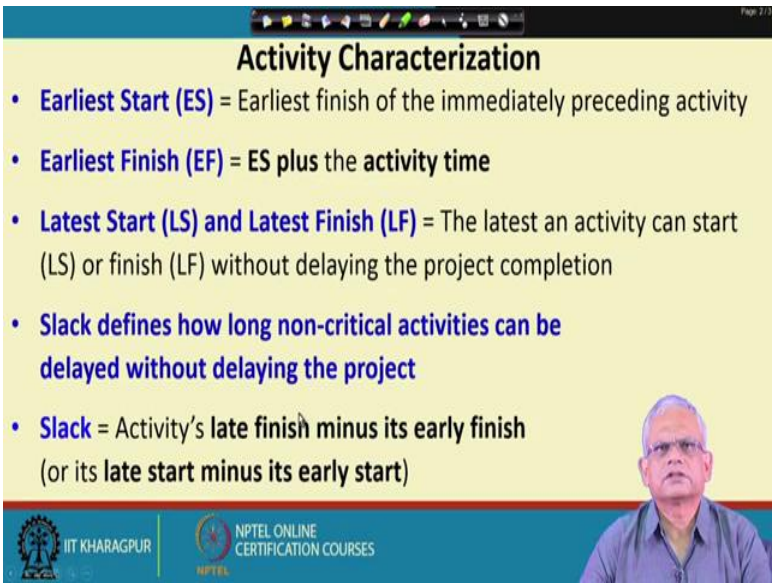

Now, let us see the activity how they are characterized on a PERT CPM chart. One is the earliest start, the earliest start that is the earliest time that a task or a activity can start is the earliest finish time of the preceding activity; only after the preceding activity completes the task and start. Therefore, the earliest start of an activity is the earliest finish of the preceding activity.

Similarly, the earliest finish of an activity is the earliest start time plus the activity time; very intuitive that earliest start is the time by which the preceding activity completes and then if everything goes well; it will start in earliest start plus the total duration of the activity which we call as a earliest finish time.

Similarly, we can define the latest start and latest finish; the slack or the float time it defines how long a non critical task can be delayed without effecting the project. And from this definition all the critical activities have zero slack time the slack is the activities latest finish time minus its earliest finish time; we will discuss these with help of an example that will make it clear.

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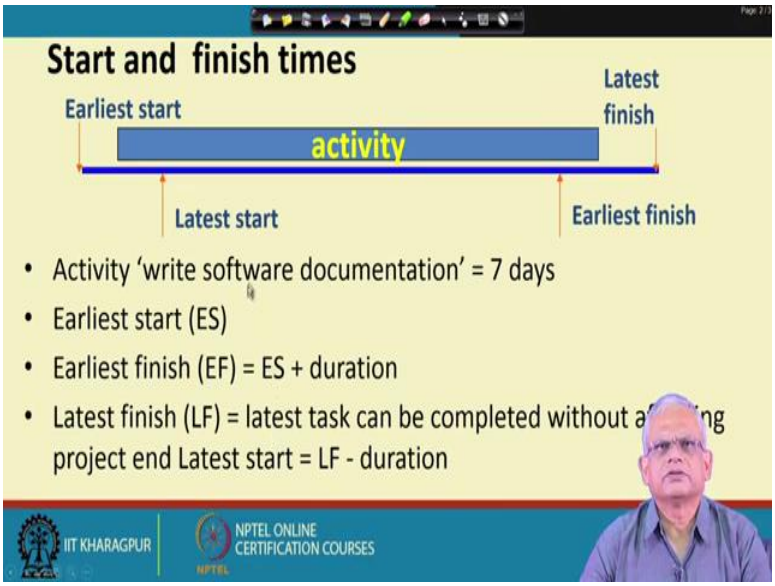

Let us look at this diagram, here the preceding activity can complete by this time and here in the PERT chart; we do not give calendar dates we just give numbers here 10, 20 and so on; which later during the resource allocation, we give the exact dates there here we do not give dates we just give numbers.

The earliest start may be a number which is a date and then this is the activity time, but then the latest finish is here. And then this plus this is the slack time or the earliest finish minus earliest start minus activity that gives the slack time. The earliest finish time is the earliest start time plus the duration.

$$\text{Earliest finish (EF)} = \text{ES} + \text{duration}$$

So, that is earliest start time plus the activity time that gives the earliest finish time;

$$\text{Latest start} = \text{LF} - \text{duration}$$

the latest finish time minus the activity time that gives the latest start time.

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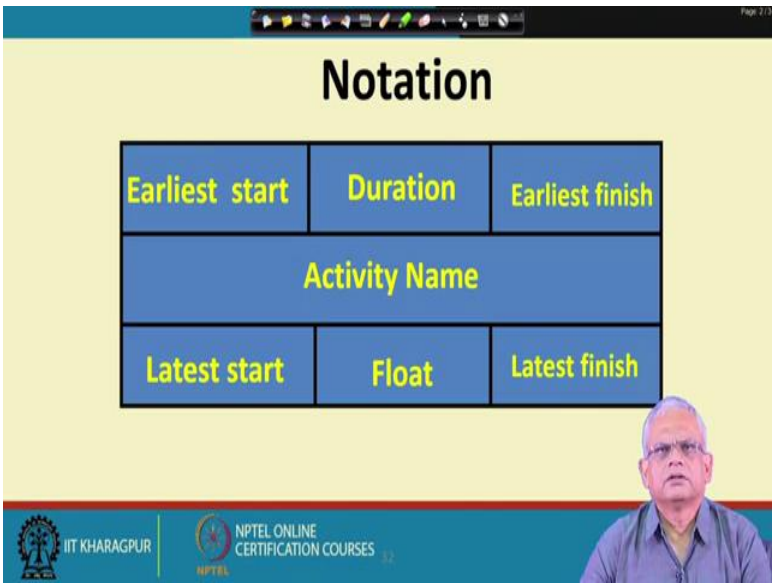

For each activity, we use this notation each node in the task network will have such a notation; where we have the activity name or the label of the activity, what is the duration earliest start, earliest finish, latest start, latest finish and what is the float time available.

This has lot of information here for every activity for the project manager; you can know that whether it is a critical activity or not. Whether it is what is the amount of float that is available, what is the duration earliest start earliest finish and so on; so there is lot of information on every node of the diagram.

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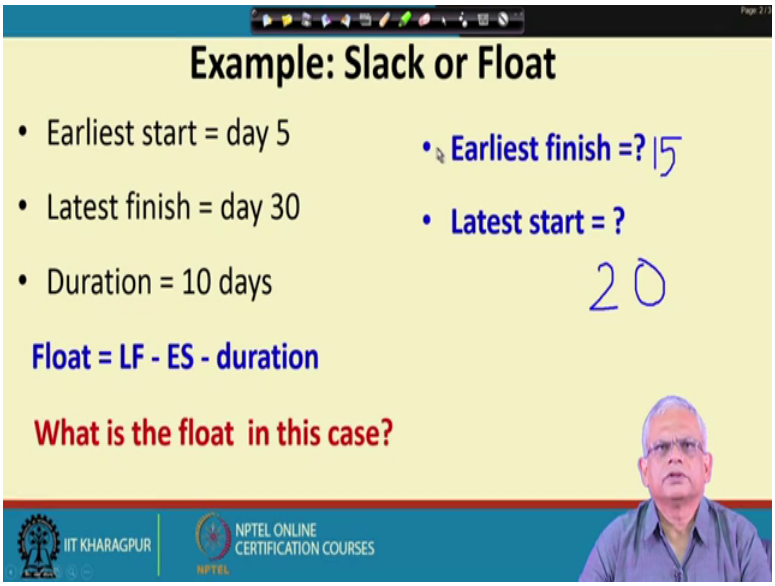

Now, let us look at an example; for a certain activity the earliest start is day 5 and the latest finish is day 30, the duration is 10 days. And we want to know; what is the float, what is the earliest finish and what is the latest start?

The earliest start is 5 that is the preceding activity can complete by day 5 and then only this activity can start and the duration is 10 days. And therefore, the earliest finish will be 15 because the earliest start was 5 and it takes 10 days the earliest it can finish is by 15. The latest finish is 10 days; sorry on day 30 the latest finish is on day 30 and the duration is 10 days and what is the latest start time? For the task to finish at most on 30 day, we must start it by 20 day 20 because it will take 10 days from then and it will complete by day 30.

So, the latest start time is the latest finish minus duration which is 20; one way to compute this float or the slack time is to have the latest start minus earliest finish which is 5 days. So, we can compute the float in this case.

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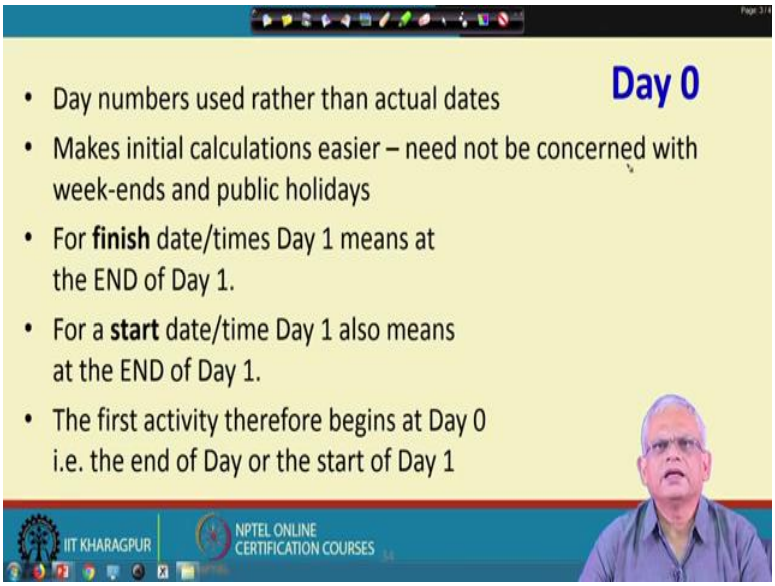

Now, one thing need to mention here that what is the day 0 for a project. We said that in the PERT we do not use really calendar dates, we do not consider which are the intervening holidays what exact date it completes here we just write days. It starts on day 0 and ends on some day; these are numbers rather than the actual dates.

This helps us in a high level scheduling where we ignore what are the intervening weekends public holidays and so on. In a lower level scheduling which we will do using a Gantt chart will take those into consideration. And here the notation is that the finish on day 1 means end of day 1 and also the start of day; the start date is day 1; that means, end of day 1 that is the notation here that if a project finish is on day 10, then it is end of day 10.

If a project start is on day 10 that is at the end of day 10 and from here; we can imagine that the project start date is day 0 that is end of day 0 or that is the start of day 1. So, that is the notation we will use the day 0 is end of day 0; start of day 1 and for every other day also we will use the finish date is end of that date and the start date is also end of that date.

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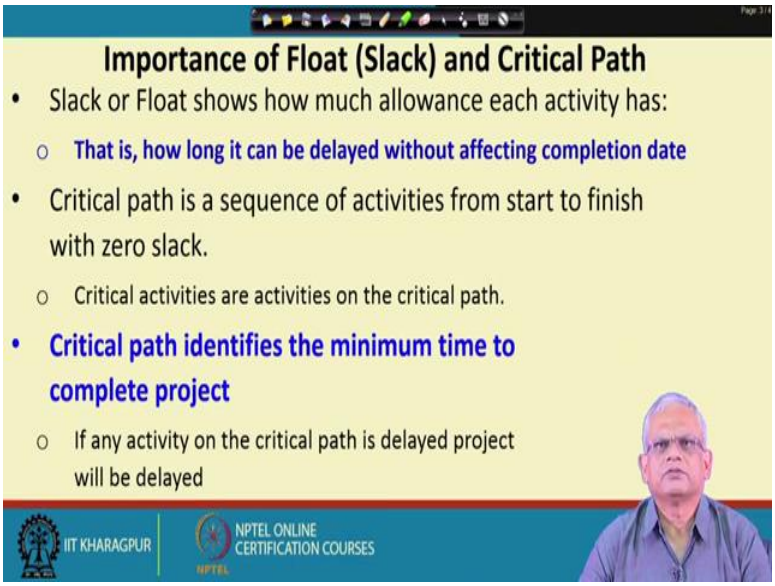

Now, let us look at the importance of the float or slack and critical path for a project manager. The slack is how much flexibility or float that activity has and this implies how long it can get delayed without affecting the project completion date. It is a very important thing for the project manager to know helps in redistributing the resources.

But then the activities that are there on the critical path are the critical activities and they have zero slack; that is the project manager has to be extremely careful with the critical activities; cannot really take out any resource from a critical path because it will get delayed. But then if for some reason the activity on a critical path is getting delayed that is a critical activity is getting delayed. And the project manager has to bring in man power from non critical activities and see that the critical activities are not getting delayed for the overall project schedule to be met.

Another thing that we must note is that the critical path identifies the minimum time to complete the project. If we can identify the critical path on a very complex task network, then we can sum all the activities there activity times and that will give us the minimum time by is the project will get completed or the project completion time; that is the best case project completion time. And the project manager will meet that time will try to meet that time that is the sum of the activity times and the critical path.

We are almost at the end of this lecture, we will take some examples and then based on the concepts developed and we will discuss a technique by which we can use the PERT CPM to compute the various parameters per project. We will stop at this point.

Thank you.

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Lecture - 29
Computation of Project Characteristics Using PERT/CPM

Welcome to this lecture. In the last lecture we were discussing about PERT CPM, we had some general ideas regarding PERT CPM. We looked at the task network by which we represent the different tasks or activities and the dependencies between them and then we distinguish between the PERT and CPM. And we said that CPM we use deterministic task times whereas, in PERT we use probabilistic task times because, tasks are uncertain, the completion times are uncertain.

And in CPM we use deterministic times and therefore, we use deterministic conclusions regarding the task parameters. Naturally, CPM is much more simpler then the PERT with statistical times. We will first look at determining various project characteristics using PERT CPM. And then we look at determining project characteristics using PERT where we use statistical times; let us get started with this.

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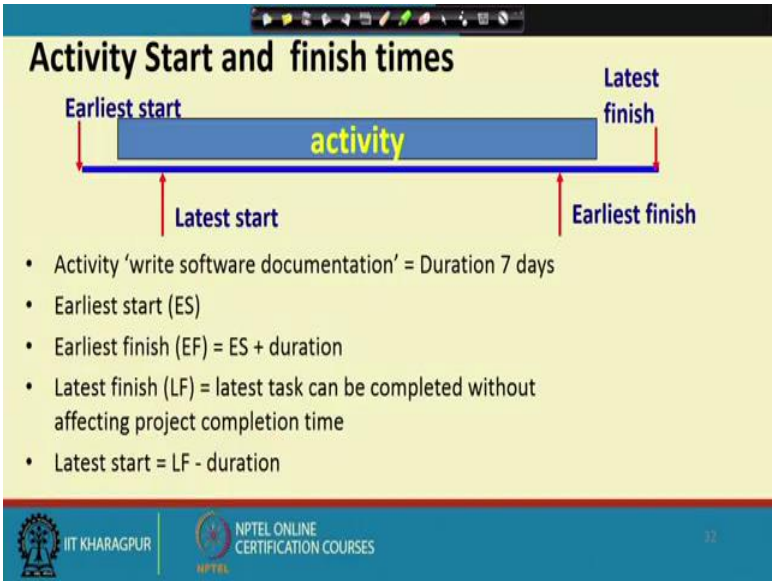

If you remember in the last lecture we were saying that the project manager during the project scheduling would like to determine various project characters characteristics and then manage the project based on these characteristics. One of the important

characteristics is to identify various activities the tasks and the dependencies among them and also to identify several attributes of the activities. One attribute is of course, the duration of the activity. So, the one that you see in the blue here this is the duration.

For example, the activity may be to 'write software documentation' and the project manager estimates the duration to be let us say 7 days. But, then once the task network is developed that is the PERT CPM network then it becomes possible to use the PERT CPM to identify other characteristics of the activity. For example, earliest start time this is you can see here the arrow mark here red arrow mark here, this is the earliest start time, this is the earliest time at which the activity can start. Similarly, the earliest finish time this is the earliest finish time based on the activity can finish.

These are important characteristics because, the project manager can deploy resources, can start other activities. And another attribute of activity that will be determined based on the PERT CPM is the latest start time. This is the time at which the activity can be started without hampering the project completion time. If the activity starts anything later than this arrow that you can see here this is the time line, the blue line that you see here this is the time line. And, if the activity started anywhere later than the latest start time then the project is going to be delayed. And latest finish time is the time by which the activity must end, if it does not end by this then the project is going to be delayed.

Naturally, the latest finish minus the latest start time is equal to the activity duration, the activity duration is shown in the blue here. And, latest finish to latest start time is the activity duration, similarly earliest start time to the earliest finish time is the activity duration. We will use the PERT CPM to compute these project characteristics and based on these project characteristics, we will also be computing the slack time. The slack time is the time that the task has for which the project manager has some liberty, if the slack is 0 and the project the earliest start time and the latest start time coincide. So, there is no flexibility for the project manager to differ the task for sometime, but then here you can see that there is a slack time.

There is the activity can start anywhere between the earliest start to the latest start and this we call as the slack time. The latest start minus the earliest start which is also equal to the latest finish minus the earliest finish, this is called as the slack time. This is the time that is available to the manager to monitor and control the project; if one activity is

getting delayed you can defer this activity little bit and put resources on the other activity.

But cannot delay this too much beyond the latest start, have to start by the latest start time. Let us look at how to determine these project characteristics and later when we discuss about project monitoring and control, we will see that how this project attributes are actually used by the project manager for monitoring and control purposes.

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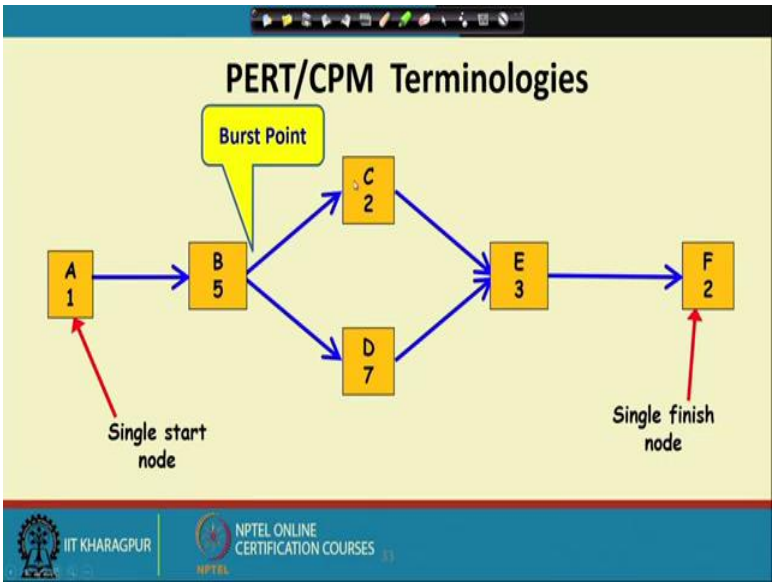

Before we get started, let us just look at few PERT CPM terminologies which we will be needing as we proceed with our discussion. We had seen the last lecture that a task network the PERT CPM is based on a task network and the tasks are here rectangles, the arrows are dependency. And we said that there has to be a single start node otherwise the methods that we give for computation of project characteristics would become very complicated.

We will need that there is a single start node, if there are multiple start nodes we will put a dummy node and make that as a single start node. Similarly, we need a single finish node, if we find that there are multiple finish nodes we will put a dummy node and we will make it a single finish node. And then we have this burst point where a task or a activity has multiple successors. Once this activity completes this two C and D can start, once B completes C and D can start and that we call as dependency and here there is a burst point multiple tasks are dependent on B, C and D both are dependent on B.

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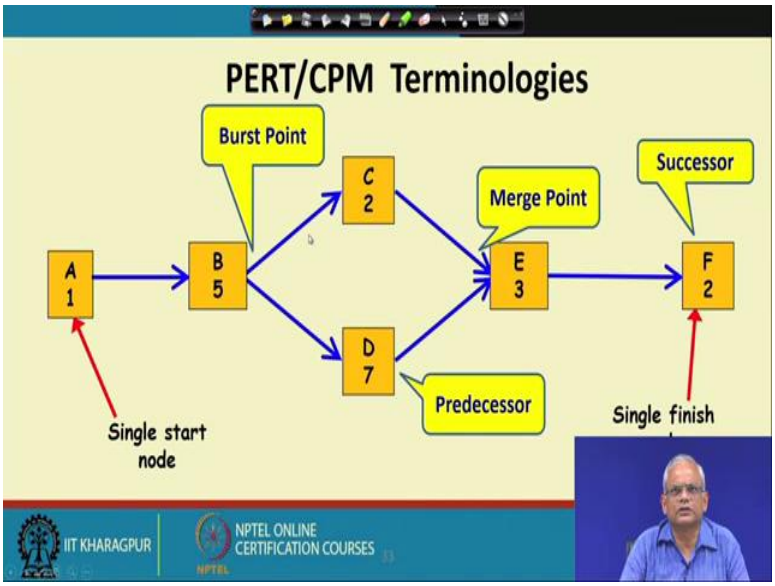

We have merge point where a task E has multiple predecessors; that means, both C and D must complete before E can start. It may so happen that C completes earlier then D, but E cannot start until D also completes both the predecessors of E must complete before E can start and that we call as a merge point. And, D is a predecessor of E, C is also a predecessor of E and F is a successor of E. These are some of the simple terminologies that will use as we discuss about the PERT CPM technique to determine project characteristics.

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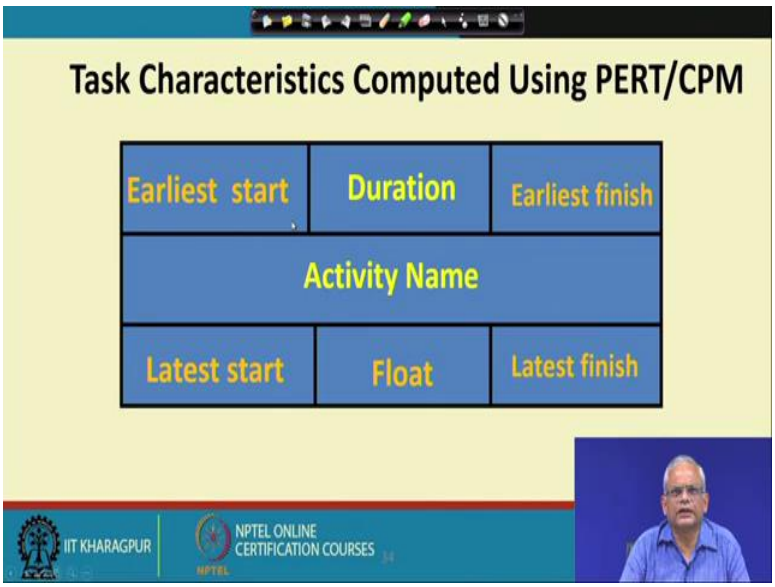

We had discussed in the last lecture that we will use this kind of notation, for a task will represent a task in a form of a rectangle. The task or activity name will be written in the middle and we would have determined the duration of the activity. Typically in weeks, 1 week, 2 week something like that or it can be in days, but rarely it is in hours or months. It is typically in days or weeks and the one that you are seeing in the red one, the earliest start, earliest finish, latest start, latest finish and the float time these will be computing using the PERT CPM technique.

We already seen the definition of the earliest start, earliest finish, latest start this is the point of time. If the activity starts anywhere beyond or after latest start then the project is going to be delayed. Earliest start at most or at the earliest the activity can start at this point and the float is also computed by our PERT CPM methodology. But, then the float is the latest finish minus latest start which is also equal to the earliest finish minus earliest start both give the float. This is the flexibility that the project manager has we also call it the slack, slack or float.

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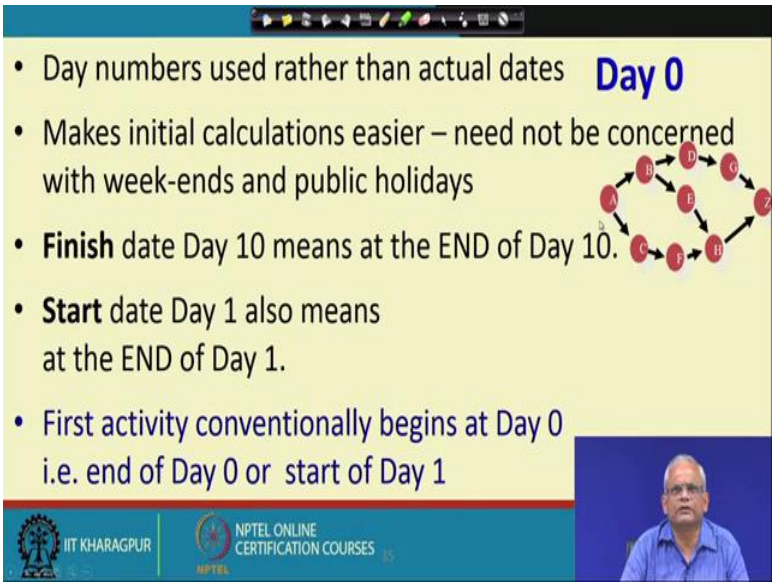

Another point we need to mention, here we just write day 1, 2, 3, 4 etcetera rather than giving calendar dates. Please remember that the project scheduling using PERT and CPM is a higher level scheduling, where we do a high level scheduling in terms of days. Starting from day 0, how many days it takes? And later the project manager does a more detailed scheduling using the Gantt chart where, we use the calendar time. The idea is

that at a very high level scheduling we make a simpler plan, where we do not take into consider week-ends, public holidays and so on.

We do not use a calendar time here, this is a high level simple schedule that we prepare and later we consider various constraints. What are the holidays and then we consider the resources whether a activity can be started based on the resource availability and that we will do using a Gantt chart. But, the initial scheduling is done using a PERT CPM chart. We will have this kind of a network here task network, the tasks and the dependencies and as I was saying that these are very old techniques developed over a century nearly a century.

And therefore, we have a divergence of notations. Sometimes we have circles, but then we said that we will use rectangles for activity name and then we will have various attributes that will also be indicated along side this for our computation of the task characteristics.

The day 0 let us try to understand what exactly is meant by day 0 or day 10 or something. If we say that the finish date is day 10; that means, it is the end of day 10 which is also a normal use in our sentences, we say that we will do something by day 10 means end of day 10. But, then the start date of day 1 means end of day 1, but then that is not really what we say. We always start at the beginning of day 1, but for that purpose we will say that we will start in day 0 which means that we will start at the end of day 0 which is also equal to the start of day 1.

Let me just repeat here that we always say that the project starts in day 0 and the implication of that is end of day 0 or start up day 1. If we write day 1 here start of the project activity is day 1 for the activity A, then it will mean end of day 1, but normally we do not do that. We start at the beginning of day 1 and for that we will use the notation day 0.

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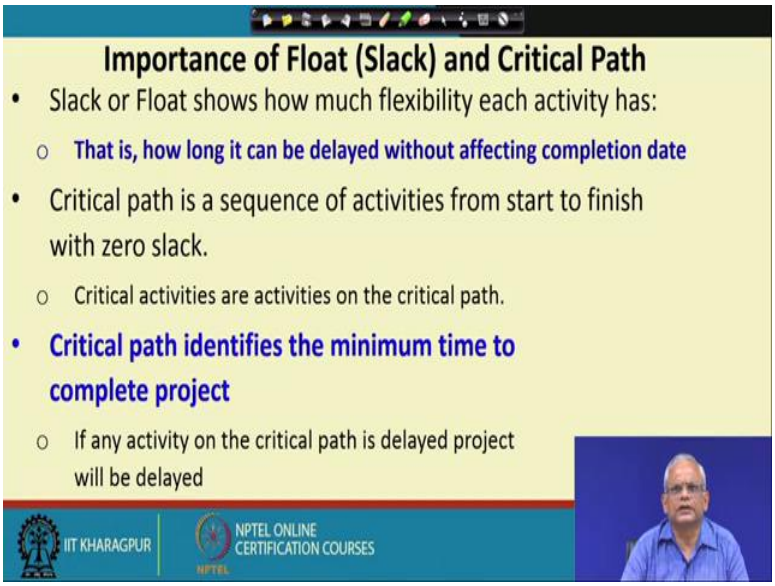

One of the important project characteristic, that is very useful to the project manager is the float time or the slack time available for various activities and also the critical path. The project manager looks at each task or activity and finds out how much flexibility he has, in particular you can find how much he can delay the activity the starting of the activity without really affecting the project completion date. This is very important for the project manager during monitoring and control, because to affectively control a project may find that some activity is getting delayed.

And, then can find out the activities which have enough slack and then may be take some man power from the activity which has enough slack, put it on activity which is getting delayed so, that there is no overall project delay. Therefore, the PERT CPM provides very important results to the project manager which are the earliest starting time, earliest completion time, latest start time, latest completion time, the float or the slack time and the critical path.

The critical path is the sequence of activities from the start task to the last task with zero slack. So, all the activities on a critical path will have zero slack, the project manager should be very careful about the critical tasks or the critical activities. That is the activities that are appearing on the critical path, the project manager gives special attention to the critical activities or the critical tasks because, any delay to the critical activities will delay the overall project. And therefore, the project manager typically

closely monitors the critical activities and at the eventuality or at the slightest indication that critical activity is getting delayed, the project manager takes corrective action.

For example: deploy a additional resources for the critical activity so, they do not get delayed. For this reason it is very important to identify the critical path and the critical activities. Not only that the critical path also identifies the minimum time to complete the project, that is if we sum all the critical activity durations that appear on a critical path we will get the project duration. Let me just repeat, if we sum the durations of all activities the critical activities appearing on a critical path then we get the project duration, that is the minimum time by which the project can be completed if the project manager monitors and controls the project well.

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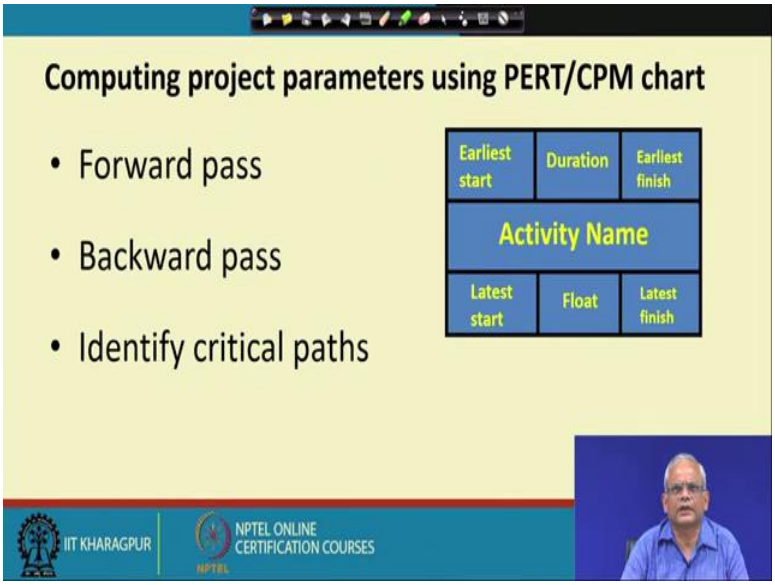

Now, let us looks at the technique that we will be using the PERT CPM using the PERT CPM we will compute the earliest start time, the earliest finish time, the latest start time, the latest finish time and the float. To start with will have the task network with activity names and their duration. All the activities would have been identified, the durations and the dependencies between the activities will be present. And we will now discuss a technique by which we will identify the other project attributes like earliest start, earliest finish, latest start, latest finish and the float.

Here we will have three steps; one we call as the forward pass where we look at the activities are the nodes appearing on the network from left to right and compute the

earliest start and the earliest finish. And then we have a backward pass start from the last node or the last activity and then slowly proceed towards the first activity. And we compute the latest start, latest finish for every activity and also we compute the float or the slack time which is the difference between the latest finish and the latest start time.

And, after having done that we identify the critical paths that is the path on which all the activities have zero slack, there may be more than one critical path. But, we mark all the critical paths typically on red color so, that the project manager gives attention to the critical path, the critical activities.

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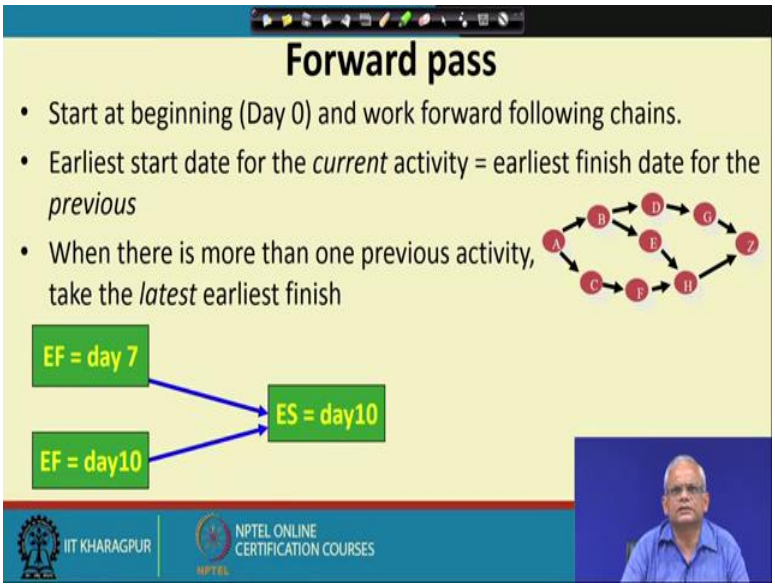

Now, let us look at the forward pass, this is the first activity using which we will compute some of the attributes. In our task network the first task will always be starting at day 0, that is end of day 0 or day 1 and then we will work forward following the chain. We will assign the earliest start date to an activity which is equal to the earliest finish date for the previous activity. We will take an example and explain how we do it, we will take a task from left to right and we will set its earliest start date which is equal to the earliest finish date for the previous. So, any task here, we look at the earliest finish date for the previous task and the earliest start date of this will become earliest finish date of the previous task.

But, then if there are more than one previous task, let us consider the example of H which has two previous task E and F then we will take the latest finish time of E and F.

The implication of this is that H can start only after both the tasks complete, even if E starts earlier H cannot start. H will start only after E and F both complete and therefore, we have to consider the latest time between all the preceding tasks to set the start time for H. This is just an example, if certain task the earliest finish time is day 7 and another task where the earliest finish time is day 10 and both are the predecessors of this task, then this task cannot start until day 10.

So, this task will complete at the end of day 10 and this task will start at the end of day 10 or that is beginning of day 11. So, we have to consider the letter of the two preceding tasks and set the earliest start time to be that. So, forward pass is simple, let us take some example and then we will see how to use this.

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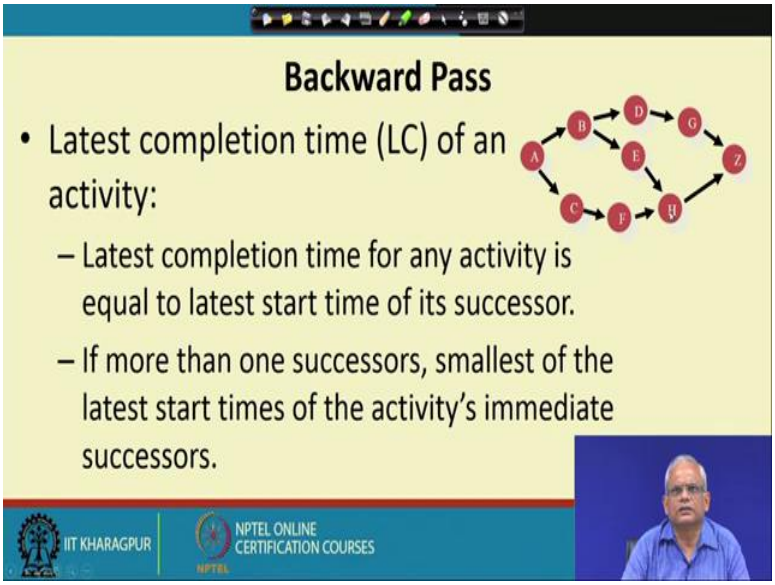

Once we have done the forward pass, we will have to do the backward pass. In backward pass we start from the last activity and here we compute the latest completion time of the activity which is equal to the latest start time of its successor. We now look at the successor of a network, successor of a task. So, if we look at B ok, let us look at a G then we look at what is the latest start time of H Z, latest start time of Z and then we set the latest completion time of G. The idea is that until if we find that Z needs to start on some date, G has to complete by that date.

So, using that concept we use the backward pass and then if there are more than one successor, the smallest of the latest start time of the activities immediate successor has to

be put. Let us look at the case of B, B has two successors D and E; now if D has certain latest start time and E has certain latest start time. Then we have to take the smallest of this two D and E and make that as the latest completion time of the B. Or, in other words B must complete before E can start let us say E has the smallest completion time let us say D has 10 and E has 7; then B has latest completion time must be 7 otherwise E cannot start again.

Here in the backward pass the rule is simple, we will take an example to consider the forward pass and backward pass. You can see the rule is extremely simple, we can do for very small or moderate sized the task networks. But, then as I was mentioning earlier nowadays there are many tools. We will discuss some of these tools, where the computation of the characteristics of the project are done automatically. The one that we do by backward pass, sorry forward pass and then followed by backward pass, identification of the critical path these are done only on press of a switch, press of a button.

So, but then it is important to understand the technique that they use to compute forward pass, backward pass and then identification of the critical path and for small task network. So, you do not even have to use a computer, we can just draw it by pen and paper and within a minute or two we can compute the task characteristics. We are almost at the end of this lecture; we will stop here and continue from this point in the next lecture.

Thank you.

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Lecture - 30
Computation of Project Characteristics Using PERT/CPM: Illustration

Welcome to this lecture. In the last lecture we had looked at the PERT CPM technique by which we compute various project characteristics, the activity characteristics, the critical path and so on. Now, we will quickly review what we discussed in a minute and then we will take some examples to illustrate how to use the methodology. Very simple we can do it in a minute or two for moderate sized networks. Now, let us look at let us review first quickly and then get started with the example.

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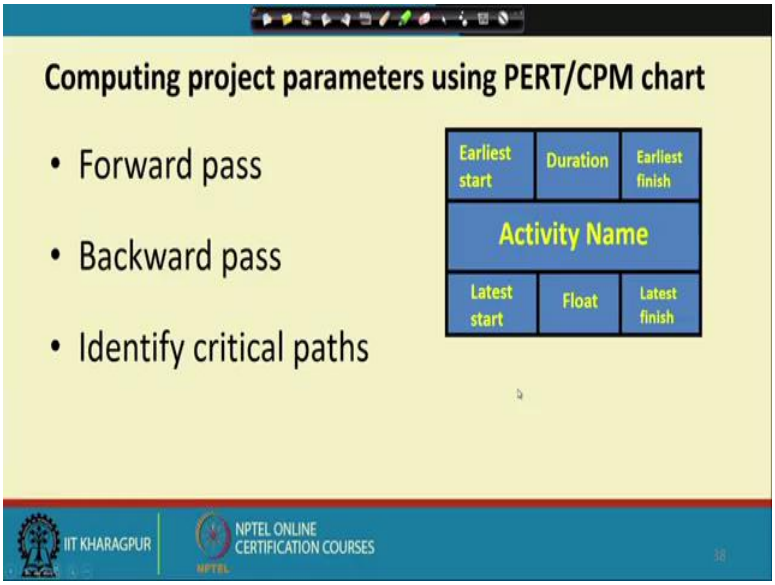

Just to recapitulate what we discussed in the last lecture, that we can compute the project parameters by using some simple methodology. We will first do a forward pass through the network; to start with we will have the activity names and the durations for all the activities and the precedence between the different activities in the form of a network. We will start with the leftmost activity or the start activity and then we will have a forward pass, where we will computer for every activity, the earliest start and the earliest finish.

In the next step we will do a backward pass, where we will start with the last activity and we will have the latest finish and the latest start computed. And, then we will compute the float which is the difference between the latest finish and the latest start. And, having done that we will identify the critical paths that is we will find out all the activities which have 0 float time. And, we will find a path through the network for which the float time is 0 and we will call it as the critical path we will mark it on red, these are of special importance to the project manager.

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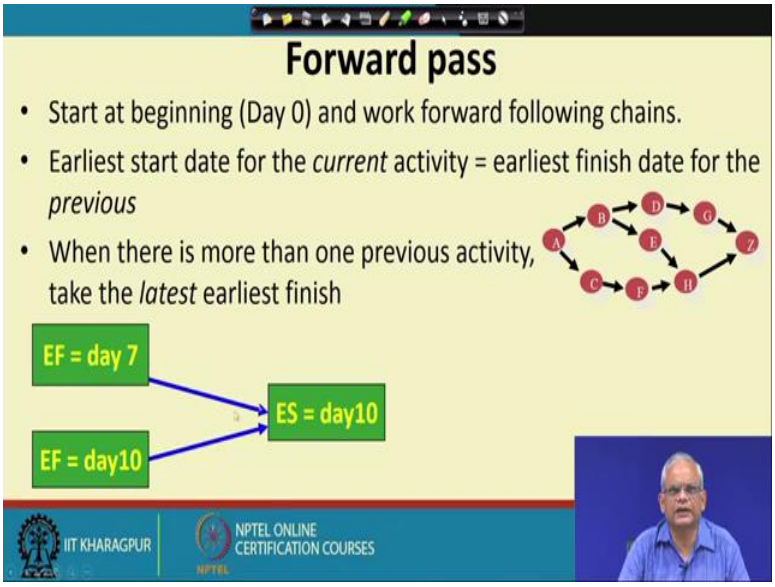

The forward pass we start from the beginning or the start task which is started day 0 and then we work forward. We will set the start date which is we will set it as the earliest finish date for the previous task. And, if there is more than one previous activity, we will take the latest finish time. There are more than one previous task and for the earliest start is the latest of the later of these two we will take 10 here.

So, the earliest start we will set as earliest finish and then for the task we once we find the earliest start, we will add the duration to get the earliest finish for that particular task.

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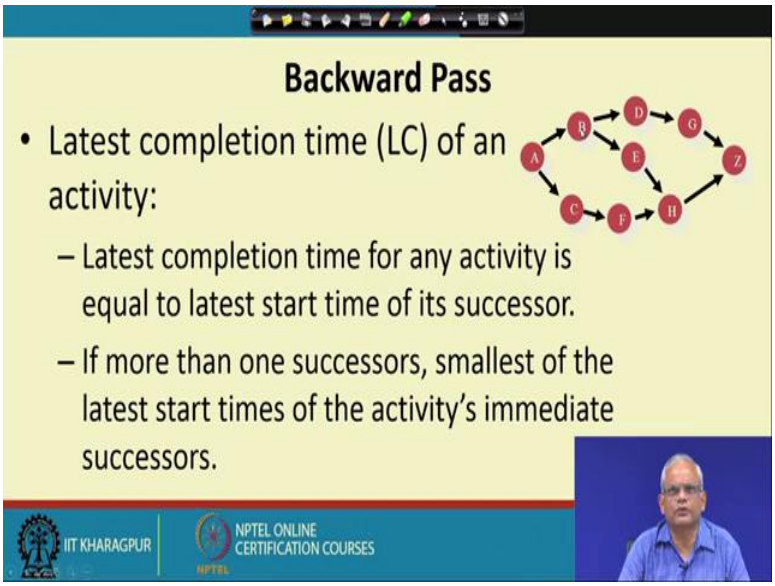

And, once having done that we will take the backward pass, start with the last task and we will compute the latest completion time which should be equal to the latest start time of its successor. The latest completion time of G should be equal to the latest start time of Z. When we take the example we will see that it is a very simple step, just a small arithmetic. The only case is that when there are more than one successor for a node, the latest completion time should be equal to the smallest of the latest start time of both the successors; this is the only thing that we must be careful.

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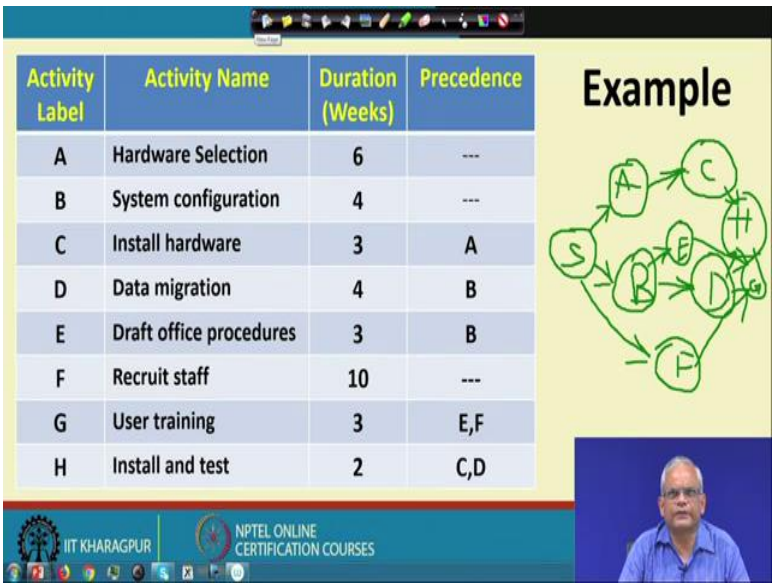

Activity Label	Activity Name	Duration (Weeks)	Precedence
A	Hardware Selection	6	---
B	System configuration	4	---
C	Install hardware	3	A
D	Data migration	4	B
E	Draft office procedures	3	B
F	Recruit staff	10	---
G	User training	3	E,F
H	Install and test	2	C,D

Now, let us take an example: here we have project where we have several activities which have been identified. Hardware selection, system configuration, install hardware, data migration, draft office procedures, recruit staff, user training and install and test. In our task network it will be difficult to write all these levels and the activity nodes and therefore, we use the label A B C D E F G H. And, the duration in weeks written here this is the estimated duration of every task. These are deterministic times we know well, but later we will see that how to use probabilistic estimates.

The precedence is also identified A B have no precedence and C has precedence of A, if we draw this A has no precedence, B has no precedence. We just draw it as it is, we have for C A is the precedence, we draw C and draw an arrow between A and C to show the precedence. And, we have for D B is the precedence draw an arrow between B and D, showing that B is the preceding activity for D until B completes D cannot start. And, since we have only we should have only a single start node which will be a day 0 we will just write as start node here, we cannot have two start nodes A and B.

So, even though there is no dependency for A and B you just draw a start node, just to have a single start node and then for E we will have B as the precedence. So, we will draw here E and for this B will be the precedence and then we will have F for which there is no precedence. And therefore, S will be its precedence and then for G both E and F are precedence. So, we will draw G here and for which both E and F are the precedence and then H for which C and D are the precedence. For H C and D are the precedence, you can draw it nicely because here there are arrows intersecting should not normally have that. And, then for H and G there are two end nodes, we will put another E node which is the successor node.

Now, once having drawn that we will apply our forward pass and backward pass.

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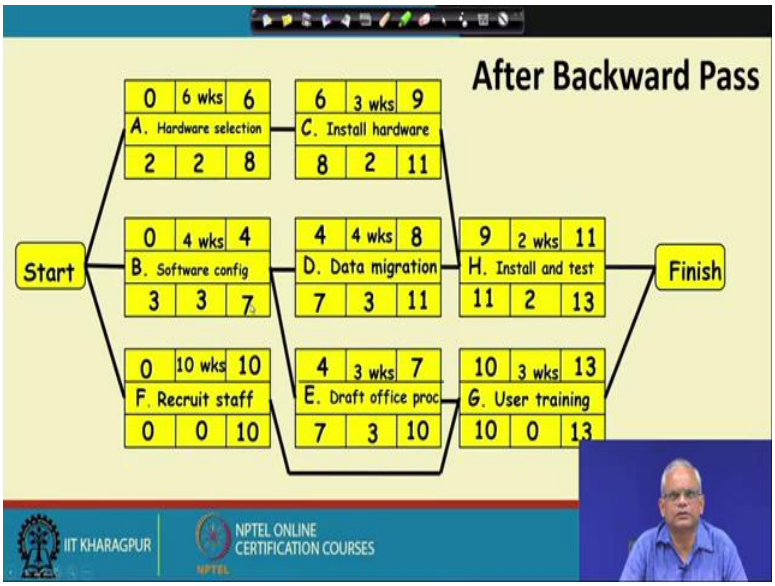

So, this is the one, if we redraw the one that were drawn in the previous slide we will have this. I have also written down here the names of the activities, but that is not necessary A B C D etcetera should be enough. So, A B and F these are the successor node at the start node, start and finish are dummy nodes, duration is 0, start date of start task is day 0. And, then A takes 6 weeks, B takes 4 weeks, F takes 10 weeks that we can get from the table that has been given and for every task we write the name of the task and also the duration.

Now, let us apply the forward pass, in the forward pass the start task starts and completes on day 0 and therefore, this can start on day 0. We will put 0 here and then this will be 6, the earliest start time is 0, the earliest finish time is 6. Similarly, this is earliest start time is 0 that is end of day 0 or start of day 1 and we put 0 here as the earliest start date because, this has completes day 0 and then we add 6 here put 6 here, 4 here and 10 here. So, that we have done here that for all these 3 tasks A B and F, we in the forward pass we first the completion time of this is 0 0 0.

And, add the duration here 6 and put 6 4 and 10 very simple arithmetic and in the next step we look at the successor here C is the successor for A and we put 6 here indicating that. The earliest completion time of task A is 6 and therefore, the earliest start time of task C in 6. And, then it takes 3 weeks we put 9 here that is the earliest completion time.

For task D the earliest start time will be 4 and earliest completion time is 8 and for task E the earliest start time is 4, earliest completion time is 7.

Now, we have just written that the earliest time start time is 6, completion time is 9 added 6 plus 3 is 9 and here earliest start time is 4, earliest completion time is 8. And, here earliest start time is 4 and earliest completion time is 7. In the next step, we will compute for H, but for H just see that it has two predecessors D and C and both have to complete for H to start. Even though D complete at 8th day end of 8th day, H cannot start on 8th; it has to start on 9th that is end of 9th day. Now, G also has two predecessors that is F and E and even though E completes at 7, it cannot start on 7 it has to start on 10.

We will put here 9 and 10 and this becomes 11 and this becomes 13. So, I have done that so, that is the end of the forward pass, all the earliest start time and earliest completion time are all marked on the tasks. Now, we will start the backward pass where we will complete the latest start time and latest finish time for all the tasks. We will start from the finish node, the finish node, the latest finish time is 13 that is both have to complete; even the earliest finish time is 13 because both have to complete. So, for the last node the earliest finish time is equal to the latest finish time which is equal to 13.

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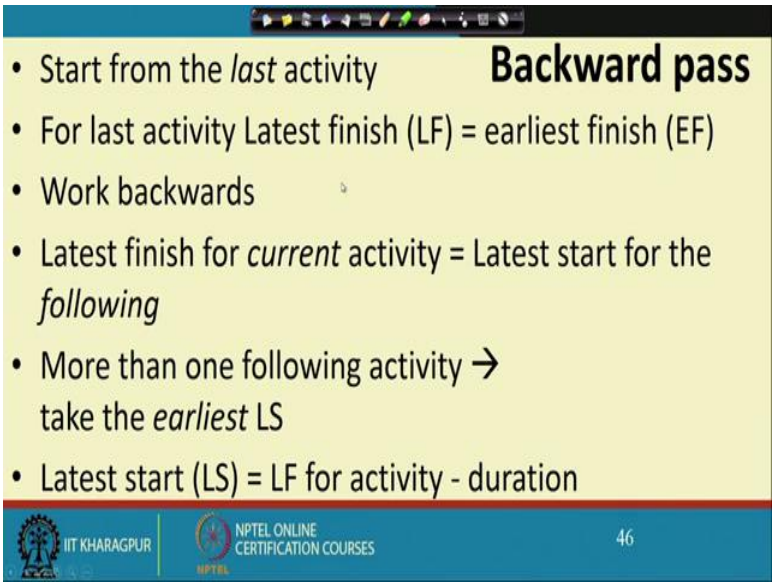

And, we will start the backward pass for the last activity we will set the latest finish time is the earliest finish time as I was mentioning from then we work backwards. The latest

finish time for a task will be the latest start time of the following task and either more than one following activity we will take the earliest start time. And, then from the latest start time, the latest finish time we will compute the latest start time by subtracting the duration.

Now, let us start the backward pass the finish here it is 13 ; that means, that both of these have to complete by 13, they have single successor. So, the thing is very simple we just write 13 and 13 here and then we have the duration is 2 weeks and we from 13 we get that here it must be 10. And, here it must be 11 subtract 2 from here sorry 2 from here we get 11 here and from 13 we get 10 here. Now, in the next thing that we compute is the slack time which is the difference between the earliest finish and the latest finish; sorry the latest finish mean earliest finish or the latest start minus earlier start both will give 2.

So, we put 2 here, we put 0 here; now let us computer for this ones, now this has a single successor which is completing at 9 sorry 11. And therefore, the latest it must complete by 11, similarly for this and for this it is 10 and we must have 10 here. So, 11 11 and 10 we have written here and then we subtracted 3 from here and got 7 and 7 minus 4 or 10 minus 7 is 3 and similarly here 11 minus 4 is 7 here and here 11 minus 3 is 8 and 8 minus 6 is 2. In the next step for this it is simple, the preceding the succeeding task for this F is G and G the completion the earliest start, sorry the latest start time is 10.

And therefore, it must complete by 10 and we put 10 here. But what about B? B has two tasks here a successor D and E, but then for both of this it is 7 so, we can put 7 here and for this we have C as the successor for A, C is the successor and 8 is the latest start time. So, we will put latest completion time for A as 8, let us do that here both were same and that is why you put 7 here. If one was 8 and one was 7 we would have put 7 here, if one was 8 and 6, we would have put 6 here and then 7 minus 4 is 3 and 3 minus 0 is 3 is the slack time.

Similarly, here 10 minus 10 is 0 and 0 minus 0 is 0, 0 slack time and for this similarly 8 minus 6 is 2, we write 2 here and then the slack time is 2 minus 0 put here 2. And, we have got all the task characteristics here, we have completed the backward pass. The next task is to identify the critical path, let us find out all the tasks that have slack time 0; here we find that F has slack time 0 and we have G has the slack time 0. So, we indicate this path as the critical path we will draw it using red colour ok.

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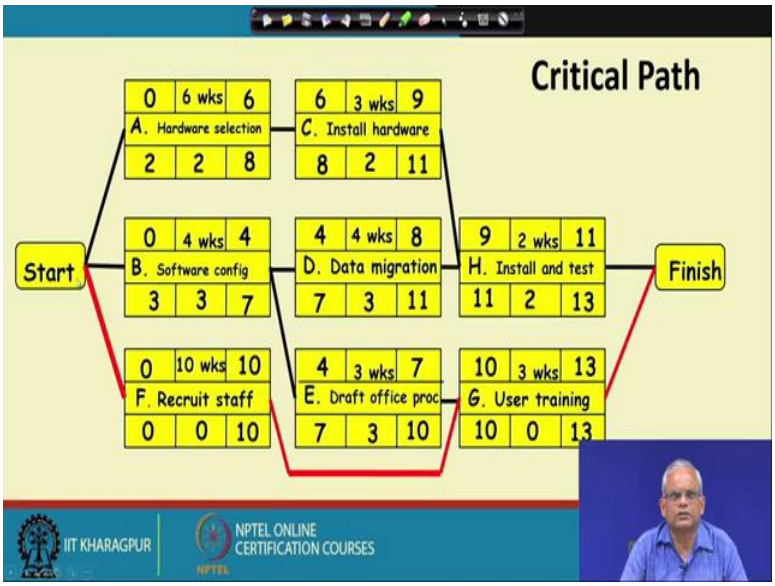

We drew it using red colour. So, this is the critical path, critical path identification becomes very simple once we identify all the activity characteristics and we can write this in the form of a table all the task parameters.

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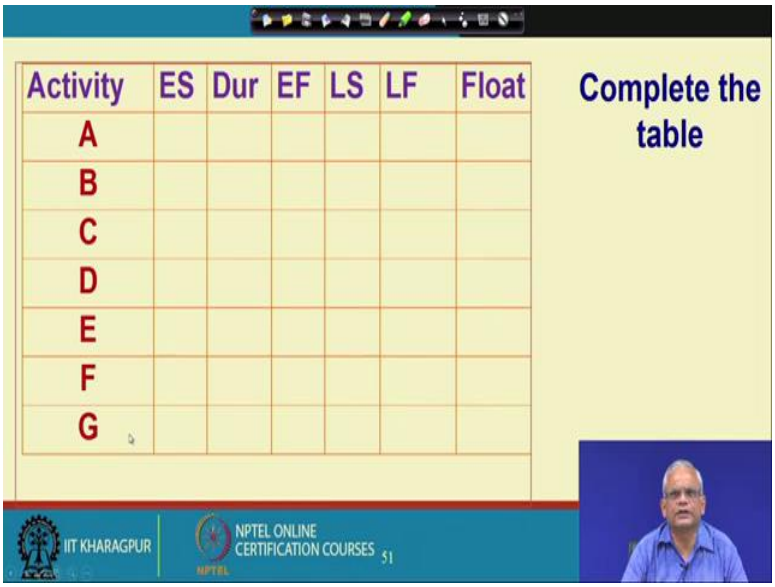

We can complete this table; we have A B C D E F G as the tasks, the earliest start time, duration, earliest finish time, latest start time, latest float time sorry, latest finish time and the float time. We can even write in the form of a table, we will just copy from the diagram; sometimes this representation also becomes useful to the project manager.

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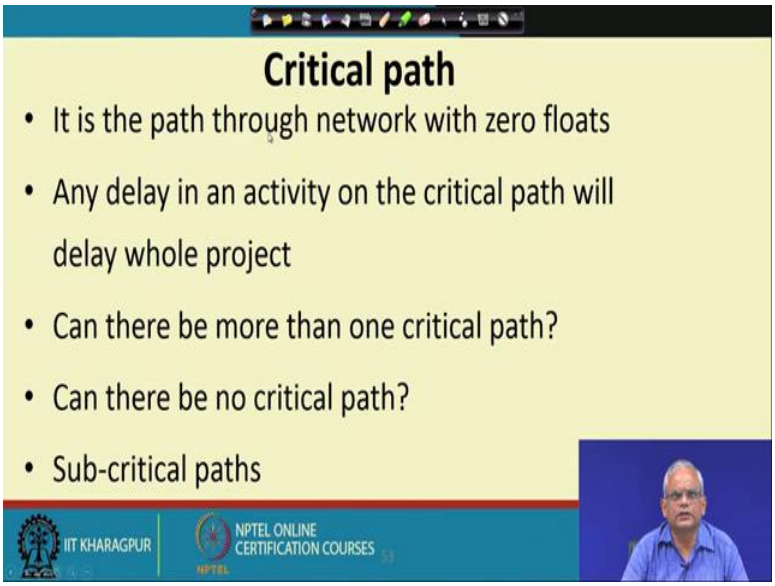

The critical activities here are the activity F and G, the critical path as we saw its the path through the network with zero float. It is very important for the project manager to pay careful attention to the critical activities because, any delay on a critical activity will delay the whole project. But, just out of curiosity some questions one is that: can there be more than one critical path for a project? The one example we discussed, we had only one critical path that is the slack time were 0 on one path and we identified that as a critical path.

But it may so, happen especially for larger task networks that there is more than one critical path, it is possible that more than one critical paths and on each path the slack time available to the activities is 0. But, is it possible that there is no critical path for a task network? The way we computed there has to be a critical path, the forward pass and backward pass. But, suppose we have a situation where the completion time is also predefined, the last task the latest completion time is predefined; it is not computed by the forward pass. And, then we can have it is possible that we can have no critical path, all tasks might have some slack time or other.

But what about subcritical paths? We identified all tasks or activities that have 0 slack time and we said that the project manager has to pay special attention to those tasks. But, what if all tasks have other than the critical path all tasks have like slack time of 10 7 etcetera, but there is one task which has slack time of 1. It is not 0, but slightly more than

0 and therefore, those tasks are also subcritical; that means, that we do not have too much laxity or too much flexibility with those tasks. The project manager also needs to consider the subcritical paths, where the activity is do not have too much of even though they do not have exactly 0 flexibility 0 slack time, but the slack time available is very less.

Especially for larger projects it may be also necessary to identify the subcritical paths, where the slack time even though is not 0 for the path, but then it may be a very small number.

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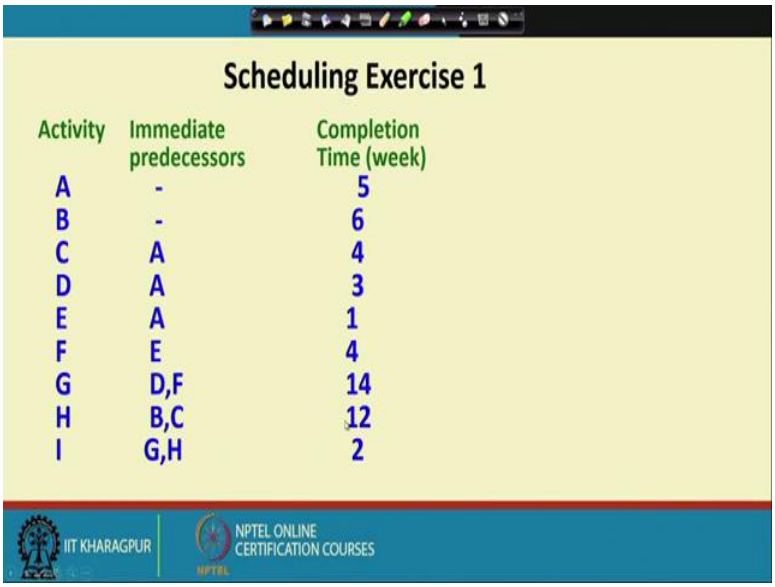

Here are few exercise; please try this out, draw the task network and then use the PERT CPM methodology to identify the project characteristics.

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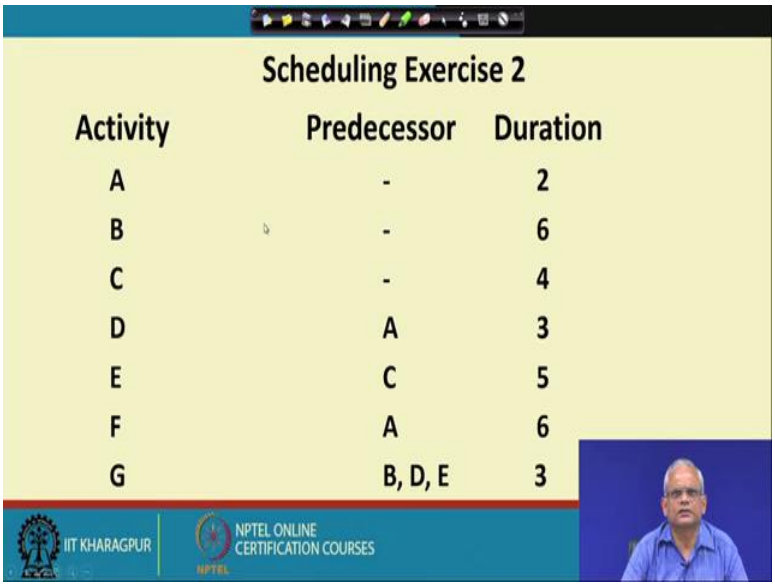

There is another exercise, the activity name, activity label, predecessor of the activity and duration. Please try this out, identify the critical path, if not look at the example that we worked out, it is very simple. You have to use the forward pass, identify the earliest start time and completion times, then start the backward pass. And, you will have the latest start time and finish time and then compute the slack and based on that you can identify the critical path. We are almost at the end of this lecture; we will stop here and continue from the next lecture.

Thank you.

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Lecture – 31
PERT, Project Crashing

Welcome to this lecture, in the last lecture we had looked at PERT CPM and how to identify determine the project characteristics, various characteristics of the activities critical path, critical tasks and so on.

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In this lecture, we will look at PERT where statistical times for tasks are considered. We will just look at an overview idea of that, we will not go into details and then we will look at project crossing. That is if the customer wants the project duration to be reduced how does the project manager go about doing it and then we look at team management. Let us start with the today's topics. So, these are the PERT Project Crashing and then team management that is the plan for this lecture.

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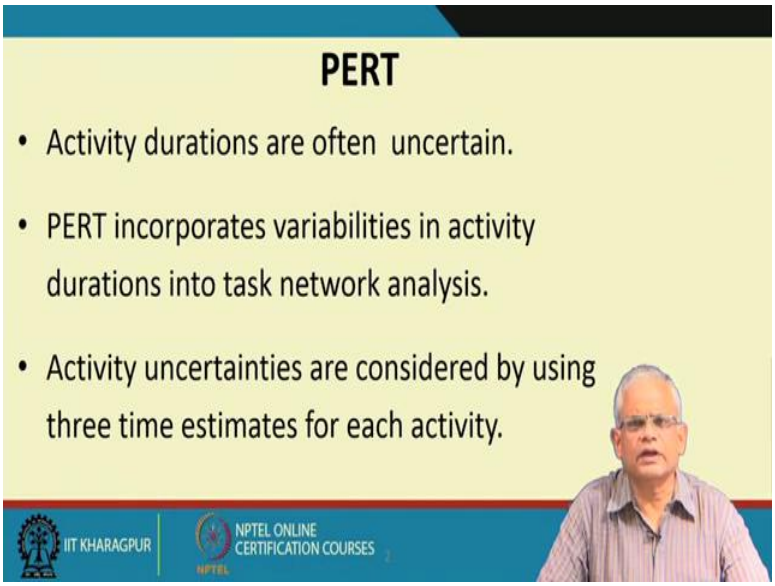

We had said earlier that in the PERT CPM, the task durations are deterministic; it is simple that all the tasks have estimated duration. But then often in development work things are uncertain; in a routine work like maintenance and so on; the durations are known quite accurately.

But in a development work often the project manager; it becomes hard to estimate the exact duration of activities. Only statistical values for activities can be given and the PERT can be used for that. Here the activity durations are uncertain and variability of the activity durations; that is probabilistic variations can be handled in PERT. Here there are three time estimates that are given to each activity; we will look at the three estimates and based on that various project parameters and the critical path and so on are determined.

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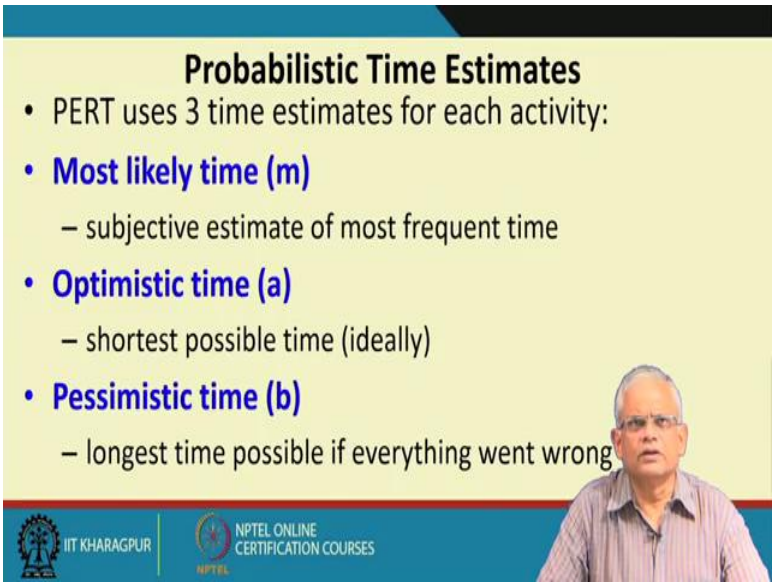

The three probabilistic time estimates for the activities are the most likely time which we indicate by m. The optimistic time that is given that everything goes alright; the task will get completed in time a, this is the shortest possible time. And then we have b which is the pessimistic time that is if things do not work out obstacles appear and so on. This is the worst case time for the activity b; b is the worst case, a is the optimistic and m is the most frequent time.

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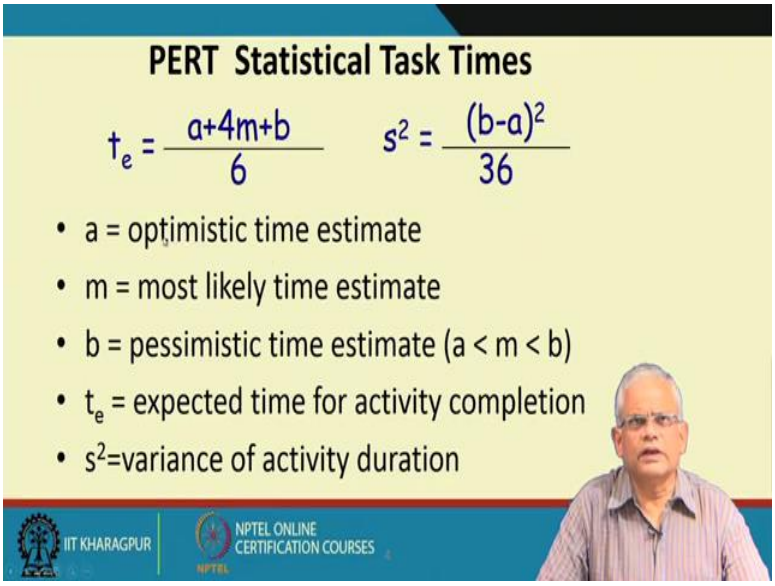

And based on that; the task estimated time is given as a plus 4 median plus b by 6

$$t_e = \frac{a + 4m + b}{6}$$

and the variance is given by

$$s^2 = \frac{(b-a)^2}{36}$$

b minus a whole square by 36. And based on these statistical parameters inferencing is done just like the way we were using in the PERT CPM.

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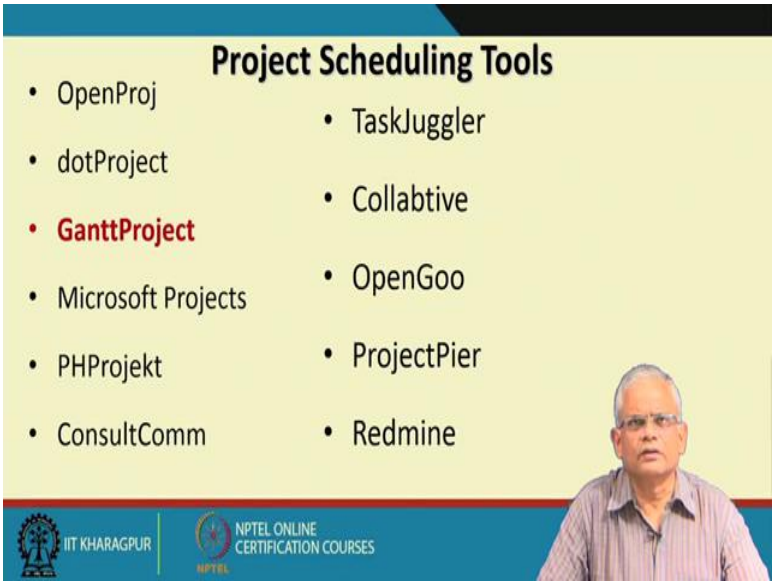

As I said that we will not go details into the statistical calculation of project times, but there are several tools available; where these can be done easily the PERT CPM and PERT can; there are large number of tools are available many are open source and some are price tools. But then some tools which we have used it is easily downloaded, takes very less space, does meaningfully well is the Gantt project. And if it is a task which is; Gantt project is typically the project manager uses alone on a desktop standalone machine.

But if there are multiple people who would like to use like the team members would like to examine their task characteristics, the tasks assigned to them, give inputs and so on then Redmine is a tool which is again open source; Gantt project in redmine. Redmine

runs on a server and there can be users and different computers whereas, Gantt project is a very simple tool, do not even need a user manual you can start right away using it.

The project characteristics are computed automatically the critical path is shown as you input. And of course, there are many other tools that are available; it will be nice as part of this course you can download some of these tools and get used to it. If you get used to one tool you will see that it becomes very easy to use the other tools.

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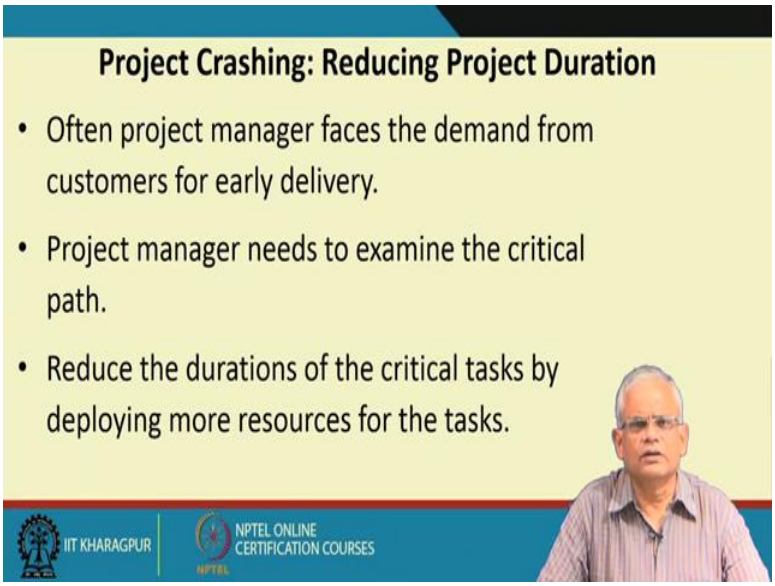

Now, let us look at project crashing that is common problem that the manager estimate certain time; let us say 6 months. And then the customer says that 6 month is too long can it be done in 5 months? Then what does the project manager do?

The project manager tries to reduce the project duration that is called as project crashing. As we have already discussed that the longest path in the task network is the critical path. To reduce the project duration, we need to reduce the time for the critical path, but then as we reduce the total duration for a critical path; we will see that there are other critical paths that appear that has to be considered. But how does the project manager reduce the duration of the critical tasks that appear on the critical path?

Of course, by deploying more resources if there was one tester testing was the task; then might deploy more testers, coding might deploy a additional coders. By assigning more resources to a task the duration of the critical task can be reduced, but then which critical

task to take up. If the requirement is let say 15 days reduction in a 6 month project which critical task to take up?

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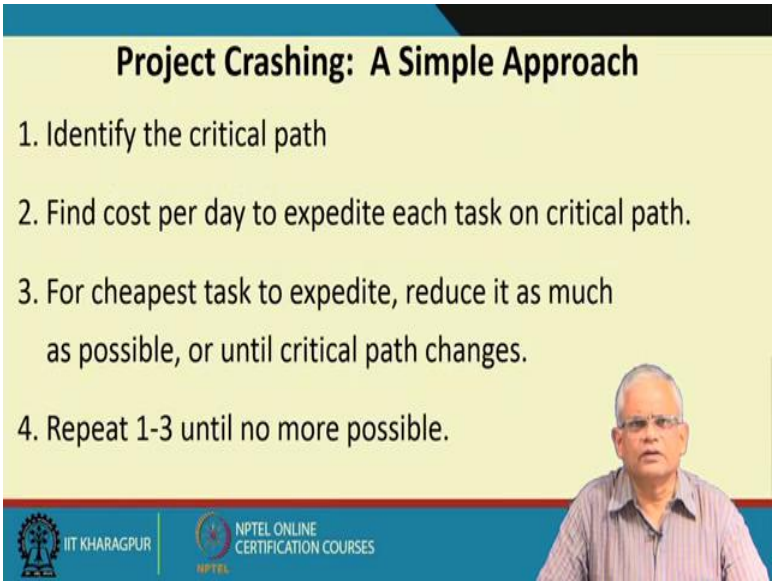

We can do that by a simple approach for project crashing. First is we must identify the critical path and then look at all the critical tasks on the critical path and then find the cost per day to expedite each task on the critical path. And then identify the cheapest task to expedite and then gradually reduce it, but then it may so happen that as we reduce it by a day or 2 day, 3 day etcetera we will find that the critical path changes.

The critical path which was let say 6 months and as it becomes 5 months 25 days; it may not remain the critical path, there maybe another path which becomes critical. And then we need to reduce that critical path the current critical path after reduction. And we keep on doing this steps until no more reduction is possible or the target is met.

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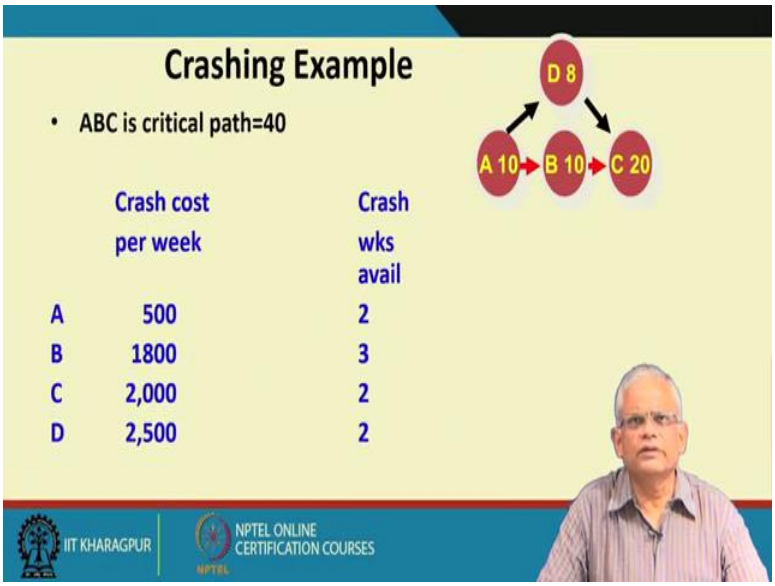

Just to give a small example of the crashing; let say a very simple projects with a four tasks A, B, C, D. The task durations are 10, 10, 20 and 8; we can easily see the critical path shown here in red the duration is 40 that is A, B, C is the critical path with 40 weeks as the duration. And then the project manager needs additional information that deploying additional resources here costs how much? The crash cost for a is 500 rupees, but it can at most be reduced by 2 weeks; it will not be possible to do it in less than 8 weeks. Task B, 1800 is the cost per week, but then it can at most be reduced by 3 weeks.

Tasks C, 2000 is the cost per week and only it can be reduced by 2 weeks and task D is 2500 per week and can at most be reduced by 2 weeks. Obviously, the project manager will choose A to reduce at first reduce it by 1 day; A becomes 9, B 10 and C 20. The critical path becomes 39, the overall project duration reduces by 1 day with an expense of 500 and the critical path does not change because the other critical path is 38.

Now, reduce it by 1 more day the critical path becomes 38, but the other path becomes 36; so still the critical path does not change. The next task to reduce possibly is 18; sorry B and the cost is 1800; now let us reduce it by 1; the critical path becomes 8 plus 9 plus 20; 37 and still remains the critical path the project duration reduces to 37. And we reduce it by one more it becomes 36 and there are two critical paths now; we have to reduce both just reducing this; does not reduce the project duration you need to reduce D as well.

So, this is basically the approach the project manager takes to reduce the project duration or the project crashing.

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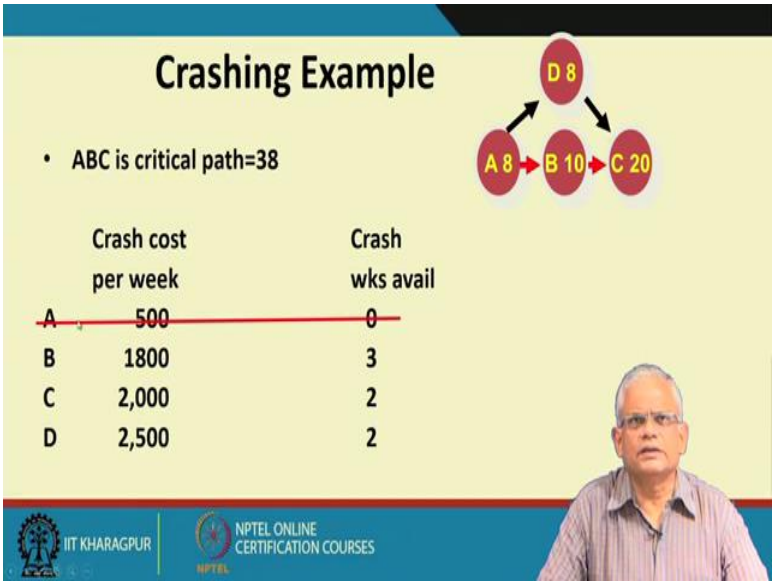

A is reduced as much as possible because that is the cheapest task to reduce and the duration for a becomes 8, the critical path becomes 38; the project duration is 38 now and the sub critical path is 36. Now, let us look at the topic of team management; how does the project manager go about doing the team management.

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To manage the project team; the project manager has to put in daily effort, has to track the team member performance on a day to basis day to day basis, motivate the team members to improve their performance. Provide feedback which can help them and if there are issues and conflicts among the team members these need to resolved; so that the overall project performance improves.

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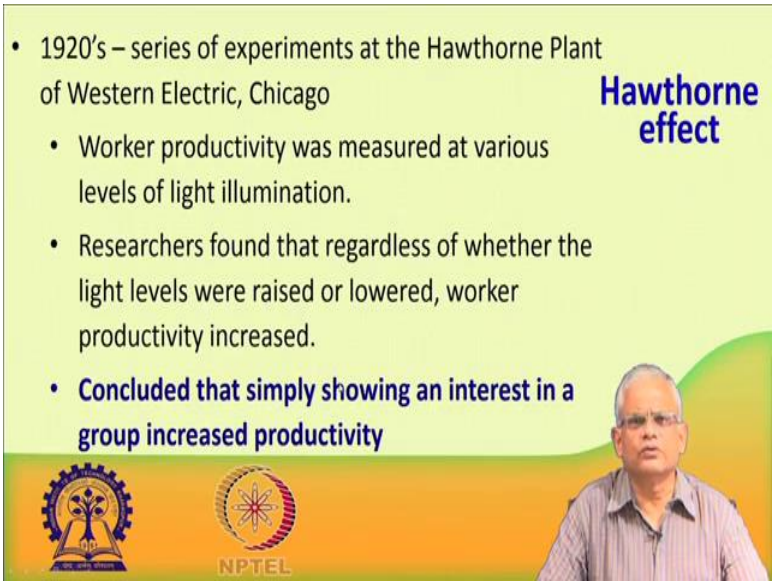

The project manager needs to be aware of some theories and observations which have been made over the last century or so; these are very fundamental results applicable even now and the project manager needs to be aware of this. One is the Hawthorne effect; way back in 1920; a series of experiments were conducted at the Hawthorne plant in western electric, Chicago. Here the idea was that how does the team performance change as the lighting conditions improve? More lights were added it was made very bright room, lights were reduced made relativity darker, but then the surprising observation the thing was actually repeated across many workers.

But something that was clearly identified was that regardless whether the light levels were raised or lowred; the productivity always increased, surprising that not only that the light level did not matter so much, but also for both cases the whether the light levels were raised or lowered; the productivity increased. What can be the cause for this? It was concluded that since the workers were under observation, their productivity was being measured; they were paid attention that is what increase the productivity.

The conclusion from here the Hawthorne effect is that; if the workers are given attention, the manager pays close attention that what they are doing their productivity increases. Simply showing an interest in a group increased it productivity.

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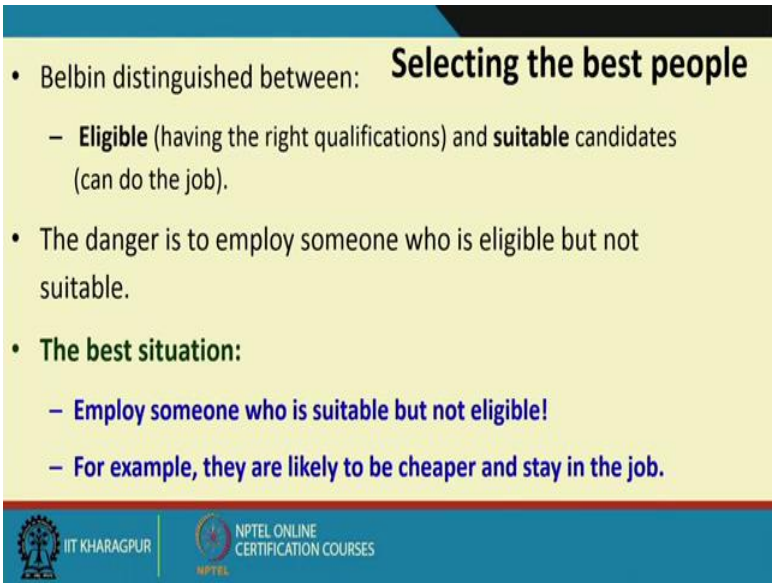

The manager has to select the best people for the project, here we must note that Belbin; he distinguished between eligible and suitable candidates. Eligible candidates are the ones who have the right qualification and suitable candidates are the one who can do the job.

May be the eligible candidate may not be suitable and a candidate who is not eligible; does not have the right qualification may be actually the suitable candidate. But the danger is to employ somebody who is not eligible sorry who is eligible, but not suitable. As per educational qualification other requirements the candidate is eligible, but then for the project is not suitable; the project suffers and also the candidate suffers, it does not help anyone.

But the best cases that employ someone who is suitable, but not eligible; they would be cheaper because they do not have the right educational qualification and they will stay on the job they will be motivated. So, this is the best case to employ somebody who is not eligible, but suitable, but then how to identify that is the; that is a problem.

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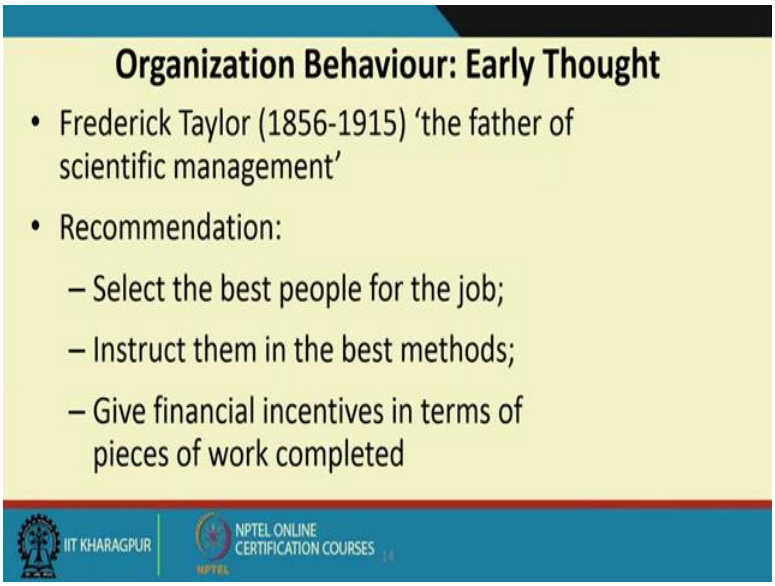

One of the early thoughts in organization behaviour was by Frederick Taylor; long back he recommended that select the best people for the job. And then instruct them in the best methods and then give financial incentives in terms of pieces of work completed.

Of course, this was for a manufacturing job for a software job it is not so well defined that what do you mean by pieces of work completed. But still let us see what he really meant that select the best people for job instruct them in the best methods and give financial incentive quite intuitive actually.

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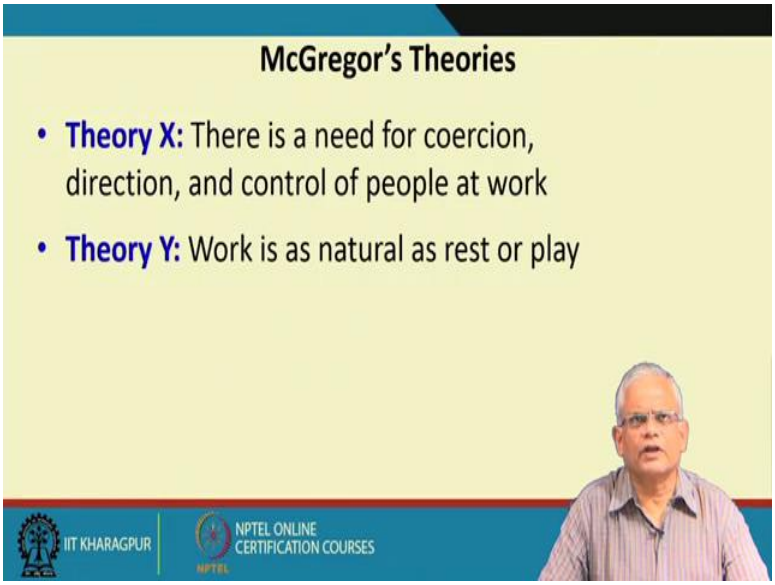

But then McGregor, he propounded two theories; theory X, there is no need for coercion, direction and control there is a need for coercion direction and control of people at work. He assumed that whenever the manager finds the worker; he needs to coerce monitor closely ask them to do the work because the workers by themselves do not work. They have a tendency to laze around not work and therefore, the manager needs to have coercive techniques, control the people.

But then the theory Y is a different type of manager who assumes that workers they love to work. Work is as natural as rest or play just needs to encourage them; these are two different ways in which managers work. One is the theory X manager; who assumes that people always cheat, they are not motivated, they will not work unless they are scolded they are forced and so on.

And theory Y is the manager assumes that the workers are efficient; they love to work. It will be very easy to spot what is the type of the manager if you visit a team and find that the day the manager is absent; if you find that everybody is relaxing feeling happy that the manager did not come and then that is a theory X manager; who is a dominating things that the people do not work, he needs to tell them to work and so on. And theory Y manager if you visit and see that even when the manager is not there; the workers are working as usual then the manager; you can think of him as a theory Y manager.

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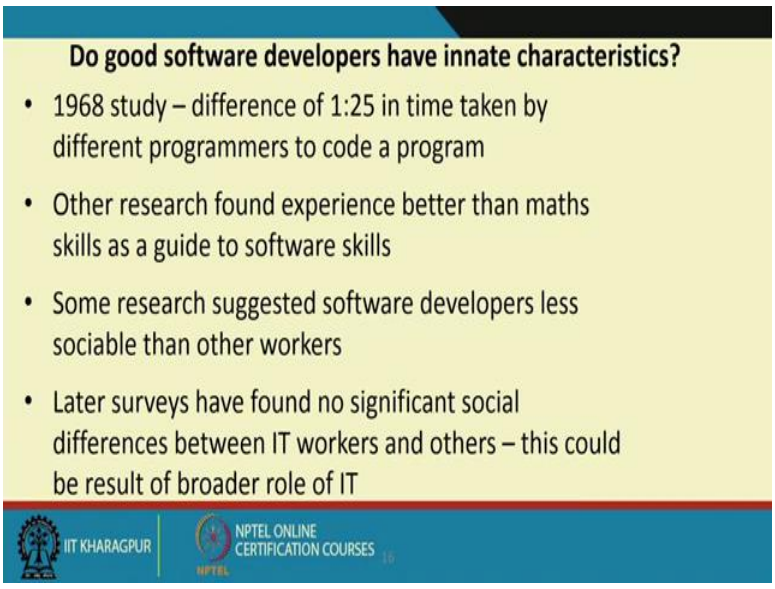

But do the software developers have some characteristics which make them do a good job; because that will give us a hint how to choose a good developer. In a very old study, 1968; the difference in time taken by different programmers to code the same program is 1 is to 25. Somebody who takes 1 hour to code a program, the other programmer takes 25 hours and these are employed by the same organization.

So, the proficiency level of the programmers vary widely that was the conclusion the study in 1968. Obviously, this can be interpreted to mean that during the selection; we need to find out who is a good coder, codes very fast writes good code. But then we also need to remember that nowadays software development is not just coding; coding is a small part of the activity may be in 1968; coding was on one of the major activities in program writing, but now coding is a small part of the activity of a software engineer.

For example, finding out which reusable libraries to use, which tools to use etcetera. So and of course, testing, design, requirements analysis and so on. So, this study is an indicator, but then we must remember that the times have changed; now we have not only coding, but other activities, but then still we can interpret this, to say that there is a wide variation in the competency of different engineers in a project. And we need to identify who is a good developer who can help the project and select the right people.

One of the study early study found that those who were good in maths; they had good software development skills, but then later it was found that this not may not be so. Another study found that the software developers are less sociable than other workers. But then later surveys found no social difference between IT workers and other; maybe when the study was done, it was the case where the programmer has to just write the code that was the major activity.

But now as you are mentioning software development has many other activities and possibly that may be the result that the developers have a broader role now and they have to be social. We are almost at the end of the time at this lecture, we will stop here and we will continue in the next lecture.

Thank you.

Software Project Management
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Lecture - 32
Team Management

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Welcome, to this lecture. In this lecture, we will discuss about Team Management, some very basic concepts and then, we will look at the organization structure.

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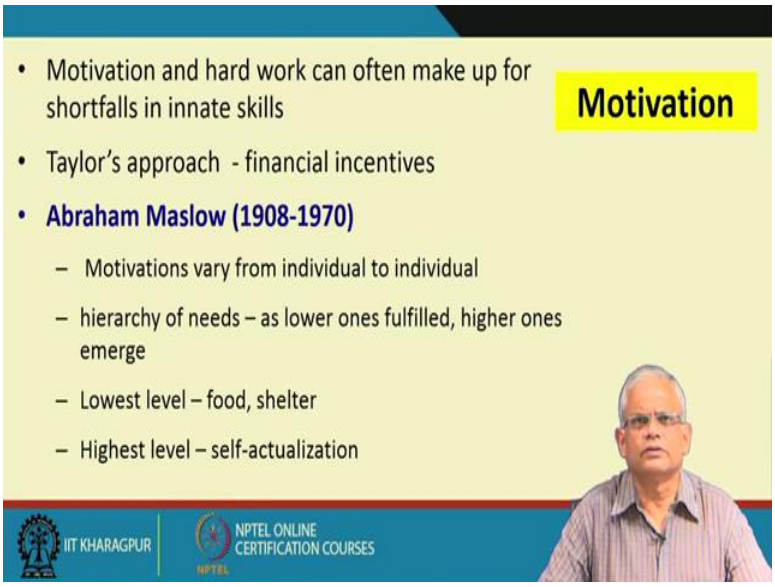

In the last lecture, we had seen some very basic theories that the manager must be aware for effective team management; today, in this lecture we will see few more concepts. The first one is with respect to motivation. We had said that software project manager sorry, the software developer has certain skills and there is a wide variance in the capability of the software developers.

But, then we must also be aware that motivation and hard work can often make up for the shortfall in the skills. We had said that some studies found that the competency or the effectiveness of the developers varies from a in a scale of 1 to 25, but then the one who is less competent, if he is very sincere, motivated and hard worker he can make up for the gap in the competency.

We had discussed about the Taylor's approach about giving financial incentive as the major motivator. But, Abraham Maslow he said that financial incentives often do not work. The motivation actually vary from individual to individual and also based on the time or the point of time in his carrier when you are giving him the incentive. For every individual there is a hierarchy of needs, as the lower items in the hierarchy are filled then the higher ones emerge.

At the lowest level the requirements are food, shelter. A worker who is not having enough food, shelter and so on, and you provide him he will feel motivated, but then the workers who were well of, employed for long time and then you give them these incentives it may not motivate them much. And, at the highest level of the hierarchy are the self-actualization. Let us look at this.

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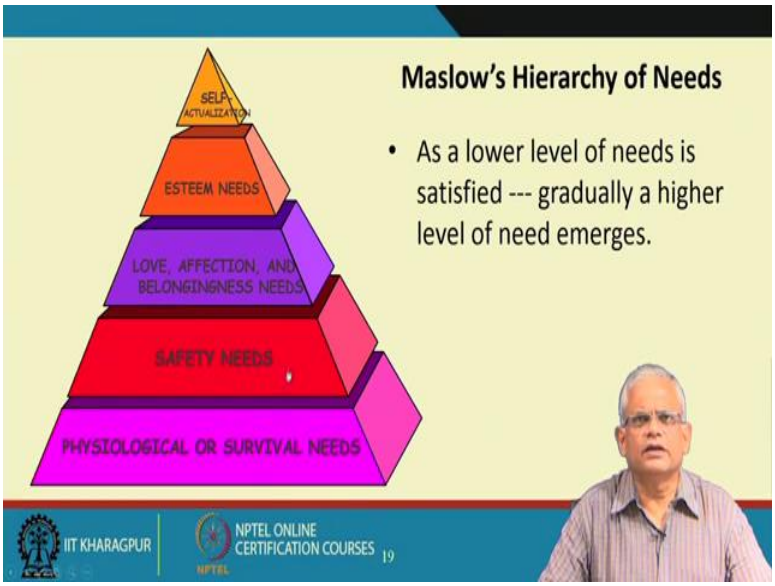

Maslow's hierarchy of needs we can see that at the lowest level of the hierarchy is the physiological or survival needs – these are food, shelter and so on. And, then at the safety needs, freedom from danger and so on. Safety then the social needs and once these are filled like belongingness, liking the organization, people liking, then the esteem needs and finally, self-actualization.

As one of the level is filled for a project member then the higher ones emerge, once the lower one is filled more of that does not motivate the team member. We need to give a higher level incentive. Let us look at these incentives are sorry this hierarchy of needs in some detail.

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The physiological needs are the need to overcome hunger, thirst, sleep etcetera. The safety needs are security, protection from danger and so on. The social needs are friendship, engaging in social activities, group membership and so on. The esteem needs are self-confidence, achievement, recognition, awards, appreciation and so on. And, at the top level is the self-actualization where all the previous needs are filled and the employee, the motivation is the desire to develop one's full potential.

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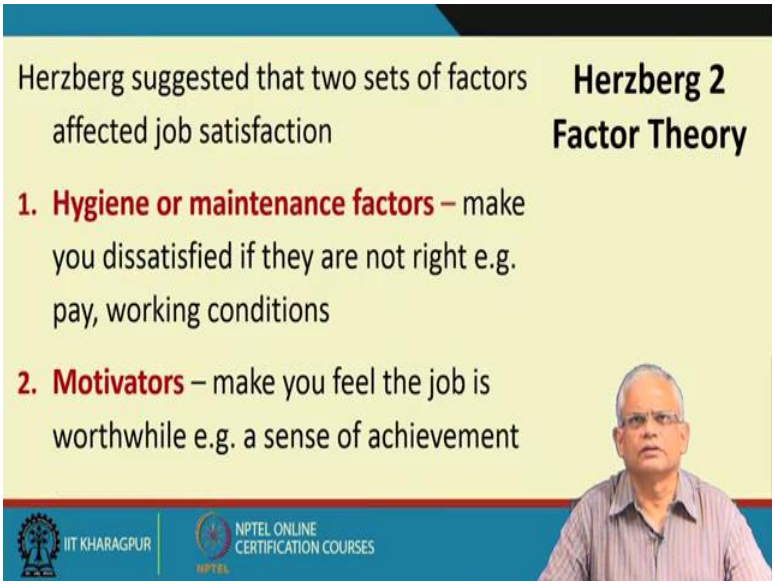

The Herzberg's 2 factor theory: here Herzberg suggested that two sets of factor affected job satisfaction. One, he called as the hygiene or the maintenance factor and if these are not right the employees become dissatisfied. For example, the pay, working conditions etcetera. And, there are several things which are motivators. These factors make it worthwhile the make the job worthwhile and there is a sense of achievement for the employees.

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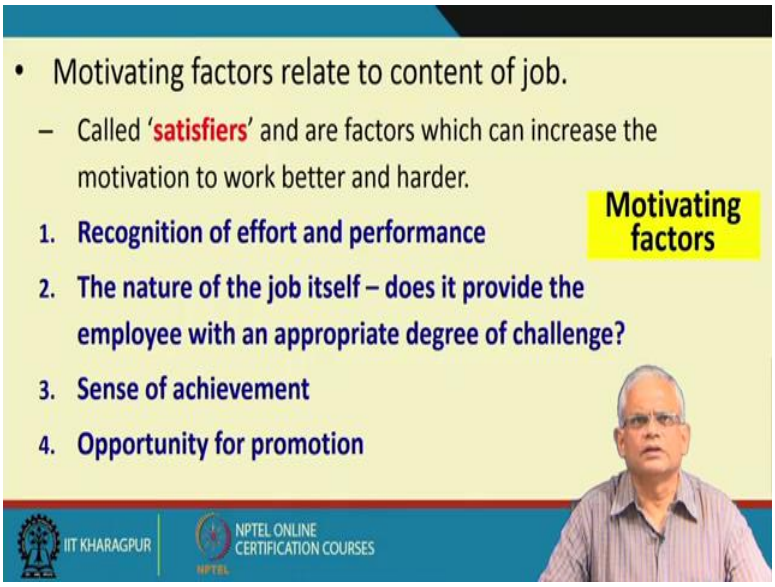

Let us look at the motive motivators, these are also called as satisfiers. Examples of these are recognition of effort and performance. The manger or the organization need to recognize employees who put lot of effort, show superior performance. Another motivator is the nature of the job itself – is it a routine job where the employee finds no challenge at all or it makes him think put his best. The sense of achievement, opportunity for promotion and so on these are the satisfiers.

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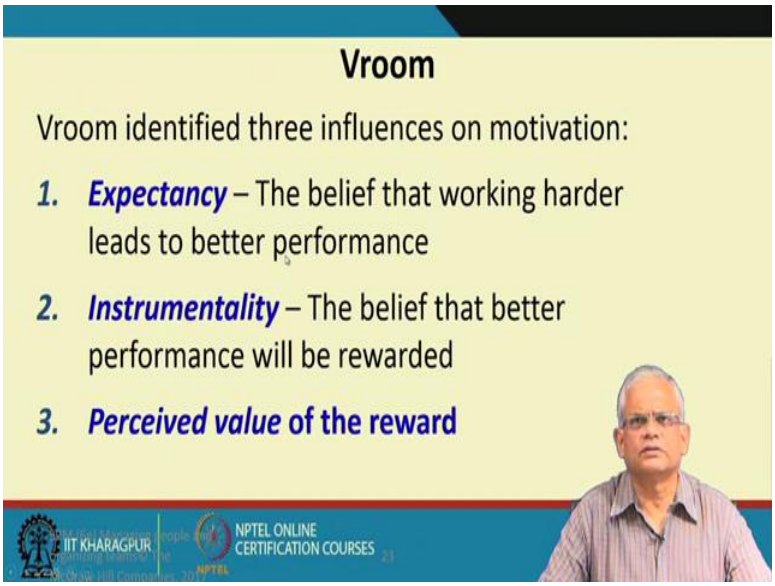

Vroom identified three influences on motivation. One was expectancy; the belief that working harder leads to better performance. Just to given example where expectancy does not hold let us say a software has a bug and different developers are put large number of hours trying to locate the bug and still are not able to find it. And, here they are demotivated because they do not have the expectancy. They do not believe that if they put another 10 hours, they would really be able to detect the bug because already hundreds of hours have being spent without detecting the bug. There is no expectancy that few hours of additional work will help them detect the bug.

Instrumentality – the belief that better performance will be rewarded. Let us say the developers are developing a software and if they implement all the features well, then this will be accepted by the customer, appreciated and so on. But, let us assume that already the customer has got hold of another software which is working quite well and since they have already given this project it is being developed, but then they may not use it actually, just because they had given it and not able to cancel, then the project team feel demotivated.

They do not have the instrumentality, that is, if they do a good software that will be appreciated because it may just gather dust and put into cold storage, nobody will use it because the customer is already using another software. So, in that case, there is no

instrumentality; just giving an example. The perceived value of reward. This is another motivator what the developers perceived the reward to be.

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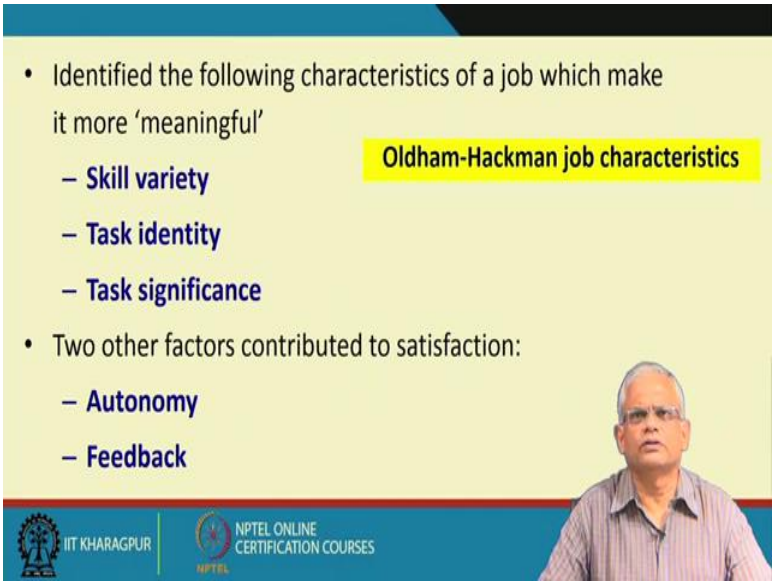

The job characteristics which make the job more meaningful and motivate the team members is given by the Oldham-Hackman job characteristic model. Here he identified three aspects of the job; one is skill variety. If the developer is trying do just one type of thing over the years, there is no skill variety and the developer loses motivation. To think of it just somebody who is doing coding over several years may not be as highly motivated, as somebody who is having skill variety, for example, a designer or an analyst who is having a much wider things to do.

The task identity ones the task is complete can the developer be identified who has contributed to develop it or is it that there several people who contributed and no one in particular can be identified about who can be given credit. Here, if the manager gives clear objective, clear work to the developers then there is a task identity and this is the motivator, that somebody is responsible for some work and he as he completes does a good work it will be appreciated that this specific work is done by certain team member.

The task significance whether the task that is assigned to somebody it is significant, whether it will be useful to the team to the organization to the customer; if it is a very insignificant work like let us say just collecting all information from various developers

and just storing it, then it may not be so motivating as doing a crucial piece of the user interface or certain I/O tasks and so on which are easily noticed and are identified.

Two other factors he identified which contributed to the job satisfaction are autonomy, is the developer has freedom to work on his own way or he is told every step. He is told the exact steps he has to follow, he cannot use his mind then he lacks autonomy and he will be less motivated. And, the feedback from the manager, that is another factor which motivates the team members.

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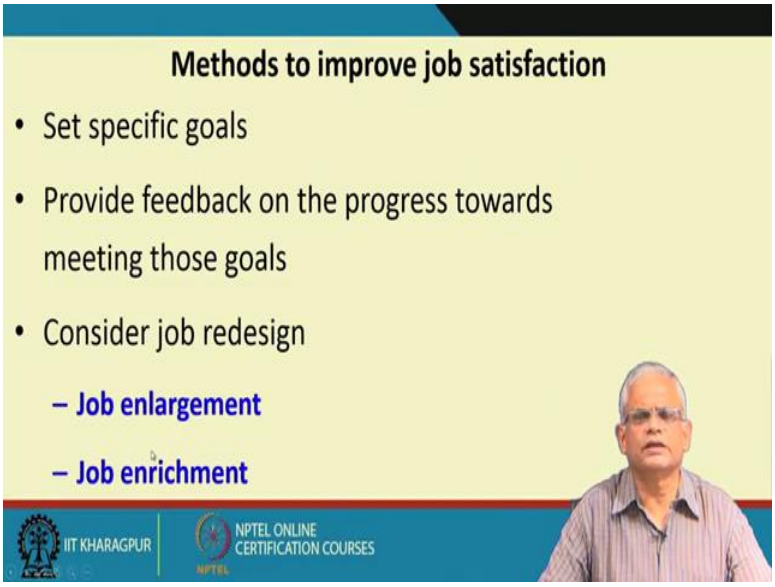

What does the project manager do to improve job satisfaction? Have to set specific goals; this clearly defines the activity that is given to a developer. Provide feedback and progressed towards meeting those goals. Consider job redesign if it is becoming too mundane, widen the job. Job enlargement, job enrichment that is give some decision power.

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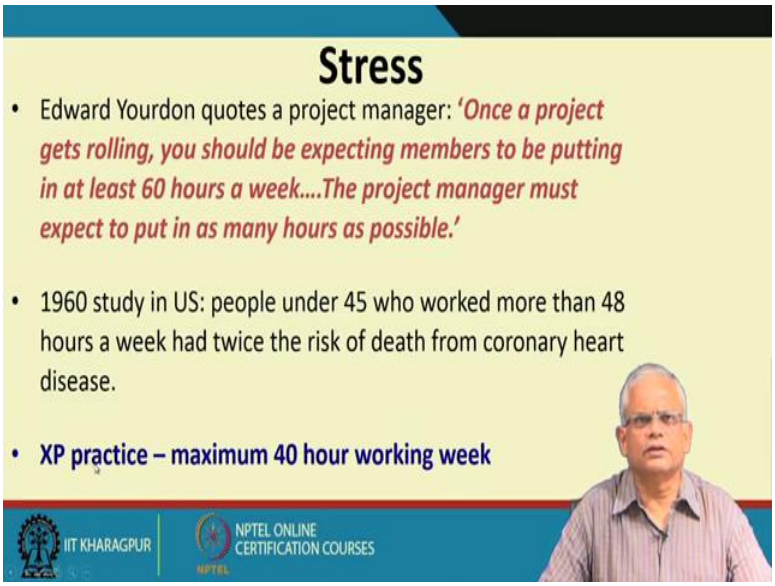

Another aspect which the project manager needs to be aware is the stress that the team members undergo. Edward Yourdon quotes a project manager saying: Once a project gets rolling, should be expecting members to a putting in at least 60 hours a week and also the project manager must expect to put in as many hours as possible. Every project when it starts there is not much stress on the team members, but as the project develops, the project team is under time pressure and there is a stress. Often they are asked to work long hours as his quoted by Edward Yourdon.

But, then as a manager will must be aware of the 1960 study in US, where people under 45 who worked more than 48 hours a week had twice the risk of death from coronary heart disease. And, many of the popular development practices like the extreme programming here it clearly identifies 40 hour working week.

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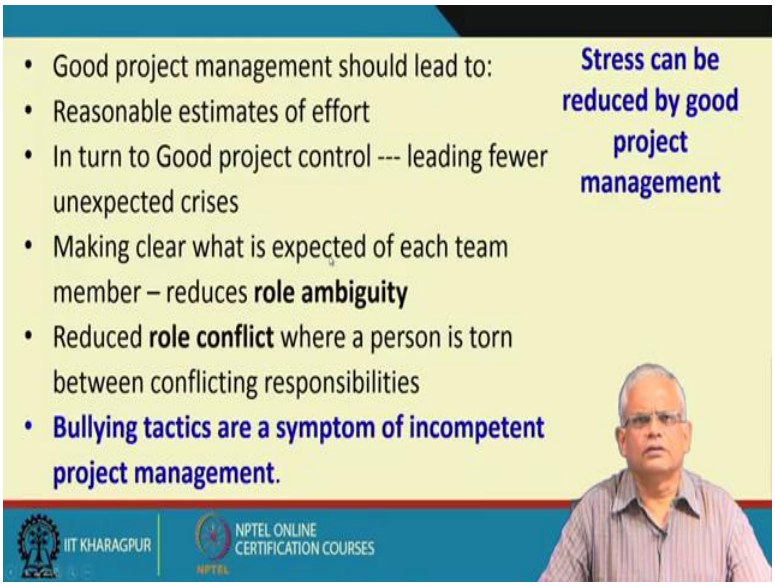

An effective project manager reduces stress on the team. If the team is under stress it is partly the project manager who is to blame. A good project manager should have reasonable estimate of the effort. The project plan should be proper. There should be good project control, so that the crises in the project are very few and what is expected of the team member should be unambiguous. If the project manager is not able to assign clear work to the team members that creates lot of stress.

Reduced role conflicts that is if a project manager assigns different duties, different work to a project team and those are conflicting. For example, he has to attained the project team member has to attained two meetings at the same time because he is member of the quality team and also he is a developer, he has to attend a review and at the same time he has to attend a quality meeting, that creates stress. And, often the project manager tries to control the project using bullying tactics we just saying the theory X managers, but then the theory X managers are incompetent project managers, there should be theory Y managers.

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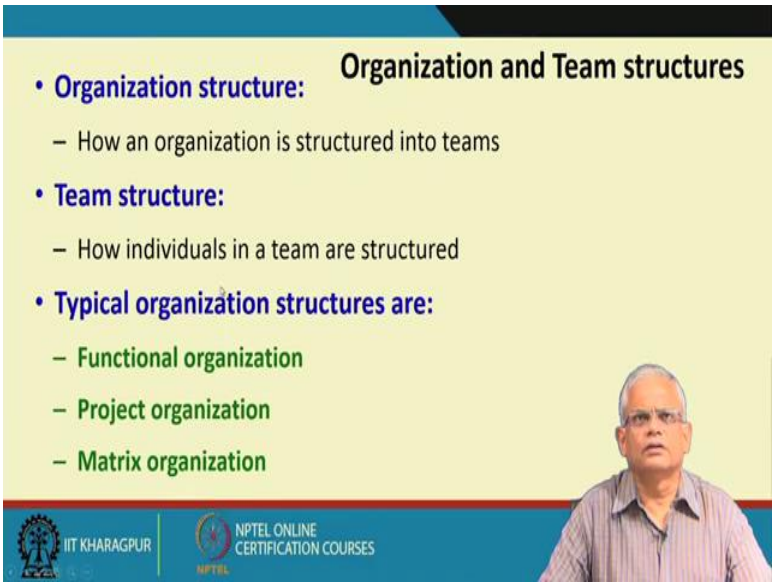

Now, let us look at the organization and team structures. In a software development organization if you visit in organization, how is the organization structured into teams? We can visit various companies and find how are these companies they have structured into teams like organization might have been might be having let us say two dozen projects and each project will be having a project team. So, how are these two dozen projects organized, whom do they report and who are the members of this team?

The other question is that how the teams individually are structured. These two aspects we call as the organization structure. The organization how is it organized in to teams and then the second thing is the individual teams, how are they structured. First let us look at the organization structure, then we will look at the team structure.

There are three basic types of organization structures. If you visit a organization you can identify that some organizations have functional organization. We will look at what is meant by functional organization. The second is the project organization and the third is the matrix organization. Let us look at three organizations structures.

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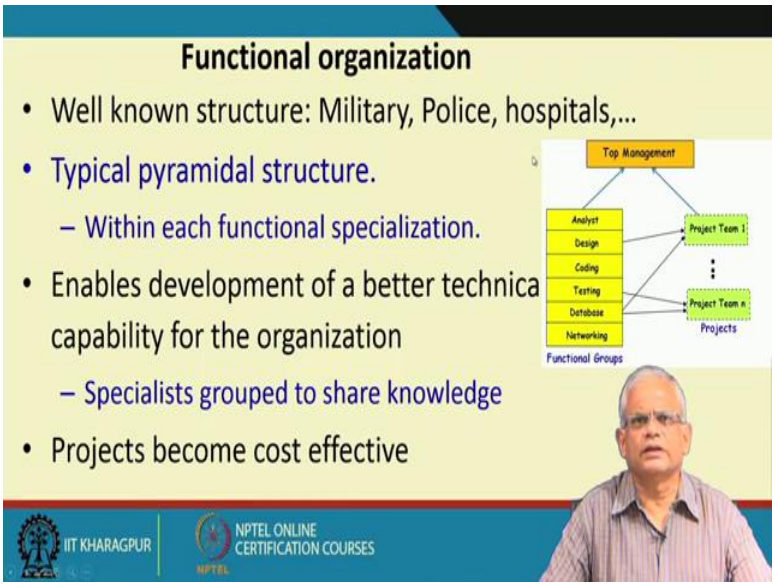

In the functional organization we have various functional groups. For example, we might have analysts several analysts under the functional group. There may be another group called as designers, another group as coders, testers, database experts, networking experts and so on.

And, the organization might have several project teams and as the project starts the manager determines who or what type of expertises needed, he will contact the respective functional group managers and then they are assigned to the project. For example, project team 1 might have some designers, database experts and as the designers complete their work, the manager may determine and he will return the designer to the functional group and then request for coders and so on.

This is a well known structure for military, police, hospitals etcetera. For example, in a hospital there may be functional group of nurses, there may be functional group of various types of doctors, specialists, pharmacists and so on. And, for each activity for each patient they might need some of these. Similar is the case for military. There are various types of functional groups here, there are doctors, hospitals, there are soldiers, there are airman and so on.

And, here it is a typical pyramidal structure with each functional specialization. Here the advantage is that since the analysts are together a team reporting to a manager. Similarly, the designers together form a team, a functional group. There is a better coordination

among them. They discuss among each other. They develop better capabilities. The specialist groups at the functional groups who share knowledge. And, also the project becomes cost effective because the project team request for personnel agent when the need arises and returns them and takes the other functional groups as happens in a hospital and so on.

But, what about in a software organization? Let us look at the other organizations and then we will compare that how does these different organizations they perform in a software development work.

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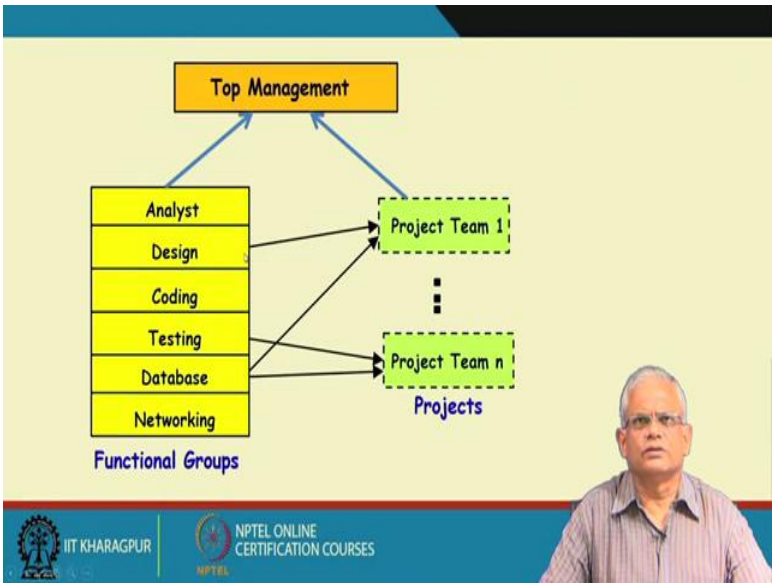

So, this is the functional organization. We have different functional groups and as the project teams are formed and the project develops they request from different functional groups and return them. And, all of them the project team and the functional team they report to the top management.

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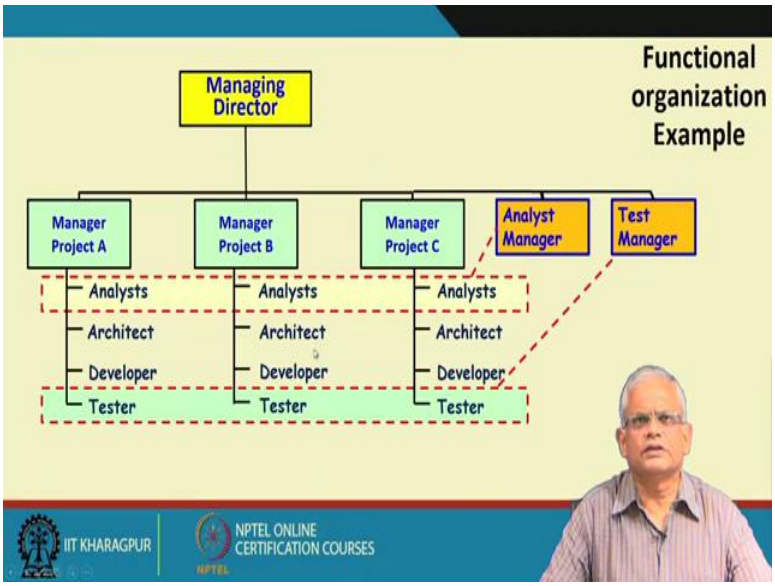

This is just a different diagram to explain what happens. The analysts they form a functional group, report to the analyst manager, and for different projects they may request analysts and the analyst manager assigns them specific analysts. Similarly, the architects are needed for different projects and they have each of these team members have two mangers to report: one is their functional manager and another is the project manager.

Similarly, the coders, the developers, the testers they report to the test manager and also to the project to which they have been assigned. And, all of them, all the managers report to a managing director.

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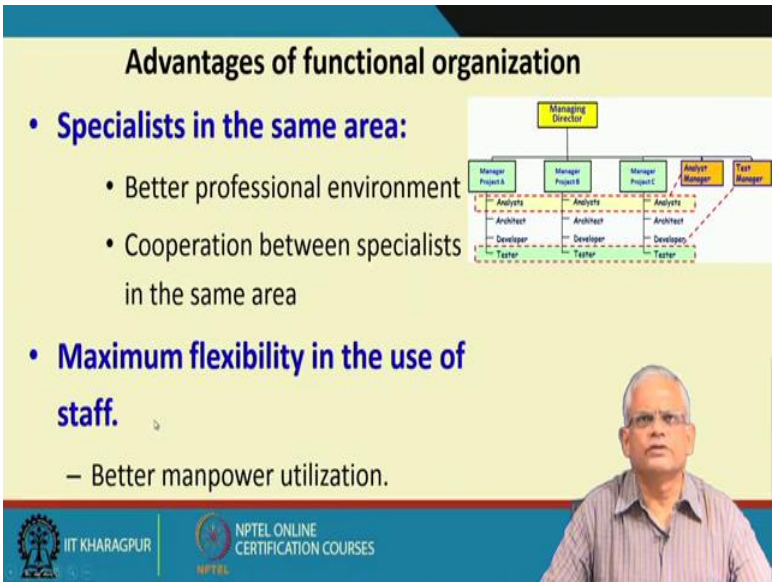

The advantage of the functional organization as we said that it provides better professional organization, better professional environment where the different designers, they can discuss with each other; the different analysts who form a functional group can share their ideas; there is a cooperation between specialists in the same area, they can consult each other and so on.

And, one of the biggest advantage of the functional organization is the flexibility in use of the staff. As you are saying that as and when required the specialists are requested to complete their work and go back to their functional group, but in a project organization as we will see shortly that the team members are assigned permanently to the project. And, therefore, somebody who is a tester, he may not have work all the time. Somebody who is a designer once design is complete what does he do?

Compare to a project organization where the team members are there throughout the project, in a functional organization they come and go and this provides better manpower utilization. We will see more of this as we proceed in the other lectures.

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The disadvantages of the functional organization is that each team member has two managers to report to. They might give different directives and there can be conflict. And, also the project manager is control over the team decreases because he might ask his team members to do something and they might say that see we have other work to do as given by our functional manager. We will just stop at this point because the lecture hour is getting over and we will continue in the next lecture.

Thank you.

Software Project Management
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Lecture - 33
Organization and Team Structure

Welcome to this lecture, in this lecture we will discuss about the Organization Structure and the Team Structure.

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In the last lecture we had said that if you visit a Software development organization; the first thing you will notice is that the organization is structured into teams, it has many projects and different teams are carrying out the projects. The way the organization is structured into teams is called as the organization structure.

On the other hand if we look at the individual teams and how the team is structured that we call as the team structure. We had started to discuss about the organization structure that is if you walk into a software development organization, what will you notice, what are the different structures that the organization might have, how the teams are organized in the software development organization.

Broadly you will notice that any software development organization will have one of the following types of organization into teams; one is functional organization, project organization and then the matrix organization.

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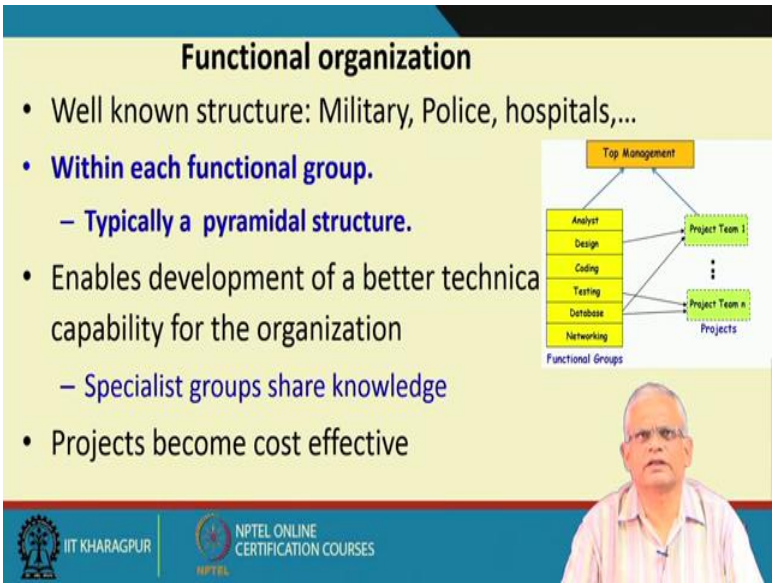

In the functional organization we will find functional group or departments. The projects come and go, the projects are started projects dissolve, but then what is permanent; is the functional groups are departments. There is a department for analyst, department for designers, coding, testing, database, networking and each one of these has a functional manager each department has a manager called as the functional manager.

Now, as the projects are formed a project manager is appointed; and the project manager plans the project and finds that at different points of time of the project needs different expertise. And then consults the corresponding functional manager, for example, might initially requires some analyst. Request the manager corresponding functional group manager for, the required number of analysts and as they complete their work they go back to their department and the manager would requires some designers from the design department.

Even though in a typical software development organization this kind of set up is bit rare, but then there are many places where this is a very well accepted structure. For example, military, police, hospitals, even educational institutes. In a military, the hospital is a department there are only doctors there; the doctors might be called up on for a

specific project. There may be administrators administration unit, there may be transportation unit, logistic units, store house, etcetera. Similar, is police and also the hospitals there are various types of specialists and these are organized into departments; for example, the surgery department, the medicine department, administration department and so on.

But if we look at the functional groups themselves; for example, let us say the design. The design department inside that if we look we will find the pyramidal structure where there is a manager is assisted by some senior designers, in turn the junior designers they report to the senior designer, the senior designer reports to the functional manager for the design unit and so on. One of the major advantage of this is that all designers they are in a department and therefore, they interact with each other, enhance their technical knowledge, educate each other and so on.

The specialist groups since the testers are all there any new tool, any new testing technology they share with each other. Another major advantage of the functional organization is that; the projects become cost effective because each time a project needs certain number of persons gets them and then they return to the respective department after they complete their work. Just imagine if they have to employ a person throughout the project duration, the designer initially he has work; but later what do he do they might ask him to do something else or he may idle.

So, as far as the cost effectiveness is concerned, functional organization is very cost effective because when a specific expertize is required it is procured and retuned as soon as they complete their work.

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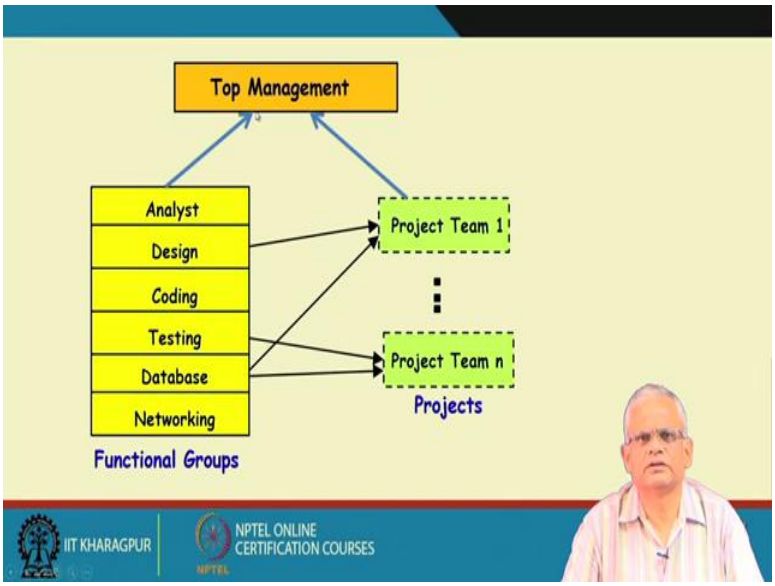

So, here the project team has taken designers and database experts who are working under project team. They would be returned as soon as they complete their work and possibly the project manager would need other expertise; and here both the functional groups and the projects, the managers here they report to the top management.

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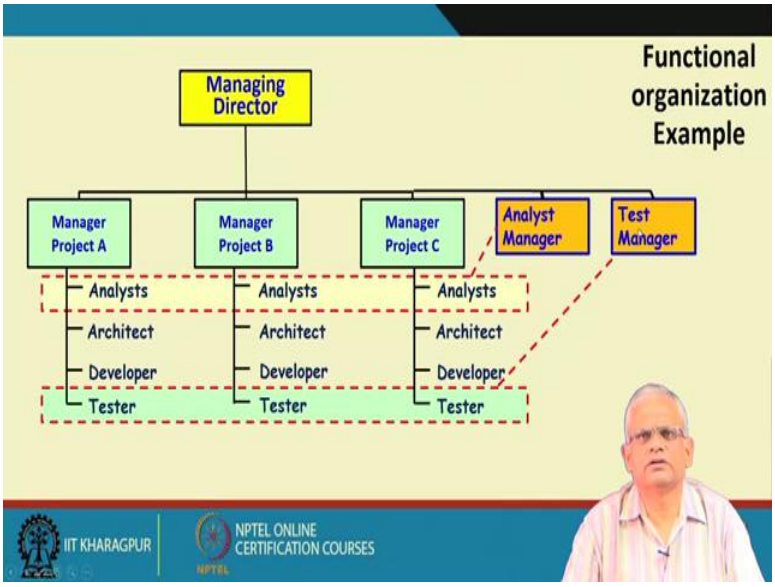

This is another view of the functional organization where each project has got some analysts, architects, developers. And they have two reporting; one is to the manager of the project and other is to the functional group manager.

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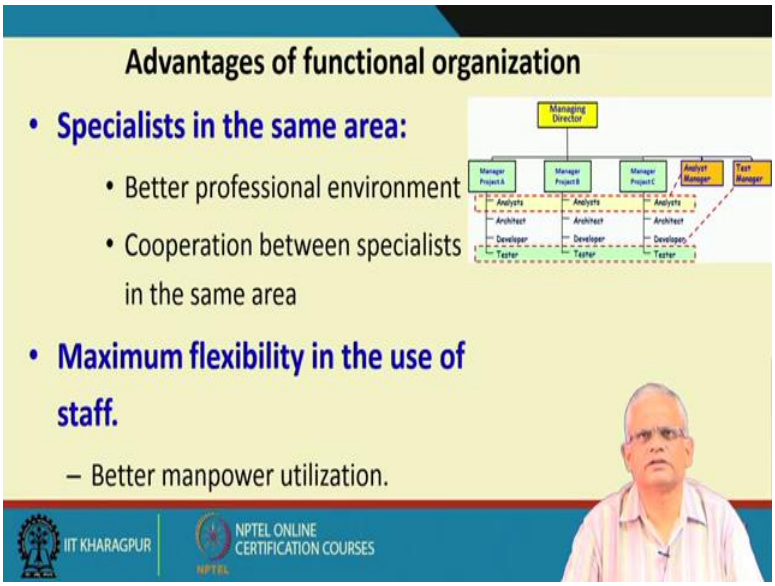

The advantages of functional organization is that better professional organization, better professional environment, cooperation between specialist; ones specialist cannot solve a problem, seeks help from the other specialist because after all they belong the same department. And possibly the biggest advantage is flexibility in use of staff, as and when manpower required get them from the department return them to the department when their work completes. The idling of the staff is minimized here, unlike we will see the other organizations.

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The disadvantage of functional organization if we think of it, one is that each employee reports to two managers; one is the project manager other is the functional manager. And what if they give contradictory directives; the functional manager wants him to work in two projects and the project manager says you cannot and so on.

So, this sometimes is a stress on the workers; the project manager control decreases because he may not get exactly the persons he needs. The functional manager assigns any person that he thinks suitable. Another big problem here is that a designer keeps on doing design, a tester keeps on testing, a coder keeps on coding from one project to another project and so on. So, there is a lack of job variety.

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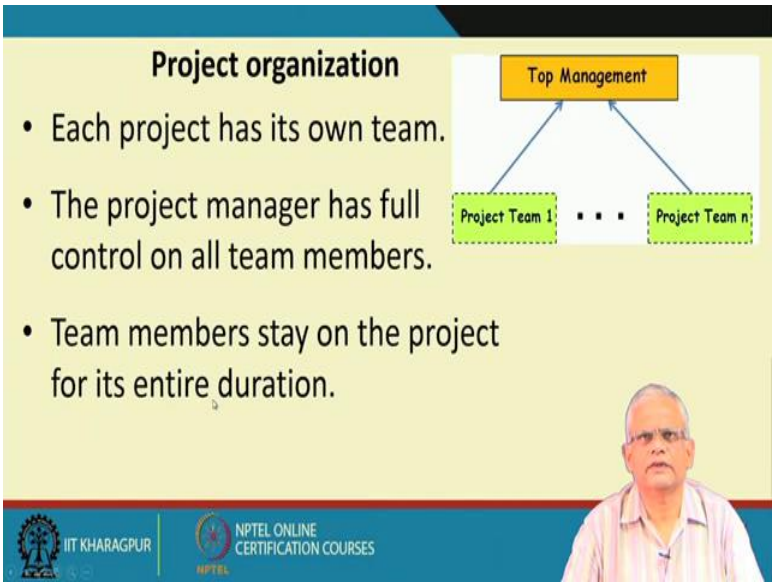

Possibly the most popular organization in software development companies is the project organization. Here when you walk into the organization you do not see the departments the department of designers, department for coders, department for testers and so on, the testing department, coding department and so on. Here what you see is the projects. The company as a whole is structured into number of projects and the workers belong to the project, they fully report to the project manager and therefore, the product managers has full control on the employees.

The team members are selected for a project and they stay on for the entire duration; but then the question is that suppose we have an analyst and then he is with some project team, initially he does requirements analysis and so on. But after that what does he do?

Ok; he also does design, also does coding testing and so on. So, here in a project organization typically each member does a variety of roles and therefore, there is a job variety here. A coder need not be coding always, he will also have option for testing, design and so on.

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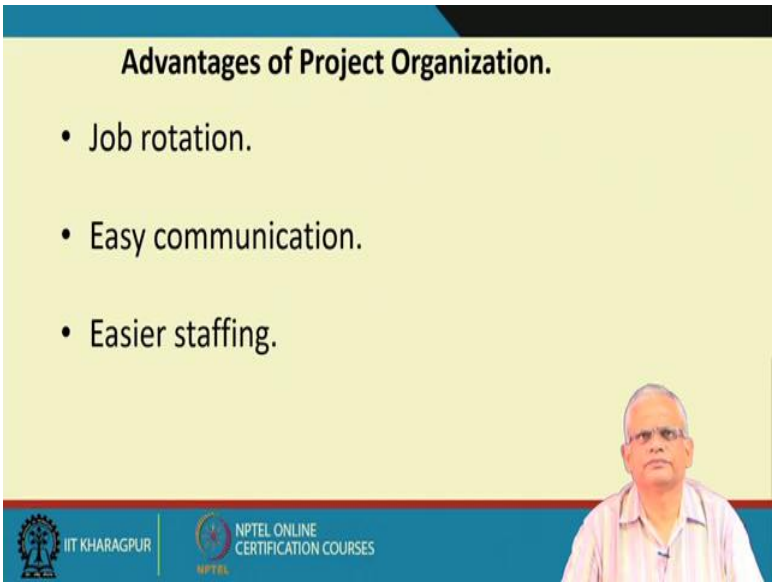

And that is a big advantage of the project organization, is the job rotation. Because there is a lot of manpower turnover in software industry. If you employ somebody as a coder he will always try to change his job and say; that may be I would like to become a designer or a analyst or a tester why should I just keep on coding all the time. That problem would arise in a functional organization, but not in a project organization because a different points of time in the same project he will be doing different activities. So, this is the big advantage in the project organization leads to job satisfaction and less manpower turnover.

The other big advantage is ease of communication. Imagine that in a functional organization the designers have designed and have gone back to their department respective department. And now the coders have done their work gone to the coding department back and the testers are working on it; and they do not understand some code some design whom do they consult, they only have some may be paper documents with them.

Communicating with them with the persons who actually designed, who actually coded may become difficult; because they might be working on an other project and when the telephone or something and try to meet and find out, they may say that we are busy on another project. So, here communications of those who do the different parts of the project become difficult.

Another big advantage of the project organization is ease of staffing. The manager has the option to select the best people for the project and they continue to stay on the project; they develop domain knowledge and the project. On the other hand in a functional organization when the project manager requests for a designer after the analysts have completed; the design manager might say that see right now we do not have designers they are all busy in respective projects and they will be given to you after 2 weeks. So, that problem does not occur in the project organization. So, this is has much easier staffing compared to the functional organization.

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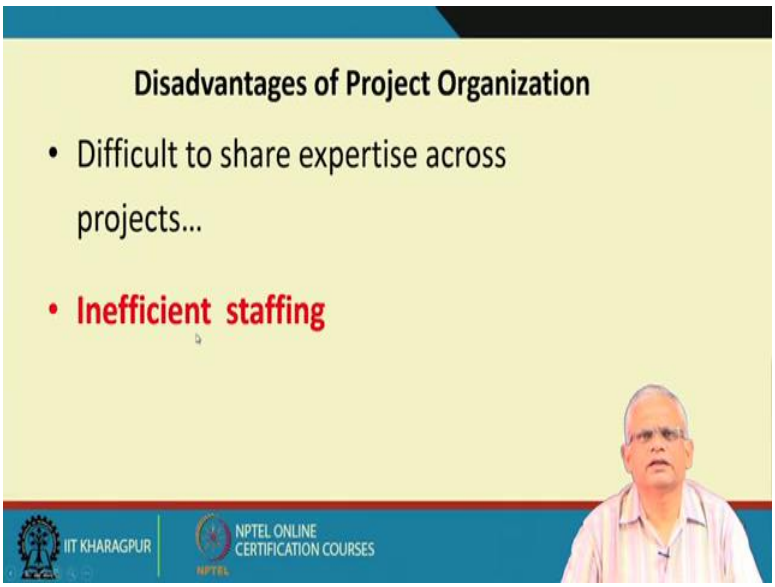

But let us look at the disadvantage of the project organization. Here there are team members continue to stay on a project, the things they learn from a project becomes difficult to share across other projects. Because the projects may run for 6 months, year and so on and they continue to work on a project and by the time they move to different project, they might have forgotten what they learnt in project.

But possibly the biggest disadvantage is inefficiency; in any project it is well known that to start with need few people in the project, in during the requirement analysis just 1 or 2 persons, during design may be 3, 4, during coding may be 10 and during testing may be 20. So, the manpower overtime in a project is not constant, it varies with time and typically given in the form of a Rayleigh distribution. But then in a project organization, the project manager appoints the project personnel at the start of the project and therefore, by definition the project organization the manpower in the project is constant.

If the project manager employees ten people, initially they have very little work, but then during design also not everybody is busy and then during coding everybody becomes stressed, there is not enough hands everybody works overtime. And during testing the situation become still worse and therefore, stress is inbuilt into the project organization; initially everybody relaxes and as the project picks up the members are under stress. Initially there is idling; later there is requirement of overworking and so on.

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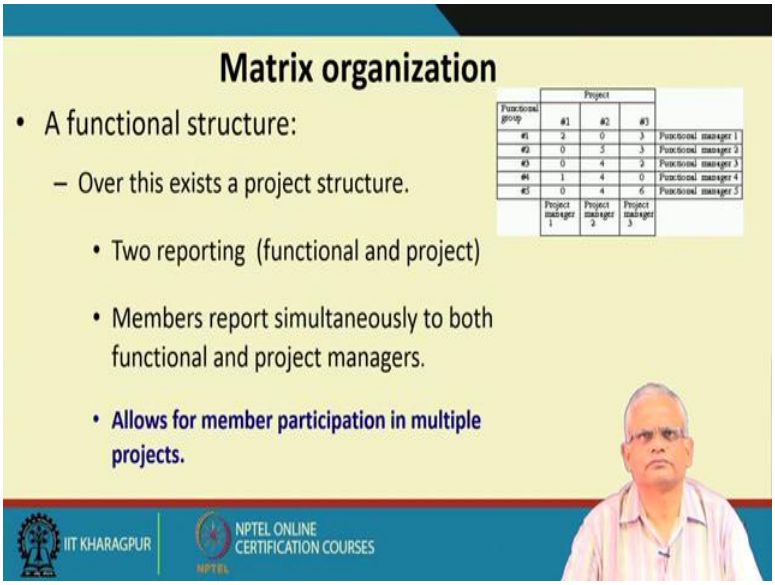

The third category of organization structure is the matrix organization. Here essentially it is a functional structure, you can find here the functional groups here, these are the departments; but then over the functional group there is a project structure you can see the projects here.

So, here again, just like a functional organization the members report to both functional group manager and the project manager. You see here that the functional group one

which may be the analyst group, there are two persons who working on project 1, no one is working on project 2 and 3 working on project 3. Here the members reports simultaneously to both functional and the project manager. But one advantage of this is that, a member in a functional group may work on two teams that is also becomes possible.

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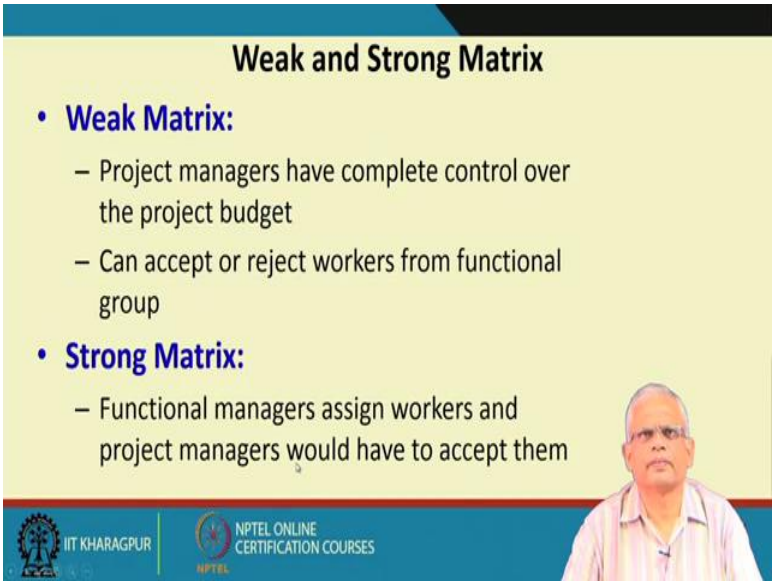

A matrix organization can be either weak or a strong matrix. In a weak matrix organization it is the project manager who has more control over the project budget, over the team members and so on. For example, he might choose a specific designer from the design department; and if the functional manager the design manager gives a different designer he may have the option to either accept or reject from the functional group.

On the other hand in a strong matrix; the functional managers have are much more powerful, they decide whom to depute for a specific project and the project managers they have to accept them.

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Let us look at the advantages and disadvantages of the matrix organization. Disadvantage is just like the functional organization here each members has multiple bosses to report, the project managers have reduced control as compare to the project organization.

And in a strong matrix organization the functional manager often finds that different projects are facing problem and some projects are comfortable. So, the functional manger would like to shift manpower. The competent designers he might take from one project half way replace him; take him to another project which is facing difficulty and so on.

So, it is observed that in a strong matrix organization there is a frequent shifting of workers in a firefighting mode. Advantages just like the functional organization seems the experts belong to same departmental group, they share knowledge and also better professional growth path; because designer will become a senior designer, become a design manager and so on.

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Now, let us look at the team structure. If you again work into different software development organization and look at the individual teams, you will find that the team is organized in either of the three ways; the democratic team, chief programmer team and the mixed organization. And we will see that each of these team structure has it is own advantages and disadvantages and depending on the project a specific team structure may be advantageous.

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First let us look at the democratic team. These are small teams; we will see that for a large team the democratic team structure becomes very very inefficient nobody would like to have a democratic structure for a large project. Here as a name implies all technical decisions are based on consensus. Everybody can give his opinion on every issue, they talk to each other and here there is a manager who provides the administrative leadership.

But the technical part is supervised by a technical leader; but the technical leader is one among the team member who is elected by rotation. And if you look at here since everybody is consulting everybody, there is lot of time is goes for discussion. One good thing is that they can brain storm come up with good solutions, but then the other thing that it wastes lot of time and specially; if the team size is large, you can see that there are for a team size N there are N square communication path. So, it may become inefficient, too much of time may be spent in talking to all the team members and therefore, it is rarely used for large teams.

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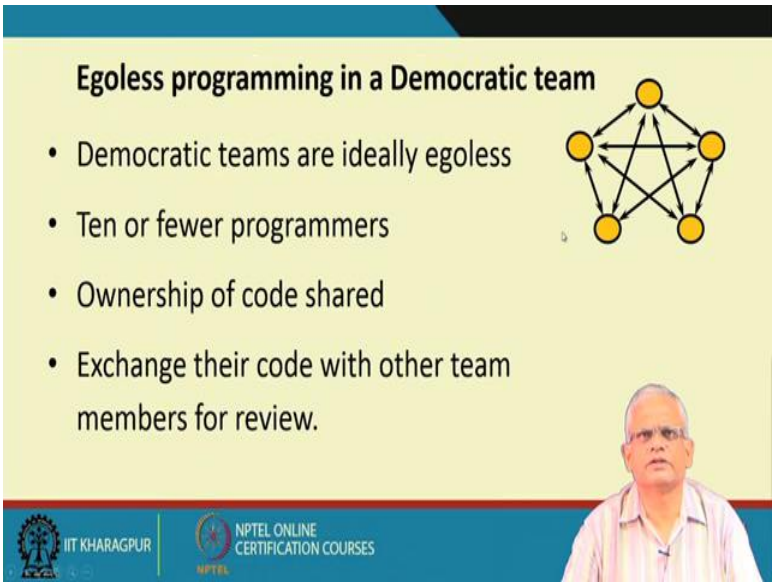

In a Democratic team ideally there can be a egoless programming environment. In a typical project every employee there, every worker has some knowledge the work that he does he is the expert. The others if you ask them that what happened to the database aspect then they may say that we do not know let the corresponding person come he will

explain. But in a democratic team these are egoless teams; that is no one owns anything there is no ownership we can't say that only he knows or one person knows.

Here the ownership is shared each team member whatever work he does is reviewed by other members. So, everybody has knowledge about every aspect of the project. This is a big advantage of the democratic team when we have egoless programming.

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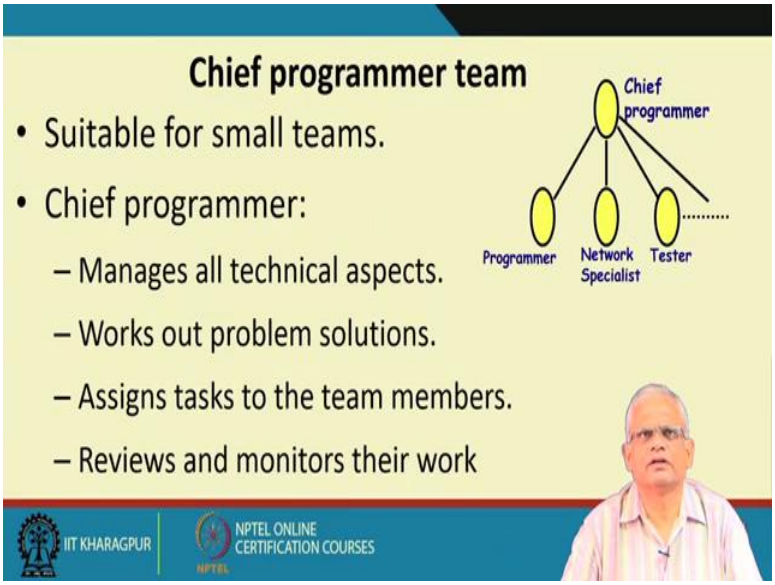

The third type of organization is the chief programmer team. This is again suitable for small teams where there is a chief programmer; he is responsible for all high level decisions, all technical decisions, works out the overall solution and gives small pieces of work to the different team members. And then reviews the work that is completed; monitors and reviews the work and then integrates them. We will see that this is an efficient organization for very simple problems; but then for challenging problems this may not be the right structure. We are almost at the end of the time; we will stop here and continue in the next lecture.

Thank you.

Software Project Management
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Lecture - 34
Team Structure (Contd.) and Risk Management

Welcome to this lecture. In the last lecture we were discussing about the organization structure and team Structure. In the team structure we were discussing that each team has some structure, the way the members do their job report to somebody there in the team and so on. We found that there are three major team organization; one is the democratic team suitable for small projects where there are about 7, 8 persons.

Rarely used for large projects, because the decisions there is collectively taken everybody consults everybody. It is suitable for research type of projects where there is lot of innovation, new ideas are required; but then it is inefficient in the sense that lot of time goes in discussions, meetings and so on.

We were looking at the chief programmer team. In the chief programmer team, these are also suitable for small projects where there is one person designated as the chief programmer. He is the expert for the project, he is in overall charge of the project, provides the high level solution, divides the project into small pieces of work, assigns to the team members, helps them the team members monitors and supervises them and then gets the work from them and integrates.

This is suitable for very simple programmers, it is a very efficient team structure for simple problems because hardly much time goes in discussions meetings and so on. Let us look at the chief programmer team first; we were already looked at the democratic team in the last lecture. We will start with the chief programmer team and then we will look at the hybrid organization.

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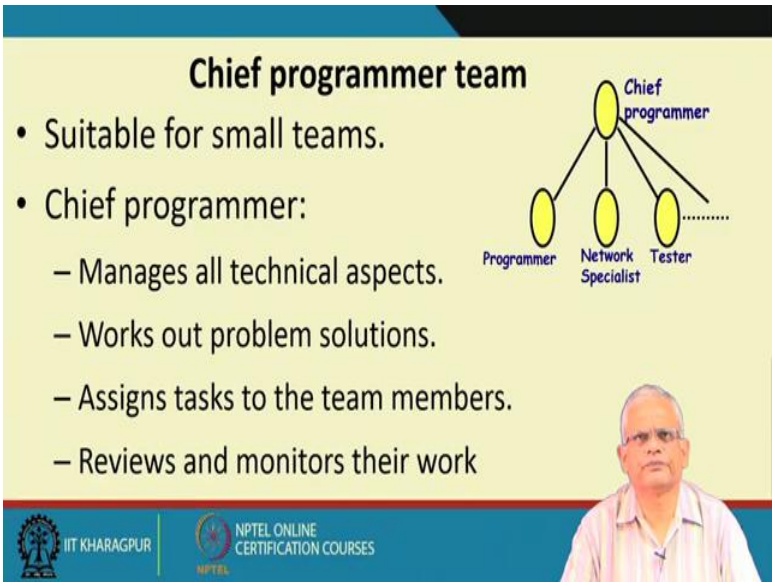

The chief programmer team as the name says we have one the programmers designated as the chief programmer. All other team members report to the chief programmer; the chief programmer is responsible for the project. He is the technical lead and also the managerial lead; all other team members report to him, he visualizes the solution, works out the design, explains the work to different team members. For example, the programmers he might explain them what coding exactly they would have to do, to the network specialist he may say what network solution is needed, to the testers he would say that how to test and so on, what aspects to test and so on.

So, here the entire project solution is with of chief programmer. The other team members have only partial views of the project, they do not fully understand what is going on; but whatever has explained the small tasks that has been explained to them they only know that and they do that. The chief programmer monitors the work, provides any help that may be needed further explanation and so on. And as the different team members complete their work, reviews the work and once these are done satisfactorily, integrates and slowly the full system develops.

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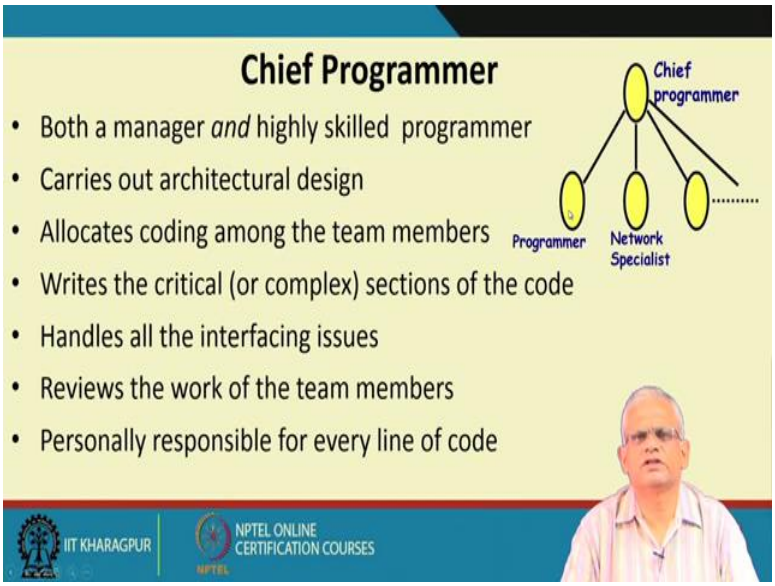

Here the chief programmer is the main person responsible; he is both a manager and a technical expert, carries out the architectural design detailed design and then allocates the coding among the team members helps the team members and the critical parts of the code himself writes, handles all interfacing issues how to integrate the different parts, reviews the work of team members, anything deficient explains them and asks them to do those things and the chief programmer alone understands the complete project every line of code, every design everything is known to the chief programmer. All of the team members have only partial views, only small parts of the project they know.

As you can see that one of the major problem here is that what if the chief programmer becomes indisposed. The project would stall, but then this is a very efficient way of doing simple projects, because hardly there is much time wasted in discussions and so on. The chief programmer does everything, effectively supervises gets all the work done on time, he knows how much time it takes because he is the expert there and therefore, this is the fastest way of doing the simple projects.

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But what about large projects? Neither the chief programmer team nor the democratic team is suitable. We said that both of these work for small projects. The chief programmer cannot supervise too many other programmers; the democratic everybody discusses with everybody and therefore, as the size of the team increases, the number of communication path increases is N square.

And quickly it becomes very inefficient for even moderately large projects. And for this reason the hierarchical team structure is used for large projects. This is as the name says this is the hierarchical organization; there are junior programmers who report to senior programmer, the senior programmer report to a project leader, the project leader reports to a senior leader and so on.

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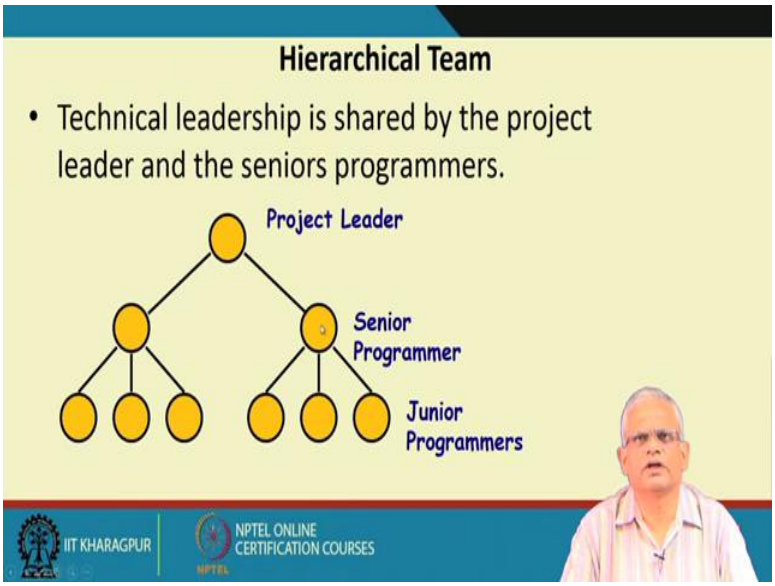

So, as you can see this is the tree structure where the junior programmers for a specific part; the project leader assigns work to the senior programmers. The different parts of the project are done by different senior project programmers, they have their own group of junior programmers and they talk to each other.

See here there is a democratic setup at the bottom level and a chief programmer set up at the top level. The technical leadership unlike the chief programmer here it is given by the project leader and the senior programmers.

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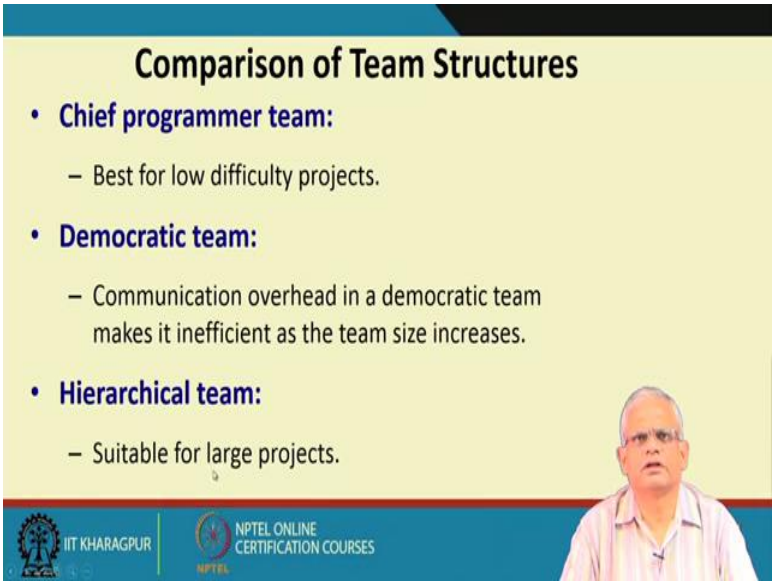

The chief programmer is the best structure for simple low difficultly projects. The democratic team, its suitable for research oriented, innovative projects, but the communication overhead increases rapidly with team size; and therefore, democratic team is also used for small projects, the hierarchical team is used for large projects. We will now discuss about the Risk management issues; how does a project manager carries out risk management?

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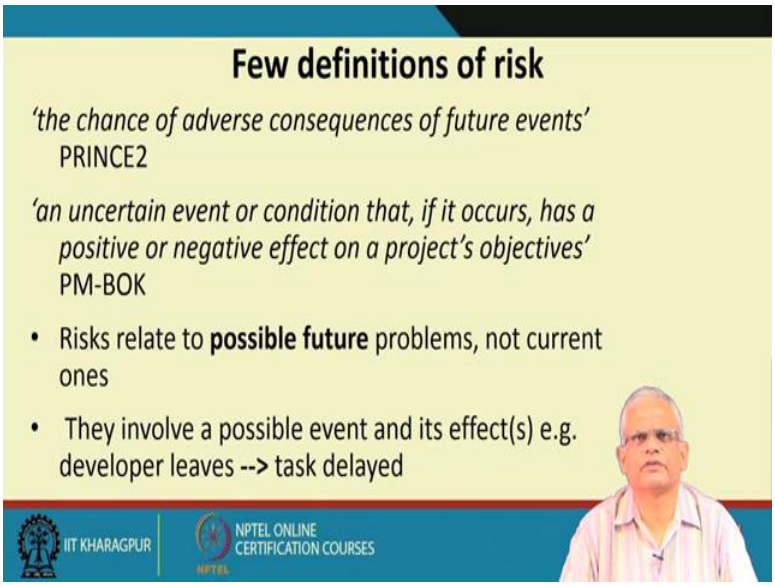

The first thing that to know here is, what is a risk? A project suffers from many types of risks; in the PRINCE 2 risk is defined as the chance of adverse consequences of future events as the project progresses many events occur and the risk is the adverse consequence of a future event.

For example, one event may be that a team member leaves that is a event and that has an adverse consequence; that who will do his work. Another future event maybe that the budget becomes very constraint, the customer faces financial difficulties and is not able to provide the funding and so on. Now, let us look at the PM-BOK definition Project Management Book of Knowledge; the definition there is an uncertain event or condition that, if it occurs, has a positive or negative effect on a project's objectives.

So, this is again similar definition about an event which may or may not occur, it is an uncertain event or condition; if it occurs has a positive or negative effect. So, there is a positive risk; the positive risk is that something good happens. For example, the

customer may say that you can take an additional 2 months because our actual product launch is delayed, for your part you may take another 2 months, so that is a positive risk.

The negative risk maybe that the customer says, that you need to complete it by one month early because our project launch has been prescheduled. But in all these definitions the risk relate to some events, that may occur in the future not the current one's. So, the risk is some event that occurs in the future and the problem that may arise because of that.

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There are many many types of risks that a project may suffer; one possible categorization of the risks is the market risk. The market risk is the project complete successfully, but then the market conditions have changed and the product is not accepted in the market; may be the competitors have come up with the better product, maybe the technology has changed and it is not no more acceptable in the market, for the specific type of technology it was being developed it has further advancement and the product is not acceptable in the market. Those are all market risks examples of market risk.

Financial risk is risk of financial commitment by the customer not being honored; the customer gone bankrupt cannot pay for the project halfway through the project and so on. The technology risk is that the project assumes some technology and then the project proceeds based on that technology; but it may so happen that better technology gets developed and comes into use before the project completes. So, the technology

obsolescence is a example of a technology risk. And then there are project risks like project personal leave unable to find replacement for a specific role, the schedule gets delayed and so on.

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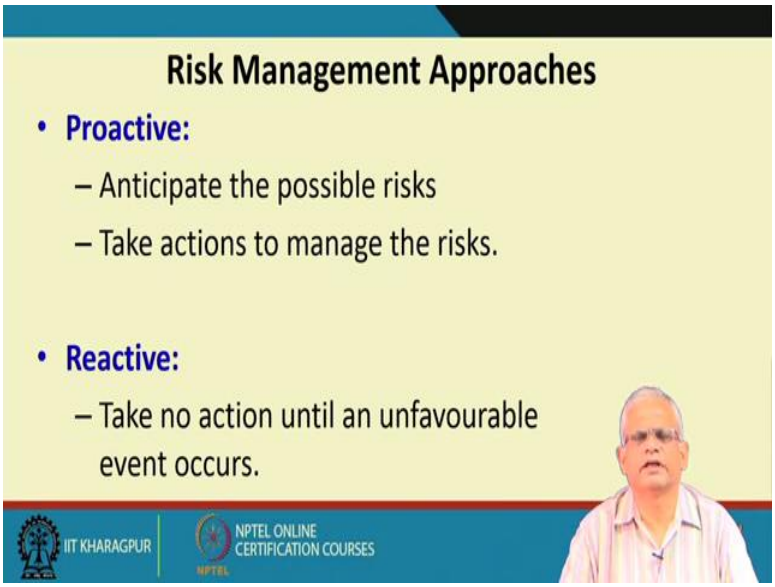

Now, let us look at the risk management approaches, there is hardly any project that does not have risk. Every project has several dozens of risks that it faces and it is a important job responsibility for the project manager to handle the risks.

And broadly two options are available to the project manager. One is the proactive approach, here the project manager anticipates sits down and thinks, what are the risks that may the project may be susceptible to, anticipate all possible risks and also plan what actions to take. If the risk actually becomes true and then as the risk becomes true, take the planned actions and manage the risks.

This is the one which is very popular among the managers the proactive approach; but then there is a reactive approach, where the project manager cannot even anticipate the risks and therefore, takes no action until the event occurs. Typically the project managers take the reactive approach that as the risk occurs because these risks are not predictable they cannot be foreseen by the project manager take no action until the unfavorable event actually occurs.

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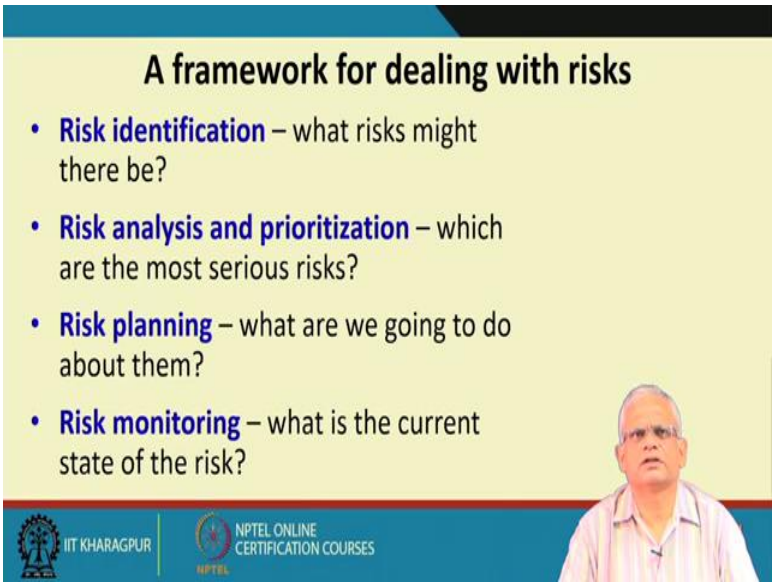

But in the proactive approach we said that in the reactive approach just handles the risk as they occur; but in the pro active approach how does the project manager go about dealing with risks? The first thing of course, is to anticipate the risks. Risk identification that is what risks can be there in for this project, notes down all the risks anticipates, finds out what technical risks can be there, what project risks like people leaving the project what will happen, schedule delayed what actions to take and so on.

So, the first step in dealing with a risk is to identify all possible risk at the start of the project. And once the risks have been identified need to analyze the risks, what will be the impact of the risk, which is a serious risk, need to prioritize all the risks; that which risks are the most damaging. And then the risk planning; what to do, when the risk actually becomes true and then as the project proceeds, risk monitoring is the risk probability becoming higher and so on, what is the current state of the risk.

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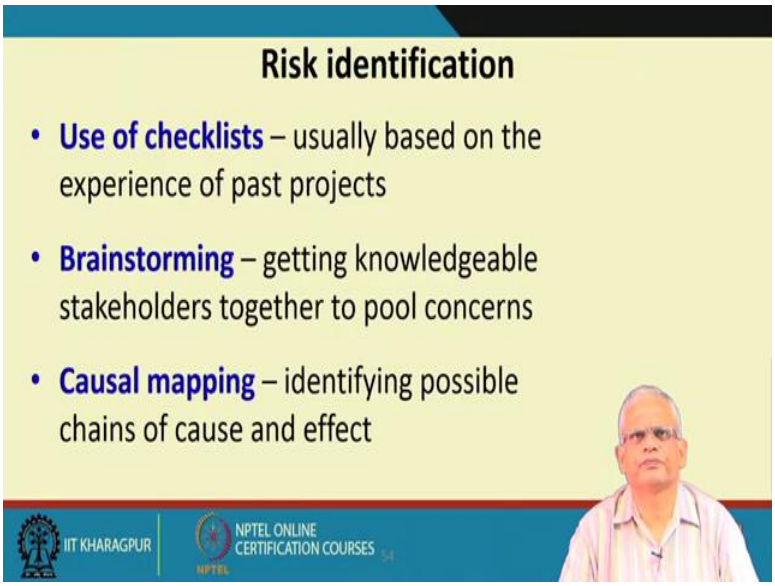

The first thing in this framework is Risk identification. How does the project manager identify the risk because these are future events and these are uncertain events; how does the project manager anticipate the risks? One is use checklist. Find out the common risks common type of risks that have occurred across different projects that he has worked or the projects that were running in the organization. And finds out what were the types of risks that the other projects faced and then checks out whether such risks are likely to be relevant for the project at hand.

The second way the risk can be identified is by brainstorming; get the team members, other stakeholders and then get their concerns and those at the risks. And the third type of risk identification is the causal mapping here identify the possible events that may occur and the affect that will of these events.

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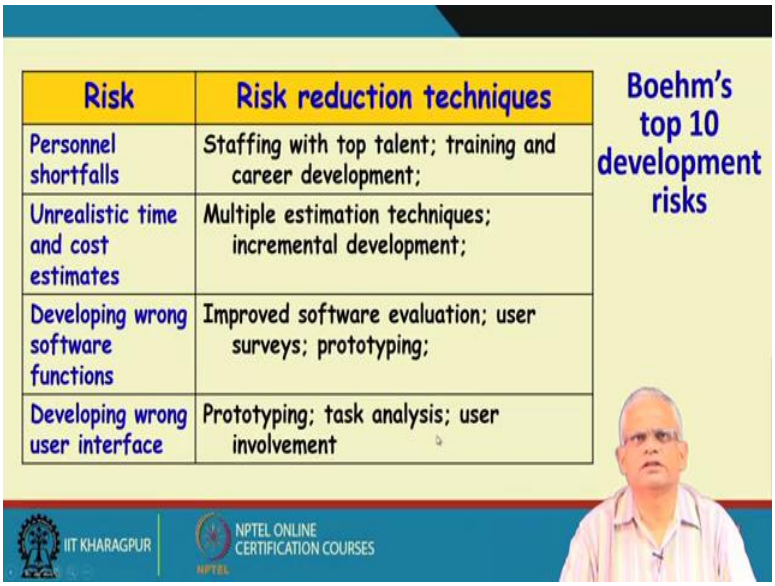

It is important for the project manager to know the different types of risks that fall on various projects and what are the possible risks reduction techniques that are suitable for these risks. Boehm has identified 10 most frequent risks that development project suffer; and therefore, this is quite in instructive for every project manager to identify what are the most frequent risks that the other projects have suffered and what are the best ways available to handle those risks.

The first risk is of course, relates to the manpower; personal short fall, lack of competent manpower in the project. We have the project personal, but they are not able to perform, the project is getting delayed this is a major risk here. And Boehm suggest the corresponding reduction technique says that staff to start with the project get the top talent and also in the areas of the project provide training and career development.

Unrealistic time and cost estimates; the project manager could not really estimate the time and cost correctly and quoted a wrong figure to the customer. And now the customer says that, since you said that you will do within this budget and within this time, why you are not able to do it. To get more realistic, accurate estimate, the project manager needs to use multiple estimation techniques. And also one way to handle it by using incremental development because, making a long term estimate is very difficult it is error prone; but a short term estimate that is for one feature at a time is a much accurate way of estimating and that those two are the risk reduction techniques.

Developing the wrong software functions; the customer wanted something and finally, the development team develops some other functions, which the customer is not happy says that, we did not want it this way and that happens very frequently. And the risk reduction technique for this is improved software evaluation; user surveys trying to find out what exactly the customer needs; prototyping and getting it evaluated by the customer these are the risk reduction techniques.

Developing the wrong user interface, this is also a risk similar to the functionality; here the interface the customer may not like; here again prototyping getting it evaluated by the customer; task analysis do a proper task analysis where identify for performing a specific functionality or task, what are the input and output that are required; user involvement in the project these are the risk reduction techniques.

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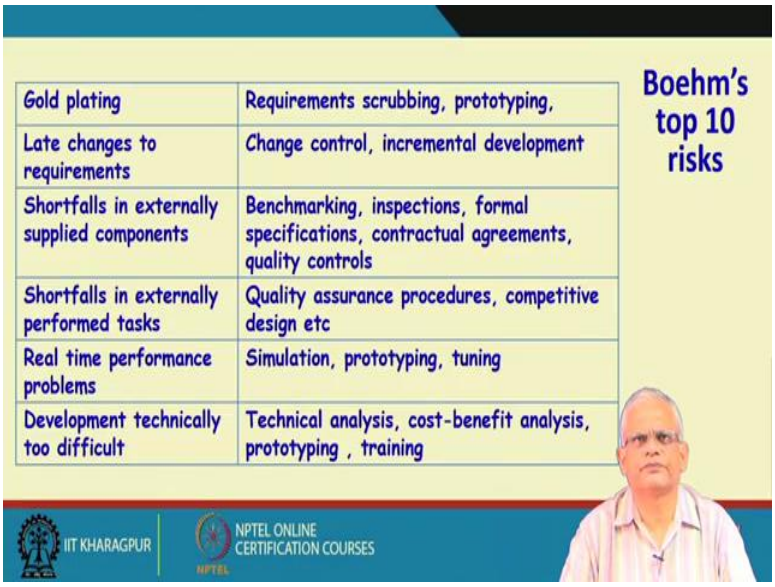

Gold plating; in gold plating the team members they think that some features may be appreciated by the customer, even though those are not really documented in the requirements. They start developing something thinking that, this work will be highly appreciated it will be very useful; but then it may so happened that the customer does not need them actually this is just a waste of time, it just delays the project, increases the cost. The risk reduction technique is requirement scrubbing, that is dropping any requirements that have been put as a later thought and are not actually required;

prototyping having the user evaluate a prototype version and say that whether it is required or not.

Late changes to requirements that is the customer may suggest some changes later; here change control if the change actually has to happen then we have to somehow have a procedure to effectively handle the change and that is called as a change control. Incremental development, so that the customer. Incrementally evaluates and the changes are not too much at the end, because the changes are troublesome; if they are to be done at the end once everything completes.

Shortfall in externally supplied components the projects do give out, contracts to other developers to other organizations and they may turn in inferior work products; here the risk handling is through benchmarking, inspection, formal specification, contractual agreement and quality control.

Short fall in externally performed tasks here it may not be a component, but a task which is performed externally; maybe let us say coding or may be testing and so on. Here the risk reduction technique is quality assurance procedures, competitive design, etcetera. Performance problems here the risk handling is through simulation, prototyping and tuning.

The development becomes technically too difficult; the risk reduction techniques are through technical analysis, cost benefit analysis, prototyping, showing it to the customer, evaluating by the team members, the prototype helps in developing the actual solution and then of course, training.

The project managers need to be aware of various important risks that the past projects are suffered and what were the risk reduction techniques. And these are the 10 most frequent risks according to Boehm and the corresponding risk reduction techniques he has identified; every project manager needs to know this. We are almost at the end of the time for the current lecture, we will stop here and continue in the next lecture.

Thank you.

Software Project Management
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Lecture - 35
Risk Management (Contd.) and Introduction to Software Quality

Welcome to this lecture, in the last lecture we had looked at some basic concepts on Risk Management we had discussed what is a risk different types of risk and a framework for risk management. Today we will discuss further on risk management and after that we will start discussing about quality management; let us get started.

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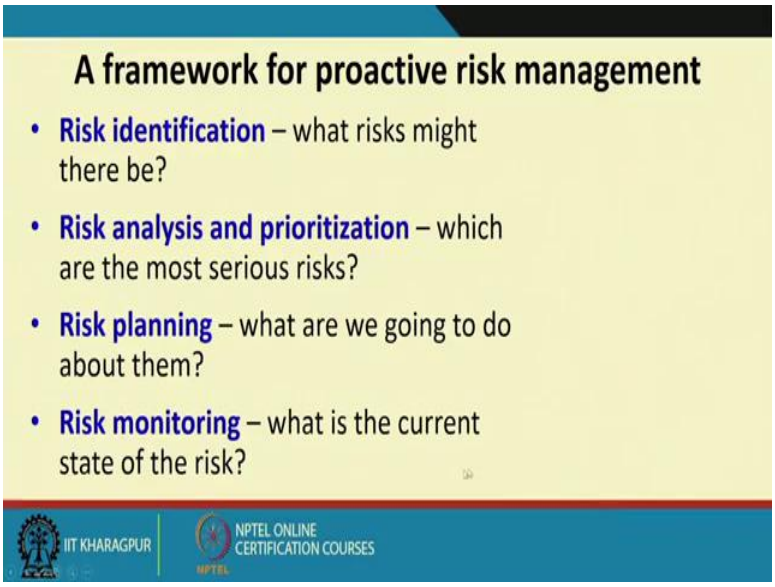

In the last lecture, we had said that there are two main ways for risk management; one is a proactive approach where we identify and make plans for a risk. In a reactive approach, we do not make any specific plans; possibly the risks are hard to anticipate we handle the risks as they appear during the project. In the proactive risk management, the first task is risk identification. We list down all the risks we had discussed some guidelines about how to go about identifying the risks and once we have a list of risks, we do an analysis of risks.

The analysis is in the form of what is the probability of the risk becoming true and what will be the cost implication of that risk. And based on that we can prioritize the risk the one which will be most damaging to the institute to the project that will be the most

serious risk; we need to make elaborate plans for that. In the risk planning we will try to see that how the effect of the risk can be either avoided or minimized.

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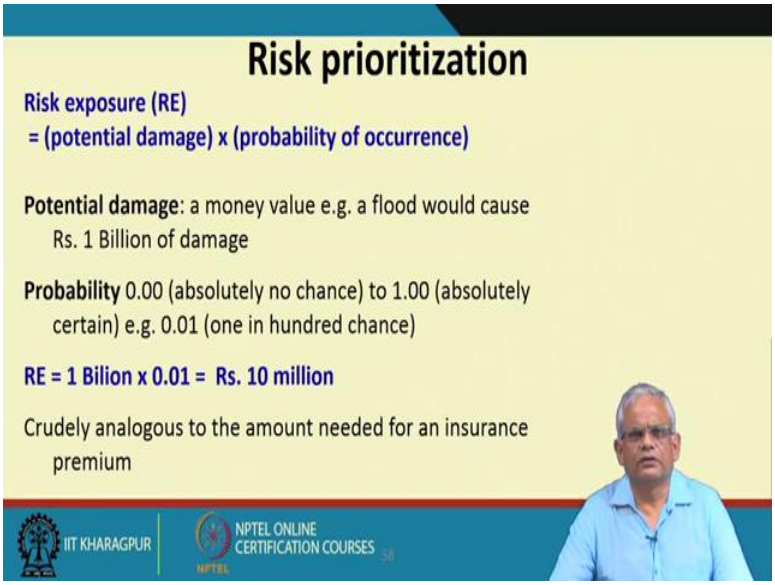

And finally, the risk monitoring where we see that what is the status of the risk. Once we have listed all the risks in a project to the project manager; we need to anticipate the risks. And once we have made the list of risks we need to analyze the risks; we mean analysis of the risk means that we qualitatively assign a probability of occurrence; it will be very hard to give a quantitative value for example, we cannot say that the occurrence of this risk is 66.5.

It will be hard and also we qualitatively identify or estimate the potential damage due to the risk. And based on this, we compute a Risk Exposure or RE for all the risks which is the multiplication of the potential damage and the probability of occurrence. The potential damage we assign a money value approximate money value for the risk.

For example, a flood would cause 1 billion of damage and the probability of the flood occurring is 0.01; that is 1 in a 100 chance a rare chance. And for this two values the potential damage is 1 billion and the probability of the risk occurring is 0.01; we can compute the risk exposure for this case is 1 billion into 0.01 which is 10 million. We can think of this as the insurance premium; the insurance is basically computed something similar way that what is the probability of an accident occurring and what is the cost

implications of the accident and that would give an indication of the insurance premium to be charged.

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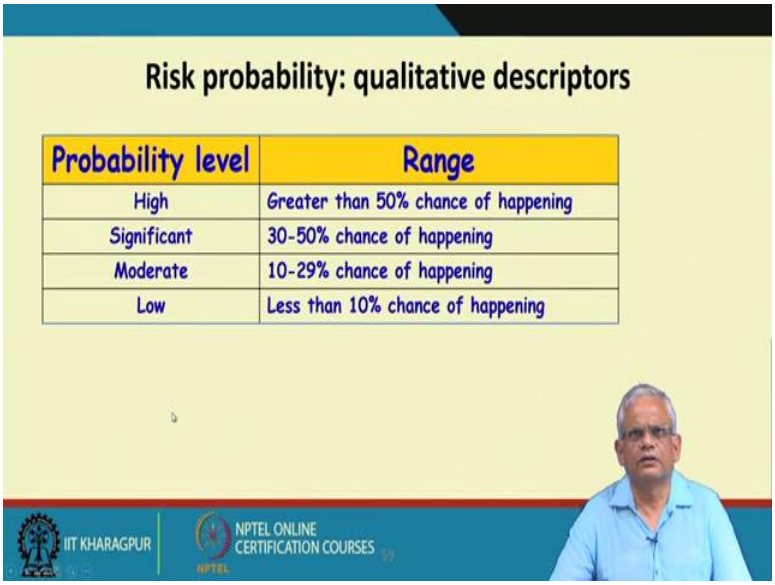

It is hard to give a quantitative value to the risks, often we can tell qualitatively that whether the chances are very high, the chances are significant or moderate and low. And based on this qualitative probability of a risk; we can give quantitative values. When we say that the risk is high it is something greater than 50 percent; we give a value greater than 50 percent based on the qualitative estimation of the risk probability. Significant is something like 30 to 50 percent, moderate is 10 to 30 percent and low is less than 10 percent.

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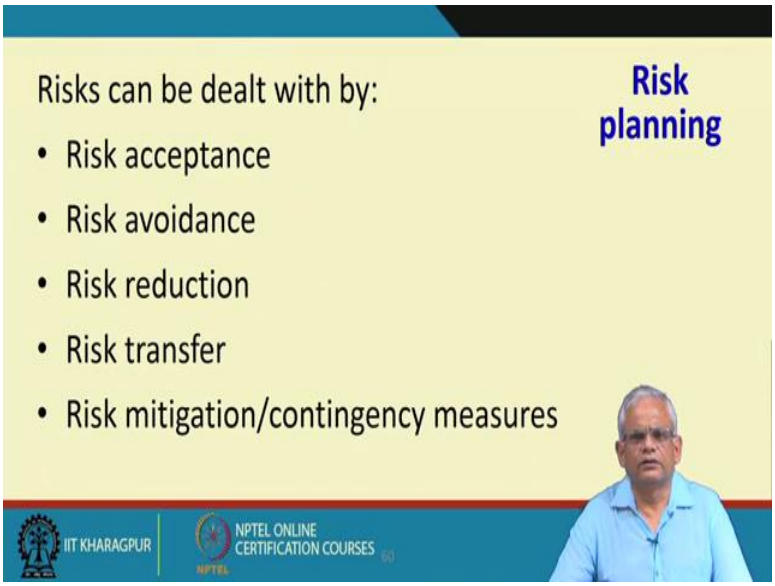

And once we have prioritized risks based on the risk exposure computed for various risks, we need to do risk planning for the most damaging risks. The various ways that risks can be dealt with a risk acceptance, risk avoidance, risk reduction, risk transfer, risk mitigation and contingency measures. In risk acceptance, we determine that the things that we need to do; to handle the risk is more expensive than the risk itself.

The risk acceptance is done for very low risk exposures; the damage will be very less and the probability is less; so this we do the risk acceptance. For these cases maybe the things that we need to do to handle manage the risk will be much more expensive than the risk itself; in that case it is better to have the risk acceptance. And even sometimes with risks with high risk exposure; it may so happen that we can do we cannot do much about that.

For example, that one of the key person leaving we cannot really employ a another person in the start the project that will be too expensive; it will be more damaging than the risk itself; so in this case we do the risk acceptance. Risk avoidance if it is possible to avoid the risk for example, if we find that one of the modules of the software is very difficult to develop and we are likely to have delay, bugs in this module; we may outsource it. So, here we have avoided the risk because we did not have capability and we gave it to some third party who have good capability of developing the module and we have avoided the risk.

Risk reduction in this case we make plans so that the impact of the risk will be less. For example, if we anticipate that some key persons might leave we might have good documentation; we might have review so that other members are aware of the work that is carried out by the key person. In this case, the new person who comes might refer to the documents already prepared and also discuss with others who have reviewed the work.

Risk transfer here we might buy insurance so that the risk is transferred to the insurance company we give premium but then the risk is now transferred to the insurance company. Risk mitigation and contingency measures we might prepare plans; if risk happens what contingency measures we can take.

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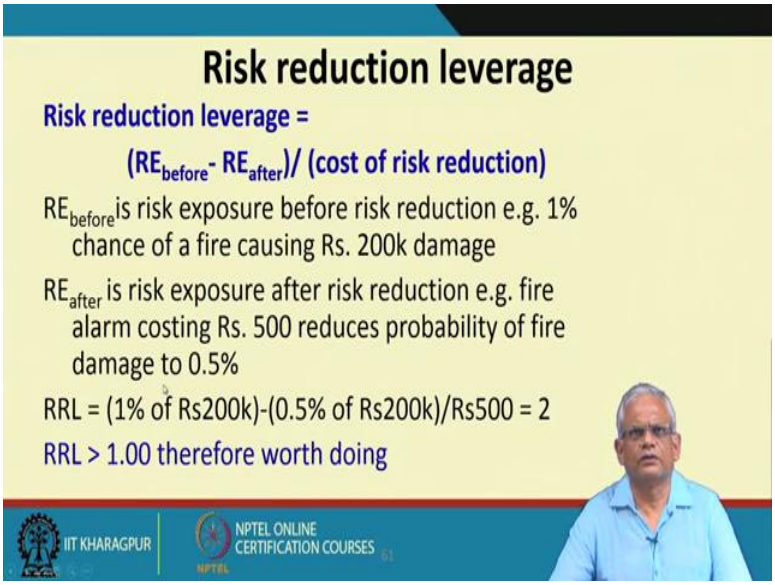

The risk reduction leverage, this is the effectiveness of a risk handling mechanism. The risk reduction leverage is a quantitative value for the effectiveness of a risk handling technique. Let us say we deploy a some risk handling technique and the risk handling technique; the cost of the risk handling technique and the reduction of the impact; we consider this here that the reduction in impact is; the risk exposure before deploying the risk handling technique minus the risk exposure after the risk handling technique is deployed.

So, this is the reduction in exposure after the risk handling technique is deployed and this is the cost of the risk handling technique. For example, that initially we had 1 percent chance of a fire causing 200 k damage.

So, the potential damage is 200000 and the probability of the risk occurring is 1 percent; so, the risk exposure before the risk handling technique is deployed is 1 percent of 200000. And now that we have installed a fire alarm which costs 500 rupees and now the probability of the damage due to a fire is 50 percent. So, 1 percent of 200 minus 0.5 percent of 200000 which is equal to 0.5 percent of 200000 which is equal to 100000; 1 percent of 200000 is 2000 minus 0.5 of 200000; so, it is equal to 100 sorry 1000 and 1000 by 500 is equal to 2.

And therefore, we benefit by deploying the risk reduction technique which is a fire alarm costs 500, but then the risk exposure reduces by 1000 and therefore, the risk reduction leverage is greater than 1. And whenever the risk reduction leverage is greater than 1, we say that this risk handling technique is worth doing it. With this discussion, we will conclude the risk management and we will discuss about Software Quality management.

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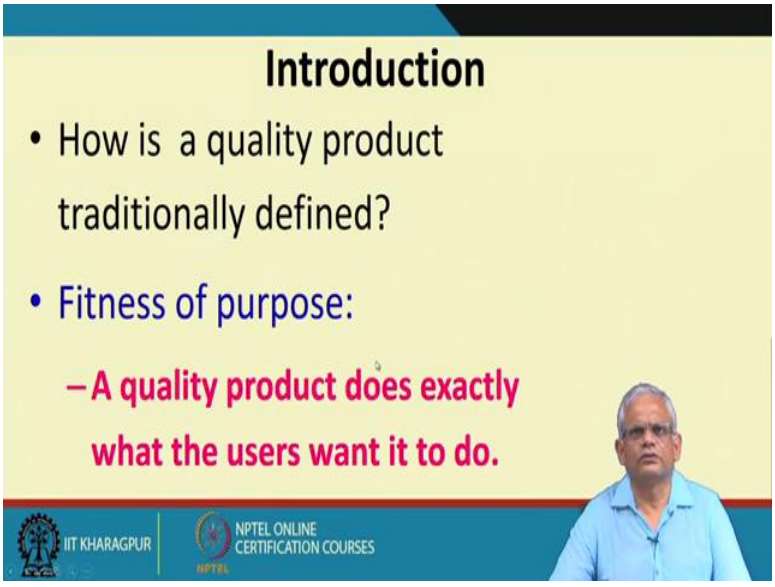

How do we define a quality product? Which one would you would you call as a quality product? Traditionally we say that a product is of quality; if it does what exactly the user wants it to do; this is the traditional definition of quality, it is called as fitness of purpose. For example, you want to buy a table fan and let us say you got one from the market.

And found that it runs smoothly and gives good wind; you say that it is a good quality product.

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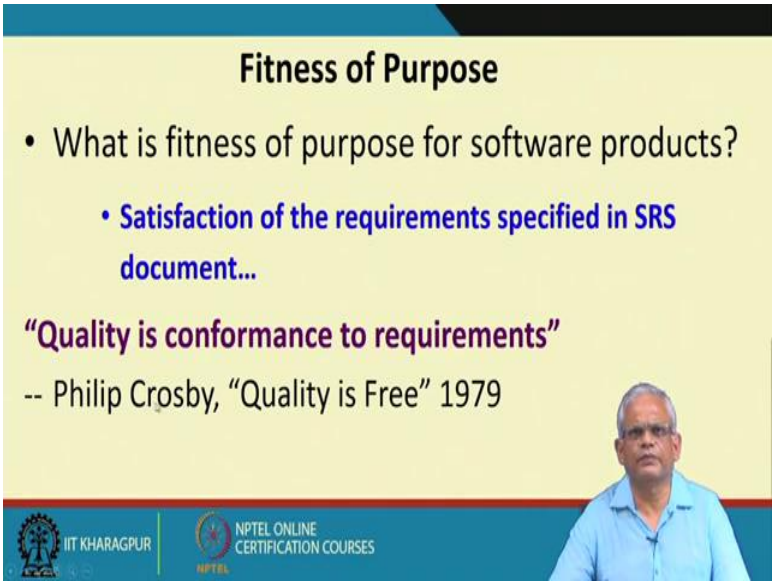

But for traditional items like a mixer, grinder, a table fan or a car, a bicycle the fitness of purpose is intuitively clear. But what is the fitness of purpose of a software product? When do you say that software product has fitness of purpose?

One way to define the fitness of purpose of a software product is to say that all the requirement specified in the SRS document have been satisfied. In other words, for a software product quality of the software product is essentially conformance to its requirements; every software is developed with some requirements in mind and as long as it meets all the requirements we might be tempted to say that it is a quality software. And its accepted by many including Philip Crosby in his "Quality is Free" book.

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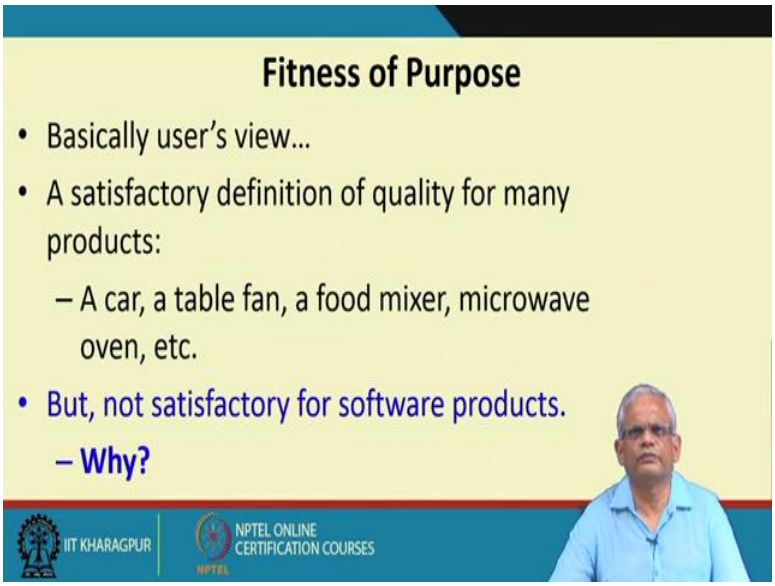

But to think of it satisfaction of the requirements is basically a users view; this is something similar to finding the quality of a car, table fan, food mixer, microwave oven etcetera. But then for software products; this is not really a good definition of quality, but then what is the difference between a table fan, food mixer, microwave oven and a software; why is it that fitness of purpose even though it is a good definition of a traditional product it is not a satisfactory definition of quality for software products?

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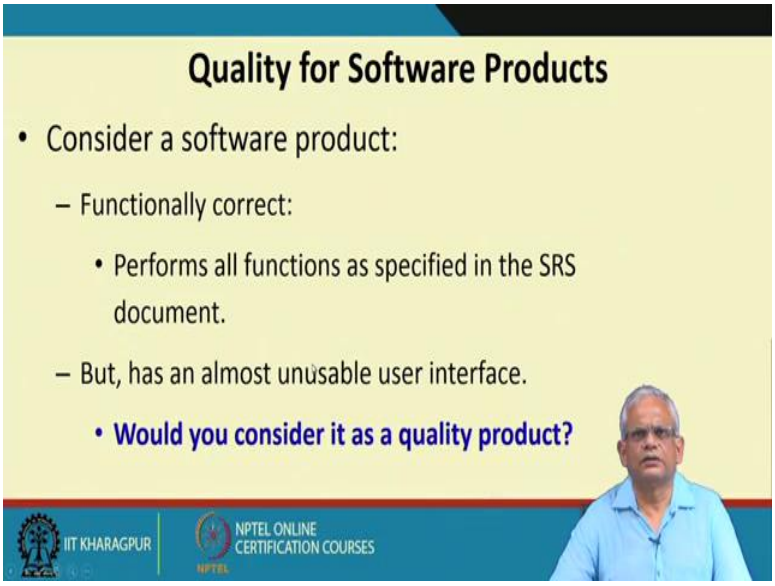

The answer is that for software product, functional correctness is only one aspect of the quality there are many other aspects. For example, what if, its functionally correct, but users find it almost impossible to use very difficult user interface; we can't call it as a quality software.

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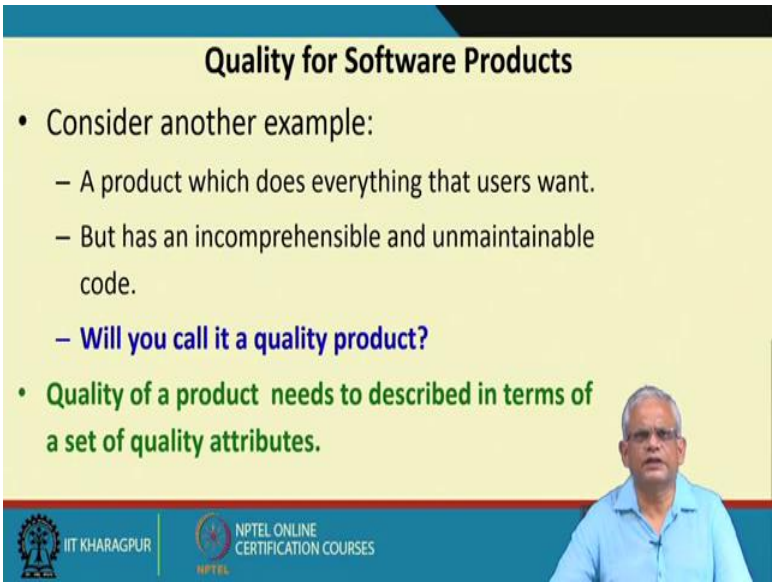

Similarly, another software it satisfies all the requirements that are written down in the requirement specification book, but then it has a incomprehensible and unmaintainable code. So, any bugs, any enhancements will be very hard to do for this software; let us say we installed a student grading software and find that after few days that there is a bug. Now, we want to maintain this software we requested that this bug be removed, but since it has a incompressible and unmaintainable code it will be hard to make any changes.

Similarly, if any enhancements are needed let us say all the functionalities were satisfied have been implemented correctly, but then we need a small enhancement, but it will be hard to do because the code is very bad we cannot call this as a quality software. So, one thing is clear that unlike other products; software products cannot be defined to be quality product based on just fitness of purpose or satisfaction of the requirements. We need to define the quality of a software in terms of a set of quality attributes. Of course, one of the will be correctness, but then there will be other attributes that need to be satisfied.

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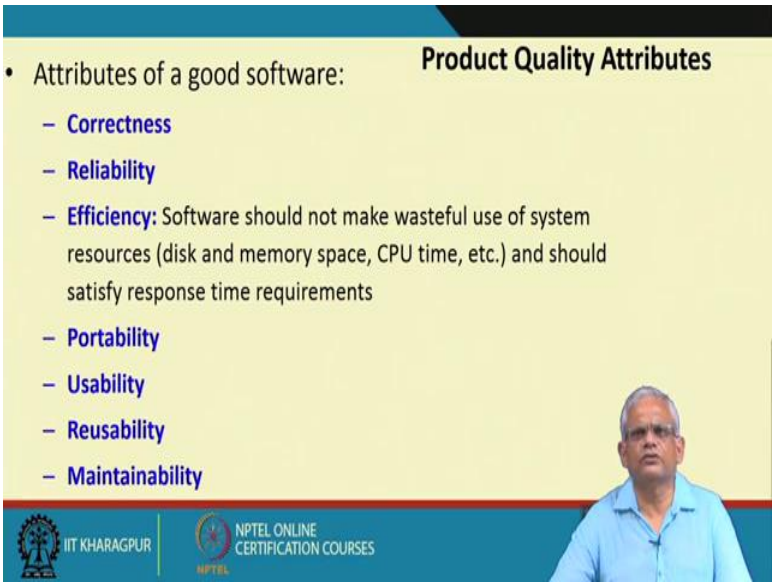

Let us look at what can be the set of quality attributes. Correctness is definitely one of the attributes for a quality software, reliability, chances of failure after uses. Efficiency it should not make wasteful use of resources for example, disk, memory space, CPU time and should satisfy the response time requirements.

Portability; tomorrow we want to run this on new hardware we should be able to easily do it that is also a quality attribute. Usability should have a good user interface the users should feel comfortable, easy to use. Reusability; if we want to extend this software, modify this for different purpose; it should be possible to do.

It should be possible to maintain the code should not be so bad that we cannot even maintain it. It should be possible to extend it, these are a possible set of quality attributes correctness, reliability, efficiency, portability, usability, reusability, maintainability, but there are can be many other quality attributes that may be considered.

Several practitioners and researchers have come up with quality models; the quality models basically try to estimate the quality of the software product based on some quality attributes; we will not really look at the quality models but we just need to be aware that based on these quality attributes; the quality models have been proposed so that given a software product we can measure the quality of the software product based on the quality model.

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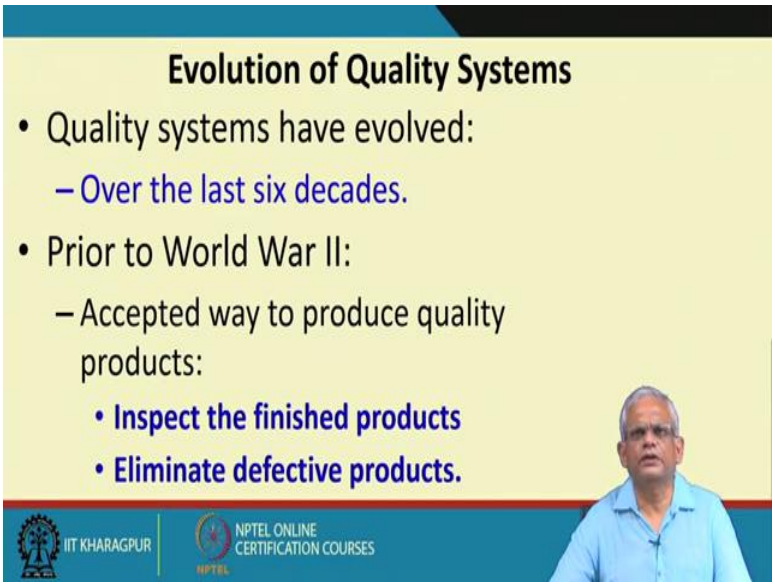

Now, let us see how the quality systems have evolved over years. In the last 6 to 7 decades; the quality system have evolved very rapidly; prior to the World War II; that is 1940s, 30s; it was accepted that to have a quality product, we must have a rigorous inspection of the finished products and then eliminate the defective products.

For example, a company manufacturing nuts and bolts; they would define a tolerance for the bolts and the nuts. And if they are beyond the tolerance just separate them out and reject them; the good product should be given to the customer and the bad one's are rejected. This was the only way that quality product where produced by various vendors before World War II; that is to rigorously inspect and test the product and then eliminate the defective product and give only the good products to the customers.

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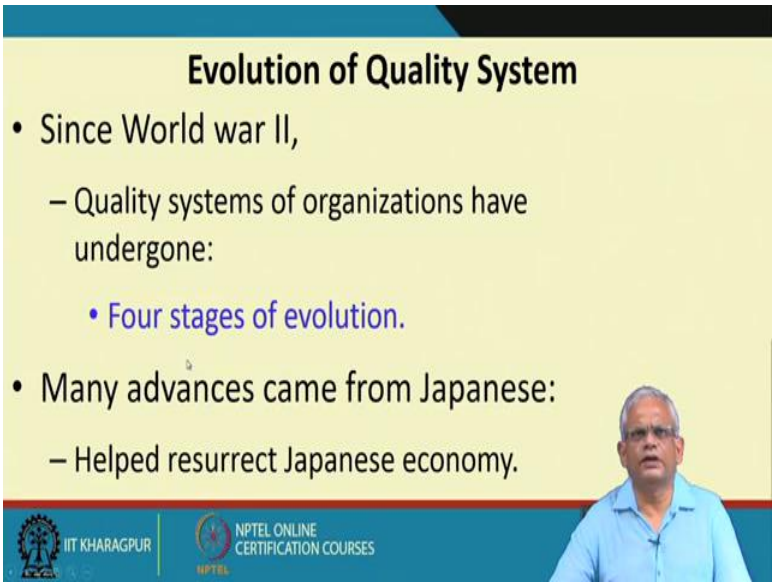

But then after the World War II, there have been four stages of evolution and may have come from the Japanese which helped resurrect Japanese economy and Japan is known to be producing very quality goods.

The initial quality efforts were testing that is identifying the defective products through testing; once that which fail the test are rejected good products are passed to the customer, bad products are rejected. But after that slowly we will see that the focus shifted from testing and rejection of the bad product to the process of manufacturing. We are almost at the end of this lecture, we will stop here and we will continue from this point in the next lecture.

Thank you.

Software Project Management
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Lecture - 36
Resource Allocation

Good morning. So, today we will take up a new chapter that is on Resource Allocation.

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We will see about little bit basic concepts of resource allocation. But what do you mean by resource's what are the different types of resources; then, we will discuss about the nature of the resources; categories of resources and then, we will see how to identify the resource requirements in a project.

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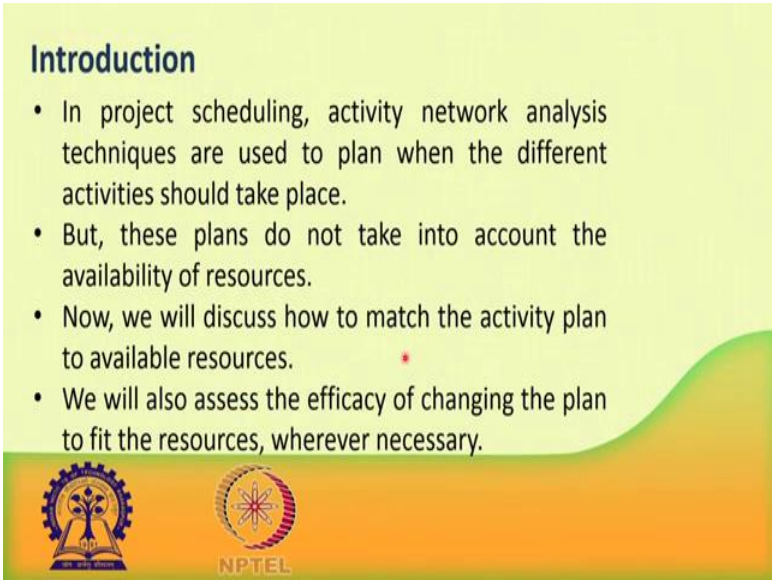

You know that in project scheduling, I hope you have already known about project scheduling. You must have used the terms critical part method or CPM PERT project evaluation and review technique etcetera those are used for scheduling a projects.

In project scheduling, normally, activity network analysis techniques they are used to plan where the different activity just take place. So, people are using activity network or the precedence networks, these are the difference what basically diagrams they will be used to make this planning regarding which activity will take when; which activity will take place when.

Now, these plans that are being made for the project scheduling, they do not take into account the availability of the resources. But actually we may make the plan if we do not take into what resources are available, then that plan will be just ideal, it will not work properly. So, now, that is why we will discuss how to match the activity plan that we have made to what to the available resources.

So, if the available resources are not sufficient enough, then we might have to change the activity plan. We might have to change the plan, then in that case we will assess the efficacy of changing the plan to fit the available resources. So, whenever it is necessary. So, we will see basically two things here; how to take into account the resources while making the activity plan and then, if the resources are not sufficient; how we can change the plan depending upon the available resources.

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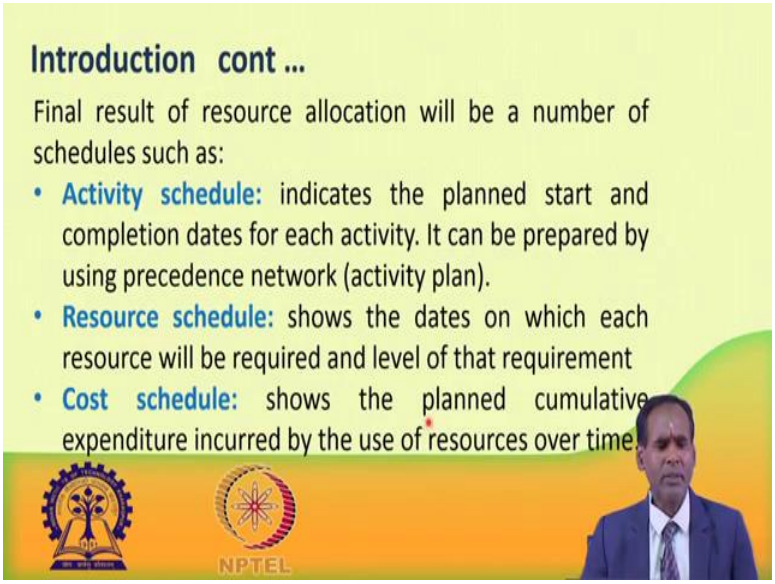

Now, let us see what is the result of the resource allocation. As we know that every phase of software development lifecycle is having an output. So, similarly here, the final result of resource allocation will be a number of resource will be number of schedules; those schedules are activity schedule, resource schedule and cost schedules. So, what do you mean by activity schedule?

So, this schedule will indicate the planned start and the completion dates for each activity. You know every activity we must have a plan when it should start and when it should be complete. So, this activity schedule will indicate the plan start date as well as the completion date for each activity and how it can be prepared? It can be prepared by using a special type of diagram or a graph, we call it as a precedence network or activity plan.

So, another outcome is resource scheduling. So, where it shows the dates on which each resource will be required and level of that requirement. So, as I have already told you in the activity plan we should also take into account the availability of the resources. So, this resource schedule will show the dates on which the different types of the resources will be required the level of that requirement.

Then, the another output of this resource allocation is a cost schedule which will show the planned cumulative expenditure; what will the cumulative expenditure that will be incurred by the use of the resources over the time the during the project development.

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Now, let us see, first of all, what you mean by resource? A very trivial what the concept we can define conventionally that resource means it is any item or a person which is required for successful execution of the project. Some resources they will be required for the whole duration of the project and some of the resources will be required only for a single activity. For example, you can take the project manager.

He is required for the whole duration of the project whereas, the programmer he will be required only for a specific activity, may be the coding. Designer will be required for only when we will perform the design and similarly, analyst or system analyst they will be required during the requirement analysis phase. So, some of the resources will be required for the whole duration of the project; whereas, some others will be used or during will be used only for a single activity or a what couple of activities.

A project manager is very much vital to the success of the project. So, it does not require the same level of scheduling as a programmer; programmer you might be required for coding, you might require say two programmers or so. But a project manager it does not require the same level of scheduling as a programmer or a system designer or a tester. So, the manager may request for the use of a programmer who belongs to a pool of resources at the program level. So, we have already seen earlier the difference between project and programme. So, the manager is responsible for one project.

So, now, but a programmer may belong to a pool of resources as a programming level. When you require the help of a programmer, you have to consult to the program head and you will release a particular programmer for your work. So, for your project. So, that is why they are the level of this programmer may not be same as that of the what the project manager.

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Now, let us see categories of resources this is very much important what could be the possible what categories or the types of the different resources. We will see the different important categories of resources are level labour, equipment, materials, space, services, time and money. So, let us see one by one. So, labour means normally this will include the members of the development project; the persons, the members of the development projects such as project manager system analyst program or system designer, tester these are all coming under this category labour. So, basically they are the members of the development project.

So, then comes equipment. So, the various computing and other equipments they are required for successful execution of the project. They will come they will come under these resources will come under equipment. For example, the servers workstations desktops, PCs, laptops, keyboards, printers, scanners, all those things are coming under the equipment and since, they require some accessories like furniture etcetera, they will also come under the category of equipment.

Materials, these are the items that are consumed rather than used. So, these items they will be consumed. So, some items are there you can use them for some period of time or for some duration. But these are the materials which will be consumed that it is after some time or when it is used. So, for example, you want to copy what some of these was softwares from one system to another system etcetera we have to use disks, CDs etcetera.

So, once they are used normally we will treat them these are consume. So, papers etcetera, these are the items those are consumed rather than being used. Space is very simple. So, you may require some office space or working space in order to execute your project where your programmers, also system designers etcetera they will sit.

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Services, some projects may require procurement of specialized services. So, along with the of the your own specialities, they may require what hiring some specialized services. For example, if you are developing a wide area distributed system, you may require scheduling of the telecommunication services. So, you may require what some help of a telecommunication services. You may require postal courier services etcetera. So, these are coming under these are also, even if they are services they will be considered as a resources and they are coming under the services resources.

Time is also consider an important what resource. So, here we consider time as elapsed time. So, the time that is elapsed. So, elapsed time it can be considered as a resource. The elapsed time can be reduced by adding more staff. So, this time requirement, the required

time if you are meeting you the deadline very soon, then the elapsed time that can be reduced by adding by hiring more staff members.

See money it is also considered as a resource very important, but normally we do not consider it as a primary resource. It is considered a secondary resource. Because this money you can you are using this money to buy other resources that is why money is also considered as a resource; but it is a treated as a secondary resource. So, it is similar to other resources in the sense that it is available at a cost, but it is that in case of the interest charges.

So, here we may in order to purchase something, we have to borrow money from the bank. We have to borrow money as a loan from the bank with some interest charges. So, that is why it is also similar to other resources and it is available at a cost. In this case this is the interest rate.

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Now, let us see how to allocate the resources to different activities. So, first step in this resource allocation is identify what are the resources, they are needed for each activity and then, create a resource requirement list. So, the first step in resource allocation is identify the different types of resources needed for each activity and create a resource requirement. We have already seen different types of resources just few minutes before. Then, we have to identify what the resource types, we have already discussed about some of the possible resource types.

So, in some cases the individuals are interchangeable within a group. For example, if you are using VB programmers and they can be interchangeably used with this software developers. So, after identifying the resource types, we have to allocate the resource types to various activities and examine the resource histogram. So, after this identifying the resource types, we have to allocate the different resource types to the different activities.

We will say how to allocate them and then, we have to prepare a diagram called as a resource histogram and then, we have to examine the resource histogram whether the resources are properly allocated or not; they have uniformly allocated or not we have to see this resource histogram. I hope this histogram you have already studied earlier in 10th or plus 2 level; basically these are some kinds of what graphs bar graphs etcetera.

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We will see as I have already told you one of the important step is identifying the resource types, I have already identified several what resource types. But now let us see in a software development project; so, what could be the possible resource requirements.

So, basically in a software project development, the followings are the resource requirements; we have to identify the hardware, we have to identify the software that will be required to develop the project and identify the supporting staff. The support staff required to carry out the project. So, basically these resource requirements are needed.

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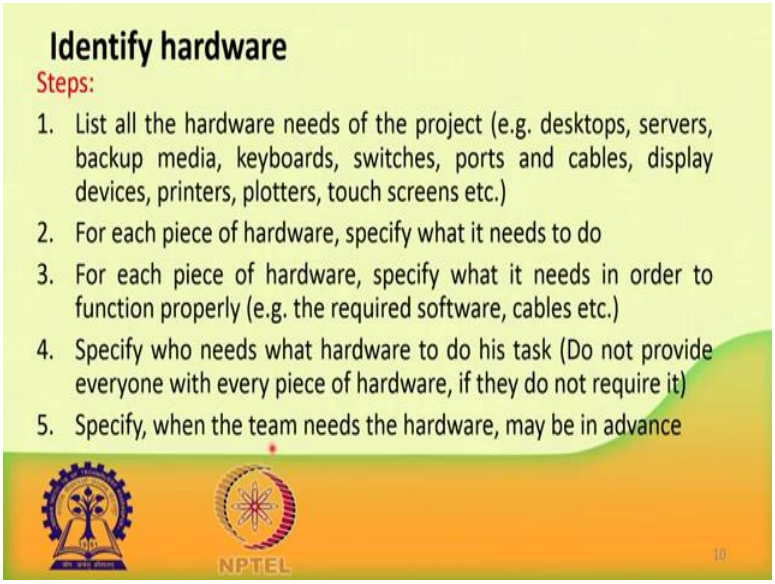

So, let us first say about the identify the hardware. What are the steps they must be followed to identify the hardware? So, first we have to see first we have to make or we have to prepare a list of all the hardware needs of the project. So, your project will require what kind of hardware; what types of hardware your project will need. First list all those things.

For example, you have to list if your project required some desktops, laptops, servers, backup media, keyboard, switches, different networking ports and cables, display units, printers, plotters, touch screens etcetera. So, examine your project requires what kind of hardware devices you will list them.

Then, for each piece of hardware specify what it needs to do. So, each piece of hardware suppose a cable what desktops server, what kind of activity it will do; what will its functionality. So, for each of the hardware specify what it needs to, then for each piece of hardware specify what it needs in order to function properly. So, first of all we should see that for each piece of hardware, what it will do; what will its objective; what will its functionality.

Then, in order to do that functionality in order to achieve that functionality. For each piece of hardware, we have to specify what it needs in order to function properly. In order to achieve that functionality, so what it needs for example? Whether the what hardware, it requires some special kind of software tools; it requires some cables for

interconnection etcetera. So, for each piece of hardware, we also specify what exactly it needs in order to properly function.

Then, we have to specify who needs what hardware to his task. So, there are several manpower several persons associated with the project. So, which person needs what kind of hardware to perform its task. See please do not provide everyone with the every piece of hardware. So, all the persons they will we have seen different types of hardware in a project, they may be required. So, do not assign do not what provide all the types of hardware to all the different persons.

Because they may not require all these things. They may require only a suppose desktop, they may not require a server. Similarly, they may require a what just display monitor, they do not require touch screen etcetera. So, that is why do not provide everyone with every piece of hardware, if they do not require it and finally, specify when the team needs the hardware may be in advance.

So, it is before what taking of this execution specify when the team, when the project needs the hardware; at what point of time if you can identify a little bit advance that will help in asking the resources from the top management. So, that is why specify when the project needs, when the team needs the hardware and if possible in advance. These are the different steps required to identify the hardware requirements of your project.

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Then, the next is how to identify the software. So, we know that in order to develop a software project many other supporting software are also needed. So, some examples are like the basic software's like operating systems, compilers, linkers, loaders etcetera they may be required for developing a project and if these basic software like operating system, compilers etcetera, they are wrongly installed or they you have not using the proper latest version, upgraded version; then, they may also cause projects.

So, different versions of software mismatched service packs and the different release of libraries, they may also create problems. If we will ignore the issues that will lead to severe problems during the software development.

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The experienced project managers they know that the software upgrades during the project, they cost the team productivity and may create serious problems. So, that is why they take proper care when to use the proper when to use the when to go for upgradations, when to use the different versions they judiciously use they judiciously decide. So, there is a need to plan for the upgrades, we should not go for the upgrades randomly. The project manager should plan for the upgrades at right time, judiciously, properly and they schedule them and the impact on the team and the project can be minimized.

So, the project manager to see that when the impact of this was upgrades on the team and the project, it can be minimized then he may go for a plan for upgrades and scheduling

them. Project managers can get a handle on software support by following the below steps let us see what the steps are.

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So, he has to list. So, let us see we are discussing now, the steps for identifying the software. The project manager first he has to list the software requirements just like listing the hardware requirements. So, for any software to be developed or in a software project, the project manager has to prepare a list containing the software requirements. Then, specify the software versions; what are the different versions that you your software will use.

Identify the upgrades and service packs; what are the different upgrades and the service packs and when you should go for them that identify. Also identify who upgrades which software; who is responsible for upgrading which software, you must ensure that because all persons they cannot be what responsible for upgrading the software. Then, specify when to upgrade which software; at which point of time here we are identifying which software needs upgrades. But then also we have to specify when the software needs upgradations and which software needs upgradation at what point of time that you have to specify.

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Let us see the first step here that is list the software requirements. The team may need we are discussing about the list of the software requirements; what kind of software we may additionally required for developing your software project. The team may need a well specified set of software products so that product inconsistency can be minimized the project or the team, they may require a well specified properly specified set of software products.

So, that the product in inconsistencies can be minimized. Upgrades can be planned properly and licensing also can be done properly. To get this set the project manager needs to manage the details of all the software to be used. So, project manager he has to properly manage the details of all the software's, they are to be used in the project. He may start by specifying the software that the team will need including the following. So, let us see what he project manager should specify? He should specify these additional softwares.

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What operating systems, he will use or what operating system the team members will use; what compilers will be used in the project; the configuration management system will see towards the end of the session about configuration management system; what kind of email what system you will in project will be used; the different software supporting software such as FTP, web browser etcetera, what will be the list of that.

Then the different libraries or the other DLL dynamic link libraries, standard templates libraries, graphic user interface library etcetera that will be used in your project. Then, office of the office software such as word processing package, excel sheet, power point presentation etcetera that will be required list of that and if any what software tools are required by your projects or such as various CASE tools, design tools such as if you know that RSA, Visual paradigm etcetera.

Testing tools such as JUnit and these what JUnit, Jumble, Selenium etcetera the testing tools and finally, the SPM software project management tools such as Gantt chart tools, Libra project etcetera. So, what kind of different software tools are required. So, that also have to be listed. So, these are the additional or the supporting what components that the project manager must prepare this list.

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Then, the next step is specify the software versions. Given the list of the existing software products, the project manager has to specify the team needs exactly which version of each of these pieces of software because there are different versions. So, which version exactly the team members needs that also we have to specify. We have to also specify when changes to these might become available. So, when changes will become available to these what versions that also the project manager has to specify.

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Then, next step is identify upgrades and service packs, you know software is not a static entity, there will be upgrades and enhancements from time to time. So, the project manager, he needs to know which pieces of software need these upgrades and when these upgrades will occur during the project. So, the project manager may have to answer the following questions for each software product, the team will use.

For example, what will the length of your project in relation with the upgrades service packs or new versions of the software products. Will the team need to upgrade whether the team request for upgradation or not? Similarly, when if yes, when the upgrade be available; when the upgrade will be available? So, these answers must be made. Accordingly, he can identify the upgrades and this service packs.

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Then, identify who upgrades which software. Please see you have to be project manager has to identify who is responsible for installing and upgrading of each piece of software. Because all persons are not responsible for that. Whether the system administrator overlay the system; administrator he is responsible or for a your particular project is the project may be a project what manager or a project developer or a system designer.

He is responsible for these upgradation that the project manager has to what identify. Make sure that the project manager has identified who does what and everyone agrees on who performs installations and upgrades. So, the project manager has identified the correct person who will make these upgrades and everyone agrees to that that which

person is doing that; who is doing the installation and upgrades, everybody must know it everybody must agree on it.

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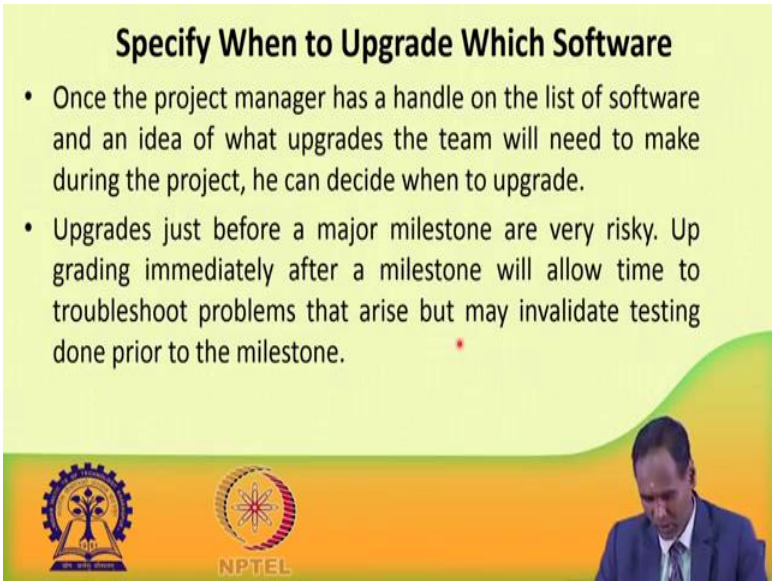

Then, specify when to upgrade which software. As I have already told you once the project manager has the handle on the list of software and an idea of what upgrades the team will need to make during the project, he can decide when to upgrade ok. So, the once the project manager has a handle on the list of the software and the idea of what upgrades; then, the team will need to make during the project that he can decide when to upgrade because we have to specify at what point of time the he has to upgrade and which software.

So, upgrades. So, normally if you will upgrade the software, just before a major milestone; then it is very risky because that what upgradations may not work properly and you may what miss the milestone, you may miss the target date, but upgrading immediately after a milestone will allow to get some time to troubleshoot the problems that arise, but may validate the testing run prior to the milestone.

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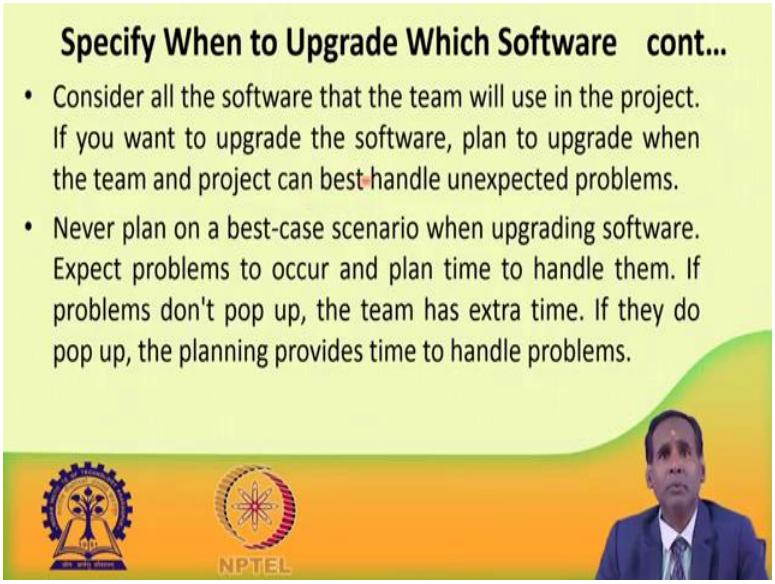

Then, specify. So, consider all the software that the team will use in the project if you want to upgrade the software, you have to plan to upgrade when the team and project can best handle the unexpected problems. So, when the problems are not there, you are expecting at this point of time problems may not occur. Then, at that point of time, you may go for the upgradation. Do not go for the upgradation before some major milestone or some what major target line. So, never a plan on a best case scenario, when upgrading the software do not be optimistic that at this point of time no problem will be there. So, we may go for the upgradation.

So, always expect that problems to occur and plan time to handle them. So, if ah. So, do not say that do not expect that problems will not occur and you should go for this what upgradation. Always anticipate that expect that problems may occur and plan the time to handle them; if problems they do not pop up, then its fine. The team will get extra time. If they do not pop up and if they pop up, then what will happen? If the problems pop up, then we have already done the planning. The planning will provide time to handle the problems. So, this is how we have to specify when to upgrade and which software.

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Then, identify the support. So, first here identify these steps are like this identify the support needed for each group; from each group of this what from each group of the different persons or these stakeholders, identify the support needed from each group. Then, specify when support will be required at what point of time so that the project can make progress without making any delay.

Then, specify how the support will be occur whether the staff will be available physically or via phone or by email etcetera, then gain commitment from each group for the support required For example, you may get the commitment through a verbal commitment or through a contract or a commitment letter and then, maintain a good relationship with the project staff. Unless you maintain a good relationship with the support staff, its very much difficult to complete the project.

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So, then after identifying the resources, we have to prepare a list called the resource requirement list first. The first step in producing a resource allocation plan is to prepare a resource requirement list which will contain the different resources that will be required along with the expected level of demand. So, this preparation of resource requirement list can be done by considering each activity present in a precedence network. I hope in project scheduling you must have studied precedence network or this activity network.

So, preparation of resource requirement, a list can be done by considering each activity present in the precedence network and identifying the resource required. We have already seen what are the different types of resources required. Now, we will prepare sample resource requirement list from a given activity network or a given precedence network.

So, you I have told you that there can be some resources that are not activity specific, but they are part of the whole project infrastructure. For example, the project manager he is required for the whole completion for the full duration of the project or required to support some other resources etcetera.

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See this is a very sample what example of a precedence network. So, here I hope for this development of precedence network, activity network we have already seen in the project scheduling. Here say these are different stages are given and here, the earliest start time, earliest finish time, latest start time and latest finish time is given.

Also what is this float I mean the free time also can be is given. Free time means by this much amount of time if the activity can be delayed the project completion date will not be affected. So, in this way a given precedence network, its shown where the different activities are there like specify the overall system. Then, what design, module A, design module B, design module C, design module D are there and similarly coding and other activities are mentioned in this diagram.

So, since you have already known this thing in the project scheduling, I have not discussing the details of this. My objective is how to create a resource requirement list from this given precedence network.

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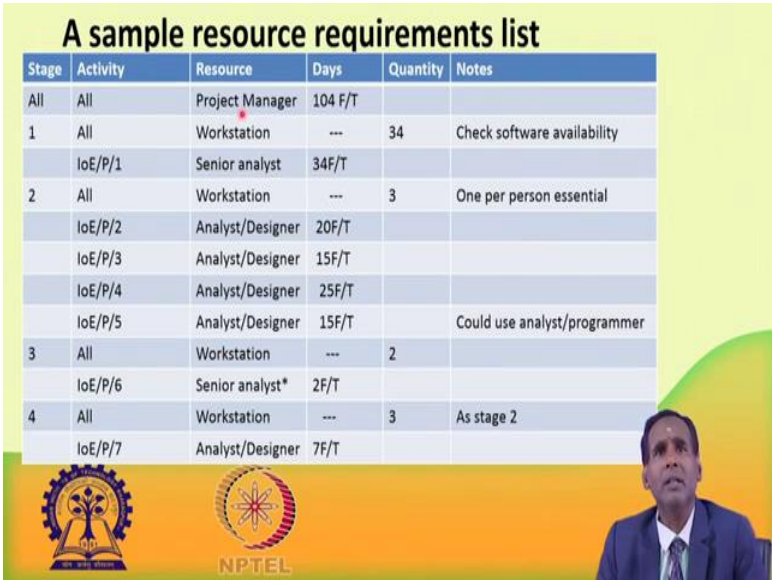

So, looking at this we have we will prepare the list like this it will see that the resource requirement list will contain these items like what are the stages, what are the activities, what are the resources, for how many days the resources are required, the quantity if any and if any notes or remarks are there you can see that.

So, we require as I have already told you whatever may be the stage, whatever may be the activity, for all the stages or the activities we require a resource person we require a resource called the project manager. He is available for the whole project and how many days are required for the total day and he can see that the total duration of the project is 104 days. So, that is why here we have written that he must be available 104 days full time.

And similarly, how many work station are there. So, for the stage one for all activities, we require some around 34 number of what work station and also for those working of the work stations, we must check the whether the associated softwares are available or not and then, comes P 1 ah. Here this is the what project activity 1; for this activity 1, how many days required you can see I think this is 34. So, we require some system see this is a stage 1 and activity 1.

This is we require 1 system or 1 senior analyst and that should be required for how many days? As I have already told you, here it is required 34 days. So, he may be required 34 full days and then, for we can go to stage 2. This is stage 2 and we require some

workstations or the resources say maybe 3 workstations one for person essential and for P 2, I mean activity 2 we can see; this is activity 1, this is activity 2. You see how many days are required? I think this is 20 days.

So, see in this case we require for activity 2, we require analyst and or designer for how many days? 20 full time days and like this you investigate all the stages all the activities, identify what are the different resources required and maybe for how many days and what is the quantity and if any note or remarks you can write here. In this way, you can prepare the sample requirement list.

So, this sample requirement list has been prepared, I using this given precedence network or by taking this precedence network. So, in this way we can prepare the resource requirement list given the precedence network ok.

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So, in this way you can prepare the resource requirement from a given precedence network. So, this is the first step before the resource allocation, you have to prepare the resource requirement list from or by using a given precedence network ok.

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So, today we have discussed about the different categories of resources. We have to explain the identification of resource requirements such as what the different types of resources are the labour and time, money, equipment etcetera. We have also presented the steps for identifying the hardware software and supporting staff, we have also discussed how to prepare a resource requirement list from a given precedence network. So, today, we have seen the basic concepts or the basic steps of the at least we have seen the first step of resource allocation. These are the references we have taken.

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Thank you very much.

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Lecture - 37
Resource Allocation
(Contd.)

Good morning. Now, now let us take up the next phases of Resource Allocation.

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We will see we have already seen identification of resources in the previous class. Now, we will see the scheduling of the resources; resource histograms; how to smooth the resources that is known as the resource smoothing; then the resource classes and how to prioritize the different activities.

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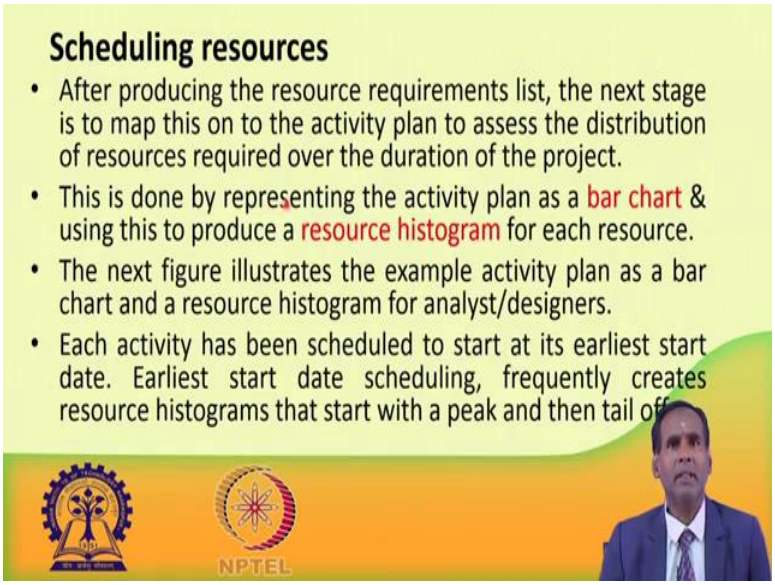

So, let us first discuss about the scheduling resources. I have already told you the first step in resource allocation is preparing a resource requirement list. After preparing the resource requirement list the next stage is to map this list into the activity plan. Why? To assess the distribution of resources required over the duration of the project.

So, this mapping can be done by representing the activity plan as a bar chat and then, using this bar chart, we can produce another diagram called as resource histogram for each resource. We will see I have taken an example in the next figure which illustrates an example activity plan as a bar chart and the corresponding resource histogram for the analyst and designers.

So, normally in this what mapping that we are doing. So, each activity has been scheduled to start at its earliest start date. I have already told you in the activity network, there are four important components; activity start sorry earliest start, earliest finish or earliest completion; then, latest start and latest completion.

So, in this kind of scheduling each activity has been scheduled to start at its earliest start time. So, let us see what is the, what problem with earliest start date scheduling. So, in case of or when you are using the earliest start dates for scheduling, then we will see it will frequently create the histograms, it frequently create resource histogram that start with a peak and then tail off; let us see how there will be so many what ups and downs will be there.

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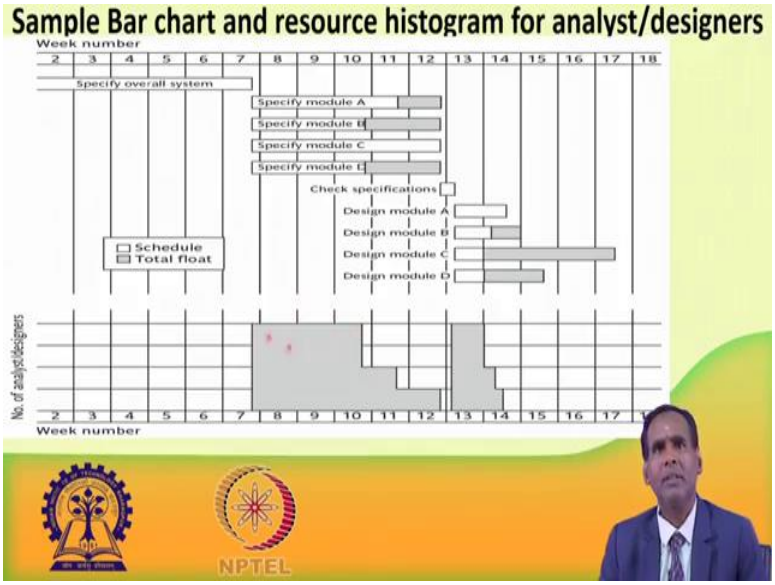

See. So, for the previous example that we have taken the earliest this activity network or the precedence network, from that activity network or the precedence network, we have created or the project manager can create a bar chart where in this you see the weeks the week numbers are written in the x axis and here, we are taken the different activities here, like in the here we have taken the specify the activity specify the overall system.

Then, what we can say specify module A, specify module B etcetera. Then, design module A, module B etcetera. So, like this you can see. So, this white space is required, what are the scheduling time and the shaded portion is known as total float. Total float means it is a measure of that time that you can see that how much time it is free and that is freely available this is slack time so that that activity can made little bit delay. So, that the whole project will not be affected, the completion date of the project will not be affected; so, float means the slack time, the free time that is available with the activity.

So, you can see that for what specify model A. So, this much time the shaded portion is what it is slack, it is free. So that means, if this starting time can be little bit delayed. So, that it will not what affect the completion of the project because this will be adjusted this free time can be adjusted so that this activity can be finished in proper time. So, you can say that for the previous diagram that we have shown, this is the sample bar chart, where the week numbers are written in the what x axis and in the y axis.

Here we have taken the different activities and the activities are consisted with the activity; what durations these timings are divided into two parts. What is the scheduling

time and this float time? Float time is the free time or the slack time available so that the project can be delayed by some amount so that the project duration the whole project duration the completion date would not be affected.

So, the upper part is this bar chart and this lower part is the called as this histogram. I hope histograms we have already discussed maybe in the matriculation or plus 2 carrier. So, this is a histogram and you can see that from this sample bar chart, we have prepared this what histograms. So, this you taken you have seen in the x axis again, we have taken the weeks and y axis we have taken the number of what developers, may be number of system analyst or the programmers or whatever that.

So, you can see that here, there will be and I have already told you this we are now scheduling the various activities. We are estimating how many numbers of manpower will be required. This scheduling is based on the earliest start date. We have assumed that the activity is started at the its earliest start time. So, then these kind of graphs you are getting you can see that if you are the scheduling is based on earliest start time, see there are so many ups and downs. There are peaks; then, suddenly it falls down. Again, there is a peak; then again, it is falling down.

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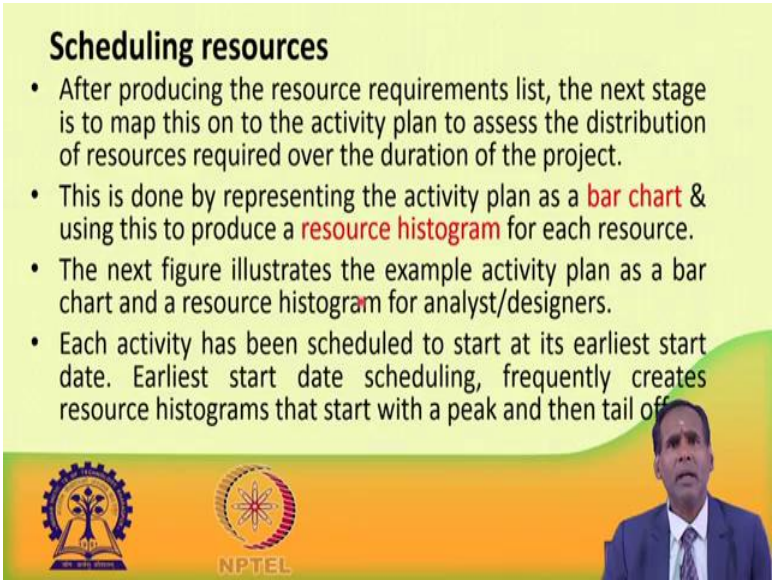

So, it is not uniform it is not even, this is the problem of what this is the problem of the scheduling based on earliest start time that I have already told you that, I have told you that this figure illustrates the example activity plan or the bar chart and then, the

histogram for the analysts designers. The upper part is the bar chart; the lower part is the what histogram. I have already told you that.

Here, this activity the activities or each activity has been scheduled according to the start at its earliest start date. The problem of earliest start date scheduling, I have already told you earliest start date scheduling. It will frequently create the resource histograms that starts with a peak and then tells up immediately that starts with a peak and then, it falls down, again It starts with a peak; then, falls down. It is not uniform, it is not even. So, this is the problem with earliest start date scheduling.

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So, changing the level of resources on a project over time particularly changing the personnel ok. So, you may change the level of resources, but, but changing the level of resources on a project over time particularly the personnel or the manpower, it generally adds to the cost of the project. You know that recruiting why because recruiting staff it also, it has also some cost. It takes some cost. Even if when staff are transferred internally, sometimes will be required for familiarization with the new project environment. So, this may also add some cost.

So, then we will see that the resource histogram shown in the previous figure, I have already shown you this is the resource histogram. So, the resource histogram shown in the previous figure, it poses it poses some kind of problems. Let us see what kind of

problems it faces? It leads to what kind of problems? It poses what kinds of problems? Some analysts or designers may sit idle for some days ok.

Let us see between the specification and design stage of the histogram, you see here. So, this is the specification phase of 2 and then, they started the design. You can see some people are idle during this see here you take this example. So, initially four people were required 1 2 3 4. Now, you see after this what analysis phase is over; now only 2 designers are used, another 2 designers are sitting idle. How many weeks? 2 weeks 2 designers are sitting idle.

So, similarly here only 1 is used. Then, I mean in 3 designers or 3 analysts, they are sitting idle and similarly here. So, this is the problem due to or since we have used the scheduling based on the earliest start date scheduling that is I have already told you the resource histogram shown in the previous figure poses some problems. The problems are due to the fact that we have used what kind of thing, we have used? The scheduling based on the earliest start date scheduling. Here, some analyst or designers may sit idle for some dates.

For example, between the specification the design phases stages of the histogram as I have told you. Initially, 4 system designers or this analyst were used; then, 2 are used and the 2 are sitting idle. Then, 1 is in the job and 3 are sitting idle. So, what will happen that it is it may not be possible that these idle persons, they will be required in another project requiring their skills for exactly those particular periods of time.

So, what will happen? So, this idle time then even if they sit idle, but their salary etcetera so that the idle time may also be charged to the project. So, that is why what we require? We have to smoothen this because see there are so many peaks and the ups and downs. So, if you can make it even we can make it uniform, then probably this idle time will be less.

So, now let us see how we can minimize the idle time; what should be the idle resource size histogram? The idle resource histogram should be smooth, should be nearly even with an initial build up and may be staged run down later on.

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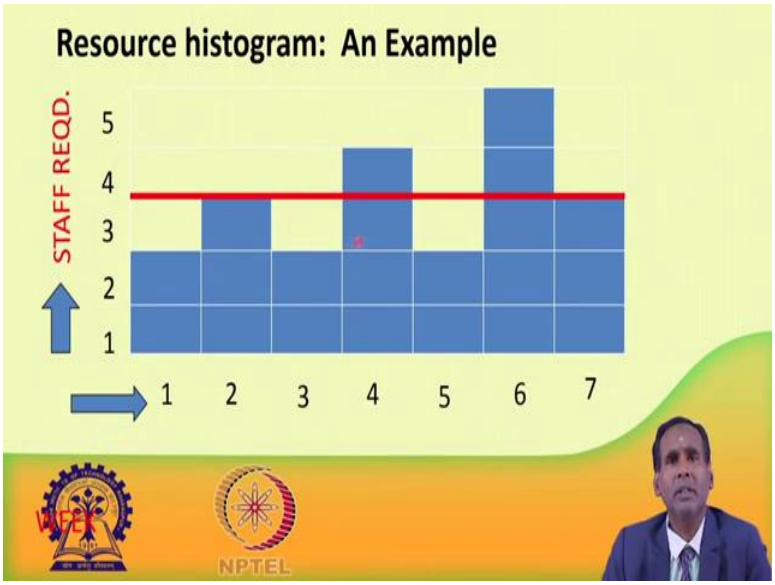

So, you see an example this is a resource histogram which is not even see so many ups and downs; peaks are there and so, this is what this is sample histogram which is not even which is not uniform. So, what we have to do? We have to smooth it. We this process we call as resource smoothing so that the idle time will be minimized.

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Like so, we have to go for resource smoothing. So, what is resource smoothing? Smoothing, it is usual difficult to get specially staff who will work odd days to fill in gaps. So, we cannot make busy all the persons always. It is very much difficult to get specialist staff who will work only odd days to fill in the gaps. There is a need of the

staff to learn about the application etcetera, it will take some time. So, staff often have to be employed for a continuous block of time.

So, we have to keep them busy staff often, they have to be employed for a continuous block of time. Therefore, it is desirable to employ a constant number of staff on a project. So, the good project manager always he will desire to employ a constant number of staff on a project, who as far as possible are fully important full time employees. So, hence, there is a need for resource smoothing ok.

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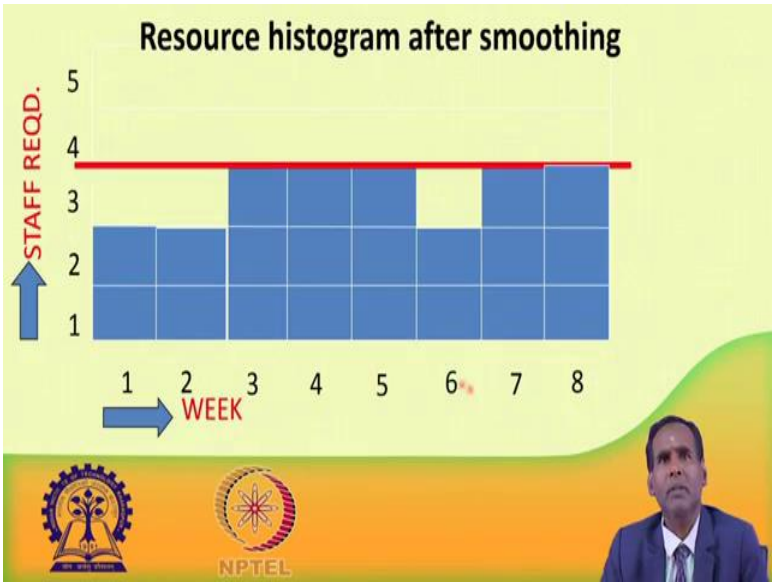

So, then that is why there is a need for resource smoothing and you see this is a example of resource smoothing. See the previous example I have taken see how many ups and downs are there. It is not even that means, here one person is sitting. So, one the person is sitting for yes; 1 person is sitting idle here, 1 unlike this, they are sitting idle. So, we have to what make it uniform, make it even, the histogram has to be smooth, the histogram has to be met even or just to met uniform see. Now, this it is more or less even and you see this person sitting idle it is very less.

So, this is the example resource histogram of smoothing the previous example here. So, this histogram is being what smoothened here and let us see how can smoothen the resource histograms.

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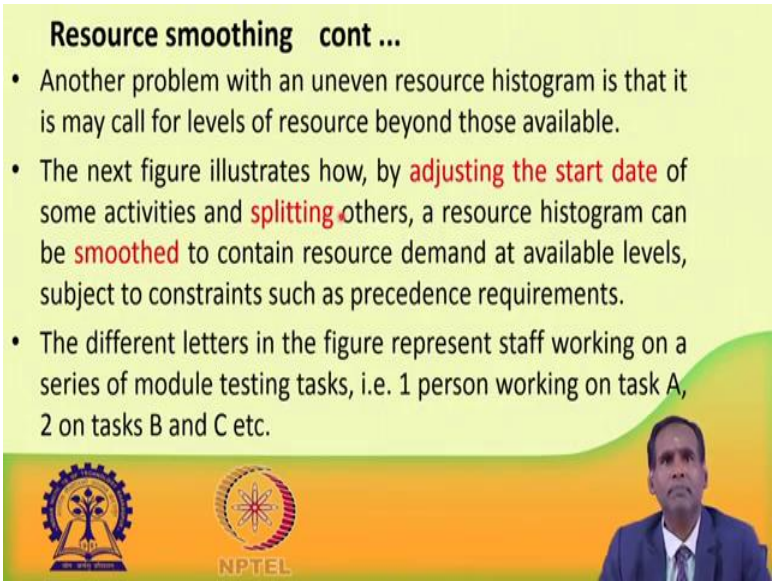

So, before smoothing this thing let us see what is the another problem with this uneven resource histogram. So, another problem with an uneven resource histogram is that it may call for the level of resource beyond those available; say few its or say 10 resources are available, but this type of histogram uneven resource histogram, it may require 15 number of what manpower. So, what to do? So, another problem with an even resource histogram is that it may call for level of resources beyond those available 10 are available, but it requires 15 resources.

So, the next the next figure, it shows how by adjusting the start of the some of the activities and splitting the other resource histogram can be smoothened to contain resource demand at available levels subject to the constraints such as precedence requirements. So, now, let us see how the need is now to smoothen the histogram. The question is how to smoothen the histograms?

So, this histograms can be smoothened by using two important what methods; one by adjusting the start date will not exactly what start from the earliest start date wherever some slack time is there, wherever some float time is there. We will avail will take advantage of the float time and we may little bit delay the starting time that is number 1; number 2 is these activities what are they are we not put in continuously wherever possible we may split some of the activities.

So, by using these two methods; that means, by adjusting the start date little bit or by delaying the start date little bit and then by splitting some of the activities, resource

histogram can be smoothened to contain the resource demand at available rates. So, what is available resources, we will only avail those many number of what manpower from the resources. Of course, subject to constraints such as the what are the precedence requirements that also has to be satisfied.

Now, let us say this what we have taken the first example we have taken this, you see this is looking very uneven. This is not smoothened. We will take the same example and here, you will see many activities they are having float times free time slack times decided portion. So, instead of starting from the very beginning, we can little bit what shift we can delay the start date so that the project completion date will not be affected because some free time is available here.

So, then we will we can minimize, then we can say that the resource requirement we can met and the resource histogram will be even or uniform. We will take the same example and we will show it here.

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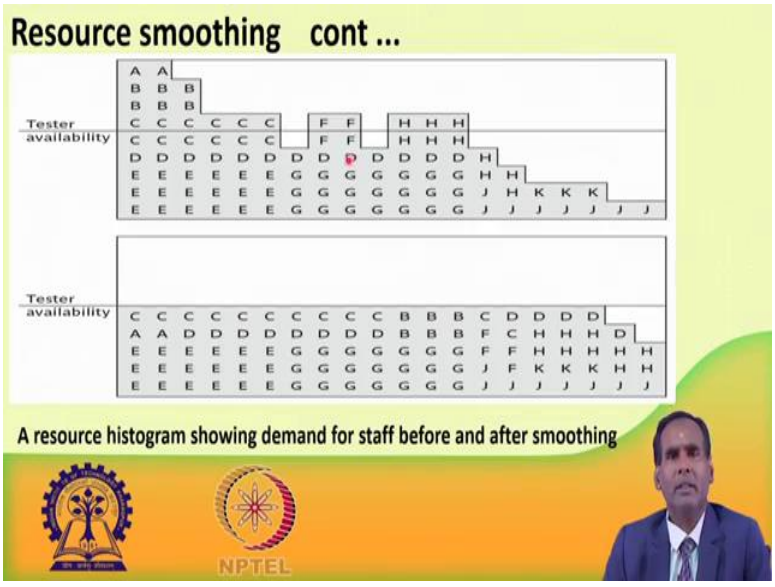

See it will take that same example. Here there or you can see that see from this you see what are the activities are there like A B C D different work activities are there and if we will take up a sample case and example of histogram, see for a project suppose the tester available like this. So, see this histogram is it is not smoothened properly. It is not a even see many times what are the resources the manpower, they are sitting just idle say it just idle. So, that is why we want to smoothen it.

How we can smoothen it? I have already told you we can follow these two important methods; we can adjust, we can delay the start date wherever float is available this slack time is available. Another is we can split some of the activities so that the resource histogram can be smoothed. Here you can see and here what A B C required the different letters in the figure represents staff working on a series of modules distinct task ok.

So, what are the staffs required in a series of what modules testing task. For example, 1 person working on task you can say. 1 person only working on staff. 1 person is working on activity A. Here one person is working on task A. Similarly, 2 persons are working on task B. 2 persons are working on task C like this it is explained and now, by delaying some of the activities and by splitting some of the activities, see this resource diagram which is not even here it is made even. So, what activities? So, you see some of the activities have started on late little bit delayed where free time is available and let see which tasks are splitted see C G are continuous here.

You can see that C G splitted. Here, you see up to this and again C is after some weeks again C is started. Then, D we can see that first these what it is it is performed here. These weeks D is performed and you see there is a break and again these performed here these. So, activities or task C and D they are splitted. So, this is the histogram which shows the demand of staff; this is before smoothing and this is the histogram after smooth after smoothing; that means, we are doing this smoothing.

The histogram is smooth smoothed by delaying some of the activities, where total float is there where start time is there and by splitting some of the activities or tasks such as C and D. So, this histogram is obtained this you can see it is much more even, it is must much more uniform than the previous one. So, another advantage you can get from this that here before smoothing, we require how many manpower? You can see 9 manpower we required my 9, but available how many? 5, but in this what you see A, B, C, D, E 5 what staff members are here. But it requires 9 staff members, but after smoothing you see now the requirement is 1 2 3 4 5.

So, now, 5 are 5 staff members are required and also how according to this resource scheduling, our resource scheduling is done by taking our resource scheduling is performed by taking those 5 staff members. You know only no extra staff members, this is the advantage of resource smoothing.

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In this figure, the original histogram was created by scheduling the activities at their earliest start dates. This one. This is done as I have told by using earliest start date scheduling. So, here you can see the resource histogram so that the typical peaked shape peaked shape caused by earliest start date scheduling and calls for a total of 9 staffs; whereas, 5 are available for project. I have already told you this you see count 1 A and 2 for B, 2 for C like this.

So, 9 staff members are required, but only 5 were available. So, that is why this is the unknown the problem of what uneven resource histogram. So, by as I have already told you we can smoothen this thing by 2 techniques; one is by delaying little bit the start of the some of the activities where float is available and another is by splitting some of the activities.

So, by delaying the start of some of the activities, it is possible to smooth the histogram and reduce the what manpower reduce the maximum level of demand for the resource and here, you can see after smooth smoothening, we require only 5 manpower, 5 designers or five testers.

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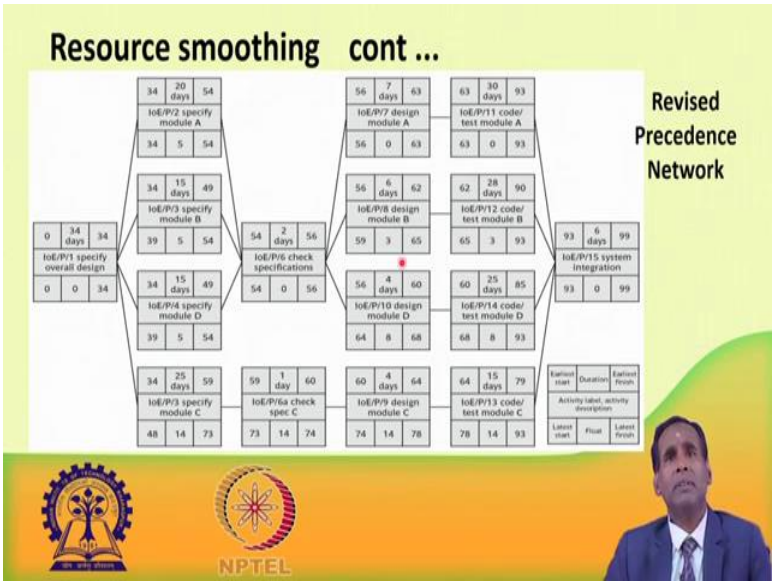

And I as have already told you by this process, we have to revise the precedence network. We have to slightly delay from the earliest start date, where total float available. So, you see so that is why the original what precedence network, we have already given earlier and this is the revised precedence network where the earliest start dates have been slightly delayed wherever float is there.

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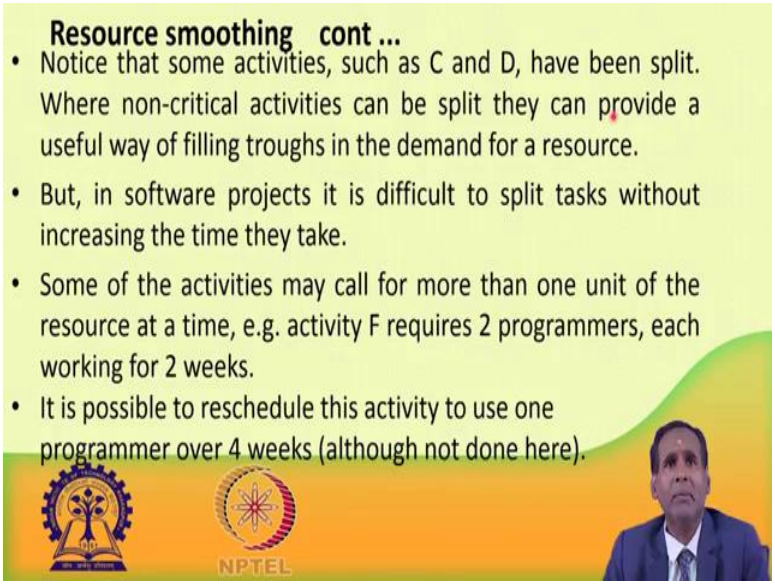

Please note that some activities such as C and D have been split. I have already told you in the what last example that here C the C it was not continuous here. It is delayed, it is splitted after see again these activities are followed then C and similarly D after this

again it is splitted and it is taken up in the other weeks. So, here C and D activities are splitted ok.

So, notice there are some activities such as C and D have been split where non-critical activities can be split they can provide a useful way of filling troughs in the demand of resource. So, whenever please remember that do not try to split the critical activities. So, only the non-critical activities can be split, where the non-critical activities can be split, they will provide useful way of filling the troughs. It will help in making even the histograms in the demand of in the demand for a resource.

But in software projects, it is very much difficult to split the tasks without increasing the time they take that is one that is another problem. Some of the activities may call for more than 1 unit of the resource at a time. For example, you see F if you will see here F requires see 2 2 persons for activity D and for 2 weeks; that means, we require if you this thing; that means, two 2 persons for activity F for 2 weeks, it can be replaced with if you are having less manpower. We can take only 1 person for activity F, what we can give for 4 weeks.

Of course, we have not shown it here, but that is another possibility that is what I am saying that activity F required 2 programmers each working for 2 weeks, but it is possible to reschedule this activity to use only 1 manpower, 1 programmer over 4 weeks that is another possibility.

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Then, we will say about the resource clashes. So, resource clashes means where same resource is needed in more than 1 place at a time. So, only 1 resource suppose that is needed in more than 1 place at the same time. So, this problem can be resolved by following one of these what methodologies.

First possible that you may delay one of the activities so that the work that one can wait for the available to be for the resource to be free to be freed; so, that can be allocated to the needed activity. You may take advantage or float to change the start date because I have already told you we can use the what you can what use the float time. So, take the advantage of the float time or the free time so that the start date can be little bit delayed.

Delaying start of one activity until finish of the other activity or the second possible that delay the start of one activity until the what previous activities finished that resource being used on puts ok. So, delaying start of one activity until finish of the other activity that resources being used on; so, but this will put back the project completion, the project completion might be delayed.

Another possibility is moving the resource from a non-critical activity. So, if a critical activity, there is targeted each appearing what we can do we can take away, we can migrate some of the resources from a little bit less critical or non-critical activity to one of the critical activity that will also this resource clashes can be solved.

Another possibility is the bringing in additional resource. If it is not possible to moving the resource from other activities or delaying the activity, it is not also possible. Then, what you can the project manager can do he can bring in additional resources he can hire additional resources to meet the target dates. Of course, this will increase the cost. So, these are some of the possible what methods solutions for resource classes.

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So, then the last activity will see it is called as prioritizing the different activities. In practice, the resources are allocated to a project on an activity by activity basis. Normally, in a real life project the project manager allocates what the resources on what allocates the resources based on activity by activity basis on an activity by activity basis. But finding the best allocation is very much time consuming and difficult.

Allocating a resource to one activity limits the flexibility for resource allocation and scheduling of other activities. Once you have allocated to some resource one activity means it will now limit the flexibility. You do not have that much flexibility to allot that resource to another activity and schedule the other activities. So, there is a need to prioritize the different activities based on their priority based on their importance; so, that the resources can be allocated to the different competing activities in some rational order. The priority must always be to what the priority must always be to allocate resources to critical path activities.

So, you must what given topmost priority for allocation of resources to the critical path activities. The next priority can go to that the activities which are most likely to affect other activities. In this way what will happen? The least priority will go to the what the activities, those are very what less of those are very less important. So, in this way the lower activities, they are made to fit around the more critical already scheduled activity.

So, in this case we are first what scheduling the highest priority activities. So, now, somewhat time gaps are there, around the critical activities. So, those gaps those free times can be what filled by the lower priority activities during scheduling.

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So, there are two main ways of prioritizing the activities; one is total float priority, another is order list priority.

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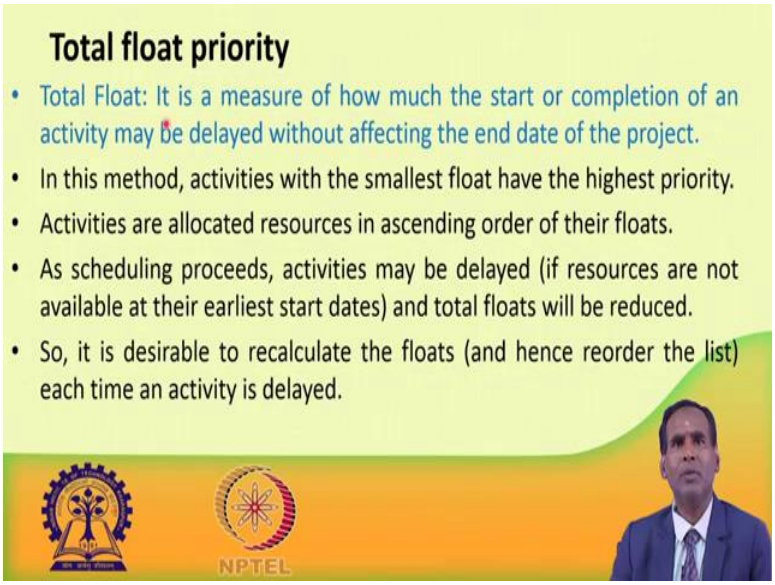

So, in total float parity, we have to see what you mean by total float. I have already told you total float means it is a measure of how much this start or completion upon activity

may be delayed without affecting the end date of the project that is some free time slack time is available. So, this float shows that by how much time, it is a measure by which says that by how much time; how much time the start or completion of an activity. It can be delayed without affecting the end date or the completion date of the project.

So, in this method; that means, in the total float priority method the activities which are having the smallest float; 0 float or 1 float, 2 float or 1 days, 2 days etcetera the float. So, the activities with the smallest float will have the highest priority because they do not have any free time. If you will delay, then obviously, project the completion date will be delayed. So, that is why in this method the activities with the smallest float. They are given the highest priority the activities are allocated resources in ascending order of their float.

So, in this method what you do? Arrange the activities according to or their floats arrange them in the ascending order of their floats as scheduling process the activities may be delayed. If resources are not available see while during scheduling or a scheduling proceeds, some of the activities may be delayed because some of the resources are not available at their earliest start dates and then, what will happen?.

It will consume from the total float and hence their total float will be reduced. So, it is desirable to recalculate the floats and hence, you have to reorder the list each time the activities delayed. So, each time the activities delayed, you have to re-compute the float because the float is changing it is consumed and the list may be changed. So, hence you have to reorder the list.

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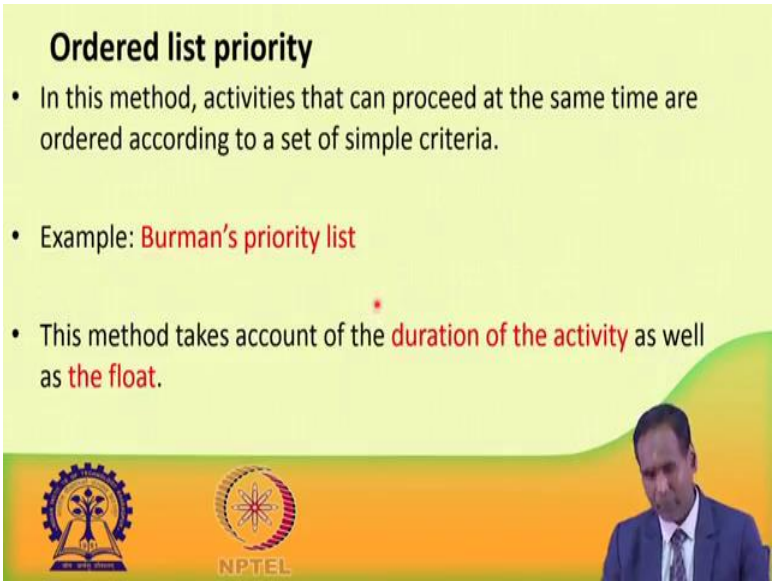

So, this is happening in total float priority. The next is order list priority. Here in these method activities that can proceed are they ok. Here activities that can proceed at the same time. So, in this method the activities that can proceed at the same time, they are ordered according to set up some simple criteria; one example is Burman priority list and in this method this method takes account of duration of the activity as well as the float. In the previous case, we have taken only float. This method takes into account both the things the duration of the activity as well as the float the total the float.

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So, in Burman's priority list, priority is given in this order first priority is given to the activities having this test what or we can say that the shortest critical activities; activities

having what very shortest what or we can say lowest float. So, shortest critical activities, they are giving first priority; then, other critical activities. Next priority go to the shortest non-critical activities; that means, what these non-critical activities having this what shortest time or so, and then non-critical activities with least float. So, non-critical activities what they are having what float is least and finally, which are the least priority will be given to the non-critical activity.

So, in this way Burman's priority list works priority given to the different activities in this way. First priority goes to the shortest critical activity. Then, next priority to other critical activities; next priority to shortest non-critical activities; next priority to non-critical activities with the least float and the least priority goes to non-critical activities.

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So, what are the alternative to resource smoothing? Resource smoothing is not always possible because deferring activities to smooth out smooth out resources, it peaks often, it puts back the project completion date. The project completion date may be delayed. So, in this case the project manager need to consider some alternative solutions such as he may have to increase the available resources resource levels in that case more cost will be incurred or he has to follow different working methods; he have to altering he may he is altering the working methods he may have to alter or change the working methods.

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Resource usage; the project manager needs to maximize the percentage of usage of resources; that means, he will he will he should what allocate the resources in such a way that it he should reduce the idle periods between the task. There is a need to balance the cost against the early completion date; also there is a need to allow for some contingency in hand.

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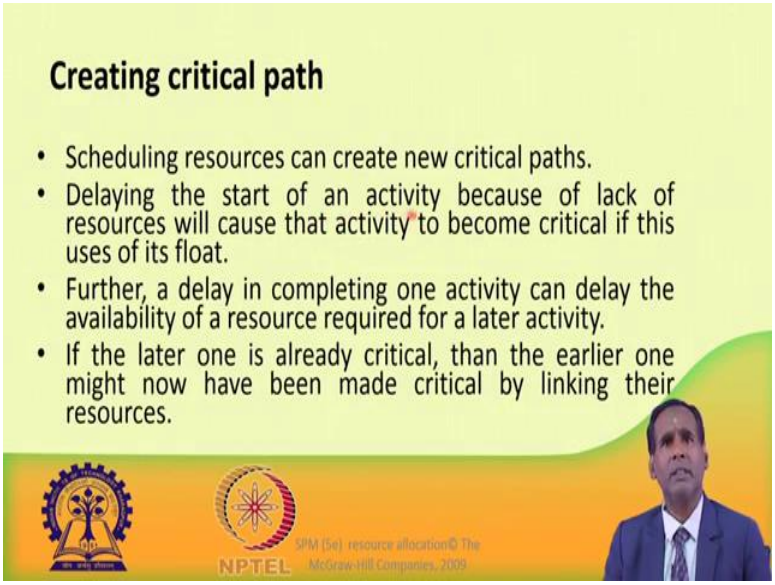

So, creating critical path scheduling the resources can create new critical paths because delaying the start of one activity because of lack of resource will cause that activity to become critical if it uses its float. So, further it delay in completing one activity can delay the availability of the resources required for a later activity. If the later one is

already critical, then the earlier one then the earlier one might now have been made critical by linking the resources. So, while scheduling the resources, it is possible that it can create a new critical paths because, what slack period the float period is changing.

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Scheduling the resources can create new dependencies between the activities. It is possible to while you are scheduling allocating resources or scheduling resources. It may create, it can create new dependencies between activities. So, that is why it is better not to add dependencies to the activity networks to reflect resource constraints. Why?

Because it will make the network very messy the activity network will be messy as well as a resource constraint may disappear during the project, but its links still remains in the activity network. So, that is why do not do not add the dependencies to the activity network to reflect the resource constraints rather you may amend dates on the schedule rather you may amend the dates on the schedule to reflect the resource constraints.

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So, we have seen these resources scheduling of resources and the resource histogram, resource smoothing. Also we have seen the fundamental concepts of resource clashes and we have also discussed the methods for prioritizing the prioritizing activities. Here we have seen two methods based on the total activity as well as these order priority list. These are the references.

Thank you very much.

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Lecture - 38
Resource Allocation (Contd.)

Good afternoon, let us see the; remaining portion of Resource Allocation. In the last class, we have seen that there are different outcomes of resource allocation.

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One is activity schedule then resource schedule and cost schedule; last class we have discussed about activity schedule and cost resource schedule. Now, today the main focus will be the cost schedules. We will see about how to allocate resources to individuals, then how to publicize the resource schedule, in what forms, then the cost schedule and what will the scheduling sequence.

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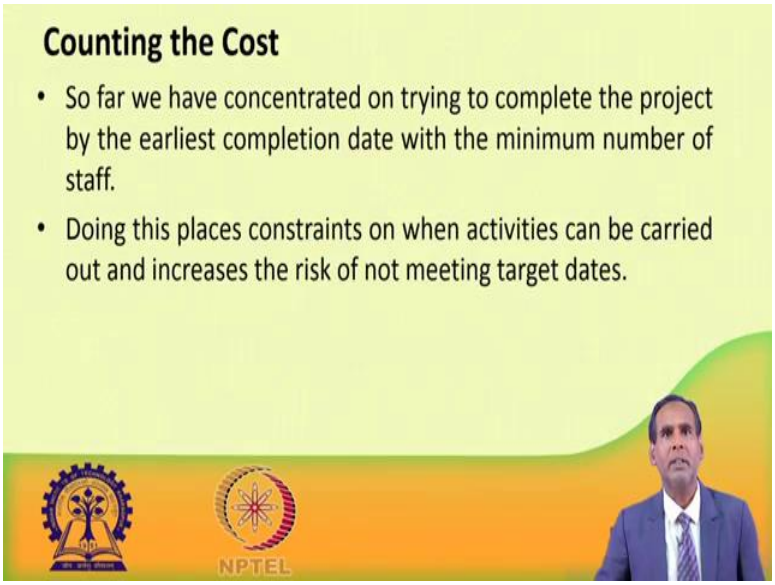

So, far we have concentrated on trying to complete the project by the earliest completion date technique with the minimum number of staffs. I have already told you the in the resource scheduling; this resource scheduling was based on the concept of earliest start date and earliest complete date. So, we have tried to complete the project by the earliest completion date where minimum number of staff is required. Doing this it will place constraints on when the activities can be carried out, this also may increase the risk of not meeting the target dates.

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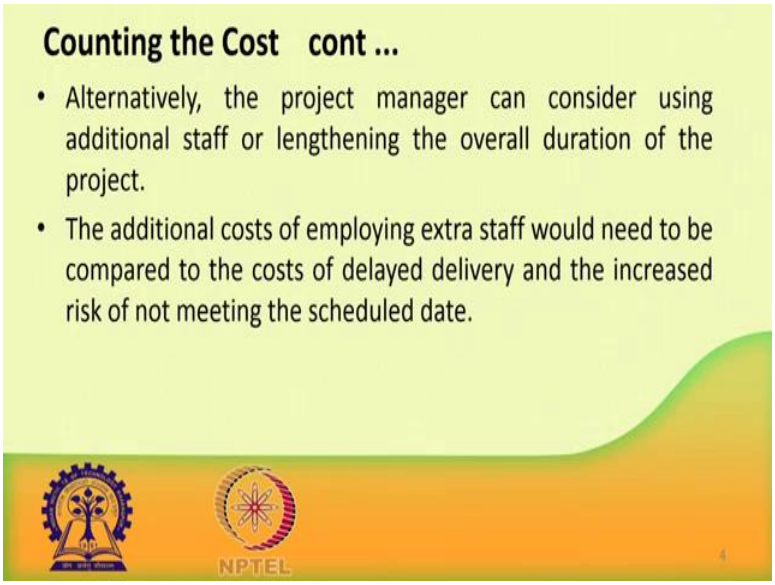

So, there are some alternatives; the alternatively the project manager can consider using more manpower, using additional staff or he may lengthen the overall duration; he may extend the overall duration of the project.

Of course, the additional cost of the employing extra staff; it would need to be compared with the cost of delay. So, if we will add some extra manpower what will the cost required; we say this if this delivery can be delayed and how much risk will be include. So, all those things they have to be compared ok. So, the additional cost due to employing extra staff that has to be compared to the cost of delayed delivery; as well as the increased risk of not meeting the scheduled date.

Then an appropriate action the project manager may take; either he will delay the delivery date or he will if it is so, urgent that you have to meet the target date, then he has to employ extra manpower so as to complete the job complete the project in time as per the prescribed schedule.

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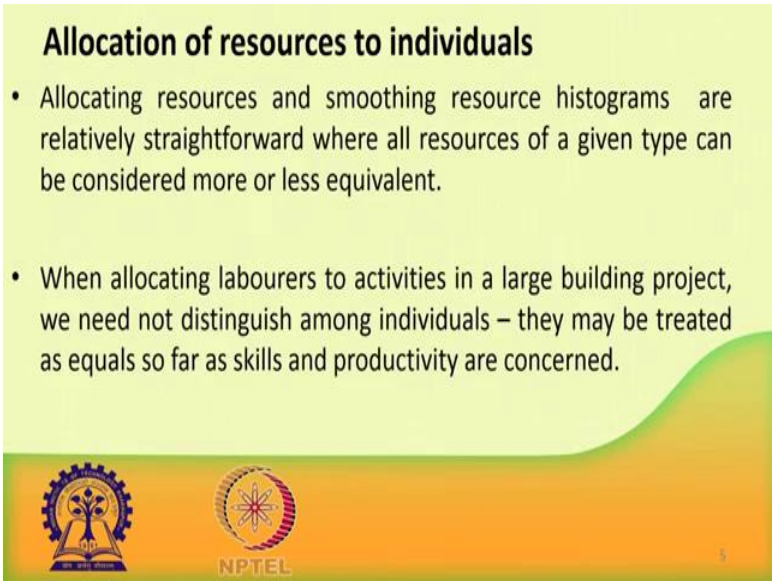

Now, let us see how the project manager can allocate resources to individuals. Allocating resources and smoothing resource histogram, they are relatively straightforward where all the resources of a given type they can be considered more or less equivalent. But actually you will see all the resources they cannot be of same important, they cannot be considered in practice they cannot be considered as equivalent.

When allocating labours to activities; let us take a small example; suppose you are building a what project and when allocating the labours to activities in a large building project. You are constructing a large building say 50 storeyed building also, there we need not distinguish among the individuals; we will treat the labours at part. We will treat the labours as what similar they will be treated at part; they may be treated as equals so far as their skills and productivity are concerned, but this is not the case while developing software projects.

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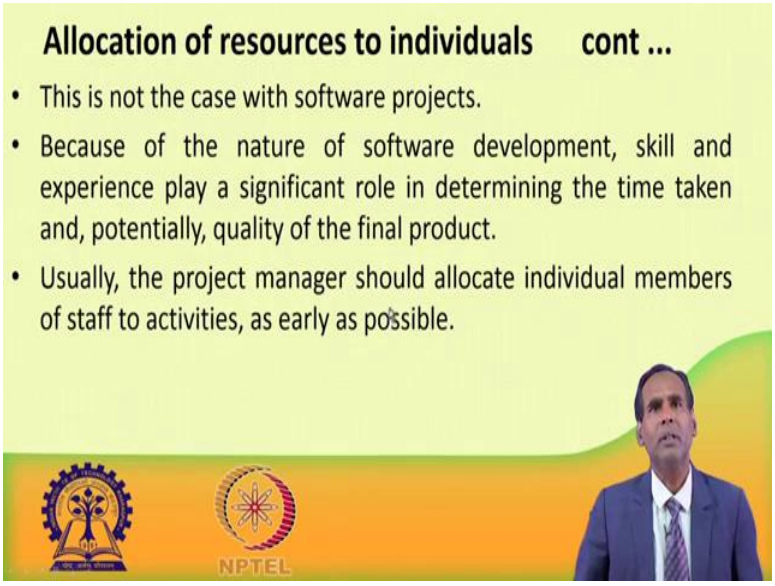

Because of the nature of the software development skill and because due to the or because of the nature of the software development; here the skill and experience of different personals, they cannot be treated at work, they cannot be treated equal.

So, skill and experience they play a significant role in determining the time taken and potentially quality of the final product. So, depending upon what the situation; we have to assign different persons, different experienced persons to different what jobs different tasks or different activities. Usually the project manager; so allocate individual members of staff to activities as early as possible so, that if some problem happens; some time will be there in the managers hand he can do the; he can take the remedial action.

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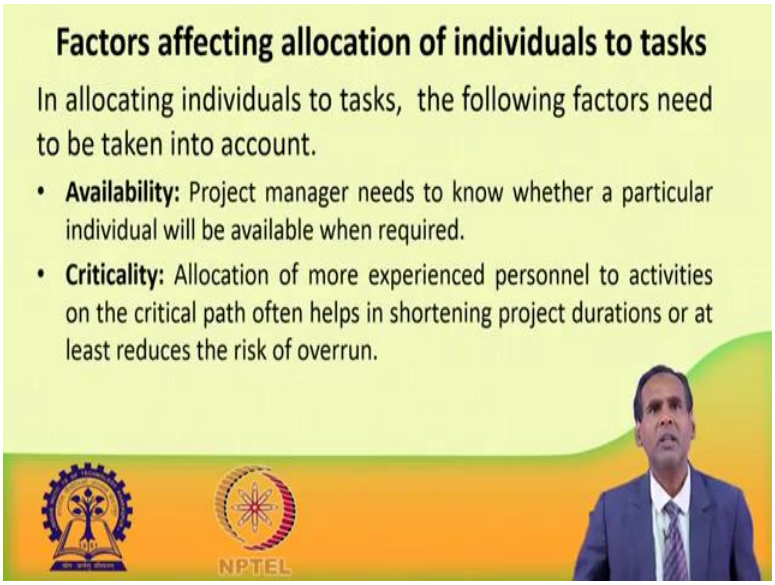

In allocating individual to tasks, several factors are there which need to be considered. The following factors may be considered while allocating the individual to tasks. Number one is availability; that means, in the project manager needs to know whether a particular individual will be available when required. See you are requiring now certain programmers, but these programmers; they are busy in developing some other projects; so during your need so they are not available. So, that is another that is one of the main important factor you have to take into account that is availability.

Whenever the project manager or whenever you as a project manager need a particular individual; you should know whether that is available or not. So, that is why it is sufficient before, in advance you have to say that for this I require 5 programmers to the central pool, so that 5 programmers can be what employed for your project in the due course of time.

So, project manager needs to know whether the particular individual will be available when required or they are busy in some other projects so they; so that they cannot be what allotted to your project immediately. Next factor is criticality, allocation of more experienced personnel to activities on the critical path often helps in shortening the project duration or at least reduces the risk of overrun.

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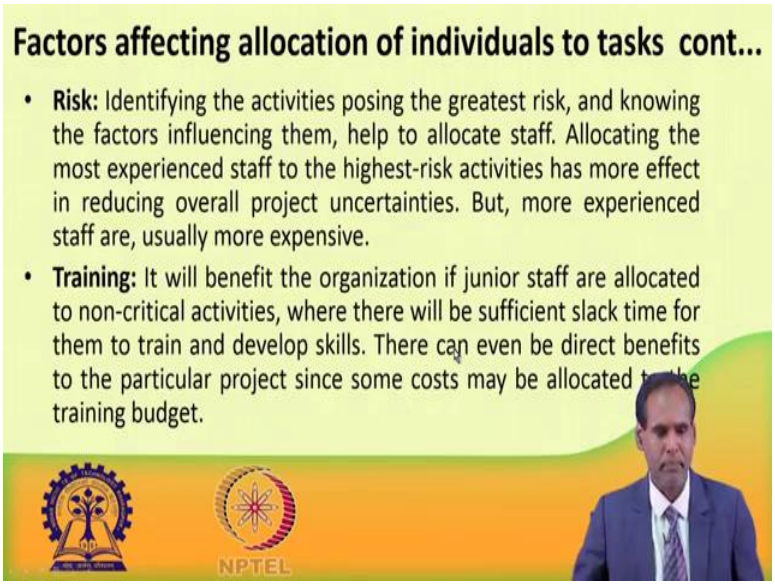

So, which are more critical activities? So, as a project manager you should allocate more experienced personnel to those critical activities this will help in shortening the project development or the project duration also this will reduce the risk of over run.

Another factor in which will; which may affect allocation of individual personnel to different task which risk. So, identifying the activities which are posing the greatest risks and knowing the factors which are influencing them will help the project manager to allocate the staff. So, you must identify which activities are posing greatest risk; as well as we have to know the factors which are influencing them, this will help in allocating the staff members to the different tasks or the activities.

So, as the rule states; allocating the most experienced staff to the highest risk activities. So, as a project manager you should allocate the most experienced staff to the highest risk activity. Because this has more effect in reducing the overall project uncertainties, the risk will be reduced; the project uncertainties will be reduced; if I will you will allot the highest or the most experienced staff to the highest risk job. So, but you know that more experienced staff; obviously, they will be more expensive; they will require more salary or more what you can say they are more fees for that; so more charge. So, more experienced staffs are usually more expensive.

Another factor that may also affect the allocation of individuals of tasks is training; it will benefit the organization if junior staff are allocated to non critical activities. So, which are very critical activities, which are very what important activities we must allot

very senior and experienced staffs to those activities. But which are little bit not so much critical, non critical activities ordinary activities you can say; so there we can apply there we can allocate junior staff members.

Because there will be sufficient float time there will be sufficient slack time or there will be sufficient free time for those activities, so that those junior staff members they can learn, they can train and they can develop their skills. There can even be direct benefit to the particular project; since some of the some cost may be allocated to the training budget. So, training is also another factor while allocating or while taking the decision regarding allocation of individuals to different tasks.

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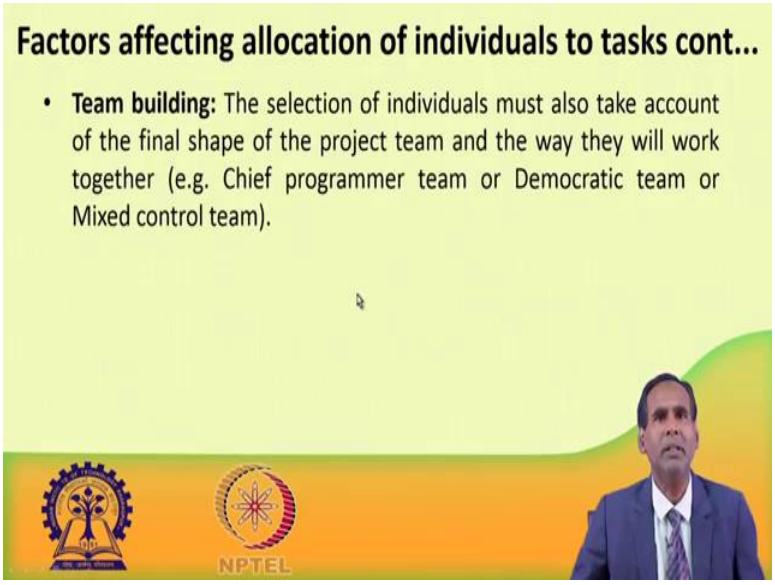

And the last one is team building; so the selection of the individuals it also take into account the final shape of the project team. How the project team its structure will look like and the way they will work together. You know that there; there could be different team structures like chief program or team structure or democratic team structure or mixed control team structure.

In chief programmer team structure you know, there will be a single programmer as the chief the boss under him there are some what programmers are working. And under each individual programmer, there are some sub some what subordinate staffs are working. So, these subordinate staffs will report to the programmer and in turn the programmers;

they will report to the chief programmer. So, this is one type of structure called as chief programmer.

Similar, another one is democratic; in democratic team structure each developer or each programmer can communicate to every other what programmer or every other developer; so there is no bar they will what report to a single boss. So, all can communicate among themselves. And another is mixed control of team; that means, here this is a hybrid team structure here both the features of single chief programmer team as well as democratic team; what the both the features are there.

Because each phase each team structure has its own advantages and the disadvantages. In order to avail the advantages of both the team structures; so a mixed control team structure is followed. So, that is why while allocating the individuals to different tasks, you must see that what will the team structure; where this individual will work whether he is working in a chief programmer team structure based project or a democratic team structure based project or a mixed control team structure based project.

So, based on this team structure or the final shape of the project; you should decide that which individual you will allow; you will allocate to that particular project. So, this is how team building also has to be taken into or the also it should be considered while allocating individuals to task ok. So, that is why the team building also plays another important role while allocating individuals to the task.

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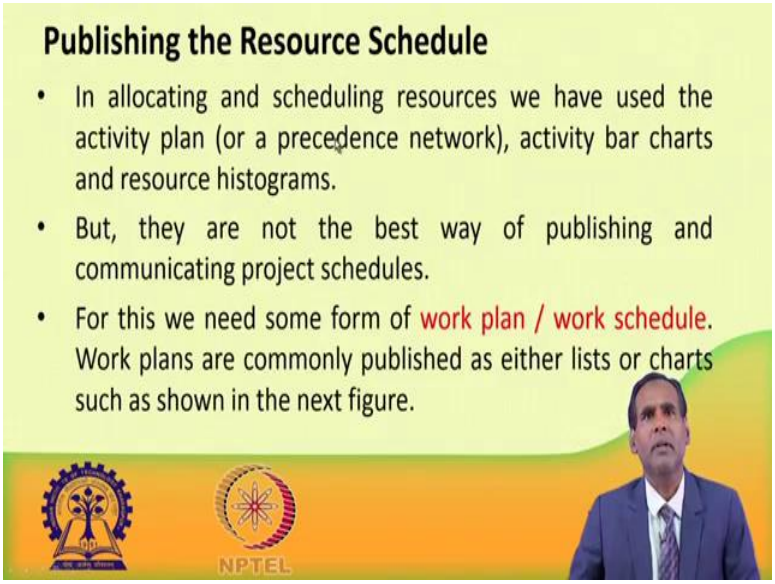

Now, let us see so far we have already seen about how to prepare this, how to allocate the resources. First we have prepared this resource requirement list and then we have mapped to a resources histogram. So, then let us see how efficiently we can publicize the resource schedule.

In allocating and scheduling the resources, we have used the; what we have used the activity plan or a precedence network. We have also used the activity bar chart as well as resource histograms. So, these three things already we have discussed in the last class how to what construct activity, plan or precedence network and a activity bar chart and a resource histogram in order to or for allocating and scheduling the resources. So, these already have seen, but these are not the best way of publishing and communicating the project schedules.

Let us see one of the; efficient way or best way of publishing and communicating the project schedules that we call as a work plan or a work schedule. So, for this; that means, for publicizing; for publishing the communicate for publishing the project schedules, we need or the project manager needs some form of work plan or work schedule. So, what is the work plan?

Work plan is a commonly work plan and it is commonly published as either a list or a chart ok. So, normally the work plans are they are commonly published either some kind of lists or charts one example I will show in the next slide.

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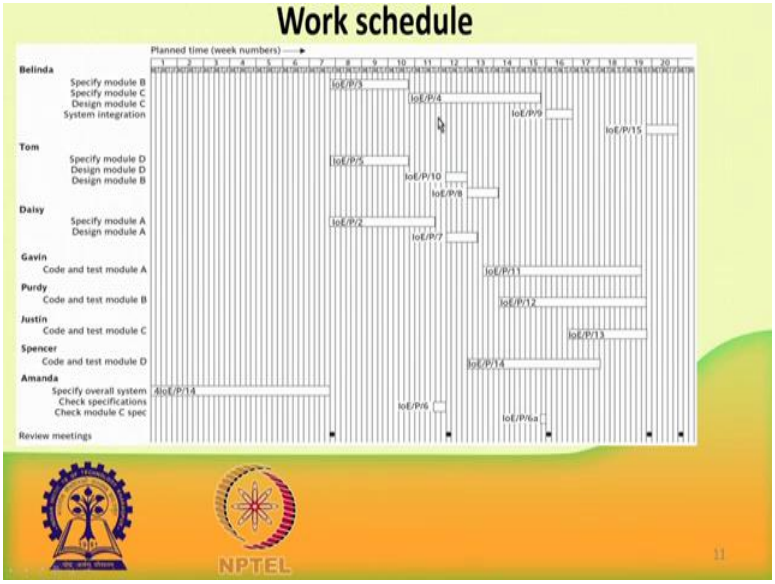

See here this is one example of a; this is the example of a work plan or work schedule we are writing who are the resources? The manpower, the name of the what developers or analyst or testers. Then what activity they are supposed to do? Then the week numbers and under week each days are given ok.

These are the activities and as you know there are two parts the scheduling part as well as the float or the slack time everything has been shown. So, this is a better way of representing or publishing the resource allocation. So, work plans or the work schedules are the best ways of publishing the schedules; the resources schedules.

So, here basically we show the name of the; the name of the employee, then or the name of the personnel. Then what activities they are supposed to do and then we will see that the weeks will show, under each week the days we are showing and here we show the progress; what the schedule we want show here. So, in this way the schedule can be shown ah; the resource schedule can be shown in the form of a work schedule.

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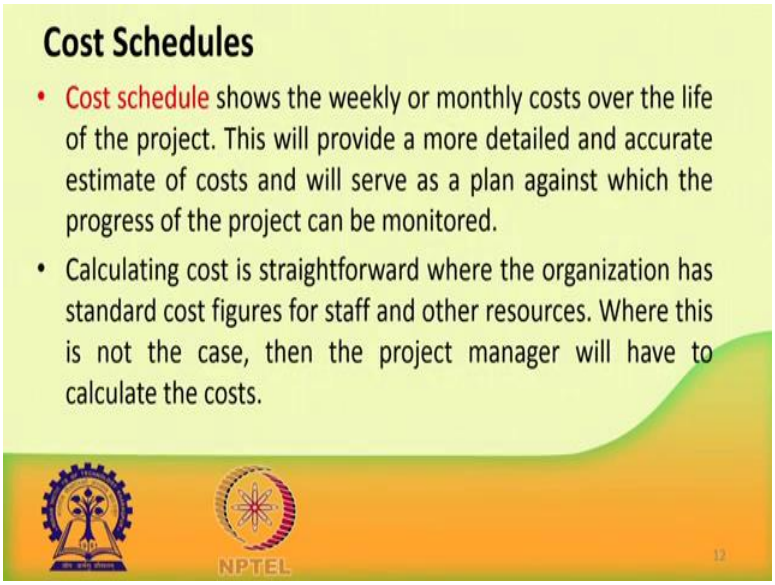

Now; so now, we have to see something about this cost schedule. We have already seen the activity schedule, we have also seen the resource schedule; now this is the time to discuss something on the cost schedule. So, cost schedules normally they show the weekly or monthly cost over the life of the project.

So, during on the life of the project what is the weekly cost or what is the monthly cost that can be shown in the form of a what figure or a diagram that we call as cost schedule; this will provide a more detailed and accurate estimate of the cost. So, this cost schedule it will give you a more detailed and accurate estimate of the cost. This will also serve as a plan against which the progress of the project can be monitored.

You want to; if you want to monitor the progress of the project then you can use the cost schedule ah. So, this can be treated as a plan against which you can measure the progress of the project against which you can monitor the progress of the project. Calculating the cost is normally straightforward where the organization has standard cost figures for different staff members and other resources. But where this is not the case, then the project manager you will have to first calculate the costs then he can or first estimate the cost, then he can prepare the cost schedule.

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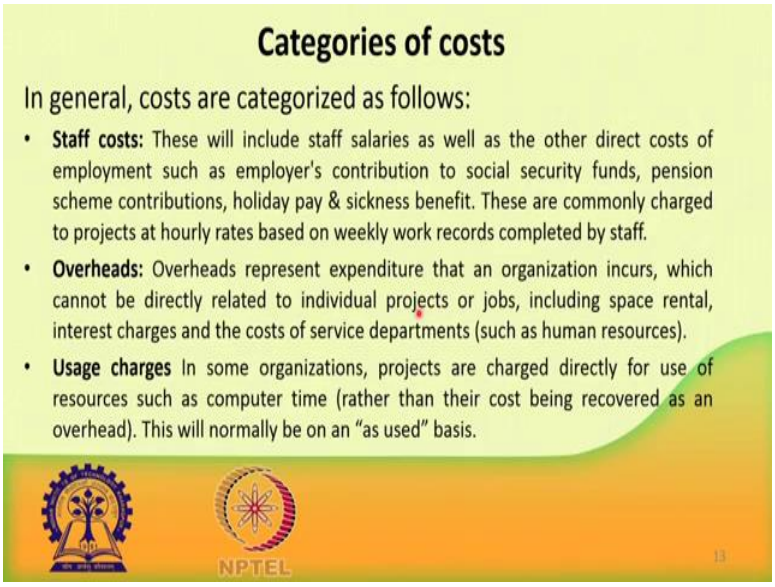

In the beginning of this what course we have I have already told you the different types of costs in different based on different what metric different types we have done. Like we have seen direct cost versus indirect cost, benefit then tangible cost versus intangible costs, fixed cost versus overhead cost; these cost we have already seen earlier.

And now let us see for might be development of software normally the cost are categorized as follows like the staff cost overheads and usage charges staff cost. Staff cost will normally include the staff salaries as well as other direct cost of the employees;

such as employers contribution to social security firms, pension scheme contributions, holiday pays and sickness benefit I mean medical benefits etcetera.

So, these are all coming under staff cost costs; these are commonly charged to the project at hourly rates based on weekly work records of the completed by this staff. And overheads these cost represent the expenditure that an organization incurs which cannot be directly related to individual projects or jobs. So, some of the examples are so these are valid ah, these remain valid for all the projects that the organization undertake.

For example, the for the space how much you are giving space rental how much rents you are paying for the building or the space, then the if you have brought some loan from the bank what is the interest charges. And the cost of the service departments such as human resources etcetera HRA department. So, all those things they cannot be what marked for a particular project; this is valid for all the projects those are undertaken by an organization; so these are all examples of overhead cost. Similarly, usage charges in some organizations projects are charged directly for use of the resources.

For example, if you are using what some computer; so then per hour you may give some money. If you are using this internet for some hours you can give what these what is the charge. If some online tools you are using for that or some online software's you are using per hour basis you have to pay something. So, these are the what charges; these are some of the cost which are charged based on the uses.

So, that is why in some organizations projects are charged directly for use of resources such as computer time etcetera or you can say internet usage etcetera and this will normally be on as used basis this charges will be made. So, these are some of the what charges these are some of the costs that the software development organization may have to incur and then based on this cost, you can prepare the cost schedule.

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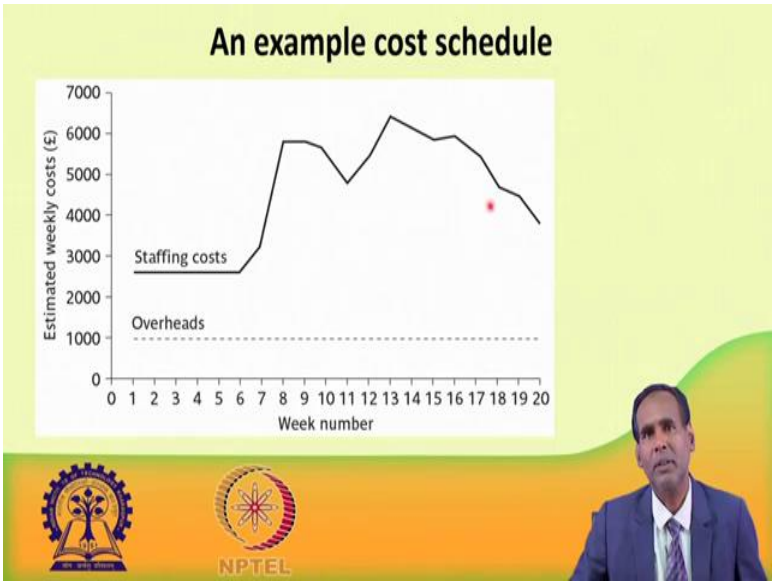

This is an example of a cost schedule; here you can say that in the x axis we have taken the weeks and y axis the estimated weekly cost. So, here we have taken first the overheads and the overheads may be charged in a fixed amount say there are total overhead cost is suppose say 1 lakh and there are 5 projects running simultaneously.

So, the overhead may be equally divided among all the five. So, maybe per project its coming 20000; so, normally this is charged as a fixed rate. So, that is why I see overhead is coming as a; fixed rate here and then the here in this project this staff cost is initially it was a constant. Then there are some what more critical jobs or so, then the more number of manpower required. So, staffing staff cost is increasing rapidly and then gradually it is coming down going up like that; so, in this staffing cost is shown here. So, here for this cost schedule; this is you can see it is given for different weeks; so the we have overheads and the staff costs are shown.

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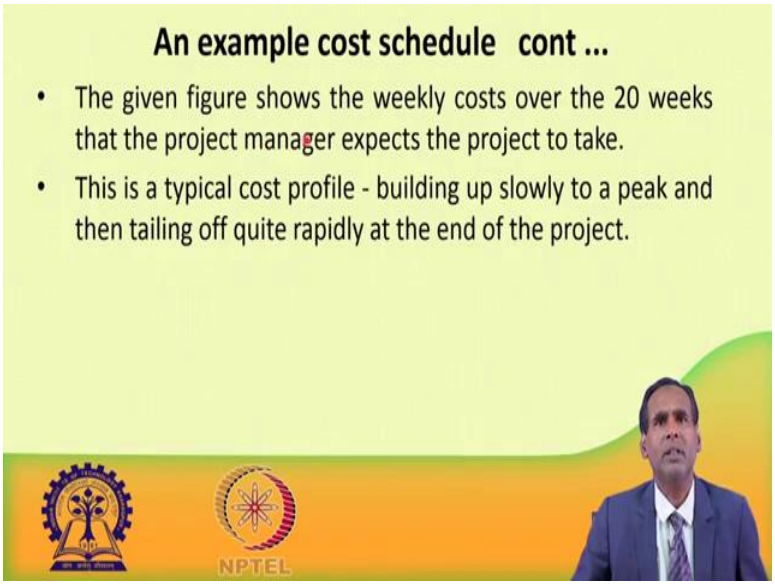

So, in that figure you see it shows the weekly cost near about for over 20 weeks; yes it is shown up to 20 weeks; this project manager expects the project to take ok. So, this figure shows the weekly costs over 20 weeks that the project manager may expect that the project will take. So, this is a very this is a typical cost profile, building up slowly to a peak you can say that this cost initially towards almost constant; then it is slowly it is going up a high amount going to a peak.

And then then tailing up quite rapidly at the end of the project and then it is tailing up you can see this is falling down rapidly ok. This is falling and towards the end the staff cost is falling down because by; by that time what we are at the verge of completion of the project. So, that is why this what staff cost is falling down. So, this is how you can prepare a cost schedule ah; maybe a weekly cost schedule for your organization taking into account the staff cost on the overheads.

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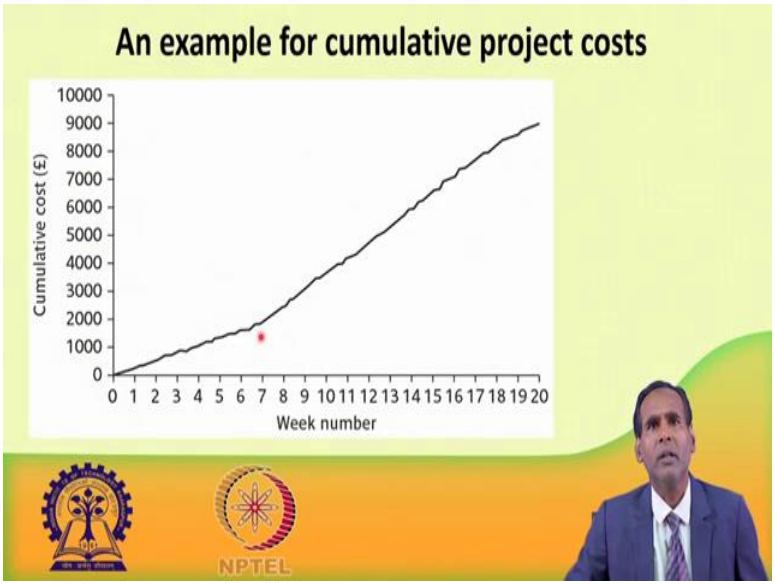

Now, this is another example; here instead of using this weekly what cost you can take into account to the cumulative project costs. So, here again in the x axis you are taking the week number and here instead of taking the just weekly cost we have taken the cumulative costs.

So, if you see the cumulative cost normally the cumulative cost slowly what it increases the number of weeks increases. So, this is another example ah for plotting the cumulative project cost in the form of a graph; so this is another example of a cost schedule. So, these cost schedules can give an indication that whether your; your progress is on the right track as you have planned or the cost is exceeding or so that or it is just as per your expectation or it is below your expectations. So, accordingly you can take remedial action looking at the cost schedule.

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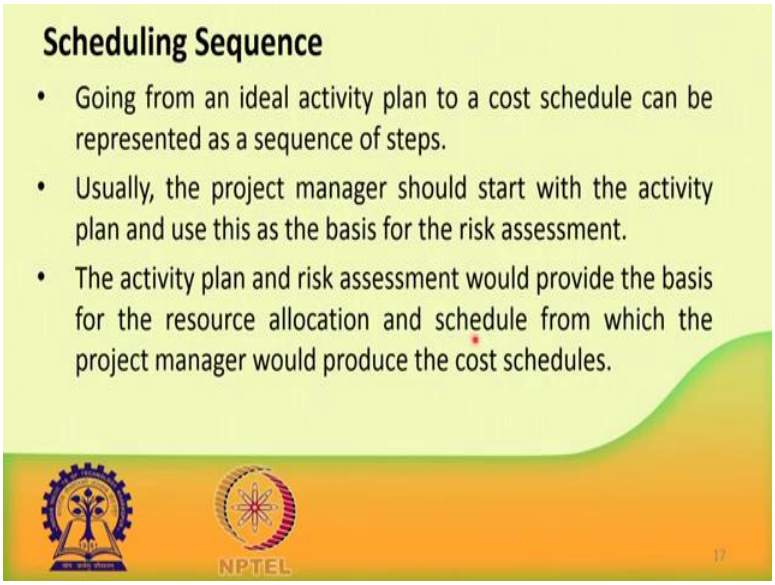

Now, let us see how to schedule the sequences. So, as I have already told you what will the scheduling sequence? First, we have already seen last class that we have to first prepare a activity plan or a precedence plan. So, going from an ideal activity plan till getting the cost schedule; it can be represented as a sequence of steps.

Let us see what are the steps you should follow from what constructing the ideal activity plan to the cost schedule; usually the project managers to start it; he should start with the activity plan. So, in first he should create or build the activity plan, then this activity plan can be used as the basis for the risk assessment just like this figure I have shown.

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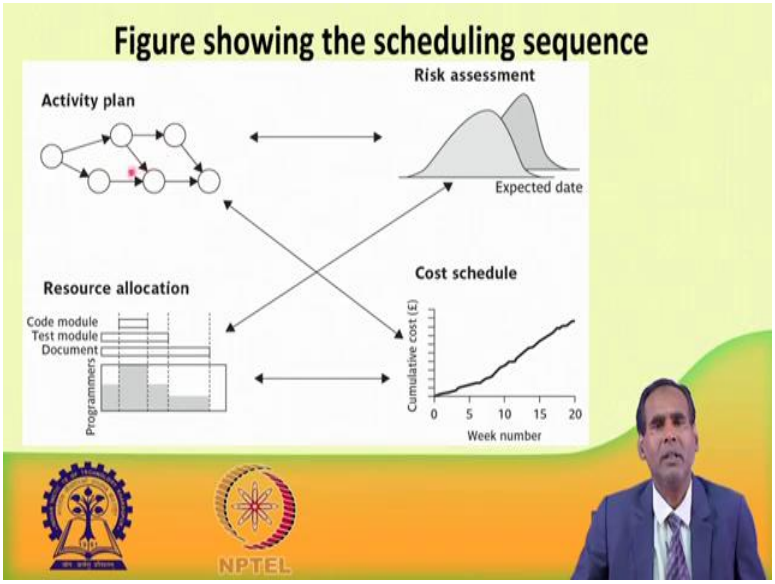

So, initially you have to prepare the activity plan; a kind of graph we have already done possible what the activity plan for our graph our problem last class. Then from this activity plan can be solved as a basis for the risk assessment. And then what will happen? Now, the activity plan and the risk assessment they would provide the basis for the resource allocation.

So, based on the activity plan you can create the what resource requirement list; then from the resource requirement list and the risk assessment ah by considering both the things you can prepare your resource allocation that we have already shown here. The activity from the activity plan the project manager can prepare the risk assessment, then the activity plan and the risk assessment will provide the basis for allocating the resources you can prepare the resource allocation; then from the resource allocation you can prepare the final cost schedule.

So, these are the sequences that you must follow for what preparing the cost schedule starting from the activity plan ok. So, the project manager should start with the activity plan and use this as the basis for the risk assessment. Then the activity plan and the risk assessment would provide the basis for the resource allocation and schedule the resource schedule. Now, from the resource allocation and the resource schedule; you can prepare the final cost schedules like this ok.

So, but you can see that these arrows are bidirectional; that means, each one affects the other one. So, if there is a change in activity plan; there will be a change also in the risk assessment and when risk you are assessing the risks; so you accordingly you might have to divide the activity plan.

Similarly, based on the activity plan you are preparing the; what based on the activity plan and the resource allocation; you are preparing the cost schedule. So, whenever there will be some change in the activity plan or some change in resource allocation; the cost schedule or the estimated cost will be different. So, if you want to again divide the cost maybe what making high values or high value high cost or low cost. Then accordingly you have to revise the activity plan and the resource allocation.

So, that is what I want to say that these components these steps you can say these are dependent on each other. If you want to make any change at any step, the other steps have to be revised accordingly.

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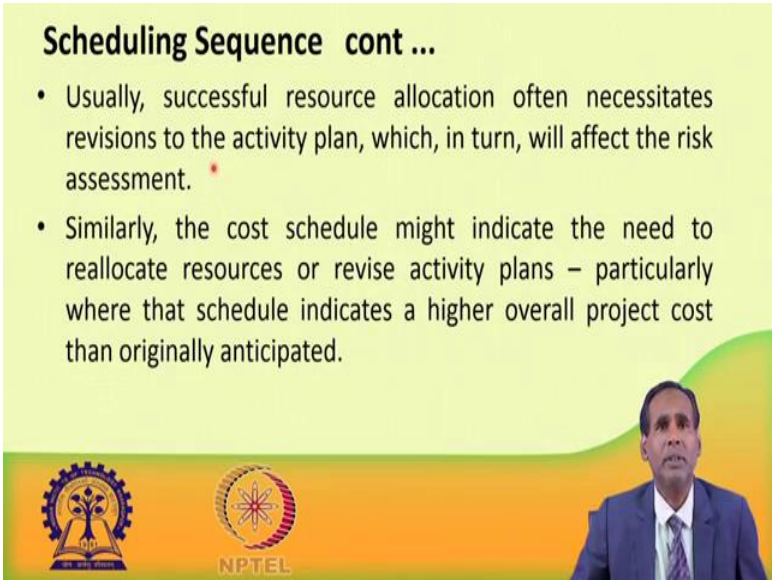

That is why I have to say that usually the successful resource allocation upon necessitates revisions to the activity plan. So, if you want to make a successful resource allocation; you it might be necessary to revise the activity plan. And if you are revising the activity plan then of course, what will happen? You have to this will affect risk assessment; you have to divide the risk assessment because the risk assessment the risk assessment may be affected.

Similarly, the cost schedule might indicate the need to reallocate the resources or revise the activity plan particularly where the schedule indicates a higher overall project cost than originally anticipated. What is saying here according to your previous plan activity plan and risk assessment resource allocation plan what cost you are getting; it is a very high value which you cannot afford. So, what you have to do? We have to now reduce the cost.

So, if you want to reduce cost then effectively you have to what revise the activity plan, you have to revise the resource allocation might be we have to reduce some of the staff members or so. So that is what I want to say here that the cost schedule might indicate that there might be a read need to reallocate the resources or revise the activity plans; When the schedule indicates a higher over overall project cost than originally anticipated.

And this high must this highest or the higher overall project cost, you cannot afford in that case you have you have to reduce you have to revise the cost schedule. And indirectly, since you are reducing the costs schedule; you have to also what since you are you are revising the cost schedule, you have to also revise the activity plan and you have to also revise the resource allocation plan. And you may have to reduce some of the what resources for the use, you might cut down some of the manpower also.

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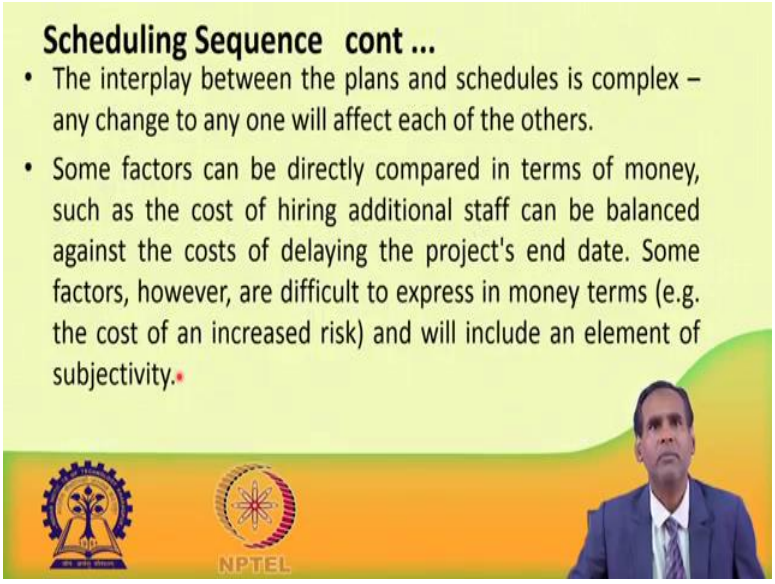

The interplay between the plans and the schedules is complex, you can see that everything is interconnected with everything else. They are bidirectional, everyone is affecting the, every component is affecting the other components.

The interplay between the plans and schedules is very much complex, any change in one schedule will affect each of the others. So, some factors of while preparing the schedules some factors can be directly compared in terms of money. So, charge cost of having additional staff it can be balanced against the cost of delaying the projection date.

So, if it is possible that the project's completion date can be delayed little bit; then we may not go for this what hiring additional staff. But if the project's completion date has to be met strictly; then we have to hire additional staff members. So, that is why some factors can be directly compared in terms of money. For example, cost of hiring the additional staff; it can be balance, it should be compared what should be the cost if the project's completion date can be extended or can be delayed.

But some factors; however, they are very difficult to express in terms of for money; for example, the cost of an increasing risks. So, if we will not meet the; what deadline what will the risk whether some customers; they will go away from our company and they may migrate to another company. So, how to measure these kinds of increase risk if I will not automate these our running activities people because nowadays everybody is going for computerization activities; so if still we are doing our activities manually, so there might be a risks that the clients may migrate the clients may migrate to other companies.

So, what risks is associated here? It is very much difficult to express the cost of this risks in terms of money and it will include an element of subjectivity; of course, these are subjective in nature, we cannot quantify them. So, some factors can be directly compared in terms of money such as the cost of hiring additional staff; which can be balanced against the cost of delaying the projects end date, but some factors; however, they are difficult to express in terms of money for example, the cost of an increase risks and the it will include of course, on element of subjectivity.

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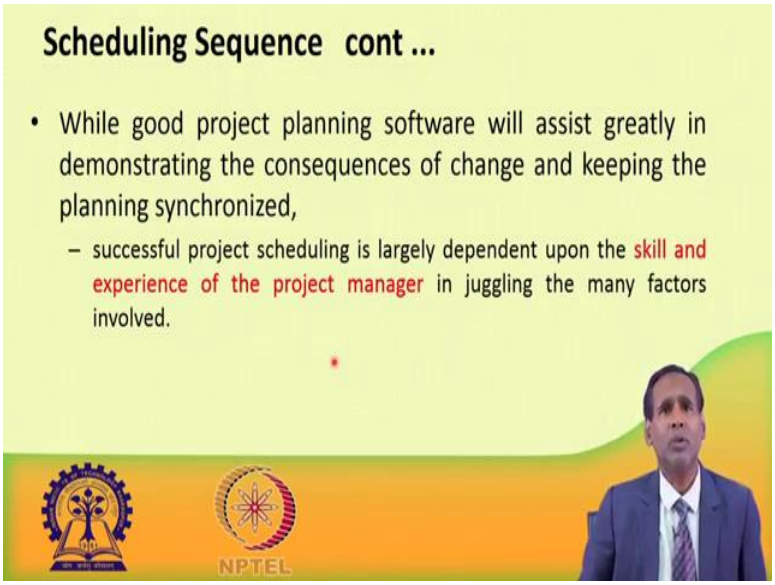

While yeah yes there are so many what the project planning software's are available. While good project planning software; they will assist greatly in demonstrating the consequences of change and keeping the planning synchronized, but successful project scheduling it is largely dependent upon the skill and experience of the project manager;

that matters a lot what is the skill what is the experience of the project manager; that will what affect a lot during this project scheduling; in juggling in the many factors that we have seen they are involved in the project development.

So, while some of the good project planning software; they will assist greatly in demonstrating the consequences of changes and keeping the planning synchronized. The successful project scheduling is totally dependent; it is largely dependent on the skill and experience of the project manager in juggling the many factors involved in the project development this is something about the cost schedule.

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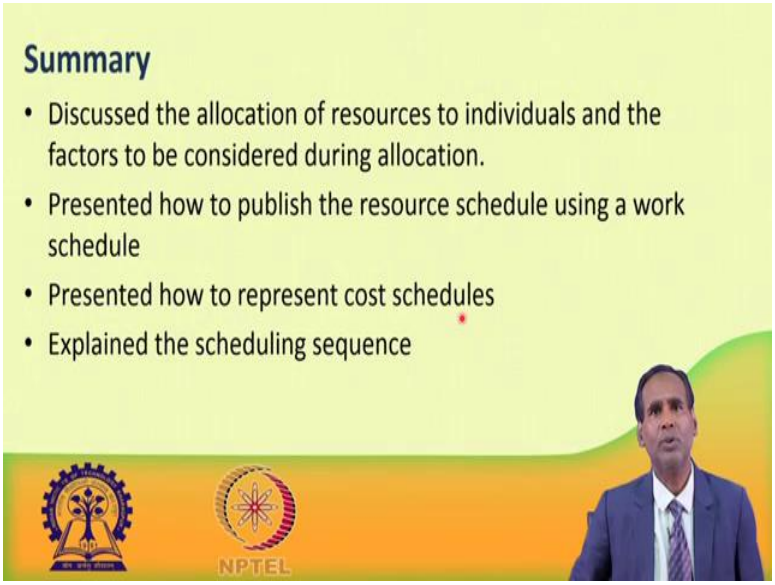

In this way, we have seen on the resource allocations the different schedules we have seen. So, we have discussed the allocation of resources to individuals and the factors to be considered during allocation.

We have also presented how to publicize, how to publish the resource schedule using a work schedule or a work plan. We can better, in a better way you can publish the resource schedule using a work schedule. We have also presented how to represent the cost schedules, we have also explained what is the scheduling sequence starting from the building the activity, activity plans to the cost schedules.

Thank you very much.

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Lecture - 39
Project Monitoring and Control

Good afternoon. Now, let us take up a new chapter that is Project Monitoring and Control. So far we have seen how to prepare schedules, prepare resource schedules, how to prepare cost schedules etcetera, how we can represent, how we can display the schedules, may be in the form of a work plan or work schedule. Now, let us see how to monitor and control the progress of a project.

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We will see little bit about introduction of the project monitoring and control, then we will see a cycle known as project control cycle for controlling the project for controlling the progress of the project, then how can see the project reporting structures what will the structure, then how to assess the progress of a project.

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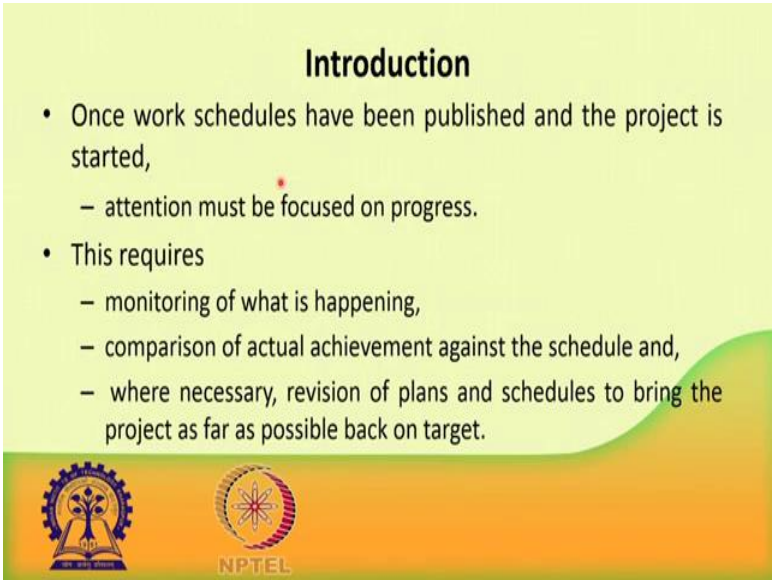

Once the work schedules have been published and the project is started, then the project manager must pay attention or must focus on the progress. I have already told you how we can publicize the outcome of this resource scheduling, may be through the help of work schedules, we can publicize the outcome of this what resource scheduling. So, once work schedules have been published, published and the project has been started, so attention must be on attention must be given on the progress of the project.

So, how to what to focus on the progress of the project? So, this requires monitoring of what is happening in the project, comparison of actual achievement against the schedule one, and wherever necessary, we have to revise the plans and the schedules to bring the project as far as the as far as possible back on target. If in schedule something is what proposed and we are running were we are running away from the schedule, then we have to revise the earlier plans so as to bring the project target so as to what so as to bring the project to the target date.

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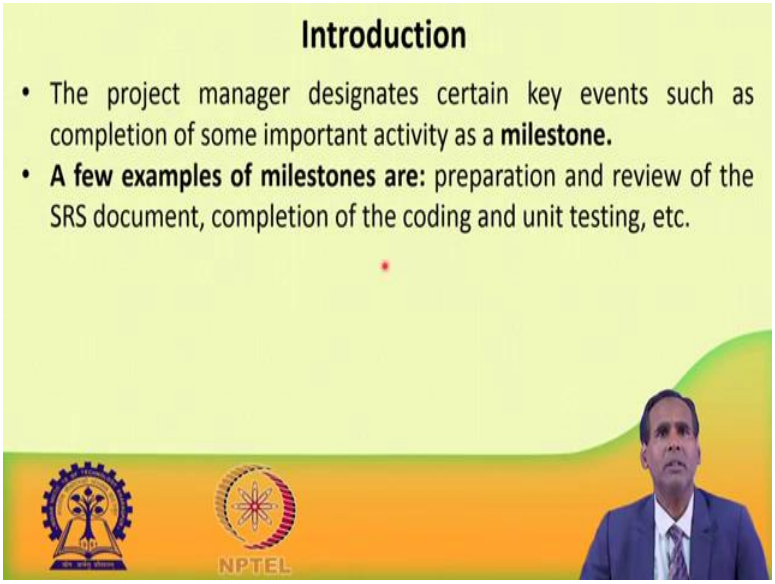

So, once the project is undertaken, and the project manager has to start monitoring the project, the during this or for the project monitoring the project manager designates some key events such as completion of some important activity might be a completion of the requirement analysis, completion of the design, completion of the coding, etcetera. So, these activities may be set as a milestone ok.

So, for this monitoring purpose, the project manager designates certain key events such as completion of some important activity as milestones. The important activity could be completion of requirement analysis, completion of design, completion of coding, completion of testing etcetera, there may be what treated as the milestones.

A few examples of milestones are preparation of review of the SRS document, completion of design, completion of coding, completion of unit testing, completion of system testing etcetera. So, these are few examples of milestones. So, then the project manager can measure the progress looking at the milestone, whether what was proposed in the schedule for the milestone and what is the actual situation right now, whether we have completed this review of this SRS document or not, whether we have completed the coding by this date, by the target date or not, accordingly you can what monitor the progress of the project.

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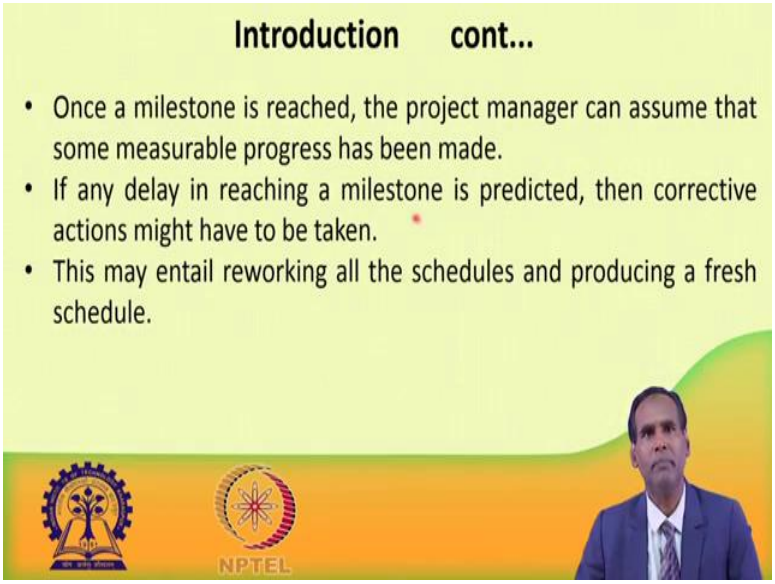

So, once a milestone is reached, the project manager can assume that yes some measurable progress has been made. If any delay is there in reaching a milestone or it is predicted that there will be some delay in reaching a milestone, then the corrective action has to be taken by the project manager. So, this may entail what, so if you are expecting that there will be some delay in reaching the milestone, then the project manager has to take some remedial action.

This may entail reworking all these schedules, so whatever schedules we have repaired earlier; you have you may have to revise them, you have to rework them, because some delay is expected and we have to producing a fresh schedule.

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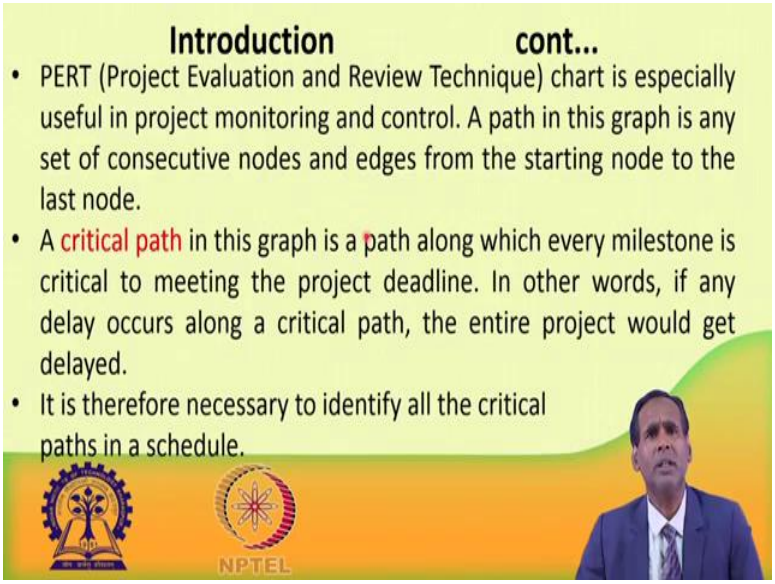

So, there are some tools there are some techniques helpful for monitoring the progress for controlling and the project. So, one such technique is PERT, PERT you have known earlier stands for Project Evaluation and Review Technique; so this strategy especially, very much useful in project monitoring and control. In PERT actually there are several paths and a path you know it is defined a path in a graph is defined as any set of consecutive nodes and edges from the starting node to the last node, this definition we have already seen in graph theory ok; also a path in this graph that means, in this PERT chart is any set of consecutive nodes and edges from the starting node to the last node.

And among all the paths there can be some path, so we call them as critical path and how define the critical path, a critical path is a path along which every milestone is very much critical to meeting the project deadline. In other words, we can define that if any delay occurs along a critical path, then what will happen the entire project would get delayed. Therefore it is very much necessary to identify all the critical paths in a schedule.

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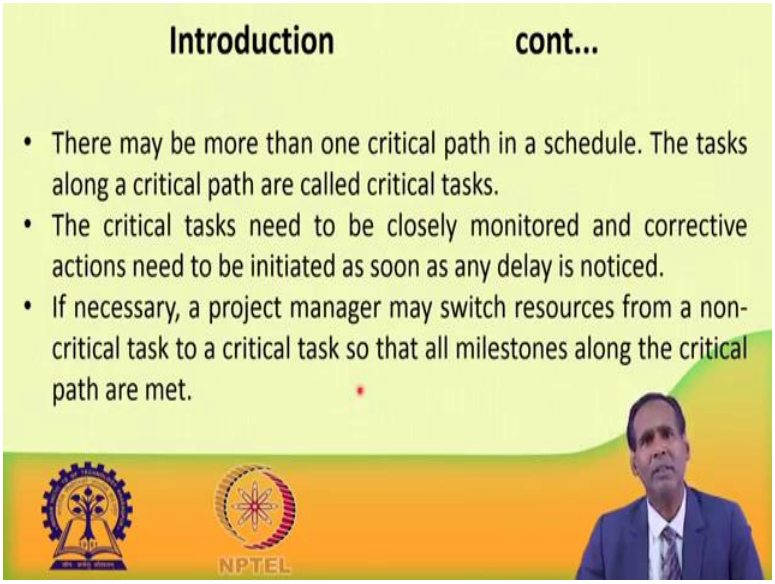

See there can be more than one critical path in a schedule. The tasks along the critical paths, so we call them as critical tasks. The critical task need to be very closely monitored, they has to be controlled, they have to be very closely monitored, because what if there will be any delay, then the critical task will meet will miss that deadlines and hence the overall project duration completion time will be delayed.

So, the critical tasks need to be closely monitored and corrective actions need to be initiated as soon as any delay is notice. So, whenever the project manager expects predicts that some delay will occur or already some delay has delay has been occurred, then he has to initiate some corrective actions in order to what bring the project into the main track into the what scheduled track or the schedule duration. So, if necessary a project manager may switch resources from a non-critical task to critical task so that all milestones along the critical path are met.

So, if a critical task it is getting delayed. So, as a what precautionary action what you can do, the project manager can divert some of this staff which are allocated to the non-critical task to the critical task, because the non-critical task you know it they have some what free times, they have some slack times or float times. So, even if there will be little bit delayed that can consume therefore, what free times or the slack times and it will not affect the overall project completion date.

So, if you find that the critical task they are getting delayed, then as a project manager you may switch some of the resources, you may divert some of the resources, some of the manpower from the non-critical task to the critical task so that all milestones along the critical path they will be met.

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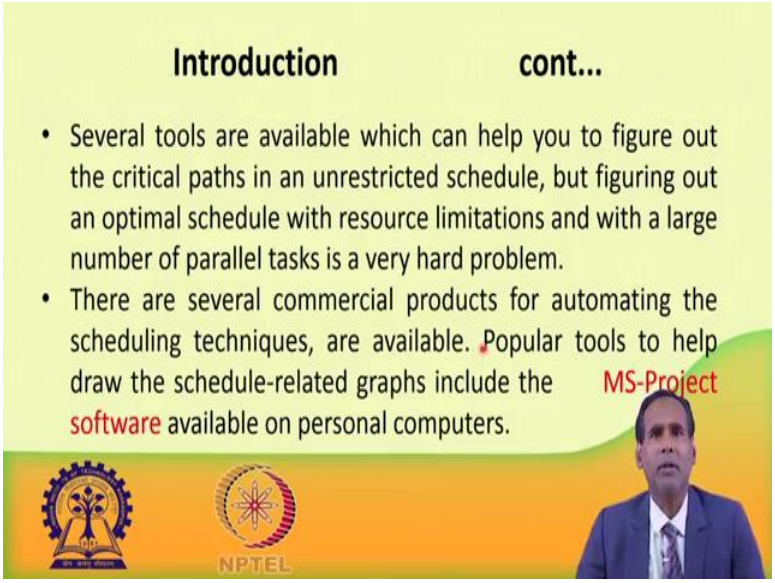

So, several tools are available for project monitoring, etcetera; so we will see some of the tool here now. Several tools are available which can help you figure out the critical paths in an unrestricted schedule, but figuring out an optimal schedule with resource limitations and with a large number of parallel task is very very hard problem.

There are several commercial products or tools for automating the scheduling sequences. For example, the popular tools to help draw the schedule-related graphs include the MS-Project software available on personal computer So, in if you are having you the personal computer is there, then there is a what software there is a you can say software MS-Project software that is available on almost all of the personal computers and this tool you can use to draw the schedule-related graphs such as CPM, PERT etcetera.

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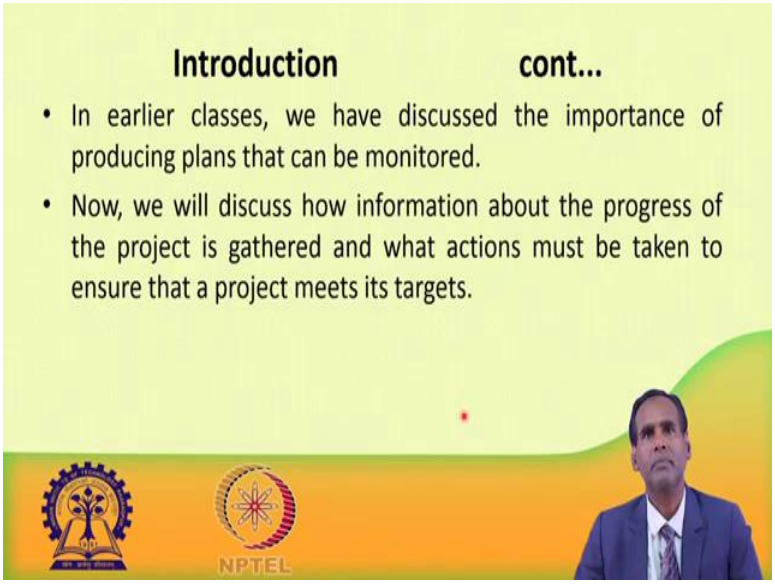

So, in previous classes we have discussed the importance of producing the plans, different plans how to produce we have seen earlier how ok. So, in the previous classes we have discussed the importance of producing plans, which can be monitored. Now, we will discuss how the information about the progress of the project is gathered.

So, how far the progress we have already made the plan, but how far the project is progressed, is it according to the what schedule mentioned in the plan or not; if not, then what action we have to take so that is why, right now let us discuss. How the information about the project of the about the progress of the project how it can be gathered and what actions the project manager must take to ensure that the project meets its targets. If he expects some delay will is to occur, then how he should take some precautionary measures, so that the delays can be avoided and the project can meet the target dates that is the subject matter of today's class.

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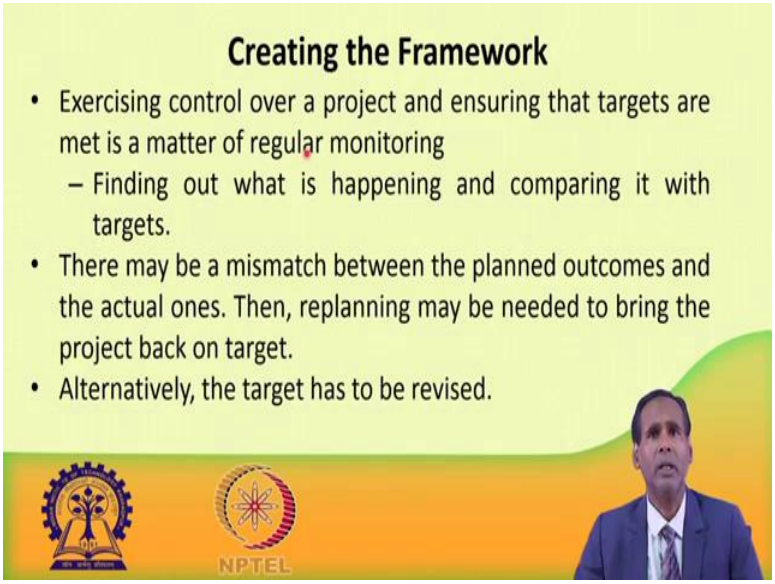

Let us see about the how to create the framework. Exercising control over a project and ensuring that targets, they are met is a matter of regular monitoring. So, a regular monitoring means finding out what is happening now and comparing it to what the target. So, what is currently happening and what should be they are at this time as per mentioned in the schedule as per the target, so compare that. Finding out what is happening and comparing with what the targets as mentioned in the schedules.

There may be a mismatch between the planned outcome and the actual ones. Then if there is a they are not matching, then what will happen we have to do replanning. Then replanning may be needed to bring the project back on target, so that some remedial actions has to be taken they have to be taken in order to bring the project, what to its a original target. So, if it is not possible replanning is not possible, then the target that is they have to be revised.

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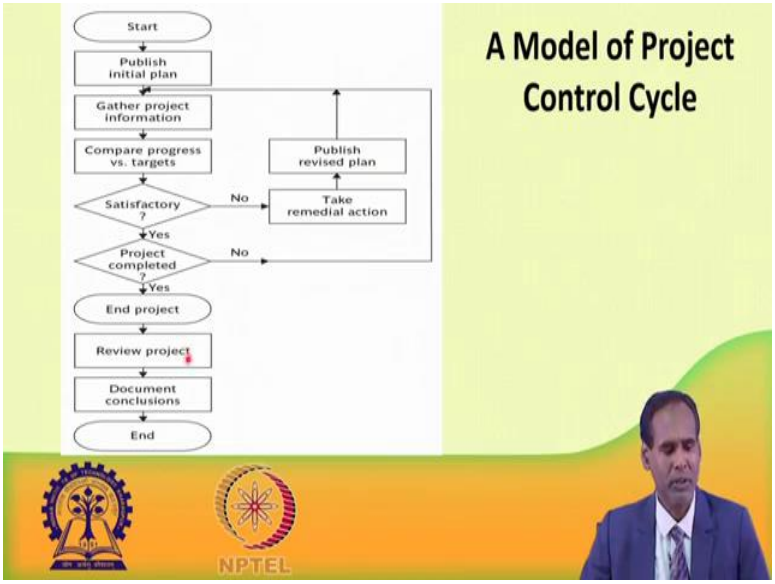

So, let us see a particular model a specific model of project control cycle. So, how to what control the progress of a project. So, first what we are doing, first we are what we are doing publish the initial plan we had we had made the initial plan as a project manager you have made the initial plan, then you publish it. Then you gather the project information regarding the progress, so gathered the project information regarding the progress of the project, then what you do compare the progress versus the targets.

So, in the initial plan the targets you have mentioned, then you have collected the current progress information compare them. Is it satisfactory that means, are you moving according to the plans is the progress, is the progress or is the progress is happening is the progress happening according to the initial plan or the scheduled plan, is the progress satisfactory, if yes, then terminate the project end the project.

If it is not sorry, yes if it is what satisfactory, then we will see whether the project is completed or not; if the progress is not satisfactory, we have to take a remedial action, then we have to revise the initial plan, then we have to publicize the revised plan and then again we have to gather the current information, the current progress information or we have to gather the project information regarding the progress or the at present progress, current progress.

And again we have to compare, again we have to see that whether it is satisfactory or not; again if it not this loop will continue and when will be satisfied that yes the progress

is according to the plan made in the schedule, then we will check whether the project is completed or some portions are left. If the project is completed we have to declare that yes project is ended, project is completed; if it is project is not completed, then again we have to go back, again we have to gather the information of the project regarding the current progress and again the loop will fall, again the loop to loop will be followed.

And after the project is ended what we have to do, we have to review the project, then we have document the conclusions, we have to draw the inference, document the conclusions, this we must note down, because if in your future we will get similar types of projects. The lessons what we have learnt earlier that will help us in executing similar types of projects in future and then we will end. So, this is how a model of the project control cycle looks, in this way you can control the progress of a project.

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So, this whatever I have explained that I have written here, you can see yourself that the project control cycle it shows ah how, once the initial project plan has been published, then project control is a continual project please see, project control it is not a one time job, it is a continual process of monitoring progress against what; against the initial against a plan and wherever necessary we have to revise the plan to take account of the definitions. It also illustrates the important steps that must be taken after completion of the project, see after completion of the project we have to review the project, then we have to document the conclusions, etcetera.

So this also so, so this framework also illustrates the importance steps that must be taken after completion of the project, so that the experience the lessons that we have gained in any one project can feed into the planning stages of the similar future projects, thus it allows us to learn from the past mistakes.

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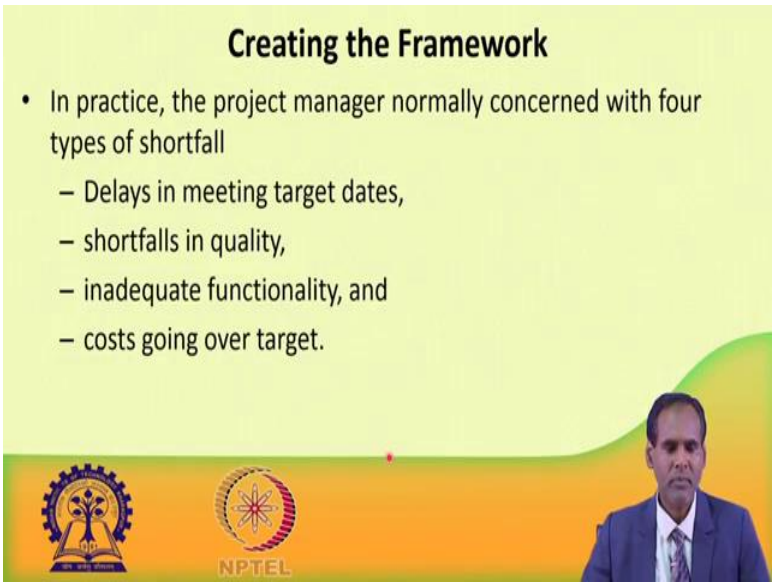

So, in practice the project manager normally concerned with a four types of shortfalls what are they, number one, delays in meeting target dates, why there is a delay in meeting the target dates that we must as a project manager you must see. Then shortfalls in quality, whether the quality as we have expected as we have what anticipated, whether that quality is maintained or there is a shortfall in that.

Similarly, inadequate functionality whether the functionality that of the final product is sufficient or as far ours what plan or it is inadequate. And finally, cost going over the project, what cost we have plan, what cost we have mentioned in the cost schedule, whether it is within that or it is over the target. So, these are the what four types of shortfalls that project manager normally concern do it during development of a project.

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Now, let us see the responsibilities whose responsibility is what, the overall responsibility for ensuring satisfactory progress on a project is the role of the project steering committee, sometimes it is known as project management board; the this is the overall responsibility that is given to the steering committee of the project management board.

The day-to-day responsibility normally it is given to the project manager, it rests it relies with the project manager and you know for every project there is a project manager. Other normal aspects of the projects, they can be delegated to the team leaders.

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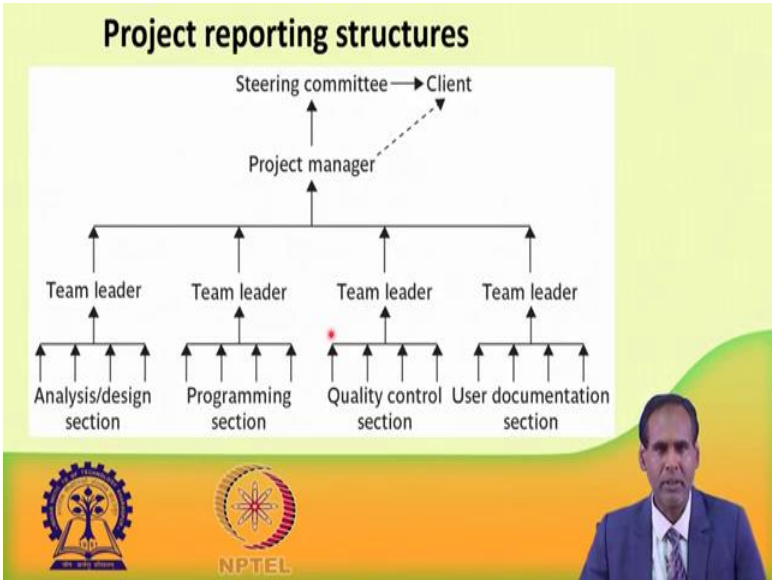

And this is how the reporting structure this looks likes. So, steering committee looks after the overall aspects of the project, overall progress of the project. Then under the steering committee, the project manager works and under the project manager there are several team leaders are there, and or not the one team leader several what other what personals are working.

Like one under one team leader, the analysis and design section their employees are there. Another team leader the programming section I mean, the programmers coders are there. Another team leader the quality control section is there, testers are there. Another team leader the user documentation employees they are there.

So, these what bottom level employees they report to the team leaders, then the team leaders they prepared this summary and they report to the project manager, project manager adds his comments, etcetera and modified this one revises the what reports given by the team leader and sends the reports to the steering committee; and steering committee forwards to the these reports forward to the clients, sometimes the project manager also can directly communicate to the client regarding the progress or the reports documents, you can directly communicate to the clients.

So, this is how the reporting structure look like. The end developers or the end personnel the end or the bottom level personnel in the structure, they will report to the team leaders

and the team leaders will report to the project manager, project manager will report to the steering committee.

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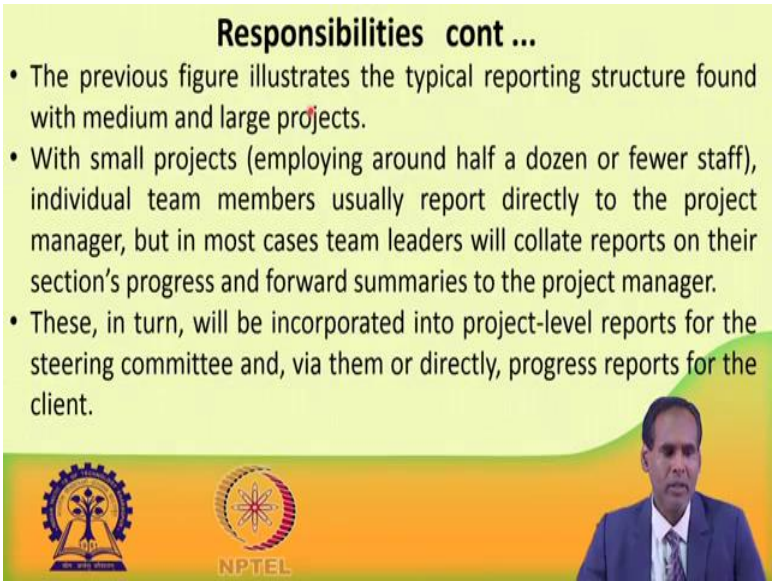

So, I have already told you that this previous figure, it illustrates the typical reporting structure, which is found with a minimum with medium and large scale projects, but for small projects we are less than half a dozen or fewer staff are there. The individual team members, they usually report directly to the what project manager, but in most cases the team leaders will collate the reports on their sections progress and then they will forward the summaries to the project manager.

Then these, in turn, I mean these reports are given by the project manager they will be incorporated into the project-level reports for the steering committee and, via them or even if in some cases directly, the progress reports for the client it will be sent. So, these progress report for the clients may be directly sent from this project manager to the clients or via this steering committee.

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So, reporting could be oral or written, see let see the different kinds of reporting. We have already shown you the reporting hierarchy, this is the reporting structure. Then how do the what staff members they report, the reporting can be oral or written, it can be formal or informal or it can be regular or adhoc. So, informal the communication is also necessary and important, but whenever any such informal reporting of a project progress, it is followed, it must be complemented by some formal reporting procedures.

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See these are the categories of reporting, like this is the oral formal regular type of reporting type and one example is weekly or monthly progress meeting, and this is the comment like while reports may be oral, formal written minutes this would also be kept.

Then oral formal adhoc type of reporting, example is end to end-of-stage review meetings. And written formal regular, example is job sheets, progress reports, etcetera. Then another type is written formal adhoc, examples are exception reports, change reports, etcetera. Oral informal adhoc, example is like discussion in canteen, social interaction, etcetera, these are some of the content. So, these are the different categories of for reporting.

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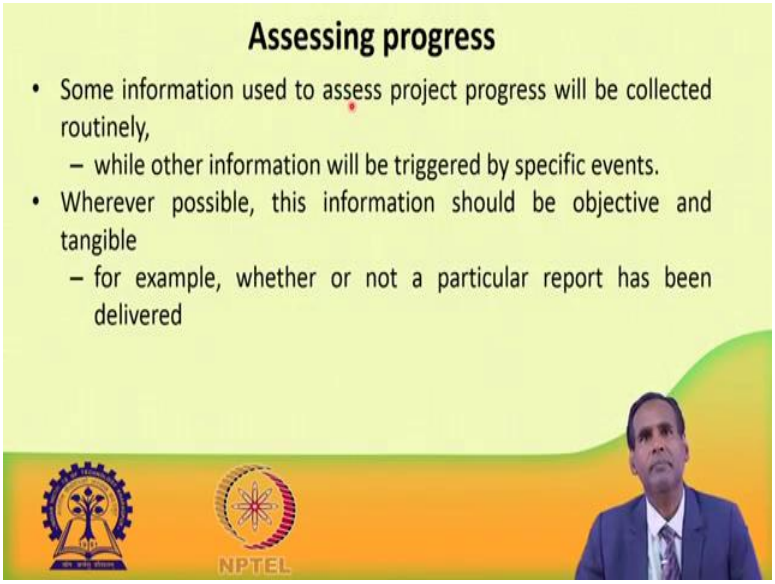

Now, let us see how to assess the progress of a project; how as a project manager, how we can assess the progress of a project. Some information are used to assess project progress or the progress of the project that should be collected routinely, while some other information will be triggered by specific events. So, in order to assess the progress of the project, some information used to assess the project progress will be collected routinely, maybe weekly or monthly also.

While some other information and they will be triggered by specific events, maybe when the requirement specification document SRS document is over, say may be you have to collect some information. When the design is over, you may collect some other

information like this, whenever some specific event is triggered, then you will collect some information.

So, now wherever possible this information should be very much objective and tangible. For example, whether or not a particular report has been delivered or not, so this kind of information, please try to make them objective and the tangible as far as possible.

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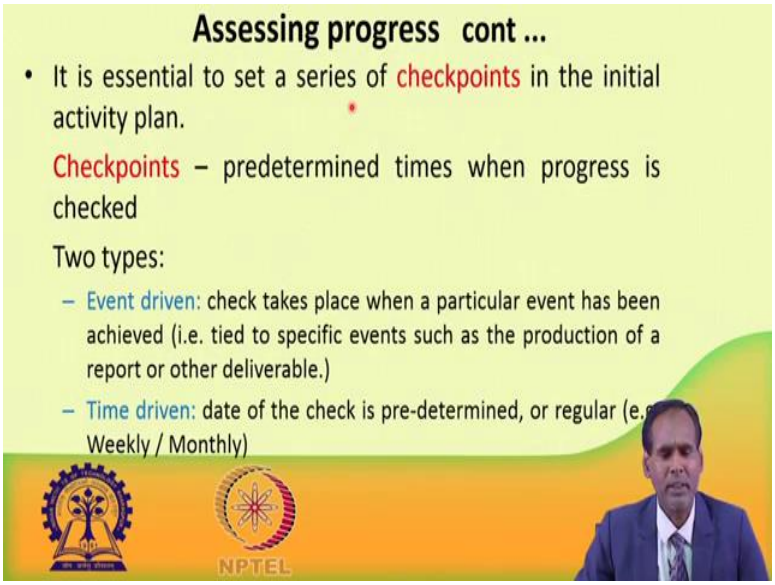

So, it is essential to set a series of checkpoints in the initial activity plan. So, in order to assess the progress of a project, it is very much essential to set a series of checkpoints in the initial activity plan. What you mean by checkpoint, checkpoints means these are some predetermined times when progress is checked. So, you have to determine some ok, you have to pick some predetermined times, when the progress is checked.

So, the checkpoints can be up for two types, event driven and a time driven. In event driven the check takes place when a particular event has been achieved. So, what could the example of a particular event, might be what the production of some report or other deliverable or what the coding is complete or testing is complete like that some event is completed, then you check whether you have achieved whatever it is mentioned in this schedule or not.

So, another checkpoint is time driven. So, here date of the check is pre-determined. So, on what date you will make the checking that is pre-determined that means, it is regular

might be weekly basis or monthly basis; you will check the progress, you will you will check the progress so that is why this is regular, this is known as time driven checkpoints.

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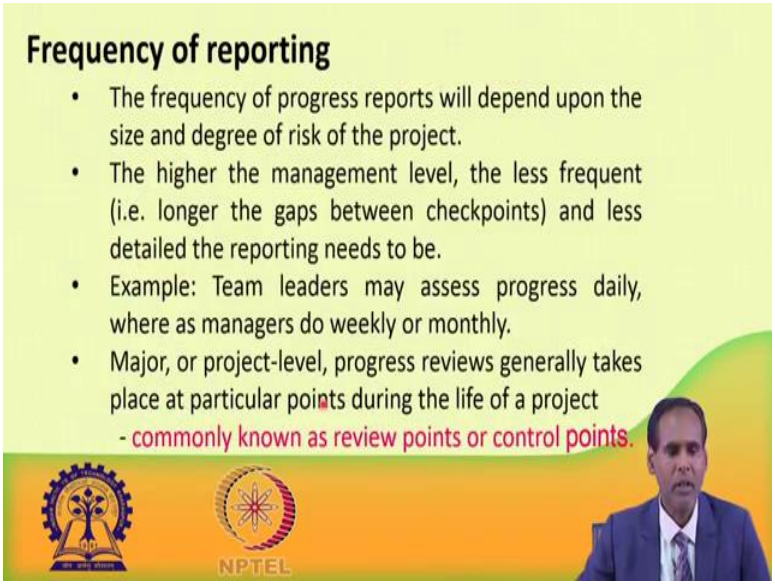

Now, let see what is the frequency of reporting, how frequently you should do the reporting. The frequency of progress reports will depend upon the size and degree of risk of the project. So, the how frequently as a project manager you should check the reports, you should prepare the reports like this.

So, the frequency of progress reports will depend on several factors such as what is the size of the project, what is the degree of risk of the project; if this is a what real time system, if this is a safety critical system, where risk is more, so then the frequency will be very what it is very frequent might be every day you have to monitor this thing. If it is a very large size project, might be again very frequently you have to check the progress. So, the as per the rule the higher the management level, the less frequent is this what reporting.

For example, longer the gaps between the checkpoint, so if higher is the management, less is the frequent reporting or longer the gaps between the checkpoints and less detailed the reporting needs to be, because the high level management they are not interested in the detailed what report, they will see the summary of the report etcetera.

But lower the management more frequent in the reporting and more detailed should be the ah reporting documents. For example, you see the team leaders may assess progress daily. So, we have already seen what the hierarchy somewhere else here.

So, at the bottom level like team leader they will require the reports more frequently might be on every day whereas, project manager may require the report may be on every week or so. So, steering committee maybe they will require the reports on every month. So, higher is the management lesser is the frequency and lesser is also the detailed reports.

Ok, this I have already told you that higher is the management level, the less frequent is the reporting as well as less detailed the reporting needs to be. In other words we can say, longer if the gap between those between the checkpoints. Example I have already told you, the team leaders may assess the progress of the projected daily whereas, the managers do that in every weekly and the steering committee may do that in every month.

Major, or project-level, progress reviews generally takes place at a particular points during the life of a project, maybe you can say that every a month ok. Major, or project-level, progress reviews generally takes place at a particular point, maybe particular point during the life of a project might be in every month, so the these points ok. So, as I have already told you the project-level progress reviews, it generally takes place they generally ok.

So, project project-level progress reviews generally takes place, when at some particular points during the life of a project maybe in one month or so, so these particular points are commonly in known as review points or control points. So, this is how the frequency of a reporting it occurs, as I have already told you. It will be what if it is a bottom level management, the frequency of reporting is more, the detailed reports will be more, but if it is a as we are moving towards the upper level of the management, high level of the management; then the frequency of the reporting will be less as well as the detailed reports, they will require a less detailed report.

So, this is about how frequently you should do the reporting in a what project organization.

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So, finally we have discussed the importance and the need of project monitoring and control in project development. We have also explained the project control cycle, so not only that after the project is completed; what we should do like project review and documents, etcetera from where we can gain some lessons, some experience which will help us in handling similar types of projects in future.

We have also presented the project reporting structures and we have also seen the frequency of the reporting, we have also explained how to assess the progress of a project. So, this we have seen in this today's class.

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We have taken from these references.

Thank you very much.

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Lecture - 40
Project Monitoring and Control
(Contd.)

Good afternoon to all of you. Now let us start the some other aspects Project Monitoring and Control.

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We will first see how to collect the progress details of the project, then we will see a special technical called as partial completion reporting, then we will see about the project review.

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So, there is a need to collect the data about the progress details, may be the achievements that you have made, the costs that have been incurred in the project etcetera that we have to collect the data regarding these details, but a big problem is how to deal with the partial completions; that means, the 99 percent completions syndrome. So, some what activities are there, they are always when we will see 99 percent complete, still 1 percent remaining.

So, how to deal with this partial completion problems. There are that some possible solutions to these problems; number one we have to control the products, control of the products, we should control the products not the activities. We should have controls on the product not the not on the activities. Similarly, if larger complex activity are there we should sub divided these complex activities into a number of sub activities.

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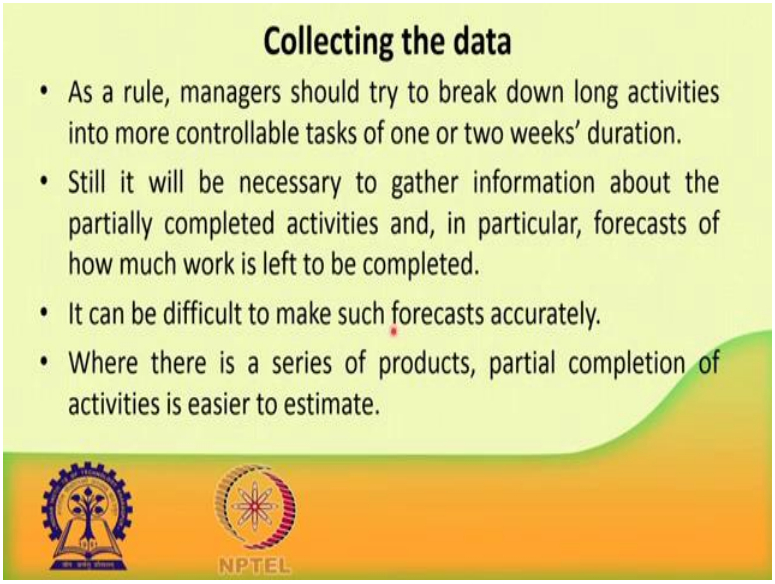

Now, let us see how to collect the data. Normally the managers should try to break down the long activities, complex activities into more controllable tasks of 1 or 2 weeks of duration, still it will be necessary to gather information about the partial completed activity. That means, the activities which are not fully completed they are partially completed, we should collect the data regarding these activities. In particular we should collect the information regarding the forecast of how much work is still left that has to be completed, but it is very difficult to make such forecast accurately in advance. So, where there is a series of products partial completion of activity is easier to estimate.

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So, where there is a some series of products then partial completion of activity is easier to estimate. Now let us see counting the number of records specifications or screen layouts produced for example, it can provide a reasonable measure of the progress. So, if you can count that what is the total number of records and how many records you have specified. The counting the number of records specifications or what is the total number of screen layouts and how many you have produced so far. Those things also can give a reasonable measure of the progress, but in some cases intermediate products they can be used as inactivity milestones ok.

So, in some cases some intermediate products can be used as inactivity milestones. For example, the first successful compilation of a program, even if it is not the completed one, but first time the first successful compilation of a program which is an intermediate, what result it can also be considered, it can also be considered as a milestone, even if it is not the final product is it is an intermediate product, but it can be considered as a milestone.

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So, now let us see about the partial completion reporting, we have to collect data regarding the partially completed activities, we have to report the status. Many organizations they use standard accounting systems. Several organizations are there they use standard accounting systems with weekly timesheets to charge stop time to individual jobs. So, they use something a timesheet, weekly timesheet in order to

determine what will the charge, what will the, what salary that the staffs they have to be paid on weekly basis. The staff time booked or to a project indicates the work carried out and the charges to the project ok.

The staff time booked to a project, it will indicate the work carried out by the staff and the charges to be to the project, but it does not what tell the project manager what has been produced or whether the tasks are on schedule or not. These weekly timesheet, it will not give any information regarding that whether what has been produced or what has been what or whether the task or on schedule or not, so that is why it is common to adopt or we can enhance the existing accounting data collection systems to meet the needs of the project control. We have, we should enhance or extend the existing accounting data collection systems.

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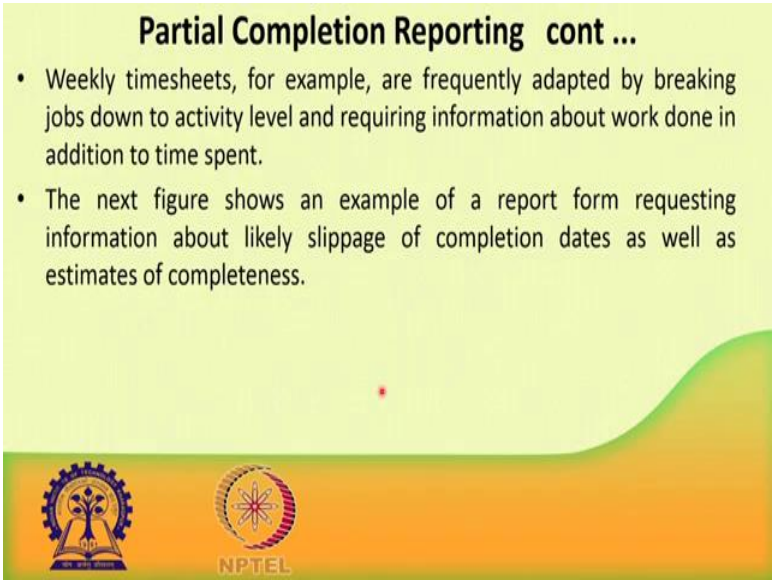

So, let us see how these weekly timesheets look like. The weekly timesheets are frequently adopted by breaking job or the task, breaking down the jobs or task to activity level ok. So, in timesheets the, what the jobs or the tasks they are broken down to activity level and they require information about work done in addition to the time spent. So, we also, they also, so it also requires information about the work that has been done in addition to the time that has been spent.

So, in next slide you will see an example of a what weekly timesheet, where we can see that this example, this figure, we will see one example of this weekly timesheet. So, here what is being shown.

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So, this figure is shows an example of a report form requesting information about likely what is the slippage, what is the likely slippage of the completion dates as well as the estimates of completeness. You can see in the timesheet is there what is the staff name the for which week you are preparing. Then mainly two types of charges are there the rechargeable hours and the non rechargeable hours, then for which projects he has worked, then what the activities he has performed, the corresponding activity codes and the description of the activities like coding and documenting etcetera, then how much hours in this week he has worked for this activity.

Like in the coding activity he has spent 12 hours, so and also it shows what is the percentage of the work completed, 30 percentage completed and what is the schedule completion, say 24 April, but what is the estimated completion same 24 April.

But for the next activity document take on hours this week he has worked 20 hours and percentage completed 90 hours. As a result the schedule completion was 6th April, but it was see it will be estimated early before 2 days before, so it 4th April. So, because 90 percent of the work have been done and it will, so that the total rechargeable hour is 32. There are some non rechargeable hours, because he has done in lieu or something, so

hours 8 week, this 8 hours, this is what you can say this is non rechargeable. So, now, this is rechargeable hour 32, if you know per hour what is the price multiply by 32. So; that means, this much what amount you have to pay, you have to pay to John Smith. So, this is how a times sheet looks like.

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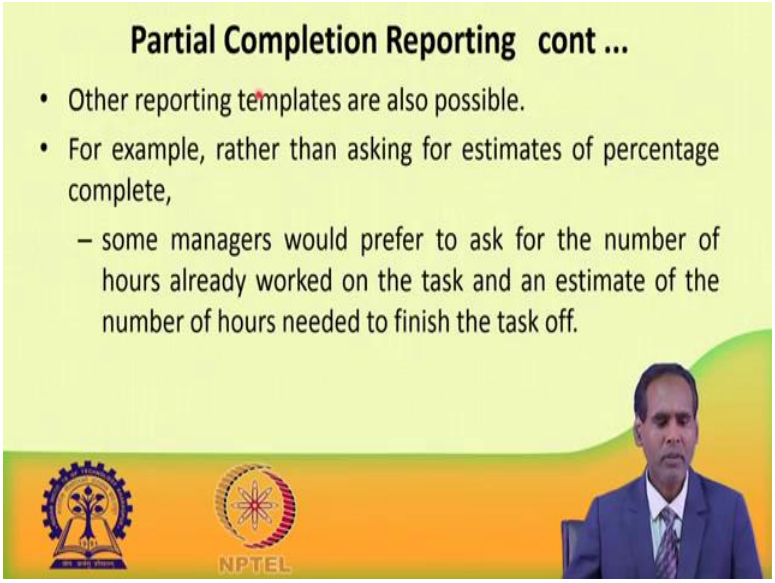

So, other reporting templates are also possible people are using. For example, rather than asking for the estimates of the percentage complete some managers may prepare to ask for the number of hours, already worked on the task and estimate and an estimate of the number of hours needed to finish the task off. Just like as here your saying 30 percent is complete, so 70 percent is left and 90 percent is complete, 30 percent is left. Instead of asking that, so the manager may ask in terms of hours, how much hours you have spent and how much hours still required to finish the tasks of.

These are alternatives to this reporting method, this what timesheet or this weekly timesheet form, there are this is this can be another alternative. Now, let us see a particular technique called as read amber green reporting method for this partial completing work, for this partial completion reporting aspect.

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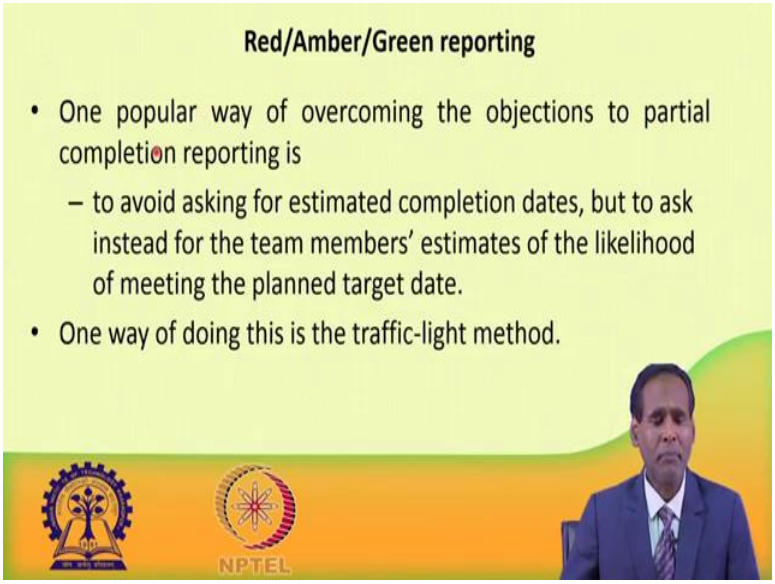

One popular way of overcoming the objections to partial completion reporting is, to avoid asking for the estimated completion dates. So, we can avoid asking the estimated completion dates, but we can ask the team members about the estimates of the likelihood of meeting the plan target date.

What is the possibility, what is the probability or what is the likelihood of meeting the plan target date? So, one way of doing this is by using a traffic light method. As you know in traffic light there are three lights are there; red, green and amber or yellow. So, those kinds of things also be used here for overcoming the problems of partial completion reporting. So, this approach works as follows, the steps are followed in this way.

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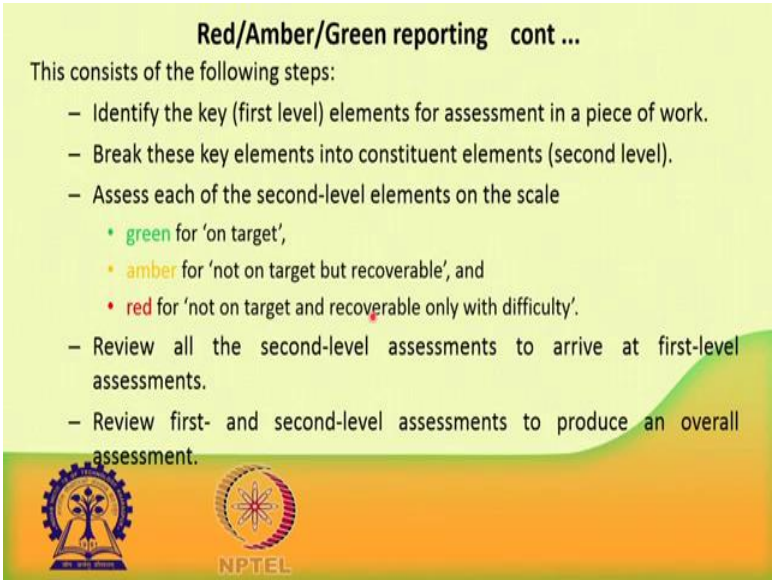

First step is identify what are the key elements or the first level elements for assessment in a piece of work. Then break these key elements which are the higher level or the first level into some constituent elements, we will call them as a second level elements. Now we have to assess each of the second level elements on the scale of these three colors; one is green color for on target; that means, the activity is on target ah. So, no problem.

So, second one is amber; that is for not on target, the activity currently it is not on target, but it is recoverable. If we can take some remedial actions we can recover, we can meet the target and the last, the final colour is red which is represents that not on target right now and recoverable only with difficulty. If we will put much more effort then it may be recovered, but it may be recovered with a difficulty. So, these are in this way we can assess each of the second level elements on these the on the three color scale

Then after assessing each of the second level element we should review all the second level assessment to arrive at the first level assessment and then we have to review the first level and second level assessment to produce an overall assessment; that means, overall assessment for the project. So, this is how we can overcome some of the problems of the partial completion reporting.

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So, following completion of assessment forms for all activities, the project manager uses these as a basis for evaluating the overall status of the project. As I have told you in the critical activity, normally it is as is classified as amber or red will require further consideration. As long as it is green no problem it can meet the deadline, it can meet the target, but any critical activity which is classified as amber or red it will require further report further consideration and often leads to a revision of the project schedule. In this case we might have to require to revise the project schedule.

Non-critical activities, they are likely to be considered as a problem if they are classified as red. So, if the non-critical activities, see critical activities amber or red means we have to reconsider, we may have; we may have to revise a project schedule, but the non-critical activities if they are green or amber, no problem so, but if they are red then also problem might be there. So, in this case we have to what, we have to revise the plan because, so this they are float they are free or the slack time may be consume. So, after some time again they may be they will be termed as critical activities.

So, non-critical activities they are also like to be considered as a problem if they are classified as red. Especially, if they are float or slack is very less and likely to be consumed in near future, so that again they will be converted to critical and then it will difficult to what meet the target lines. So, also we have to take remedial actions to bring the project, to get the project to the right track to the schedule.

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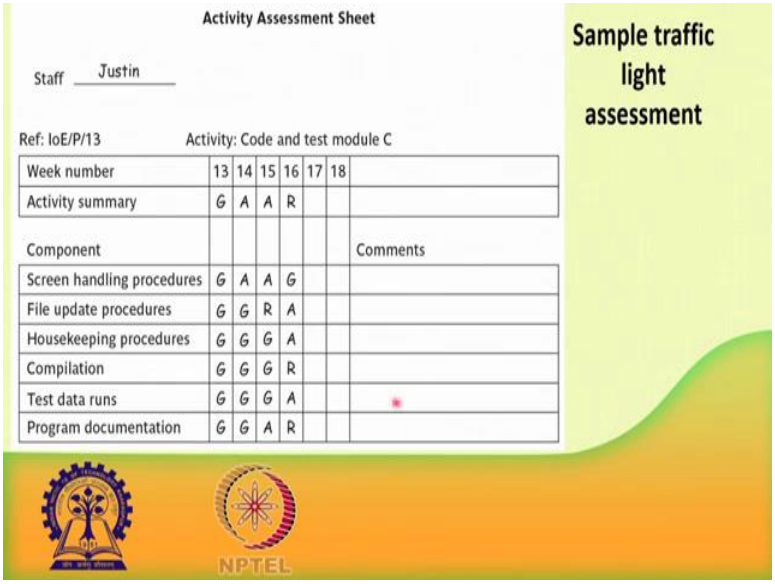

This is a sample traffic light assessment you can see here. So, its that the name of the staff say Justin and reference number which project, activities per code and test module of C. These are the week number 13, 14, 18 etcetera, you can see that. Let us see first the components first component is screen handling procedures, its you can see this in what green and also file update procedure also green, so these are all green. So, we can say that activity summery at the end of 13th week its green.

But at the end of 14th week this screen handling procedures it is amber; that means, what amber means we have already told you. Amber means it is not on target, but it is recoverable. So, it shows that it is amber, so it can be recovered, but file update procedures it is red, again these are green and program documentation is amber. So, overall assessment is that it is. Sorry this is overall assessment here in this case is, at the end of 14 it is amber. Similarly or end of 15, at the end of 15th week you can see different components of different values green, amber or red accordingly the activity summery is rated as is assessed as A.

And for this you see which is one of the main important thing is compilation, this is a critical activity, this is red also program documentation is red. So, as a result at the end of 16th week the activity summary is assessed as the red. That means, now we have to do something; otherwise it will go beyond the what schedule, so we have to take some remedial actions, so that we may what revert back, we may meet the what target lines,

but with lot of difficulty. This is how the sample light asses assessment based on the red, amber, green scales this works.

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Then we will take another important technique called as review. So, from a managers perspective review of work products is very much important, it is an important mechanism for monitoring the progress of the project and ensuring the quality of the work products. Every project each developed through iterations over a large number of work products. What could be the work products? The work products could be requirement documents, design documents, project plan document, code documents, test plans etcetera.

So, each of these work products they can have large number of defects in them, because human beings are preparing this thing, so there is a possibility that each product may contain a large number of defects in them due to mistakes committed by the development team members which are human beings. That is why it is necessary to eliminate as many defects in this work products to realize a product of acceptable quality.

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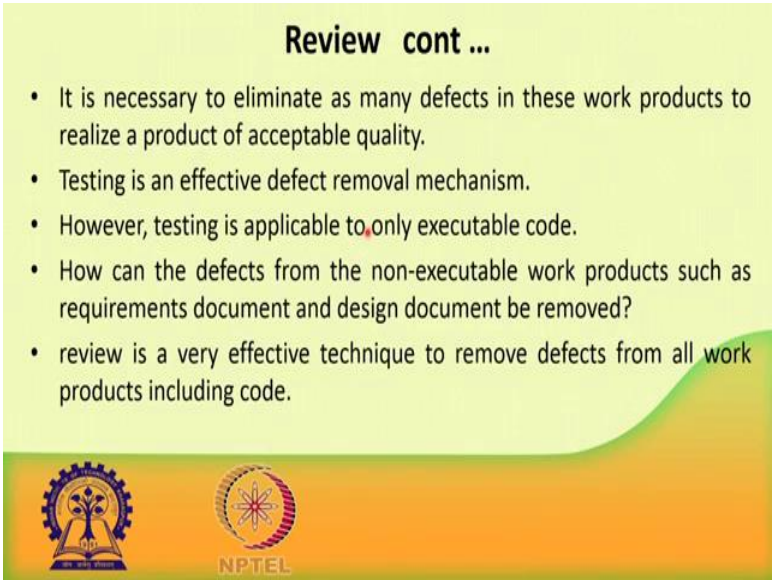

Testing is an effective defect removal mechanism no doubt, but; however, testing is applicable to only executable code. Very much difficult to apply this testing to this requirement documents or the design documents. So, how can the defects from the non executable work product such as requirement documents, design documents be removed. So, for this review comes to be very much handy, its an useful technique for removing and this defects. So, review is a very effective technique to remove defects from all work products even including code.

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In fact, review has been acknowledged to be more cost effective in removing defects as compared to testing. Early review techniques they focused on code and systematic review techniques were developed for this specific purpose, but over the years the review techniques have become extremely popular and have been generalized for the use with other work products also.

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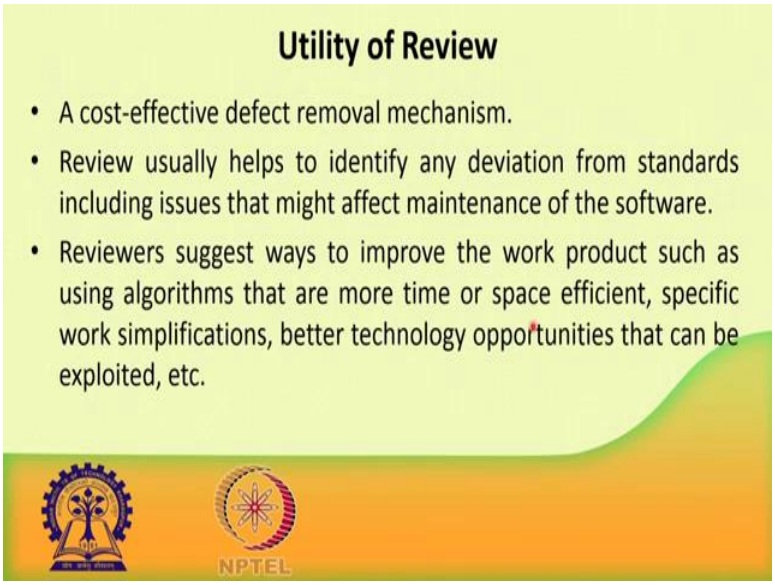

This is a very cost effective defect removal mechanism. Review usually helps for what? review usually helps to identify any deviation from the standards including the issues that might affect maintenance of the software. So, anything else, so this for review it can be used to help identify any deviation from the given standards. Reviewers suggest ways to improve the work product such as using, so these the persons associated with the review, normally we call them the reviewers.

The reviewers they suggest different ways to improve the quality of the work product; such as using efficient algorithms which are more time and space efficient using specific work simplifications, using better technology opportunities that can be explored etcetera. So, these reviewers can suggest during the review process and hence the quality of the product may be improved.

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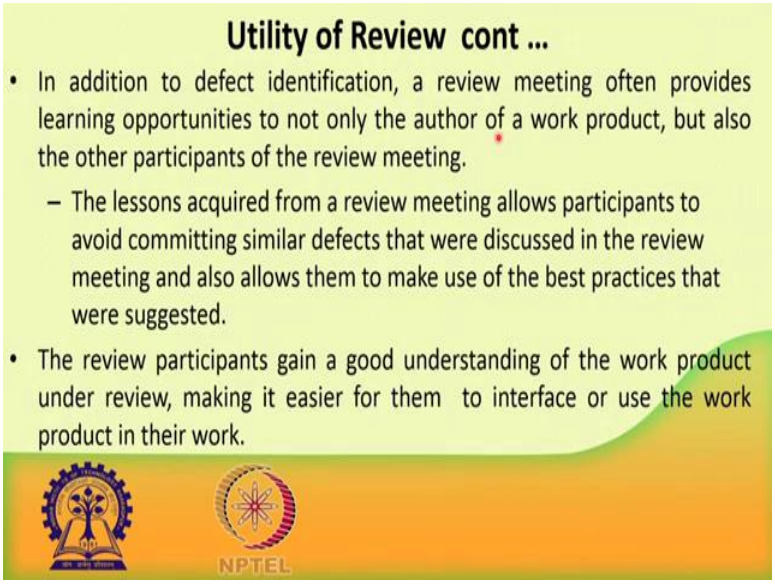

So, in addition to defect identification a review meeting upon provides learning opportunities to not only author of the work product, but also the other participants of the review meeting, it can give good experiences, it can give you good lessons. So, the lessons which are acquired from a review meeting, it will allow the participants to avoid committing similar types of defects, when similar types of what projects will come in near future. The review participants also gain a good understanding of the work product which is under review, making it easier for them to interface or use the work product in their work with other components.

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So, now let us see candidates work products for review. What things can be what reviewed? So, all interim and final work products they are usually candidates for review, they can be reviewed. Usually the work products considered to be suitable candidates for review are as follow, let us see what work products, what products can be sent for review. So, like this SRS document, software requirements specification documents, user interface specifications, design documents.

Then different types of design documents; such as architectural design documents, high level design documents and this detailed design documents, test plans and the what test cases those have been designed, then project management plan, configuration management plan. All those things can be reviewed, these are the possible candidates for review.

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Now, let us see what is the review roles what are, which persons are associated in review, what roles are given during the review process. In every review meeting a few key roles are needed to be assigned to the review team members. So, the some of the review are these roles are moderator, recorder and reviewer. So, some, so for a work product, so the project manager will assign one person as the moderator, maybe one person as the recorder and there can be a group of persons they will be designated as reviewers.

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So, let us see what will the role of the moderator what he will do, what is the job of moderator. So, moderator he plays the key role in the review process. The principal responsibility job moderator, it include scheduling and convening the different meetings, distributing the review materials to the reviewers, leading and moderating the review sessions and ensuring that the defects are tracked to closure. So, these are so; that means, whatever the suggestions are given by the reviews they have been addressed ok. So, he ensures that the defects are tracked to closure. So, these are the responsibilities of the moderator.

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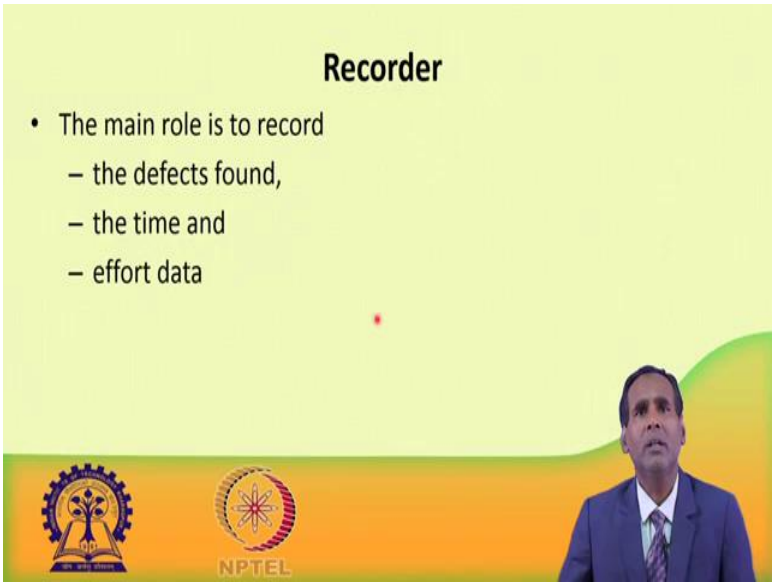

Similarly, the jobs of; the jobs of recorder are like this the main role is to record as a names of the recorder, he will record the following things, he will record the defects found, he will record the total time spent and he will record the effort that they have put in conducting this review.

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And reviewer what is the role the. So, we have already seen about the moderator, we have seen this recorder then the role of the reviewer. The review team members normally they review the work product, as their names are review. So, these review team members will review the candidate work products and they will give specific suggestions to the author of the work product about the existing defects and also they will also point out how to improve the work product, how to eliminate the defects, how to improve the work products.

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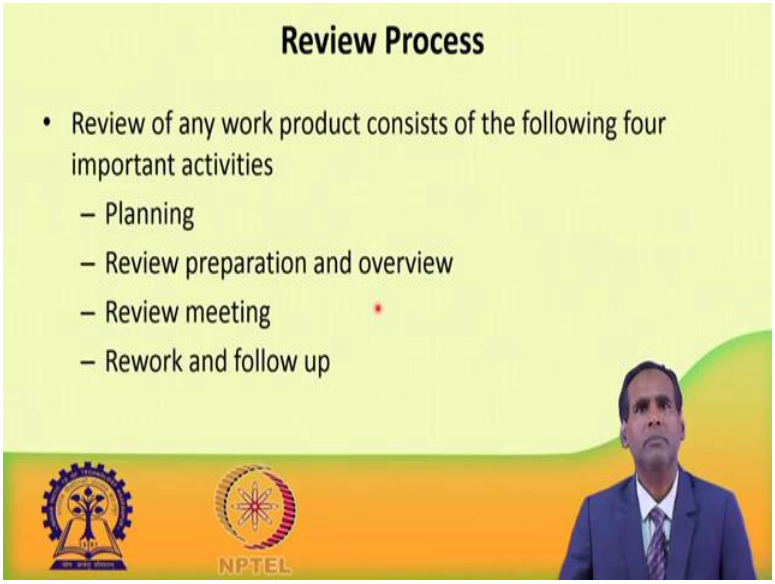

Now, let us see how the review process works? The review of any work product consists of the following 4 important activities. First you have to do the planning, then review preparation and an overview, then usually review meetings and finally, you have to rework on the suggestions and follow up. These steps, these activities can be put in the form of a figure like this.

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So, the candidate work product it will be supplied at the input. Then first activity is planning, you have to prepare the planning. In planning stage the project manager will

appoint one moderators and one recorder, as well as you will form a committe of what review members. Normally 5 to 7 members of the members will be selected as the in the review team. Then the output will be this what review learn on the schedule and then it will be a preparation meeting. The review moderate, the moderator appointed by the project manager, he will what convene a brief meeting for preparation, he what documents has been submitted by the author, the work product or the author of the work product first submit some documents.

So, these documents in this first brief meeting the moderator he circulates among the review team members. The review team members individually they carry out review for on this submitted work product, then they whatever the outputs it is mentioned in a document called as the reviewers log, then the actual review meeting takes place. Here the reviewers give the suggestions on the work products to the what author, then the author responds to the issues raised by the reviewers, other reviewers also participate here, then describe write down all the details.

And finally, the document prepared its called a defect log and this defect log has been will be handed over to the author. Then the author sees what are the issues raised, then he will rework he will revise the work product, he will address all the issues raised by the reviewers then he will prepare a rejoinder a review, saying what are the points raised by the issue what reviewers and how he has addressed all those things.

Then again a final meeting will be convened, the reviewers will investigate, they will look what whether how the author has addressed the issues, how he has corrected, how he has incorporated the suggestions he will check. The moderator will finally, see that yes all the defects which have been there in the earlier version now they have been removed they will, then they will prepare the summary report. In the summary report they will see the details of the defects, the time spent etcetera that will be that will mentioned in the summary report.

So, in this how, you know in this way the review process model works. These are the different activities carried out in the review process and these are the interim or intermediate products and the summary reported the final product of this review process.

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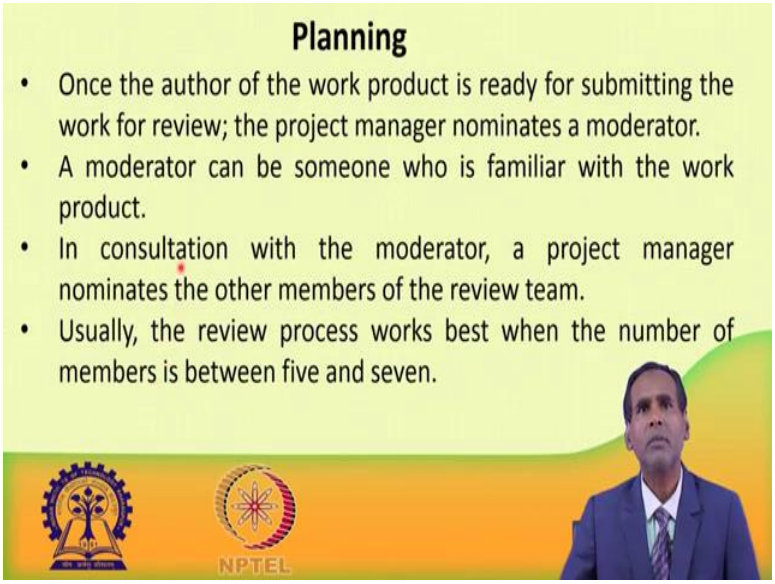

So, what I have already told that has been explained here all these stages like planning and this preparation, review meeting, rework, I have already explained in the diagram, they have been mentioned here.

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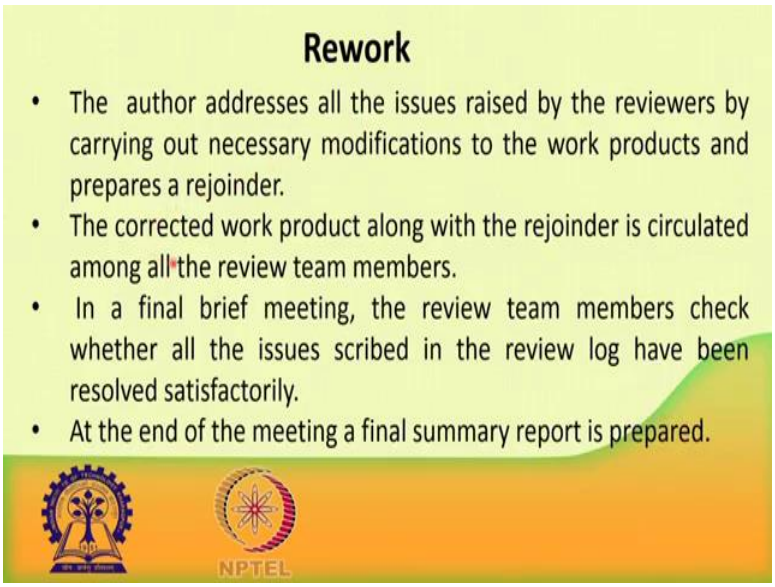

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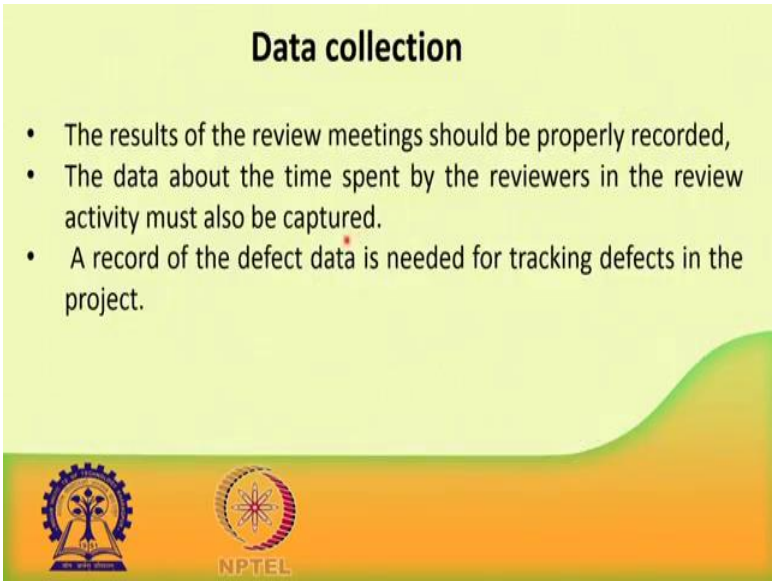

And then final step is data collection. In data collection what we are doing, the results of the review meetings should be properly recorded. So, because we are human beings after some days we may forget everything else, the documents may be or they discussing, the results of the discussion may be lost. So, the results of the review meeting this should be properly recorded. Not only that the data about the time spent, how much time you have spent by the reviewers in the review meeting that also have to be captured.

A record of the defect data is needed for tracking defects in the project. Why will record all the details of the defect data? Because a record of the defect data is needed for tracking the defects in the project. So, in order to what material, what error free, defect free the product, so this record of the defect data it will be needed, it will be helpful for tracking the defects in the project.

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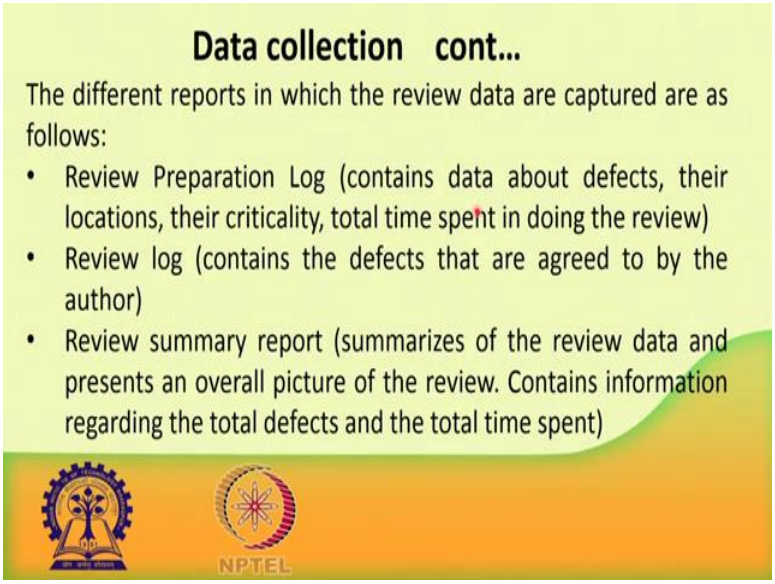

So, now, let us see what kind of reports are prepared the during the review, the different reports in which the review data are captured as follows; first one is review preparation log, which contains the data about the defects, their locations at which point they are identified they are located. Their criticality how much critical they are, whether they are very much serious or they are what normal or things like that they are criticality, the total time spent in doing the review etcetera.

So, all those things are mentioned in the review preparation log. So, then the review log, it contains the defects which are agreed to by the author. The author has agreed that yes these defects are there and I will remove in the next revision. So, review log it will contain the defects that are agreed to by the author, then the review summary report as its name suggest summary. So, this report it is summarizes the review data and it represents an overall picture of the review.

So, the outcome, the summary of the outcome of the review meeting is documented here. It also contains the information regarding the total defects present into our product and how much time its spent, the total time spent in this review process. So, these are the different documents, these are the different reports they are prepared during the review process, may be review preparation log and review log are intermediate what documents report or reports or as the review summary report is the final report.

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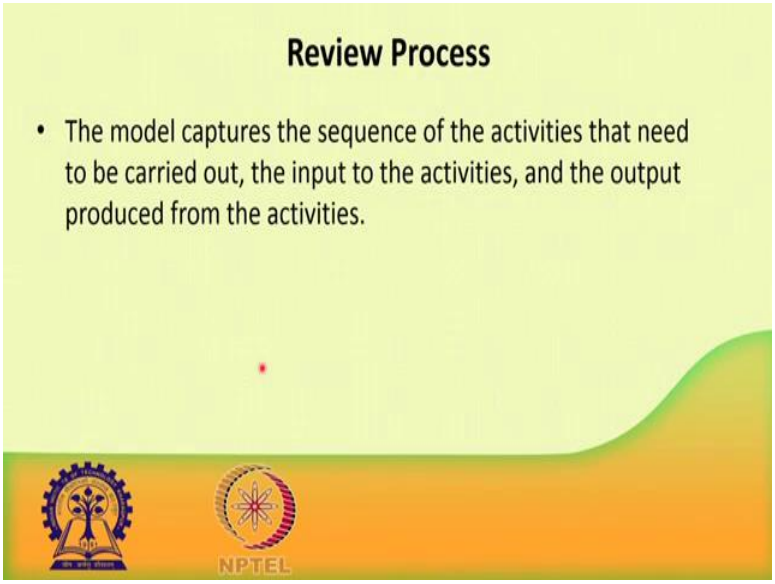

So, the model that we have proposed it captures the sequence of the activities that need to be carried out and the input to the activities and the output produced from activities, so everything. So, we have shown what is the input to this process. So, inputted the work product, these are the intermediate products such are the review team and the this schedule, this reviewers log, the defect log these are the intermediate products and the final product is the summary report ok.

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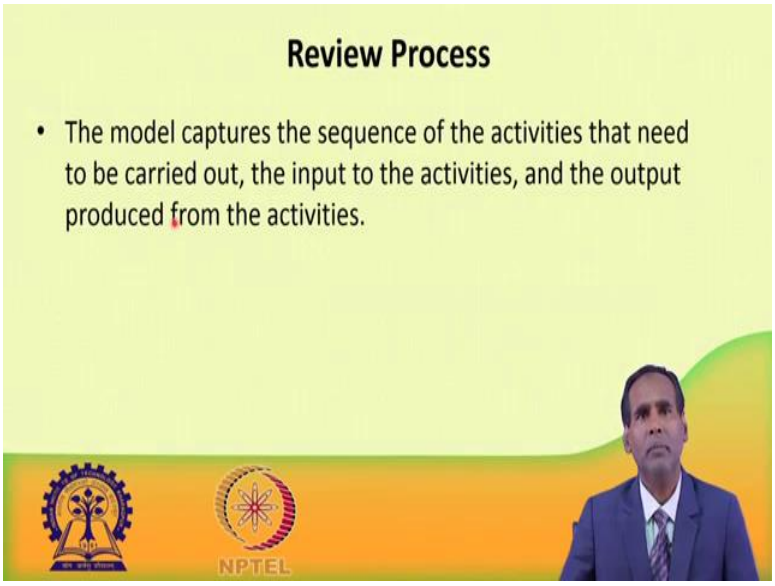

So, here we have seen that this. So, this review process model, it captures the sequence of the activities. 4 activities we have shown here that need to be carried out for the review process. Now the input to the activities that is the candidate work product and the output produced from the activities. So, there are some intermediate outputs as I have already told you and that is the final output that is the summary report.

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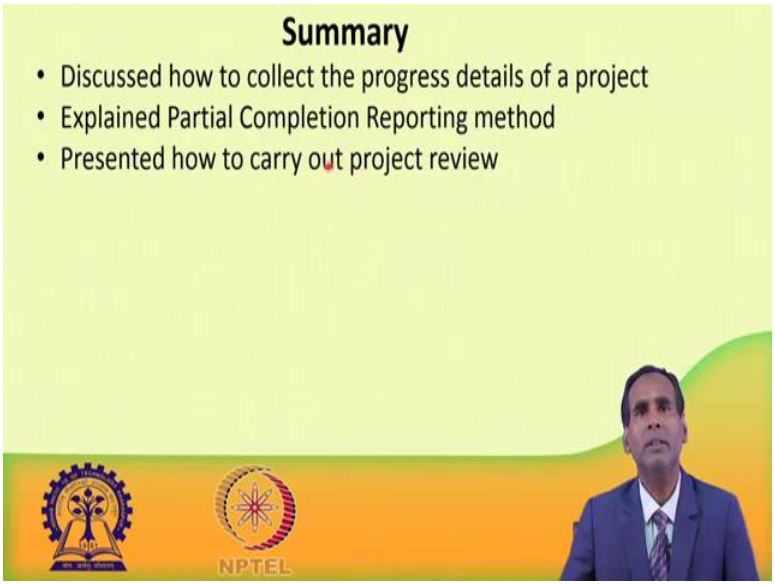

So, in this lecture we have discussed how to collect the progress details of a project. We have also explained the partial completion reporting method. We have seen how the drawbacks of the partial completion reporting method can be overcome using the what the red, amber, green method. Also we have presented how to carry out to the project review, what are the activities of project review, what are the input and outputs of this project review. We have seen the some of the logs that will be used like project review.

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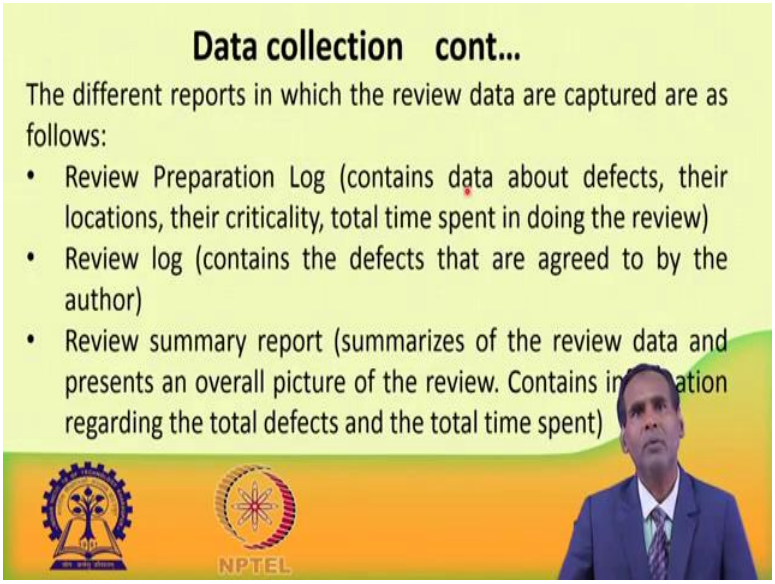

We have seen some of the what logs like review preparation log, review summary report log which are the intermediate outputs of this review process and the final output of the review process in the review summary report. So, these are the outputs of the review process. So, we have seen how to carry out the project review. So, this is all about this project review.

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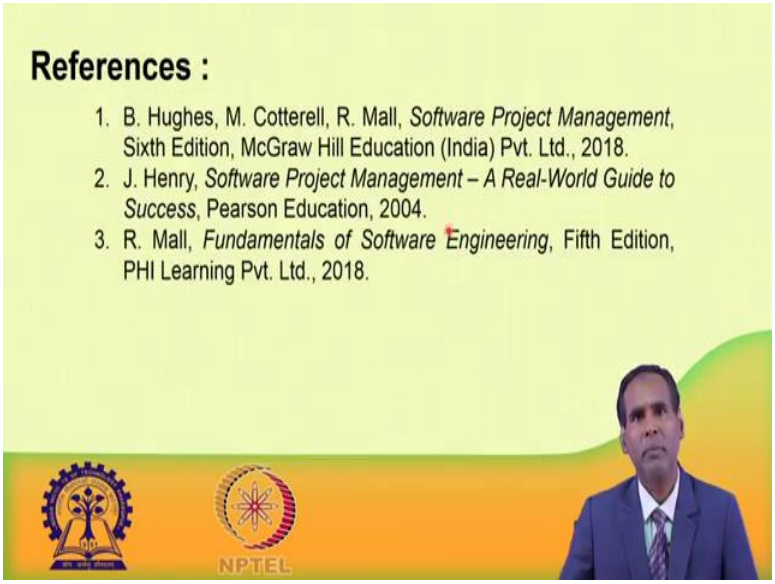

These are the references.

Thank you very much

Software Project Management
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Lecture – 41
Project Monitoring and Control (Contd.)

Good morning to all of you. Last class we have discussed about Project Review Technique. Today, we will discuss about Source Code Review.

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We will see two types of code review that is code walkthrough and code inspection. Also, we will discuss something about a relatively new technique called as cleanroom technique.

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So, the yesterday we have discussed that all the interim and final work products they are usually the candidates for a review, and particularly the following work products they are suitable candidates for review. Those are like requirements specification documents or SRS document, user interface specification and design documents. Architec different design documents are such as architectural design, high-level design, detailed design documents, also source codes obvious, this can be also reviewed.

Then various test plans on the test design test cases, they can also be reviewed. Also, the various project management plans, the configuration management plan etcetera, they can also be reviewed. So, right now we will discuss about how the source codes they can be reviewed.

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We have also seen that yesterday review is a very effective technique to remove the detail defeat detects from the source code. And in fact, a review has been acknowledged to be more cost effective in removing the defects as compared to testing. Over the years various review techniques have become extremely popular and they have been generalized to be used with other work products also.

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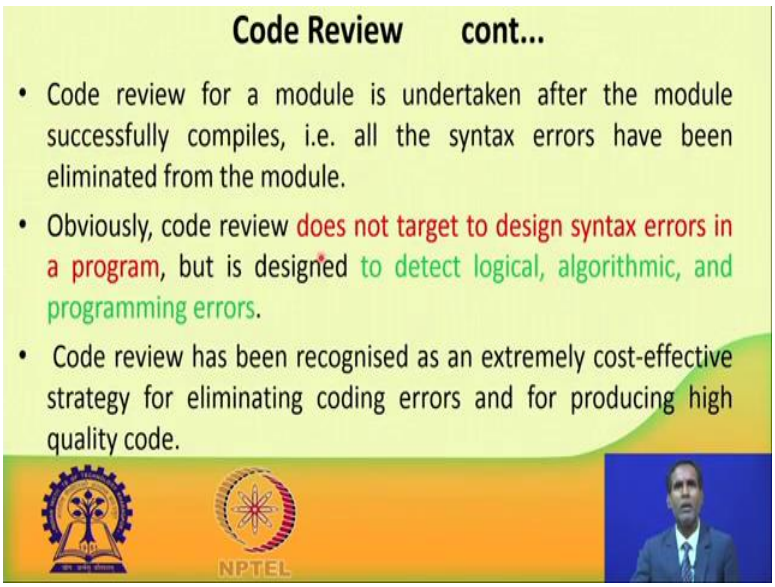

So, now let us see when the code review should start. The code review for a particular module it is undertaken after the module is successfully compiled that is all the syntax

errors that have been alimented eliminated from the module. Obviously, you see that code review does not target to design the syntax errors. Please remember, code review it does not target to design the syntax errors in a program. So, it does not target to syntax or design syntax errors in a program. It is designed or it is targeted to detect logical errors, algorithmic errors and programming errors. That should be kept in mind.

The code review has been recognized as an extremely cost effective strategy for eliminating the coding errors and for producing high quality code. So, here it will be a cost effective, much cost effective than this a what testing techniques while detect eliminating the or and detecting the errors.

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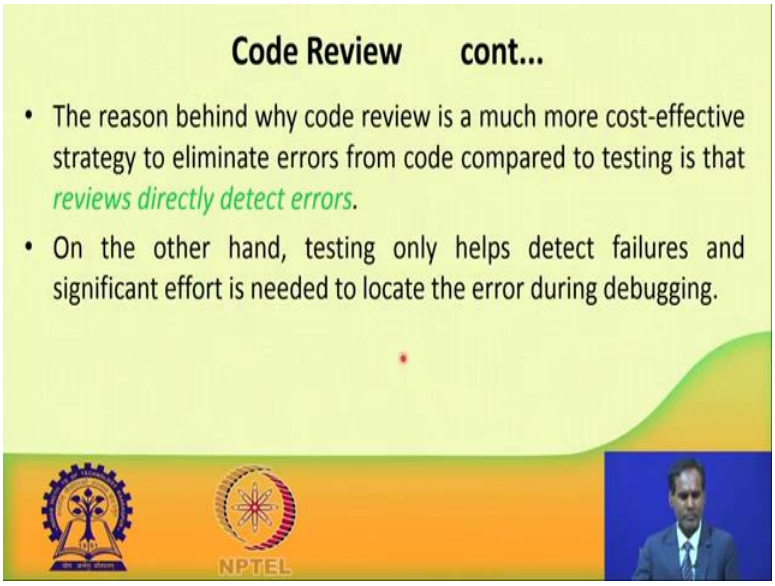

Now, let us see what is the reason why code review is so much cost effective than the testing technique. The reason behind a why code review is a much more cost effective strategy is to eliminate the errors from code compared to testing is that review directly detect the errors. Let us see what is happening in testing, why in a review it is so much cost effective, because review is that directly detect errors while testing they only help detect the failures and the significant errors.

By using the testing technique you can only detect the failures and the significant, and you can only detect the failures. But in case of this review you can directly what detect the errors. So, in and here in case of testing you have to spend a lot of effort to locate the errors during debugging.

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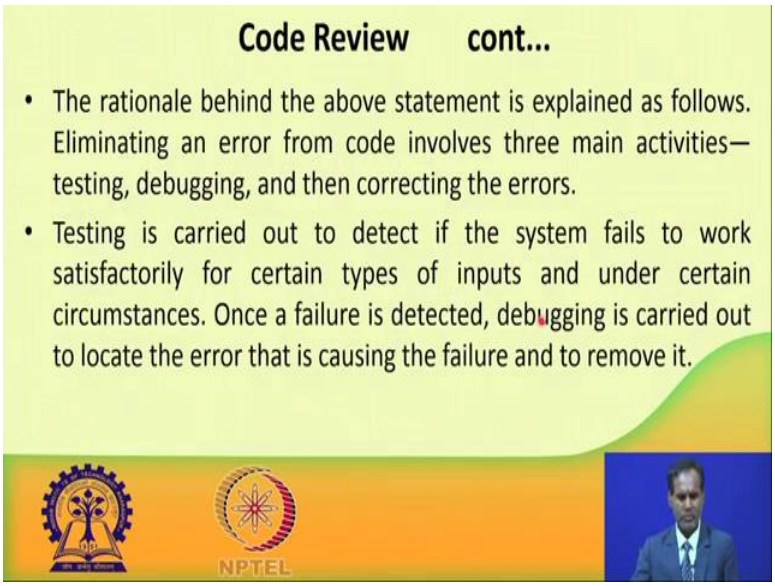

The rationale behind the above statement is explained as follows. Why it is cost effective? What is the rationale behind this? You see the eliminating an error from code it involves 3 main activities, one testing, then debugging and then correcting the errors. So, a 3 activities are there where in case of eliminating errors from the code.

Testing is carried out to detect if the system fails to work. So, normally testing is carried out why? It is carried out to detect if the system fails to works for certain types of inputs and under certain types of circumstances, and when a failure is detected once a failure is detected, then debugging is carried out to locate the error that is causing the failure and then you can remove it. But in a what a code review you can directly detect the error, so that is why this is much more cost effective than the testing process.

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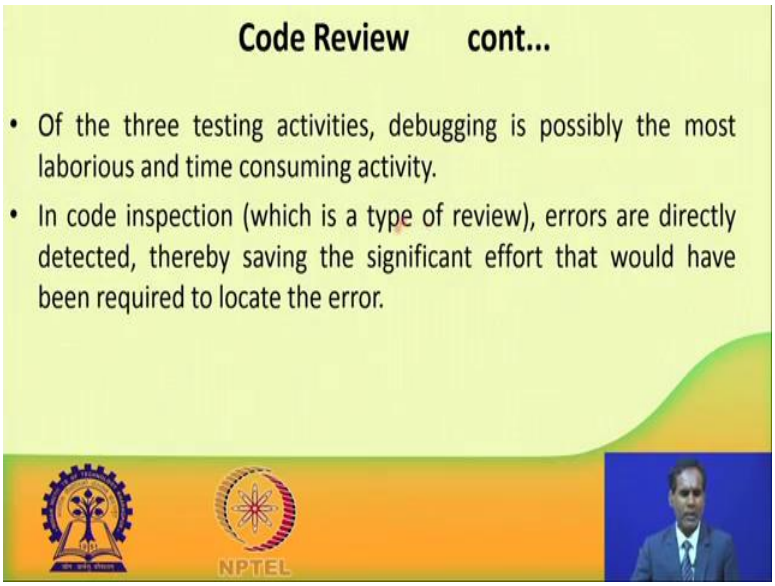

So, of this as I have already told you there are 3 activities are called what carried out to due while you are eliminating the errors that is testing debugging and correcting the errors. So, out of these 3 testing activities, debugging is the most lubricous and time consuming activity.

So, we will see that in code inspection which is a particular type of review, we will see that there are two types of review, one is code walkthrough, another is code inspection. In code inspection what is happening the errors are directly detected. So, here in what testing what it is only help you in detect the errors which is again a time consuming and lubricous activity, but in code inspection the errors are directly detected thus saving the significant effort that is required to locate the error. So, that is why a review is much more cost effective than the testing technique.

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Normally, there will be two types of code review techniques, one code walkthrough and the other is code inspection.

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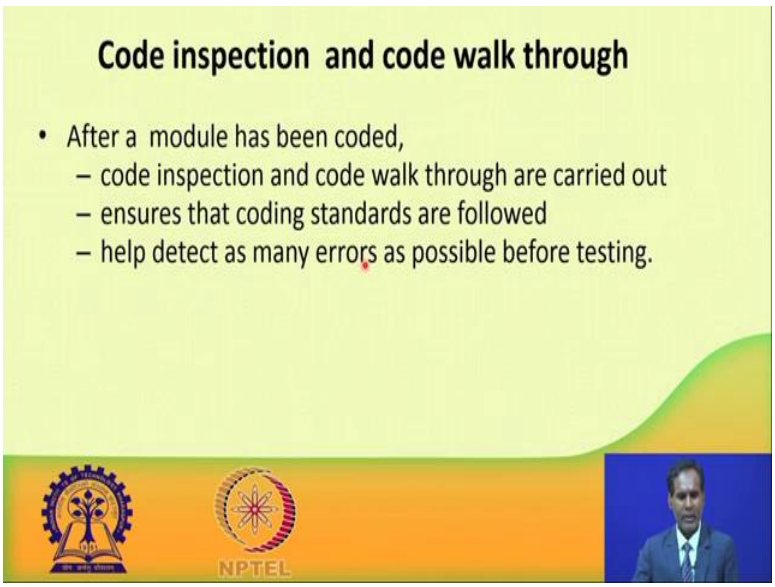

Let us see about this things. So, as I have already told you that when this what code inspection or code walkthrough can start after it is what compiled, after a module has been coded, then code inspection and code walkthrough can be carried out.

So, code inspection and code walkthrough they ensure that the proper coding standards that have been followed like the standard errors you have used and what proper variables

etcetera that you have used it ensures the various coding standards that the programmer has used in the software that he is developing, it. They also help to detect as many errors as possible before testing. So, before this system testing or integration testing the this code review techniques that is code inspection and code walkthrough they help detect as many errors as possible before the system testing and the integration testing starts.

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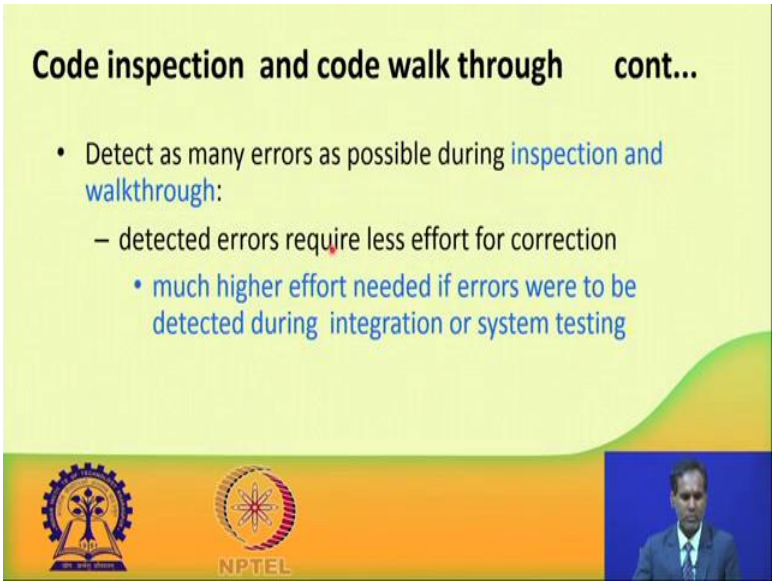

So, this techniques they detect as many errors as possible during a inspection and code require, ok, the code inspection and code review, they detect as many errors as possible during this process. They also, and these detected errors require less effort for correction. So, after this detection we must correct these errors. So, the errors detected during inspection and walkthrough they require less effort for correction, otherwise it would require much higher effort if these errors was would be detected during the integration and system testing.

So, we will summarize that the errors detected during a code inspection and walkthrough, they require less effort for correction, in comparison to the errors if they could have been detected during what integration or the system testing level.

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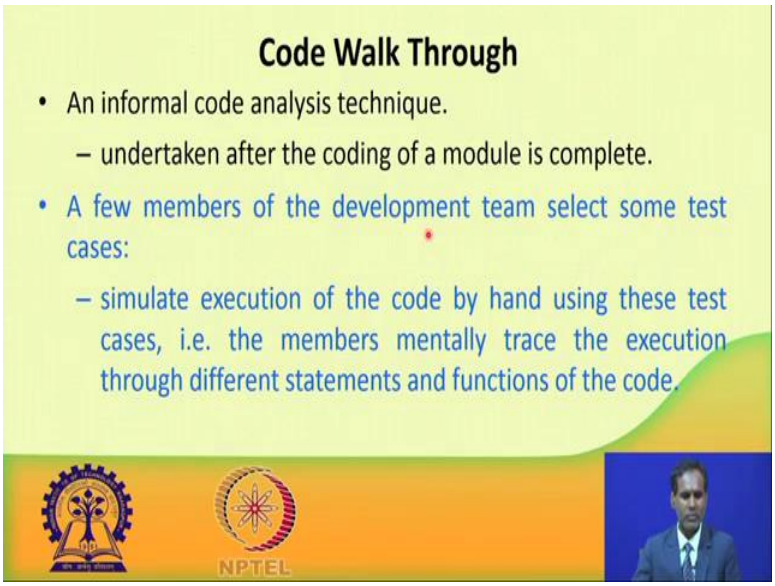

We will say first the code walkthrough. Here, this is basically an informal code analysis technique. So, code walkthrough is considered as an informal code analysis technique. It is not a formal code analysis technique, it is a informal code analysis technique which is undertaken after the coding of a module is complete. So, after one module or one unit its coding is over, then you can go for this code walkthrough which is an informal code analysis technique. Now, let us see how this code walkthrough process system works.

So, a few members of the development team select some test cases. So, a small what committee maybe formed and a few members of the development team and they select some test cases, and those test cases then they simulate execution of the code by using those test cases. So, they first select some test cases, then simulate the execution of the code. How? Not by computer, by hand, by mentally, by hand using these test cases that is these members, this development team members they mentally trace the execution through or different statements and a functions of the code. So, this is how this code walkthrough works.

So, that is why its name is code walkthrough. You are, just as if you are working through the a code you are executing the code by hand or you are mentally testing the execution that is why this is known as a code walkthrough.

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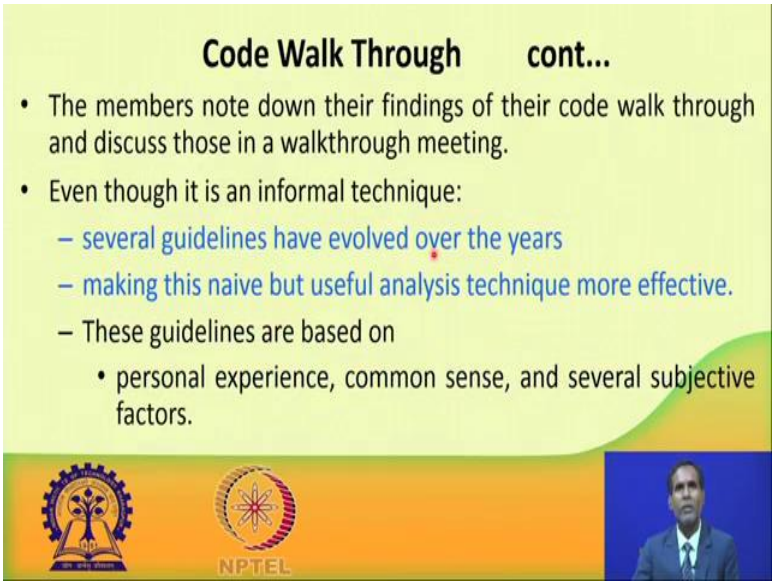

The members the while they are executing the code by hand or by mentally, they note down their findings of their code walkthrough and discuss those in a walkthrough meeting. So, in that discussion they will say what kind of what defects that they have found or what could be the cause. So, those details they discuss in the walkthrough meeting.

So, even though this code walkthrough is a in is an informal technique, but several guidelines have already evolved over the previous years, so which makes a or which make this naive approach, which make this naive, but this is a very useful analysis technique and this is a more effective technique. So, these guidelines those have been evolved over the years, they are based on what? They are based on the personal experience of the programmers, they are common sense and several subjective factors.

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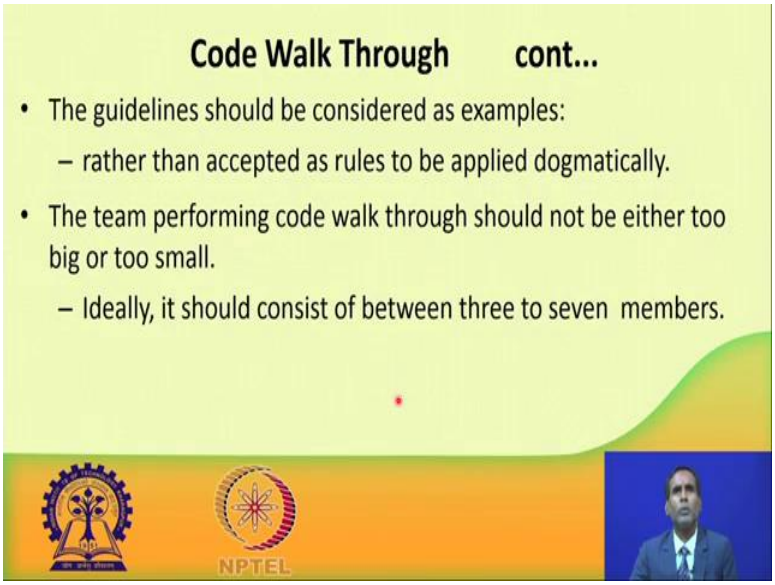

The guidelines should be considered as example please say. So, as I have already told you during a or in these guidelines have evolved over the years, these guidelines are not actually very a hard and fast that you have must have to follow. So, rather these guidelines should be considered as examples, rather than accepted as rules to be applied dogmatically. These are not that hard and fast you have to follow exactly, they can be considered as only examples.

The team performing the code walkthrough should not be either too big or too small. So, let us say a how many what members should be there in their code walkthrough committee. It should not be a either a too big or too large. Ideally, it should consist of between 3 to 7, 7 members.

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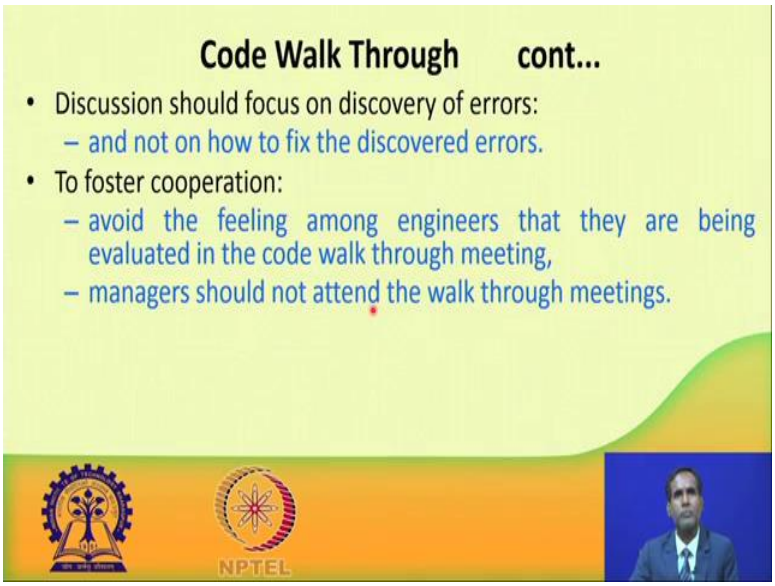

Now, let us see in this code walkthrough meeting a what should be focused, the member should focus on what. Discussion in the code walkthrough meeting should be focus on only the discovery of errors, they should not discuss how to fix the discussion discovered errors.

The objective is, the focus is only on discovering the errors, but not on how to fix the discovered errors. So, to faster cooperation and the members should avoid on filling that they are being avoid, they are being evaluated in the code walkthrough through meeting. You know they should that fear not be there that somebody is evaluating them in the meeting, so that is why managers should not attend the walkthrough meetings, only the developers or the programmers, their peers they may attend the walkthrough meeting. So, that this fear should not be there, that they are being evaluated in the walkthrough meeting, that is why managers or draft able managers they should not attend the walkthrough meetings.

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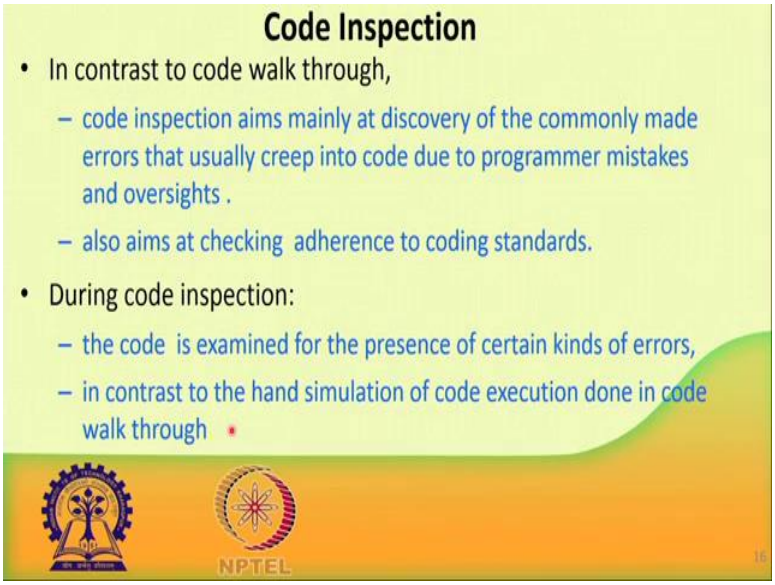

Now, let us see, we have seen about code walkthrough what is code inspection. In contrast to code walkthrough on code inspection aims mainly at the discovery of the commonly made errors that usually creep into code due to programmer mistakes and over sites. I am repeating again.

So, in code walkthrough what is happening? That programmers they execute the what test cases by hand or by mentally trace the execution of the code but here in code inspection the objective is discovering ops of commonly made errors, because the human the developers are human beings and they may make some common mistakes.

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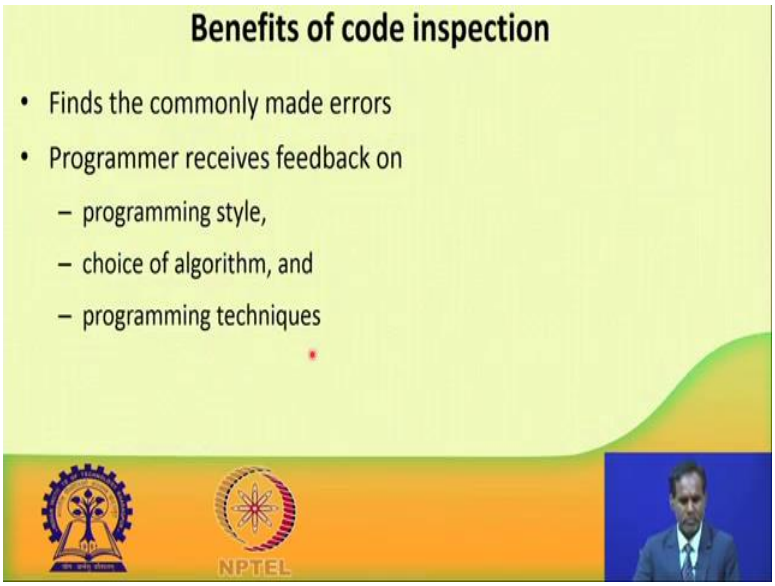

So, the objective the aim of code inspection is to discover those commonly made errors that usually creep into the code. Why? Due to the programmer's mistakes and their oversights. This code walkthrough also it aims at checking adherence to coding standards for the proper coding standards like a proper header files global variables etcetera you have used properly or not. So, it also what checks that.

So, another objective of code war, another objective of code inspection is that to check adherence to the coding standards. So, now, let us see what is for performed during code inspection. During code inspection the code is examined for what? For the presence of certain really certain common kinds of errors.

So, in code inspection, the code is examined, it is checked for the presence of certain common kinds of errors that the programmers might commit the mistakes due, they might do, they might commint those errors due to some oversight issue. So, but in case of code walkthrough what is happening? You are the programmer or the team members they perform some handy hand simulation of the code execution, that is done in code walkthrough. But in code inspection, these code is examined for the presence of some commonly made errors.

What are the benefits of code inspection? I have already told you in code inspection the objective is to find the commonly made errors, so several commonly made errors which are committed by the programmers they are detected here. The programmer also

receives; so, let us see what are the side benefits, the side benefits of code inspection is that the programmer receives a feedback on his programming style, the choice of algorithm or the algorithm you have used and the pre the programming techniques he has used, you can get a feedback on these things, ok. So, these are the side benefits of code inspection.

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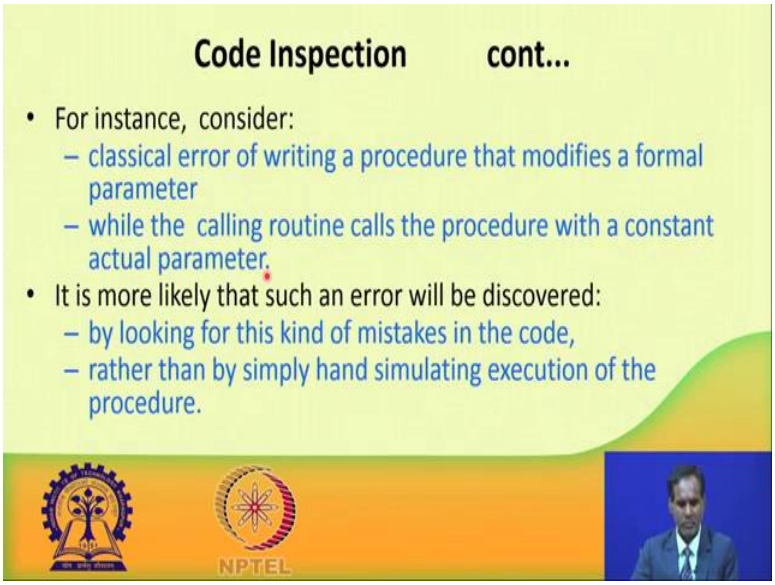

And now let us take a simple example of code inspection. So, for instance let us consider the classical error of writing a procedure that modifies a formal parameter, you have written a procedure and that is modifying a formal parameter, when while the calling routine calls the procedure with a constant actual parameters. So, there is a you are calling a routine and the calling routine calls a procedure with a constant actual parameter and you are trying to modify a formal parameter.

This is obviously an error, and this error can be detected by this code inspection. So, it is more likely that such an error will be discovered by looking for this kind of mistakes in the code, you while you will inspect the code probably it will come to the eye of the what developer or so.

So, by looking for this kind of mistakes in the code, this types of errors can be discovered, rather than by simply hand simulating execution of the code. So, for this kind of errors code inspection is the better one, rather than the code walkthrough where you are simply hand simulating the execution of the procedure.

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So, a good software development companies, they collect statistics of errors committed by their engineers, ok. From their previous projects what kind of errors the engineers have committed, in similar kinds of projects, they collect all those information, they collect all those statistics and record in a file. So, the good software development companies, they collect the statistics of errors which are committed by their engineers and then they identify the types of errors most frequently committed.

So, out of all those errors those are statistically recorded then they identify which kind of errors the programmers, they are most frequently they are committing. Then, what they do? They prepare a list, they prepare a list of a common errors made by the programmers and this list of common errors can be used during the code inspection. So, they can prepare just a checklist and this checklist can be prepared from the previous experience of the programmers.

so then this checklist can be given and to the what programmers while doing code in inspection for the other types of projects, then this list checklist of the common errors it can be a used during code inspection for searching for possible types of errors. So, now, let us see, let us look out about some commonly made errors which are there or which the a programmers might commit during the development of a software.

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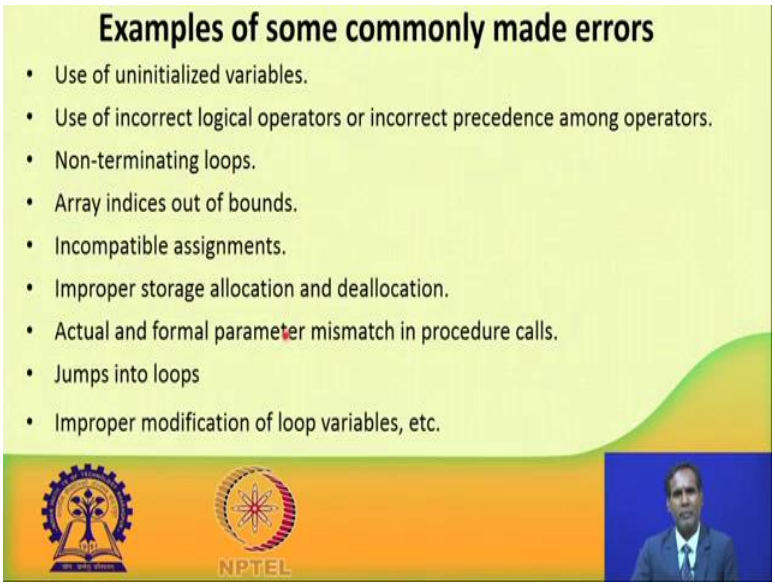

So, these are some of the examples of some commonly made errors. Like use of un initialized variable. The programmer has used a variable, but he has forgotten to initialize it. So, that can be searched during code inspection. Similarly, use of incorrect logical operators or incorrect president precedence among operators, the programmer due to what oversight he has used some incorrect logical operators or incorrect precedence among the operators he has used that also can be detected during code inspection.

And non-terminating loop here that, the programmer suppose has written a loop like a while do, while or for, but that is not terminating he has forgotten to give a what put the termination, the condition for termination. So, then the loop will be a what leading to what infinite or the for loop will be will be leading to for infinite loop. So, those kind of non-terminating loops. These errors also can be detected by your code inspection.

Similarly, array indices out of bounds. You have written an array maybe a 100 element array, but you are searching an array or an array which position is more than 100. So, array indices which are out of bounds these errors also can be detected. Then incompatible assignments, you know that in assignment step statements left hand side of the left hand side must correspond to the right hand side. If the left hand side is what integer and right hand side is a floating point variable then this becomes incompatible assignments. So, those type types of errors also can be detected by code inspection.

Similarly, improper storage allocation and de-allocation. You know that memory allocation and de-allocation like malloc and dealloc etcetera, so if you have improperly done this storage allocation and de-allocation or some mistakes, you have put some errors by mistake have put that also can be detected during code inspection.

Similarly, actual and formal parameter mismatch in the procedural calls. While you are writing a procedure call like a some the number of factual parameters should match with the number of formal parameters. The types of the form actual parameter should match with the type of the formal parameters.

So, if there is any mismatch among them then also that means, any mismatch between the a number of actual parameters to the formal parameter, number of formal parameters or the types of actual parameters with the types of formal parameters. So, then these kinds of mismatch also can be easily find out, can easily be found out during this code inspection and similarly jumps into loops. You have use some go to statements or so, and those are eventually leading to the loops. So, jumps into loops which is not desirable. So, this is also a commonly what made data error, what, one of the commonly made errors by the programmers. You can use these code inspection techniques to detect those things.

Then, improper modification of loop variables you have used a loop variable inside your loop and you should not actually a modify, so you are, if you are improperly modifying a loop variable. Then, what will happen? This will be an error and this error also can be detected by using code inspection. So, these are some of the commonly made errors that a programmers might do due to some common mistakes or due to their oversight. So, these types of errors can easily be detected using code inspection.

So, in the for these types of things it is better to go for code inspection rather than going for this code walkthrough. Similar examples, so you can as a programmer you can also experience, you might have experience. So, all such commonly types of errors can be detected by using code inspection techniques.

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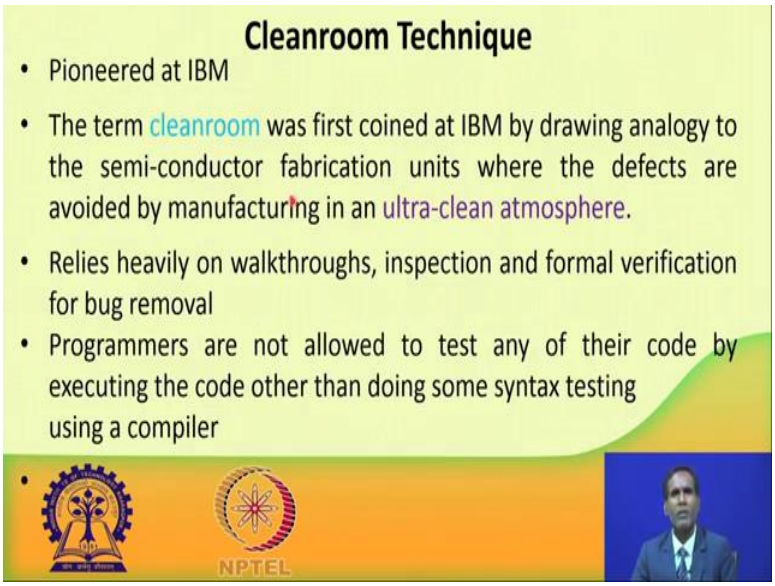

We will take up a little bit of a new technique called as cleanroom Technique. So, this cleanroom technique is basically pioneered at IBM and let us see why they have use this term cleanroom. How this was coined? The term cleanroom was first coined at IBM, ok. This term first coined at IBM. How? They have drawn some analogy with what they have drawn, they have drawn an analogy to the semi-conductor fabrication units.

So, in the IBM they were trying to establish an analogy to the semiconductor fabrication units, where the defects are avoided. How? By manufacturing in an ultra-clean atmosphere. They are trying to not allow the defects. They are trying to avoid, they are trying to what avoid the defects. How? They are thinking that if the what semi-conductor fabrication it can be manufactured in an ultra-clean atmosphere then probably the defects will not arise at all. The defects can be avoided from the beginning. So, that is why this name clean they came, and they have seen from this and they have an, they have made an analogy with the same with the software development and then that is why they have given the term cleanroom testing or this cleanroom technique.

So, this technique cleanroom till, this technique that the cleanroom technique it relies heavily on these various review techniques we have seen, ok. So, this cleanroom technique relies heavily on code walkthroughs, code inspection and formal verification for bug emoval. So, here the programmers are not allowed to test any of their code by executing the code, ok. The programmers are just name suggest they will try to avoid the

defects. So, here programmers will not be allowed to test any of their code by executing the code. They have to use what walkthroughs, inspections or formal verification for bug removal for what detecting the bugs and removing the bugs. They cannot use. What? They are not allowed to test any of their code by what executing the code.

They may do only some exceptions like the accepting doing some syntax testing testing they may due to using a compiler, but normally they are not allowed to what test their code by a code execution. They can only perform walkthrough, code walkthrough inspection and formal verification for bug identification and bug removal. This is what is the objective of cleanroom testing or cleanroom technique.

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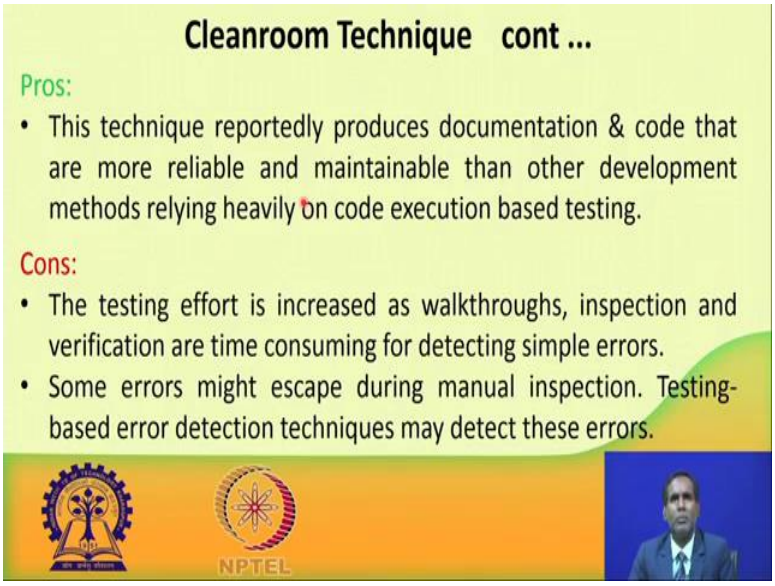

Now, let us quickly see what is the advantage of cleanroom technique. This cleanroom technique reportedly produces documentation and code that are more reliable and a maintainable. So, this technique produces reportedly produces various documents and code. Those are more reliable, also they are maintainable than other development methods which are reliable relying heavily on code execution based testing.

So, the technique as it produces a various documents and code, and these are more reliable and maintainable in comparison to other methods which are based on execution based testing, but the disadvantage is that the testing effort is increased as walkthroughs, inspection and verifications are time consuming for detecting this simple errors.

You know that code walkthrough; so, how many inputs you will take to for hand simulation? You it will take a much long amount of time. You have to inspect the code line by line. Again, it is a time consuming process. Also, formal verification, that is also another time consuming process. So, that is why the testing effort is increased to a large extent as all these techniques, walkthroughs, inspections and verification are very much time consuming for detecting this simple errors and the cleanroom technique is based on only these what code review un-formal execution techniques.

Another problem is that some errors might test still escape doing manual inspection. You see all these walkthrough or inspection etcetera they are manual techniques. So, some errors might escape from the eyes of the team members or the programmers during manual inspection, because after all a we are a manual human beings we may commit mistakes. So, some of the errors might escape during manual in inspection. So, those errors, those escaped errors can be maybe detected using the error base detection techniques.

So, these errors which can be expect what escaped during this manual inspection, they can be detected easily, these escaped errors maybe detected easily if you will use some testing based error detection techniques. So, that is another disadvantage of cleanroom testing.

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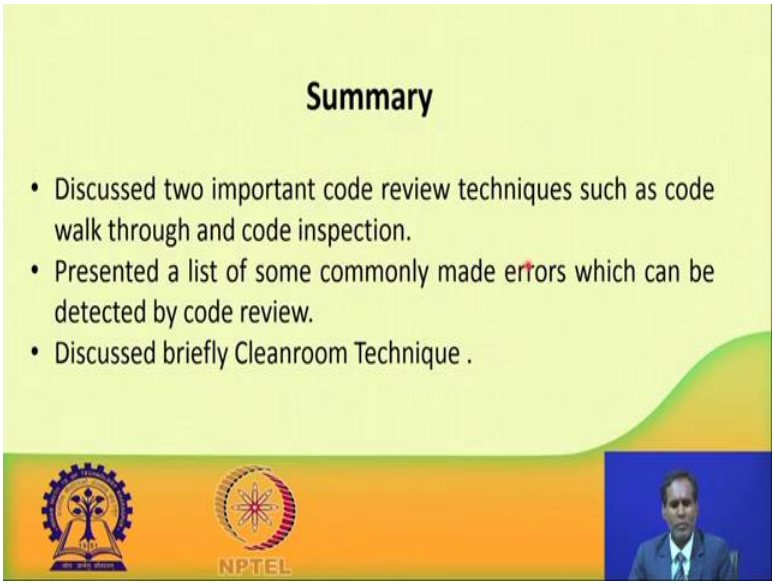

So, today we have discussed two important source code review techniques. Those are code walkthrough and code inspection. We have also presented a list of some commonly made errors which can be detected by what a code review, particularly by a code inspection. We have also discussed briefly about cleanroom technique which is which heavily relies on code structured walkthrough and or code walkthrough, code inspection and formal verification technique.

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We have taken the materials from these books, ok We have taken the materials from these books.

Thank you very much.

Software Project Management
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Lecture - 42
Project Monitoring and Control (Contd.)

Good morning to all of you, now we will take up how to visualize the work schedules.

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We will see we can visualize the work schedules might be a different work charts such as the Gantt charts, slip chart, time line chart and then we will see how to monitor the costs.

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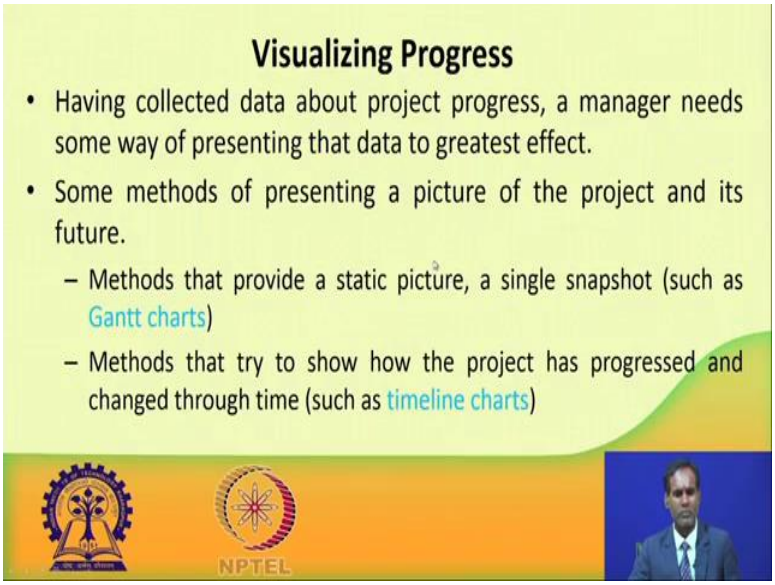

Let us see how to be visualize the what progress of a project. So, we have seen last class how to collect the data about the progress of a project. So, after collecting the data about the progress of a project, the manager he needs some way of presenting that data to the greatest effect. So, there are some methods for presenting a picture of the project and its a future. So, some methods that provide a static picture such as a single snapshot for example, Gantt chart there are some other methods which tried to show how the project has progressed and changed through time that they represent the dynamic aspects of the system and example in the timeline charts. We will see first the Gantt chart then we will see slip line chart and then we will see the timeline charts.

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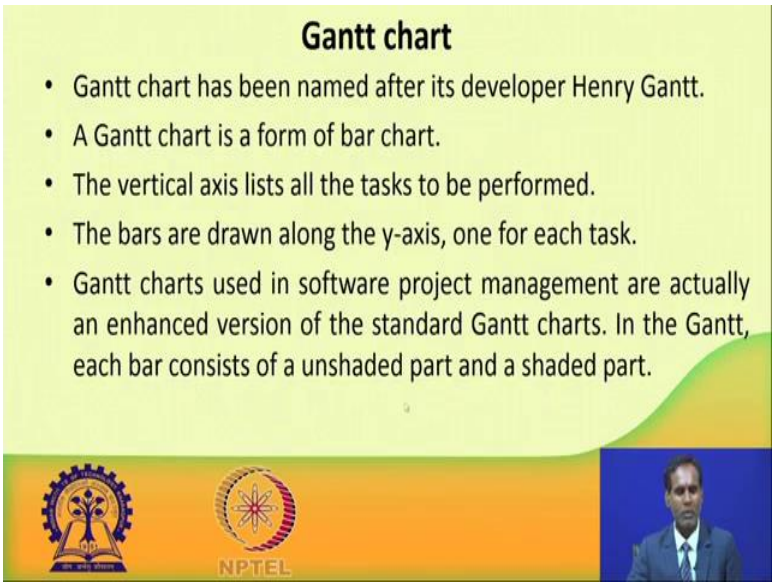

Gantt chart has been named after its developer Henry Gantt, this Gantt chart is a form of bar chart that we have already seen earlier also in your matriculation or plus we must have seen about a what is a bar chart. So, a Gantt chart is a special form of bar chart. Here the vertical axis it lists the tasks to be performed, the bars are drawn along the y axis and the 1 bar for each task the Gantt charts are used in software project management, they are actually an enhanced version of the standard Gantt charts and in the Gantt chart as we have already seen that each bar consists of two parts one unshaded part and a shaded part we will see in the example.

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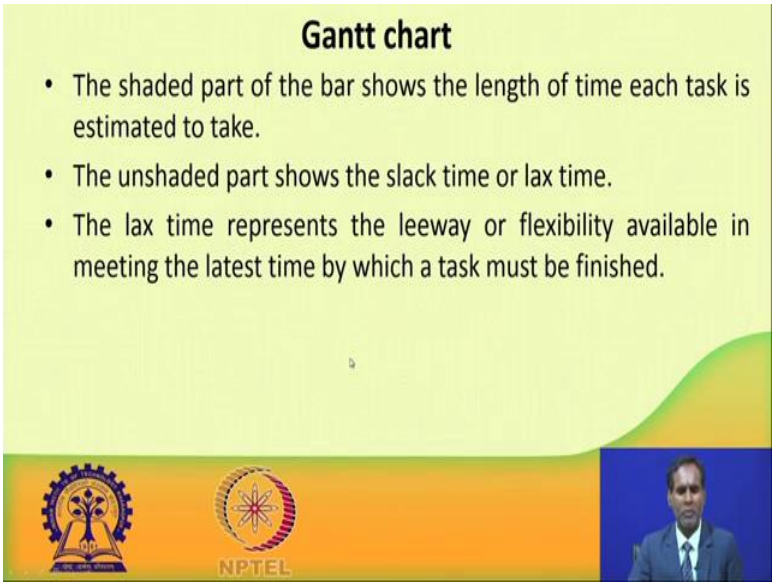

Now, let us see what the shaded part represents and what the un shaded part represents. The shaded part of the bar shows length of the time each task is estimated to take. So, each task is estimated how much time it will take so, that time is represented at the shaded bar shaded part and the un-shaded part so, the slack time or the lax time.

I have already discussed what is a slack time or float time or a free time in the last class, the un-shaded part represents the slack time or the lax time the what is the lax time we can say that? The lax time represents or the slack time this slack time or the lax time represents from the leeway or the flexibility that is available in the in meeting the latest time by which a task must be finished. So, here some free time is available you can little bit delay the task so that, it will be what it can be finished before it can be completed before its target time.

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A Gantt chart is a special type of bar chart that we have already discussed, where each bar represents an activity and the bars normally they are drawn along a timeline. The length of each bar is proportional to the duration of the time that is planned or expected for the corresponding activity. The Gantt chart representation of a project schedule is helpful how let us see how Gantt chart is helpful. The Gantt chart representation of a project schedule is helpful in planning the utilization of the resources, how we can plan the utilization of the resources for that you can use a Gantt chart.

So, but those who have I think you have already seen CPM PERT earlier. So, while PERT chart is useful for monitoring the timely progress of the activity. If you will compare between Gantt chart and PERT chart you can see that Gantt chart is helpful in planning the utilization of resource while PERT chart is very much useful in monitoring the what timely progress of the different activities.

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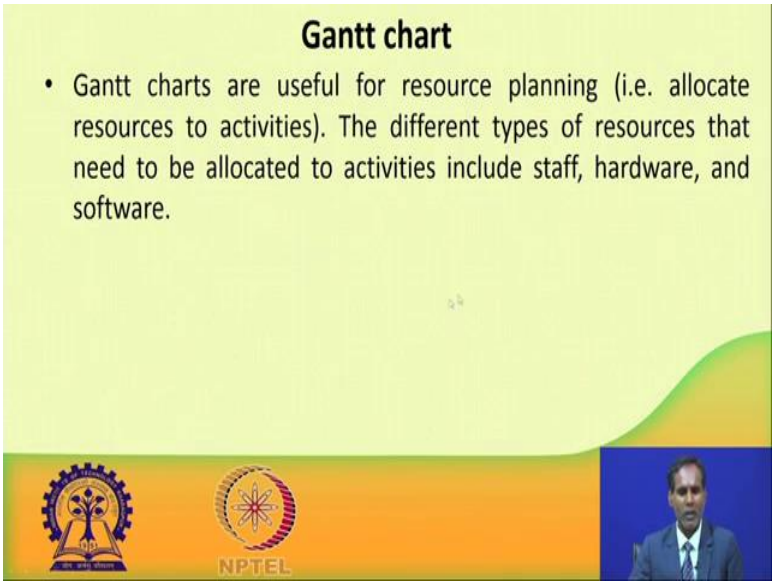

Gantt charts are also useful for resource planning how to allocate the resources to the activities based on what. So, these Gantt charts are very much useful for resource planning that is how to or during allocation of resources to activities. The different types of report resources you know that need to be allocated to various activities, you have already discussed in last class they can be the resources could be staffs, could be hardware, could be software, could be other equipments, could be what space even could be time these are various what resources.

How the different types of resources that can be allocated the different types of resource resources that need to be allocated to activities such as staff, hardware and software these allocation can easily be represented using Gantt charts.

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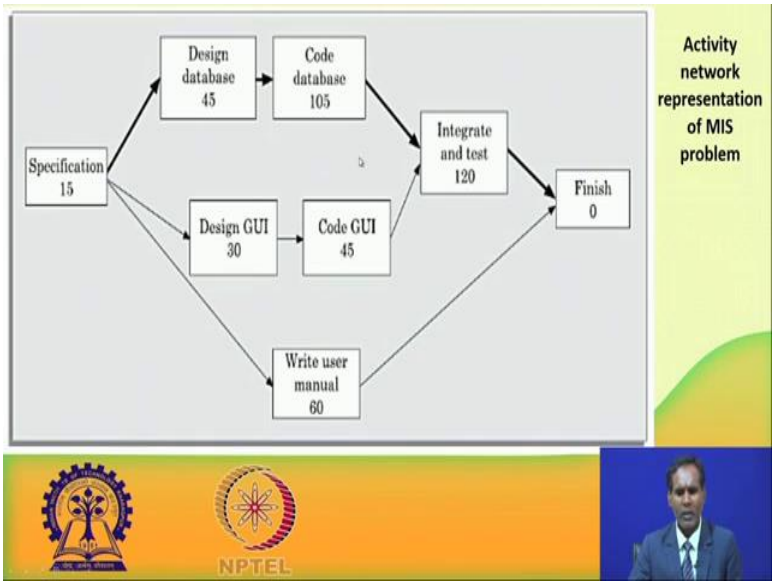

So, normally the Gantt charts you can easily construct if from the activity network or from the precedence network. So, first step is that you construct the activity network or the precedence network then from that activity network or the precedence network you can easily consider you can easily construct the Gantt chart.

So, this is a sample what activity network for a MIS problem where the various activities and their expected times they will take to what for their the these activities how much time they are expected to take that is also written like requirement specification will takes 15 days, designing database will take 45 days, designing GUI will take 30 days and then writing manual will take 30 days then what code database will take 105 days coding GUI part will take 45 days integrate and testing will take 120 days and then complete.

So, you can say that in this activity network the dependencies are also shown like before coding design of database should be there, before coding GUI design of GUI must be completed and before integration and system test this coding of database and GUI should be completed. So, in this way if a given activity network ok. If in this way if an activity network is given you can easily construct the Gantt chart you see.

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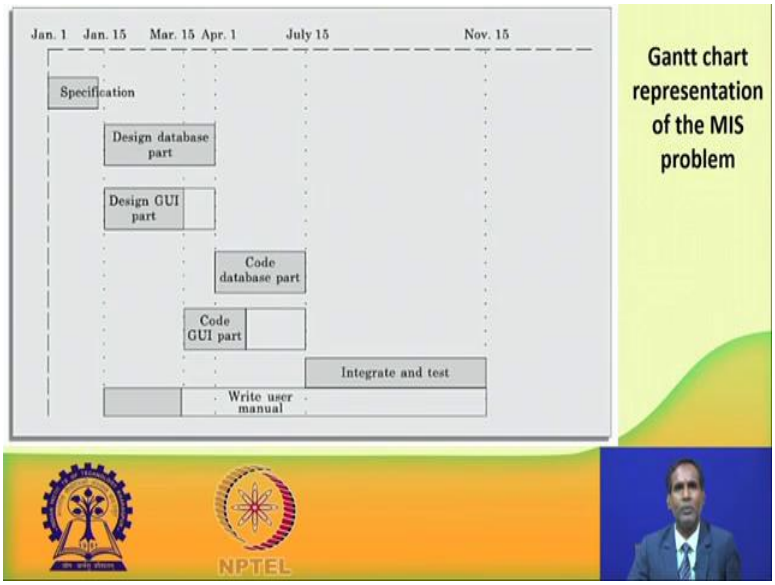

So, now you see you see first we will be we have already seen what is the first? First one is the specification, requirement specification that will take 15 days. So, accordingly we have seen that, we have started with specification and suppose the starting date is January 15. So, it will take how many days we have already seen it will take 15 days. So, up to January 15 it will continue then as we know that only after specification is completed we can start 3 activities parallelly what are they? Design database, design GUI and write user manual.

So, you see now after January 15. So, from January 15; that means, 15 days after the specification January specification is started in January 15 January 1st it will take 15 days, then 3 activities can be started what parallelly designing database, design GUI and write user manual see all those 3 activities have been started parallelly from January 15 and you have to show the timings.

So, design database will take 45 days, design GUI will take 30 days and writing man user manual to 60 days see. Design part should take how much days its take 45 days so; that means, January 15 to what may be what February 15 and then you will take that up to what you can say April 1 or so, then design the GUI part tool should take how many days it should take what 30 days. So, here its I think January 15 to what see it is taking up to March 15.

So, I think this what days are little bit wrong you can correct accordingly. So, this should take from January 15 it should take how many 45 days. So, count what is January 15 to 45 days accordingly please write down this one might be a small mistake here. So, similarly designing the GUI part it takes 30 days so; that means, the January 15 to, it will take what maybe February 15 or so. So, accordingly please change the date here and then what you see this will take how many days writing user manual it will take 60 days may be January 15 to what march 15 or so, see exactly what the 60 days and accordingly you can proceed.

So, then after this you can after this designing database, you can do the coding the database it will take 115 105 days. So, coding will starts from this see after this 105 days coding can only start and then coding GUI part can start coding GUI part it can start after what after these design GUI part is over. So, here design GUI part is over from here. So, then you can start the coding in GUI part and finally, you can do the integration and testing after this what code database and code GUI part is over. So, code database is over only it takes maximum days 105 days. So, even if your code GUI part it takes less than 45 days, but it has to wait because integration and test is testing, but can start only after this code data base part is there.

So, you can see this is the slack time or the flex time because this time is free. So, if sufficient resource is not available, then you can little bit delay the code GUI part in order to consume this free time part. And you see write user manual takes 60 days, but here see this the here this much time is what free time is there and if sufficient user is staffs are not there resources not there you can also delay what this write user manual so, that it can be it will not affect the end date, it will project can be finished in its appropriate target date. So, this is how from this activity network we are able to construct the Gantt chart for this MIS problem.

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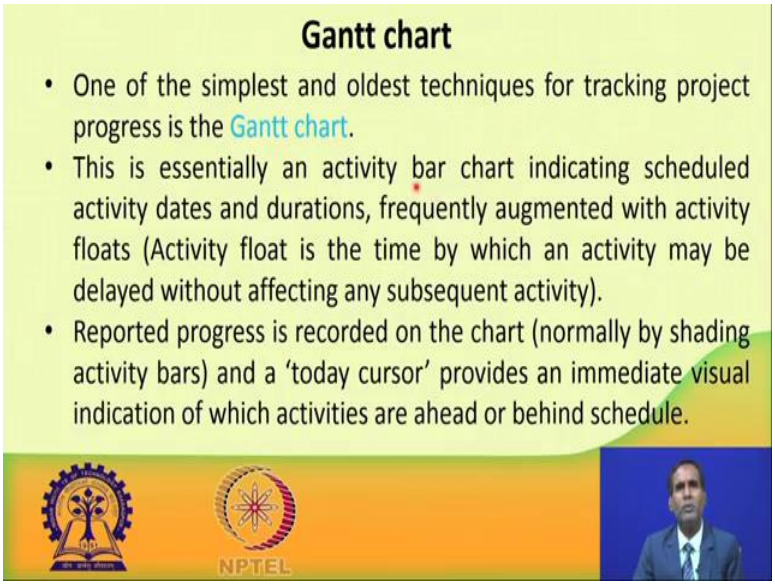

Now, as we have already told you that Gantt chart is one of the simplest and oldest techniques for tracking the project program progress. This is essentially an activity bar chart which indicates the scheduled activity dates and the durations. The frequently augmented with activity floats I have already shown you how you can represent the floats these un-shaded parts are the float or times or the slack times. We have already I have only told you float that is float or activity float is the time by which an activity may be delayed without affecting any subsequent activity.

So, the reported progress is recorded on the chart normally by shading the activity bars ok. The progress is represented by this bars or the shaded bars and un-shaded part is this what slack time or the what free time. So, the reported progress is recorded on the chart normally by shading the activity bars and today cursor provides an immediate visual indication of which activities are ahead and which activities are behind the schedule. So, this is how you can prepare the Gantt chart.

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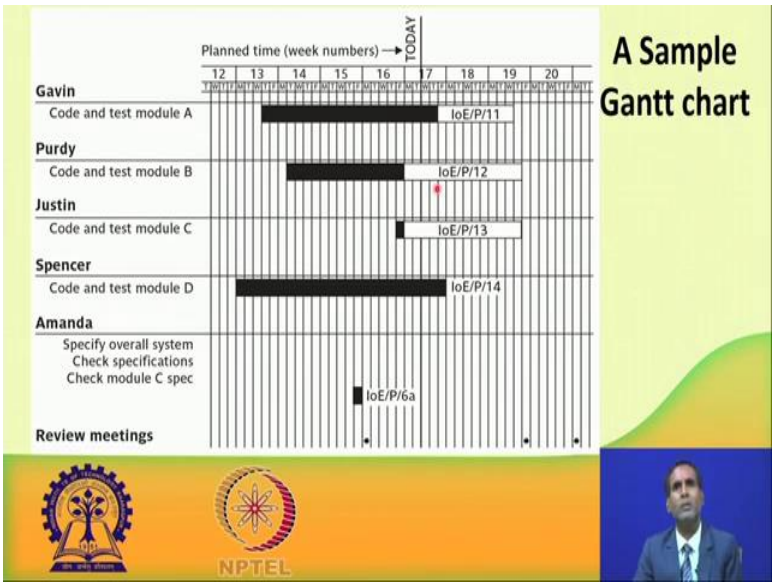

Another example also has been shown here. So, here these are the what different activities, we have taken coding and testing this is another a plan. This is another example coding and testing for module A coding and testing for module B and like code and test module C, code and test module B etcetera and these axis x axis we have taken the weeks where week 12, week 13 week 14 etcetera then the days of those weeks and this shaded parts said that code and test module A takes a this much period and this un-shaded part is the free time.

If we are on today at this point; that means, at 17th weeks we can say that in module A is ahead of the schedule also you can see that in today this is the 17th week and we have also done some extra; that means, code and test module D is also ahead of the planned schedule, but if you look at the other two activities code and test module B and C you see this is the today's date this one is the today's date.

So; that means, we are behind the planned schedule, we are lagging behind the planned schedule for this code and test model B and C. So, accordingly you have to take remedials what action how you can what meet the time maybe by hiring more what manpower or so. So, you should try to meet the what a target line, this is how you can prepare a sample Gantt chart.

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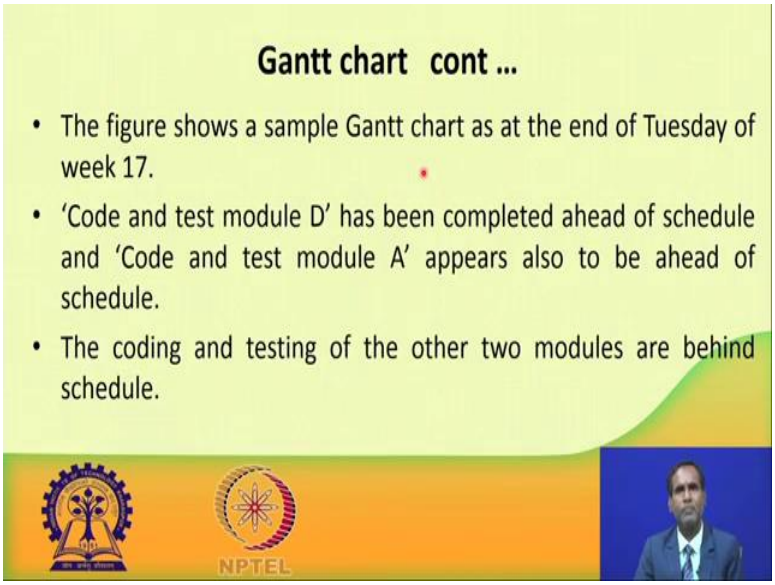

This already I have explained ok.

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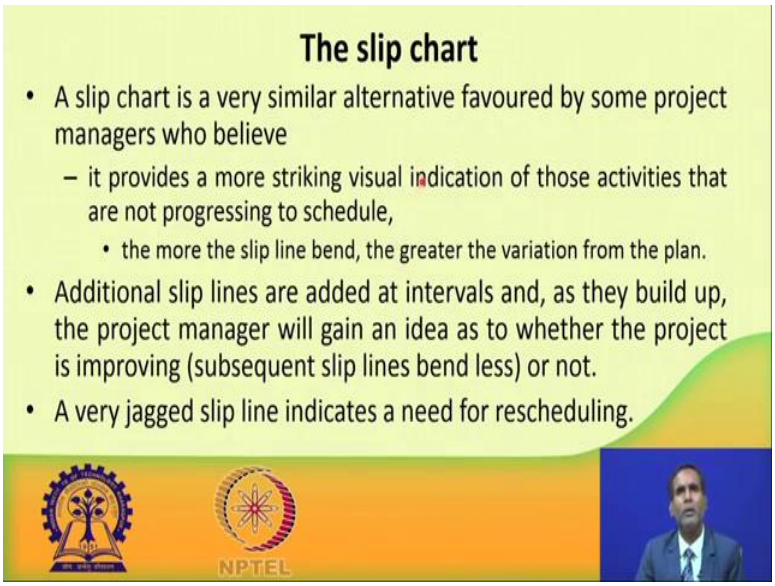

Now, let us see the other type of what the chart that is called a slip chart. A slip chart is a very similar alternative which are favoured by some project managers its a similar alternative to Gantt chart. And so here these as you can see that is a very similar alternative this these are favoured by some project managers who believe that it provides a more striking visual indication of those activities that are not progressing to schedule. So, this slip chart it can provide you a more striking visual indication for those activities

which are not progressing as for the schedule, that is you can see that the more the slip line bend you can say that greater the variation from the plan, let us take a small example say.

Like this similar to your Gantt chart, but you see here if this is today you see some what you are connecting and these what activity the progress of different activities through somewhat lines. So, more it is bend; that means, more there is a variation from the schedule, had it been straight line then we are what moving as per the schedule as per the plan. If there are more bendings if the slip line is more bended; that means, is more variation more deviation from the what the planned schedule which should not there. So, you should take remedial action to overcome this problem.

That is what we are saying here that this chart it will provides a more striking visual indication of those activities that are not progressing to the schedule and more the slip line bend just like this is the slip line, if it is more bend it indicates that the greater the variation from the plan and hence you have to take remedial action. So, additional slip lines also they can be added at intervals and as they build of the project manager will get an idea as whether the project is improving; that means, the subsequent slip lines bend less or not.

So, here you have put one, then after take the remedial actions afterr two weeks another again you draw the what slip line C whether this is becoming more straight or not. See if more straight means we are meeting the target lines we are as far the schedule more bending means it is more deviation from the schedule.

So, periodically after every 1 week 2 week draw the what slip lines and observe whether you are moving into the planned schedule or not. If you are not moving towards the planned schedule you are if you are deviating then take remedial action such as a higher more man power etcetera so, that you can meet the target dates and you can move as far the schedule ok. So, a very jagged slip line it indicates a need for rescheduling if the slip line is very much jagged, then it indicates that there is a need for re scheduling the activities.

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Next one is the timeline chart. So, let us see one of the drawback of this Gantt chart and slip chat. So, one drawback is that they do not show clearly the slippage of the project completion date through the life of the project. So, these two diagrams we have discussed Gantt chart and slip chart, they do not show the slippage of the project completion date through the life of the project. So, analysing and understanding the trends the project so far allows us to predict the future progress of the project. If we will there should be some mechanism for analysing and understanding the trends in the project. So, far happen and this will allow us to predict what will happen in future. It will help in predicting the future progress of the project.

For example, suppose if a project is behind the schedule because so, far productivity has not been as high as I assumed at the planning stage, then it is likely that the scheduled completion date also it will be pushed back even further unless action is taken to compensate for or improve the productivity. The timeline chart is a method its a technique for recording and displaying the way in which targets have been changed throughout the duration of the project I have already told that Gantt chart is a static figure it represents the static aspect of the project, but timeline chart it represents the dynamic aspects how the progress is moving with respect to the time.

So, its a it represents the dynamic aspect. The timeline chart is a method of recording and displaying the way in which the targets have changed during what the development of the project.

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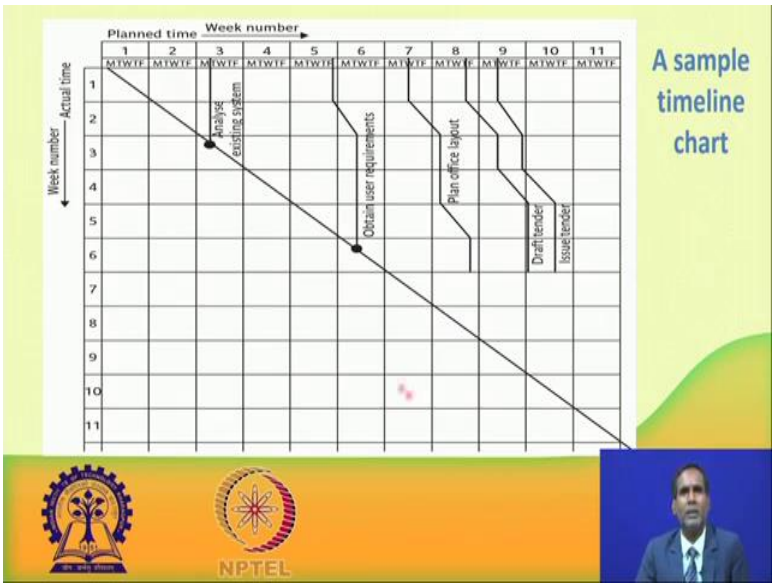

Will take a simple example. See this is a what sample timeline chart we are here we have taken the wig in this y axis we have taken the week numbers and here in the down if you will see the actual time, it will go downward this represent the actual time and in x axis also we are taking the what plan time in week number in y axis we are taking the actual time in terms of the week numbers and in x axis you are taking the plan time also in terms of the week number. So, these are say there are 11 weeks here and here there are also 11 weeks and we see how we are progressing how the project is progressing this can be visualized from this graph let us see what is saying.

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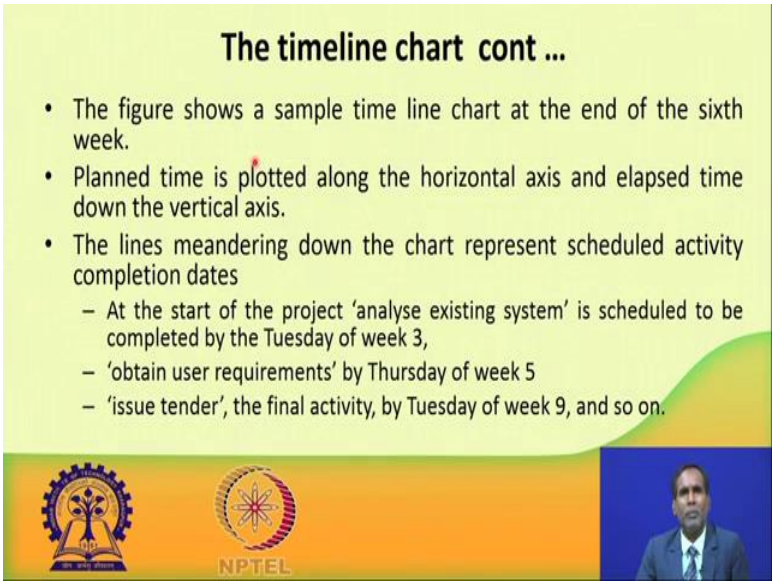

This figure given in the previous slide that is the timeline chart, it is show the sample timeline chart at the end of the 6th week you see at the end of the 6th week. It is a 4-5 it is the 6th week suppose we want to see this is the progress at the end of the 6th week we want to know what is the progress that is shown here and the plan time which plotted along the horizontal axis and the elapsed time along down the vertical axis. So, here x axis is planned a time y axis this actual time. The timeline and the lines mean meandering down the chart is represent the scheduled activity completion date.

So, the this line these are meandering the lines meandering down the chart represent the scheduled activity completion dates. So, this it is meandering, so, they represents the completion dates. So, at the start of the project if you will see the analyse existing system is scheduled to be completed by the Tuesday of week 3. So, this is this what analyse the existing system, it is expected to be completed in the 3rd week on the Tuesday and then at the start of the project as I have already told you the analysing every system with scheduled to be completed by the Tuesday week 3 and obtain user requirements by Thursday of week 5, it was initially planned that week 5 this activity should be finished by what thursday of week 5 and then these issue tender etcetera by Tuesday of week 9 issue tender where is the issue tender? This is about issue tender it was initially planned to be completed in week 9 maybe Tuesday ok.

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But the at the end of first week the project manager review the progress review the target dates and leaves them as they are that is the lines are therefore, drawn vertically downwards from the target dates to the end of week 1 on the actual time axis. So, what we have seen? At the end of 1 week you see it at the end of the 1st week at the end of 1st week see lines are just straight forward it was what see this is the 1st week actual one this 1st week, just straight forward we expected that every activity will meet their dead line then what happened?

At the end of the week 2 then the project manager he could have observed that obtained user requirements will not be completed until Tuesday of week 6. So, this obtained user requirements you can see that after this first week he has just seen that this obtain user requirements it is getting delayed, it cannot be finished this it will take some more days might be what week 6 it will take one more week. So, that is why it said that at the end of week 2 he decides that he felt that obtained a user requirements cannot be completed until what Tuesday of week 6 and hence he therefore, extends that activity line diagonally to reflect this.

So, that is why what has done up to 1st week see it is straight then he found that it cannot be finished by Tuesday of 6. So, that is why it has bended the he has what made bent the line up to what this the 6th week Tuesday and then again now it is what moving straight forward ok. And similarly what other activities also they are expected to be delayed and

accordingly he had made them as bended after this 2nd week. So, he has bended all these lines for other activities ok.

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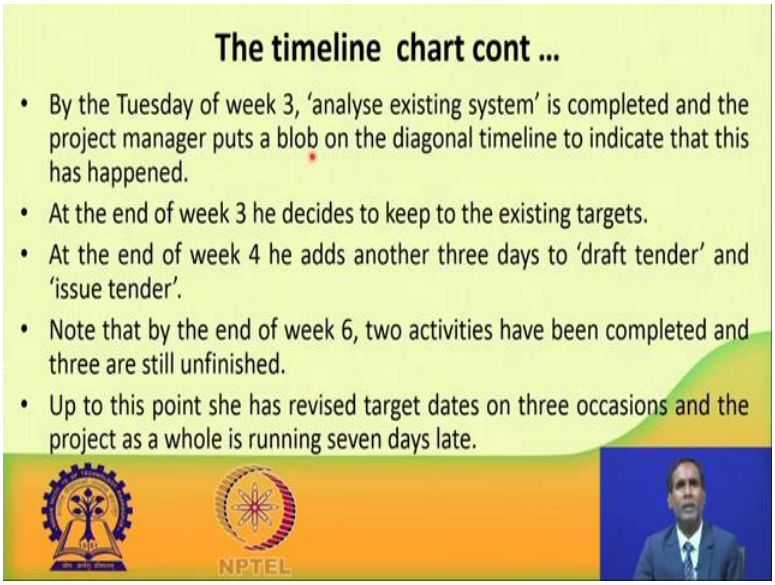

So, then by the Tuesday of week 3 analyse the existing system is completed. So, by the use of 3 see 3 this one is finished. So, they have made a circle in order to mark that the activity is completed and then at this what time at the end of at the by the Tuesday of week 3 analyse the existing system is completed and the project manager puts a blob on the diagonal life timeline in order to indicate that this has already completed. Then at the end up week 3 decides to keep the existing targets he has tried to meet the existing targets and at the end of week 4 he adds another three days to draft tender and issue tender.

So, then he has seen that. So, again at the end of week 3 he has seen that this thing are not able to be finished the target dates again they will be delayed, see again after this again it is dealing to this. So, whenever he expressed that there will be further delay again he will bend the line and every after every week you will see the progress if it is delaying, then you will accordingly bend the line. Then what note that by the end of week 6, two activities have been completed and this 3 are still unpinning not completed. So, this is week 3, first activity is completed then at the end of week 6 he has seen that this is also completed, but you see this is originally it was supposed to be finished in 5th week.

But since it is expected that it cannot be finished in a week 5 the line has been bended and it is expected that it will be over in 6th week maybe by Tuesday. So, accordingly this has been bended. So, at the end of 6th week the project manager again he reviews the progress and he has observed that two activities first two activities are being finished and still three are not finished. So, in after every week again he will what review the progress and if again it is expected that these activities will be delayed accordingly he will bend the lines and when it will completed again he will mark a what dark circle in order to indicate that those activities have been finished.

So, at the end of 6th week the manager observes that the first two activities are finished and other activities are still they are not completed, they will take some more time to be completed. So, periodically he will review the progress of these activities and then if they are not moving as per the plan or the schedule, then he will be in bend the lines as and when required. So, this is how this timeline chart it represents the dynamic aspects. Dynamically the lines are just what keeping unbending as it is expected that this will take more time the lines are being bended.

So, it represents the dynamic aspects how the project is progressing it represents the dynamic aspects. So, gantt chart represents the static aspect whereas, the timeline chart it represents the dynamic progress of the project. So, now, we have observed that by the end of week 6, two activities have been completed and the three activities are still unfinished. Up to this point he has revised the target dates on the three occasions and the project as a whole is running seven days late.

It will see on a as a whole the project is running almost to 7 days late it was it should be finished in 9th week, but it is moving towards what a 10th week. So, almost 7 days the whole project is delayed the timeline chart is ok.

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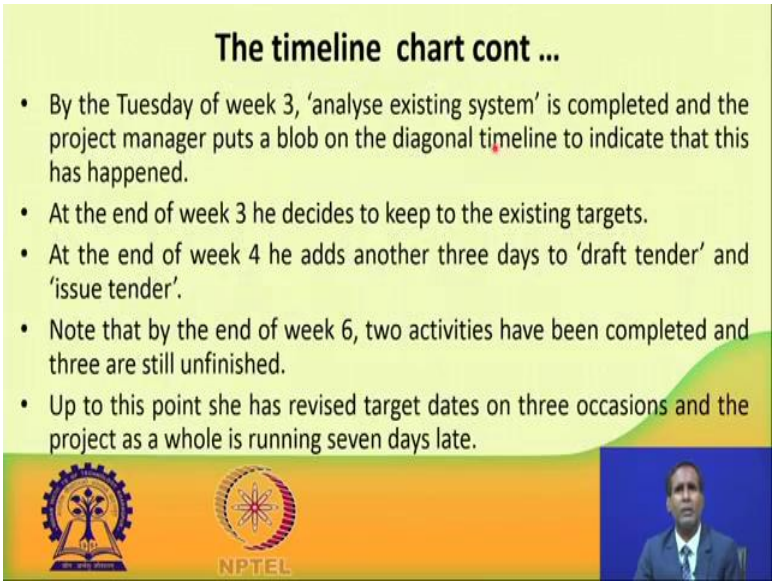

So, this we have seen the timeline chart is very much useful both during the execution of a project as well as the part of the post implementation review. Analysis of the timeline chart and the reasons for the changes they can indicate the failures in the estimation process or other errors that might be avoided with that knowledge ok.

So, if you will analyse the timeline chart analysis of the timeline chart and the reasons for the changes why you have made the changes they can indicate the failures in the estimation process, they can or the other errors that might be there and with that knowledge these what errors or failures they can be avoided or these changes also changes in the timelines etcetera they can be avoided in future. These errors also can be avoided in future by using by analysing the timeline chart.

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We will quickly say about the cost monitoring. A project could be late because of the staff originally committed, they have not been deployed at the right time. And so, in this case the project will be behind the time, but under budget it will be behind time, but still it is under budget. A project could be on time, but only because additional resources have been added ok. A project can be completed on time, but only when we will hire additional resources of course, this will be over budget you have to spend some more money. So, that is why you have to need to monitor both the achievements whether you are completing in time or not and the cost that we are spending.

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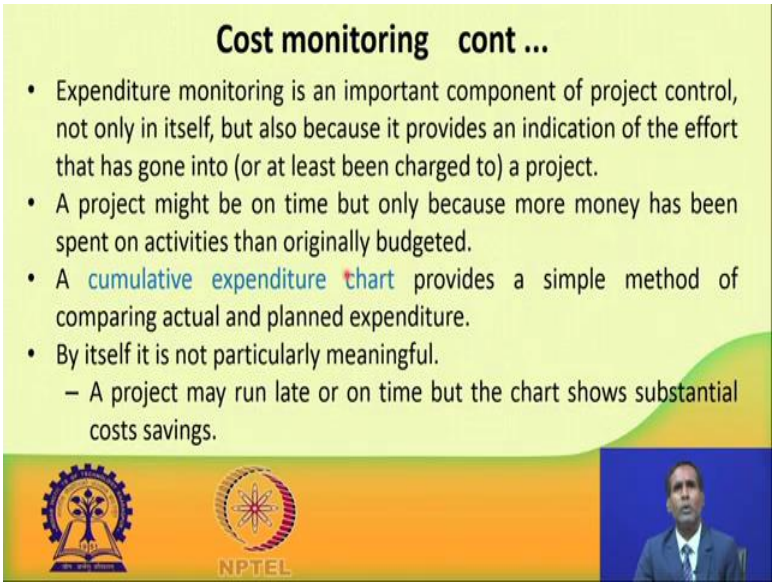

So, expenditure monitoring is an important component of project control, it not only in itself, but also because it provides an indication of the effort that has gone into or at least that has been charged to a project. A project might be on time, but only because more money has been spent on the activity that originally budgeted. So, we have originally expected this much money will be spent, but since you have what moving late, you might have to add more manpower and hence you have to what spend more money.

So, in that case the project might be on time, but on the cost of what are the cost of additional what money. A cumulative expenditure chart you can draw which will be provide a simple method of comparing the actual cost or the actual expenditure and the planned expenditure by itself it is not particularly meaningful, but a project made on late or on time, but the chart shows the substantial cost savings.

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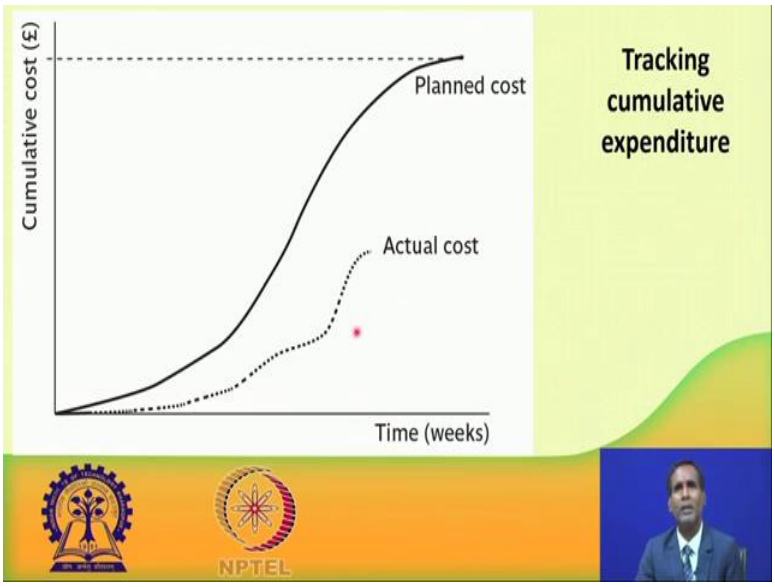

So, what is the substantial cost savings that the chart can show this a sample example of the what the cumulative expenditure chart. So, here in the x axis you are taking the time weeks and y axis we have taken the cumulative cost, you can see the actual cost was estimated sorry the planned cost was estimated to be what this, but the actual cost is moving like this. So, this shows the comparison between the what actual cost and planned cost you can track the cumulative expenditure by looking at this expenditure and by looking at this graph and accordingly you can take remedial action.

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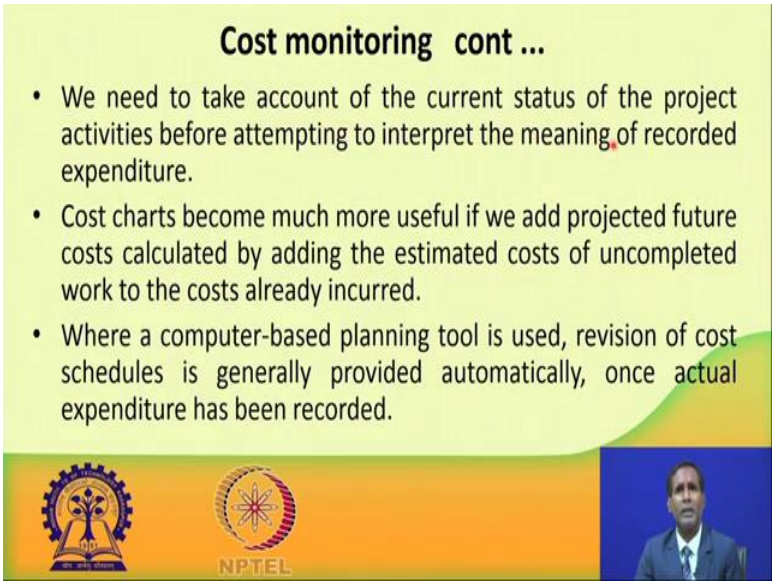

We need to take account of the current status of the project activities. First we should take account of the current status of the project activity, before attempting to interpret the meaning of the recorded expenditure the cost charts they become much more useful if we add the projected future cost ok. So, these costs that have been developed there can become much more useful if we can add the projected future cost which are calculated by adding the estimated cost of the uncompleted work to the cost which have already incurred.

Where a computer based planning tool is used, revision of cost schedules is generally provided automatically if you are using tools software for what these what the cost schedule the revision of cost schedule is generally provided automatically once the actual expenditure has been recorded.

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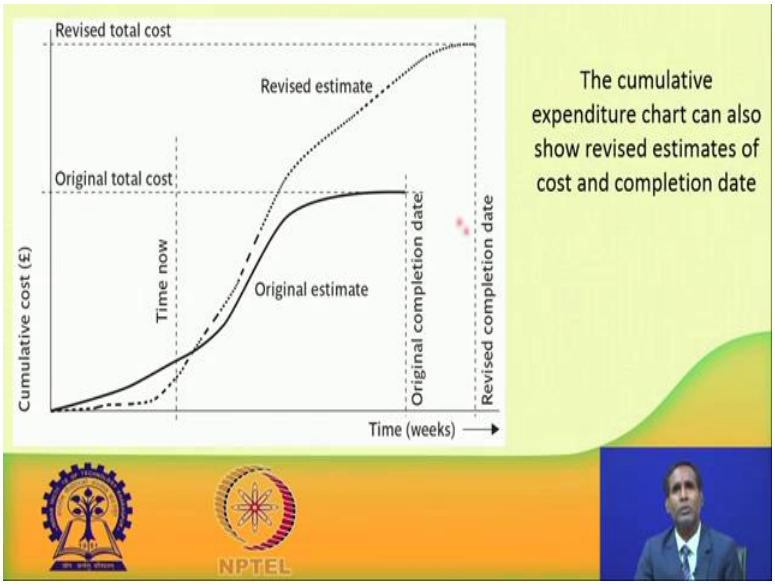

So, there is another diagram you shows this what the cumulative expenditure, here we are taking the time in weeks and here the cumulative cost. You can see that the cumulative expenditure chart it can also show what is the revive what are the revised estimates of the cost and on the completion dates. So, this was the original estimate and then since the progressive the during the progress of the project, you might add more manpower or some additional cost.

So, now the cost is being changed. So, this is the this diagram this line show the dotted line show the revised estimate, you can compare this what original cost estimate versus the revised estimate similarly this was the what original completion date and since you are lagging behind and this is the revised completion date, as the progress as the project progresses you can draw the revised completion date the you can find out the revised completion date.

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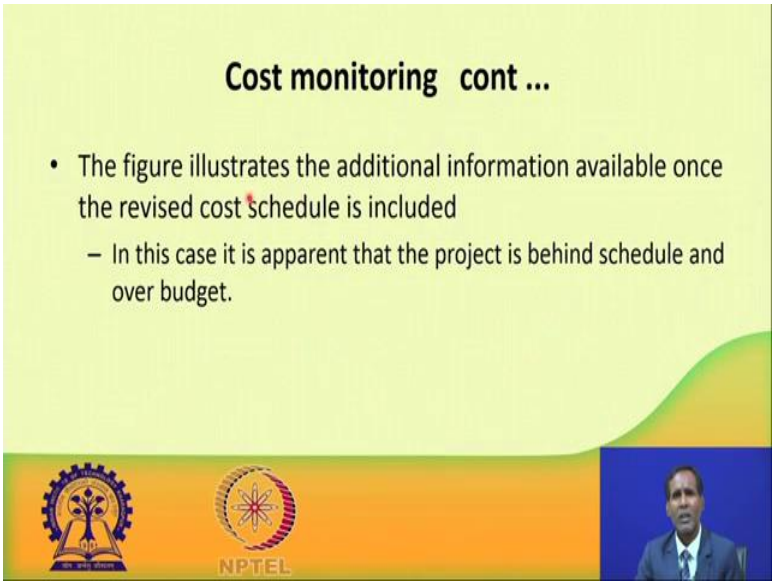

You can see that this figure illustrates the additional information available once the revised cost schedule is included. In this case it is you can see that the project is behind schedule and over budget. Please see the project is what the time original completion with time, but it is revised completion t. That means the project is behind the schedule. Similarly the original cost was this and revised cost is this; that means, say you are spending it is expected that more money has to be spent. So, in this case it is very much apparent that the project is behind the schedule and over the budget.

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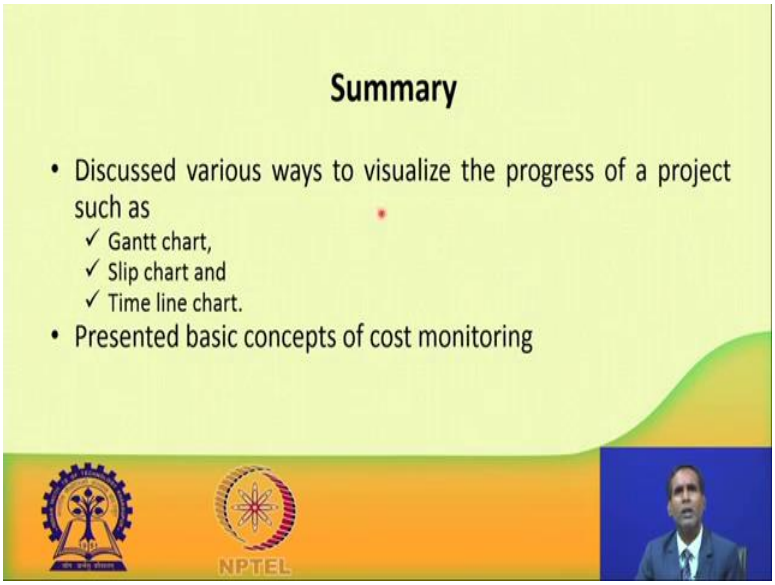

So, we have discussed the various ways to visualize the progress of a project by using Gantt charts, slip charts and timeline charts we have also presented the fundamental concept basic concepts of cost monitoring. How can use cumulative cost chart to expect or to visualize the progress of the project with respect to time and the cost.

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These are the references.

Thank you very much.

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Lecture - 43
Project Monitoring and Control (Contd.)

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Good morning. So, now, let us see the another aspect of Projector Monitoring and the Control. We will see about something a different concept called as earned value analysis and then we will see the fundamental concepts of a baseline budget.

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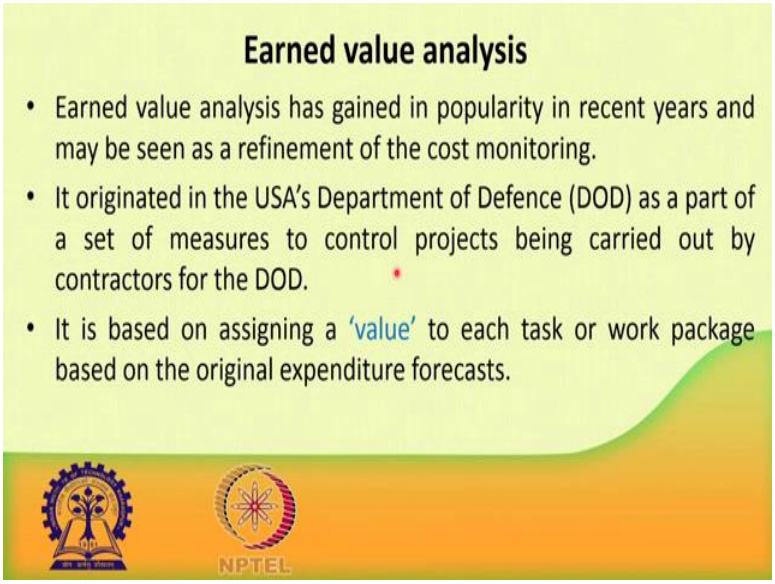

So, yesterday in the last class we have discussed about cost monitoring. So, earned value analysis nowadays has gained in popularity and it can be viewed as a refinement to the cost monitoring process that we have discussed in the last class. So, this earned value analysis it was originated in the USA's department of defence as a part of a set of measures to control different projects which are being carried out by contractors for the department of defence. This concept is based on assigning a value we have to assign basically a value to what to each of the tasks or the work flow packages. You have studied the WBS work breakdown structure, there you have seen how a job is divided into several work packages.

So, this early earned value analysis a concept this analysis is based on assigning some value to each task or work package in case of a WBS based on and the original expenditure forecasts or based on the original expenditure prediction.

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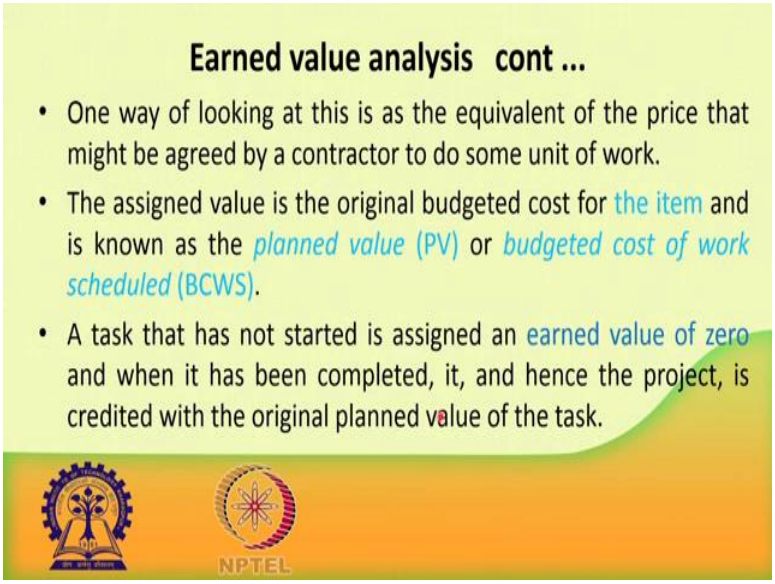

So, let us see an analogy with this earned value analysis. So, this earned value analysis it is similar as viewing and the price that might be agreed by a contractor for doing some unit of work. The assigned value is the original budgeted cost. So, that value will be assigned it to a task or the planned what the forecasted the cost we can say the estimated cost that the assigned value is the original budgeted cost for a particular item or a particular task we can say and it is known as planned value or it is also known as budgeted cost of work schedule or BCWS.

So, planned value can be considered as the value which is the original budgeted cost for some item or for some a task. Now a task that has not started yet it is normally assigned an earned value of zero ok. So, a task that is. So, far not started a task that has not started yet is assigned initially an earned value of zero and then when it has been completed, then the project this task as well and hence the project is credited with the original planned value of for that task.

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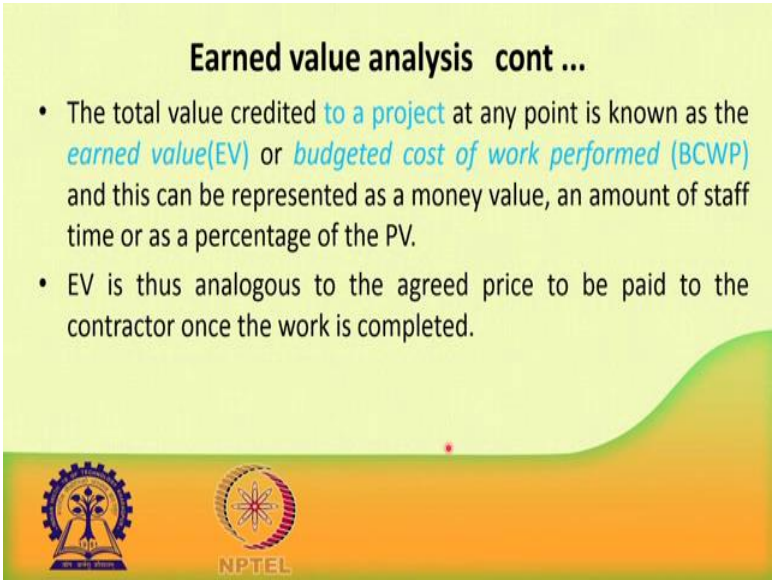

The total value credited to a project at a point of time is known as the earned value ok. So, we have already seen about what is this planned value, the total now we will see is something about earned value. Earned value is the total value which is credited to a project at any particular point of time this is known as earned value, this is otherwise are known as budgeted cost of work performed or BCWP. So, this value can be or this earned value be represented just we can think of as an analogy like this. So, this value can be repented as a money value or an amount of staff time or a percentage of PV. In any of the either ways the EV can be represented maybe in terms of money or an amount of staff time maybe in terms of personal months or so, or as a percentage of the PV.

So, similarly just we have seen analogous to PV present and that value we have already seen an analogous to this planned value, now similarly let us see an analogous to the earned value. Earned value can be thought of as the agreed price that has to be paid to the contractor once the work is completed.

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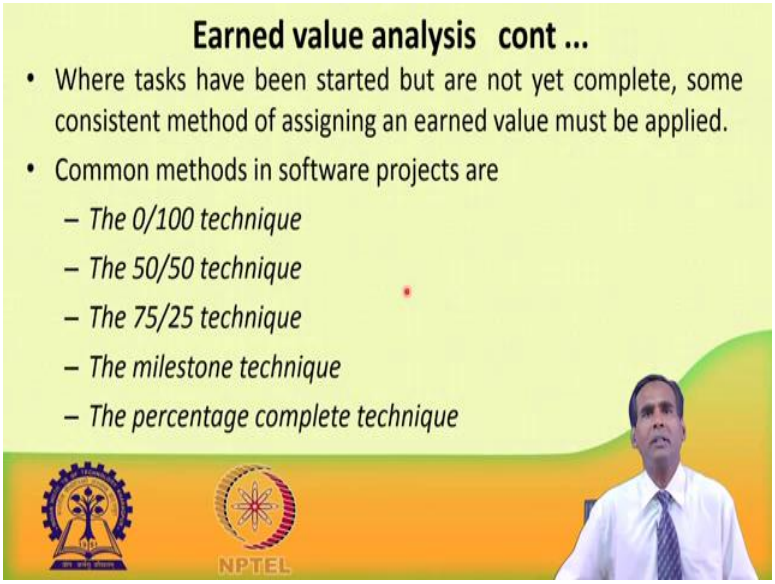

So, now let us see in case of when a work is started how we can assign these what values the earned values. See there are different techniques when the task has been started, but it is not completed so far, then how to assign the earned value. So, where tasks have been started, but are not yet completed then some consistent method of assigning an earned value should be applied. In software project development these are the common methods which are applied for the tasks which have been started.

But so, far they have not been completed in order to assign an earned value this techniques are 0 by 100 technique, 50 by 50 techniques, 75 by 20 technique 75 by 25 technique, milestone technique and a percentage complete technique.

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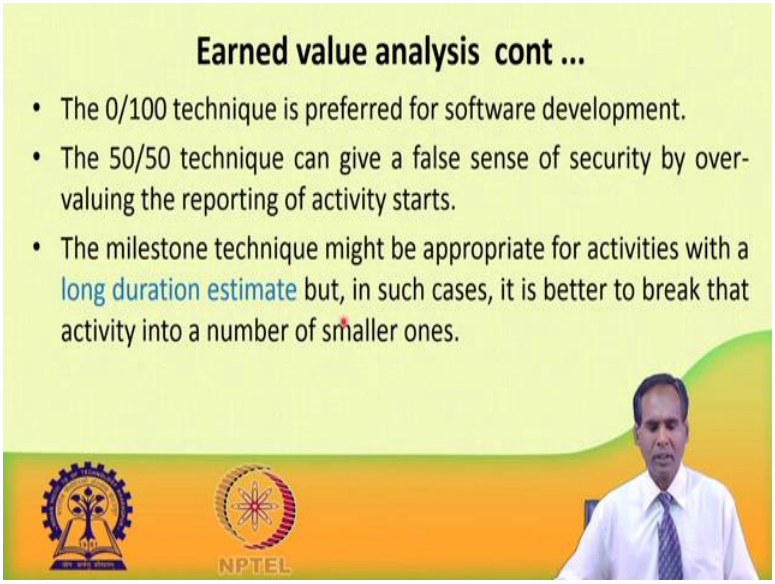

Now let us see one by one these above techniques. First we will before going to the techniques let us see some common general things on these techniques, you can say that the 0 by 100 technique is normally preferred for a software development we will see why it is more useful. If you will use 50 by 50 technique this then this may give you a false sense of security by over valuing the reporting of activity starts say it is not so far say completed 50 percent but the what developer may say that we have done 50 percent.

So, they may ask for the payment for the 50 percent. So, it might give a false sense of a security by just over valuing over rating the reporting of the activity starts. The milestone technique is a suitable for the activities with a long duration estimate. So, for the activities we are having long duration activity long duration estimates, in those cases a milestone technique is more appropriate. But while you are applying milestone based technique say, it is better to first to break the activity into a number of smaller tasks, then you can apply milestone based technique for this earned value analysis for assigning the earned value.

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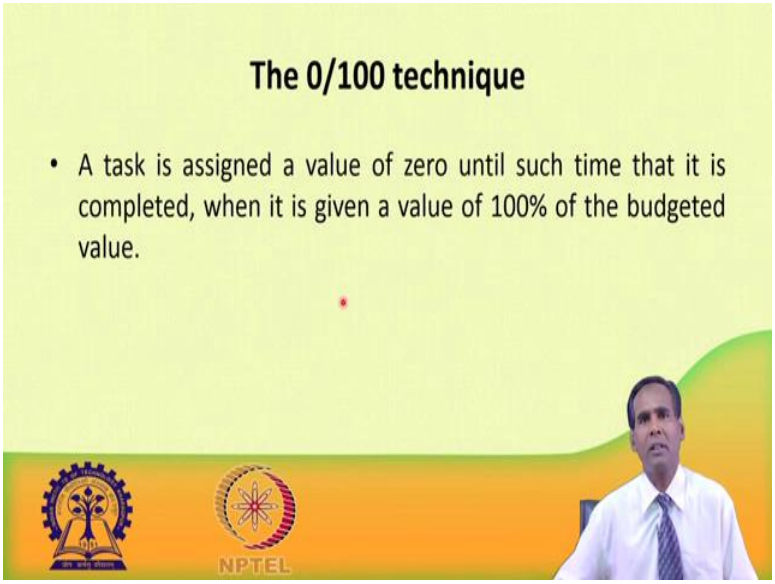

Now, let us see the first technique that is 0 by 100 techniques. So, in this technique a tasks is assigned a value of 0 first initially. So, in this technique a task is assigned a value of 0 until what time? Until such time that it is completed. So, until it is completed no value is given. So, a task is assigned a value of 0 until such time that it is completed, but when it is and when it is given a value of 100 percent of the budgeted value. So, when the tasks is completed, then only totally 100 percent the value of 100 percent of the budgeted value is assigned to that tasks this is known as 0 by 100 technique.

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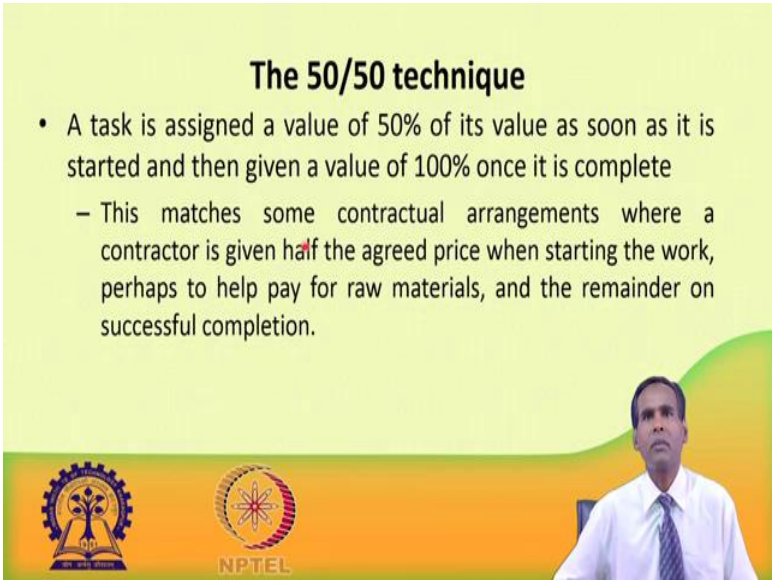

50 by 50 technique says that a task is assigned a value of 50 percent of its value as soon as it is started. So, as soon as the tasks started, a value of 50 percent of the total value is assigned and the rest 50 percent is assigned once the task is completed and they are given a value of 100 percent once it is completed. So, this matches some contractual arrangements where a contractor is given half the agreed price when starting the work, suppose you are considering a building construction.

So, the contractor has been what agreed to some price and then the 50 percent of the price you have to give when he starts because he might have to what a pay for raw materials and the pay for labour works and so, like that. So, first 50 percent you will you have to pay as soon as he starts the work and the rest can be paid upon successful completion of that project. So, this is something known as 50 by 50 technique as I have already told you this might work for building construction other projects etcetera.

But software development this may not work because how to measure that exactly they have done what some 50 percent what will be the 50 percent that will difficult to what compute. So, that is why here it may not be suitable and for software normally software development we use 0 by 100 technique.

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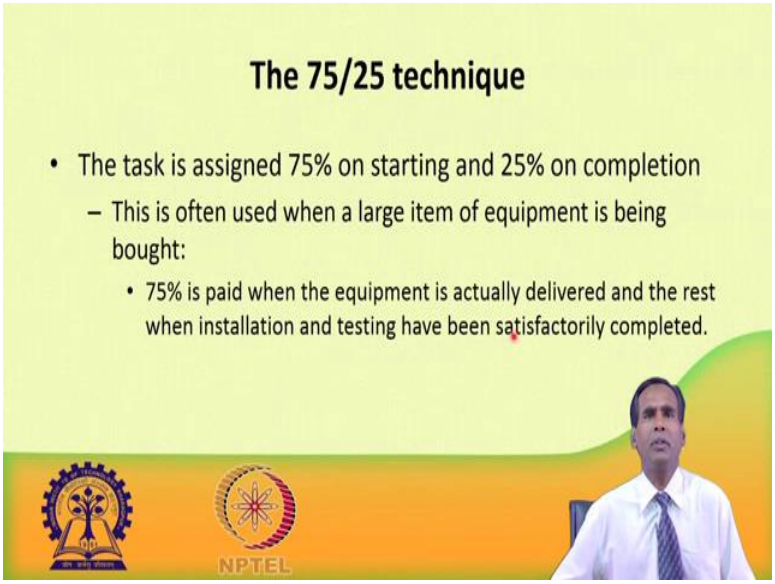

So, it is just a similar to 50 by 50 technique there is another technique called as 75 by 25 technique. So, here the task is assigned is 70 percent on starting as soon as the project is started. So, 75 percent value is assigned and the rest of 25 percent is assigned on

completion of the project. So, this is often used when a large item of equipment is being bought. So, for example, you are purchasing a very large generator or a very vast a huge server you are purchasing. So, while you are purchasing you the what contractor he may say that, I want 75 percent because I have to immediately know what purchase and give it and the rest 20 percent may be given on successful installation ok.

So, this technique may be used when a large item of equipment such as big generators big what servers etcetera or clouds etcetera they are being bought. So, on in the in those cases 75 percent of the total value is paid when the equipment is actually delivered and the rest of 25 percent may be paid after successful installation and the testing.

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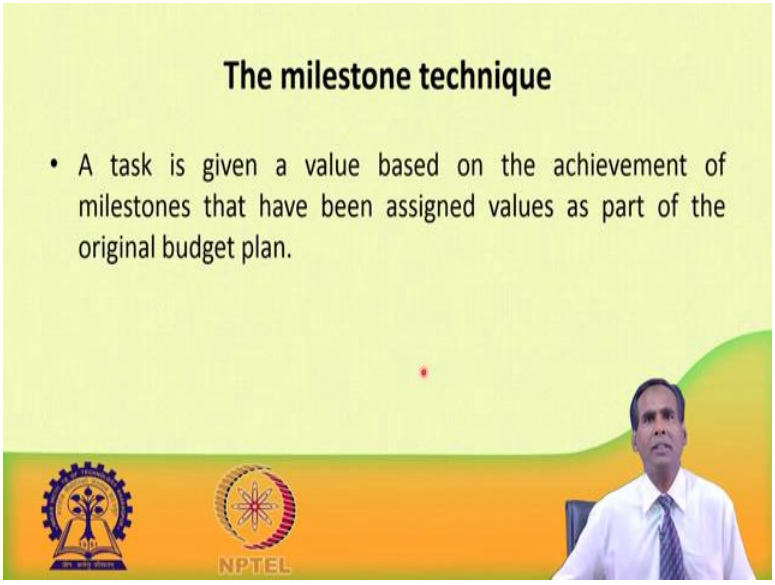

So, next technique is called as milestone techniques. So, here a task is given a value based on the achievement of some milestones. So, here we do not use that what a 50 by 50 or 75 by 20 etcetera the value they assigned based on achievement of some milestones what could the milestones? The milestones could be suppose requirement analysis over SRS is successfully prepared 1 milestone. Design is your for we have successfully prepared the data flow diagrams, UML diagrams etcetera design is over.

Another milestone could be that coding is over, another milestone could be that testing is over. So, normally here the value will be assigned based on achievement of some milestones. So, in this technique a task is given a value, it is assigned a value based on

the achievement of the milestones that have been assigned values as part of the original budget plan.

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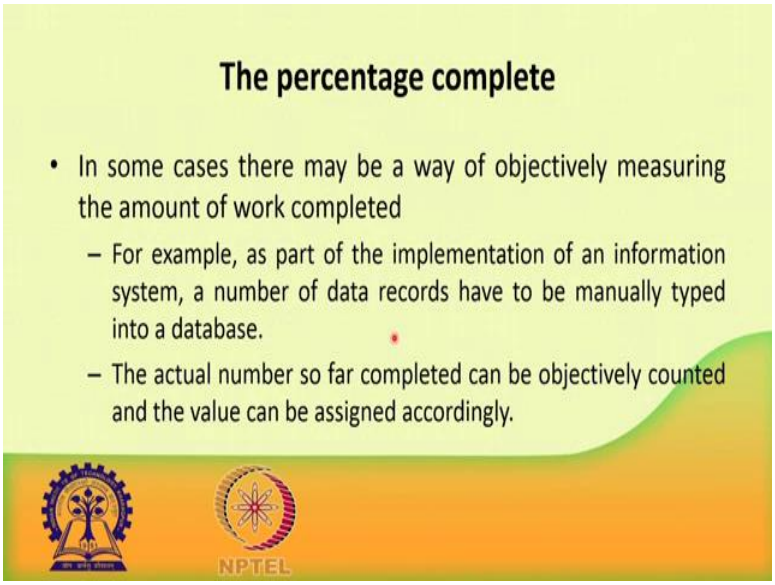

The next is the percentage complete. As its name suggests percentage complete here we have to count we have to measure the amount of work completed, then we can say that what is the percentage of completed 10 percent 20 percent 50 percent like that. In some cases there may be a way of objectively measuring the amount of work completed how much what is the total work and so far what percentage of work we have completed let us take a let us take an example. As part of the implementation upon information system, a number of data records have to be manually typed in the database. Say in an institution there are 10,000 students and for each for each student you have to create a record.

So, and those are what student records has to be typed manually. So, that and they will be saved in a database. So, in this case the value and can give in terms of percentage. So, after 1 month, you say that only what to the 1000 students record has been entered; that means, 10 percent is completed. So, based on that 10 percentage data entry we can assign a value. After say 2 months we have seen 20 percent 2000 records have been completed; that means, say 20 percent work has been done.

So, basically we are counting here the number of data records and based on this what amount of work or the number of records have been completed we can assign some value this is something known as the percentage complete. So, as I have already told you in

this example, suppose as part of the implementation of an information system a number of data records have to be manually typed into the database, then the actual number so, far it is completed can be objectively counted, it can be objectively measured and then the value can be assigned accordingly. This is known as something this percentage complete method.

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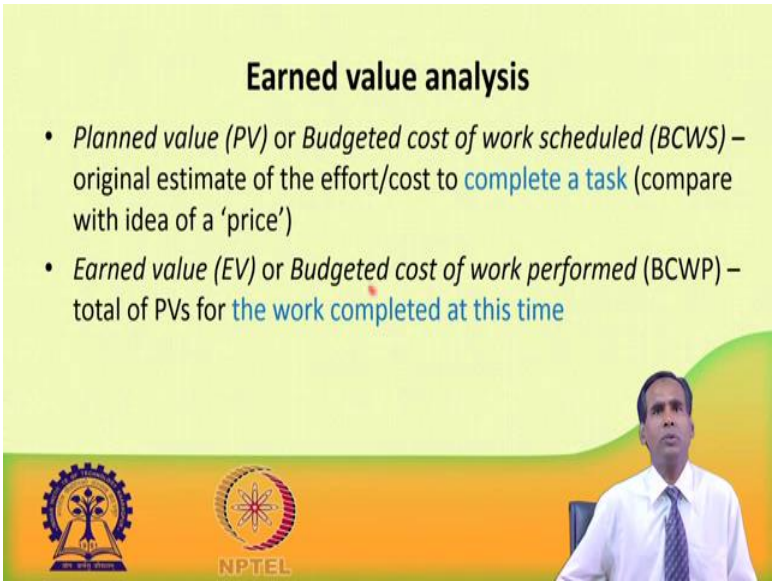

Now, I have already told you two important terms, planned value and earned value. So, planned value it can be considered as that the original estimate of the effort or the cost to complete a task. So, in order to completed a task what is the original estimate may be in terms of a effort or person months or a cost this is known as what planned value or PV, where earned value or budgeted cost of work performed this may be considered as the total of the planned values for the whole work completed at this time ok. So, earned value may be considered as the total of the PVs for all the task for the whole work completed at this time this is known as earned value. So, with this you will move further.

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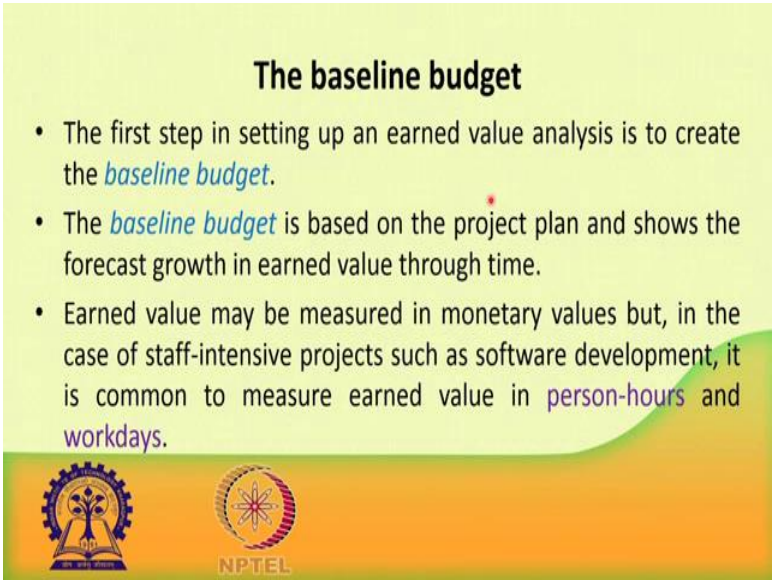

So, the we will used those terms earned value and planned value etcetera in the subsequent slides now let us say about this baseline budget. What do you mean by a baseline budget? So, the firsts step in setting upon earned value analysis is which create a baseline budget

So, we are discussing now on the earned value analysis, the first step in setting up or in preparing the earned value analysis its to create a baseline budget. So, what is a baseline budget? The baseline budget is based on the project plan ok. So, we have already discussed planning earlier and so, this baseline budget is based on the project plan it shows the forecast growth in earned value to time. So, as time passes what is the forecast growth what is the estimated growth or what is the predicted growth in earned value this is represented by the baseline budget.

So, the earned value let us see how it can be measured. Earned value can be measured in monitor values, but in case of staff intensive projects were number of staffs are working such as software development, it is usual to measure the earned value in person hours or person months and in workdays. So, basically earned value will see it can be represented it can be measured in monetary value such as rupees, but for software development organizations normally people use this are people will present the earned value in terms of person hours or person months and in terms of the work days.

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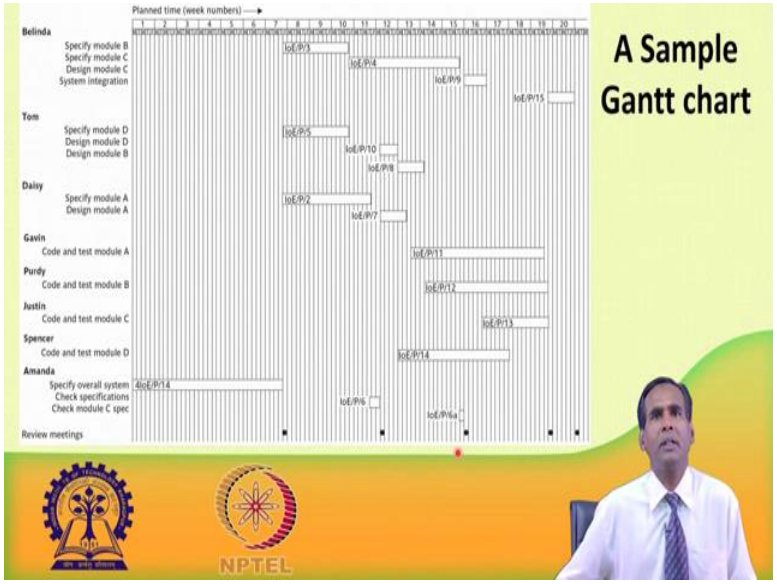

Now, let us see how it can prepare a baseline budget. So, a baseline budget it can be prepared easily if we Gantt chart is already given to you and Gantt chart we have already seen earlier this is a simple Gantt chart is shown this Gantt chart already also we have shown earlier in the last classes. So, here you can say that here in the x axis the weak numbers are taken and here the resources have been considered and here is the showing that from which day of the week it has been started and two parts you can see in these part in the bars.

The one is the shaded part, another is unshaded part and the un shaded part represents the float or the free time or the slack time this we have already seen while discussing about the Gantt chart. Now let us see how you can prepare a baseline budget from thee from given a Gantt chart.

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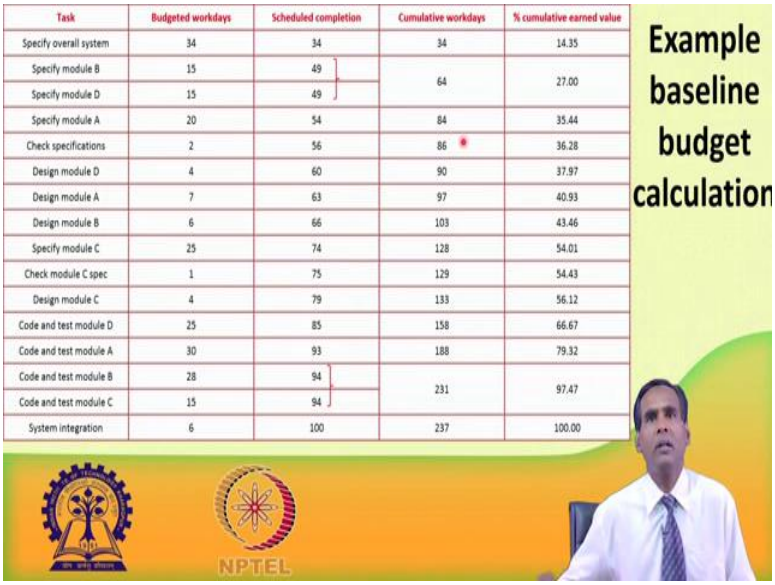

Task	Budgeted workdays	Scheduled completion	Cumulative workdays	% cumulative earned value
Specify overall system	34	34	34	14.35
Specify module B	15	49	64	27.00
Specify module D	15	49		
Specify module A	20	54	84	35.44
Check specifications	2	56	86	36.28
Design module D	4	60	90	37.97
Design module A	7	63	97	40.93
Design module B	6	66	103	43.46
Specify module C	25	74	128	54.01
Check module C spec	1	75	129	54.43
Design module C	4	79	133	56.12
Code and test module D	25	85	158	66.67
Code and test module A	30	93	188	79.32
Code and test module B	28	94	231	97.47
Code and test module C	15	94		
System integration	6	100	237	100.00

See we have to see that we want to calculate these what earned value and we want to first prepare the baseline and budget. So, first you have to list what are the task, the task are already shown here yes the tasks are already shown here on the left hand side. And then what is the budgeted work days? What if the planned work days? You can see from this what Gantt chart the different planned worked the planned the budgeted work days are given like first one is that specify the overall system. So, specify the overall system you can see it is taking 34 days starting from the beginning specify the overall system it takes how many days that it a 34 days.

So, that we have to lists here. So, specify the overall system it takes 34 days and scheduled completion it will start from 0 so; obviously, it will be taken what 0 plus 34 days and cumulative work days also 34 days. So, now, then specify module B and you can see from the diagram above that specify module B and the specify module D also specify module A these three activities these three tasks are started parallely.

These are what a performed parallely after these 34 days of this specify overall system is over. So, this specifying module B takes how many days? 15 days specifying module D also 15 days and specifying module A 20 days. So, that we have to write here that specify module will B 15 days, specify module D 15 days, specify module A 20 days now what we this is the budgeted workdays its planned work days. Now the scheduled completion will be what?

The first activity says specify overload system have started from 0 th date. So, scheduled completion in 34 days module. Specified module B and module D can be started only after this overall system specification over. So, after 34 days you have started how many days it takes? Budgeted work days 15 days. So, 34 plus 15 is 49 this is D and specifying module D and they also started parallely along with B.

So, specifying module A will also take comment at 15 days, when it can be started after this 34 days. So, 34 plus 15 is 49. And now you can see what is the cumulative work days? So, cumulative work days first work job is over on 34 days then next this second job takes 15 days. So, 34 for 15 is how much 49 and next; that means, to specify module D also takes 15 days.

So, 49 plus 15 is 64. So, up to the third activity the cumulative work day is 64 days. So, similarly the fourth one is specified module A this take 20 days after what? After the first job is completed because these three are running parallely we just started parallely. So, after 34 days plus 20. So, scheduled completion is 54 days. So, now, what is this work what cumulative work days? This activity takes 20 days.

So, 64 plus 20 is equal to 84 days. So, cumulative workdays is equal to 84 days. In this way you can find out see the budgeted work days is already available in the Gantt chart then you can find out the scheduled completion and then you can add the what values appropriately, then you will get the cumulative work days. So, up to; so, now, you can say that, the total cumulative work days for completing all the tasks is coming to be 237 days.

Now, we should find out what is the cumulative or percentage of cumulative earned value. So, total cumulative days is 200; total cumulative work dates required is 237 days, now we will take into account the first activity it takes 34 days. So, what is the percentage? 34 divided by 237 into 100 which will come to 14.35 percentage. So, similarly for this second and third activity the cumulative work days is 64 and the total cumulative work days is 237. So, 64 by 237 into 100 is coming to be 27 percentage. Similarly for the fourth activity you can see the cumulative work days up to 430 to each is how much it is 84 total cumulative days is 237. So, 84 by 237 in to 100 is coming to be 35.44 percentage.

So, in this way you can compute the percentage of cumulative earned value. So, I have already told you what is this baseline budget. Baseline budget is the what based on the initial project plan the budgeted plan and the shows the forecast growth in earned value through time as time passes, what will the forecasted rate growth in earned value that is shown in baseline budget and we have seen for this sample example. This is the forecasted the growth this is the forecasted growth in earned value.

So, in this way you can get the percentage of cumulative earned value and of course, finally, the total earned value will be 100 percentage. So, in this way you can prepare the baseline budget for a given project. So, if the Gantt chart is given preparing this will be much more easier preparing the baseline budget or calculating the baseline budget will be much more easier.

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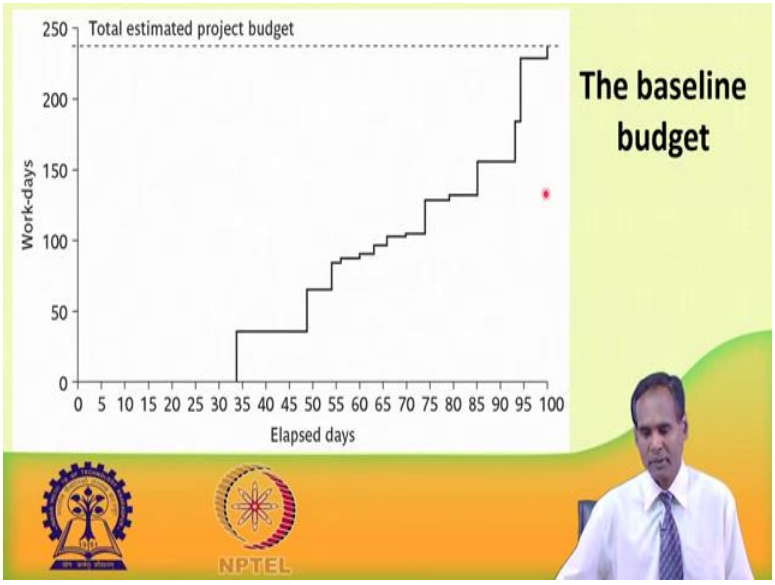

The same thing has been shown in the form of a graph. The same thing we can say from this data, from the baseline budget, from this given data this can be represented as a what graph, let us analyze the graph let us interpret the graph.

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So, in this budget out of the techniques we have given. So, you see 0 by 100 technique is used for crediting the earned value to the project what do you mean by 0 by technique in this context? You can see that the project is not expected to be credited with any earned value until their 34.

When the activity specified overall system is to be completed you can say that. The first task is expected to be completed on 34 days. So, till 34 days no value is assigned. So, this means we are using 0 by 100 technique initially no value is assigned when the job is completed then we assign the full value what is the full value here? 34 days because I have already told you earned value can be represented can be expressed in monetary terms in what personal months or in work days.

So, here the work days is taken to account. So, earned value the initially no earned value is given 0 value is given and when the work is completed this job is completed on 34 days. So, on 34 days the value 34 working days value 34 is assigned to this earned value that is what I am saying here you can see also in the graph. This is 30 up to 34 see the value is 0 and from 34 then the value goes how much on 34 days.

The cumulative this work days means the cumulative work days. So, this is the elapsed days the actual days passed. So, on 34 days the cumulative work day becomes 34; that means, earned value of 34 is assigned here and now as the time proceeds from these values you can. So, next is what 64 and that this values can be given when on forty nine.

So, when a 49 days passed here you can see 49 here. The value is what? We are giving to like this. So, you can see. So, when 49 then this is 64. So, that is when it is a 49 days you can see that this is something what 64 it will be given. So, in this way you can give you can assign the different values from this table you find out the different what cumulative work days and you can assign the value.

So, up to 34 no value is given 0 and on the day of 34 full value is given that is 34 is given here. So, in this way you can see that here the 0 by 100 technique is used for assigning earned values. This I have already told you that this project is not expected to be credited with any earned value until day 34, when the activity first activity that is specified overall the system is to be completed.

This activity was forecasted to consume 34 person days and hence it will be therefore, credited with 34 person days of earned value when it has been completed that I have already told you when it is completed 34. So, you see earned value of 34 is assigned here. The other steps in the baseline budget chart coincide with the scheduled completion dates of other activities. So, other steps in the baseline budget chart, they coincide with the scheduled completion dates of other activities. So, then this is running perfectly.

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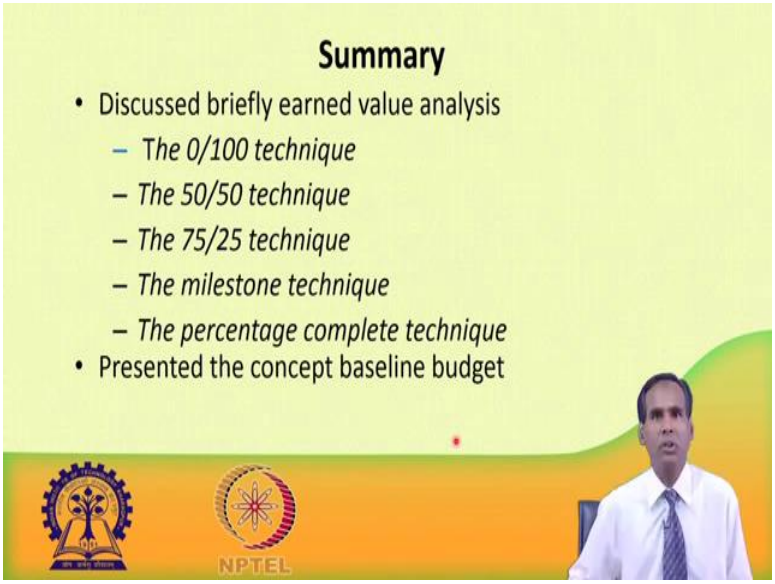

So, now today we have seen about the 0 by 100 technique where initially 0 values assigned and the full value is assigned once the job is completed, then in 50 this is most suitable for the software development on projects and 50 by 50 technique here once the

work is started the job is started 50 percent value is assigned. And the rest 50 percent is assigned after the work is job is completed and in this is suitable for like a building constructions etcetera for paying about labour charges, for purchasing raw metals etcetera.

The 75 by 20 technique it is used in this case where you want to purchase what a large equipment or so. So, they are initially you have to when the you have when the contractor supplies the equipment then you can what assign 75 percent or 75 value you may assign and rest 25 value may be assigned to the earned valued or next rest 25 percent payment may be made after successful installation and testing. And in milestone based technique the value may be asked based on achievement of some milestones of SRS is completed. So, some value in design is completed, some value assigned coding is completed, some value may assign and the testing is completed, this is another milestone some other value some value may be assigned like this.

And in the percentage complete technique here basically here you have to count here you have to measure the work progress and the example we have seen that in a you suppose you are developing a database, where you have to enter 1000s of student records see after the what end up 1 month, if the total records entered to be is 100 sorry 10,000 and you have entered only 1000; that means, a 10 percent what data entry has been made.

So, accordingly the value can be assigned. We have also presented the concept of a baseline budget. So, the concept of a baseline budget will help in the what earned value analysis. So, in the next class we will see how these PV values EV values baseline budgets they will be helpful in monitoring the progress of the project.

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So, this thing we have taken from this book, you can refer this book for further details.

Thank you very much.

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Lecture - 44
Project Monitoring and Control (Contd.)

Good morning, now let us start this how to monitor the earned value.

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Last class we have discussed about earned value analysis; now let us see how to monitor the earned value.

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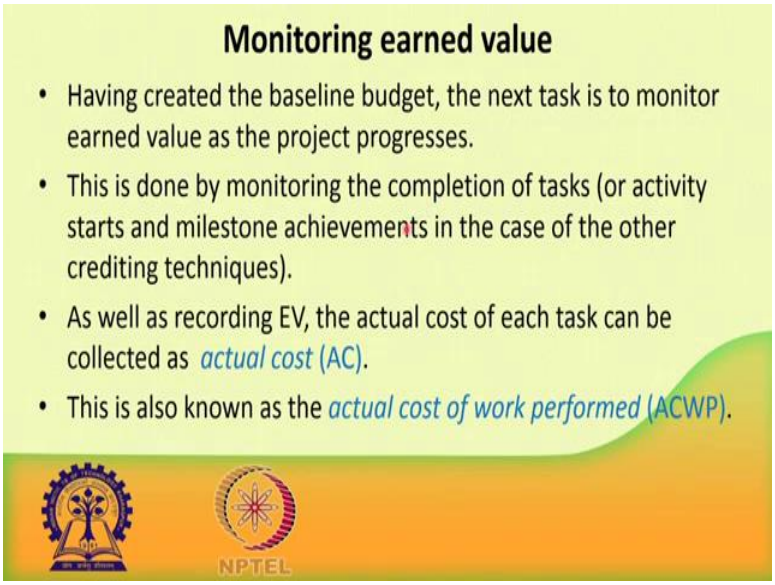

So, after creating the baseline budget; we have already discussed base line budget in the last class. So, after creating the baseline budget the next job is to monitor the earned value as the project progresses through time. So, this can be done by monitoring the completion of the tasks ok; this can be done by monitoring the completion of task or activity starts or milestone achievements in the case of the other crediting techniques.

So now, we have seen that how the earned value can be assigned, how a value can be assigned to the earned value. So, as well as recording the earned value the actual cost of each task also can be recorded; it can be collected which we call it as actual cost or AC. So, similar to a recording EV; the earned value, the actual cost of each task can also be collected this is known as actual cost or AC; this actual cost is also known as actual cost of work performed.

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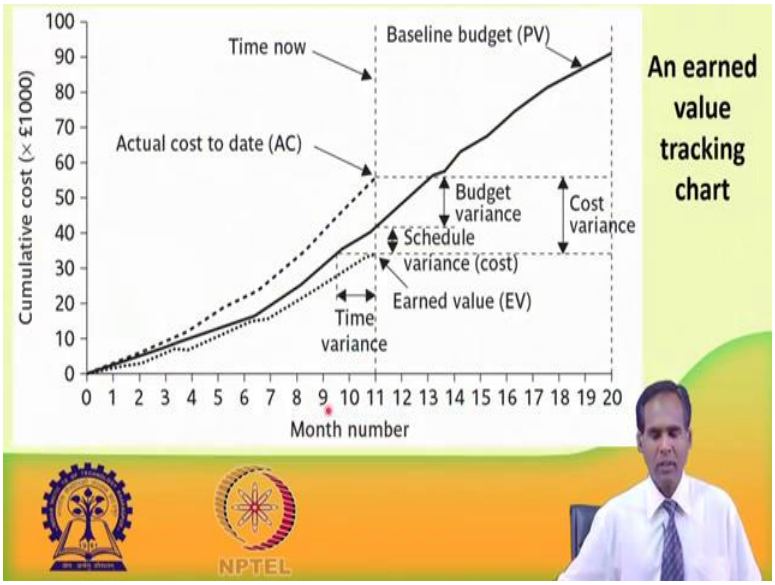

Now, we can use a chart called as earned value tracking chart; this chart can be used for monitoring earned value, we will see that three different lines are shown here; this initial this dotted line represents the earned value.

This middle line; solid line represents the baseline budget which is also known as the planned value and this upper dot line; it represents the actual cost to date up to date. So, suppose this is the time now this middle line this vertical line is the time now or today's time. So, given this we want to measure various parameters; you can see that in x axis; we have taken the month number and y axis we have taken the cumulative cost may be in terms of 1000s. So, you can see that the bottom line which represents the earned value and acts; this is the today's point.

So, and it was the earned value, but what was the planned value? You can see the corresponding planned value is here. So, this difference; that means, what is the planned value and what is the earned value difference between this; it is known as what you can say scheduled variance which may be expressed in terms of for cost.

Similarly, this difference this is the actual cost and this is; this what we can say earned value. So, the difference between the actual cost and this earned value; this can be known as cost variance. And similarly, here you can see that the work is planned to be finished at this time, but you can see that it is finishing actually; initially, it was planned to finish at near about somewhat a 9.25 months or so and actually it is finished at what 11 months;

so this difference almost 11 minus 9.25 maybe 1.75; so this difference is known as time variance.

So, basically this earned value tracking chart here, it is showing us three variances; the schedule variance; sorry four variances schedule variance, cost variance, budget variance and time variance. So, let us see how you can compute these variances so that we can monitor the progress of a project.

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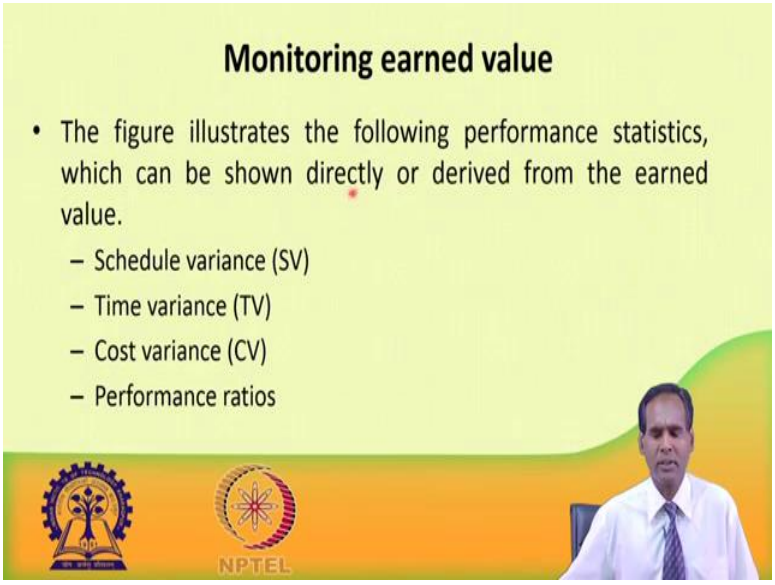

I have already told you this figure illustrates the following performance statistics which can be shown directly or they can be derived from the earned value. Those are schedule variance, time variance, cost variance and we will see some performance ratios; so those all these variances have been shown here. So, this is the; schedule variance that is difference between PV and EV and this is the cost variance; difference between actual cost and this is what you can see this is; this you can say earned value and this is the time variance.

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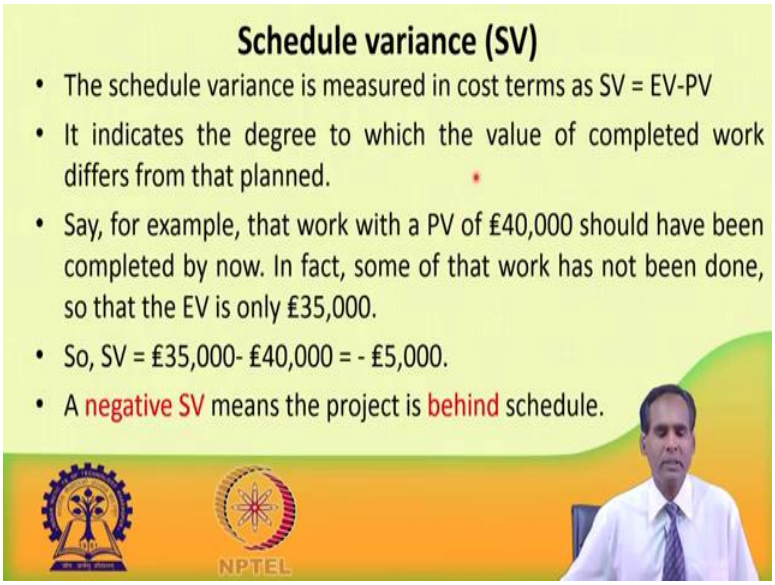

Now, let us define these terms mathematically; the schedule variance is measured in terms of cost. Mathematically we can say that schedule variance is equal to earned value minus planned value.

$$SV = EV-PV$$

What does it indicate? It indicates the degree to which the value of completed work deffers from the planned, what was the planned value and what are the earned value?

So, it will indicate the degree to which the value of the completed work; that means, the earned value; it is differing from what from the planned value. Let us take a small example; suppose a work with a PV of planned value of 40000 pound should have been completed by now; by today a work with planned value of 40000 pounds should have been completed.

In fact, what had happened? Some of some portion of the work has not been completed; so far has not been done so that its earned value as in today is only 35000. So, what does it mean? So, schedule variance we can compute like this earned value minus planned value. Earned value, we are saying 35000 minus planned value is already we have shown 40000 that is minus 5000 pounds, so; that means, it is negative. So, what a negative S V represents? A negative schedule variance means the project is behind the schedule, here you can see actually why it is not 40000? Because some portion of the work has not been done and what does it mean? That means the project is lagging behind the schedule.

So, a negative S V means the project is lagging behind the schedule. So, this I have already shown in the graph that schedule variance usually EV minus P V. So, this is schedule variance; so this is the; what EV, this is the P V. So EV minus PV, it gives what this schedule variance and you can see that a negative; that means, it is what it is lagging behind is this is less. So, this is lagging behind the planned schedule; this is lagging behind the schedule.

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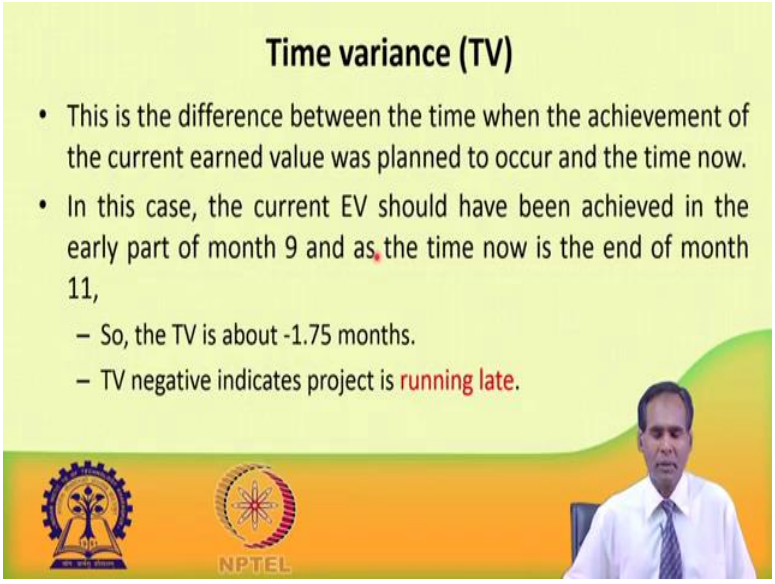

Now, let us see about time variance. So, how time variance you can calculate? So time variance is defined as the difference between the time when the achievement of the current earned value was planned to occur and what is the time current now?

So it is defined as the time when the achievement of the current value was planned to occur, it was originally planned to occur and the time now. So, what you can see here? When to occur and what is the time. So, you can see that this is the current time and if you will see; it was planned to be finished at this time, this is roughly 9.25. So, in 9.25 months it was planned to complete but it is as on today; it is 11th month; it is almost completed on 11th month. So, time variance will be how much? 9.25 minus 11. So, coming to be almost 1.75 minus 1.75; so that is what we do show here.

So, in this case as I have already told you; current EV should have been achieved in the early part of month 9; early part of month 9 means around 9.25, but it is completed on 11 month 11. So, time variance is 9.25 minus 11 coming to be minus 1.75 months; so again

here it is negative. So, negative time variance indicates what? It indicates the project is running late; the project is running late. You can see here that the project it should be completed what by 9.25 months, but so it is completely 11; that means, the project is running late the project is running slowly; it is running late.

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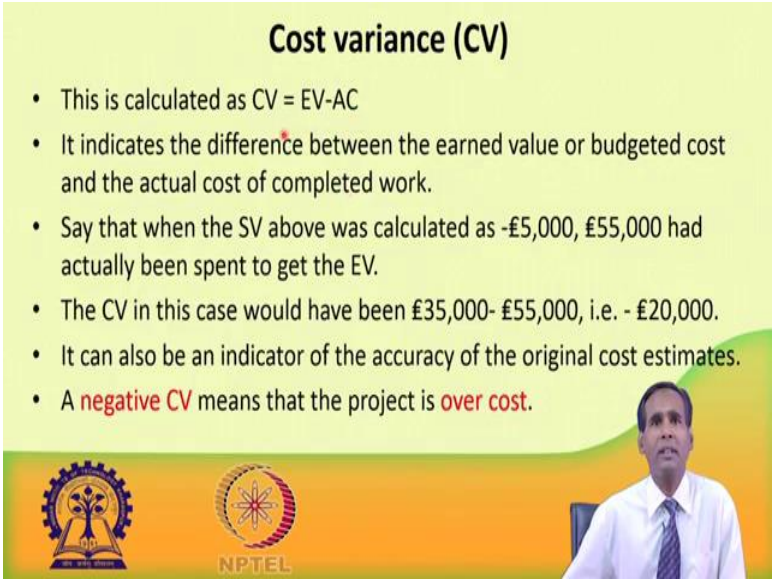

Next variance is cost variance as its name suggests cost variance; it can be calculated as what? We denote it as CV and is calculated EV minus AC from the graph you can see cost variance.

$$CV = EV-AC$$

What is the actual value on today? This point AC and what is the earned value? Earned value is which one? This one, this is the earned value. So, EV minus AC it will give you the cost variance that we have shown here. This is calculated as EV minus AC; what does it indicate? It indicates the difference between the earned value or the budgeted cost and the actual cost of a completed work ok.

So, cost variance is calculated; is defined as the difference between the earned value or the budgeted cost for the completed work and the actual cost of the completed work. So, now so that means, CV is equal to earned value for the completed work minus actual cost of the completed work. Now let us take a small example the previous example let us take; so say that when the S V above was calculated minus 5000 for the previous

program for the previous problem; we have already seen S V is coming to be minus 5000 pound. So, say that the S V above was calculated minus 5000 and actually 55000 pound had been spent to get this EV. So, to get this earned value actually 55000 pound have had been spent.

So, now what will this CV then accordingly? So, CV in this case would be how much? 35000, which is the EV value minus 55000; what is the actual cost? So, this is coming to be how much? So, 35000 minus 55000 coming to be minus 20000 pound; so you can see that again it is negative. So, a negative means what? A negative CV means that the project is over cost, the cost that is budget you have planned; so, it is the it is costing more than that; so it is over cost.

So, a negative CV means the project is over cost had it been positive; that means, the project is below cost it is under control. So, here as we have seen that here negative cost variance means it represents that the project is over cost. It can also be an indicator of the accuracy of the original cost estimate. How accurate was your original cost estimate that can also be indicated by the CV. So, CV also indicates the accuracy of the original cost estimates.

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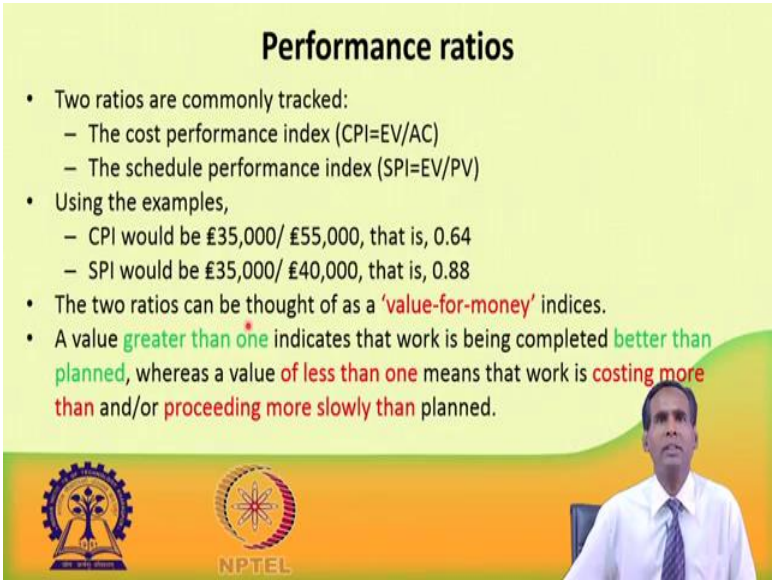

Now, with this three terms cost variance, schedule variance and time variance; we can measure, we can compute some performance ratios. So, we will see two important performance ratios; here two ratios are commonly what tracked here. One is this cost

performance index known as the CPI, this is defined by as EV divided by AC earned value divided by actual cost.

$$CPI=EV/AC$$

Similarly, another parameter is the scheduled performance index; in short we write as SPI which is calculated as earned value divided by the planned value.

$$SPI=EV/PV$$

Now, let us take again the previous example and let us calculate CPI. So, CPI would be how much as these definition? CPI is equal to EV by AC. So, what is the value of EV and AC you can see? In the earlier example, we have seen that EV is equal to here you can see EV is equal to 35000 and AC actual cost is how much? 55000.

So, this is equal to how much? 35000 by 55000 that is a ratio is coming to 0.64. Similarly, what is the SPI? SPI is equal to EV by PV earned value by planned value. So, what is the earned value? Again it is 35000, but what is the planned value? We have already seen earlier planned value is 40000; planned value is 40000; so 35000 by 40000 coming to be 0.88. So, now let us see what is the interpretation, what is the inference? The two ratios, they can be thought of as some value of money indices; they can be treated as that as if this represent the indices of value for money ok. The two ratios can be what considered as can be thought of as value of money indices.

Now, let us see if the value there are two possibilities the values may be less than 1, the values may be greater than 1. A value greater than one will indicate that the work is being completed better than the planned ok. If the values are greater than one, then this indicates that the work is being completed better than the planned whereas, a value of less than one will indicate that the work is costing more than it is planned or you can say earned or the work is proceeding more slowly than it is planned.

So, it is desirable that it should be the value should be greater than one; if it is; that means, it we are moving better. But if it less than one, then two possibilities will be there; the cost is over shooting the project is over cost than what planned and the project is moving very slowly than planned ok. So, in this way these ratios will help in monitoring the progress of the project.

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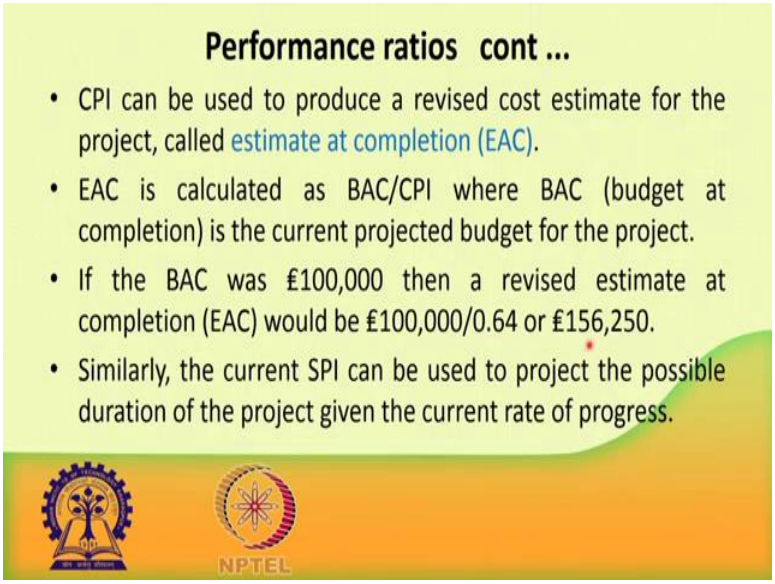

So, the CPI value that we have already seen; here CPI is equal to how much EV by AC or let us see what it is used. CPI can be used to produce a revised cost estimate for the project called estimate at completion ok. The CPI value it can be used to generate a revised cost estimate and updated cost estimate for the project we call this as estimate at completion or a EAC. And EAPs, EAC can be calculated mathematically can be represented mathematically as BAC by CPI.

EAC = BAC/CPI

CPI we have already known earlier and BAC represents budget at completion; that means, it will say that what is the budget at completion? That is BAC.

So, EAC is equal to BAC by CPI; so here. So, BAC is the what current projected budget for the project. So, as you have seen that EAC is equal to that BAC by CPI, where BAC which transfer budget at completion; it is the current projected budget for the project if BAC was 100000. Let us again take the previous example or we can take also this as a may be a fresh one; if the BAC was 100000 pound; then a revised estimate we can prepare by computing EAC; where EAC will be equal to BAC by CPI. So, what is the BAC value? BAC is estimated to be 100000 divided by CPI we have already computed CPI in our earlier case; it is 0.64.

So, 100000 by 0.64 is coming to be this; that means, say initially it was what initially it was the current project budget was almost BAC was 100000, but then as we progress projected as the progress; as the project is progressing; we can revise it we can estimate it observing the current progress and as CPI is equal to 0.64, then you can see that now it is exceeding the 100000 value.

Now, it is 156250; that means, we require more cost, we require more money to complete the job. So, similarly the current value of SPI you can calculate by using the formula you know what EV by PV. Similarly, the current SPI can also be used to project the possible duration of the project given the current rate of progress. You can also what use SPI to what; you can also use the current value of SPI to the or for the project the possible duration of the project given the current rate of progress. So, what is the; what possible duration that it will take to complete the revised duration that it will take to complete the project that also can be computed if you will use this what current SPI value.

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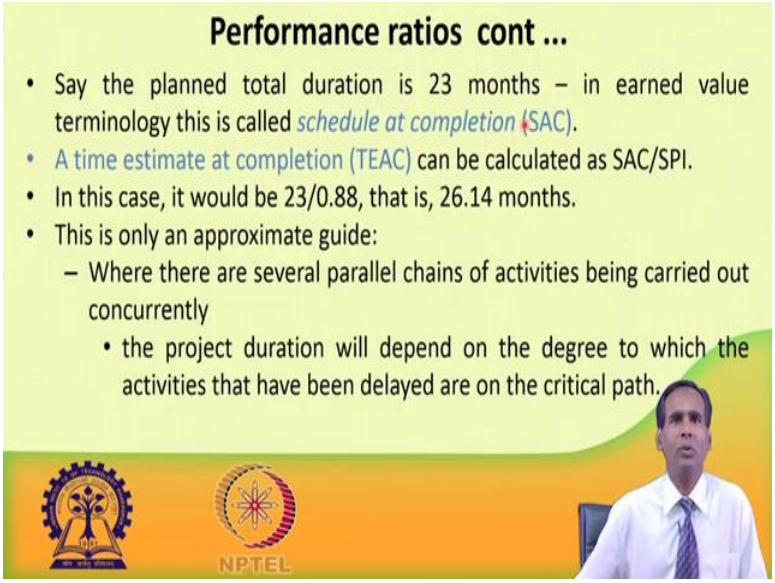

Now, let us see take another performance parameter. Suppose the planned total duration is 23 months; initially it has planned that it will take 23 months to complete the job. In earned value terminology this is called as scheduled at completion ok; say that the planned total duration is 23 months. In earned value terminology; it is called as schedule

at completion using this term scheduled at completion, another parameter can be calculated which known as a time estimate at completion.

What is the estimate for this completion; a time estimate completion a time estimate at completion which is in short is written as TEAC; it can be calculated as SAC by SPI.

TEAC= SAC/SPI

What is the SAC? We have already seen schedule at completion, SPI is that schedule performance index. I think SPI you have already seen here SPI is the schedule performance index.

So, this will be equal to how much? This would be equal to see SAC is given is 23 months and SPI you have seen in the previous slide SPI is coming to be 0.6 0.88; SPI 0.88. So, then this ratio is coming to be 26.14 pounds; that means, from the current progress; we can measure that the work cannot be completed in 23 months; it will require another 3.14 months extra; that means, the revised duration for completing the project would be 26.14 months.

So, this is only an approximate guide; where there are several parallel chains of activities being carried out concurrently. Then; this can be this is not a hand hard and fast rule; this is only an approximate guide, but where there are several parallel chains of activities are being carried out concurrently, then the projected duration will depend on to the degree; to which the activities that have been delayed to which the degree to which the activities have been that have been delayed are on the critical path. How many of them are in the critical the activities we are delayed how many of them are on the critical path; so that factor will also decide that what will be the revised duration.

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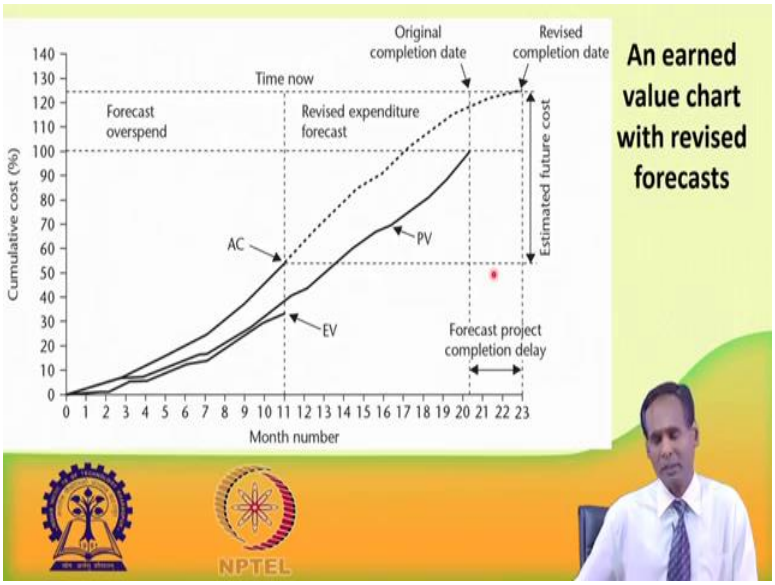

So, once as you have computed the SPI etcetera then you can revise the forecast. As I have already told you that the CPI can be used to revise the cost estimate. Similarly, SPI can be used to divide the project duration; so once CPI and SPI are done; they have found out; you can also what revise the whole estimate see. After computing the CPI and SPI the earned value chart; it can be revised the other parameters can be revised, you can prepare a revised earned value chart.

So, this is the example of an earned value chart with revised forecast. You can see here that this is; this vertical line is the today's time or the today's date. So, this was the earned value and this is the actual value; you can see that. So, today this is the actual value, but based on this planned value and based on the current progress, you can estimate that you can again forecast that the actual cost will be now increasing and may come to this.

So, similarly the time it was previously estimated according to planned value the work will be completed here. But as the progress is there as you are observing the progress, now the project will be completed almost at this time, so this gap we call as the forecasted project completion delay. Similarly, this was the original plan regarding the cost, but actually it is estimated revised estimate for the expenditure is this. So, this we call them as the revised expenditure forecast. So, this was the original completion date you can see and this is the revised completion date; so this gap we call that the forecasted project completion delay.

And similarly, as we have already seen that this was the actual cost; this was the estimated actual cost; this is the estimated future cost what actually you can say that this is the what? This value is the we can say sorry this value as the actual cost; you can see this the actual cost and now it was the revised cost; this value is the revised cost. So, this gap we call as the estimated future cost this is the; revised expenditure forecast. So, in this way if you know the CPI and SPI; you can prepare a revised earned value chart. From the revised earned value chart, you can estimate the forecasted project completion delay; as well as you can know the estimated future cost.

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This earned value analysis has not again up to this we have seen; this also we have completed ok. The earned value analysis has not yet gained much what acceptance universal acceptance ok. Why? Let us see what is the reasons. Earned value analysis has not yet get universal acceptance for use with software development projects. It might have worked perfectly for or it; this earned value analysis might have been worked for other projects. But earned analysis has not yet gained universal acceptance for use with the software development projects.

What is the reason; let us see that perhaps largely because of the attitude that whereas, a half built house has value reflected by the labour and material that have been used. But a half completed software project has virtually of no value at all; it is very much true that a building you have given for contract.

So, a half build building has some value is it not it? In terms of what the labour and material that have been used, but a software unless it is fully completed fully installed and tested nobody will use it does; it does not have any value at all. So, that is why this earned value analysis has not gained what universal acceptance to be used in software development projects ok.

So, this is the so; this is to misunderstand you purpose of earned value analysis which is a method for tracking what has been achieved on a project ok. So, that is why this is to misunderstand that the purpose of the earned value analysis which is normally carried out or which is a method for trafficking what has been achieved for a on a project; which is measured in terms of the budgeted cost of completed tasks or products.

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Now, let us quickly take a small example; so, suppose there are three tasks to be performed in a project specifying a model which what it is planned to take 5 days coding module; coding the module it is planned to take 8 days and testing the module is planned to 6 days. So, now let us say we will observe the progress on the beginning of the 20 dead date; 20 date or 20th day. So, you can say planned value will be how much? 5 plus 8 plus 6; so 19 days. If everything, but testing is not completed if everything is completed, but the testing is left if everything, but testing is completed then EV becomes how much? So, 5 plus 8 there is 13 days.

So, now we will compute schedule performance we know the formula for schedule performance schedule variance is equal to EV minus PV; so, 13 minus 19 which is negative; so minus 6 and schedule performance indication SPI is equal to EV by P V. So, EV is equal to 13 by 19 is equal to 0.68.

So, here you can see that SPI is negative as well as a scheduled performance indicator is less than 1. So, both the things suggested that actually we require schedule value schedule variance positive and schedule performance indicator should be greater than 1. So, since SPI is negative, as well as SPI is less than 1, this indicate that the project is lagging behind the schedule.

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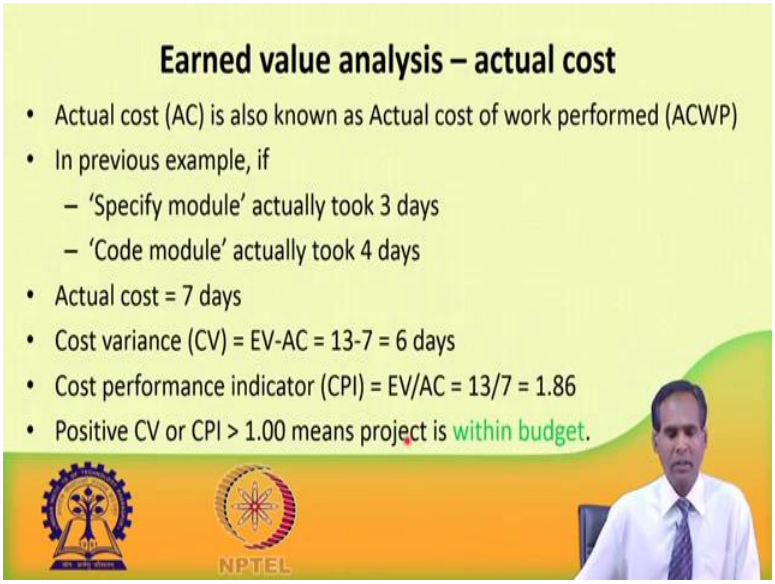

Now, let us use actual cost in this example; we know that actual cost is also known as actual cost of work performed or ACWP. So, let us again consider this previous example; suppose in this previous example specify module actually took 3 days; it is planned for 5 days, but actually it has taken 3 days. Similarly coding module actually took 4 days, but it was planned for 8 days. So, now the actual cost will be how much? 7 days, I have already told you these cost means also you can use the work dates; so actual cost will be 7 days. So, now you can perform can compute cost variance CV which is equal to EV minus AC.

So, a EV we have already what calculated earlier that is 13 days and minus AC; actual cost is 3 plus 4, 7; so it is 6 days. Again now it is positive; a good symbol and cost

performance indicator or CPI is equal to EV by AC; 13 by 7 is equal to 1.86. So, you see both the cases cost variance is positive as well as cost performance indicator is greater than 1. So, this indicates that the project is within budget; within budget ok. So, positive CV or CPI is greater than 1 means the project is within a budget; it will not exceed the planned budget; so this is under control.

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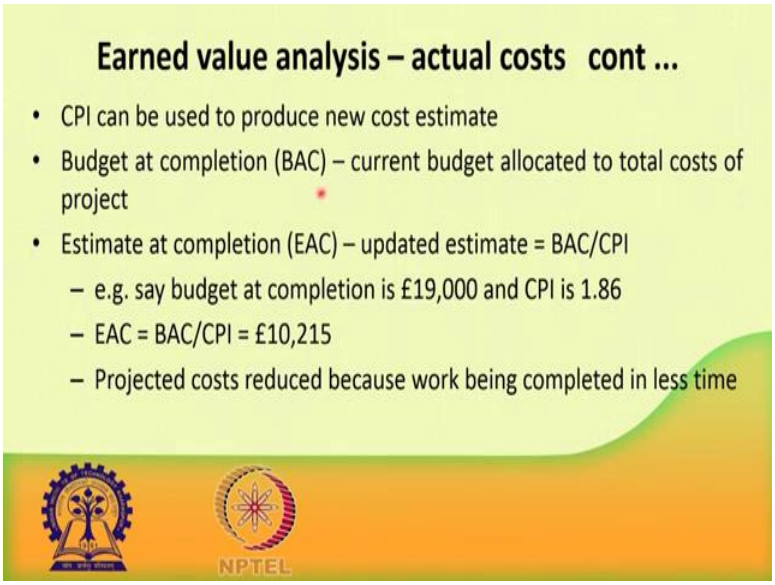

So, CPI can be used to produce new cost estimates that we have already shown in the earlier slide; CPI can be used to produce new cost estimates. So, budget at completion now let us see what is the budget at completion. So, budget at competition I have already told you this represents the current budget allocation to total cost of a project. So, and estimate at completion this also we have defined earlier this is the updated estimate which is equal to BAC by CPI. So, for example, say the budget at completion is given you a as 19000 and CPI is given as suppose 1.86, then EAC can be computed as BAC by CPI is equal to 10215.

Now you can see that it is the budget it is reduced, the project cost is reduced project projected cost is reduced because work being completed in less time; is not it? Because you can see that budget at completion estimated 19000, but now it is reduced to 10215. The projected cost is reduced because the work being completed in less time, the work it is completed in less time that is why this has happened. is not it?

And we have already seen earlier that the work is what it was planned it is what taken let us see cost performance indicator is 1.86 and why? Because this you has planned to be over completed in 8 5 plus 8; 13 days, but actually it is taking 4 plus 3 7 days; so that is why it is what it is taking less days. And since your how we are computing that CPI is equal to 1.86 previous value we have used here this 1.86 you have used. And hence and the budget at completion even if it is estimated 9000 and we are finishing the work much early that is why the cost is reduced. So, the projected cost is reduced because the work being completed in less time.

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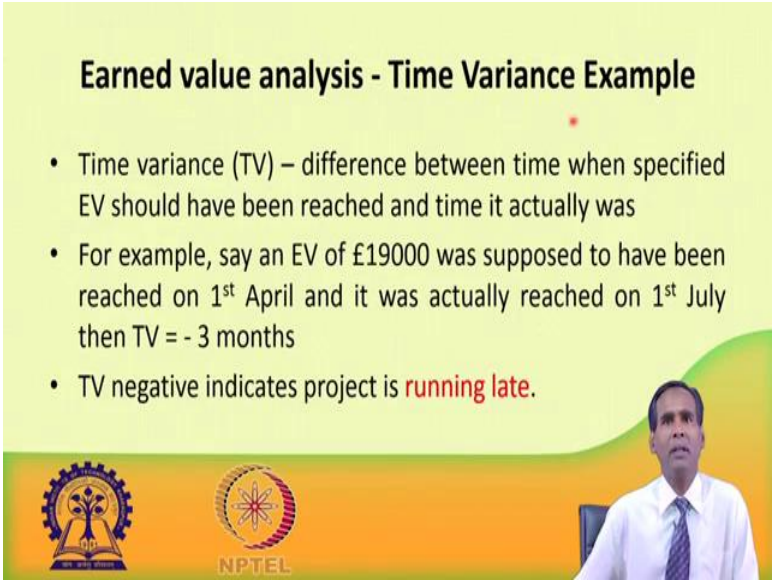

So, let us again explain this time value; time variance as an example. So, and I have already told you time variance means the difference between the time when specified EV should have been reached and the time it actually was what is the current time. For example, say on a EV of 19000 was suppose this EV. So, we have expected that this 19000 pound; for example, say an EV of pound 19000 was supposed to have been reached on 1st April; it should have this job should have been done by 1st April. And it was actually found that it was reached it was completed on first July, then TV is equal to how much? 1st April minus 1st July you can say.

So, fourth month this is seventh month that is it will be minus 3 months. So, negative symbol indicates that TV negative indicates the project is running late ok. So, this is the

conclusion this is the inference that your TV is coming negative and the TV is negative means; it indicates that the project is running late.

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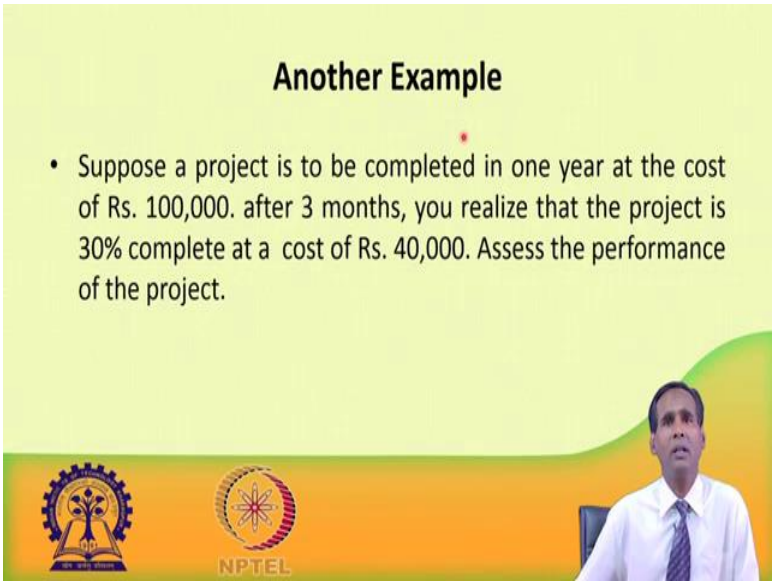

Let us quickly take another 2 minutes this another small example. Suppose a project is to be completed in 1 year it has to be completed in 1 year at the cost of rupees almost 100000. After 3 months, you realize that the project is only 30 percent complete at a cost of 40000, then assess the performance of the project how can assess let us see.

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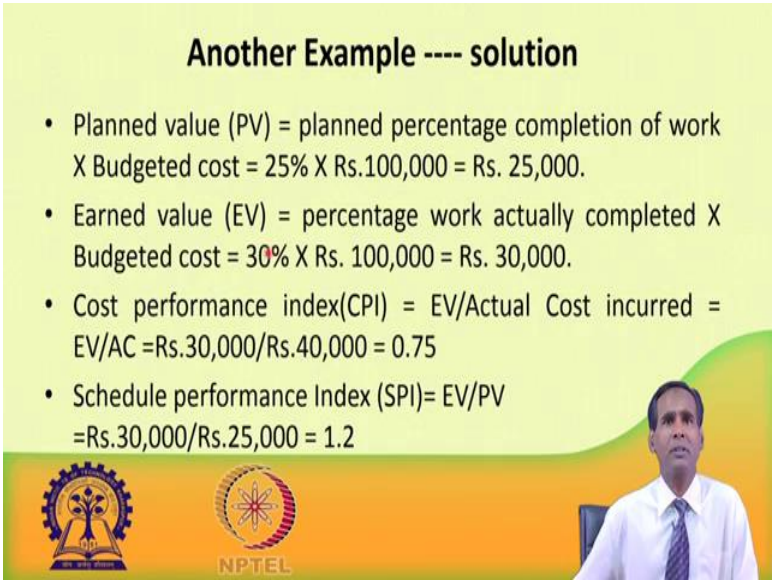

We have to find out the planned value, the budgeted cost the sorry the earned value budgeted cost, CPI, SPI etcetera. You can know the planned value what is the planned value? See planned value it is given here, how much? 100000, but we are evaluating when? After 3 months.

So, a year is 12 months, after 3 months means 25 percentage. So, planned value means the planned percentage completion of work into budgeted cost. Budgeted cost is 100000, the planned percentage of completion is after 3 months means it is 25 percentage to 25 percent in 100000 coming to be 25000. Earned value I have already told you earned value also can be represented as a percentage of EV.

So, within the problem it is that how much? Only 30 percentage completed here. So, here earned value EV is equal to percentage work actually completed into budgeted cost. So, what percentage of work was completed? Is 30 percent into what is the budgeted cost? 100000; this is coming to be 30000. Now, you can easily compute CPI and a SPI. So, CPI is equal to what? EV minus actual cost EV divided by actual cost; EV is equal to 30000 and actual cost is you have already given in the problem actual cost is 40000.

So, this 30000 by 40000 is coming to be 0.75 and schedule performs index SPI is equal to at EV by PV; EV we have already seen 30000, PV we have sorry EV by PV. So, EV is computed to be 30000 PV is computed to be 25000; so 30000 by 25000, coming to be 1.2. So, here you can see that CPI is less than 1 and SPI is greater than 1.

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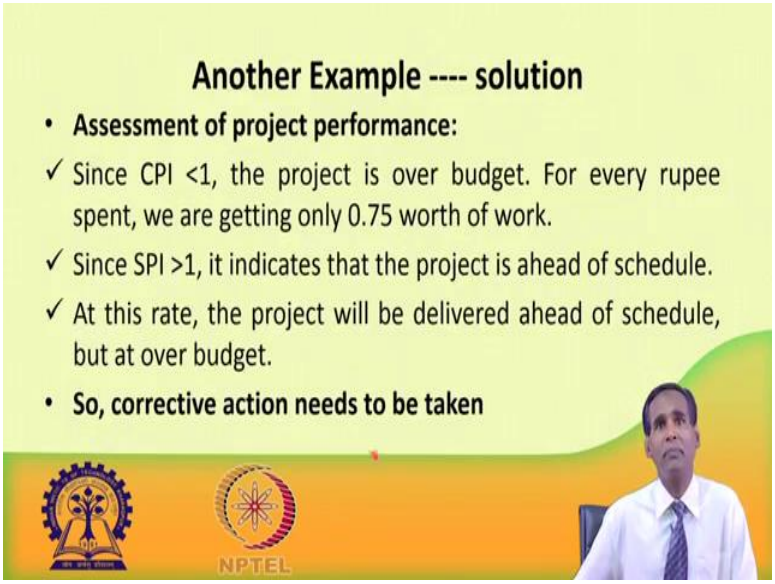

So, what is the conclusion we can draw? So, the assessment of project performance is as follows since CPI is less than 1, it is infer that the project is over project because CP is CPI less than 1, it should be greater than 1. So, CPI since it is less than 1 it is estimated that the project or; it can be infer that the project is over budget.

Here you can see that for every rupee spent we are getting only 0.75 worth of work; see the CPI 0.75; its meaning that for every rupee spent, we are getting only 0.75 worth of work. And you have already see SPI is greater than 1; it is 1.2; since SPI is greater than 1, it indicates that the project is ahead of the schedule; we are not lagging behind the project is ahead of the schedule ahead of the schedule.

So, if the project is moves the; if the project moves at this rate. So, if the project moves at this rate or the progress is made at this rate, then the project will be delivered ahead of the schedule much before of the scheduled time the project will be delivered. The project will be delivered ahead of the schedule, but at over budget but at over cost the cost will be more. So, since the cost will be more we have to do something.

So, here are corrective action needs to be taken since the money you know; it will take more cost even if the schedule it will be delivered early. So, even if the project will be delivered much early than the what planned schedule but the project is what; it is exceeding the cost it is at over budget. So, necessary corrective action needs to be taken in order to what reduce the budget or in order to reduce the cost.

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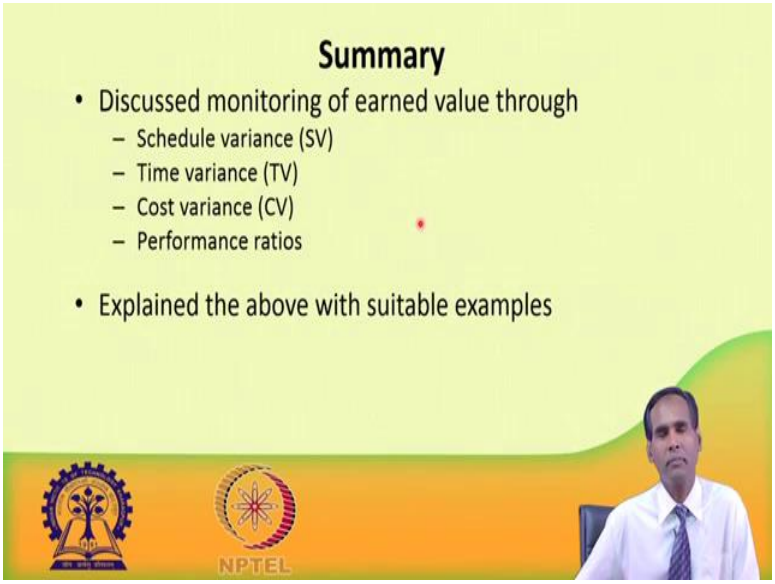

So, these things already we have seen. We have already seen these things we have discussed this monitoring of earned value through different techniques, different parameters such as schedule variance time variance cost variance and performance ratios. We have explained the all the how to calculate the above parameters; now we have explain the above with suitable examples ok.

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We have taken from these references.

Thank you very much.

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Lecture - 45
Project Monitoring and Control (Contd.)

Good afternoon, last class we have discussed about cost monitoring and monitoring earned value. Now, let us see how we can prioritize the, we can prioritize the monitoring activities.

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So, prioritizing the monitoring first will see; then will see a if a project is getting late how to get the project back to the planned tracked; to the original target.

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So, far we have assumed that all the aspects of a project; it will receive equal treatment in terms of degree of monitoring applied. That means, during monitoring we are treating all the activities the similar footing. We have assumed that all the what activities during monitoring they will receive equal treatment, but actually in practice that does not happen some activities they require a more priority, some activities they require less priority.

Now, let us see how we can assign priorities to the different activities. For example, which are critical path activities they should given maximum priority highest priority, but which are activating what which having less what float less then we can give priority to the activities with no free float. Then activities is less than a specific fold float and a we can also give some priority to high risk activities and we can also consider activities which are using critical resources.

So, in that way we will assign some priorities depending or to the different activities depending upon certain parameters.

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We will first see about critical path activities. We know that if there is any delay in an activity lying on the critical path then that will; obviously, cause a delay in the completion of the whole project. So, critical path activities they have to be given very high priority during for close monitoring; otherwise the project the end a completion date of the project will be delayed and we cannot meet the target line.

So, critical path activities; that means, activities lying on the critical path must be given very high priority.

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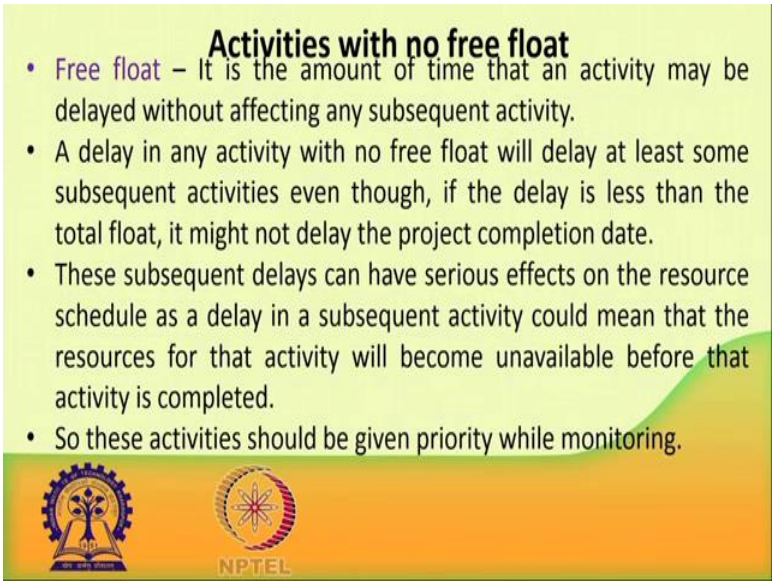

Similarly, activities which are having no float; activities with no free float they also should be given some priority during monitoring; we have already discussed what is free float in some of the earlier classes.

Free float it represents the amount of time that an activity may be delayed without affecting any subsequent activity. So, we can there is some free time available with some activities such that if a some urgency occurs; that activity can be a little bit delayed. So, that that free time can be consumed.

So, we define free float as it is a slack time or a free time. So, it is amount of that free time that an activity it can be delayed, so that the other subsequent activities will not be affected; they are target dates their completion dates will not be affected. Now, you see a delay in an activity which has no free float; it will delay at least some subsequent activity later on.

If the delay is less than the total float value, then what will happen? It might not delay; if it is less than the total float, then it will it might not delay the project completion date. But if delay is more than the total float then it will affect the other activity, it will delay the project completion date of the it will delay the project completion date for the whole project.

Now these subsequent delays can have serious effects; so in such a way if one activity is affecting another activity another the next activity. If we are changing this what their execution dates; the subsequent activities they are delayed, then in turn the end date of the whole project it will be affected and it will be delayed.

So, that is why this subsequent delays it can have serious effects on the resource schedule also because it delay in subsequent activity it means that the resource for that activity; it is now become now unavailable for that activity completed. And other activities they will be now waiting for the resources held up by that activity which do not have any free float, but that is being delayed. So, these activities should also be given priority during the monitoring process.

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Next is activities with less than a specified float, say you are saying that a float is one week. And if the activity is delayed by more than 1 week; what will happen? Obviously, the subsequent activities that deadlines will be missed. So, that is why we should give also some priority to the activities with the less than a specified float maybe 1 week or what 2 weeks or like that. So, if any activity has very little float maybe 1 week or less than that; then it might use up this float before the regular activity monitoring brings the problem to the project managers attention.

And hence; what will happen? This activity which is have been less than a specified float; if the whole free time is consumed due to delay then what will happen? It will become an actor what critical activity and this will create problem. So, that is why it is a common practice to monitor closely those activities which are less than or which have less than a specified float maybe a 1 week of a free float or like that.

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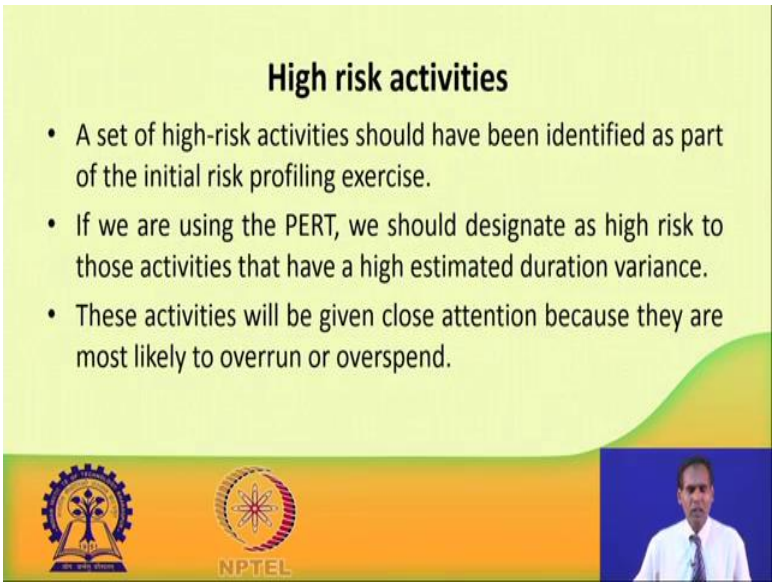

Similarly, high risk activities also given some priority during monitoring; so, you know in several cases that a set of high risk activities should have been identified. So, the activities user of cost which are of risky if any what dead line is missed or so then a severe consequence will occur. So, these; so, first we have to identify the set of high risk activities and as part of the initial risk profiling exercise, I hope you have already read or will read on this risk handling methodologies.

So, the so first we have to identify a set of high risk activities and then and you see you have already known about PERT. So, if you are using PERT; then what will happen? You can easily identify which are of high risk activities. We should designate high risk to those activities that have a high estimated duration variance.

So, the activities which are having high; what variance they will be treated as what most risky and you should give some priority to what while monitoring them. So, these activities will be given very close attention; they should be very closely monitored because they are most likely to overrun the schedule or they might what most likely to overspend the budget might be more. So, you should give a sufficient attention; you should give more priority in handling these activities said while the monitoring these activities.

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Then activities using critical resources; see there are some resources which are very critical and those critical resources are very less. So, the activities which are using these critical resources, they have to be also monitored very carefully. So, activities can be also critical because they are very expensive just you can consider the example of case of specialized contract programmers. And those contract programmers are very much expensive and they are very much expensive.

So, the activities those are using those specialized contract programmers; they are very much critical. These contract programmers are hired for only few days; if within those few days say 1 month, you cannot complete the job then they cannot stay beyond that 1 month or if they will stay then again they will charge a very high amount.

So, that is why activities they can be also critical because they are expensive. So, staff or other resources might be available only for a limited period. As I have already told you the a specialized contract programmers they may be available for an activity maybe for 1 month. If you do not finish your work within 1 month; then they will not stay or even if they will stay they will what charge very high amount.

So, that is why; so you have to give what special priority to them while you are monitoring you should closely monitor their progress. So, staff or other resources might be available only for a limited period; especially if they are very a controlled outside if they are controlled outside the project team.

Now in any event an activity that demands a critical resource requires a high level of monitoring. So, if a project if there are some activities that those demand a critical resource; they should be monitored with a high priority.

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Now, we will see the another aspect for this project monitoring that is getting the project back to the target. Suppose you have made a plan, but the project is not moving according to schedule it is lagging behind. So, how what steps we can take to bring the project back to the original target? So, almost any project it will be subject to delays and unexpected events that is not under our control. So, one of the tasks to the project manager; so here we will see that one of the task of the project manager is to recognize when this is happening; that means, you when this delay is occurring, when these unexpected events are occurring ok.

So, that is one of the important task of the project manager to identify when these unexpected events or these delays they are happening. So, and with the minimum delay and disruption to the project team he should attempt to mitigate the effects of the problem. So, he should try somehow to mitigate to handle these delays on unexpected events. In most cases, the project manager or at least initially what we can say; he should try to ensure that this scheduled project end date remains unaffected; whatever happened maybe in between he should tried that the project end date, it should not be affected; it should remain non effected. There should not be any change in this.

So, this he should take utmost care to the see that the project end date it remains unchanged. So, this can be done by several ways; we will see some of the ways how we can whatever problem might happen in between how we can try to see that the project end date still it is met it is unaffected.

This can be done by shortening the remaining activities say what activities so far we have already executed and what are left. So, time is less; so what we can do? We can shorten some of the remaining activities, we can reduce their execution time or we can shorten the overall duration of the remaining project. So, what is the total project say 1 year and a by 6 months if the 50 percent of the activities are not over; we have to reduce the overall duration of the remaining project. So, these are different ways by following which we can get the project back to the original schedule or the original target.

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So, now but this is not always the most appropriate response to disrupt your plan. Always it is not actually suitable or appropriate to disruptive to a plan. So, there is a little point also in spending considerable sums in overtime payments. It is also not what its very little required that always you will a higher manpower or you will employ your existing employee in overtime basis to speed up a project this also may not be what viable every time. And particularly if the customer is not overly concerned with the delivery, if he accepts that delivery date can be little bit late or there is no other valuable

work for the team members in other what activities; then why you will employ them in overtime and why you will pay extra money? Not required.

So, there are, but let us see how we can get back or how the project can get back to the original target; so two main strategies are there. So, one is shortening the critical path, two, altering the activity precedence requirement; let us see first the first one how to shorten the critical path length so that we can what try to get back to the original target date.

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The overall duration of a project is determined how? It is determined by the current critical path. So, speeding; so if you will try to just speeding up the noncritical path activities; that means, activities lying on other noncritical paths then it will not solve the purpose; it will not bring forward a project completion date.

There are several ways in which this problem can be solved; number 1 by adding resource, you can add more resources especially more staff members of course; it will lead to more cost. Then increase use of concurrent resources; if some things some activities can run parallelly give concurrent or assigned concurrent resources; maybe concurrent or what manpower or concurrent computers or concurrent output devices or concurrent input devices so that some activities can run parallelly.

Reallocate staff to critical activities. So, bring the take away some of the staffs from noncritical activities and reallocated them to the activities lying on the critical path and then reduce the scope otherwise you can reduce what is the original scope. Now, we see it is not possible to achieve all the scopes. So, we can reduce this scope and the last alternative is reduce the quality. If it is not possible to what achieve the desired quality with the limited amount of time; you will be forced to reduce quality so that the project can be completed on the specified date on the original date.

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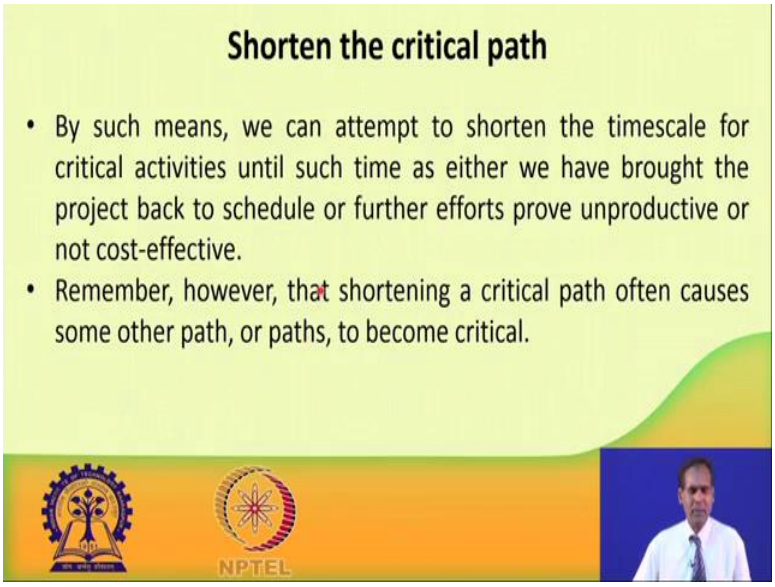

By such means so now, let us see how to shorten the critical path. So, by this means we can attempt to certain the timescale, to reduce the timescale for the critical activities until such time as we have other either we have brought the project back to schedule or further efforts to be unproductive or non cost productive effective. So, these ways we can follow so that the critical what path time or the time required for the executing the critical activities that it will be now reduced.

So, these things should be continued till either we have brought the project back to the original schedule original target or if we will prove that or if we can justify that; if we will further do this it will be unproductive or it is not cost effective, then we must stop this activity here. So, remember that a shortening a critical path upon causes some other path or paths to become critical.

So, when you are shortening a critical path; then it can happen that some other path may become critical, ok, or either only one path or more than one path also become critical during the process of shortening a critical path. So, you have to handle this case accordingly.

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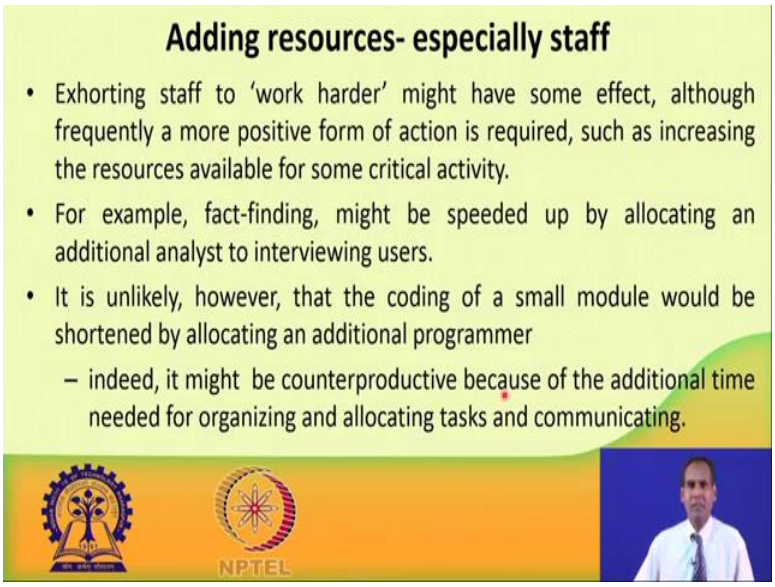

Now, let us see the second option that is this adding resources; especially the staff members. So, exhorting staff to work harder might have some effect; although frequently a more positive form of action is required such as increasing the resources available for some critical activity.

So, whenever you have exhorting the staff to work harder; it might have some effect definitely and more positive form of action is required; such as you have to increase the resources available for some critical activity. So, what other resources are required you have to increase those resources.

For example, suppose you are doing the requirement analysis; you are conducting the fact finding a process, collecting the requirements; then this process can be or this activity can be speeded up by what allocating an additional analyst to interview users. You know the different fact finding techniques like interview and a questionary, on site observation, record review etcetera. So, if you are using interview method for collecting the facts then what you can do? In order to speed up the process, you can assign, you can

allocate another analyst staff to help in the interview process; so, we can meet the deadline.

So, it is unlikely; however, that the coding of a small module would be shortened by allocating an additional programmers. See, one programmer is doing the coding and if you will see that say that I will speed up the coding process; I will allot another program or two programmers will do, that will not help in speeding up the activity. And it might be rather counterproductive because the additional time that is needed for organizing and allocating tasks and communicating that will be more, so you should not do this kind of thing.

So, you have to judiciously think which activities can be what speed up can be speeded up by allocating additional staffs that you have to think judiciously.

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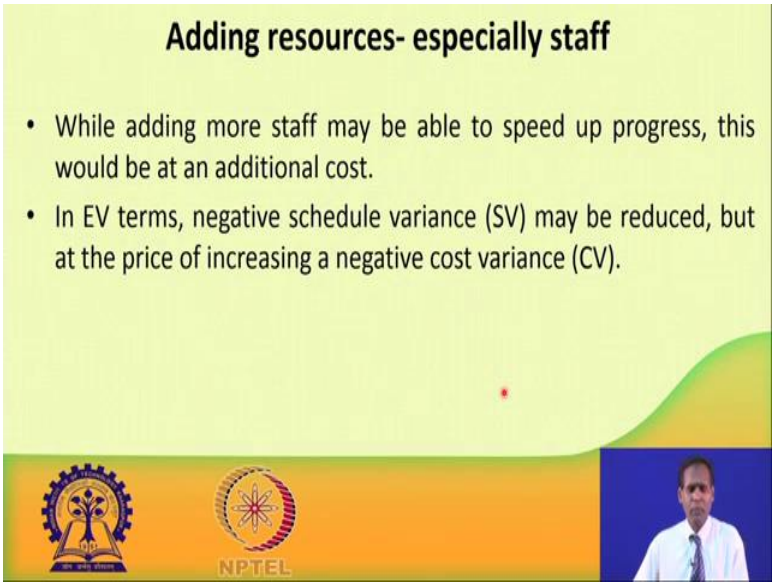

So, while adding more staff maybe able to speed up the progress; this should be at an additional cost this I have already told you. You can take more staff members, but; obviously, it will take more cost. We have already last class discussed the terms like EV that earned value, PV planned value, SV schedule variance, CV cost variance etcetera. So, if we will express this thing in EV terms then a negative schedule variance may be reduced. If it is negative schedule variance can be reduced, but at the price of increasing negative cost variance.

So, if we will add a staff members then mathematically we can say that this negative schedule variance it may be reduced, but at the cost of increasing a negative cost variance CV.

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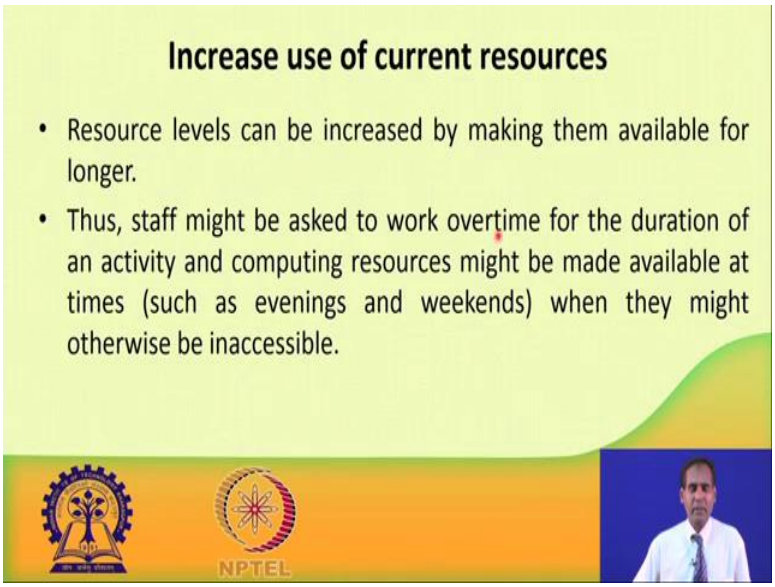

Now, increase a use of concurrent resources. So, a resource levels can be increased by making them available for longer. So, the resource levels what you can do? You can also increase them and by making them available for a longer period of time.

Thus the staff might be asked to work overtime if the staff is working daily say 8 hours; then what you can do? He may be requested to work overtime maybe another 2 hours or 4 hours for the duration of a activity. And the competing resources also you should provide at during the overtime; they must be are made available at those times such as evenings or weekends or holidays etcetera; when they might otherwise be inaccessible.

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Another thing is that, a reallocate staff to critical activities. The project manager may consider allocating more efficient staff to activities on the critical path or swapping resources between critical and noncritical activities. So, there are two major activities as you know critical activity or non noncritical activity. So, if you observe that after some days the project is getting delayed; the critical activities they are getting delayed what you can do? You can transfer some of the staff members, you can swap some of the; either you can transfer or swap some of the staff members from the noncritical activities to the critical activities so that you can take an attempt to meet the deadlines of the critical path activities.

When a project is actually executed, the critical path may change as the actual durations of activities will vary from the original estimates and staff allocations maybe adjusted to reflect this; this very correct that when the start activities are executing. Initially you have a drawn the critical path, you have identify what is the critical path at the plan.

But actually when the project is running the activities are running due to the actual running times execution times; there might be changes in the durations of the critical activities and hence the critical path may be changed. So, if the critical path is changed accordingly you have then that will be what different from the original estimates and the staff allocation that you have originally made that may be changed.

So, accordingly you have to change this plan and you have to accordingly alert accordingly estimate the resources, accordingly alert the what staffs to the critical activities based on this change in the critical path durations.

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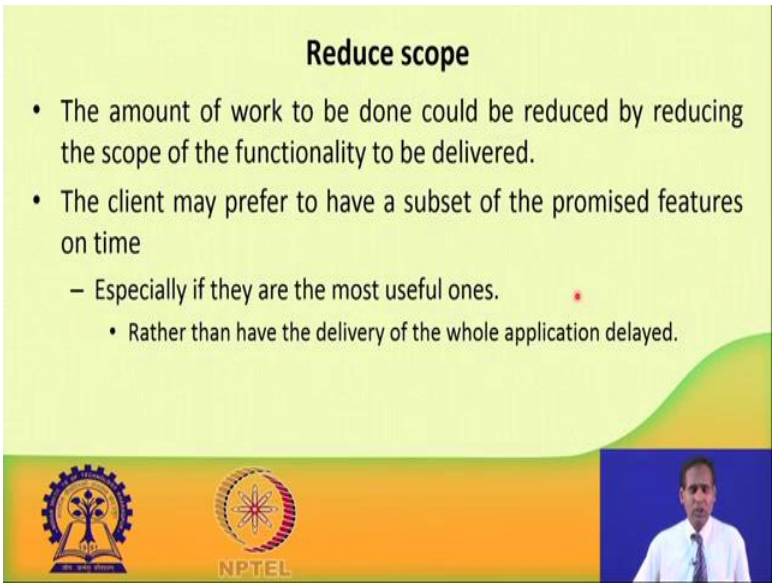

So, this I have already told you about reduce scope; the amount of work to be done could it be reduced by reducing the scope of the functionality to be delivered. If you see that the actually the project is getting delayed; we cannot meet all the, we cannot achieve all the, what planned scopes. So, what we can do? Then we can reduce the scope, the amount of the work to be done it can be reduce by reducing the scope, but the client may prefer to have a subset of the promised features on time.

So, the client definitely he may want to have a subset of the promised features on time; that means, that you have agreed during the software requirement specification; so, especially if they are the most useful ones rather than have the delivery of the whole application delays. So, instead of making or delivering the whole application delayed; the customer may want that the some subset of the promised features may be delivered on time.

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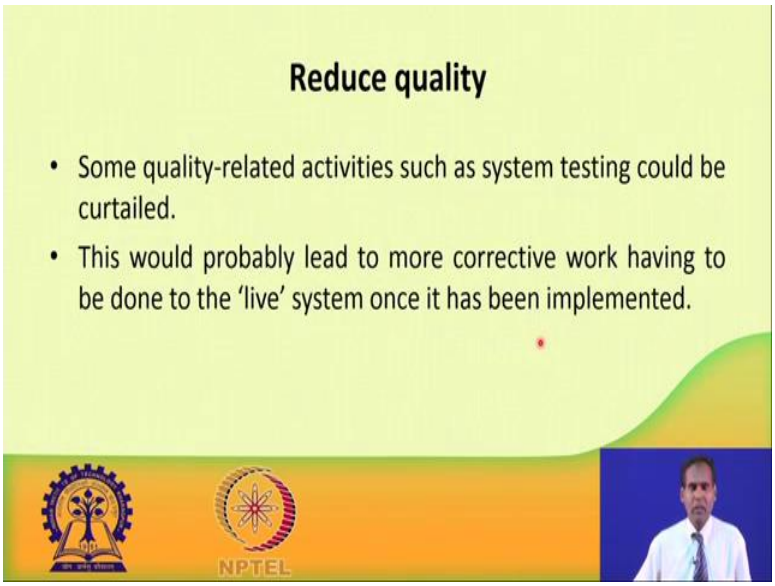

And one option is also there to get back the project onto the original track; that is reducing the quality. Some quality related activities such as system testing etcetera could be curtailed because there is a very less time. And this would probably lead to more corrective work having to be done to the 'live' system; once it has been implemented. So, by reducing the quality also we can try to what get back to the original target.

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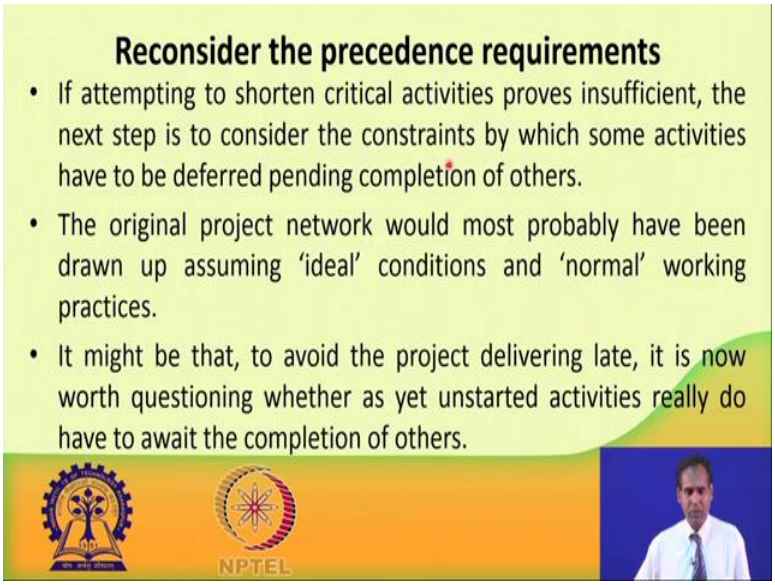

But you see that the customers sometimes may not be agreed if you are reducing the quality, if we are skipping the system testing the customer may not be agreed. So, there

is a what; there is a doubt or there is a what, you can say dilemma what to do in this case. And next this is the last option if all these what, thing what or the if you are not able to reduce the critical activities; so, this is the last option that we have to reconsider the precedence requirements.

So, if we are attempting to shorten critical activities it proves in sufficient what we should do? The next step is to consider the constraints by which some activities have to be deferred; some activities have deferred pending completion of others. So, some activities in the activity network we have already identified the activities. So, if the critical activity we cannot reduce the or shorten the critical activities or it will be proven to be insufficient; then we have to consider the other constraints by which some activities they have to be deferred pending until completion of the others.

So, the original project network would most probably have been drawn up assuming ideal conditions on normal working practices; while you have considered the while you will have drawn the what original project network or the activity network; there we have assumed two important things that the project is running under ideal conditions and normal working practices are followed, but I in practice that will not happen. So, that is why we have to; what we have to reconsider the precedence requirements. It might be that to avoid the project delivering date; it is now worth questioning whether as yet unstarted activities really do have to await the completion of others.

So, is it possible that to avoid the end project delivery date in order to avoid the end project what; the late in the end project delivery date what we can question that whether it is possible that the unstarted activities can they really await till the completion of others?

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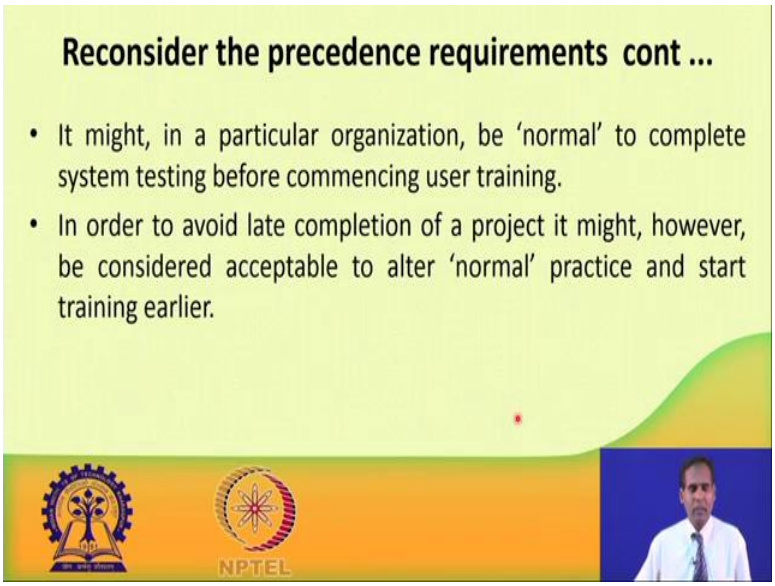

So, it might be in a particular organization on that the normal it; it may be normal to complete system testing before commencing user training. Let us see one of the example how we can reconsider the precedence requirements. Normally, the system testing must be or should be conducted before commencing the user training. First complete the system testing then you can give training to the users.

But if we are having what so; that means, this user training is dependent on the system testing. After system testing; then start the user training, but if you do not have time; what you can do? In order to avoid late completion of the project, what can be done? That the; we have to alter the normal practice that start the training first and then you can what after this training is completed then you can perform the system testing, you can alter the normal practice, you can reconsider change the precedence requirements.

So, now, we will assume that training is not dependent on system testing so that a first system testing can be performed; sorry first this training can be performed and then system testing can be done.

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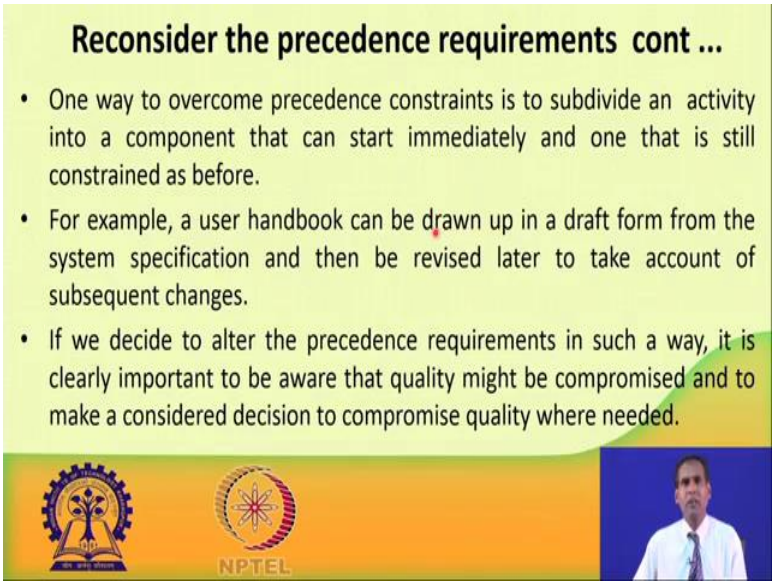

So, now one way to overcome the precedent constraint is to subdivide an activity into a component, which can start immediately and the one that is still what constraint before see. So, one way to overcome the precedence constraints requirements is that we can divide the activity into a component, which can start immediately and one that is still constrained at before; it is under constraint.

So, for example, suppose a user has a; user handbook can be drawn in a draft; form from the system specification. And then be revised later to take account of this subsequent changes what we can do here? You can see that one first; we want to prepare a user manual, what we can do? We can prepare a draft form; from where? From the, what system specification and then that can be revised later on to accommodate the subsequent changes.

If we decide to alter the precedence requirements in such a way that; it is clearly important to be aware that a quality might be compromised; so, if you will altered the precedence requirements in some cases what will happen? The quality might be compromised and to make a considered decision to compromise quality where needed. So, we have to take a judicious decision whether we can compromise with the quality or not. Because we are changing the precedence requirements due to the changing the precedence requirements; this what quality might be compromised.

So, whether the; a customer can accept this a compromise with the quality or not? We have to take a judicious decision whether we can compromise the quality when we are changing the precedence requirements.

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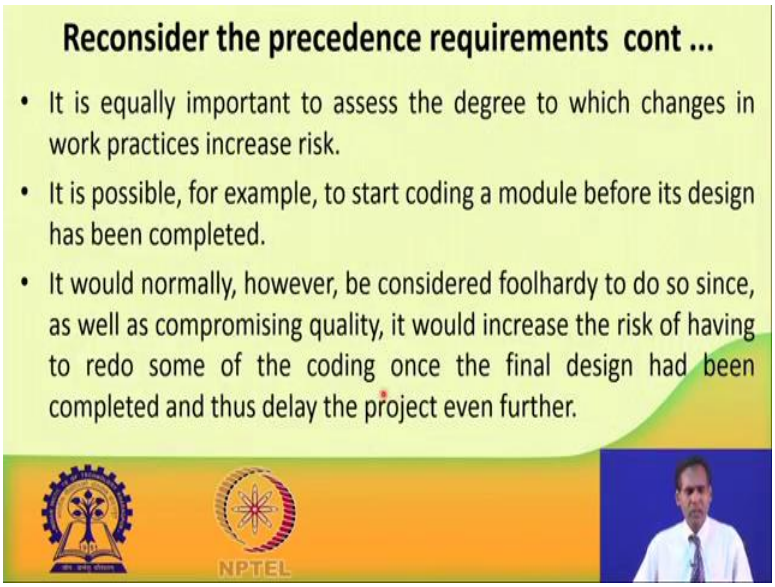

It is equally important to assess the degree to which changes in work practices increase risk. So, during this what, re-considering the precedence requirements; now it is also equally important to assess the degree to the extent to which the changes in the work practices it increase the risk. It is possible for example, to start coding a module before its design has been completed. See normally, normal practice say that first we should do the requirement analysis, then design, then coding, then testing. So, if somebody said that the first we will start coding and later on we will do the testing; then you see what is the; what problem will occur.

So, it would normally; however, be considered foolhardy to do since as well as compromising quality here it will be a; it will be foolhardy as well as it might be you might compromise with the quality. And there will also risk involved because if you are doing coding first, then design then if there will be the changes on the final design; then what will happen? You have to do you have to redo so many things in the code, so which is not advisable.

So, that is why before what changing the precedence requirements; please assess what is the degree to which changes in the work practices they increase the risk. So, here starting

the code before design it is a risky thing and you have to re do so much so many works if the design is changes. So, though and the quality also maybe what compromised. So, that is why a while a doing or making these changes in the precedence requirements or changing the normal practices; judiciously think and then take the action.

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So, now, let us summarize what we have studied how to get back on track. First negotiate renegotiate the deadline with the client, if the kind client is happy yes we can still wait some days we can tolerate a few more days delay; then no problem. If he does not agreed, if negotiation is not possible regarding the deadline then if not possible then try to shorten the critical path by what working overtime or by allowing your staffs to work overtime. By re-allocating staff members from the noncritical or the less pressing work to the critical activities or by what buying or hiring more staff. Also you can reduce the scope of the work to get back the project on track.

Another option is that we can reduce the quality so that by compromising the; with quality; we can finish the project on time. And the last option is that reconsidering the activity dependencies wherever possible you can change alter the activity dependencies. You can reconsider the activity dependencies by overlapping the activities so, that the start of one activity it does not have to wait for completion of another and by splitting the activities. Also by splitting the activities you can change the activity dependencies or

these, what did by splitting the activities you can also what alter the dependency among the activities.

In this also in this way also you can get back or the project can be the project can be brought back to the original target or the original track. It can be possible to meet the target line by following these options or by following these methods.

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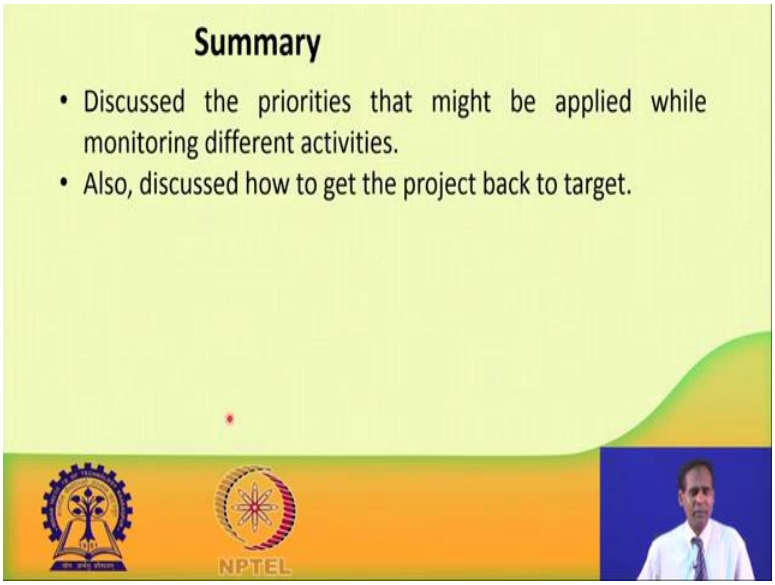

So, in this class we have discussed the priorities that might be applied while monitoring different activities such as we should give a top priority to the critical path activities and we should also give priorities to the activities having no float and etcetera. And we have also discussed the project how; we can back to the target different options I have just told.

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We have taken the references here and.

Thank you very much.

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Lecture - 46
Project Monitoring and Control (Contd.)

Good afternoon. Now let us discuss another important aspect of Projective Monitoring and the Control that is known as change control and software configuration management.

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First let us discuss about change control, then we will go to Configuration Management.

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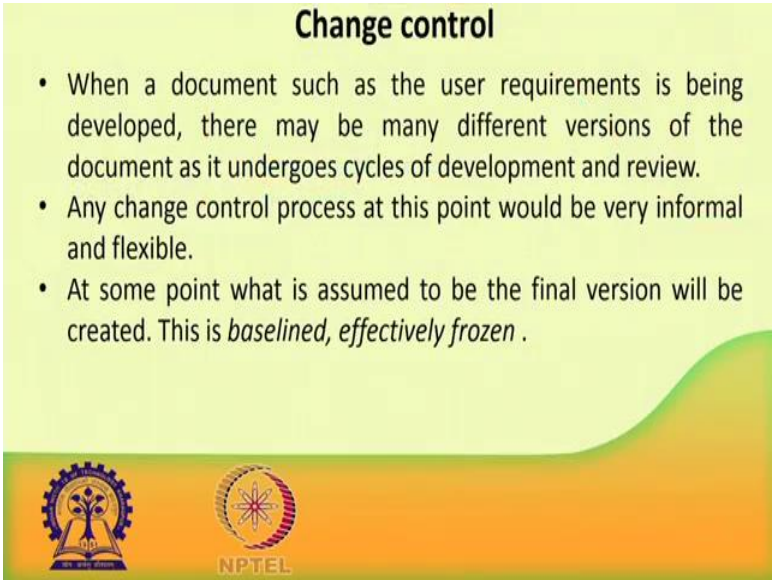

So, when documents such as a user requirement etcetera it is being developed, there may be many different versions of that document, because as it undergoes the cycles of or the different phases of software development process and review. Any change control process at this point would be very informal and it will be flexible and later on we will see the details.

At some point what is assumed to be final version that has to be frozen, at some point or some point will come when it is assumed that this document is final it is the final version or that has to be created; then we must what freeze that document development process there. So, this will be treated as the baseline version and this is all most effectively frozen at this point.

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Baseline products are normally the foundation for the development of further products or further documents; for example, if you want to develop what the interface design documents, then you can develop it from the baseline user documents.

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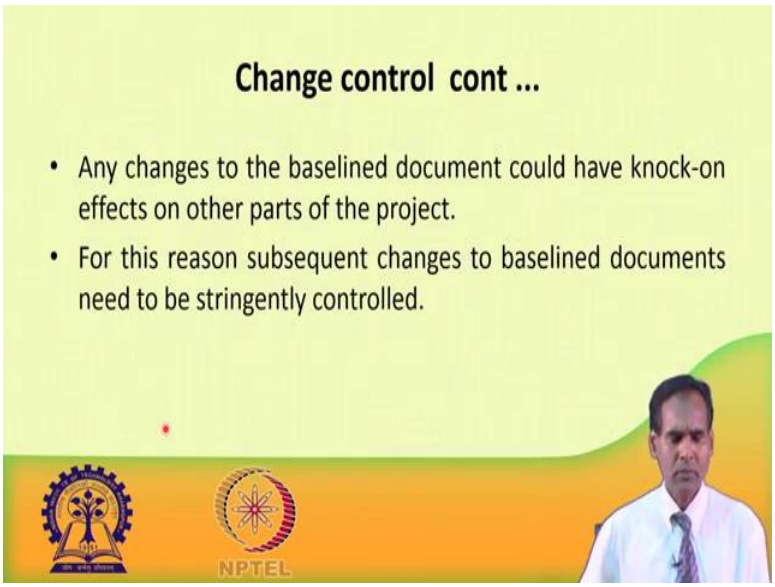

Now, any change to the baseline document it may create what a knock-on effects on other parts of the project; that means, the other parts of the project might be affected, if you will make any changes to the baseline document, but sometimes these changes are

required. So, that is why what for this reason, the subsequent changes to the baseline documents they have to be very stringently controlled and monitored.

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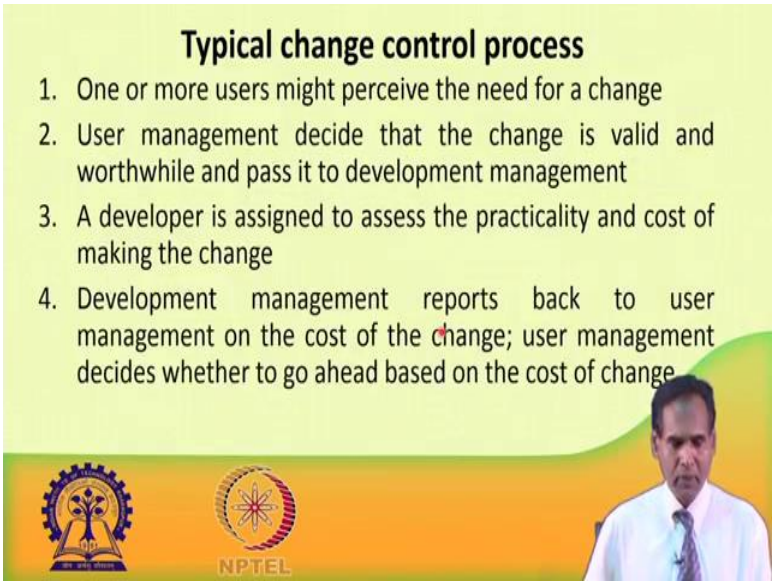

Now, let us see what are the typical change control process, what steps are followed in the change control process. First, one or more users they might perceive that there is a need for a change; then if they perceive that there is a need for the change. Then the user management, they have to decide that the change that is perceived by the users that is thought by the users it is valid and worthwhile and then they will pass it to the develop management. Next a developer from the development group, he will be assigned to assess what is the practicality and what is the cost of making the change; if it is practically feasible then they may go for to the next step.

Similarly, they will evaluate how much cost will be involved with the change. Then the develop management, they reports back to the user management that, yes this is the cost that will be required for the change and then the user management has to take a decision whether to go ahead with the change based on the cost. If the cost is high, they may not go further or if they cost which acceptable with them, they may give green signal yes you can go ahead for the change to the development team.

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Now, then after the user management is agreed, then one or more developers they are authorized to make copies of the components they have to be modified. Then the copies are modified by these one or two developers; then after initial testing, a test version is released to the users for conducting acceptance testing by the users. Then the users they conduct the acceptance testing, when the user satisfied with the acceptance testing, yes the change is ok; then the operational release of that product or document is authorized, it is released and then the master configuration items they are updated with these new changed one's.

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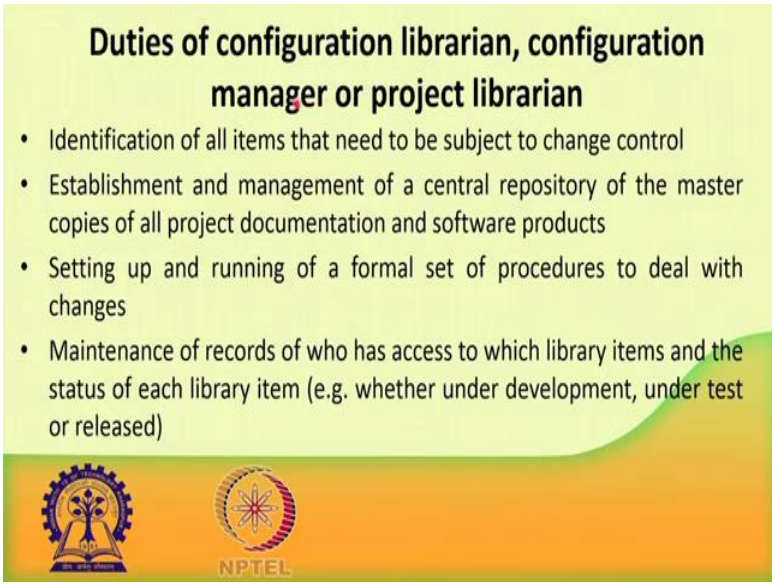

So, during this change control process, so in every organization normally one staff is there known as the configuration librarian. Let us see what can be the duties of the configuration librarian. So, first he should do the identification of all the items that need to be subject to change control; what items, what documents that need to be undergo a change control. He will also responsible for establishment and management of a central repository of master copies. So, all the master copies will be what stored in a central repository; so it is the job of the configuration librarian to for making arrangements for the establishment and management of a central repository of the master copies of all the project documents and software products.

Then he is also responsible for setting up and running up a formal set of procedures to deal with the changes; he should set up and run a formal set of procedures. Also, he is responsible for maintenance of the records of who has access to which library items and when and what is the status of each library item; like whether it is under development, or under test or it is released. So, those kind of things will be maintained by this configuration librarian.

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Now, let us see the important aspect that is the software contribution management. So, normally changes in a project, it can take place in any of the work product such as requirement document, SRS document, design documents, coding test plans, etcetera and this may be due to many regions such as the bug fixing, changes on account of the work

simplification and considering the efficiency issues, etcetera; you might have to what change these various documents.

So, change management can be done actually manually, it can be done manually by a designed configuration librarian. In the last slide, I have told what are the duties of a configuration librarian, the configuration librarian can manually what do the change management. But the manual change management process gets overwhelmed it very much difficulty and time consuming; when we consider the changes taking place on all work products.

So, when all the work products you have to in a project there are several work products and if you want to change all the work products; then it is very much difficult to do it manually. Also when there are multiple variants of the product were there, multiple variant, several variants of the product are there and then several persons may simultaneously work on these multiple variant they may try to change, then this will create problem. So, that is why manual change management is very much difficult for this, you have to go for automating it.

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So, in this situation, a systematic software configuration management process called as SCM, it with appropriate tool support needs to be deployed. SCM also the software configuration management is concerned with tracking and controlling changes to a

software. So, software configuration management, it is concerned with how to track and how to control the different changes made to a software.

In any systematic development and maintenance environment; various work products may require continually change such as a code, design document and SRS documents etcetera, associated with the software; they may require continual, they may continually change during the development phase as well as maintenance phase and you have to systematically manage these changes.

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So, in a team development environment what will happen, each member of the development or maintenance team they would be decide they will be assigned to handle some modification requests. So, both from the development team as well as management team, maintenance team, so some members will be assigned to handle the modification request. Therefore, every work product would have to be accessed and modified by several members.

So, every work product they have to be accessed first, whether it is now possible to change it or not; if it is possible that you can make a change it is viable, then it can be what modified by several members. But if several members simultaneously they want to modify a single document, then problem will be there, it will be it may lead to inconsistency. In such a situation, unless you use a proper configuration management

system, then several problems can appear. We will see what problems can appear, if you will not use a proper configuration management system.

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So, let us see what is configuration management, as I have already told you that configuration management deals with, it concerned with tracking and controlling the changes to a software. So, we can define configuration management like this, it is the set of activities through which the configuration items are managed. We will see what are configuration items in the next slide. So, configuration management is the set of activities through which the different configuration items they are managed as well as maintained as the product undergoes it is lifecycle phases.

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Now, let us see what are the context in which configuration management is necessary. So, two important phases will use configuration management during the development phase as well as during the maintenance phase. During the development phase, the work products they may get modified as a development activities are carried out, so you have to what you go for configuration management.

Similarly during the maintenance phase, the work products may change due to various types of enhancements and adoptions that are carried out including the what bug fixing etcetera. So, for this also you may have to go for changing the, what you may have to go for changing the configuration. So, change also configuration management is also necessary at this point of time. So, thus the state of the work products they continually change both during the development period as well as the maintenance period.

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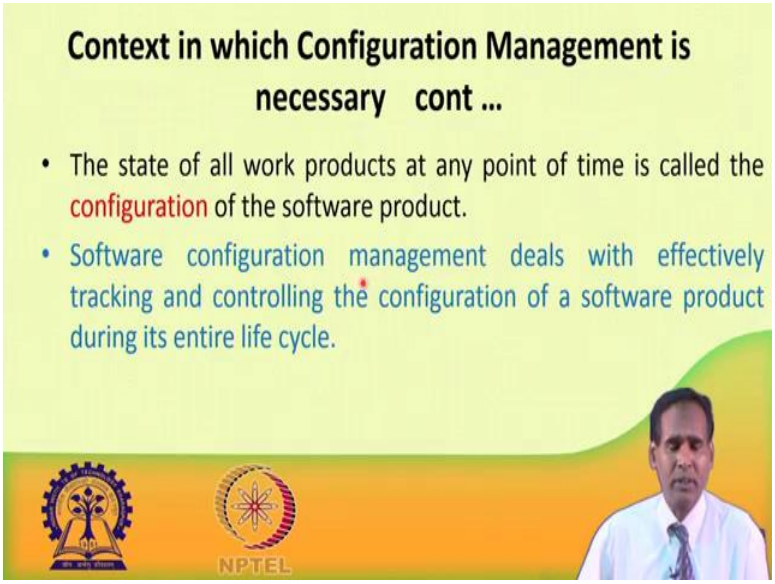

Now, let us see what do you mean by configuration? The state of all work products at any point of time is called the configuration ok. The state of all the work products such as SRS document and these who can say that design documents, coding, test plans, etcetera are the state of all work products at any point of time is called the configuration of the software product. Software configuration management deals with I have already told you that, software configuration management deals with effectively how to track and how to control the configuration of a software product during its entire lifecycle.

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For effective configuration management, it is necessary to deploy a configuration management tool. There are many configuration management tools available out of them, some are open software, that is there you can get them free of any licensing fees and others are commercial tools; towards the end I will give a list of some commercial tools or some free open source tools for software configuration management. So, configuration management practices it includes two important things; version control and the establishment of the baselines.

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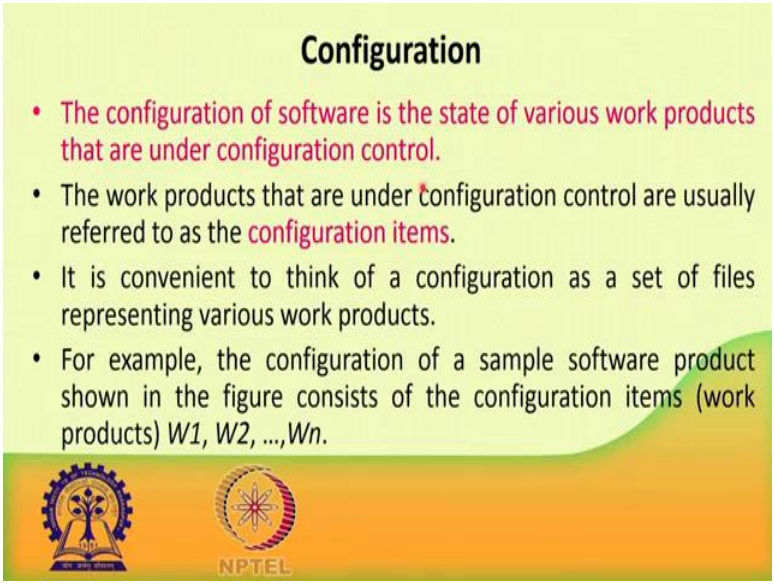

Now, let us see some of the important terminologies associated with configuration management. First configuration, I have already told you that, configuration of software means it is the state of the various work products those are under configuration control ok.

So, configuration means, configuration of software means it is the state of various work products such as SRS document, design documents, codes or codes test plans, etcetera that are under configuration control. The work products that are under configuration control are usually referred to as configuration items ok. So, these work products which are under configuration control they are termed as contribution items. It is very much convenient to think up a configuration as the following; you can consider them as a set of files representing various work products this may be thought of as configuration.

For example the configuration of a sample software product which we have shown in the next slide, it consisting of the following configuration items or the work products like this.

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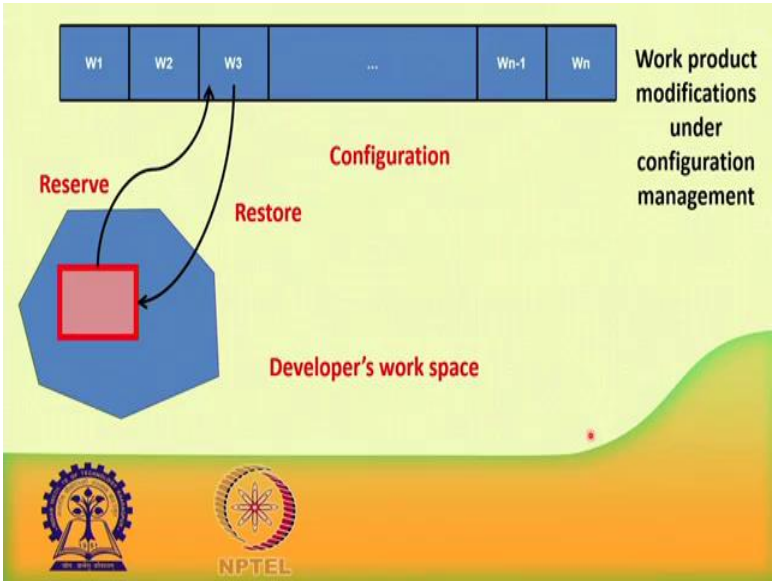

This is a says work product. So, W1 it is divided into so many configuration items, here the configuration items are W 1 W 2 W 3 up to W n etcetera.

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Now, let us see the next term that version; as development and the maintenance activities are carried out on a software product, it is a configuration keeps changing. So, I already

told you during development period as well as maintenance period, so whenever they are carried out on a software product, it is configuration that is one or more of the configuration items they keep on changing. It is often necessary to refer to the configuration that existed at certain point of time; we want to know that, what is the configuration say before two days like that.

So, very often it is necessary to refer to the configuration that existed that the configuration of the software that existed at certain point of time. For example, say we want to refer to the last week's configuration of the software or last year's configuration of the software. So, in that way we might have to refer the configuration at some point of time. Therefore, a version means it is a configuration that existed at certain point of time.

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In very briefly we can say, that a version is a configuration that existed at some point of time at certain point of time. So, more technically versioning is a numbering scheme that help us, identify a specific configuration at a certain point of time, ok. So, technically we can say that versioning is means it is a numbering scheme. Why it is required? That helps us identify a specific configuration at a specific or at a certain point of time. How this can be achieved, how this numbering scheme can be achieved, how this versioning or numbering scheme it can be achieved? This can be achieved by a configuration tool by tagging the files representing or yes representing the configuration items with the version name.

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As a software is released and used by the customers; many errors are discovered, and then enhancements are made to what improve, but the functionalities. So, a new release of the software, then several releases of the software come; a new release of the software is an improved system which is intended to replace an old one. So, you know that in many versions are coming, many releases of the software coming may be Microsoft; so a new release of the software it is an improved system which is intended to replace what the older one. So, usually a product is described as version m and release n or simply as a version m dot n; where m refers to the version and n refers to the release.

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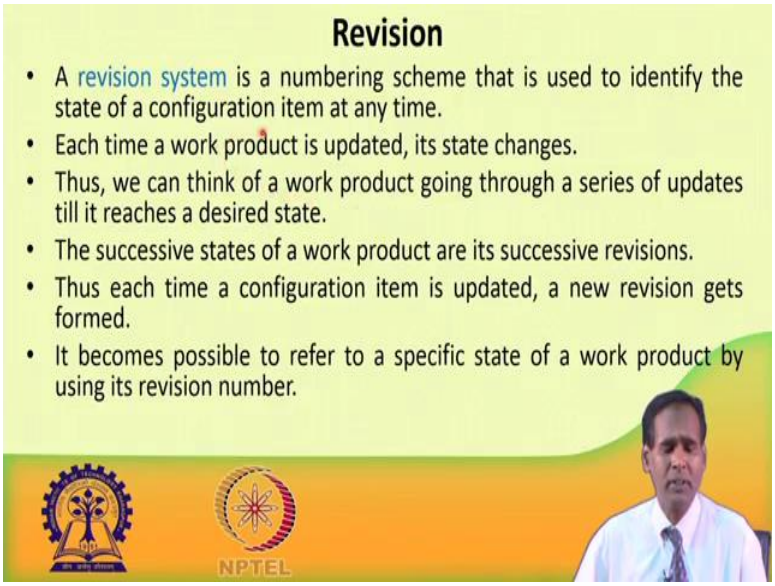

Now, let us quickly see about what is a revision. A revision system is a numbering scheme that is used to identify the state of a configuration item anytime. So, because you must be clear about the difference between what version and revision; a revision system is also a numbering scheme that is used to identify the state of a configuration item at any time. Each time a work product is updated, it is state; obviously, it changes. So thus, we can think of a work product which is going through it is series updates till it reaches a desired state. So, till reaching a desired state a work product may go through a series of updates. So, the successive of states of a work product are it is successive revisions.

So, whenever you are making any updation, updates, so the successive states of a work product they are it is a successive revisions. So, thus each time a configuration item is updated, any time any configuration items such as code or design document or a SRS document is updated a new revision gets formed. Then it becomes possible to refer to a specific state of a work product by using it is revision number. Every time you are when you are making upgradation, a number you are assigning it becomes possible to refer to the specific state of a work product by using it is what revision number.

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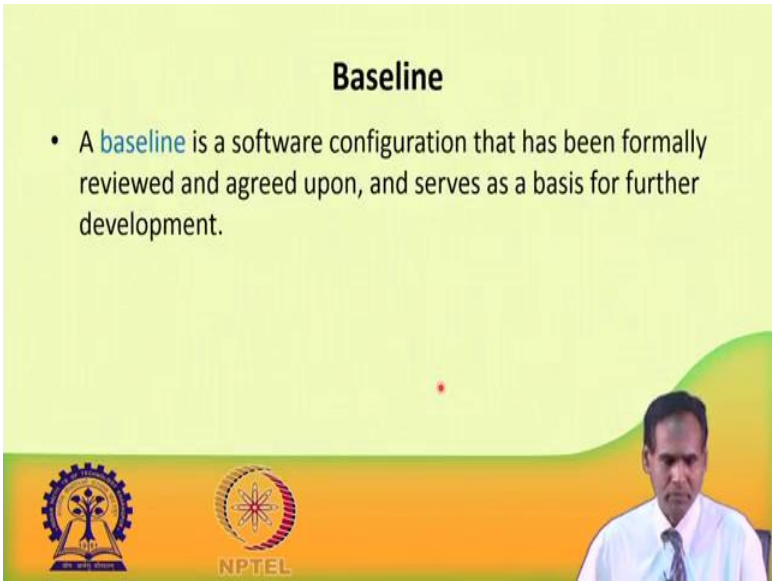

Now let us see what is a baseline? Baseline is a software configuration that has been formally reviewed and agreed upon and it serves as a basis for further development. So, here the what, first this a software configuration that has already been formally reviewed by the developer also the users have seen in it and they have agreed on, both of them

have agreed upon it and then it can serve as a basis for further development of the software products.

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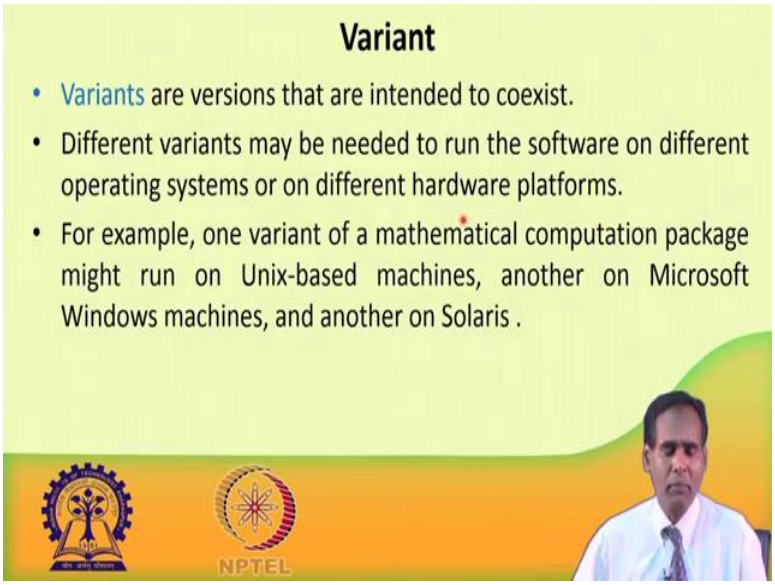

Variants; variants means they are the versions that are intended to coexist. So, these are the different versions they are intended to coexist will. And different variants may be needed to run the software on different operating system ok. It might be required that different variants, they might be required to run the software on different operating systems or on different hardware platforms, etcetera or maybe on different browsers.

So, let us take a small example that one variant of a mathematical computation package might run on Unix-based machines, another may run on Microsoft windows machines, another variant may run on Solaris etcetera. So, this could be one example. Another example could be those who have used LaTeX, so LaTeX normally it is used in where in Unix version; but another variant is they are, if you are using windows version the other variants is called a MiKTeX. So, that is a text processing software. So, you can use in Unix variant what LaTeX and for windows the variant is with MiKTeX.

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So, variants may also be required to be created when the software is intended to be used with different levels of sophistication of the functionalities; for example, for novice users it is the novice version, for enterprises it is enterprise version and for experienced versions you can say or for professionals the version may be professional version.

So, variants are often created during the operation phase, during the development phase, and as and when the software products with overlapping functionalities they are required, and then you can create different variants. Even the initial delivery of this software might consist of several versions. So, whenever you are initially, first time you are delivering, the initial delivery of software might consider up several versions and more variants may be created later on.

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Now, let us see what are the purpose of SCM why what software configuration management is used, what is the purpose. So, there are several purposes, some important purposes I have written here that why software configuration management is required; first, problems associated with concurrent access.

So, one configuration item, simultaneously many users or many developers they are trying to access then, it will lead to inconsistency. So, in order to avoid the problems associated with concurrent access, we should go for software configuration management. Similarly undoing the changes, several times what is required, we may have to we have done something wrong thing and we want to undo it. So, in order to undo the changes, this, what software a configuration management is required. Another is system accounting, in order to keep track of that who had made the change, when he had made the change and why he had made the changes.

So, to keep track of all the details, that is you that we call in technology the system accounting. So, that can be another reason why SCM is required. Similarly handling the different variants and helping fix bugs in them, I have already told about variants, how to handle the variants, how to tackle the variants and how to what help the, how to what fix the bugs in the variants, for them also you can use this software configuration management.

Similarly, accumulated determination of the project status, what is the current project status, we want to know the accurate what status of the project. So, for that, you can also use the software configuration management. And similarly preventing unauthorized access to the work products, see only the project manager he will give the permission to the developers to access the work products.

Without the permission anybody or anybody else cannot actually access this. So, if anybody is trying to access the work products unauthorized way, then that will be prevented; if you are having software configuration management. So, this is another reason that preventing unauthorized access to the work product this can be achieved by using software configuration management. So, these are the different reasons purposes of using software configuration management, these things have been explicitly what explained here, I am skipping this thing, these are easy things you can read there yourself.

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Now, we should see that what is the, what we have seen so far, we have first seen that, what is the need of change control, why change control is at all required. Then we have also seen that the steps of the change control, how the change control can be achieved in an organization. We have also discussed the change control procedure; we have also explained the concept of software project management, what is software sorry what is software configuration management, what is the need, what are the purposes. We have

also discussed what the context in which it is necessary, in which context, what the software configuration management is necessary.

For example I have told that it is required at the development phase also, it is required at the what maintenance phase, also what are the different purposes just I have already told you that, the important purposes for which software configuration management we required. We have also defined various terms, the common terms which as a what as a student you must know that what are the different terminologies used in configuration management such as what is the configuration, what is a configuration item, what is a baseline item and what is revision, what is a release.

So, those kind of basic things also we have discussed here. So, actually next actually we must see about, we have seen now the basic concept of configuration management; then we must see about how to do it, how to what process, you should follow what, for doing this configuration management. I have already told you doing configuration management manually by a configuration librarian is very very difficult, you must require you have to automate it.

So, what are the tools there available for automating this. So, I will tell that some of the tools are there for doing this and some tools are freely available, some tools are what commercial tools; for example, freely available tools will use SCCS and RCS. So, those are details of the tools we will discuss in the next class; but first we will discuss about what are the steps they must be followed for achieving this or for making software configuration management and those steps will discuss in the next class.

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And these things we have mainly taken from these two books ok.

Thank you very much.

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Lecture - 47
Project Monitoring and Control (Contd.)

Good afternoon. Now, let us see the remaining portion of software configuration management.

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We will see first the configuration management process, then we will see some of the tools available for configuration management, and then at last we will see some other project management tools.

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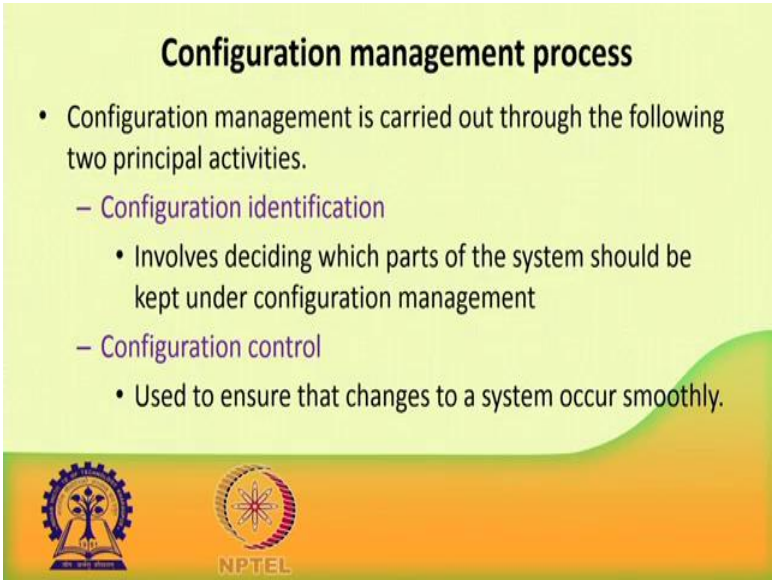

So, we have already discussed the fundamental concepts of configuration management, various terminologies of configuration management in the previous class. Now, let us see how configuration management is carried out. Configuration management is carried out through the following two principle activities, one configuration identification, then two, configuration control. In this process configuration identification, it involves deciding which parts of the system should be kept under configuration management, which items will be kept under what configuration management such as your SRS document, code, design documents, test plans, etcetera. And in configuration control this process is used to ensure that the changes made to a system, they occur very smoothly.

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We will first see about configuration identification in detail, and then we will see about this configuration control. So, the project managers normally they classify the work products associated with a software development process into three main categories. So, the work products could be as I have already told, there could be SRS document, design document, codes, test plans etcetera or the design test cases they are the work products. They can be divided into three categories, controlled products, pre-controlled products and uncontrolled products.

So, let us see what are controlled work products, controlled work products are those that are put under configuration control, that means, the work products which can be put under configuration control, these are known as control work products. The team members must follow some formal procedures to change these; they have to follow some accepted procedures, formal procedures to change this. We will discuss the procedure. Then pre-controlled work products are those work products which are not yet under configuration control, but eventually they will be under configuration control may be in near future.

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And then last one is this uncontrolled work products. These are the work products, which will not be subject to the configuration control; they will not be subject to the configuration control. These controllable work products and pre-controlled work products and pre-controlled work products together, they are known as controllable work products.

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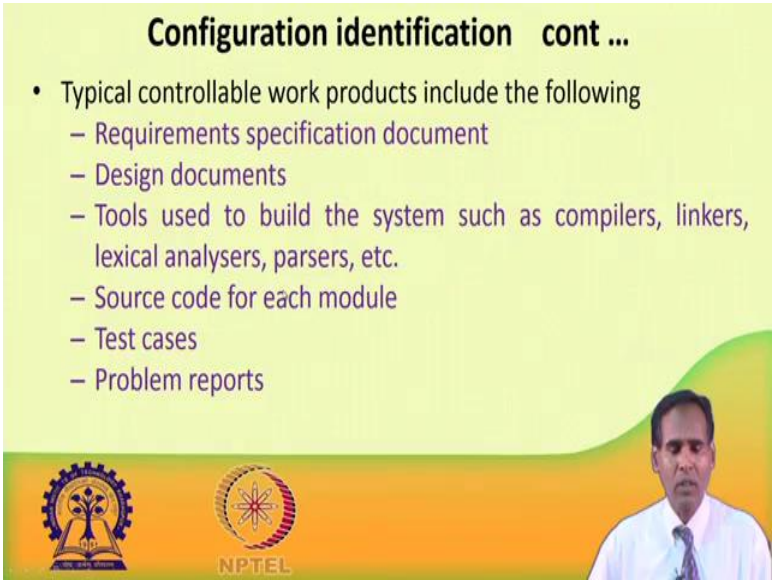

Then let us see what work products can be coming under which work products may come under controllable work products like as I have already told you requirement

specification document, design documents such as DFDs, UML diagrams etcetera, the tools which are used to be in the system such as compliers, linkers, loaders, lexical, analyzers, parsers, etcetera, they can come under control level work products. Similarly, the source code for each module that can be also considered as a controllable work products; the design, test cases, test plans etcetera, they will be also coming under controllable work products. And finally, the problem reports they can also be what; they can also come under the controllable work products.

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Next we will see about configuration control, how to perform configuration management or how to perform this configuration control. So, configuration control is the process of or configuration control is the part of configuration management system, which most directly affects the day-to-day operations of the developers. The configuration control it allows only authorized changes. So, those persons which have being authorized might be a project what steering committee or the project manager.

So, the configuration control, it will allow only those persons only authorized the changes made by those authorized persons to the control objects, and it prevents any unauthorized changes. Anybody else whoever want we cannot make change. So, they have to get permission either from the project manager or from the steering committees. So, that is why configuration control it allows only the authorized changes to be controlled and it prevents the unauthorized changes. The project manager can give

permission, can give authorization to some specific members who will be able to change or assist the specific work products, and others they cannot make any unauthorized change or any unauthorized access.

In order to change a controlled work product such as a work code, etcetera, a developer he has to get a private copy of the module through a operation called as a reserve operation. He cannot just do the change on the master copy. He has to first request or he has to first reserve for a private copy; and once his request is approved, he will be given a private copy where you will make the changes, then that copy will be tested if everything is what successful. If the project committee of the project manager approve the changes, then only that master copy will replaced by this developed private copy, this is how the configuration control, this process it works.

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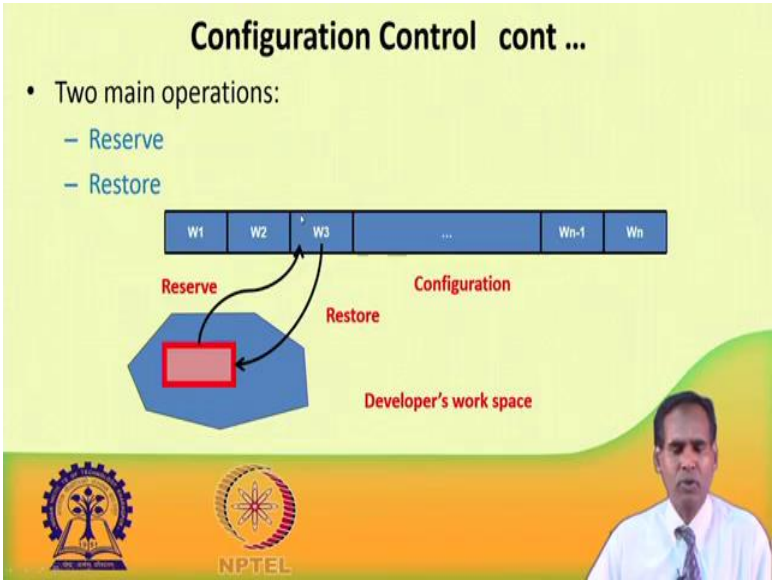

So, let us explain how this configuration controls this process work. As I have already told you, there will be two main important operations in configuration process, one reserve; another is restore. You see, so suppose this is the developer's work space. One developer wants to make to some changes to the what configuration item W 3. So, what he has to do, he has to send a reserve message to this project manager or to the steering committee requesting that I want to make some changes in the configuration item W 3. So, if his request is approved by the project manager or by the steering committee, then a private copy of the W 3 this configuration item will be given to the developer. Then the

developer will make any sort of changes whatever he wants; what is justifiable he will make the changes on W 3.

He will, please remember he will not directly do in the masters copy. A copy of W 3 will be given as the private copy. The developer will make the changes in this private copy and then this what, changes what he has made in the private copy that will be sent to the project manager or the steering committee, we call it a committee as a committee called as CCB. I will just show you, so that committee then he will investigate the various changes various aspects of the changes if they are satisfied, then they will approve the changes.

And after that, if everything is approved, yes, the changes are correct, then the, what developer will replace this master copy W 3 with the developed or the yes with the developed copy or with the private copy through a operation called as restore operation. So, through reserve operation, he can reserve a copy of W 3. And he will be issued a private copy and then with the restore operation, then he; and if the changes are approved, then the developer can restore, these privately developed this private copy, it can be put in the masters copy place, that means, the master copy will be replaced with this private copy if that is approved by the CCB. So, this is how the configuration control process it takes place by using two different important operations reserve and restore.

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So, this thing I have explained here that the configuration management tools allow only a one team member to reserve a module at any time. So, because we have already seen the problem of what concurrent assess, it will not allow more than one person at a time to do the reserves, so that is why configuration management tool allows only one team member to reserve a particular module at any time. Once that work product is reserved, it does not allow anybody else to reserve the same module until the reserve module is restored. So, in this way, we can prevent or the tool can prevent more than one developer to simultaneously reserve a module, thus the problems which are associated with the concurrent access that can be taken care of.

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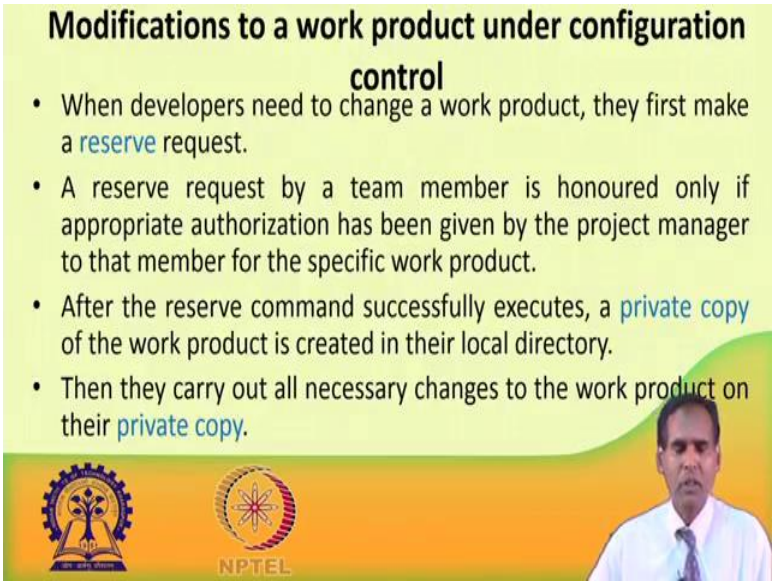

Now, let us see how the modifications to a work product are made under configuration control. I have already told you that when a developer needs to change a work product, he first makes a reserve request. Then that reserve request is honoured only if the appropriate authorization has been given by the project manager; otherwise this request will not be honoured, ok. So, a reserve request by a team member is honoured only if the appropriate authorization has been given by the project manager to the member for that specific work product.

After the reserve command successfully executes, a private copy of the work product or the configuration item is created in his local directory. And then; and the project or the developer he can carry out the necessary changes to the specific work product on the

private copy only not only master copy, he can make any changes in the private copy only.

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Then once the developer has satisfactorily a completed all necessary changes, then the changes are need to be restored in the configuration management repository. However, restoring the changed work product to the system configuration, it requires approval; it requires permission of a board called as change control board. The change control board is usually constructed by taking members among the development team members. Now, for every change that needs to be carried out, this CCB reviews the different aspects of the changes made to the controlled work product and certifies certain aspects they endorse certain comments about the changes.

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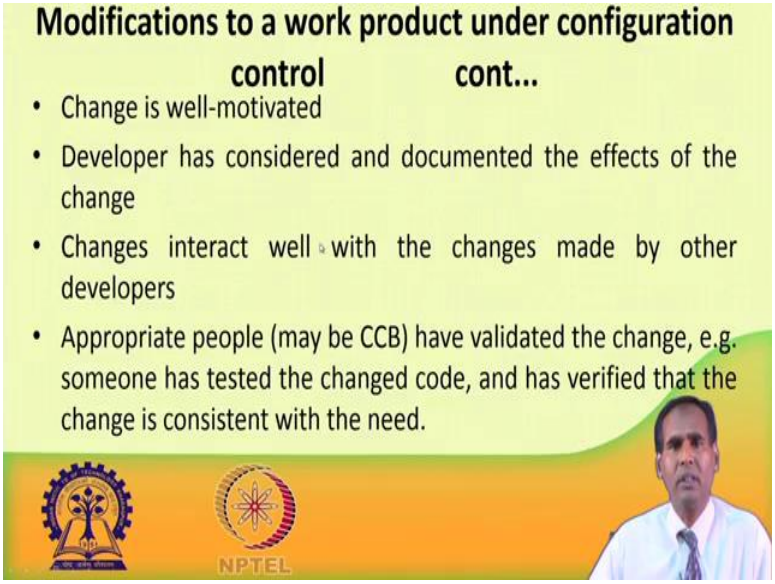

They may give comments in these aspects that change is well-motivated. The developer has considered on document the what will the effects of the change the developers has also already considered and documented, whether there is a adverse effect or only positive effects will be there, if any negative effect also the developer has already mentioned. So, the committee is aware of that. Then the changes interrupt well with the changes made by other developers. So, what changes particular developer has made, whether those changes are what interacting well with the other the changes made by the other developer that also the committee will see, and then the appropriate people may be from the CCB, they have validated the change, they have tested the change, they have validated the change.

For example someone has tested the changed code and has verified that the change code is consistent with as per their need. So, if they verify and the committee or the CCB satisfied with every aspect of the changes made in that what work product, then they will approve and they will give the permission, and then the what software developer, he can replace, he can restore the master copy with these changed copy.

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So, now let us see in the CCB will be there. Normal, with the changed control board is seldom a group of people, normally for smaller project the project manager himself if the he test the role of CCB. Except for very large projects, the functions of the change control board are normally discharged by the project manager or by some senior member of the development team. So, once the CCB reviews the changes to the module and approves all these changes, then the project manager updates the old configuration item with the new one through a restore operation.

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A configuration tool does not allow a developer to replace a work product in the configuration with his local copy unless he gets an authorization from the CCB. So, unless the CCB authorizes, unless the CCB approves the changes even if the software developer has got the private copy, but he cannot upload it, he cannot restore this modified copy. So that is why configuration control tool, it does not allow any developer to replace work product in the configuration with his local copy unless he gets an authorization or permission from the CCB. Therefore, incompletely modified or improperly modified or inappropriately modified work products, they cannot be updated in the configuration.

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So, we have last class we have already seen about what version and we have also seen release. We have already seen how the versions can be managed. The software what configuration management we have seen let us quickly see out how the different releases can be managed; so, release management. The release management process it systemizes the work carried out by developers to provide a new release of a software and on the part of the users to smoothly obtain and use a new release. So, see two important things you can see that the real release management process is the systemizes two things, the work carried out by the developers, how to provide a new release of a software, and for the users how they can smoothly obtain, and use a new release this can be provided by the release management process.

The release process should involve what see now that this release management becomes very much important after the users getting the facility of easy downloading of the releases from the internet. Then, so that is why this release management has become very much important. The release process should involve what minimal effort on the part of the developer. See two important stakeholders are there, developer as well as the user. This release process should involve minimal effort as much as less effort on the part of the developer to upload a new release of software; similarly this for the users this process should involve minimal effort for downloading it and installing it.

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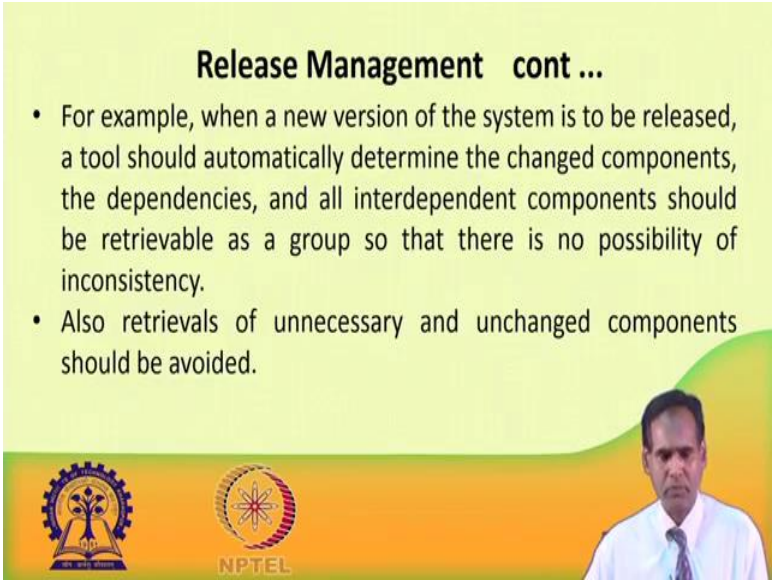

For example, when a new version of the system it has to be released, the tool should automatically determine what will be the changed components, the dependency among them, and all the interdependent components, they should retrieval as a group, so that there will be no possibility of inconsistency. Also retrievals of unnecessary and unchanged components should be avoided. So, whenever there will be retrieval of some unnecessary components some unchanged components that should be avoided by the release management tool.

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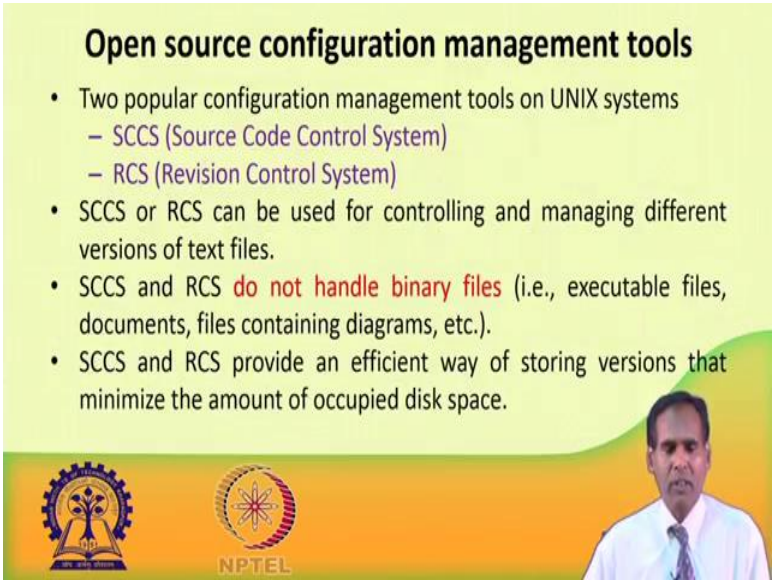

Now, let us see as I have already told you, there are several configuration management tools are available; out of them some are free, that means, open source tools and some now somehow some are commercial tools. So, today we will discuss what some open source tools. So, two popular configuration management tools on UNIX systems are there may be these are open source, one is SCCS, it is named as source code control system. Another is RCS that is that stands for revision control system either SCCS or RCS they can be used for controlling and managing different versions of text files.

Please remember they can be used for managing different text different versions of text, files, they do not handle binary files. For example, executable files, documents, file changing diagrams, etcetera they cannot be handled by SCCS and RCS. SCCS and RCS they provide an efficient way of storing the different versions that minimize the amount of the occupied disk space. So, these tools they provide some efficient way for storing versions that will minimize the amount of occupied disk space.

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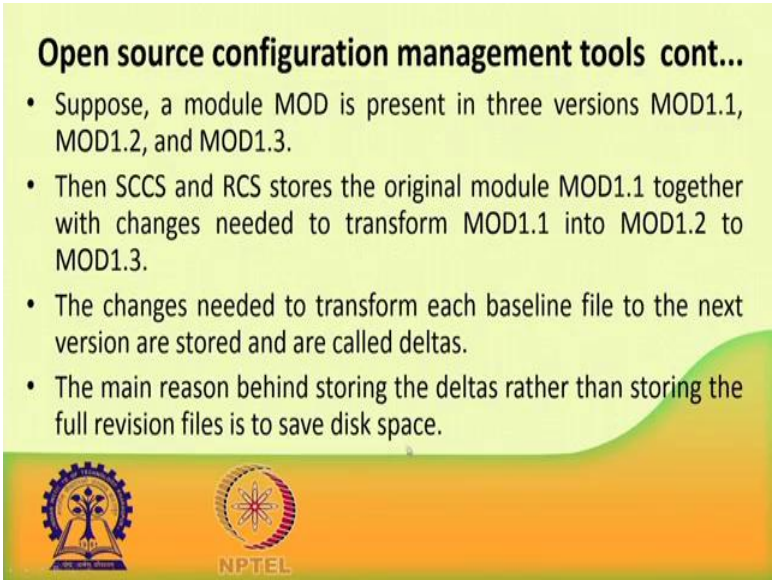

Now, let us see suppose a module mod is present in three versions. For example, a module mod is present in three versions may be module 1.1, module 1.1, module 1.3. Then the tools SCCS or RCS they store the original module what module 1.1 together with the changes needed to transform from module 1.1 into module 1.2 to module 1.3. Please see they will not store all the versions module 1.1, module 1.2 and the module 1.3, they will store the original module, the baseline module, module 1.1. Along with that they will store the changes which are required to transform from module 1.1 to module 1.2 and from module 1.2, module 1.3. So, basically they will store the baseline version module 1.1, and the changes required to move to module 1.1, module 1.3.

The changes needed to transform each baseline file to the next version are stored, and are called deltas. So, the changes which are needed to transform from the baseline module may be from your module 1.1 to module 1.2. So, these changes which are needed to transform each baseline file to the next version, they are stored and these changes are called deltas. The main reason behind storing the deltas rather than storing the full revision files is to save the disk space.

So, if you while store all these versions module 1.2, module 1.2, module 1.1, module 1.2, module 1.3, the disk storage requirement will be very high. Both the files store only the baseline version module 1.1 then the changes required to transform into module 1.1

module 1.3, then the required disk space will be very less. So, RCS and SCCS do or follow this approach.

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The change control facilities provided by SCCS and RCS. What facility they provide, they include how to the ability to incorporate restrictions on this setup individuals, who can create new versions. So, proper what, authorization is required, and facilities for checking the components in and out, that means, who can reserve one can reserve among the who can perform the restore operations, ok.

So, who can perform and who can perform reserve and restore operations and how they can perform, these facilities are provided by this tools SCCS and RCS. Individual developers they check out to the components and modify them, ok. The individual developers they check out the components and modify them. Then after they have made all the necessary changes to a component, and after all the changes have been reviewed and approved by CCB, then they can again check in what the changed module into the SCCS and RCS, and the master file is replaced with this what changed module.

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Now, let us see about some of the features of SCCS. I have already told you that SCCS is a configuration management tool; it is available on UNIX systems and it helps control and manage text files only. It also implements an efficient way of storing the versions; it minimizes the amount of occupied disk space by storing only the baseline version and the changes required for transformation, not all the what, revisions.

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Suppose a module is present in 3 versions this I have already told you like module 1.1 module 1.2 and module 1.3. And SCCS stores the original on the baseline version 1.1.

Along with the changes those are needed to transform it into module 1.1 and module 1.3 and these changes or modifications they are known as deltas.

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So, access control facilities provided by SCCS, they include that facilities for checking components in and out. Individual developers they can check out components and modified them. And after they have changed the module as required and the module has been successfully tested and verified by CCB, then they can check in the changed module into the SCCS. The master copy now will be replaced with this modified copy.

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Revisions are normally generated by numbers in ascending order like revision 1.1, revision 1.2, 1.3 etcetera. It is also possible to create variants of a component, ok. It is also possible to create different variants of a component by creating the what by creating a fork by using fork in the development history ok. It is also possible to create different variants of a component by creating a fork in the development history. So, these are the configuration management tools. Many other configuration management tools are there. You can find out from other books on internet.

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So, now I will just conclude with this I will give some examples of other project management tools. So, these two are RCCS and SCCS are explicitly they are used for software configuration management, but there are other project management tools to support other activities such as drawing Gantt chart, etcetera for the CPM for handling, CPM PERT for dealing with CPM PERTetcetera, there are other project management tools.

So, I have listed here some other project management tools to perform some other project management related activities. For example, Gantt project is one of the freeware ok. For example, Gantt project is one of the freeware. It is GPL-licensed project management software. It is a very open source software that runs under the, what windows operating systems, Linux operating systems and MAC operating systems. There is another tool called as Microsoft project. It is the basic project management

software from Microsoft Corporation. The, and you can see this is available if you are using Microsoft what system; Microsoft software you know computer then you can see that this Microsoft project is already there.

Advanced capabilities of Microsoft project the advanced capabilities are supported through Microsoft office project server and Microsoft office project as access software. The advanced capabilities, they are supported through what Microsoft office project server as well as Microsoft office project web access software. So, if you are having Microsoft office, then you can see in your computer that this Microsoft project software is there, which you can use for project management related activities.

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So, another software is there called as Primavera project management software which is a widely used suite of project management software. So, you can see there are two versions here SureTrack is one which is the entry level software, and Primavera 6 is the advanced software. These software's you can use also project management related activities. Using SureTrack's Project KickStart a project manager can define the different project phases, establish the goals, anticipate the obstacles and delegate the different assignments. So, these are only basic things I am just giving the information. You can see the details from the books and the internets.

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So, let us just compare the different features of the project management tools. Like see Gantt project, it can handle scheduling project scheduling if you have known CPM and PERT etcetera. So, Gantt project supports scheduling. It also supports resource management. I have already told you for what representing resource management, Gantt project is one of the best software and it is open source software.

Similarly, Microsoft project it supports portfolio management, portfolio management you have already what discussed initially. So, its supports project management. It is also web based. It also supports scheduling activities such as through PERT CPM and PERT. Also cost management is supported through Microsoft project. And resource management we have already discussed resource management; resource management is also support is supported through Microsoft project, but this is not an open source software when you are what licensing Microsoft package, then you will get this tool.

Then primavera SureTrack is this also supports portfolio management it is also an web based what software. This supports scheduling activities through CPM, PERT etcetera, and it also supports cost management it also supports resource management, but this is not on open source software. So, these are few of the, what project management tools we have listed here. Many other what projects are there for example, another project management tool is called as Libre project. So, you can also use Libre project for the different project management related activities ok.

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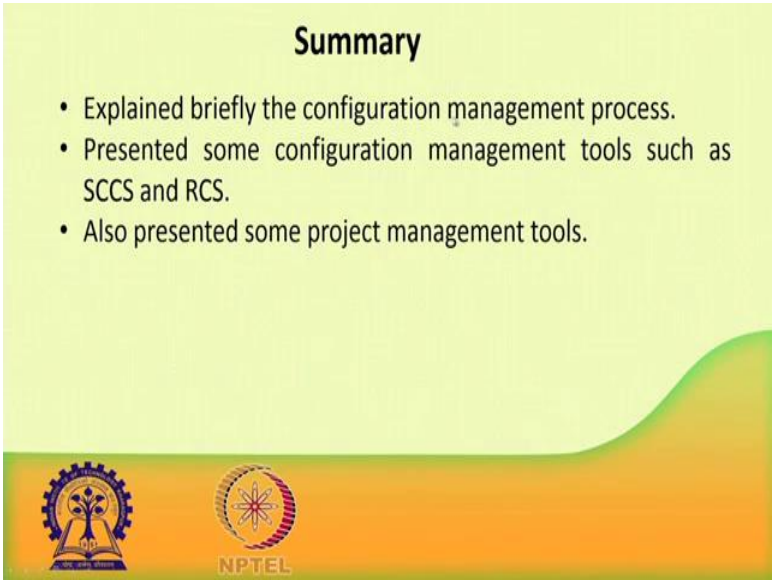

So, today we have discussed in this class the detailed steps for configuration management process. We have already seen, in this class we have seen the details of configuration management process. We have presented some configuration management tools such as SCCS, RCS etcetera. We have also presented some other project management tools such as Microsoft project and primavera and these what Gantt project Libre project etcetera.

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We have taken the represents from these books.

Thank you very much.

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Lecture - 48
Contract Management

Good morning. Today we will take up a new chapter on Contract Management.

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So, first we will see little bit introduction about managing the different contracts then we will discuss today what are the different types of contracts that one might have to what see in case of any organization.

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You know that many organizations they choose to obtain or to get their software externally, given their limited because their capability for developing new and reliable software is limited. So, getting it from outside from external suppliers it seems sensible it is viable.

These organizations think that, it is more cost effective that to employ outside software developers for new development. And then reduced group of their in house software developments staff members they can maintain and the support existing systems, they will try to purchase it from outside from a what third party from a supplier. Then they have some in house staff members less staff members, few staff members they can be deputed for maintaining and supporting the what, existing system.

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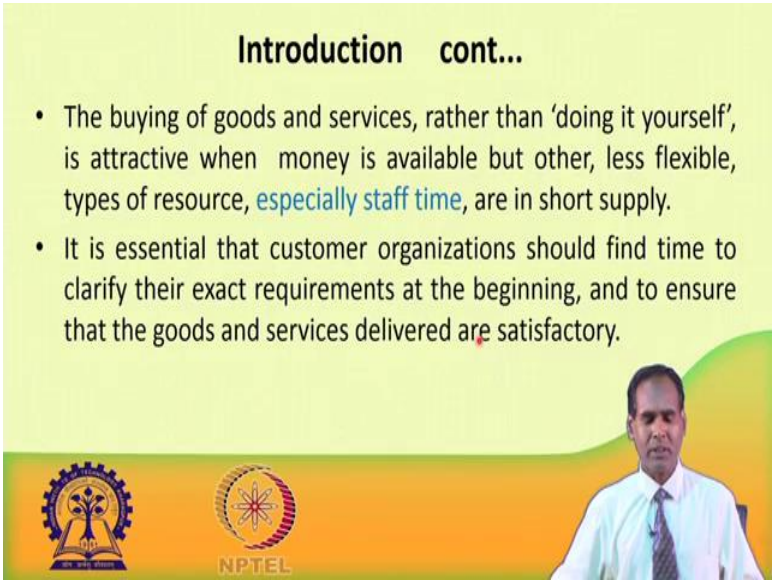

Buying of goods and services, rather than doing it yourself is attractive under the following situation. When money is available, but other resources especially staff time they are in short in short supply; you are very having very less number of staffs, but you are having money in that case it is better to buy the goods and services from the market from the suppliers. It is essential that the customer organizations they should find time to clarify their exact requirements at the beginning and to ensure that the goods and services delivered are satisfactory.

See even if you will give contract to a third party to develop software, it is you who know about the detailed requirements for your organization. So, that is why the customer organization they have to find out what are their exact requirements at the very beginning. And those requirements they have to inform to the developer, then only the developers can develop a good and good quality what product or a good quality service.

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Potential suppliers they are likely to be more accommodating before and contract is signed than afterwards. See this is normal tendency this suppliers they are likely to be a very much accommodating before any contract is signed but, once after signing these thing they are becoming very much casual they may not listen the complaints or request from the clients. So, especially if the contract is up for a fixed price then this problem will come accurately. So, thus as much as forethought and planning are needed with an acquisition project, same thing also it is also required with internal development, ok.

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So, as much forethought and planning as you require, in case of acquising a project same thing almost what similar forethought and planning is also required for internal development. Now let us see the different types of contracts, there are different classifications based on certain parameters. The first classification we will see so, according to this classification there are three types of contracts; one is the bespoke system, another is off the shelf package, another is customized off the self or popularly this is known as COTS software.

So, bespoke system is a system, which is created specially for one customer for one organization because, depending up on their requirements their requirements are very much specialized. So, the third party or the software, the developer, the supplier may create a special software for that customer only. So, bespoke system means these software they will be created or the facility the service that will be created specially for one customer. And, off the shelf package means just it is what generalize software are available in the market you just go to the market and brought as it is.

So, off the shelf package, but as it is also known as shrink wrapped software. Actually this off the shelf what component you see very much applicable in the case of hardware, you suppose you want to develop what a motherboard. So, what you have to do? Go to the market see what are the different components are available, purchase all those components and the ICs, then on a board you just assembly all those things.

So, here what we are doing? Going to the market bought the components as it is and then we are assembling. Similarly in the off the shelf of a package, so we will see several generalized what packages, software packages are available and we will bought them as it is, then we may what link them we will interface them. Then another category is customized off the shelf or COTS software, this is basically a core system is which is customized to meet the needs of the client.

So, this is a basically what the; it is a core module kind of thing and it is a core system all the fundamental facilities are there; and then you purchase it and customize it, according to your requirements according to what just to meet the needs of the client. You can have seen just government have does in this housing schemes they prepare only core houses and then, the purchaser when he will purchase the core house, then he will what customize it according to his needs. So, he may put marbles or he may put different what

different arrangements for A Cs, different arrangements for lightening etcetera. So, this is that similarly in customized off the shelf cots software, you will just go and purchase a core system then that can be customized to meet the needs of the client; either you can do yourself the customization or the supplier he will customize the software according to your own need. So, this is known as something customized of the self or COTS software.

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So, where an equipment is purchased, is referred to as a contract for the supply of goods ok. So, whenever an equipment or any hardware device it is purchased, normally it is referred to as a contract for this supply of goods; so, equipment is coming under the goods. But when you are going for software with the supply of software this is known as supplying a service, in case of what equipment we call it as supply of goods, but in case software we call it as supply of a service.

That means you have the what, vendor has to develop the software or write the software for you or it can be granting of a license so, or granting of a permission to the user, to use the software which remains in the ownership of the supplier. The supplier has the ownership of the software; he may give you license; he may give you permission to use this for a particular period of time depending on your contract. So, this is known as supply of service.

So, let us see again that if you are going to purchase the equipment, normally we call it has a what contract for the supply of goods and when you are going to purchase this software we call as supplying a service or granting a software license.

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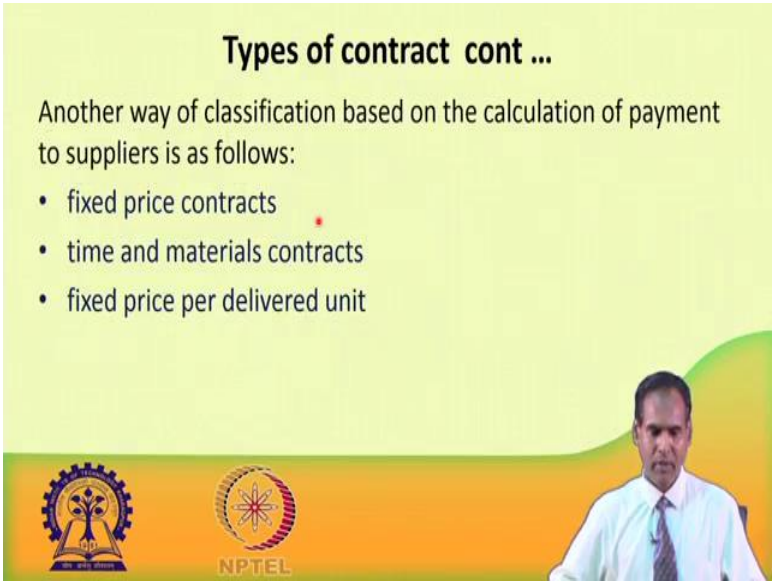

Now, let us see another classification; so, there is another way of classification, based on the calculation of a payment to the suppliers. How you are calculating the payment that has to made to the supplier. According to this there are three categories of a contracts fixed price contracts, time and material contracts, then fixed price per delivered unit.

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Let us first see about what is the fixed price contracts. So, in case of a fixed price contracts, the price I just name suggest the price is fixed in case of this fixed price contracts the price is fixed when the contract is signed. So, at the time of signing the contract, the price is fixed. And here the customer knows that if there are no changes, in the contract term. So, in the contract owner signing that are some terms are some causes are mentioned. So, if there are no changes in the contract terms in near future then this price, then they will pay on completion of this contract, there is no change in the price.

So, the customers requirement has to be fixed at the outset. So, in this type of contracts the customer at the very beginning he has to fix the what, requirement. The customer's requirement has to be fixed, at the outset that is the detailed requirements analysis ok. The detailed requirement analysis and specification it must have to be done before this signing the contract or at the beginning of the contract. And it will be very difficult that change it later on because this is based on this fixed price based on the requirements the customer who will sorry the developer who will quote the price.

So, once the development is under way, then the customer cannot change their requirement please see this is the drawback of this type of contract. Once the development is underway then the customer cannot change their requirements without renegotiating the price of the contract. If you want to change some of the terms, if you want to change some of the requirements, then you have to again renegotiate the price with the customer with the developer, he may charge some extra money for what incorporating the new requirements you are saying. So, this is something about the fixed price contracts.

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Let us see what are the advantages of the contracts, here the customer knows that what is the expenditure this budget is fixed I have to pay. Suppose 1 lakh rupees for the surface and supplier is motivated to be cost effective also this supplier is motivated to cost effective he will give what a known value of the price it will be cost effective.

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Let us see what are the drawbacks there are several disadvantages of this contract; first of all supplier will increase price to meet contingence absorb the risks. Because, supplier knows it is fixed, price is fixed later on he cannot what change or something. So and

during the development of projects several weeks may come; and in order to observe the weeks what you will do you will put some contingencies in each quotational quoted what quoted price. So, you will you will definitely try to increase the price to meet the contingencies or to absorb the risks.

And next is difficult to modify the requirements this I have already told you. In this type of contract the customer has to what perform the requirement analysis and inform all the requirements at the beginning later on you cannot change; if you will change then the what supplier will ask more money.

Cost of changes likely to be higher; so, once the what price is fixed and next you say that few changes then the developer will charge high price to you. So, cost of changes likely to be higher and threat to system quality; obviously, since the price is fixed then the developer will not be motivated to give what high quality service; because, you know whether I will put what kind of service my price is fixed. So, it is a threat to the quality of the system. So, these are some of the drawbacks or the drawbacks you can see from the books or from the internet sources.

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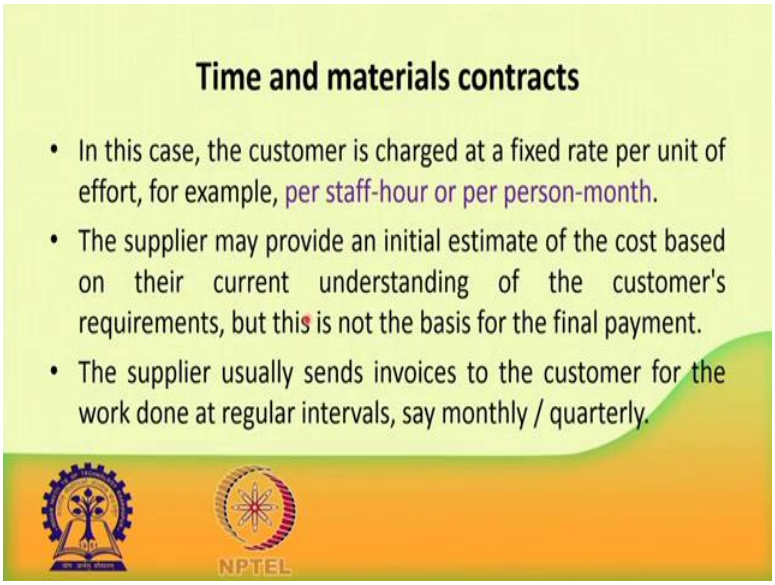

Then, next one is time and material contracts. So, in this case the customer is charged at a fixed rate per unit of effort, ok. So, in this kind of contract the customer is charged at a fixed rate, rate is fixed per what, per unit of effort, ok. So, in this case the customer is charged a at a fixed rate per unit; you put may be per staff hour or per person month.

And the supplier may provide an initial estimate of the cost ok, this initially the supplier will provide a base cost the supplier may provide on initial estimate of the cost based on what based on the current on the standing of the customers requirement.

So, looking at the what, customers requirement, the supplier may provide an initial estimate of the cost. But this is not the basis for the final payment, this is not the final payment the supplier then what will do he will usually send invoices to the customer for the work done at regular intervals say monthly or quarterly. So, he will send the supplier will send what bills invoices to the customer, for the work done for that period at regular intervals may be monthly or quarterly.

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Now, let us see what are the advantages of this approach? The advantages are that it is easy to change requirements.

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Now, since the price is not fixed, the charge is; the price is fixed or can say that the customer has to pay some fixed rate per unit of effort, may be per person hour or per person month this charge will vary now. So, the customer is charged at a fixed rate per unit of effort; so, if you want to add some new requirements. So, then again how much per staff per hour required or how much person month is required accordingly it will be multiplied by the price. So, there is no problem in what change changing the requirements. So, it is easy to change the requirements in case of this method, but and another advantage is that lack of price pressure can assist product quality.

Now, the customer the what, developer is not under pressure for this what price because, he knows not if the customer will say some new requirements they have to pay more. So, they will concentrate on improving the product quality which has not the case with the previous approach that is the fixed price contracts.

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Now, let us see the drawbacks. Drawback is customer liability; the customer absorbs all the risk in the fixed price contract. The supplier absorbs the risk and that is why he keeps some contingency and he may make more and he may demand more money. But in this case the risk will be absorbed by the customer, if any change customer in the requirements customer will say that these are the changes for that the developer will charge extra price.

So, all those risks that has to be what absorbed by the customer; so, that is why this is the customers liability. The customer absorbs all the risks associated with the a poorly defined or changing requirements or adding new requirements everything that has to be borne by the customer; and the lack of incentive for supplier to be cost effective. So, there is a very lack of incentives for the suppliers to be cost effective.

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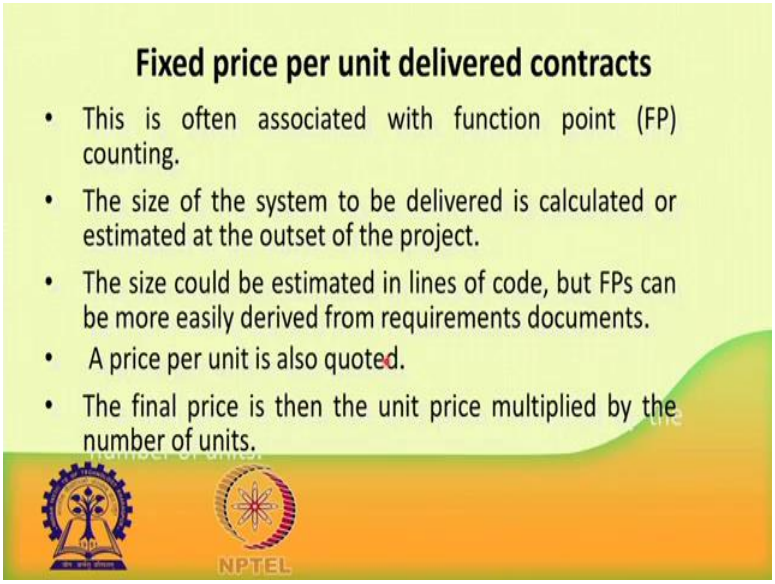

Then the next category is fixed price per unit delivered contracts. So, now, let us see what is meant by this; so, this is often associated with a function point counting; and here this say that fixed price per unit delivered contracts. And this approach is associated with a function point counting, while discussing about the cost estimation we have already discussed the function points LOCs (Refer Time: 15:55) etcetera.

So, this technique is associated with a function point counting; here the size of the system that has to delivered is calculated or estimated at the outset of the project. So, at the very beginning, what will the size of this system that has to delivered is estimated at the very beginning or at the outset of the project. The size could be estimated in terms of lines of code, you can estimate the size in terms of LOC or SLOC, but we know SLOC has several drawbacks. So, and function points removes some of the drawbacks of this SLOC. So, that is why a FPs or function points they can more easily derived from the what, users requirements or from the requirements documents. Then a price per unit is quoted, ok.

So, based on this function points a price per unit is quoted, the final price is then the unit price multiplied by the number of units. So, how many so now, price per unit is quoted; you see your software will require how many units then the total price will be the unit price multiplied by the total number units its very simple.

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Let us take a small example, suppose you have to develop a software and the charge is as follows. So, up to 2000 function points the design cost is 200 rupees per function point. And coding cost is a per function point 700 rupees and the total cost of development is 900 rupees per function point.

Similarly, the function points greater than 2000 and less than 2500 design cost is 240 coding cost per function point is 760 and say total each 1000 rupees. In that way for different ranges of function point counts, the unit prices for the design cost as well as the coding cost are given and from these the total cost is found out. Now let us see you suppose you require to develop may be a software, which has function points which is estimated that it will contain say 2700 of a function points, ok.

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Suppose the company that produce this table ok; so, here the company that produce this table. In fact, charges a higher fee per function point larger systems you can see, up to 2000 is less, then for higher function points you see the value is the cost is increasing.

Now, suppose you require a system and that has to be implemented and the supplier for implementing this system it estimates that there will 2700 FPs ok. Suppose you have to you require a system the you have told you requirements to the developer, the supplier; the supplier has estimated that the system that has to be implemented it will contain 2700 FPs. And then he will charge for this system based on this calculation based on these rates. So, now, 2700 FPs means you see you can first avail up to 2000 then for 500 another rate. Then for remaining 200 this slob third slob will be applied.

So, then what will be total price? The total charge that supplier will make or he will quote for the customer will be how much 2000 into because, now I have broken 2700 means in order to fit in this slob two for 2000 this charge will be applicable. For five hundred another this charge will be applicable and remaining 200 function points this charge will be applicable. So, now, this will be how much, 2000 into rate is 900 plus 500 into 1000 plus remaining 200 will be charged at the rate of 1040 this.

So, this will what, this will come to or this is coming to 25, 08,000. So, for developing a system which will be implemented at the customer site having or which has estimated which is estimated to contain 2700 function points will be approximately what priced 25,

08,000 based on what this fixed price per unit delivered counts. So, this is here we have chosen FP, we can also chose for LOC, but LOC has several drawbacks. So, you can use FP in place of LOC for this type of contract that is for fixed price per unit delivered contracts.

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The scope of the application can grow during development of course, during development you may require some more scope some more requirements. So, this scope of the application can grow during development it, but you see it would be unrealistic for a contractor to be asked to quote a single price for all the stages of a development project. So, if you are want software, so for the all the stages of the development project it is very much difficult the contractor cannot quote a what value it is unrealistic.

See, how can they estimate the construction effort needed when the requirements are not yet established; you do not know what will be the requirements. If requirements are evolving so, you the customer can the supplier cannot quote the full value for a single price for all the stages rather if this kind of things are there. Then for different what slobs for different how many function points are there, even if it is increasing it can be accommodated some to this ranges or still this can be extended.

So, if your software development consist of many stages then what you can do in since it is difficult for the supplier to give a single price; in this case one approach can be or one approach would be to negotiate a series of contracts. Then what you can do, you can

make a series of contracts each covering a different stage of system development for each stage some price you can negotiate. So, what could be the possible stages for each of the stages you can negotiate what some prices you can negotiate a series of contracts and then the prices can be charged to the developers sorry to the customers.

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What are the advantages in this approach? You can see that the customer understands ok, customer understanding of how price is calculated because see everything is given may be in the form of a table or what unit wise this is given. So, how the price is computed? It is very simple mathematics. So, the advantage that customer can understand how the price is calculated. So, comparably between different pricing schedules, he can compare easily between the different pricing schedules. Emerging functionality can be accounted for if a new functionality you want to add, that depending up on the what situation or the time it can be easily added. And the charge it can be easily calculated by this supplier and supplier incentive to be cost effective of course, the supplier incentive here it will be much more cost effective.

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But the drawbacks are that difficulties with software size measurement, may need independent FP counter. So, how to what measure the software size? We have seen the advantage or drawbacks of LOC; and how the drawbacks of LOC can be removed we have seen also in FPs, but FPs also have some drawbacks. So, that is why difficult it will find some difficulties while software measuring the software size you may need some independent function point counter.

Similarly, changing as oppose to new requirements how do you charge see if you add a new requirements its easy we can use the per unit. So, or this range can also be extended. So, adding new requirements it is easy yeah; that means, the supplier can easily calculate the extra charge. But what is about changing requirements? The requirements is already there, you want to little bit changes then how you can calculate the charge. So, this is difficult in this approach.

So, these are the different what this is these are the different contracts based on another classification ok. So, this is a different classification of contracts or these are the different types of contracts based on the other another classification how the payment is calculated.

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So, there is still another type of classification, another way of classification based on the approach that is used in contract selection is as follows. So, this classification is based on the approach that is used for contractor selection. So, number one is open tendering process, then restricted tendering process and then negotiated procedure or this negotiated procedure we call it as single tender or singly quotation process.

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Very simple the open tendering process so, here any supplier can bid the response to the invitation to the tender ok. So, basically first the client he will float invitation to tender or

he will what display he will advertise some request for proposals popularly know as RFP. So, any supply in open tendering process any supplier can what submit the bid in response to the invitation to tender or request for proposal. Here all tenders must be evaluated in the same way that is one important thing, how many tenders have submitted by different what bidders? So, all the tenders must be evaluated in the same way. And the government bodies may have to do this by local or international law.

So, what are the government bodies? They have to what follow this open tendering process while purchasing equipments or what services by this way. And the government bodies may have to follow this way by what observing the local and international laws; with a major project this evaluation process can be time consuming and expensive. So, this is what drawback that for major projects this evaluation process can be very much time consuming and expensive. Because, for publication of these what RFP or what advertisement you have to give some days minimum one month then for what you have to evaluate technical proposals then cost evaluation then you have to finally, award the contract unlike this there is according to the laws that is the timing gaps are given. So, this will take huge amount of time.

So, if you do not have sufficient amount of time then this method may not be suitable. Similarly where the, what client is a public body, an open input tendering process may be compulsory. So, for normally government organizations and public organizations or public bodies open tendering process normally it is a compulsory.

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Then we will see the next one that is the restricted tendering process. So, in this case the bids are received only from those suppliers who have been specifically invited. So, please mark the difference, here anybody else can supply the bid can quote the bid. But here the bids are received only from those suppliers who have been specifically invited.

You know in your locality also say these are the 5 or 10 what suppliers. You will send your RFP request for bids only to those suppliers; and when you receive only those ten to whom you have send this request for proposal their bid will consider others cannot apply. So, that is why this is a you can say that special case of open tendering process where everybody cannot submit, but to whom we have to invited to submit the bid they can only submit and you can evaluate their bids.

Unlike the open tendering process at any time the customer can reduce the number of suppliers being considered at any stage. So, in open tendering you cannot reduce the number of suppliers, but in open what restriction tendering process that is why the name is restricted. At any time you can restrict you can reduce the number of suppliers which are being considered any stage.

So, normally you have to say minimum perhaps six number of what these request you have to send you have to invite the bids from minimum 6 number of what firms or suppliers. Then how many are coming then you have to see you have to evaluate and then you have to what select the appropriate supplier for offering the contract.

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Then the last one is know as negotiated procedure or single tendering process. So, because see open or restricted tendering process may not be suitable in some particular sets of circumstances particularly when time is not there let us take two examples here.

Suppose there is a fire that has occurred and that in your computer center and that has destroyed some of the ICT equipments. Then what is now your major objective the major objective is how to get replacement the equipment up and running as quick as possible.

And there may not be sufficient time to embark on a lengthy tendering process; I have only told you these what open tendering or restricted tendering this take what a long amount of time because that time gap you have to maintain as per rule and you these things you require urgently. So, in this case you do not have time to go for this lengthy tendering process. So, what to do? So, the solution is this single tendering process or negotiated procedure.

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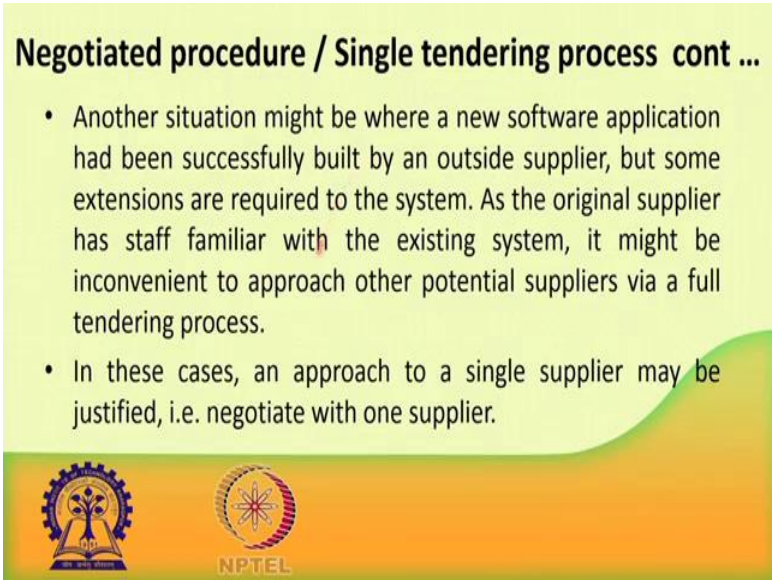

Let us quickly take another example, where a new software application you have just recently built or you have just recently you have obtained it which was supplied by outside supplier. Now you require some extensions to this newly purchased system. So, you know as the original supplier has this staff similar with or familiar with the existing system they can do it easily.

But it will go no I will go for the tendering process, then you see even if you get what firm with less price, but they are not conversion they are not familiar with the software. So, you will get so many technical problems, so that is why in this case also it is inconvenient to approach other potential suppliers by a filtering process. So, the solution is you should go for the single tendering process, who has developed who has supplied the what, software to you should approach that one following this single tendering process.

So, but in this cases so, in this cases an approach to a single supplier may be justified that is you have to negotiate with only one supplier. So, that is why you have a single tendering process may be here from where you have supplier, you have purchased this supplier from which supplier you may negotiate with that supplier with some original price.

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So, negotiated procedure or the single tendering process there is a drawback of this. So; however, approaching a single supplier a could expose the customer to charges of favoritism; you may get complaints that you have shown favoritism to that particular single person and you have awarded the contract so, you may get that sort of criticism.

So, whenever you are going for single tendering process, you have to give full justification, clear justification, proper justification, why you have gone for single tendering and two such examples I have already shown. So, with a proper justification you can go for single tendering process, so that the drawbacks of what restricted tendering or open tendering process can be avoided mainly the timing is used out. If there is no time very less amount of time you should go for single tendering process with a clear justification or with a proper justification.

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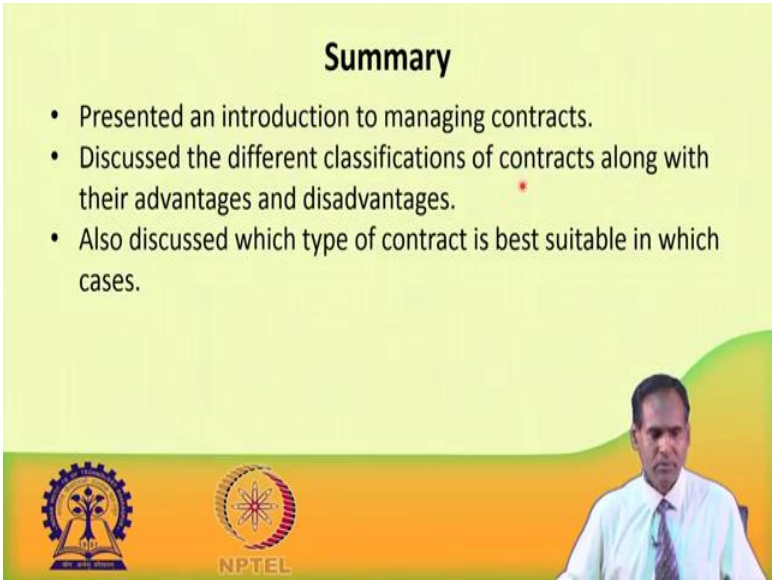

So, in this class we have discussed a little bit of introduction to managing contracts. We have discussed the different classifications of contracts along with their advantages and disadvantages. We have also discussed which type of contract is best suitable under what circumstances.

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So, this is the reference we have taken this material; and then we will stop here.

Thank you very much.

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Lecture - 49
Contract Management (Contd.)

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Good morning, so, now, let us see the remaining portion of this contract management. We will first see what are the different stages in placing a contract, then we will see some of the terms associated with what contract tendering process or the contract placement process so, we will see the contract checklist. Then we will see this contract management how to manage the contracts.

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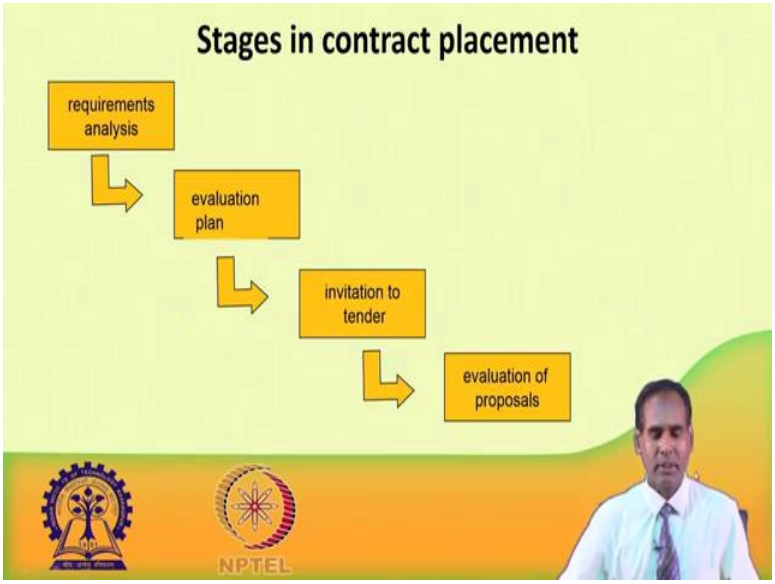

These are the different stages in contract placement; first you have to analyze what are your requirements then only you can initiate the what contract process then you will think of how to evaluate the contracts. If you will invite the contracts how to evaluate the contract you have to think of and then after these after making this plan then you can send invitation to tenders or you can send the RF request for proposals RFCs and then you will receive RFCs or this RFCs require request for proposals from different vendors and after receiving then after receiving the proposals, you have to determine what mechanism you will use to evaluate the different proposals.

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So, now let us explain all those steps in detail all those steps in detail. So, first one I have already told you before thinking of something for giving some contract you have to yourself internally you have to analyze the requirements. After analyzing the requirements, you specify what exactly your requirements are. You document you prepare a document for your requirements, you write down all the requirements in a document that we call as SRS software requirements document.

So, normally this requirements document should contain these sections, there should be introduction section that should contain the introduction to this proposal or what are for what is your problem that needs to be automated then description of the existing system and the current environment. Then what is the future strategy or what are the future strategy or plans.

One section should be there the system requirements if you will develop if you will get this what system what are the requirements out of this something might be mandatory compulsory and something might be desirable and then what are the possible deadlines when you are expecting that the firm will give you the product; may be the whole product by one date or phase wise we can give the products in incremental manner and then in any additional information required from the bidders that also we have to put. So, in the requirements document so for the contract you have to prepare the requirements document and the requirement document must contain these sections.

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So, the requirements will include what? The requirements that you will mention in the requirements document should include functions in software or what are the functions of the software within what and what will the inputs and outputs with all necessary inputs and outputs that should be included. Then standards to be adhered to the software must meet which kind of standard; ISO standard or that WTO standard World Trade Organization standard etcetera. So, the requirements should also contain these standards to be adhered to. Then other applications with which software is to be compatible your once after getting this application it should be what compatible with what other software. Suppose you are trying to what automate.

So, previously this railway reservation system was manually and when it has a given a proposal for automating, it you have to see it has to be compatible with what other things might be see this railway reservation system while you are booking this has to be compatible with this gateway of the banks because the money that you have to pay it has to be done through bank gateway. So, it has to be compatible with that. So, similarly quality requirements or the response times so, what kind of quality you require, how much the response time should be and what should be the throughput all those things must be clearly mentioned in the requirements document.

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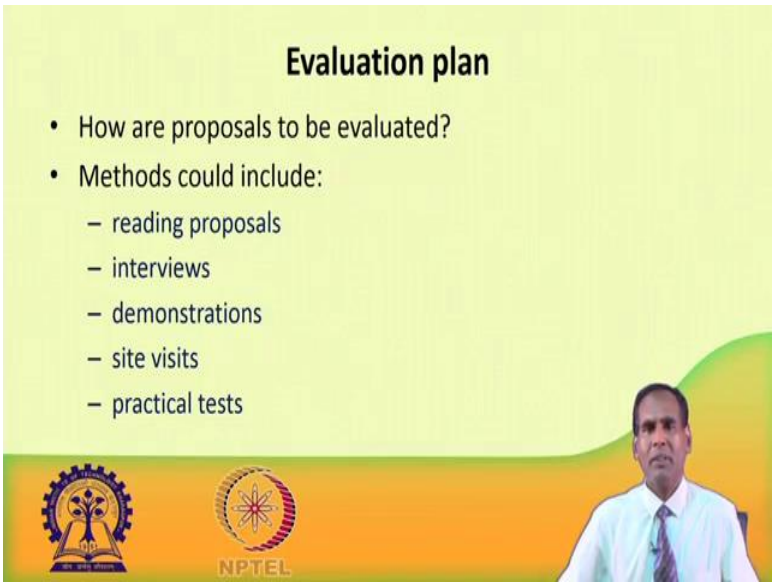

Then second phase is evaluation plan. You have to think of when the proposals will come from different vendors how to evaluate. So, you have to think how are the

proposals to be evaluated. So, there are different methods that you can use for evaluating the what proposals, you have to plan for them, like just reading the proposals see which one is the best proposal. You can conduct some interviews invite all the what suppliers take an interview from with each of them or demonstrations if you are just purchasing a new hardware or a software call the parties the software the suppliers say them give a demonstration each may be all are present or you may confidentially by one by one you call and let them give their demonstrations.

Another one is site visits what has to be automated. So, what you can do you an tell them that site they may visit your site or you can visit their sites what kind of things they are doing. Based on that you can evaluate and last one is practical test you can tell them they should what give some tests somewhat online test can online demonstration kind of thing they will come to your place you say that we want this give that say test and shows how this is working. So, those kind of things those kinds of methods can be used to evaluate the proposal submitted by different vendors.

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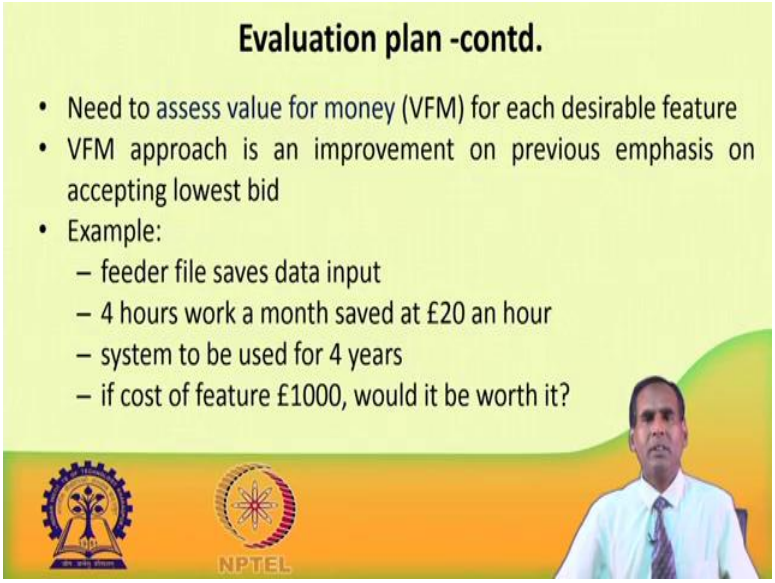

So, need to assess, so, in case of the evaluation plan while you are making the evaluation plan while you are making the evaluation plan there is a need to assess value for money. There is a need to assess the value for money or you what you call VFM for each desirable feature. So, for each desirable feature you have to there is a need to assess the value for money and this actually VFM approach is an improvement on previous

emphasis on accepting lowest bid. You see earlier days how the proposals are evaluated just financially. The supplier who has quoted the lowest amount that was selected he was awarded the contract he was getting the contract.

So, VFM approach is an improvement of this traditional approach which was based on what this lowest bid and some of the examples are they are like a feeder files saves the data input or 4 hours work a month saved if 4 hours are worked a month, then it may save some 20 pound an hour or. So, system to be used for 4 years that is a constraint you put the system to be used for 4 years or if the cost of a feature is 1000 pound then would it be worth it. So, like this kind of considerations it can take. So, basically this evaluation approach is an improvement of the previous approach which was based on emphasizing the lowest bid.

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Next step is invitation to tender. So, after what deciding that this plan will use for evaluating the proposals then you can what invite a tenders from different vendors. So, note that the bidder is making an offer in response to ITT. So, when we will issue ITT invitation to tender or the we call it as RFP request for proposal. So, in response to the ITT or the response or this RFP what request for proposals, then bidders different firms they can what send offers, they can quote, they can send proposals to you and acceptance of offer creates a contracts they then when a when these proposals will be evaluated after

that which will be what selected. So, that will be awarded that formula would be contract acceptance of the offer creates a contract customer may need further information.

So, what information the firms have mentioned in that proposals that may not be sufficient. So, the customer may need further information. In that case the supplier has to be informed and what information you require more, they have to give those information and then the evaluation of the proposals may take place. The disadvantage that different technical solutions to the same problem may exist. So, for what do you want to solve, the same problem for the same problem different vendors may provide different technical solutions. Then how to evaluate?

It is little bit difficult to evaluate, because for the same problem different technical solutions are quoted by different what vendors. So, but in case of lowest what this lowest bid this problem does not arise, which is having the lowest amount he will award the he will be awarded the contract. But in this case where for the same problem different technical solutions are there it is difficult to evaluate.

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Then in during this what invitation to tender, proposals will be submitted and we have to what know one important document called as Memoranda of Agreement or in short, we call as a MoA. So, here the customer asks for technical proposals. So, the proposal actually can be divided into two parts; one the technical flow part another the financial part. So, the customer may ask for the technical proposals.

Then the technical proposals will be submitted by the firms with these technical proposals are examined and discussed, may be examined through a committee internal committee may be some experts you may involved in that committee, then the agreed technical solution in the MoA. So, the technical solution which is agreed by all that will serve as the MoA that is the MoA. Then tenders are then requested from suppliers based on MoA.

So, after this preparing the technical proposal which is now agreed by all and this will serve as the MoA, then technic the tenders are requested from suppliers. Then you advertise it tenders are requested from suppliers based on this MoA, because this contains all the technical details, and this is these are agreed by all then the tenders are judged on price.

So, because now technical proposals are now fixed, they are frozen now the tenders will be judged based on the price, because every vendor has to what quote these technical proposals. They have to provide this technical what details as mentioned in the MoA, then tenders are requested from the suppliers based on the MoA and then after receiving the tenders they are judged based on the price, because now as a uniformity on the technical proposals technical proposal or technical contents have been frozen now. So, then the fee could be paid for technical proposals by the customer, but so, what charge or what amount has been spent in preparing the technical proposal that amount may be paid by what the customer.

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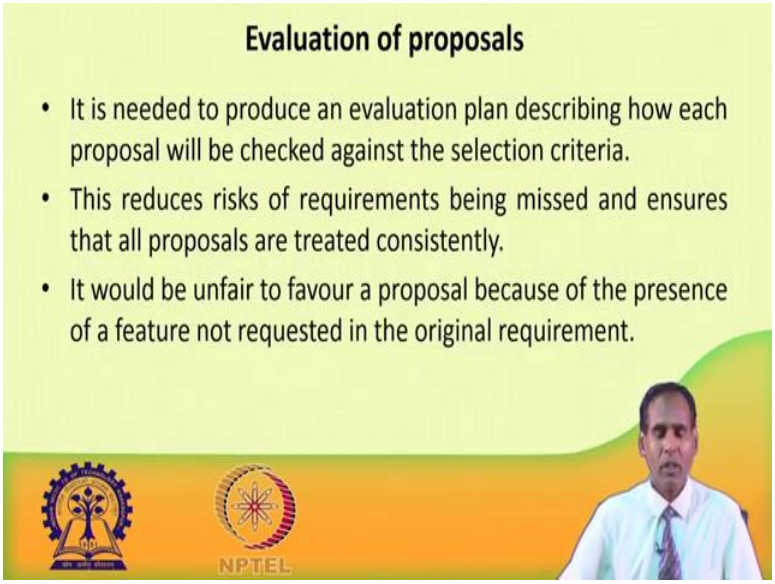

The last step is evaluation of proposals. So, here it is needed to produce an evaluation plan describing how each proposal will be checked against the selection criteria. So, based on what criteria you will select that which is the best firm and the contract will be awarded to that firm. So, that is why it is needed to produce an evaluation plan describing how each proposal you can check, how each proposal will be checked against the selection criteria. So, this will reduce the risks of requirements being missed and ensures that all proposals are treated consistently. So, if you will do you will finalize the selection criteria this will reduce the risk of any requirements being missed and it will ensure that the proposals are treated consistently. There is no consistent in these cases.

It will be unfair to favour a proposal, because of the presence of a feature not requested in the original requirement. So, if in the original requirement or in the MoA one what a feature it is not there, but somebody a quotation a supplier has mentioned that feature and based on that if that proposed feature if you are selecting that vendor then this will be unfair, because it has not mentioned in the original requirement or in the MoA. So, that is why it will be unfair to favour a proposal, because that a feature is not present in the original requirement and since supplier has mentioned in a proposal you have selected him it will be unfair.

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An application could be of three types. We have already seen in the last class that an application could be bespoke. That means, for a particular user with the off the shelf may be generalized, or it is off the shelf, but it is customized. In the case of off the shelf packages, the software itself could be evaluated and it might be possible to combine what to combine some of the evaluation with the acceptance testing.

But with the bespoke development it would be a proposal, that is evaluated while the COTS this software it may involve the elements of both; that means, elements of bespoke as well as elements of the what off the shelf package ok. Thus, different approaches would be needed because COTS approach uses the best features the features of bespoke and off the shelf methods. So, that is why this so, that is why thus different approaches would be needed ok.

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Now, let us see what are the steps of the evaluation of the proposals. how you will evaluate the proposals, what are the what is the process. The process of evaluation may include the followings; first you have to scrutinize the submitted proposal documents.

Then you may have to conduct interview with the suppliers or their representatives, then you may request them to if it is not clear they may give some demonstrations, they may do some what site visits or he may do go for site visits and also he may go for the practical test, you say them, you show us all those things here practically. So, he may getting conduct some practical test and based on this practical test who can show the best results you can award the contract to them.

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So, the proposal documents provided by the suppliers they can be scrutinized to see if they contain all the features as mentioned in the MoA, which are satisfying all the original requirements. If they have missed some of the points then clarification might be sought over certain points. It is important to get a written agreed record of these clarifications, do not accept something verbally or orally.

It has to be what agreed and recorded the customer might take some initiative by taking the minutes of the meetings the customer what he might take the initiative by what preparing or taking minutes of the meetings and then writing afterwards to the suppliers to get them to confirm their accuracy. So, after recording the minutes, you can send them to the suppliers in order to get them to confirm their accuracy.

A supplier could in what the final contract document attempt to exclude any commitment he may what attempt to exclude some of the commitment to any representations made in the pre contract negotiation. That is the term and terms of the contract needed to be scrutinized for this, but where there is an existing product it should go for demonstration.

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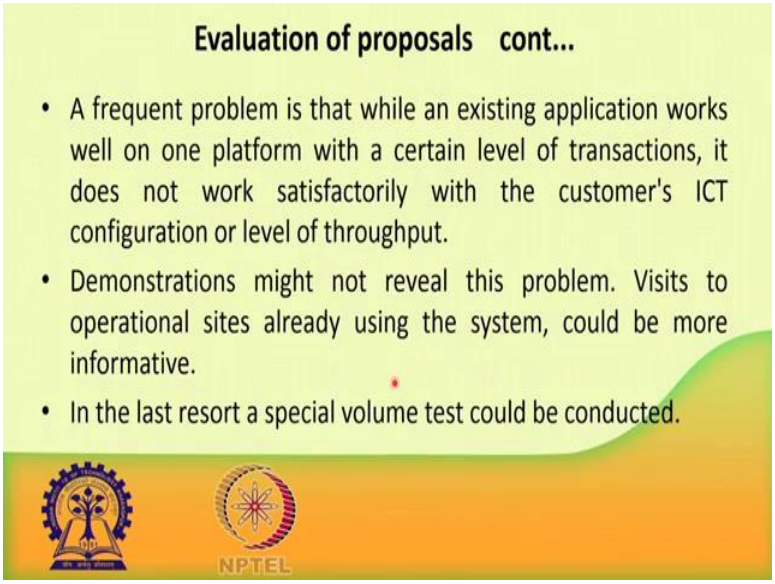

A frequent problem which that while an existing application works well on one platform with a certain level of transactions it does not work satisfactorily with the customers ICT configuration or level of throughput. So, one what existing application, it is working on one platform may be windows, but while you are taking to the customer site and that is based on Unix. So, that what application may not work or the throughput may not be same or the response time may not be same.

So, demonstrations might not reveal this problem. So, these kind of things demonstration might not this kind of problem problems demonstrations might not reveal. So, these things can be what better what evaluated if you will make visits to the operational sites, visits to operational sites already using the this system could be more informative and you can get more idea which will help you in selecting the firm. In the last resort a special volume test could be conducted. So, similarly at last a volume test you should conduct and then you can what see which proposal or which yes which proposal passes this volume test and that may be selected for awarding the contract.

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So finally, a decision has to be made to award the contract to a supplier. The reason for following so, now, let us see why will follow such a structured and objective approach to do the evaluation, because we want to demonstrate that the decision that has been made that is impartial that is unbiased. A project manager please remember a project manager he cannot be expected to be legal expert.

It needs advice. So, there might be some legal what affairs. So, the project manager cannot be a legal expert he needs advice but ensure that the contract reflects the true requirements and expectations of supplier and client. So, whatever the contract you are finally, awarding that should reflect the true requirements of the client and the expectations of both the suppliers and the clients.

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So, let us quickly see some of the what contract checklist or the some of the terms associated with the tendering process. So, you must give proper definition to the terms such as what do what is who is a supplier, what does what is meant by a supplier, who is an user, who what is an application proper definition should be there in the tender. Similarly the form of agreement you must specify for example, is this a contract for a sale or for a lease kind of thing or this is a license, just a permission to use a software application, can the license be transferred to another what party or another what person. So, all those forms of agreement must be what properly explained.

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Then you can show that another term is goods and services to be supplied. So, as an organization you might require some goods or services. So, will see two things that equipment and software to be supplied. So, if you are what hiring or if you are purchasing some equipment and software, then this should include one actual list of the individual pieces of equipment; what are the individual pieces of equipment to be delivered? Do not write just server exact model number with exact specification and the different what components of the server that has to be clearly mentioned. So, it has to be complete with this specific module number version numbers etcetera.

So, similarly if you are going for what purchasing or hiring some services, this should cover such things as whether the service requires some training and whether that will be provided by the firm or not. Documentation, installation, conversion of existing files during automation, maintenance agreements, for how many years the supplier will provide free maintenance may be 1 year 2 year 3 year etcetera. Similarly, warranty and transitional insurance arrangements for how many years the firm will be at the insurance. So, all those things must be specified while you are going for what purchasing goods and services.

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Similarly, what is who is the owner of the software ownership of the software that also has to be clearly mentioned, can the client sell the software to others is it possible, can the software can the supplier sell software to others, could specify that customer has

exclusive use, if he is yes you have to see otherwise see there are two possible whether the supplier can sell or cannot sell.

So, it could specify that the customer has exclusive use or not. Does the supplier retain the copyright or it is given to the customer that has to be also clearly mentioned. Where supplier retains the source code may be a problem if the supplier returns the source code may be a problem because if the supplier goes out of the business to circumvent a copy of code could be deposited with an escrow service.

So, this is a serious problem if now your relationship good and if next time the supplier does not give the contract then he will create problem and the source code with him you are just having the using the application. So, you cannot what do your jobs you will be in serious problem, because any maintenance etcetera you cannot easily do because the source code is lying with the supplier.

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Similarly, environment for example, where equipment has to be installed what is the environment, who is responsible for various aspects of the site preparation like the power supply back up facilitation etcetera this is who is responsible that has to be clearly mentioned. Similarly customer commitments for example, providing access supplying information and approving the intermediate results etcetera. So, customer what are the customer commitments that is also should be properly mentioned in the contracts.

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Acceptance procedures, that good practice it accepted delivered system only after user acceptance test. So, you have to carry out the user acceptance test. So, when the user acceptance test will be there, who will be who will perform the user acceptance test all those things should be clearly mentioned all those things should be clearly mentioned in the contract.

Part of the contract would specify such details as the time that the customer will have to conduct the tests ok. So, part of the contract would specify such details as the time that the customer will have to what conduct the test, deliverables upon with the acceptance test they depend on and the procedure for signing of the testing what is the procedure for signing of the testing as completed all those things must be clearly mentioned in the contracts.

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Similarly what standards to be met? The product should met ISO standards or the product should met that WTO standard that World Trade Organization standard etcetera. That also has to be clearly mentioned in the contracts. Then project and the quality management, the arrangements for the management of the project must be agreed ok. The arrangements for the management of the project must be agreed by the what client as well as the supplier.

These include the frequency and nature of for the progress meetings and the progress information to be supplied to the customer. So, all those things should be clearly mentioned there and timetable of activities like this will provide a schedule of when the key parts of the project should be completed. The different milestones when they will be achieved a clear-cut timetable a what, unambiguous time table a complete time table of this different activities should be mentioned in the contract.

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Then on the price and payment method; obviously, the price is the most important item in your contract. So, what also needs to be arranged is when the payments are to be made. Whether the payment will be made a monthly or whether the payment will be paid payment will be made quarterly or half yearly or yearly or when some milestone is achieved. Some deliverable they have made then only we will make the payment or on increment basis that has to be clearly mentioned. Payments may be tied to completion of some specific task say when task A is completed then you release say 10 percent when task B is released completed then what release another 20 percent like this. So, payments can also be tied to the completion of some specific task or events.

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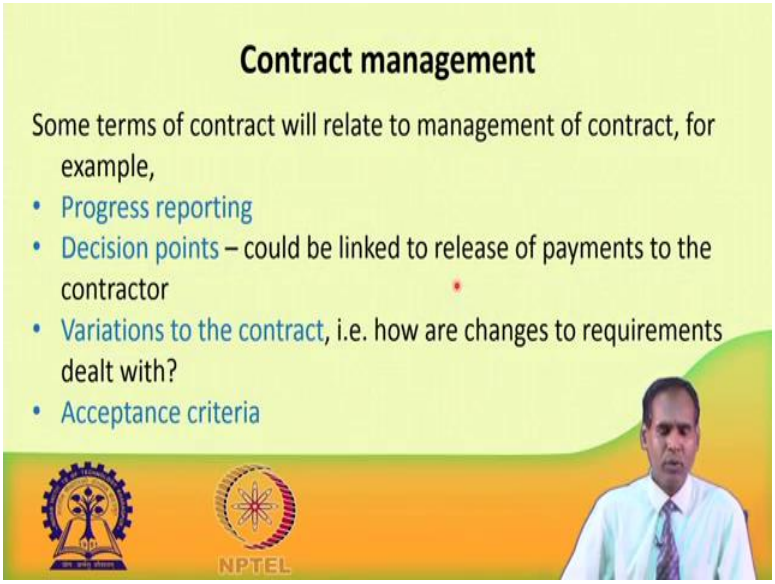

Then we will quickly see about the contract management. So, some terms of contract will relate to management of a contract ok. So, some terms of the contract will relate to the management of the contract for example, about progress reporting, how frequently you can report the progress. That should be clearly mentioned.

Some decision points decision points may be some of these what you can say decision points could be some of the milestones, it could be linked to release of payments to the contractor; that yes testing is completed release say 50 percent, then installation is completed another 25 percent like this. So, decision points or these what we can say milestones they could be linked to release of payments to the contractor. So, when the variations to the contract that is how are changes to requirements dealt with.

So, these were the original what requirements mentioned, after some days some changes are made to requirements. So, how these changes to the requirements will be dealt with by these what supplier that also should be mentioned. So, that is a part of the contract management. Similarly what should be the acceptance criteria as I have already told you should go for user acceptance test and if it is successful then you can accept you can accept otherwise you can reject. So, what is the acceptance criteria that also should be clearly mentioned.

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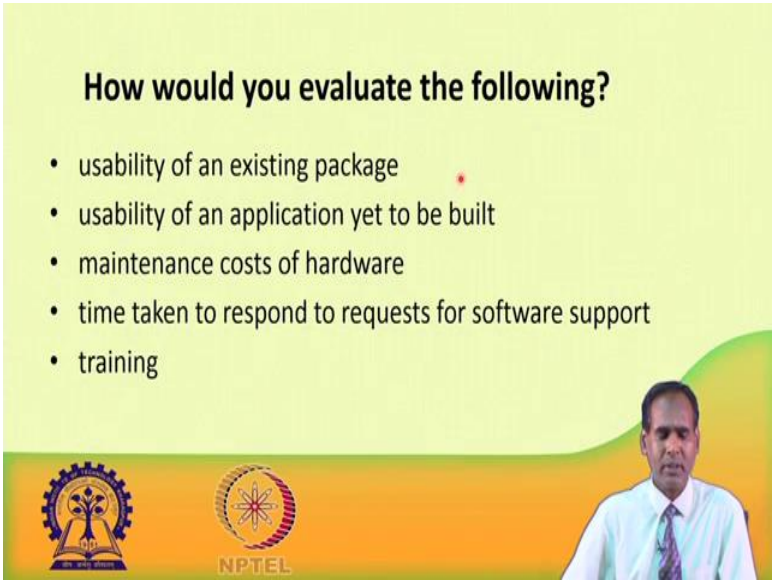

How do you evaluate the followings like, what is the usability of the existing package? A package is there what is adjusting what is the usability of the existing package? Similarly, usability of an application yet to be built an application it is first application is existing what is the usability? Similarly, an application is yet to be developed, what will be its usability? Maintenance cost of hardware this software will not run alone it will request some hardware. So, what maintenance cost will be required for the hardware?

Similarly, what time will be taken to respond to requests for software support? So, you are facing some problems and you are informing the supplier who was delivered the software how much time they will take to respond to your query to a request for providing the details what service or the support. So, that time also is very much important.

Similarly training in order you are purchasing a new software your end users do not know how to run it so, you require some training. So, that training should be include as a part of the what contract they should give free training to the users and that fee they might be charge in the what their proposal, but training should be free for what educating the users how to use the software. So, also all those things you should evaluate while what looking at the contracts while evaluating the contracts.

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Contracts should the contracts should include some agreement about how the customer supplier relationship is to be managed is very much important the contracts, it should include the agreements that how what is what should be the relationship between customer and supply relationships and how it will manage.

For example, if decision points or milestones and what should be the decision points based on where the payment will be released. Some of the quality reviews what should be the quality of the software or the intermediate products there as given by the supplier when I change to requirements if, because in what organization changes will must occur. So, if there is a for every change, they will say that we require money very much difficult. So, minor changes they should do it freely and whenever there will be some major changes how they will charge the price. So, that also should be clearly mentioned in the agreement.

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So, today we have discussed the different stages in contract placement. Then we have discussed the typical terms of a contract, the checklist of the what contracts. We have also explained some terms of a contract which relate to the management of contract. So, these are the references we have taken.

Thank you very much.

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Lecture - 50
Project Close Out

Good afternoon. Today, we will discuss the last phase of a Software Project Management cycle that is Project Close Out or project closure; how to close the project.

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Today, we will discuss first the little bit introduction of project closure, then what are the possible reasons for project closure, then what is the process to be followed for closing a project, then performing a financial closure and finally, publishing the project closure report.

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So, you know that every project must, it will come to an end, sometimes today or another day. This is the last phase of a project management lifecycle. It is the responsibility of the project manager to decide the appropriate time to close a project, when the project will be closed that is the responsibility of the project manager to decide.

Normally, project closure activities can be divided into two types administrative closure and contract closure. Let us see what is administrative closure.

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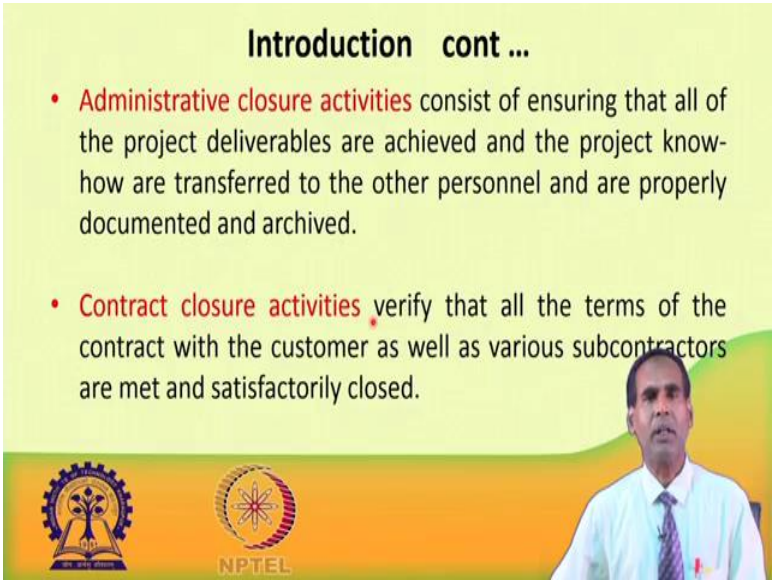

Administrative closure activities; they consist of ensuring that all the project deliverables they are achieved and the project know-how are transferred to the other personnel and are properly documented and archived ok. As its name suggest administrative closure activities so, these activities consist of ensuring that all the project deliverables such as requirement specifications, your design documents codes, test plans and the modules tested modules, etcetera. They are properly archived and the project know-how; that means, the project what knowledge project details, etcetera are these are transferred to the other personnel, which are associated with project and these project know-how they are properly documented and archived.

Then the other activity is known as the contract closure activities. So, these activities verify that all the terms and conditions of the contract with the customer as well as various subcontractors they are met. All those terms and conditions they have been successfully met, and the project and their satisfactory closed.

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Now, let us see why we require to close a project? What are the possible reasons for project closure? There are two main reasons for closing a project; one, all the project goals the objectives they have been successfully accomplished. So, the project is completed, now we have to close the project, other one, that it has been found that the project is unlikely to achieve its stated objectives. What are the possible mentioned objectives or the goals it is very much difficult to achieve them it is unlikely that we can

achieve them and hence, we have to prematurely terminate the project. So, these are the two possible reasons why a project needs a closure.

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Now, let us see I have already told you there are two reasons; one that project is successfully completed, another is project is not successfully completed and it is unlikely that it will achieve its mentioned objectives and goals and hence, we have to prematurely terminate. Let us see, some of the possible reasons, why we have to prematurely terminate the project before its due date.

Many reasons are there to prematurely terminating a project, but few important reasons we will discuss below. Number one lack of resources, the project requires some resources and those resources we do not have right now. We have started the project, but after the starting project, we fail that we cannot, what accommodate the resources might be due to financial deficits or some technical problems we do not have the lack of resources, we do not have the sufficient man power etcetera. So, in that case we have to prematurely terminate the project.

Similarly, changed business need of the customer; the customer was requiring something, but in the meanwhile after the starting of the project there is no need. So, that because suppose, he wants to the customer has given a project to develop a inventory management system. And after posing the status then or after posing the after starting project, they felt that we will give this development of the inventory control system to a

third party then so, the business need does not now, it is does not occur the it is there is a change in the business need of the customer. So, in that case you have to forcibly terminate the project.

Number three the perceived benefits accruing from the project may remain what they may not no longer remain valid. For example see while starting the project you are expecting that this much benefits, I will get; if I will what automate it. But in the meanwhile several competitors came into the market and some competition products they come into the market. So, you are afraid of that this much benefit I may not get. So, in that case this, what project has to be prematurely terminated.

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Similarly, next possible reason is incomplete requirements, if you do not know your all the requirements; requirements are incomplete then even if in assuming that we can clearly state the requirements within say few days after the starting of the project. We have started the project, but even if some days have passed, we are not able to define the requirements; we are not able to clear what the customer is not able to clearly specify the requirements.

So, in that case the project has to be prematurely terminated. Similarly, changes to the regulatory what policies. So, your organization is bounded by the rules and policies regulations of a government and if a government, while you have started there was no such what binding policies, but after the project has been started government has

imposed some policies, some restrictions. Such as suppose, you have planned to develop a product which is based on satellite communication and then that we will release to the market. And suppose, government has imposed a rule no, nobody can use a satellite communication based product in this country, then what will happen? You have to forcibly, prematurely terminate the project even if you have started it.

Then some key technologies used in the project have a become obsolete during the project execution. What are the key technologies during the starting of the project what key technology are there and after starting of the project after few months or few years. Now, new technologies have come up and these then techniques now, they are being outdated. For let us take an example; you have suppose started with development of a project, we using 4 G. Now, you see now after using few days 5 G is coming. So, once 5 G will come to the market then; obviously, 4 G will become obsolete and if I have started your project keeping in a mind that I will use 4 G then that will be obsolete and nobody will purchase your product. So, you have to forcibly prematurely terminate the project.

Similarly, risks becoming unacceptably high, while you have started the project, you have known that some minor risks, what lower risks will come. But after starting the project within few days you see the risks are becoming very much high, may be technical risks are there or financial risks are there or operational risks are there. So, you cannot tolerate all those risks. So, in that case also you have to terminate the project. So, these are some of the possible reasons, why you have to terminate a project before it's time; that means, you have to prematurely terminate a project.

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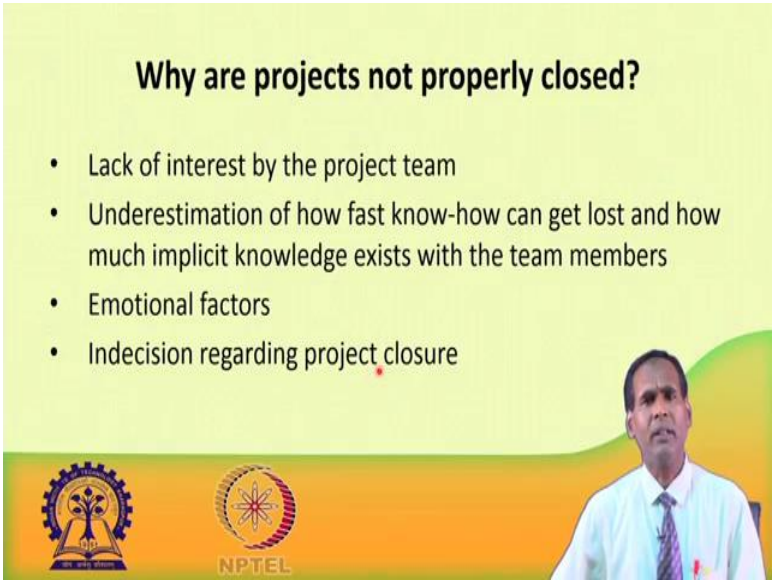

Then let us see why it is required to properly close the program or projects and why people are not properly closing the projects? Why? Due to somewhat reasons might be due to negligence or so they are not properly closing, after these we will see, what are the problems it will not properly close the program.

So, let us first see why are projects not properly closed number one lack of interest by the project team members. The project team members, after either successfully completing the project or prematurely terminating the project, they do not have any interest in that project. So, due to lack of interest they do not officially what or properly close the what, project. Then under estimation of how first know how can get lost and how much implicit knowledge exists with the team members.

See finally, after developing the project you gain some knowledge and you know after the what, project is over this gain, what knowledge gain; It will be simply, you will forget, you will lose, you will loss those knowledge. So, now and you are under estimating ok, I have that knowledge now, I have lost it. So, you are underestimating that or you are not imagining that how quickly the, know how or the knowledge, it will get lost.

So, due to underestimation of how fast, how quickly the know, how only the knowledge regarding the project it can get lost, people may what, they may not do the proper closing of the project. Also, how much implicit knowledge exist within the team members, you

are underestimating that and as a project manager, that is why you may not properly close this, what project this is another reason.

Another reason is emotional factors suppose a project is continuing for three years, four years, five years. Within those four five years, we are having strong relationship among your teammates and you are having some emotional, what relationship among your teammates and you do not want that. So, quickly you will lose these emotional relationships. So, due to the emotional factors you do not want that no, this much quickly, I will not properly close this project.

Another reason is in decision regarding the project closure. As a project manager we are not able to take a decision, when to project the close the project, you are making the late. So, because you are in dilemma whether the close, whether to close the project or today or tomorrow or after few dates so, due to this delay you are not able to take a decision so, that is why due to this indecision you are not able to properly close the project. So, these are some of the what reasons, why the projects are not properly closed.

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Now, let us see these things, these points what I have explained, these have explained in this slides so in detail that you can read yourself.

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Now, let us see about what are the problems if project closure is not becoming proper, if a projects are not closed properly what kind of problems we may face? Number one time and cost overrun; see if a project termination is delayed what will happen the project as a cost center runs up expenditure in the meanwhile leading to cost overrun. You are nothing, no work you are doing, because project is already completed, but since you have not officially have closed the project.

So, some expenses will still continue. So, these will be what these cost in that way it this will lead to cost overrun, also the project duration appears to be longer than what it should actually be. Already, you have finished within four years the project, but another two three months you are not able to officially properly close it so; that means, the project duration it will be counted as four years three months. So, unnecessarily the project duration it appears to be longer than what it should actually be.

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Next problem is locking up what valuable human and other resources. Your project is over, but since you are not officially closed it the human beings, the staff members and other resources such as computer, etcetera these are being held up. Those resources or human beings these staff members otherwise, they would have what assigned to some other project.

So, unnecessarily you are locking up the valuable human beings or the staff members and other resources, because you have not properly closure the officially mark the closure of the project. So, this is another problem that may occur or if you are not properly closing the project.

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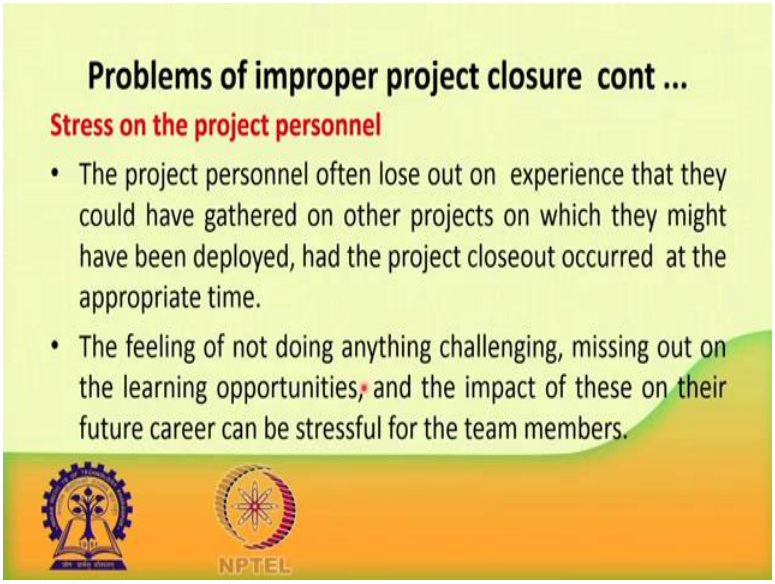

Then this stress on the project personnel is another problem. So, the project personnel often lose out on experience that they could have gathered on other projects on which they might have been deployed. Had the project closeout occurred at the appropriate time? So, if the project would have closed at the appropriate time, these staff members could have been what allocated they could have assigned to some other projects and they could have what gain experience there. So, unnecessarily they are losing their experience.

So, and the feeling of not doing anything challenging and missing out on the learning opportunities, sitting ideal and the impact of these on the future carrier can be very stressful for the team members and they will be unnecessarily stressed. So, this is another problem that you will get, that you will observe, if the projects are not properly closed.

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Now, let us see; so, quickly some of the issues associated with project termination. The problems with project termination are two-fold; one emotional problems, two intellectual problems. The emotional issues, it can concern both the team members as well as the customer or the clients.

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Now, let us see what will happen or what these emotional issues they suggest. The emotional issues ok, the emotional issues are that the team members may experience; the emotional issues that the team members may experience include what the uncertainties

and apprehensions concerning their assignment to the next project. So, if the project is not properly closed, then the emotional issue the emotional thinking will come among the team members that they may not be what they may not get another suitable project. There will be some, they will apprehend that they may not be assigned to the next suitable project.

So, this may manifest as general loss of the interest in work and lack of enthusiasm to perform the remaining project work. So, in that way what will happen? What are the remaining project work they will not do sincerely, they will not have interest and they will what they will lack of enthusiasm to do work for the next projects. Similarly, there can also be diversion of attention, if the project is not officially close they will sit idle, the staff members will sit idle and the team members their attention will be diverted. They will pay more attention to issues such as getting reassigned to a project of their choice and the project work can take a and then the main objective the project work, it may take just a back seat, they will just lose interest, they will have diverted attention, if they will sit idle.

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On the client side, there can be a sudden change in attitude and the loss of interest in the project that is another important issue associated with the project termination. The client may even change the personnel dealing with the project, since you are not officially have closing or terminating the project. So, the client may even change the personnel who is

dealing with the project and thereby, causing further disconnect and difficulties in project closure. So, the official project closure will be further, what we will face so, further difficulties and problems.

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Now, we will come to the next problem. Next one is intellectual problems; the intellectual problems may include hand what handling some sensitive issues. Let us take what sensitive issues may come. So, when a project is to be prematurely terminated the terms of the contract. And the list of deliverables need to be renegotiated once it is premature terminated what terms what conditions, we are mentioned in a contract and what list of deliverables, they were plan previously, they need to be renegotiated. And also even when some deliverables and tasks that are considered to be not necessary at any more; however, before dropping these, it needs to be verified by the client ok.

So, even when some deliverables and tasks that are considered. There are not necessary any more, but before dropping, these it requires to be verified from the client. Also, the closure decision, it has to be effectively communicated to all stakeholders since, you are what closing the project may be successfully, after successful completion or premature termination, these decision has to be effectively communicated all the stakeholders may be this project manager, developers, designers, clients, etcetera to all of these stakeholders, you have to pass on these decision, effectively.

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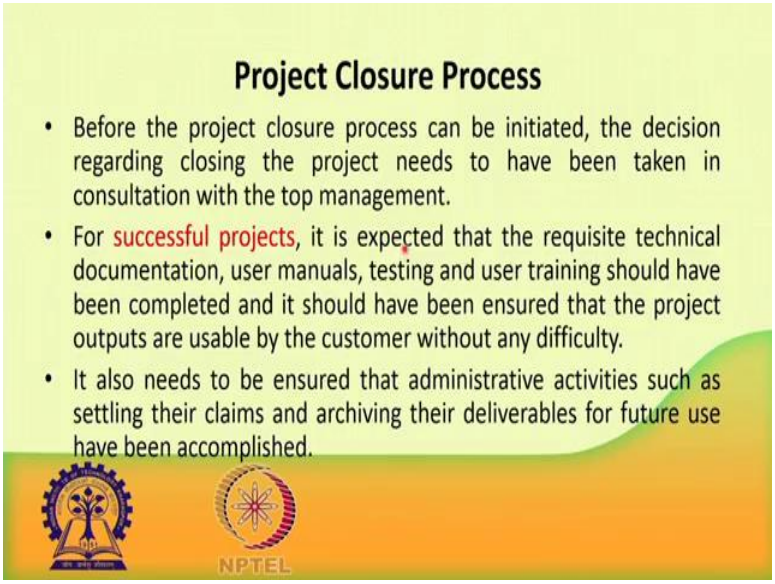

The learn now let us see, how will, what how will do this project closure first? What is the project closure process, what steps you must follow to closure a project, but before that let us see some fundamental things. Before the project closure process can be initiated the decision regarding closing the project, it needs to have been taken in consolation with the top management. You are not competent enough to what close the project you have to get a permission approval from the top management.

For now, there are again two possible cases; you might want to close a project after successful completion of the projects or after premature termination; now, let us these two what cases individually. For successful projects it is expected that the requisite technical documents, the user manuals, testing documents, user training, etcetera should have been completed perfectly. And it should have been ensure that the project outputs are usable by the customer without any difficulty.

So, two things you can see here that for successful projects, all the documents they have been completed successfully and you should ensure that the project output or the final outcome that is usable satisfactorily by the customers by the clients without any difficulty. It also needs to be ensured that the administrative activities such as settling their claims, archive what archiving the deliverables for future use, they have been also accomplished successfully.

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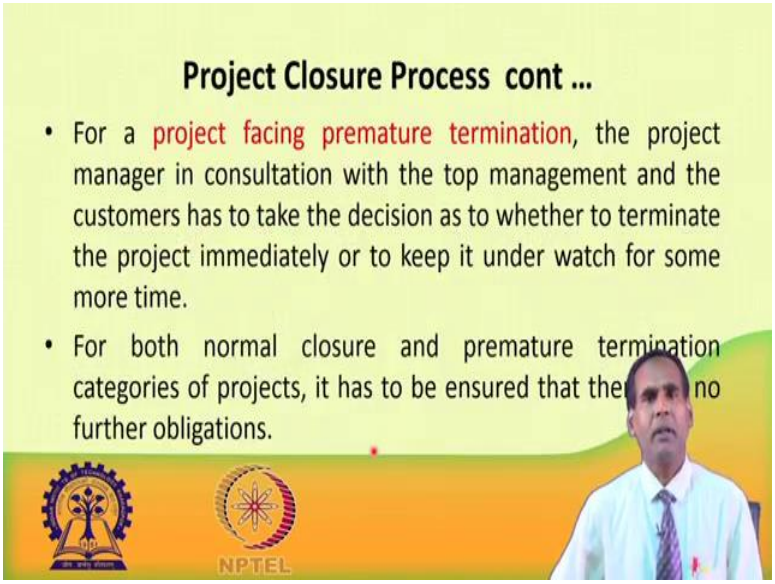

Now, let us see about the second case that is project those who will be terminated in premature. For a project facing premature termination, the project manager in consolation with the top management and in consolation with the customers, he has to take the decision as whether to terminate the project immediately right now, he will what terminate the project or he may take some time to keep it under watch for some more time.

So, this decision the project manager has to take in consolation with the top management as well as the customers, whether to terminate the project immediately or we can observe the some more days and then will close it. For both the normal successful closure or the premature termination categories of the project, it has to be ensured that that there is no further obligations. In both of the cases there are no further obligations pending, you must as a project manager you must ensure that there is no further obligations pending whether it is a normal closure or it is a premature termination.

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Now, let us successfully; now let us quickly see, what are the steps they have to be followed for project closure process; number one; we have to get the client acceptance. You have to get the acceptance from the client that we are going to close the process, getting client acceptance, then archiving the project deliverables, whatever the project the outcomes the deliverables such as your SRS document, design documents, code and this test cases, etcetera all those things you have to archive properly. Next preserve the project know-how the project details, the project knowledge, the experience that you have gained out of the project you have to preserve properly.

So, that they can be used in similar projects in near future, then performing a financial closure, all the financial documents, balance sheets, etcetera in income expenditure statement, etcetera every what the financial aspect must be close must be closed properly. There should be a financial audit and the financial obligations should be over year or you must close the financial detail. There must be financial closure that has to be performed before the financial, before the final closure.

Then performing post implementation project review yes, whether the project has been successful completed or you are pre maturely terminating. So, you must have to conduct a post implementation project review. You call it as a post mortem post mortem analysis, because by this process you can know what are the what plus points of your development, what are the negative points and what are the what lessons that you have

learnt from the what project development and these things you can use for getting benefit in similar types of projects. This experience can be used, their help can be taken in the similar projects in near future to get the benefit, and then you have to prepare a post implementation review report.

So, after carrying out these post implementation analysis or this review you have to prepare a report and archive it and finally, you have to release the staff, which staff members we are associated with development of this project. After the official closure of the project, you have to as a project manager you have to release them and, it is your duty to see that they are successfully allocated to other similar projects, depending upon their experience and skill.

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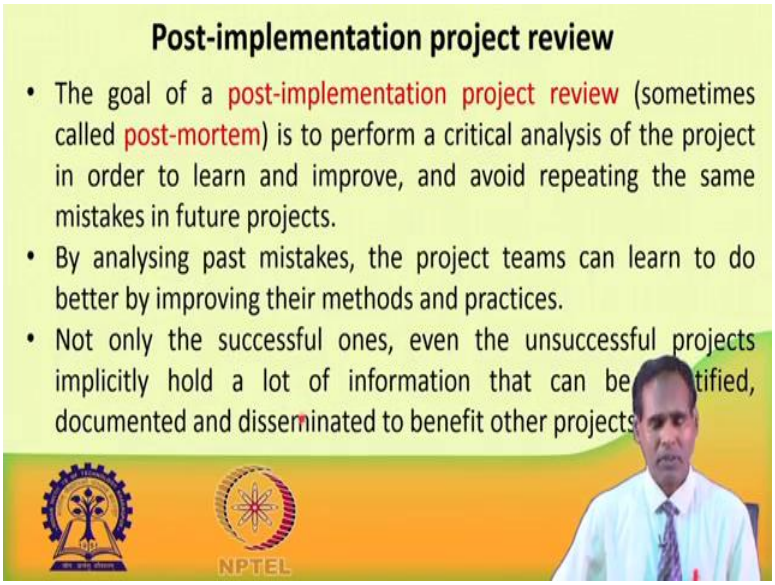

Those things I have explained here in the couple of slides. I am skipping those things you can these are easy things you can read those yourself. But I want to say something about this post implementation project review, because this is very-very important. The goal of a post implementation project reviews as I have already told sometimes, you call as post mortem analysis it is carried out to perform a critical analysis of the project. Why? In order to learn and improve and avoiding that repeating the same mistakes in future projects. You can learn many things, you can improve how can what steps you can take so that you can improve in your similar projects and you can avoid the mistakes what you have done in this project, you can learn from the mistakes and avoid the similar

mistake mistakes in a future projects. By analyzing the past mistakes the project teams they can learn to do better by improving their methods and practices.

So, not only the successful projects, but also the from the unsuccessful projects, they can also implicitly hold a lot of information that can be identified, documented and disseminated to get benefit from the other similar projects in near future.

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Now, let us quickly see, what are the steps of post implementation project review; what steps you must follow for the post implementation project review. First step; you have to conduct projects surveys to collect various types of information. So, after the project is over you should conduct some surveys to collect various types of information. For these you may carry out, you may prepare some questionnaires and collect the information from the stakeholders. So, information such like gather performance issues and these organization structure, etcetera those kind of information you can collect from the survey. Then you collect of you collect some objective information or some project metrics from this review.

For example the cost metrics, say schedule metrics, quality metrics, those metrics you can call it during this post implementation project review. Then hold a debriefing meeting. So, before making the final project review meeting hold a debriefing meeting called as the preparatory meeting, where the project manager will there all and some senior persons or senior management persons may be there, analyze what you have

happen and what are the advantages you have got, what are the experiences and what are the lacunas, then yeah, what arrive at a consistent decision. And after this you can hold the final project review meeting and then discuss all those, then see what are the experiences you have gained, what advantages, what are the skills you have learnt and what are the mistakes that you have committed. So, all those things just analyze then prepare a report based on this post implementation review.

This report you archive and these findings will help in improving the similar types of projects in near future. These report will also give some of the recommendations for the similar types of projects in near future. If you will follow those recommendations definitely you will get more profit more benefit out of these similar types of projects in near future and finally, after what preparing the review report you have to publish the report you have to send it to all the stake holders. So, that they will analyze, they will see what are the what plus points of, what team members, what are drawbacks what are the mistakes they have done and these what findings they can use in the near future projects of similar type and nature.

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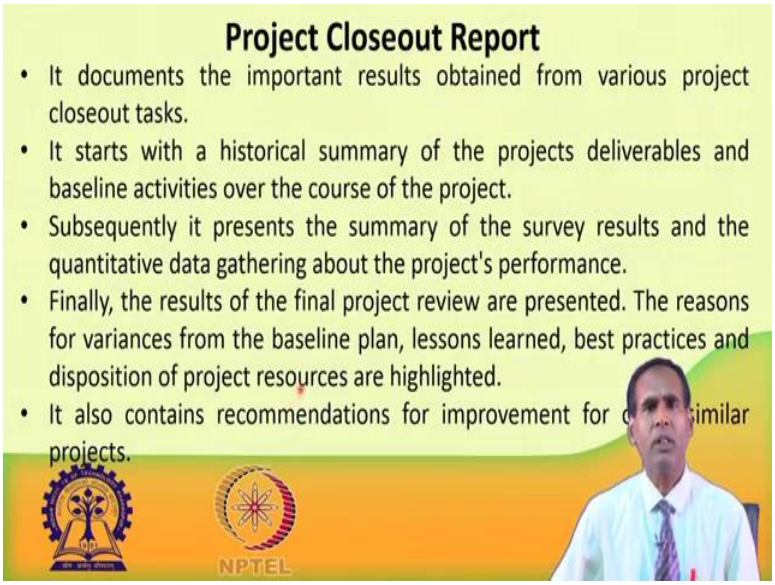

So, then as I have already told you the outcome of these post implementation project review will be what report ok, it will be a report. We have also seen that after the project is closed out you have to prepare a final report, this is known as the project closure report. Now, let us see it contains what? This project closeout report it is documents the

important results, which are obtained from the various project closeout tasks from the various project closeout activity or task what results you have got, this document will contain that.

First, it is starts with a historical summary of the projects deliverables, what are the project deliverables? A historical summary will contain and the baseline activities over the course of the project that it will be contained, then subsequently, it will be present the summary of the survey results, ok. So, I have already told you before this you have already conducted a survey.

So, from this survey what you are getting. So, then subsequently it will be present the summary of the survey results and the quantitative data, those are gathered about the projects performance ok, regarding the performance details also it will contain.

Finally, the results of the final project review meeting, I have already told you here, you have to carry out a final project review meeting what are the findings what are the results they must be contained here finally, the results of the final project review meeting are presented in this document. The reasons for variances from the baseline plan, we have already prepared the baseline plan. What are the reasons that what are the reasons for which there are variances, there are what diversions from the baseline plan and also the lessons that we have learnt, the best practices that we have followed and the what disposition of a project resources, etcetera, they are also highlighted in this report.

It also contains recommendations for improvement for other similar projects. So, from this experience; from this project, what experience you have got you can prescribe some recommendations for improvement of similar types of projects. Those recommendations also will be what written in this closeout report so that, you can get benefit of these in the next similar types of projects.

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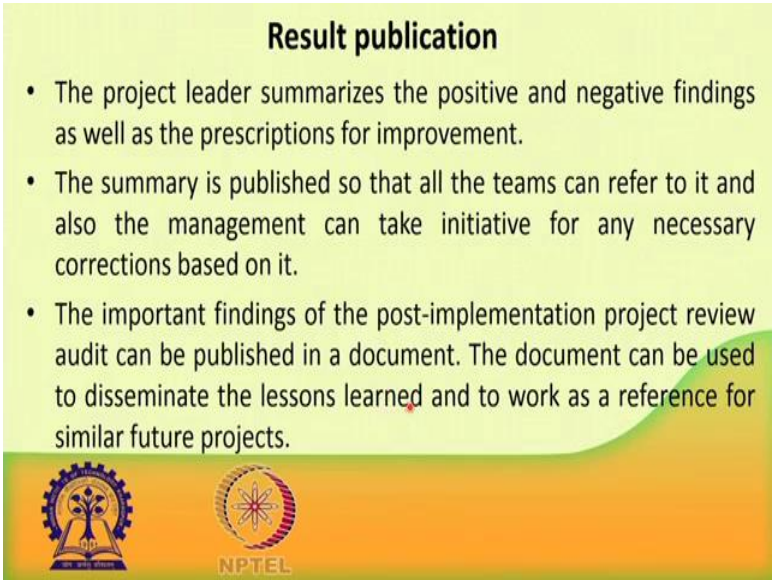

So, then finally, you have to publicize the result; the project leader he will summarize the positive and the negative findings as well as the prescriptions or the recommendations for improvement. The summary is published so, that all the team members can refer to it and also the management can take initiatives for any what necessary corrective actions based on this.

The important findings of the post implementation project review audit, it can be published in a document. The document can be used to disseminate the lessons which are learnt and to work as a reference for similar future projects. So, whatever the important findings that have to be archived, that has to be published in a document and that document has to be archived. The document it can be so, the document it can be used to disseminate the lessons that we have learnt from the previous project and it can be used, you can under the to work as a reference it can be used as a reference for similar types of projects in near future.

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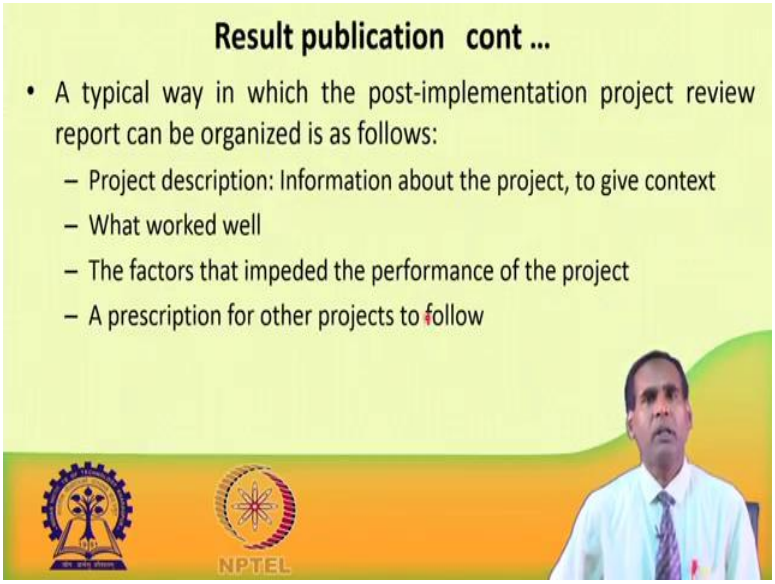

A typical way in which the post implementation project review report can be organized as follows; this is a typical structure. First you have to write down the project description, the information about the project, then to give the context in which it will be used, that has to be written then what worked well, what are the good things positive things, what are the advantages, ok.

The positive things they have to written here, then the factors that impeded the performance of the project ok, which factors have impeded the performance of the project that will be written the negative aspects also will be written. The mistakes that you have committed, they have also to be written here, then a prescription for other projects to follow. So, out of this project what you have learnt the prescription the recommendation that you will give to other similar projects in near future they should follow. So, those prescriptions or those recommendation also should be written in this what project review report.

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Finally, the final step in project closure is releasing staff. So, now, the project is officially closed, you have to release the staff. So, this is the final step of the project closure to process, this is the last meeting before the project team members disburse to different other projects. And the project manager should ensure that the team members who had working for these project, they are assigned to proper projects according to their expertize and their skill and this meeting is also a ground for celebration for celebration before the a team members.

This would be before the team members disperse to different projects and for recognizing what are the exceptional performance by the team members, then you must recognize here and recognizing the experience and proficiency gained by the team members. Because if different team members, they have a different experience, they have different proficiency, you must recognize the experience and proficiency gained by the different team members. They as a project manager you must experience, you must recognize the experience and the proficiency which are gained by the different team members, you must recognize them you if possible you can reward them.

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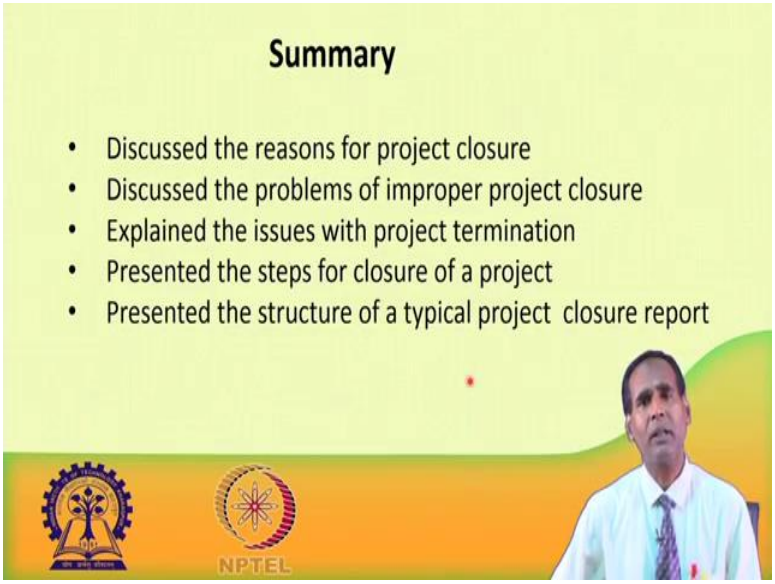

So, these are the; so, the final step is release the what team members and the as a team manager ensure that these team members should be assigned to other different projects, depending upon their experience and skills. So, today we have discussed the reasons for project closure, why project are becoming why there is a need to what close a project. We have also discussed the problems of improper project closure, what problems will face, if you improperly what close the project.

We have also explained the issues with project team in termination, we have presented the steps, they have to be carried out for closure of a project. We have also presented the structure of a typical project closure report what should be the contents and what it should contain and how we can get the benefit in similar projects in near future these are the things we have discussed today.

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This is the reference we have taken from this book, these contents we have taken from these books.

Thank you very much.

Software Project Management
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Lecture - 51
Software Quality Management

Welcome to this lecture. In the last lecture, we had discussed about risk management and then we had started discussing about Software Quality Management. We had discussed about some very basic concepts about software quality. For example, we discussed what is a quality software. And we said that fitness of purpose is a good definition of quality for traditional products, but for software products the fitness of purpose is defined as the satisfaction of the requirements document.

But then, this is only a basic requirement for a quality product, one of the attributes of a quality software, but a software product is said to be a quality one, when it has several other desirable attributes in addition to the correctness or fitness of purpose. And we had discussed about the various attributes, quality attributes and we said that there are several quality models which define that what are the quality attributes and how to measure them given a software product.

After that we had just started discussing about the evolution of the quality systems over the years. Let us proceed from that point. We will continue with the evolution of software quality.

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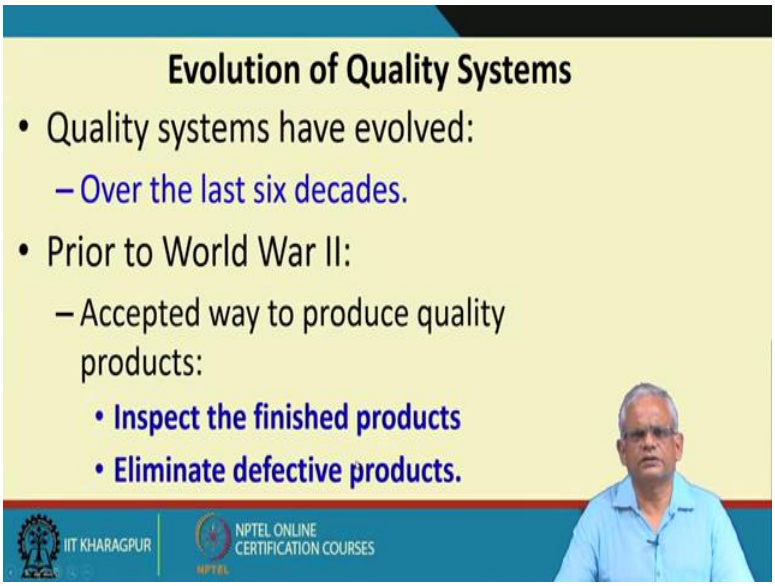

We had seen in the last lecture that before World War II. The only way quality products were produced is by testing, to test a product rigorously, identify all the bad products and eliminate the defective ones. This was the only way that quality was used in manufacturing.

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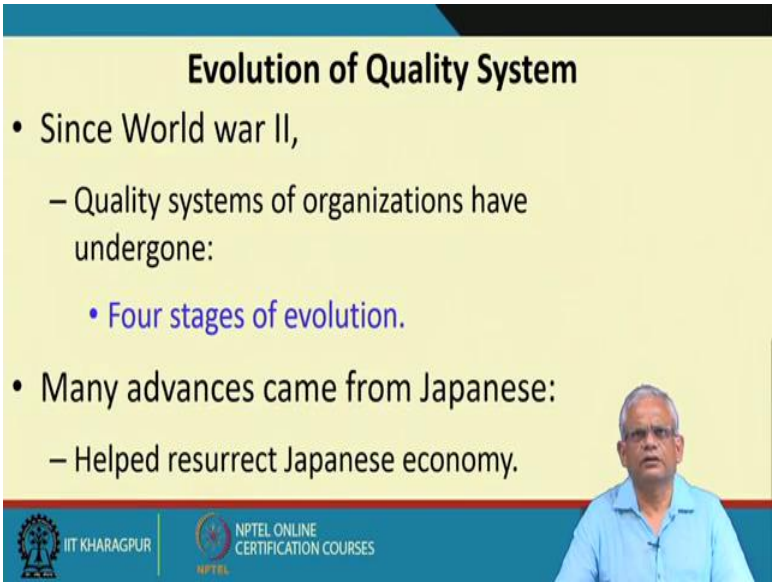

But since then, the quality systems have rapidly evolved and there are four stages of evolution and many advances originated by the Japanese.

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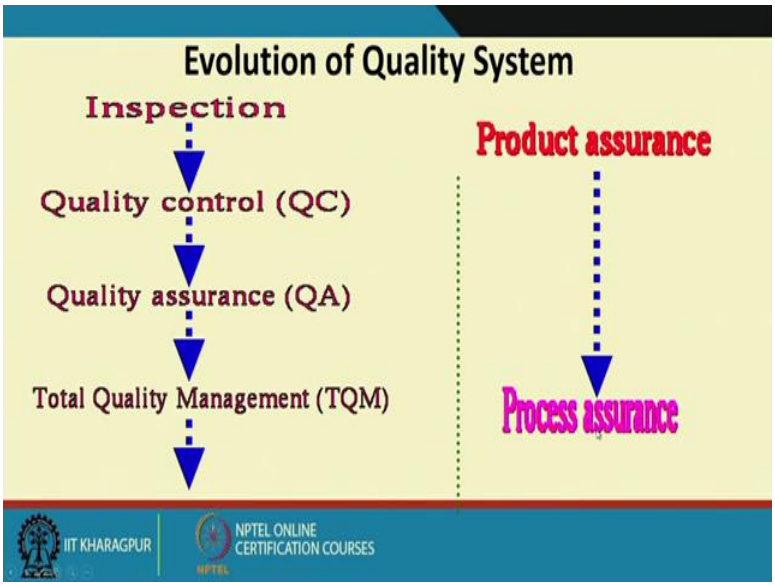

The initial inspection and testing was superseded by quality control. We will see what is quality control and statistical quality control and later the quality assurance techniques evolved. And since then the total quality management techniques evolved and still we are further developments in this area. As you can see that there are various stages of growth of quality because the customers are becoming quality conscious and it is very important for an organization to produce quality products.

And the manager has a very important role in an organization producing quality software. If we look at the different developments that have occurred here from inspection to total quality management, we will see that initially there was too much of emphasis on the product, the product was tested thoroughly, but later we will see that we do not really look at the product much we look more at the process.

The basic assumption is that the process if it is good it is bound to produce quality products. We need not look at the product and reject the bad products that need not be the emphasis, the emphasis should be on having a good process. Over the time that the quality systems have evolved the emphasis has shifted from product assurance to process assurance.

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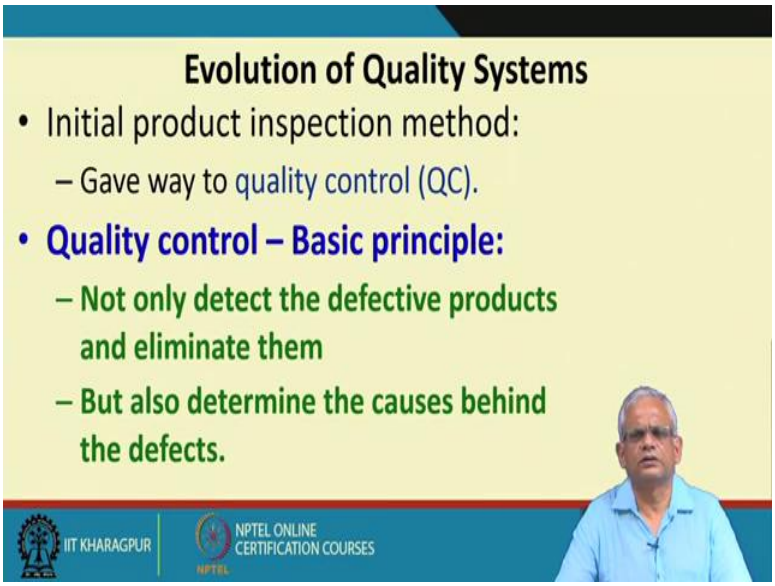

The initial technique deployed for quality was testing and inspection and later the quality control technique or QC started appearing. The basic principles of the quality control is that not only test the manufactured products detect and eliminate the bad products, but also determine what causes the defects. That is look at the process of manufacturing.

If it is the case that the rejection rate is high for a nuts and bolt manufacturer, just check that what is causing this defect, is it that the temperature at which it is done is really expanding the cost and the cost is not tolerant to the temperature and that is causing the defect, then use a different cost. So, the quality control principle, as you can see that gradual shift is there to the process, yes do the testing, inspection and testing.

That is still there it is one of the basic techniques even in the modern quality system testing is there, eliminate the defect product, but also look at the process; why the defective products have arise in the first place. This form the basic quality control principles, but then we had the statistical quality control.

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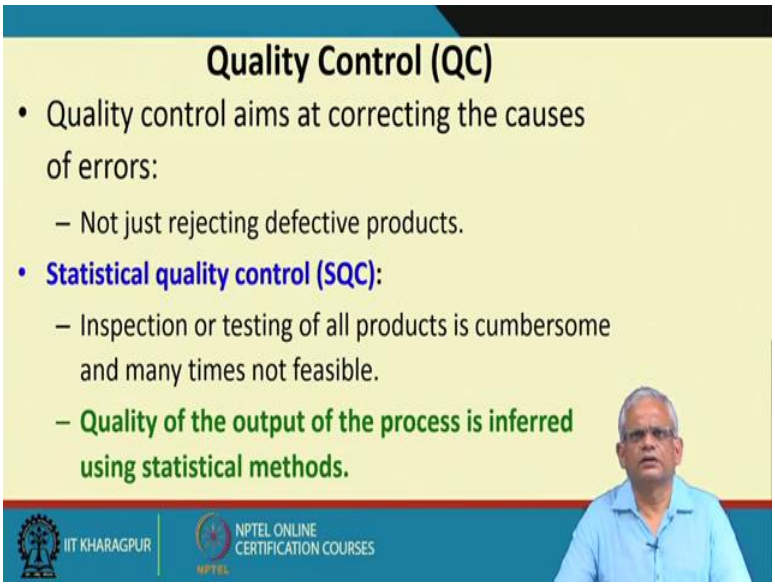

The quality control not only in targets to reject the defective products, but also aims at correcting the causes of the manufacturing process that was resulting in the defecting products and slowly the quality control techniques evolved into statistical quality control SQC, where the idea is that, many times testing each and every product is difficult, can we use statistical techniques to make inferences about the quality of the product and also make variations to the process and check whether the quality is improving or not.

So, the statistical quality control techniques, essentially use statistics to make inference about the quality of the output produced, the quality of the process that is produced and then make changes to the process and see whether the quality is improving.

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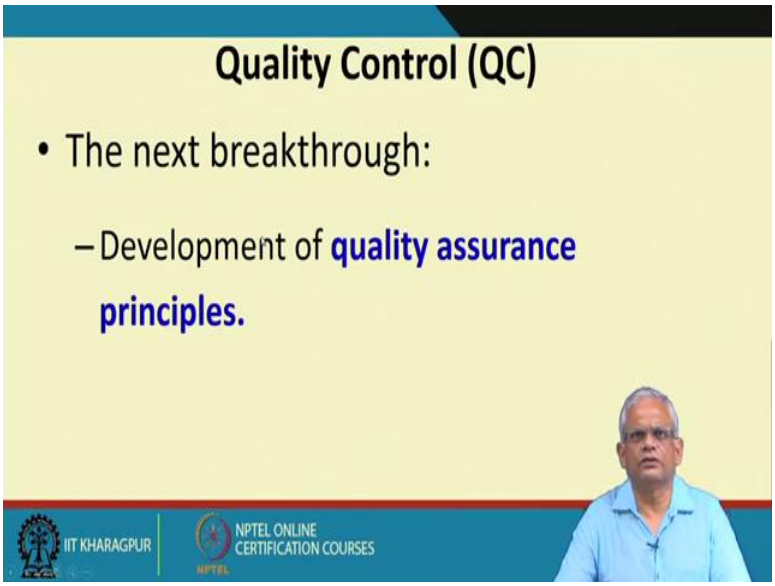

And after quality control the next breakthrough was quality assurance.

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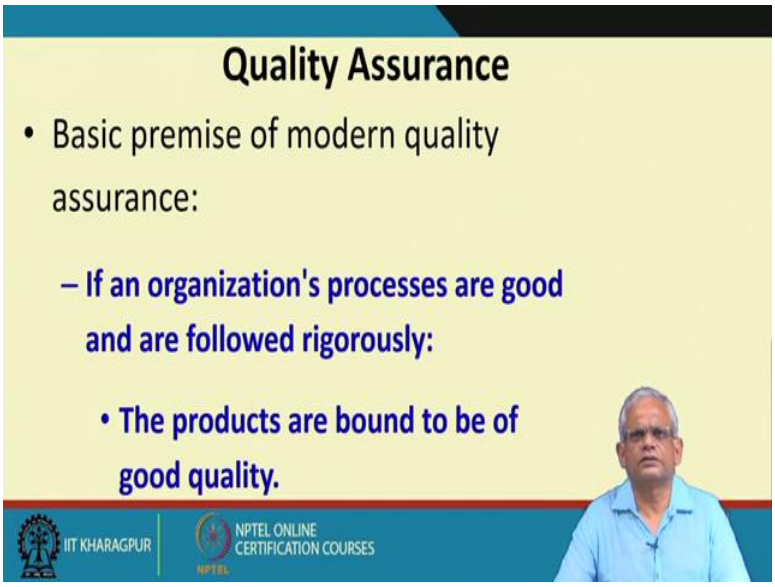

The main assumption in quality assurance is that to produce quality product, looking at the finished product and then doing testing is not that important. The major focus of the quality team should be on the process. If a good process is in place good products are bound to result, if an organizations manufacturing process is good and is followed rigorously then the products are bound to be of good quality. Just looking at the products rejecting bad products is not a good way of ensuring quality.

Rather the team, the quality team needs to concentrate on having a good process and then seeing that the process is followed rigorously. In that case we the products are bound to be good.

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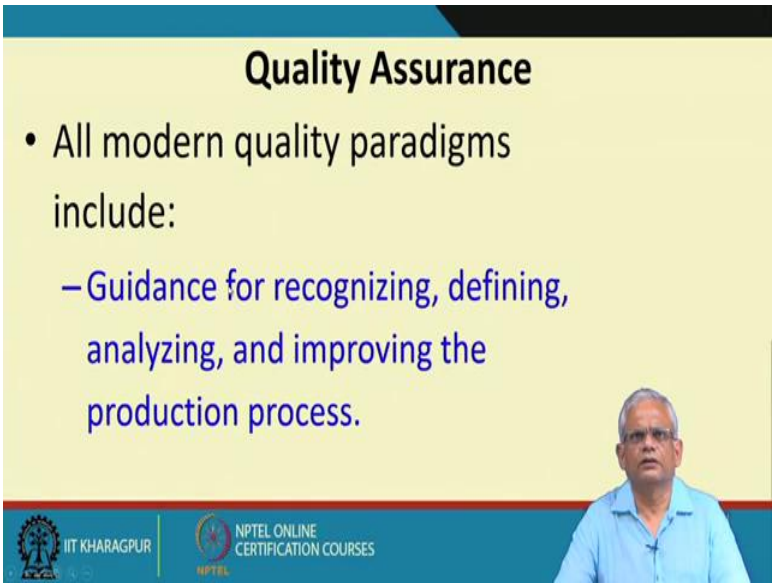

The modern quality paradigms are based on the quality assurance principles and we will see that there is guidance for recognizing, defining, analysing and improving the product process. So, see the shift from the product to the process. The quality assurance techniques have shifted from just testing based techniques to recognizing, defining, analysing and improving the production process.

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And even that further evolved into the total quality management principles. This was the next step of evolution. Here make process measurements that is how effective is the process. Is it producing good products? Is it efficient? And other desirable characteristics of the process, how effective is the process and then make changes to the process, so that the process improves and the focus here is continuous process improvement through process measurements.

We need to have metrics collected about the process and then see where there are problems and then change the process itself and process continually improves.

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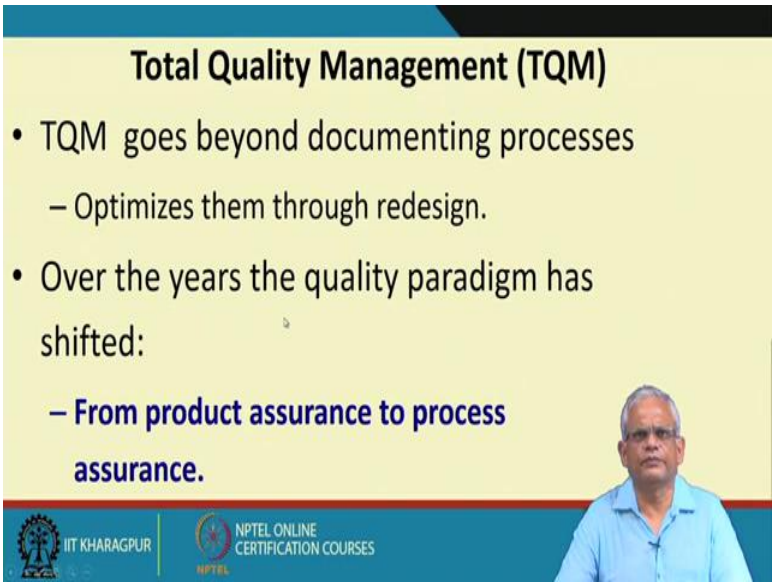

So, this is the total quality management principle. Here it is not just documenting a good process, but then here the emphasis is to as the projects run to collect statistics, the metrics regarding the process performance and find the bottlenecks in the process and optimize them through this redesign. But one thing is clear that over the years the quality paradigm has shifted from product assurance to process assurance.

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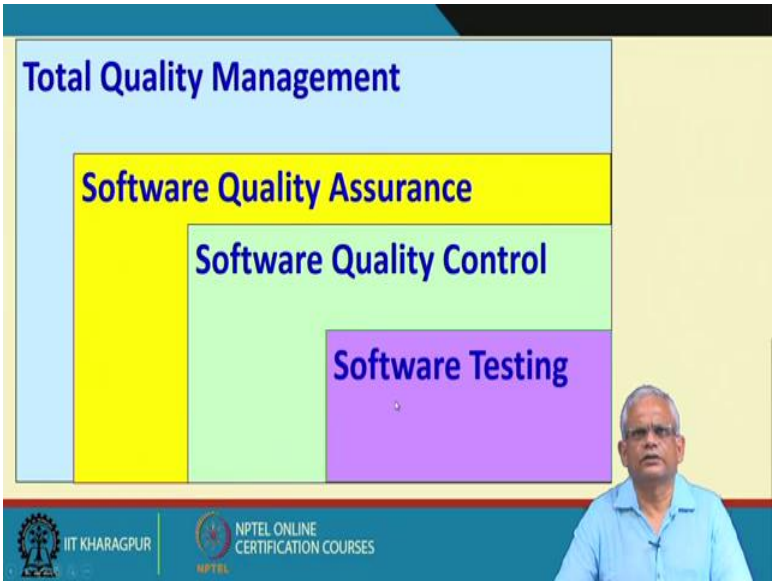

Initially software testing was the primary means of manufacturing good quality products. It is the basic technique, it is still there, but then we have the we had the quality control

as the next development. It incorporated software testing no doubt, but then the defective products were analysed to identify what was the problem in the process that was resulting in the bad products and modify the process. That is the software quality control.

In software quality assurance, the basic premise is do not focus on the product testing, focus on having a good process, document the process and then follow it rigorously, then the product is bound to be good. That was the next development you can see the shift from testing to testing is a smaller activity and analysing the process the shift to process here and then finally here software quality assurance have a good process, good products are bound to follow.

And then the next step of development is total quality management where not only have a documented process to start with a good documented process, but through process measurements continuously improve the process. So, this diagram emphasizes that software testing is still there, it is part of it is a smaller part now. There are many other things that happen.

For example in software quality control, we not only do the testing and inspection, but also we look at the defects analyse the defect and find out why the defects are arising software quality assurance these things are still there, but then we also have documented process. And total quality management we have processed measurements, find out the bottlenecks and continuously change the process.

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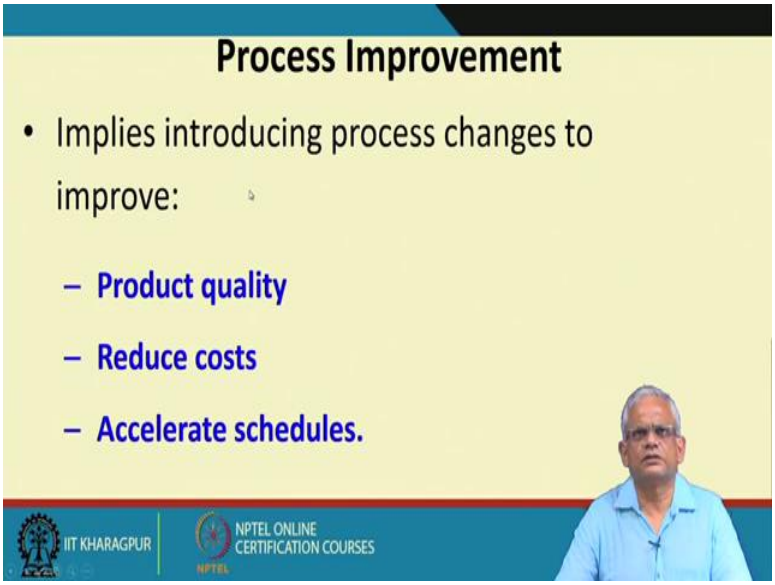

The modern quality systems emphasize on process improvement and here the changes are made to the process to improve the product quality, reduce costs and accelerate the schedules. We find where are the process bottlenecks which are causing the product quality to be poorer, resulting in higher cost, delays and so on and fine tune the process to improve the product quality, reduce cost and accelerate the schedule.

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One thing that is well accepted now is that a good process is required to produce good product. But we must remember that given a process a manufacturing plant, product manufacturing plant can run the process again and again and each time good quality products will come out, no doubt about that for traditional manufacturing.

Let us say we look at a iron manufacturing company it consumes iron ore and other raw materials produce good quality of iron may be flat sheets or rods. And once the plant has been set up and process is followed like getting good quality raw material and so on, it will keep on continuing to produce good quality products once the process has been set up. But for software development and other design activities, we cannot say that we set up a good process and always good products will come out there are other factors.

For example, let us say the complexity of the problem. The process that we heard worked well and good quality products came out for accounting software. But let us say we are doing something very different in a telecom domain and maybe we cannot replicate that

process to have a good product in telecom. And also, the capability of the designers this is the important fact, important factor in producing good quality.

Even though the process is there, but then if the designers and coders are inherently not good, the product cannot be good. So, just making this point here that for traditional products having a good process is enough to have good quality products, but for software development having a good process is only part of the story, there are other factors which affect the quality.

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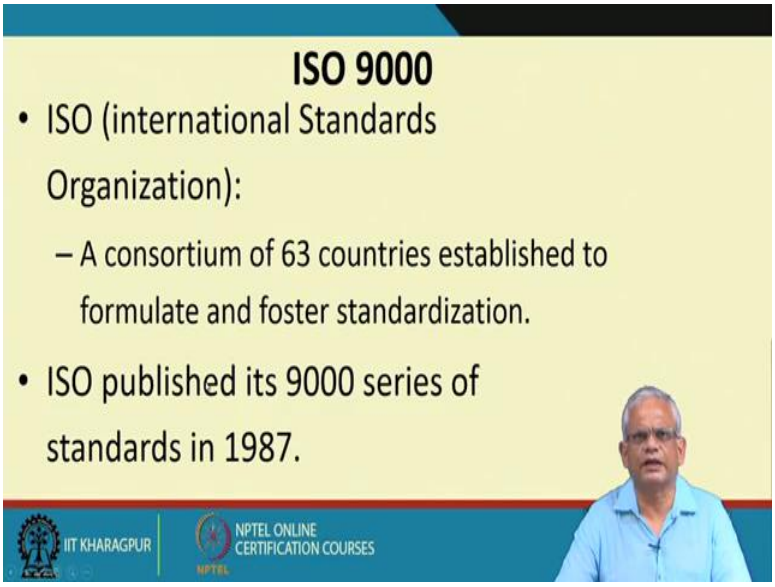

One of the early standards in quality is the ISO 9000 series of standards. ISO stands for International Standards Organization it is a consortium of a large number of countries established to formulate and foster standardization. The ISO 9000 series of the quality standards were published quite some time back in 1987. These are basically sets of documents which are guidelines for the process.

They do not really look at the product, it is not a product specific, it is based on the process. And ISO 9000 came up with three documents, the ISO 9001 standard, 9002 standard and ISO 9003 standard.

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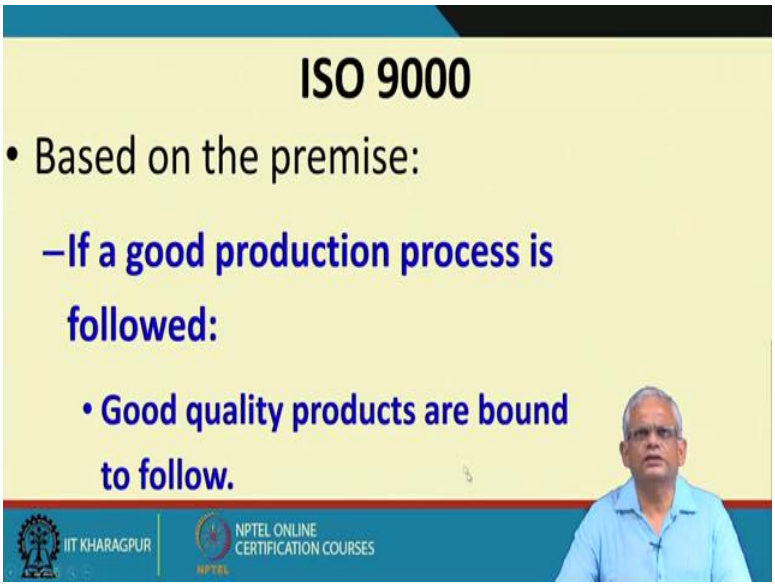

The ISO 9000 is based on the quality assurance principles that if a good production process is followed good quality products are bound to follow, you can see that this is essentially the quality assurance principle and ISO 9000 is based on the quality assurance model.

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Now, let us see what is the difference between ISO 9001, 2 and 3. ISO 9001 is applicable to organizations engaged in design, development, production and servicing of goods. So, here for the manufacturing there is initial design and the design is used for development

and then production and then servicing. And therefore, this is the right standard for software development organizations because they do these requirements design, develop, test and service.

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But there are the other standards 9002, it is applicable to organizations who do not do the product design, but are involved in production and servicing. For example that there is a consultant organization who designed the plant. The company here did not design the plant, there is a consultant organization which designed the plant and once the plant was set up and the process was there then it just continued producing goods and then servicing.

So, here the plant and the process is purchased and the company only manufactures. And the 9002 series of standards is applicable to such organizations and of course, it is not applicable to software development organizations because here the development organization does requirements design, coding, testing, everything. It is not that it just gets the design and code and just keeps on manufacturing.

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ISO 9003 standard, this is another document that was produced by ISO it is applicable to organizations which do only inspection and testing of products.

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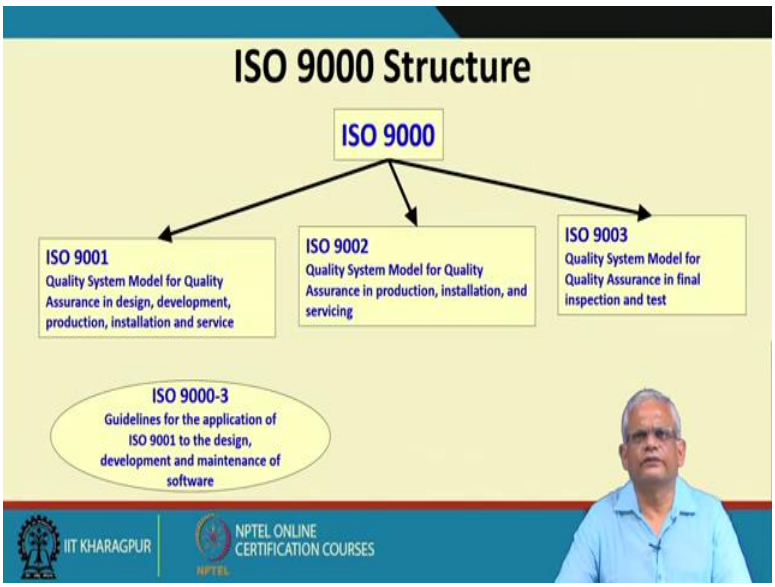

This is just pictorially represented here that 9001 here, the applicable to organizations which do design, development, production, installation and service and software companies come under 9001, but then 9002 and 9003 standards are applicable to 9002 is applicable to those who do production installation and servicing whereas, 9003 is only those who do the inspection and test.

But 9001 is a very generic standard applicable to many types of industries, those who produce design and produce goods but then software is different. Here there is no raw material that is consumed. Even as the software is being developed its not visible, only when the software is finally, running you can see it.

And therefore, software needs a very different interpretation of the guidelines given in the ISO 9001 document and this was felt because it became very difficult to apply the clauses of 9001 to the software industry. And ISO later came up with a new document called as ISO 9000 part 3, which provides the guidelines for software design development and maintenance. This was specific to the software.

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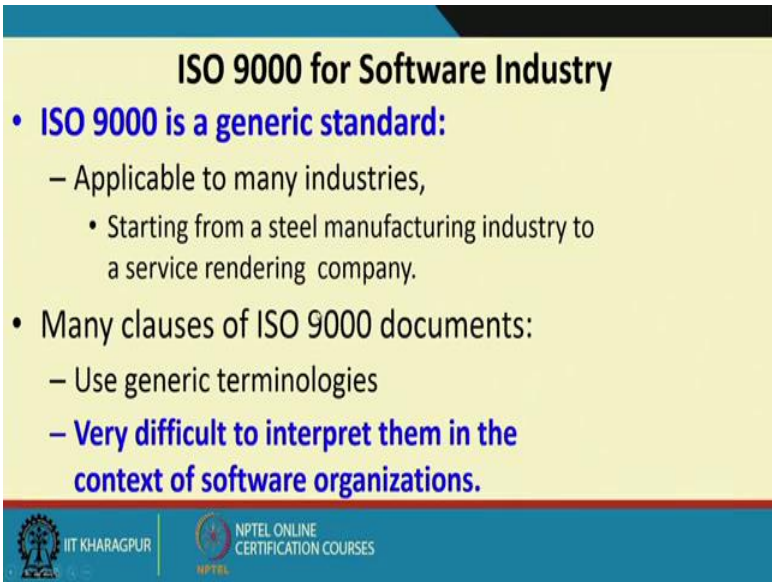

We must understand that ISO 9000 is applicable to a wide variety of industry. It is a generic standard starting from steel manufacturing to service industry and so on. And ISO 9000 documents are written using generic terminologies which are difficult to interpret for software organizations and therefore, ISO came up with a new document which is ISO 9000 part 3 which provide interpretation of the ISO 9001 for the software industry.

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The software industry is very different from the other industries.

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For example, during software development the only raw material consumed is data, if we can think of some raw material which is basically the data or inputs. Whereas, the other types of development, manufacturing and development they consume lot of raw material. For example, iron ore, coal, limestone and so on. And therefore, many of the clauses of the 9001 pertains to raw material controller, input controller and that is not applicable to software industry. These clauses are not relevant for software organization.

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And therefore, ISO came up with a new document applicable to the software industry. It recognized that there is a radical difference between software and other product and came up with a new document which is ISO 9000 part 3, few years later the original is, original documents were published in 1987 and the ISO 9000 part 3 came up in 1991.

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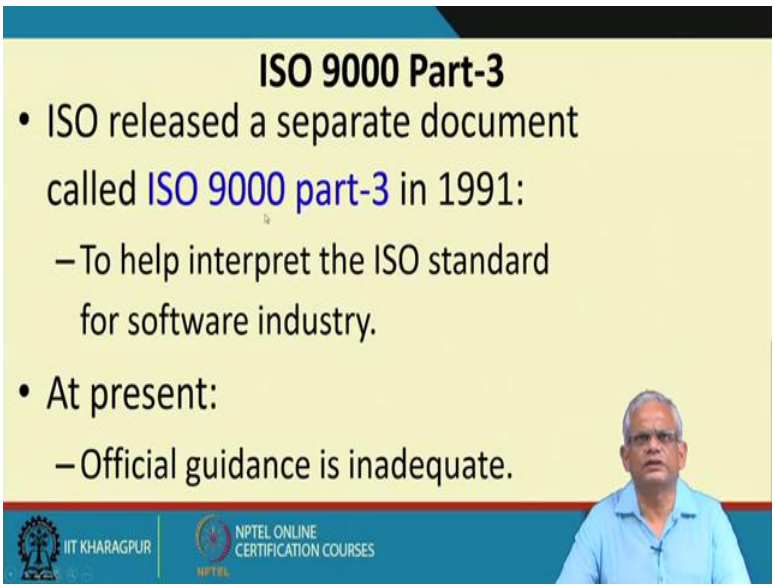

It helped to interpret the ISO standard for software industry, but still the official guidance is inadequate and it is interpreted by various consultants.

We will stop here. Almost at the end of the time and we will briefly look at what is the requirement for ISO 9001 for the software industry, what areas it emphasizes and what is the model that it proposes. And then the subsequently we will look at the software engineering institute, capability maturity model. We will stop now.

Thank you.

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Lecture – 52
ISO 9000

Welcome to this lecture. In the last lecture we had looked at a quality system and it said that there are some popular quality systems. The guidelines for which I have been formed, one of that is ISO 9000 and another one is SEI CMM. These two we will discuss in some depth and then we will look at the personal software process and the six sigma concepts. Let us discuss about the ISO 9000.

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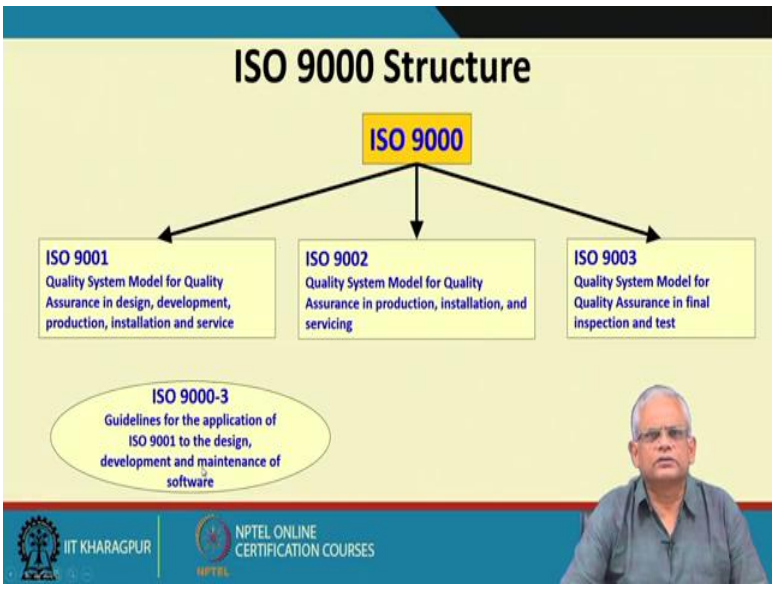

In the last lecture we had said that the ISO 9000 is a series of guidelines to set up a quality system, it came up with a series of volumes. ISO 9001, 9002 and 9003; initially came up with these three volumes in 1987. The ISO 9001 it discusses the guidelines for the quality system model, for quality assurance in design development product installation and service. This is a generic set of guidelines applicable to a wide spectrum of industry and including software industries.

ISO 9002 on the other hand it is a set of guidelines for setting up a quality system model, for quality assurance in production, installation and servicing. ISO 9003 is the quality system model for quality assurance and final inspection and test. But, then as we were

mentioning in the last lecture that the 9001 is applicable to a wide spectrum of industries and software is very different software industries working is very different from other industries.

And, there were difficulty in interpreting the ISO 9001 clauses for software industry; due to this purpose ISO came up with new volume, a new document the ISO 9003, part 3. This interprets the ISO 9001, the context of software industry is specific to the software industry, is guidelines for application of ISO 9001 to the design development and maintenance of software. Our discussions here will be formed on the ISO 9000 part 3 document.

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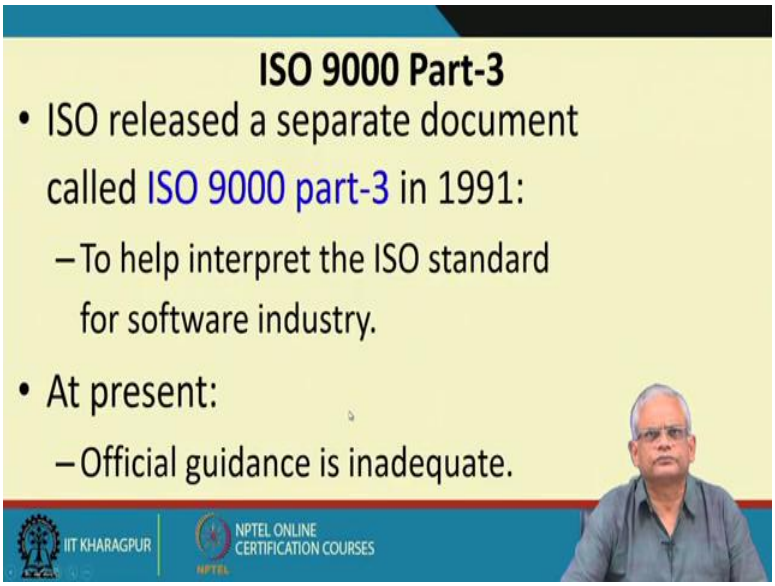

The ISO 9000 part 3 document, it interprets the ISO standard for software industry. But, then still the official guidance provided in setting up the quality system is still inadequate. And therefore, there are many quality consultant who provide help in setting up the quality system according to ISO 9001.

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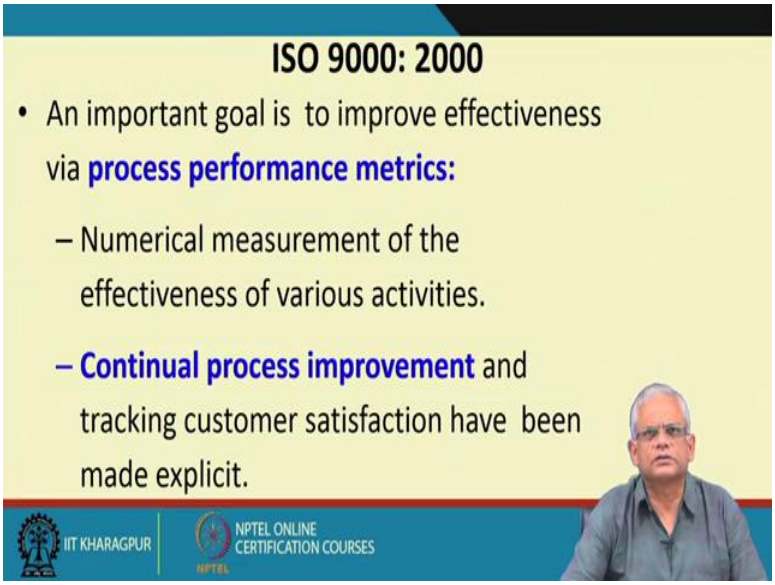

Later in the year 2000 ISO 9000 came up with a new revision. Here the important goal was to improve the effectiveness of the process by collecting performance metrics, this was not there in ISO 9001. It identified many activities, where numerical measurements need to be taken, numerical measurements or the effectiveness of the process activities.

And these measurements would be used to improve the process and also tracking of the customer satisfaction is made explicit. As you can see that ISO 9001, the original one in 1987 the main goal was quality assurance whereas, ISO 9000 the 2000 revision; it is towards total quality management, where there are process performance measurements and based on that continual process improvement has been addressed.

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But before we discuss about the details of ISO 9000 quality system, let us first discuss why it is necessary to get ISO 9000 certification, what benefits macro to an organization if we gets the ISO 9001 certification. The first thing is that a software organization to participate in a bid for some work, sometimes it is necessary to have a ISO 9000 certification. Without this they would not qualify to even participate in the bidding process. This is a good enough reason for many organizations to go for the ISO 9000 certification.

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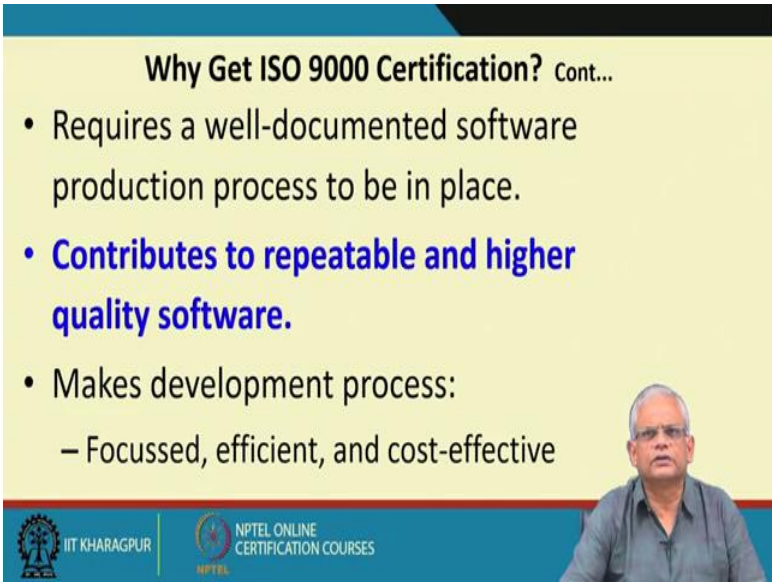

But then there are other benefits of getting the ISO 9000 certification. The ISO 9000 guidelines, the mandate that well documented software production process should be in place. And this yields benefit in the quality of the project completed, the cost effectiveness, meeting the schedule and so on.

And therefore the organizations not only get a chance to participate in the bids, but also their quality of the work, cost effectiveness, meeting schedules etcetera improves. In other words, it contributes to repeatable and higher quality software.

Over the next few slides we will be using the term repeatable. Let me just tell one sentence about what we mean by repeatable. Repeatable means that an organization can repeat its success on a project in similar projects. Let me just repeat what I said; the term repeatable means that if an organization had a success in one project it can repeat its success in similar projects. If the organization does not have a disciplined process,

documented process it may succeed in one project, but then even in a similar project it may fail, because it is not repeatable.

The success might have come due to some individual efforts, some very enterprising energetic individuals who might have worked hard to make one project success. But, then another similar project, it may be a failure if there is well documented software process, software production process is not in place.

And therefore, ISO 9000 helps in repeating success in one project to another project and also results in higher quality software. It fine tunes the development process because this gets documented reviewed. Therefore, the development process becomes more focused, efficient and cost effective.

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But one thing let us be clear before we discuss about the ISO certification. ISO 9001 certification is not really awarded by the ISO; the ISO does not in fact certify. It does not really issue any certificates the ISO by itself does not issue any certificates. The ISO by itself does not accredit, approve or control the certification bodies. There are certification bodies in every country who award the ISO certificates, but then ISO not even accredits, approves or controls the certification bodies. Every country has its own certification body which awards the ISO certificate.

But, then if we think about it there can be wide variation in the requirements to get ISO 9001 certification across countries and also it becomes sometimes very important that who had certified. For example: an Indian company may also get certified by a European certification body. Since, there is a wide variation of ISO certification it becomes important who had certified.

But, then even though ISO does not certify, it develops the standards, documents them and it also guides the certification bodies to have a uniform accreditation and certification. But the certifying bodies they can relax certain of these overlook and give still certifications.

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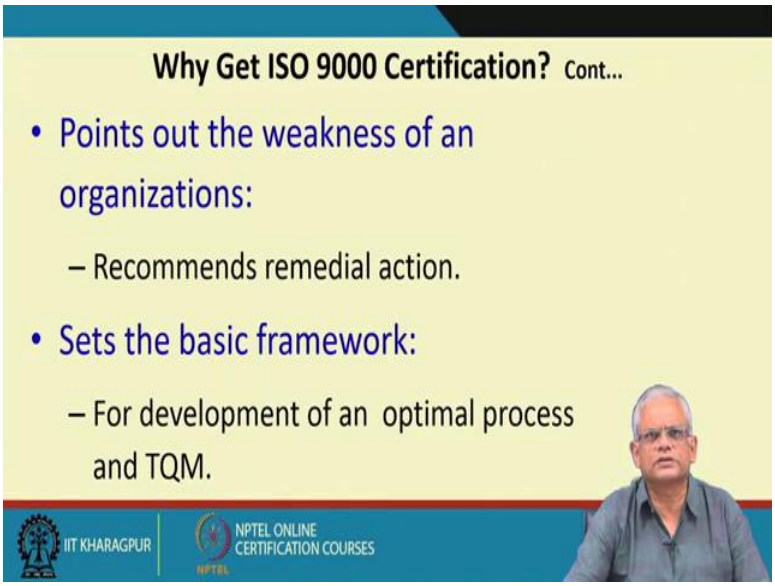

The ISO 9000 certification also points out some weaknesses of an organization and recommends a remedial action and sets the basic framework for a continually improving process.

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We have so far seen that what are the benefits of getting ISO 9000 certification for a software industry. And also some issues regarding the certification itself. For example, we said that ISO by itself does not award certificates, it just produces the guidelines, the standards and every country has its accreditation body to whom the companies apply and get certification. But, the next question that comes to mind is suppose a software development company wants to get ISO 9000 certification; what does it have to do?.

Naturally, it has to apply to an ISO 9000 registrar for registration. Every country has a set of registrars identified by the accredition body of that country and the registration body, the registrar who carries out the registration process. And there are several stages of this, let us just have a look at what are the stages at a very overview of this.

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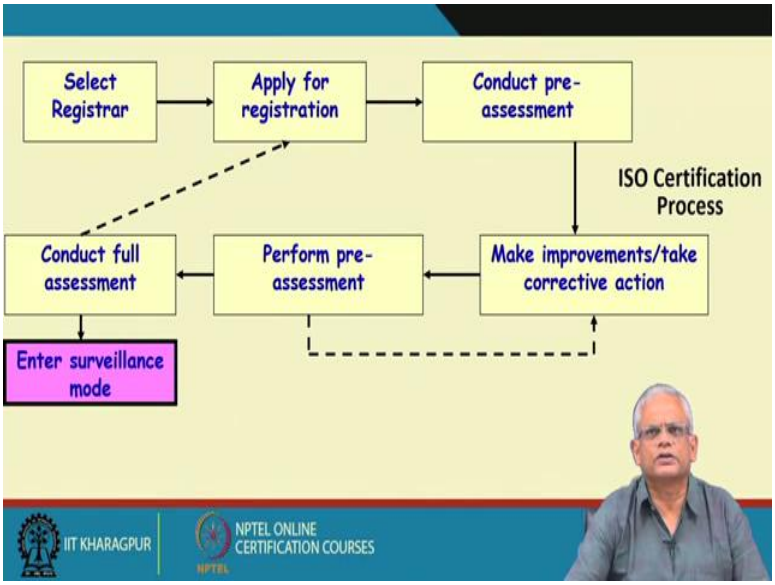

Every country as we said that there are several registrars. The company needs to identify registrar and typically a company takes the help of a consultant who sets up the companies documents, processes, helps in selecting the registrar and then they apply to the register for registration, the online application process and then provides the required documents. The registrar in turn looks at the uploaded documents and suggests some improvements and corrective action.

And this again the certifying sorry the registrar again looks at the corrected documents and this process may go on until the documents that have been uploaded are to the satisfaction of the registrar. And after this the registrar performs a full assessment that is the quality system, the process and so on. And if it is the success certificate is awarded and the company enters into a surveillance mode that is every few years it needs to again get reviewed for retaining the ISO 9001 certificate.

If the registrar is not satisfied then again suggest some corrective actions and fresh registration application need to be made. But, one thing please note here as you discussed that the entire ISO certification process is only based on the documents; the process documents, the quality system documents and so on.

We never mentioned anything about the product. So, the registrar does not even look at the product. In other words the ISO 9000 certification is not given to a product of a company, it is given to the company or the organization as a whole and not to any specific product. It is given to the process, the manufacturing process or the development

process. And the assumption is that a good process should result in a good product, but the product are not even looked at to give the certificate.

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The first stage is the application stage, apply to registrar for registration, the pre assessment stage where the registrar makes a rough assessment of the organization based on the documents uploaded.

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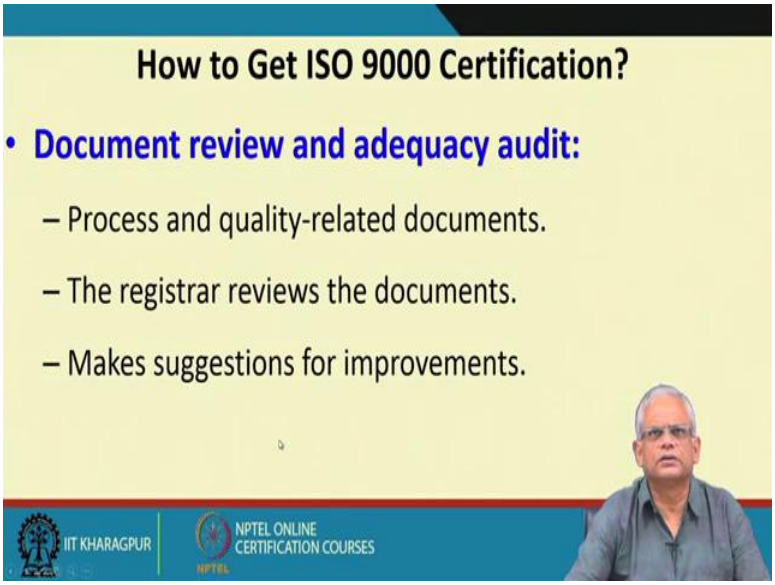

Reviews the uploaded documents and the adequacy audit, the process and quality related documents are uploaded. The register reviews the document and makes suggestions for improvements.

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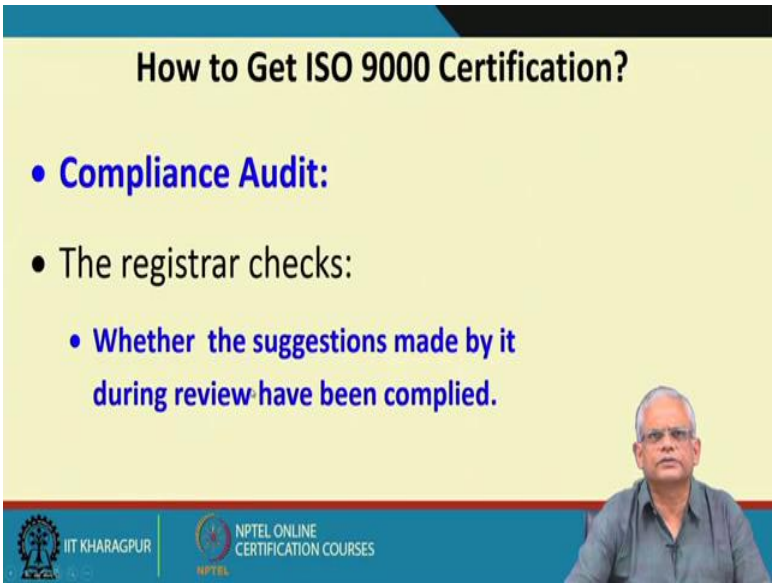

And once the company revises the document as per the registrars suggestions, compliance audit is done by the registrar. And this goes on until the registrar is satisfied whether the suggestions made by the registrar is a complied that is reviewed by the registrar.

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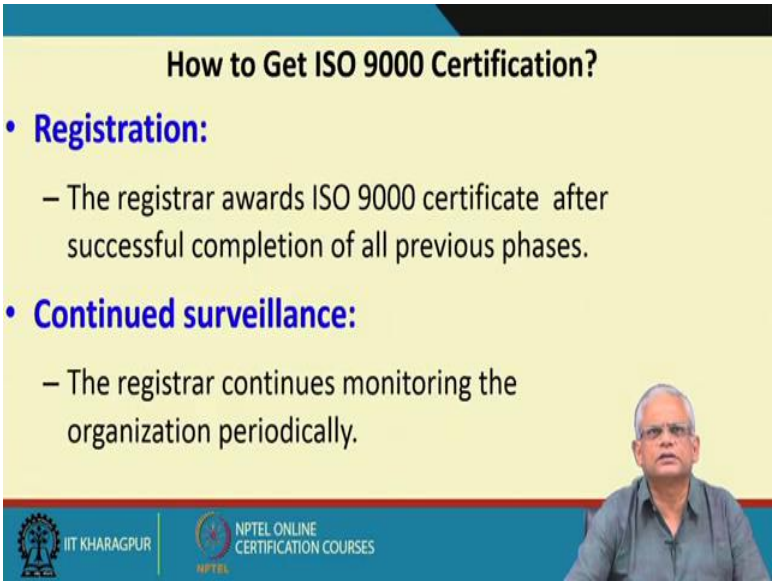

And once the registrar is satisfied, it awards the ISO 9000 certificate. But, then it is not given permanently, requires continuous monitoring by the registrar for the organization to retain the ISO 9000 certification.

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Since the certification is given based on the process of the organization, the quality system of the organization and not on the product. It is alright if the company uses the certificate in its corporate advertisements, that the organization has been certified to ISO 9000, but it cannot advertise its product. It cannot say that its products are as per ISO 9000, because the certificate is given to its process not to the product. And, if any organization uses the ISO 9000 certificate in its product advertisement then its certificate may be withdrawn.

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So, far we have looked at some basic things about ISO 9000, what benefits it provides, how to get the certification and so on. Now, let us look at what are the ISO 9000 requirements, what does the company have to do, what are the specific aspects of its process that it must clearly state in the process document. So that, it will be awarded the certificate. The first thing is the management responsibilities, the management should be committed to the quality of the products and also the quality system should be independent of the development system.

This is required that the quality system that is the people who are checking quality, they should not be working under the product manager. This is one of the requirement that they have to be independent. The product manager, the quality manager they should report to the top management only. It should not be the case that the quality manager reports to the project manager, in that case the quality may get compromised and this is one of the requirements in the quality system.

The quality system itself how it is formed, the specific documents it maintains and so on. Contract review; an organization during any of its project execution may take help of subcontractors, the subcontractors do some part of the work.

And, in this case whether the review is made of the terms and also when a organization enters into a contract with a party for to develop a software is the contract reviewed properly, is it understood what needs to be done by what time and so on, who reviews it;

these things must be detailed here. The main idea here is that unless the contract is reviewed properly and the work is accepted, then the project may not succeed.

Design control: as the design is made, it undergoes many changes due to various reasons; is the design controlled so that, the latest design is available and you can easily find out what changes had occurred to it before it and so on. In plain words this means configuration control that is using a tool a configuration management tool to keep the design documents under configuration control.

Document and data control: here all the documents and data must be under configuration control. Just see here that configuration control is really emphasized or in other words without using configuration control an organization may not get the certification. This is a very important requirement that configuration management must be used. As part of this course we will be discussing about the configuration management. Purchasing must be reviewed, control of customer supplied product again configuration control, product identification and traceability.

This is again configuration management, the product or the software as it is being developed undergoes many changes. We should be able to identify what is the latest product, what changes occurred earlier and so on. The process control: the process document must be under configuration management, because the process may change and the process document should be under configuration control.

Inspection and testing: testing is a very important requirement to get a quality product. So, both inspection and testing, the organization has to specify what are the steps of inspection, where all inspection occurs. And what are the stages of testing? Unit integration and system testing. Control of inspection and measuring test equipments, if there are some test equipments are used how are these calibrated, this also needs to be discussed.

We just looked at a very brief overview of the ISO 9000 requirements. The specific items that the organization must emphasize and also give its process for this in the documents to be uploaded. This is just a highlight, but there are many other requirements for ISO 9000, these are just the highlight requirements, some of the very important requirements. We will just look at few more requirements in the next lecture and then we will proceed further.

Thank you.

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Lecture - 53
ISO 9001 (Contd.) & SEI CMM

Welcome to this lecture.

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In the last lecture, we had said that ISO 9000 is a very important quality standard. There are lots of benefits to an organization who gets ISO 9000 certification. We had discussed about what benefits accrues to an organization and how to apply for the registration and get the ISO registration and finally, awarded the ISO certificate and after that we were discussing about the major requirements that the organization must fulfil.

We had identified several important points; now let us proceed from there. It said that some of the important requirements are the management responsibility; how is the management setup; are they committed to the quality; how do they do it; the quality system itself that how is the quality system setup; is it independent of the development system; contract review – before entering into contract is the contract reviewed for the capability schedule and so on; design control; document data control; control of supplier product; control of customer supplied product; product identification; process control and all these we said that configuration management is key to this.

Without configuration management we cannot have control of the documents traceability the changes that have occurred over time and so on. And, therefore, one of the major requirement of ISO 9001 is to use a configuration management tool and inspection and testing is another major requirement, have to identify at what stages inspection occurs and what are the types of testing that are undertaken and also if there are some test equipments are used.

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How are these calibrated? There are few more requirements. Inspection and test; control of nonconforming product that when there are rejections or defects how are these kept out of a circulation to the customers. If it is found that one of the product is having defect then how is it ensured that the same product is not given to another customer, and again here it means use of configuration management tool because in configuration management tool we can easily identify where a defect has been fixed and get the latest software.

Corrective and preventive action; how are these corrections and preventive action taken. Handling, storage, packaging, preservation and delivery; control of the quality records. It is not that the quality is the quality system just does inspection and it just approves but it must maintain the quality record. The quality system must be audited, who audits the internal quality system that must be clearly identified.

Another important requirement of the ISO 9000 requirement is that periodic training should be given to the developers. The developers must undergo periodic training because the skill changes frequently that needs to be a skill up gradation and this has been identified by ISO 9001. The document must also identify at what frequency training will be given, in what mode, how and so on. Servicing – how will servicing be provided; statistical techniques – what statistical techniques will be used and so on.

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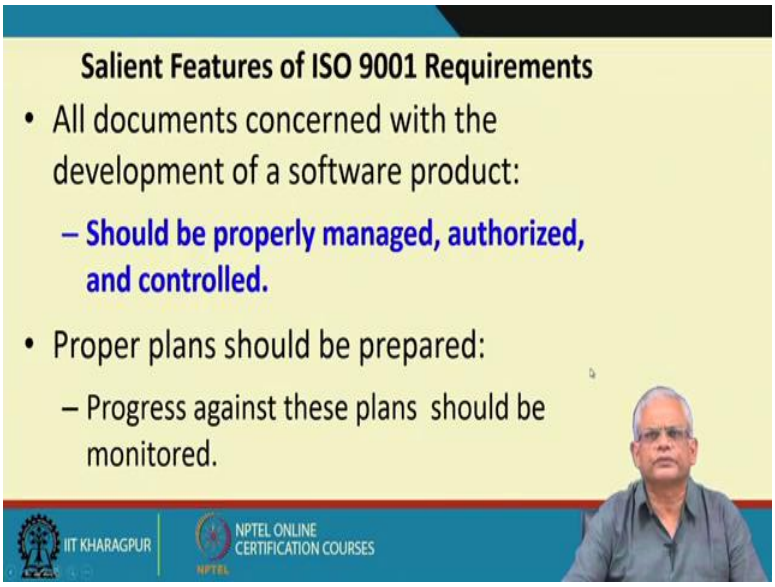

Now, let us look at some of the points some of the requirements more elaborately. As we mentioned that we are not going to look at all the requirements that will be too much for a short discussion. We are just getting a overall idea about the ISO 9000 certification process, the major requirements and so on. I am not really going into the nitty gritty of every clause that needs to be satisfied.

One of the major requirements is that all documents that are produced during developed must be properly managed, authorized and controlled. That means, who is authorized to change the document should be identified and enforced that is controlled and without a configuration management tool the authorization system will not work because somebody else may change it. The configuration management tool enforces the authorization and who makes change and so on are noted here.

Project planning is a important requirement of ISO 9001. Proper plan should be made and progress against these plans will be monitored. So, planning, monitoring and control

should be practiced by the organization without which ISO 9000 certification will not be given. It mandates planning, monitoring and control. So, proper project manager has to be there who has to do the planning monitoring and control.

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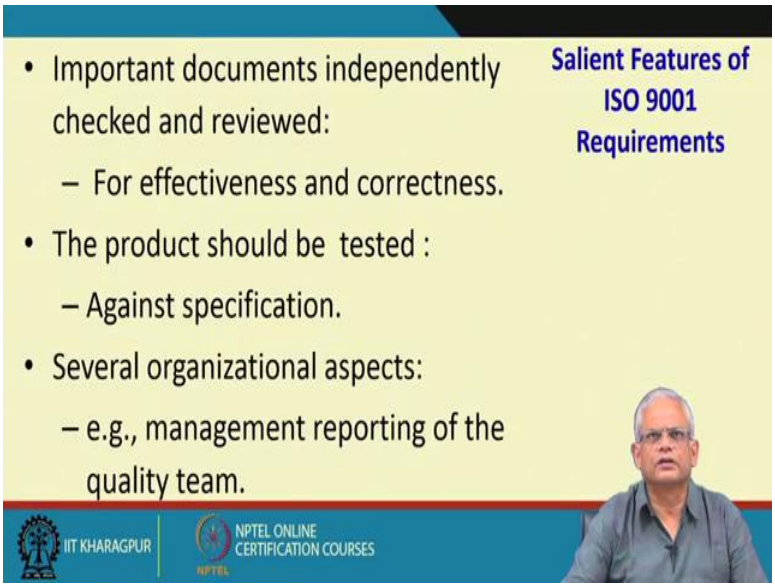

Review of documents is another important requirement. Testing against specification, unit integration and system testing must be there and also there are several organisational aspects like how whom does the quality system report. Does it report to the project manager, in that case certification will not be given. The quality system needs to report to the top management and not to the project manager. This is one of the major requirements of the ISO 9000 certification process.

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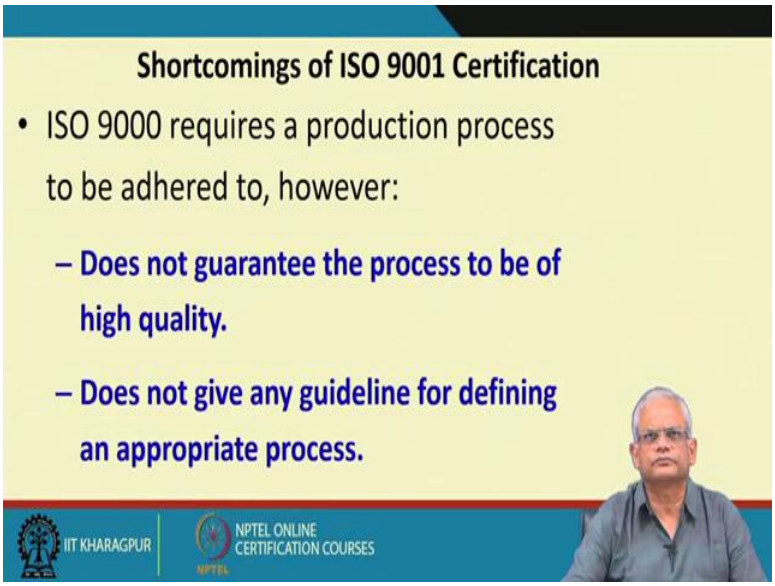

So far, we discussed about some of the important points regarding ISO 9000 certification process we said that the registrar looks at the documents, the process document, the quality document and so on and based on that awards the certificate. This we can interpret to say that the organisation has a good document, good setup document. It has proper production process quality system documented.

But, then one of the short coming of the ISO 9000 certification is that it does not provide any guarantee that the organization actually follows those documents because it is given on the documents the documents that reviewed they are made to have all the requirements of the ISO 9000 but whether the organization actually following that across all projects that is not monitored by ISO 9001.

And, also it says that what are some of the points that are required, but how do have that process is not there in the document and that is also very difficult because it is applicable across many industries and even in the software industry there are various types of projects for which different processes may be required and that is the reason why the consultants make a roaring business to help the organization set up guidelines to help them define an appropriate process on which the certification will be obtained.

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Also we had mentioned at the beginning of our discussion that the ISO 9001 certification process is given by an accreditation agency and it is not given by the ISO and there are various degrees of rigour with which the accreditation agencies they give the certificate some may be very flexible, in some country it may become very easy and so on. There is no international accreditation agency exists.

So, when ISO 9000 certificate given by different accreditation bodies may mean different and therefore, let us say an Indian company or some other company or wants to bid for the project in Europe, they may need the certification from a European accreditation agency like (Refer Time: 11:48) or something. Since these are given by accreditation bodies across different countries that are likely variation in the norms of ordering the certificates.

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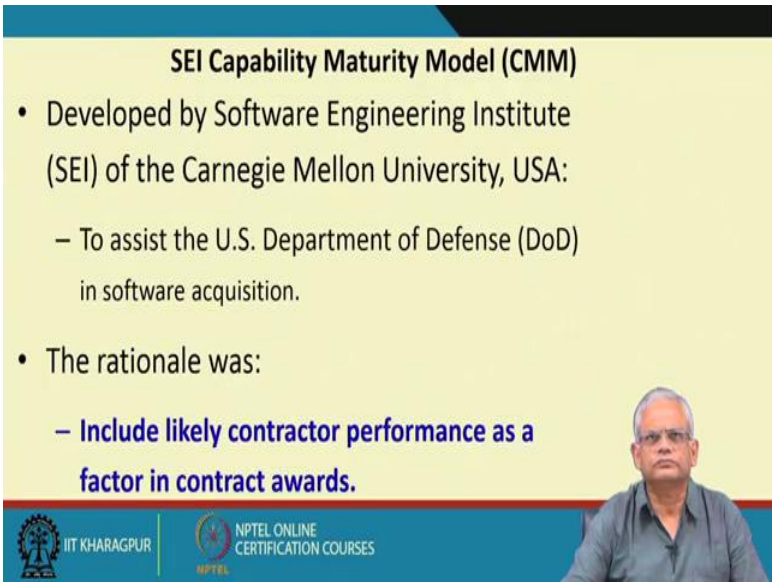

With this brief idea of ISO 9000 which is very important for a project manager let us look at another quality system standard which is the SEI capability maturity model or the CMM. The SEI stands for the software engineering institute at the Carnegie Mellon University in USA, itdeveloped a capability maturity model which in fact, in the quality system the set of quality system guidelines, but then this was done by the Carnegie Mellon University in USA on the request from Department of Defence in the USA. The Department of Defence in USA is one of the largest purchaser of software. It takes the help of many contractors or software developers to get its software.

But, then it was facing a problem that some of the developers used to give poor quality work them some of them even could not complete the work and so on and with that purpose it gave this project to the Carnegie Mellon University to develop a quality system guidelines which can be checked by the Department of Defence to see if the vendor organization has a good quality system and it can deliver the software. Or, in other words, the US Department of Defence by using the CMM model can assess the contractor performance before awarding a contract.

Since the work gets delayed money wasted if the work is awarded to an organization who cannot complete the work or does a poor job. It is very important for the Department of Defence before awarding the contract to make an assessment of the

contractor to check whether you it can be awarded the contract for software development and this was precisely done by the CMM.

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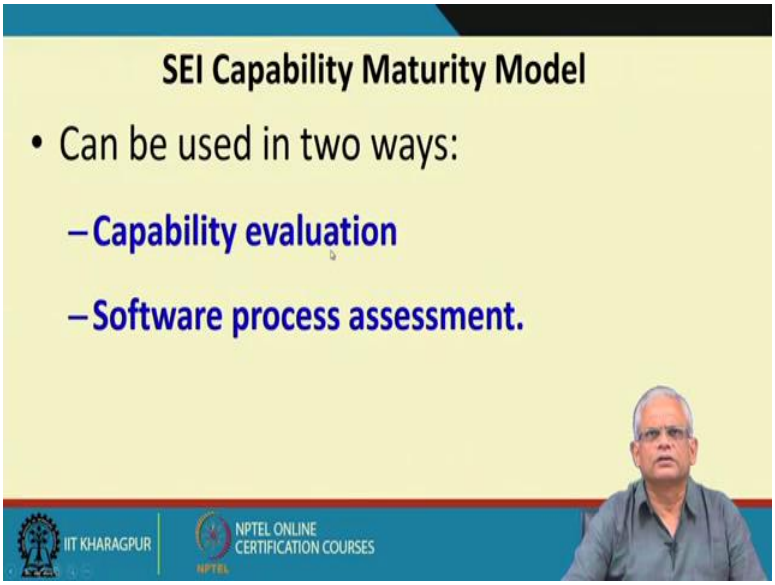

But, then the CMM can be used in two ways, one is capability evaluation. This was used by the department of Defence to evaluate the capability of the vendor that whether it can actually develop a good quality software and deliver but then the CMM also is useful for doing a software process assessment. In the software process assessment it is not done by the organization awarding the contract and checking the contractor likely performance, but here a software development organization checks its own quality system.

When a contract awarding authority contract awarding organization evaluates the capability of a vendor it is the capability evaluation, the CMM also supports software process assessment where an organisation can assess its own capabilities to check where all it has to improve to come up with a good quality software.

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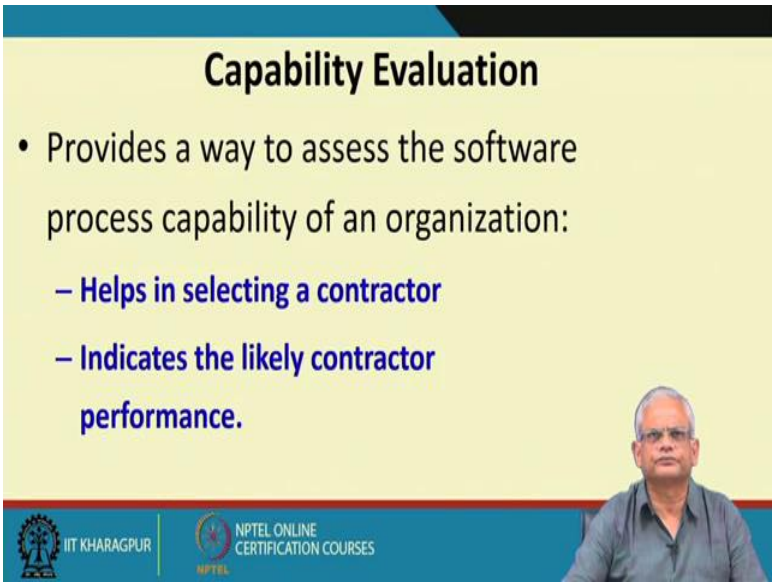

The capability evaluation can be used by an organization to select a contractor. The CMM model can be used by the contract awarding authority and go to the contractors premises and carry out the capability evaluation and that would give the likely contracted performance for a project that is awarded to the organization and based on the capability evaluation the organizer the organization awarding the contract can decide whether to award the contract or not.

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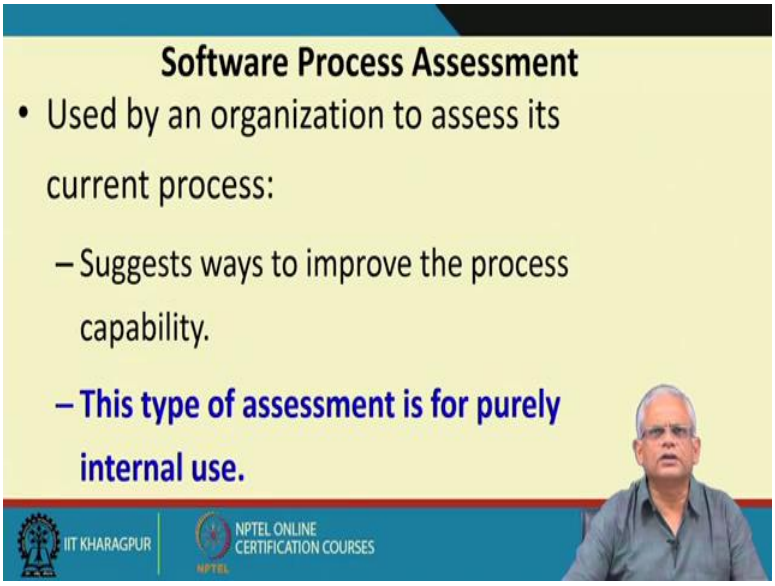

The other use of the CMM model is software process assessment. Here it is used by an organization to assess its own process the objective of this is to find out how to improve its own process and naturally this assessment is purely for internal use because the organization is assessing itself with the objective of improving its short comings. And, therefore if it says that we have been assessed at SEI CMM, level 5 and so on that would mean very little because we have assessed themselves with the objective of improving where it is improvements are required and therefore, it is not very meaningful to claim that they have the level 5 capability.

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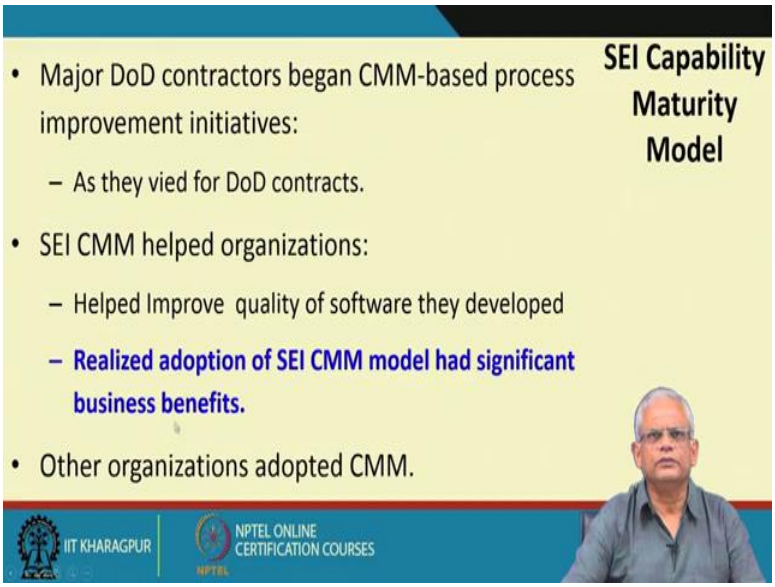

But, as the DoD the Department of Defence of United States, it awarded the project to the Carnegie Mellon University who developed the CMM and then the department of DoD enforced the capability evaluation, then all the contractors for the Department of Defence they used the CMM on themselves before the capability evaluation, so that, they can win the contracts at the Department of Defence. So, that helped them to get the contract, but also it helped the organizations themselves.

It was found that it helped definitely helped them to improve the quality of the software they developed. They could meet the timelines more frequently and they could develop the software within the required cost and because the organisations realized that using the SEI CMM was beneficial to them also not only winning the contract but also it had benefits to themselves in developing better quality software within time and cost

effectively they adopted the SEI CMM model and even other organizations who were not really contractors of the Department of Defence they started using the CMM for this purpose that it will help them come up with good quality software cost effectively and within time.

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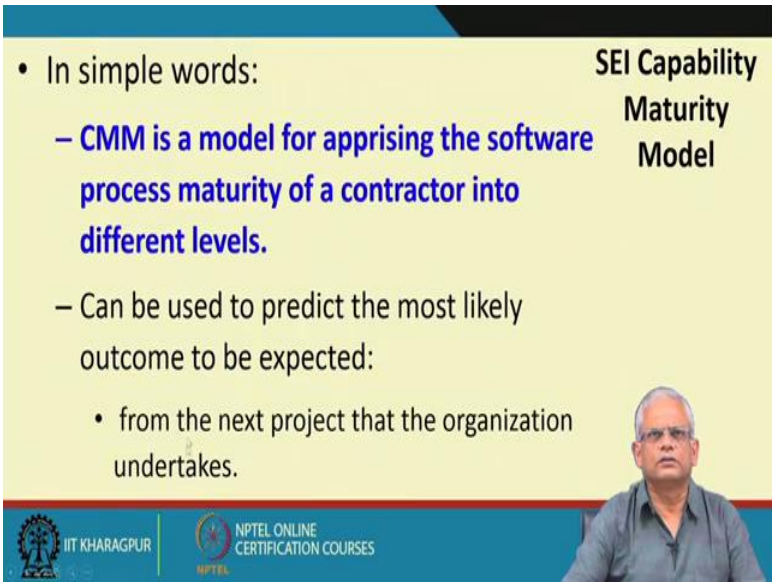

Now, let us look at the SEI CMM model itself. So far we have been discussing how it originated, what are its benefits, its uses – two major uses of the CMM, capability evaluation and process assessment and so on. Now, let us look at the SEI CMM model itself.

In simple words the CMM model can be used to appraise the process maturity of an organization into different levels and the implicit assumption here is that an organization using a good process produces good software. So, here we classify the organizations into different levels based on the process they are following and it has been found that this class actually is an indicator, this class of the contractor or the class of the vendor is a likely indicator of the performance when a project is awarded.

If it is at level 1, it is likely to produce poor quality work schedule slippage, higher cost compared to a level 5 organization who is likely to produce work that is cost effective within time and higher quality.

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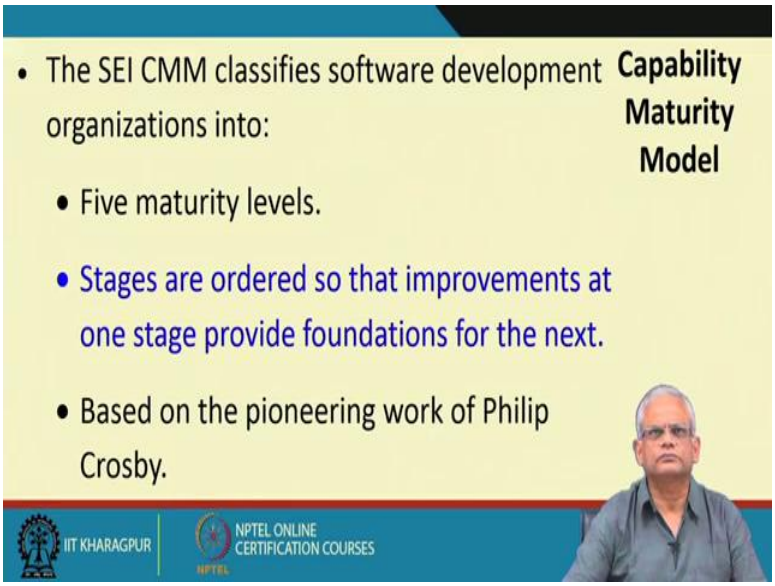

The CMM actually classifies organisations into five levels called as five maturity levels of the process that is how mature is the process and not only that that it classifies them into five levels. It also tells that what the organization needs to do to improve its process from one stage to another. If an organization is assessed at level 1, it clearly says that what it needs to do, what aspects it must improve and how to go to the level 2 and from level 2 to level 3 and so on.

But, then from level 1 it can go to the level 2 and not directly to level 3 without going to level 2 because the requirements of level 2 are a pre-requirement for level 3 without meeting the requirements for level 2 it cannot directly go to level 3. So the requirements of level 3 also have pre-requirement of level 2; requirement of level 4 needs the pre-requirement or that the organization is already satisfying level 3 and level 3 requires already satisfied level 2 and so on.

And, the CMM model if we really dissect it we will find that it is largely based on the work of Philip Crosby who had published a book called is Quality is Free. There also we have five levels of organizational process maturity and so on. But, then the CMM model has structured it and identified the specific questions to be asked based on which the organisation can be assessed and so on.

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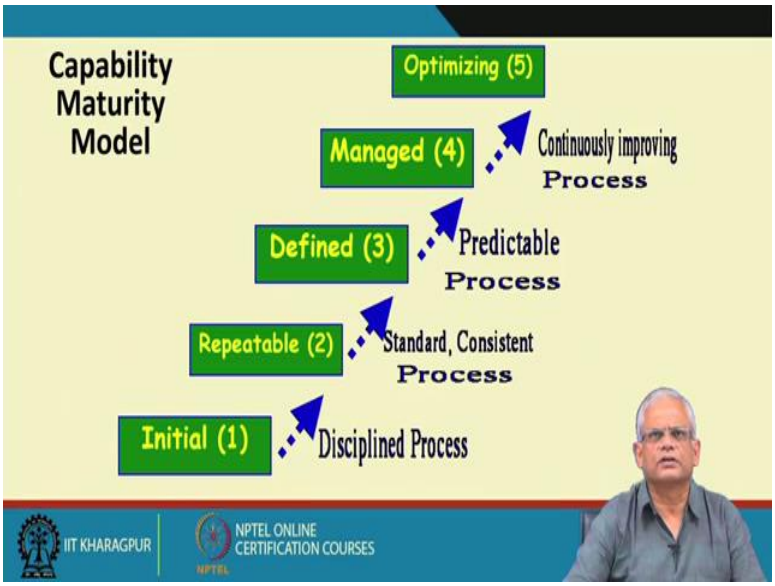

The five levels that the CMM identifies are the following. The lowest level is the initial, this is level 1. Every organization by default is at initial level and then is the repeatable level or the level 2. The third level is the defined level, the fourth is the managed level or level 4 and the fifth level is called as optimizing. This is the continuously improving process.

We are already at the end of this lecture we will stop here, and in the next lecture we will discuss about the specific requirements that the organization must meet to be able to a to be able to qualify for a specific level of the SEI CMM model. We will stop here.

Thank you.

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Lecture – 54
SEI CMM (Contd.)

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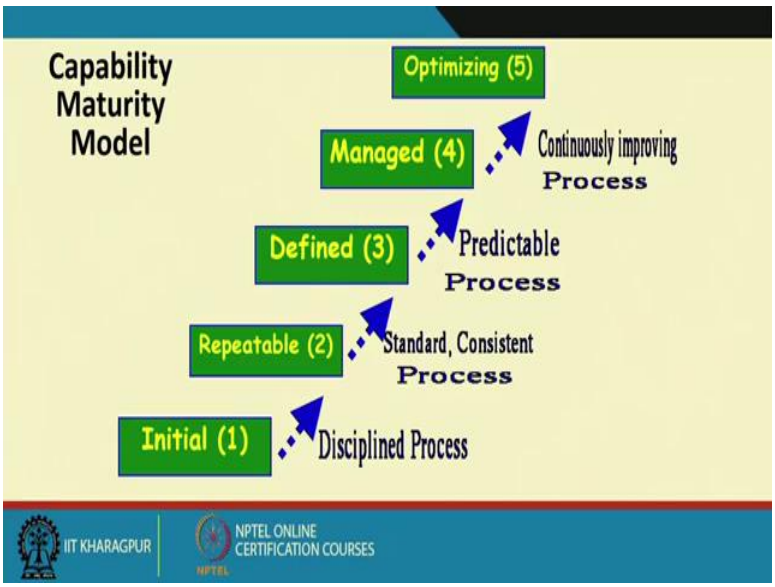

Welcome to this lecture. In the last lecture, we were discussing about the SEI, CMM quality model. We had said that this was set up by the Software Engineering Institute at the Carnegie Mellon University, on a request from the department of defense. The department of defense wanted to assess the capability of its vendors the likely performance of the vendors before awarding the contract. But then it had significant benefits for the organizations themselves. It helped them to improve the quality of their work, meet the schedules and also develop in a cost effective manner.

And therefore, it is widely accepted, world wide it helps the organizations to improve its quality in stages. We are just discussing this model here where the level 1 is called as the initial model. All organizations by default are at initial, even if they do not have any documented process they do ad hoc build and fix for all these they are at initial level, every organization by default is at the initial or level 1.

And to go to the level 2, they must have a disciplined process. A disciplined process is that they must be using some process. They must be having project plans made,

monitoring against the plan, there must be a proper project manager who makes plans, monitor against these plans, configuration management of the documents and so on. And then it goes to the repeatable level. This is level 2, the implication of the repeatable as we are mentioning is that an organization can repeat its success on one project on similar projects in the future.

That is if we did a job that was awarded to it well, then it can repeat its success on that project on similar projects. The third level is called as defined level. In the defined level, the process must be documented in the repeatable it is a standard consistent process. In the repeatable level, it is a disciplined process and then in the defined level the process must be documented, so that the process is used consistently by all members.

In the initial level the process there is no disciplined process, everybody is free to use their own process, even build and fix there are no process, only build and fix, that is also would qualify for initial. In the repeatable there must be a disciplined process that is the process must be followed by different developers, but there may be variation in the process, their understanding a process may vary.

But in level 3 there is a standard consistent process, only then they get to level 3 and to get a standard consistent process, the process has to be documented, it has to be defined what exactly is the process. Here in the disciplined process and the repeatable, the project management is there and also the process is disciplined, everybody have the process in their mind, but then there can be variation across the different developers. But in the level 3, there is a standard consistent process and this can be achieved only when the process is documented.

In the level 4, its necessary to have a predictable process that is the process performance needs to be measured and then whether the goals that are there are met this is identified and corrective actions are taken if the goals are not met. For example, let us say one of the process metric or process measurement is how effective is the review where most of the bugs identified in the review process or they got identified during testing and that measurement is tracked and if there are very few bugs which are identified during the review and most of the bugs got identified in the testing.

Then it was identified that why the review was not effective and therefore, it was enforced more thoroughly may be the review was not done properly and that is why most

of the bugs they escaped to the testing. From the managed go to the optimizing, the organization must have a continuously improving process.

Now, let us see the same example that if there are bugs identified, the number of bugs identified in the review versus the number of bugs identified in the testing if it was identified that review was not very effective in a managed stage it would be investigated that why the review was not effective. Maybe it was not followed rigorously, may be the people who reviewed they just were lacks could not review properly and so on.

Whereas in the optimizing level, it was also identified that whether the review process itself is not proper; does the review process needs improvement and the process may be changed here. So, in the managed stage the metrics are collected, the process metrics are collected to investigate and find out why the targets were not met, what was the short coming on the part of the team members who were using the process.

Whereas, in the optimizing level it is also identified that does the process itself needs improvement and then the improvements are made to the process. So, that is the overall idea here in the capability maturity model that the initial all organizations quality, this is also called as a chaotic organization where everybody all developers are free to develop their own way chaotic activity, that is the hallmark of the initial level.

In the repeatable processes are followed, but then there can be variation across the process different team members. The project management is followed. And then the next level three is the defined process where the process is defined and documented. The managed level, the process performance metrics are collected and these are used to identify why the targets are not met, why certain things were not working certain activities were not satisfactory, but the process is not changed based on the metrics.

Whereas, in the optimizing the process continuously improves. The documented process every time necessary it keeps on improving.

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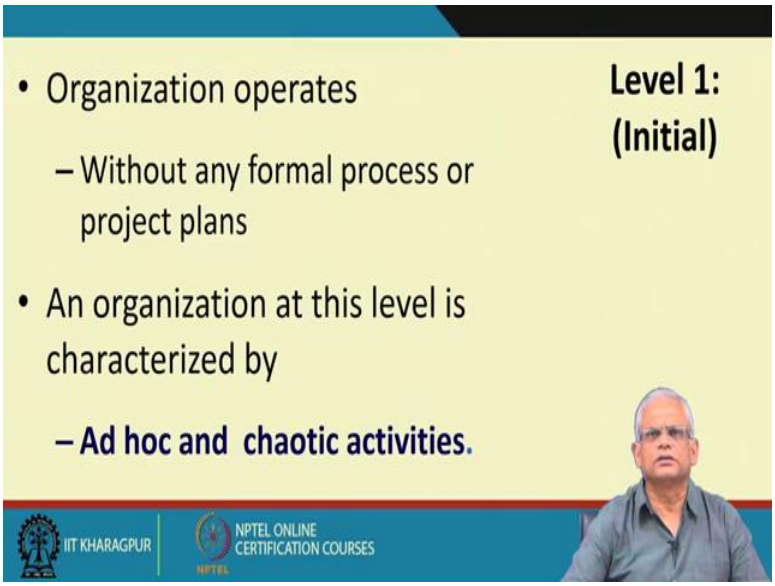

The level 1 is the initial level. Here every organization by default qualifies to be level 1 there are no process requirement, no project plans required. So, every organization just developers got together developing and that is will qualify the level 1 or the initial. The hallmark of a level 1 organization if a organization if qualified at level 1, then the projects are done ad hoc, the developers are free to do whatever ad hoc and chaotic activities this is the hallmark of a level 1 organization.

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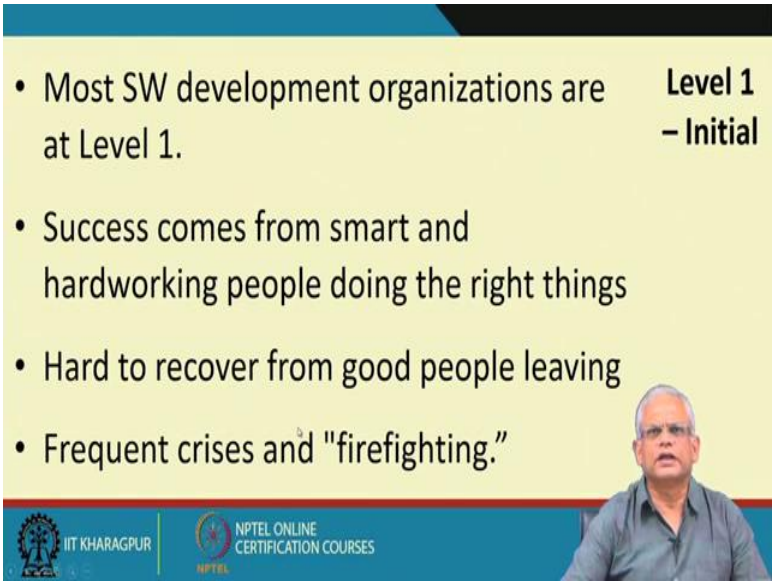

But the thing is that many software organizations they are at level 1. They, the developers there are free to do whatever way they want to develop software. Success comes from smart people and hardworking people doing the right things. If a level 1 organization has a success in a project its largely attributed to some of the team members who had worked overtime, who are doing smartly getting some things right and so on.

But then this kind of organization get can get into big problem if during the project some of those good people leave, then the project there will be a total failure. And this kind of organization during the project tenure there are frequent crisis and the good people the calm down in a firefighting mode, let us say some module is not working, the good people the smart people the those are there in the team. They take up those activities help out and bring the project to success.

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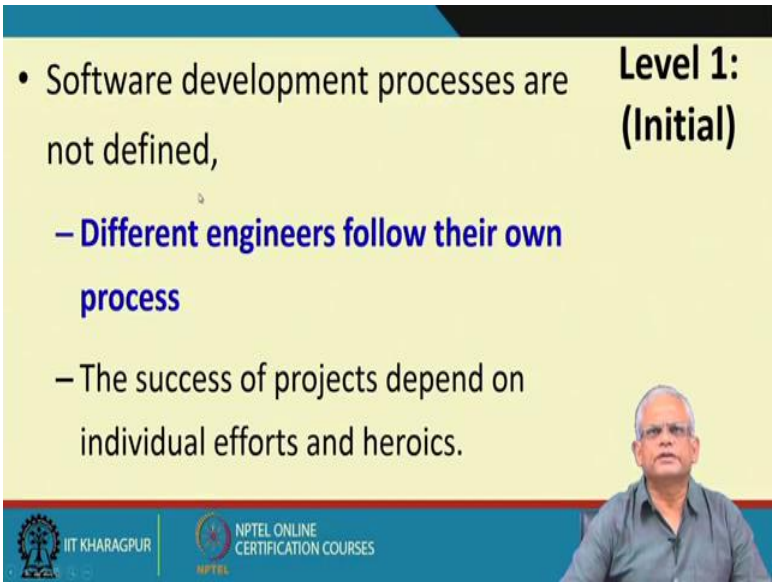

Here there are no specific process followed. Different engineers have their own understanding of their process, they may not follow any process and the success of the project depends on the individual effort and heroics.

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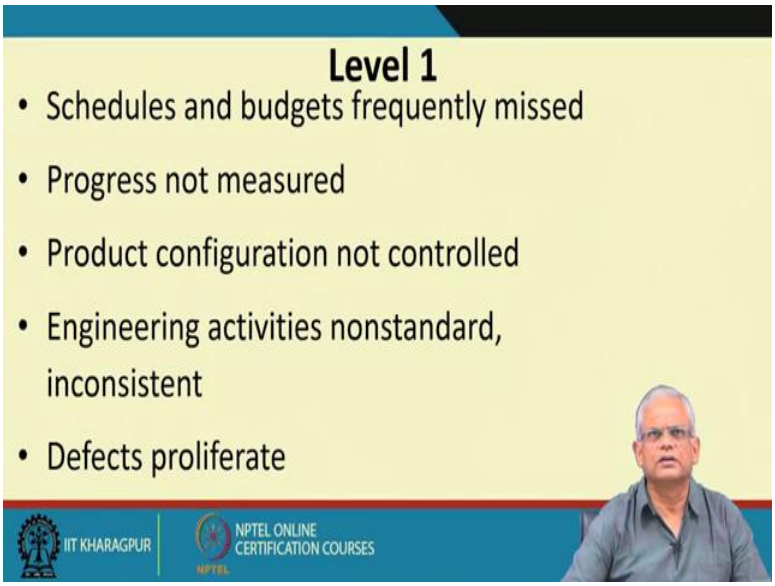

The schedules and budges are missed. Progress is not measured, there are no plans made there are no progress measurement. And that is one of the major reason why the schedules are missed because to start with and not made any plan that what work will be complete by what date and so on. Also the product configuration is not controlled.

It is likely that buggy software where the bug was fixed, but still the buggy version was forwarded to a customer. The activities that the developers take a nonstandard inconsistent and there are large number of defects in this kind of software developed by a level 1 organization.

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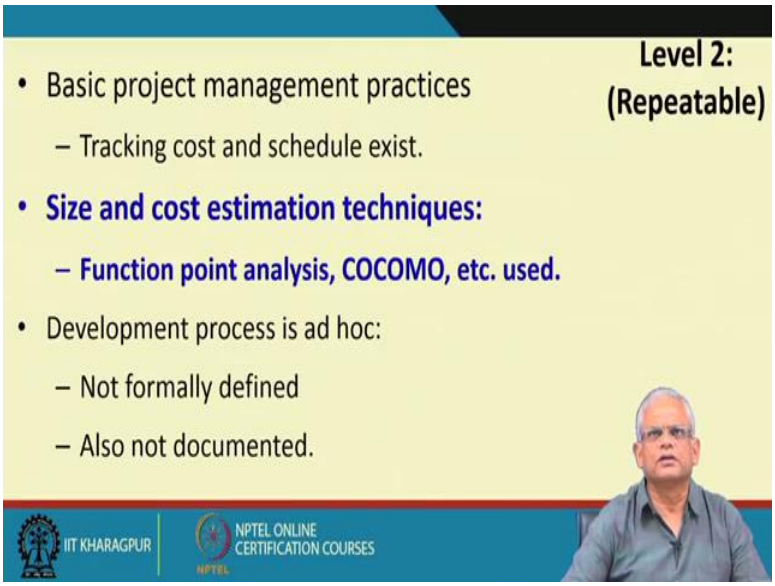

The level 2 is called as a repeatable level because here the basic process framework is in place and some of the basic requirements are in place and therefore, an organization at level 2 if it has a success in a project it can repeat its success on a similar project. If it developed a payroll software successfully for some organization and another payroll software it can with success develop or some other organization.

One of the major requirement of the level 2 is project management practices that is making the basic plans and then monitoring against the plan, tracking the cost and schedule. Size and cost estimation techniques are done by the project manager function point analysis, COCOMO etcetera these are used. But again here the developers though they have their process, they do not use a build and fix.

They have their process, but these are not formally defined. There may be variation of the process that are used by different developers and also these are not documented.

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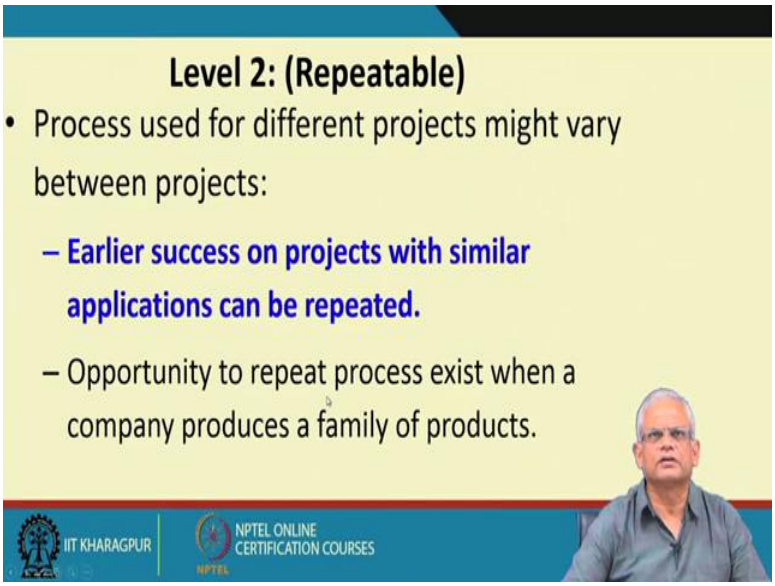

Also the process that is used for different projects may be different and here the success on a project can be repeated on a similar project and if a organization produces a family of products then a level 2 organization, is it is possible for a level 2 organization to repeat its success on one product across the family of products.

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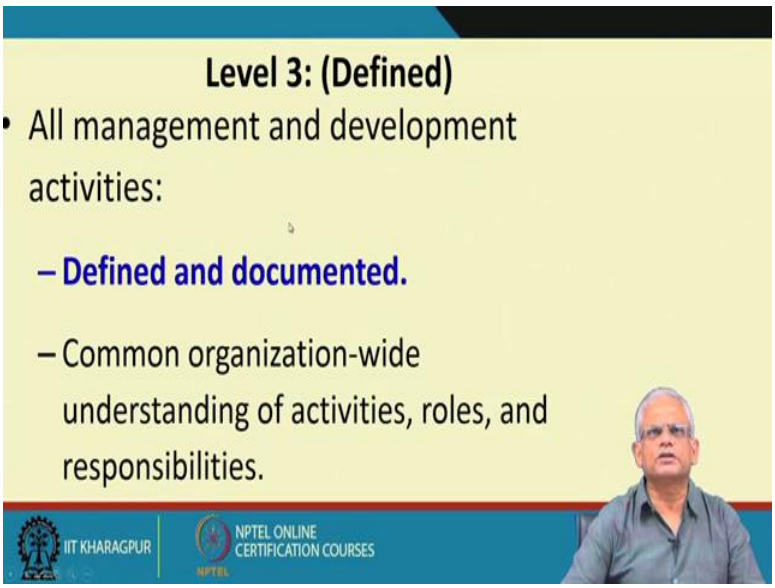

In level 3, the process is defined and documented and therefore, there is a consistent process that is followed by all developers. There is a common organization wide

understanding of what are the activities to be done at what time, what are the roles, what are the responsibilities.

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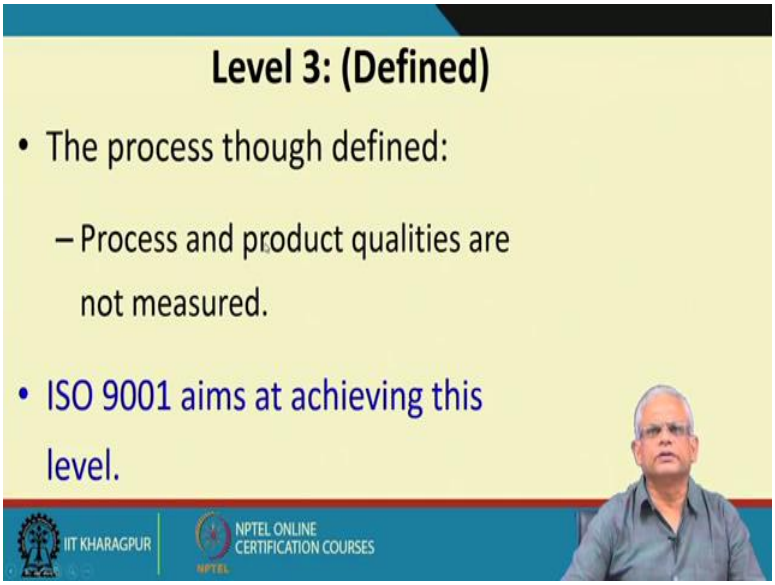

But then even though the process is defined. But then there are no process measurements are done and roughly ISO 9001 is at level 3. But ISO 9001, the 2000 version that targets for level 5, but the original ISO 9000, 1987 is roughly at the level 3 of the SEI, CMM model.

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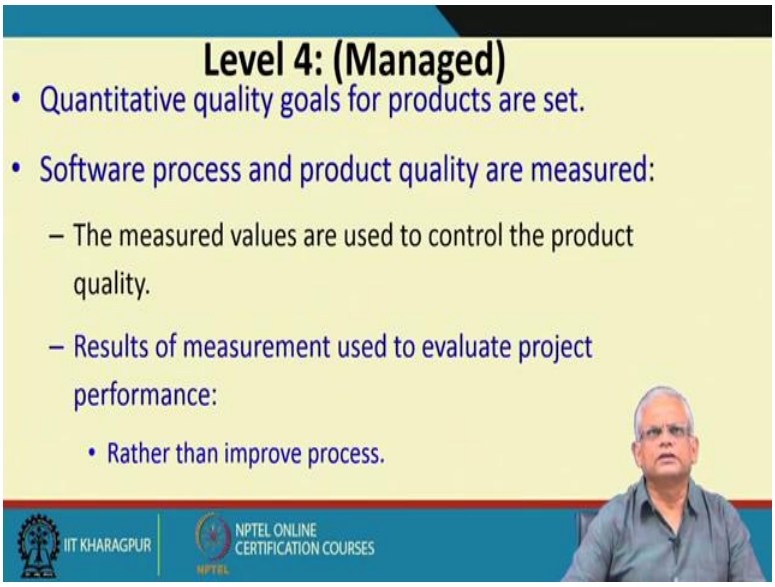

In the level 4 here, there are quantitative quality goals for the product are set. For example, that how many bugs per 100 lines or 1000 lines should be there in the final product. After all the testing is complete all the development and testing is complete all the development and testing is complete what is the quality goal for the number of bugs per 1000 line or what percentage of bugs to be identified during the inspection process and what percentage to be identified by the testing process?

Here these kind of process and product qualities are measured and if this fall short, then these are used to improve the project performance that is some things were not done properly. And it was used to give this feedback to the team that you did not do a proper review or a proper testing was not done. This has to be done more rigorously. So, the project performance is enhanced and not the process itself. The no change to the process is made, but then the process that exist are used more rigorously to improve the performance.

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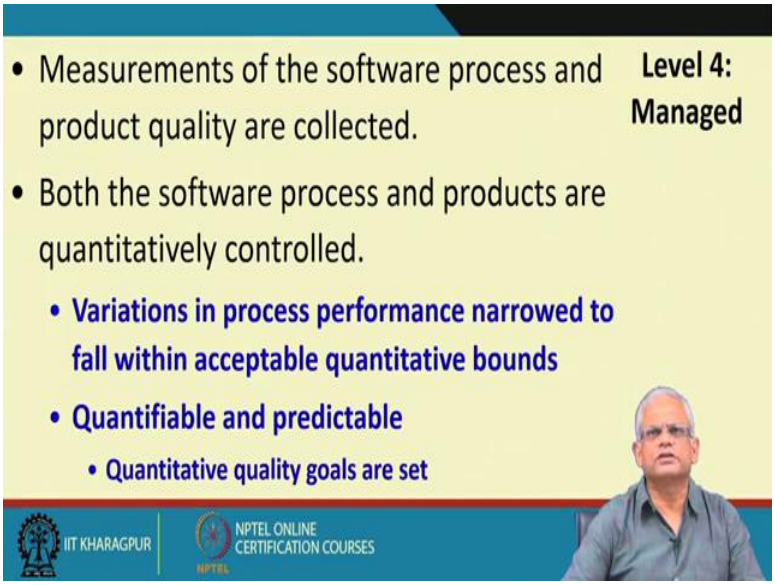

And level 4, the process and product quality are collected and the software process and product are quantitatively controlled and variation of the process performance is narrowed to fall into acceptable quantitative bounds. That if let us say during inspection 50 percent of the bugs must get detected and should not go to the testing stage, then more or less all projects the same thing happens that is all projects do the inspection

rigorously. Here quantitative quality goals are set and therefore, the quality is quantifiable and predictable.

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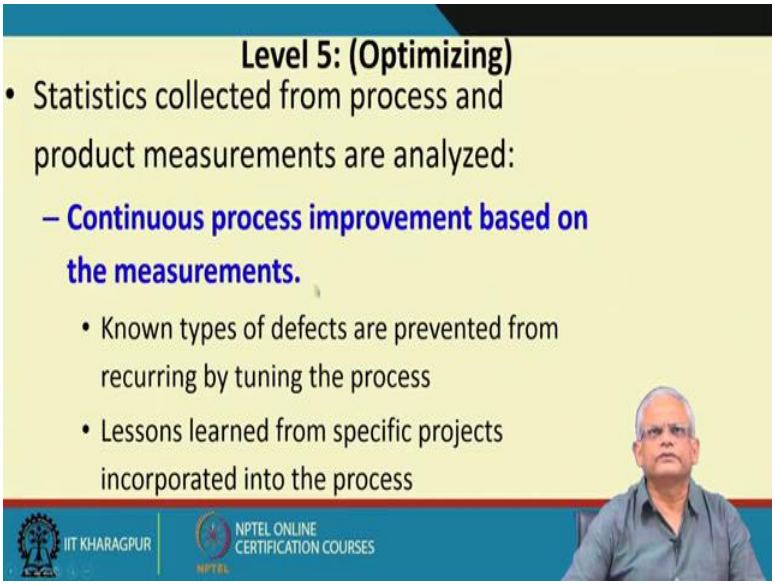

In the level 5, the measurements are used not only to enforce the process is followed rigorously, but also it is identified whether the process itself needs improvement because it may so happen that the review is done rigorously, but then the number of defects getting detected during the review is not satisfactory. Then the review process may need to be improved and therefore, the process keeps on improving continuously based on the measurements.

If there are some known problems with the process, then the process is fine-tuned and the lessons learned into some specific project learn from specific projects these are incorporated in the process, so that these good things that some project used and got very positive results, these can be used organization wide and therefore, the process itself is changed.

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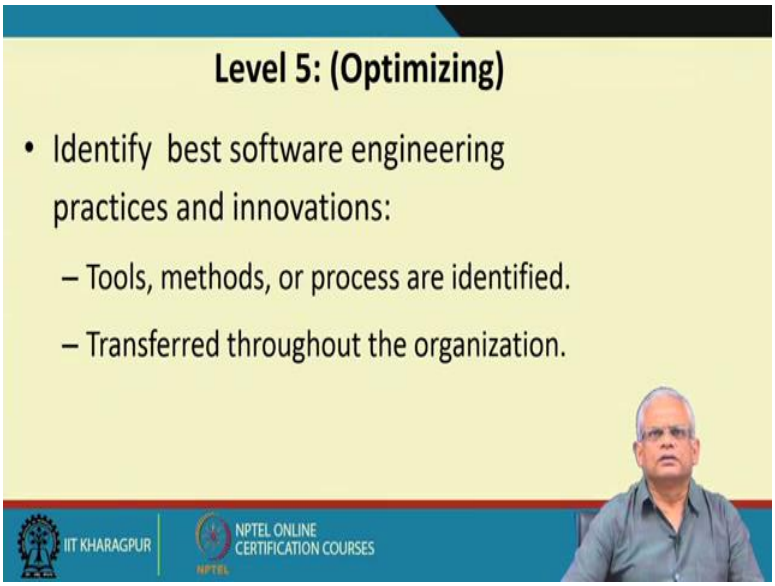

Not only that, there are lot of innovations and improved practices are being reported across different organizations, in the journals, magazines and so on. These tools methods and processes are identified, there is a special team here in a level 5 organization whose work is to find out if there are improved methods tools and processes which are reported then these are incorporated into the process itself and these are transferred throughout the organization.

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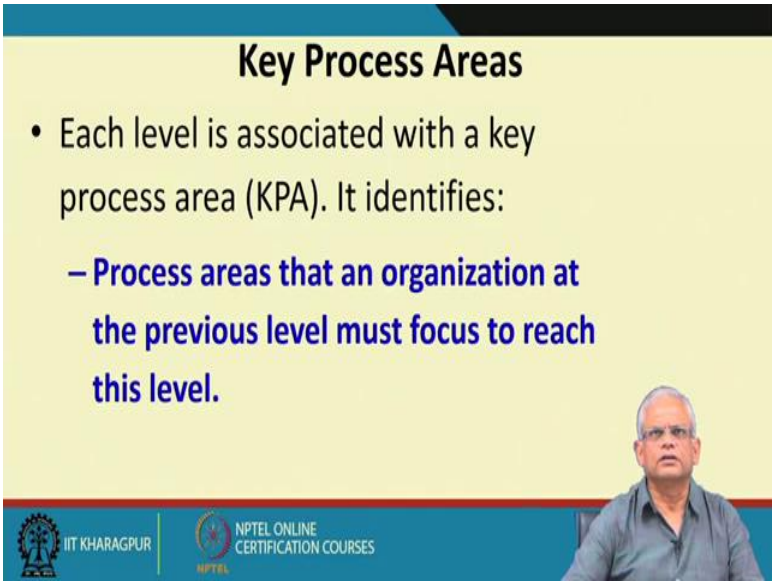

Now, let us discuss about another important aspect of the SEI CMM which is the key process areas. As you might have already noted that the ISO 9000 is a go, no go scenario, either a organization qualifies for ISO 9001 or it does not. But here in the SEI CMM when organizations get assessed at certain level and then it tries to go to the next level, and there are set of things that have been identified that organization at one level to go to the next level and that is called as the key process areas, some part of the process that needs to improved.

The KPA is a important aspect of the CMM model. An organization assessed at certain level it can go to the next level if it satisfy some key process areas. These are the process areas that an organization at some level must focus to reach the next level or let us say level 3 KPA, it identifies the process areas which a level 2 organization must focus to reach to the third level.

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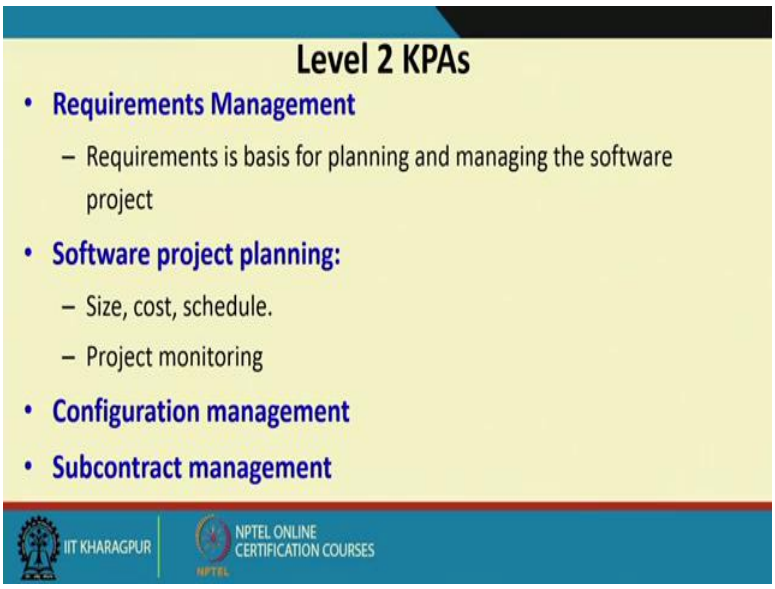

Now, let us see what are the KPAs for the different levels. For level 1 there are no KPAs because every organization is by default in level 1 and KPAs are those process areas which an organization must focus to come to that level. The level 2 KPAs are the process areas which a level 1 organization must focus to come to level 2. These are the KPAs requirements management, the requirements document must be written and managed.

If the software project planning, there has to be a project manager plans have to be made estimates have to be made, based on that plans to be made and based on the plan the

project monitoring has to be done. The third requirement is configuration management. The configuration management allows a document to be controlled, only the authorized persons can change it and also it will be possible to identify the latest version of the document easily and also it will be possible to go back to a previous version and subcontract management.

So, these are some of the process areas that a level 1 organization must focus to come to level 2. Must have proper requirements document, because requirements are the basis for planning and managing. There has to be a project manager who does project planning, project estimations and monitoring and control based on the plan.

Configuration management must be used. Without configuration management an organization will not satisfy level 2. And also subcontract management for awarding work to a vendor from the project some part of the work may be given to a vendor, but then the work that is given to the vendor must be properly identified, reviewed and the item that are completed by the vendor must be properly tested to comply with the requirements that was given to the contractor.

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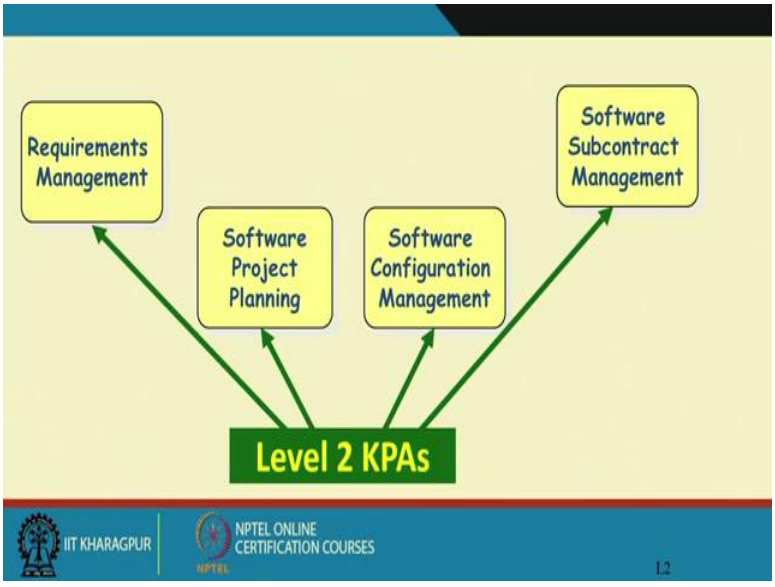

We can pictorially describe what are the KPAs per level 2, one is the requirements management. The requires, the requirements must be obtained and this must be put under configuration management because requirements keep on changing. Software project planning monitoring and control, this is another KPA software configuration

management. This is also an important KPA and subcontract management, this is another KPA.

We are almost at the end of this lecture. We will look at the KPAs for the other levels of the SEI CMM in the next lecture and we will move on from there.

Thank you.

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Lecture – 55
SEI CMM (Contd.)

Welcome to this lecture. Over the last couple of lectures, we have been looking at the software quality management issues and if you remember we had said that for effective quality management, an organization must have a quality system. And there are several quality standards, quality system standards that are available used throughout the world. We had discussed about ISO 9001 and then we had been looking at the software engineering institute capability maturity model.

We had looked at the 5 models and what are the quality system that is required at the different levels. The 5 levels of SEI CMM and the quality system requirements at each of the level. And also we had said that the CMM gives a step wise approach to improve the quality at an organization and this is known as the KPAs or the Key Process Areas.

An organization at a certain level that wants to improve its quality standard to the next level, will have to incorporate certain key process areas. It has to focus on improving some key process areas that is clearly defined by SEI CMM. In the last lecture, we had just started looking at the key process areas, now let us continue from there.

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We will first look at the key process areas and then we will compare the CMM with ISO 9001 and after that we will look at the personal software process.

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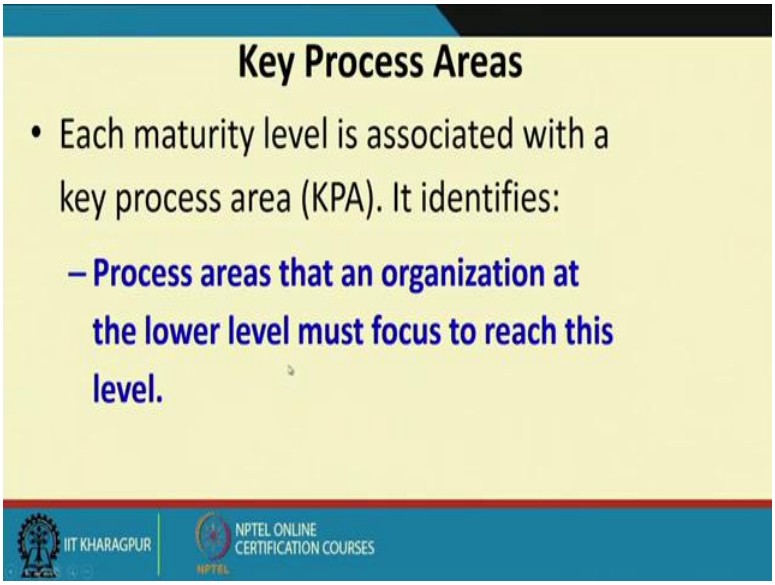

The key process areas are defined for every maturity level of the SEI CMM. The KPA identifies the process areas that an organization at a lower level must focus to reach this level, that is suppose an organization is assessed at level 2 and it has to, it needs to improve its process its quality system to level 3. Then it must look at the KPA is identified for level 3 and focus on those.

In that sense, the KPA is a very important concept here in the CMM and it gives a step by step approach to improve the quality at an organization.

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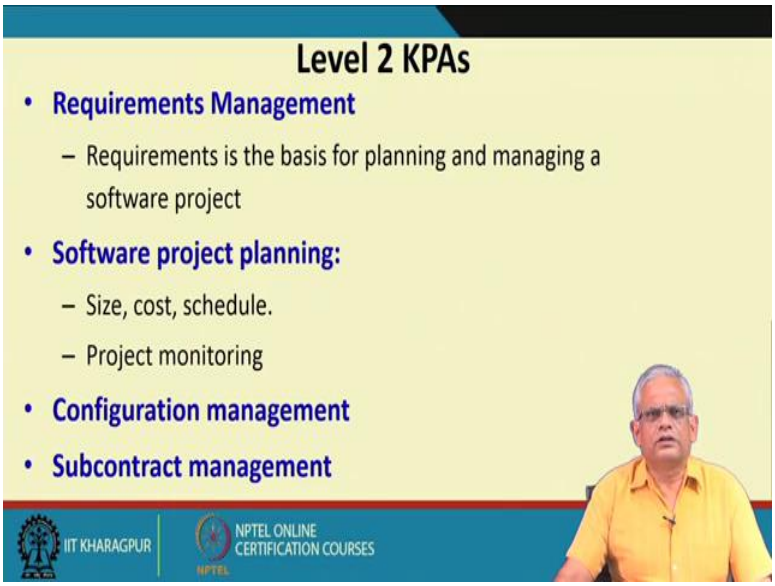

For the level 1, there are no KPAs because every organization by default is at level 1, it does not need even a process. So, there is no process areas, key process areas identified for level 1. Now, let us look at the level 2 KPAs that is if an organization is assessed at level 1, what are the process areas it must focus to improve its quality system to level 2.

The first thing identified in the KPAs is requirements management, that is the organization must have proper requirement specification and also how does it manage the requirements that is the changes in the requirements how are these controlled. This is important because requirements is the basis for planning and managing a software project that is identified by the CMM, requirements is a fundamental document, is a basic document which is required for many purposes including planning and managing a software project.

Another aspect process area which is highlighted is the project planning and monitoring. So, this is basically the software project management. Unless a organization has a project manager who estimates the size cost, makes plans according to the size cost, develops a schedule and then monitors the progress, the organization cannot be assessed at level 2, one of the emphasis of the level 2 process is the project management. It is identified that project management is one of the basic aspects of a project without which the project is hard to become success.

Another issue that is process area that is identified is configuration management. A configuration management tool must be used, so that various documents can be put under configuration control, the requirements document, the design document and so on. So that, the changes to this can be authorized that is who changes it is not that anybody can change, an authorized person can change the document and also it should be possible to identify the latest configuration or the latest version of the document and also to be possible to find out what changes have occurred so far and so on.

The ones that are the process areas that are identified for level 2 are some of the basic aspects of a software project without which it is very difficult to go to the higher levels, the going to level 3, level 4, etcetera without incorporating this basic process areas would be counter productive. The other process area is subcontract management.

Every project at some point or other needs to give out work, well defined pieces of work to other vendors and how are these specified is the contract clearly specified what is the work to be done and when getting the delivery is it checked against those specification. So, those are subcontract management. So, the level 2 KPAs are 4 important KPAs. There are other KPAs, but we have just highlighted here most important ones, requirements management, software project planning, configuration management and subcontract management.

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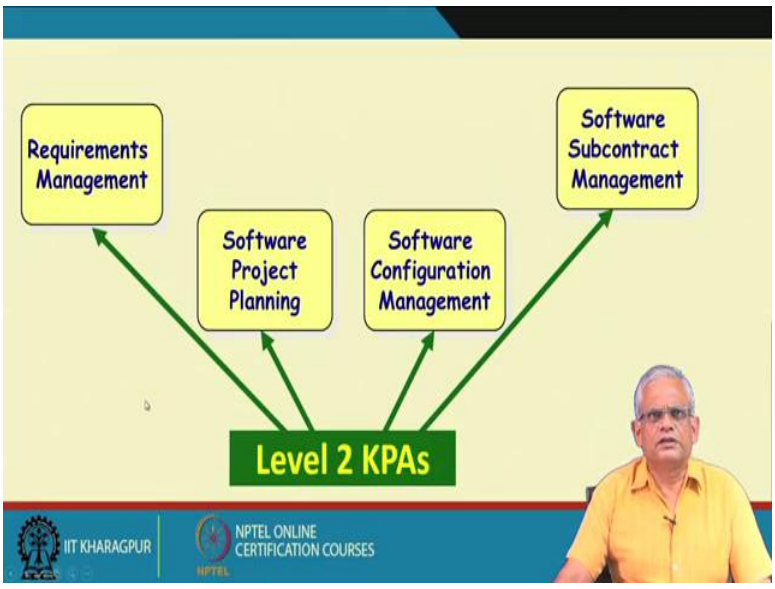

Just represented this in a diagrammatic form here. The level 2 KPAs are 4 process areas requirements management, software project management, that is software project planning monitoring and so on, software configuration management and software subcontract management.

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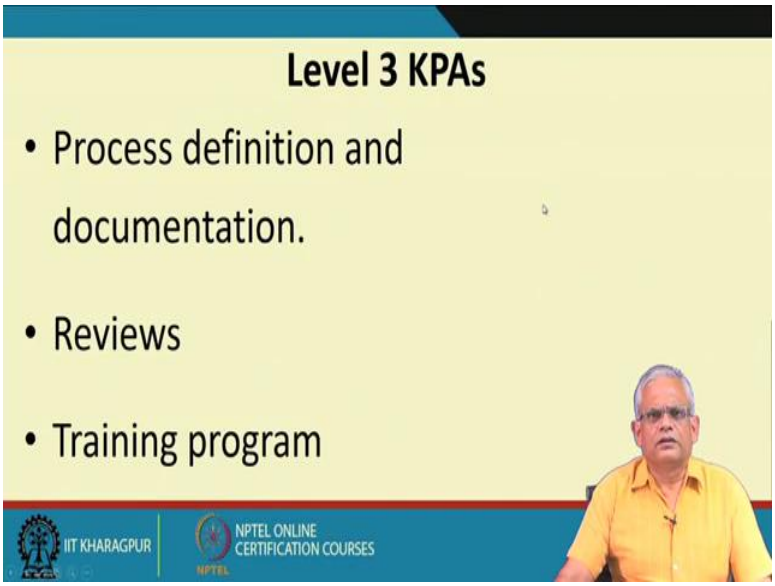

The level 3 KPAs are the ones are the process areas which a level 2 organization must focus and incorporate in order to improve its process to level 3. The level 3 KPAs are process definition and documentation. In level 2 different developers might be using their own process, but there is no standard understanding of which process is being followed, there is no documented process available in the organization whereas, in a level 3 organization the process that will be used by all developers is clearly defined and it is documented in the form of a process handbook.

Reviews are carried out at various stages, the requirements are reviewed, design is reviewed, code is reviewed and so on. And the third process area is training program it is identified that the skills for software development is continually changing, and unless the developers keep up with the latest developments it would be heard for a project to succeed and therefore, periodic training program is another issue which is highlighted by the level 3.

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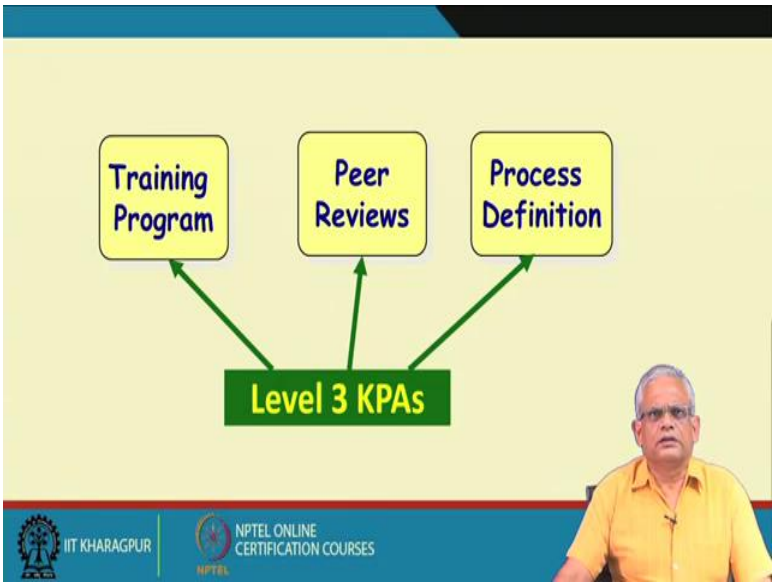

Represented this in the form of a diagram. 3 important level 3 KPAs training program, peer reviews of various documents and process definition and documentation.

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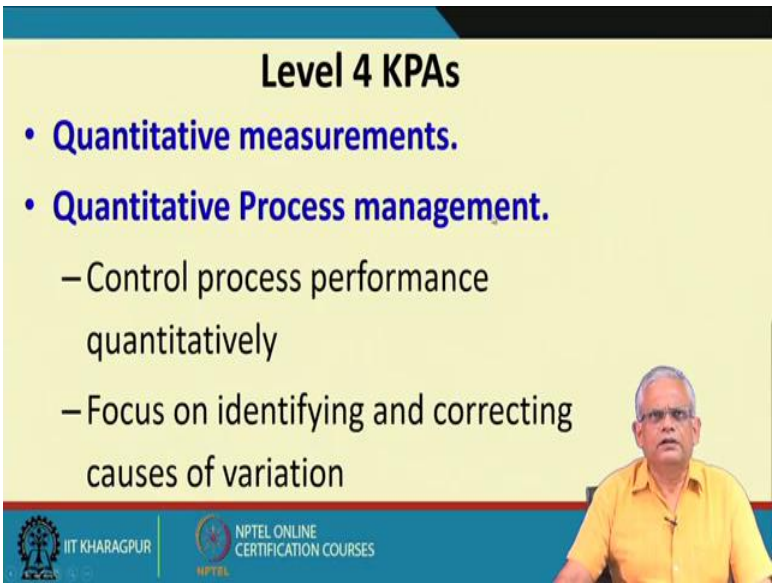

The level 4 requires measurements of the process performance, both the process and product metrics are obtained and there is a quantitative process management, that is based on the measurements and the process and the product, if there is a deficiency somewhere then it is identified why this is occurring.

May be the development team is not strictly practicing some process and therefore, it focuses on identifying and correcting the causes of variation. If the defect rate that is identified after testing is high, then possibly the testing was not proper, may be the testing was not done rigorously and therefore, the project manager identifies that what caused the metrics to miss the specified values and then it takes a counter measures. So that, the process variation is reduced.

In summary here, for level 4 KPAs process and product measurements and using this measurements quantitative process management is practiced here. If there is a deficiency in any of the measured attributes it is identified and corrective action is taken, but one thing must point out that the corrective action here does not really imply changing the process. For example, the testing is not effective because certain types of tests were not documented in the process document that is not identified actually at level 4.

The level 4 only tries to find out why there is a variation in the process that is sometimes there are more defects for example, and then identifies what action needs to be taken by the project manager to correct that deficiency.

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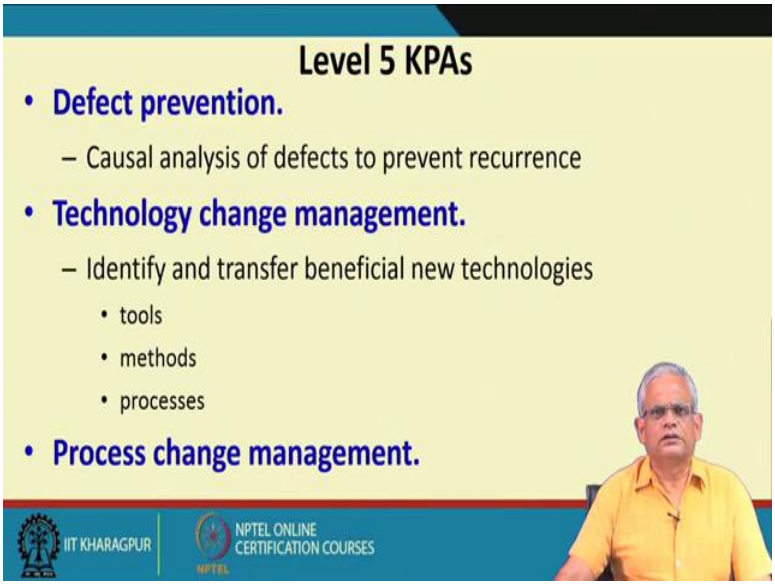

The level 5 KPA on the other hand, it one of the focus is defect prevention that is identifies what is causing the defect and then if necessary change the process. The level 4 assumes that the process is all right, no changes requires only the way it is enforced or practiced is causing the process variation which is causing the defects whereas, in the

level 5 the defects are analyzed their causes and if necessary the process would be changed.

The other process area that is identified as the level 5 KPA is technology change management. In the area of software development the technology changes rapidly and therefore, there must be a group in the organization whose sole work is to identify latest and beneficial new technologies and transfer these into the process and organization wide. The different tools that are become available new methods, new processes, these are all identified and incorporated into the process.

The other process area that is identified is process change management. The process continually changes in a level 5 organization and therefore, the process itself must be put under configuration management. So that, the latest process can be identified. There can be different versions of the process which are applicable to different teams and therefore, the process changes need to be managed, otherwise what is the latest process, latest state of the process would be difficult. The process itself is put under configuration control.

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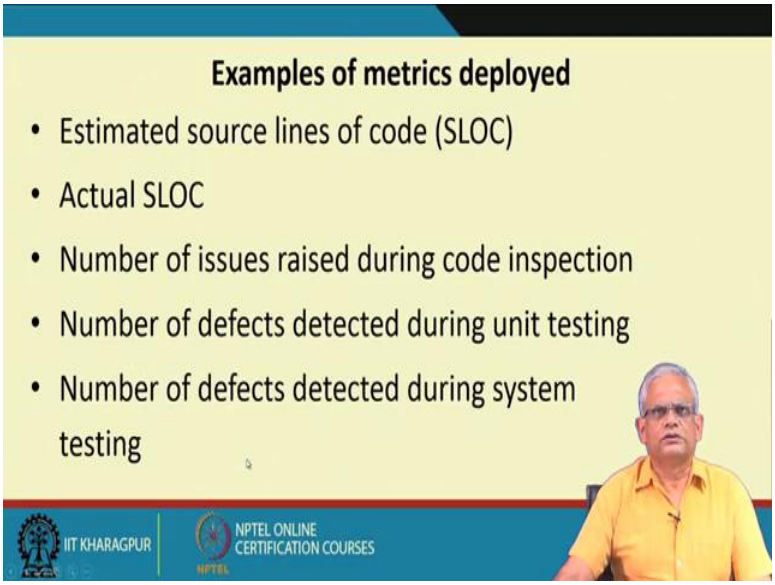

In the level 4 KPA, we had identified quantitative measurement and the process in product and based on this the project manager identifies deficiencies and determines what is causing the variation and takes corrective action. Some of the metrics that are deployed, here there are few examples. The estimated source lines of code and the actual lines of code, the estimated source lines of code and the actual source lines of code.

Across project, the estimated lines of code and the actual lines of code the variation must be bounded if for some project the estimated source lines of code is very different from the actual source lines of code and the project manager needs to introspect and identified what causes the variability, what aspects were not considered, what went wrong and so on.

The number of issues raised during code inspection, the number of defects detected during unit testing, the number of detected defects detected during system testing these are also important metrics. The number of issues raised duuing code inspection these are much smaller compared to the defects detected during unit and system testing; that means, the code inspection is not working properly.

And similarly if the number of defects detected during system testing is too many, then possibly the unit testing and code inspection did not work properly.

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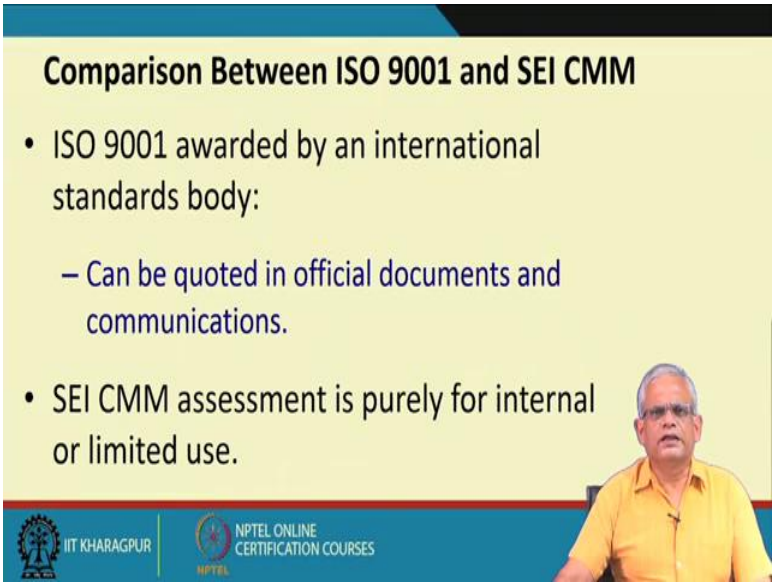

So, far we have looked at two quality standards the ISO 9001 and SEI CMM. Let us quickly compare these two standards. The first thing to note is that ISO 9001 is awarded by an international standards body, but Software Engineering Institute is not a standards body. It has just made some recommendations to the department of defense and which we are using and therefore, the ISO 9001 can be quoted in official documents and communications and the this can be accepted where these are given.

On the other hand the SEI CMM assessment is purely for internal use that is an organization wanting to improve its process uses the SEI CMM assessment or an organization trying to select a vendor uses its own CMM assessment on the organization.

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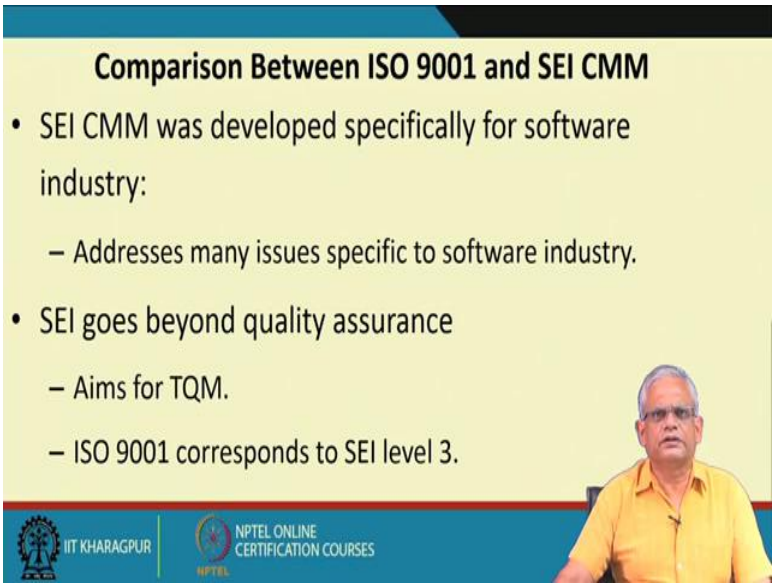

The second issue is that the SEI CMM is very focused on the software industry whereas, the ISO 9001 is used across various industries and therefore, the SEI CMM addresses many issues which are specific to the software industry which are not really taken care by the ISO 9001. And also the SEI CMM targets for total quality management.

unlike the ISO 9001, the original ISO 9001 document which was proposed in 1987, it targets only quality assurance that is the level 3 of SEI. So, roughly the ISO 9001 that was brought out in 1987 corresponds to the level 3 of the SEI CMM, but of course, the ISO 9001 the 2000 version, the one that came out in 2000 version and later they also target for TQM.

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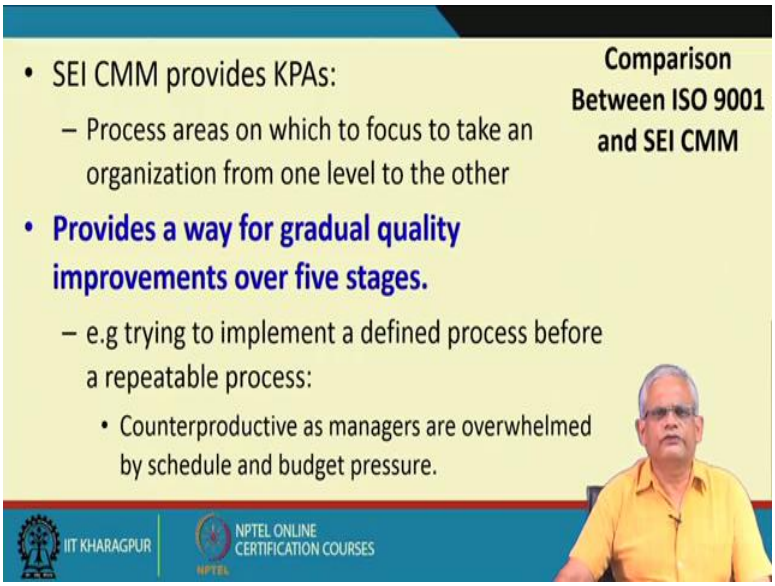

Another major difference is that the ISO 9001 provides a go, no go solution that is an organization either qualifies for ISO 9001 or does not. Whereas, the SEI CMM it provides a step by step approach by which an organization can improve its quality, that is starting at level 1 or very poor quality, very poor quality system. It can reach level 5 through a step by step approach.

It provides a way for gradual quality improvement over 5 stages and also it clearly identifies at each level what is the most pertinent or important process area that it needs to focus, that is if a level 2 organization without really focusing on level 3 KPAs it tries to focus on level 4 KPA then it will be counterproductive.

Let us say a level 1 organization without really practicing the project management, configuration management, requirements management etcetera it tries to have a defined and documented process. It will not be counter product, it would be counterproductive because the managers would be overwhelmed by schedule and budget pressure because the basic project management aspects are not there, the configuration management is not there and having a process and using it will not benefit much.

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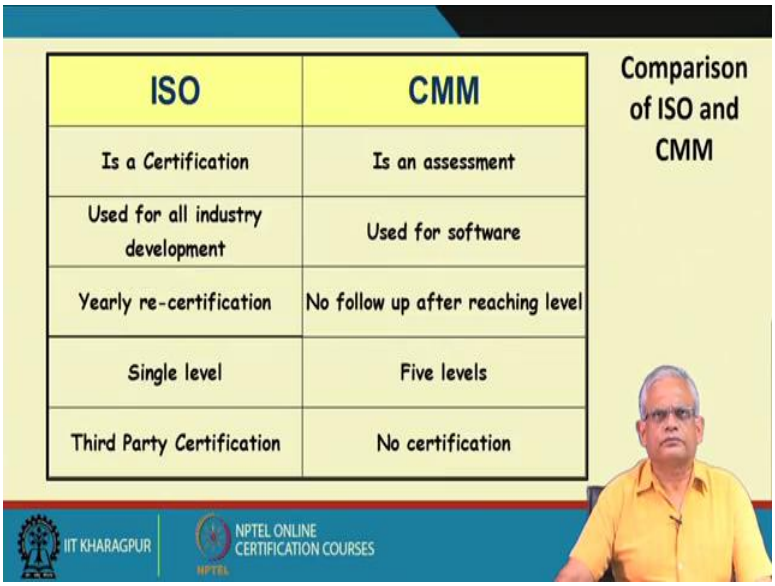

So, this is a tabular comparison, ISO is a certification whereas, CMM is just an assessment. ISO is generic used across all industries. CMM is used for software. ISO requires yearly recertification, whereas, in CMM it is not necessary that it has to be reassessed ISO either pass or fail whereas, CMM are 5 levels through an organization can improve its quality system. ISO is actually a third party certification, ISO does not give the certification as we had said.

It is only the accreditation agencies per specific country they provide the certification whereas, CMM there is no certification actually.

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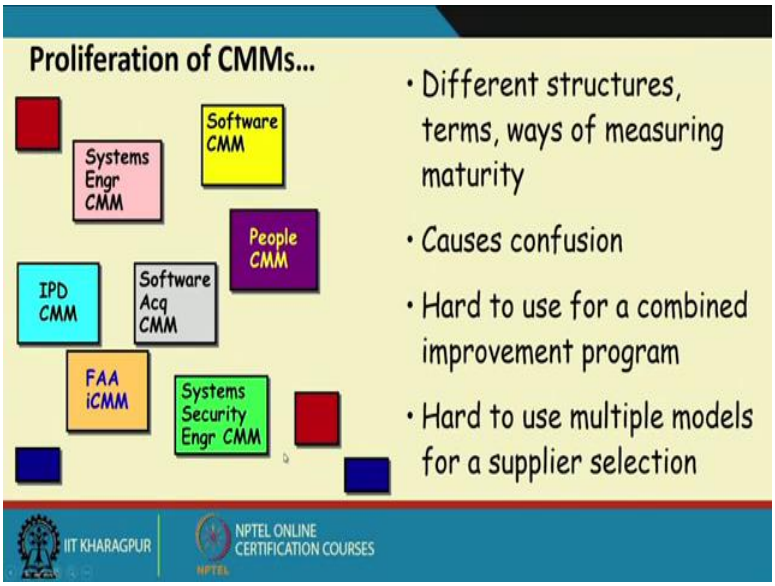

But then over the years lot of CMMs came into place. CMM for software development, CMM for systems engineering that is both hardware and software combined developments, CMM for software acquisition, people CMM or the CMM for people management security engineering CMM etcetera. These are all pertinent these are all relevant to an organization, but then each had its different structure, different terms were used the different ways of measuring the maturity which caused confusion.

And when an organization tries to use CMM as a whole for all its activities including acquisition and so on it was very hard because these had different terminologies and different ways of measuring the maturity. And also if multiple areas an organization had to select supplier it was very difficult.

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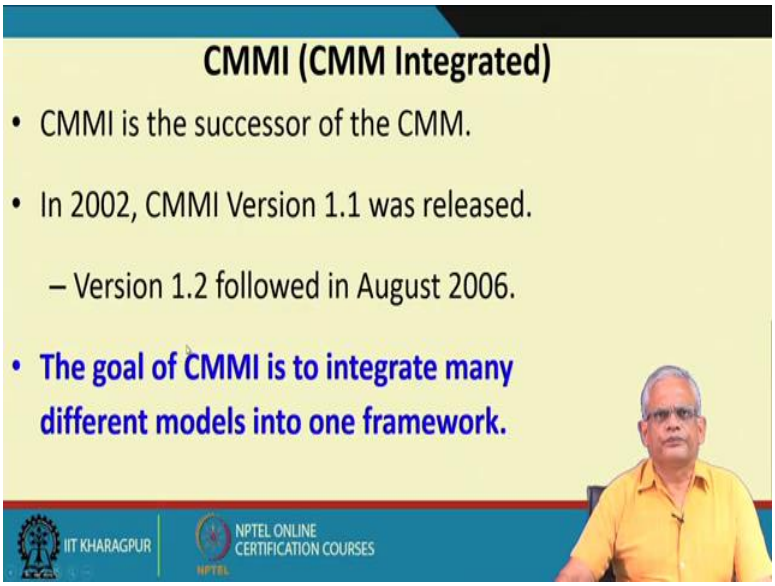

And with this the CMMI was proposed in 2008, 2002 and later it has been revised several times. The main idea here is integrating large number of CMMs that an organization would have to deal with into a single one.

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But then there are 3 flavours of CMMI, one is for development where there is which is applicable for software engineering, system engineering collaborative teams and acquisition of suppliers, acquisition from suppliers. There is CMMI for service industries, these are used across different industries from taxi aggregators to call centers,

they use CMMI for service and then CMMI for acquisition which is used for acquiring products and services.

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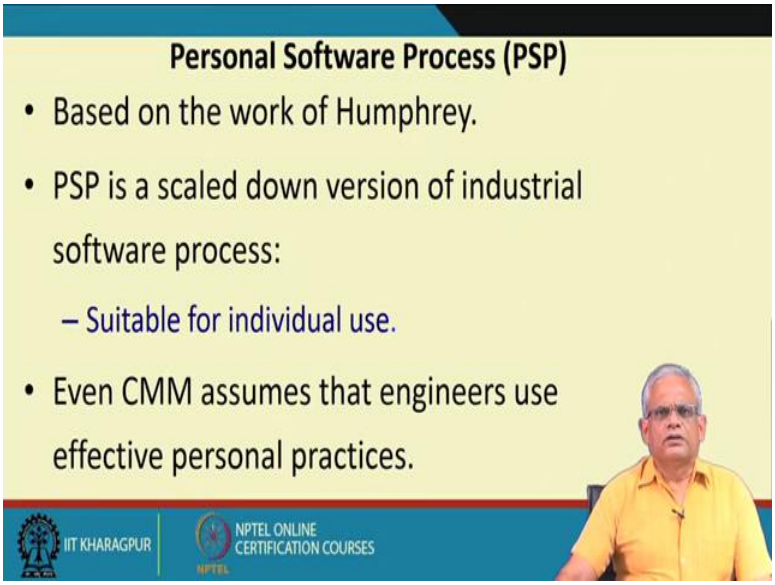

We are already at the end of the lecture time. We will just stop here. And in the next lecture we will discuss about the personal software process or PSP.

Thank you.

Software Project Management
Prof. Rajib Mall
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur

Lecture – 56
Personal Software Process (PSP)

Welcome to this lecture, over the last couple of lectures we had discussed some basic concepts in Software Quality Management and we had looked at quality standards. The ISO 9001 and SEI CMM. We had also compared these two standards, which will give us an insight in as to which one is more beneficial to use for a specific organization.

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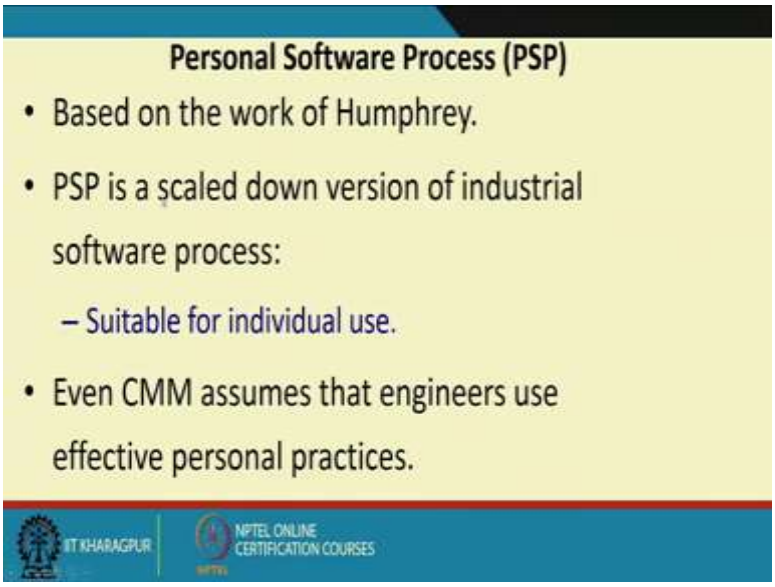

In this lecture we look at the Personal Software Process. The ISO 9001 and SEI CMM are applicable to organizations, these look at the process that is being followed at the organization and then they make recommendations on the quality issues of this process. But the need for a personal software process was identified because, every individual uses his own personal process. These are actually scaled down version of the industrial software process and are used by individuals, unlike the SEI CMM and the ISO 9000 which are used across an organization across all projects in an organization.

Here it focuses on the individuals working style or the individuals process, ISO 9000 and CMM they just make an assumption then individuals have effective personal practices. But then how do the individuals improve their personal practices starting with poor

practices, how can they improve their practices over stages and that is the focus of the personal software process or PSP.

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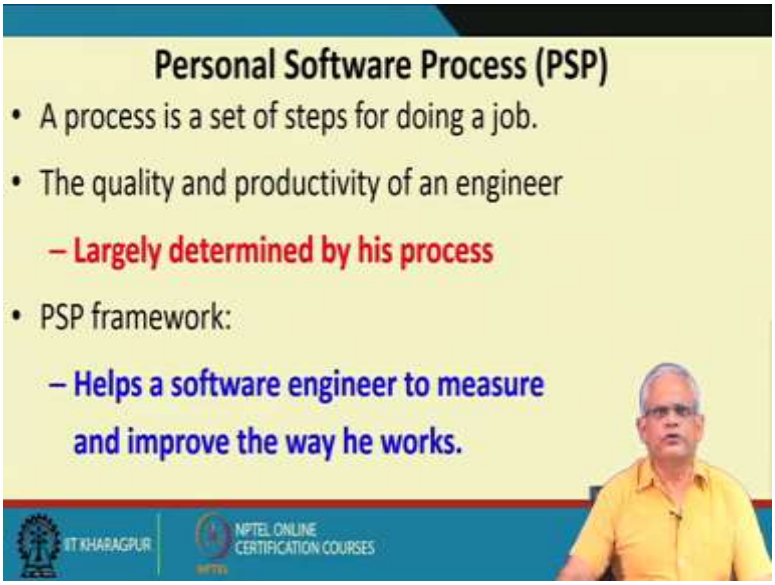

As we know a process is the set of steps that somebody uses for doing a job. For example, a developer is given the task of coding a module, how does he go about? Does he just understand the module requirement and then start coding, does he look at the module make an estimate of the code, the coding time and then really keep the time and then check whether he is according to the plan that he has made, does he have a coding process how to code.

So, these are the ones that are targeted by the PSP. The quality and productivity of a engineer is largely determined by his process, the PSP framework it helps a software engineer to measure and improve the way he works.

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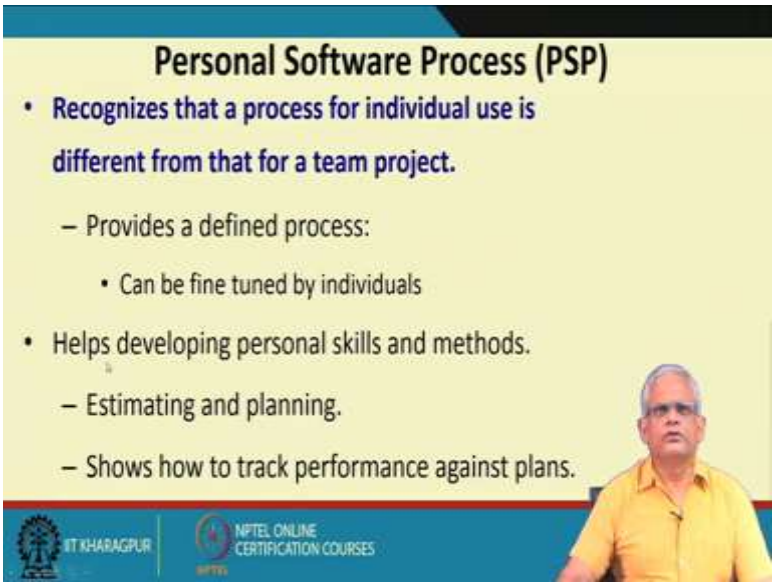

The individuals they have their different styles of working and it provides a defined process for the individuals and of course the individuals can finetune and continually improve their own process. It helps individuals to incorporate certain personal skills and methods. For example, they can do estimation and planning for their own work the project manager does estimation and planning for the project.

But what about the developers do they do any estimation of their own work that is assigned to them. Do they plan out how to go about doing that work and do they really track the performance against the plan they made.

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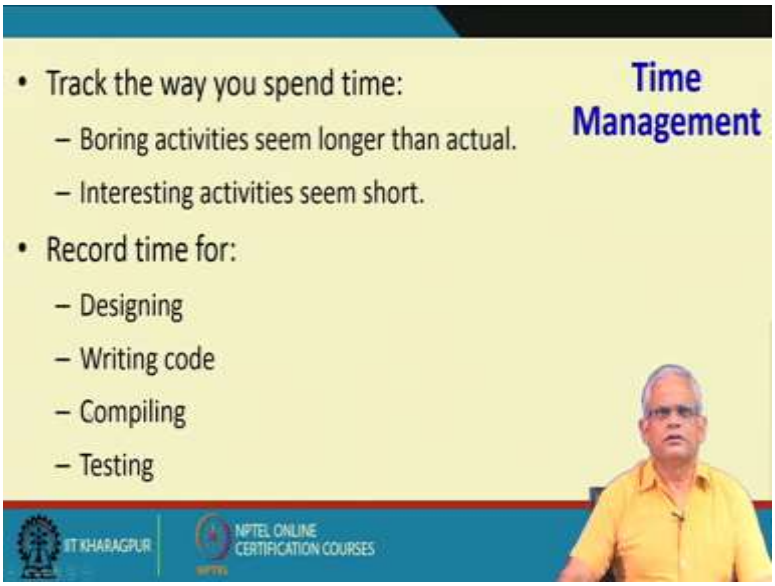

One of the most basic issues that has been pointed out by the personal software process is the time management. Unless the time is measured it becomes very difficult or very erroneous if somebody tries to do a boring activity which he does not really enjoy, then it will appear to be much longer than it is. For example, somebody let us say does not like to design he likes to code and he is asked to design even if he spends just few hours, he may say that I have been doing the design activity for a long time and therefore the time needs to be measured.

Similarly, interesting activity seem very short, somebody who enjoys doing coding and he has coded for let us say eight straight hours and you ask him that how long did he code he will say that no I just spent very little time in coding, I just done it all without spending too much time. For this reason it is necessary to record the time for designing, writing code, for compiling and removing the syntax errors and then testing and removing the bugs.

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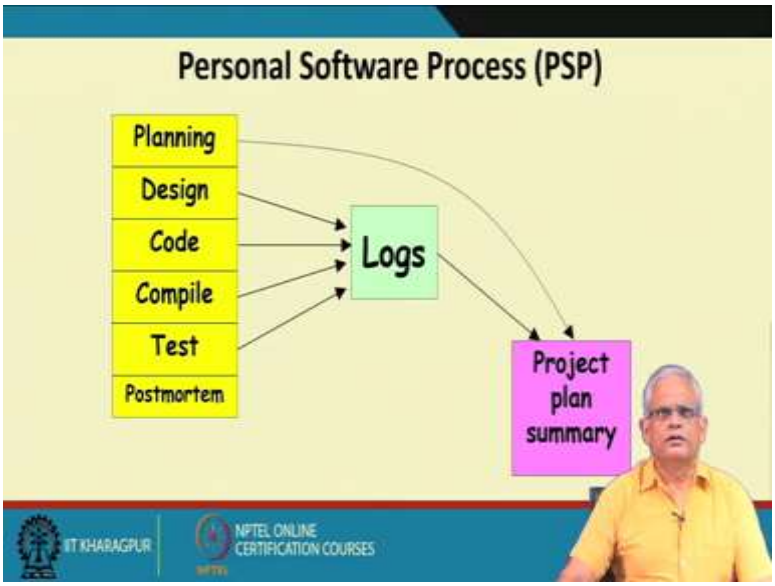

Plans have to be made by the individual if he is given a unit to code, he has to plan how is he going to do it has to write down the plan for the design must record the time taken in the log and the plan must be ok. That is compared later with the log for the coding how much time really it took compilation to remove syntax errors, how much time does it took test to remove the bugs, how much time does it took and these are all compared with the initial plan made and a project plan summary is developed.

And then a postmortem is used to identify what went right what did not go ok, how it needs to be improved and so on. So, that is the basic of the PSP.

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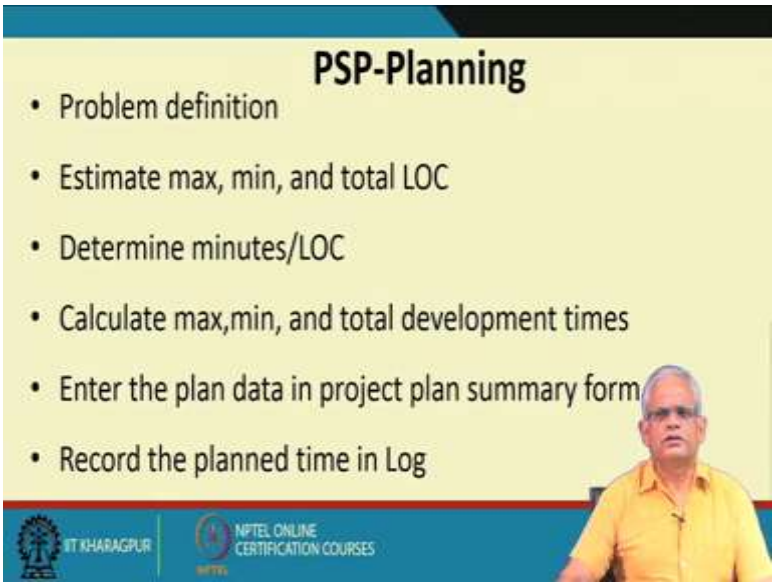

Other than the time management and keeping records or logs, the other aspect which is highlighted by the PSP is Planning. Starting with the problem the problem definition need to estimate what is the size of the work, what is the maximum minimum and estimated lines of code and from this must determine how much minutes it will take.

If a individual knows his coding speed that is how many minutes per lines of code, then he can determine the maximum minimum and total are the estimated development time. And these must be entered in a project plan summary form and once starts working must record the actual time the plan time and the actual time must be recorded.

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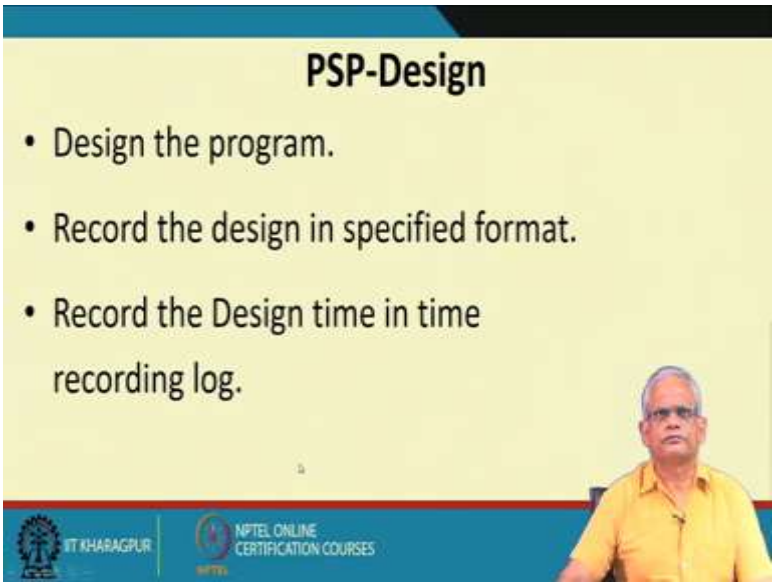

The design the specific format in which the design to be done. The developer must have his own standard which is of course final output must be compliant with the required standard of the team. But the way he goes about designing the specific format he must identify and then record the design time the time recording log.

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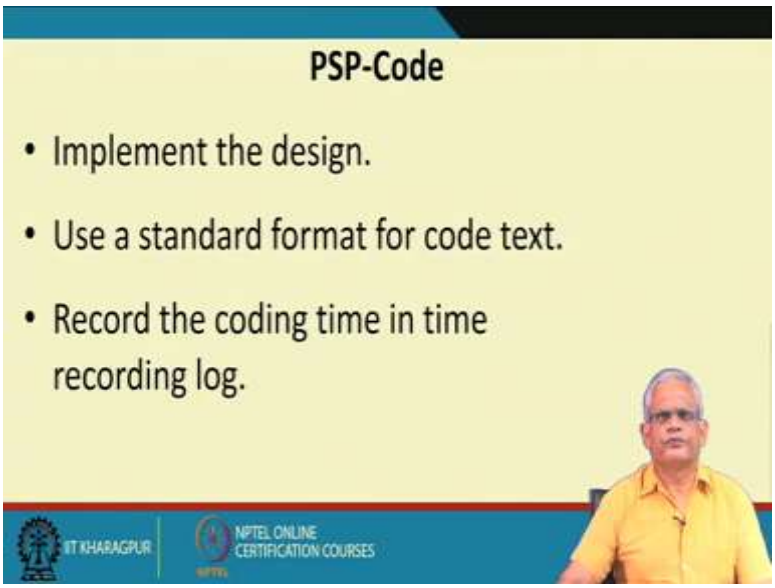

For the coding must implement the design and should have his own way to go about coding his own process some developers they first write the function prototypes and then identify the variables and define them and then write the code. Some other developers

may feel comfortable in first writing a simple version of the function and slowly enhancing that adding more and more variables parameters and so on. But finally, the way the text is laid out that also must be for that must be standard and the coding time must be recorded in a time recording log.

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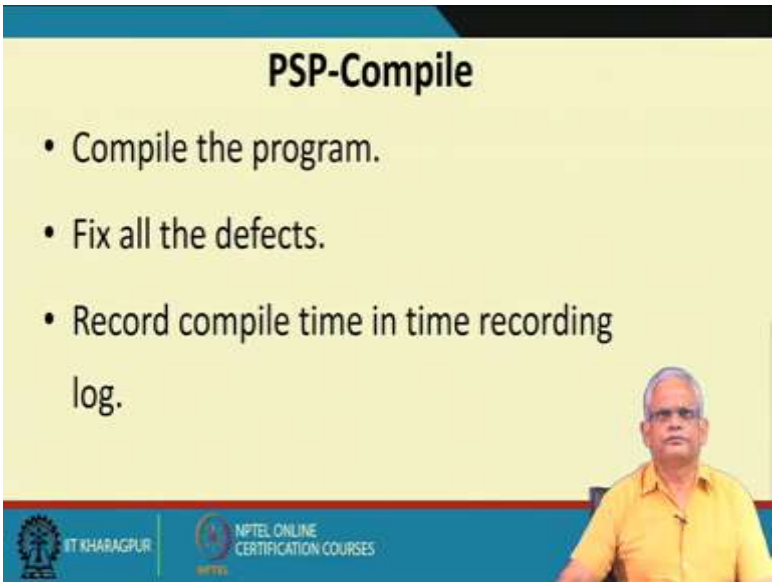

The Compile to remove all syntax errors, each time the program needs to be compiled and the syntax errors identified must be fixed until the program compile satisfactorily and this time is recorded in a log.

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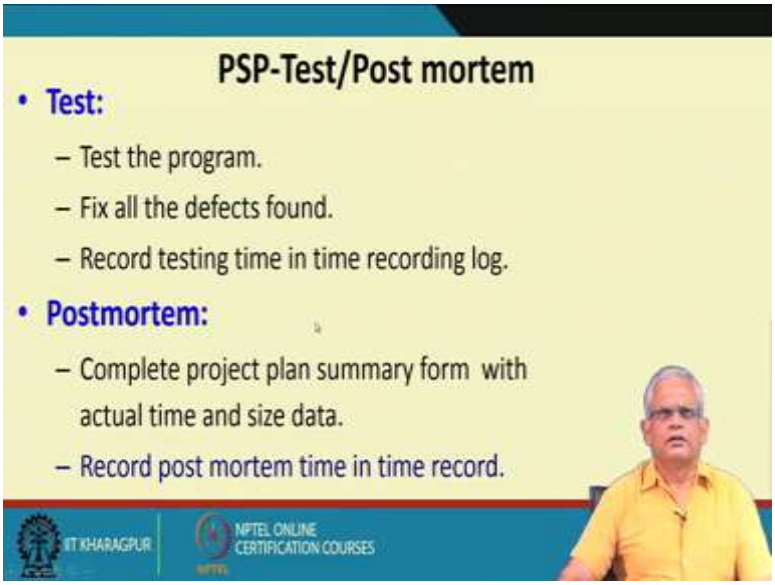

Then the testing here all the bugs have to be identified and fixed and the time is recorded in a log. And finally the postmortem where the project summary data that is the plan time and the estimated time are analyzed and then the observations are written down in postmortem.

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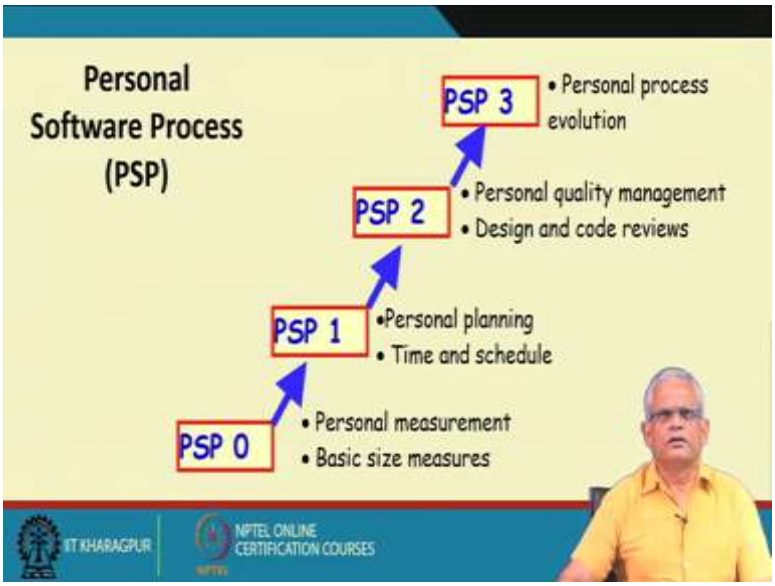

If we represent it schematically in the PSP there are four levels, PSP 0, PSP 1, 2 and 3 at the 0 level only the measurements are done, that is the personal measurement that is the how much time somebody takes for a line of code. It is time per lines of code the basic size measures that is given a work does he estimate that how much time will it take how many lines of code and so on.

In the PSP 1 level the basic planning is done, the personal planning that how to go about doing the work when to do which one this is planned and there is a schedule that by this time I must do this and so on. In the level 2 of PSP the personal quality management and the design and code review, the developer himself reviews his design and code and is written in the form of a log. And in the PSP 3 the personal process evolution and not only the personal process has been defined, but the opportunity to improve the process is identified and the process is continually changed at the level 3.

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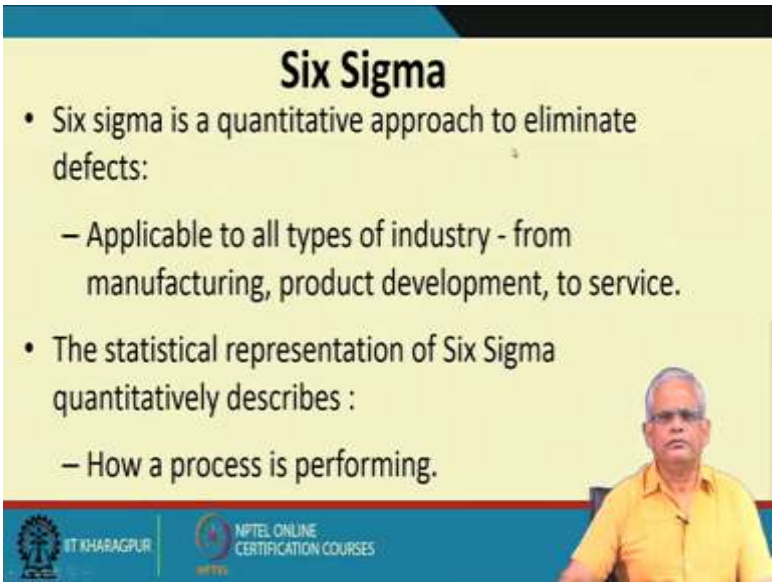

Now, let us look at very briefly at Six Sigma, the six sigma initiative is a quantitative approach to eliminate defects. Again this is applicable to all types of industry from manufacturing, product development, service. It is applicable to all industries and it incorporates statistical representation of the defects and based on that one can conclude that how a process is performing.

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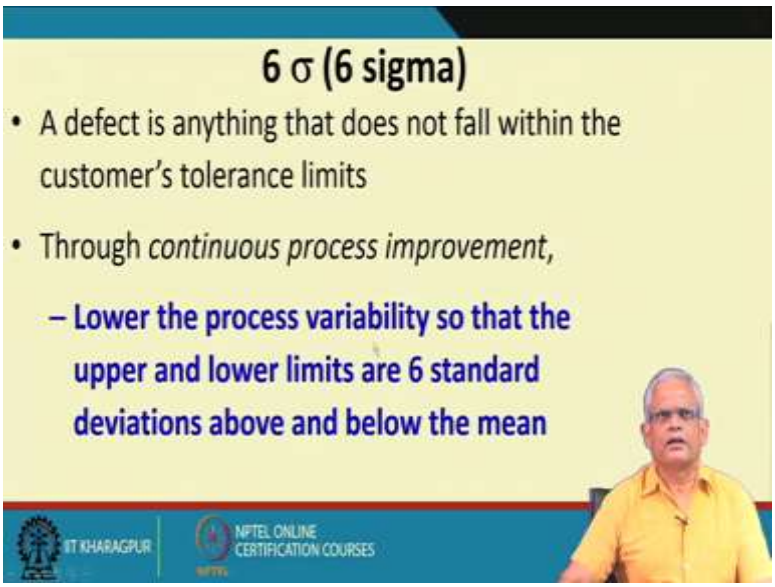

A defect is defined in six sigma is anything that does not fall within customers tolerance limit. If something is a defect, but the customer is OK with that that is not really a defect.

In the six sigma it argues for continuous process improvement. So, that the variability in the process the upper and lower limits that are set this fall within six standard deviations above and below mean. We will not go into the detailed mathematical computations how to define the six standard deviations and so on.

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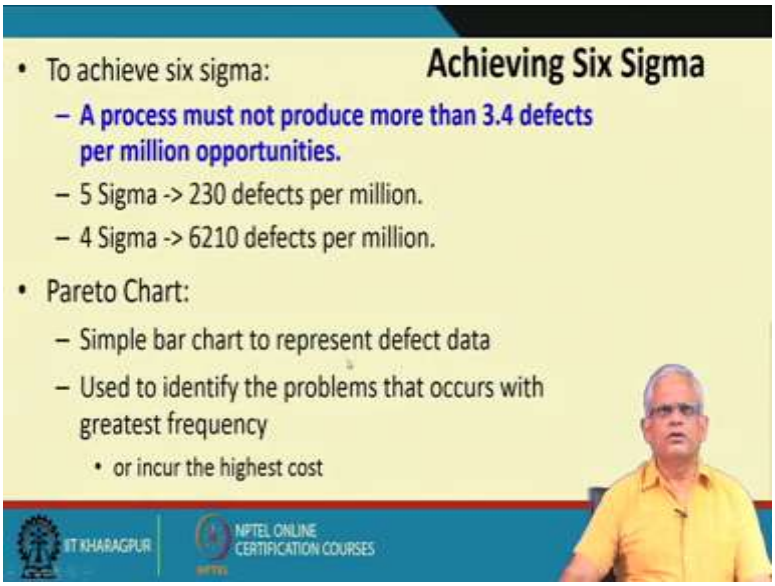

But we will just say that to achieve six sigma a process must not produce more than 3.4 per million opportunities. If we look at definition of a 5 sigma or 4 sigma can see that the number of defects acceptable are much higher, a six sigma process must not produce 3.4 defects per million opportunities. Here the defects are represented using a Pareto chart and the defects that occur with the greatest frequency or that incur the highest cost are identified to take corrective action.

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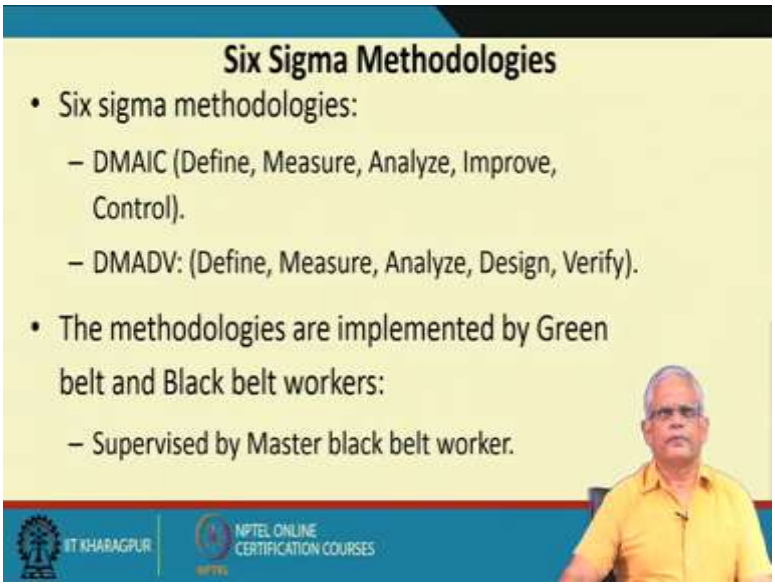

Several Six Sigma Methodologies are defined like define measure analyze improve and control DMAIC. Define measure analyze design and verify a DMADV and these are implemented by green belt and black belt workers supervised a master black belt worker. We will not go into the nitty gritty of six sigma methodologies we do not have that much time, in this course we just looked at a very overall idea of the six sigma.

We will now start looking at the software reliability issues. Now reliability is a very important issue in software, the customers they look for highly reliable software they would not buy software which is not reliable.

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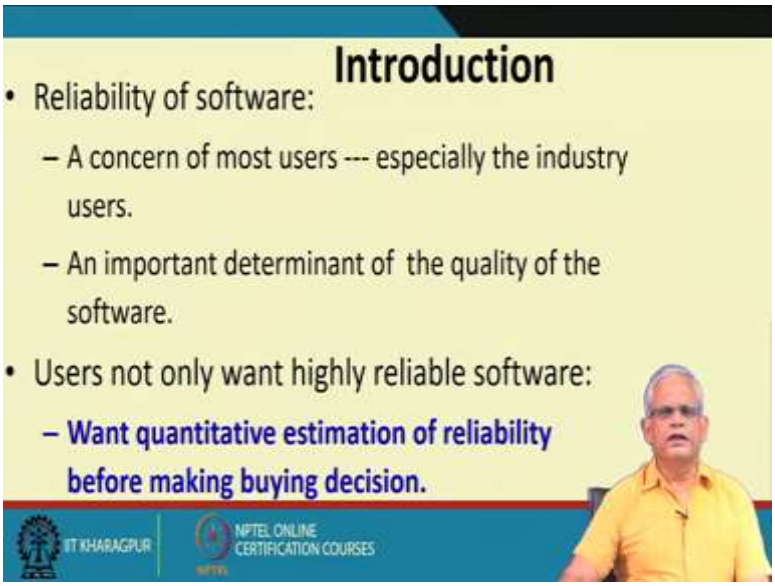

And therefore, it becomes a very important issue with the project manager to understand the nitty gritty of reliability and ensure that the final developed software has high reliability. Reliable software is a concern for all users, especially if it is used for some industry application it is very crucial and also reliability is a important determinant of the quality of the software. A software having poor quality sorry poor reliability cannot be said to be a quality software.

The users not only want highly reliable software, but they need a quantitative estimation to be given to them about the reliability before they can make any buying decision.

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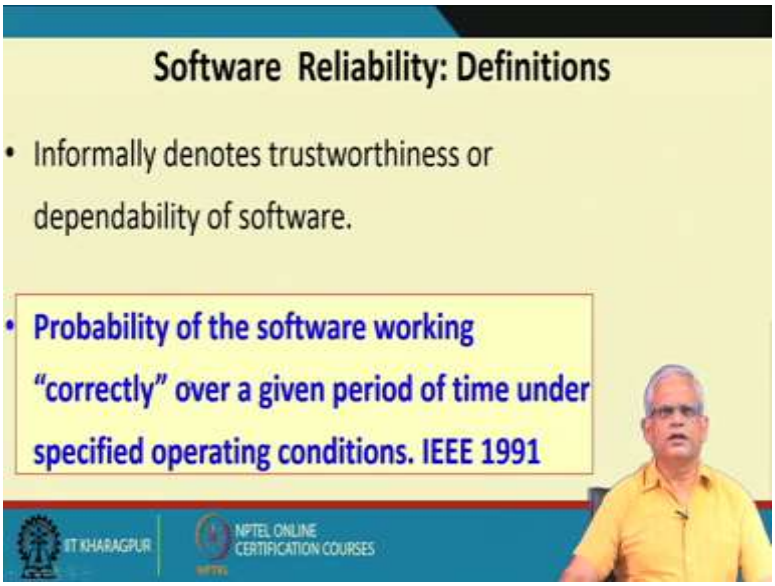

But how do we define Software Reliability? Informally we can say that the reliability is basically the trust worthiness of the software or the dependability of the software. Intuitive definitions trust worthiness, dependability. But can we give a more formal definition of software reliability. IEEE document in 1991 defines reliability as the probability of software working correctly over a given period of time under specified operation conditions. So, here it is defined in terms of the probability of the software working correctly over a given period of time, under specific operating conditions.

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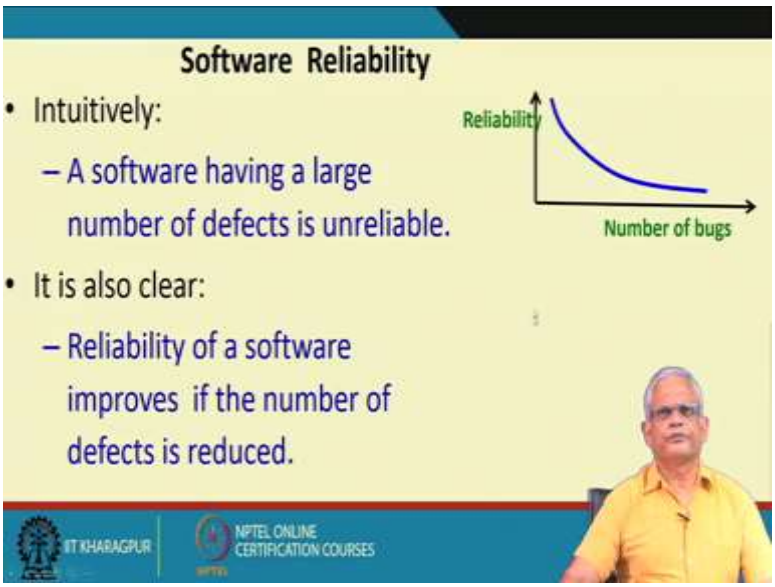

To get some intuitive idea about software reliability, a software that has large number of defects is clearly unreliable. And also we know that if we remove some bugs that is as the bugs are identified and fixed, the reliability improves. When there are large number of bugs the reliability is low. But as the number of bugs reduces the reliability improves.

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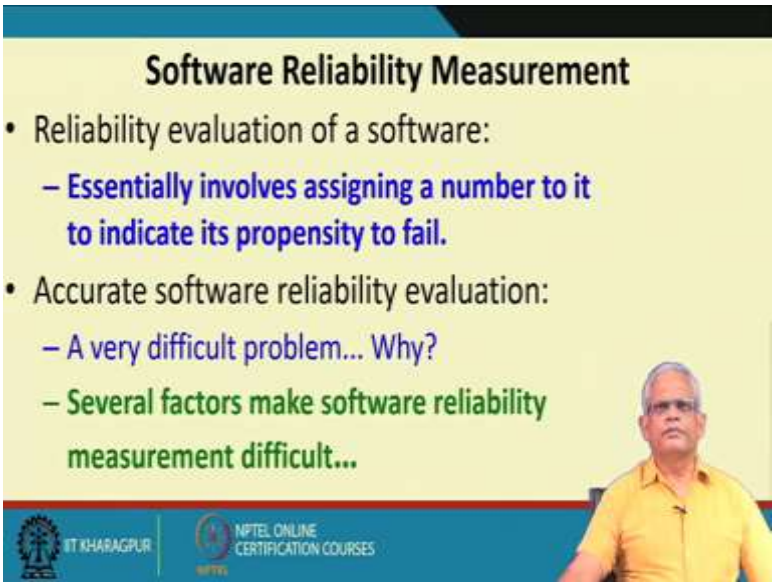

For Reliability Measurement we need to give a number to the software, that will identify or indicate it is likelihood of failing. For hardware it is a relatively simple issue how to compute the reliability of hardware you keep on operating the hardware, make observations and the failure and that gives the reliability of the hardware. But for software it is a difficult problem, for hardware just put it on use for a long period of time and then just give the reliability measurement of the hardware.

But for software it is difficult let us first understand what is causing or making the software reliability evaluation a much more harder problem than software. There are actually several factors several important factors. We will look at four five of them which make software reliability measurement much harder than hardware reliability measurement.

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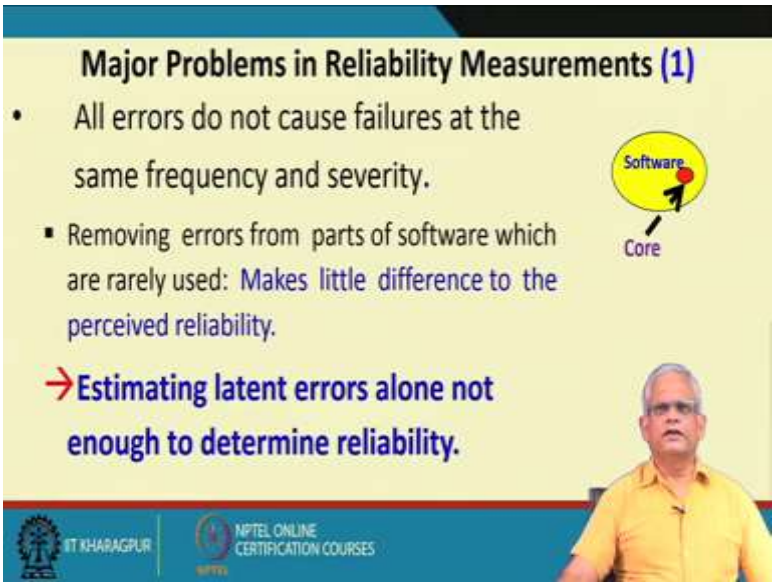

One issue is that if there are hundred bugs in a program, all bugs do not cause failures in the same frequency and same severity. Some of the bugs they keep on making the software fail every now and then. Whereas, some other bugs they rarely make the system fail. We can think of a software as it operates there are some areas of the program code which get executed again and again, very frequently some part of the code gets executed.

And therefore, if that part of the code has a bug this will normally show very frequently, but if some other part of the code which is rarely used has a bug this will not appear so frequently. If this is the software some part of the code gets executed very frequently this is called as the core of the program. For every program we can identify the core, there are tools available called as profilers which can identify the core and therefore if bugs are present here in the core they will cause it to fail very frequently.

So, if there are ten bugs present in the core or let us say just one bug present in the core, the software may exhibit much higher failure rate than another software which has let us say hundred bugs in the noncore part. And for this reason just estimating the number of errors that are remaining in the software is not good enough to predict how frequently it will fail. We must also identify whether it is in the core part or in the noncore part.

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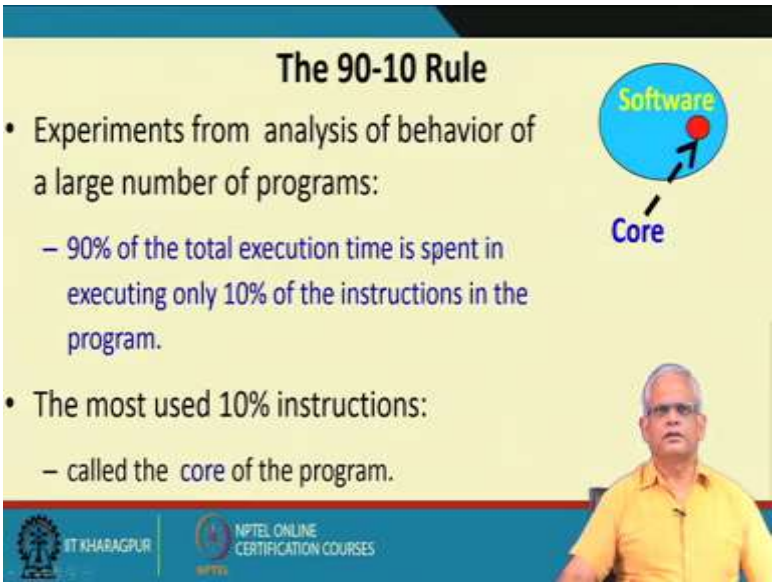

The 90-10 rule says that 90 percent of the execution time of a program is spent in executing 10 percent of the code. For any program this can easily be identified by a profiler tool for Unix the tool name is p r o f prof and we can easily identify the core of a program. And if the bugs are in the core of the program the reliability will be very low it will fail very frequently. But even if much larger number of bugs are present in the noncore part of the program the program may not fail so frequently.

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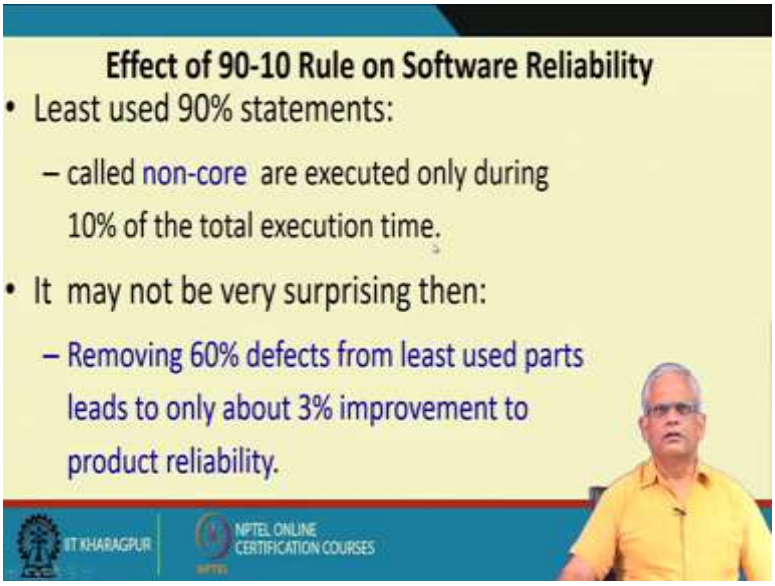

Study shows that removing 60 percent defects from the least used parts leads to only about 3 percent improvement in the reliability. But if these are used from the if they are removed from the most used parts or the core of the program then the reliability will improve by a significant factor. Just imagine that if there are hundred bugs in the least used parts that is noncore and you remove 60 of them the reliability improves by only 3 percent.

So, to estimate the reliability of a software just knowing the number of bugs is not good enough there are several other factors which make the reliability evaluation much more complicated. We look at these issues in the next lecture, for this lecture we are almost at the end of the lecture hour, we will continue from this point in the next lecture.

Thank you.

Software Project Management
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Indian Institute of Technology, Kharagpur

Lecture – 57
Software Reliability – I

Welcome to this lecture. In the last lecture we had looked at some very basic aspects of Software Reliability. We had said that quality and reliability are closely related and the software quality to large extent is determined by it's reliability.

If a software is of not if poor reliability we cannot say that it is a good quality and with that intention we started discussing about some very basic aspects of software reliability. Like what do we mean by Reliability? How do we formally define reliability? And then we were discussing about why software reliability is much difficult to measure why the software reliability study is very different from hardware reliability study and so on.

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Now, let us continue from that point onwards. We will first briefly look at why software reliability measurement is a much harder problem than hardware reliability measurement or measurement of the reliability of any other equipment, structure and so on. And then we will define some metrics for software reliability measurement and then we will say that to meaningfully estimate software reliability. We have to use the two approaches

which is reliability growth modeling and statistical testing and we will see the pros and cons of both these approaches.

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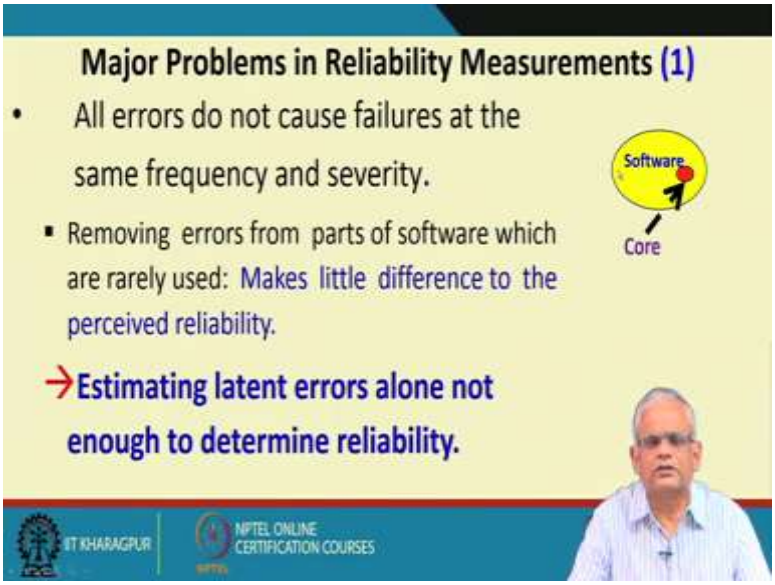

We have been saying that measurement of software reliability is a hard problem. There are many factors which contribute to making this hard problem, the first issue is that a software will have many errors and one would expect that if we can estimate the number of errors we can derive the reliability from that. But unfortunately all errors do not cause failures at the same frequency and severity. Actually there is a very very wide difference with the frequency with which different errors called system failure and also the severity of the system failure.

There are some errors which rarely show may be after a million hours of uses may show up once. Whereas, another error may show up even in the first minute of users, the error which is occurs very rarely if we remove that error it will make a very little difference to the reliability of the software the perceived reliability. But the one which appears again and again in the form of failures that would result, if we can eliminate that error it will result in significant improvement of the reliability of the software.

From this argument it is clear that even if we are somehow been able to estimate a number of errors remaining in software, that is not enough to estimate the reliability of the software. If this is the software then as the code executes some code some part of the code that executes almost all the time, the other part does not execute as frequently and

this part marked in the red is called as the core of the program those statements are executed again and again.

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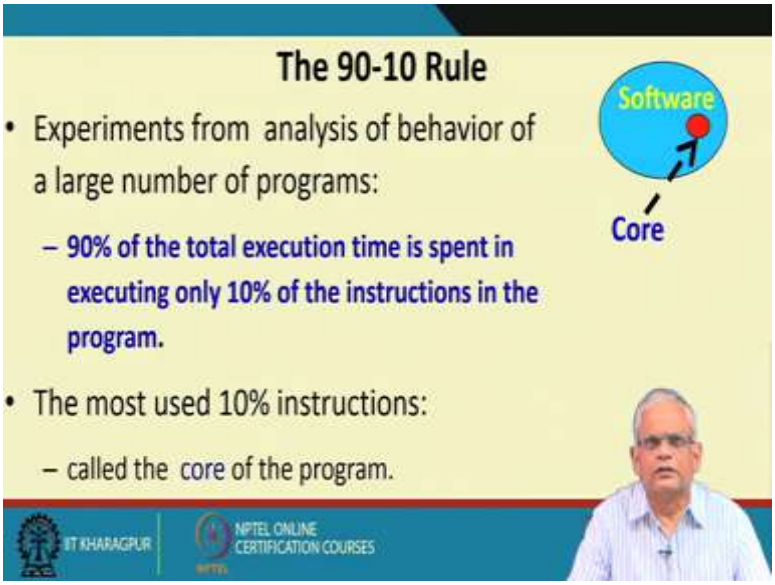

The 90-10 rule says that 90 percent of the execution time is spent only in 10 percent of the instructions of the program, the rest 90 percent of the instructions execute only 10 percent of the time and 10 percent of instruction execute 90 percent of the time.

And therefore, if we remove a error appearing in the core part that will result in a significant improvement of reliability. Whereas the one which is very rarely used would make hardly any difference to the reliability.

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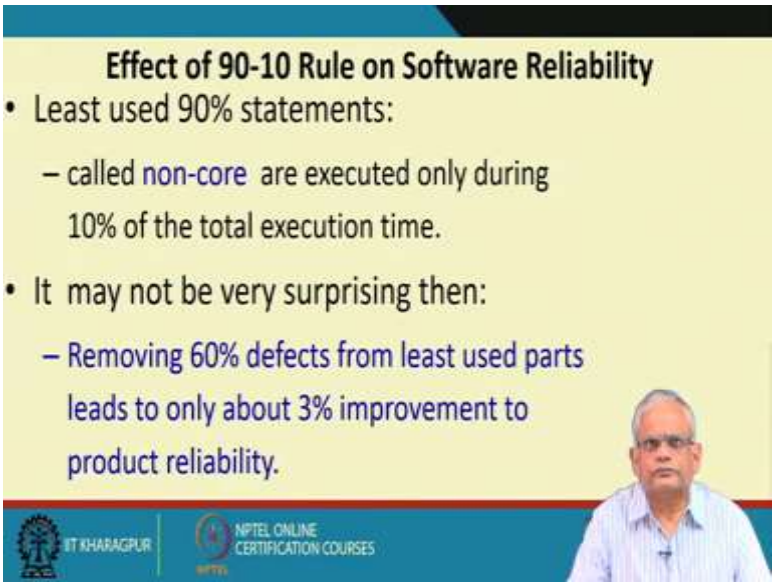

Some studies suggest that if a software has let us say thousand bugs and if we remove six hundred of those bugs which are there in the noncore part. It results only 3 percent improvement to the reliability, very very surprising result, that removing almost more than half the bugs results in only 3 percent improvement. Because those bugs happen to be in the noncore part and this issue makes reliability estimation from the number of latent error estimation not very feasible.

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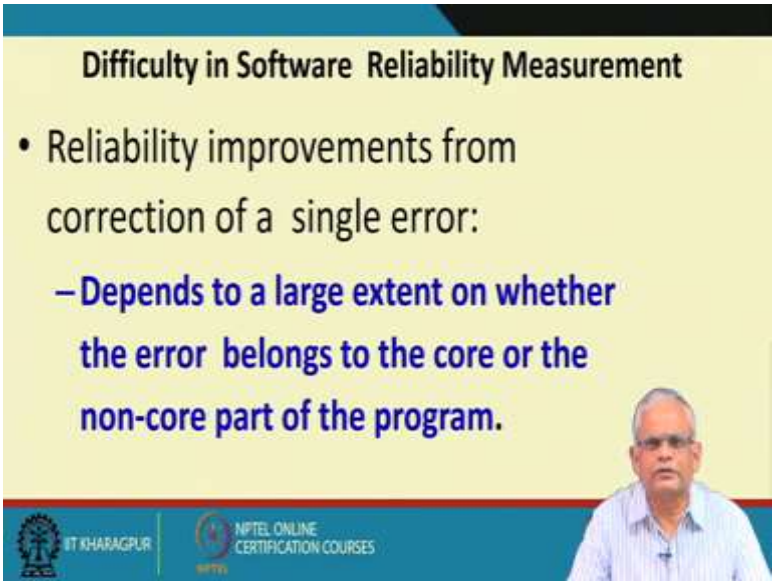

The reliability improvement from correction of a single error depends to a large extent whether the error belongs to the core part or the noncore part of the program, this is one issue which we need to keep in mind when we think of software reliability.

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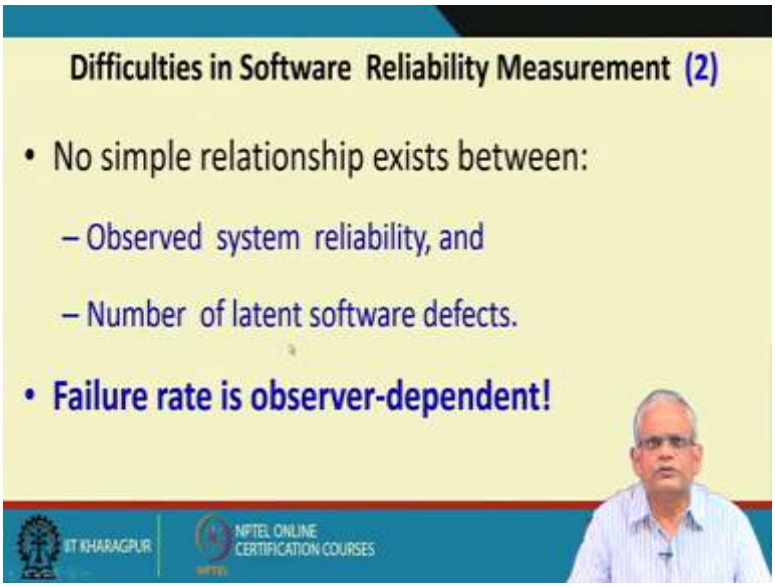

Another issue so this is still in the first issue, because the frequency with which a error causes failure depends on the part of the program at which the error lies. Therefore, the system reliability the observed system reliability and the number of latent software defects, there is no simple relationship otherwise you can derive or empirically propose a expression in which you should be able to input the number of latent software defects and get the reliability. But this is the main reason why we cannot do that.

Now, let us look at the second issue, the second issue is the failure rate is observer dependent, a software which appears to be very reliable to one person may appear to be very unreliable to another person. Now, let us investigate the reason behind this.

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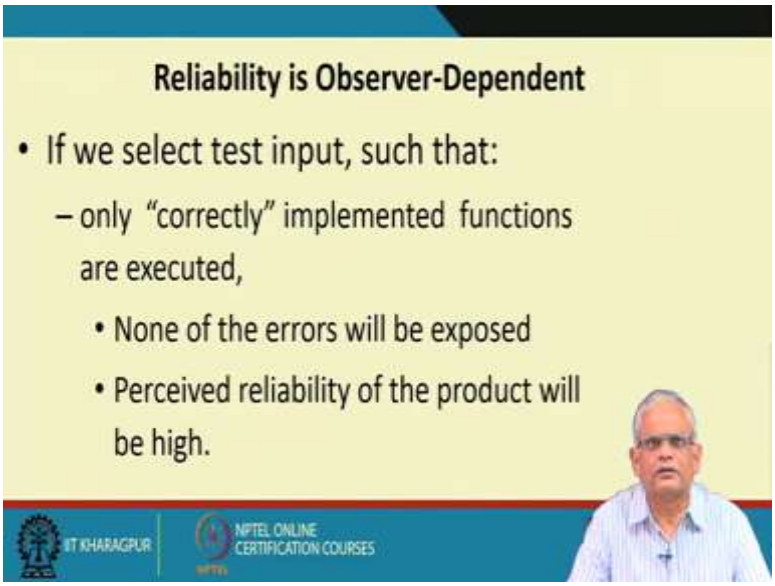

Let us say one user selects inputs or executes the software. So that only the correct functionalities are executed a software has hundreds of functionalities and let us say each user is interested in executing only part of the functionality. Let us say a library software the students or the members are interested to issue the book, return book, query book and so on. The librarian would be executing the functions create member, delete member create book, collect statistics and so on.

The accounts department would again be using the same software, but executing very different functionality. Like what is the income from fee collection membership collection how much is from the fine collection how much grant is received how much books have been procured this year and so on.

So, every user every type of user use different functionalities of the same software. Now, let us say one type of user they are executing functionalities which are correctly implemented functions and they find that all the functions they execute work perfectly all right and they would give the opinion that the software is really good highly reliable. On the other hand for another category of users almost every function they execute displays failure error and so on and they would form the opinion that the software is of poor quality it is unreliable.

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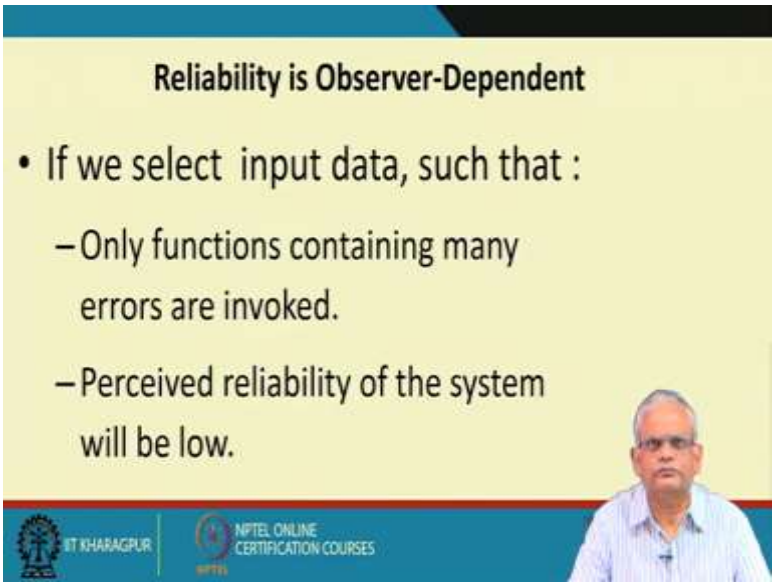

And therefore, for the same software two different users can have two different opinion about it's reliability. The opinion of a user about the reliability of a software we call as the perceived reliability. So, the perceived reliability can be low for one group of users and the perceived reliability can be high for another group of users.

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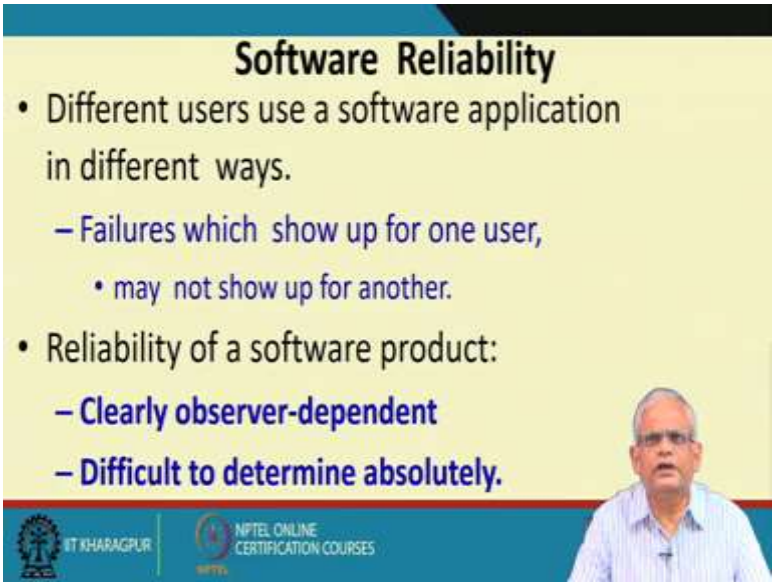

We are just mentioning this point the different users of a software application, they use the software in different ways they are interested in different functionalities and therefore the failure which shows for another one user may not use so for another user.

So, different users can come up with different reliability numbers for the same software and therefore it is clearly observer dependent. But again if we have to give one absolute value how do we do it, which user will take into account. Because we will get different numbers for different users, we will see how to handle that problem, but then it makes reliability estimation complicated because the reliability has different observers give different reliability numbers to the same software. And it makes it difficult to give a absolute reliability number a single reliability number to the software.

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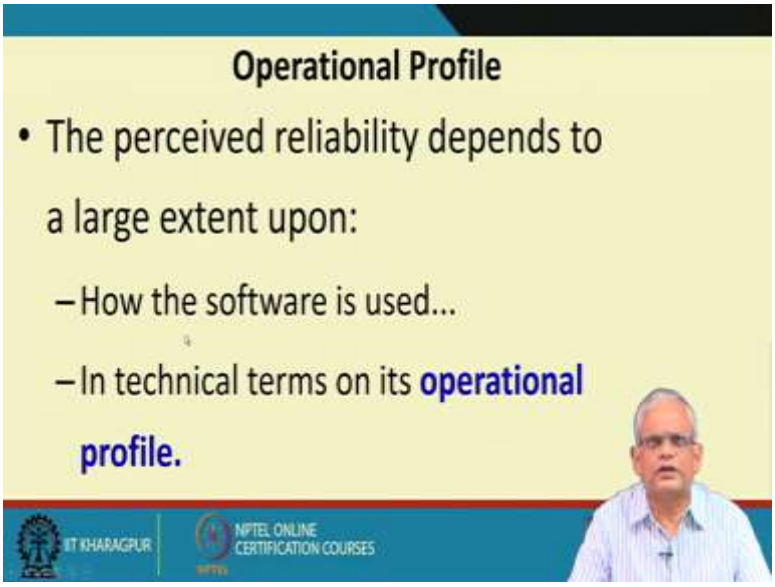

The way it is done that is to give a single reliability number is to consider the operational profile of the software. And the if you have to give a single number we will have to give it based on it is operational profile. Since the perceived reliability depends to a large extent on how the software is used or in other words it is operational profile. Let us first define what is operational profile and then we will see how to give a single reliability number to a software.

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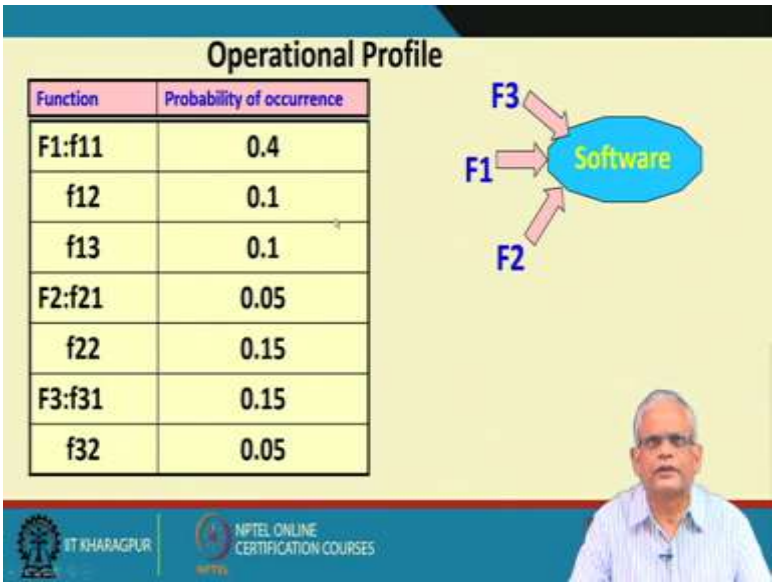

Let us say a software has many functionalities F1, F 2, F3 etcetera and also each of this functionalities will have different scenarios of operation. Let us say this is a ATM software, bank ATM and then one function is withdraw cash, one is the main line scenario that is, go there do everything correctly give password insert the amount to be withdrawn and cash is dispensed come out.

The second scenario is that the password may be incorrect or the card may be rejected or may be that the account does not have enough balance. So, that is another scenario one more scenario can be that the ATM does not have enough money to dispense and so on.

So, the same functionalities can have different scenarios of operation. We can observe the execution of these different functions by the users, we can take a log of that as the different users use a log gets collected and we analyze which function and which scenario gets executed how many times. So, that is function and the probability of occurrence we might have these three functions F1, F2, F3 and we may determine that one scenario the first scenario F11 gets executed 40 percent of the time, F1 2 the second scenario of the F1 function gets executed 10 percent of the time and F1 3 the third scenario for F1 get executed 10 percent of time.

The first scenario of F2 gets executed 5 percent of time and the second scenario gets 15 percent of time. The first scenario third use case the third function gets executed 15 percent of time and the second scenario of the third function gets executed 5 percent of

time. And this we call as it is operational profile of the software. Let me just repeat here that the operational profile of a software is constructed by observing the way the software is used by various users over a long enough period of time and then finding what is the rate at which the different functions get executed. The probability that the first scenario of F1 gets executed is forty percent and so on, this we call as the operational profile of the software.

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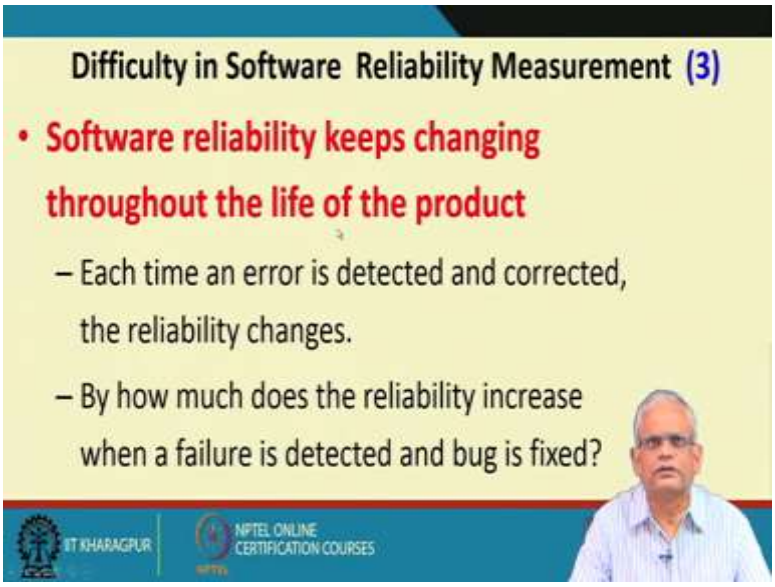

And once we have the operational profile that is the average system users of all types of users and therefore based on the operational profile. We can give a single reliability number to a software and that we will discuss a little later how to go about giving a single reliability number to a software based on it's operational profile. And we call that a statistical testing.

Now, let us look at the third issue which makes reliability measurement difficult, this problem is that the reliability of a software keeps on changing throughout it is life cycle. Because each time there is a failure, the software is debugged, the error is detected and corrected and therefore the same type of failure will not occur.

But then the improvement of reliability is not fixed some error fixes may cause large improvement of reliability, some may cause less improvement in reliability and so on. But typically as the software gets used, the errors are expressed in failures and these are corrected the reliability keeps on improving for the software. Unlike hardware where the

reliability more or less remains constant, in software the reliability changes and typically improves.

And therefore, becomes very difficult to give a reliability number to a software, because if we give it today, then after a year the same number will not hold or may be after a few days the same number will not hold. So, it becomes difficult to give a number reliability number, reliability estimation value which can remain valid for sometime.

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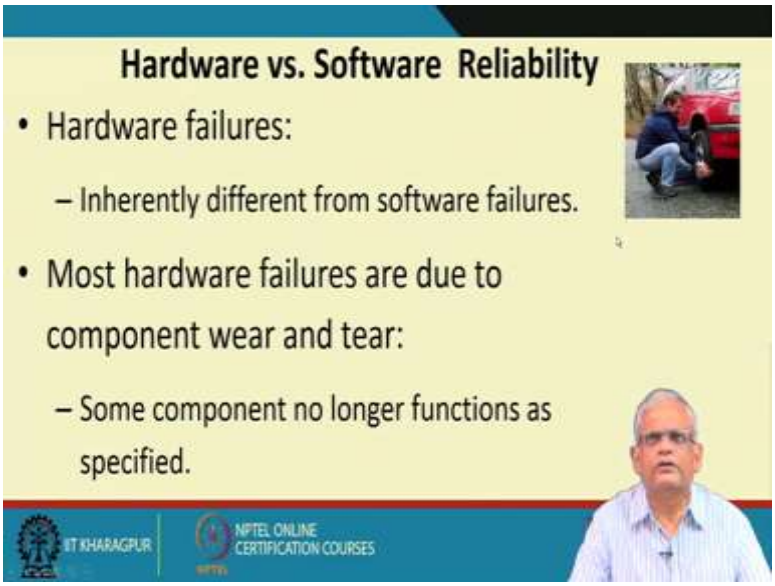

If we compare hardware and software reliability there are some issues which are very different in hardware and software. The first thing is that the type of failure that occur in hardware is very different very different from the failure that occurs in software. In hardware the failures occur due to component wear and tear, if it is a automobile the failures that typically occur are let us say the tyre punctured the tyre worn out replace the tyre the battery become old, replace the battery and so on.

So, if we analyze hardware failures like a car or a mixer grinder, we find that the failure is due to use of the system and some components, they wear and they break and the failures are largely due to component breaking due to use or aging. But in software there is no such thing as wear and tear, here this is a car the wheel has worn out punctured and it can be replaced and the failure here is due to wear and tear. Whereas, in software there is no wear and tear, this is one major difference between software failures and hardware failures.

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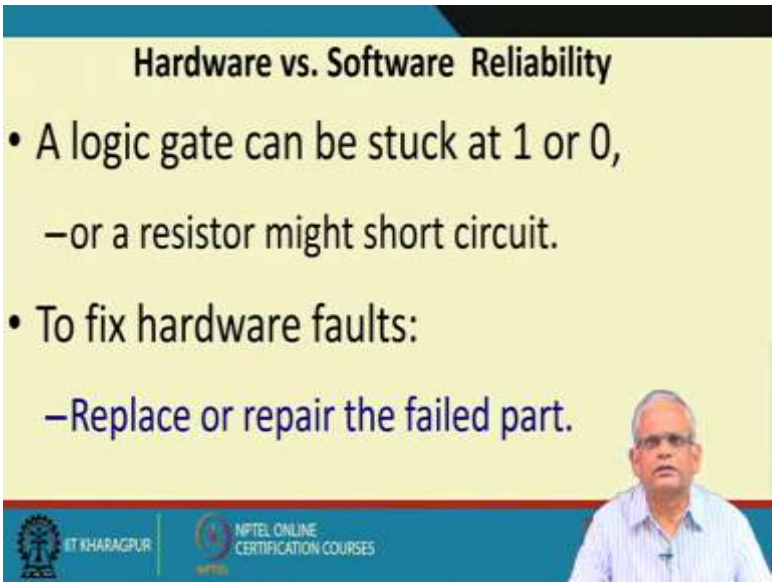

If it is a logic circuit a logic gate can become short circuited or open circuit and therefore it may be 1 or 0 in all types of hardware. If there is a failure typically find out which component is failing and replace that part and then the reliability is back to what it was. A reliability estimation can be done for a hardware system, a number can be given and there can be failures. But then on failure we replace a part and the system is back to the previous reliability.

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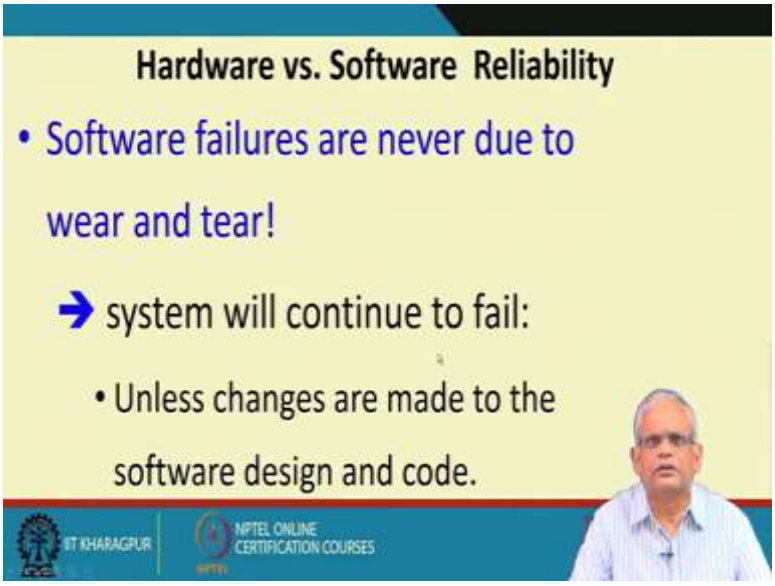

But software the failures are not due to wear and tear, the failures are due to bugs and here the only way to correct the failure is to remove the bug. And once we change the code and remove the bug the reliability changes, the system continuous to fail unless changes are made to the design and code.

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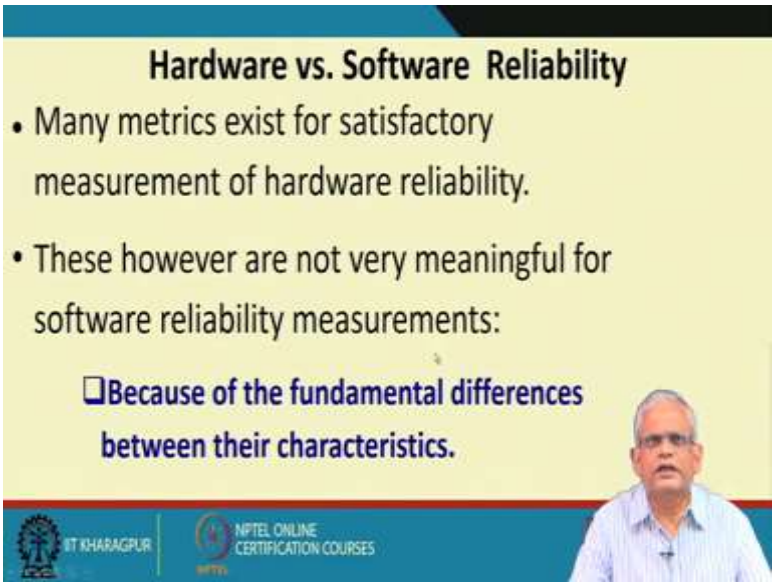

And once we change the design the reliability changes typically the reliability improves. Now, the metrics that are used to measure hardware reliability, they are basically based on observing the number of failures over a period of time long enough time and then noting the number of failures and giving a reliability value. But for software these metrics will not be very meaningful because, the failure types are very different here the each failure results in reliability improvement in software. Whereas, the hardware the reliability is maintained.

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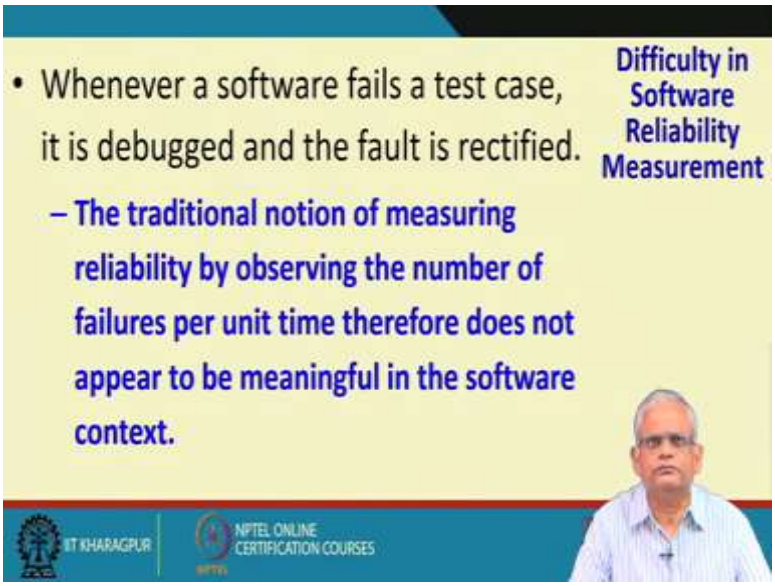

Whenever a software fails at test case there is a failure it is debugged the fault is rectified and therefore the traditional notion of measuring reliability by observing the number of failures per unit time does not appear to be meaningful in the software context.

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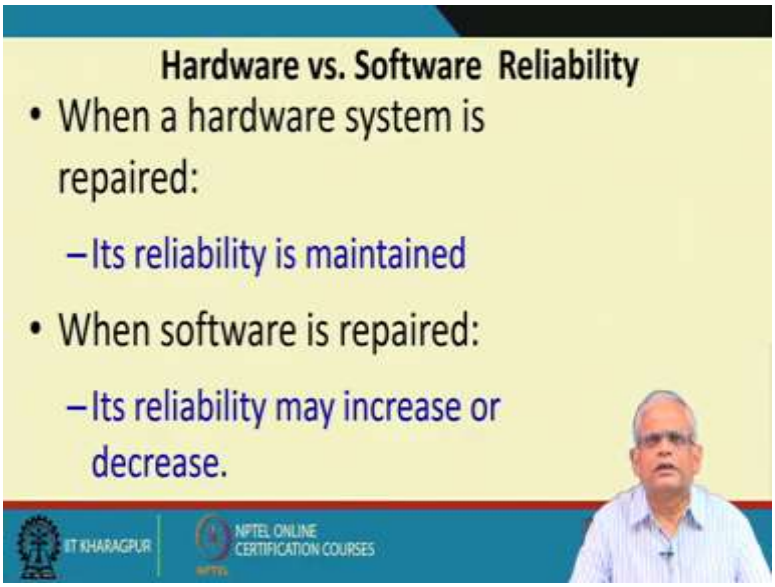

In hardware system once there is a failure, the component is replaced and the reliability is maintained. On the other hand, when software is repaired it is reliability may increase in that we have removed a bug or it may decrease also, because one fix error fix may create new bugs, let us say we fixed one bug. And let us say four five new bugs got

introduced because that error fix was not proper it fixed that bug, but then there were some other bugs which resulted due to that bug fix.

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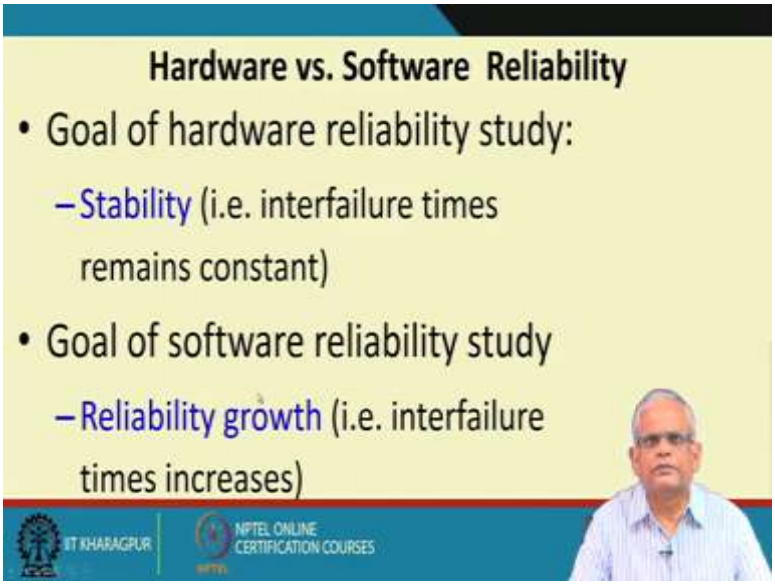

So, the we can say that the goal of hardware reliability study is stability that is the reliability is maintained the inter failure times remain constant, once the failure conditions are removed.

On the other hand the goal of software reliability study is reliability growth that is typically the inter failure times decreases the inter failure times decrease. I am sorry the there is a reliability growth the inter failure times increases that is, it does not fail as often because bugs are being removed. So, there is a reliability growth or inter failure times increases.

So, the study of hardware reliability stability, reliability stability, whereas the goal of software reliability is reliability growth the inter failure times increases. Whereas, in hardware the inter failure times remain constant.

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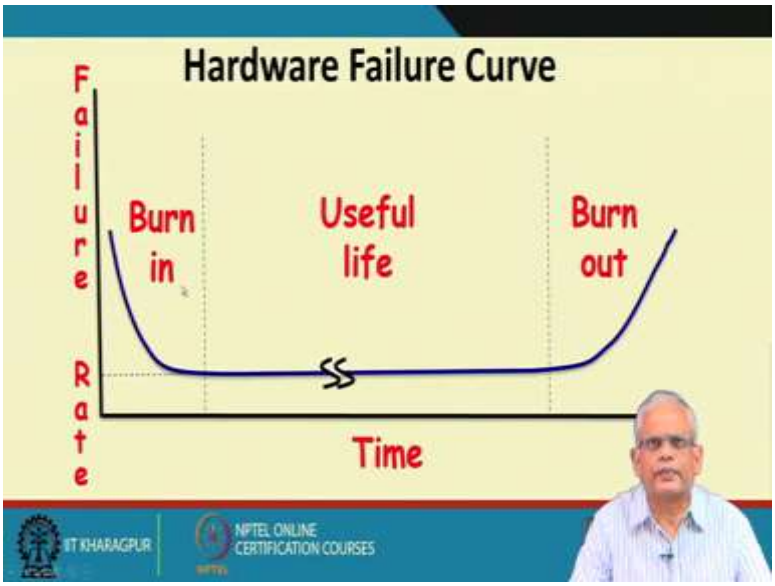

So, we can see behavior of hardware and software failure in a diagram called as the bathtub curve. We are almost at the end of this lecture we will continue from this point in the next lecture.

Thank you.

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Lecture – 58
Software Reliability – II

Welcome to this lecture. In the last lecture we were discussing the difference between hardware reliability and Software Reliability. We said that hardware reliability measurement is a relatively simple problem, because the failures there are due to wear and tear of components. And, once we replace the broken component, the reliability comes back to its previous reliability or the reliability is maintained.

Whereas, in software the failures are not due to wear and tear, these are due to bugs and to correct the failure it is debugged and the error is corrected. And therefore, the reliability increases or the inter-failure times increases. And therefore, the reliability keeps on changing in case of software whereas, in case of hardware the reliability is maintained.

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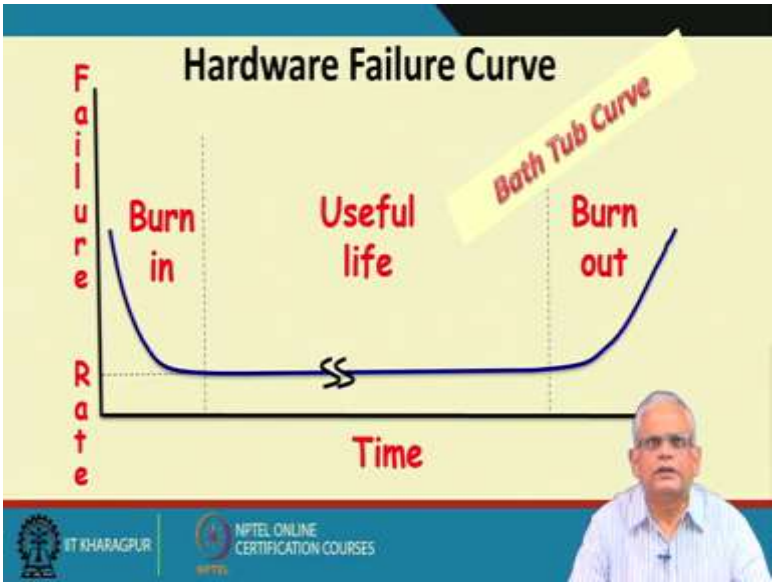

Now, let us look at this issue a little deeper. This is a typical behavior of a hardware system, here we have plotted the failure rate that is number of failures per unit time with the time of users. For a automobile or something it may be let us say for a period of 30 years or something and for a simpler system it may be 1 year and so on. Now, every

hardware system it shows this kind of behavior that initially the failure rate is very high and then with time it decreases. It may be automobile, it may be a mixer grinder, it may be a refrigerator, it may be a television set.

Any hardware system initially shows very high failure rate and the failure rate keeps on decreasing and then for a long time the failure rate is maintained. The inter-failure times remain constant over a long period of time that we have shown the symbol. And, this is called as the lifetime of the system; when the failure rate starts to increase. And, this is called as the bathtub curve, it is a typical model of any hardware system reliability. Initially the weak systems, the weak components each hardware system is made up of many components. Initially the weaker components they fail and may be the interfaces among components they fail recurring interfaces.

So, this is called as the burn in where the weaker components and weaker interfaces among components they fail and then they are repaired and reliability becomes high, stable and maintained in this rate for a long time. Typically, the initial part is covered by the manufacturer's warranty otherwise the people will not buy the hardware system because, it is showing a very high failure rate.

Typically, all good manufacturers cover this part by the manufacturer's warranty because of the burn in. The reliability is maintained and at the end of the lifetime, this point is the lifetime the failure rate starts increasing. And, here most of the components are worn out and the major components start failing.

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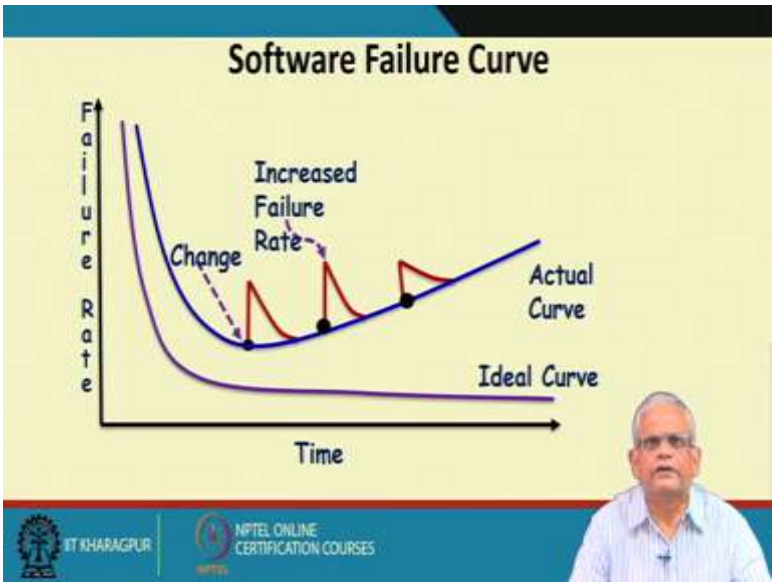

In contrast to a hardware failure curve, the software failure curve appears like this. It should actually the ideally it should appear like this violet curve, that as the time passes. The different problems or failures are reported and these are corrected and typically the failure rate is high of the for the software, any software. Once, these are released and these are fixed, as they are fixed the reliability improves and the failure rate comes down and then it slowly becomes better and better. Initially, there is a steep fall because the frequently occurring failures or the failures in the core part of the software get noticed and they are repaired.

And, each of those repair causes a huge improvement in the reliability and later slowly the more esoteric or rarely used parts of the software errors are found and those are fixed and they little improve the reliability. But, actually the curve is not like this, the actual software failure curve is that initially there is a rapid increase in reliability or decrease in failure rate. But, then this is not smooth curve throughout and the curve I have just shown few of this once here, but throughout this there are glitches like this. Each time there is a failure reported and these are corrected, there is a jump here because of a fix causing other bugs to appear.

Of course, sometimes the fix may be very proper, but sometimes a fix, bug fix may cause new bugs to arise and therefore, the reliability decreases after a fix. And, then after sometime the reliability decreases; initially the reliability increases or the failure rate

comes down and then there is temporary glitches, these are due to the bug fix creating new bugs. But just see the trend here, the unmistakable trend here is that the reliability is decreasing over time. Why is that? The reason is that because of many bug fixes slowly the structure of the software becomes poor. And therefore, the reliability becomes poor, many ad hoc fixes degrade the software structure and therefore, it gets poorer reliability.

We can see that the failure curve for software is very different from the bathtub curve for hardware systems. In other words the behavior, the reliability behavior of hardware and software are very different.

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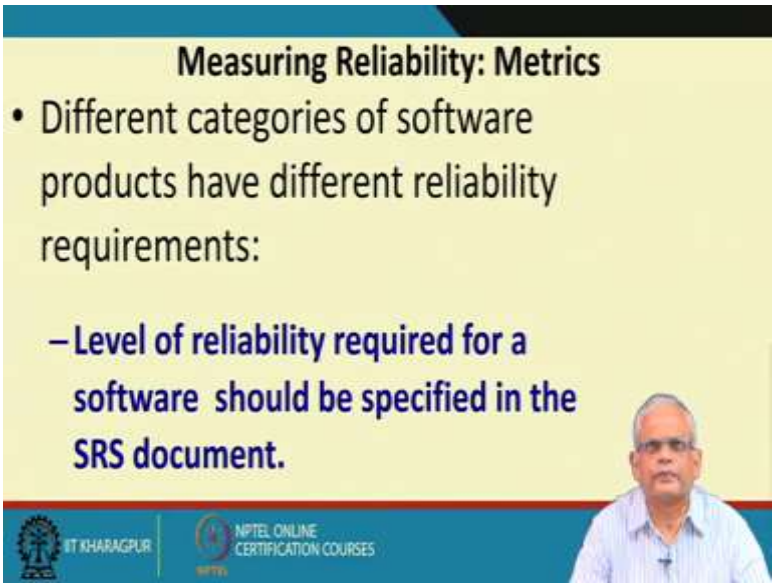

Now, let us see how do we measure reliability because, the user should be interested. They may specify the level of reliability for a software, typically for safety critical software they would definitely give a reliability figure. And, even the other users they are concerned about the down time, the frequency with which failures appear and so on and they may specify a failure rate for a software, they would like to install on their systems. But, using what metrics? Let us look at that issue that what are the metrics for measuring reliability.

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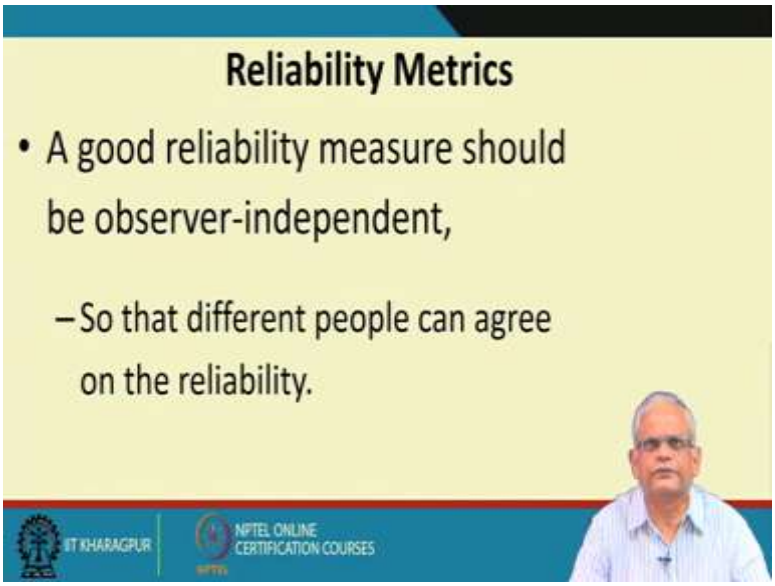

Of course one very basic issue with software is the different types of users use the system differently. And therefore, we cannot give different ratings to the same software. We need to give one reliability rating and that should be observer independent and all users should agree on that rating.

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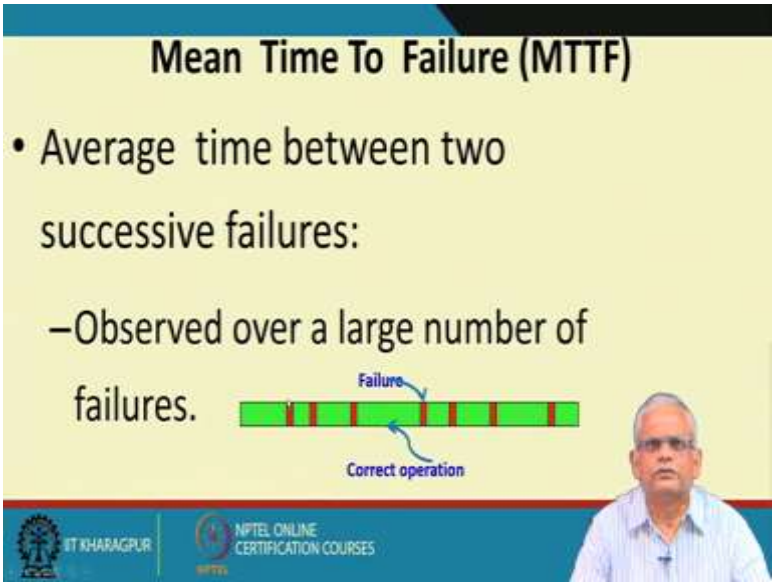

Now, let us see the reliability metrics for hardware, would they be useful for software. The mean time to failure is very popular hardware reliability metric, it is the average time between two successive failures or the inter-failure time. The mean inter-failure

time that is mean time to failure, here the software sorry the hardware this is a hardware metric we are trying to discuss whether you can use it for software. For a hardware system it is put to use for a long time and the and the failures are observed over this time and the average time to failure is computed from this.

For example, let us say this is the time of uses, there are correct operations after quite some time there is a failure. For a car may be the wheel has worn out and then again correct operation and then there is a failure, may be the battery has run out; replace the battery. And, then again there is a correct operation and then there is a failure may be the wire has got heated and shorted. So, replace the wire and again correct operation. The time between one failure to another failure is the inter-failure time and we can measure the inter-failure time for different failures and then compute the mean of that, that we will call as the mean time to failure.

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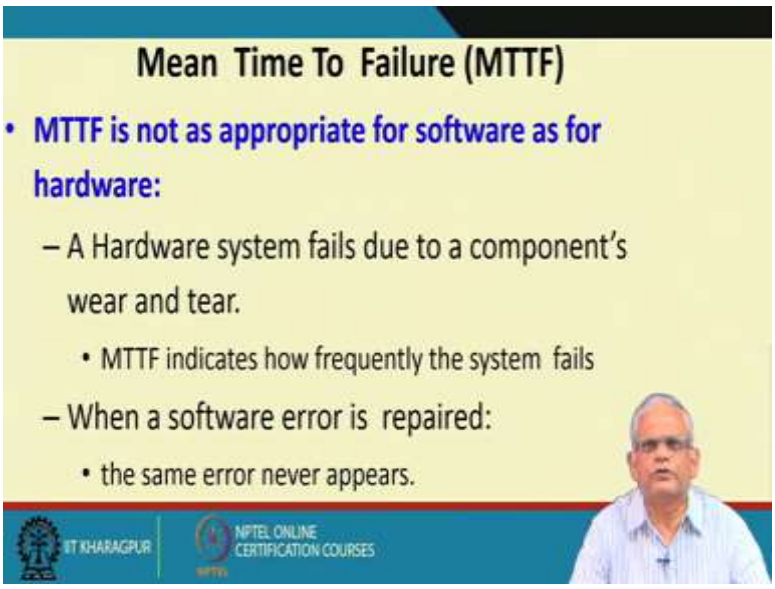

But, mean time to failure would not be very appropriate for software as per hardware because each time there is a failure, the bug is fixed and the rate of failure changes. For hardware it is a very reliable metric because the reliability is maintained and its meaningful to observe it for a long period and measure its reliability. But, for software reliability keeps changing, it improves typically, but then there can be glitches or the software may degrade due to a bug fix, the reliability may degrade and therefore, it either

increases or decreases. So, the mean time to failure where we observe the inter-failure times not be a really very appropriate metric.

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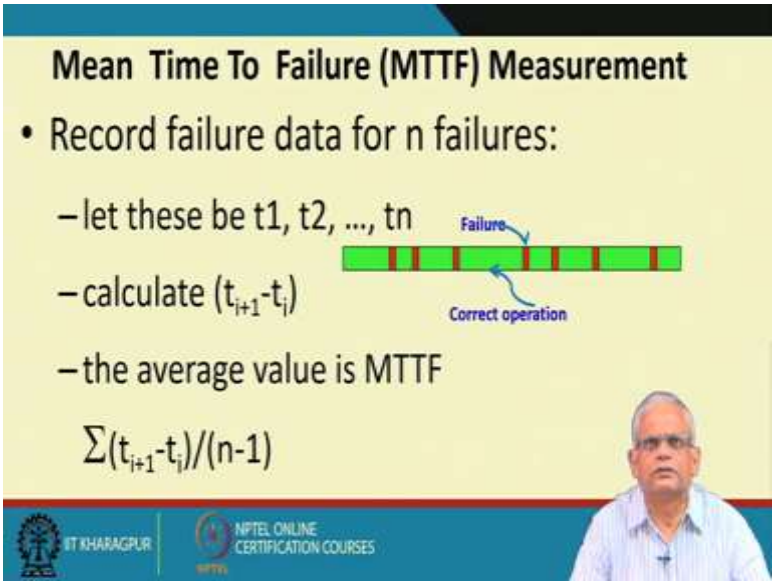

For a hardware system typically the failure times are recorded, let us say the failures occur at t 1, t 2, t n and then the inter-failure times are computed. And, then they are summed the total number of inter-failure times and divided by the total number of failures. So, computing the MTTF is straight forward. MTTF = $\Sigma(t_{i+1}-t_i)/(n-1)$

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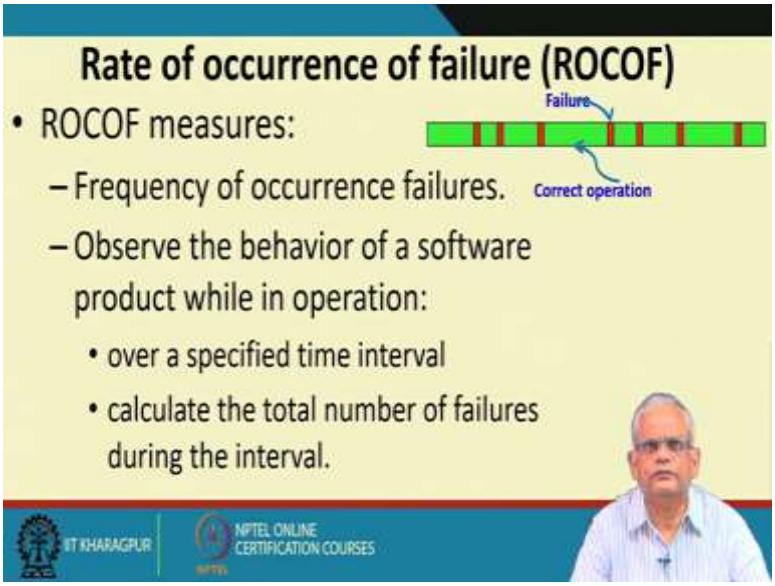

But, what about rate of occurrence of failure, this is another metric. Here the frequency of occurrence of failure is obtained. We observe the time period, long enough time period and find how many failures have occurred and then from that we find the number of failures per unit time. So, that is called as the rate of occurrence of failure. We just count how many failures and then the total time of uses and then find the number of failures per unit time or the rate of occurrence of failure.

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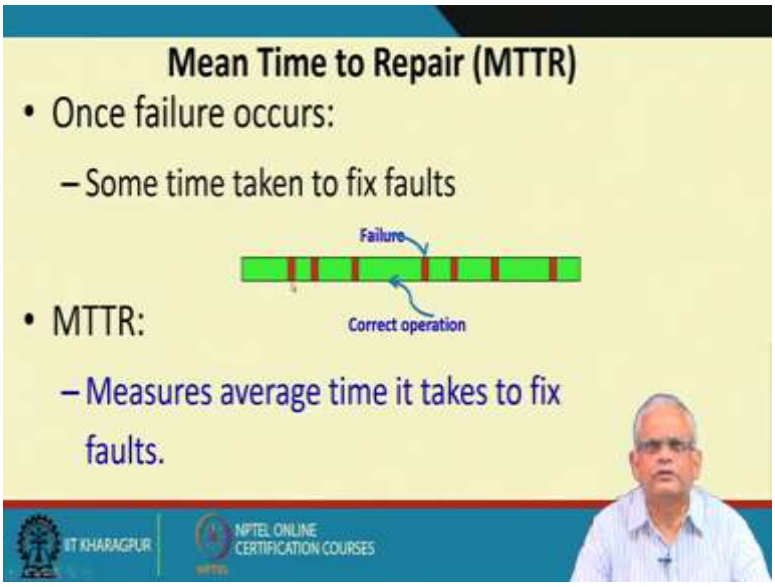

But, another issue is that once a failure occurs, it takes some time to fix the bug. There is a repair time because each time there is a failure, there is a repair time. For a hardware may be need to identify which component has failed and then replace the component. And for software debug, find out the bug, correct the code; but in both cases hardware and software there is a time to repair which is the time to fix the bug. So, each time there is a failure some time is needed to fix it, that we call as the mean time to repair.

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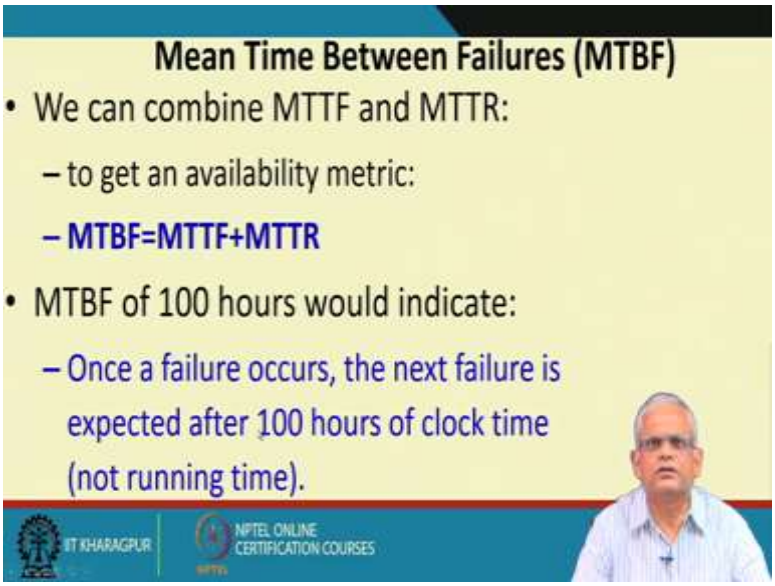

If we consider mean time to repair, in the previous case we just considered mean time to repair is 0. But, if we really consider the time to repair then the mean time between failure is the mean time to failure plus the mean time to repair. This is the time a system is not used, mean time to repair and mean time to failure is the time the system is used. And, the mean time between the failure is once the failure occurs when is the next failure expected that is MTBF is MTTF plus MTTR. MTBF=MTTF+MTTR

If we say that MTBF of 100 hours; that means, that once a failure occurs the next failure will occur after 100 hours and this includes the downtime of the software. Once a failure occurs the next failure is expected after 100 hours of clock time, that is including the downtime and it is not the run time of this software.

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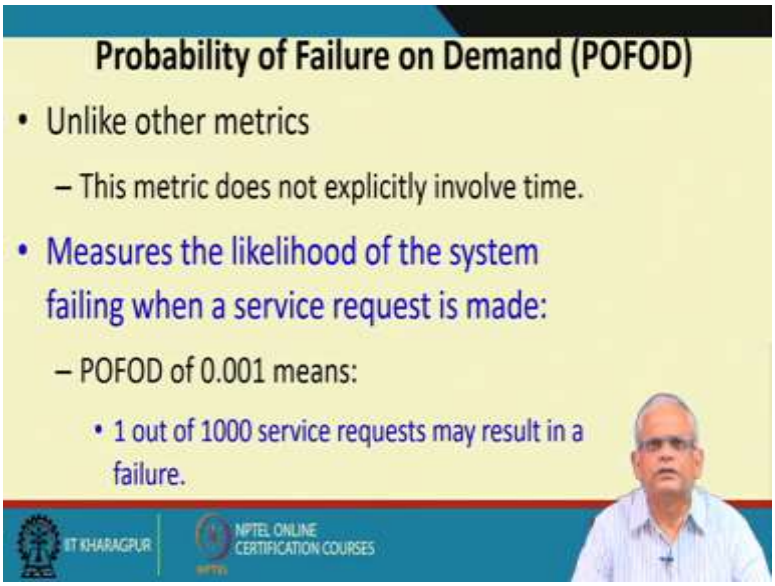

There is another metric, the probability of failure on demand; unlike the MTTF and rate of occurrence of failure, the probability of failure and demand does not involve a time. We do not measure it over a interval. Here we just keep on executing the software with different functionalities and found out, if we invoke the functionality with different data test input for 1000 times. How many times did the software fail?

If it is the probability of failure and demand is 0.001 that is on; that means, that if we executed the functionalities of the software 1000 times, then only one failure was observed. This is a more meaningful metric for software, because unlike a hardware which is continuously used, a software does not run unless a function is invoked. If the user invokes let us say each function, it runs for a fraction of a second or something and waits for the next input. So, the system is not running continuously unlike a hardware, let us say car is running for 10 hours, it is continuously the hardware is being used.

Whereas, here the software each time the user invokes a functionality, the software runs for a small time and then keeps waiting for the next input. And therefore, observing the software reliability over a period of time is less meaningful because, the question comes is that how frequently were the functions execute. Because, the software is not really running all the time, much of the time it was waiting for the user input. So, the probability of failure on demand is more appropriate metric for software.

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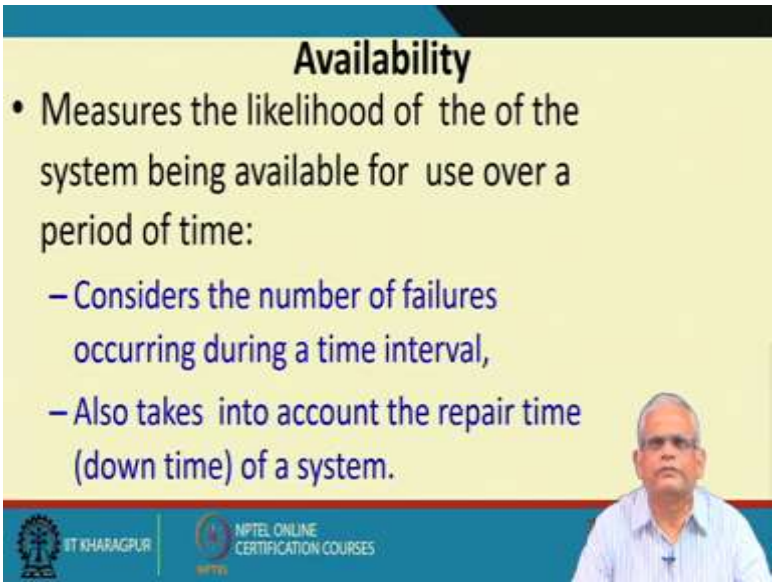

We can also define another metric called as availability which is basically characterizes the likelihood of the system being available for use to the user over a period of time. And of course, it will consider the failure time and the time to repair which we call as the downtime. And, based on this we can give a availability number that is a user once trying to use the software how likely that the software is available for use. Because, it may be undergoing, it might have failed, it might be undergoing debugging and so on.

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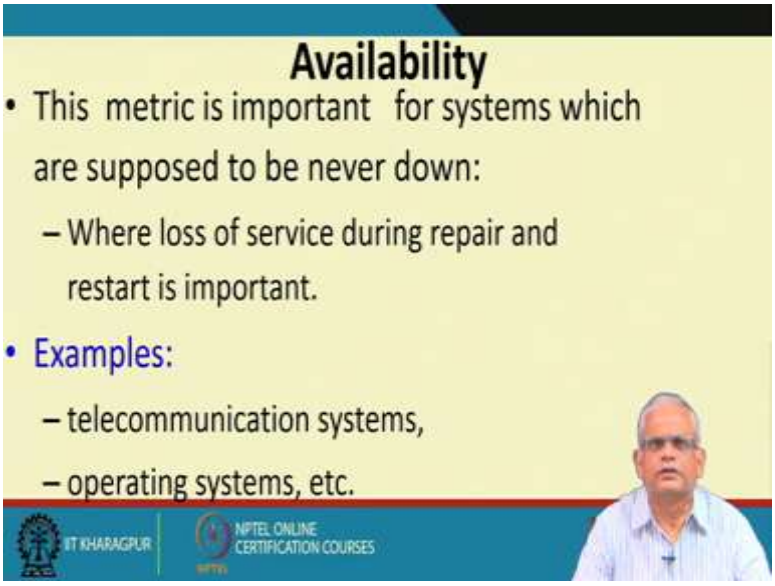

The availability metric is important for systems which are not suppose to be down, they run continuously. For example, an operating system or a telecommunication software.

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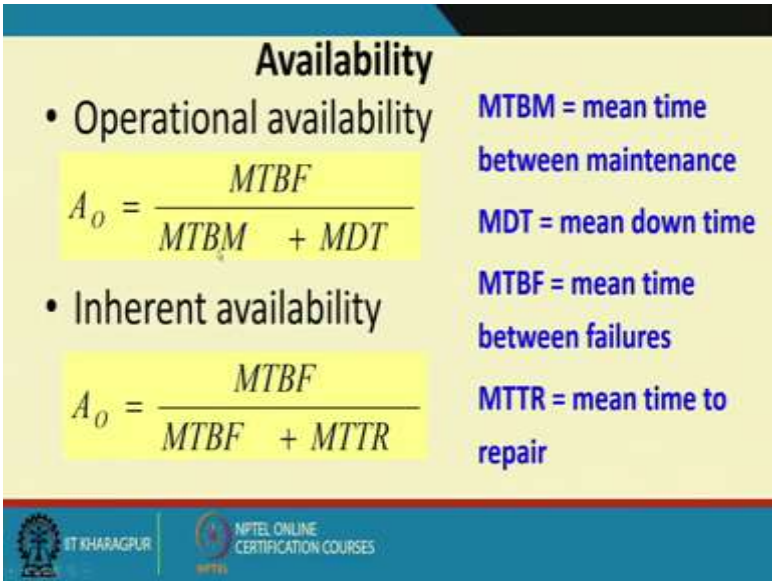

But, then we have two notions of availability: one is the operational availability and the other is inherent availability. In the operational availability we just consider the mean time between maintenance and the mean down time, whereas, the inherent availability is the mean time between failure and the mean time to repair.

$$\text{Operational availability} = \frac{MTBF}{MTBM + MDT}$$

$$\text{Inherent availability} = \frac{MTBF}{MTBF + MTTR}$$

So, here we just consider the time between failure and the repair time, whereas here we consider the mean downtime. The downtime is different from the meantime to repair, in the sense that this is the time to repair.

But, then there may be preventive maintenance, there may be initialization study and so on why the system is down. So, the mean downtime is a more general measure of the downtime which not only the repair, but also the system may be down due to various reason including preventive maintenance, routine initialization etcetera. So, the operational availability is the actual availability to the user whereas, the inherent

availability is a ideal notion of availability; where we do not have preventive maintenance etcetera. The only downtime is due to the repair time.

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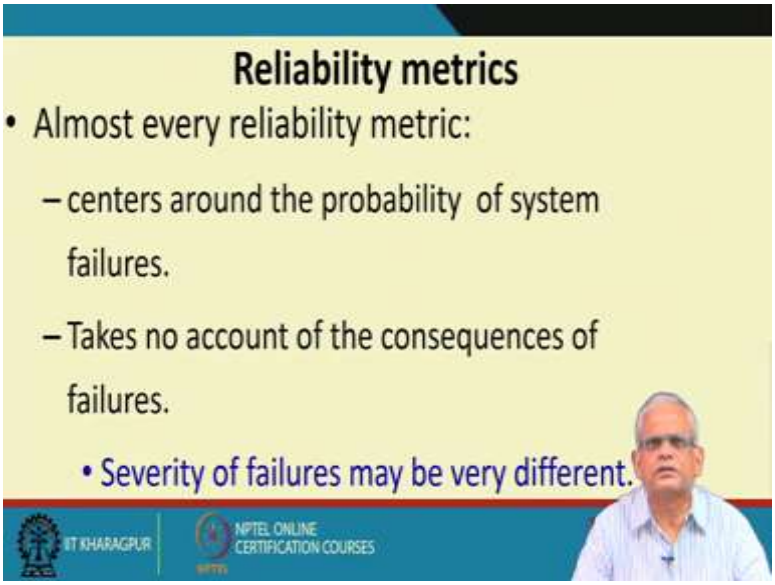

But, if we just look back into all the metrics that we discussed, all the metrics are centered around the probability of the system failure. And, they do not take any account of the consequence of failure, but in reality some failures are very severe, they just the system hangs. We need to reboot, work is lost and so on. Whereas another function, another failure may be that the current system, the current function invocation did not work, but the system did not hang; just need to try different function or the same function the second attempt it may work and so on.

So, the severity of different failures may be different. If we just consider all failures occurring over a period of uses that may not be very proper. We cannot compute all types of failure and then come up with a reliability metric, because some may be very insignificant failures. And, if we only count the severe failures like cross type of failure that also will not be proper. So, how do we go about handling this problem? The different failures have different severities.

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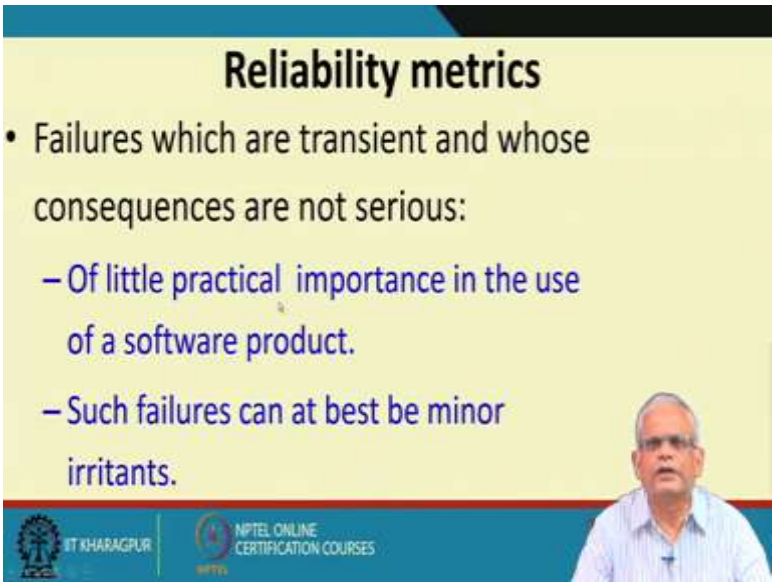

Some failures are transient and consequences are not serious and they are of little practical importance. At best they may be just minor irritants may be the mouse did not work, but next time the mouse worked and so on.

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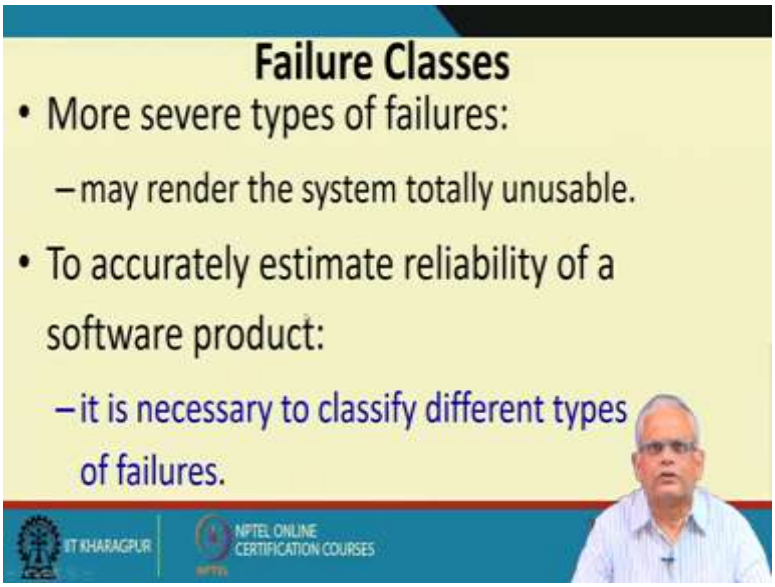

How do we handle this problem? We handle this problem by defining a failure class, different classes of failure.

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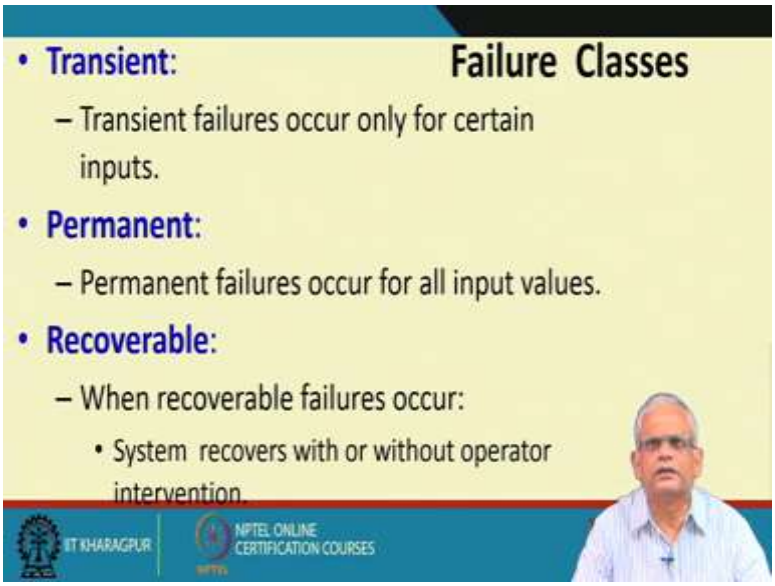

We classify the different types of failure. Transient: this class of failure they occur only certain inputs, permanent failure they occur for all input values, some function they fail for any input value. But, let us say only fails per one input value when we give only 100 it fails, for all other it work satisfactory; then we call it as transient. Permanent it fails for all input values, recoverable that the system did not hang. We could recover either the user himself or with operator could recover and again without rebooting we could use the system.

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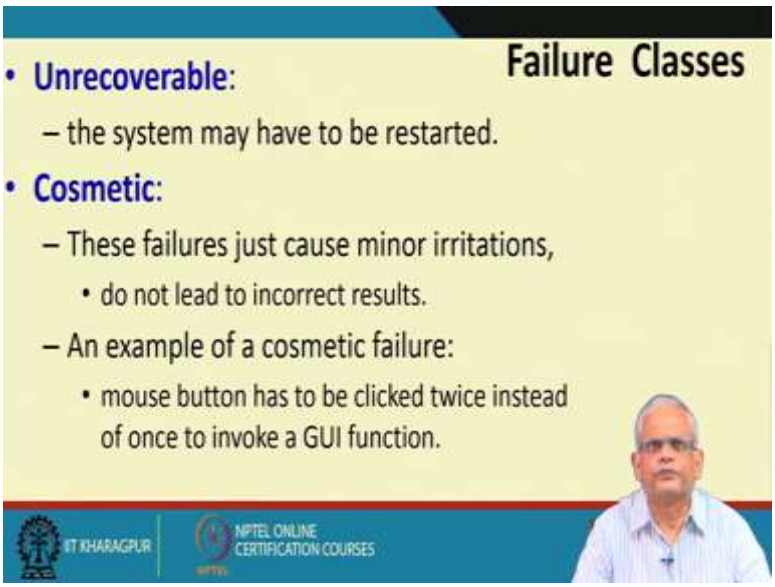

Unrecoverable the system might have hanged, we might have to restart the system those are unrecoverable. And, the cosmetic are the ones which are minor irritants, they do not really cause incorrect results or something, but just minor irritant may be the user need to sometimes press the mouse button twice. So, we can measure the reliability for each class of these failures and that will give a better idea of the reliability.

Maybe the reliability by considering the unrecoverable and the permanent of time type of failure or may be ignoring the cosmetic failure we can measure the reliability and so on. So, far we have looked at the reliability metrics and some very basic issues and then in the next lecture we will discuss about how to go about measuring the reliability of a software. We will stop at this point.

Thank you.

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Lecture – 59
Software Reliability – III

Welcome to this lecture. In the last lecture we had discussed about some issues in Software Reliability measurement.

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In particular we discussed why software reliability estimation is a much harder problem compared to reliability estimation of other systems. We looked at the specific characteristics of software which make the reliability estimation difficult, we also discussed about few mtrics which can be used to estimate the software reliability.

This lecture let us first look at software reliability estimation techniques. We will basically discussed two main techniques for software reliability estimation; one is reliability growth modelling and the other is statistical testing let us get started with that.

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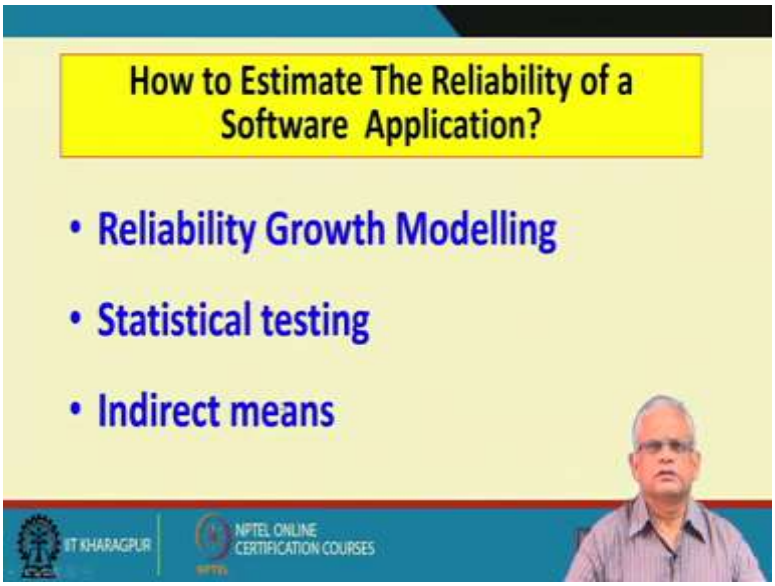

So, the problem that we are trying to address is that given a software application how do we go about estimating the reliability of the application? The application has been developed, installed and running. Our task is to estimate the reliability of the application, we need a quantitative value for the reliability of the application.

One of the options is reliability growth modelling. We just try to map the failure rate with a reliability growth modelling curve and based on that we can say that what is the reliability of the software after elapse of sometime at the movement so on. We will look at this technique reliability growth modeling as a way to estimate the reliability of a software application. We will look at statistical testing this is another way to estimate the reliability of the application.

The third technique is indirect ways of measuring the reliability here we do not really look at the failure rate and so on and measure the reliability, but based on the characteristics of the software. For example, various metrics of the software its complexity matrix number of calls per use cases per use case and so on.

So, based on a set of metrics that are collected for the application, we can predict the reliability. We will not look at the indirect means we will look only at the reliability growth modelling and statistical testing.

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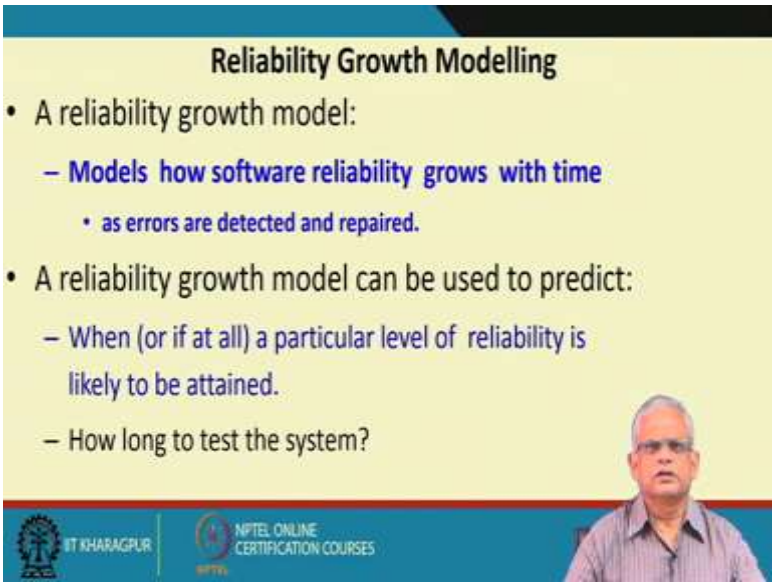

Let us look at reliability growth modeling to start with. A reliability growth model is a graphical model of the reliability growth of the software with time. As different errors are detected and repaired the reliability of the software changes and we model over time how the reliability would grow.

The use of such a reliability growth model is that we can say that what will be the reliability of the software after some time. We can also use this model to predict that given a quantitative requirement of the reliability of the software how long will the testing need to go on. Because as testing goes on, more and more bugs are detected and fixed given that a certain level of reliability is given for the application. We can predict how long how much longer to test is it 2 weeks, a month, 2 month this is another use of a reliability growth model.

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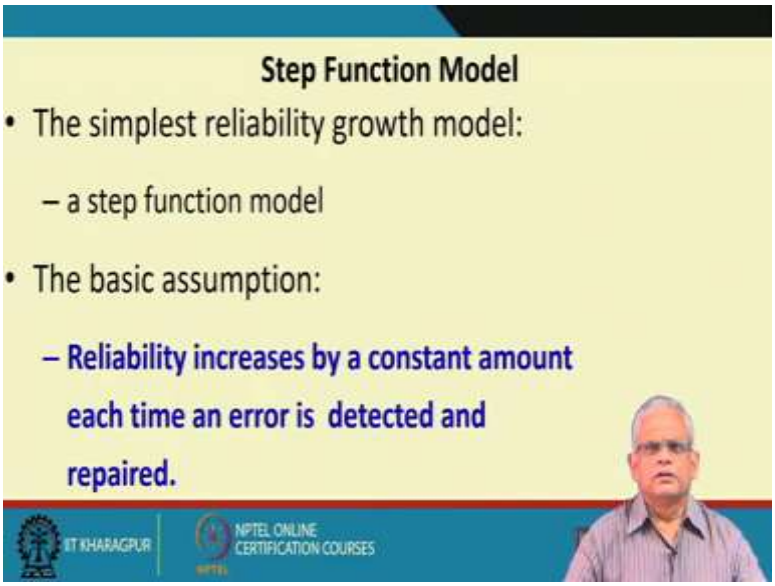

The simplest growth model is the step function model its quite old model very simple, but then it is also intuitive and forms a background for other more sophisticated reliability growth models. Here the basic assumption is that each time a failure occurs the corresponding error is detected and fixed and the number of errors in the software decreases therefore, the reliability increases.

The simple assumptions made by this model is that all error fixes are ideal and error fix permanently fixes the error there are no side effects of that no other bugs are introduced by that and also each time an error is fixed, the reliability increases by a constant amount. Very simplistic model we know already based on our intuitive knowledge of the software reliability that reliability growth on an error fix is not constant.

Initially the bugs that are there on the core of the software get exposed they cause failure because those are the ones which cause frequent failures and they are the ones to get detected first. And, fixing the bugs in the core part increases the reliability by a rather larger amount compared to fixing a bug that is there in a remote part of the software. So, the assumptions are too simple, but then this can be used to predict reliability with some approximation.

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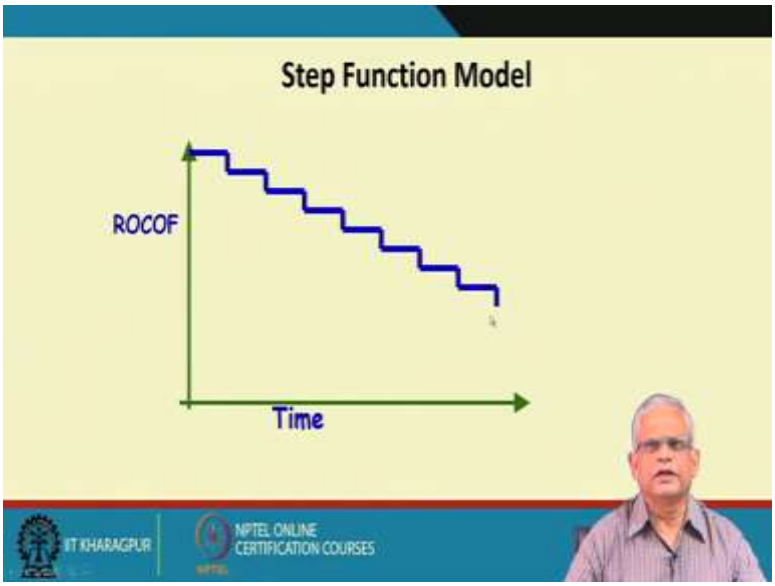

The graphical representation of the model is this step function, the rate of occurrence of failure on the y axis and time and the x axis and each time there is a failure, the rate of occurrence of failure comes down reliability improves. And, based on how frequently are errors discovered and so on we can fit onto this model and predict the reliability of a given application, this is the simplest model.

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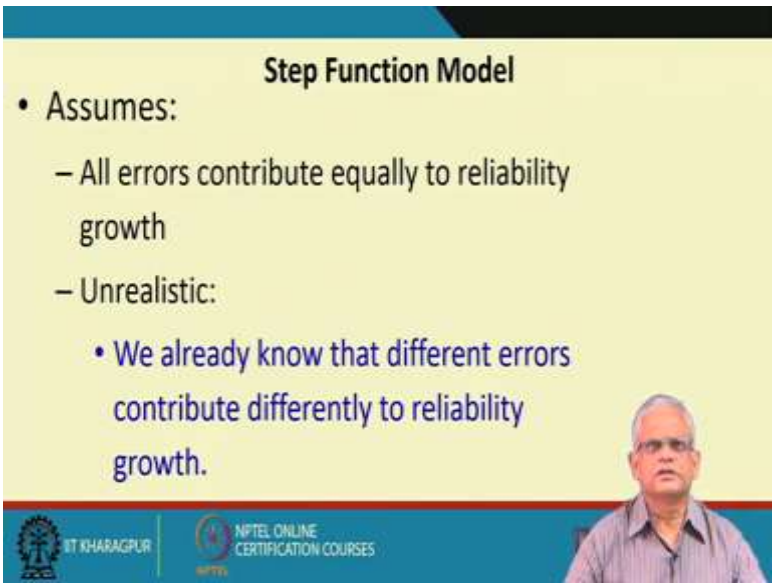

Here one of the implicit assumption is that all errors contribute equally to the reliability growth we know from our basic understanding that different errors contribute differently to reliability growth. And therefore, the results of this model will not be very accurate.

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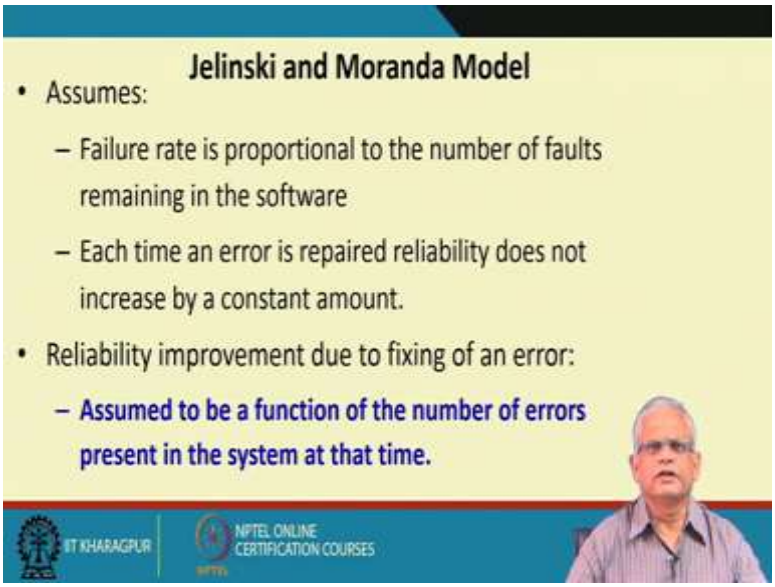

Another basic model is the Jelinski Moranda model again proposed long time back. Here the basic assumption is that the failure rate is proportional to the number of faults remaining in the software. Again implicit assumption being that each error is responsible to cause failure at the same rate.

So, the failure rate is proportional to the number of faults remaining in the software and each time an error is repaired reliability does not increase by a constant amount. We can see here since the failure rate is proportional to the number of faults remaining in the software. If n is the number of faults, the failure rate is the function of n, it is proportional. So, let us say k n and the failure rate after bug has been detected and corrected is n minus 1, k n minus 1.

We can see that the improvement in the reliability is not constant here and is a function of the number of bugs. Here initially the improvement in reliability will be low on a bug fix because out of n bugs we fixed one bug. So, basically proportional to 1 by n, the, but towards the end if there are two bugs and we fixed one of that the reliability growth will be higher this is the assumption here.

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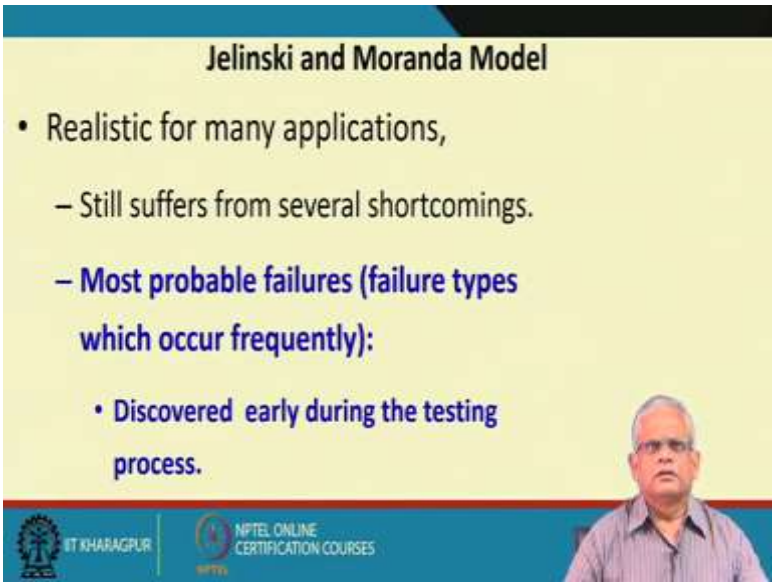

Its better than the step function model here each bug fix contributes differently to the reliability growth, but then still there are several short comings. As we are saying that intuitively we know that the bugs that are there in the core part get discovered early contribute to larger growth of reliability.

But then as time passes by the bugs that are there on the noncore part of the software get detected and these contribute to lower increase in reliability. And therefore, initially the reliability growth should be high and later part the reliability growth should be slow down, but here this model does not do that.

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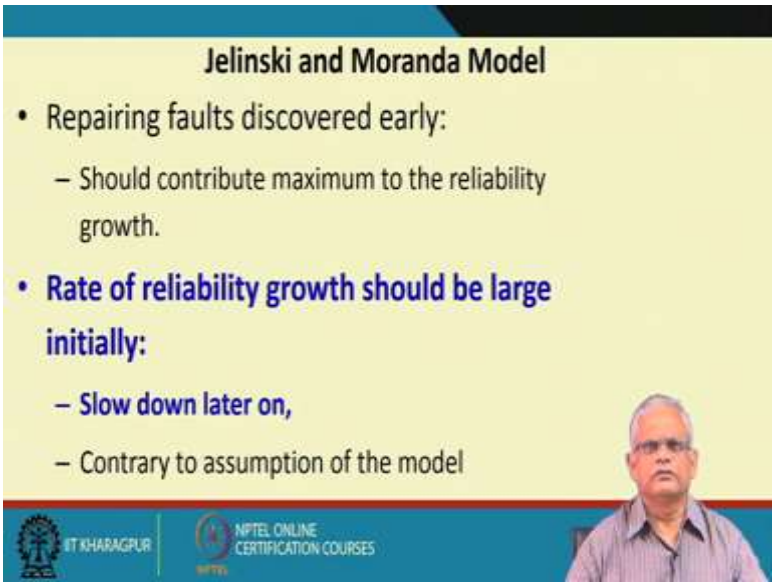

Initial faults that are detected and corrected should contribute maximum to the reliability growth and the rate of reliability growth should be large initially and slowed down later on and this is contrary to the assumption of this model. And therefore, this model will not really give very accurate results.

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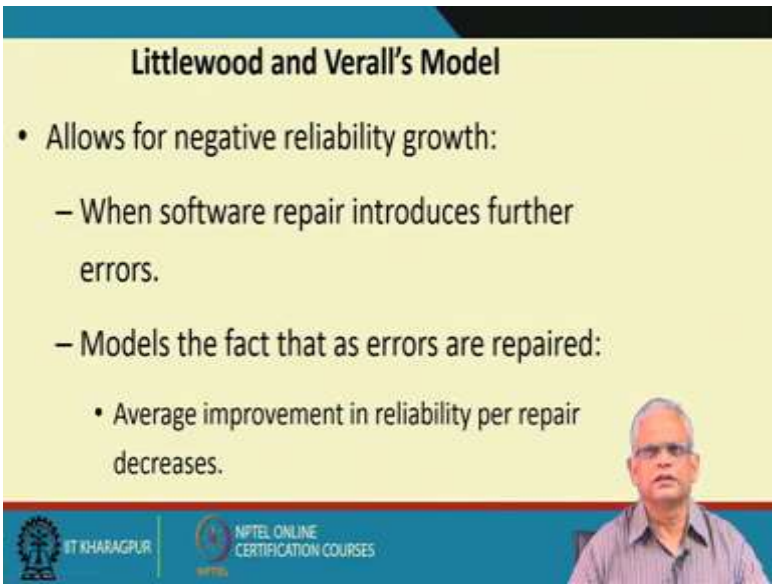

Now, let us look at the Littlewood and Verall's model, in both the models we have seen So far, the step function model and the Jelinski Moranda model both it was assumed that the bug fixes are ideal, that is when a bug is fixed error is reduced. But in reality on a

bug fix more errors may get introduced that bug may get fixed, but it may cause more error to get introduced. And therefore, the number of bugs may actually increase. And therefore, on a bug fix the reliability may actually decrease there is a negative reliability growth this really happens practically and this is taken care in the Littlewood and Verall's model.

Negative reliability growth may occur sometimes when the repair introduces further errors. And here as we can see that the growth in reliability is not constant and also the average improvement in reliability per repair decreases because sometimes there are negative reliability growths.

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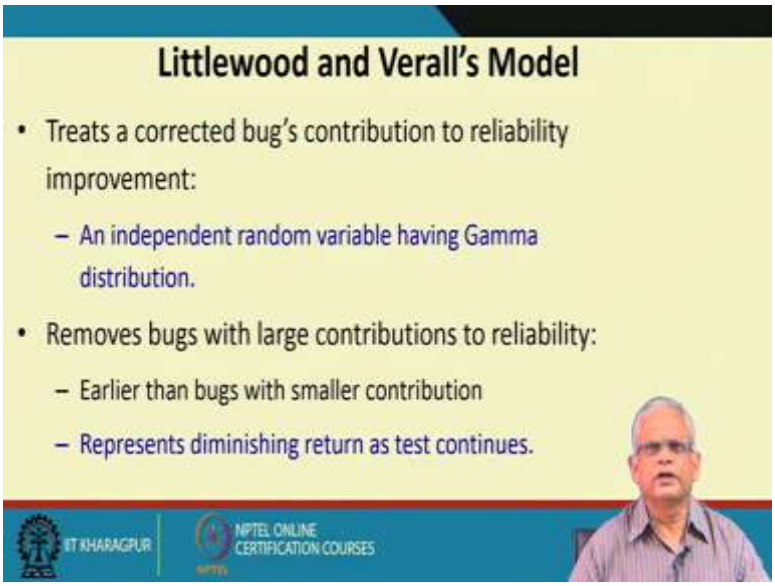

To do this the model treats a corrected bugs contribution to the reliability as a independent random variable having gamma distribution. And here the bugs with large contributions to the reliability are considered initially these the bugs that are initially detected and removed these contribute to higher reliability growth than the bugs that are detected later.

So, this matches with the intuitive understanding of the reliability growth. It represents diminishing return as the test continues and this model of course, matches with the intuitive idea of the reliability growth. And therefore, would give a more accurate result than both the step function model and the Jelinski Moranda model.

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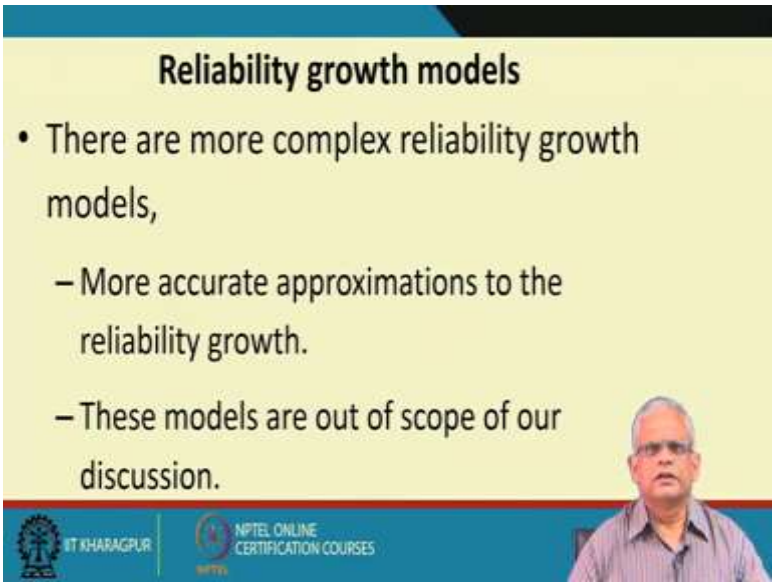

There are several dozens of reliability growth models that have been proposed based on these simple reliability models. These provide more accurate approximations of the reliability growth, but then ours being a project management course we are more interested in getting the basic ideas on the reliability growth the basic models rather than really discussing very sophisticated models and spending time on that. So, these more sophisticated models are out of scope of our discussion.

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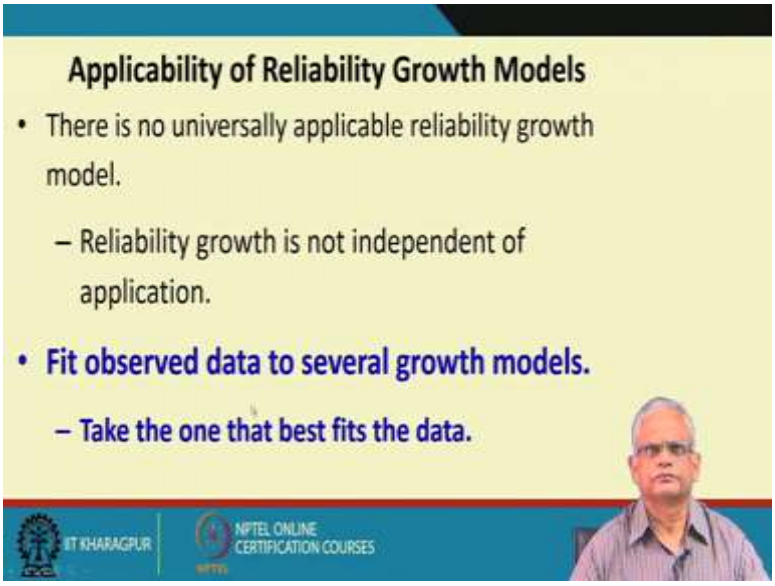

But given that there are several dozens of models, each making some simplifying assumptions and some realistic assumptions and so on. How do we determine which model to use, is it that there is a best model which will be used for all applications that will give the best result? No not really because the reliability growth is dependent on the specific application that we are considering.

The bug distribution the number of functions supported the complexity of the software and so on. And therefore, we cannot straight away say that this model is suitable for this type of software, but what we can do is given the failure data over a time period a considerable time period we can try to plot it and match with these growth models.

The one it matches most accurately that will be the one to be used and the best fit model is taken and based on that we can say that after elapse of certain time what will be the reliability of the software.

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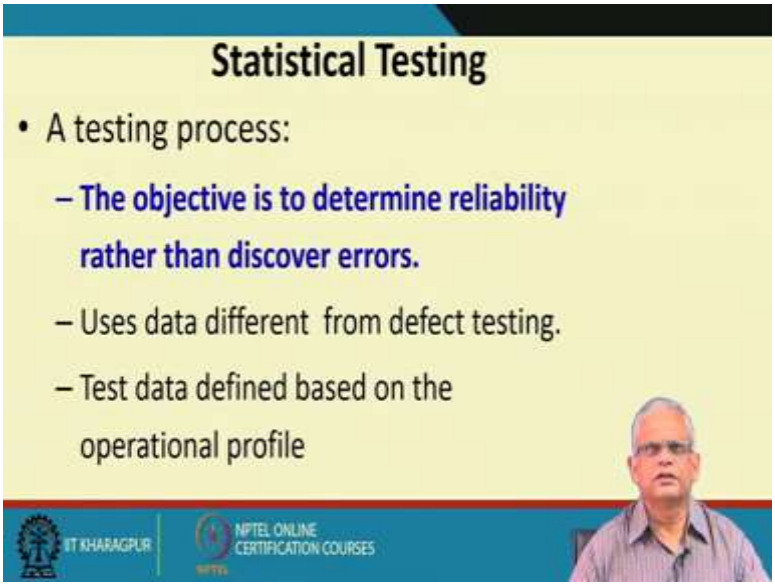

Now, let us look at another technique the statistical testing technique, this is a testing process no doubt, but then here the objective is not to detect bugs. Here the objective is to determine reliability and if some bugs get detected that is not the focus, but then that is good that some bugs get detected, but here the objective is to determine the reliability of the software. And naturally the test case design here is different than the defect testing. One thing we can say that the test data here these are designed based on the operational profile of the software.

Since the test data is defined based on the operational profile of the software this gives a numerical value for the reliability of the software which is agreeable to most users. We had said that the reliability of a software is observer dependent, different observers may have a different estimation of the reliability of the software. And we had said that it will be confusing if we give different reliability values for the same software for different classes of users we need to give only one value.

And the value that we can give is the average of all the users and that is captured here in the operational profile, this is the average rate at which the different functions of the software are invoked considering all the users.

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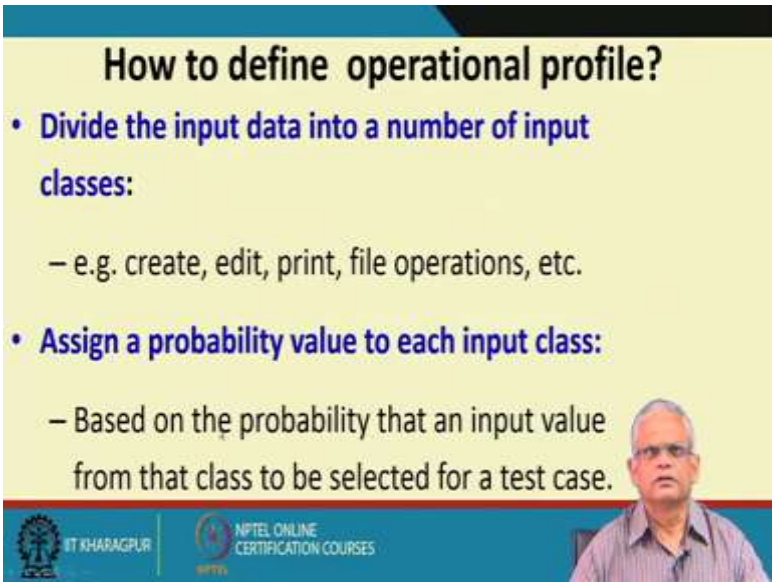

We had discussed about the operational profile in one of our earlier lectures. Here we look at the different functionalities like a let us say word processing software may have the functionalities create document, edit document, print file operation etcetera. And then we find out the test data which correspond to create, edit print file operation etcetera. And we observe how the software gets used by different users.

Maybe we can write a small log collecting software which will keep on collecting the log of uses by various users of the software and from there we can find out what is the rate at which these different functions are invoked by different users. And based on this log we can assign a probability value to each input class, each input class basically belongs to some of the operation. And then we will have all the functionalities of the software and

probability of an input value getting selected from that class that will form the operational profile.

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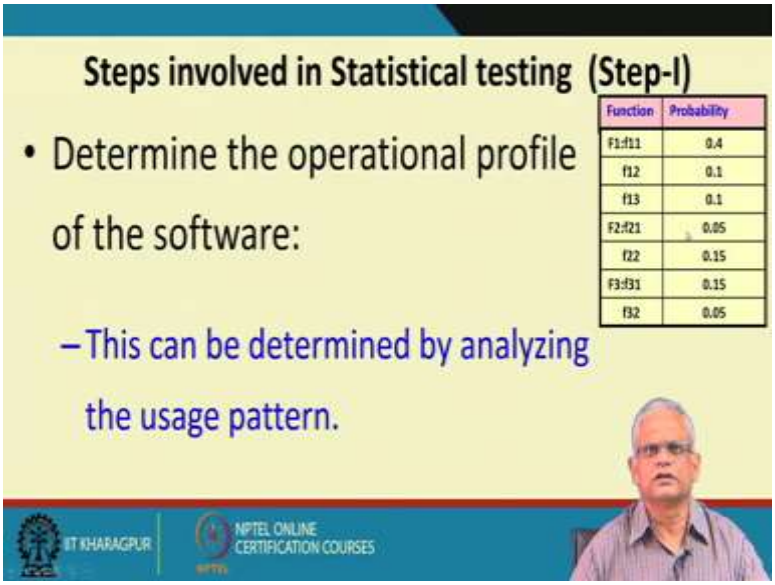

So, here in statistical testing the first step is to determine the operational profile of the software, the user's pattern can be captured by a log and then we can analyze to assign probabilities to each of the functions supported by the software.

Typically we will have various functions supported by the software and also different scenarios for the functionalities and then we give a probability of the function being invoked. This forms the operational profile definition of the software.

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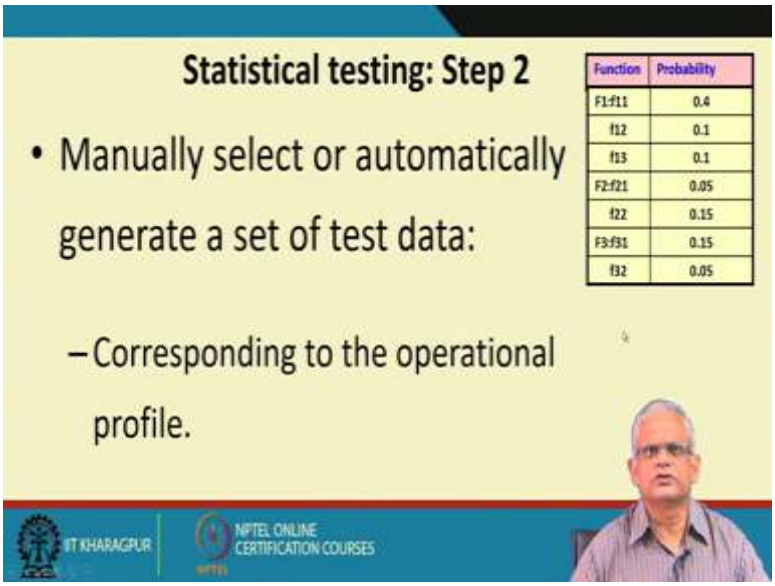

The second step is to generate the test data; here the test data should correspond to the operational profile. What we mean is that if 40 percent 0.4 is the probability of the first scenario of F1 functionality, then in our test suite, we should have 40 percent of the test cases where F1 is invoked on the f11 scenario.

Similarly, 0.1 is the probability of the f12 scenario of functionality F1 and therefore, 10 percent of the test cases should correspond to the f12 scenario of the F1 functionality. As you can see here that if we analyze the test suite we should find a correspondence with the actual uses of the software. In the defect testing we do not do this we just anticipate where defects might be there and we design test cases, here we do not do that. Here based on the user's pattern we generate test data.

If f11 scenario of the F 1 functionality has a probability of 0.4 in our test suite design, we design 40 percent of the test cases with test values that correspond to the f11 scenario of the F1 functionality.

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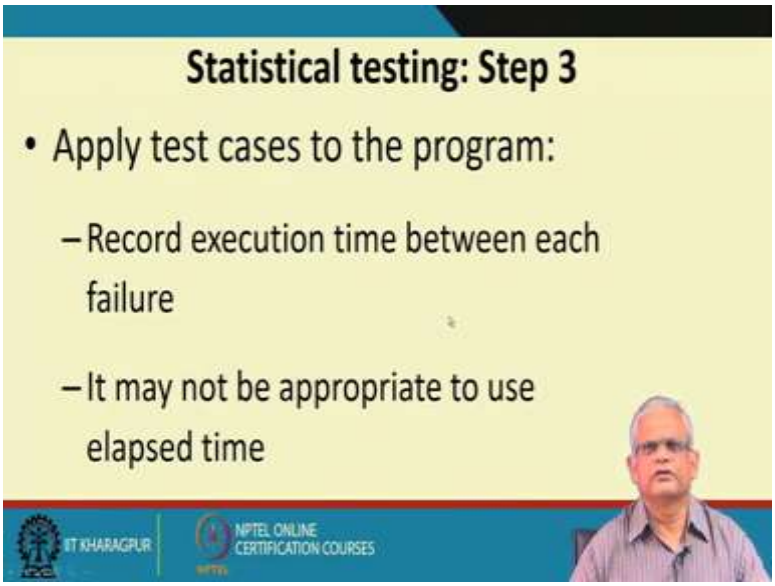

And in the step 3 actually carry out the execution of the software using the test cases and find out the failures. But then measuring the clock time may not be the right thing to do we should actually measure the execution time. As we are discussing earlier, that the software runs very fast. Each functionality it typically completes in milliseconds, but the human input may be let us say you can give a input in several seconds or minutes.

And therefore, most of the time the software idles on each input it runs for a very short time and then idles for the next input. Therefore we should have some instrumentation some means by which we collect the actual run time information of the software and that we plot on our failure data, we consider that as the time of failure not just the clock time. Using clock time we will result in a very inaccurate reliability estimation.

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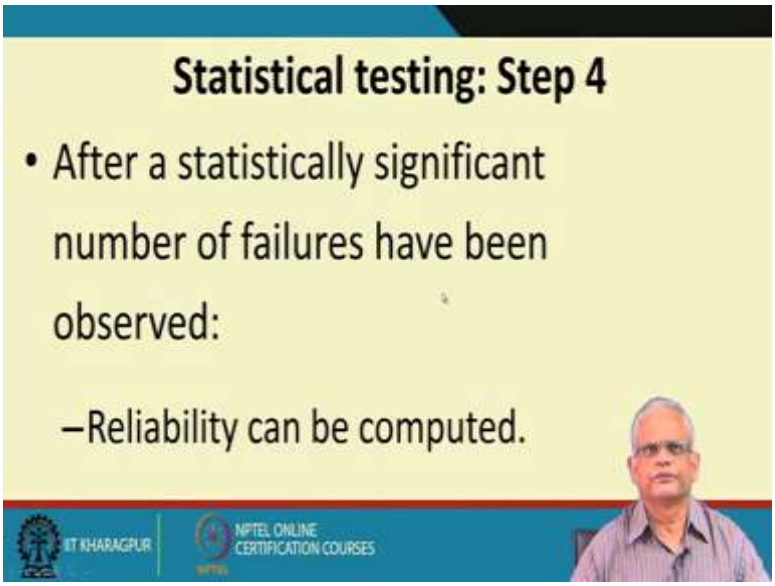

And finally, after statistically significant number of failures have been observed the reliability can be computed that is how many functionalities were invoked and how many failures where observed and so on.

Here the result is likely to be agreeable to most of the users because the test data have been defined based on the operational profile. We are at the end of this lecture we will stop here and continue in the next lecture.

Thank you.

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Lecture – 60
Software Testing

Welcome to this lecture. In the last lecture, we had discussed about how to estimate the reliability of a given software application. One way we had said is a reliability growth modeling. We can consider a model; there are several models have been proposed, we discussed only three models. So, these models are basically graphical representation of the failure rate over time.

And, for the specific software application that we are interested we just collect its failure data over time and then try to plot it and see that which model is the most accurate which fits this our failure data most accurately. And, based on that, we can predict the reliability of the software after certain time at the present time and so on.

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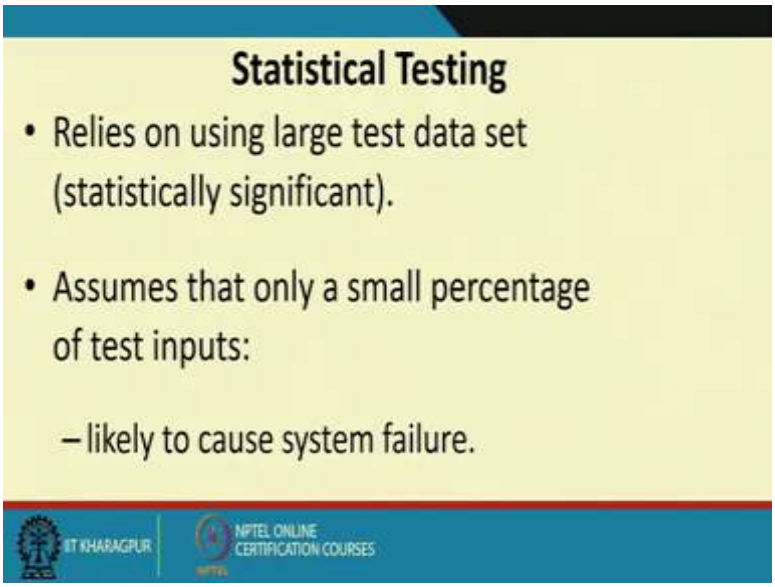

The second approach we discussed was statistical testing. In statistical testing, we use the operational profile and design test cases. Even though name is testing, here, the objective is not to detect bugs. The objective of statistical testing is to estimate the reliability of the software. Here we design the test cases such that the test data correspond to the frequency of invocation of the functions by the users. If the users on the average invoke

some functionality 40 percent of the time, then our test data should actually have 40 percent test data corresponding to that functionality.

The other important thing is that, it relies on using large test data because as the name says it is a statistical technique, we should have a significantly large data based on which we can infer and predict the reliability after certain time. If we use only handful of data that will give a very wrong result and also we assume that only a small percent of test input is likely to cause system failure. If there are system failures occurring in almost every test cases, then of course, the prediction of reliability will not only be poor, but it will be very gross it won't be accurate.

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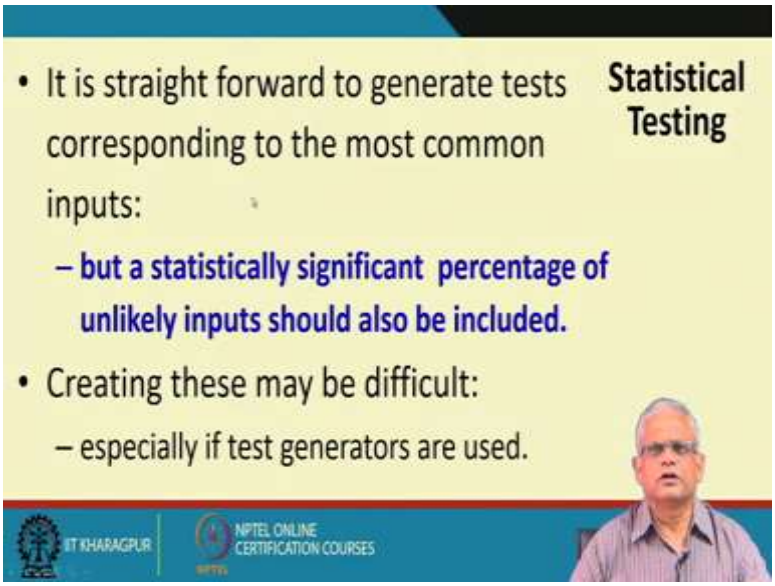

But, the complexity of using statistical testing is that how do we generate test data because we need a significantly large number of data. It should be statistically significant number of data and not only that if we give data that are most likely to be input by the user that is ok, but then we must have a significant percentage of unlikely input data should be also included.

These basically correspond to the corner cases that is the input value that the users give on normal operation that is fine, but then we should also include the input values or the combination of the input values which the users normally do not give, but may sometime give and those are the ones which actually cause failures. Those are called as the corner test cases.

But, creating those test cases are actually the challenge. The normal values that we can easily generate, but the unlikely values that becomes a big challenge and the statistical testing, the accuracy of the result depends largely on whether we have taken care to have enough corner cases, significant percentage of these corner cases.

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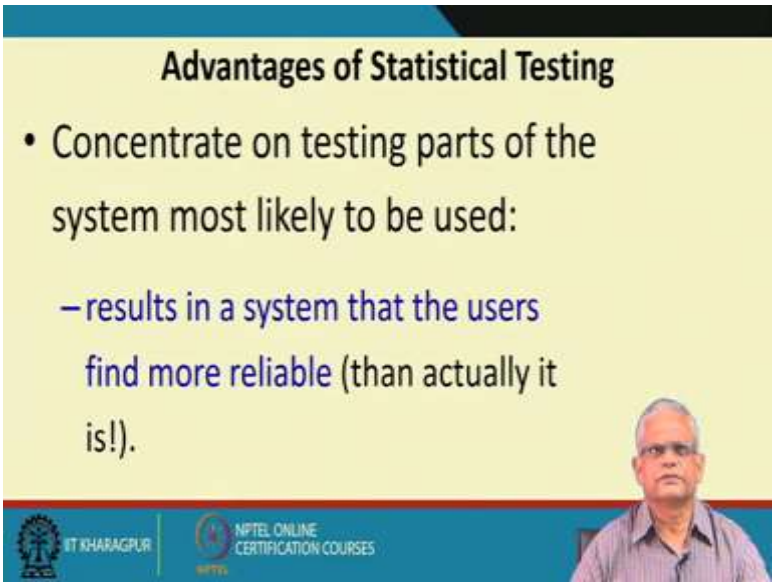

Here we give a large number of input for the ones which are frequently used by the average user and therefore, the average user will find that the system is portrays his estimation or his understanding of the reliability of the software. And, here the idea is to test the parts that are frequently used with more test cases and therefore, those are the bugs in those areas are removed.

The frequently used functions they were fine because lot of test cases have been designed for that and therefore, not only that we get a reliability estimation, but also significant number of bugs are removed and that too from the frequently used functions. And, therefore, the reliability of the software appears to be higher compared to when the bugs are removed uniformly from all functions. Here we are concentrating on testing the function which are most frequently used.

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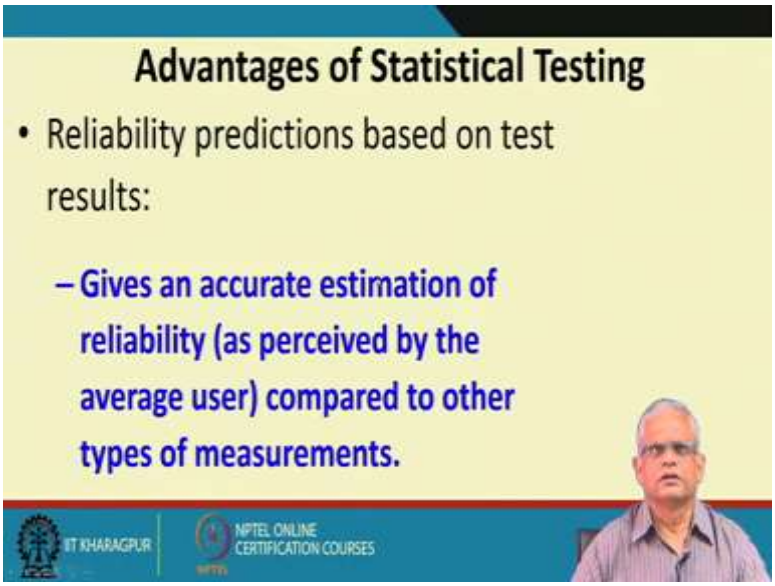

And, here we define the test data based on the operational profile and therefore, it gives a accurate estimation of the reliability as perceived by the average user compared to the other type of measurement because the test data is generated based on the operational profile.

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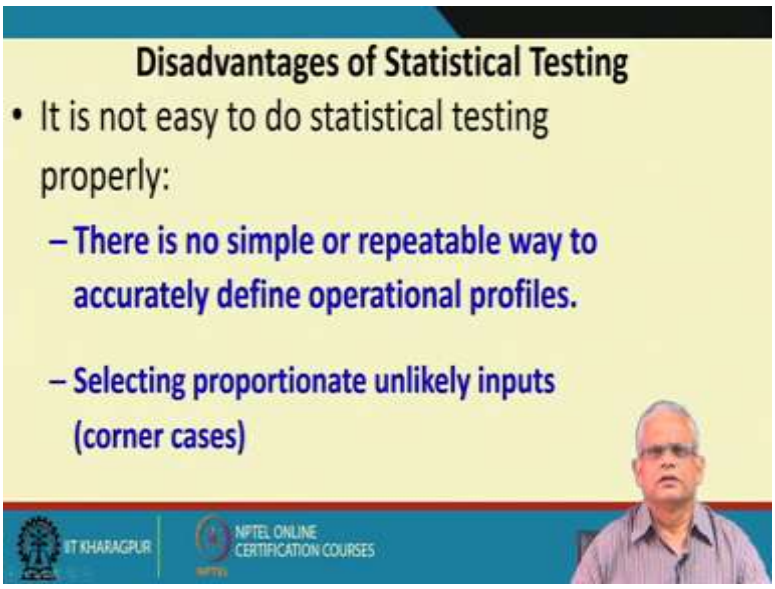

But, if we think of the disadvantages of the statistical testing one is that the results to a large extent depend on how do we define the operational profile and also how do we design the test cases based on the operational profile. Even if we define the operational

profile by a laborious study where we collect the logs across a large number of users; in actual users scenario of the software, but then even if we have the operational profile correctly the first challenge is to define the operational profile. It has to be collected from the actual users situation and also a large interval of users.

The second problem is that even if we have the operational profile correct, how do we design the test cases. Designing the test cases that are rather usual test values is not difficult, but selecting a proportionate unlikely input or the corner cases is a big challenge and if we do not do this the results would not be accurate.

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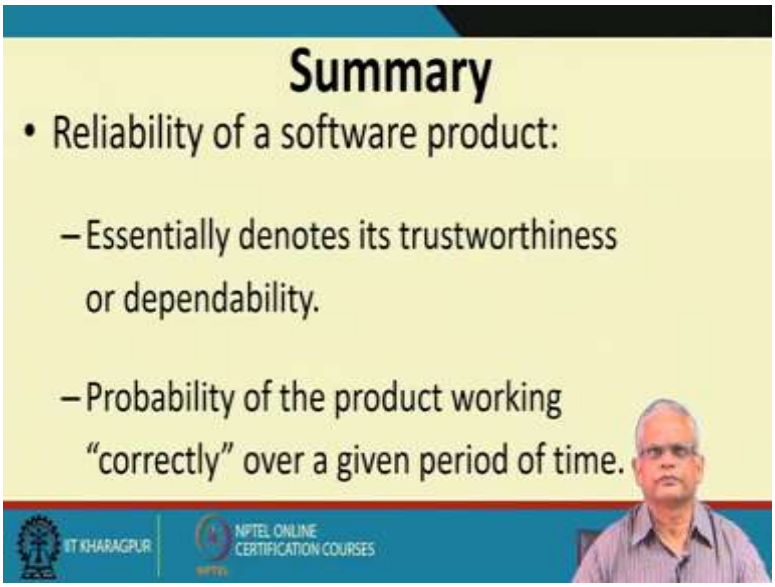

In the summary of our reliability study we can say that the reliability of a software can be defined as the trust worthiness or dependability. More accurately this is a very overall definition trust worthiness or dependability is their intuitive idea of the reliability of a software. More accurately we can say that the probability of the product working correctly over a given period of time.

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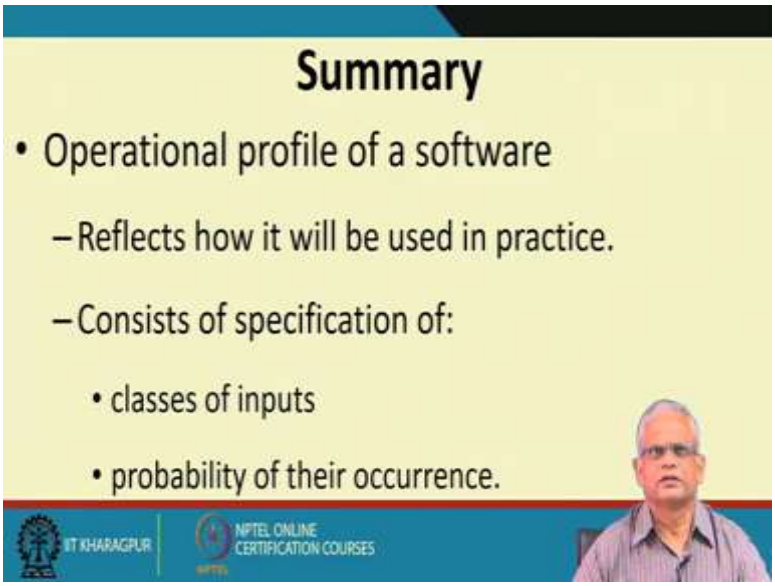

And, we had seen the problems in software reliability measurement. It is seen that the reliability changes as bugs are detected, unlike the hardware and not only that it is observe dependent. The reliability of a software depends on who is using and therefore, we had said that we must use the operational profile to estimate the reliability.

Our test cases should be designed based on the operational profile of the software then we can get a result of the reliability which is agreeable across most of the users and the operational profile is based on the different classes of input and the probability of their occurrence.

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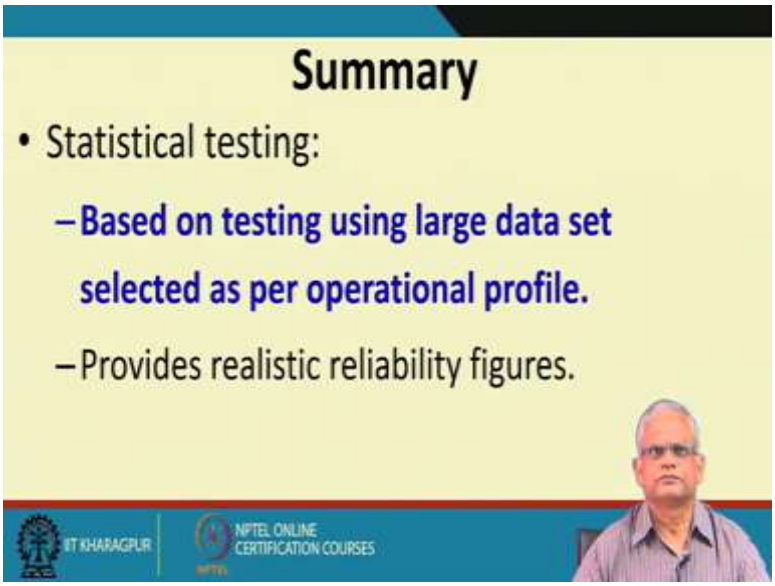

And, the statistical testing is based on testing using a large data set. Tens of thousands or millions of data, test data which are designed based on the operational profile to get a very realistic figure. We will now discuss about software testing. The project manager needs to also understand the software testing issues because finally, getting a good quality product testing is a major contributor.

And, to manage the software, so that the software is a good quality the project manager needs to understand some of the basic issues in software testing. With that motivation, we will discuss some issues on software testing.

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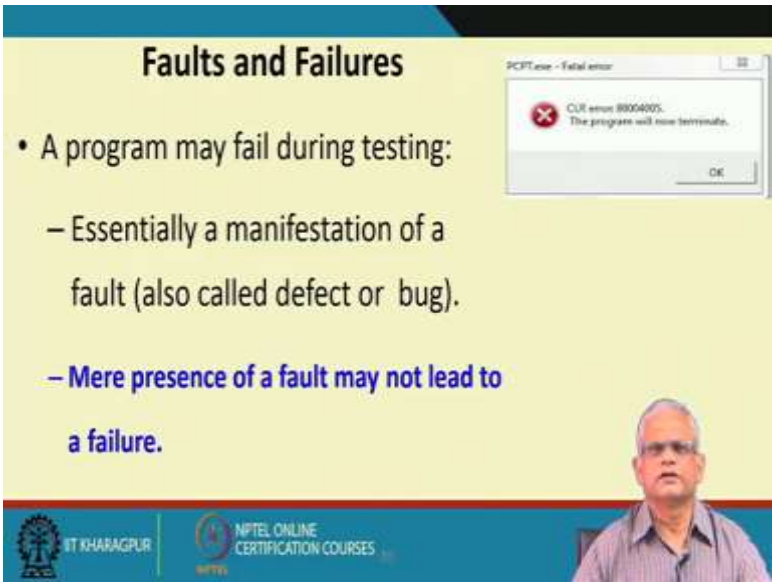

The first thing we will discuss is the difference between a fault and a failure. When a program is being tested it may fail. For example, on a given input it may display fatal error, the program will not now terminate. What we are observing here is not a bug, if we gave the input test input and we observed that this message came that the program will now terminate fatal error. We are observing the failure not the error.

So, what we observe on testing is a failure, but the failure is caused by a defect, bug or a fault. In other words, we say that a failure is a manifestation of a fault. There is a bug somewhere in the code and that cause the failure, but we observed the tester the failure. But, then, do every bug in the software cause a failure? No, there may be bugs which may never cause a failure. We will just discuss about that little later. You can also think about it that how can a bug be present, but not cause a failure.

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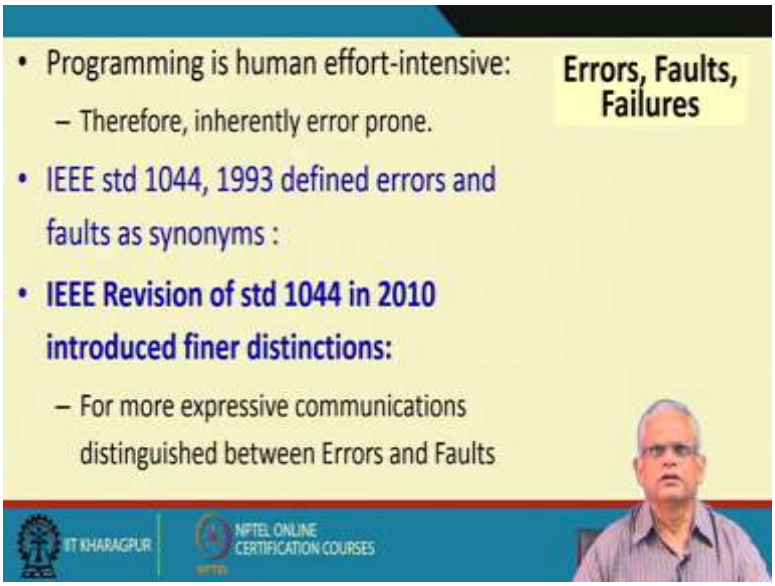

Let us distinguish between error, fault and failure. Programs, even though now a lot of automation automated tools are available, still programming is largely effort intensive. In 1993, IEEE standard 1044 errors and faults were considered to be synonyms. But, in 2010 revision finer distinctions were proposed, because there was a need to be more expressive in communications, we need to distinguish between error and fault because coding, designing etcetera are laborious manual effort and people do mistakes. And, therefore, we must know what is a error, mistake, fault, failure because these are the terms that are frequently used in this area.

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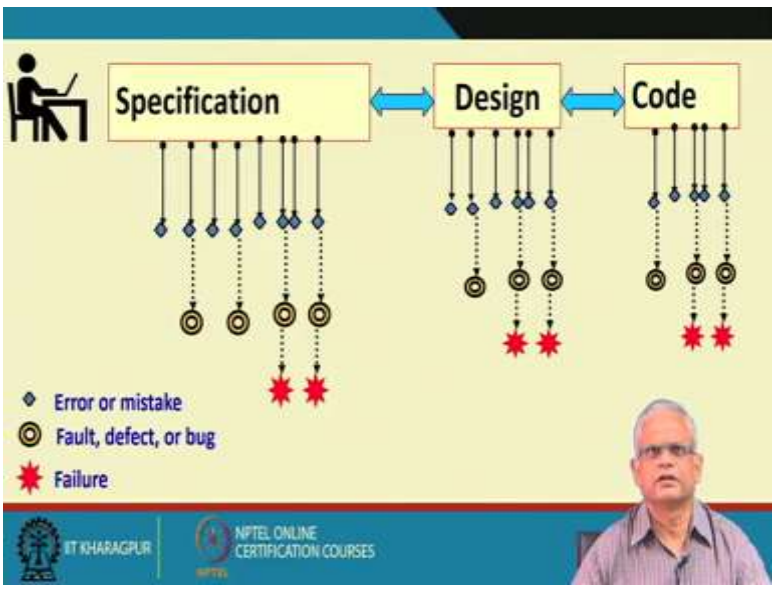

To understand we will just see this picture here the programmer here is typing using his laptop or desktop and initially he does the specification, design and then code. During the specification may make several mistakes. The mistakes we represented using this blue rectangles, but many mistakes they do not create problem, they are not bugs. Some of them are bugs, but then all bugs do not create failure, some bugs create failure.

Similar is the design and the code. The code the programmer may make mistakes or errors, but all errors are not faults. Let us say the programmer made a mistake instead of writing a is equal to b made a mistake wrote a equal to c, but then it so happened that b and c had the same value and therefore, it is a not a bug, it is not a fault. But, it was a mistake on the part of the programmer, he should have written a equal to c, but he wrote a equal to b.

But, then some of the faults may cause failure. There may be a fault let us say erroneously wrote that on some condition b is equal to 2 into b, but then let us assume that condition never occurs for the normal inputs. Instead of writing b into 3 written b b is equal to b into 2. There was a mistake committed and that is a fault, but then that does not cause failure because those inputs are not given during the actual uses.

Anticipating that such conditions may occur the programmer had given, but that condition never occurs and therefore, that bug does not express does not manifest as a failure.

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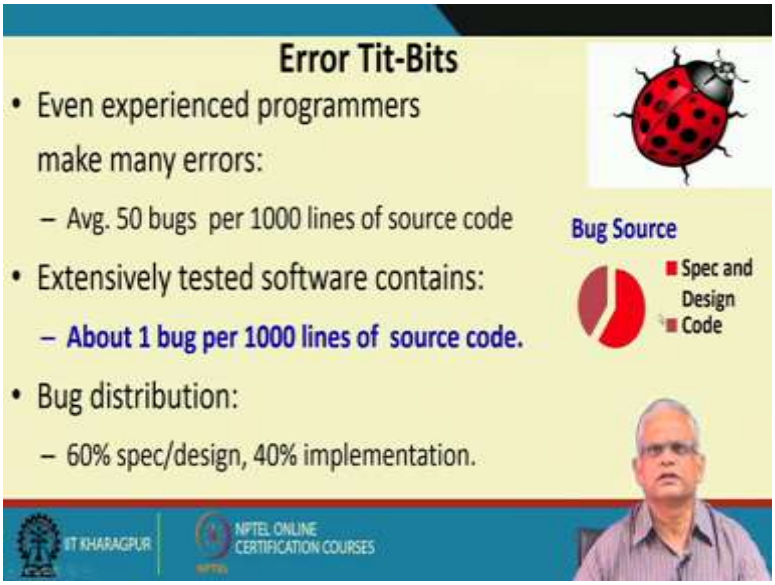

All programmers however proficient make mistakes and many of these mistakes they are basically bugs remain in the code and even with the most experienced programmers the data collected shows that average there are 50 bugs per 1000 lines of source code, is a large number. Even after testing thoroughly by professional testers, we do not really eliminate all the bugs.

Out of the 50 bugs that are there after the programmer completed programming by the time the testing is over most of the bugs are detected, but still 1 bug per 1000 lines of source code is typically present, is a study saying that even after though testing, 1 bug per 1000 line of source code is still there.

For some applications this may be acceptable, but let us consider a banking application where there is a bug and when the bug gets triggered into failure, there may be significant financial loss or may be a nuclear power plant, 1 bug per 1000 line of source code is not acceptable because those are safety critical systems, can cause huge damage if there is a failure.

But, what is the distribution of the bug? Typically majority of the bug that remain after development pertain to the specification and design, 40 percent of the bugs are based on the coding mistakes. Specification and design contribute the larger percentage of bugs about 60 percent; 40 percent bugs are present due to coding mistakes.

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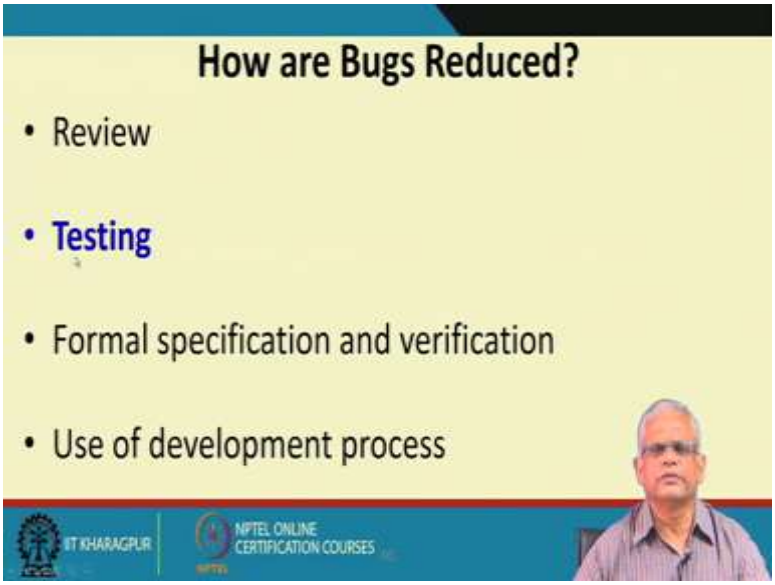

But, given that program development is a manual activity, bugs are inevitable because people do mistakes after all, how does one go about reducing the bugs to the minimum? One technique is review; review the work at various stages that is very well accepted technique to reduce the bugs. Testing – this is possible the most important technique, most effective technique to reduce the number of bugs.

Formal specification and verification this can also be used to reduce the bugs. And, even use of a proper development process, developing good quality software then also reduce the number of bugs. So, as we can see that there are various ways in which we can remove the bugs and testing remains as one of the most promising ways of reducing the bugs.

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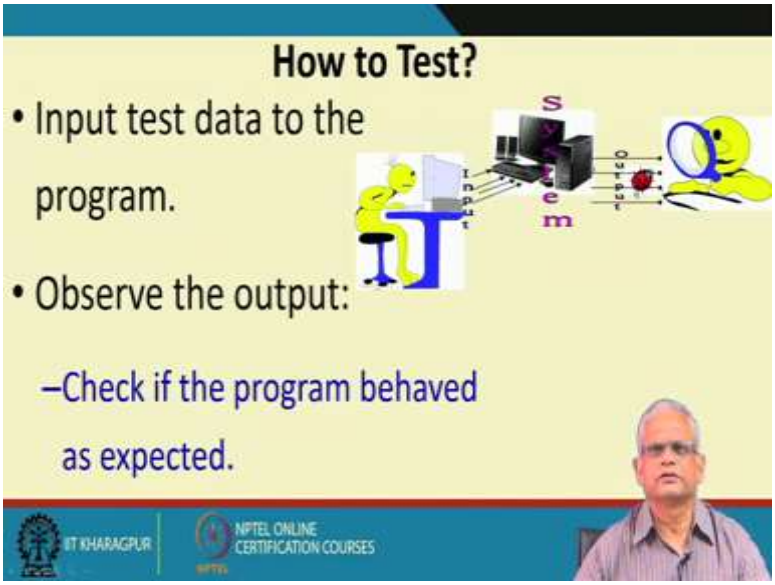

But, given a software how does one go about testing? To test any software we need to give it input data and then observe the output, and see if the generated output matches with the expectation we have for the given input. That is, observe the output and check if the program behaved as expected; is the system software to be tested we give input and observe if the output is as we expected otherwise we say that there is a failure.

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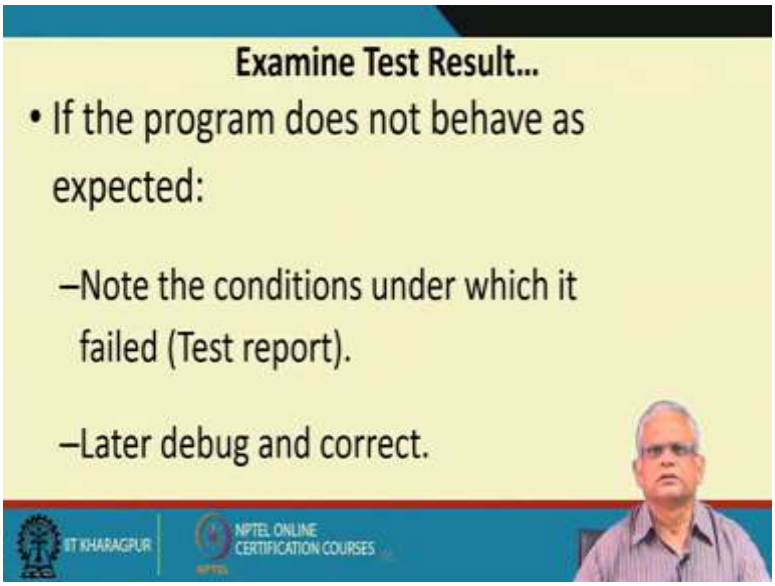

If the program does not behave as expected, we find what are the test data we gave and under what condition we gave the test data that is this software was at what state when we gave the test data. And, this we note it down the condition and the test data and this is called as the test report which contains all the conditions under which different data were given, the output produced and whether it was a failure or not.

And, these are taken up later by the developers. They reproduce the failure, debug and then debug and then correct the software.

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Out of all development activities, testing consumes the largest effort. Largest man power among all roles, companies invest heavily on testing, give much more effort to testing than design, coding or specification. And, of course, that means, that if you walk into a company you are more likely to meet a tester because number of testers are much more than other roles, which also implies that more job opportunities in testing because the companies need large number of testers.

For a typical software, 50 percent of the development effort is spent on testing, but then if we look at the time for which testing is carried out it is about 10 percent of the development time. If it is a 1 month project, then may be hardly 3 – 4 days are given for testing, the rest are for specification, design, coding and so on.

But, then 50 percent of the development effort is spent in 10 percent of the time how is it possible? It is possible because the testers can work parallely there is lot of parallelism in the testing activities. We can deploy a large number of testers to carry out testing at the same time, they test the different functionalities, different test data and so on.

On the other hand, in design we can have only few designers designing it. We cannot really put large number of designers hundreds of designers designing it. There is typically a couple of designers. Similarly, coding we cannot deploy thousands of coders. We need to assign at least some module to each developer, each coder and therefore, the parallelism in other development activities are less and that is why they take more time even though less effort is given. During testing significantly larger effort is done in a shorter time.

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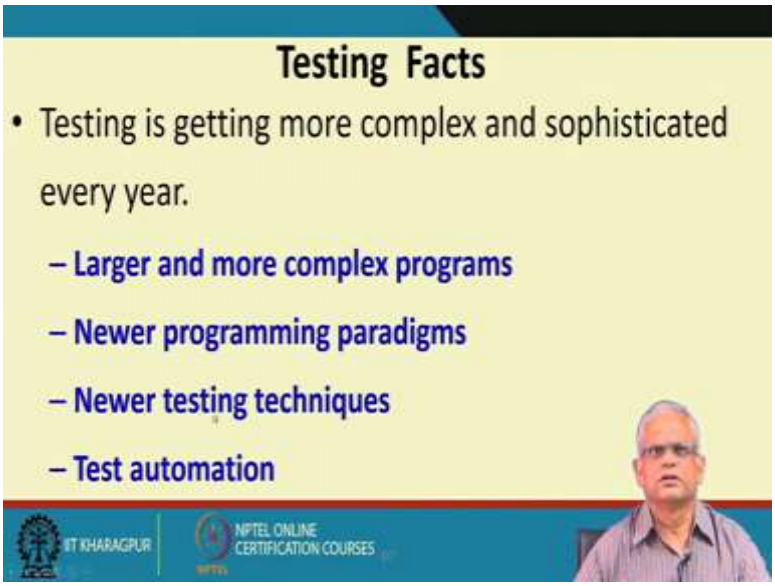

The testing techniques are evolving very rapidly that becoming more complex and sophisticated. The main reasons are larger and more complex programs; newer programming paradigms; test automation and for this and also new testing results. So, these all contribute to making testing a very challenging work.

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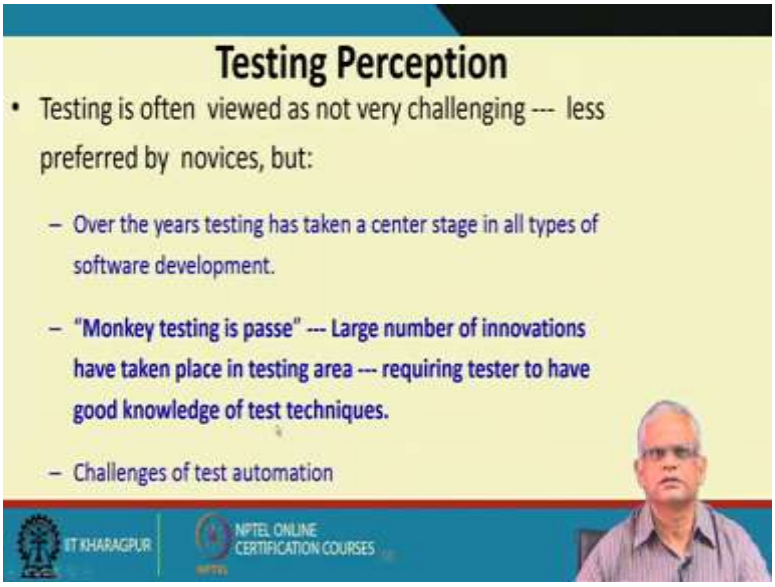

Testing is a actually very challenging even though the old time view was that testing is only a routine work, but now that has changed because testing has become very challenging. Earlier about 50 years back the testing was largely monkey testing where

only the random values were given, but now large number of innovations have taken place. Many testing techniques testing tools etcetera and the tester needs a good knowledge of these techniques.

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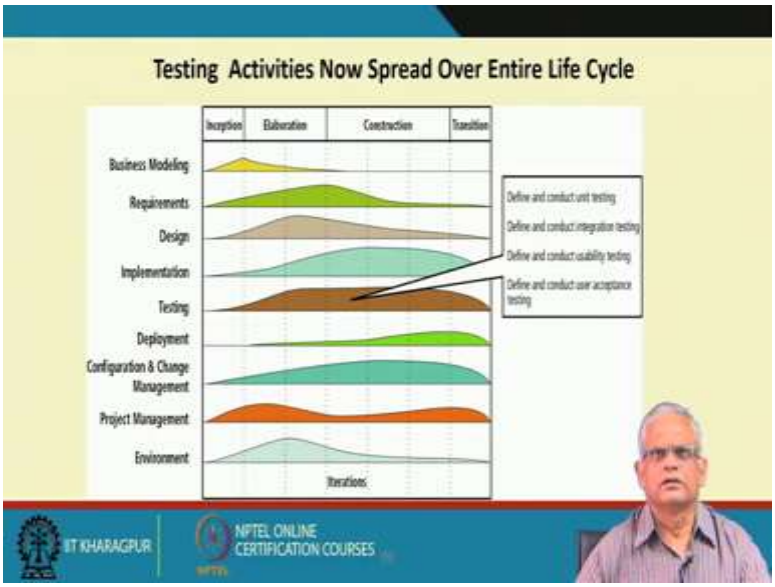

In reality, how long does a testing go on? As we can see on the unified process representation here the testing is actually carried out over the life cycle. At different time of the life cycle the test cases are defined, unit testing is conducted, integration testing defined, conducted, usability testing, system testing is defined and conducted.

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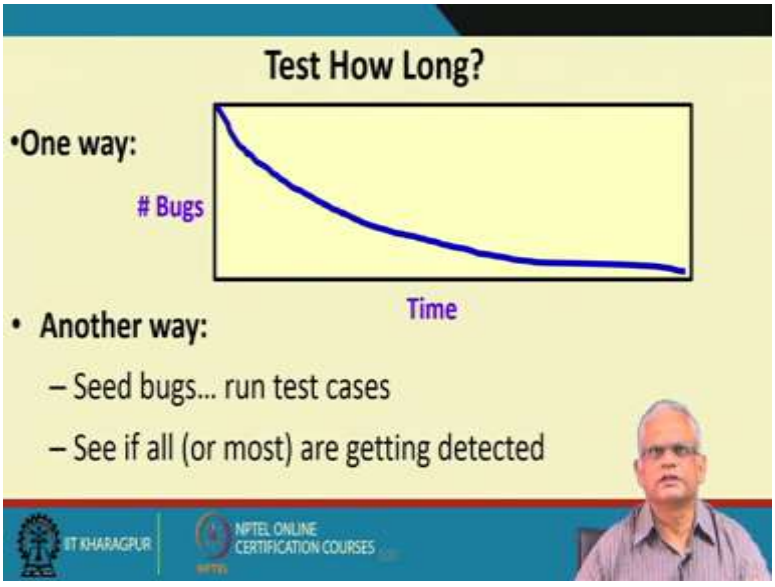

But, then one important question is to test how long, because as more and more test data defined bugs get detected. One way is that we observe the number of bugs that are getting detected over unit time and as testing progresses, we find that the number of bugs detected per day is decreases and then we have a situation where testing takes place for a day or a week and no bug is detected and that is the time when we may stop testing.

The other way is to seed bugs. The manger introduces bugs into the program without telling the developers and testers and then observes how many of the bugs that he had introduced are getting detected. If he finds that most of the bugs have got detected, then the manger knows that it is the time to stop testing.

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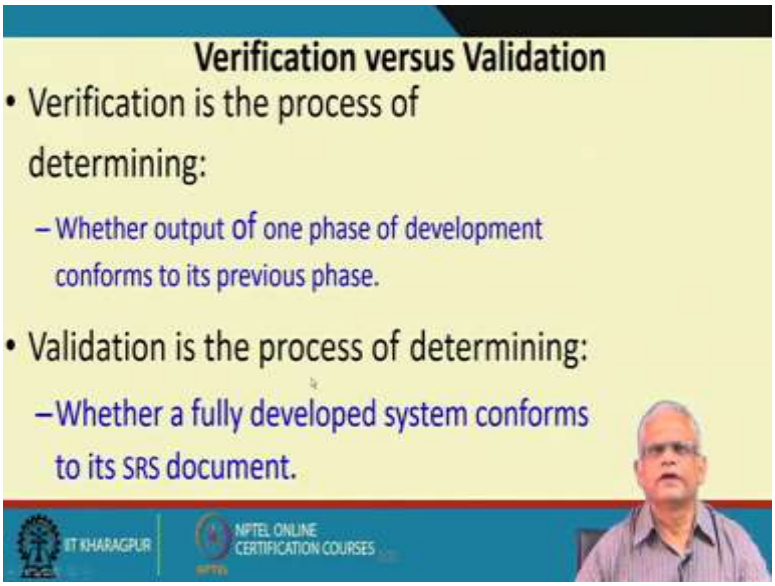

Another terminology that manger needs to be aware is the difference between verification and validation. In verification, we determine when whether output of one phase of development confirms to the previous phase. For example, if the design results are consistent with the specification whether all specification have been taken care properly.

Similarly, for coding whether all the design modules that were designed have been taken care properly and that is called as the verification whereas, validation is the process of determining whether a fully developed system confirms to its specification. So, that means, the validation is carried out on the completed system whereas, verification are done as the development proceeds to check whether the development is preceding

correctly, that is, one activity produces result that are consistent with the output produced by the previous activity. We are running out of time for this lecture and we will stop here.

Thank you.

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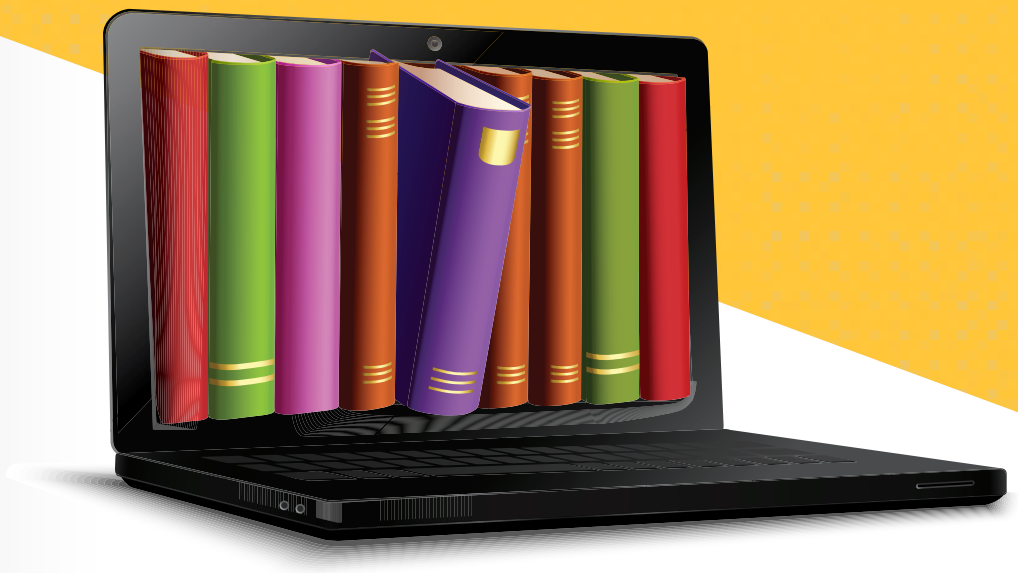

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